

Society of General Internal Medicine

23rd Annual Meeting
Boston, Massachusetts
May 4-6, 2000

ABSTRACTS

LEARNER/TEACHER CENTERED RESEARCH

A CROSS-SECTIONAL DESCRIPTIVE STUDY OF MENTORING RELATIONSHIPS FORMED BY MEDICAL STUDENTS. EM Aagaard, KE Hauer, Division of General Internal Medicine, University of California San Francisco, San Francisco, CA

PURPOSE: To describe the prevalence and characteristics of mentoring relationships among medical students and to determine medical student factors predictive of having a mentor.

METHODS: We surveyed 232 third and fourth year medical students attending the University of California, San Francisco during the 1998-1999 academic year. The survey included questions regarding student demographics and the presence and characteristics of mentoring relationships. Data were analyzed using descriptive statistics, Chi-square, Fisher's exact, and Mann-Whitney rank sum tests.

RESULTS: One hundred fifteen of 161 third year (71%) and 117/141 (83%) of fourth year students completed the survey, for an overall response rate of 77%. Although 96% of students rated having a mentor as important or very important, only 36% actually had mentors. Students most commonly met their mentors during inpatient clinical clerkships (21%), through research activities (21%), or by seeking a specific mentor based on similar interests (24%). Fourth year students were more likely to have mentors than third year students (44% vs. 28%, $p = 0.01$). Student age ($p = 0.4$) and gender ($p = 0.5$) did not predict having a mentor. African American and Latino students were 1.5 times more likely to have a mentor compared with students of other backgrounds (50% vs. 33%, $p = 0.06$). Students who performed research prior to medical school were 2.3 times more likely (39% vs. 17%, $p = 0.01$) and students who performed research during medical school were 2.9 times more likely (44% vs 15%, $p < 0.001$) than students who did not do research to have a mentor. Students who ranked themselves highly compared to the rest of the class ($p = 0.01$), and students with strong career interests in research ($p < 0.01$), academic medicine ($p < 0.01$), or primary care ($p = 0.03$) were all more likely to have a mentor. Students with a specialty interest in a surgical field ($p = 0.02$) and neurology ($p = 0.005$) were also more likely to have a mentor.

CONCLUSION: Only one third of medical students formed mentoring relationships during medical school even though nearly all felt that mentors are important. Students that found mentors tended to have well-defined career goals, perceive themselves as performing well during medical school, and perform research. These findings may highlight a group of students who are best able to develop longitudinal relationships with faculty. Future study is needed to evaluate mechanisms to promote mentoring relationships for students with different or less well-defined career aspirations.

CORRECTIONS HEALTH CARE: AN INNOVATIVE OPPORTUNITY FOR MEDICAL STUDENT EDUCATION. LJ Adams, L Sander, Division of General Internal Medicine and the Denver County Sheriff Medical Department, University of Colorado Health Sciences Center, Denver, CO

PURPOSE: A medical student educational collaboration has been developed between the University of Colorado School of Medicine and the Denver County Sheriff Medical Department. Educational content and logistical considerations in setting up such programs are described.

METHODS: With the increased emphasis on community-based teaching for medical students, the Internal Medicine Department sought additional primary care teaching sites. The School of Medicine utilizes primary care physicians in the required three-year Primary Care Curriculum that places students in primary care practices for one afternoon per week. This has required the identification of a very large number of community providers. Approximately 40% of the 390 primary care preceptors are internists. The internists at the Denver County Jail responded to this need and have precepted two students per year since fall of 1995. The Denver County Sheriff Medical Department serves approximately 2300 inmates, male

and female. A full range of medical problems are seen, similar to a typical internal medicine practice, e.g., hypertension, diabetes, acute and chronic infections, HIV/AIDS, headaches, musculoskeletal problems, hepatitis.

Subsequently, we developed a two-week elective in Correctional Health Care for fourth year medical students. The course objectives are: 1) Increase clinical skills in general internal medicine; 2) Obtain skills in dealing with manipulative or drug-seeking patients; 3) Gain knowledge about the interface between health care and the legal system; 4) Learn about issues in correctional health care- HIV, TB, and incarcerated juveniles; 5) Gain skills in assisting patients with significant social problems which can compound their medical problems; and 6) Learn about health care delivery in a specialized system.

The logistics of developing clinical rotations will be discussed from both the correctional institution and medical school perspective including faculty appointments, course development, overcoming negative attitudes, and security concerns.

RESULTS: Preceptor and course evaluations for both Primary Care and the Correctional Health elective are uniformly positive, as good or better than similar courses. The Correctional Health Care elective has proven popular with students pursuing primary care and psychiatric careers.

CONCLUSION: The corrections setting provides an excellent primary care teaching site while exposing students to the unique needs of a medically under-served population as well as introducing correctional health care as a career option.

FACTORS INFLUENCING INTERNAL MEDICINE PROGRAM DIRECTORS' DECISIONS ABOUT APPLICANTS. LJ Adams, S Brandenburg, M Blake, CT Lin, Division of General Internal Medicine, University of Colorado Health Sciences Center, Denver, CO

PURPOSE: We sought to evaluate what factors were most useful to residency program directors in making decisions about applicants.

METHODS: In conjunction with the Association of Program Directors in Internal Medicine (APDIM), we conducted a survey of all 407 program directors. A response rate of 79% was obtained via the APDIM list serve, direct e-mails, letters, and faxes. Programs self-described on three variables: type of program (university, community), size, and geographic location. They then rated the usefulness of Dean's letters, personal statements, transcripts, application form, USMLE scores, Chairman's letter, and actual interviews for making decisions about interview invitations and ranking applicants for the match. Respondents could choose between highly/moderately/ mildly/not useful. In addition, a text box for comments was associated with each group of questions.

RESULTS: The items that were rated highly/moderately useful for interview decisions were: USMLE Score (94%), Dean's letter (87%), Transcript and Application Form (85%), Chairman's Letter (79%) and Personal Statement (61%). Items that were rated highly/moderately useful for ranking decisions were: Applicant Interview (96%), USMLE Score (93%), Transcript (83%), Dean's Letter (82%), Chairman's Letter (77%), Application Form (74%), and Personal Statement (57%). For interview invitations, community programs were more likely to rate USMLE Scores as highly useful (59%) vs. University programs (41%). Similarly, for ranking decisions, community programs ranked USMLE Scores as highly useful (60%) vs. 40% for university programs. In preparing their match lists, University programs were more likely to rank transcripts as highly useful (54%) vs. community programs (46%). University programs also rated Chairman's letters as highly useful (19%) more often than community programs (13%).

CONCLUSION: USMLE scores are consistently important since they provide programs with objective comparisons between applicants. Community programs find them even more useful than university programs in both interview and ranking decisions. While Dean's letters are seen as useful, many programs noted that they invite candidates for interviews prior to their release. The considerable variability in quality of these letters was noted as was the redundancy between Dean's and Chairman's letters. While personal statements often reveal unique individual characteristics, they are seen as the least useful information. The interview is an opportunity to assess personal characteristics. Once applicants are invited, the interview is critical for determining suitability of candidates for a program.

LEARNER CENTERED CONTENT AND EVALUATION OF PSYCHOSOCIAL ROUNDS BY INTERNAL MEDICINE RESIDENTS. RS Adler, Division of General Medicine, Mt. Sinai School of Medicine, New York, NY

PURPOSE: A learner centered psychosocial program for internal medicine residents which focused on personal awareness was started in 1996. The hypothesis was that a learner centered program would be a more relevant patient specific learning experience and would also expose residents to all the topics covered in teacher centered curriculum driven psychosocial programs. This report describes the content and resident's evaluations of the program.

METHODS: All consecutive cases presented by internal medicine residents at psychosocial rounds from July 1996-June 1998 were analyzed for major themes. The themes were classified based on a published proposed core curriculum for physician personal awareness developed by the Working Group on Promoting Physician Personal Awareness of the American Academy on Physician and Patient. This curriculum addresses all aspects of patient centered interviewing and psychosocial medicine while emphasizing the development of humanistic values through personal awareness. To ensure that the cases presented were personally chosen by the residents, the only criteria used by residents in selecting their patient was that they had to have thought about the patient while they were at home. The presentations were made in groups of 6 and were scheduled separately for PGY 1's, 2's, and 3's during each resident's 8 wks/yr on ambulatory block time. Each resident met individually prior to their presentation with the study author who took notes during the individual and group sessions which were then analyzed for content. Only the major themes are reported here. All the residents evaluated their experience of the psychosocial program for relevance and learning experience using a 1 to 5 scale with 1 = poor and 5 = outstanding.

RESULTS: Of the 316 presentations that were analyzed, 124 were made by PGY 1's, 113 were made by PGY 2's, and 79 were made by PGY 3's. The major different themes presented were as follows: difficult patients 38% (n = 120), conflict between resident's and patient's core beliefs and attitudes 18% (n = 58), anger at patients 15% (n = 47), sociocultural influences 11% (n = 35), dying patients 9% (n = 30), family of origin issues 3% (n = 10), gender issues 3% (n = 9), mistakes 2% (n = 6). The program was highly regarded with mean ratings for relevance of 4.0 and learning experience of 3.9. There were no significant differences in the ratings between PGY 1's, 2's, and 3's.

CONCLUSION: In a learner centered psychosocial program, residents present patients which focus on all the major personal awareness curriculum themes. Residents are learning simultaneously about themselves and their patients. Evaluation results demonstrate a high level of program relevance and acceptance.

EFFECTIVENESS OF COMMUNICATION BETWEEN RESIDENTS AND GERIATRIC PATIENTS FROM A MULTICULTURAL BACKGROUND: A SURVEY OF RESIDENT'S ATTITUDES AND BEHAVIORS. M Ahmad, E Eckstrom, Dept of Medicine, Legacy Health System, Portland, OR

PURPOSE: As our population ages, primary care providers will be increasingly expected to provide optimal health care to the older population with a unique spectrum of health and social issues. Most residency training programs offer limited opportunities to learn and practice effective communication skills with geriatric patients. The scope of learning is even further restricted for multicultural communication pertaining to the elderly. The purpose of this study is to examine resident attitudes and behaviors toward communication of health issues with elderly patients, including those from the dominant culture and first generation immigrants.

METHODS: A 22-item questionnaire including both open and closed-ended questions was distributed to all 50 residents of the Legacy Internal Medicine Training Program. Descriptive statistics and paired t-tests were used for data analysis (SPSS 8.0).

RESULTS: 82% of residents completed the survey. 47% of respondents agreed that successful communication with elderly patients requires focused learning but thought that communication with geriatric patients follows the same basic principles regardless of the patient's cultural origin. Only 3% of residents believed that time spent communicating with elderly first generation immigrant patients is at the expense of somebody else's visit time. 47% strongly agreed that a patient's explanation of their illness is influenced of his/her culture, but only 10% of residents routinely ask for sociocultural data. 42% of residents ask about treatment expectations from patients from the dominant culture, compared to 35% if the patient is a first generation elderly patient ($p < 0.01$ for difference). More residents attempt effective communication with geriatric patients from dominant cultures compared to non-dominant cultures (65% vs. 50%, $p < 0.009$ for difference). 58% agreed or strongly agreed they had sufficient formal training to provide effective communication to elderly patients of the dominant culture and 12% agreed or strongly agreed they had sufficient training to provide effective communication to elderly patients of minority cultures ($p < 0.001$ for difference).

CONCLUSION: Residents agree there is a need for structured learning about effective communication with elderly patients across cultures. However, they also believe the health information they deliver transcends culture. This indicates that residents need to identify and work on cultural blind spots. Not asking about treatment expectations or limited attempts at effective communication across cultures suggests dialogue with our elderly patients may benefit from educational interventions targeted at specific communication skills deficits.

CONFERENCE ROOM VERSUS EXAM ROOM DISCUSSIONS IN THE CLINIC: RESULTS OF TWO RANDOMIZED CONTROLLED TRIALS. RJ Anderson, E Cyran, L Schilling, CT Lin, G Albertson, L Ware, J Steiner, University of Colorado, Denver, CO

PURPOSE: To delineate the optimal role for attending physicians (AP) and optimal site for AP-houseofficer (HO) interactions in a HO-based primary care clinic.

METHODS: Initially, 227 patient encounters were randomized to either exclusive HO-AP conference room discussion or to HO-AP conference room discussion plus AP seeing the patient (randomized controlled trial, RCT-1). Subsequently, 186 patient encounters were randomized to either exclusive HO-AP conference room discussion or to exclusive HO-AP exam room discussion with the patient present (RCT-2). Post-visit, questionnaires were obtained from patients (related to visit satisfaction) and from HO and AP (related to AP teaching, AP diagnostic and therapeutic contributions and HO comfort and autonomy).

RESULTS: Patient demographics and self-rated health were comparable in all study groups in both RCTs. Patient overall visit satisfaction, satisfaction with medical care and feeling of reassurance with seeing their doctor(s) as well as HO and AP assessment of AP contributions to teaching, diagnosis and therapy did not differ in RCT-1. In RCT-2, patients involved in exam room encounters significantly more often felt more comfortable with the discussion ($p < 0.001$) and would prefer to listen to the discussion in the future ($p < 0.001$) than those in the conference room group. In RCT-2, patient overall visit satisfaction and satisfaction with their medical care were comparable with exam room and conference room presentations. HO and AP assessment of AP contributions to teaching, diagnosis and treatment were nearly identical when conference room and exam room presentations were compared in RCT-2. APs felt that 83% of exam room encounters were beneficial to patient care. A small but significant ($p < 0.001$) per cent of HOs (10%) but not APs (0%) were made uncomfortable by exam room presentations and some HOs (11%) felt exam room presentations diminished their autonomy.

CONCLUSION: Patients and APs perceive that exam room presentations are beneficial and exam room presentations do not decrease patient rating of overall visit satisfaction. Also, AP contributions to teaching, diagnosis and therapy are maintained in the setting of exam room presentations. Some HOs perceive that exam room presentations cause discomfort and diminish autonomy.

INTERNS' VS. SENIORS' ATTITUDES TOWARDS MANAGED CARE AND UNDERSERVED POPULATIONS. TM Bailey, BC Williams, JE Jensen, JO Wooliscroft, Internal Medicine, University of Michigan; Ann Arbor VAMC, Ann Arbor, MI

PURPOSE: To test the hypothesis that incoming internal medicine interns have more favorable attitudes towards managed care and caring for underserved populations than senior residents.

METHODS: As part of a project to develop curricula in Medicaid Managed Care, we surveyed all incoming interns and graduating seniors at a large academic medical center in Family Medicine, Pediatrics, and Internal Medicine between February and June, 1999. The 34-item questionnaire was modified from a previously published instrument; item responses were 5-point Likert-like scales. Question domains included managed care, practice guidelines, capitation, Medicaid, success in a managed care environment, and the likelihood of serving underserved populations.

RESULTS: A total of 51 incoming interns and 60 graduating seniors completed questionnaires, for an overall response rate of 64%. There were no significant differences between intern and senior responses for 28 out of 34 items. For example, 58% of residents agreed (response "agreed" or "strongly agreed") that managed care emphasizes cost over quality. Most residents (76%) agreed that capitation encourages the withholding of tests, and expressed favorable attitudes towards practice guidelines (62%). 84% of residents stated that acquiring best evidence for the treatment of medical problems is important to success in a managed care environment.

Some differences between seniors and interns were observed. Seniors were somewhat more likely than interns to agree that practice guidelines improve clinical outcomes for patients (52% vs. 33%, overall $\chi^2 = .11$); that Medicaid reimbursement is the greatest barrier to participation in Medicaid managed care (72% vs. 45% of ($\chi^2 = .06$), and that assessing the health needs of a defined population is important (25% vs. 8%, $\chi^2 = .04$). Fewer seniors (67%) than interns (72%) feel it is likely they will serve the underserved ($\chi^2 = .04$).

Comparisons between departments were not possible due to limited sample size.

CONCLUSION: Recent inclusion of managed care curricula into medical schools does not appear to have resulted in differences among resident cohorts in attitudes towards managed care in most areas. Secular trends (affecting both groups) or residency training (paralleling changes in medical school training) may account for the lack of difference. Some managed care concepts (e.g., practice guidelines and evidence-based medicine) were relatively well accepted by residents, whereas others (e.g., caring for defined populations) were not, suggesting areas for curricular emphasis in residency training programs.

A COMPARISON OF PRECEPTORS' AND STUDENTS' RATINGS OF STUDENTS' SKILLS IN A LONGITUDINAL PRECEPTOR PROGRAM. CJ Baltimore, MM Belanger, A Perla, KM Mazar, University of Massachusetts Medical School, Worcester, MA

PURPOSE: To evaluate the correspondence between preceptors' ratings and medical students' self-ratings of student performance.

METHODS: First and second year students who participated in a longitudinal preceptor program and their preceptors rated student performance in several areas, including interviewing, initiative, physical exam skills and problem solving. Students also indicated their self-confidence and comfort level in these areas; preceptors rated whether the student appeared confident/comfortable. Self and preceptor ratings of performance and confidence/comfort level were calculated for each

student, and compared using paired t-tests and Pearson correlations; 38 pairs of ratings were analyzed for year one, 36 for year two.

RESULTS: Preceptors tended to rate students higher than students rated themselves. For year one, the mean of student self-ratings of skills was approximately three quarters of a standard deviation less than the mean of preceptors' ratings of the same skills; for year two students, student ratings were a full standard deviation less than preceptors' ratings. Both differences were statistically significant ($p < .000$). With respect to confidence/comfort in skills, student ratings were also lower than preceptors' ratings of apparent confidence/comfort. For year one, student ratings were approximately one half of a standard deviation lower, for year two, student ratings were more than one half a standard deviation lower. Again, mean differences for both years were statistically significant ($p < .01$). Correlations between student and preceptor ratings varied: for year one, the correlation between students' and preceptors' skills ratings was .33 ($p < .05$), while for year two, the correlation was .07 (not significant). The correlation between students' confidence ratings and preceptors' ratings of students' confidence/comfort was .41 ($p = .01$) for year one, and .33 ($p = .05$) for year two.

CONCLUSION: The finding that the students in this study tended to rate themselves lower than their preceptors has several possible explanations. Students' experience with self-assessment may be limited; their understanding of performance criteria may be incomplete; medical students may be critical of their own performance, preceptors' may tend to be lenient. In addition, preceptors may rate students based on a very small number of observations or no observations. While the current study does not allow us to determine the reason for the results found here, it does highlight the fact that it is important to look carefully at such ratings. Training of students and preceptors in the specific items, explicit definitions of performance criteria, and repeated opportunities for evaluation are recommended. In addition, frequent direct observations by preceptors are advised.

ATTITUDES OF INTERNAL MEDICINE RESIDENTS TOWARD OUTPATIENT DIABETES CARE AT A LARGE MUNICIPAL HOSPITAL. CS Barnes, JP Doyle, RP Hayes, DN Thompson, CB Cook, IM El-Kebbi, DC Ziemer, DL Gallina, CD Miller, WT Branch, Jr, LS Phillips, Medicine, Emory University; Center for Clinical Effectiveness, Grady Memorial Hospital; Education, Georgia State University, Atlanta, GA

PURPOSE: Although most patients with diabetes are managed in primary care sites, we have little understanding of the attitudes of their providers. Such information is particularly important to guide the training of resident physicians who will be the primary care practitioners of the future.

METHODS: To meet this need, we developed a 34-item questionnaire to assess 1) barriers to delivering care as perceived by internal medicine residents in an outpatient primary care clinic at Grady Hospital, that primarily serves urban African-American patients, and 2) thresholds for glucose control at which residents would start or intensify diabetes medications.

RESULTS: A total of 153/185 residents completed the survey (83% response rate); their mean age was 28 years; 29% were women; 35%, 37%, and 28% were in their first, second, or third year of training; and 26% indicated plans to practice in a primary care setting. There was no consensus that time or staffing were inadequate, or that patient psychosocial problems were not typical of other patient populations. However, the residents saw barriers in several areas: 1) System: 82% had problems because charts and data were unavailable; 2) Patients: although 77% felt that diabetes risk factors were similar to those in other populations, 62% said that patient literacy levels limited care, and 40% felt that patients did not view diabetes as a serious problem; 70% said that diabetes control could not be improved unless patients made lifestyle changes; 3) Disease: 87% felt that diabetes is not too complex to be managed in primary care sites, and 71% said their diabetes training was adequate; 4) Physician: 75% said they were not reluctant to be aggressive in diabetes management despite the presence of other medical problems. However, 88% were not sure that glucose levels measured during clinic visits could be used to guide therapy, and their stated thresholds for intensification of therapy varied widely: fasting glucose 139 ± 26 mg/dl and HbA1c $7.4 \pm 0.7\%$ ($M \pm SD$); 80% said they would appreciate feedback on their diabetes performance.

CONCLUSION: Barriers to outpatient diabetes management reported by internal medicine residents include both typical municipal hospital problems and probable sociocultural misperceptions. Although training was perceived as adequate, many residents do not understand how to use glucose levels to inform clinical decision-making. New approaches to clinical education will be needed to improve diabetes management.

REWARDING EXPERIENCES OF A PRIMARY CARE EXTERNSHIP. D Barnett, D Rudy, G Talente, C Griffith, J Wilson, University of Kentucky, Lexington, KY

Early clinical experiences are becoming a common method to expose students to medical practice at the beginning of their curriculum. What is unclear is what students find most or least rewarding in these early clinical experiences. Our early clinical experience was a primary care externship in which ninety first year medical students spent one week observing a volunteer community based primary care physician. After the externship each student completed a survey which included open-ended questions regarding their experiences. Responses were coded for prevalent themes. The most common themes in response to "my most rewarding experience" were interactions with patients (41.6%), observing the physician interact with patients (36.0%), interactions with physician (10.1%), applying knowledge (10.1%), and helping patients (4.5%). In response to "my least rewarding ex-

perience" the common themes included, downtime when not interacting with physician or patients (22.5%), difficult patient interactions (16.9%), only being able to observe the physician (14.6%), "nothing" (11.2%), poor interaction with physician (5.6%). Other responses included writing assignments, long distance commute, and aching feet. There were no correlations between student responses and their demographic data. We examined the associations between these responses and students' ratings of their level of understanding of primary care and time spent in various activities. Students that felt that observing the physician interact with patients was the most rewarding experience were more likely to feel that the externship improved their understanding of what a physician does ($r = 0.22$, $p < 0.05$). These students also tended to work with preceptors of a younger age ($r = 0.21$, $p < 0.05$). On the other hand, students that felt that the most rewarding experience was patient interactions were less likely to feel that they had a good understanding of what being a physician is all about ($p < 0.05$). Students who felt that the most rewarding aspect of the externship was helping patients were more likely to have spent a significant amount of time observing the physician at other sites other than the office and hospital i.e., free outreach clinics ($r = 0.39$, $p < 0.0001$). These students were also more likely to want to go back to the preceptors' office during clinical years ($r = 0.20$, $p < 0.05$). The preceptors of these students tended to be older and women ($r = 0.30$, $p < 0.05$). Understanding what students view as most and least rewarding experiences will help structure externships so that future experiences are positive. Positive interactions with physicians and patients may give students a better understanding of primary care. If the goal of a preclinical externship is to provide a better understanding of what a physician does, students may benefit more by spending most of their time observing the physician rather than interacting with the patients on their own.

ASSOCIATION OF INTERN WORKLOAD WITH LENGTH OF STAY AND WITH PATIENT SATISFACTION. MJ Bittner, EC Rich, RL Recker, PD Turner, RL Rospond, MW Lubeley, Medical Department, VA Medical Center; Department of Internal Medicine; Center for Practice Improvement and Outcomes Research, Creighton University, Omaha, NE; College of Pharmacy, Drake University, Des Moines, IA

PURPOSE: We analyzed data from July 1998-June 1999 at the Omaha VA Medical Center D see if intern workload affected resource utilization and patient satisfaction. **METHODS:** We studied patients of Creighton medical interns assigned to non-intensive care unit inpatient floors. We used length of stay (LOS) as a measure of resource utilization. We studied satisfaction with the American Board of Internal Medicine's Patient Satisfaction Questionnaire (PSQ). This ten-item instrument assesses physicians' interpersonal skills. Each item has five responses, from 1 (excellent) to 5 (poor). We analyzed responses from patients who completed the questionnaire in a face-to-face interview the morning after admission. For patients admitted after 5 pm, the interview occurred the second day after admission. Using multiple linear regression, with natural logarithm of LOS as the dependent variable, we assessed the effect of intern workload (census size, number of admissions during the 24-hour admitting period) and controlled for severity of illness (Fiscal Year 99 DRG weight) and intern factors (gender, experience, faculty rating of humanistic qualities, in-training examination score). We performed a similar regression analysis with PSQ as the dependent variable.

RESULTS: Among 434 patients, LOS was 6.1 ± 7.3 days (mean $\pm$ standard deviation) and skewed to the right (skewness 4.0). Satisfaction scores were 1.9 ± 0.8 , on a scale where 1 was best and 5 was worst. Intern census was 4.7 ± 1.9 and admissions 2.0 ± 1.2 . In the LOS model, adjusted R^2 was 0.14 ($p < 0.001$). DRG weight was associated with longer stay ($p < 0.001$), but male intern gender was associated with shorter stay (mean 7.2 days for women, 5.5 for men, $p = 0.048$). In the satisfaction model, adjusted R^2 was 0.015 ($p = 0.061$). Male intern gender ($p = 0.015$) and a larger census ($p = 0.038$) were associated with greater patient satisfaction, but more admissions ($p = 0.023$) was associated with less satisfaction. There was a trend for more satisfaction among patients of interns with better ratings of humanistic qualities ($p = 0.054$).

CONCLUSION: In a setting with controlled intern workload, workload variables were not significant in explaining resource utilization. A model of satisfaction incorporating workload variables explained only 1% of variance in satisfaction. Factors associated with satisfaction included male intern gender, fewer admissions, and larger censuses. There was a trend for more satisfaction with interns who had better ratings of humanistic qualities.

TIME SPENT BY INTERNS WITH HOSPITALIZED PATIENTS: ASSOCIATION WITH PHYSICIAN CHARACTERISTICS, PATIENT LENGTH OF STAY, AND PATIENT SATISFACTION. MJ Bittner, EC Rich, RL Recker, PD Turner, RL Rospond, MW Lubeley, Medical Department, VA Medical Center; Department of Internal Medicine; Center for Practice Improvement and Outcomes Research, Creighton University, Omaha, NE; College of Pharmacy, Drake University, Des Moines, IA

PURPOSE: We asked whether internal medicine interns who spend more time with their patients are more efficient in using resources and whether they have higher patient satisfaction ratings.

METHODS: We studied patients of Creighton medical interns assigned to non-intensive care unit inpatient floors at the Omaha VA Medical Center July 6, 1998 through June 30, 1999. We used length of stay (LOS) as a measure of resource utilization. We studied satisfaction with the American Board of Internal Medicine's Patient Satisfaction Questionnaire (PSQ). We analyzed responses from patients who completed the questionnaire in a face-to-face interview shortly after admission. For a week while awake and in the hospital, each intern carried a pager that

went off, at random, an average of 3.2 times an hour. Whenever the pager went off, the intern made note of the person with whom he or she was in contact. We also obtained DRG weights and information on these intern factors: census size, number of admissions during the 24-hour admitting period, previous postgraduate training, faculty ratings of humanistic qualities, and in-training examination scores.

RESULTS: Of the 25 interns, 64% were men; 32% were in the categorical program, 16% in medicine-pediatrics, and 52% in a preliminary year. Four reported previous postgraduate training. Preliminary interns spent less time with patients (5.3% vs. 16.0%, $p = 0.009$). The fraction of time spent with patients was higher with more experience before starting the internship ($r = 0.67$, $p < 0.001$) and better examination scores ($r = 0.42$, $p = 0.036$). Among the 90 patients with PSQ data whose interns carried the workload pager the day they were admitted, the mean PSQ was 1.80 (s.d. 0.67). Their mean LOS was 7.63 days (s.d. 10.0). Interns caring for them had a mean of 2.5 admissions (s.d. 1.5) and a census of 5.1 (s.d. 2.1). The mean proportion of contacts with patients was 11.6%. LOS was related to DRG weight (Pearson $r = 0.22$, $p = 0.37$) but there was no significant correlation with admissions, census, or proportion of contacts with patients. Neither was there a significant correlation of PSQ with those three variables.

CONCLUSION: As in previous studies, internal medicine interns spent less than 15% of their time with patients. Preliminary year interns spent even less time with patients. Time spent with patients was not correlated with patient satisfaction or length of stay. Nor were these outcomes related to intern census or admissions.

DETERMINING RESIDENT NEEDS FOR OPTIMAL LEARNING IN CONTINUITY CLINIC ENVIRONMENTS. JL Bowen, SS Desai, RA Harrison, Medicine, OHSU, Portland, OR

PURPOSE: Research in the area of resident continuity clinic experiences has focused on resident satisfaction. Satisfaction instruments are often constructed without resident input and may not ask questions that explore the residents' underlying needs to enhance learning. Needs assessment instruments ask questions of residents that probe for key underlying issues important to residents. This study is designed to assess residents' needs for optimal learning in continuity clinic settings using a critical incident technique.

METHODS: We convened five focus groups of internal medicine residents. All 29 OHSU continuity residents were invited to participate in this study on a voluntary basis. Consent was obtained before audio-recording each confidential session. Each session, facilitated by the investigators, consisted of two to six resident physician volunteers who were asked to recall and describe a memorable "good" day and "bad" day in clinic. The audiotapes were transcribed, and independently analyzed and coded by each investigator. The coded transcripts were then compared and discussed, generating a thematic coding schema. Disagreements were discussed and resolved by consensus. OHSU IRB approved this study.

RESULTS: Sixty-eight percent of eligible participants completed the study. Seven major themes were identified. Although residents addressed teaching issues (e.g., preceptor style, availability, role modeling) and learning issues (e.g., autonomy, patient-centered learning), the themes of clinic efficiency, workload, and physical space often dominated the narratives. Problems with teamwork and communication were emphasized, especially the importance of teamwork and consistent pairing with support staff. Residents also described the impact of both positive (e.g., self-confidence, control, predictability, empowerment) and negative (e.g., intimidation, isolation, vulnerability, guilt) feelings on their learning experiences.

CONCLUSION: The critical incident technique is an effective method for uncovering the conscious and unconscious needs of residents in continuity clinic. Clinic inefficiency distracts the residents' ability to focus on their education. In order to optimize residents' learning experiences in their clinics, clinic process issues and the affective domain of resident education needs as much attention as typical internal medicine curricular issues. Information gathered from our qualitative study can be used to create quantitative assessment tools that better reflect the needs of residents in continuity clinics.

NATIONAL SURVEY OF INTERNAL MEDICINE RESIDENCY PROGRAM DIRECTORS OF THEIR FIRST YEAR EXPERIENCE WITH THE ELECTRONIC RESIDENCY APPLICATION SERVICE. S Brandenburg, L Adams, CT Lin, M Blake, M Lemenger, Division of General Internal Medicine, University of Colorado Health Sciences Center, Denver, CO

PURPOSE: Determine the impact of the Electronic Residency Application Service (ERAS) on Internal Medicine Residency programs, specifically on number of applicants, screening, interviews, ranking, and quality of final match.

METHODS: A Web-based survey was developed based on consultation with Association of Program Directors in Internal Medicine (APDIM) and ERAS officials and a review of the literature. The survey was designed as a web page to allow ease of response and data collection. The survey consisted of descriptive questions about the program (geographic region, size of program, and type of program-university or community), questions that could be answered on a Likert scale, and open fields that invited individualized comments. It was then sent to program directors via the APDIM list serve. In order to increase the response rate, this was followed up with personalized e-mails, letters, and faxes that were sent to non-responders. The quantitative data was collected in an Access database and is being analyzed using univariate and multivariate statistics.

RESULTS: The final response rate was 81 percent. ERAS was well received with 86 percent reporting they found overall screening of applicants easier. The results show that the overall number of applicants varied greatly (49 percent reported more, 33 percent reported less). There was a wide distribution as well regarding the

number of IMG applicants, but there was a tendency to have fewer IMG applicants (48 percent reported less). Most programs interviewed the same number of applicants, but 35 percent interviewed more. Most programs also ranked the same number of applicants, with 33 percent ranking more, and 13 percent ranking less. The quality of final match was rated the same as previous years by 47 percent, better by 38 percent, and worse by 15 percent. Qualitative comments reveal a high level of satisfaction with ERAS, specifically the efficiency of the process and the ability to communicate with applicants by e-mail. Specific recommendations for improvement included: more versatile filters, improved readability of scanned data, and increased use of shortcut keys to facilitate review of applications.

CONCLUSION: The transition to ERAS was successfully accomplished in internal medicine. However, lessons learned will be useful for both continuously improving the ERAS process and for guiding new specialties as they make the transition to ERAS.

DOES ADDING A DESIGNATED TEACHING ATTENDING FOR STUDENTS IMPROVE THE INPATIENT LEARNING EXPERIENCE? RC Brooks, M McNeil, L Marts, MM Brooks, D Macpherson, University of Pittsburgh, Pittsburgh, PA

PURPOSE: Teaching to learners of varying levels in traditional ward attending rounds is a challenging task that runs the risk of satisfying no one. We hypothesized that adding teaching rounds dedicated to students would improve (1) the perceived quality of the educational experience for students and housestaff, and (2) teacher and learner satisfaction, without negatively impacting ward team dynamics.

METHODS: Student Teaching Attending (STA) sessions were added to traditional Ward Attending (WA) rounds during the third year Internal Medicine clerkship for the 1998-1999 academic year. STA rounds replaced a didactic lecture series and were similar in organization to traditional ward rounds except that only students were present. Students, housestaff, and WA's were surveyed prior to the introduction of STA sessions to document baseline attitudes. These same groups were surveyed again, along with STA's, during the 1998-1999 academic year.

RESULTS: Baseline survey response rates were: students 25 (18%), housestaff 11 (10%), WA's 33 (48%). For the 1998-1999 academic year, response rates were: students 121 (88%), housestaff 81 (56%), WA's 45 (46%), STA's 18 (82%). After the addition of STA rounds, students felt more strongly that they achieved their educational goals during the rotation (38% strongly agree in 1998-1999 vs 16% baseline, $p = .033$), while housestaff opinions did not change significantly ($p = .47$). Students who experienced STA rounds felt more strongly that the rotation increased their interest in a career in Internal Medicine (50% strongly agree vs 24%, $p = .016$). We had anticipated a shift in the focus of ward rounds towards housestaff and away from students. Contrary to expectation, students, housestaff, and WA's perceived the success of ward rounds in meeting individual learner needs as similar between the two years. WA's felt more strongly that they enjoyed teaching rounds after STA sessions were added (89% strongly agree vs 67%, $p = .016$), while enjoyment was less improved for students ($p = .071$) and stable for housestaff ($p = .50$). There was no measured impact on perceived team dynamics.

CONCLUSION: The addition of a supplementary teaching attending for students during the Internal Medicine Clerkship improved student and teacher satisfaction with the clerkship and enhanced career interest in Medicine without impacting the quality of ward attending rounds or team dynamics.

USING PRACTICE GUIDELINES TO TEACH DISEASE MANAGEMENT SKILLS TO PRECLINICAL STUDENTS. AS Brown, DS Anderson, HM Szerlip, Community Health Sciences, Tulane University School of Public Health; Medicine, Tulane University of School of Medicine, New Orleans, LA

PURPOSE: This study was designed to ascertain the feasibility of introducing second year medical students to clinical practice guidelines and to determine if they could use these guidelines to manage a simulated patient (SP) with newly diagnosed Type 2 diabetes mellitus.

METHODS: Half the second year class was divided into 18 groups of 4 students each and assigned an SP with hyperglycemia. The other half of the class served as controls. Both groups attended lectures and small group on diabetes in their pathophysiology course. In addition to the hyperglycemia, the SPs gave a history of smoking, were overweight, had elevated BP's, hypercholesterolemia and proteinuria. The students were given ADA guidelines as well as several well-done clinical studies on the management of diabetes. The students' task was to manage the SPs appropriately. BP's and laboratory parameters changed depending on therapeutic interventions. At the end of the semester, the SP's charts were audited to determine what interventions were instituted. All second year students were examined using an OSCE, which included a case of a young type 1 diabetic who required counseling. The performance on this station was compared between the control and experimental groups.

RESULTS: All groups obtained glycated Hgb, fasting lipids, urine for albumin and ophthalmology consults. Podiatric exams were documented on all charts. All groups initiated ACE inhibitors. Hyperglycemia was treated with a sulfonylurea by 54%, metformin (24%), both (12%) or acarbose (6%). ASA was prescribed by 72% of the students. All groups counseled the SPs on smoking cessation and an appropriate diet. On the diabetes OSCE station the students who participated in the study scored significantly better than the students who did not participate (66% vs 59% $p < 0.001$).

CONCLUSION: Second year students with minimal faculty input can successfully use practice guidelines to manage patients with diabetes. The knowledge ob-

tained from this innovative teaching methodology can be transferred to similar but non-identical situations. This problem based approach may be an ideal way to teach disease management. In addition, it is hoped that introducing practice guidelines to unbiased preclinical students will allow future clinicians to be more accepting of clinical guidelines.

THE ODYSSEY OF MENTORING IN INTERNAL MEDICINE: A NEEDS ASSESSMENT OF INTERNAL MEDICINE RESIDENTS. MM Budev, DG Litaker, HL Thacker, M Hewson, General Internal Medicine; Education Division Office, The Cleveland Clinic Foundation, Cleveland, OH

PURPOSE: As Odysseus entrusted his son to his friend/teacher Mentor, modern day academic medicine informally entrusts the guidance of young physicians to the tutelage of experienced educators. In the past, efforts at describing factors of mentorships have been poorly defined and lacking empirical justification. This needs assessment is aimed to identify: 1) the perceptions of internal medicine residents regarding mentoring, 2) their overall mentoring experience 3) possible impacting factors, and 4) their willingness to participate in a formal mentoring program.

METHODS: A cross sectional survey was conducted of all internal medicine residents at a tertiary care institution. The respondents voluntarily completed a 29-item questionnaire asking for perceptions on various aspects of mentoring. The responses were scored using a 5 point Likert scale (1 = not important to 5 = important). Descriptive analyses, using frequency and means were performed.

RESULTS: 96 of 133 surveys were returned (72%). Respondents consisted of 74% (n = 71) males with 47% PGY-1, 21% PGY-2, and 30% PGY-3, and 2% PGY-4 (med/peds). A majority (75%) of residents felt a mentoring relationship had been important to their careers. 60% residents indicated never having a mentoring relationship, although 57% of these individuals were only into the first 4 months of training. The most important limiting factor cited was the absence of a formal mentoring program (X = 4.8). Paucity of staff interest also contributed to the perception of scarce mentor availability (X = 4.17). The majority of respondents (91%) expressed interest in having a formal mentoring program. A majority of respondents believed that mentoring impacts careers, and that an individual has to be mentored to become a mentor (88% respectively). The most important criteria for mentor selection were: being a role model (X = 4.25); high professional achievements (X = 3.48); serving as a networking source (X = 3.40); and high academic rank (X = 3.29). The least important criterion was mentor gender (X = 1.95). In fact, the data further supported this showing equal numbers of male and female residents had selected female mentors.

CONCLUSION: The current mentoring environment has changed with the addition of fellows who serve as liaisons between experienced attendings and residents, and an emphasis on service rather than teaching. Our data reveals a strong resident desire for a formalized mentoring program. This could overcome some of the constraining factors in initiating and nurturing mentoring relationships. The data also indicates that gender is not a limiting factor. We believe formal mentoring will not only benefit the mentee and the mentor, but academic medical institutions as well.

RESIDENT SURVEY ON DOMESTIC VIOLENCE. MM Budev, D Abood, C Henry, General Internal Medicine, The Cleveland Clinic Foundation, Cleveland, OH

PURPOSE: To determine the knowledge base, attitudes toward, and experience with domestic violence among Internal Medicine (IM) and Medicine/Pediatric (Med/Peds) residents at the Cleveland Clinic Foundation. Other secondary objectives included determining if educational predictors, personal experience contribute to making residents more likely to suspect domestic violence (DV) in their longitudinal clinics.

METHODS: A survey consisting of twenty-eight questions was distributed over a four-week period to members of the Cleveland Clinic Foundation's IM and Med/Peds residency program. The Chi-Square test, Fisher's Exact test, Mantel-Haenszel Chi-Square test, and Wilcoxon rank sum test were used where appropriate, using version 6.12 of SAS.

RESULTS: Approximately 68 out of 121 residents completed the survey, yielding a response rate of 56%. Distribution of residents responding to the survey included 43 male (63%) and 25 female (37%) residents. Of the respondents, 32% were PGY-1, 24% were PGY-2, 41% were PGY-3, and 3% were PGY-4 Med/Peds residents. Ninety-one respondents felt that DV was an important issue to address in residency training. Only 27% of respondents recalled formal DV training in their residency. In fact, 63% of respondents felt that routine DV screening should be implemented in their outpatient history and physical exams. Yet, only 35% of residents suspected abuse in their longitudinal clinic patient population. Of the residents who suspected abuse, 88% questioned the patient, 48% felt comfortable with questioning the patient, 30% felt their training had prepared them for the discussion, and 34.8% were able to make the appropriate referrals. When those residents who reported having received DV training during their residency were looked at independently, they were more likely to be able to offer the appropriate referrals to victims of DV (p = 0.02). They may also have been more likely to suspect DV in their clinics, although this association did not reach statistical significance (p = 0.054). There was no significant association between gender, age, level of training, country of origin, medical school location, or having a friend or family member who was a DV victim, and suspecting DV in the longitudinal clinic. Even those residents who had previous training in DV issues were not more likely to answer more than three questions correctly in the general knowledge section.

CONCLUSION: Despite the heightened awareness about DV in the healthcare community, surprisingly few residents recalled formal education on the topic during their training program. Those who did recall any formal training may have been more likely to suspect DV in their longitudinal clinic patients and were more likely to be able to offer victims of DV the proper referrals. Further research is needed to determine the most effective clinical training strategy in this area.

MEMORABLE CASES OF PHYSICAL EXAMINATION "FINDS" AND "MISSES": PERCEPTIONS OF INTERNAL MEDICINE RESIDENTS. JB Bundrick, JTC Li, Internal Medicine, Mayo Clinic, Rochester, MN

PURPOSE: To assess and characterize the perceptions of internal medicine residents regarding their most memorable "finds" and "misses" on physical examination of their patients.

METHODS: In May 1999, forty-one first year residents in an internal medicine training program at the Mayo Clinic participated in an objective structured clinical examination (OSCE). Immediately afterward, the residents were asked to complete a brief, anonymous questionnaire which surveyed their perceptions regarding the relative importance of clinical exam skills and asked them to record their most memorable physical examination "find" and "miss", how they felt about it, and what effect it had on the overall care of the patient.

RESULTS: All residents completed at least a portion of the survey. When asked to rate their physical examination skills, 56% felt that they needed some degree of improvement. Thirteen residents (32%) believed that it was possible to complete the residency program with *major deficiencies* in physical exam skills and yet still maintain a reasonable degree of overall clinical competence. Twenty-four residents recorded their impressions of a memorable exam "find", while only eleven of this group could recall an example of a notable "miss". While the discovery of a positive finding was judged to have helped expedite the diagnosis and management of the case 87.5% (21/24) of the time, a corresponding delay in diagnosis/appropriate therapy with a missed finding could be recalled in only 18% (2/11) of the cases.

CONCLUSION: Most internal medicine residents at the end of their first year of training seem to view the physical examination as a significant contributor to their overall clinical competence and recognize a need for self-improvement in this area, but a significant minority do not seem to share this perception. Although a considerable number of residents can easily recall a case in which they discovered an important physical exam finding which seemed to expedite patient care, only half as many could remember a single case of a *missed* finding, and these were generally judged to have had no detrimental impact on the management of the case. While the phenomenon of psychologic denial may play a role in these perceptions, it is also quite possible that they simply reflect the recognition of multiple diagnostic "cues" in the majority of cases seen by the internist (such that missing one of them will not usually cause harm). At the intern level, there is also possibly an awareness that missed findings will be shortly thereafter detected by others in the "supervisory safety net". These potential explanations remain to be explored, as does the intriguing possibility that perhaps these perceptions may be countered by an educational emphasis upon critical findings and patient scenarios which are not subject to these criteria.

DO EXCELLENT TEACHERS MAKE EXCELLENT ROLE MODELS? E Caiola, A Hull, DG Litaker, M Hewson, L Copeland, Department of General Internal Medicine, The Cleveland Clinic Foundation, Cleveland, OH

PURPOSE: The availability of excellent physician role models (RMs) benefits many aspects of medical education. Attributes of RMs are multifaceted, but are strongly related to teaching skills. Few studies have explored the relationship between routinely collected resident evaluations of attending-physician (AP) teaching and identification as a RM. We sought to determine if AP teaching evaluation scores distinguished resident-identified RMs from their colleagues and to further define specific teaching characteristics most closely associated with status as a RM.

METHODS: The 118 Medicine housestaff at our institution were anonymously surveyed to identify their current RMs in Internal Medicine. Our validated Teaching Effectiveness Instrument (TEI) database was used as the source for the teaching evaluation scores of both resident-identified RMs and other attending-physicians (APs). The database contains teaching evaluations of 215 Internal Medicine APs (mean 10.6 evaluations/AP) and provides a composite TEI score as well as scores for its 15 individual items. Primary domains of the TEI include Learning Environment (LE), Teaching Strategies (TS) and Teaching Medical Knowledge and Skills (TMK). At the end of each rotation, residents rate APs on a 5-point Likert scale (5 = excellent teacher, 1 = poor teacher) for each item, generating an overall mean score. Mean composite and domain scores of RMs were compared to other APs using Student's t-test. Individual items were compared between RMs and other APs using ANOVA.

RESULTS: Seventy-three (62%) residents returned the survey. Ninety APs (42%) were identified at least once as a RM. The overall mean evaluation score of RMs was 4.34 compared to 4.05 for other APs (mean difference 0.29, p < .001). RMs scored higher than other APs in each category, with significant mean difference in the LE (0.31), TS (0.30), and TMK (0.27) domains (p < .001 for each). The individual items with the greatest mean differences included "Organizes time to allow for both teaching and care giving" (0.54), "Asks questions that promote learning" (0.49), and "Gives clear explanations" (0.47) (p < .001, for each item respectively). Interestingly, the item with the smallest mean difference between the two groups was "Observes and coaches my clinical skills" (0.32, p < .04).

CONCLUSION: This study suggests that routinely collected teaching evaluation scores can differentiate RMs from other APs. Excellent teachers are also identified as excellent role models by behaviors and skills that are learnable and modifiable. These findings suggest specific competencies to improve overall effectiveness of APs and are potentially important for faculty development and promotion.

DEFINING THE "HIDDEN CURRICULUM": DOES MEDICAL STUDENTS' ETHICAL EXPERIENCE ON CLINICAL ROTATIONS VARY BY SPECIALTY? SL Clever, KA Edwards, C Feudtner, CH Braddock, III, Internal Medicine; Medical History and Ethics; Pediatrics and RWJ Clinical Scholars, University of Washington; Center of Excellence in Health Services Research and Development, VA Puget Sound Health Care System, Seattle, WA

PURPOSE: To determine whether medical students' willingness to challenge team members regarding ethical issues varies by specialty.

METHODS: We surveyed all medical students attending an evening informational seminar about applying for residencies. The questionnaires asked the students about instances of perceived unethical patient treatment they had seen or committed; how comfortable they felt challenging members of their teams in different specialties about ethical issues; what attributes of the team contributed to their discomfort; and how they prefer that ethical issues be addressed. We used a logistic regression to examine the relationship between their level of comfort raising perceived ethical concerns and specialty rotation.

RESULTS: One hundred and three students attended the meeting; 76 surveys were returned (74%). Fifty-three percent of respondents were women. Compared to medicine, pediatrics, family medicine, psychiatry, ER and rehabilitation rotations, students were significantly less willing to challenge ethical issues on their surgery and OB/GYN rotations ($p < 0.001$ in both cases), with 48% and 24% rating themselves as either "uncomfortable" or "very uncomfortable" on those clerkships respectively. Of the 29 examples offered by the students of unethical behavior they had seen or committed, 41% occurred on the surgery rotation. Of those who ranked reasons for their unwillingness to confront team members, 82% cited difficult personalities and 79% being too low on the hierarchy as "important" or "very important" causes. Sixty percent of students wished to have the team talk together to resolve these issues; 47% would have liked the attending to address these issues.

CONCLUSION: The goal of ethics education is to train physicians who can recognize, analyze and resolve ethical issues. The last of these skills may require physicians to speak out against the prevailing culture within their institution or team. The results of this survey suggest that at the University of Washington there are specialty rotations on which students are frequently exposed to what they perceive as unethical patient care. Distressingly, it is on these same rotations that they feel most uncomfortable challenging other team members about ethical issues. Many students would like a team-based, attending-led intervention to address these issues. Developing interventions with the specialty departments that were identified as problematic may help improve the "ethical milieu" on those rotations.

QUALITY: HOW WELL ARE INTERNS DOING WITH OUTPATIENT CARE? J Cohen-Kogan, EE Reynolds, KJ Kovath, JA Shea, Medicine, University of Pennsylvania Health System, Philadelphia, PA

PURPOSE: Medical record auditing has been used to evaluate quality of care and has been recommended by the ABIM for resident evaluation. This study's purpose was to collect baseline information about medicine interns' adherence to national guidelines regarding preventive health and management of common outpatient medical conditions, and to examine documentation differences between male and female interns, and male and female patients.

METHODS: Charts of 497 general medicine patients seen 7/1/98-2/15/99 by 44 medicine interns at one of five outpatient practice sites were randomly selected (12 charts per intern) for retrospective review. Adherence to recommendations for screening, immunizations, counseling, and management of hypertension, diabetes, asthma, and depression was categorized as high (>70%), intermediate (40-69%) or low (<40%). Differences in performance by intern and patient gender were examined with Student t-tests.

RESULTS: Mean patient age was 48.5 years, 31.2% were male, and the mean number of visits per patient was 2.4. Forty-three percent of patients had hypertension ($n = 212$), 18% had diabetes ($n = 91$), 12% had asthma ($n = 58$), and 11% had depression ($n = 53$). High compliance screening items included documentation of weight (95%), blood pressure (99%), tobacco use (78%), alcohol use (74%), and mammography (71%); intermediate compliance items included cervical cancer (64%), colorectal cancer (53%) and lipid screening (62%). Recommendations for immunizations were moderate for pneumovax (51%) and influenza (54%) but low for tetanus (19%). Except for counseling about smoking cessation (59%), documentation was low for all counseling behaviors including exercise (27%), hormone replacement (24%), and sexually transmitted diseases (28%). Compliance was generally low for recommendations related to management of specific conditions. Notable exceptions were that hypertensive patients were asked about tobacco use (73%), and adherence was intermediate to high for many items related to diabetic management (screening lipids, ACE-I use, foot exam). Intern gender was related to recommendations concerning women's health; female interns were more likely to counsel women about hormone replacement ($p = .001$), and screen for cervical ($p = .038$) and breast cancer ($p = .017$). There were few differences related to patient gender for items applicable to both women and men.

CONCLUSION: There is variability in intern adherence to national recommendations regarding screening, counseling and immunizations. Future efforts could examine within and between intern variability, explore the use of profiles as an educational feedback tool for residents, and describe the effect of training on profile performance.

PHYSICAL DIAGNOSIS MORNING REPORT. J Connaire, A Ishani, K Baum, K Worzala, D Berg, University of Minnesota

BACKGROUND: The ability to perform and interpret the techniques of physical diagnosis is pivotal to the efficient and effective practice of primary care medicine. Although there have been several curricula developed to teach these skills to student-physicians, there have been few interventions for residents-in-training. As such, we developed a morning report-based program of teaching the residents physical diagnosis.

CURRICULUM DESCRIPTION: One to two morning reports per week have been transformed into physical diagnosis teaching rounds. During these one-hour sessions, the chief medical residents bring specific patients with incontrovertible findings to morning report. A faculty member with interest and expertise in physical examination demonstrates the findings, discusses and demonstrates various techniques to improve the detection of the findings and then allows the residents to practice on the patient. Review of sensitivity and specificity of the techniques is integrated into the presentation. Evaluation of this intervention has been by an OSCE using physical examination stations, one performed early in the academic year (pre-intervention) and another scheduled for late in the academic year (post-intervention). The Stations consist of common physical examination findings including torus palatinus, olecranon bursitis, Tinel's for cubital tunnel syndrome.

RESULTS: At present, approximately 40 residents in the University of Minnesota Internal Medicine Residency program have experienced this intervention. The findings demonstrated have included those of tricuspid regurgitation (presence of Rivero-Carvalho maneuver, a positive Vitums' sign, a blinking earlobe and expansile liver), spastic hemiparesis (positive Oppenheim's, Chaddock's and Hoffman's signs with the presence of crossover reflexes and clonus) and various entrapment neuropathies-cubital tunnel, ulnar tunnel, meralgia paresthetica (Tinel's sign). The pre-intervention 38 station, physical examination-based OSCE was administered to 41 residents-in-training and revealed an overall detection rate for of 29.7% for these classic physical findings. The OSCE will be repeated in 1-2 months on the same residents.

CONCLUSION: We describe an intervention to teach physical diagnosis to residents-in-training, using a teaching session extant at most residency programs: morning report. Thus this is a paradigm to teach physical diagnosis in a patient-centered, time-efficient and fun fashion.

PREDICTORS OF PRIMARY CARE CAREER CHOICES BY YEAR AND STAGE OF TRAINING: A NATIONAL SURVEY. MT Connelly, AM Sullivan, AS Peters, N Clark-Chiarelli, N Zotov, N Martin, SR Simon, JD Singer, SD Block, Department of Ambulatory Care & Prevention, Harvard Medical School and Harvard Pilgrim Health Care, Boston, MA; Department of Human Development, Vanderbilt University, Nashville, TN; Harvard Graduate School of Education, Cambridge; Division of Psychiatry, Brigham and Women's Hospital, Boston, MA

PURPOSE: To explore whether role models, encouragement, and demographics are similarly associated with career choice at different stages (medical school vs. residency) and periods (1994 vs. 1997) of training.

DESIGN: A split-panel design with two cross-sectional telephone surveys of fourth-year students and residents in 1994 and 1997, including a panel of fourth-year students interviewed in 1994 who were interviewed as residents in 1997. Participants: A national probability sample of 307 4th year students in 1994, 219 4th year students in 1997, 645 2nd year residents in 1994 and 494 3rd year residents in 1997. Of the 4th year students interviewed in 1994, 241 (78.5%) were re-interviewed as 3rd year residents in 1997.

RESULTS: Having a primary care role model was a stronger predictor of primary care career choice (PCCC) for residents (OR 18.0, 95% CI 11.2-28.8 in 1994; OR 43.7, 95% CI 24.4-78.3 in 1997) than students (OR 6.3, 95% CI 4.0-9.9; no variation by year). Likewise, peer encouragement was more predictive of PCCC for residents (OR 5.4, 95% CI 3.3-8.9 in 1994; OR 16.6, 95% CI 9.7-28.4 in 1997) than students (OR 2.0, 95% CI 1.2-3.1; no variation by year). Encouragement by residents was associated with PCCC among students (OR 2.3, 95% CI 1.5-3.5, no variation by year). Socio-emotional orientation was also associated with PCCC (for students: OR 3.7, 95% CI 2.5-5.5 no variation by year; for residents: OR 2.3, 95% CI 1.6-3.3 in 1994; OR 4.9, 95% CI 3.1-7.8 in 1997). All predictors remained significant in multivariate ordinal logistic regression models. The effect of peer encouragement increased with time for residents in multivariate models. Gender was the sole demographic characteristic associated with PCCC in multivariate models (OR 1.6, 95% CI 1.0-2.4 for students; OR 1.5, 95% CI 1.1-2.1 for residents); this borderline association did not change by year.

CONCLUSION: Predictors of PCCC differ for residents and students. Primary care role models are strongly associated with choosing primary care careers, particularly for residents. Encouragement from peers, and, in the case of students, from residents, predicts PCCC. After adjusting for encouragement and socio-emotional orientation, women are only marginally more likely to choose primary care. Changes in the academic health center that address the differential impact of these factors may affect primary care production.

A RANDOMIZED CONTROLLED TRIAL OF RESIDENT TRAINING IN SMOKING CESSATION: APPLICATION OF BEHAVIORAL THEORY AND PRACTICE WITH STANDARDIZED PATIENTS. J Cornuz, J-P Humair, L Seemaster, H Stalder, A Pecoud, Institute of Social and Preventive Medicine, Lausanne University, Lausanne; Department of Community Medicine, Geneva University Hospital, Geneva; Outpatient Clinic, Lausanne University Hospital, Lausanne, Switzerland

PURPOSE: Smoking cessation interventions are effective in primary care but many physicians lack the necessary counseling skills. We designed a physician training program in smoking cessation based on behavioral theory and use of standardized patients. We measured its effectiveness on the quality of smoking cessation counseling and 1-year cessation rates in a university primary care center.

METHODS: 35 residents from the general medicine outpatient clinics of Lausanne and Geneva (Switzerland) were randomized to attend either a two half-day training program in smoking cessation or a control intervention on management of hyperlipidemia. Experimental residents learned to provide counseling that targets smokers' motivation to quit and practiced these skills in role plays with standardized patients. We assessed the quality of residents' counseling by interview of 251 consecutive smokers (mean age: 36 years, 60% male, mean number of cigarettes/day: 20). We measured how often residents used 14 counseling strategies and computed a score based on the number of strategies used (1 point per strategy, range: 0-14). After 1 year, we mailed a questionnaire to assess smoking status, using point-prevalence abstinence and counting patients lost to follow-up (26%) as smokers.

RESULTS: At baseline, intervention and control smokers did not differ in demographic factors, smoking behavior, stages of change distribution or medical history. In consultations with smokers, trained residents had a higher mean score than control residents (4.0 vs 2.7, $p = .01$). Residents of the intervention group were more likely to offer help to smokers (23% vs 7%), give self-help materials (14% vs 1%), advise how to cope with cessation (16% vs 6%), ask the time of the first cigarette (18% vs 11%), discuss the benefits of quitting (21% vs 12%), and arrange follow-up to discuss smoking (7% vs 3%). At one year follow-up, smoking cessation rate (13% vs 7%, $p = .04$) and proportion of smokers willing to quit (94% vs 80%, $p = .02$) were higher among smokers attended by residents trained in smoking cessation than those visiting control residents. Consumption tended to be lower for smokers consulting trained residents (median number of cigarettes/day: 16 vs 20, $p = .07$).

CONCLUSION: Training residents in smoking cessation counseling with an approach based on behavioral theory and practice of counseling with standardized patients was effective to increase the quality of counseling, smokers' motivation to quit and smoking abstinence at 1 year.

WHAT DO ATTENDING PHYSICIANS CONTRIBUTE IN AN HOUSEOFFICER-BASED AMBULATORY CONTINUITY CLINIC? E Cyran, G Alberston, L Schilling, CT Lin, L Ware, J Steiner, RJ Anderson, University of Colorado, Denver, CO

PURPOSE: To ascertain the impact of attending physician supervisors in an internal medicine housestaff-based ambulatory continuity clinic.

METHODS: Self-administered questionnaires were completed by residents (R) and attendings (A) after individual ambulatory encounters. These questionnaires assessed A contributions to teaching, and to diagnosis (DX), therapy (RX), and health care maintenance (HCM). All encounters occurred at least six months into the academic year such that the presence of an A was not mandatory.

RESULTS: Overall, 428 encounters were studied. R assessed that A made teaching points in 82% of cases and contributed to DX in 56%, RX in 61% and to HCM in 19% of cases. R assessment of A contributions significantly exceeded A self-assessment of their own contributions with regard to teaching ($p = .001$), DX ($p = .001$), RX ($p = .019$), and HCM ($p = .035$). Both R and A assessment of A contributions to teaching, DX, RX and HCM decreased significantly ($p < .05$) and progressively with each increasing year of R experience. Primary care and categorical R assessed A contributions to teaching, DX, RX and HCM comparably. However, A felt that they made more contributions to DX, RX and HCM in categorical than in primary care houseofficers ($p < .05$). Male and female R assessed A contributions to teaching, DX, RX, and HCM comparably. However, A self-assessment of teaching, DX, RX and HCM contributions were generally greater in male than in female R ($p = .003$ to $.10$). Both R (4%) and A (7%) felt that there were rarely differences between R and A with regard to DX and RX. In 8% of encounters, either R or A felt that A evaluation of the patient was needed. When A evaluation was deemed necessary, both R and A felt that A made more contributions to DX ($p = .001$) than when A did not see the patient.

CONCLUSION: Attending physicians have a significant impact with regard to teaching, DX, RX and HCM in a housestaff-based clinic. This is often underestimated by attending physicians and declines with advancing years of houseofficer training. Personal involvement of A usually involves assistance with diagnosis. The declining contributions of A to DX, RX and HCM with increasing R experience suggests opportunities for emphasis on other educational aspects of ambulatory practice.

PUBLICATION BIAS IS THE MAIN ETIOLOGIC FACTOR IN RETRIEVAL BIAS. C Daniels, VM Montori, D Dupras, Department of Medicine, Mayo Clinic and Foundation, Rochester, MN

PURPOSE: In a retrospective study, we have previously shown that internal medicine residents prefer to retrieve articles with positive results five times more often than articles with negative results. Positive studies were retrieved more frequently

from high-impact journals. We set out to prospectively determine if this finding (retrieval bias) is due to their preference for positive results (retrieval bias *per se*) or if it is a consequence of publication bias.

METHODS: The six-abstract tool is a collection of six published abstracts of controlled clinical trials, 3 with positive and 3 with negative results, matched with 6 invented ("false") abstracts identical to its match except for the direction of the results and conclusions (i.e., the published abstract had positive results, the "false" abstract had negative results). Thus, each pair member was identical in journal of publication, authorship and topic. We tested this tool for the ability of sophisticated readers to distinguish the published abstract from its "false" match (phase I). After validation, we asked 30 internal medicine residents to choose a member of each published/"false" pair for presentation (phase II). This sample size was calculated to find a 20% difference with 80% power.

RESULTS: In phase I, the six-abstract tool provided indistinguishable pairs of published/"false" abstracts of both published-positive/"false"-negative and published-negative/"false"-positive pairs: after 162 observations, 90 observations correctly identified the nature of each pair member (55.6%; $p = 0.249$). In phase II, residents showed no statistically significant preference for positive studies: of 180 observations, 106 preferred the positive study of a pair and 74 preferred the negative study of the pair (58.9 vs. 41.1%; $p = 0.15$). There was no statistically significant predilection for the published versus the "false" version of the abstracts.

CONCLUSION: Internal medicine residents prefer to retrieve positive studies more as a consequence of publication bias than retrieval bias. This finding should stimulate journal editors and researchers to publish all good quality trials: readers are similarly interested in positive and negative results.

THE EXPERIENCE OF MED-PEDS GRADUATES IN RESEARCH CAREERS. M Davis, M Farrell, S Lavoie, Robert Wood Johnson Clinical Scholars Program, University of Chicago, Chicago, IL; Internal Medicine and Pediatrics Residency Program, Yale University, New Haven, CT; Departments of Internal Medicine and Pediatrics, Virginia Commonwealth University, Richmond, VA

PURPOSE: Residency programs in internal medicine and pediatrics (med-peds) offer multidisciplinary clinical training, but little is known about how med-peds graduates combine their clinical work with research pursuits. We conducted a mail survey in order to characterize the ways in which med-peds-trained physicians pursue research careers.

METHODS: We developed a self-administered survey regarding med-peds graduates' standard sociodemographic and professional characteristics, as well as their research experience. Surveys were mailed to all graduates of U.S. med-peds residency programs in Spring 1999, based upon master files from the American Boards of Pediatrics and Internal Medicine, with support from the Medicine-Pediatrics Section of the American Academy of Pediatrics and the American College of Physicians—American Society of Internal Medicine. Recipients were asked to complete the survey if they had "ever done research of any kind, either during or after your med-peds residency."

RESULTS: As of December 1999, 55 med-peds graduates with experience in research had returned the survey; 33 (60%) remain active in research. Mean age of the sample was 40 years (range 29-55); mean year of graduation from residency was 1992 (range 1977-1998). 34 (62%) respondents hold academic appointments; 12 of 13 serving as associate professor or professor are active in research. Med-peds clinician-researchers in our sample reported a mean of 16 hours/week spent in research activities, but 11 out of 33 had no protected research time. Of 12 respondents whose residency programs required research, 8 are still active in research; those who completed residencies without a required research component appeared less likely to be active vs. inactive. Post-residency research training (either in a generalist or subspecialty fellowship), as compared to no formal research training, was significantly associated with continuing activity in research ($p < 0.001$ by Fisher's exact test). Of 22 (40%) respondents no longer active in research, 19 said that they preferred clinical work; lack of mentorship was cited as a concern by only 2 respondents.

CONCLUSION: Preliminary findings from the first wave of a survey of med-peds physicians suggest that several continue to pursue research careers following their clinical training. However, med-peds clinician-researchers are bringing their multidisciplinary perspective to research questions with little protected time. These findings may be helpful guides for medical students considering med-peds careers, med-peds residents with research aspirations, and directors of academic units who want to bring the med-peds perspective to research questions.

PREDICTING SCHOLARLY PRODUCTIVITY BY INTERNAL MEDICINE CLERKSHIP DIRECTORS. DM Elnicki, P Hemmer, A Albritton, R Kovach, R Wong, M Udden, M Battistone, West Virginia University, Morgantown, WV

BACKGROUND: The scholarly, clinical and educational productivity of academic physicians is coming under increasing scrutiny. Clerkship directors (CD) have a prominent role in medical schools, and determining factors associated with scholarly productivity may enhance their career development.

METHODS: The internal medicine CD's of US medical schools were surveyed in 1999. Questions concerned teaching responsibilities, inpatient and outpatient clinical activity, committee service, administrative aspects of the clerkship (e.g., number of students, sites used, assistance) and the use of an article on expectations of/for CD's in negotiations with department chairs. The number of peer reviewed articles published and grants funded as principal investigator in the previous 48 months were used as indicators of scholarship.

RESULTS: The response rate was 90% ($n = 110$). Of those responding: 72% were men, 45 was the mean age and 58% had completed a fellowship. CD's received: (mean) 22% salary support for while spending 28% of their time on the clerkship. They spent 3 half days weekly in clinic, 3 months on inpatient services and had been clerkship directors 6.5 years. The mean number of articles was 2 (range 0–20) and grants was 0.7 (range 0–4). Only 38% had any grants, so they were considered as a dichotomous variable in the comparative analysis. In univariate analysis, the factors associated with publishing articles ($p < .05$) were: male gender (2.7 v .9), having 3 or less outpatient clinic sessions weekly (2.8 v 1.3), fellowship training (3.0 v 1.2), institutional faculty development program (3.4 v 1.8), teaching in other courses ($F = 5.9$) and using the "expectations article" (3.7 v 1.8). The variables associated with grants were identical. Clerkship characteristics and inpatient clinical activity were not associated with scholarship. In multivariate analysis, a model using the teaching, fellowship training, outpatient clinic and the "expectations article" had $R^2 = .22$ for articles published. Regarding grants, a model using teaching, outpatient clinic, faculty development, gender and "expectations article" had $R^2 = .35$.

CONCLUSION: Several factors were noted to be associated with increased scholarship. The explanation for the gender differences was unclear. A threshold effect for outpatient clinical activity was noted, which was not seen for inpatient work. Faculty development, either through a fellowship or a formal program was beneficial. This national sample of internists gives insight to factors influencing scholarship which should be considered by those responsible for the career development of clerkship directors.

CAN AN EVIDENCE-BASED APPROACH RESUSCITATE SKILL IN PHYSICAL DIAGNOSIS? MJ Fagan, RA Griffith, LCW Obbard, CJ O'Connor, Division of General Internal Medicine, Brown University School of Medicine, Providence, RI

PURPOSE: To determine if the introduction of a curriculum in evidence-based physical diagnosis (PD) into a third year medical student clerkship in internal medicine could improve student knowledge, skill, and confidence in PD.

METHODS: We developed an evidence-based PD curriculum based on 8 articles selected from JAMA's Rational Clinical Examination series. The topics included central venous pressure, splenomegaly, low back pain, ascites, carotid bruit, systolic murmur, deep venous thrombosis, and hypovolemia. Each weekly, one-hour session included a discussion of the article emphasizing the sensitivity and specificity of PD findings and a practice session with an inpatient. Students at one hospital site were the intervention group, and students at 3 other clerkship sites served as controls. At the end of the clerkship, all students completed a PD quiz based on the articles, a skills assessment using standardized patients, and a survey about confidence in their own PD skills. Confidence scores were calculated by summing the responses to the 7 point Likert scale survey questions, with 7 indicating "very confident" and 1 indicating "not confident". The intervention group also received a pre-clerkship assessment of PD skill using standardized patients.

RESULTS: For the first academic quarter of 1999–2000, 12 intervention students and 19 control students completed the quiz, skills assessment, and survey. Intervention students scored significantly higher on the end-of-clerkship skills assessment compared to control students, mean correct score ($\pm$ SD) 92.8% ($\pm$ 6.8%) v. 51.4% ($\pm$ 12.5%), $p < 0.001$. Intervention students also scored higher on the end-of-clerkship skills assessment than they did on their pre-clerkship skills assessment, mean correct score ($\pm$ SD) 92.8% ($\pm$ 6.8%) v. 40.5% ($\pm$ 12.3%), $p < 0.001$. Compared to controls, intervention students reported greater confidence in their PD skills related to the 8 topics in the curriculum, mean confidence score ($\pm$ SD) 41.6 ($\pm$ 4.6) v. 36.4 ($\pm$ 4.8), $p = 0.005$. In contrast, there was no significant difference in the confidence scores of the intervention and control groups for 4 PD topics not covered in the curriculum, mean score ($\pm$ SD) 18.9 ($\pm$ 2.4) v. 18.2 ($\pm$ 3.2), $p = 0.52$. There was a non-significant trend toward higher mean quiz scores in the intervention group compared to controls, mean correct score ($\pm$ SD) 69.5% ($\pm$ 9.3%) v. 65.5% ($\pm$ 7.3%), $p = 0.18$.

CONCLUSION: We conclude that our evidence-based PD curriculum was successful in improving student skill and confidence in specific PD domains. Medical educators attempting to improve physical diagnosis skill should consider adopting an evidence-based approach.

USING PATIENTS WITH CANCER TO EDUCATE RESIDENTS ABOUT GIVING BAD NEWS. NJ Farber, A Friedland, BM Aboff, DB Ehrenthal, T Blanchetta, Dept of Medicine, Christiana Care Health System, Wilmington, DE

PURPOSE: Communication skills, including the communication of bad news to patients, have been taught to residents using a variety of methods. Standardized patients have become increasingly used; however, the use of actual patients with life-threatening illnesses has not been studied in resident education. We therefore conducted a workshop with residents using such patients.

METHODS: Four patients with cancer from the Wellness Community of Delaware were recruited to participate in a 2 hour resident workshop on giving bad news to patients. The curriculum was based on work by Urban and Rabatin (1), with patients using their own diagnoses and histories as the role play scenarios. Residents were instructed to give bad news as they usually do, then feedback from the patients, discussion about the role play, and about the method as described by Buckman (2) ensued. A second role play using the learned concepts was conducted, and a final debrief then occurred. Prior to the workshop, residents com-

pleted an 11 item questionnaire about actions to be taken during the communication of bad news, with a similar post-test being administered at the end of the workshop. The residents answered the items using a 4 point Likert scale (strongly agree to strongly disagree), and demographic questions were included. Pre and post tests were compared using paired student's T tests, while the impact of demographic data on responses was analyzed via analysis of variance (ANOVA).

RESULTS: Twenty-five residents participated in the workshop, with 15 (60%) completing both pre and post tests. Attitudes toward ensuring hope is conveyed to patients ($p < 0.05$), starting the discussion by ascertaining the patients' understanding of the condition ($p < 0.01$), and encouraging the patient to express his/her feelings ($p < 0.01$) all significantly improved with the use of the workshop. Most of the improvement occurred in residents with previous training.

CONCLUSION: A role play workshop using untrained patients with cancer can significantly improve residents' attitudes about giving bad news. Volunteers are without cost to the institution, and have a positive influence in such workshops. Residents who are previously trained in giving bad news may hold counter-productive attitudes, and therefore all residents should participate in workshops using role play by actual patients about giving bad news.

1. Urban SY, Rabatin JS. Communication between health professional and patient when the patient is diagnosed with cancer. *J Gen Intern Med* 1998; 13 (Suppl 1):86.
2. Buckman R. *How To Break Bad News*. Baltimore: Johns Hopkins U Press, 1992.

DEPRESSION IN A RESIDENTS' CONTINUITY CLINIC. M Fato, J Janosky, M Wurm-Schaer, The Western Pennsylvania Hospital, Pittsburgh, PA

PURPOSE: Although depression is one of the most frequent psychiatric disorders seen in the primary care setting, research suggests that patients with depression are seriously underdiagnosed and undertreated. Since approximately 40% of depressed adults receive mental health care in primary care settings, we wanted to explore the prevalence and treatment of this disorder in our residents' continuity clinic.

METHODS: After review of the literature, variables were identified and used to develop an instrument that served as the basis for data extraction from patients' charts. One hundred twenty-two (122) charts were selected for review and data collection. Two PGY3 medical residents conducted the chart reviews. Comparisons were made between those identified as depressed and those not identified using either a t-test or Chisquare test.

RESULTS: Of the 122 charts reviewed, 23 (19%) had the diagnosis of depression noted on the patients' active problem list. Twenty of the patients with depression (87%) were receiving some form of treatment, either medication or medication with counseling, and 3 (13%) had no documented treatment. Only 5 (22%) of the diagnoses of depression were substantiated by the use of an instrument or appropriate DSM4 criteria.

CONCLUSION: Although the prevalence of depressive disorders in the primary care setting is purportedly in the 5% to 9% range, a higher prevalence rate was observed in our residents' continuity clinic (19%). The accuracy of this prevalence rate is questionable given the limited use of instruments to substantiate the diagnosis of a depressive disorder. However, the residents predominantly followed recommended treatment guidelines for depression. This study raises concerns regarding the accuracy of diagnoses and corresponding documentation issues. Additionally these results have broader implications regarding medical error.

TEACHING HEART AUSCULTATION TO INTERNAL MEDICINE AND FAMILY PRACTICE TRAINEES: DOES IT WORK? B Favrat, J-J Goy, A Bloch, A Jaussi, A Pecoud, Medical Outpatient Clinic, University of Lausanne, Switzerland

PURPOSE: Proficiency in physical diagnostic skill seems to decline. Few data have examined this question and generally by using recorded cardiac events from patients instead of testing directly the bedside diagnostic skills. Objectives: 1. To compare the cardiac auscultatory skills of physicians in training in comparison with skilled cardiologists by using real patients to test bedside diagnostic skills 2. To evaluate a teaching program at bedside in order to improve auscultatory diagnostic skills.

METHODS: In an academic primary care center, 10 physicians (internal medicine and family practice trainees) and two skilled academic cardiologists, blinded to the patients' characteristics, examined 13 patients and documented their findings on a questionnaire. They should find 33 cardiac events and 13 diagnosis. Cardiac sound, murmurs and diagnosis were previously determined by an independent skilled cardiologist using echocardiography. The physicians participate in a 45 minute-course, once a week for 5 months. After the course, they listened again consecutively the same patients. The percentage of correct diagnosis was the main outcome of interest. **RESULTS:** The experts were the most skillful with 62% of correct diagnosis and heard 100% percent of diastolic murmur. The physicians gave only 21% of correct diagnosis and detected 62.5% of diastolic murmurs. 2) After the course the mean percentage of correct diagnosis was 35% ($p < 0.05$, paired t-test), a relative increase of 54%.

CONCLUSION: The bedside diagnostic skills of internal medicine and family practice trainees are effectively low but could be improved by a course focusing on bedside teaching. A randomized clinical trial is needed to confirm these results.

MANAGED CARE: THE LEARNER'S PERSPECTIVE. JG Frohna, Internal Medicine and Pediatrics, University of Michigan, Ann Arbor, MI

PURPOSE: To examine residents' views of what managed care knowledge and skills are important for future practice.

METHODS: A 27-item survey was administered to residents in internal medicine residency programs, with items designed to address aspects identified by educators as essential for practice in a managed care environment. Knowledge/skill and the perceived importance of the items were assessed with a five-point Likert scale (1 = low knowledge/importance, 5 = high knowledge/importance). Factor analysis was used to identify empirically coherent dimensions within the set of items. Factors were analyzed to determine differences based on a number of variables: university versus community program, year of training, managed care penetration, continuity clinic location, resident career plans, and resident attitude towards managed care.

RESULTS: Five distinct factors were identified. Residents reported the greatest knowledge/skill for Medical Decision-Making and Evidence-Based Medicine & Informatics. The largest gap between importance and knowledge/skill was seen for factors most essential to practice in a managed care environment: Non-Medical Aspects of Health Care; Health Systems Finance, Organization, Delivery; and Quality Improvement and Utilization Review.

Item	Knowledge/ Skill	Importance	Mean Difference
	Mean (SD)	Mean (SD)	Mean (CI)
Factor 1: Medical Decision-Making	3.5 (0.5)	4.5 (0.5)	1.0 (0.9, 1.0)
Factor 2: Non-Medical Aspects of Health Care	3.0 (0.6)	4.1 (0.6)	1.1 (1.0, 1.2)
Factor 3: Health Systems Finance, Organization, Delivery	2.7 (0.9)	4.0 (0.9)	1.3 (1.2, 1.4)
Factor 4: Quality Improvement and Utilization Review	2.2 (0.8)	3.7 (0.8)	1.5 (1.4, 1.6)
Factor 5: Evidence-Based Medicine and Informatics	3.3 (0.8)	4.2 (0.6)	0.9 (0.8, 1.0)

There were no major differences for the five factors between program variables. A more positive attitude towards managed care correlated with a larger perceived importance for all factors, except Evidence-Based Medicine & Informatics.

CONCLUSION: Curricular development needs to focus on the key areas related to managed care practice. Managed care curricula need not be adapted for different settings and learners. Taking the learner's perspective into account can result in improved managed care curricula, which will ultimately enrich the education of our trainees.

THE ONE-MINUTE PRECEPTOR MODEL OF FACULTY DEVELOPMENT: RESULTS OF A RANDOMIZED CONTROLLED TRIAL IN INTERNAL MEDICINE RESIDENTS. SL Furney, AN Orsini, KE Orsetti, DT Stern, LD Gruppen, DM Irby, Department of Internal Medicine, University of Michigan, Ann Arbor, MI; Department of Medical Education, University of California, San Francisco, CA

PURPOSE: The "One-Minute Preceptor" (OMP) model of faculty development is a widely used method of teaching improvement, but its effects have not been rigorously evaluated. Our purpose is to study the effect of this model on internal medicine residents' teaching skills.

METHODS: We conducted a randomized, controlled trial of the OMP model on residents assigned to inpatient services. The intervention was a one-hour session incorporating lecture, role-play and discussion. Residents' use of teaching behaviors was evaluated by resident self-report and learner (intern & medical student) report pre and post-intervention. Surveys used multiple items with Likert scales to assess each of the five domains taught in the OMP model. Data was analyzed using t-tests for means.

RESULTS: Twenty-eight residents were assigned to the intervention group and 29 to the control group. Survey response rates were 90% for residents and 80% for learners. Residents assigned to the intervention reported statistically significant changes in all rated behaviors with $p < 0.05$. Eighty-seven percent of residents rated the intervention as "useful or very useful" on a 1-5 Likert scale. The results of learner surveys show significant improvements in at least one question from each domain except "teach general rules". No difference was seen in measures of overall effectiveness. See Table.

Mean Change in Learner's Ratings of Resident Teaching Behaviors

Domain	Item	Control	Intervention
Get Commitment	Ask for diagnosis	-0.11 0.11	0.14
	Involve in decision-making	-0.17 0.20	0.01
Probe Knowledge	Asked for my reasoning	-0.17 0.11	0.08
	Evaluated my knowledge	-0.19 0.13	0.03
Teach General Rules	Taught general rules or pearls	-0.14 0.08	0.16
Feedback	Gave positive feedback	0.02 0.24	0.17
	Explained why I was correct/incorrect	-0.14 0.12	0.08
	Offered suggestions for improvement	-0.13 0.53	0.003
	Gave feedback frequently	-0.20 0.61	<0.01
Overall	Overall teaching effectiveness	-0.14 -0.01	0.32

CONCLUSION: The OMP model is a brief, easy to administer intervention that improves residents' teaching skills. Further study is needed to determine the duration of the effect and to optimize the impact of the intervention. We conclude that this intervention is an effective way to improve clinical teaching in medical residents.

MEDICAL RESIDENT'S CONFIDENCE IN TREATMENT DECISIONS AND IMPACT ON PATIENT SATISFACTION IN RESPONSE TO CONSUMER DEMANDS IN THE CLINICAL ENCOUNTER. KA Galt, EC Rich, BL Houghton, PD Turner, M Monaghan, Center for Practice Improvement and Outcomes Research; Department of Medicine, Creighton University, Omaha, NE

PURPOSE: Direct-to-consumer pharmaceutical advertising has increased pressure on physicians to accommodate marketing-inspired patient expectations for drug therapy. The purpose was to assess resident confidence in prescribing appropriate therapy while satisfying a patient in response to a patient treatment request.

METHODS: A cross-sectional written survey was administered to 50 internal medicine residents in their 1st-3rd year. Resident characteristics studied were gender, year of training, and USMLE admission score. Residents self-rated their confidence in prescribing appropriate therapy in response to patient specific requests and in achieving a satisfactory patient visit from the patients point of view on 2 separate 5-point scales (1 = not very confident; 5 = very confident) for each of 6 scenarios. The 6 scenarios represented consumer requests based upon information from a newspaper article, an internet resource, a television ad, a magazine ad, prior experience, and testimonial from a relative. Resident's self-rated attitudes (perceived accuracy, accessibility, applicability of information, frequency of resource use in practice) toward medical knowledge resources using a 5-point scale. Medical knowledge resources evaluated were the internet, drug reference manuals, electronic medical text, peer reviewed research, and pharmaceutical sales representatives.

RESULTS: No differences in responses were found in their confidence with either treatment or satisfaction based on gender, year in training, or USMLE score. No relationship was found between confidence in the accuracy of all 5 medical knowledge resources studied and confidence in treating or satisfying a patient. A significant correlation was found between ready availability of the internet with confidence in prescribing for 5 of the 6 scenarios (newspaper article, internet resource, television ad, magazine ad and prior experience). A significant correlation was also found between availability of all 5 medical knowledge resources and resident confidence in prescribing for the newspaper article scenario. A negative correlation was found between frequency of use of pharmaceutical representatives and confidence in treatment success and patient satisfaction for several of the scenarios.

CONCLUSION: Ready availability of the internet is associated with resident confidence in responding to a specific patient request for treatment. Greater interaction with pharmaceutical sales representatives may be associated with lower resident confidence in ability to manage patients with specific treatment requests.

ELECTRONIC OR PAPER? A SURVEY OF INTERNAL MEDICINE RESIDENCY PROGRAMS. R Germany, L Cation, Medicine, University of Illinois College of Medicine, Peoria, IL

PURPOSE: Internal medicine residency programs have traditionally used costly paper residency brochures to provide information on their programs to applicants. The advent of the Internet allows residency programs to post program information on their websites with ease and minimal expense. Now that the National Residency Matching Program requires that all internal medicine residency programs use the Electronic Residency Application Service (ERAS) for applications, we examined how many programs have Internet websites and how many are still distributing paper residency brochures.

METHODS: A list of the United States internal medicine residency programs was obtained from the AMA Directory of Graduate Medical Education 1999-2000 edition. Each residency program was contacted by phone or email between September and December 1999 and a survey was completed. The survey determined if the program had a paper brochure and/or an Internet website. Websites were then examined for their content.

RESULTS: Of the 380 internal medicine residency programs listed, we were able to contact 347 (91%). Of these 347 programs, 293 (84%) had a program website, 159 (50%) had a paper brochure, 134 (39%) had both, and 9 (3%) had neither. Regarding website content, 225 (77%) of websites contained the program phone number, 179 (61%) had an email address listed, and 168 (57%) had an email direct link. Website program information included the specific yearly rotation schedule (77%), conference schedule (64%), and resident clinical research opportunities (57%). Other common features included vacation (62%), salary (56%), and community links (44%). Sixty-eight percent of websites had a picture of the residency program director and 53% had a picture of the department chairperson. Seventy percent of websites had a reference to ERAS in their application section but only 35% had a direct link to ERAS. Only 19% of websites had a reference to the program's ABIM Certification Exam first-taker pass rate.

CONCLUSION: Our study reveals that of the internal medicine residency programs we were able to contact, most have an Internet website with information on their program. The reasons for this are many and include cost and ease of updating information. The content of program websites varied widely but most had information on their curriculum, benefits, and a way to contact the program. With the boon of the Internet and the move to the ERAS system, most internal medicine

residency programs are abandoning the traditional residency brochure in favor of an Internet website.

TEACHING MEDICAL DECISION MAKING TO STUDENTS MAY "TRICKLE UP" TO MEDICAL RESIDENTS. *RM Golub*, W Wallace, Medicine, Northwestern University Medical School, Chicago, IL

BACKGROUND: For the last 5 years Northwestern University has had a formal M1/M2 preclinical Medical Decision Making (MDM) curriculum. During 90 hours of contact time the entire class of 175 students studies clinical epidemiology, study design, literature critique skills, biostatistics, decision analysis, cost-effectiveness analysis, psychology of decision making, and the practice of evidence-based medicine. We have previously reported that students rarely hear these concepts mentioned by house staff during their clinical clerkships.

PURPOSE: We hypothesized that, because of the intensity of the MDM curriculum, our students actually use the terminology more frequently than the house staff.

METHODS: 79 students completing their junior medicine clerkship over seven 6-week rotations were asked to quantify the number of times they heard 18 specific MDM concepts mentioned by their interns or residents during that rotation. They included use of probability in MDM, heuristics and bias, test characteristics, calculating post-test probability, searching/critiquing literature, decision analysis, and cost-effectiveness analysis. The scoring system was 0 = never, 1 = once, 2 = twice, 3 = 3 times, 4 = 4 or more times. For comparison, at the end of two 4-week rotations 70 interns and residents completed a questionnaire asking the number of times they heard the same concepts mentioned by students. The mean results of each group were compared using pooled t-tests after adjusting for the difference in rotation lengths. The responses were also recoded as 'never' vs 'ever', and the groups were compared using χ^2 tests.

RESULTS: Mean reported student use exceeded reported resident use for 5 of the 18 concepts: "use of decision trees to clarify clinical problems" (1.1 vs 0.4), "use of probability in making decisions" (3.0 vs 1.8), "effective ways of searching medical literature" (2.8 vs 1.1) at the .05 significance level; "critiquing medical literature" (2.0 vs 1.5) and "effect of sensitivity and specificity on interpretation of test results" (3.4 vs 2.9) at the .10 level. Mean resident use did not exceed student use for any of the concepts. When analyzed as never "vs ever", student use exceeded resident for "use of decision trees to clarify clinical problems" (41 vs 21.5%) and "effective ways of searching medical literature" (79 vs 51%) at the .05 level; student use was less than resident only for "sources of bias in published studies of test performance".

CONCLUSION: (1) Students report rarely hearing MDM concepts mentioned by house staff. (2) The students appear to be more familiar than house staff with a number of these concepts, including some related to clinical epidemiology, literature evaluation, and decision analysis. (3) This apparent "trickle up" may provide a unique opportunity to motivate house staff to learn these concepts.

TEACHING MEDICAL DECISION MAKING TO INTERNAL MEDICINE RESIDENTS MAY HAVE INDIRECT IMPACT ON THEIR MEDICAL STUDENTS. *RM Golub*, Medicine, Northwestern University Medical School, Chicago, IL

BACKGROUND: For the last 5 years Northwestern University has had a formal M1/M2 preclinical medical decision making (MDM) curriculum; the entire class studies clinical epidemiology, study design, literature critique skills, biostatistics, decision analysis, cost-effectiveness analysis, the psychology of decision making, and the practice of evidence-based medicine. We are now extending this into the clinical clerkships, where we have previously shown that students rarely hear house staff mention these concepts.

PURPOSE: Since teaching from house staff to students is central in reinforcing content and role-modeling the importance of these MDM concepts, we measured the impact of a new house staff educational program on how often students hear these concepts discussed during the medicine clerkship.

METHODS: 79 students completing their junior medicine clerkship between March and September 1998 were asked to quantify the number of times they heard 18 specific MDM concepts mentioned by their interns or residents during the previous 6 weeks. They included use of probability in MDM, heuristics and bias, test characteristics, calculating post-test probability, searching/critiquing literature, decision analysis, and cost-effectiveness analysis. Scoring was 0 = never, 1 = once, 2 = twice, 3 = 3 times, 4 = 4 or more times. This comprised the baseline (pre-intervention) test group.

In August and September 1999, PGY-2 Medicine residents attended 3-day retreats (12 hours of interactive lectures with homework), addressing topics similar to the student MDM curriculum but in a very intensive format. 16 students completing their medicine clerkship in November 1999 were then given the same questionnaire as the baseline students (post-intervention test group). Pre- and post-intervention mean results were compared using t-tests. They were also recoded as 'never' vs 'ever' and compared using χ^2 tests.

RESULTS: The student post-intervention mean scores were improved for "measures of risk and association" (.8 vs 1.6), "use of probability in making medical decisions" (1.8 vs 2.8), and "study design" (.6 vs 1.5) at the .05 significance level, and for "critiquing medical literature" (1.5 vs 2.3) at the .10 level. Another 8 concepts showed improvement that was nonsignificant, possibly due to limited power. Only 6 of 18 concepts did not show improvement. χ^2 results were similar.

CONCLUSION: (1) Students report rarely hearing MDM concepts mentioned by house staff. (2) Since this type of MDM exposure may be critical to reinforce the preclinical concepts and to validate them by role-modeling, it is important that the house staff understand and overtly use them. (3) Given the increase in student reporting of house staff use of these concepts, the present intervention of an intensive house staff retreat may help to achieve this goal.

THE IMPACT OF A COMMUNITY-BASED PRIMARY CARE EXPERIENCE ON CAREER CHOICE. *MS Grayson*, M Klein, K Franke, K Hudson.

PURPOSE: Studies of the effect of medical school courses on ultimate career choice have had varied results. We studied the effect of taking a first year community-based primary care course on the choice of a primary care vs. nonprimary care residency.

METHODS: Participants were all first year students at New York Medical College from the academic years of 1988-89 through 1993-94 (N = 1086). Students were selected via lottery for participation in an office-based primary care course, Introduction to Primary Care (IPC). Since the students wanting the course exceeded the available spaces, they were broken into three groups: 1) selected for IPC (IPC), 2) wanted, but not selected (W), and 3) not interested in course (NI). The study outcome measure was the choice of a primary care residency, Internal Medicine, Pediatrics or Family Medicine(PC) vs. a nonprimary care residency (nonPC). A relative risk for PC was calculated by student group.

RESULTS: The percentage of students within each group choosing a PC residency was 48% IPC, 38% W, and 40% NI. In a 2×3 analysis, the relative risk for choosing a primary care residency was 1.26 (1.06-1.50) for IPC vs. W and 1.22 (1.00-1.48) for IPC vs. NI. There was no significant difference in residency choice between students who wanted the course (W) vs. those that did not express an interest (NI). When data were collapsed to compare IPC students vs. all other, the relative risk of PC residency was 1.25 (1.07-1.45). Even after controlling for class year, sex, age and race, the relationship between IPC status and residency choice remained significant.

CONCLUSION: Participation in a first year community-based course was associated with an increased likelihood of selecting a PC residency. Moreover, students who were interested in the course, but not given the opportunity to take it, were no more likely to enter a PC specialty than students not initially interested in the program.

LONG-TERM OUTCOMES OF HARVARD MEDICAL SCHOOL'S NEW PATHWAY PROGRAM: ENHANCEMENT OF HUMANISM IN MEN AND LEADERSHIP IN WOMEN. *RP Greenberger-Rosovsky*, AS Peters, SD Block, GT Moore, Department of Ambulatory Care and Prevention, Harvard Medical School, Boston, MA

PURPOSE: To determine if an innovative curriculum—the New Pathway (NP) at Harvard Medical School—that centered on small group problem based learning promoted interest in humanistic medicine in men, an area more stereotypically associated with women, and enhanced leadership in women, a role more historically dominated by men.

METHODS: Long-term follow up of a randomized controlled trial. Telephone interviews in 1998 of 100 (50 NP, 50 Traditional; 69 men, 31 women) 1989 and 1990 Harvard graduates. Descriptive analysis of self-reported behaviors and attitudes related to humanistic medicine and leadership.

RESULTS: NP men differed from participants in the Traditional program on two attitudinal measures: NP men rated their preparation to practice humanistic medicine (mean = 21.1 versus 16.1, $p < .0001$) and their confidence in managing patients with psychosocial problems (mean = 29.6 versus 26.4, $p < .05$) higher than Traditional men. On a behavioral measure, 61.3% of NP men and 42.1% of Traditional men engaged in careers in primary care or psychiatry; the difference was not statistically significant. The women in the NP and Traditional programs differed on two behavioral measures: 37% of NP women but none of the Traditional women were chief residents ($p < .01$). NP women also spent a greater proportion of their time in administration than Traditional women (9.3% versus 4.2%), but the difference was not statistically significant. However, NP and Traditional women did not differ in their confidence to take a leadership role.

CONCLUSION: Differences between NP and Traditional men on measures of humanistic medicine and between NP women and Traditional women in leadership areas suggest that small group problem based learning enabled men and women to enhance characteristics less traditionally and stereotypically associated with their own gender.

EFFECT OF STUDENT INVOLVEMENT ON PATIENT PERCEPTIONS OF AMBULATORY CARE VISITS: A RANDOMIZED, CONTROLLED TRIAL. *TW Gress*, JA Flynn, L Simonson, SD Sisson, T Thompson, HR Rubin, FL Brancati, Internal Medicine, Marshall University, Huntington, WV; Internal Medicine, Johns Hopkins University, Baltimore, MD

PURPOSE: Although the importance of integrating medical education into ambulatory care is widely acknowledged, its effect on patient satisfaction is uncertain, since prior studies in this area have relied on observational designs subject to selection bias.

METHODS: We conducted a trial in which outpatients (N = 134) in a general group practice were randomly assigned to see the attending physician alone (N = 66) or the attending physician plus a medical student (N = 68). In this latter group, students performed a patient history and presented it to the attending phy-

sician in the examining room. Finally, patients were co-examined. Randomization was stratified by attending physician. Patient perceptions of the encounter were assessed by telephone interview. Ratings of the visit and self-assessment of overall health status were obtained using 5-point Likert scales (1-Excellent to 5-Poor). Interviewers were blinded to group assignment for questions concerning the main outcomes.

RESULTS: The two patient groups (attending physician alone vs attending physician plus student) were similar in regard to age (60 vs 58), sex (51 vs 49% male), race (47 vs 48% white), education level (27 vs 38% with college or greater), and self-rating of health status (2.8 vs 3.0; all $p < 0.10$), but differed by insurance status (86 vs 94% with insurance; $p = 0.02$). Attending physicians spent similar time with patients in both groups (25.0 vs 22.5 minutes; $p = 0.21$). Patients rated the office visit overall in favor of the attending only group, although this difference was not statistically significant (61 vs 48% excellent; $p = 0.16$). Thoroughness of the office visit (56 vs 53% excellent; $p = 0.72$) and rating of the attending physician (80 vs 85% excellent; $p = 0.44$) were similar in both groups. Among those that rated their attending physician as excellent, 73% rated the overall office visit as excellent in the attending only group compared to 55% in the attending plus student group ($p = 0.04$). After the blinded assessment, interviewers were then unblinded and posed additional questions to the 68 patients in the attending plus student group. Despite their high level of overall office visit and attending physician satisfaction, 30% would decline a similar arrangement for future visits, and 17% would not recommend a similar arrangement to a friend or family member.

CONCLUSION: We found subtle differences in patient satisfaction with the office visit that deserve further investigation. In addition, there was a significant number of patients that did not want to see a medical student at future office visits. Ambulatory medical education interventions should target improving patient satisfaction in this group.

FIVE YEAR DOCUMENTATION OF THE ASSOCIATION OF EXCELLENT CLINICAL TEACHING WITH BETTER STUDENT EXAMINATION PERFORMANCE. CH Griffith, JC Georgesen, JF Wilson, University of Kentucky College of Medicine, Lexington, KY

PURPOSE: Despite our cherished beliefs, little quantitative evidence exists that students learn more from better teachers. Preliminary research documenting an association of better teaching with more learning has been limited by small sample of students, teachers, and few teaching evaluations, making measurement of excellent teaching imprecise. We hypothesized that third-year medical students working in their medicine clerkship with our "best" teaching attending physicians and residents would perform better on clerkship examinations and on USMLE 2, and we proposed to document this with a larger sample of students and residents over many more academic years than in prior reports.

METHODS: We collected data from 5 academic years 1993-1998 regarding 380 third-year medical students and their randomly assigned internal medicine attending physicians and supervising internal medicine residents. Outcomes of interest were student scores on post-clerkship NBME Subject Examination in Medicine, and the long-term outcome of USMLE 2. Attendings and residents were confidentially evaluated by students at the end of each month's rotation (prior to students assigned a clinical grade), on 16 and 9 item (respectively) 5 point Likert-scale evaluation forms (coefficient alpha both >0.90). One teaching rating for an instructor was the mean score across all 16 or 9 items of the form, and their overall teaching rating was the mean evaluation of all the students they precepted in the study period. "Best" teachers were defined as those in the top 20% of all student evaluations in the 5 year period (amounted to a score of >4.6 of 5). Instructors with less than 5 overall evaluations were excluded. "Worst" teachers were those in the bottom 20% of evaluations.

RESULTS: Using multiple regression approaches, controlling for prior student academic achievement with USMLE 1 Score, independent predictors of better post-clerkship NBME Score were exposure to a "best" attending (adjusted mean score 477 vs. 455, [about $\frac{1}{3}$ of a S.D.], $p = 0.03$) and "best" resident (adjusted mean of 478 vs. 454, [about $\frac{1}{4}$ of a S.D.], $p = 0.01$). Exposure to a "best" attending was associated with better USMLE 2 Scores (adjusted mean 204 vs. 198, [about $\frac{1}{3}$ of a S.D.], $p = 0.02$).

CONCLUSION: Excellent clinical teaching is associated with better student learning outcomes, both short-term (post-clerkship exams) and long-term (USMLE 2). As academic medical centers are pressured to justify the costs of the teaching mission, further studies of the outcomes of teaching are needed.

CONTROLLED TRIAL OF A WEB BASED VIRTUAL REALITY PROGRAM IN TEACHING PHYSICAL DIAGNOSIS SKILLS TO MEDICAL STUDENTS. JA Grundman, RS Wigton, D Nickol, Gen Int Med, University of Nebraska Medical Center, Omaha, NE

PURPOSE: Teaching physical diagnosis skills lends itself particularly well to multimedia and virtual reality (VR) presentation, but the efficacy and efficiency of this approach has not been tested. We compared the achievement of students who used a web-based multimedia tutorial (consisting of interactivity, pictures, and virtual reality simulating the examination of the eye and ear) with that of students who used a printed tutorial presenting the same information written by the same authors.

METHODS: Student scores on a pretest covering information about the eye and ear were used to divide the first-year medical student class at the University of Nebraska into two equal groups. Group A ($n = 60$) used the written tutorial for the ear material and the multimedia tutorial for the eye material. Group B ($n = 61$) did the reverse. We measured knowledge gained with a multiple-choice post-test

that included both paper (40 questions) and web-based (10 questions) sections. We recorded time by recording check-out and check-in times for the written tutorial and log-in times for the computer program.

RESULTS: We found that students who used the multimedia version scored higher on the post-test than those using the written version. Scores for the eye test showed a larger gain (15.9 vs. 13.4 of 25, $p < 0.001$), than for the ear. (16.1 vs. 14.5 of 25, $p < 0.03$). These effects persisted after controlling for the time spent and the pre-test score: ($p < 0.001$, ANCOVA). Students using the multimedia versions spent more time than those using the written versions (mean time = 44.1 min vs. 33.4 min, $p < 0.01$). The correlation between the time spent and the post-test score, however, was higher for those using the multimedia tutorial than for those using the written tutorial. Regression analysis found three independent predictors of the post-test score: use of the multimedia version (rather than the written tutorial), time spent on the computer section, and pre-test score. 117 of 121 students (97%) returned post-surveys: 78% preferred the multimedia version; 63% rated the multimedia learning more effective.

CONCLUSION: In conclusion, this controlled trial with 121 students at one institution found that computer learning incorporating VR and interactivity resulted in increased learning as well as increased time on task. The increase in learning exceeded what would be expected from the increased time alone. Differences in effect size suggest computer learning benefits some areas more than others. These results are promising and suggest that computer learning contributes uniquely to learning physical diagnosis.

EFFECT OF SUBSPECIALTY TRAINING ON CLINICAL PRACTICE PREPARATION. RA Harrison, Oregon Health Sciences University, Portland, OR

INTRODUCTION: Few studies have evaluated the effectiveness of subspecialty consult rotations as an educational method for preparing general internist for practice. The primary objective of this study is to determine to what extent subspecialty education during residency training prepares graduates for general internal medicine practice. The secondary objective is to assess the impact of subspecialty consult rotations on graduates' sense of being prepared for practice.

METHODS: Surveys were sent to all 92 OHSU graduates (1995-1998). Demographic questions assessed type and location of practice. Graduates were asked to indicate if they had taken the rotation or not. Using a 1-5 Likert scale (unprepared to prepared) each subspecialty was assessed in three areas; 1) general knowledge, 2) diagnosing common disorders, and 3) management of disease. Graduates who responded with a 4 or 5 were considered prepared. They were asked to describe their exposure to each subspecialty and to identify other areas that they wished they had received more training during residency.

RESULTS: A total of $\frac{61}{92}$ (66%) graduates responded to the survey. Thirty five percent were associated with academics. Over half spent greater than 50% of their time in an outpatient practice. Of graduates who rotated through cardiology, rheumatology, gastroenterology, infectious disease, 83% or higher reported being prepared. Nephrology scored higher in general knowledge (90%) compared with diagnosis (73%) and management (73%). Geriatrics, pulmonary, endocrinology, and hematology/oncology had a comparatively smaller proportion of graduates reporting they felt prepared (47%-79%). Graduates reported being least prepared in Neurology (24%-45%). Graduates who did not take a specific rotation reported feeling less prepared in nearly all subspecialties compared with those who completed rotations. Graduates reported that they had too little exposure to endocrinology, pulmonary, geriatrics and neurology. Orthopedics, neurology, womens' health and dermatology were most frequently cited as needing more exposure during training.

CONCLUSION: The majority of recent graduates believe their subspecialty training experiences has prepared them for their current practices yet there are several areas needing improvement. Graduates who did not do a consult rotation were less likely to feel prepared. This study demonstrates that the subspecialty consult rotation is an important method for assuring that graduates feel prepared for practice. Ongoing challenges include deciding how subspecialty material will be taught to all residents and how predominantly hospital-based medicine programs will be able to meet the needs of graduates in primary care.

RETROSPECTIVE PRE-POST EVALUATION METHODS FOR ASSESSING FACULTY DEVELOPMENT OUTCOMES. DS Hatem, S Barrett, M Hewson, R Smith, D Steele, American Academy on Physician and Patient, McLean, VA; University of Massachusetts Medical School, Worcester, MA

PURPOSE: Measuring the effect of faculty development and medical interviewing outcomes is challenging. We developed an innovative method to evaluate an experiential, learner-centered, skills based national faculty development course on teaching medical interviewing.

METHODS: We administered a 29 item evaluation questionnaire at the conclusion of the AAPP's national faculty development course, asking participants to assess their skills prior to and at the conclusion of the course. Areas assessed included interviewing, clinical teaching, and self-awareness skills. Four items not explicitly taught in the course were clustered to form a "dummy" score. Responses were on a 7 point Likert scale (1 = not at all confident, 7 = completely confident). Paired t-tests were used to analyze pre-post differences in skills and linear regression was used to determine whether participant baseline variables predicted post course skills assessment.

RESULTS: 65 of 79 course participants completed evaluations (82%). Mean pre-post differences were significantly different for skills in: clinical practice (pre 4.5,

post 5.0); interviewing (pre 5.0, post 5.6); teaching (pre 4.1, post 5.1); and self awareness (pre 4.3, post 5.5) at the $p < .001$ level. "Dummy" scores also improved but the magnitude of change was less than for other skill areas. There was no overlap in 95% confidence intervals of the teaching and self-awareness skills change scores compared to the other 3 skills areas. Self assessments of confidence prior to the course were taken into account, regression analysis revealed that "years teaching medical school" was additionally predictive of post course interviewing, self awareness ($p < .05$) and clinical skills ($p < .001$), but not teaching skills.

CONCLUSION: The retrospective pre-post method of evaluation demonstrated clear positive outcomes of the AAPP's Teaching Medical Interviewing Faculty Development Course. While the "halo" effect of the course partially accounts for some of the positive outcome (as demonstrated by the changes in the "dummy" variables), pre-post changes in skill areas most unique to faculty development (teaching skills) and those most unique to the AAPP course (self-awareness skills) demonstrated greater change scores whose confidence intervals did not overlap with "dummy", clinical, or interview scores, more common components of participants everyday practice. The lack of predictive power of years teaching medical school for post course teaching skills is likely a reflection of the learner-centered teaching methods used in the course. Whether true skills change follows this confidence change is the focus of future research.

IMPROVING PHARMACOTHERAPY DECISION-MAKING IN PRIMARY CARE: INFLUENCE OF A PHARMACEUTICAL MARKETING CURRICULUM. B Houghton, P Turner, E Rich, K Golt, Department of Medicine, Creighton University; Creighton Center for Practice Improvement and Outcomes Research, Omaha, NE

PURPOSE: Pharmaceutical costs have been identified as the single largest driver of the current trend in medical cost increases, accounting for nearly half of the trend (HSC, RWJ Foundation, 9/21/99). We developed and implemented a curriculum to educate medical residents on the marketing techniques used by the pharmaceutical industry and strategies to improve cost-effectiveness of pharmacotherapy.

METHODS: The curriculum was delivered as 4—1 hour lectures in a 6 week period to 1st, 2nd, and 3rd year internal medicine residents at Creighton University. Specific learning objectives addressed pharmaceutical marketing, professional position statements on physician interactions with industry, use of evidence-based information resources for appropriate therapeutic decision making, and strategies to address patient requests for specific medications (e.g., inspired by direct-to-consumer marketing). We evaluated the intervention by survey through a pre-posttest design. The surveys assessed residents' knowledge of pharmaceutical marketing techniques, knowledge of professional position statements on industry interactions, attitudes toward and use of information sources (including psr's), and confidence in managing patients' requests for specific drugs. We also surveyed prescribing for three common ambulatory problems (supraspinatus tendinitis, mild hypertension, and acute cystitis). A relative value index was used to calculate the cost of prescribing for each resident. Bivariate (Dependent t Test, Sign Test) and multivariate (Repeated Measures ANOVA) statistical analyses were used to evaluate the effects of the intervention.

RESULTS: Fifty of the 59 residents completed both the pre- and posttest survey. There were no significant differences in the pre-posttest scores regarding residents' knowledge of pharmaceutical marketing techniques ($p = .21$), professional position statements on industry interactions ($p = .32$), and the cost of prescribing ($p = .25$). Residents did report that following the intervention, they were more aware of the availability of clinical/drug reference manuals ($p = .012$), could more easily apply the pharmaceutical information from Internet/Worldwide Web medical resources ($p = .004$) and peer-reviewed research/review medical journals ($p = .008$) to their clinical practice, and more frequently used pharmaceutical information from electronic medical text CD ROMs ($p = .035$). Residents also reported that they were more confident in managing patients' requests for specific drug prescriptions received through direct marketing on the Internet ($p = .035$).

CONCLUSION: A focused curricular intervention can result in modest changes in resident attitudes relevant to pharmacotherapy decision making.

FACULTY DEVELOPMENT IN AMBULATORY TEACHING: RESULTS FROM A NATIONAL CONFERENCE. I Houston, JM Clark, D Kern, P Alguire, D Boulware, J Bowen, W Branch, R Esham, G Ferencick, R Kahn, R Horowitz.

PURPOSE: To evaluate the first in a series of national conferences designed to improve faculty development in ambulatory teaching on a national level, and to compare the experience of hospital versus community-based attendees.

METHODS: The General Internal Medicine Faculty Development Project: Generalist Faculty Teaching in Community-Based Ambulatory Settings, funded by HRSA, is a collaborative effort of the major organizations in internal medicine. (Bowen et al. AJM: 107, 193-197) Teams from internal medicine teaching programs are selected to attend one of three national conferences based upon a submitted plan for ambulatory faculty development. Teams included an administrative team leader, a community-based teacher, and other key faculty members. Participants attend facilitated team meetings, plenary sessions, and 4 of 17 workshops related to educational skills and program implementation.

RESULTS: Sixty-two teams (259 persons) were selected to attend the first conference in December, 1999. Response rate for the evaluation was 91%. Overall, 1% would not recommend the conference to a colleague, 9% would recommend as a satisfactory experience, 42% as a good experience, and 48% as an outstanding ex-

perience. Mean ratings for individual workshops ranged from satisfactory to outstanding. Workshops with the highest attendance were "Teaching in the outpatient setting: precepting skills" and "How Doctors Learn: Tips for Teaching Faculty to Teach." The "precepting skills" workshop received the highest rating (70% outstanding). Participants who identified themselves as being primarily community-based (CB) ($N = 85$) rated the conference similarly to hospital-based teachers (HB) ($N = 130$) ($p = 0.4$). On a scale listing prior training in 24 teaching skills, on average, CB teachers were trained in 4.5 fewer teaching skill areas, compared with the HB teachers ($p < 0.001$). CB teachers were less likely to have protected time for their faculty development project compared with HB teachers (odds ratio 0.49, 95%CI 0.24-0.98), but supervisor support and salary support rates were similar. Participants rated the likelihood that their group will accomplish the objectives of their plan with a mean = 4.2 (SD 0.73) on a likert scale from 1 = Not Likely to 5 = Extremely Likely.

CONCLUSION: This collaboratively implemented national faculty development project on ambulatory teaching succeeded in soliciting acceptable faculty development plans from teams composed of hospital and community-based teachers from 62 teaching institutions. Despite disparities in previous training and protected time, both CB and HB teachers rated the conference highly and were confident in the potential accomplishments of their teams. We will assess the accomplishments of individual teams through prospective follow-up.

A CRITICAL APPRAISAL OF THE CRITICALLY APPRAISED TOPIC. MR Huber, VM Montori, Internal Medicine, Mayo Clinic, Rochester, MN

PURPOSE: The purpose of this study is to describe the Critically Appraised Topics (CATs) currently available on the Internet.

METHODS: We searched the World Wide Web using the string "critically appraised topic" in the meta-search engine Metacrawler (www.metacrawler.com—using Altavista, Infoseek, Webcrawler, Thunderstone, About, Excite, Lycos, Looksmart, GoTo, and DirectHit search engines). The last search date was November 28, 1999. We also used personal and institutional websites that list evidence-based medicine web pages (Netting the Evidence, www.ceres.uwcm.ac.uk, members.tripod.com/jepling/EBM.htm) and institutional and commercial medical websites (www.hon.ch/MedHunt/, www.medweb.emory.edu/MedWeb/, www.health-atoz.com, healthweb.org). We obtained 105 hits of which only 6 websites featured CATs of interest to general internists. One of us (MRH) collected data from each CAT including number of authors, each author's academic degree or training level, the existence of a clinical scenario, the existence of a focused clinical question and the number of stated components—if all components were present we considered them "well-built", type of question (i.e., therapy, diagnosis, prognosis, harm), the existence of a search strategy description, date of searching, list of databases searched, number of articles included in the appraisal, existence of a statement describing the article's validity, results and applicability, an expiration date and if the CAT had expired based on it, and if the CAT had been peer-reviewed before publication.

RESULTS: A total of 236 CATs were identified. Most CATs are single-authored (94.5%) by housestaff or junior faculty. Only 18% of CATs described the search strategy followed to find the appraised evidence. Sixty-four percent of CATs found described a single article. Of all CATs with expiration date, 88% had expired based on it. Most CATs did not fulfill the stringent validity criteria for systematic reviews of the literature or the less rigorous validity criteria set forth by the Health on the Net Foundation or by the Discern project.

CONCLUSION: CATs are readily available but potentially biased, incomplete and outdated sources of medical information on the Internet. Evidence-based medicine educational activities that include publishing CATs on an institution's Internet website need to be mindful of the high standard expected of medical information on the World Wide Web by patients and healthcare providers. At the time of this evaluation, CATs do not meet those standards and should only be shared within the setting in which they were created.

INCREASING FACULTY PRESENCE WITH A HOSPITALIST MODEL: IMPACT ON INPATIENT TEACHING EFFECTIVENESS. JM Huddleston, GJ Hanson, K Offord, JB Bundrick, HJ Schultz, Department of Medicine, Mayo Clinic, Rochester, MN

PURPOSE: To assess inpatient faculty teaching effectiveness in the setting of the hospitalist and traditional academic models of inpatient practice.

METHODS: At the end of each month for the first half of 1999, a 5-point Likert-type survey was administered to residents on general medical services. This questionnaire was broken into categories assessing the effectiveness of teaching, feedback, collegiality, allowance of autonomy, and efficiency of daily rounds. Residents were asked to rank-order the importance of feedback and collegiality, quality of teaching, autonomy, and efficiency of daily rounds in the overall scheme of their perception of satisfaction on general medical services.

RESULTS: There was a 54% response rate (104/180 surveys). Residents prioritized quality of educator (34%) in ranking contributing factors to their satisfaction with a hospital rotation. This was followed by efficiency of morning teaching rounds (18%), type of patient population (15%), team dynamics (13%), collegiality (11%) and finally autonomy (8%). In areas demonstrating quality of education, faculty with increased presence on the wards were perceived to demonstrate physical exam ($p < 0.05$), cite literature ($p < 0.01$), discuss differential diagnosis ($p < 0.01$), and provide more formal teaching ($p < 0.001$) than those in the more traditional model of inpatient teaching. Perceptions were similar for areas of collegiality: giving regular feedback ($p < 0.01$) and comfort with asking faculty questions

($p < 0.001$). In terms of autonomy for the residents, they felt that those faculty with increased presence on the wards allowed them to present their management plans in detail ($p < 0.01$) and allowed more overall autonomy ($p < 0.05$). Overall, residents were more satisfied with the faculty spending more time on the wards ($p < 0.001$). Finally, these residents thought that the quality of patient care being delivered was better when faculty were present more often ($p < 0.001$).

CONCLUSION: Evolution of our inpatient teaching services to a hospitalist model raised many concerns about preservation of resident autonomy in the face of increasing faculty presence on the wards. Our survey revealed that these concerns were unfounded. Faculty practicing in the hospitalist model were perceived by the residents as more effective teachers, more collegial, more efficient and allowed more autonomy. The results also indicated that increased faculty presence on the wards improved residents' overall satisfaction with the general medical service and their perception of the quality of patient care.

IS THERE FREE LUNCH? RESIDENT AND FACULTY PERCEPTIONS AT THE TURN OF THE CENTURY. TM Hulgán, MC Wilson, WB Applegate, General Internal Medicine, Wake Forest University School of Medicine, Winston-Salem, NC

PURPOSE: Recently, the debate over appropriate interactions between pharmaceutical representatives (PR), physicians, and physicians-in-training has intensified. Before addressing whether we should develop departmental policies governing these interactions, we explored our physicians' perceptions.

METHODS: A 2-page questionnaire utilizing dichotomous, 5-point ordinal, and open-ended responses was mailed to all Internal Medicine residents and clinical faculty. We collected the surveys in a number-coded, anonymous fashion over a two-month period during the Fall of 1999. Non-responders received two follow-up mailings.

RESULTS: Response rates were 99% (83/84) for housestaff and 87% (90/103) for faculty. Most residents (84%) and faculty (76%) trained at medical schools that allowed contact between residents and PR, and a majority of respondents believed that attempts to limit interaction would be unrealistic. While 60% of faculty and 37% of housestaff felt that contact with PR can influence the prescribing patterns of other physicians, only 44% and 22%, respectively, felt that their own decision-making had ever been influenced by these interactions. Seventy percent of residents felt that both "trinkets" (e.g., pens, mugs) and free lunches at conferences were definitely acceptable gifts. Only one-third of faculty agreed. Thirteen percent of faculty and 54% of residents felt that overnight/weekend trips for conferences sponsored by drug companies were acceptable gifts. Although very few residents or faculty believed that open interaction with PR lessened the quality of the training program, 49% of residents actually felt it made the program more attractive to applicants. Seventy-five percent of housestaff felt that there was no ethical conflict-of-interest created by PR interaction, as compared to only 32% of faculty. Subspecialty faculty were more likely than generalists to have section lunch conferences hosted by PR (83% vs. 24%), to have received research support from drug companies (76% vs. 27%), and to have received drug company honoraria for educational presentations (87% vs. 36%). More generalists (65%) than subspecialists (31%) felt that involvement of PR with our residents caused ethical conflicts-of-interest.

CONCLUSION: These results suggest that many of our physicians believe that a "free lunch" still exists. Further, our physicians-in-training appear to be either more desensitized to the influence of PR than their faculty, or they underestimate the impact of these marketing efforts. Generalists have less interaction with PR than subspecialists, and were twice as likely to acknowledge the possibility of a conflict-of-interest. Based upon these results, we will initiate a new case-based educational initiative that addresses what is known and unknown about PR interactions.

BARRIERS TO AN EFFECTIVE RESIDENT EVALUATION SYSTEM. AL Hull, H Copeland, General Internal Medicine; Center for Medical Education Research and Development, Cleveland Clinic, Cleveland, OH

PURPOSE: Competence evaluation of residents serves several purposes: guide professional development; assess achievement; and document failure. Although an important process, previous studies show that feedback is often overlooked and is sometimes perceived to be futile. We undertook a study of an Internal Medicine resident evaluation process to better understand and improve current evaluation practice.

METHODS: A review of the literature and interviews with staff and residents were used to develop and pilot-test a 33-item questionnaire. Residents were asked to indicate the frequency of actual evaluation practices, and their opinions about the system effectiveness and barriers to the system in both in-patient and ambulatory settings using a 5-point Likert scale. The questionnaire was mailed to 102 categorical and primary care track Internal Medicine residents. The resident's annual performance evaluation was linked to their response to determine possible bias in responses.

RESULTS: 48 responses (47%) were received. No difference in annual performance ratings were found between responders and non-responders ($p = .57$). Residents reported a total of 10.3 ± 4.8 written evaluations, but only 3.1 ± 3.1 face-to-face feedback sessions. Using a 5-point scale residents reported less frequent direct observation of patient interaction, especially in the in-patient setting ($2.6 \pm .8$ vs $3.1 \pm .9$, $p < .001$), and less frequent direct feedback from staff ($2.7 \pm .9$ vs 2.7 ± 1.0) in both settings. Residents rated the in-patient setting as having greater barriers to effective evaluation than the ambulatory setting ($3.3 \pm .5$ vs $3.0 \pm .6$, $p < .001$). The most frequent in-patient barriers were: not providing timely feed-

back; no face-to-face review with the staff; not enough time observing residents; too many different staff interacting with the resident team; no time devoted to evaluation; and no clear expression of performance expectations. The most frequent ambulatory barriers were: not providing timely feedback; no face-to-face feedback; no time devoted to evaluation; no clear suggestions for improvement; and no clear expressed expectations.

CONCLUSION: Many of these barriers to an effective evaluation system can be addressed by improved time organization (setting a specific time for staff-resident evaluation sessions—both to assess performance and to provide timely feedback, suggesting areas for improvement early in the rotation, rounding with the team to observe resident performance) or faculty development (setting clear expectations of performance). This study provides insight into how evaluations can be made more effective in the in-patient and ambulatory settings.

IMPORTANT DIFFERENCES IN THE EDUCATIONAL EXPERIENCES OF INTERNATIONAL AND US MEDICAL SCHOOL GRADUATES PRIOR TO INTERNSHIP. EA Jacobs, M Clay, G Vachon, M Lemon, Department of Medicine, Cook County Hospital, Chicago, IL

PURPOSE: As of 1998 International Medical Graduates (IMGs) constituted 27% of all residents and 40% of internal medicine residents in the United States. Pre-residency evaluation has shown that IMGs perform as well as US medical graduates (USMGs) on tests of medical knowledge and are proficient in history taking and physical examination. However it is not known whether IMGs have had similar educational experiences as USMGs in all areas important for their preparation for internship in the United States.

METHODS: To answer this question, we surveyed a convenience sample of 60 first-year internal medicine residents at an academic hospital and a public hospital in Chicago, IL. The 54-question survey assessed resident's previous educational experiences in medical school and residency, attitudes towards certain patient problems, and confidence in their abilities to address those problems.

RESULTS: Compared to the 28 USMGs, the 32 IMGs were significantly less likely to have had formal coursework in how to take a sexual history (41 vs. 89%; $p < 0.01$), how to recognize domestic violence (19 vs. 68%; $p < 0.01$), the impact of culture on health (45 vs. 79%; $p < 0.01$), how to discuss the process of dying with patients (19 vs. 82%; $p < 0.01$), and how to assess patients for substance and alcohol abuse (41 vs. 94%; $p < 0.01$). Most IMGs and USMGs had training in interviewing skills (84 vs. 96%) and psychiatry (91 vs. 100%). Few in either group had formal coursework in how to manage patients' pain (34 vs. 39%). Despite these differences, IMGs were as confident in their abilities to address these issues as USMGs. They also endorsed certain attitudes more often than USMGs reflecting discomfort in discussing personal issues with patients, lack of awareness of the signs and treatment of domestic abuse, and cultural differences in the care of dying patients.

CONCLUSION: There were important differences in the educational experience of international and US medical school graduates prior to internship in this study. IMGs were significantly less likely than USMGs to have had formal skill training in several areas essential to the practice of internal medicine in the United States. These differences should be addressed in residency curriculum given the large and important role IMGs play in US Graduate Medical Education.

DOES PRACTICE SETTING INFLUENCE RESIDENT EDUCATION? AB Jotkowitz, P Strachan, S Kim, JH Oh, Division of General Internal Medicine, Long Island Jewish Medical Center, New Hyde Park, NY

PURPOSE: Many medicine residency programs have started to use community-based practices as ambulatory care training sites for their residents. However, there are not many studies evaluating the influence of the use of community base practices on resident education. The purpose of this study is to determine whether the setting for ambulatory care training influences resident education.

METHODS: We conducted a survey of all internal medicine residents at a tertiary care hospital. The survey included questions regarding training year, ambulatory care training site, average number of patient seen per session, satisfaction with the ambulatory experience, diversity of diagnosis and socio-economic background, independence managing their patients, satisfaction with teaching, evidence based medicine, alternative medicine, cost effective medicine, preventative medicine, exposure to the business aspects of medicine, and having adequate time with their preceptor. A preliminary survey was piloted on interns for reliability. Those questions that reached reliability score (alpha) above 0.75 were used in the main survey. The main survey results were analyzed by using Kruskal-Wallis Test.

RESULTS: Three questions (cost effective medicine, preventative medicine, and exposure to the business aspects of medicine) were eliminated during the pilot phase because they did not reach a reliability score of 0.75. The main survey was sent to 33 residents and 28 responded (85%). Twenty-one percent of residents practiced in medical clinic, 64% in community based practices and 14% in full-time faculty practice. Most residents (83%) saw 5 to 6 patients per session. Using a Likert scale of 1 to 5, the average score for satisfaction with the ambulatory experience was 4.5, diversity of diagnosis 4.1, diversity of socio-economic background 3.1, independence managing their patients 4.2, satisfaction with teaching 4.3, evidence based medicine 3.7, alternative medicine 2.1, and having adequate time with their preceptor 4.3. Statistically significant differences were seen in questions regarding independence managing their own patients and diversity of diagnosis between residents from medical clinic and other training sites.

CONCLUSION: This study suggests that residents who practice in traditional medical clinic setting see a greater diversity of patient diagnoses and have more independence of managing their own patients. These results are concerning in view of the recent trend towards training medicine residents in community based sites. Further multicenter studies are needed to confirm our results and to evaluate the effect of ambulatory care practice setting on the resident education.

INVASIVE ALLERGIC FUNGAL SINUSITIS: A PATIENT WITH AN "ASYMPTOMATIC" INVASION. E Karcic, AA Karcic, T Mir, Nassau County Medical Center, East Meadow, NY; State University of New York at Stony Brook

LEARNING OBJECTIVE: 1). To be able to recognize allergic fungal sinusitis (AFS), a common and underdiagnosed condition.

CASE: A 28 year old man presented with yet another flare up of chronic sinusitis (he also had a history of allergic rhinitis), complaining of impaired taste and smell, itching in ears, and minimal epistaxis—mainly on the right side—after blowing his nose. The physical exam revealed a hyperemic nasal mucosa smeared with bright red blood and covered with yellowish crusts. The right inferior turbinate was pink and swollen. Investigations: skin prick test for *Curvularia lunata* was positive, white blood cell count was 7,700, with 9% eosinophils (with an absolute eosinophil count of 704), radio-allergo-sorbent test for *Curvularia lunata* was positive, total serum immunoglobulin E was 1926 IU/ml, serum eosinophilic cationic protein was 80 micrograms/l. A CT scan of the sinuses revealed pansinusitis and a soft tissue process (with multiple areas of calcification) in the right maxillary sinus eroding through the supero-medial wall into the right orbit. The patient was treated surgically. Analysis of debrided tissue revealed on histology large amounts of eosinophil-rich mucoid material, necrosis and bone remodeling. The bacterial smears and tissue cultures were negative. The silver impregnation stain revealed fungal short hyphal elements. Tissue fungal cultures grew *Curvularia lunata*.

DISCUSSION: Although responsible for many cases of chronic rhino-sinusitis, AFS is heavily underdiagnosed. We present a case with some unusual features. Our patient had AFS, which tends to affect patients with significant immune deficiencies (our patient was immunocompetent). The invasion into orbit (usually a disastrous complication) was discovered only as accidental finding on a CT scan. Positive radioallergosorbent test (RAST) results, skin test results, or presence of serum precipitins for fungal allergens, in combination with fungal culture results that match that particular fungal species, are crucial in making a diagnosis of allergic fungal sinusitis. Some believe that allergic fungal sinusitis may be a trigger of chronic rhinosinusitis in up to 93% of cases. Taking into consideration the lack of specificity of nasal eosinophilia, and the ubiquity of fungal colonization of the nose, it has been proposed to apply the term AFS only to those patients with chronic rhinosinusitis that fulfill the above criteria. Therapy is surgical.

CARBAMAZEPINE DIMINISHING THE EFFECTS OF WARFARIN AND LEADING TO MESENTERIC EMBOLISM IN A PATIENT WITH ATRIAL FIBRILLATION. E Karcic, AA Karcic, Departments of Geriatrics and Internal Medicine at St. Louis University School of Medicine and Nassau County Medical Center, East Meadow, NY

LEARNING OBJECTIVE: 1). To know drugs that decrease warfarin levels

CASE: A 66 year old woman, nursing home resident, with history of chronic atrial fibrillation, hemiplegia secondary to a stroke, congestive heart failure, hypertension and seizure disorder, presented to the hospital with worsening nausea and vomiting persisting for 3 weeks. Her medications included digoxin, warfarin and enalapril. The physical exam showed hemiplegia. Standard admission work up revealed an INR of 2.3 and a high serum phenytoin level of 42 mg/dl. The patient had phenytoin toxicity, so phenytoin was stopped and the patient observed. Two days later the patient developed generalized clonic-tonic seizures. At that time the phenytoin level was 22 mg/dl. It was decided to give the patient loading doses of carbamazepine oral suspension in order to prevent further seizures. The following evening the patient suddenly developed acute abdomen. The INR was 1.4. The patient developed mesenteric infarction secondary to embolism, as revealed by urgent exploratory laparotomy. The patient expired the next day.

DISCUSSION: The event was brought on by inadequate anticoagulation secondary to a drug-drug interaction between warfarin and carbamazepine that resulted in a drop in INR. This, in our patient with atrial fibrillation, allowed for cardiac thrombi formation and their embolisation to the mesenteric circulation. Drug-drug interactions with warfarin are among the most common interactions seen in clinical practice. The following drugs decrease warfarin level and predispose to thrombosis (remember the acronym "DECREASE"):

Diuretics: spironolactone
Estrogen containing oral contraceptives
Carbamazepine, Corticosteroids
Rifampin
Ethchlorvynol
Alcoholism, Aminoglutethimide
Sucralfate
Etc: barbiturates, griseofulvin, and vitamin K (broccoli and bananas).

IMPROVING RESIDENTS' AND ATTENDING'S ATTITUDES AND KNOWLEDGE OF MEDICATION COSTS. LM Korn, S Reichert, T Simon, EA Halm, Dept. of Medicine, Mount Sinai Hospital, New York, NY; Dept. of Medicine, Englewood Hospital, Englewood, NJ; Dept. of Health Policy, Mount Sinai Hospital, New York, NY

PURPOSE: The costs of prescription drugs are high, vary by insurance plan, and often influence compliance with therapy. In a baseline study, we found that, while most physicians (MDs) felt it was important to prescribe cost-effectively, most were unaware of actual costs to patients, never received any formal training about costs, and lacked easy access to such information. This study evaluates the impact of an educational intervention designed to foster cost-conscious prescribing attitudes and improve knowledge of drug costs and insurance coverage.

METHODS: We administered a written survey, assessing attitudes toward drug prescribing and knowledge of drug costs, before and several months after a multifactorial intervention. Study participants were internal medicine houseofficers (HO) and general medicine attendings in 4 hospitals, affiliated with 2 residency programs. The intervention included a 45-minute interactive teaching conference at all 4 hospitals and distribution (in person and by mail) of a pocket guide, listing the average wholesale price of 100 drugs commonly used in primary care. Matched t-tests were used to compare pre- and post-survey responses, using MDs as their own controls.

RESULTS: We received completed pre-intervention surveys from 203 of 281 MDs and post-intervention surveys from 146 of 207 MDs (71% response rates). Eighty-percent of respondents were HOs and 20% were attendings. Of the 146 MDs who completed the post-survey, 85% received the pocket guide and 46% attended one of the teaching conferences. Over half (54%) reported using the pocket guide once a month or more, and 55% rated it as moderately or very useful. Nearly all MDs (92%) wanted to see the guide updated and distributed yearly. Among the 105 MDs for whom there was matching pre- and post-survey data, MDs after the intervention were more likely to ask patients about their out-of-pocket drug costs (26% vs. 18%, $p < .002$), somewhat more likely to agree that drug costs were important (94% vs. 85%, $p < .07$), and slightly less likely to feel unaware of drug costs (77% vs. 81%, $p < .11$). After the intervention, MDs were more likely to report considering the cost of drugs when prescribing for patients with Medicare (76% vs. 62%, $p < .007$) or no insurance (99% vs. 89%, $p < .004$). Of the 33 drugs surveyed, accurate estimates of cost improved significantly ($p < .05$) for 8 (24%) of the medications and marginally ($p < .11$) for an additional 4 (12%) drugs.

CONCLUSION: Our multifactorial intervention favorably changed physicians' attitudes about cost-effective prescribing and modestly improved overall knowledge of actual drug costs. The pocket drug cost guides were well-regarded and frequently used by over half the MDs.

HOW IS IN-HOSPITAL CARE IN JAPAN EVIDENCE BASED? H Koyama, K Matsui, Y Noguchi, T Shimbo, T Fukui, General Medicine and Clinical Epidemiology, Kyoto University Hospital, Kyoto, Japan

PURPOSE: Recently, the importance of evidence-based medicine has been gathered considerable attention. However there have been no reports on this issue in Japan thus far. We performed retrospective study to elucidate how in-hospital care in Japan is evidence based.

METHODS: By reviewing discharge summaries, two independent physicians chose the most important problem of each patient and up to 5 most important interventions concerning it. When there were disagreement, they reached the agreement through discussion based on the predetermined rules. The quality of evidence was graded according to that of Agency for Health Care Policy and Research (1993). The search for evidence was performed using MEDLINE.

RESULTS: Among 300 summaries 234 were eligible for review, and we found 145 kinds of diseases or conditions as the most important problem for each admission. In total, 328 kinds of interventions had been performed. Among them, 46% were supported by randomized controlled trials.

CONCLUSION: This result is consistent with those of previous reports from UK (53%) and Canada (57%).

EVALUATION OF A NEW ACADEMIC HOSPITALIST PROGRAM: A QUALITATIVE ANALYSIS OF MEDICAL STUDENT AND RESIDENT COMMENTS. S Kripalani, A Pope, D Dressler, MV Williams, Internal Medicine, Emory University, Atlanta, GA

PURPOSE: With the implementation of a new hospitalist program (1/2 of the ward teams supervised by hospitalists, 1/4 by primary care internists, and 1/4 by subspecialists) at a large, inner-city public hospital, we surveyed medical students and residents to qualitatively determine the influence of hospitalists and other factors on their satisfaction with the clinical rotation.

METHODS: At the end of each ward month, housestaff and students completed a detailed questionnaire which provided a blank page for "any additional comments." Out of the 673 surveys received, 232 included written responses on this page. One author (SK) reviewed them to identify recurring themes. Two authors (SK and DD) then independently classified the comments into these themes, judging each item to be positive, negative, or mixed. A blinded third reviewer (AP) resolved disagreements. All reviewers were blinded to the student or resident completing the survey and to the supervising faculty.

RESULTS: The following 8 themes surfaced in order of decreasing frequency: educational environment, interaction with attending, hospital structure and ancillary services, team relationships, autonomy, workload, quality of care, and patient interaction. Concordance between the two primary reviewers was good, $\kappa = 0.68$. Good teaching rounds with the attending best determined a positive educational environment. Learning was felt to be inhibited by excessive patient volume, large workload, insufficient time to read, lack of ancillary services, and attendings who did not encourage discussion. As expected, over 85% of comments about excessive workload belonged to interns and residents. Ancillary services were viewed

negatively by 88%, with interns speaking out most often. By contrast, 94% of comments about *patient interaction* were positive and usually came from 3rd-year students and interns. *Team relationships* were important and positive at all levels, particularly when the evaluator felt a balance of autonomy and support from the attending. Hospitalists received more positive *attending evaluations* (79%) than did general internists (61%) and subspecialists (56%). Comments focused on their enthusiasm, excellent teaching, role-modeling, involvement in patient care, and use of evidence-based medicine. However, a few residents felt their *autonomy* was at risk, and others preferred attendings with more experience.

CONCLUSION: Not surprisingly at this public hospital, poor ancillary services and excessive workload recurred as themes contributing to a negative experience on the wards. Hospitalists were perceived to improve team relationships, raise the quality of care, and foster a better educational environment, despite occasional concerns about resident autonomy.

THE CLINICAL COMPETENCY EXERCISE: A PILOT PROGRAM FOR EVALUATING RESIDENTS MERGING STANDARDIZED PATIENTS WITH WRITTEN QUESTIONS. D Kuo, AC Smith, Department of Medicine, Morristown Memorial Hospital, Morristown, NJ; UMDNJ-New Jersey Medical School, Newark, NJ

OBJECTIVE: Traditional methods of resident evaluation are imperfect because they are either too subjective or based on limited observation. Comprehensive evaluation programs are difficult to implement in community hospital training programs because of insufficient resources. We designed a Clinical Competency Exercise (CCE) to complement the more traditional methods of evaluation offered at our internal medicine residency program.

PROGRAM DESCRIPTION: Morristown Memorial Hospital is a 575 bed community teaching hospital with 30 internal medicine and 8 med/peds residents. The CCE required one-half day in mid-May for PGY-1 and PGY-2 residents. Half of the exercise comprised 12 minute encounters with Standardized Patients (SPs) trained through a program based at UMDNJ-NJMS. The other half comprised 15 minute blocks of short-answer questions. Each participant rotated through all five SP stations and all five written response stations. For the SPs, we selected five scenarios which we felt would best test the clinical skills of a PGY1 or 2 internal medicine resident: divulging bad news to an HIV patient, diagnosing meningitis, managing a manic patient, diagnosing latent alcohol abuse, and counseling a non-compliant hypertensive patient. Residents were observed and evaluated by the SP on tasks such as history taking, physical examination, and communication. To compose the questions used for the written portion of the CCE, we enlisted the help of a staff cardiologist, an infectious diseases specialist, and two of the general internal medicine faculty. All questions were reviewed by general internal medicine faculty for clarity, fairness and appropriate level of complexity. Following the CCE, all participants were asked to complete a post-CCE evaluation form and provide verbal feedback on the experience. Summary evaluations consisting of performance scores from both parts of the CCE as well as individualized commentary by the CCE coordinator (DK) were provided to each participating resident.

RESULTS: 13 residents participated in the CCE (7 PGY1s and 6 PGY2s). Seven attending staff were directly involved with various stages of the program. Residents felt that the CCE was most useful in assessing their skills in patient communication and generally found the SP stations to be useful, realistic, and reasonably difficult. Written questions were felt to be somewhat more difficult than expected. PGY1's performed slightly better as a group compared to PGY2's. Overall, residents felt the experience to be somewhat useful. The program cost a total of \$1820 (about \$91 per resident for a group of 20 residents).

CONCLUSION: In our first year experience, the CCE was a useful multidimensional assessment of residents' knowledge and clinical skills which did not require a large number of staff. Timing of the CCE needs to be convenient for all involved to maximize participation and minimize distraction. Collaboration with a pre-existing SP program reduces some of the organizational challenges. We plan continue to repeat this experience again the following academic year for returning PGY2's and new PGY1's.

ASSESSING DEFICIENCIES IN WOMEN'S HEALTH EDUCATION AMONG MEDICAL STUDENTS: A KNOWLEDGE-BASED SURVEY. D Kwolek, J Wilson, D Grigsby, College of Medicine, University of Kentucky, Lexington, KY

PURPOSE: To guide curricular changes in women's health at our institution, we administered a knowledge-based survey to medical students covering gender differences in coronary artery disease (CAD); detection, triage, and treatment of domestic violence (DV); cancer in women; and other women's health topics.

METHODS: At the end of the 1998-1999 academic year, 296 first-, second-, and fourth-year medical students were given the 18-item anonymous short-answer survey; the response rate was 69%. Results were analyzed with multiple regression approaches, using the general linear model with medical education level and gender as categorical variables.

RESULTS: In CAD evaluation, 63%, 39%, and 1% of fourth-year students correctly identified at least one gender difference in CAD risk factors, CAD presentation, and CAD prognosis, respectively. More female students than male students correctly identified gender differences in CAD presentation ($p < 0.05$), and number of correct answers increased with level of training. In DV assessment, fourth-year clinical students listed more than one factor related to DV detection more often than first- and second-year pre-clinical students ($p < 0.004$), but did not perform better listing factors in the triage or treatment of DV ($p < 0.956$). Female students listed more correct factors related to the detection, triage, and treatment of DV than male students across all levels of clinical experience ($p < 0.016$). Stu-

dents' ability to list the top three causes of cancer death in women and three cancers associated with smoking improved with level of experience ($p < 0.0001$); the total percentage of students correctly answering the former was 27%. Students overall underestimated the incidence of urinary incontinence and Alzheimer's disease in women, but overestimated the incidence of depression and morbidity associated with hip fractures in women.

CONCLUSION: Our survey identified many topics in women's health that need curricular attention throughout undergraduate curriculum. Gender differences in CAD and domestic violence triage and treatment are areas of particular need. Reasons why male students may be less knowledgeable on the subjects tested should be studied and remedied to provide better care for all patients.

THE RESIDENT AS PRIMARY CARE PROVIDER IN THE TEACHING CLINIC: WHAT IS THE PATIENTS' PERSPECTIVE? GC Lamb, KIU Boyle, Medical College of Wisconsin, Milwaukee, WI

PURPOSE: To explore the patient's perception of primary care received from a resident working with assigned faculty versus a student-faculty pair or faculty alone. **METHODS:** The study design was a cross sectional survey of patients attending two academic hospital based Internal Medicine clinics, one a predominantly faculty practice with resident and student participation (FWest) and the second being predominantly resident practices with faculty supervision (FEast). Two focus groups, one from each clinic, were recruited to identify issues relating to the teaching encounter. A survey was constructed using 7 questions derived from the focus groups; 5 questions from the Medical Outcomes Study (MOS)-9 item visit rating form; the MOS-6 item general health survey; and 22 questions from the Components of Primary Care Index (CPCI) which includes scales for continuity, interpersonal communication, coordination of care, knowledge of the patient and advocacy. All patients attending the clinics during a twenty day study period were eligible for the study. Surveys were given to the patients at the time of arrival in the waiting room and collected upon departure.

RESULTS: Surveys were completed by 361 patients. Overall satisfaction was high, 4.5 on a 5 point Likert scale. However, among patients seen by residents, satisfaction and CPCI scores were significantly worse than either the student/faculty pair or the faculty alone. In contrast, among patients seen by a student, ratings were comparable or better than that of faculty alone. Resident patients perceived themselves as sicker (general health = 3.1) vs. faculty patients (3.65) and a greater number were indigent (general assistance/T19 = 33%) vs. those of students (12%) or faculty (7%). However no correlation between ratings and health status, insurance status or site of care was noted.

CONCLUSION: These results suggest that in this institution, resident participation in care is perceived more negatively than that of students or faculty care alone. Clearly the patient populations differ even though there was no apparent correlation with health or insurance status. Other hypotheses might involve resident attitudes, competing agendas, or level of faculty involvement.

	Residents (n = 138)	Student/faculty (n = 127)	faculty (n = 96)
	mean	mean (p)	mean (p)
Satisfaction	4.37	4.53 (0.24)	4.65 (.0004)
Continuity	4.4	4.62 (.005)	4.54 (.09)
Knowledge	4.2	4.6 (.0001)	4.4 (.11)
Communication	4.36	4.6 (.01)	4.64 (.008)
Coordination	4.12	4.57 (.0001)	4.59 (.01)
Advocacy	4.3	4.66 (.0002)	4.59 (.01)

PELVIC EXAMS ON ANESTHETIZED WOMEN FOR PRACTICE: A NATIONAL SURVEY OF FOURTH YEAR MEDICAL STUDENTS. LS Lehmann, JC Weeks, EF Cook, B Meehan, Brigham and Women's Hospital; Dana-Farber Cancer Institute; Harvard Medical School, Boston, MA

PURPOSE: Although medical students may initially learn how to perform pelvic exams on paid volunteers, they often achieve proficiency by practicing on anesthetized women. Practicing pelvic exams on women without consent is a violation of patients' right to self-determination. We sought to determine the prevalence of medical students' performing pelvic exams under anesthesia for practice and to assess students' attitudes toward the need for explicit informed consent.

METHODS: We surveyed 1500 randomly selected, fourth year, US medical students using a 44 item self-administered questionnaire. Descriptive statistics were used to characterize students' experiences and attitudes about performing pelvic exams under anesthesia. Logistic regression was used to identify variables associated with beliefs that consent was implied or that explicit consent should be obtained before students perform pelvic exams under anesthesia.

RESULTS: After one month, 387 completed surveys have been returned and an additional 142 had invalid addresses (preliminary response rate 28%). Among respondents 50% were women and 74% were ages 25-29. Among the first 387 respondents, 90% had performed a pelvic exam on an anesthetized patient. Among those who performed exams under anesthesia, the main reasons for the exam were: to palpate an abnormality (64%) and to practice pelvic exams (29%). Over half (58%) of students believed consent for pelvic exams under anesthesia was implied by patients being in a teaching hospital. However, only 9% believed that women in teaching hospitals for surgery understand that medical students perform pelvic exams under anesthesia. 66% of students believed written or verbal consent for students to perform pelvic exams under anesthesia should be ob-

tained, but only 12% of students reported a physician had obtained consent for the student to perform a pelvic exam. Students who believed that explicit consent was necessary, were less likely (OR 0.26, 95% CI 0.15–0.45) than other students to believe that consent was implied by patients being treated in a teaching hospital. Students who believed pelvic exams under anesthesia were important to helping them become proficient at performing pelvic exams were more likely to believe consent was implied (OR 2.3, 95% CI 1.3–4.1). Student's age, gender and race were not significant predictors of believing written or verbal consent should be obtained.

CONCLUSION: Approximately one-quarter of US medical students have performed pelvic exams under anesthesia for practice. Explicit consent for these exams is rarely obtained. Respect for patients' autonomy, however, requires that students or supervising physicians obtain explicit consent to practice pelvic exams on anesthetized women.

BEHAVIORAL SCIENCE CONSULTATIONS IN A PRIMARY CARE INTERNAL MEDICINE RESIDENCY PROGRAM. BR Leslie, L Adams, L Cyran, J Rifkin, S Kick, Division of General Internal Medicine, University of Colorado Health Sciences Center, Denver, CO

PURPOSE: To describe the types of behavioral science consultations requested by primary care residents and faculty in continuity clinics.

METHODS: In September 1999, our primary care residency program hired a full-time behavioral scientist to provide consultation to residents and faculty at three continuity clinics. On days that the behavioral scientist was present in the clinics, a daily log was maintained to track the frequency and type of consultation requests from faculty and residents. This log included the following data: date of referral, source of referral (faculty or resident initiated), and type of problem (i.e., psychiatric diagnosis, communication issues, behavioral change). This log was used to track referrals during scheduled individual precepting times with residents and during times that the behavioral scientist was precepting in continuity clinics.

RESULTS: In the first four months of the behavioral science program 172 patient cases were presented to the residency behavioral scientist. 64 of these were discussed during scheduled individual precepting times with residents, while 107 were presented during clinic precepting times when the behavioral scientist was available to residents and faculty. Initially, faculty were more comfortable in requesting consultation from the behavioral scientist than residents. This quickly changed as the residents became more familiar with the role of the behavioral scientist. Individual precepting appointments that were scheduled with all residents provided a non-threatening opportunity for initial consultations. To date, approximately 20% of the referrals are from faculty. The most common types of resident referrals in order are: depression, somatic disorders, thought disorders, bipolar disorders, and difficult physician-patient interactions. Faculty referrals in order are: depression, PTSD, eating disorders, differential diagnosis, and evaluation of suicidal patients. This data is preliminary, and the n is small, but can be used to identify training, clinic, and resource needs for the residency.

CONCLUSION: Primary care residents in Internal Medicine identify a number of patient psychosocial problems that are presented to the behavioral scientist. The types of problems presented suggest content areas that a formal curriculum and individual precepting would need to address. Most residents report that individual precepting around specific cases is a useful learning experience, and are eager for additional behavioral science teaching during block curriculum times. Faculty may have different needs for the behavioral scientist in the clinics. This preliminary data along with survey data from recent residency graduates will be used to design a formal behavior science curriculum.

USE OF A WEB-BASED SURVEY INSTRUMENT TO GATHER RAPID NATIONWIDE FEEDBACK ON THE ELECTRONIC RESIDENCY APPLICATION SYSTEM (ERAS). CT Lin, WJ Adams, S Brandenburg, M Blake, M Lemenger, Division of General Internal Medicine, University of Colorado Health Sciences Center, Denver, CO

BACKGROUND: 1999 was the first year that internal medicine participated in the Electronic Residency Application Service (ERAS). We surveyed the Association of Program Directors of Internal Medicine (APDIM) to understand the impact of this change on their residency programs. Gathering information from a network throughout the U.S., Canada and Puerto Rico is problematic given 407 internal medicine residency programs. We assumed that programs would be "web savvy" since they had just participated in ERAS, a web-based system. Therefore, we decided to administer the survey on the Internet. We will report our experience in using the Internet for survey research.

METHODS: A survey instrument was developed with input from APDIM and ERAS leadership. The web survey instrument was distributed by APDIM to all members of their e-mail list-serve. The e-mail message contained a direct Web link to the survey. The survey took approximately 10 minutes to complete since most items required only a mouse click to respond. Text boxes were provided for those wishing to provide additional comments. The completed survey responses were automatically collated in an electronic spreadsheet for analysis. This avoided the need for data entry. Subsequently, personalized e-mails were sent to non-responders since we found that many programs were not on the APDIM list-serve or had not responded. Finally, surveys were mailed and faxed to remaining non-responders. **RESULTS:** Forty-nine responses to the list-server generic e-mail request were received by end of the first day of the survey (12% response rate). By end of the first week, we received 93 responses (23%). A personalized e-mail reminder was sent

after one month, which resulted in another 62 responses that week (cumulative 36%). Mailed surveys of non-responders and those without known e-mail addresses increased the total response to 54%. Finally, personalized fax follow up of non-responders gained another 27% (a total of 81% response rate). Of the 332 responses received, 119 (36%) were submitted by fax or mail and were then entered on the web site by our staff, a time consuming process.

CONCLUSION: Web-based survey of an electronically sophisticated population is a novel technique and did garner an initial rapid response from a nationwide audience. A combined and repetitive approach improves response rate substantially. Despite using the ERAS web-based technology, over one-third of respondents preferred to fill out a paper survey and mailed or faxed their responses.

PHYSICIAN ATTITUDES TOWARD CLINICAL E-MAIL COMMUNICATION IN AN ACADEMIC MEDICAL CENTER. CT Lin, L Ware, G Albertson, E Cyran, L Schilling, J Steiner, RJ Anderson, University of Colorado, Denver, CO

PURPOSE: E-mail is increasingly used as a communications medium in medical practice. We wished to understand how physicians would respond to adopting e-mail as a communication tool with colleagues and with patients.

METHODS: Mail survey of 1100 practicing faculty physicians affiliated with the University of Colorado Health Sciences Center.

RESULTS: We received 342 responses (31%) to the first mailing. Ninety-four percent of respondents noted that they currently use e-mail. Ninety-two percent currently communicate with colleagues by e-mail. Physicians were more likely to communicate by e-mail with colleagues about informal "curbside" consults (66%) than about formal consults (43%). More than 80% of respondents cited benefits of e-mail with colleagues: eliminates "phone tag", faster, more convenient, and more efficient. Potential risks of e-mail with colleagues; inability to discuss complex issues, lack of confidentiality and possibility that message would not be seen. Thirty-five percent of respondents currently communicate with patients by e-mail. Male and female physicians were equally likely to e-mail patients. Forty-nine percent of internists, 34% of surgeons and 23% of pediatricians currently e-mail their patients ($p = 0.009$). Primary care physicians were more likely to e-mail patients than specialists (52% vs. 33%, $p < 0.025$). Though not significant, physicians who graduated from medical school after 1980 showed an increasing trend to e-mail patients more than those graduating before 1980 (39% vs. 33%). Physicians with greater than 50% direct patient care e-mailed patients at about the same frequency as physicians with less patient care (38 and 35% respectively). Of those who communicate with patients, 91% correspond with 10 or fewer patients. Forty-nine percent of respondents would be interested in communicating by e-mail with patients in the future. More than 50% of respondents cited potential benefits of e-mail communication with patients as eliminating "phone tag", able to print documentation, faster, more convenient and more efficient. Most commonly cited risks of e-mailing patients were inability to discuss complex issues, concern about patient overuse, potential for misunderstanding, lack of confidentiality, and possibility that message would not be seen.

CONCLUSION: Most clinicians (94%) currently use e-mail. Many (85%) communicate with colleagues via this tool, but few (35%) communicate with patients. Collegial e-mail mostly concerns informal "curbside" consults, and is perceived to bypass "phone tag", be faster and more efficient. However, e-mail with patients is perceived to be riskier, with concerns about complexity, confidentiality and patient overuse. Internists and primary care physicians were more likely to e-mail patients.

THE WIRELESS KEYPAD SYSTEM: AN INNOVATION IN TEACHING LARGE GROUPS. ML Malone, A Choithani, R Kane, F Sims, E Danto-Noccon, M Sincaban, Sinai Samaritan Medical Center, University of Wisconsin Medical School, Milwaukee, WI

PURPOSE: Technologic advances have enabled us to allow each member of an audience to interact with the speaker by answering questions posed by the speaker via wireless keypads. These keypads signal a computer and give the audience immediate feedback regarding the entire audiences' responses. How does this system compare to standard lecture in short-term educational outcomes?

METHOD: Prospective randomized study of participants of three sessions of keypad format ($n = 135$) vs. four sessions of standard lecture format ($n = 89$) to assess audience performance on mid-presentation evaluations, written post-tests, and evaluation forms.

RESULTS: There was no difference in the mid-presentation attentiveness scale ratings between the keypad format and the lecture (mean score = 75.9 of 100 vs. 74.2). There was no difference in post-presentation evaluations of the two formats for: speakers' overall performance, enjoyment of the format, and participants' assessment of their learning. Participants of the lecture scored higher on post-test scores compared to keypad participants (75.2% correct vs. 62.3%, $p = < 0.001$).

CONCLUSION: Our first attempts at using this expensive innovation in teaching technology, compared to standard lecture, failed to show improved teaching outcomes. Perhaps more experience in creative use of this innovation may lead to better outcomes.

THE EFFECTIVENESS OF A MEDICAL INFORMATICS WORKSHOP SERIES FOR INTERNAL MEDICINE RESIDENTS: A CONTROLLED TRIAL. RS Mangrulkar, CT Whelan, BC Williams, General Medicine, University of Michigan Health System; General Medicine, Ann Arbor VAMC, Ann Arbor, MI

PURPOSE: To assess the impact of a medical informatics workshop series for internal medicine residents on the knowledge, attitudes and skills in evidence-based medicine (EBM).

METHODS: Every other month, from July 1998 through February 1999, second and third year house officers participated in an 8-hour workshop series during their ambulatory care rotations ($n = 31$). The seminars took place weekly in a computer laboratory, and focused on advanced MEDLINE searching skills and EBM computer resources. Residents in the control group ($n = 42$) did not participate in the workshop.

A one-hour written assessment was administered to all residents in March 1999. Content focused on choosing an information resource and performing effective searches of the resource. Question formats included (1) self-assessment of attitudes, behaviors and skills, and (2) demonstration of these skills in clinical scenarios. Reported items were validated through semi-structured interviews with 5 content experts.

RESULTS: At baseline, the two groups were similar demographically. Participation in the seminar series was associated with increased familiarity with many Internet-based medical information resources. In addition, the intervention group reported increased use of certain Internet sites (e.g., Cochrane database, ACP Journal Club's Best Evidence, MD Consult) during their clinical work (all $p < 0.02$).

When given a specific clinical scenario, the intervention group was more likely to use Cochrane ($p = 0.001$) to assist with a clinical decision, and less likely to use a textbook ($p = 0.05$), consistent with differences in the groups' evaluation of the scientific soundness of these resources ($p = 0.005$ and $p = 0.04$, respectively). Nearly all other resources were judged equally sound and utilized equally between the two groups. The frequency of selection of computer-based resources also did not differ between the groups. Textbooks, professional consultation and MEDLINE were the resources sought most often.

The intervention group was more familiar with many effective MEDLINE search strategies, including the use of methodologic search filters ($p < 0.01$), but did not report any increased frequency of their use during their clinical work. However, when asked to complete partial literature searches, residents in the intervention group used more effective search strategies ($p = 0.04$).

CONCLUSION: Participation in this hands-on medical informatics workshop series resulted in improved awareness and use of many computer-based information resources, and measurably improved the effectiveness of searching skills. The content, format and evaluation tool for this curriculum could be adapted for other residency programs considering implementation of medical informatics instruction.

DEVELOPMENT OF A COMPREHENSIVE ASSESSMENT TOOL TO MEASURE EVIDENCE-BASED MEDICINE COMPETENCY USING A CONCEPTUAL FRAMEWORK OF CLINICAL PROBLEM SOLVING. RS Mangrulkar, CT Whelan, BC Williams, General Medicine, University of Michigan Health System; General Medicine, Ann Arbor VAMC, Ann Arbor, MI

PURPOSE: Most test instruments of evidence-based medicine (EBM) competency focus only on critical appraisal skills. We have developed a more comprehensive assessment tool, based on the paradigm of clinical problem solving advocated by Guyatt and Sackett.

METHODS: This conceptual framework describes six steps to fill an identified knowledge gap: (1) categorizing answerable clinical questions (CAT) (2) choosing an information resource (RES) (3) effectively searching the resource (SCH) (4) critical appraisal of a particular source (CRI) (5) placing the results in the context of existing knowledge (CTX) and (6) applying the results to the patient (APP). Each item in the original 16-question instrument was designed to test one of these domains. Questions contained many sub-items, with 89 total responses. The number of questions per domain was CAT-2, RES-3, SCH-2, CRI-6, CTX-2, and APP-1. Items were of two types: (1) self-assessment of attitudes, behaviors and skills, or (2) demonstration of these skills in clinical scenarios. Questions in this latter category were either multiple-choice, single word response, or phrase response questions.

The instrument was administered to 73 medicine residents and 5 content expert faculty. Semi-structured interviews with the faculty were held to establish (1) face validity (2) content validity and (3) accuracy of multiple choice and single word response questions. Response categories for open-ended questions within the RES and SCH domains were generated by consensus on a 20% response sample. Interrater agreement for categorizing all responses using the developed categories was calculated among three reviewers.

RESULTS: In general, face and content validity of this instrument was felt to be high by the five content experts. Agreement among the experts for answers to the factual questions was also very high (average agreement = 98%). One question in the CAT domain was discarded because of discrepancies in the answers between experts. Four sub-items within one question in the CAT domain were felt to have two correct answers, resulting in a scoring change. Finally, one sub-item within the CRI domain was discarded as a result of confusion in the wording of the stem. Qualitative analysis of the open-ended questions resulted in excellent agreement in classification of responses (average agreement range from 86.3% to 95.1%).

CONCLUSION: The resulting 15-item test instrument (with 87 sub-items) can be used to evaluate the effects of the implementation of EBM curricula for different learner levels. This instrument provides a more comprehensive, reliable and valid assessment tool of EBM knowledge, attitudes, and skills than what has currently been published.

RESIDENT'S EXPERIENCES OF MISTREATMENT DURING RESIDENCY TRAINING. RM Mejia, AP Diego, M Aleman, M Petracci, S Irigoyen, ER Casal, Medicine, Hospital de Clinicas Jose de San Martin; Instituto Gino Germani, Universidad de Buenos Aires, Buenos Aires, Argentina

PURPOSE: The Residency training formally appeared in the country in 1944. Since then it has acquired the prestige of the best formal medical education but also the tradition of an arduous and stressful activity. Shortness of resources and an authoritarian style has prevailed in the country's recent history and consequently in the hospital institutions. In this milieu mistreatment should be considered an issue of concern. The purpose of this study is to explore mistreatment experiences reported by residents during their training.

METHODS: Design: Focus group interviews. Setting: Ambulatory care center from Hospital de Clinicas, Buenos Aires University. Population: Internal Medicine and General Medicine residents (PGY2 and PGY3) volunteered to participate in two 90 minutes sessions coordinated by two of the authors. Anonymity was guaranteed. Sessions were tape recorded, transcript and analyzed by two of the investigators. Participants were invited to define by their own mistreatment and openly worked out personal episodes. Measurements: Qualitative, consensus review of the content of session transcripts.

RESULTS: Personal mistreatment has been perceived by almost all the participants. Mistreatment often took the form of public humiliation and belittled. Although task for punishment were commonly infringed. High concern was elicited by the obligation to work in impaired personal conditions as a result of sleep or food deprivation. They also expressed as with high frequency been obliged to work in non medical tasks replacing nurses or as ancillary services as stretch-bearers. Threats of physical harm and of being slapped, pushed or hit were rare.

Respondents believe that the origin of the mistreatment is in the relative powerlessness of residents, who are on the lowest rung of the medical hierarchy. In their opinion this situation is well known and tolerated by authorities. Senior residents also exert mistreatment to their lower degree companions.

CONCLUSION: This study reveals that this group of residents perceive mistreatment and hostility in their training environment. Cultural determinants and shortness of resources may be responsible for this situation. Further research as well as educational interventions designed to prevent and address these issues can have positive impact on the experience of medical residents.

TEACHING RESIDENTS TO TEACH—THE RAT SURVEY. E Metallos, J Beddel, J Gross, R O'Keefe, J Arnstein, Montefiore Medical Center, Bronx; Urban Family Practice, Beth Israel Medical Center; Columbia University Health Center, New York, NY

PURPOSE: Departments of Medicine and Medical Schools rely on residents to teach interns and medical students. Most residents, however, have little formal training for this role. In 1996 Montefiore developed the Resident As Teacher (RAT) Project. RAT is a mandatory full day of interactive seminars designed to help PGY 2's become better teachers, evaluators, and feedback givers. This survey was conducted to evaluate the impact of RAT on residents' beliefs, skills, and attitudes towards teaching.

METHODS: The instrument consisted of two surveys—a Pre-test and Post-test—administered and analyzed in a blinded manner. The Pre-test was given before RAT. The Post-test was given after the residents had completed at least one rotation where they were responsible for teaching interns/medical students, an interval of 2-4 months after the RAT intervention.

RESULTS: Thirty-six PGY-2's participated in the study, 30 returned Pre-tests (83%) and 34 returned Post-tests (94%). Before the intervention 93% of the participants reported they would like to become better teachers. However, prior to RAT, only 43% reported they knew how to achieve this goal. Prior to RAT, 43% described their previous teaching experience as limited (i.e.: "a topic presented on rounds") and 50% as moderate (i.e.: "math tutor in high school"). None described their prior experience as substantial. Of note, 90% reported they were unsure of their current role as teacher and reported little or no previous help developing their teaching style. After the RAT intervention, residents demonstrated improvement in three of the questions posed (on a Likert scale of 1-5, where 1 = strongly agree, 3 = neutral, 5 = strongly disagree): **1**) "I consider myself a teacher" ($p = 0.02$), **2**) "I feel I know how to become a better teacher" ($p = 0.04$) **3**) "I feel comfortable giving feedback and evaluation to medical students and interns" ($p = 0.05$). There was a trend towards improvement after RAT for five questions with $p < 0.1$, though they did not reach statistical significance: **1**) "The Department of Medicine relies on me to teach" **2**) "The Medical School relies on me to teach" **3**) "Giving feedback is an integral part of teaching" **4**) "I would like to be a better teacher" **5**) "I know how to achieve the characteristics which I feel make excellent teachers". After RAT 89% reported they had learned useful tools at the workshop to develop their teaching style, 73% reported making changes in their approach to teaching, and 73% reported feeling more prepared for their role as a teacher.

CONCLUSION: The RAT intervention has a positive impact on residents' beliefs, skills, and attitudes towards their role as a teacher.

ELDER ABUSE: A RESIDENT SURVEY. JS Miller, S Sandhu, B Larive, Internal Medicine, Cleveland Clinic Foundation, Cleveland, OH

PURPOSE: 1. To determine the amount of formal education Internal Medicine residents receive about elder abuse. 2. To determine any demographic trends among the residents in regards to elder abuse. 3. To determine how many patients have treated an abused elder.

METHODS: A 15 question survey was distributed to Internal Medicine residents over the course of two weeks at a daily noon conference. The survey was anonymous and was collected at the end of each conference. Answers to the questions were given in frequencies and percentages. Comparison groups were looked at to determine differences among the groups. P-values were calculated using Chi-square, Fishers Exact, or Mantel-Haenzel tests, as appropriate. P-values were considered significant if they were less than or equal to 0.05.

RESULTS: 64(49%) of the residents responded. 67% were male. 24 were postgraduate year 1 (PGY 1), 20 were PGY 2, and 20 were PGY 3. There were 37 foreign medical residents as compared to 27 residents born in the United States (U.S.). There were no significant differences comparing gender of the residents. There were significant differences comparing PGY 1, PGY 2, and PGY 3 residents. The PGY 3 residents felt they had treated an abused elder more often than PGY 2 and PGY 1 residents (52.4% vs. 26.3% vs. 12.5% $P = 0.014$). When comparing U.S. trained residents vs. Non-U.S. trained residents, the former had heard of Adult Protective Services more often than the latter (88% vs 64% $P = 0.034$). Also, U.S. trained residents feel they report domestic violence more often than Non-U.S. trained residents (30% vs. 3.7% $P = 0.016$). Residents who have been taught about elder abuse during inpatient service months were more likely to treat abused elders as compared to residents who have not been formally taught (53.8% vs. 23.5% $P = 0.033$).

CONCLUSION: Many studies have shown physicians report a low percentage of elder abuse cases. This study concluded that third year residents were more likely to treat abused elders than first or second year residents. Also, residents formally taught about elder abuse during their inpatient service months were more likely to treat a case of elder abuse. However, the number of residents aware of and treating abused elders is still extremely low. The survey showed that more formal education is needed during a residents' training in order to recognize and treat elder abuse.

THE CLINICAL DECISION-MAKING CONFERENCE: A QUALITATIVE ASSESSMENT OF LEARNERS' EXPERIENCE. VM Montori, CC Tabini, JO Ebbert, Internal Medicine, Mayo Clinic and Foundation, Rochester, MN

PURPOSE: To explore the learners attitudes towards the incorporation of evidence, patients' preferences, physicians' preferences—including expert opinion, and the constraints of reality in decision-making in the context of the Clinical Decision-Making Conference.

METHODS: We interviewed participating residents using a semi-structured questionnaire. The questions covered both the structure and format of the conference, the learner's level of comfort with the decision-making steps, and the likelihood they will use these steps in real-time decision-making. Interviews were conducted by the same investigator until the point of saturation was reached. The interviews were recorded and the transcripts analyzed independently by the three authors for recurrent themes. An interpretative theory was then assembled and offered to the participating residents for feedback and comments.

RESULTS: Seventeen PGY-1 residents were interviewed for an average of 15 minutes per interview. The residents identified the skills of integrated decision-making as "difficult to master", and requiring "more formal teaching sessions" and "more opportunities to practice" them. The process was felt to be "time-consuming" and "difficult to implement on a regular basis". Converting knowledge gaps into clinical questions and searching the medical literature databases were the skills they were most comfortable with and more likely to use in real-time clinical practice. Critically appraising the evidence and bringing it back to the bedside were the most challenging steps for our learners. The conference was felt to be "a lot of work" but "an excellent exercise". Our learners felt most comfortable with information derived from first-hand critical appraisal. They considered this a time-consuming process. Under time pressures, they were most comfortable relying on expert sources. At the PGY-1 level, our residents were not comfortable incorporating their opinions into clinical decision making but were comfortable integrating patients preferences.

CONCLUSION: The Clinical Decision-Making Conference was a challenging exercise for our learners. Its success was offset by the relative unfamiliarity of the learners with the skills needed and the time demands of their clinical duties. Our learners are more comfortable with and more likely to ask questions and search literature databases than to critically appraise the evidence and bring it back to the bedside. Time availability seemed to be the main determinant in choosing to obtain information from primary sources versus expert sources.

HOW WELL DO ACADEMIC GENERAL INTERNISTS PRACTICE EVIDENCE-BASED MEDICINE (EBM)? K Mukohara, MD Schwartz, Division of Primary Care, New York University, New York, NY

PURPOSE: Academic general internists are increasingly expected to teach EBM to students and residents. However, it is not known how they actually use evidence in their own patient care.

METHODS: We conducted a national, web-based survey of academic internists as part of a randomized trial of an electronic journal club, to assess use of evidence in practice, attitudes toward EBM, and ability in critical appraisal. Thus far we have 123 (9%) responses to an e-mail invitation recently sent to 1375 academic internists.

RESULTS: Mean age was 40.1 years and 40% were female. Most have teaching responsibility for residents (95%) and students (84%). Half, 50%, teach EBM to medical students or residents. When faced with clinical questions, they reported

finding an answer 73% of the time. They were most likely to seek answers from colleagues (93%), Medline (84%), paper textbooks (79%), and research summaries like ACP Journal Club (57%). In a typical month of practice, they reported using published evidence in 65% of patient encounters. They actually read published evidence to guide clinical decisions for 30% of patients. They were confident in critical appraisal skills (mean = 3.2 on 4-point, 14-item scale) and had positive attitudes about the role of evidence in patient care. Those who teach EBM reported using evidence in their clinical practice more often (69% vs. 61%, $p = .08$), were more confident in their critical appraisal skills (3.4 vs. 3.0, $p < .001$), and read more journals/week (4.5 vs. 3.8, $p = .007$) compared with those who do not teach EBM. However, they reported similar success rates in finding answers to questions arising in practice.

CONCLUSION: Academic general internists, especially teachers of EBM, appear to use evidence in their clinical practice most of the time, and are successful finding answers to questions using a variety of resources. It is reassuring that teachers of EBM seem to practice what they preach.

COPING WITH CERTAINTY: PERCEIVED COMPETENCY VS TRAINING AND KNOWLEDGE IN END-OF-LIFE CARE. PB Mulligan, D Weissman, C Von Gunten, B Ambuel, J Hallenbeck, Medical Education Research, Michigan State University, East Lansing, MI; Hematology, Medical College of Wisconsin, Milwaukee, WI; San Diego Hospice, San Diego, CA; Medicine, Stanford University, Palo Alto, CA

PURPOSE: Evaluate the end-of-life care competencies taught and learned in Internal Medicine Residency Programs participating in an interinstitutional study designed to improve end-of-life care.

METHODS: Prospective interinstitutional study in which 30 Internal Medicine Residency programs 1) completed profiles of their end-of-life (EOL) care teaching practices and 2) administered EOL knowledge and self-assessed competency assessment instruments to their residents and faculty. Programs were recruited by the American Board of Internal Medicine. Residency program directors identified the EOL education offerings within their institutions, and the means used to evaluate this teaching. Performance on the knowledge examination is represented by the percentage correct score. Ratings on the self-assessed competency assessment ranged from "1" (need further basic instruction) to "4" (competent to perform independently). The alpha value, representing a measure of the reliability of the instruments, was .62 for the knowledge test and .95 for the self-assessed competency instrument. Distribution of training levels among the 1,139 participants was 29% PGY1, 44% PGY2-4, 27% faculty.

RESULTS: Ethics and pain assessment and treatment constituted the EOL topics most often represented as required instruction, in just over half (54%) of programs. Non-pain symptom treatment and education about hospice care were the EOL topics least often presented, in either required or elective form. The most consistent source for programs to assess their residents' competencies in EOL care were faculty ratings. Performance on the knowledge examination revealed small but statistically significant increases over training levels (PGY1 mean = 50.5%, PGY2-4 = 55.2%, to faculty mean = 60.7%) Self-assessed confidence moved from the PGY1's mean rating (2.6) indicating they could perform EOL practices with supervision to the PGY2-4 (3.2) and faculty (3.5) mean ratings corresponding to judgments that they could perform practices independently ($F = 272.9$, $p < .000$.)

CONCLUSION: The study's findings speak to the willingness of Internal Medicine Residency programs to critically assess their EOL educational offerings. Although the study did find statistically significant increases across training levels in EOL knowledge, the relatively small magnitude of change and the overall knowledge mean score (55.2%) indicate considerable gaps remain in the EOL knowledge base. Despite these gaps, the mean value of the ratings of self-assessed competencies of the residents and faculty reflect their judgment that they are prepared to carry out EOL care responsibilities independently.

A SYSTEMATIC APPROACH TO CENTRAL VENOUS LINE INSERTION. IL Nip, M Haruno, Medicine, University of Hawaii Internal Medicine Residency Program, Honolulu, HI

PURPOSE: With increasing technology, medicine is becoming more invasive in its practice. With this comes the mandated responsibility for residency programs to prepare graduates to perform a core set of procedural skills. We describe a 1.5-hour workshop, which provides standardized, pedagogic training in one of these core procedural skills—insertion of the central venous line (CVL).

METHODS: Nineteen interns (76%) of the University of Hawaii Internal Medicine Residency Program were surveyed using a self-reported questionnaire regarding their attitudes and knowledge of CVL insertion. They identified discomfort with procedural technique and knowledge. Instruction was described as haphazard with no standardized, uniform training in existence. In collaboration with faculty of the Department of Anatomy, we executed a multi-disciplinary workshop with the goals of improving resident attitudes about CVL insertion and improving clinical knowledge. Curricular content included informed consent, indications and contraindications for CVL insertion, technique, complications, and documentation. Instruction was via interactive discussions, practical skills demonstrations, and a cadaver demonstration.

RESULTS: Comparison between pre- and post-questionnaires revealed improvement in intern comfort with informed consent, insertion technique, recognition of proper placement, knowledge of complications, and writing a proper procedure note ($p < 0.05$). Comparison between pre- and post-objective assessment showed significant knowledge gain ($p < 0.05$).

CONCLUSION: All participating interns agreed that they enjoyed the workshop and recommended it to be continued for future interns. Our data demonstrates that a systematic approach to CVL insertion has a positive impact on intern attitudes and knowledge.

EFFECT OF ALCOHOL RESTRICTION ON BLOOD PRESSURE: A META-ANALYSIS OF RANDOMIZED CONTROLLED TRIALS. Y Noguchi, T Morimoto, S Nagata, M Sekimoto, M Takemura, K Hira, T Fukui, Dept of General Medicine and Clinical Epidemiology, Kyoto University Graduate School of Medicine, Kyoto, Japan

PURPOSE: To ascertain whether alcohol restriction lowers blood pressure in hypertensive and normotensive individuals.

METHODS: An English-language MEDLINE search from 1966 to January 1999 was conducted. Bibliographies of review articles were also searched. Trials that had randomized allocation to control and alcohol reduction groups, monitored by timed alcohol consumption, with outcome measures of systolic and/or diastolic blood pressure were selected by blinded review of the methods section. Two investigators extracted data independently, using purpose-designed forms, and discrepancies, if any, were resolved by discussion. Q statistics was used for a test of homogeneity and fixed-effect-model was used for weighted-pooled estimates of net blood pressure change.

RESULTS: Seventeen trials that met our inclusion criteria did not show heterogeneity. The mean baseline alcohol consumption was 49.7 g/d in 13 trials with 1365 hypertensive subjects and 52.8 g/d in 4 trials with 929 normotensive subjects. The mean net reduction in alcohol consumption was 30.0 g/d in hypertensive trials and 22.3 g/d in normotensive trials. The net decrease (95% confidence interval) in blood pressure was 4.0 mm Hg (3.0–5.0 mm Hg) for systolic and 2.2 mm Hg (1.3–3.0 mm Hg) for diastolic in the hypertensive trials, and 2.2 mm Hg (1.3–3.0 mm Hg) for systolic and 1.8 mm Hg (1.0–2.7 mm Hg) for diastolic in normotensive trials.

CONCLUSION: Based on our analysis, alcohol restriction should be included in the recommendations for a non-pharmacological approach to the control of blood pressure in individuals with the habit of moderate alcohol drinking.

DOES AMBULATORY CARE TRAINING SITE INFLUENCE PATIENT EXPOSURE? JH Oh, P Strachan, S Kim, AB Jotkowitz, Division of General Internal Medicine, Long Island Jewish Medical Center, New Hyde Park, NY

PURPOSE: Many medicine residency programs have begun to use community-based practices as ambulatory care training sites for their residents. However, there is a paucity of studies evaluating the influence of community base practices on resident education. The purpose of this study is to determine whether residents practicing in different ambulatory training sites care for patients with similar complaints.

METHODS: We conducted a survey of all internal medicine residents (PGY-2 and 3s) at a tertiary care hospital. The survey included questions regarding training year, ambulatory care training site, average number of patients seen per session and how frequently they encountered the 20 most common presenting symptoms from the National Ambulatory Medical Care Survey (NAMCS), a probability sample survey conducted by the Centers for Disease Control on patients' office visits since 1973. The 1996 NAMCS dataset was used to generate a list of the most common presenting symptoms seen by internists in outpatient setting. Sampling weights were applied to generate national estimates of the total number of visits. Resident survey results were analyzed by using Kruskal-Wallis Test.

RESULTS: The survey was sent to 33 residents and 28 responded (85%). From these 50% were PGY-2s. Twenty-one percent of residents practiced in medical clinic, 64% in community based practices and 14% in full-time faculty practice. Eighty-three percent of residents saw 5 to 6 patients per session. The most common presenting symptoms in the NAMCS were: general medical exam 8.4%, progress visit 6.7%, cough 3.9%, hypertension 3.8%, chest pain 2.4%, back symptoms 2.3%, stomach/abdominal pain 2.3%, diabetes mellitus 2.2%, blood pressure test 2.1%, throat symptoms 2.0%, shortness of breath 1.7%, headache 1.7%, tiredness 1.5%, vertigo/dizziness 1.4%, upper respiratory infection 1.4%, rash 1.4%, medication 1.2%, nasal congestion 1.0%, low back symptoms 1.0%, and neck symptoms 1.0%. Residents in all settings perceived seeing the top 20 presenting symptoms more frequently than the national estimates. Once adjusted for multiple analysis there were no statistically significant differences in the frequencies of the most common presenting symptoms between all three training sites.

CONCLUSION: This study suggests that residents see the most common presenting symptoms in similar or greater frequency than the national estimates, but no differences were seen among the training sites. Further studies are needed to determine if these symptoms correlate to similar diagnoses and the educational impact of training in different ambulatory sites.

THE OBJECTIVE STRUCTURED CLINICAL EXAMINATION FOR BASIC NEUROLOGICAL EXAMINATION AS AN EFFECTIVE FORMATIVE ASSESSMENT TOOL FOR UNDERGRADUATES. H Onishi, Y Oda, S Emura, S Yamashiro, S Koizumi, General Medicine, Saga Medical School, Saga, Japan

PURPOSE: To formulate and investigate feasibility of the objective structured clinical examination for basic neurological examination (Neuro-OSCE) as a formative assessment tool for undergraduates.

METHODS: We formulated a 31-item Neuro-OSCE with 3 point scale (0, 1 or 2)

based on the four-minute neurologic exam (Goldberg). In 1998, we administered it to 104 medical students in senior year. Responses to our questionnaire from 103 students provided the information on their self-learning behavior and self-evaluation. For statistical analysis, Mann-Whitney's U-test and Spearman's correlation test were used.

RESULTS: Cronbach's coefficient alpha was 0.71 indicating fair internal consistency. Mean score for the entire students was 53.6 $\pm$ 6.1 (SD). Female students marked significantly higher score than male students ($P < 0.001$). Higher score was associated with better self-evaluation ($p = 0.013$) and longer self-practicing time ($p = 0.011$) but not with knowledge-based-self-learning time ($P = 0.84$).

CONCLUSION: We designed Neuro-OSCE for the assessment of skills in psychomotor domain. As expected, higher score predicted longer self-practicing rather than knowledge-based-self-learning time. Extracting essential items from complex neurological examinations enabled undergraduates to self-evaluate and to master the skill proportional to the length of practice. To verify the external validity of our Neuro-OSCE neurology specialists should be involved.

A CURRICULUM IN WOMENS' HEALTH FOR INTERNAL MEDICINE RESIDENTS. MA Onwudiwe, Department of Medicine, Wright State University, Dayton, OH

PURPOSE: To develop and implement a curriculum in women's health for internal medicine residents in a community hospital based university program.

METHODS: Participants are first year internal medicine residents in their ambulatory block month ($N = 10$). The curriculum content was determined by a needs assessment and consists of the following five units: 1) Introduction to Women's Health, 2) Selected Gynecological Conditions, 3) Psychosocial Issues, 4) Obesity and Eating Disorders, 5) Menopause, Osteoporosis and Hormone Replacement. The didactic sessions start with a brief case presentation by each resident, this is followed by an in-depth discussion of the topics of the day with expected resident participation. All necessary reading materials are handed out at the beginning of the month. Each three hour didactic session ends with role-plays with the instructor taking the role of the patient and one resident the doctor. This enables the residents to put into immediate practice what they have just learned. There is also a half-day visit to a domestic violence prevention center. Clinical practice of women's health is expected to occur when residents see female patients in their continuity practices. To expand the scope of this curriculum beyond these limited topics both residents and faculty have been encouraged to incorporate into everyday teaching and learning how various conditions differ in women. During the last session of the month each resident is responsible for presentation of a topic in women's health not covered in the curriculum. This well researched topic is presented to the instructor and other resident(s). Learners are evaluated by the completion of a standard evaluation form used in the program. Evaluations are based upon residents' performance on a pre- and posttest, participation during the didactic sessions and the quality of the end of the month presentation. Learners evaluate the curriculum by filling out a survey.

RESULTS: The curriculum was well received by all residents. All residents surveyed either agreed or strongly agreed that because of the knowledge gained during the month they would be able to evaluate, manage or appropriately refer female patients who presented to them with the conditions discussed. The mean posttest scores were 80%, an improvement over the mean pretest score of 62%. Residents stated particularly that the role-plays helped improve their understanding. A survey of these residents in their third year regarding their perceptions on the quality of women's health education in this program is planned.

CONCLUSION: The curriculum has improved residents' knowledge in selected women's health topics. Using this same curricular model, topics could be adapted to meet the learning needs of other residency programs.

SCREENING FOR COLORECTAL CANCER: DOES PRACTICE SETTING INFLUENCE RESIDENT BEHAVIOR? YL Oppenheim, JH Oh, AM Eichorn, AB Jotkowitz, Division of General Internal Medicine, Long Island Jewish Medical Center, New Hyde Park, NY

PURPOSE: The increased focus on ambulatory care in Internal Medicine residencies has necessitated the use of community based practices as training sites for residents. However, there is a paucity of data regarding the influence of community versus hospital based training on resident practice. The purpose of this study is to determine whether setting of ambulatory training influences resident practice patterns when screening for colorectal cancer.

METHODS: We conducted a survey of all internal medicine residents at a tertiary care hospital. The survey included questions about resident training level, gender, ambulatory care setting and practice patterns regarding colorectal cancer screening. Endpoints on interest were based on current screening recommendations and consisted of appropriate age of screening (50 years old) and methods of screening of asymptomatic individuals. Appropriate methods of screening were defined as the use of home based fecal occult blood testing, flexible sigmoidoscopy, colonoscopy or barium enema but not digital rectal exam or office based guaiac testing after digital rectal exam. We also surveyed resident response to a positive home-based fecal occult blood test. Appropriate response to a positive home based fecal occult blood test was defined as performing a colonoscopy only. Data was analyzed by logistic regression to determine independent predictors of the endpoints.

RESULTS: Surveys were sent to 55 residents. Responses were received from 31 residents, of whom 29% were first year, 32% were second year and 39% were third year residents. A total of 55% practiced in community based practices while 45% practiced in hospital based practices. Seventy-four percent of residents started screening at the appropriate age and 58% did not have an upper age limit for

screening. Eighty-seven percent of residents utilized inappropriate initial screening tests in addition to or in place of the recommended screening tests. Seventy-seven percent of residents did not perform colonoscopy in response to a positive home based fecal occult blood test. There was a strong trend toward hospital based setting being a predictor for appropriate age of screening (OR = 9.03, CI = 0.95 to 86.04) and of performing colonoscopy after a positive home fecal occult blood test (OR = 9.19, CI = 0.88 to 95.61).

CONCLUSION: In our study, residents who practiced in hospital based settings more closely followed the guidelines for colorectal cancer screening. If confirmed in further studies, these results are concerning given the current trend toward ambulatory training in community based offices. Further studies are needed to determine the influence of ambulatory care training setting on the quality of resident education.

CORRELATION BETWEEN THE AMERICAN BOARD OF INTERNAL MEDICINE IN-TRAINING EXAMINATION AND MONTHLY IN-HOUSE EXAMINATION IN PREDICTING RESIDENT SCORE PERFORMANCE. Gay Peress, Department of Medicine, Queens Hospital Center of The Mount Sinai School of Medicine, Jamaica, NY

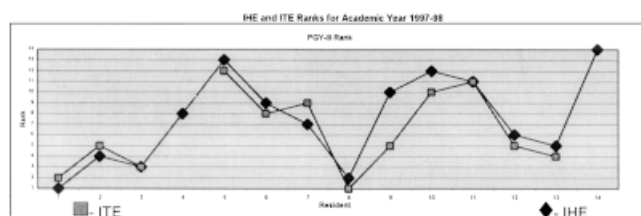
PURPOSE: To review the results obtained on a series of monthly In-House examinations (IHE) given, in all levels of postgraduate training, by our internal medicine training program and compare them to the results our residents achieved on the American Board of Internal Medicine In-Training Examination (ITE).

METHODS: Each year, medical residents at all levels of postgraduate training are given monthly in-house examinations (IHE) in each of the different medical subspecialties. Each examination consists of multiple choice questions that tests skills in data recall, data interpretation, diagnosis and/or treatment, and clinical decision making, as it pertains to the particular subspecialty being tested for that month.

Results of ITE and IHE examinations for academic years 1997-98 and 1998-99 were reviewed and residents' scores in each post-graduate (PGY) class were compared amongst themselves and a numerical rank was assigned to each resident (i.e., the highest scorer received a rank of one; the second highest scorer a rank of two, etc.). Each resident was assigned two ranks: one based on the score achieved on the ITE, and the other based on the yearly average score for the monthly IHE. A total of six ranking groups were, therefore, obtained for each academic year, corresponding to the PGY-I, PGY-II, and PGY-III IHE and ITE scores.

In each academic year, the rank on the IHE for each resident was compared to his or her ITE rank, in each PGY class.

RESULTS: Using the Spearman Rank Test, a high concordance (ρ) was found between the numerical rank the residents achieved on the IHE compared to that of the ITE. In both academic years, this concordance was highest in the PGY-III classes, with the $\rho = 0.92$ in 1997-98, and the $\rho = 0.70$ in 1998-99, and was statistically significant ($p > 0.0001$ and $p > 0.002$, respectively).



Concordance for the PGY-I and PGY-II classes in 1997-98 was $\rho = 0.52$ and $\rho = 0.84$ ($p > 0.054$ and $p > 0.0001$), respectively, and in 1998-99, was $\rho = 0.67$ and $\rho = 0.64$ ($p > 0.013$ and $p > 0.027$), respectively.

CONCLUSION: The IHE is a fairly reasonable tool to predict which residents will score highest and which will score lowest on the ITE, especially in the PGY-III classes.

DOES RESIDENT INVOLVEMENT IN A MANAGED CARE PRACTICE AFFECT PATIENT SATISFACTION? R Pirkle, DR Campa, C Grudzen, G Gildengorin, M Rabow, EJ Perez-Stable, Medicine, University of California, San Francisco School of Medicine, San Francisco, CA

PURPOSE: Managed care organizations often exclude residents from providing care to their enrollees in part because of the perception that residents provide care that is less than satisfying. We set out to test whether there is a difference in satisfaction between patients in a predominantly managed care practice receiving primary care through resident-faculty teams versus full-time faculty alone.

METHODS: A 15-item patient satisfaction questionnaire (PSQ) was administered to patients at a community-based academic practice via telephone by 2 medical students blinded to the purpose of the study. The questions were derived from 4 previously validated instruments and presented in a 5-point Likert scale format. They addressed the accessibility and convenience of care as well as the physicians' thoroughness and competence, and communication and interpersonal skills. Two full-time faculty preceptors worked with six residents to form management teams consisting of one resident and one faculty preceptor per patient (cases). Randomly selected comparison patients (controls) were matched by primary faculty physician, age, and gender.

RESULTS: 66 cases and controls were interviewed over an 18-month period. De-

mographic characteristics of the two groups were similar: 74% in capitated managed care plans, 37% women, 34% White, 28% Latino, 25% Asian, and 5% African American. There were no significant differences in mean scores between patients seen by resident-faculty teams vs. faculty alone for all areas evaluated (see Table).

Question #/content area	Mean score resident-faculty team (SD)	Mean score faculty alone (SD)	P value
1) Wait time for appt.	2.7 (1.2)	2.7 (1.1)	.88
2) Wait time in clinic	2.8 (1.2)	3.2 (1.3)	.09
3) Time spent with MD	2.3 (0.9)	2.6 (1.2)	.19
4) Thorough/competent	2.1 (1.0)	2.0 (0.9)	.65
5-14) Interpersonal skills	2.1 (0.8)	1.9 (0.7)	.26
15) Overall satisfaction	2.2 (0.9)	2.0 (0.9)	.19

Scoring scale: 1 = excellent; 2 = very good; 3 = good; 4 = fair; 5 = poor

CONCLUSION: Our results imply that patients are as satisfied receiving primary medical care through resident-faculty teams as they are from faculty alone. They also suggest that managed care organizations could allow residents, under faculty supervision, to be primary-care clinicians for their members without sacrificing patient satisfaction.

RESIDENTS' PERCEPTIONS OF EDUCATIONAL NEEDS IN WOMEN'S HEALTH AND GENDER-BASED MEDICINE COMPARED TO ACTUAL KNOWLEDGE ASSESSMENT. HG Pursley, JF Wilson, CH Griffith, DS Kwolek, Division of General Internal Medicine, University of Kentucky, Lexington, KY

PURPOSE: The American Board of Internal Medicine has stated that women's health and gender-based medicine should be included in internal medicine residency curricula.

METHODS: Two instruments were used in order to evaluate perceived curricular adequacies and core knowledge in women's health. We designed and administered to 53 internal medicine residents a 17-item questionnaire using a 5 point Likert scale. This survey rated the extent to which medical residents felt that their residency training addressed gender differences in subject matter such as presentation of coronary artery disease, cardiac risk factors, and signs of physical abuse. Also administered was a short answer questionnaire to evaluate actual knowledge of gender differences in risks, presentation, and prognosis of coronary artery disease; and factors related to the detection, triage and treatment of domestic violence. The results of the Likert scale and the short answer questionnaire were then compared using chi-square analysis.

RESULTS: Only 5 of the 53 residents felt there were no inadequacies in women's health curriculum. Ninety-four percent and 96%, respectively, of residents felt that gender differences in presentation of coronary artery disease and cardiac risk factors were covered adequately. Seventy-four percent of residents rated their curriculum adequate in the area of signs of physical abuse. A marginal association existed with gender ($p < 0.09$), in that females related more inadequacies in this aspect of the curriculum (physical abuse) than males. Of those residents who had rated the curriculum adequate in regards to cardiac disease, 26% were unable to describe two gender differences in risk factors, presentation, and prognosis of coronary disease. Of those residents who rated adequacies in the signs of physical abuse curriculum, 26% were unable to describe even two factors important in the area of detection, triage and treatment of domestic violence cases. However, 87% of residents did know whether or not to report domestic violence to authorities.

CONCLUSION: Residents' perceptions of adequacies in internal medicine women's health curriculum include cardiac disease ($p < 0.05$), and less so signs of physical abuse. However, there exists a gap to be filled between perceptions of curriculum adequacies in these areas and actual knowledge in these domains.

MENTORING IN MEDICINE: KEYS TO SATISFACTION. RA Ramanan, RS Phillips, RB Davis, W Silen, JY Reede, Division of General Medicine, Beth Israel Deaconess Medical Center, Harvard Medical School; Faculty Development and Diversity, Harvard Medical School, Boston, MA

PURPOSE: Successful mentoring has been shown to promote career satisfaction in non-medical fields, but occurs less frequently among women and minority populations. Our aim was to identify factors associated with having a mentor and being satisfied with mentorship in medicine.

METHODS: We distributed 8,093 questionnaires (response rate 47%) to full-time assistant professors, instructors, fellows and house officers of Harvard Medical School (HMS). We collected data on demographic factors (sex, ethnicity, academic rank and commitment to a career in academic medicine) and characteristics of the mentoring relationship (personal communication, professional development, academic guidance, research and skill development). We used logistic regression to identify factors significantly associated with having a current mentor and being satisfied with the relationship.

RESULTS: Of 3913 respondents, 38% were women, 72% caucasian, 16% Asian American, and 5% were underrepresented minorities (URM). Most respondents (66%) reported a current or past mentor; 55% of respondents identified a current mentor. Multivariable analysis showed factors associated with having a current mentor were being female [AOR 1.2 (95% CI 1.0, 1.4)], planning a career in academics [2.8 (2.3, 3.3)], and being an instructor [1.3 (1.1, 1.6)], fellow [11.6 (9.2, 14.8)] or house officer [2.2 (1.7, 2.9)]. Of the respondents, 63% reported satisfaction with current mentoring. In multivariable analysis, fellows were most satisfied

with current mentoring [3.0 (2.5,3.5)]. Satisfaction was independent of sex or URM status. Characteristics of mentoring which were associated with satisfaction included keeping in touch regarding progress [3.8 (2.8, 5.1)]; advice on professional decisions [2.9 (2.0,4.2)], research [2.7 (2.0,3.8)] and career plans [1.6 (1.0,2.7)]; not abusing power [2.0 (1.5,2.6)]; helping with funding [2.0 (1.4,2.7)]; and taking a respectful attitude toward work [1.8 (1.3,2.5)]. For women, additional factors associated with satisfaction were: accounting for cultural issues [2.5 (1.2,5.6)], discussing pitfalls in academic growth [2.3 (1.1,4.9)] and developing communication skills [2.1 (1.1,3.9)]. No additional factors were identified among URM.

CONCLUSION: Most junior faculty and trainees have had a mentoring relationship while at HMS, and 63% report satisfaction with mentorship. Women and URM had equal levels of satisfaction as their male and majority counterparts. Fellows were most likely to be satisfied with mentoring. Keeping in close touch with mentors and receiving useful advice were key attributes of satisfaction. Factors identified in this study may help guide the development of mentorship programs.

PEER AND SELF EVALUATION IN A FIRST-YEAR INTERVIEWING COURSE. DW Rudy, M Fejar, CH Griffith, III, JF Wilson, Internal Medicine; Behavioral Science, University of Kentucky, Lexington, KY

PURPOSE: Peers may provide valuable feedback to medical students learning interviewing skills. However, even in non-grade situations, students may be hesitant to verbally critique their colleagues. We assessed the use of a written anonymous peer- and self-evaluation of videotaped interviews during a first-year interviewing course.

METHODS: Videos of student's interviews with standardized patients were viewed by groups of 5 to 10 students along with faculty preceptors. Students were asked to evaluate their peers and themselves in the areas of knowledge of structure, techniques, and style of the interview using a 15 point scale as well as providing written comments. Faculty completed an identical form. Written comments were classified as either positive or negative. The ratios of positive to negative comments were also assessed.

RESULTS: Because there was little variation in individual responses across the three categories (reliability: self 0.86, faculty 0.79, peer 0.91), these results were averaged and the composite scores analyzed. The written comments were quantified as either positive or negative. * Statistically different between the 3 groups.

CONCLUSION: Peers generally rated their colleagues higher than the students did themselves or did the faculty, and provided very few negative comments, even in a non-graded situation. Peers did give their colleagues a "feedback sandwich" of 4 positive to 1 negative comments, which is what is often suggested as appropriate in the literature. Students were more critical on their own performances and were more likely to give themselves negative rather than positive comments. Peer evaluations using an open-ended format for feedback may be more beneficial than quantitative "grades." Students' self-perceived areas of weakness may be useful for instructors to develop further educational interventions.

	Composite Scores*	Positive Comments*	Negative Comments*	Ratio Pos/Neg
Self	12.05 (1.53)	0.53 (0.85)	1.58 (1.39)	0.34
Peer	12.91 (0.89)	2.44 (0.89)	0.57 (0.40)	4.28
Faculty	12.41 (1.03)	4.82 (3.23)	0.82 (1.21)	5.88

BEYOND CLEVER NIHILISM: TEACHING RESIDENTS EVIDENCE-BASED MEDICINE (EBM). MD Schwartz, AL Kalef, K Mukohara, Div. of Primary Care, NYU Medical Center, New York, NY

PURPOSE: Can clinical epidemiology training turn Primary Care (PC) residents off to EBM as they learn the limits of published evidence, turning them into clever nihilists?

METHODS: We surveyed 2nd year PC residents (PGY2) before and after a 60-hour, 6-week course in EBM using an instrument designed to measure "clever nihilism," self-reported critical appraisal (% of maximum score on 13 items) and electronic searching skills (% of maximum score on 17 items). The 6-item, 4-point nihilism scale, confirmed by factor analysis (with Cronbach alpha = 0.7), addressed distrust of research methods and of applicability of published studies, and difficulties of practicing EBM.

RESULTS: Over the last 2 years, 14 residents and 38 PC faculty were surveyed. Compared with the pre-course survey, after the course residents had a decrease in nihilism while improving their self-reported critical appraisal and electronic searching skills. Compared with PC faculty, residents after the course had higher scores for critical appraisal (85% vs. 71%, $p = 0.002$) and electronic searching skills (83% vs. 68%, $p = 0.001$), but had similar levels of nihilism.

CONCLUSION: PGY2 PC residents completing an intensive course in clinical epidemiology improved their perceived critical appraisal and electronic searching skills without becoming more nihilistic regarding EBM. In fact, nihilism decreased after the course. It remains possible, however, that less intensive interventions could breed clever nihilism among our trainees and long-term impact on use of evidence in practice is unknown.

	Pre-Course	Post-Course	p value
Nihilism (4 = most nihilistic)	2.5	2.2	0.01
Critical Appraisal Skills (out of 100%)	63%	85%	<0.0001
Electronic Searching Skills (out of 100%)	70%	83%	0.0007

COST AWARENESS AMONG MEDICAL RESIDENTS. SJ Shaikh, PH Mehta, NB Patel, Internal Medicine, New York Methodist Hospital, Brooklyn, NY

PURPOSE: Managed care has revolutionized the practice of medicine. Learning cost-effective medicine without compromising the quality of care is indispensable for successful medical practice especially in contemporary managed care era. In teaching hospitals, residents often order many laboratory and radiological tests, prescribe medications, decide length of hospital stay and frequency of follow-ups in clinics, which requires rational, scientific and managed approach for patient care. So we decided to survey cost-awareness of this modalities among residents in our teaching hospital.

METHODS: We conducted a cross-sectional survey of sixty postgraduate (PGY-1, PGY-2 and PGY-3) medical residents at our medical clinic. They were given a multiple choice questionnaire listing cost of twenty four different laboratory and radiological tests, cost of hospital beds, cost of clinic visits and cost of common medications used. The data were analyzed statistically and used to find out their projection and comparing intergroup variability.

RESULTS: Correct costs for any test were chosen only by 1.6% to 35.5% of respondents for twenty four given items. Most of the cost were underestimated except cost for regular bed in hospital per day was overestimated by 12% of its correct cost and cost for aspirin by 600%. Less than 5% of respondents knew the correct cost for common tests like echocardiography (1.6% of residents) and cost of first clinical visit (3.4%). Cost of items like urine analysis, echocardiography and first clinic visit were underestimated by >90%, >75% and >90% of respondents respectively. Highest rate of correct responses was for cost of regular bed (35.5% of responders), chest x-ray (35.5%), MRI of head with contrast (33.8%) and cost of ICU bed per day in hospital (30.5%). There was no statistically significant difference or variation for correct responses between PGY-1 (least experience), PGY-2 and PGY-3 (most experience) residents.

CONCLUSION: This results bring out several serious issues about cost awareness. First, costs estimated by most of the residents in this study were highly inaccurate as compared with the actual costs indicating great lack of knowledge. Second, there was the consistent tendency among residents to underestimate the costs, with few exceptions. Finally, noteworthy conclusion was the absence improvement in cost awareness at higher levels of training (PGY3 or PGY2) where residents are preparing to enter the world of practice. Degree of illiteracy was equivocal but not equal in all areas tested, there was variable and statistically significant difference among extent of correct response among given group. While we do not advocate a financial basis for clinical decisions, it is clear that major educational efforts to rectify this lack of awareness should be seriously considered.

THE ROLE OF HOSPITALISTS IN MEDICAL EDUCATION. JA Shea, Y Wasfi, KJ Kovath, DA Asch, LM Bellini, Medicine, University of Pennsylvania, Philadelphia, PA; Medicine, University of Colorado Health Sciences Center, Denver, CO; Leonard David Institute of Health Economics, University of Pennsylvania; VAMC, Philadelphia, PA

PURPOSE: To determine how many internal medicine residency training programs have hospitalists on staff, describe the role of hospitalists in teaching activities, and define attitudes regarding hospitalists.

METHODS: All program directors of accredited internal medicine training programs were surveyed by mail starting in April 1999 about the presence of hospitalists, attitudes about hospitalists, and participation in teaching activities by hospitalists. The response rate was 57.6%.

RESULTS: There was no apparent response bias. Overall, 50.5% of the programs employed hospitalists. General attitudes about hospitalists were positive. For example, the majority agreed that hospitalists are more familiar with practical aspects of inpatient care (58.4%), patients are satisfied with the inpatient care that they receive from hospitalists (54.0%), and they expect the use of hospitalists to increase over the next few years (85.2%). Most disagreed that hospitalists needed more training beyond IM residency (53.1%). Most also thought that use of hospitalists disrupted continuity of patient care (54.1%). Attitudes were significantly more favorable for program directors that had hospitalists. For the 109 programs with hospitalists, nearly all reported that hospitalists participated in teaching of medical students (80.2%) and residents (84.5%). Other educational activities in which they participated include attending rounds (74.7%); management conferences (53.5%); journal club (48.5%); resident report (58.6%); and curriculum development (55.6%). The majority agreed that hospitalists are viewed as good educators (78.5%), have led to improved housestaff supervision (52.2%), are more accessible to housestaff (62.2%), and are respected (72.3%). They are less certain that hospitalists had an impact on housestaff's consideration of length of stay (37.8%) or costs of tests and procedures (37.4%), or will impact on their future ordering behaviors (24.7%).

CONCLUSION: Half of the medicine programs have hospitalists and nearly all participate in teaching. Attitudes regarding hospitalists are generally favorable though there is not consensus regarding their impact on residents' future behaviors. Given trainees' substantial exposure to hospitalists, there may be a role for faculty development targeted at improved hospitalist teaching and feedback methods. Future studies should examine what, and how, students are taught by hospitalists and whether this teaching differs from that provided by other attendings. Studies should also look in-depth at hospitalists' relative contribution to residents' education and the impact on their behaviors.

ARE OSCEs WORTH THE EFFORT? FACULTY AND STUDENT PERCEPTIONS OF AN OBJECTIVE STRUCTURED CLINICAL EXAMINATION. *SR Simon*, C Hamann, SW Fletcher, Department of Ambulatory Care and Prevention, Harvard Medical School and Harvard Pilgrim Health Care; Department of Medicine, Massachusetts General Hospital, Boston, MA

PURPOSE: To describe the views of faculty members and second-year medical students after completing an Objective Structured Clinical Examination (OSCE) at the end of a year-long course in physical examination and diagnosis.

METHODS: Design: Anonymous written survey administered at the completion of an OSCE in April 1999.

Setting: Harvard Medical School.

OSCE Format: Sixteen 6-minute stations that tested students' abilities in history-taking, physical examination, and differential diagnosis. Each station included 2 minutes for faculty-to-student feedback on performance.

Participants: 130 second-year medical students and 156 faculty members

Measurements: 8 Likert-type scaled items for students and 6 items for faculty members, assessing the OSCE overall, components of the OSCE, and the relevance of the OSCE to the year-long course.

RESULTS: All students and faculty members completed and returned the survey forms; responses were missing for 3% of items. Ninety percent of students agreed that the OSCE was a fair evaluation of the skills that were taught in the course, and 98% agreed that the OSCE helped them to identify skills that needed improvement. Ninety-nine percent of students agreed that the feedback they received from faculty examiners during the OSCE was helpful. Overall, 97% of students agreed that the OSCE was a worthwhile experience, but only 9% thought the OSCE should contribute to their grade in the course. All faculty members agreed that their participation in the OSCE was worthwhile; 70% strongly agreed and 30% agreed. Ninety-one percent of faculty members agreed that the station they supervised was a fair evaluation of students' skills and 56% agreed that their stations were difficult for students. Fifty-seven percent agreed that the OSCE should contribute to students' grade in the course.

CONCLUSION: Faculty members and second-year medical students perceived a feedback-laden OSCE to be a valuable component of a year-long physical diagnosis course. However, most students were resistant to having the OSCE contribute to the grading process of the course.

A CONTROLLED TRIAL OF ALTERNATIVE APPROACHES FOR TEACHING PROCEDURAL SKILLS TO RESIDENTS. *MK Singh*, RD Cebul, I Solti, C Thomas, RS Wigton, Case Western Reserve University at MetroHealth Medical Center, Cleveland, OH; University of Nebraska, Omaha, NE

BACKGROUND AND OBJECTIVES: Proficiency in procedural skills is increasingly important in primary care, yet little is known about how best to teach these skills. We examined the comparative effect of 3 pedagogical approaches on new interns: 1) performance rates of ABIM-required procedures over their first 10 months; and 2) changes in their scores on an objective test of knowledge.

DESIGN, SETTING, SUBJECTS: Controlled (Firm System) trial among all interns in 3 parallel groups at an urban teaching hospital, from 7/98-6/99. In prior work, we have documented equivalent distributions of patients and conditions across firms. **INTERVENTIONS:** In their first 6 months, CDROM interns (N = 8) completed a CD-ROM text and video guide on 7 procedures: arthrocentesis (Arth), paracentesis (Par), thoracentesis (Thor), lumbar puncture (LP), n-g intubation, and ABG and central line placement. CDROM+LEC interns (N = 8) also received lectures on each procedure. CONT interns (N = 10) underwent traditional training and served as the control group. **MAIN MEASURES:** Interns completed an 89-item knowledge test during orientation and again in 6/99. Their performance of 4 procedures (Arth, Par, Thor, and LP) through April, 1999 was determined by administrative (i.e.; lab) data and chart abstraction (to reliably link the MD to the procedure). Cross-group differences in performance rates were adjusted for in-patient months of opportunity to perform these hospital-focused procedures. We also examined cross-group changes (post-pre) in test scores.

MAIN FINDINGS: At baseline, interns across the 3 groups were similar in demographics, previous training and procedural experience, track within the residency, and knowledge. All interns completed both the pre- and post-tests. All groups improved in knowledge from pre- to post-test (mean % correct: pre = 58.0%, post = 65.6%; $p < 0.001$), although there were no cross-group differences in improvement ($p = 0.38$). Of 396 procedures performed during the 10 months, interns (vs. others) performed 72, over 140.5 intern in-patient months (0.51 procedures/MD-month). Intensity of intervention was associated with performance of procedures: rates were 0.61, 0.60, and 0.44 in the CDROM+LEC, CDROM, and CONT groups, respectively. The CD-ROM interns performed procedures at a higher rate than the control group ($p = 0.04$), although the CDROM+LEC interns did not perform more than the CDROM interns ($p = 0.84$).

CONCLUSIONS: The CD-ROM guide is a promising, standardized approach for teaching procedural skills to medical interns. Further investigations should examine timing of the teaching, longer term effects, the incremental value of other teaching, and the impact of its use on appropriateness and outcomes of procedures.

TEACHING PHYSICAL EXAMINATION OF THE HEART AND CIRCULATION: A CONTROLLED TRIAL OF TWO METHODS. *CA Smith*, AS Hart, LS Sadowski, JM Riddle, PM Clarke, PS Ganschow, ED Mason, W Sequeira, Y Wang, Medicine, Cook County Hospital/Rush Medical College; Internal Medicine, Rush Presbyterian St. Lukes Medical Center, Rush Medical College, Chicago, IL

PURPOSE: To determine whether a structured program of bedside teaching can improve the procedural and perceptual skills of medical residents in performing the cardiovascular examination.

METHODS: A firm-based controlled trial was performed comparing three groups. Participants consisted of residents rotating on the inpatient medicine services over a two-month period. The two intervention groups received bedside teaching sessions (three 2-hour sessions over 4 weeks) with identical content but different teaching styles: (1) a traditional bedside teaching method, "demonstration and practice" (n = 26); (2) an innovative method, "collaborative discovery" (n = 24); and (3) a control group, which received usual teaching (n = 25). Outcome measures were scores on procedural and perceptual components of an objective structured clinical examination (OSCE) using two patients. Participants were tested before and after the intervention.

RESULTS: There were no significant differences between groups at baseline. For procedural skills, adjusted mean differences on a scale of 35 were: collaborative discovery vs. control, +3.3 points (95% CI, 0.7 to 5.9; $p = 0.01$) and demonstration and practice vs. control, +3.9 points (95% CI, 1.3 to 6.4; $p = 0.004$). For perceptual skills, adjusted mean differences on a scale of 65 were: collaborative discovery vs. control, +4.2 points (95% CI, 0.6 to 7.8; $p = 0.03$) and demonstration and practice vs. control, +1.2 points (95% CI, -2.3 to +4.7; $p = 0.5$).

CONCLUSION: Both educational interventions improved procedural examination skills for the cardiovascular examination. One intervention, collaborative discovery, also improved perceptual skills.

"ONLINE" VS. "LIVE" LECTURE ON SCREENING IN PRIMARY CARE: A RANDOMIZED TRIAL OF SENIOR MEDICAL STUDENTS. *A Spickard*, N Alrajeh, J Gigante, Internal Medicine; Engineering; Pediatrics, Vanderbilt School of Medicine, Nashville, TN

PURPOSE: To compare the impact of an online lecture versus a live lecture on students' knowledge, time, and satisfaction.

METHODS: Setting: Senior medical student primary care clerkship based at an academic institution with multi-state practice sites.

Design: Prospective randomized trial of students to an identical 90 minute case-based lecture on screening in primary care "online" (audio of the class instructor with power point slides) or "live" from the same instructor in a conference room.

Measures: Survey data of baseline items including career choice, computer skills, access to computers, and attitudes about the use of computers for education were obtained. Improvement in knowledge about and application of screening principles measured by a pre- and a post-lecture exam. Both exams contained a 12-point scale of different multiple-choice and discussion (clinical case scenarios) questions that have been used in the rotation for two years. The post-lecture exam was taken on average 2 weeks after the lecture. A priori, a 25% improvement in pre- to post-exam scores was considered a "clinically significant" improvement in knowledge. Survey data was used to monitor students' time to complete the lecture, satisfaction, and narrative comments about the experience.

RESULTS: Sixty students (30 in each group, mean age 26, 39% female, 98% response rate) completed the study. There were no differences between the groups on all baseline items and on the pre-lecture exam (6.5 on the 12-point scale in both groups). Forty-four percent of the online group and 54% of the live group showed a 25% improvement on the post-lecture exam ($p = 0.4$ for the between-group improvements). Students in the online group saved 53 minutes in completing the lecture (less driving time to access the lecture and less time listening to the lecture). Seventy-eight percent of the online students versus 93% of the live students were satisfied or very satisfied with their experience ($p = 0.1$). Satisfaction was independent of baseline student computer skills and attitudes. Many students offered creative suggestions and some offered a few warnings about integrating online teaching into primary care clerkships.

CONCLUSION: This study demonstrates that senior student clerks improved their knowledge about screening equally by an online and a live lecture of similar material. Students gave various reasons for preferring one format to the other, but overall, students were satisfied with either venue, even considering that the traditional live lecture required almost an additional hour to complete. Further studies of the effectiveness and efficacy of the online lecture will elucidate its role in clinical clerkships.

RELIABILITY OF CLINICAL OBSERVERS' & STANDARDIZED STUDENTS' RATING OF FACULTY RESPONSES TO UNPROFESSIONAL LEARNER BEHAVIOR. *M Srinivasan*, K Lane, R Seshadri, D Litzelman, Indiana University School of Medicine, Indianapolis, IN

PURPOSE: Learner professionalism has become an increasing focus in medical education. Educators have reported that a faculty development course increased their ability to effectively address unprofessional learner behavior (ULB). However, instruments to objectively evaluate faculty response to ULB have not been described.

METHODS: We developed a 22 item instrument of behaviors that educators potentially employ in addressing ULB. 4 standardized students (SS = 4th year medical students) & 7 clinical observers (CO = 2MD, 2RN, 3MSW) trained >10 hours to recognize, categorize & record these behaviors as present or absent. SS trained 6 hrs further to portray consistently provocative, unprofessional behaviors. 20 live objective structured teaching evaluations (OSTE), in which volunteer educators interacted with SS who behaved unprofessionally, were used to assess inter-rater reliability of the CO & SS, using % agreement with the majority response & kappa scores. 10 prerecorded OSTEs were used to assess intra-rater reliability at 0 & 1 month to determine the % answers scored consistently per rater. Questionnaire

items were classified as low [1-<2], moderate [2-<4], or high [4-5] inference behaviors by the developers of the ULB faculty development workshop.

RESULTS: In each 5 minute interval of the 20 live OSTE, raters observed 6 ± 2 educator behaviors in reaction to ULB. Raw agreement was high among all 11 raters ($86 \pm 7\%$), while kappas varied with the inference level of the behavior. For instance, 5 of 6 low inference behaviors had kappas $>.4$; specifically, clarification of words or actions [.66], acknowledging learner's emotions [.63], interrupting learner [.53], repeating disrespectful words [.46], rescheduling discussion time [.43]. Only 1 of 12 moderate inference behaviors (assessing learner's motivating behavior [0.51]) & none of 3 high inference behaviors had kappas $>.4$. Agreement rates within CO & SS were similar. Observed frequency & inter-rater kappa were not statistically associated ($r = .16$, $p = .50$). Based on 10 pre-recorded OSTEs, 7 CO & 2SS rated each behavior identically $86 \pm 0.1\%$ of the time at 1 month.

CONCLUSION: Accurate feedback on faculty interactions in addressing ULB is essential for improved faculty development. Raters are able to rate low inference faculty behaviors more reliably than higher inference behaviors. Overall test-retest agreement using our instrument is high. Traditionally, educators have felt that faculty behaviors in response to ULB were difficult to observe and categorize. Our preliminary evidence suggests that, with extensive training and inclusion of items worded as low inference behaviors, educators' strategies for handling unprofessional learner behavior can be reliably evaluated.

PREDICTORS OF PUBLICATION PRODUCTIVITY AMONG GRADUATES OF PRIMARY CARE RESEARCH FELLOWSHIPS. JF Steiner, P Curtis, BP Lanphear, KO Vu, Medicine, University of Colorado, Denver, CO; Family Medicine, University of North Carolina, Chapel Hill, Chapel Hill, NC; Pediatrics, University of Cincinnati, Cincinnati, OH

PURPOSE: To identify characteristics of primary care research fellows and their training that are associated with productivity in publication early in their careers. **METHODS:** Retrospective cohort study of 1988-97 graduates of 25 National Research Service Award primary care research fellowships funded by the Health Resources and Services Administration. A 40-minute, self-administered survey was sent to participants to assess the characteristics of their training program, current position, and publications (defined as original research reports, editorials, commentaries, and reviews) since fellowship. A current curriculum vitae (CV) was requested to corroborate self-report of publications. Standard bivariate analyses and multiple logistic regression models identified predictors of publishing an average of ≥ 1 paper/year (a common standard for academic promotion) since the end of fellowship training.

RESULTS: $149/215$ program participants (67%) completed the survey. Of them, 111 (76%) returned a CV. Respondents were 39 ± 5 years of age; 51% were male and 84% were white. Clinical disciplines were GIM (35%), family medicine (24%), general pediatrics (31%) and other (10%). Respondents had completed fellowship 4 ± 2 years prior to the survey. They reported publishing a total of 3.2 ± 4.5 first-authored papers and 2.9 ± 4.7 co-authored papers since fellowship; 45 (31%) published an average of ≥ 1 paper/year. The correlation between self report and CV was $r = 0.91$. Predictors of publishing at this rate are shown in the Table. The C-index for the multivariate model was 0.81. Characteristics unrelated to publication productivity included: age, race, years since training, an advanced degree (e.g., MPH), research training model (apprenticeship vs. independent project), and writing a grant during fellowship.

Variable	Bivariate OR (95% CI)	Multivariate OR (95% CI)
Male	4.6 (2.1 to 10.1)	4.2 (1.7 to 10.5)
Spent $\geq 40\%$ of fellowship conducting research	4.5 (2.1 to 9.9)	4.5 (1.8 to 11.3)
Had significant mentor	4.5 (1.5 to 13.9)	4.5 (1.3 to 15.4)
GIM (vs. other discipline)	2.4 (1.2 to 5.0)	1.4 (0.6 to 3.5)

CONCLUSION: The time allocated to research and the presence of a significant mentor during fellowship were associated with publication productivity early in the career of these primary care fellows. These associations may be due to structural characteristics of the training programs or to the choices made by fellows during and after their training. In either case, evaluation of fellowship outcomes may assist in the design of more effective training programs.

PRINCIPLES AND PENS: ATTITUDES AND PRACTICES OF MEDICINE HOUSESTAFF TOWARD PHARMACEUTICAL INDUSTRY PROMOTIONS. MA Steinman, MG Shlipak, SJ McPhee, Dept. of Medicine, University of California, San Francisco; Dept. of Medicine, Veterans Affairs Medical Center, San Francisco, CA

PURPOSE: The involvement of the pharmaceutical industry in residency training programs has engendered much debate, yet little is known about the perceptions and behaviors of housestaff. We studied the attitudes and practices of internal medicine residents toward gifts from the pharmaceutical industry.

METHODS: Confidential surveys were mailed to all first- and second-year medicine residents at a large university-affiliated residency program. Attitudes about nine types of industry promotions (e.g., inexpensive trinkets, dinner lectures) were scored from 1 to 4 on a Likert scale from "very inappropriate" to "very appropriate." Promotions were then grouped by cost and educational value, and summary scores calculated as the mean of the numerical responses within each category. Residents also reported their practices, perceptions of influence, and attitudes about institutional policies.

RESULTS: Ninety percent of residents (105 of 117) completed surveys. We observed a wide variability of attitudes about different types of gifts, ranging from 92% who considered it appropriate to accept a pocket antibiotic guide to 15% who stated it was appropriate to receive luggage. More than half of residents made a distinction (a difference between category summary scores ≥ 1 standard deviation of the overall sample mean) between appropriate and inappropriate promotions based on their cost. On the contrary, only 15% distinguished the appropriateness of promotions by their educational value. Behaviors were often incongruent with attitudes: every resident who considered free pens and conference lunches inappropriate ($n = 18$ and 13 , respectively) had accepted these gifts, and half of those who felt industry-sponsored recreational events were inappropriate ($n = 36$) had either participated in such events or intended to do so. Most respondents (61%) stated that their prescribing practices were not affected by industry promotions or contacts, whereas only 16% believed other physicians were free of influence. Nonetheless, more than two-thirds of residents agreed that it is appropriate for a medical institution to have rules on industry interactions with residents and faculty. **CONCLUSION:** Residents distinguish the appropriateness of industry promotions primarily on their financial, rather than educational, value. While viewing themselves as less susceptible to influence than other physicians, more than two thirds of housestaff support the existence of policies that govern their interactions with pharmaceutical industry representatives. Residency program leaders should address these attitudes in creating educational and policy initiatives about house-staff-industry interactions.

THE ROLE OF MEDICAL SCHOOLS IN SELECTING AND GRADUATING STUDENTS WITH UNPROFESSIONAL CONDUCT. DI Stern, RE Oshel, TP Hofer, JO Woolliscroft, MR Schwarz, Internal Medicine, Ann Arbor VAMC, Ann Arbor, MI; Internal Medicine, University of Michigan; Division of Quality Assurance, Department of Health and Human Services; China Medical Board of New York

PURPOSE: Since the late 1950s, researchers have documented that socialization into the values, attitudes and behaviors of medicine can occur during medical school. Are there differences between medical schools in the degree to which their graduates will behave professionally once in practice? If so, are there differences due to selection (admission variables) or due to socialization (medical school variables)?

METHODS: The National Practitioner Data Bank (NPDB) is a federally-mandated collection of all malpractice payments and certain adverse actions against licensed physicians in the US since September, 1990 (156,369 events). For the purpose of this study, only the 1200 adverse actions for unprofessional conduct were included. Malpractice payments and all other adverse actions (e.g., substance abuse, felony) were excluded. Each action was linked with the medical school from which the physician graduated. Medical school characteristics were obtained from Association of American Medical Colleges resources. We used Poisson regression with estimation of an extra Poisson component of variation to model the variance in the school rates of reporting. An offset representing the number of graduates in practice from each medical school, during the 10 years the NPDB has been in existence, was included in the model.

RESULTS: The rate of unprofessional conduct identified in graduates is approximately two actions per thousand graduates. There is one outlier school, with a rate of nearly one action per fifty graduates (excluded in subsequent analyses). The mean number of reports per medical school is 7.9. The number of reports varies significantly across schools, with a standard deviation of 1.3 counts (95% CI [1.2, 1.5]). The percent of instate admissions per instate applicant explains about 12% of the variance at the school level. Stated another way, a 10% increase in the instate acceptance rate is associated with a 15% increase in the future rate of actions for unprofessional conduct. Other measures (public/private, US News rank, overall selectivity) did not contribute significantly to the model.

CONCLUSION: Medical schools graduate students with varying degrees of future professional conduct. Some schools have no graduates with reported adverse actions for unprofessional conduct, while others have as many as one action per fifty graduates. Some of this effect appears to be related to the admissions process, but much of the variability remains unexplained. Further study is needed to uncover the institutional variables that predict future unprofessional conduct.

HOME CARE TRAINING DURING RESIDENCY: A NATIONAL SURVEY. CM Stoltz, LG Smith, JH Boal, Division of General Internal Medicine, Hospital of the University of Pennsylvania, Philadelphia, PA; Department of Medicine, The Mount Sinai Medical Center, New York, NY

PURPOSE: The aging of the American population and shorter lengths of hospital stay have prompted an increase in the demand for home health care. Internists should be able to recognize the home care needs of their patients, activate these services, and feel comfortable supervising the home health care team. Residency remains one of the only opportunities for formal training in this area and several governing bodies have supported the inclusion of home care education in internal medicine residency curricula. The purpose of this study was to determine the extent to which internal medicine residents in the United States receive instruction in home care during their training. The areas of focus included didactic sessions (lectures) about home care and resident participation in house calls.

METHODS: A national survey of program directors of all accredited internal medicine residencies in the United States was conducted in 1997. Using a four question survey instrument, data were collected regarding the extent to which residency programs provided education via didactic sessions (lectures), house calls,

or both. Demographic information about the residency programs was also collected and analyzed. The chi-square test was used for analysis of nominal variables.

RESULTS: Of the 397 programs surveyed, 312 responded, yielding a response rate of 78.6%. Sixty-eight percent of programs included home care instruction in their residency curriculum, consisting of house call(s), lecture(s), or both. Half of the programs provided opportunities for residents to make house calls with physicians or nurses, but only 25% of all residency programs had a mandatory house call training experience. Less than half (45.5%) of programs included lectures about home care in their curriculum. Of those that did offer lectures, the mean duration was 3.2 hours/year. Residency programs with a primary care track were more likely to include house calls and/or lectures in their curriculum ($X^2 = 7.44$; $p = 0.006$).

CONCLUSION: Nationwide, internal medicine residents receive limited training in home care. In fact, one in three internal medicine residency programs offer no instruction whatsoever in home care. Absence of this curricular piece may leave future internists inadequately prepared to coordinate and supervise their patients' home health care and may leave them out of step with the changing needs of their patients.

MORNING REPORT: AN ANALYSIS OF CURRICULAR CONTENT AND COMPARISON TO NATIONAL GUIDELINES. JM Sweet, LJ Cation, Department of Medicine, Uniformed Services University of the Health Sciences, Bethesda, MD

PURPOSE: To describe the format and educational content of morning report at a successful internal medicine residency and compare the content to recent curriculum guidelines published by The Federated Council for Internal Medicine (FCIM) Task Force on the Internal Medicine Residency Curriculum.

METHODS: A review was performed of all morning reports over three years at a tertiary teaching hospital. Morning report topics were categorized by clinical condition, organ and system competency, and inpatient or outpatient venue. The educational content of morning report was analyzed and compared to the curriculum guidelines created by the FCIM.

RESULTS: Of 583 consecutive morning reports at the end of the three-year study period, 331 (57%) focused on inpatient cases and 252 (43%) focused on outpatient cases. The FCIM Curriculum Task Force organ and system competencies discussed most often were infectious disease (14%) and cardiovascular illness (14%). The organ and system competencies most likely to be discussed in the context of inpatients were infectious disease, cardiovascular illness, and pulmonary medicine. The competencies most likely to be discussed in the context of outpatients were rheumatology, general internal medicine, and endocrinology and metabolism. With the exception of general internal medicine topics, between 60–86% of all organ and system competency focus areas were the subjects of a morning report by the end of the study period.

CONCLUSION: Depending on format and educational content, morning report can be a valuable educational conference and a key component of an internal medicine residencies' overall curriculum. The FCIM report can serve as a tool in creating, implementing, and assessing a program's curriculum.

THE EFFECT OF EARLY CLINICAL EXPERIENCES AND 1ST YEAR MEDICAL SCHOOL TRAINING ON STUDENTS VIEWS OF PHYSICIAN COMPETENCIES. G Talente, D Rudy, D Barnett, C Griffith, J Wilson, Division of General Internal Medicine, University Of Kentucky, Lexington, KY

BACKGROUND: As the new millenium begins, there has been discussion about the competencies new physicians will need to be successful. Another question is how do we integrate new competencies with the traditional physician skills and qualities being taught in medical school? Are we accomplishing this now? We looked at the competencies that first year students reported as important following an externship with a primary care physician early in the first year and again upon completion of the year in order to evaluate the effect of their early training on their views.

METHODS: Ninety seven first year students were asked to answer the question, "The competencies needed to practice medicine in the 21st century are?" The question was first presented to them following a weeklong observation at a community practice in October of the first year. The question was repeated upon completion of the first year. At this time they were also asked if their views of important 21st century competencies had changed. Responses were analyzed by content domain, correlated with demographic data and compared over time.

RESULTS: Similar competencies were reported on initial and follow up questioning: knowledge 48% and 44%, compassion 47% and 50%, communication skills 46% and 48%, life long learning 20% and 13%, application of new technology 28% and 27%, computer skills 13% and 18%, business skills 18% and 7%, and understanding of insurance systems 11% and 7%. Twenty-one students believed their view of important 21st century competencies changed during their first year of school. For these students, communication skill was recognized more frequently after completion of the first year (29% post externship vs. 57% post first year, $p < .018$). Business skills were mentioned less frequently (43% vs. 10%, $p = .005$). There was no correlation between responses and demographic data except for a tendency for students from small towns to find business skills important.

CONCLUSION: Early medical school training and clinical experiences appear to reinforce the traditional competencies of knowledge, compassion and interpersonal skills with much less effect on students' appreciation of less traditional

skills they will need when they enter practice, and even a negative effect on their appreciation of business skills.

THE IMPACT OF AUTOPSIES ON CLINICAL EDUCATION OF RESIDENTS. K Taneda, EJ Boyko, DC Martin, SD Fihn, H Itoh, S Ohbu, T Fukui, Y Matsui, HSR&D Center of Excellence, VA Puget Sound Health Care System; Medicine, University of Washington, Seattle, WA; General Medicine, Kyoto University Hospital, Kyoto; Internal Medicine, St. Luke's International Hospital, Tokyo, Japan

PURPOSE: The autopsy has been recognized as a valuable educational tool that helps residents understand the link between clinical findings and pathology. Although recommended for accreditation of teaching hospitals, the autopsy rate has fallen over time. We analyzed the impact of autopsies on resident clinical education.

METHODS: The data were collected at a 520-bed private teaching hospital in Tokyo from 1993 through 1997. The performance of autopsies at this institution is actively promoted by teaching staff and administration. Participants included 89 1st post-graduate year (PGY) and 88 2nd PGY residents from 4 departments: internal medicine, surgery, obstetrics & gynecology, and pediatrics. The number of autopsies attributable to each resident and results of standardized resident performance evaluations by teaching staff in the same academic year were compared using linear regression analysis. Admission score (AS) is the result of the exam taken by medical students to qualify for residency positions.

RESULTS: Average numbers of autopsies and deaths per resident were 3.6 and 8.2 for the 1st PGY and 5.5 and 12.5 for the 2nd PGY, respectively.

Competence areas evaluated by faculty (1st PGY)	Number of autopsies Slope (p-value)	No. of autopsies adjusted for AS Slope (p-value)
Medical knowledge	-.039 (.207)	-.051 (.106)
Problem solving	-.027 (.406)	-.046 (.166)
Decision making	-.071 (.017)	-.084 (.007)
Overall	-.050 (.088)	-.066 (.034)
Sum of above four (2nd PGY)	-.182 (.089)	-.237 (.035)
Medical knowledge	-.015 (.509)	-.029 (.246)
Problem solving	-.046 (.056)	-.055 (.033)
Decision making	-.026 (.336)	-.032 (.271)
Overall	-.025 (.294)	-.039 (.131)
Sum of above four	-.105 (.250)	-.155 (.110)

CONCLUSION: A negative association existed between the number of autopsies obtained by residents and performance evaluations. It is possible that a higher autopsy rate led to a higher discovery rate of missed diagnoses or incorrect pre-mortem management decisions, thereby leading to lower performance ratings for residents. Although implausible, another interpretation of these data is that higher clinical competence was related to a lower autopsy rate.

EFFICIENT GROUP PROCESS TECHNIQUES IN DEVELOPING AN INPATIENT TEACHING CURRICULUM. KM Taylor, K Welch, SK Yuh, MM Yeh, JM Watt, Internal Medicine, St. Joseph Mercy Hospital; Center for Statistical Consultation and Research, University of Michigan, Ann Arbor, MI

PURPOSE: Effective curriculum development requires extensive time and manpower utilization. The purpose of this study was to develop teaching strategies for our inpatient teaching service using time-efficient qualitative and quantitative group process techniques.

METHODS: The residency review committee for internal medicine requires that 4.5 hours per week of inpatient rounding be dedicated to explicit teaching apart from patient management rounds. Fifteen internal medicine faculty who comprise 85% of the total inpatient rounding schedule responded to an electronic mail (email) inquiry: "What specific teaching strategies would you employ during these 4.5 hours of inpatient teaching time?" Using the Delphi technique the list of ideas were reported in a second email in which the faculty were asked to clarify their ideas and list the strengths and weaknesses. A list of 13 teaching strategies was developed from this iteration. The faculty were then asked to use the Nominal Group Technique to score the strategies by impact and feasibility using a 10 point scale (1-least, 10-most).

RESULTS: Thirteen teaching strategies were compared by impact (I), feasibility (F), and total score (S) of I and F. There is a significant difference between the teaching strategies in terms of their mean scores on I ($p < .001$), F ($p < .001$) and S ($p < .001$). Case based teaching strategies had the highest I and S score. Five teaching strategies were significantly different from case based teaching using the Mean and Standard Deviations of the total score. Methods with the highest impact also had the highest feasibility with six strategies demonstrating a significant correlation using the Pearson Correlation Coefficient. Five strategies were distinguished by significantly higher total scores and correlation coefficients including: case based teaching, team teaching, primary care case discussions, ABIM board review questions, radiology rounds and feedback sessions.

CONCLUSION: Nominal group and Delphi techniques can be used to identify teaching strategies for curriculum development. Our study identified five teaching strategies by these qualitative and quantitative group process techniques which our faculty have implemented in their inpatient teaching sessions. This process

allows all group members to participate in a complex problem solving process in a time-efficient manner.

FELLOWSHIP TRAINING AND CAREER OUTCOMES FOR ACADEMIC PRIMARY CARE PHYSICIANS. JS Taylor, RH Friedman, AS Ash, JL Speckman, MA Moskowitz, Family Medicine; General Internal Medicine, Boston University, Boston, MA

PURPOSE: The generalist disciplines of family medicine, internal medicine, and pediatrics have increasing numbers of academic fellowship positions and programs. We examined the association between fellowship training and academic career outcomes among primary care faculty.

METHODS: In 1995, we conducted a cross-sectional study of 500 full-time academic primary care physicians from a national sample of 24 randomly selected US medical schools. The self-administered questionnaire included items regarding fellowship training. Primary outcomes were grant submissions and funding during the most recent 2 years, career refereed publications, rank (being a full or associate professor), and salary. Secondary outcomes were preparation for and commitment to research, self-perceived research and teaching skills, career aspirations, and job satisfaction. All results were adjusted for age, race, gender, years since first faculty appointment, and institution.

RESULTS: Among study respondents, 234 physicians had completed a fellowship, while 266 had not. Fellowship-trained physicians devoted almost triple the amount of time at their current job to research (17% vs. 6%, $p = 0.0001$), and in the past 2 years, were 4.8 times more likely to have submitted a grant ($p = 0.0001$) and 4.2 times more likely to have had a grant funded ($p = 0.0001$). They were also more likely to have had any refereed publications (OR = 3.8, $p = 0.0001$) and to have a higher rank (OR = 1.9, $p = 0.02$). Fellowship-trained physicians felt more prepared for research ($p = 0.0003$), more committed to research ($p = 0.0001$), had better self-reported research skills ($p = 0.0001$), and were more likely to aspire to full professorship ($p = 0.02$). The groups did not differ with respect to salary, self-perceived teaching skills, or job satisfaction. Among faculty with fellowship experience, those with at least 1 year of research training during fellowship were even more academically productive with grant submissions (OR = 1.9, $p = 0.02$), grant funding (OR = 2.9, $p = 0.0003$), and publications (OR = 2.4, $p = 0.02$). In addition, those with extensive research training were 2.3 times more likely to have a higher rank ($p = 0.03$), although salary remained comparable. All secondary outcomes were significantly better among those with more research training, except for job satisfaction which was equivalent.

CONCLUSION: In our study of academic primary care physicians, fellowship-trained faculty submitted more grants, were funded more often, and published more than their non-fellowship trained peers. They also held higher academic ranks, although salaries were similar. Among faculty with fellowship experience, additional research training as a fellow was associated with higher productivity and rank.

TEACHING MEDICAL STUDENTS THE FOCUSED HISTORY AND PHYSICAL: ARE THE RED FLAGS FLYING? PA Thomas, JH Shatzler, Dept. of Medicine, Johns Hopkins University School of Medicine, Baltimore, MD

PURPOSE: Medical students in ambulatory settings are encouraged to develop focused history and physical examinations of common patient problems. A potential drawback of teaching the focused history and physical to early learners of medicine is that attention to checklists may result in the curtailment of diagnostic decision-making. The purpose of this study was to evaluate the impact of an evidence-based clinical practice guideline approach to low back pain on ambulatory clerks' diagnostic decisions.

METHODS: All 108 students taking a required 4 week ambulatory clerkship in one academic year were included. 80 students assigned to one regional site by residential zip code received the intervention of a 2-hr. small group instruction, using the AHCPR Clinical Practice Guideline for Acute Low Back Pain. 28 students assigned to a second regional site did not receive the intervention and were used as controls. Evaluation consisted of student performance in a standardized patient examination at completion of the clerkship. A low back pain case, developed as a possible pathologic vertebral fracture, contained "red flags" in the duration and quality of the back pain, and the finding of vertebral point tenderness. Data-gathering was assessed during the 15 minute encounter by per cent checklist items done. The student post encounter note was scored for: documentation of 5 red flag items in history, documentation of spinal point tenderness in physical, extending the differential diagnosis beyond regional back pain, and inclusion of further diagnostic testing. Measurement of differences in interventions were analyzed by effect size calculations, $d = \text{effect size}$.

RESULTS: Students taught the clinical practice guideline scored higher on SP checklist performance than controls (mean .64 vs. .60, $d = 0.4$, with no difference in interpersonal scoring between the two groups. Students taught the guideline recorded the same number of red flags in their progress notes, 3.0 vs. 2.7, and 83% reported the finding of point tenderness on examination. Students taught the guideline were less likely however to extend the differential diagnosis of the patients' problem beyond regional back pain (30% vs. 36%), and less likely to consider diagnostic testing at the time of the visit (32% vs. 39%). ³⁴/₁₀₈ students appropriately extended the differential diagnosis. These students had similar SP checklist scores as those who did not (.63 in both groups), but recorded more red flags into their post-encounter notes (3.2 vs. 2.7, $d = 0.4$).

CONCLUSION: The use of an evidence-based clinical practice guideline for low back pain did not improve student identification of a potential high risk low back

pain patient. Students using abbreviated history and physical approaches need to be impressed with the significance of positive findings in their assessments.

RESIDENTS' ATTITUDES AND CHARACTERISTICS ASSOCIATED WITH PHARMACOTHERAPY COST. P Turner, K Galt, E Rich, B Houghton, Creighton Center for Practice Improvement and Outcomes Research; Department of Medicine, Creighton University, Omaha, NE

PURPOSE: Direct-to-consumer and -physician pharmaceutical advertising has increased pressure on physicians to make pharmacotherapy treatment decisions upon marketing rather than the best cost-effective medical treatment option available. The purpose of this study was to assess the association between medical residents' attitudes and characteristics with the cost of prescribing.

METHODS: We surveyed medical residents at Creighton University regarding their attitudes toward pharmaceutical information resource use, knowledge of professional position statements on physician interactions with industry, attitudes towards pharmaceutical representatives, and resident demographics (i.e., gender, ethnicity, USMLE scores, and residency year). Attitudes towards pharmaceutical information resources were assessed by residents' perceived accuracy and use of the Internet/Worldwide Web medical resources, clinical/drug reference manuals, electronic medical text CD-ROMs, peer reviewed research/review journals, and pharmaceutical sales representatives. Knowledge of professional position statements was evaluated with items developed from the literature (i.e., AMA, ACP). Attitudes towards psr's related to the usefulness/accuracy of information and residents' interactions with psr's. The cost of prescribing was measured with three clinical case scenarios of common ambulatory problems (supraspinatus tendinitis, mild hypertension, and acute cystitis). Academic internists and pharmacists with extensive experience in treating each problem served as consultants in developing each case. Four choices of drug treatment for each scenario were offered based on current recommendations of the literature, with widely varying costs but equal efficacy. A relative value index was used to calculate the cost of prescribing for each resident. Bivariate and multivariate (ANOVA with Tukey HSD post hoc comparisons) statistical analyses were used to evaluate the relationship between resident attitudes/characteristics and the cost of pharmacotherapy treatment.

RESULTS: Fifty of the 59 residents completed the survey. Year in residency was significantly associated with prescribing costs ($p = .025$), with 3rd year residents having lower prescribing costs than 1st year residents ($p = .019$). Residents more confident in the accuracy of information in clinical/medical reference manuals were associated with higher prescribing costs ($p = .037$). There were no other significant relations between residents' resource use, knowledge of professional position statements, or attitudes towards psr's associated with the cost of prescribing.

CONCLUSION: The year in residency training and type of pharmaceutical information resource utilized may be associated with prescribing costs.

THE USE OF PSYCHOTROPIC MEDICATIONS AMONG MEDICAL RESIDENT PHYSICIANS. JA Underwood, KA McGarry, Department of Medicine and Psychiatry, Brown University School of Medicine, Providence, RI

PURPOSE: To investigate the self-administration of psychotropic medications among resident physicians for the symptoms of depression, anxiety and sleep disorders.

METHODS: An anonymous survey was distributed to all medical residents at the internal medicine residency program at Rhode Island Hospital. The survey had three separate sections, one each for depression, anxiety and sleep disorders. The residents were asked if they had self-prescribed for the above diagnoses with any prescription or non-prescription medications. The duration of treatment, the acquisition of the medications and whether they were under the care of a clinician was also asked.

RESULTS: Seventy-nine (63%) of the 125 medical residents responded. Of the three disorders, residents self-treated for the symptoms of a sleep disorder most often. Twenty-five residents (32%) used at least one medication for the symptoms of a sleep disorder. Only one resident was prescribed a medication. The most commonly used medication was diphenhydramine, which was most frequently used during night float and emergency room rotations. The majority of the residents used a sleep agent for less than or equal to one month. Thirteen residents (16%) used at least one medication for the symptoms of depression. The most commonly used medications were selective serotonin reuptake inhibitors and St. John's Wort. Only two residents (2.5%) were under the care of a clinician. Three (23%) of those who self-treated for depression took a medication for greater than or equal to one year. The majority of the residents obtained the medications from a sample closet. Five residents (6%) used at least one medication for the symptoms of anxiety. Only one resident (1%) was under the care of a clinician. The most commonly used medications for anxiety were beta-blockers and alcohol, both self-prescribed.

CONCLUSION: This study suggests that medical residents often self-prescribe for sleep disorders, depression and anxiety. The ethical implications of this practice warrant further consideration. The lack of clinician input is concerning considering the possible impairment of a residents' judgment while treating patients.

HOW DO INTERNS PERCEIVE THE PHYSICIAN'S ROLE IN END-OF-LIFE CARE? WA Ury, CS Berkman, FP McKeegney, MG Pignotti, Medicine, Saint Vincents Hospital; Graduate School of Social Services, Fordham University, New York, NY

To assess how interns perceive their role in different aspects of caring for dying patients and their families and how this is related to educational experiences, and

self-perceived skill and comfort in various aspects of end-of-life care. A self-administered questionnaire was distributed during orientation in 1998 and 1999 ($n = 115$, response rate 96.7%). Measures included: 1) the role of the physician in 9 aspects of end of life care; 2) amount of formal education, observation of attending physicians and clinical experience with pain management, giving bad news, and family, prognosis and advance directive discussions; 3) comfort in giving bad news, managing pain, and having discussions about prognosis and advance directives; 4) skill in discussing prognosis and caring for dying patients; and 5) demographic and personal experience with death and dying. A large majority of interns rated pain management (96.5%) and giving bad news (93.9%) as very important to the physician's role. About three quarters rated helping patients with emotional concerns related to illness (79.1%), fears of death (75.7%) and maintaining hope (73.9%) as very important, and fewer thought responding to emotional needs of family members (62.6%) or patient concerns about family well-being (60.0%) was very important. Family bereavement (44.3%) and religious concerns of patients (29.6%) were least likely to be rated as very important. The importance of the types of physician roles was not associated with skill, comfort, educational experiences (formal education or observation), demographics or personal experiences with dying, with the exception of:

Importance of helping with: emotional needs of family	Significant Associations ($p < .05$) # observations of: advance directives ($p = .001$), family discussions ($p = .03$), bad news delivered ($p = .01$)
family bereavement	# observations of family discussions ($p = .001$)
giving bad news	skill in caring for dying patients ($p = .01$)
pain management	skill in caring for dying patients ($p = .05$), comfort in managing pain ($p = .04$)

These findings indicate that physician involvement in emotional concerns of dying patients is not seen as very important by one quarter of young doctors, and that even fewer view attention to family issues as very important. Skill in caring for dying patients was associated with medical (giving bad news and pain management), but not with emotional, spiritual or family concerns that are viewed with great importance by patients. Role modeling (through observation) may be an important factor in how young physicians perceive the physician's role in end-of-life care. Further research is needed to better understand what factors determine a physician's perception of his/her role in end-of-life care.

CAN EDUCATION ABOUT PALLIATIVE CARE IMPROVE THE OPIOID PRESCRIBING PRACTICES OF MEDICAL RESIDENTS? *WA Ury, M Rahn, V Tolentino, J Yoon, DP Sulmasy, Medicine, Saint Vincents Hospital, New York; Medicine, New York Medical College, Valhalla, NY*

PURPOSE: To assess whether a case-based palliative care curriculum improved the opioid prescribing practices of medical residents, we performed a prospective review of the charts of medical residents' as well as a comparison group of neurology and rehabilitative medicine (neuro/rehab) residents.

METHODS: The curriculum consisted of 10 one-hour case-based seminars that addressed attitudes, skills and clinical practice issues. It included two seminars on pain management that emphasized the side effects of meperidine, the need to prevent constipation and the additive effect of nonsteroidal anti-inflammatory drugs (NSAIDs). Consecutive billing records of patients who received an opioid during hospitalization on the medical ($n = 600$) and neuro/rehab ($n = 300$) services from eight month periods before (1/1/97-4/30/97) and after (1/1/99-4/30/99) the implementation of the curriculum on the medical service were reviewed. The data abstraction protocol was validated by standard chart review of a random subsample of 45 charts (concordance = 89.9%). Three outcomes were measured: 1) percent opioid orders for Meperidine in non-gastrointestinal (non-GI) patients (excluded because of a longstanding belief that meperidine is the opioid of choice in biliary disease); 2) percent opioid orders accompanied by a bowel regimen; and 3) percent opioid orders accompanied by adjuvant nonsteroidal anti-inflammatory drugs.

RESULTS: The results represent a preliminary analysis of 80.3% ($n = 723$) of the total number ($n = 900$) of charts calculated by Fisher's exact method (power of 80% to detect a 10% difference at $\alpha = .05$, two-tailed).

CONCLUSION: Although preliminary, this data suggests that a palliative care curriculum can change the opioid prescribing practices of medical residents, over and above secular trends among all house officers. Larger samples and further research are needed to develop a better understanding of these findings and to learn how palliative care education can improve patient care.

	Medicine		Neurology and Rehabilitative Medicine	
	% change from pre- to post period	significance	% change from pre- to post-period	significance
Received				
Meperidine (non-GI) patients	-5.1	$p = .034$	-0.5	$p = .919$
Bowel Regimen	12.6	$p = .002$	15.8	$p = .024$
Non-steroidal	4.2	$p = .16$	5.9	$p = .10$

EVALUATION OF A BLOCK CURRICULUM IN EVIDENCE-BASED MEDICINE FOR INTERNAL MEDICINE RESIDENTS. *EW Vogel, MCP Hahnemann University, Philadelphia, PA*

PURPOSE: To evaluate the implementation of an evidence-based medicine (EBM) curriculum for internal medicine housestaff in a large, urban, university-based residency program.

METHODS: An eight-week EBM curriculum was developed, and implemented over one year with second-year residents ($N = 42$) in their two four-week ambulatory care blocks. Topics covered in the curriculum included developing clinical questions from patient encounters, using the best quality information sources, searching MEDLINE and the Internet for information, and critical appraisal of various types of journal articles. Residents also gave presentations in the last week of the curriculum. The curriculum was evaluated with written pretests and posttests for each of the four week blocks, pretest and posttest MEDLINE and Internet searches based on clinical scenarios, as well as an independent MEDLINE search at the end of the curriculum. Also, a survey of EBM attitudes and abilities was administered to the participants at the end of the academic year, and responses were compared to a baseline survey that was completed at the beginning of the year. Third year residents who were one year ahead of the curriculum participants also took the survey, and served as a quasi-control group.

RESULTS: For each of the four-week blocks, residents' mean written test scores increased significantly (pretest 49%, to posttest 86% for block one, 42% to 57% for block two, $p < 0.001$ for each), and residents' computer searching skills improved markedly. Residents felt the curriculum was useful for them, and were generally enthusiastic about participating. At the end of the year, participants reported increased use of research journals and the Internet, compared to use before the course and to the quasi-control group. Participants also had more confidence in an internist's ability to use the literature to answer patient care questions instead of relying on consultants, compared to baseline. They reported increased skills in assessing the relevance and validity of journal articles. There was no improvement in their skills in choosing which journal articles are important to read. Residents demonstrated only a slight improvement in their ability to detect design flaws in a simulated journal article.

CONCLUSION: A block curriculum in EBM for internal medicine residents resulted in improved short-term knowledge, MEDLINE and Internet searching skills, and overall self-reported attitudes toward the use of research literature in patient care. Although residents reported improved skills in critical appraisal at the end of the year, there was less improvement in objective measures in this area. The curriculum content and design can potentially be adapted for other programs that are considering formally teaching EBM to their residents.

PATTERNS OF PARTICIPATION IN A FACULTY DEVELOPMENT WORKSHOP. *EW Vogel, DG Smith, RS Eisenstaedt, Medicine, MCP Hahnemann School of Medicine, Philadelphia; Medicine, Abington Memorial Hospital, Abington; Medicine, Temple University School of Medicine, Philadelphia, PA*

PURPOSE: We sought to identify patterns of participation in a faculty development workshop (FDW) and to elicit attitudes that might explain why faculty chose to participate.

METHODS: All 93 members of the department of medicine at a northeastern urban medical school were invited to attend a FDW. Demographic data were compiled for participants (P) and compared to data for those who did not participate (NP). Teaching reputation (TR), strong, average, or weak, was estimated by a formula using 3 years of survey data from medicine residents. A self administered questionnaire asked faculty to assess the relative importance of various work responsibilities and was used to compare P versus NP. Finally, a second survey was sent to NP asking them to rate various explanations for their decision. The FDW consisted of a 2 hour noon session held in a convenient location on 6 consecutive Wednesdays so that faculty could choose a preferred date. The workshop was led by a senior faculty member experienced in FD.

RESULTS: Forty-eight (52%) of 93 faculty attended the workshop; all completed their survey. Of the 45 NP, 19 (42%) returned surveys. Demographic data showed no difference between NP who returned surveys and those who did not. P were more likely to be women ($p < .05$), more likely to be assistant professor or lower in rank ($\chi^2 = 9.313$, $p < .01$), and more likely to be generalists than sub-specialists ($p < .01$). There was no relationship to TR. There was no difference in how priorities were assigned to teaching, patient care, and administrative responsibilities, but NP ranked research a more important responsibility than P ($p < .05$). NP ranked "too busy" as the most important reason for their decision (mean = 4.75, 5 = extremely important, 1 = not at all important), versus "inconvenient time" (3.81), "not enough advance notice" (2.81), "unaware of opportunity" (2.67), "would not have improved my skills" (2.00).

CONCLUSION: This study suggests that faculty who choose to avail themselves of FDW opportunities are more likely to be women, generalists, of lower academic rank, and less involved in research. Limitations of this study include the imprecision in measuring teaching skills and teaching reputation, and the single institution sample source and single FDW/FDW leader offered. These observations, if confirmed, should be considered in both the design of future FDW programs and in efforts to analyze the impact of FDW on outcome measures such as ratings of teacher effectiveness or ability of FDW to modify learner behavior or enhance learner knowledge.

THE EFFECT OF AN ITERATIVE THINKING CURRICULUM ON PRESENTATION SKILL. JG Wiese, L Tierney, Jr., Dept. of Internal Medicine, Univ. of California at San Francisco, San Francisco, CA

PURPOSE: This study objectively evaluates a curriculum designed to improve the spoken case presentation.

METHODS: In a prospective, blinded study, we evaluated the effects of a curriculum to improve presentation skills. Sixty-four third-year medical students at three different hospitals were videotaped presenting a standardized case before and after the medical clerkship. Thirty-two students received a four-week curriculum, while the other thirty-two received the standard ward instruction. Two independent evaluators blinded to the case and control assignments reviewed the before and after videotapes. Each tape was scored on a 1 to 30 scale for quality, time, and accuracy. Students and their respective residents and attendings were surveyed as to the students' communication, history taking, and physical examination skills.

RESULTS: The average time of the spoken case presentation at the beginning of the clerkship was similar in both groups (intervention group: 7 min. 59 sec.; control group: 8 min. 10 sec.; $p > 0.50$). When compared to the control group, students who received the curriculum reduced their presentation time by 45 percent (intervention: 5 min. 12 sec.; control: 9 min. 29 sec.; $p < 0.001$). The quality of presentations improved in both groups, but was significant only in the intervention group (intervention quality score: 17.1 to 26.2 $p < 0.001$; control 20.3 to 22.7; $p = 0.06$). Students taught the curriculum, but not those who received standard instruction, reported increased abilities in communication, history taking, and physical examination skill (performance composite index; intervention: 17.5 to 22.0 $p = 0.0001$; control: 18.4 to 19.0; $p = 0.21$). Attending physicians and residents reported improvement in both groups, although the greater improvement was seen in the students given the curriculum. The reviewers' assessment's had excellent agreement ($r = 0.87$).

CONCLUSION: A curriculum based upon early diagnosis generation and the iterative thought process improves the efficiency and quality of presentations, and augments overall ward performance.

MEDICAL STUDENTS' PERCEPTIONS OF ADEQUACY OF THE MEDICAL CURRICULUM REGARDING GENDER-BASED MEDICINE. L Williams, D Grigsby, J Wilson, D Kwolek, College of Medicine, University of Kentucky, Lexington, KY

PURPOSE: The College of Graduate Medical Education has stated that many women receive incomplete and poorly coordinated care for their routine and comprehensive health concerns, in part due to deficiencies in physician education. This study reports on a survey designed to determine the perceptions of medical students on the extent to which gender differences in medical topics are addressed by the curriculum.

METHODS: The survey, a 17-item Likert scale with 4 = excellent, 3 = good, 2 = adequate, 1 = poor, and 0 = not at all, was given to 180 first-, second-, and third-year students, with a response rate of 85%. Data were analyzed with multiple regression using the general linear model, with variables measuring the adequacy of the curricula as dependent variables and with the level of medical education and gender as independent variables.

RESULTS: As the students' level of education increased, students rated the coverage of sex differences in given health issues more favorably (year 2 and 3 were significantly different from year 1, $p < .0006$, $.0001$, respectively). For example, year 1 students gave an average rating of 1.59 for "female and male differences in preventive health behavior" whereas year 2 and year 3 students gave ratings of 2.00 and 2.68, respectively ($p < .0001$). In addition, male students rated the coverage more favorably than female students, on average ($p < .0048$). For example, the average male rating for coverage of "female and male differences in non-reproductive physiology" was 2.28 whereas female students gave an average rating of 1.72 ($p < .001$). Likewise, men rated coverage of "signs of physical abuse" as 2.24 whereas women rated it as 1.55 ($p < .0028$).

CONCLUSION: Although the average rating for coverage of health issues increased as the students' level of education increased, the overall average rating is less than adequate, and the coverage of no health issues is rated as good or excellent. Changes in the medical school curriculum at our institution are needed to better educate future physicians in the gender-based care of patients.

DO WE STILL NEED A TEXTBOOK FOR MEDICAL STUDENTS ROTATIONS?: A SURVEY OF ELECTRONICALLY AVAILABLE MEDICAL LITERATURE. JL Wofford, MM Wofford, PH Pride, WP Moran, JR Kimberly.

PURPOSE: We explored the on-line availability of full-text review articles suitable for accompanying the SGIM-CDIM third year clerkship curriculum in ambulatory Internal Medicine.

METHODS: We performed a Medline search of review articles (limited to English language, publication type—review and years—1998–99) for 7 syndrome and 3 disease SGIM-CDIM curriculum topics. Two faculty members determined citation relevance to third year clerkship students and practicing generalists based on the citation title, journal, and number of references.

RESULTS: 336 relevant citations were identified from 175 journals (131 for medical students) with only 11 journals with >4 relevant citations. For the 3 disease curricular topics the number of relevant citations were 108 (diabetes), 40 (depression), 88 (COPD) compared with a mean number of citations of 13.9 (range 6–34) for the 7 syndrome curricular topics. 244 were judged relevant to ambulatory In-

ternal Medicine students and 92 to practicing generalists but not to students. 25.0% (61/244) of citations relevant to medical students were syndrome-oriented compared with 42.4% (39/92) of citations relevant to practicing generalists. 28.9% (97/336) of all relevant citations were available in electronic, full-text version (free - 5.1%, 17/336); through Ovid™ - 20.2%, 68/336%; or through MDConsult™ - 6.0%, 20/336%. Only 29.5% (72/244) of citations relevant to students were electronically available as full-text articles.

CONCLUSION: Disease-oriented review articles are more numerous than syndrome-oriented review articles. Electronically available or not, the small number of syndrome-oriented review articles may make it difficult to avoid the use of textbooks for an ambulatory internal medicine clerkship. The contrast in availability of full-text reviews for clinical syndromes versus those for diseases likely reflects the specialization within academic medical centers.

ELECTRONIC PUBLICATIONS: CURRENT STATUS AND PHYSICIAN OPINION. SM Wright, WT-C Tseng, K Kolodner, Departments of Medicine, Johns Hopkins Bayview Medical Center, Baltimore, MD; Southern California Permanente Medical Group, La Mesa, CA; Johns Hopkins University School of Medicine, Baltimore, MD

PURPOSE: Many medical journals are currently offering physicians the option to subscribe electronically, allowing the reader to access publications via the Internet. In addition to assessing the current status of electronic publications, a study was conducted to better understand the opinions and attitudes of physicians regarding electronic journals.

METHODS: Data collection from books and websites was performed to determine which biomedical journals offer a full-text version electronically and the specific characteristics of this format. Also, a three-page questionnaire was delivered to all active physicians at the Johns Hopkins Bayview Medical Center that explored features of electronic journals that make them distinct from printed journals.

RESULTS: Hundreds of biomedical journals offer full-text versions of their printed journal electronically. Rarely, access is offered free of charge. Most journals with an electronic format offer access either: a) as a bonus to those who subscribe to the print version, b) at an additional cost to those who subscribe to the print version, or c) as a stand-alone subscription. An increasing number of electronic journals offer the electronic format in a form that is identical to the printed version (as a down-loadable pdf file). Of the 314 physicians surveyed, 255 (81.2%) returned a completed questionnaire. The mean age of the respondents was 40.7 years, 65% were male, 20% were housestaff, and 63% were from the Department of Medicine. Twenty-six percent of respondents thought that electronic journals would lower the quality of the medical literature and 25% believed that the prestige of authorship would be lessened. Seventy to eighty percent of physicians responded that electronic journals would: i. decrease clutter in their offices and homes, ii. be more environmentally friendly than the current system, iii. make it easier to locate research reports that they had read, iv. offer the benefit of linkage to other related articles. Seventy-four percent of physicians were concerned about losing the convenience of being able to read their printed journal anywhere. In multivariate analyses, the independent variables that were associated with positive attitudes towards various aspects of electronic journals included: female gender, faculty member (vs. house officer), fewer publications, better computer skills, and more frequent internet use, [all $p < 0.05$].

CONCLUSION: The availability of full-text online versions of biomedical journals is steadily increasing. Physicians responded favorably to the many potential values and applications of electronic publications but are most concerned with the loss of convenience that printed journals offer.

A NATIONAL SURVEY OF INTERNAL MEDICINE RESIDENCY PROGRAM DIRECTORS REGARDING "PROBLEM RESIDENTS". DC Yao, SM Wright, Division of General Internal Medicine, Johns Hopkins Bayview Medical Center, Baltimore, MD

PURPOSE: Limited data are available on "problem resident", which is defined by the American Board of Internal Medicine as "someone who demonstrates a significant enough problem to require intervention." A questionnaire was developed to survey internal medicine program directors to gain an increased understanding of the issues related to "problem residents", including their identification, contribution of underlying causes, and management.

METHODS: We recently mailed a confidential survey to all 406 accredited internal medicine residency programs in the United States, and recruitment is ongoing.

RESULTS: Preliminary results from the 247 (61%) surveys already returned show that the prevalence, during academic year 1998 to 1999, of "problem residents" in internal medicine residency programs ranged from 0% to 39%, with the mean of 7% (SD = 5.8). Only 6% of program directors reported an absence of "problem residents" in their training program. Program directors stated that "problem residents" often (half of the time or more frequently) have insufficient medical knowledge (49%), poor clinical judgment (47%), and inefficient use of time (44%). Stress and depression are the most frequent underlying issues for "problem residents" (41% & 25%); cognitive dysfunction and substance abuse are felt to be infrequent. Sixty percent of program directors agree or strongly agree that it has been difficult to convince "problem residents" of their deficiencies because of lack of honest and accurate written evaluations. Furthermore, while 76% of program directors reported that attending physicians made verbal comments that helped them recognize "problem residents", only 41% included these comments in their written evaluations ($p < 0.01$). Chief residents and program directors most frequently discover the "problem residents" (83% and 77%); "problem residents" rarely come forward and identify themselves. In managing "problem residents", program direc-

tors feel that more frequent feedback sessions (63%) and an assigned mentor for structured supervision (55%) are the two most helpful interventions. Measures such as regular meetings with the program director (94%) and timely evaluations from attending-physicians (85%) are helpful to prevent deterioration in the performance of the "problem residents". When asked about their comfort level in the identification of "problem residents", diagnosing underlying causes, and management of the issues involved, program directors had decreasing levels of confidence with each progressive step in the process (89%, 76% and 58%, respectively).

CONCLUSION: The vast majority of internal medicine residency programs have "problem residents". The presenting characteristics and underlying issues are diverse, suggesting that an individualized approach is needed for each case.

ASSESSING AUSCULTATION SKILLS OF RESIDENTS AND CARDIOLOGISTS USING THE OBJECTIVE STRUCTURED CLINICAL EXAM. EB York, MR Thomas, KG Thomas, DM Dupras, HJ Schultz, JT Li, Internal Medicine, Mayo Clinic, Rochester, NY

PURPOSE: The objective structured clinical examination (OSCE) has been used as a tool to evaluate clinical competency. The external validity and degree to which an OSCE can be used to assess cardiac auscultation skills, however, is unclear. The purpose of this study is to objectively assess the OSCE as a tool to evaluate the cardiac auscultation skills of first year internal medicine residents, using academic cardiologists as a gold standard.

METHODS: Our OSCE consisted of four cardiac auscultation stations utilizing patients with the following diagnoses: 1) Aortic Regurgitation (AR1), 2) Aortic Stenosis (AS), 3) Aortic Regurgitation (AR2) and 4) Aortic Regurgitation plus Aortic Stenosis (AR/AS). The results of 42 first year categorical internal medicine residents and 6 practicing academic cardiologists were compared. Testing was performed on two consecutive days. The same patients were used for each station and both groups received the same instruction. Grading was performed in a blinded fashion.

RESULTS: Residents significantly identified aortic regurgitation or aortic stenosis alone better than combined aortic regurgitation/stenosis. (AR1 81% correct vs. AR/AS 12% correct, $p < 0.0002$, AS 83% correct vs. AR/AS 12% correct, $p < 0.0005$, and AR2 40% correct vs. AR/AS 12% correct, $p = 0.0008$). Cardiologists performed significantly better than residents in diagnosing aortic regurgitation plus aortic stenosis (Cardiologists 83% correct vs. Residents 12% correct, $p = 0.008$). In addition, cardiologists agreed with standard selection technique and with each other 100% on 3 stations (AR1, AS, AR2) and 83% on the fourth (AR/AS).

CONCLUSION: Results suggest that the cardiac auscultation OSCE is an externally valid tool and that patient selection is important. Specific diagnoses such as aortic stenosis/regurgitation alone or in combination, should be considered in the design to produce the desired degree of difficulty.

DO MEDICAL RESIDENTS SCREEN FOR DOMESTIC VIOLENCE IN WOMEN THEY SEE IN A CONSULTATIVE SETTING? C Zlotnick, RO Powrie, E Joyner, S Nygaard, K Rosene-Montella, Brown University School of Medicine, Providence, RI

BACKGROUND: Medical providers are ideally situated to identify and intervene on behalf of women who are experiencing domestic violence. Although, awareness of the frequency and importance of domestic violence (DV) has been increasing in the community, few recent studies have examined how this increased awareness has impacted domestic violence screening by physicians.

METHODS: Copies of all consults performed by 50 different third year internal medicine residents participating in an ambulatory and inpatient general medicine consult service at a large woman's hospital between 1995 and 1999 were obtained. 6-16 consults were randomly selected for each resident. These consults were reviewed for the presence of somatic complaints that are believed to be markers of potential DV. These consults were then reviewed for documentation of inquiries regarding whether the woman was or had been a victim of DV.

RESULTS: In total 674 consults were reviewed. 92.7% met our criteria for being appropriate for domestic violence screening. 50% of these patients were pregnant. In only 9 of the 674 consults reviewed was the presence of absence of DV noted. In none of these cases was DV present.

CONCLUSION: Third year residents on consultative services at our center are not documenting DV screening as part of their evaluation of women with somatic complaints that have been associated with DV. Determining why this is so and how best to address it should be a priority for individuals interested in improving the health of women.

PATIENT-CENTERED RESEARCH

A META-ANALYSIS OF EFFECTS OF GARLIC ON SERUM LIPID PROFILES. RT Ackermann, CD Mulrow, G Ramirez, University of Texas Health Science Center, San Antonio, TX

PURPOSE: To assess whether oral garlic supplementation improves serum lipid profiles in human subjects.

METHODS: We searched 1800 English and non-English citations through December 1999 using 11 electronic databases (e.g., MEDLINE, EMBASE, the Cochrane Library, NAPRALERT), article bibliographies, manufacturers, and experts. Search terms included: aglio, allium sativum, garlic, garlic extract, garlic oil, knoblauch, kwall, kyolic, and 17 additional related chemical terms. Selection criteria were randomized trials of at least 4 weeks duration that measured lipids and compared garlic with placebo, no garlic, or another active agent. Two independent physicians abstracted data from trials. Lipid outcomes were pooled using standardized mean differences.

RESULTS: Of 43 trials meeting selection criteria, 1 had clearly adequate randomization processes; 33 used double-blind designs; and 4 had >20% drop-outs. Trial interventions included standardized dehydrated garlic preparations ($n = 22$), aged-garlic extract ($n = 4$), garlic oils ($n = 6$), raw garlic ($n = 1$), various non-standardized dried preparations ($n = 8$), and combination products ($n = 2$). Pooled analyses of placebo controlled trials showed that garlic preparations significantly reduced total cholesterol by 9.9 mg/dl (CI 1.7 to 18.2) at one month and by 20.4 mg/dl (95% CI 12.4 to 28.5) at 3 months. Combining only those studies that evaluated standardized dried preparations showed total cholesterol reductions of 14.1 mg/dl (CI 1.7 to 18.2) at 1 month and 27.4 mg/dl (CI 18.2 to 36.5) at 3 months. Six trials had six-month outcomes; pooled analyses showed no significant reductions in total cholesterol. Patterns of pooled analyses of triglyceride and low density lipoprotein cholesterol paralleled total cholesterol analyses, but were not statistically significant. No significant changes in high density lipoprotein cholesterol levels were found.

CONCLUSION: Randomized trials consistently show that garlic leads to modest short-term reductions in total cholesterol. Data regarding long-term effectiveness, superiority of particular preparations, and relevance to actual clinical outcomes are inconclusive. Unclear randomization procedures and inadequate blinding limit applicability of these findings. Additional trials of longer duration are warranted that better control for these and other methodological shortcomings.

ANTIBIOTIC PROPHYLAXIS FOR ENDOCARDITIS PREVENTION: A DECISION AND COST-EFFECTIVENESS ANALYSIS. Z Agha, RP Lofgren, JV VanRuiswyk, General Internal Medicine, Medical College of Wisconsin, Milwaukee, WI

PURPOSE: Antibiotic prophylaxis for bacterial endocarditis (BE) is recommended by the American Heart Association for patients undergoing invasive dental procedures. The cost-effectiveness of this strategy is not clear.

METHODS: We compare antibiotic prophylaxis versus no-prophylaxis for a hypothetical cohort of patients (age 42 years) with native heart valves undergoing an invasive dental procedure. A standard cost-effectiveness analysis from a societal perspective is conducted. Effectiveness is measured as quality-adjusted life years (QALYs) saved by the prophylaxis program. The model included direct and indirect medical care costs and cost of productivity losses. All future costs and benefits were discounted at 3%. Values for model parameters were extracted from published literature. Estimates for antibiotic efficacy were derived by conducting a meta-analysis of available data. All parameters were subjected to extensive sensitivity analysis.

RESULTS: Under base-case assumptions, antibiotic prophylaxis (amoxicillin 2 gm po) for 10 million patients undergoing a dental procedure would prevent 33 cases of BE and 5 deaths, while 9 lives would be lost due to fatal antibiotic side effect. As a result there would be net loss of 4 lives but a net saving of 289 QALY due to cases of BE prevented. The incremental cost-effectiveness (CE) ratio for prophylaxis was \$245,549/QALY. The analysis was most sensitive to changes in estimates of antibiotic efficacy. If antibiotic efficacy is raised from 30% (base-case) to 64% efficacy (best-case), the CE ratio is \$20,323/QALY. The percentage of BE that is attributable to dental procedures also changed the results. If this percentage is raised from 11% (base-case) to 29% (best-case), the CE ratio is \$50,418/QALY. Similarly, the population incidence of BE affected the results. If incidence is raised from 38 cases/million (base-case) to 99 cases/million (best-case), the CE ratio is \$49,943/QALY. Prophylaxis with clarithromycin 0.5 gm po had similar results to amoxicillin, but ampicillin iv was not cost effective and led to a net loss of life & QALY due to higher incidence of fatal side effects.

CONCLUSION: Antibiotic prophylaxis is not cost-effective as a public health policy. In our analysis if antibiotic efficacy is assumed to be 53% or higher, then prophylaxis can be considered as cost-effective, i.e., CE ratio less than \$50,000/QALY. Due to the lack of good human data on antibiotic efficacy we recommend that decision for prophylaxis be individualized. Patients that are highest risk for endocarditis, such as patients with prosthetic valves, cyanotic congenital heart defect, and prior endocarditis should be candidates for prophylaxis. Patient risk for antibiotic side effect should also be evaluated.

SMOKERS ELIGIBLE FOR A CLINICAL TRIAL: CORRELATES OF NOT RETURNING FOR RANDOMIZATION. JS Ahluwalia, K Richter, N Thomas, M Mayo, K Resnicow, University of Kansas School of Medicine, Kansas City, Kansas

Recruitment and retention of minority participants into clinical trials is an important component of research and often the rate-limiting step towards successful completion. Participants who are eligible for, and express interest in, enrolling into a trial will sometimes not return for their randomization visit. Factors associated with failure to return for randomization are largely unknown.

We compared 287 eligible African American (AA) smokers who enrolled, but did not return for randomization (NR), to 500 AA smokers who returned and were randomized (R) to participate in a study comparing culturally sensitive educational

materials to usual care materials for smoking cessation AAs. The 500 participants who returned for randomization received brief counseling and 8 weeks of transdermal nicotine patches free of charge. An analysis was conducted on variables potentially associated with not returning for randomization. As shown below, those not returning for randomization ($n = 287$) were significantly younger, were more likely to be male, smoked fewer cigarettes, were less likely to have been told to quit in the past year, and less likely on planning to quit in the next 30 days. In addition, they were less educated, had greater heavy drinking, were more likely to have been proactively recruited (participant was approached), and were less likely to have their own transportation.

Variables	NR	R	p-value
Age (yrs)	39.8	42.9	<0.001
Male	48.1	39.8	0.024
Cigs per day	18.1	19.8	0.018
Initiation of Smoking	17.1	7.5	.275
Lower tar	44.3	61.2	<0.001
Plan to Quit	78.4	91.6	<0.001
Told to quit in past year	53.3	67.6	<0.001
Heavy Drinking	42.9	25.8	<0.001
≥HS Education	61.3	75.6	<0.001
>6 th grade Literacy	59.2	71.8	<0.001
Proactively recruited	76.9	34.8	<0.001
Regular Care	31.7	54.0	<0.001
Health Insured	25.5	35.6	0.007
Have their own transportation	36.9	49.2	<0.001

* continuous variables expressed as means and categorical as percent (%)

Potential participants who were eligible for randomization, but did not return, differed in a number of ways from those who did return for randomization. Better understanding of these factors may allow researchers to target recruitment efforts, potentially resulting in enhanced accrual and retention, and therefore, generalizability.

WHOLE GRAIN CONSUMPTION AND RISK OF TYPE 2 DIABETES MELLITUS AMONG MEN.

WS Akbarov, MR Singer, RC Ellison, Fellow of General Internal Medicine, Boston Medical Center and Boston University School of Medicine; Chief Nutritionist, Section of Preventive Medicine and Epidemiology; Chief, Section of Preventive Medicine and Epidemiology, Boston University School of Medicine, Boston, MA

PURPOSE: To estimate the effect of specific dietary variables, especially whole grain consumption and glycemic index (GI), on the development of Type 2 diabetes mellitus (DM).

METHODS: We used the Framingham Heart Study dietary assessment data collected in 1966-69 from men, 45-65 years of age, who had no diabetes at baseline. Diabetes was defined as fasting blood glucose >140 mg/dL, physician diagnosis of DM, or use of oral hypoglycemic agents. Glycemic index was defined as the ranking of a food based on its glycemic response compared with a standard reference food. We used Cox proportional hazard model to examine the relation between dietary variables and risk of DM.

RESULTS: Among 732 participants during 20-year follow up, 81 incident cases of DM were documented. Most dietary variables showed no relation with the development of DM. Of 133 participants who consumed any whole grains, 7 (5.3%) developed DM, versus 74 (12.4%) of 599 participants who did not consume whole grains. After adjusting for age, alcohol intake, Body Mass Index (BMI), smoking and total calories, the relative risk (RR) of DM for whole grain consumers was 0.45% (95% CI, 0.20-0.98). GI and other dietary variables (total carbohydrates, total fat, total fruits and vegetables) showed no relation to DM. Comparing the highest to lowest quartile of GI, the adjusted relative risk of diabetes was 0.97 (95% CI, 0.83-1.13). The only other lifestyle factor significantly affecting the risk of diabetes was BMI, RR = 1.19 (95% CI, 1.14-1.25) for each unit increase in baseline BMI.

CONCLUSION: Our study indicates an inverse relation between whole grain consumption and subsequent development of DM. We found no support for the hypothesis that high glycemic index increases the risk of DM, however, our results further support the direct relation between increased BMI and DM.

COST-ANALYSIS OF SCREENING PRIOR TO HEPATITIS A VACCINATION. JA Andrioli, Section of General Medicine, Saint Vincents Medical Center, New York, NY

PURPOSE: To analyze whether universal or selective antibody screening prior to Hepatitis A vaccination leads to decreased costs by avoiding unnecessary vaccination of persons already possessing immunity. To develop a selective screening algorithm by identifying patient characteristics or risk factors whose presence predicts likely possession of HAV immunity.

METHODS: This cross-sectional study included 134 adult patients who met ACIP criteria for receiving the Hepatitis A vaccine. Each participant's risk factors for prior exposure to Hepatitis A were assessed by physician questionnaire. Variables included age, sex, immigration status, race/ethnicity, sexual orientation and unsafe practices, number of past sexual partners, history of other sexually transmitted diseases or illicit drug use as well as several other possible factors. Hepatitis A immune status was then determined by testing for HAV IgG antibody. Once data

collection was complete, decision analysis was utilized to develop a selective screening algorithm. Finally, simple cost-analysis using average local vaccine and lab costs was performed using three vaccination strategies: universal vaccination without screening, universal screening with vaccination in those lacking immunity, and selective screening using the proposed algorithm.

RESULTS: Of the 134 participants, positive HAV IgG antibodies demonstrating immunity were found in 34 (25%). Of all the variables assessed, only three were associated with significantly higher rates of immunity. These characteristics, which include being an immigrant to the United States, being age forty or older, or having a history of either HIV or Hepatitis B, had rates of immunity of 53% (RR 3.3, 95% CI 1.9-5.7), 42% (RR 2.1, 95% CI 1.2-3.6) and 43% (RR 2.3, 95% CI 1.3-4.1), respectively. Analysis of the 100 individuals remaining when immigrants were excluded again showed that significantly higher rates of immunity are only found in individuals age forty or older 43% (RR 5.6, 95% CI 2.3-13.7) or in those who have a history of HIV or Hepatitis B 32% (RR 3.3, 95% CI 1.4-8.0). Cost-analysis comparing the three vaccination strategies demonstrated that cost reductions of up to 10% could be achieved using the selective vaccination algorithm.

CONCLUSION: Selective screening for immunity prior to Hepatitis A vaccination leads to decreased costs by avoiding unnecessary vaccination of those already possessing immunity. Individuals who should be screened include immigrants from high-risk countries, persons age forty or older, and those with a history of either HIV or hepatitis B.

ROLE OF PRAYER IN THE TREATMENT OF ARTHRITIS: IS THERE AN ETHNIC DIFFERENCE IN PERCEPTION? DC Ang, SA Ibrahim, CJ Burant, LA Siminoff, K Kwok, Medicine, Louis Stokes Dept. of VAMC; University Hospitals of Cleveland; Case Western Reserve University, Cleveland, OH

PURPOSE: Observed differences in health care utilization between African-Americans (AA) and Whites may reflect differences in health-related attitudes/beliefs. Previous reports have suggested that religious beliefs may impact health care seeking behavior. In a cohort of older male veterans with chronic knee and/or hip pain, we examined ethnic differences in patients' perceptions of the role of prayer in the management of their arthritis.

METHODS: A cohort of 285 patients aged 50 or older (45% AA and white 55%) with moderate to severe knee and/or hip pain of more than six months duration were surveyed. To assess patients' perceptions of the role of prayer in their own management of arthritis, we used a component of Efficacy of Specific Treatments Scale (Bill-Harvey) "Do you consider prayer to be very helpful; somewhat helpful; or not at all helpful" and whether they have actually tried prayer as a form of therapy for arthritis. An assessment of religion included religious denomination as well as 4 items to assess religiosity (i.e., frequency of praying, religious service attendance, importance of religion, and self-perception of religiosity). Other measures we examined included severity of their arthritis (Lequesne Scale); Arthritis-Specific Functional Health Status (WOMAC); Comorbidity (Charlson); and Fatalism (Powe).

RESULTS: AA and white patients in this cohort were comparable with respect to age (67 ± 9 vs. 68 ± 9 , $p = 0.53$), Lequesne score (mean 11 vs. 12, $p = 0.09$) and WOMAC score (mean 17 vs. 18, $P = 0.25$) and Fatalism (mean score, 1.98 vs. 2.04; $P = 0.37$). AA reported higher Religiosity (Mean score 76.5 vs. 70.21, $P = 0.005$) and were more likely to perceive prayer as "helpful" in the management of arthritis compared to whites (crude odds ratio 3.70, 95% CI 2.32 to 5.90). This difference persisted after controlling for severity of arthritis, Fatalism, religious denomination and other demographic, clinical and psychosocial covariates (adjusted odds ratio 2.44, 95% CI 1.27 to 4.70). Furthermore, we found that compared to whites, AAs were more likely to have actually tried prayer as a form of therapy for their arthritis (74% vs 55%, $P = 0.002$) and were more likely to be Baptist (60% vs 11% $P = 0.000$).

CONCLUSION: In this cohort of AA and white male veterans with moderate to severe chronic knee and/or hip pain, AA patients were more likely to perceive prayer as "helpful" in management of arthritis and to have tried prayer as a form of therapy compared to whites. These ethnic differences in patients' perceptions of the role of prayer in the management of arthritis may explain disparities in health care utilization.

ADHERENCE WITH ANTIRETROVIRAL THERAPY IN HIV-INFECTED DRUG USERS: HOW DOES SELF-REPORT COMPARE WITH ELECTRONIC MONITORS? JH Arnsen, RW Grant, PA Demas, MN Gourevitch, D Buono, H Farzadegan, EE Schoenbaum, Montefiore Medical Center, Bronx, NY; Johns Hopkins School of Hygiene and Public Health, Baltimore, MD

PURPOSE: To compare self-reported (SR) and electronically monitored (MEMS) adherence with antiretroviral therapy, and to describe the relationship between adherence and viral load.

METHODS: Patients (pts) were recruited from an ongoing longitudinal study of HIV-infected current and former drug users. Over a median study period of 170 days, adherence with all antiretrovirals (mean = 3 drugs per pt) was assessed by SR and MEMS at each of 6 monthly interviews. HIV viral load was also assessed monthly.

RESULTS: To date, 303 SR estimates and 281 MEMS estimates have been obtained from 62 pts. Pts are 39% female, 58% Hispanic, and 87% unemployed; their mean age is 43 yrs. 31% of pts reported cocaine use and 37% reported heroin use during the study period. The median length of HIV infection was 7 yrs, and the median CD4 count was 324 cells/ml. 81% of pts were currently taking a

protease inhibitor. Mean SR adherence (% adherence = doses taken/doses prescribed $\times$ 100) for the week preceding each interview was 77%; mean MEMS adherence for the same time period was 53% ($p < 0.0001$). Though SR and MEMS were correlated ($r = 0.05$, $p < 0.01$), SR adherence was overestimated by a mean of 33% (range = 0–93%). Both SR and MEMS were significantly associated with viral load, but viral load was undetectable in 85% of pts with $>90\%$ MEMS adherence compared to only 58% of pts with $>90\%$ SR adherence. Similarly, viral load was undetectable in 57% of pts with 80–90% MEMS adherence, compared to only 29% of pts with 80–90% SR adherence. Further analysis of MEMS data revealed that pts took no medications at all on 38% of monitored days; that pts took the correct number of medication doses on only 37% of monitored days; and that pts took all medication doses within 25% of the correct dosing interval on only 25% of monitored days. Factors associated with poor MEMS adherence included female gender (mean % adherence = 31% for women v. 61% for men, $p < 0.05$), and active cocaine use (mean % adherence = 38% for active users v. 60% for former users, $p < 0.05$). Active heroin use, and use of alcohol at least several times per week, were also associated with poor MEMS adherence ($p = 0.1$).

CONCLUSION: Self-reported adherence overestimates electronically monitored (MEMS) adherence, and high self-reported adherence is less predictive of virologic suppression than high MEMS adherence. A substantial proportion of patients are unable to adhere to correct dosing intervals, though they may take the correct number of doses. Treatment of substance abuse, particularly cocaine abuse, may increase adherence in HIV-infected drug users. More research is needed to understand the relationship between poor adherence and female gender.

INCIDENCE OF DEEP VEIN THROMBOSIS IN PATIENTS WITH INDETERMINATE STUDIES FOR PULMONARY EMBOLISM. A Arshad, S Klein, D Chandoy, Medicine; Radiology, Westchester Medical Center, Valhalla, NY

PURPOSE: The accurate diagnosis of acute pulmonary embolism is an important clinical problem. The estimated incidence in the United States is 300,000–600,000 and when untreated the mortality exceeds 30%. Spiral CT is a relatively recent non invasive test that makes it possible to depict contrast enhanced pulmonary arteries down to the segmental and sub segmental levels and is increasingly being used to diagnose PE. The utility of the Ventilation Perfusion Scan (V/Q scan) in making a diagnosis of PE has already been established based on several multicenter studies including the PLOPED study. However 20–30% of patients with non high probability V/Q scans may in fact have PE based on PLOPED data. It is critical to identify these patients with indeterminate diagnostic V/Q or CT scans and multiple tests including contrast venography, serial non invasive doppler testing for DVT, pulmonary angiograms or blood tests to detect coagulation activation have been recommended.

METHODS: In a retrospective review from 1/1/99 to 6/30/99 a total of 168 adult patients who underwent either a V/Q scan or Spiral CT for a clinical suspicion of PE were studied. Results of lower extremity Doppler studies were obtained for patients falling in the low to intermediate probability groups and further analyzed.

RESULTS: On review we found of 168 patients undergoing V/Q or CT 63 (37.5%) had a normal result, 20 (12%) had a high probability result and 85 patients (50.5%) undergoing a diagnostic study to establish PE had a low to intermediate probability result. 5 people went on to have diagnostic pulmonary angiograms of which 4 were negative for PE. Of the 80 patients who did not have angiograms 71 had dopplers of the lower extremities of which 18 were positive and 53 were negative for DVT. We found 25% of patients with indeterminate studies had evidence for DVT prompting anticoagulation.

CONCLUSION: Venous thromboembolic disease remains the third most common cardiovascular disease and one of the leading causes of sudden death in the US. Many patients with intermediate probability studies performed for a clinical suspicion of PE may have a concomitant DVT and this allows appropriate treatment for PE without its definitive diagnosis. In our Study we found that about 25% of patients with indeterminate studies had DVTs and went onto anticoagulation.

CARE OF PATIENTS WHO REQUIRE MECHANICAL VENTILATION AFTER HEMATOPOIETIC STEM CELL TRANSPLANTATION: AN ASSESSMENT OF MORTALITY PREDICTION RULES. PB Bach, JS Groeger, Epidemiology and Biostatistics; Anesthesiology and Critical Care, Memorial Sloan-Kettering Cancer Center, New York, NY

PURPOSE: Patients who require mechanical ventilation (MV) after hematopoietic stem cell transplantation (HSCT) have a high probability of death. We conducted a study to determine the accuracy of previously published characteristics said to predict mortality of these patients, and to estimate the long term survival of patients who recover from this condition.

METHODS: Prospective evaluation of predictive characteristics on a multi-center inception cohort of 226 HSCT patients requiring mechanical ventilation. Estimation of predictive power of characteristics using (Bayesian) combination of prior probability of mortality generated from a systematic review of the literature and likelihood ratios of predictors as determined from the prospective multi-center cohort.

RESULTS: Expected mortality of patients requiring mechanical ventilation after HSCT is in the range of 82% to 96% based on our review of 14 published studies. We assessed ten predictive models that were statistically significantly associated with death in the original studies. Of these, only the presence of combined hepatic and renal dysfunction was strongly associated with death, with a post-test probability ranging from 99% to 100% (Table contains sample analyses). Other reported

predictive characteristics were not useful. Of patients who recover, six month survival ranged from 25% to 88%.

Predictor	Prevalence	Prior probability of death	Likelihood ratio	Post-test probability of death
Age $\geq$ 40	64%	82%–96%	0.92	80%–95%
Vasopressor use	23%	82%–96%	1.29	88%–98%
Hepatic and renal injury	25%	82%–96%	3.34 to ∞	99%–100%
$>$ 4 days ventilation	38%	82%–96%	1.58	92%–99%
BMT vs. PBST	85%	82%–96%	1.35	89%–99%
APACHE II $\geq$ 29	22%	82%–96%	1.21	86%–98%

CONCLUSION: Expected mortality of patients requiring mechanical ventilation after HSCT is high, ranging from 82% to 96%. However, survival for those who recover is substantial. Of models available for identifying which of these patients are likely to die, the presence of combined hepatic and renal dysfunction is the only one associated with a very high probability of death (99% to 100%). The presence of this characteristic may be sufficient to justify withdrawing life sustaining care from patients with this condition.

RACIAL AND ETHNIC DIFFERENCES IN HYPERTENSION MEDICATION USE. DW Baker, JJ Sudano, Center for Health Care Policy and Research, Case Western Reserve University and MetroHealth Medical Center, Cleveland, OH

PURPOSE: To determine whether there are significant differences in medication use for hypertension (HTN) between non-Hispanic whites, African-Americans, and Hispanics after adjusting for socioeconomic status and other confounding variables.

METHODS: This analysis used the publicly-released database from the 1992 Health and Retirement Study (HRS), a national probability sample of community-dwelling adults age 51–61 years old with a 82% participation rate. The HRS obtained information on income, education, health insurance coverage, health status, and health behaviors. The population for this study consisted of age-eligible participants who responded "yes" to the question: "Have you ever been told that you have hypertension or high blood pressure?" The dependent variable was self-reported current medication use for HTN (yes/no). The main independent variable was participants' self-reported race and ethnicity. Only whites, blacks, and "Hispanics" were analyzed. The "Hispanic" group was heterogeneous with 54.8% Mexican-American, 9.9% Puerto Rican, 12.5% Cuban, and 22.9% other. Differences in HTN medication use by race and ethnicity was analyzed with chi-square tests. Logistic regression was used to analyze the independent relationship between race, ethnicity and HTN medication use after adjusting for demographics, socioeconomic status, health status, health insurance, and health risk behaviors. All analyses were adjusted for sampling weights and the complex survey design using the survey modules in STATA.

RESULTS: A total of 2450 whites, 939 blacks, and 345 Hispanics reported a history of hypertension. Of these, 63.6% of whites, 72.6% of blacks, and 52.5% of Hispanics reported that they were currently using HTN medications ($p < 0.001$). The lower rate of HTN medication use among Hispanics was consistent regardless of gender, insurance coverage, or health status. Compared to whites, the adjusted odds ratio for medication use for Hispanics was 0.59 (95% CI = 0.45–0.76) and 1.58 (95% CI = 1.26–1.97) for blacks. Other variables associated with HTN medication use included older age, female gender, worse self-reported health, and higher body mass index. Individuals who were uninsured or had ≥ 2 positive responses on the CAGE were less likely to take HTN medication. The results were similar for Mexican-Americans, Puerto Ricans, and Cubans, and there were no differences in HTN medication use according to the language of interview (Spanish vs English).

CONCLUSION: Differences in HTN medication use between whites, blacks and Hispanics, particularly the markedly lower rates among Hispanics, are not explained by differences in demographics, socioeconomic status, insurance coverage, health status, or health behaviors. Additional interventions targeted at physicians and the Hispanic community are needed to control HTN for this growing population.

RISK OF MAJOR DECLINE IN SELF-REPORTED HEALTH AMONG UNINSURED ADULTS IN LATE MIDDLE-AGE. DW Baker, JJ Sudano, EA Borawski, A Dor, J Albert, Center for Health Care Policy and Research, Case Western Reserve University and MetroHealth Medical Center, Cleveland, Ohio

PURPOSE: To determine whether people age 45–61 years old with ≥ 1 episode of being uninsured between 1992–96 are more likely to have a major decline in self-reported health than individuals who had private insurance throughout this period.

METHODS: This study used data from the Health and Retirement Survey (HRS), a nationally-representative longitudinal project initiated in 1992. Participants are interviewed every 2 years to obtain information about income, health insurance coverage, chronic diseases, self-reported health, physical difficulties, and health behaviors. We defined a major decline in self-reported overall health between 1992 and 1996 as a change from a) excellent/very good/good fair or poor, or b) a change from fair poor. Logistic regression was used to determine whether individuals with ≥ 1 episode without health insurance were more likely to have a major decline in self-reported overall health after adjusting for age, gender, race, marital

status, employment, income, years of school completed, chronic health conditions, 1992 self-reported health, smoking, alcohol use, and obesity.

RESULTS: Of 12,652 participants, 9241 had private insurance (PVT) and 1980 were uninsured (UNS) in 1992; 1431 people with public insurance or unknown insurance status were excluded. A total of 11.2% were lost to follow-up and 3.1% died, leaving 9698 people for analysis. Loss to follow-up and mortality rates were similar for PVT and UNS. Of those with PVT at baseline, 526 (6.5%) became uninsured by the time of their 1994 or 1996 interview. A total of 2160 people had ≥ 1 episode without health insurance from 1992–96, and 7538 were continuously insured. Between 1992–96, 19.4% of people with ≥ 1 episode without insurance had a major health decline compared to 8.8% of the continuously insured ($p < 0.001$). The risk of a major decline was similar regardless of the number of episodes without insurance (1, 2, or 3) or when the episode occurred (i.e., 1992, 1994, or 1996). After adjusting for the covariates listed above, the odds ratio for a major health decline was 1.64 (95% CI 1.39 to 1.95) for people with ≥ 1 episode without health insurance compared to those who were continuously insured.

CONCLUSION: Periods without health insurance independently increase the risk of a major decline in self-reported health for adults age 45–61. Transitions between private insurance and being uninsured are common in this group; cross-sectional studies therefore underestimate the number of older Americans who are at risk from being uninsured. Changes in health or social policies that affect the number of uninsured adults age 45–61 may greatly affect the number of elderly in poor health and lead to higher Medicare costs as this cohort reaches retirement age.

QUANTIFYING DIAGNOSTIC PERFORMANCE OF BIOMARKERS TO DIAGNOSE ACUTE MYOCARDIAL INFARCTION IN THE EMERGENCY DEPARTMENT (ED). *EM Balk, JP Ioannidis, J Lau, Division of Clinical Care Research, New England Medical Center, Boston, MA; Department of Hygiene and Epidemiology, University of Ioannina School of Medicine, Ioannina, Greece*

PURPOSE: Acute myocardial infarction (AMI) is the leading cause of death in the United States. Much research has been focused on the early diagnosis and treatment of AMI. We performed meta-analyses to assess the accuracy of various biomarkers for diagnosing AMI in the ED.

METHODS: We conducted a systematic review of the English-language literature published between 1966 and January 1999. We accepted prospective and retrospective studies that enrolled adult patients presenting to the ED with symptoms suggestive of acute cardiac ischemia and included diagnostic evaluation of biomarkers for AMI while in the ED setting. The diagnostic reference standard, AMI, was generally defined by WHO criteria. Meta-analyses were conducted to quantify diagnostic test performance by generating summary receiver-operating characteristics (ROC) curves and by independently combining sensitivity and specificity using a random effects model.

RESULTS: The following table presents the numbers of studies and patients eligible for meta-analysis of specific biomarkers to diagnose AMI in the ED, the ranges of AMI prevalence in eligible studies, and the summary sensitivities and specificities. Despite the large heterogeneity of inclusion criteria (note the large AMI prevalence ranges) and generally moderate to poor study quality, test performances were generally consistent and were not associated with AMI prevalence, inclusion criteria, or test threshold.

Biomarker	# Studies (Subjects)	AMI Prevalence ¹ , %	Sensitivity, % (95% CI) ²	Specificity, % (95% CI) ²
		%	%	%
CK	Presentation	12 (3,195)	(7–44)	37 (31–44)
CK	Serial	2 (786)	(26–43)	(69–99) ³
CK-MB	Presentation	18 (6,268)	(6–42)	44 (38–49)
CK-MB	Serial	14 (11,777)	(1–53)	84 (75–90)
Myoglobin	Presentation	13 (2,048)	(12–53)	46 (39–54)
Myoglobin	Serial	4 (642)	(23–37)	93 (84–97)
Troponin T	Presentation	5 (1,413)	(6–53)	47 (35–60)
Troponin T	Serial	2 (1,440)	(5–6)	(79–94) ³

¹Range from studies.

²95% confidence interval. ³Range of values from studies.

CONCLUSION: Single measurements of biomarkers at presentation to the ED have inadequate sensitivity for AMI although they have high specificity. Serial measurements greatly increase the sensitivity for AMI while maintaining their excellent specificity. For serial biomarkers to achieve a high sensitivity testing needed to be done to at least 4 hours (h) after presentation for CK-MB, to 2–4 h for Mb, and to 6 h for Tp T.

IS PERCEPTION OF BREAST CANCER RISK RELATED TO HEALTH AND PSYCHOLOGICAL STATUS? *MB Barton, S Moore, JD Allen, KE Emmons, SW Fletcher, Department of Ambulatory Care and Prevention, Harvard Pilgrim Health Care and Harvard Medical School; Adult Oncology, Dana-Farber Cancer Institute and Harvard School of Public Health, Boston, MA*

PURPOSE: Many women overestimate their risk of breast cancer. The impact of health status and psychological status on risk estimates has not been studied.

METHODS: A survey was mailed between April and December 1999 to women 40 years of age and older enrolled in a large managed care plan. All women were

scheduled to have screening mammograms in the coming month, and had no history of breast cancer. We asked them to estimate their risk for breast cancer using standardized questions, including: "Imagine 100 women exactly like you: how many would develop breast cancer some day?" Psychological and health status were assessed with the Hopkins Symptom Checklist Anxiety and Depression subscales, and the SF-12.

RESULTS: We mailed 447 surveys and had a response rate of 56%; 227 eligible women returned completed surveys. Mean age of respondents was 55.5 yrs (range 41–86 yrs). Race was 82% Caucasian, 6% African American, 4% Hispanic, 2% Asian and 5% other. 33% reported a family history of breast cancer. Women's estimates of their lifetime risk of breast cancer ranged from 0 to 90%; the most frequent response was 10%, and $\frac{1}{2}$ estimated their risk between 0 and 20%. $\frac{1}{10}$ th of the women estimated their lifetime risk of breast cancer at 45% or higher. Health status was significantly associated with perceived risk (see table, $p = 0.05$).

Health Status	Perceived Breast Cancer Risk (mean %)
Excellent	15
Very good	25
Good	24
Fair	29
Poor	33

Neither psychological status nor age was related to perceived breast cancer risk. Twice as many non-white as Caucasian women reported their risk at $\geq 45\%$ (20% versus 10%, $p = 0.03$). Women with a history of breast cancer in a first-degree relative, and those with lower educational attainment estimated their risk of breast cancer higher than did other women.

CONCLUSION: Women participating in mammography screening in a managed care setting estimated their own risk of breast cancer to be between 0 and 90%; the most common estimate was near the population average risk. In contrast with previous studies, age was not significantly associated with perceived risk. Non-white women and those with poor health, relatives with breast cancer, and lower education gave higher estimates than others.

SMOKING STATUS AND DEPRESSIVE SYMPTOMS IN THE KENTUCKY HEALTH SURVEY. *PE BASS, JF Wilson, CH Griffith, University of Kentucky (UK), Lexington, KY*

BACKGROUND: Cigarette smoking is associated with lower education and socioeconomic status and smokers are more likely to have a history of major depression than are non-smokers. The purpose of this study was to examine differences in health related quality of life among smokers, former smokers, and non-smokers.

METHODS: The data are drawn from the 1997 Kentucky Health Survey administered by the UK Survey Research Center. The survey included 677 Kentuckians who were 18 years of age or older. Participants were contacted during October 1997 by telephone using a random digit dialing method. The response rate was 56% of all eligible respondents. The margin of error $\pm 4\%$ at the 95% confidence level. 92% of those surveyed were white and 5% were African-American; 56% of respondents were female; 45% had some education beyond high school and 18% did not complete high school. By self-report, 30% resided in a rural community and the median income was \$30,000. A mental health score as well as role functioning, social functioning, physical functioning, pain perception, and health outlook were calculated using the MOS-20 format and compared to self reported smoking status. The effects of self-reported smoking status on these MOS-20 scales were examined using multiple regression analysis controlling for age, sex, education, and rural status.

RESULTS: 29.8% were current smokers; 24.1% were former smokers; and 45.8% were non-smokers. 30% of current smokers and 29% of former smokers began smoking before the age of 15. In comparison to those who never smoked, current smokers were younger, less well-educated, and predominantly urban dwellers. Current smokers had worse mental health scores (71.7 vs. 75.8 vs. 76.9 $p = .01$) when compared to former and non-smokers. Current smokers also had significantly poorer health outlook (62.9 vs. 65.1 vs. 69.1 $p = .03$) compared to former and non-smokers. No significant interactions were found when comparing smoking status to gender or rural living. Decrements in health related quality of life scores were demonstrated for persons initiating smoking before the age of 15 ($p < .05$ for all scales). These decrements ($p < .05$) are more pronounced for females who were teen smokers than for males in all of the scales except for physical functioning. In the mental health scale, the decremental difference between those who initiated smoking before the age of 15 was three for males and 14 for females. Similarly the decremental difference in health outlook was 9 for males and 23 for females.

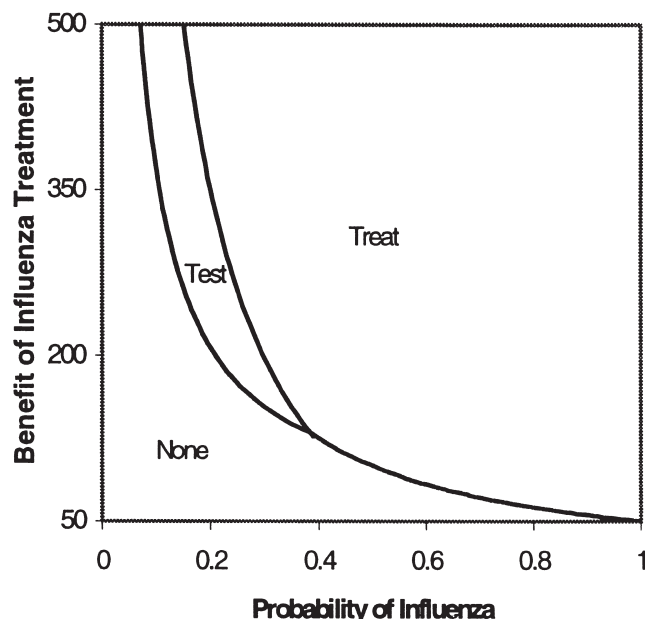
CONCLUSION: Current smokers report more depressive symptoms and poorer health outlook than non-smokers. Former smokers are more similar to those who never smoked in health related outlook than they are to current smokers. Regardless of current smoking status, people who initiate smoking before the age of 15 have a poorer health related quality of life. Females who initiate smoking before age 15 appear to have a much poorer health related quality of life than their male counterparts.

UTILIZING INNOVATIONS IN INFLUENZA MANAGEMENT: THE CLINICAL AND ECONOMIC TRADEOFFS OF DIAGNOSTIC AND TREATMENT STRATEGIES. *SG Blittz, P Cram, M Chernew, AM Fendrick, Internal Medicine, University of Michigan Health System, Ann Arbor, MI*

PURPOSE: Influenza and other viral respiratory diseases are common conditions with significant clinical and economic burden. In the absence of effective treatment, the ability to distinguish influenza from influenza-like illness (ILI) is not critical. The recent introduction of neuraminidase inhibitors (NI), well-tolerated drugs that can shorten the course of influenza, may alter our approach to this disease. The clinical and economic tradeoffs of reserving treatment for patients with confirmed influenza compared with treating patients with febrile respiratory illnesses who may have influenza merits exploration.

METHODS: We created a decision analytic model to determine the incremental cost of true influenza treated with NI for patients presenting with ILI. Three strategies were evaluated over the range of probability of true influenza infection: 1) no testing or treatment (NONE); 2) all patients undergo rapid testing for influenza virus (sensitivity .70, specificity .95, cost \$20) and NI (cost \$50) prescribed for individuals with a positive test (TEST); and 3) all patients receive NI (TREAT). The cost of adverse effects of therapy was included.

RESULTS: The figure illustrates which strategy is preferred at any given probability of influenza and assigned benefit of NI therapy for patients with true influenza. Areas above the curves, designated by a specific strategy name, demonstrate where the magnitude of benefit of NI therapy exceeds incremental cost of treating an additional patient with true influenza. The domain where TEST is the preferred strategy is very narrow and does not substantially broaden with improvement of test sensitivity.



CONCLUSION: For patients with ILI, the decision to prescribe NI only for patients with documented influenza or for those with a compatible clinical picture should be driven by the likelihood of influenza and the value that the physician and patient place on the opportunity to ameliorate the disease. Efforts to improve the predictive value of symptoms for diagnosing true influenza will promote optimal utilization of new therapy.

CARING FOR THE URBAN HOMEBOUND PATIENT: A PROSPECTIVE COHORT STUDY. J. Bogal, L. Adler, E. Halm, R. Villongco, S. Mun, D. Muller, Department of Medicine; Department of Geriatrics, Mount Sinai School of Medicine, New York, NY

PURPOSE: Physicians, having moved over the past century from the home into the office, nursing home, and acute care hospital, have left many patients who are unable to come to the doctor's office cut off from care. We undertook a study of patients enrolled in a primary care program for homebound persons who have difficulty accessing traditional outpatient services.

METHODS: We studied a prospective cohort of 602 consecutive patients enrolled in the Mount Sinai Visiting Doctors program, which serves patients throughout Manhattan, from July, 1997 to October, 1999. Doctors and nurse practitioners performed a detailed, standardized intake during the initial home visit which included data on demographics, clinical problems, and mental and functional status.

RESULTS: Mean age of the patients was 81.7 years and ranged from 25 to 110. Thirty-six percent were White, 33% were Latino, and 30% were African American. Most patients had Medicare (84%), 52% had Medicaid and 6% were uninsured. Half of all patients were widowed, and 27% lived alone. The most common reasons for referral to the program were; inability to leave the home (68%), terminal illness (7%), refusal to go to the doctor (7%), and noncompliance with care (4%). Other reasons included caregiver stress and frequent hospitalization. Functional status was significantly limited in most patients. Forty percent of patients were bed-bound and an additional 39% of patients were unable to ambulate without assistance. Fifty percent of patients had a Karnofsky score of 40 or less, indicating dis-

ability requiring special care and assistance. Most common problems were difficulty walking (80%), incontinence (61%), hypertension (47%), dementia (39%), osteoarthritis (28%), diabetes (23%) constipation (22%), poor vision (19%), poor hearing (17%), pain (17%), caregiver stress (11%) and insomnia (11%). Among the 117 patients (19%) who are able to walk nearly all had psychiatric illness. Fifty-one percent had dementia, 21% had a major depressive illness, 8% had anxiety, and 5% had psychosis. Another 6% of patients were blind.

CONCLUSION: Most patients referred to our homebound program were elderly, had multiple comorbidities and severe functional limitations. A subset of patients, however, had psychiatric or psychosocial reasons for being homebound. The serious functional limitation in this cohort underscores the need for programs to be developed to meet the needs of this very vulnerable, disabled, but often unseen, population.

MEDIASTINAL ADENOPATHY AS AN UNUSUAL PRESENTATION OF REACTIVATION TUBERCULOSIS. R. Borra, V. Nagarajan, Internal Medicine, New York Methodist Hospital, Brooklyn, NY

PURPOSE: 1) Recognition of Mediastinal adenopathy as an unusual presentation of Reactivation tuberculosis.

2) CT chest and gallium scans are sensitive to recognize reactivation TB.

3) Utilization of Polymerase Chain Reaction (PCR) for rapid confirmation of tuberculosis with species identification including multi-drug resistant(MDR) strains to enable early diagnosis and control measures.

METHODS: 34 year old Asian Indian man presented with low grade fever, general weakness, myalgias, sore throat and dry cough which was worse on extension of the neck for 3 months. He was PPD +ve 2 yrs ago but did not take prophylaxis. He was afebrile without lymphadenopathy and the examination of heart, lungs, abdomen, extremities, joints was normal.

RESULTS: Lab data : WBC 6600/ul; Neutrophils 67%, Lymphocytes 17%, Monocytes 10%, Hb/Hct 12.6/36.8; Platelets 222000/ul; ESR = 74 mm/hr. Chemistry results and chest X-ray were normal. Gallium scan showed increased uptake in mediastinum. CT chest revealed multiple mediastinal adenopathy, 2 x 2 cm, in retrocaval, precarinal, and subcarinal spaces with a 5 mm nodular density of the left upper lobe of the lung without hilar adenopathy. Biopsy of the mediastinal lymph node showed confluent necrotizing and non-necrotizing granulomas. Repeat smear from the lymph node was +ve for acid fast bacillus, although culture was -ve. The diagnosis of tuberculosis was made by DNA probe. He responded to standard anti-tuberculous treatment.

CONCLUSION: Mediastinal lymphadenopathy is an unusual presentation of reactivation tuberculosis. Even though PPD was +ve, Tuberculosis was not suspected because of lack of loss of weight or pulmonary symptoms, with a normal physical exam and chest X-ray. C.T. chest and gallium scan are sensitive and indicated when clinical and chest X-ray findings are -ve and when reactivation is suspected. CT scan differentiates active and inactive healed lesions. Increased uptake on Gallium scan indicates active disease and serial scans are used for monitoring disease activity and response to treatment.

PCR: Rapid confirmation of tuberculosis with species identification and MDR strains can be done in few hours by PCR enabling early treatment and control measures. It is comparable in sensitivity and specificity to culture requiring less than 10 organisms for detection compared to the 10,000 necessary for AFB smear. PCR does not differentiate between live and killed organisms and patients on therapy may remain PCR +ve for an unknown period despite mycobacterial sterilization. DNA probes directed against RNA may be capable of detecting only live bacilli. Rapid diagnosis of TB meningitis is made by PCR of CSF and detection of tuberculoesteric acid in CSF. Rapid culture methods within one to two weeks is possible by BACTEC and SEPTI-CHECK methods.

HOW WELL DOES THE "EVIDENCE" APPLY TO A COMMUNITY COHORT OF ADULTS WITH ATRIAL FIBRILLATION? PRELIMINARY RESULTS. A. Bossio, ES Holmboe, Internal Medicine, National Naval Medical Center, Bethesda, MD

PURPOSE: Randomized controlled trials (RCTs) are considered the gold standard evidence for evaluating the efficacy of medical therapies. Five important RCTs have demonstrated the efficacy of warfarin therapy to prevent stroke in patients with atrial fibrillation (SPAF Study, AFASAK Study, BAATAF, VA Stroke Prevention Study, and CAFA Study). However, practicing physicians are limited to the RCT's published exclusion criteria when assessing the generalizability of the treatment's benefit for their own patients. The purpose of this study is to assess the degree of generalizability of evidence-based medicine, based on published RCTs, to a cohort of patients with atrial fibrillation in a general internal medicine clinic.

METHODS: Retrospective descriptive cohort study. Patients with a diagnosis of atrial fibrillation on or after 1 January 97 were identified from an Internal Medicine clinic database. Medical records were abstracted for patient demographics, Charlson comorbidity score, medications, and any complications attributable to anticoagulation. Finally, we noted whether the patient possessed any specific exclusion criteria for each of the five RCTs. Exclusion criteria were obtained from any published manuscripts related to each of the five RCTs.

RESULTS: A total of 48 charts have been abstracted. The mean age was 76 years (range 46 to 94 years) with 30 patients older than 75 years. Forty-four patients (92%) were either receiving aspirin or warfarin, with 27 patients (56%) on warfarin. Only 12 patients (27%), however, would have been excluded from all five trials, and only five of these 12 patients had a medical contraindication to warfarin. Twenty patients (42%) had a Charlson comorbidity score of 2 or greater, but 12 of

these 20 (60%) patients were still eligible for at least one trial. There was a modest positive correlation ($r = 0.27$) between eligibility for any RCT and treatment with warfarin and a modest negative correlation ($r = -.30$) with age and warfarin use. There was no association between warfarin use and Charlson comorbidity score ($r = -0.04$).

CONCLUSION: These preliminary data found that the majority of patients in a teaching clinic would have been eligible, based on published criteria, for at least one RCT studying the efficacy of warfarin therapy to prevent stroke in patients with atrial fibrillation, and that advancing age, but not comorbidity, was associated with less warfarin use.

DOES HAVING SOMEONE TO TALK TO MAKE A DIFFERENCE?: SOCIAL SUPPORT AS A PREDICTOR OF HOSPITAL LENGTH OF STAY. C Boutin Foster, ME Charlson, Division of General Internal Medicine, New York Presbyterian Hospital, New York, NY

PURPOSE: Despite convincing evidence linking poor social support with poor health outcomes, questions about social support are not routinely asked as part of the medical interview. The purpose of this study was to determine the prevalence of poor social support among hospitalized patients and to determine the impact of poor social support on length of stay and cost of hospitalization.

METHODS: During a six week period, patients admitted to the internal medicine service were asked to complete a questionnaire designed to assess the psychosocial needs of hospitalized patients. Social support was determined by asking patients whether or not they had someone who they could talk to about important things in their lives. Patients were also asked to rate their self-perceived health status. The burden of comorbid illness was determined by the Charlson comorbidity index. Diagnosis related groups (DRG) were abstracted from patient charts. Variation in the length of stay was determined by calculating the difference between estimated length of stay for that DRG and the patients' actual length of stay. Total cost of hospitalization was obtained from the hospital's computerized data base.

RESULTS: Of the 317 patients interviewed, 12% reported that they did not have a source of social support. Their mean age was 63 years, 54% were female, and 30% were African-American or Latino-American. Patients who did not have a source of support were more likely to describe their health as poor ($P = .01$). The most common reasons for admission among patients with poor support were for cardiovascular, pulmonary, and HIV related illnesses. The mean length of stay for patients who were admitted for these conditions and who lacked a source of support was 13 days as compared to 7.5 days for those patients who had a source of support. ($P = .001$). Patients with poor social support stayed in the hospital an average of 5 days longer than expected for their DRG. The total hospital cost for these patients was on average \$3,000 higher.

On multivariate analysis, after adjusting for age, race, sex, functional status and Charlson comorbidity score, poor social support remained as a significant predictor of length of stay for patients admitted with cardiovascular, pulmonary, and HIV related illnesses. ($P = .02$)

CONCLUSION: Having someone to talk to is an important factor which physicians may often overlook. Approximately one out of ten patients admitted did not have a source of social support. These patients were more likely to view their health as poor and stayed in the hospital longer than expected. Innovative methods of identifying patients with poor social support and interventions which enhance social support may improve patient outcomes as well as decrease health care cost.

ESTIMATING HIP FRACTURE MORBIDITY, MORTALITY, AND COSTS. RS Braithwaite, NF Col, LK Hirota, JB Wong, Division of Clinical Decision Making, Informatics, and Telemedicine, Tufts New England Medical Center, Boston, MA

PURPOSE: Hip fractures lead to approximately 300,000 hospitalizations and 50,000 deaths annually in the US, but published estimates of the long-term nursing home care rates (6% to 60%) and costs (\$8.1 to \$15 billion) vary widely. Our purpose was to estimate the morbidity, mortality, and costs of a hip fracture.

METHODS: We developed a Markov model to estimate the long-term effects of hip fractures on 80 year-olds. The model included short- and long-term mortality and morbidity attributable to hip fracture (permanent disability and need for hospitalization, rehabilitation, nursing home, and home care services). Morbidity estimates were based on a meta-analysis of 14 US studies and data from the 1997 National Hospital Discharge Survey. Costs used were those directly attributable to hip fracture and were based on Medicare reimbursement rates and home care costs. We adjusted all medical costs to US \$1998 using the Medical Care component of the Consumer Price Index.

RESULTS: Among 10 studies examining outcomes between 6 and 12 months after hip fracture, our meta-analysis suggested that $14 \pm 5\%$ of patients ($n = 3067$) required long-term nursing home care. In 4 studies, $34 \pm 11\%$ ($n = 1259$) experienced permanent disability. Overall, hip fracture decreased life expectancy by 23% (1.7 years) with most of this mortality (18%) occurring within the first 6 months. Sixteen percent of that life expectancy is spent in a nursing home (0.9 years, on average). Our model estimated a lifetime attributable cost to hip fracture of \$73,200 with \$25,300 occurring during the first six months. The most important factors influencing costs (in order of importance) were the 1) long-term nursing home rate, 2) daily nursing home cost, 3) hospitalization costs and 4) inclusion of economic value for home care delivered by unpaid caregivers.

CONCLUSION: Hip fractures result in significant mortality, morbidity, and costs. Most of the attributable expenses result from functional losses in the activities of daily living, leading to long-term nursing home or supplemental home care. Based

on this analysis, the estimated lifetime cost for the 311,000 hospitalized hip fractures in the US in 1997 may exceed \$19 billion. These results emphasize the importance of screening and treatment interventions to decrease the incidence of hip fracture, especially as the US population ages.

QUALITY OF LIFE, WORK AND ALCOHOL USE AFTER LIVER TRANSPLANTATION. DM Bravata, I Olkin, AE Barnato, EB Keefe, DK Owens, Medicine, VA Health Care System, Palo Alto, CA; Medicine, Stanford University School of Medicine, Stanford, CA; Statistics, Stanford University, Stanford, CA

PURPOSE: Approximately 7500 liver transplantations are performed around the world at an average 1-year cost of \$280,200 per transplant. Alcoholic liver disease (ALD) is the second leading cause of liver failure necessitating transplantation. An understanding of recipients' quality of life (QOL), employment and alcohol-use outcomes is essential for counselling pre- and post-transplantation patients. Our purpose was to evaluate patterns of QOL, employment, and alcohol use among liver transplant recipients with ALD and other etiologies of liver failure (non-ALD). **METHODS:** We identified 5473 potentially relevant articles using structured MEDLINE and Embase searches and 32 additional references from articles' bibliographies. We included studies if: they reported an assessment of quality of life (QOL), employment or alcohol consumption; reported either pre- and post-transplantation data or had a comparison group; and were written in English. We combined studies to calculate summary proportions, odds ratios, and performed a sign-test to evaluate the direction (positive or negative) of the effect of transplantation on QOL. **RESULTS:** Of the 5505 identified articles, 89 studies reporting data on 6099 transplant recipients met our inclusion criteria. We found significant post-transplantation QOL improvements ($p < 0.05$) in Karnofsky, Sickness Impact Profile, and Nottingham Health Profile scores; physical health, sexual functioning, daily activities, general QOL, and social functioning; but not psychological health. Employment among ALD vs. non-ALD recipients was 29% vs. 59% pre-transplantation and 33% vs. 85% at three years post-transplantation ($p < 0.00001$ for differences between groups at each interval). Non-ALD recipients using alcohol had lower rates of employment than those who abstained; however there was no such association for ALD recipients. Although, there was no difference in the proportion of ALD and non-ALD recipients reporting alcohol use post-transplantation (4% vs. 5% at 6 months, and 17% vs 16% at 12 months), the non-ALD recipients were more likely to drink moderately ($p < 0.0001$), whereas ALD recipients were more likely to drink excessively ($p < 0.05$). The odds ratio for alcohol use among those patients with less than 6 months of pre-transplantation abstinence was 7.8 (95% confidence interval: 4.0-15.3).

CONCLUSION: Liver transplantation provides clinically important increases in functional status in both ALD and non-ALD recipients. Prior to transplantation and at long-term followup, substantially fewer ALD recipients are employed than are non-ALD recipients. Less than 6 months of abstinence from alcohol pre-transplantation is a strong predictor of post-transplantation alcohol use.

SOCIODEMOGRAPHIC DIFFERENCES IN THE USE OF EVIDENCE-BASED THERAPIES FOR MEDICARE BENEFICIARIES WITH DIABETES IN MANAGED CARE. AF Brown, AG Gross, PR Gutierrez, SR Starr, MF Shapiro, CM Mangione, Department of Medicine, UCLA, Los Angeles, CA; MCP Hahnemann University, Philadelphia, PA; RAND, Santa Monica, CA

PURPOSE: To determine whether the use of evidence-based therapies for primary and secondary prevention of cardiovascular disease and nephropathy in older persons with diabetes enrolled in Medicare managed care (MC) is influenced by sociodemographic characteristics.

METHODS: Telephone interviews and clinical examinations were performed on a random sample of subjects with diabetes cared for in a Medicare managed care plan that contracts with 17 provider groups in Los Angeles. Current medications were inventoried at the study's clinical examination. Patients were questioned about sociodemographic characteristics and health status. The evidence-based therapies evaluated included primary and secondary prevention with cholesterol lowering medications overall and HMGCoA reductase inhibitors specifically; aspirin use in all patients and for those with a history of coronary heart disease (CHD); beta-blocker use in persons with a history of myocardial infarction (MI); and angiotensin converting inhibitor (ACEI) use. The influence of age, gender, race/ethnicity, income, Medicaid, and education on use of each therapy was tested with a series of logistic regression models, adjusting for health status using the SF-12 and for comorbid illness.

RESULTS: Clinical examinations were completed in 308 of 466 eligible persons (66%). Mean age 75 ± 5 years; 47% women; 51% white, 22% Latino, 17% African American, and 6% Asian Pacific Islander; 55% earned under \$20,000 per year; 8% received Medicaid; and 28% had not graduated from high school.

Use of Evidence-Based Therapies—Odds Ratio (p).

	Age ≥ 75	Female	Medicaid	Income $< \$20K$	<HS Grad
Chol meds	0.5 (p = .008)	2.2 (p = .007)	—	—	—
HMGCoA (1 st)	0.5 (p = .03)	2.5 (p = .003)	—	—	—
HMGCoA (2 nd)	—	—	—	0.5 (.02)	—
Aspirin (1 st)	—	—	—	—	0.4 (.006)
Aspirin (2 nd)	—	—	—	—	0.2 (.003)
ACEI (1 st)	—	—	0.3 (p = .01)	—	—

Use of evidence-based therapies did not differ by race/ethnicity. Use of cholesterol medication or beta-blockers for secondary prevention did not differ by sociodemographic characteristics.

CONCLUSION: In this cohort of Medicare MC patients with diabetes, all of whom had the same pharmacy benefit, there were several sociodemographic differences in the use of evidence-based therapies. While women more often received cholesterol-lowering medications, older, poorer persons and those with less education were less likely to receive several recommended therapies.

THE CHALLENGE OF PROMOTING INFORMED, SHARED DECISION-MAKING ABOUT PSA TESTING AMONG AFRICAN-AMERICAN (AA) MEN. RC Burack, J George, J Coombs, Barbara Ann Karmanos Cancer Institute; School of Medicine, Wayne State University, Detroit, MI

PURPOSE: Given the increased risk of prostate cancer among AA men and the uncertain benefit and potential harms of screening, we sought to develop an educational brochure which advocated neither for nor against testing but rather promoted shared decision-making by providing information and encouraging discussion with one's doctor.

METHODS: The target population was AA, predominately Medicaid-eligible men enrolled in two health care organizations. The 12-page brochure was tailored to the needs and preferences of these men based upon results from a telephone interview and a series of focus groups. To then evaluate the brochure we telephoned 200 eligible practice visitors 40 to 75 years of age, of whom 104 completed a baseline interview. The brochure was then mailed to these men, 58 of whom also completed a post-brochure interview. Those completing versus not completing the second interview did not differ in measured factors.

RESULTS: Half are over the age of 60 and have fewer than 12 years of formal education. Knowledge of prostate testing was limited. At baseline 37% identified either rectal exam or PSA test as included in testing and 73% were uncertain about the possibility of a false positive test result. Attitudes toward testing were highly favorable. Over 85% believed that "testing saves lives" and that "all AA men should be tested". At baseline, 35% had made a testing decision: 30% to be tested and 5% to talk to their doctor. Factors independently associated with having made a decision include age <60, recognizing possibility of false test results, and belief that one's doctor recommends testing (a factor closely correlated with favorable test attitudes). Among the 58 men completing pre and post interviews, 97% agreed the brochure was valuable and easy to understand and 74% believed it provided the right amount of information about pros and cons. The knowledge score increased from 2.5 to 3.2 on a 5-point scale ($p < 0.001$) but there was no change in the favorable pre-brochure testing attitudes. Testing decisions did change. The proportion of men choosing to be tested increased by 20%. No man decided to not be tested.

Decision	%Pre	%Post
Be tested	33	53
Talk to doctor	3	9
No decision	64	38

CONCLUSION: We conclude that a carefully developed and targeted "non-advocacy" brochure may increase knowledge but does not consistently reduce (and may increase) enthusiasm for testing among men with pre-existing and strong beliefs about its value.

DOMESTIC VIOLENCE DETECTION: LESSONS LEARNED FROM PREGNANT AND NONPREGNANT VICTIMS. RA Buranosky, KA Dowd, BH Hanusa, SH Scholle, Division of General Medicine and the Center for Research on Healthcare, University of Pittsburgh, Pittsburgh, PA

OBJECTIVE: Routine screening and counseling for domestic violence (DV), defined as physical, emotional, and sexual intimate partner abuse, is recommended in primary care. The purpose of this study was to assess the effects of this screening on pregnant and nonpregnant victims.

METHODS: Women who were counseled for DV in a university obstetrics/gynecologic clinic from 6/98 to 3/99 were identified from forms completed after a woman had been counseled ($N = 69$). Of the 69 victims identified, medical charts were available for 50 (72.5%). Data on demographics, abuse, health care use, physical and mental health problems, and social work referral patterns were abstracted from medical and billing records.

RESULTS: Of the 50 women in the study, 26 (52%) were pregnant. Detection of DV occurred more frequently at return vs new patient visits (54% vs 20%). Counseling rarely occurred for emotional abuse alone (10%), yet was present in 76% of the cases. The predominant reason for receiving counseling was physical abuse (84%). Of the women with data available on the last incidence of abuse ($N = 34$), 27 (79%) had been abused within the past month; yet, 63% were not living with their abuser. In addition, only pregnant victims (6/14) reported new-onset abuse. Pregnant women were also more likely to be referred to social work for reasons other than DV (47.6% vs. 21.4%, $FE = 0.16$), such as substance abuse and depression, with DV subsequently being detected by the social worker. Of the total number of victims referred to social work for DV ($N = 27$), 48.1% did not see a social worker the day of detection. Depression (54%), tobacco use (46%), illegal drug use (16%), and history of STD's (64%) were equally prevalent in both pregnant and non-pregnant patients.

CONCLUSION: Return clinic appointments with questioning at each visit may increase the detection of DV, especially new-onset abuse. Since most victims were abused by partners with whom they did not live, one cannot assume that a woman who does not reside with her partner is safe. The time between DV detection and the social work visit places a woman and sometimes a fetus at risk for further abuse. Social work referral for psychosocial reasons, such as depression and substance abuse, may yield increased rates of detection of domestic violence.

DETERMINANTS OF ALCOHOL ADVICE OR TREATMENT AMONG AT-RISK DRINKERS IN THE OUTPATIENT SETTING. RESULTS FROM THE ACQUIP STUDY. ML Burman, MB McDonell, SD Fihn, KA Bradley, HSR&D, Center of Excellence, VA Puget Sound Health Care System; Department of Medicine, University of Washington, Seattle, WA

PURPOSE: Little is known about factors that lead primary care providers to counsel patients about their drinking. We examine rates of alcohol-related advice and treatment reported by male at-risk drinkers who receive primary care in the VA, and identify factors associated with at-risk drinkers' reports of not receiving any alcohol-related advice or treatment in the past year.

METHODS: We surveyed patients followed at seven VA General Internal Medicine Clinics participating in the VA Ambulatory Care Quality Improvement Project (ACQUIP). At-risk drinkers, identified using a validated augmented CAGE questionnaire, were sent the ACQUIP Drinking Practices Questionnaire (DPQ) which includes the Alcohol Use Disorders Identification Test (AUDIT) and three questions about alcohol-related advice or treatment. Other data collected included demographic characteristics, socioeconomic status, patient reports of health problems, questions about health-related quality of life (SF-36), and patient satisfaction. Site investigators indicated whether a standard alcohol-screening program had been instituted. **RESULTS:** The DPQ was returned by 3,891 at-risk drinkers (57% response). Respondents were primarily male (98%) and Caucasian (62%), with a mean age of 61. At least one symptom of dependence in the preceding year was reported by 819 (21%) respondents. Among DPQ respondents, 700 (18%) indicated that in the past year they had been advised by their primary care provider to drink less, 661 (17%) had been advised to quit, and 156 (4%) had received alcohol treatment. Excluding those treated in the past year, a total of 784 (21%) received advice to change their drinking. Advice was more common among patients who were younger, reported less education, were unmarried, had lower income, reported liver disease, or smoked. Multivariate logistic regression revealed that the following groups of at-risk drinkers were significantly more likely to report not receiving any alcohol-related advice in the past year: patients who drank <14 drinks/week (OR 1.99), those without symptoms of alcohol dependence (OR 3.33), who did not self-report a drinking problem (OR 4.22), who did not smoke (OR 1.32), and who reported their health as very good or excellent (OR 1.62). Implementation of clinic-based screening with the CAGE questionnaire was not significantly associated with receipt of advice to modify drinking behavior (OR 1.12).

CONCLUSION: While primary care providers appear to be focusing their alcohol counseling on those at-risk drinkers with the most severe problems, the number of patients receiving alcohol counseling continues to be a small proportion (21%) of those who might benefit from it.

INCREASING PREVALENCE OF TRIMETHOPRIM-SULFAMETHOXAZOLE RESISTANT ESCHERICHIA COLI. W Burman, P Breese, H Batal, T MacKenzie, PS Mehler, Denver Health Medical Center, Denver, CO

PURPOSE: Acute uncomplicated cystitis is one of the most common community acquired infections in women. Empiric treatment with trimethoprim-sulfamethoxazole for uncomplicated cystitis has been a widely utilized strategy. However, there has been increasing resistance to this antibiotic in *Escherichia coli*, the most common cause of these infections. The purpose of our study was to determine the trends in this resistance among *E. coli* uropathogens and to ascertain if there are risk factors, which predict this resistance in acute cystitis.

METHODS: There were two parts to the study: Part A was a retrospective study of antibiotic resistance in 786 urine cultures from the microbiology computer system of all patients with *E. coli* ($\geq 10^5$ /ml). Part B was a prospective observational study performed at multiple different clinic sites at Denver Health. All patients with symptoms typical of uncomplicated cystitis seen between July 1998 through September 1998 had a questionnaire completed. Treatment details were at the discretion of the care provider. All patients with an isolate resistant to trimethoprim-sulfamethoxazole were phoned and asked to return to clinic where a repeat urine culture was obtained and a second questionnaire was completed.

RESULTS: The rate of resistance increased from 2% in 1981 to 26% in 1998. Factors associated with a resistant strain were a urinary tract infection in the past six months (OR = 2.5 [1.0-6.1]), and travel outside the United States in the past six months (OR = 6.0 [1.3-28.6]). Hispanic ethnicity, compared to persons of other racial/ethnic backgrounds and age less than 3 years ($P < 0.01$) also predicted a higher rate of resistance in the retrospective study. Rates of resistance were still greater than 15% in patients without any of these risk factors. The rate of trimethoprim-sulfamethoxazole resistance was similar between inpatient versus outpatient isolates ($P = 0.35$). Resistance to trimethoprim-sulfamethoxazole predicted resistance to at least one other antibiotic in these urine cultures.

CONCLUSION: There is increasing resistance of *E. coli* in cystitis to a commonly used antibiotic. This is not strictly a nosocomial problem. Prior use of antibiotics is one of the significant risk factors for this resistance, but rates were high even in patients without risk factors. This will likely require closer attention to local antibiotic susceptibility data and a change in empiric treatment strategies.

SHOULD WOMEN BE SCREENED FOR ACTIVATED PROTEIN C RESISTANCE PRIOR TO INITIATING ORAL CONTRACEPTIVES? A MARKOV ANALYSIS. SA Call, JC Barrett, MD Rousculp, *RM Centor*, Medicine, University of Alabama at Birmingham, Birmingham, AL

PURPOSE: Activated protein C resistance (APC-R), of which Factor V Leiden accounts for 95% of cases, results in an increased risk of thrombosis. This risk is increased dramatically among patients with APC-R using oral contraceptives (OCPs). Screening of women intending to initiate OCPs provides a potential means of preventing avoidable veno-thromboembolic events (VTEs), though it remains unproven whether screening is effective in a population and at what cost.

METHODS: We constructed a Markov model to compare two strategies: screening for APC-R before initiating OCPs versus no screening. Outcomes consisted of VTEs, number of pregnancies, number of spontaneous abortions, and deaths. The model assessed cost per VTE averted. We assumed a cohort starting age of 20 years, and tracked the cohort for five years. Using available data, we made base case assumptions on: APC-R prevalence (3.7%), sensitivity/specificity of screening tool (0.90/0.90), cost of screening (\$45), and relative risk of VTE given APC-R positive plus OCP use (35). Specific published population estimates of APC-R prevalence were utilized in sensitivity analysis. We also used a broad range of costs representing different institutions and reimbursement levels.

RESULTS: Under base case assumptions, the cost of screening per VTE averted was \$106,635. There were 125 excess pregnancies and 33 excess spontaneous abortions per 10,000 women screened. The model was highly sensitive to assumptions of disease prevalence and cost.

Population	Prevalence	Cost/VTE Averted (for each cost estimate) VTEs averted per 10,000 women			
			\$10	\$45	\$150
US-Asian	0.45%	1.4	\$72,464	\$326,087	\$1,086,957
US-Black	1.23%	2.1	\$48,544	\$218,447	\$728,155
US-General	3.70%	4.2	\$23,697	\$106,635	\$335,450
US-Caucasian	5.27%	5.6	\$17,889	\$80,501	\$268,336
England	8.86%	8.7	\$11,468	\$51,606	\$172,018
Family Hx	12.0%	11.5	\$8,726	\$39,277	\$130,890
Sweden	14.7%	13.8	\$7,241	\$32,585	\$108,617

CONCLUSION: Results suggest that screening for APC-R before initiating OCPs may have a favorable cost-benefit ratio in selected populations. While the cost of screening appears prohibitive among Asian and African Americans, screening of Caucasian populations and of those with family histories of VTEs may be reasonable. The excess pregnancies and spontaneous abortions must also be considered.

ANTIRETROVIRAL CARE FOR WOMEN AND INDIVIDUALS WITH INJECTION DRUG USE (IDU) RISK BEHAVIOR. R Campo, W Jordan, J De Hovitz, J Stansell, D Purdom, D Butcher, D Wolde-Rafael, P Landsman, *P Duong*, L Markson, University of Miami, Miami, FL; Oasis Clinic, Los Angeles, CA; State University of New York, Brooklyn, NY; University of California at San Francisco, San Francisco, CA; Kansas City Free Health Clinic, Kansas City, MO; Chase Brexton Health Services, Baltimore, MD; Outcomes Research & Management, Merck & Co., Inc., West Point, PA

PURPOSE: Evaluate uses of protease inhibitor (PI) and virologic response of women and individuals with IDU risk behavior in clinical practice.

METHODS: HIV ~ PRACTICE Cooperative is a random sample chart review of 599 adult patients in 6 HIV clinics. Patients with ³ 6 months documented clinic follow-up and first ever prescribed PI after 5/1/1996 were included for analysis if baseline vRNA was >500 copies/mL. PI regimens were classified per DHHS guidelines as "preferred" [i.e., a PI or ritonavir + saquinavir] with 2 reverse transcriptase inhibitors] or "other" PI regimens. PI start date was the index date; a "simultaneous start" occurred when regimen agents began within 14 days of the index date. Antiretroviral therapy (ART) stopped prior to this date was defined as treatment experience. Multivariate logistic regression analyses were used to assess odds of patients receiving preferred PI regimen and odds of achieving suppressed vRNA (vRNA < 500 copies/mL) for female and IDU patients, adjusting for age, race, ART experience, baseline CD4. The suppression model also included simultaneous start and Guidelines preferred PI regimen variables.

RESULTS: In 285 patients on PI, 54.7% were on a preferred PI regimen (62.7% of female patients (n = 67) and 55.2% of IDU patients (n = 58)). Among patients on a PI regimen, odds of receiving a preferred regimen was 1.78 (CI 0.95-3.3; p = 0.07) for female versus male and 0.8 (CI 0.42-1.58; p = 0.5) for IDU versus non-IDU risk groups. In 218 patients with vRNA measures: 51.4% achieved suppression; 50% of female and 41.5% IDU patients. Odds of achieving suppressed vRNA was 1.08 (CI 0.53-2.2; p = 0.84) for female versus male, and 0.79 (CI 0.36-1.72, p = 0.55) for IDU versus non-IDU risk groups. Significant predictors of viral suppression were use of Guidelines preferred PI regimen (OR 2.46; CI 1.3-4.9; p = 0.009) and simultaneously started regimen (OR 3.69; CI 1.4-9.7; p = 0.008).

CONCLUSION: Among patients prescribed a PI regimen, odds of achieving suppressed vRNA did not appear to be influenced by the patients' gender or IDU risk behavior.

CHARACTERIZATION OF FOOD-B12 MALABSORPTION: NEW PERSPECTIVES ON A COMMON DISORDER. R Carmel, *L Aurangzeb*, Medicine, NYM, Brooklyn, NY

PURPOSE: Food B12 malabsorption (FBM), in which absorption of food-bound B12 is inadequate despite normal absorption of crystalline B12, seems to account for 30-40% of all low B12 levels. However, little is known about it and it frequently remains unrecognized.

METHODS: We determined patient characteristics and gastric markers in FBM. Food-B12 absorption was measured in 202 subjects by the egg yolk B12 absorption test (EYBAT), a food variant of the Schilling test. These subjects included healthy volunteers and patients with unexplained low B12 levels. Demographic data and common laboratory tests were obtained. 167 subjects underwent testing for Helicobacter pylori infection, 158 had serum gastrin and parietal cell antibody determination and 133 had serum pepsinogen I and II measured. Statistical methods were applied to determine the association between the various findings and FBM, defined as <2% excretion in the EYBAT.

RESULTS: FBM was present in 84 out of 202 subjects (not a true prevalence because of a disproportionate number of subjects with known low B12 levels). The major findings, each significantly associated with FBM, were as follows. (1) Latin Americans and blacks had lower EYBAT results and had severe FBM (EYBAT excretion <1%) more often than whites and Asian Americans (p = .0001). (2) Malabsorption was more common in the elderly with an inverse correlation between age and EYBAT (p = .02). (3) H. pylori infection was associated with severe FBM in 29 of 37 subjects (78.4%, p = .0001). (4) FBM was associated with higher gastrin levels (p = .0001), lower pepsinogen I levels (p = .01) and lower pepsinogen I:II ratios (p = .0001). All four significant associations were independent of each other by multivariate analysis.

CONCLUSION: The findings in our survey indicate that the elderly and minorities such as blacks and Latin Americans are at higher risk for FBM than others. These demographic trends may partially explain why the elderly are at high risk for B12 deficiency. The gastrin and pepsinogen associations support the accepted link between FBM and gastritis. The high prevalence of H. pylori infection that we found in severe FBM further suggests that at least some of these gastric disorders may be amenable to treatment. Defining guidelines for treatment in such scenarios may have strong clinical implications because FBM is the most common malabsorptive disorder associated with low B12 levels.

PATIENTS WHO DESIRE PRAYER: CAN PHYSICIANS IDENTIFY THEM? *D Castro*, LK Loo, LE Skoretz, DL Stottlmyer, S-J John, J Buenjeria, Medicine, Loma Linda University, Loma Linda, CA

PURPOSE: Spirituality and prayer are important to many people. Two-thirds of the American public believe physicians should talk to their patients about spiritual issues and pray with those who request it, yet only 10% of patients recall their physician doing so. Interested physicians might be more willing to incorporate spirituality as an aspect of health care if they can discreetly identify receptive patients. Our objective was to determine in different settings how often patients desire prayer with physicians and to identify the characteristics of these patients.

METHODS: General Internal Medicine patients were surveyed equally from a University, Veterans, and County Hospital (50% outpatients and 50% non-ICU inpatients). 735 patients were invited to participate and 600 agreed to do so (18% refusal rate). The survey instrument and medical record review included common patient demographics, measures of physical health, functional status, social support, quality of life, medical prognosis, and various religious and spirituality assessments. Patients also indicated whether or not they desired prayer with their physicians, and if so, whether they desired this "now" (during the immediate outpatient clinic visit or inpatient hospitalization) or in the "future" (sometime later).

RESULTS: Seventy-one percent of patients desired prayer at some time with their physicians. Surprisingly, the following variables were NOT predictive of patients' desire for prayer with their physician: patients' perceived quality of life and state of health, functional status, degree of social support, Charlson's comorbidity score, number of medications, hospital type, or inpatient vs outpatient status. After multivariate analysis, only gender (female > male, 83% vs. 63%, p = 0.023) and some spirituality variables were statistically significant predictors for patients' desire for prayer with their physician. The most significant predictor (p = 0.003) was a simple, single item 1-9 self-rated spirituality scale (see Table).

	NOW	FUTURE	NEVER
Patients Desiring Prayer	43%	28%	29%
Patients' Spirituality Score	8.0 (SE ± 0.12)	6.8 (SE ± 0.17)	5.4 (SE ± 0.23)

CONCLUSION: To our knowledge, this is the largest study to examine patients' desire for prayer among such a diverse population of General Internal Medicine patients. We expand upon previous studies and found that 71% of patients desire prayer with their physicians. Variables associated with Do-Not-Resuscitate and end-of-life medical decisions were NOT associated with patients' desire for prayer. Physicians interested in more accurately identifying patients receptive to prayer may find this simple 1-9 self-rated spirituality scale a helpful instrument.

SCREENING FOR OCCULT CARBON MONOXIDE EXPOSURE AT AN URBAN CLINIC. A Chandra, *MS Cline*, DA Phillips, CC Tang, H Young, S White, LJ Cardozo, Internal Medicine, Wayne State University/Detroit Medical Center, Detroit, MI; Emergency Medicine, Wayne State University/Detroit Medical Center, Detroit, MI

PURPOSE: The objective of this study was to identify the prevalence of elevated carboxyhemoglobin (COHgb) levels in an inner city population presenting to a university based internal medicine clinic.

Carbon monoxide (CO) poisoning accounts for the most toxicological deaths per year in the US. Symptoms of exposure to carbon monoxide are often vague and may mimic other illnesses, such that the diagnosis of exposure is often missed. No previous studies have utilized carbon monoxide screening in an outpatient clinic setting.

METHODS: Between the months of October and January, a prospective, descriptive pilot study was done in a urban, general internal medicine clinic in Detroit, Michigan. Subjects 18 years of age or older were voluntarily enrolled at time of their scheduled appointments, and indicated their smoking status and time since last used. They then underwent breath analysis using a Vitalograph carbon monoxide detector, and COHgb levels were reported in parts per million. Levels higher than 15 PPM for nonsmokers (NSM) and 23 PPM for smokers (SM), reported by Stewart et al. as elevated for Detroit, initiated referral to an emergency department. Standard statistical measures were calculated.

RESULTS: Eighty-one patients were sampled. The mean for NSM was 2.4 PPM (95% CI 0–6.7), with a median of 2 and a mode of 2 PPM. For SM, the mean was 10.4 PPM (95% CI 0–22.2), with a median of 11 and a mode of 14 PPM. One smoker had a value of 23 PPM 15 minutes after smoking 2 cigarettes and repeat evaluation was normal.

CONCLUSION: This pilot study suggests that there is a low prevalence of elevated COHgb levels in an urban outpatient clinic setting. Further investigation with larger sample size and power is needed to determine if CO screening is useful in a clinic.

ATRIAL FIBRILLATION AFTER IMPLANTABLE CARDIOVERTER DEFIBRILLATOR INSERTION: INCIDENCE AND RISK FACTORS. S. Chhabra, G Turitto, Medicine, SUNY HSC at Brooklyn, NY

PURPOSE: Candidates for implantable cardioverter-defibrillator (ICD) insertion are usually characterized by the presence of severe heart disease, which would make them prone to develop atrial fibrillation (AF) following surgery. A high incidence of post-operative AF (up to one third of cases) was reported in the era of epicardial ICDs. On the other hand, the incidence and risk factors for AF in the recipients of endocardial ICDs are less well characterized. AF is an important cause of inappropriate ICD discharges, since it may be associated with fast ventricular response and misdiagnosed as a ventricular tachyarrhythmia by single chamber ICDs. The identification of patients at especially high risk for AF after ICD implantation may justify the prophylactic use of a dual chamber system, capable of discriminating between supraventricular and ventricular tachyarrhythmias. The purpose of this study was to evaluate the incidence and risk factors of AF in a cohort of patients undergoing ICD implantation at our institution.

METHODS: 8 consecutive patients who had transvenous ICD implantation between 4/94 and 7/99 were included in this retrospective review. These patients were followed at 3-month intervals, at which time ICD was interrogated. The characteristics of patients with or without AF during the follow-up were compared.

RESULTS: Mean follow-up was 21 ± 15 months. Fifteen patients (17%) developed AF during follow-up, while 73 did not. AF was associated with inappropriate ICD therapy in 13 out of 15 patients. The two groups with or without AF did not differ significantly regarding: duration of follow-up (24 ± 16 vs 20 ± 15 months), history of previous AF (13% vs 11%), age (67 ± 12 vs 64 ± 10 years), gender, type of underlying heart disease, history of antiarrhythmic therapy (mostly indicated by the presence of frequent ICD shocks during follow-up), left atrial size, presence and severity of mitral/aortic regurgitation. Left ventricular ejection fraction was depressed in both groups, but to a lesser degree in patients with AF, compared to those without AF (39 ± 10 vs 29 ± 13 , $p < 0.01$).

CONCLUSION: 1) AF is a frequent phenomenon in first 2 years following endocardial ICD implantation, even though its incidence is less than that reported in earlier studies utilizing epicardial systems. 2) This arrhythmia is a common cause of inappropriate ICD shocks due to fast heart rate, suggesting the use of prophylactic beta-blocker therapy may be indicated in this population. 3) In our study, we were unable to identify significant differences between ICD patients with or without AF. In a compromised population, such as that of ICD recipients, conventional clinical or echocardiographic risk factors of AF appear to be frequently present, and may have a limited usefulness for stratification into subgroups at higher or lower risk of AF.

ILLNESS COURSE IN OLDER PATIENTS WITH HEART FAILURE. MH Chin, JX Zhang, PJ Rathouz, Medicine; Health Studies, University of Chicago, Chicago, IL

PURPOSE: We aimed to determine changes in health perception and functional status in older patients with heart failure (HF) over 4 years, and the impact of prior health status on subsequent illness course.

METHODS: We utilized 1991–94 data from the Medicare Current Beneficiary Survey, a database that combines Medicare claims with yearly longitudinal surveys of 14,000 beneficiaries. We identified 1,096 patients 65 years or older in 1991 with an ICD-9 diagnostic code of HF. The surveys contain a 5-point Likert scale (excellent to poor) question on health perception, ADLs, and IADLs. We determined the overall pattern of illness course over 4 years, and quantitatively modeled transitions in health status between 1991 and 1992 using ordinal logistic regression (death as worse outcome) and transition state matrices.

RESULTS: Changes in Health Perception, ADLs, and IADLs Over 4 Years (% of Patients)

	Steady Decline	Steady Improve	Fluctuate	Constant
Health Perception	28	6	49	17
ADL	29	6	39	26
IADL	28	9	49	14

Regression models adjusting for age, gender, and race, and subsequent transition matrices confirmed these general trends, and suggested threshold levels of health status beyond which prospects of recovery were significantly less. For example, 70 year old women who rated their health perception as "very good" in 1991 had a 53% chance of being "very good" or "excellent" in 1992, compared with 24% for those who rated their health as "good" in 1991.

CONCLUSION: Many older patients with HF have worsening health status over time. Measures of prior health status can help predict chances of functional recovery.

DOES EARLY BREAST CANCER INFLUENCE UTILITIES RELEVANT TO HORMONE REPLACEMENT THERAPY (HRT)? JM Clark, GL Chen, HS Sacks, DN Rose, EB Bass, AW Wu, Johns Hopkins University, Baltimore, MD; Mount Sinai School of Medicine, NY, NY; Long Island Jewish Medical Center, New Hyde Park, NY

PURPOSE: The decision to take HRT has become increasingly complex with the introduction of selective estrogen receptor modulators. Therefore, it is even more important to understand how women, including those with a history of breast cancer, view outcomes associated with HRT. Our aim was to measure the utilities for HRT-related health states in peri- and postmenopausal women with and without breast cancer.

METHODS: We interviewed peri- and postmenopausal women from general medicine (GIM) clinics and those in breast clinics with a history of early stage breast cancer at 2 tertiary care centers. We excluded women with high levels of depression or anxiety. Women used a computerized program (U-titer) with a standard gamble technique to assign utilities to their current health and HRT-related health states including: coronary artery disease—acute myocardial infarction and chronic angina; hip fracture—acute and long-term; breast cancer—newly diagnosed and long-term follow-up; HRT use—withdrawal and unexpected bleeding; and menopause—acute symptoms and chronic effects.

RESULTS: The first 54 women interviewed had a mean age of 63.6 years (SD 8.8), 56% were married, 62% white, 32% African American and 40% had a high school degree or less. There were no significant differences between the women from the 2 clinics. 62% of the women have ever taken and 27% were currently taking HRT. Utilities for current health and the chronic health states are shown below.

Health States	Breast Clinic N=18	GIM Clinic N=36
Median (Interquartile Range)		
Current Health	.98 (.93–1.0)	1.0 (.91–1.0)
Stage I breast cancer, follow-up	.96 (.87–1.0)	.98 (.91–1.0)
Cyclic HRT	.99 (.93–1.0)	1.0 (.94–1.0)
Chronic menopausal problems	.99 (.94–1.0)	.95 (.90–1.0)
Chronic angina (class III)	.90 (.73–.93)	.93 (.56–.99)
Post hip fracture, poor result	.84 (.75–.96)	.70 (.13–.93)

Although we have not yet reached the projected sample size of 300 that will give 80% power to detect a difference of 0.1 in utilities between groups, the utilities for all acute and chronic states were similar for women in both clinics ($p > .10$).

CONCLUSION: Physicians should not assume that a history of early stage breast cancer changes how women view HRT-related health states.

BONE DENSITOMETRY IN AN URBAN MULTIETHNIC POPULATION. E Cohen, R Monderer, I Rafique, L Freeman, J Arnsten, Medicine; Nuclear Medicine, Montefiore Medical Center, Bronx, NY

PURPOSE: Most published data on osteoporosis in postmenopausal women has been derived from cohorts of largely white women. In recent years studies have looked at bone mineral density (BMD) and osteoporosis in black and Mexican-American women, but little has been published in other Latina women. Common medical wisdom describes Latinas as being relatively protected from osteoporosis. This ongoing study examines BMD data in an ethnically diverse cohort, which includes large numbers of Caribbean Latinas and blacks.

METHODS: Women referred for outpatient dual X-ray absorptiometry BMD-testing between 2/98 and 2/99 had their medical records reviewed retrospectively. Clinical data obtained included BMD results, demographic data, menopausal history, use of drugs commonly prescribed to prevent and treat osteoporosis, and comorbidities likely to effect use of such therapies.

RESULTS: To date 147 medical records have been reviewed. The cohort included 40.1% whites, 30.6% blacks, 25.9% hispanics, and 3.4% others. The mean age of whites (69.2 yrs.) was older than hispanics (64.3 yrs.) and blacks (64.3 yrs.) ($P < 0.05$ for whites vs. others). Rates of use of any osteoporosis therapies (estrogen, alendronate, calcium) were not significantly different between racial groups (whites 44.8%; blacks 38.6%; and hispanics 31.6%), with a trend toward higher use in whites. More than half the women were aged 60–79 years; among this group hip BMD was significantly higher in blacks than in whites ($P = .0002$), and

LS-spine BMD was somewhat higher in blacks than in whites ($P = .06$). In hispanics aged 60–79 years neither hip nor LS-spine BMD were significantly different from whites. (See Table)

BMD	Hispanics (n = 17)	Blacks (n = 24)	Whites (n = 33)
Hip BMD ($\pm$ SD)	0.76 ($\pm$ 0.13)	0.85 ($\pm$ 0.15)	0.69 ($\pm$ 0.15)
LS-spine BMD ($\pm$ SD)	0.75 ($\pm$ 0.14)	0.90 ($\pm$ 0.15)	0.81 ($\pm$ 0.17)

CONCLUSION: While common wisdom holds that hispanics are relatively protected from osteoporosis, these data suggest that older hispanic women in this largely Caribbean Latina population had BMD closer to that of whites than that of blacks. These results support the need for more research on the epidemiology, prevention, and treatment of osteoporosis in diverse Latina populations.

SAFETY AND EFFICACY OF AGGRESSIVE TREATMENT OF PREOPERATIVE HYPERTENSION. SL Cohn, S Chhabra, S Chhabra, Medicine, SUNY Downstate Medical Center—Kings County Hospital, Brooklyn, NY

PURPOSE: The optimal antihypertensive medication and level of blood pressure for surgery are unknown. Beta-blockers are felt to be potentially beneficial in this setting whereas dihydropyridine type calcium channel blockers have been implicated in adverse effects. We reviewed data from a series of 314 consecutive hypertensive patients seen in our Preoperative Medical Consultation Clinic to examine the safety and efficacy of aggressive preoperative blood pressure reduction.

METHODS: Using an arbitrary preoperative BP goal of $160/100$, we previously reported lowering BP by 14% in the subgroup not meeting this criteria.¹ The mean BP's initially and prior to surgery for the entire group were $157/92$ and $146/86$. We randomly selected half (157) of these patients and reviewed their records to determine if they had 1) perioperative cardiac complications; 2) sustained long-term BP control; 3) met JNC VI criteria.

RESULTS: 143 patients underwent surgery. There were no perioperative deaths or myocardial infarctions. The mean duration of follow-up was 32.8 months. Only 2 patients were noted to have cardiac events during this follow-up period—1 silent MI (on EKG) and 1 requiring CABG. Three other patients died from noncardiac causes (malignancies). Preoperatively, calcium channel blockers, beta blockers, and ACEI/ARB's were used by 65, 21, and 30% respectively. For long-term follow-up, calcium channel blocker use decreased to 60% while the other drugs increased to 24 and 45% respectively.

112 patients had at least one subsequent BP measurement charted after discharge. This subgroup had initial, subsequent preop, and long-term follow-up mean BP's of 155.8/91.8, 145.6/86.2, and 151.6/82.8 respectively. Although the effect of short-term treatment preop was statistically significant ($p < .005$), it lost significance for long-term control ($p = \text{NS}$); however, long-term BP's remained lower than those at initial presentation (diastolic $p < .0001$, systolic $p = .09$). Less than one-third of the patients met JNC VI criteria for systolic BP control (< 140), but 76.8% met the diastolic criteria (< 90) at long-term follow-up compared to only 52.7% initially.

CONCLUSION: We demonstrated the ability to significantly lower BP preoperatively in a short period of time. This aggressive approach did not result in any untoward events regardless of the level of BP or class of drug used. Additionally, institution of this regimen may have resulted in better long-term control.

¹Cohn S, Chhabra S. Use of a Preoperative Clinic for Aggressive Treatment of Hypertension. *JGIM*. 1999; 14S: 20.

IMPACT OF A PERSONALIZED DECISION SUPPORT AID ON MENOPAUSAL WOMEN—PRELIMINARY RESULTS FROM A PILOT STUDY. NE Col, LK Hirota, JM Fortin, RJ Goldberg, BE Bond, RK Orr, AM O'Connor, Medicine, Tufts-New England Medical Center, Boston; Medicine, University of Massachusetts Medical Center, Worcester, MA; Surgery, Spartanburg Regional Medical Center, Spartanburg, SC; Faculty of Health Sciences, University of Ottawa, Canada

PURPOSE: To pilot test a personalized decision support aid (DSA) on menopausal women's knowledge and decision-making regarding hormone replacement therapy (HRT), lifestyle changes, and preventive screening behaviors.

METHODS: We developed decision support software based on published risk models that provides women with personalized information about their menopausal symptoms, risks for coronary heart disease, breast cancer, osteoporosis, and pulmonary embolism, the impact of HRT on these risks, other treatment options, and strategies for risk reduction. Peri- and postmenopausal women between the ages of 45 and 70 without breast cancer or any terminal disease were recruited through hospital fliers, advertisements, and clinician referral. Participants completed a brief questionnaire about their lifestyle, family, and medical history. This information was entered into a software program that generated a 35–45 page printed personalized health report (the DSA) that was mailed to the patient. Patients later completed a written questionnaire that assessed the impact of the DSA on decision making, attitudes, and intended health behaviors.

RESULTS: Of the 81 women who have been enrolled in this ongoing pilot study, 13 have completed responses to date. Over half of the respondents reported that the DSA changed the way they thought about HRT and 42% reported that it helped them make a decision. The DSA changed the way many women thought about their chances of getting breast cancer (54%), osteoporosis (46%), and heart disease (31%). The majority of respondents (75%) reported that they planned to make changes in their lifestyle based on the DSA, and 25% reported already having implemented substantial lifestyle changes. The DSA increased both motivation

and confidence in their ability to improve their diet, lose weight, exercise more, and quit smoking. Seventy-five percent of the women who were not already receiving regular screening for colorectal cancer reported being more likely to get screening sigmoidoscopy. This pilot study is limited by small sample size, lack of a control group, and a focus on process measures.

CONCLUSION: These pilot data suggest that a personally tailored DSA may help women make informed decisions about HRT, change their risk perceptions, and motivate behavioral change. This pilot study is being followed by a randomized, controlled trial to determine the long-term impact of this DSA.

IMPACT OF A RISK MANAGEMENT PROGRAM FOR MENOPAUSAL WOMEN ON KNOWLEDGE, DECISIONAL CONFLICT AND SATISFACTION WITH CARE. MT Connelly, CL Christiansen, OJ Banyon, D Rusinak, C Canning, TS Inui, Department of Ambulatory Care and Prevention, Harvard Medical School and Harvard Pilgrim Health Care, Boston, MA; Health Services Research and Development, Edith Nourse Rogers Memorial Veterans Hospital, Bedford, MA; School of Public Health, Boston University, Boston, MA; Informed Health Decisions, Inc., Montclair, NJ

PURPOSE: To determine whether a comprehensive risk management program for women in the perimenopausal years will improve satisfaction with care, affect perception of knowledge and lower conflict about hormone replacement therapy (HRT) decision making.

DESIGN: Randomized, controlled trial of a year-long risk management program to support best health practices for women in the menopause transition

Participants: Women, age 40–65, who were members of a managed care organization for at least one year

RESULTS: A total of 1,231 women enrolled in the trial. Half were randomly assigned to usual care; half participated in the risk management program that consisted of a personalized risk report, 3 newsletters, access to nurse counselors, customized care guidelines for primary care providers, and decision aids addressing healthy lifestyle behaviors and decisions about HRT. All participants completed 3 outcome measures during the year.

Of the perimenopausal and postmenopausal women who completed baseline and follow up surveys, those exposed to the risk management program were significantly more likely to report at the conclusion of the study satisfaction with overall menopause care (68% vs. 55% ($\chi^2 = 11.8$, $p = .001$). They were more likely to be satisfied with discussions they had had with health care providers about menopause management ($\chi^2 = 5.9$, $p = .02$) and about the risks and benefits of HRT ($\chi^2 = 150$, $p = .001$).

On most dimensions of self-reported knowledge, such as knowledge about ways to lower the risk for heart disease and methods for managing menopausal symptoms, the risk management participants were significantly more likely to describe themselves as having a good or great deal of knowledge compared with those in the usual care arm. A difference was not demonstrated when both study arms reported a high level of knowledge about a topic, e.g., the importance of not smoking (98% vs. 96%, $\chi^2 = 3.7$, $p = .06$). No differences were found between intervention and usual care in their level of conflict about the decision regarding the use of HRT.

CONCLUSION: Participants in a year-long risk management program designed to support women in the menopause transition were more satisfied with their menopause care and reported that they were more knowledgeable than the usual care arm about menopause-related subjects. Conflict regarding the HRT decision was not diminished by this intervention. These data support the benefit of such care models for this population.

EFFECT OF RACE AND USUAL SOURCE OF CARE ON RECEIPT OF PREVENTIVE SERVICES AS A MEASURE OF ACCESS TO CARE. GM Corbie-Smith, JP Doyle, EW Flagg, Division of General Medicine, Emory University School of Medicine; Center for Clinical Effectiveness, Grady Memorial Hospital, Atlanta, GA

PURPOSE: While having a usual source of care (USOC) is considered a marker of access to care, a more important marker of access may be receipt of preventive health services. Racial disparities in health and access to care have been well documented, however little is known regarding the impact of race on use of preventive services when USOC is taken into account. We conducted these analyses to determine the impact of having a USOC on racial differences in the receipt of adult preventive services.

METHODS: We used the 1996 Medical Expenditure Panel Survey (MEPS), a household survey of 24000 non-institutionalized individuals, for these analyses. MEPS is a nationally representative sub-sample of households that participated in the 1995 National Health Interview Survey. The main outcome, access to care, was measured by receipt of the following adult preventative services (according to the USPSTF guidelines): mammography, PAP smear, breast exam, blood pressure screening, cholesterol screening and influenza vaccine. As our primary analytic technique, we created logistic regression models for each access measure, controlling for a usual source of care, insurance status (private, Medicaid, Medicare, or any type), age, education, and employment status. In racial comparisons, we report only results for Hispanics and Blacks with white race as the reference group. **RESULTS:** In unadjusted analyses having a USOC was significantly associated with receipt of all preventive services. Minorities were less likely to report having a USOC. Compared to Whites, Hispanics were less likely to receive breast exams (OR = 0.6), flu vaccines (OR = 0.6), blood pressure (OR = 0.3), and cholesterol screening (OR = 0.6). Blacks were significantly less likely to report receiving flu vaccines (OR = 0.4). In multivariate models, having a USOC independently pre-

dicted receipt of adult preventive services and significantly reduced racial differences in receipt of all preventive services. In fact, with the exception of flu vaccines in Blacks and blood pressure screening in Hispanics, after controlling for having a USOC, minorities were as or more likely to receive most preventive services.

CONCLUSION: For ethnic minorities, a usual source of care may be an especially important predictor of the use of adult preventive services, and in some cases ameliorates racial disparities in receipt of these services.

CLINICAL FEATURES, NOT RACE OR EDUCATION, PREDICTORS OF TRIAL PARTICIPATION. GM Corbie-Smith, CM Viscoli, WN Kernan, LM Brass, PM Sarrel, RI Horwitz, Emory University, Atlanta, GA; Yale University, New Haven, CT

PURPOSE: Including representative numbers of minorities is critical to the external validity of disease prevention trials, and necessary for subgroup analyses in these populations. While government and private funding agencies have supported inclusion of minorities in clinical research, investigators have voiced concern about difficulties in recruiting minorities. We undertook these analyses to determine to what extent race, social class and clinical characteristics were predictors of clinical trial participation.

METHODS: The Women's Estrogen for Stroke Trial (WEST) is a randomized, double blind, placebo controlled trial testing whether estrogen treatment, compared to placebo, reduces the risk of death or recurrent stroke in postmenopausal women with a recent transient ischemic attack or first non-disabling stroke. Using a community-based strategy of recruitment, a comprehensive network of hospitals and neurologists was assembled to identify potential study subjects. During a consent visit all patients were asked to describe their race, marital status, education level, employment status and family income. We describe the results of 1181 women who met study eligibility criteria: 652 women who agreed to participate in the trial and 529 who completed the consent visit but refused participation. We restricted our analysis to Whites and Blacks as the numbers of Hispanics and other races were too small to make meaningful comparisons. Unadjusted comparisons were made using the chi-squared test for categorical variables and the two sample t-test for continuous variables. Logistic regression models were constructed to determine predictors of participation in the trial.

RESULTS: The WEST study sample is 84% (547) White, 13% (82) Black, 3% (18) Hispanic and 1% (5) other racial groups. Participation rates were not different based on race (Black 56% vs. White 54%) or education (in both enrolled and refused groups 71% had high school education or less). In contrast to analyses by race and education, we found participants more often had a prior history of estrogen use (27.1% vs. 16.1%, p value < 0.001) and hysterectomy (44.6% vs. 27.2%, p value < 0.001). For both ethnic groups, estrogen use, hysterectomy status, less severe stroke and less cognitive impairment were strong predictors of participation in the trial. Compared to Whites, Black enrollees had more co-morbidity and lower socioeconomic status. In multivariate models, independent predictors of participation were hysterectomy and prior estrogen use. Race was not an independent predictor of participation.

CONCLUSION: In a clinical trial with a comprehensive recruitment strategy, neither race nor education predicted enrollment. Instead, clinical characteristics related to the intervention were important predictors of trial participation.

CLINICAL PREDICTION OF DEEP VEIN THROMBOSIS IN SYMPTOMATIC PATIENTS. J Cornuz, WA Ghali, D Hayoz, M Depairon, B Yersin, Department of Medicine, University Hospital, Lausanne, Switzerland

PURPOSE: Before encouraging the widespread implementation a new deep vein thrombosis (DVT) clinical prediction rule developed by Wells and colleagues, we sought to evaluate its predictive performance relative to physicians' own predictions of DVT probability.

METHODS: In a prospective study, 305 consecutive patients referred to the vascular laboratory with clinically suspected DVT underwent: 1) clinical stratification into low, moderate, or high probability of DVT on the basis of both the physicians' and Wells predictions, the latter of which was performed by a clinical nurse; 2) ELISA D-dimer testing with >500 ng/ml as cut-off for a positive result; and 3) duplex ultrasound scanning. The clinicians and nurse were blinded to each others' assessments, and to D-dimer results. The performance of the physician and Wells predictions was expressed by the area under the receiver operating characteristic (ROC) curve.

RESULTS: Ninety-two patients (30%) had DVT (54 proximal, 36 confined to the calf, and 2 non-localized cases diagnosed based on 3 month follow-up). According to physicians' and Wells predictions, 31% and 40%, 46% and 43%, and 23% and 17% were classified as low, moderate, or high risk, respectively. Of the patients classified as low risk, 13% and 12% had DVT, respectively, for physician and Wells predictions. Of the patients classified as moderate risk, 23% and 31% had DVT, whereas for those classified as high risk, 62% and 69% had DVT, respectively. Agreement between the physician and Wells estimates was low (weighted kappa 0.31) suggesting that the physicians were not using the Wells method for their own estimates. The areas under the ROC curve are 0.724 and 0.734 for physicians' and Wells predictions, respectively. Among patients for whom D-dimer testing was available (87%), 4 out of 78 DVT patients had a level under the cut-off of 500 (3 distal and one proximal DVT) with an overall negative predictive value for D-dimer testing of 93%. The negative predictive value for the combination of both a low score and a D-dimer level below 500 was 96% and 100% for physicians' and Wells predictions, respectively. A strategy of not performing ultrasounds in pa-

tients with both a low risk Wells prediction and a D-dimer level below 500 could have avoided 30 of the 97 ultrasounds (31%) performed in patients at low probability.

CONCLUSION: The pretest probability assessed by the Wells prediction rule was slightly better than the physicians' predictions. The performance of the Wells predictions was also notably superior when used in combination with ELISA D-dimer measurements, allowing physicians to confidently avoid ordering ultrasounds in some patients. These results suggest that the Wells prediction rule should be taught to physicians and formally implemented.

THE PREVALENCE, PREDICTORS AND HEALTH STATUS CONSEQUENCES OF ORAL HEALTH LIMITATIONS IN PERSONS WITH SEVERE MENTAL ILLNESS. CK Crews, AJ Davidson, LA Crane, PS Mehler, JF Steiner, General Internal Medicine, Denver Health; Preventive Medicine and Biometrics, School of Medicine, University of Colorado Health Sciences Center; General Internal Medicine, University of Colorado Health Sciences Center, Denver, CO

PURPOSE: Persons with severe and persistent mental illness have high rates of medical comorbidity. Patients frequently note oral health as one area of medical care where their needs are not met. To provide optimal care to this vulnerable population, primary care providers must understand their unique needs. We interviewed a sample of persons with severe mental illness to determine the prevalence of self-reported oral health impairments and their effect on the participants' health status.

METHODS: 160 psychiatric outpatients underwent a personal interview assessing medical conditions, psychiatric diagnosis, and health status using the Medical Outcomes Study Short Form-36 (SF-36) survey. We assessed psychiatric severity using the Brief Psychiatric Rating Scale (BPRS) which rates psychiatric severity from 0 (none) to 6 (extremely severe) in 18 symptom areas. Oral health was assessed using the Oral Health Impact Profile-screening version, which allows respondents to report the frequency of 14 oral health problems on a 5-point scale from 'never' to 'very often'.

RESULTS: 51% of our sample were men, 55% were Caucasian, and 54% diagnosed with schizophrenia. The mean age was 42 and the mean BPRS score was 22.9, reflecting moderate psychiatric impairment. 89% of respondents reported having at least one oral health problem. 58% reported having at least 5 oral health problems. The most common complaints were self-consciousness (64.3%), embarrassment (57.9%), and discomfort eating foods (57.1%) because of problems with teeth. In bivariate analysis, Hispanic ethnicity was associated with worse self-reported overall oral health ($p = .02$). In addition, the total number of dental problems reported by the participants was inversely related to their scores in all eight domains of the SF-36 ($r = -.23$ to $-.42$; all $p < .001$). After controlling for sociodemographic factors, psychiatric illness, and medical conditions, the total number of dental limitations remained significantly associated with lower patient ratings of perceived health in 6 of the 8 SF-36 domains: bodily pain ($p = .0004$), general health status ($p = .0004$), vitality ($p = .004$), role limits-emotional ($p = .004$), social functioning ($p = .05$) and mental health status ($p = .009$).

CONCLUSION: Persons with severe and persistent mental illness have markedly elevated rates of oral health limitations compared to the general population. These problems are associated with worsened self-perceived health status. Addressing oral health may be an important way to improve the overall health of this population.

THE IMPACT OF PODIATRIC PROBLEMS ON THE HEALTH STATUS OF PERSONS WITH SEVERE MENTAL ILLNESS. CK Crews, AJ Davidson, LA Crane, PS Mehler, JF Steiner, General Internal Medicine, Denver Health; Preventive Medicine and Biometrics, School of Medicine; General Internal Medicine, University of Colorado Health Sciences Center, Denver, CO

PURPOSE: We interviewed a sample of persons with severe and persistent mental illness to determine the prevalence of self-reported podiatric impairments and their effect on the participants' health status.

METHODS: 160 psychiatric outpatients underwent a personal interview assessing medical conditions and health status with the Medical Outcomes Study Short Form-36 (SF-36) survey. We assessed psychiatric severity with the Brief Psychiatric Rating Scale (BPRS) which rates psychiatric severity from 0 (none) to 6 (extremely severe) in 18 symptom constructs. Podiatric health was assessed using 10 items from the National Health Interview Survey and an additional item addressing foot pain.

RESULTS: Fifty-one percent of our sample was male, 55% were Caucasian, and 54% diagnosed with schizophrenia. The mean age was 42 and the mean BPRS score was 22.9, reflecting moderate severity of psychiatric illness. Eighty-four percent of patients surveyed reported at least one podiatric problem and 56% reported at least 3 problems. The most common problems were foot pain (47.9%), nail disorders (33.6%), and foot infections (30%). These conditions were much more common than 1990 NHIS general population reports of nail disorders (5.7%) and foot infections (5.0%). In bivariate analysis, the total number of podiatric problems reported by the participants was inversely related to their scores in seven of the eight domains of the SF-36 (all $p < .01$). After controlling for sociodemographic factors, psychiatric illness, and medical conditions, the total number of podiatric limitations remained significantly associated with lower patient ratings in 3 of the 8 SF-36 domains: bodily pain ($p = .02$), vitality ($p = .02$), and general health status ($p = .02$).

CONCLUSION: Persons with severe and persistent mental illness have markedly

elevated rates of podiatric problems when compared to the general population. These problems are associated with worsened self-perceived health status. Addressing podiatric health may be a successful way to improve the overall health of this population.

RACIAL, ETHNIC AND SOCIAL DIFFERENCES IN THE OUTCOMES OF HIV INFECTION IN THE U.S. WE Cunningham, RD Hays, N Duan, RL Beckman, KC Heslin, RM Andersen, TT Nakazono, SA Bozzette, MF Shapiro, UCLA, RAND, UCSD and the HCSUS Consortium, Los Angeles, CA

Although blacks and drug users have been found to underuse highly active anti-retroviral therapy (HAART), it is not known whether this results in poorer health outcomes. To examine this issue we followed from 1996 to 1998 a cohort of 2267 adults in the HIV Costs and Services Utilization Study (HCSUS), a nationally representative study of persons under care for HIV, and examined predictors of survival. The population represented was 34% aged 18-34, 22% female, 51% white, 31% black, 15% Hispanic, 50% male-to-male sexually exposed, 23% drug use exposed, 19% heterosexually exposed, 8% blood product exposed, 24% less than high school educated, 19% with annual income less than \$5,000, 21% with CD4 <50, and 20% uninsured. By January 1998, 261 persons (12%) had died, and in bivariate analysis certain groups had a greater relative risk of death: blacks (RR = 1.73, $p < .001$) and Hispanics (RR = 1.48, $p < .04$) compared to whites; those exposed to HIV through blood product transfusion compared to those male-to-male sexually exposed (RR = 3.17, $p < .0001$); those with less than a high school education compared to college grads (RR = 2.09, $p < .002$); those with incomes \$10,001-25,000 (RR = 1.79, $p < .001$) compared to >\$25,000; those with Medicare (RR = 2.20, $P < .003$) and Medicaid (RR = 2.15, $p < .02$) compared to privately insured; and those with CD4 counts <50 (RR = 8.58, $p < .01$) compared to > 500. In Cox proportional hazards regression models, blacks continued to have an elevated relative risk of death (RR = 1.60, 95% CI = 1.08-2.39, $p < .03$) when controlling for CD4 count, age, gender, HIV exposure, education, income, and insurance status. However, the relative risk of dying by January 1998 for blacks (RR = 1.41, 95% CI = 0.96-2.08) was reduced, after accounting for the 33-fold reduction in the relative risk of death from being on HAART (RR = 0.03, 95% CI = 0.01-0.06) as of a median date of October 1997. Only persons with low CD4 counts and persons exposed to HIV by blood products continued to have an elevated relative risk of death in the final multivariate model. Our findings provide nationally generalizable evidence that minorities and low socioeconomic groups have worse survival than other groups and that the worse survival of blacks is largely explained by lack of treatment with HAART, and not simply by CD4 count, socioeconomic factors, or insurance status. To improve health outcomes for blacks and other disadvantaged groups with HIV infection, improved delivery of effective treatment is indicated.

HEALTH UTILITY SCORES APPLICABLE TO DECISION MODELS THAT INCLUDE CLOTTING EVENTS. S Cykert, NW Phifer, C Hansen, Division of General Internal Medicine and Clinical Epidemiology, University of North Carolina School of Medicine, Chapel Hill; Internal Medicine Program, Moses Cone Hospital, Greensboro, NC

PURPOSE: Clinical controversy currently exists regarding preventive therapies that effect an individuals' risk of experiencing a thromboembolic event. Three particular areas for which clinical decisions aren't clear include hormone replacement in women with coronary artery disease, the use of tamoxifen for the prevention of breast cancer, and anti-coagulant therapy for asymptomatic patients who carry the Factor V Leiden gene. This study was performed to define health utility scores for clotting states relevant to the creation of decision models in this field and to identify demographic groups for which these utility scores vary significantly from the mean.

METHODS: We recruited a stratified random sample of 160 individuals aged 50 to 75 yrs from church groups, health fairs, and medical clinics in urban areas of North Carolina and Florida. We administered a structured questionnaire to obtain demographic information and to assess patient preferences using a standard gamble approach. Standard gamble results were converted to a utility scale for which 1.0 = perfect health and 0 = death. Bivariate analysis was performed.

RESULTS: Sixty-three% of respondents were women and 41% African-American. The average age was 59 yrs. 19% had less than a high school education, 39% were high school grads, and 42% were educated beyond high school. Utility scores were as follows:

Uncomplicated DVT .86
DVT with post-phlebotic syndrome .62
Pulmonary embolus .81
Stroke with left hemiparesis in 5 yrs
(but functional) .53
Stroke with right hemiparesis
and aphasia in 5 yrs .32

Higher age and African-American race predicted significantly higher utility scores for the severe stroke outcome.

CONCLUSION: The above utility scores can be used in decision models that are designed to answer the questions elucidated in the introduction. With the exception of age and race in severe stroke, very little variation exists when these scores are analyzed by demographic groups.

UTILITY SCORES THAT DESCRIBE PERCEPTIONS OF DEBILITATED HEALTH STATES DEMONSTRATE CONSISTENT RACIAL VARIATION. S Cykert, NW Phifer, A Walker, Division of General Internal Medicine and Clinical Epidemiology, University of North Carolina School of Medicine, Chapel Hill; Internal Medicine Program, Moses Cone Hospital; Department of Sociology, University of North Carolina at Greensboro, Greensboro, NC

PURPOSE: Recently, our group described marked racial differences in perceptions of debilitated health states as measured by health utility scores. The initial study was limited by small numbers and a single geographic location. The study has been expanded to confirm the validity and generalizability of the initial results.

METHODS: We recruited a stratified random sample of 160 individuals aged 50 to 75 yrs from church groups, health fairs, and medical clinics in urban areas of North Carolina and Florida. We administered a structured questionnaire to obtain demographic information and assess patient preferences using a standard gamble approach. Standard gamble results were converted to a utility scale for which 1.0 = perfect health and 0 = death. Bivariate and multivariate analyses were performed.

RESULTS: In the bivariate analysis, African-American(AA)as compared to white(W)race predicted significantly higher utility scores for the 3 debilitated health states reported in our previous study. The specific categories of debility measured were inability to perform activities of daily living [AA .43, W .20], tolerance of only bed to chair activity [AA .33, W .18], and permanent nursing home placement [AA .37, W .14].The multivariate results confirmed that race remained the strongest predictor of utility scores for these outcomes while gender exerted some influence. Age, self-reported health, education, income, marital status, and health insurance had no influence on these utility scores.

CONCLUSION: We conclude that ethnic differences likely affect perceptions of physical debility. We derive 2 potential implications from these data:

- (1) To avoid unintended bias against under-represented groups, subgroup or sensitivity analyses of utility scores that account for differences according to gender and race should be included in decision analysis methods.
- (2) The described differences in health perceptions may, in part, explain variations in health care participation, e.g., lack of interest in cancer screening compared to desire for aggressive interventions at the end of life.

TESTICULAR CANCER SCREENING: LET'S GET THE BALL ROLLING. A Dean, SUNY Upstate Medical University, Syracuse, NY

BACKGROUND: Testicular cancer accounts for approximately 1% of all male cancers. It is the most common cancer which affects young males ages 15 to 35. While testicular self-examination is the gold standard of screening for testicular cancer, various studies have demonstrated insufficient evidence to justify routine screening. Prevalence of routine screening is low for testicular self-examination. There is scarce data in regards to physician attitudes and promotion of screening. There have not been large-scale studies on U.S. university campuses relating to screening and education.

METHODS: A survey of five questions was distributed to a convenience sample of 100 university male students at Cornell University in Ithaca, New York in 1999. The survey was comprised of the following questions: 1. Age; 2. Have you ever performed testicular self-examination for the evaluation of testicular cancer? 3. If yes, do you perform testicular self-examination monthly? 4. Has a doctor ever educated you on how to do testicular self-examination? 5. Has a nurse ever educated you on how to do testicular self-examination?

RESULTS: A sample of 100 males ages 17-49 were surveyed. Eighty percent of the sample was between the ages of 17 and 22. Testicular self-examination was performed by at least 42% of these college males. Only 9% performed testicular examination on a monthly basis, however. In terms of education, 28% stated that a physician educated them on testicular self-exam while 14% of the respondents were educated by nurses.

CONCLUSION: This study done at an American northeastern university found that 42% of the sample ever performed the testicular self-examination. This percentage is higher than reported in European studies of similar samples. This pilot study shows the need for further study of this important health problem.

THE EFFECT OF SEXUAL ORIENTATION ON MULTIPLE DIMENSIONS OF HEALTH: A POPULATION-BASED STUDY OF WOMEN. AL Diamant, C Wold, K Spritzer, L Gelberg, Dept. of Health Services, UCLA, Los Angeles and The Los Angeles County

PURPOSE: To better understand lesbian health, for which there is dearth of validated information, we used population-based data to assess lesbian's health behaviors, health status and their access to and use of health care.

Our study population was drawn from a population-based sample of women; the 1997 Los Angeles County Health Survey. Participants reported their sexual orientation and the gender of their sexual partners during the preceding year (male, female or both). These analyses include 4,661 women: 4,592 heterosexual women and 69 lesbians including 18 women who identified as heterosexual but reported female sexual partners only. We calculated adjusted relative risks to assess the effect of sexual orientation on aspects of health.

Lesbians were more likely than heterosexual women to report any alcohol consumption (RR 1.20, CI 1.05, 1.33), to drink heavily (RR 6.5, CI 2.26, 13.85) and to

use tobacco products (RR 1.69, CI 1.25, 2.12). During the prior year, lesbians were more likely to lack health insurance (RR 1.34, CI 1.08, 1.56) and more likely to have experienced difficulty seeing a physician (RR 1.59, CI 1.21, 1.93). During the preceding two years lesbians were less likely to have had a Pap smear (RR 0.61, CI 0.43, 0.83) and a clinical breast exam (RR 0.65, CI 0.46, 0.88).

In this first population-based study of lesbian health, we found that lesbians were more likely than heterosexual women to have poor health behaviors, and less likely to receive preventive health care services. These findings support our hypothesis that sexual orientation/behavior is an independent predictor for health behaviors and receipt of care, and supports the need for the increased systematic study of lesbian health issues.

COST-EFFECTIVENESS OF COLORECTAL CANCER SCREENING IN THE ELDERLY. RS Diffus, RM Ness, RW Klein, AM Holmes, Department of Medicine, Vanderbilt University; Quality Scholars Program and GRECC, Nashville VAMC, Nashville, TN; Department of Medicine and School of Public and Environmental Affairs, Indiana University; Medical Decision Modeling, Inc., Indianapolis, IN

PURPOSE: One-time colonoscopic screening (CS) has been recommended as a cost-effective approach to the primary and secondary prevention of colorectal cancer (CRC). Whereas one-time CS is most cost-effectively performed around age 50, elderly patients frequently present to primary care physicians having never had prior screening. We sought to address the question: Among elderly patients without a prior history of CRC screening, what is the cost-effectiveness of one-time colonoscopic screening?

METHODS: We constructed a discrete-event simulation model of the natural history of colorectal neoplasia in women and men. Probability estimates for the risk of CRC, the incidence of colorectal adenomas and cancers, the mortality from CRC, the diagnostic performance of CS, the complication rates of CS and surgery, and the direct costs of CRC diagnosis and treatment were summarized from the literature. Quality adjusted life years (QALYs) were computed as the measure of effectiveness using utilities as the weights. We used published utility estimates for the outcome states of CRC we previously determined using an instrument specifically designed to measure utilities for outcome states of CRC. We created separate hypothetical cohorts of 100,000 persons who survived to the target age without prior colonoscopy or previously diagnosed colorectal adenomas or cancer for each gender and age group (60, 65, 70, 75, 80, 85, and 90 years of age). We modeled one-time colonoscopic screening for each gender-age group to examine the cost-utility compared to no screening. The screening strategy included the removal of all identified polyps, resection of all identified CRCs and CS surveillance for all patients with adenomas or CRCs. We discounted costs and QALYs at 3% per annum.

RESULTS: Compared to no screening, screening CS among 60, 70, and 80 year old women:men was associated with an absolute reduction in CRC incidence of 3.77%:3.98%, 3.14%:3.14%, and 1.48%:1.18%, respectively. In addition, screening CS was associated with an absolute reduction in overall deaths related directly to CRC or its diagnosis and treatment among 60, 70, and 80 year old women:men of 1.66%:1.73%, 1.48%:1.55%, and 0.96%:1.00%, respectively. Screening CS was cost-saving for men and women at ages 60, 65, and 70; the cost per QALY was (women:men) \$1,604:\$2,104, \$9,143:\$11,560, and \$133,000:\$106,800 for ages 75, 80, and 85 respectively, and screening at age 90 was less effective and more costly than not screening.

CONCLUSION: One-time CS for CRC prevention, among patients without prior CS or known colorectal adenomas or CRC, remains cost-effective through age 80 for men and women.

EFFECT OF AGE ON COLORECTAL CANCER SCREENING OUTCOMES. JG Dolan, LM Miller, Medicine, University of Rochester, Rochester, NY; National Association of Chain Drug Stores, Alexandria, VA

PURPOSE: Current colorectal cancer screening guidelines recommend regular screening for average risk patients beginning at age 50 and continuing indefinitely until contraindicated by failing health. The purpose of this study was to determine the effects of patient age on the outcomes of currently recommended colorectal cancer screening programs for average risk patients.

METHODS: We used computer simulation to estimate, for average risk patients aged 50 through 80, the anticipated rates of cancer, likelihoods of major side effects (death, major bleeding, and intestinal perforation), and false positive rates associated with the 5 screening programs endorsed by the 1997 guidelines published by the American Gastroenterological Association: annual fecal occult blood tests (FOBT), flexible sigmoidoscopy (FS) every 5 years, annual occult blood tests and flexible sigmoidoscopy (F&S), double contrast barium enema (BE) every 5–10 years, and colonoscopy (CS) every 10 years (*Gastroenterology* 1997;112:594)

RESULTS: Age significantly affected the programs' relative performance on all outcomes except mortality; these results are summarized in the table below, which shows the estimated probabilities (p) of these outcomes associated with the best and worst program in each category. BE is the most effective program in reducing cancer risk for patients between 50 and 65. At 65, F&S becomes associated with the lowest cancer rates & becomes progressively more effective than the other programs as patients grow older. At all ages, FS is the least likely to cause major side effects and CS the most likely; as patients age, the differences between these 2 programs, and among all the programs, become progressively smaller. At all ages, FS is least likely to result in false positive results while F&S is the most likely. The

differences between these 2 programs, and among all the programs, become progressively smaller as patients age.

AGE 50		AGE 50 Worst		AGE 55 Best	
Cancer (p)	BE (0.0128)	FS (0.0287)	0.0159	F&S (0.0231)	
Side Effect (p)	FS (0.0007)	CS (0.0156)	0.0149	FS (0.0008)	
False Positive (p)	FS (0.0043)	F&S (2.171)	2.167	FS (0.0037)	

AGE 50		AGE 55 Worst		AGE 80 Worst	
Cancer (p)	FOBT (0.0361)	0.013	F&S (0.0298)	FOBT (0.0483)	0.0185
Side Effect (p)	CS (0.0073)	0.0065	FS (0.0006)	CS (0.0036)	0.003
False Positive (p)	F&S (1.0372)	1.0335	FS (0.0025)	F&S (0.4195)	0.417

CONCLUSION: Patient age significantly impacts the results of colorectal cancer screening. Because these age-related differences in expected outcomes could affect the choice of optimal screening program for many patients, this effect should be taken into account when making screening recommendations for individual patients. Screening guidelines should also adjust for these differences by including sequences of age-appropriate screening tests in future recommendations.

A MATHEMATICAL MODEL TO ADJUST WARFARIN DOSE DURING CHRONIC ANTICOAGULATION. AS Dunn, CB Schechter, B Lotzman, JL Halperin, Medicine; Community Medicine, Mount Sinai School of Medicine, New York, NY

PURPOSE: Clinicians typically adjust warfarin doses based on their clinical experience to maintain the international normalized ratio (INR) within a therapeutic range. Practitioners are usually only moderately effective at achieving an INR within the target range. We developed a mathematical model to predict warfarin adjustments and compared the model with the usual care provided by clinicians. **METHODS:** Visits to a cardiology anticoagulation clinic from January 1997 through June 1999 were examined. For each visit we abstracted from the database the patients' INR, warfarin dose (expressed as mg/week), and the number of days since the preceding visit. Patients were included if their target INR was 2.0–3.0. Visits were excluded from analysis if the patient was noncompliant or taking an incorrect dose, if warfarin had been recently omitted, the current or previous INR or the warfarin dose was missing, or the time from the preceding visit was <3 days or >42 days. We then fitted a fixed-effects log-linear model which included a patient-specific term that represents all idiosyncratic aspects of that patients' responses to warfarin which remain constant over time, and an error term unique to each patient and each visit. Bayes theorem was used to estimate the idiosyncratic aspects of that patients' responses to warfarin. Graphical exploration of residual plots showed no evidence of major problems with model fit.

RESULTS: A total of 267 patients were identified, 172 (64%) of whom had a target INR of 2.0–3.0. Exclusion of visits with missing data reduced the sample to 146 patients. The average age was 66 years, and 43% of patients were female. The performance of the model was analyzed both to achieve an INR between 2.0 and 3.0, and to achieve a wider "safe" INR between 1.5 and 4.0. If the model was used, the probability that the actual INR on the next visit would be between 2.0 and 3.0 was 52% and within the range of 1.5–4.0 was 91%. These probabilities increased over subsequent visits to 54% and 93%, respectively. These numbers compared favorably to the experience with usual practice, where 54% of INR's were within the 2.0–3.0 range and 89% of INR's were between 1.5 and 4.0.

CONCLUSION: We have developed a Bayesian model of warfarin dosing to predict a warfarin dose from the current INR and the previous dose of warfarin. The model was as effective as usual practice at achieving an INR within the goal range and better than usual practice at maintaining a safe INR. The model is currently undergoing validation in other settings. If validated, use of this model may decrease the resources needed to determine warfarin dose and reduce the proportion of very high and very low INRs.

EXPLORING WOMEN'S MOTIVATION TO CONTROL WEIGHT: A LIFE PHASE AND RACIAL ASSESSMENT OF THE STAGES OF CHANGE FOR WEIGHT LOSS IN WOMEN. EC Dwamena, FK Ahmed, BA Dwamena, Medicine, Michigan State University, East Lansing; Medicine, University of Michigan, Ann Arbor, MI

PURPOSE: Few studies have systematically assessed women's motivation in weight control. We sought to assess the readiness of women to control their weight and to determine differences in readiness based on life phase (late adolescence, young adulthood, middle age, advanced age), race and socio-economic status.

METHODS: We performed a mailed survey of a random sample of women 18 to 74 years old seen the previous year (1998) in a large multi-specialty faculty group practice. Demographic variables, weight, height, dietary fat consumption, smoking status, readiness to control weight and physical activity were measured with items from standardized instruments. Multivariate logistic regression was used to assess predictors of readiness to control weight. Body mass index (BMI), age, race, education, employment status, smoking status, and presence of diabetes, hypertension and hypercholesterolemia were independent variables while readiness to control weight was a dependent variable.

RESULTS: In all, 670 (56%) of 1200 women responded to the questionnaires. Almost 50% of women were either overweight (n = 181, 27%) or obese (n = 135, 20%). Overweight women (adjusted odds ratio (OR) = 0.51, p = 0.04), obese women (OR = 0.43, p = 0.02) and black women (OR = 0.35, p = 0.03) were less

likely to indicate readiness to to control weight than normal weight women and white women respectively. Women who had never smoked were more likely to report actively trying to control weight than current smokers (OR = 2.57, $p = 0.01$). There was no difference in readiness to control weight among women of different life phases once adjustments were made for BMI. Although 102 (76%) of obese women claimed to be actively trying to control weight, only 34 (25%) had been engaged in physical activity for at least 1 month and only 24 (18%) reported consuming a low fat diet.

CONCLUSION: Black women and women smokers were significantly less likely to report actively trying to control weight than white women and non-smokers respectively. This may present unique interventional and/or research opportunities. There is a large discordance in women's reported readiness to control weight and their reported physical activity and dietary fat consumption. Elucidation of the reasons for this discordance would be helpful in the fight against obesity with its attendant morbidity and mortality.

RACIAL AND GENDER DIFFERENCES IN QUALITY OF LIFE AMONG ACUTE MYOCARDIAL INFARCTION SURVIVORS FROM COMMUNITY HOSPITALS. EC Dwamena, RE Watson, FK Ahmed, Medicine, Michigan State University, East Lansing, MI

PURPOSE: Previous studies of quality of life in acute myocardial infarction (AMI) have been limited to participants of clinical trials. We sought to examine the quality of life of survivors of AMI in 5 hospitals in 2 mid-Michigan communities, 3 and 6 months after AMI, and to examine potential racial and gender-based differences.

METHODS: We performed 3- and 6-month interviews in 241 out of 393 consenting patients prospectively identified during hospitalization for AMI between January 1994 and April 1995. Demographic variables and quality of life after myocardial infarction (QLMI) were measured with items from standardized instruments. The mean (and standard deviation) at 3 months and 6 months for each of 5 domains of QLMI (emotions, restrictions, symptoms, confidence and self-esteem) were calculated for different groups. Best quality of life score = 7 and worst score = 1. Appropriate t-tests were performed to compare between and within group differences.

RESULTS: With the exception of symptoms at 3 months, means for all domains of QLMI were >5 . The means improved from 3 months to 6 months for restrictions (5.47 to 5.65, $p = 0.006$), symptoms (4.8 to 5.63, $p < 0.001$), and confidence (5.11 to 5.3, $p = 0.001$). There was no difference between the overall mean 3-month and 6-month scores for emotions and self esteem. In subgroup analyses, blacks reported slightly worse emotions, and whites reported slightly improved emotions (see Table). Blacks had a significantly lower score than whites at 3 months in restrictions (4.03 vs 5.42, $p = 0.001$) and confidence (4.53 vs 5.11, $p = 0.04$) as well as in emotions, confidence (4.00 vs 4.9, $p = 0.001$) and symptoms at 6 months. Females had scores similar to men in restrictions, confidence and self-esteem both at 3 months and at 6 month. Other gender and racial differences are presented in the Table.

	Symptoms mean (standard deviation)			Emotions mean (standard deviation)		
	3 months	6 months	p value	3 months	6 months	p value
Blacks	4.38 (0.85)	4.88 (1.42)	0.04	4.90 (1.12)	4.55 (1.38)	0.13
Whites	4.77 (0.78)	5.64 (1.02)	0.001	5.37 (1.14)	5.48 (1.12)	0.11
p value	0.06	0.01		0.10	0.001	
Males	4.87 (0.76)	5.67 (1.09)	0.001	5.52 (1.39)	5.58 (1.12)	0.27
Females	4.64 (0.84)	5.53 (0.91)	0.01	5.07	5.13 (1.11)	
p value	0.03	0.23		0.02	0.02	

CONCLUSION: The quality of life for this community cohort of patients with AMI was good and improved from 3 months to 6 months. However, there were significant racial and gender differences that warrant further elucidation.

PREVALENCE AND CLINICAL IMPLICATIONS OF HYPOCALCEMIA IN TRAUMA PATIENTS. A Ebrahim, M Marshall, MH Rashid, NYMC/WMC; Lincoln

PURPOSE: Although calcium homeostasis in the critically ill patients in the medical and surgical Intensive Care Units has received some attention, disturbances of Calcium level after musculoskeletal trauma in the otherwise healthy individual have not been widely reported. The present study aims to explore the prevalence of hypocalcemia in trauma patients and its relationship to the severity of trauma and length of hospital stay.

METHODS: Total and corrected calcium levels were prospectively evaluated upon admission and followed until normalization or discharge for all patients admitted to the Trauma service at Westchester Medical Center over a period of one month between July/August 1999. Along with serum calcium, serum magnesium, Phosphorous, albumin, BUN/Cr, amylase, daily blood transfusions and crystalloid administration were also noted from the day of admission till either discharge or normalization of serum calcium.

RESULTS: A high proportion of patients (44/56 = 79%) were noted to be hypocalcemic. 26 of the 44 were hypocalcemic at admission whereas 18/44 developed hypocalcemia during hospitalization. For patients who were normocalcemic at admission and subsequently became hypocalcemic, the average time to nadir was 19 hours post admission. Majority of the patients recovered spontaneously, the aver-

age time to recovery was 3.8 days. The vast majority had no identifiable cause of hypocalcemia.

The average minimum corrected calcium level for those patients staying <10 days ($n = 27$) was 8.06 compared with 7.71 for those staying >10 days ($n = 29$) and this difference was statistically significant [p value < 0.005]. Ca levels correlated significantly with the Injury Severity Score (ISS). The ISS is a nationally recognized system for assessment of severity of trauma, ranging between 1-75, 1 being the least amount of trauma, 75 being incompatible with survival. It is categorized into ≤ 8 (minor trauma), 9-24 (moderate trauma) and ≥ 25 (severe trauma). The average minimum Ca level was 8.14 for patients with an ISS ≤ 8 compared to 7.4 for patients with an ISS ≥ 25 , and this difference was statistically significant [p value < 0.005]. None of the patients with an ISS ≥ 25 were normocalcemic. 3/66 patients died during hospitalization, all three were hypocalcemic at admission, only one normalized prior to death.

CONCLUSION: Although the etiology remains unclear, a high incidence of hypocalcemia has been seen in patients with musculoskeletal trauma. Therefore, trauma should be included in the differential diagnosis of hypocalcemia in the hospitalized patient.

UNDIAGNOSED DIABETES AMONG VETERANS AFFAIRS MEDICAL CENTER OUTPATIENTS: PREVALENCE AND RISK FACTORS. D Edelman, LJ Edwards, TK Dudley, AC Harris, EZ Oddone, Center for Health Services Research in Primary Care, Durham VA Medical Center, Durham, NC

PURPOSE: Advocacy groups recommend screening all adults over the age of 45 for diabetes. Our objectives were to determine the prevalence of unrecognized diabetes among VA Medical Center outpatients, and to identify associated risk factors that may define the optimal target population for diabetes screening.

METHODS: We contacted by mail all veterans age 45-64 who used the Durham Veterans Affairs Medical Center between 1/98 and 3/99 and asked if they had diabetes. All respondents who did not report having diabetes were eligible for the screening study. We enrolled subjects at the time of an outpatient visit to the Medical Center. We obtained demographics, a structured medical history, height and weight for all subjects. We screened patients by using an initial random Hemoglobin A1c (HbA1c) measurement, and then obtaining follow-up fasting plasma glucose (FPG) for all subjects with HbA1c $\geq 6.0\%$. We defined a case of unrecognized diabetes as either HbA1c $\geq 7.0\%$ or FPG ≥ 126 . Logistic regression was performed to determine independent factors associated with undiagnosed diabetes and adjusted odds ratios.

RESULTS: We enrolled 1253 subjects. The prevalence of unrecognized diabetes was 4.5% (95% CI, 3.4, 5.7). Independent risk factors for diabetes identified by logistic regression were: self-reported hypertension (adjusted OR = 2.5, $p = 0.004$); obesity as defined by weight more than 20% above ideal body weight (adjusted OR = 2.2, $p = 0.02$); and self-reported family history of diabetes (adjusted OR = 1.7, $p = 0.06$). Non-caucasian race (a known risk factor for diabetes) was not associated with the new diagnosis of diabetes in our sample. Considering obesity, family history, and hypertension as 3 risk factors for unrecognized diabetes, the table shows the prevalence of unrecognized diabetes in patients with each number of risk factors ($p = 0.001$ for trend):

Number of risk factors	Prevalence (%)	95% CI
0	1/183 (0.5%)	0.0%, 3.5%
1	14/418 (3.3%)	1.9%, 5.7%
2	24/461 (5.2%)	3.4%, 7.8%
3	17/171 (9.9%)	6.1%, 15.7%

CONCLUSION: There is an opportunity to find patients with unrecognized diabetes in a medical center setting. Assessing risk factors prior to blood testing will improve the specificity, with little cost in sensitivity, of screening for diabetes in a medical center setting. The target population which optimizes any potential value of diabetes screening is patients with at least 1 of 3 risk factors for diabetes. Hypertension is strongly associated with unrecognized diabetes.

CHARACTERISTICS OF PERSONS WITH DIABETES WITH HIGH EMERGENCY DEPARTMENT UTILIZATION IN SOUTH CAROLINA 1997. LE Egede, D Zheng, DT Lackland, MD Silverstein, General Internal Medicine; Epidemiology and Biostatistics, Medical University of South Carolina, Charleston, SC

PURPOSE: The prevalence of self-reported diabetes in South Carolina (SC) in 1997 of 5.7% was higher than the national average of 5.4%. The economic burden of diabetes including emergency department (ED) costs was estimated to be \$73 million nationwide in 1997, which represented a 250% increase in cost compared to 1987. We analyzed high ED utilizers in the state of SC to identify characteristics of diabetic person within the group.

METHODS: Cross-sectional study using statewide ED data in 1997 of 102,811 patients with >1 ED visit. High utilization was defined as >1 ED visit with primary discharge diagnosis of diabetes. Non-urgent visit was defined as an ED visit that did not result in hospitalization. Only whites and blacks were included. Multiple

logistic regression was used to assess characteristics of diabetic and non-diabetic high utilizers, controlling for month of initial visit during the year.

RESULTS: VARIABLES	DM (n = 698)	Non-DM (n = 102,113)	ADJUSTED OR	ADJUSTED P-VALUE
MALES	44%	37%	1.40	<0.0001
BLACKS	68%	50%	2.40	<0.0001
MEAN AGE (YRS)	52	34	1.04	<0.0001
SELF-PAY	17%	26%	1.00	
MEDICARE	44%	18%	0.94	0.43
MEDICAID	23%	27%	1.40	<0.0001
PRIVATE	17%	29%	0.82	0.004
CHARGES	\$514	\$383	—	NS

CONCLUSION: Diabetics with high ED utilization were more likely to be older, male, black, and insured by Medicaid or commercial insurance. Interventions to reduce high ED utilization in diabetic persons in SC should target older black males in the Medicaid program.

BETTER METHODS ARE NEEDED FOR SCREENING AND MANAGEMENT OF ALCOHOL PROBLEMS IN PRIMARY CARE. JM Eisenman, NV Dawson, C Thomas, R McCormick, T Parren, D Einstadter, RD Cebul, Case Western Reserve University at MetroHealth Medical Center, Cleveland, OH

PURPOSE: We evaluated physician chart documentation of an alcohol history (screening) and recognition and treatment of alcohol problems as part of an ongoing randomized (firm) trial of alcohol screening and management in a primary care practice housed in a county hospital.

METHODS: Randomly selected charts (100 women and 100 men) from each of three firms and a faculty practice were reviewed for documentation of alcohol history, problem recognition and treatment (education, counseling, referral) during the 1997 calendar year. Levels of chart documentation of current alcohol problems were evaluated by gender, age, race and economic status (census tract median household income).

RESULTS: Overall, 47%₆₀₀ (60%) patients had documentation of an alcohol history. No differences in screening, recognition of problems or treatment across firms or faculty groups were noted. Screening methods included Quantity and Frequency (95%), CAGE (2%), AUDIT (0%), and Other (3%). 41%₄₇₈ patients (8.5%) were documented to have a current alcohol problem (34/260 or 13% of men and 7%₂₁₈ or 3% of women). Among men with documented alcohol problems, 12 were between the ages of 18–39 and 22 were between 40 and 64—no men 65 or older were identified as having an alcohol problem. No differences were noted in documentation by race or economic status. Only 23%₄₁ (56%) patients with documented alcohol problems received treatment.

CONCLUSION: Widely available, validated alcohol-screening instruments were almost never used. The lack of documented alcohol problems in patients 65 or older probably represents an underestimate. Less than two thirds of patients who had an alcohol problem identified were treated. These findings suggest that better methods for screening and management of alcohol problems in primary care settings are needed.

RURAL-URBAN DIFFERENCES IN ACCESS TO REPERFUSION FOR ACUTE MYOCARDIAL INFARCTION. EF Ellerbeck, KK Engelman, CA Hornberger, MC Mosier, PA Howard, Department of Preventive Medicine, University of Kansas Medical Center, Kansas City; Kansas Foundation for Medical Care, Topeka; Department of Pharmacy Practice, University of Kansas Medical Center, Kansas City, KS

PURPOSE: To describe the relationship between proximity to tertiary care hospitals and access to emergent reperfusion for acute myocardial infarction (AMI).

METHODS: We identified a random sample of 1,441 Medicare patients hospitalized for AMI in Kansas between 1%₉₆ and 5%₉₇. Using zip code data, we estimated the distance from the patients' residence to the nearest tertiary care hospital (i.e., hospital capable of performing coronary artery bypass grafting (CABG) or emergent angioplasty). We obtained data on patient demographics, clinical characteristics, and treatment from hospital claims data and hospital medical records.

RESULTS: We identified 11 tertiary care hospitals located in 3 metropolitan areas in Kansas. The median distance to the nearest tertiary care hospital was 36 miles with 25% of patients living more than 75 miles away. Patients living within 36 miles (urban patients) were more likely to present within 6 hours of onset of pain (44% vs 38%; $p = 0.02$). Patients residing more than 36 miles from tertiary care (rural patients) were more likely to receive care at two different hospitals (47% vs 20%; $p < 0.001$). Rural and urban residents were equally likely to receive thrombolysis, but urban residents were more likely than rural residents to receive emergent angioplasty (25% vs 8%; $p < 0.001$). In a multivariable model, rural residence, age, Charlson comorbidity score, and time to hospital presentation were associated with receipt of emergency reperfusion (either thrombolysis or emergent angioplasty). The odds ratio for emergent reperfusion for rural patients was 0.49 (95% CI: 0.38–0.63).

CONCLUSION: Many Kansas residents suffering from AMI do not live in close proximity to tertiary care medical centers. Due to their decreased access to emergent angioplasty, these patients are much less likely than urban patients to receive emergency reperfusion.

PREVALENCE OF ELEVATED BLOOD PRESSURE AMONG HOMELESS PERSONS IN AN EMERGENCY SHELTER. PJ Ellis, RI Jahiel, Primary Care Residency Program, Yale University School of Medicine, Waterbury, CT

PURPOSE: To assess the prevalence of high blood pressure among homeless persons in an emergency shelter.

METHODS: To provide medical care to the homeless in our community, primary care residents and faculty began providing on-site medical care at an emergency overnight homeless shelter clinic during one-half day per week in May, 1997. A resident-faculty team treated persons with acute complaints and scheduled follow-up for patients with chronic medical problems into one of two hospital-based clinics. We performed a retrospective chart review of all patient encounters from May 1997 through March 1999. We counted patients as having high blood pressure if they had systolic blood pressures (SBP) ≥ 140 mm Hg or diastolic blood pressures (DBP) ≥ 90 mm Hg during one or more visits. For patients with multiple encounters, we calculated the mean blood pressure over up to three visits.

RESULTS: Over a 23-month period, 202 patients made 341 visits. Thirty-six of 163 adults (22%) had high blood pressure. The percentage of adults in specific age groups having high blood pressure was 0% (0/30) in those aged 18–29 years, 19% (20/106) in those aged 30–49 years, 61% (14/23) in those aged 50–64, and 50% (2/4) in those aged over 65. Among patients with high blood pressure, the mean overall SBP was 163 mm Hg (S.D. 23) and mean overall DBP was 97 mm Hg (S.D. 13). Twenty of these 36 patients (44%) had a SBP ≥ 140 mm Hg or DBP ≥ 90 mm Hg (Stage 1), 10 patients (28%) had SBP ≥ 160 mm Hg or DBP ≥ 100 mm Hg (Stage 2), and 6 patients (17%) had SBP ≥ 180 mm Hg or DBP ≥ 110 mm Hg (Stage 3). Twenty-seven of 36 patients (81%) were taking no antihypertensives when seen initially. Nineteen of 36 patients (53%) reported no history of hypertension. Nineteen of 26 patients (73%) who were scheduled for a follow-up appointment at a hospital based clinic kept an appointment. Overall, 19 of 36 patients (56%) with high blood pressure followed-up with a physician after discharge from the emergency shelter.

CONCLUSION: Among homeless persons seeking medical care in an emergency shelter, high blood pressure readings were common, especially in those over 30 years. Most persons with high blood pressure did not report a history of hypertension and were not taking antihypertensive medications. Although many patients were lost to follow-up after discharge from the shelter, a majority of homeless persons actually kept their scheduled appointments. Clinicians working in homeless shelter clinics should consider checking blood pressure in all adults, especially those over age 30. Because many homeless patients with high blood pressure may not follow-up after discharge from emergency shelters, clinicians should consider alternative ways to care for these patients.

COST OF DYSPNEA IN ACUTELY ILL CANCER PATIENTS WITH IMMINENT DEATH. CP Escalante, CG Martin, S Gao, P Horace, M Levett, V Ho, E L Lee, S Fernandez, T Grover, L Elting, General Internal Medicine; Health Services Research, University of TX M D Anderson Cancer Center, Houston, TX

PURPOSE: Dyspnea is a terminal symptom in patients (pts) with advanced cancer (ca) and is the fourth most common complaint of pts presenting to the Emergency Center (EC) at our institution. Previous studies at our institution suggest that approximately 20% of ca pts with dyspnea die within 2 weeks (imminent death) of their EC visit. We conducted a prospective study of 269 ca pts reporting to our EC with a complaint of dyspnea from 3%₆₆–12/97. The purpose of this analysis was to compare dyspnea-related cost in ca pts who survived ≤ 2 weeks from their EC visit to those with longer survival both from the institutional and societal perspectives.

METHODS: The median age was 58 years (range, 18–92) and 170 (63%) were female. 32% had breast ca, 30%, lung ca and 38%, other ca. Most, 74%, had metastases at presentation and 65% had progressive disease. The most common causes of dyspnea were pneumonia and pleural effusion. 143 (53%) were admitted to the hospital and their median length of stay was 7 days (range 1–35 days). 47 (17%) pts died imminently. 5 were admitted to hospice from the EC. Cost was estimated from the payors' perspective and included hospital and clinic (h/c) charges (chg) and professional (prof) fees. Chgs unrelated to dyspnea were excluded by clinician investigators.

RESULTS:

	Mean pt h/c chg (\$)	Mean pt prof chg (\$)	Mean h/c chg/day of life (\$)	Mean prof chg/ day of life (\$)
Died ≤ 2 wks	13,854.09	2,021.14	2,009.70	293.19
Lived > 2 wks	17,413.66	3,517.91	61.90	12.51

Compared to the conventional \$50,000 per year of life threshold for cost-effective interventions, the total cost/day of life for pts who survived ≤ 2 weeks is \$40,550 per year of life and \$27,160 per year of life for pts who survived > 2 weeks.

CONCLUSION: Ca pts with dyspnea who die imminently incur significant dyspnea-related cost. We have previously demonstrated that it is possible to identify pts that will die imminently (Escalante, ASCO #2228, 1999). These pts may enjoy greater benefit from hospice and palliative care at a lower cost at the end of life.

POORLY CONTROLLED DIABETES AMONG PATIENTS UNDERGOING CARDIOTHORACIC SURGERY IS ASSOCIATED WITH HIGHER MORTALITY AND RESOURCE UTILIZATION. CA Estrada, S English, J Wynn, C Brooks, Department of Medicine, Brody School of Medicine at East Carolina University, Greenville, NC; Clinical Information and Support Office, University Health Systems of Eastern Carolina, Greenville, NC

PURPOSE: Diabetes mellitus is a risk factor for morbidity and mortality after open heart surgery. Poor perioperative glycemic control is associated with an increased rate of sternal wound infection. Our objective was to compare mortality and resource utilization among diabetic patients with and without tight perioperative glycemic control.

METHODS: Descriptive study of 709 patients with diabetes mellitus who underwent cardiothoracic surgery (coronary artery revascularization, valve repair or replacement, other). We defined patients as having diabetes if they fulfilled any of the following: International Classification of Disease code 250 (ICD-9-DM), glucose >200 mg/dl, glycated hemoglobin >6.5%, ketones in blood or urine, or use of insulin or oral hypoglycemic agents. We defined poor glycemic control as: glucose <60 or >250 mg/dl at any time during hospitalization.

RESULTS: The mean age was 64 (SD 10.1), 57% were male, and 27% were non-white. Poor glycemic control occurred in 52%. Among patients with poor glycemic control the length of stay was 2.5 days higher (95% CI 1.4, 3.6), charges were \$7,330 higher (95% CI \$4,700, \$9,970), and mortality was 4% higher (95% CI 0.5%, 7.7%).

	Glycemic Control Good	Glycemic Control Poor	p value
n	342 (48%)	367 (52%)	
Glucose, mg/dl	156 ± 23	209 ± 42	<0.001
Length of Stay, days	8.2 ± 5.1	10.7 ± 8.9	<0.001
Total Charges, \$	30,700 ± 12,100	38,100 ± 21,800	<0.001
Mortality	4.4%	8.4%	0.02

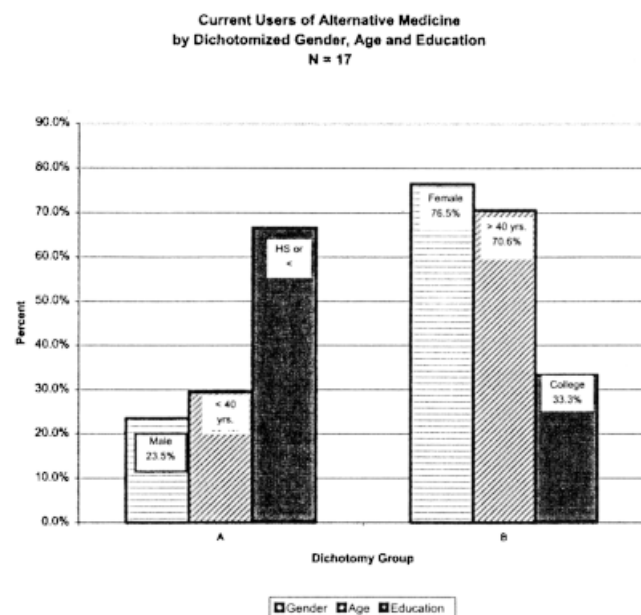
CONCLUSION: Poor perioperative glycemic control in patients with diabetes undergoing cardiothoracic surgery is an indicator of higher mortality and increased resource utilization. An opportunity may exist to improve the quality of care in the perioperative period.

PATTERNS OF ALTERNATIVE MEDICINE USE IN AN INNER CITY MEDICAL CLINIC. FO Fawehinmi, L Capps, U Uwoma, D Alao, A Olatosi, N Marsh, D Miller, A Ashford, Department of Medicine, Harlem Hospital Center, New York, NY

PURPOSE: Few studies have addressed the use of Alternative Medicine among patients in a primary care setting, so this study is examining the characteristics of patients using alternative medicine in the general medicine clinic of Harlem Hospital Center.

METHODS: A self-administered questionnaire is given to all English-speaking patients attending the clinic who are willing to participate in the study. A total of 250 patients are to be enrolled. To date 85 patients have completed the questionnaire. Alternative medicines were defined according to accepted categories.

RESULTS: 78.8% of the patients are black and only 22.4% have college education. Only 17% of the patients are current users of alternative medicine and about two-thirds had never tried any form of alternative medicine. The most popular was herbal medicine (23.5%) distantly followed by acupuncture (3.5%). The only other modalities reported were nutrition (2.4%) and chiropractic (1.2%). The typical user is middle aged and female. More than half of current users expressed satisfaction. The reasons why some patients discontinued alternative medicine use was not explored.



CONCLUSION: The use of alternative medicine is not common in this predominantly African-American population. This lends strength to earlier observations that the use of alternative medicine is more popular among white, middle class populations.

COMPLIANCE OUTCOMES IN A CLINIC DISPENSING PHARMACEUTICAL SAMPLES. RP Ferguson, N Smolyak, V Popat, YY Zhu from the Department of Medicine, The Union Memorial Hospital, Baltimore, Maryland

STUDY PURPOSE: To evaluate compliance outcomes in a clinic relying heavily on pharmaceutical samples.

METHODS: The clinic under review focused on patients who were uninsured and had few resources to pay for prescribed medications. Many were hourly employees with minimal health benefits. Donated pharmaceutical samples were the major source of medications that were dispensed at clinic visits without cost to the patient. Patients identified were those receiving chronic treatment for either hypertension, diabetes and/or hyperlipidemia. A log was established for each patient and the clinic chart was reviewed after the first scheduled follow-up appointment. The minimal criteria for satisfactory outcomes was a return within 48 hours of the appointed time for the first follow-up visit, and treatment that was not influenced by pharmaceutical sample supply.

RESULTS: Ninety-eight patients, including 25 males and 73 females with a mean age of 51.6 were included in the study. Thirty-five percent had satisfactory outcomes; 65% unsatisfactory (p = .002), including 28% missed appointments, 19% late returns and 25% prescriptions changed to fit sample supply at the follow-up visit.

CONCLUSION: In a medical clinic dispensing pharmaceutical samples for patient benefit there was a high likelihood of poor outcomes, at least when treating chronic conditions. Although we acknowledge that many factors can cause non-compliance, we believe that reliance on samples in this setting was a contributing factor.

UPPER BODY FUNCTION AMONG BREAST CANCER SURVIVORS. TS Field, CA Cadoret, J Cernieux, P O'Mara, M Granlund, JH Gurwitz, R Sillman, M Prout, G Pransky, T Colton, Meyers Primary Care Institute; Fallon Healthcare System, Worcester; Department of Medicine, Boston University, Boston; Department of Family Medicine, University of Massachusetts Medical School, Worcester, MA

PURPOSE: The success of advances in breast cancer treatment and screening has produced a large cadre of breast cancer survivors. Several studies have suggested that breast cancer survivors may have long term problems in activities requiring upper body function. The extent of the functional disabilities associated with breast cancer is critical to policy-makers and to those making therapeutic decisions. However, the available information has been based on relatively short follow-up times and coarse measures of functional status.

METHODS: In a retrospective cohort study, we interviewed enrollees in the Fallon Community Health Plan, including both breast cancer survivors who were disease-free 5 to 12 years after initial diagnosis and treatment and randomly selected women of the same age with no history of breast cancer. The standardized interview included components of the Nagi disability scale, the Upper Extremity Function Scale, the MOS-SF12, work history, exercise experience, recent symptoms, comorbid conditions, and demographics.

RESULTS: 271 survivors of breast cancer (73% response rate) and 208 comparison women (59% response rate) were interviewed. The two cohorts were highly similar in terms of age, race, education level, and comorbidity. Women who had survived breast cancer did not differ from comparison women on responses to the Nagi disability scale [the odds ratio (OR) for any response of "very difficult" or "can't do this" was .95, 95% confidence interval (CI) .59-1.5, controlling for age, comorbidity, and education level]. Breast cancer survivors had slightly, but not significantly, higher scores on the MOS-SF12 physical and mental health scales and they did not report higher rates of upper body symptoms. However, women who had survived breast cancer were somewhat more likely to score in the lowest decile on the Upper Extremity Function Scale (OR 1.8, CI .92-3.6) and they were significantly less likely to report improved ability to carry out work, either at or outside of home (OR .43, CI .28-.67).

CONCLUSION: Breast cancer survivors do not report higher levels of disability than women of the same age. However, they are less likely to report improvements in ability to work over time than women of their age and there is a trend toward reports of worse ability to carry out activities that require use of their upper extremities.

SCREENING FOR ALCOHOL PROBLEMS IN PRIMARY CARE: A SYSTEMATIC REVIEW. DA Fiellin, MC Reid, PG O'Connor, Yale University School of Medicine, New Haven, CT

PURPOSE: Primary care physicians can play a unique role in recognizing and treating patients with alcohol problems. Our goals were to synthesize the literature on the accuracy of screening strategies for alcohol problems in primary care and to evaluate this literature according to methodological standards for diagnostic-test research.

METHODS: We searched the MEDLINE database using specific MeSH terms and keywords. Eligible studies (1) were performed in primary care (2) were published in English in peer reviewed journals between 1966-1998 (3) employed a gold standard, and (4) reported the performance characteristics (e.g., sensitivity and specificity) for at least one screening method for alcohol problems. Two reviewers appraised all eligible articles for the accuracy of the screening strategy and pertinent methodological content using pre-specified coding criteria. The methodological standards were: a description of patient spectrum (demographics and comorbidity), use of eligibility criteria and reporting of participation rate, avoidance of work-up bias, avoidance of review bias, and analysis in pertinent clinical subgroups.

RESULTS: Thirty-eight studies were identified. These studies evaluated the Alcohol Use Disorders Identification Test (AUDIT) ($n = 9$), the CAGE questions ($n = 15$), the Michigan Alcoholism Screening Test (MAST) ($n = 8$), a two-question screen ($n = 3$), general screens ($n = 4$), quantity-frequency (QF) questions ($n = 6$) and clinical indicators ($n = 7$). Eleven studies screened for hazardous or harmful drinking, while 27 screened for alcohol abuse and dependence. The AUDIT had increased accuracy relative to other screening methods in identifying subjects with hazardous or harmful drinking (sensitivity 51–97%, specificity 78–96%), while the CAGE had the best operating characteristics for detecting alcohol abuse and dependence (sensitivity 43–94%, specificity 70–97%). The AUDIT and CAGE were consistently more accurate than QF questions, clinical indicators or other screening methods. The studies inconsistently adhered to methodological standards for diagnostic-test research. Overall, 8% provided a description of patient spectrum (demographics and comorbidity), 61% provided eligibility criteria and participation rates, 79% avoided work-up bias, 35% avoided review bias, and 53% performed an analysis in pertinent clinical subgroups.

CONCLUSION: We conclude that despite methodological limitations, the literature supports the use of formal instruments such as the AUDIT and CAGE over other screening strategies to increase the recognition of alcohol problems in primary care. Future research on screening for alcohol problems in primary care will benefit from increased adherence to methodological standards for diagnostic-test research.

PRIMARY CARE PATIENTS WITH HIGH UTILIZATION OF INPATIENT SERVICES: PREDICTING MORTALITY. DA Fiellin, JM Levine, KE Brown, M Chawarski, PG O'Connor, WH Sledge, Yale University School of Medicine, New Haven, CT

PURPOSE: Primary care patients with frequent hospitalizations may benefit from intensive services such as case management. Prediction of mortality in these patients can help clinicians and case managers identify patients in need of increased preventative or palliative services. The purpose of this study was to investigate mortality in patients with high utilization of inpatient services cared for in a primary care clinic (PCC).

METHODS: We performed a prospective observational analysis of mortality in the PCC of an academic medical center. We identified adult PCC patients through an administrative database. Clinic records of patients who had ≥ 2 hospitalizations and had received care in the PCC during the prior year were extracted for baseline medical diagnoses and the administrative database was queried for baseline inpatient and outpatient resource utilization. Mortality was determined at one year.

RESULTS: The 135 patients had a mean age of 58 years, $84/135$ (62%) were female and $48/135$ (36%) were white. Scores on the Charlson comorbidity index (CCI) were 0 (20%), 1 (21%) and ≥ 2 (59%). Overall these patients had a mean of 2.9 (2–11) hospitalizations, 1.9 (0–33) emergency department (ED) visits and 6.4 (1–59) PCC visits during the prior year. Eighty-one percent had an identified primary care provider. Mean hospital costs were \$29,864 (\$8412–\$109,815) during the same period. The mortality rate was $18/135$ (13%) over the one year follow up period. In bivariate analysis there was no difference between patients who were subsequently dead or alive in proportion female (67% vs. 62%), or prevalence of diabetes (44% vs. 37%), coronary artery disease (22% vs. 30%), congestive heart failure (11% vs. 26%), or cancer (39% vs. 21%) ($p > .05$ for all). Mean number of hospitalizations (3.1 vs. 2.9), and ED visits (1.4 vs. 2.0) were similar at baseline ($p > .05$ for all). Patients who died were older (67 vs. 57 years), had higher mean scores on the CCI (3.1 vs. 2.1), had fewer visits to the PCC (3.5 vs. 6.8) were less likely to have an identified primary care provider (56% vs. 85%) and had higher hospital costs (\$48,301 vs. \$27,027) ($p < .05$ for all). On logistic regression analysis age, no identified primary care provider and cost predicted mortality.

CONCLUSION: We conclude that one year mortality in a cohort of PCC patients with frequent hospitalizations can be predicted on the basis of baseline age, identified primary care provider and cost. These factors can help identify patients who may benefit from intensive services and increased efforts to establish primary care.

SUBSTANCE USE AND PSYCHIATRIC DISORDERS IN PRIMARY CARE PATIENTS WITH HIGH UTILIZATION OF INPATIENT SERVICES. DA Fiellin, JM Levine, KE Brown, WH Sledge, PG O'Connor, Yale University School of Medicine, New Haven, CT

PURPOSE: Primary care patients with frequent hospitalizations often have significant comorbid medical disorders that can benefit from increased services to decrease morbidity and mortality. However, less is known about the prevalence of comorbid substance use and psychiatric disorders in this population. The purpose of this study was to investigate the prevalence of substance use and psychiatric disorders in patients with high utilization of inpatient services cared for in a primary care clinic (PCC).

METHODS: We performed a cross-sectional analysis of PCC patients from an academic medical center. We identified adult PCC patients through an administrative database. Clinic records of patients who had ≥ 2 hospitalizations and had received care in the PCC during the prior year were extracted for clinician diagnoses of

substance use and psychiatric disorders. An administrative database was queried for baseline inpatient and outpatient resource utilization.

RESULTS: The 135 patients had a mean age of 58 years, $84/135$ (62%) were female and $48/135$ (36%) were white. Substance use or psychiatric disorders were seen in $64/135$ (47%) of patients. Substance use disorders were present in $39/135$ (29%) of patients, including $16/135$ (12%) with alcohol use disorders, $26/135$ (19%) with tobacco use disorders and $9/135$ (7%) with illicit drugs use. Psychiatric disorders were seen in $39/135$ (29%) of patients, including $26/135$ (19%) with depression $11/135$ (8%) with anxiety, $9/135$ (4%) with schizophrenia, $2/135$ (2%) with bipolar disorder and $9/135$ (7%) with an unclassified psychiatric disorder. Comorbid substance use and psychiatric disorders were seen in $14/125$ (10%) patients. In bivariate analysis there was no difference between patients with and without substance use or psychiatric disorders in proportion male (29% vs. 27%), white (42% vs. 30%), or mean number of hospitalizations (2.9 vs. 2.9), hospital costs (\$31,819 vs. \$28,101), visits to the ED (2.6 vs. 1.4), or visits to the PCC (6.9 vs. 5.8) during the prior one year period ($p > .05$ for all). Patients with substance use or psychiatric disorders were younger (52 vs. 64 years), and were less likely to have an identified primary care provider (74% vs. 88%) ($p < .05$ for all).

CONCLUSION: We conclude that substance use and psychiatric disorders have a high prevalence in a cohort of PCC patients with frequent hospitalizations but are not associated with increased resource utilization. Efforts to establish primary care providers and increased recognition and treatment of these disorders may help decrease morbidity and mortality in this population.

MARIJUANA USE IS NOT ASSOCIATED WITH HEAD, NECK OR LUNG CANCER IN ADULTS LESS THAN 55 YEARS OF AGE: RESULTS OF CASE COHORT STUDY. DE Ford, HT Vu, C Hauser, KJ Helzlsouer, JC Anthony, Medicine, Johns Hopkins, Baltimore, MD; Epidemiology; Mental Hygiene

PURPOSE: Marijuana smoke contains carcinogens. Several small case series and one potentially flawed small case-control study have provided evidence that marijuana is associated with head and neck cancer. We conducted a large case-control study to address this issue.

METHODS: 164 consecutive incident cases of head, neck or lung cancer in adults less than 55 years in the Baltimore region were recruited from 4 hospitals between 1994 and 1999. 526 individuals less than 55 years from the Baltimore Epidemiologic Catchment Area study based on a random household sample served as the controls. Cases and controls completed identical assessments of exposures via self-report direct computer entry. Lifetime and current use of marijuana, tobacco, and alcohol were assessed with detailed questions of use by 5–10 year intervals of their lives, by weekend and weekday consumption and by mode of administration of the substance.

RESULTS: Control patients were younger (44 vs 49 years, $p < .001$), more likely to be female (62% vs 34%, $p < .001$) and less likely to be white (55% vs 69%, $p < .01$). Substance use was commonly reported in both cases and controls (lifetime means, 125,000 tobacco cigarettes; 27,000 alcoholic drinks and 8,700 marijuana joints). Consistent with other studies, tobacco and alcohol use was associated with these cancers. Ever use of marijuana (66% controls vs 60% cases, $p = .17$) and lifetime marijuana joints (8,135 vs 8,946 joints, $p = .64$) were not associated with cancer. Marijuana use everyday for one month or more was also not associated with those cancers with or without adjustment (19% controls, 12% cases, OR = .74 95% CI .38, 1.43). Age of first use of marijuana, depth of inhalation of marijuana and use of pipe vs joint for marijuana was not related to these cancers. Although power was limited, marijuana use was not associated with cancer for those who never used tobacco. Adjusting for all sociodemographic factors, family history of these cancers, lifetime tobacco use and lifetime alcohol use did not change the relationship between marijuana and cancer.

CONCLUSION: The balance of evidence from this, the largest case-control study addressing marijuana use and cancer to date, does not favor the idea that marijuana as commonly used in the community is a major causal factor for head, neck or lung cancer in young adults.

PATIENT PREFERENCES FOR COMMUNICATING RISK ESTIMATES. JM Fortin, LK Hirota, BE Bond, AM O'Connor, RJ Goldberg, NF Col, Medicine, New England Medical Center, Boston, MA; Faculty of Health Sciences, University of Ottawa, Canada; Medicine, University of Massachusetts Medical School, Worcester, MA

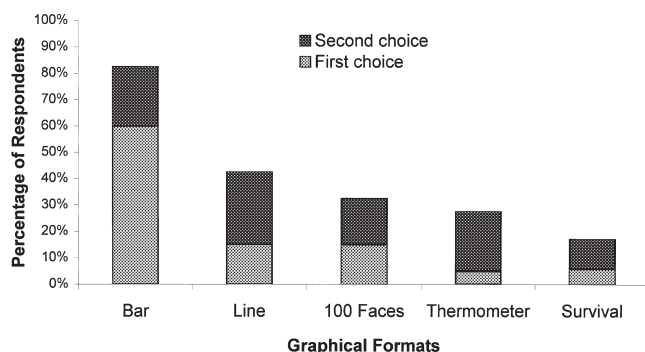
PURPOSE: Patients increasingly seek more active involvement in health care decisions, but little is known about how to communicate complex risk information to patients. Our objective was to elicit patient preferences for the presentation and framing of complex risk information.

METHODS: Peri- and postmenopausal women were recruited through hospital fliers and a community health fair. Eight focus group discussions and 15 one-on-one interviews were conducted, where women were presented with risk data in a variety of different graphical formats, metrics, and time horizons. Risk data were based on a hypothetical woman's risk for coronary heart disease, hip fracture, and breast cancer, with and without hormone replacement therapy. Participants'

preferences were assessed using likert scales, ranking, and abstractions of focus group discussions.

RESULTS: The 40 participants' mean age was 51 years, 50% were non-Caucasian, and all had completed high school. 83% of women preferred bar graphs over line graphs, 100 representative faces, thermometer graphs, and survival curves. Lifetime risk estimates were preferred over 10 and 20-year horizons, and absolute vs. relative risk estimates were the preferred metric. Few participants liked the concept of number needed to treat.

**Rankings of Graphical Formats
Containing Risk Information**



CONCLUSION: Although there are many different formats for presenting and framing risk information, simple bar charts depicting absolute lifetime risk estimates appear to be the preferred format for communicating health risks to patients.

INCOME AND ALTERNATIVE MEDICINE USE. D. Foster, RS Phillips, RB Davis, D Eisenberg, Harvard Medical School and Beth Israel Deaconess Medical Center, Boston, MA

PURPOSE: Although higher income is thought to be associated with greater use of alternative medicine (AM), few data exist to examine this relationship in detail.

METHODS: To examine the effect of income on AM use, we utilized data collected in a nationally representative telephone survey of adults conducted between November 1997 and February 1998. The data were weighted to adjust demographic features for aggregate discrepancies between sample distributions and US Census Bureau population distributions. Respondents were presented with a list of common medical conditions and asked if they had experienced each in the previous twelve months, asked about their use of conventional medical services, and asked about their use of 20 AM therapies during the prior 12 months. Questions included out-of-pocket expenses for AM. For analytic purposes, income was grouped approximately into quartiles (income <\$20,000; \$20,000–29,999; \$30,000–49,999; >\$50,000).

RESULTS: Of those eligible, sixty percent (2049) completed the interview. We found that those in the lower income groups were more likely to be female, non-white and have fewer years of education ($p < .05$ for each). In addition, lower income was also associated with worse reported health status, more medical conditions, and less health insurance coverage ($p < .05$ for each). AM use varied by income quartiles (lowest 43%, 37%, 44%, highest 48%) ($p < .05$). The average annual out-of-pocket expenditure by users of AM increased with income quartile (\$265, 440, 321, 505 $p < .05$). Use of specific modalities was similar across all income groups except that the highest income group reported higher rates of use of chiropractic and massage. The odds ratio for use of AM in the highest income group compared with the lowest was 1.2 (95% CI .93–1.6) and increased to 2.0 (95% CI 1.5–4.8) after adjusting for other demographic variables and medical conditions associated with AM use. When examining chiropractic and massage, a similar pattern emerged. For chiropractic, the odds ratios for use in the highest group increased from 1.4 (95% CI .89–2.1) to 2.0 (95% CI 1.0–3.8) when adjusting for medical conditions and demographics. For massage, the odds ratios for use in the highest group increased from 1.6 (95% CI 1.0–2.4) to 3.4 (95% CI 2.0–8.3). Whether one had personal medical insurance did not affect rates of use of alternative medicine across the income groups.

CONCLUSION: Higher income is associated with increased use of AM overall. After adjusting for other demographic variables and medical conditions, these differences become more dramatic. While income is a barrier to AM use, still, among the lowest income group (<\$20,000), 43% of respondents used AM and spent on average over \$250 annually.

MEDIATING EFFECTS OF PATIENT-CENTERED AMBULATORY CARE ON CHEST PAIN AFTER MYOCARDIAL INFARCTION. AM Fremont, PD Cleary, JZ Ayanian, Div of General Internal Medicine, Brigham & Women's Hospital; Department of Health Care Policy, Harvard Medical School, Boston, MA

PURPOSE: Previously, we found that patient-reported problems with non-technical aspects of care during hospitalization and discharge for an acute myocardial infarction (AMI), such as providing information, emotional support, and discharge

planning, increased the likelihood of chest pain one year later. We now examine whether this effect is mediated by patients' experiences with post-MI outpatient care.

METHODS: We studied 2272 patients discharged with a diagnosis of AMI during 1996–1997 in 20 New Hampshire Hospitals. Surveys asking about problems with care and health, including chest pain, were mailed to patients 1, 3, and 12 months after discharge; 1346 (59.2%), 1046 (46.0%), and 964 (42.4%) enrolled patients responded, respectively. Patients' reports about their care were used to compute a problem score (0–100), with higher scores indicating more problems. Patients were then categorized into either a low problem group (score in the bottom 3 quartiles) or a high problem group (top quartile score) for their hospital and subsequent outpatient care, respectively. Clinical data were obtained from discharge abstracts. Outcome analyses adjusted for chest pain one month post-MI, sociodemographic factors, comorbidities, prior hospitalizations, treatment type, hospital size, and transfers.

RESULTS: Most patients had relatively few problems with either their hospitalization (mean score = 15) or the first 3 months of post-MI outpatient care (mean score = 12) but the ranges were large (0–85 and 0–75, respectively). In adjusted analyses, patients in the high problem hospital care group had more problems with subsequent ambulatory care, such as not receiving sufficient information from outpatient providers (OR = 0.3, 95% CI 0.2–0.5) or not having complete confidence or trust in these providers (OR = 0.4, 95% CI 0.2–0.6) than patients in the low problem hospital care group. However, both groups were equally likely to have seen a primary care physician or cardiologist and enroll in cardiac rehabilitation. Patients in the high problem group for both their hospital and subsequent outpatient care were twice as likely (OR = 2.2 95% CI 1.1–4.5) to report chest pain one year post-MI as patients in the low problem group for both stages of care, after adjustment for potential confounder variables. In contrast, patients in the high problem hospital care group and low problem ambulatory care group were equally likely (OR = 1.0, 95% CI 0.4–2.4) to have chest pain at one year as those who reported few problems with both stages of care.

CONCLUSION: Although negative hospital experiences predispose AMI patients to worse long-term outcomes, these effects can be offset by more positive experiences with subsequent ambulatory care.

DIETARY AND EXERCISE PATTERNS OF HISPANIC ADOLESCENTS WHO HAVE LATENT TUBERCULOSIS. L. Friedman, L Gil-Trejo, M Hovell, C Sipan, J Berg, A Catanzaro, N Kelley, Departments of Pediatrics and Medicine, University of California; Center for Behavioral Epidemiology, San Diego State University, San Diego, CA

OBJECTIVE: Childhood obesity affects 25% of all North American children. Little is known concerning obesity patterns in Mexican-American teenagers. Some of the factors contributing to obesity include eating patterns and exercise behavior. The purpose of the current analysis is to examine the role of exercise and eating behaviors of Mexican-American adolescents.

DESIGN: These results are from a larger study examining the impact of a behavioral counseling intervention on adherence to preventive tuberculosis treatment in Hispanic adolescents.

RESULTS: Preliminary data were available from a sample of 242 male (55%) and female (45%) Hispanic adolescents in San Diego County. Subjects were questioned about their level of acculturation as measured by the Bi-Dimensional Short Acculturation Scale for Adolescents (BAS), place of birth, dietary and exercise behaviors and perceived body image as part of the baseline interview for the clinical trial intervention. The average age of the study participants was 16 years with 64% born outside the U.S., primarily in Mexico. On average, 0.4% of the subjects reported eating five or more servings of fruits and vegetables per week; 23% reported exercising at least 3 days/week. In addition, while only 26.9% of the sample perceived themselves as overweight, 49.6% were trying to lose weight. Exercise in this sample of Hispanics is also far below the norm (63.8%) as compared with the Youth Risk Behavior Surveillance of 1997.

CONCLUSION: These results indicate that Hispanic adolescents may be at higher risk for obesity and diseases such as type II diabetes that are influenced by sedentary lifestyle and poor eating habits. Further analysis will relate diet, exercise, and other sedentary activities, such as TV-watching to degree of acculturation, at which time the relationship between acculturation and exercise behaviors will be examined.

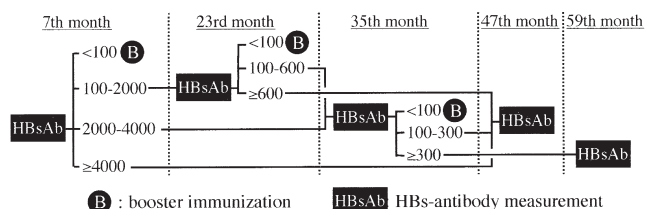
OPTIMAL TIMING OF SERUM HBs-ANTIBODY MEASUREMENT FOR BOOSTER IMMUNIZATION OF HEPATITIS B. N. Furutani, Y Kasuga, K Suzuki, Y Naito, I Ohkubo, M Niya, K Fujise, M Kobayashi, Department of General Internal Medicine, The Jikei University School of Medicine, Kashiwa-shi, Chiba, Japan

PURPOSE: There are no definitive guidelines for an optimal immunization schedule after primary immunization. We used HBs-antibody level to estimate the appropriate timing of next HBs-antibody measurement for booster immunization.

METHODS: 136 HBs-antibody-negative student nurses who receive immunization for hepatitis B virus. 129 with ≥ 100 mIU/ml HBs-antibody were evaluated at 7th month, 92, 44, and 16 were evaluable at 23rd, 35th, and 47th month, respectively. HBs-antibody was measured at 7th, 23rd, 35th, and 47th month from the first immunization of primary immunization. We settled booster immunization value of HBs-antibody at <100 mIU/ml. To evaluate the further necessity of HBs-antibody measurement, likelihood ratio of ≥ 100 mIU/ml was calculated in each hexitle stratum of 100–300, 300–600, 600–1200, 1200–2000, 2000–4000, and ≥ 4000 mIU/ml of HBs-antibody for predicting 23rd month from 7th month, 35th

month from 7th and 23rd month, and 47th month from 35th month HBs-antibody measurement.

RESULTS: Likelihood ratio was infinity at 2000–4000 stratum and over in predicting 23rd month from 7th month, ≥ 4000 stratum in predicting 35th month from 7th month, 600–1200 stratum and over in predicting 35th month from 23rd month, and 300–600 stratum and over in predicting 47th month from 35th month, respectively.



CONCLUSION: We could schedule next HBs-antibody measurement for deciding booster immunization as below:

INTRACORONARY STENT USE IN BLACK AND WHITE PATIENTS: DEMOGRAPHIC CHARACTERISTICS AND IN-HOSPITAL OUTCOMES. M. Garg, D. Hallas, J.L. Vacek, The Mid America Heart Institute, Saint Luke's Hospital, University of Missouri, Kansas City, MO

PURPOSE: Stent use for coronary angioplasty (PTCA) is increasingly employed and very successful. However, most reported data is for white patients, or does not address race-related selection or outcome differences. We analyzed the patient characteristics and in-hospital outcomes of 339 stent procedures, 328 (97%) in white patients and 11 (3%) in black patients from 1991 to the present. Seventy-eight patients received 2 stents and 7 patients 3 stents. Most stents were Palmaz-Schatz.

There was no significant differences in gender distribution, mean age, mean ejection fraction, % of patients of age >70, % with EF <40, diabetes, unstable angina, multivessel, CAD, prior MI, or acute MI PTCA between the black and white patient groups. However, a greater % of black than white (91% vs 58%) was hypertensive. Stents were part of a similar % of single vs multivessel procedures in both groups, and a similar number of lesions were intervened upon in both groups. Vessel site (LAD vs RCA vs CIRC vs graft) distribution was similar in both groups, as were procedural success rates (96% for whites, 100% for blacks).

In-hospital complications were infrequent in both groups, with no significant difference seen in non fatal MI, CABG, death, need for repeat PTCA, stroke, or vascular complications. Total in-hospital charges were the same for both groups.

CONCLUSIONS: Although data is limited, it appears that black patients benefit from stent use for PTCA similarly, as do whites. Our data would suggest that continued study of the use of these devices is indicated in various patient subgroups. The high incidence of hypertension in black patients coming to stent implantation deserves further investigation.

PATIENT CHARACTERISTICS AND IN-HOSPITAL OUTCOMES OF CORONARY ANGIOPLASTY IN ASIANS COMPARED TO WHITE PATIENTS. M. Garg, S. Chowdry, J.L. Vacek, Mid America Heart Institute-Saint Luke's Hospital, University of Missouri, Kansas City, MO

PURPOSE: Asians have high morbidity and mortality from coronary artery disease. Coronary angioplasty (PTCA) has been shown to be an effective means of revascularization and improving prognosis in selected patients. Important differences may exist in application and outcomes for patient (pt) subgroups. Little data compares PTCA for Asians and white patients. We compared patient characteristics and in-hospital outcomes of 3,447 procedures in white patients vs 58 in Asian patients.

The Asian patients were younger with mean age of 55.6 vs. 62.8 in whites ($p < .0001$) and lower % of Asian patients were of age >65 (20% vs. 47%, $p < .0001$). More Asian patients were diabetic (41% vs. 21%, $p = .0009$). White patients were more likely to present with unstable angina (53% vs. 35%, $p = .01$). There was no significant difference in gender, mean EF, multivessel CAD, prior MI, prior CABG or incidence of hypertension between the two groups. More multivessel (48% vs. 11%, $p < .0001$ and multisection procedures (60% vs. 36%, $p < .0001$) were done in the Asian patients with similar success rates as in whites (98% in Asians, 96% in whites). Usage of devices other than standard PTCA balloons including stents was similar in both groups.

Asian patients had a higher incidence of NQNI (7% vs. 1%, $p = .001$). Other in-hospital complications including Q-wave MI, CABG, repeat PTCA, stroke, vascular complications and death were infrequent and similar in both groups.

CONCLUSION: At our center Asian patients who underwent angioplasty were younger, had greater incidence of diabetes mellitus and underwent multivessel and multisection angioplasty more often than the white patients with equal procedural success and in-hospital survival.

SOCIOECONOMIC STATUS AND SURVIVAL FOLLOWING ACUTE MYOCARDIAL INFARCTION. A. Georgaklis, K.J. Mukamal, M.A. Mittleman, Department of Medicine, Beth Israel Deaconess Medical Center, Boston, MA

PURPOSE: Socioeconomic status (SES) affects the incidence of cardiovascular disease. How SES relates to long-term prognosis following acute myocardial infarction (AMI) is less clear. We therefore assessed the relationship between SES and long-term survival among early survivors of AMI enrolled in the Determinants of Myocardial Infarction Onset Study.

METHODS: We followed 1935 early survivors of AMI interviewed between August 1989 and September 1994 a median of 4 days following AMI. The National Death Index was searched to identify patient deaths through December 31, 1995. Median annual household income for patient zip codes were obtained from the 1990 US Census. We divided annual income levels into quartiles with cutpoints of \$29204, \$37466, and \$44734. We used Cox proportional-hazards regression to estimate the effect of income on all-cause mortality, adjusting for age, sex, previous MI, hypertension, diabetes, obesity, thrombolytic use, use of cardiac medications, congestive heart failure, ventricular tachycardia, physical exertion, alcohol consumption, and current and former smoking.

RESULTS: The table shows an inverse association between household income and total mortality. We observed a similar trend between educational attainment and mortality. The effect of income on mortality persisted after further adjustment for educational attainment and race.

Income Quartiles	1st	2nd	3rd	4th	P (Trend)
Number of Subjects	453	484	458	480	
Adjusted Hazard Ratios	1.00	0.98	0.73	0.68	0.02
95% CI Reference	0.69–1.40	0.50–1.06	0.45–1.01		

CONCLUSION: We found an overall inverse relationship between SES and all-cause mortality among early survivors of AMI. Further research on the mechanisms that underlie this finding may suggest strategies to reduce the observed inequality in outcome following AMI.

ASSOCIATION OF PERIOPERATIVE ANEMIA WITH CARDIAC EVENTS IN ORTHOPEDIC SURGERY. K. Gilbert, B.J. Larocque, Department of Medicine, University of Western Ontario, London, ON, Canada

PURPOSE: Many clinicians believe that there is a relationship between perioperative anemia and cardiac events. Nevertheless, there is a paucity of literature that supports this assertion. Given the ongoing concern about the use of blood products, further evaluation of this association is warranted. The purpose of this study was to compare post-operative hemoglobin in patients who suffered cardiac complications to that of patients without cardiac complications, with the hypothesis that hemoglobin would be lower in those patients who had a cardiac event.

METHODS: We identified 32 patients who underwent orthopedic surgery at one of two London, Ontario hospitals, and who had a perioperative cardiac event, defined as unstable angina, myocardial infarction, acute pulmonary edema or death. For each of these cases we attempted to identify three controls, matched for age, sex, and pre-operative Canadian Cardiovascular Society (CCS) angina grade. The lowest post-operative hemoglobin value obtained for these two groups of patients was compared using a paired t-test.

RESULTS: The lowest mean post-operative hemoglobin was 88.7 g/L in the cases and 94.9 g/L in the controls, a difference of 6.2 g/L (95% CI: 12.1–0.27, $p = .041$). The cases and controls did not differ significantly with respect to age ($p = .94$), lowest intra-operative systolic blood pressure ($p = .24$), highest intra-operative heart rate ($p = .21$), pre-operative hemoglobin ($p = .33$), or CCS angina grade (identical), indicating good matching.

CONCLUSION: The results suggest that there is an association between perioperative anemia and cardiac events in patients having orthopedic surgery. Nevertheless, it is not known whether increasing the hemoglobin by transfusion will have an impact on cardiac event frequency. We look forward to future research that will examine this issue.

"CHOOSING HEALTHPLANS ALL TOGETHER" A GAME TO ASSESS CONSUMER VALUES AND PREFERENCES FOR HEALTH INSURANCE. S.D. Goolal, A.K. Biddle, G. Klipp, M. Danis, Internal Medicine, University of Michigan, Ann Arbor, MI; Health Policy and Administration, University of North Carolina, Chapel Hill, NC; Clinical Bioethics, National Institutes of Health, Bethesda, MD

PURPOSE: As purchasers search for ways to control costs, consumer and patient control of healthcare decisions is weakening, resulting in enormous public frustration. Efforts to include consumers in allocation and insurance decisions face significant obstacles of apathy, voluminous and complex information requirements, and emotional barriers to making tradeoffs between competing needs for limited resources. In a low-income population, we evaluated an innovative exercise designed to overcome these obstacles to consumer involvement in allocation decisions.

METHODS: CHAT (Choosing Healthplans All Together) was designed to measure laypersons' values and preferences for health insurance features within the constraint of limited resources. CHAT uses 50 pegs, each of which represents 2% of a monthly health insurance premium. These pegs are distributed by subjects amongst a gameboard of insurance options that includes a range of choices: degrees of care management, breadth of coverage, restrictions on access and choice, and inclusion/exclusion of some services. Repeated cycles of the game incorporate role-playing to foster appreciation for the consequences of insurance choices and the perspectives of other group members. Groups of subjects were recruited

from community and ambulatory care sites in North Carolina. Pre-and post-game written questionnaires assessed participants' experiences of the exercise.

RESULTS: Data have been analyzed for 78 persons in 6 groups to date (Total will be ~300 persons in 30 groups).

Demographics		Health and Insurance	
Mean age (range)	42.9 (19–66)	Health Status (%fair/poor)	28.6
Women (%)	21.0	Chronic/Serious Illness in Family(%)	28.2
Caucasian (%)	20.0	Hospitalized last 6 mos. (%)	21.8
Income <35k (%)	94.8	Dr. visit last 12 mos. (%)	70.3

In the postgame evaluation, the vast majority of study participants found CHAT enjoyable (85.7%), easy to understand (93.4%), easy to do (93.4%), and informative (90.8%). The majority of participants indicated that they had learned a lot from playing the game (64.5%) and were motivated to learn more about health insurance (71.8%). More than 85% of participants stated that they would be willing to abide by the group's chosen healthplan.

CONCLUSION: CHAT is an enjoyable, easily understood, informative and motivating process which relatively disadvantaged, low income groups of persons without healthcare expertise can use to design healthplans that are cost-conscious and acceptable to them.

EMERGENCY ANGIOPLASTY WITH AND WITHOUT STENTING FOR ACUTE MYOCARDIAL INFARCTION. MS Gowda, JL Vacek, GD Beauchamp, JH Okeffe, BD McCallister, SB Laster, Cardiology, University of Missouri-Kansas City; Cardiology, Mid America Heart Institute, Kansas City, MO

PURPOSE: The role of stent use during emergency angioplasty is not well defined. **METHODS:** Risk factors and outcomes associated with emergency coronary stenting (CST) versus emergency percutaneous transluminal angioplasty (PTCA) alone were analyzed for 571 consecutive patients who presented with an acute MI during the period from April 1995 to December 1997; 390 pts received PTCA (68%) and 181 received stents (32%).

RESULTS: A greater proportion of patients selected for PTCA were over the age of 70 yrs (35%) compared to patients who received CST (23%, $p < .01$); however there was no significant difference in age >75 (18% vs 13%), gender (19% vs 18%), hypertension (47% vs 53%), diabetes (19% vs 17%), prior CABG (10% vs 11%), prior PTCA (22% vs 24%), multiple vessel disease (73% vs 70%), ejection fraction $<35\%$ (21% vs 20%), cardiogenic shock (4% vs 3%) or history of prior MI (25% vs 28%) for PTCA vs CST, respectively. In-hospital complications such as CABG (7% vs 6%), recurrent MI (1% vs 2%), repeat PTCA (10% vs 9%) and stroke (0% vs .5%) were not significantly different between the PTCA and CST groups. Overall in-hospital mortality was 9.6% with a strong trend for a lower mortality in the CST group vs PTCA (6% vs 11%, $p = .07$). CST in pts with 3-vessel disease have fewer in-hospital deaths than PTCA (6% vs 13%, $p = .04$). The length of hospital stay was similar in both groups (7.0 days vs 7.2 days). Multivariate analysis identified correlates of early mortality; (odds ratios; confidence limits): cardiogenic shock (6.82; 2.44–18.55), age >70 (2.65; 1.42–2.50), creatinine $>1.5\text{mg/dl}$ (4.04; 1.96–8.13), 3-vessel disease (2.22; 1.21–4.12) and CST (0.73; 0.34–1.47).

CONCLUSION: In this non-randomized study, the use of coronary stents in pts undergoing PTCA for AMI appears to result in a decrease in early mortality ($p = .07$), especially in pts with 3-vessel disease ($p = .04$).

EFFICACY OF SILDENAFIL CITRATE AFTER TREATMENT OF REVERSIBLE CAUSES OF ERECTILE DYSFUNCTION (ED). AI Guay, Center for Sexual Function, Lahey Hitchcock Clinic, Northshore, Peabody, MA

PURPOSE: Erectile dysfunction (ED) is a common medical condition affecting an estimated 30 million men in the United States. The objective of this study was to determine the efficacy and safety of sildenafil when either medication changes or lifestyle modifications were initiated in patients before treatment with sildenafil in a sexual dysfunction clinic.

METHODS: The study comprised 714 men diagnosed with ED of various etiologies with a broad range of concomitant medical conditions treated either by medication changes or adjustment of social risk factors before treatment with sildenafil for ED. Patients were instructed to take a 50-mg dose of sildenafil approximately 1 hour before sexual activity, not more than once in a 24-hour period and at least twice a week. Dosage could be increased to 100 mg or decreased to 25 mg based on tolerability and efficacy. Patients recorded the number of successful intercourse encounters in a patient diary for 6–8 weeks.

RESULTS: The primary etiology of ED was determined to be organic in 68% of men, psychogenic in 7%, or mixed in 24%. The incidence of an organic etiology increased with age. The predominant associated risk factors were hypertension (39%) and patients taking multiple medications (34%). The percent of patients taking multiple medications increased with age. Patients that were hypogonadal (36%) had concomitant medical conditions plus ED. Overall, there was an 82% successful intercourse rate with sildenafil treatment in the patient population. A 50-mg dose was effective for 66% of the patients with successful intercourse. There were no significant differences between patients given testosterone replacement and sildenafil or sildenafil alone. Common adverse events—flushing (20%) and headache (17%)—due to sildenafil treatment were mild to moderate in nature and resulted in $<2\%$ discontinuation.

CONCLUSION: The rate of successful treatment with sildenafil in men with ED and concomitant medical risk factors is higher when medication changes and lifestyle modifications are initiated before treatment with sildenafil.

THE MEMORY IMPAIRMENT SCREEN (MIS), IMPROVES DETECTION OF DEMENTIA AND OUTPERFORMS A MENTAL STATUS TEST IN AN ETHNO-CULTURALLY DIVERSE PRIMARY CARE SETTING. SR Hahn, G Kurlansky, H Buschke, M Katz, J Verghese, M Sliwinski, RB Lipton, Medicine, Albert Einstein College of Medicine, Bronx, NY; Neurology, AECOM

PURPOSE: To compare the operating characteristics a brief Memory Impairment Screen (MIS) with unaided physician detection and with a longer standard dementia screen—the Blessed Information Memory and Concentration Test (BIMC)—in a racially and ethnically diverse, urban primary care clinic. **Background:** Dementia is commonly undetected in primary care. The currently employed BIMC and Mini-Mental Status test (which have a correlation of .83) require 8 and 12 min respectively. The MIS is a 4-item, free and cued recall task requiring two, 1-minute tests separated by a 2 min distraction task (i.e., continuation of the medical interview). It has been validated in community and clinic populations.

METHODS: Subjects were 54 consecutive patients, age 70 and older, during scheduled visits to 31 physicians and nurse practitioners in an urban hospital clinic. Subjects received the MIS in English (38% African American; 29% non-Hispanic Caucasian; 4% Hispanic) or Spanish (29% Hispanic). Clinician assessment of dementia was recorded prior to examining MIS results. A criterion-standard, comprehensive neurological and cognitive evaluation including the BIMC was conducted separately by a geriatric neurologist.

RESULTS: Of 54 eligible patients 98% completed the MIS and 89% completed the comprehensive evaluation. The comprehensive evaluation confirmed dementia in 8 subjects (17%), 40 were not demented. The operating characteristics of the MIS, unaided physicians, and the BIMC are shown in the table. The MIS had superior sensitivity, specificity and positive predictive value. With sensitivity held constant at .88, the MIS had a specificity of .95 compared to .71 for the BIMC. The MIS was dramatically better than unaided physician detection. BIMC scores indicate more severe dementia in clinician-detected patients (27.0) compared to patients with undetected dementia (16.7, $p < 0.001$).

	Sensitivity	Specificity	Positive Predictive Value
MIS	.88	.97	.78
Unaided Clinician	.25	1.0	1.0
BIMC	.63	.87	.50

CONCLUSION: The MIS is a new, substantially briefer, screen for dementia with excellent operating characteristics that are superior to those of a standard mental status test, the BIMC. Compared to the unaided physician, use of the MIS should dramatically increase detection of dementia, particularly of mild and early dementia that may be more amenable to treatment. Because it is briefer, it should be more acceptable to providers.

PATIENT TRUST IN A PRIMARY CARE PHYSICIAN: WHAT IS IT AND CAN IT BE RELIABLY MEASURED? MA Hall, E Dugan, B Zheng, D Levine, Public Health Sciences, Wake Forest University School of Medicine, Winston-Salem, NC

PURPOSE: It is critical to have an accurate understanding of what patient trust in physicians consists of, what influences this trust, and what are the consequences of trust.

METHODS: A conceptual model of patient trust was developed and items to measure trust were generated, reviewed and pilot-tested. Items were grouped into four dimensions of trust: loyalty, honesty, competence, and confidentiality. Twenty-four candidate items were then pilot tested in a national random-digit-dial survey of 989 adults. Data collected included: relationship length and frequency of contact, how physician was selected, satisfaction with care, medical history and health status, demographics of the subjects and their physicians, desire to switch doctors or obtain a second opinion, preferences for seeking professional help and making medical decisions, and related patient, physician, and situational factors. Factor structure, internal consistency, and construct and predictive validity were assessed.

RESULTS: Factor analyses showed a single-factor solution for 12 items. Internal consistency (reliability) was .92 (Cronbach's alpha). Trust in a physician is a concept that can be reliably measured, however, trust does not consist of multiple dimensions, as we hypothesized. Physician trust was correlated with satisfaction with care (.50). The following factors were correlated with physician trust, demonstrating predictive and discriminant validity: length of relationship, number of contacts, choice in selecting physician, physician chosen based on recommendation, seriousness of treatment, having sought a second opinion (–), past dispute with this physician (–), and desire/intention to switch doctors or obtain second opinion (–). The following factors were not significantly related to physician trust: patient or physician demographics, patient education or income, having a chronic or serious condition, being worried about health, willingness to seek professional care or rely on expert judgment, or preferred role in making medical decisions.

CONCLUSION: Patient's trust in their primary care physicians can be reliably measured. Physician trust is distinct from satisfaction. The factors that predict physician trust relate to physician characteristics, the nature and origin of the relationship, and the seriousness of treatment, rather than to patient characteristics.

tics. These findings suggest that choice in selecting physicians can strengthen trust.

ASSESSING HEALTH CARE UTILIZATION IN OSTEOARTHRITIS: THE LIMITATIONS OF USING A POPULATION IDENTIFIED SOLELY THROUGH ADMINISTRATIVE DATA. LR Harold, RA Yood, W Straus, SE Andrade, JI Reed, J Cernieux, B Lewis, M Weeks, JH Gurwitz, Meyers Primary Care Institute, Worcester, MA; Merck & Co., West Point, PA; Fallon Healthcare System, Worcester, MA

PURPOSE: Administrative databases, created primarily for fiscal purposes to track health care utilization of enrollees in health insurance plans, are increasingly being used for epidemiologic and health services research on large populations. This investigation was conducted to compare health care utilization in a population identified solely through an administrative osteoarthritis diagnosis, with a group of patients in whom this diagnosis had been validated.

METHODS: We identified all enrollees in a group-model HMO with documentation of at least one health care encounter associated with an osteoarthritis diagnosis during the period 1994–1996, and who continued to be enrolled in the health plan for a one-year period following the health care encounter date. This population was comprised of 10,740 individuals. From this population, we randomly selected 700 individuals, whose medical records were abstracted by trained nurse reviewers utilizing a structured data collection instrument to ascertain information relevant to the diagnosis of osteoarthritis. Pairs of physician reviewers evaluated the abstracted information and rated the evidence for the presence of osteoarthritis according to three levels (definite, possible, and unlikely). All persons rated as having definite osteoarthritis were included in the validated group ($n = 443$). Health care utilization was assessed in both the administrative osteoarthritis diagnosis group and the validated group across the following domains: 1. non-inpatient care associated with an osteoarthritis diagnosis; 2. relevant radiographic studies (e.g., spine, shoulder, hand, hip, knee, and foot x-rays); 3. relevant surgical procedures (e.g., arthroscopic procedures of the knee or hip, or hip/knee replacements); 4. relevant medication dispensings (e.g., NSAIDs, non-acetylated salicylates, opioid analgesics, and intra-articular steroids).

RESULTS: For all domains, levels of utilization were higher among the validated group, as compared with the administrative osteoarthritis diagnosis group. In the validated group relative to the administrative osteoarthritis diagnosis group, there were 375 versus 327 non-inpatient osteoarthritis-associated health care encounters per 100 person-years, 133 versus 107 relevant radiographic studies per 100 person-years, 3.16 versus 2.06 surgical procedures per 100 person-years, and 514 versus 344 relevant medication dispensings per 100 person-years.

CONCLUSION: Estimates of health service utilization for a population with osteoarthritis identified solely through administrative data may represent substantial underestimates of actual utilization.

STAGES OF CHANGE FOR WEIGHT LOSS AMONG RURAL AFRICAN-AMERICAN WOMEN. DS Hawkins, PP Hornsby, JB Schorling, Internal Medicine; Health Evaluation Science, University of Virginia, Charlottesville, VA

PURPOSE: Obesity is a prevalent public health problem in the United States, especially among rural African-American women, and causes increased morbidity and mortality. The purpose of this study was to determine if the transtheoretical stages of change model was generalizable to weight loss intention among overweight and obese rural African-American women and to identify important predictors of the stages of change.

METHODS: The study was conducted in two rural counties in central Virginia. A population-based sample of 200 women under the age of 40, a subset of a larger study of health behavior, completed questionnaires concerning weight loss behavior and beliefs about weight. The women were classified into three body mass index (BMI) categories: BMI <25 (32%); BMI >25 and <30 (27%); BMI >30 (41%). Ordinal logistic regression was used to predict the stages of change for overweight and obese women.

RESULTS: A total of 135 women were overweight or obese (68%) and were classified into a stage of change. Overall, 30% of respondents were in the precontemplation stage, 15% in the contemplation stage, 48% in the preparation stage, 4% in the action stage, and 3% in the maintenance stage. Education, what friends think about the respondents' weight, body mass index, and a scale of the positive aspects of weight loss were significant ($p < .05$) predictors of the stage of change in a multivariate ordinal logistic regression model.

CONCLUSION: The prevalence of overweight and obesity was very high among these young adult women. Several predictors of stage were the same as those found in studies of other health behaviors, and these results provide support for applying a stages of change model to weight loss intention among rural African-American women. Two predictors in particular—significance of what friends think about weight and the pros scale—have implications for health education initiatives and social support in weight loss interventions.

HIV TESTING AMONG HIGH-RISK HOMELESS WOMEN—GOOD NEWS FROM LOS ANGELES COUNTY. B Herndon, AM Kilbourne, M Wang, M Lee, S Asch, R Andersen, L Gelberg, Dept. of Medicine, Greater Los Angeles VA Healthcare System, Los Angeles, CA; Dept. of Medicine, VA Pittsburgh Healthcare System, Pittsburgh, PA; Dept. of Health Services, UCLA School of Public Health; Dept. of Family Medicine, UCLA, Los Angeles, CA

PURPOSE: The potential for HIV testing to lead to improved health outcomes has never been greater, yet little is known about utilization of this service among populations at highest risk for HIV. The aim of this study is to determine the prevalence and predictors of HIV testing in a population of urban, poor, predominately minority homeless women in Los Angeles County.

METHODS: Women were interviewed in shelters and meal programs in Los Angeles County to create a probability cluster sample ($n = 974$). Bivariate analysis of patient characteristics, contact with health and social services, and indications for HIV testing identified variables with $p < 0.1$, and these were entered into a multiple regression model. Additionally, stepping algorithms with random subsampling for cross-validation were used to develop a more parsimonious multiple regression model. Samples were weighted and cluster analysis was performed on final models.

RESULTS: Approximately 70% of the sample reported receiving an HIV test in the past year. After adjustment for cluster effects, only a history of drug abuse or dependence and having a regular source or care were significant predictors of HIV testing in the past year. Sixty-five percent of respondents reported currently having a regular source of care. Neither income, public benefits, health insurance status, nor having a case manager predicted receipt of HIV testing. Similarly, none of the indications for HIV testing were statistically significant predictors. Of note, 15% of respondents reporting a pregnancy in the last year also reported that they did not receive an HIV test. There were no significant negative predictors of HIV testing in the past year.

CONCLUSION: The rates of HIV testing in this probability sample of women are among the highest reported for any population studied and are consistent with prior research indicating that individuals at highest risk are more likely to be tested. However, there is a significant minority of at-risk women with specific indications for testing, including pregnancy, who are not being reached. Our study indicates that policies focused on providing a regular source of care are likely to have the greatest impact on serostesting prevalence. However, the fact that our analyses revealed only two statistically significant predictors demonstrates that our current understanding of homeless women's use of this critical service is incomplete. Additional research is needed to better understand access to and utilization of HIV testing in this high-risk population.

ASSOCIATION OF BIRTHPLACE AND REGION OF RESIDENCE WITH HYPERTENSION PREVALENCE AMONG BLACK PARTICIPANTS IN THE THIRD NATIONAL HEALTH AND NUTRITION EXAMINATION SURVEY (NHANES III). LS Hicks, DG Fairchild, EF Cook, JZ Ayanian, Division of General Internal Medicine, Brigham and Women's Hospital; Department of Health Care Policy, Harvard Medical School, Boston, MA

PURPOSE: African Americans (AA) have among the highest rates of hypertension (HTN) worldwide, but Afro-Caribbeans and Africans have among the lowest rates of HTN. Regional differences in risk for cardiovascular mortality among AAs have also been noted. Despite these differences, the prevalence of HTN among Black American immigrants (BAI) compared to U.S. born AAs has not been well examined.

METHODS: Using data from NHANES III (1988–94) we examined the prevalence of HTN among 3,867 self-designated Black participants ages 30–79. Participants were classified as BAI if they were born outside of the U.S. Participants born in the U.S. were classified as southern AA (SAA) if they resided in the South census region, and northern AA (NAA) if they resided in the Northeast, Midwest or West census regions. HTN was defined by an average systolic blood pressure >140 , a diastolic blood pressure >90 , a previous diagnosis of HTN, or a history of treatment for HTN. Using NAA as the population standard we determined age-adjusted prevalence rates of HTN by sex among the three groups. We also used logistic regression to adjust for previously identified HTN risk factors and demographic characteristics of each group.

RESULTS: Of the 3,867 Black participants 1,612 (42%) were NAA, 2006 (52%) were SAA, and 249 (6%) were BAI. Approximately 54% of each group were female. NAA and SAA participants were more likely to be greater than age 60 compared to BAI participants (31%, 29% and 17% respectively, $p = .002$).

The table below compares the crude and age-adjusted prevalence rates.

	Crude prevalence (%) of HTN			Age-Adjusted prevalence of HTN		
	NAA (1612)	SAA (2006)	BAI (249)	NAA	SAA	BAI
Men	48.4	51.3	40.7 ⁺	48.4	51.4	47.1
Women	49.8	55.0*	34.4*	49.8	54.0*	38.4**

⁺ $P < 0.05$ vs. SAA men, * P value < 0.03 vs. NAA women, ** P value < 0.001 vs. NAA women and SAA women

SAA women remained at increased odds of HTN (OR 1.69, $P < .001$) compared to NAA women even after adjustment for body mass index, daily alcohol intake, and daily sodium intake.

CONCLUSION: In this large national cohort, SAA women but not men had a higher prevalence of HTN relative to NAAs. BAIs tended to have a lower prevalence of HTN when compared to AAs, though not significantly so. Future studies with larger numbers of BAIs may help to determine if this group differs in risk of HTN relative to AAs and to assess the influence of job-related stress, racism, diet, and other non-genetic factors in the development of HTN among Blacks in the U.S.

OVERALL HEALTH STATUS AND QUALITY OF LIFE AMONG BLACK AND WHITE DIALYSIS PATIENTS. LS Hicks, PD Cleary, AM Epstein, JZ Ayanian, Division of General Internal Medicine, Brigham and Women's Hospital; Department of Health Care Policy, Harvard Medical School; Department of Health Policy and Management, Harvard School of Public Health, Boston, MA

PURPOSE: Black patients on dialysis have lower mortality rates and withdrawal rates for failure to thrive than age-matched White patients. However, there has been little examination of racial differences in health related quality of life (HRQL) or self-reported overall health (OH) among these patients.

METHODS: To assess OH and HRQL, we surveyed 1,392 patients (response rate = 82.9%) age 18–54 approximately 9 months after they initiated dialysis for end-stage renal disease (ESRD) in five states and the District of Columbia. Respondents rated their overall health for the most recent month on dialysis. Questions adapted from the Kidney Disease Quality of Life (KDQOL) Instrument assessed patients' emotional health, physical activity, energy level, social activity, and burden of ESRD on daily life. All scales ranged from zero (poorest health/greatest burden possible) to 100 (best health/least burden possible). Available medical records (84% of each group) were reviewed for comorbid illness. Linear regression was used compare adjusted OH and HRQL by race.

RESULTS: Among both the 721 Black patients and the 671 White patients, 53% were female and the mean age was 42 years. Black patients reported better OH (mean score 65.0 vs. 60.0) and higher energy levels (mean score 47.2 vs. 41.3) (both $P < 0.001$). There were no differences between groups in the other four KDQOL domains. Black race remained significantly associated with OH and energy level after adjusting for comorbid illness, body mass index, perceived quality of care, amount of psychological and social support available, income and employment status. After adjustment, black men reported a lower burden of ESRD on daily life.

Characteristic	Black Women (N = 384)	White Women (N = 354)	Black Men (N = 337)	White Men (N = 317)
Overall health	65.6 ⁺	58.6	67.1**	60.7
Energy	46.2 ⁺	39.2	48.9*	43.6
Emotional Well-being	63.4	61.7	68.1	66.0
Physical activity	33.1	30.2	39.9	41.0
Social activity	57.9	57.6	62.4	57.7
Burden of ESRD	37.6	34.4	40.1*	35.1

* $P < .05$ vs. white men, ** $P < .001$ vs. white men, ⁺ $P < .001$ vs. white women

CONCLUSION: Black race is associated with a better OH and energy levels among dialysis patients, even after controlling for numerous potential confounders. Future research into determinants of overall health and HRQL for each racial group are needed to understand these racial differences.

ATTITUDES TOWARD HORMONE REPLACEMENT THERAPY AMONG WOMEN WITH A HISTORY OF BREAST CANCER. LK Hirota, BE Bond, AM O'Connor, JK Erban, NF Col, Medicine, New England Medical Center, Boston, MA; Faculty of Health Sciences, University of Ottawa, Canada

PURPOSE: Many women with breast cancer are menopausal and experience severe menopausal symptoms. Because breast cancer is more common among older women, many are at elevated risk for coronary heart disease and osteoporosis. To better counsel menopausal breast cancer survivors, we explored their decisions, attitudes, and expectations concerning the use of hormone replacement therapy (HRT).

METHODS: Women who had been treated for breast cancer at a tertiary hospital were mailed a questionnaire about their decisions, attitudes, and expectations concerning HRT. Subjective data were collected using a 5-point likert scale and the decisional conflict scale.

RESULTS: Of the 71 contacted, 51 returned a completed survey (72%). Their mean age was 52 years (range: 32–73), mean time since initial diagnosis was 6 years (range: 0–35), and 80% were postmenopausal. A majority of women were undecided about taking HRT (70%), 4% were already taking or definitely would take HRT, and 26% would definitely not take HRT. Nearly half felt the decision was hard to make. Many reported not knowing the benefits (26%) or the risks and side effects (41%) of HRT. Over half felt they needed more advice and information about the choice. Reasons cited for why they would take HRT included lowering their chances of heart disease (52%) and osteoporosis (48%), looking and feeling younger (36%), relieving menopausal symptoms (34%), and improving their sex life (32%). Reasons reported against taking HRT included the chances of breast cancer recurrence (80%), weight gain (20%), menstruation returning (20%), and vaginal bleeding (15%). Their doctors' recommendation, either for or against, weighed heavily in the decision. Eighty-four percent of women expected that HRT would lower their chances of heart disease and osteoporosis, and over a third were moderately to extremely worried about not receiving HRT's protection against these chronic diseases.

CONCLUSION: Many breast cancer survivors are undecided about the difficult decision to take HRT and need more advice and information from their clinicians about its risks, benefits, and side effects. In making this decision, women with breast cancer appear to place the most importance on their risk for breast cancer recurrence, but also seriously consider their risks for heart disease and osteoporosis, as well as their doctors' recommendation. Menopausal symptoms do not play as large a role in decision-making as might be expected.

PROSTATE-SPECIFIC ANTIGEN TESTING IN A COMMUNITY-BASED POPULATION: DIAGNOSTIC PERFORMANCE AND OUTCOMES. RM Hoffman, M Adams-Cameron, WC Hunt, CR Key, FD Gilliland, Albuquerque VA Medical Center; University of New Mexico Cancer Center, Albuquerque, NM; University of Southern California, Los Angeles, CA

PURPOSE: We created a prostate-cancer screening surveillance system to determine the diagnostic performance and outcomes of prostate-specific antigen (PSA) testing in a community-based population.

METHODS: We obtained all PSA testing data, including results, patient demographics, and date of testing from the major laboratories in the Albuquerque metropolitan area from 1995 through 1997. Medical records abstractors from the New Mexico Tumor Registry (NMTR) obtained data on all benign prostate biopsies at these laboratories. Data for men ≥ 40 years were linked with the NMTR to exclude prevalent cases of prostate cancer and to identify incident cases of prostate cancer. The NMTR also provided data on date of cancer diagnosis, staging, and treatment. The diagnostic performance of PSA testing was determined by calculating sensitivity, specificity, likelihood ratios, and the area under the ROC curve (AUROC).

RESULTS: We identified a cohort of 41,261 subjects without previously diagnosed prostate cancer who underwent PSA testing. The median age was 61 (range 40–107) and 62% were white, 29% Hispanic. Overall, 2,574 subjects (6.2%) were biopsied, including 33% of the 5,298 men with PSA values ≥ 4.0 . Biopsies were obtained from 44% of men 50–59 with elevated PSA levels, but only 23% of men ≥ 70 . Cancer was detected in 944 subjects (2.3%), 758 (80%) had localized and 49 (5.2%) had distant stage disease. The cancer detection rate ranged from 0.2% for men in their 40s to 3.8% in men ≥ 70 . PSA levels were significantly higher in cancer cases (median = 6.6) than non-cases (1.1), $P < 0.0001$. Among subjects undergoing biopsy, the sensitivity for PSA ≥ 4.0 was 78% and the specificity was 40%, the AUROC was 0.64 (SE = 0.01). Likelihood ratios (95% CI) for PSA levels $< 4 = 0.56$ (0.49, 0.64), $4-10 = 1.0$ (0.9, 1.1), $> 10-20 = 1.6$ (1.3, 1.9), and $> 20 = 5.4$ (3.9, 7.5). Sensitivity increased with age, from 68% in men in their 40s to 83% in men ≥ 70 . Conversely, specificity decreased from 48% to 32%. Seventy-four percent of the patients with localized cancers received aggressive treatment (surgery and/or radiation). However, the proportion of patients receiving aggressive treatment decreased from 82% in men 40–59 to 56% in men ≥ 70 .

CONCLUSION: In a community-based population, PSA testing provided only fair discrimination between men with and without prostate cancer. Very high PSA levels (> 20) were needed to substantially increase the likelihood of cancer. PSA testing appeared to have less clinical utility for men ≥ 70 because they were less likely to undergo biopsy or have aggressive treatment for localized cancers.

THE ROLE OF SPIRITUAL CARE IN THE GENERAL MEDICINE CLINIC. S Holmes, MW Rabow, Medicine, UCSF, San Francisco, CA

PURPOSE: To compare the views of seriously-ill out-patients with those of their primary care providers (PCPs) regarding the place of spiritual care in the primary care medical encounter.

METHODS: Two self-administered surveys on spiritual care were developed and pre-tested. One survey was distributed to all currently active PCPs in the General Medicine Practice (GMP) at UCSF. The other was distributed to GMP patients enrolled in a study of end-stage CHF, COPD, and cancer. Non-responders received two subsequent survey distributions.

RESULTS: Of the 38 patient surveys, 26 were returned complete (68%). The average age of the patients was 64. 73% were female and 73% were White. 65% considered themselves "Quite a bit" or "Very much" spiritual. While there was no gender difference between patients' desire for spiritual care, significantly more men reported their providers had asked if they would like a referral to a chaplain ($p = 0.01$).

Of the 84 PCP surveys distributed, 67 were returned complete (80%). The average age of the PCPs was 33. 55% were female and 61% were White. 61% were residents; the remainder were fellows or faculty. Only 30% of PCPs considered themselves "Quite a bit" or "Very much" spiritual. Compared to their male colleagues, female providers considered themselves to be more spiritual ($p = 0.05$) and believed it more important to attend to patients' spiritual concerns ($p = 0.03$). Female PCPs more frequently asked whether patients were part of a spiritual community ($p = 0.05$) and prayed more often for patients ($p = 0.001$). Faculty and fellows were more likely than residents to ask patients if they were part of a spiritual community ($p = 0.0001$), ask about the importance of spirituality ($p = 0.001$), offer to discuss spiritual concerns ($p = 0.001$), and offer a chaplaincy referral ($p = 0.01$). This difference persisted even when controlled for age and number of weekly clinical sessions practiced.

Similar percentages of patients (60%) and PCPs (68%) believed that it is important for PCPs to attend to patients' spiritual concerns. However, only 19% of patients had ever been asked by their PCP if they would like to discuss spiritual concerns and 19% had been asked if they would like to speak with a chaplain. 68% of patients wanted their PCP to pray for them, but only 23% reported that their PCP had ever offered. When asked what hinders them from addressing spirituality, 82% of providers cite a lack of time, 37% do not feel competent, and 21% do not feel it is part of their job.

CONCLUSION: Seriously-ill patients and PCPs in an urban, academic, general internal medicine clinic agree that spiritual care has an important role in the medical encounter. However, discrepancies exist between this stated belief and spiritual care practices, especially among male providers and residents.

SOCIO-ECONOMIC STATUS, HEALTH BEHAVIORS AND THE UTILIZATION OF RECOMMENDED AGE-GENDER-SPECIFIC PREVENTIVE SERVICES. CA Hornung, CJ Mundt, WP McKinney, Dept of Medicine, University of Louisville, Louisville, KY

PURPOSE: To evaluate socio-economic status characteristics and health behaviors (e.g., smoking status) associated with compliance with age-gender-specific preventive services recommended for individuals over the age of 50 by the U.S. Task Force on Preventive Services.

METHODS: Secondary analysis of the 1997 Behavioral Risk Factor Surveillance Survey (BRFSS). Age-gender-specific indices were created for individuals over the age of 50 involving 8 preventive services for women and 5 preventive services for men. Each index was normed to reflect the percent of the age-gender specific recommended preventive services utilized by an individual (i.e., percent compliance).

RESULTS: Women reported receiving a mean of 57% (SD = 16%, Median = 63%) of the recommended preventive services compared to 46% for men (SD = 29%, Median = 40%; $p < 0.001$). Clinically unimportant differences in the percent of services utilized were found between White Non-Hispanic (53%), Black Non-Hispanic (51%), Hispanic (48%) and Other (51%) individuals ($p < 0.001$). Gender differences were consistent across racial-ethnic groups. Both income and years of education are positively related to the percent of services received ($p < 0.001$) for both men and women and show a range from 37% for men with less than an elementary education as well as men with incomes of less than \$10K per year to only 52% for men who graduated from college and men with annual incomes greater than \$75K. The mean percent of recommended preventive services received by women was about 10% higher at each level of annual income and educational attainment (range: 47% to 63%). Nevertheless, gender, income and educational attainment were only weak predictors explaining less than 10% of the variation in the utilization of recommended preventive services. However, of particular interest is the relationship between smoking status and percent of recommended preventive services received ($p < 0.001$). While current smokers were least compliant (men 35%, women 50%), self-reported former smokers utilized more of the recommended age-gender-specific preventive services (men 49%, women 61%) than individuals who claimed they never smoked (men 46%, women 56%).

CONCLUSION: These findings confirm that most people, particularly men, those with little education and low incomes receive only about half of the preventive services recommended for their age and gender. However, individuals who report they have quit smoking utilize a greater proportion of preventive services recommended for their age and gender than do current smokers and those who never smoked. The health message about the hazards of smoking apparently convinced former smokers of the benefits of healthy behaviors and these 'converts' have become the most compliant in the of utilization of recommended preventive services.

ETHNIC DIFFERENCES IN PREFERENCES FOR TREATMENT AMONG UNTREATED DEPRESSED INDIVIDUALS. TK Houston, DE Ford, L Cooper-Patrick, Johns Hopkins Medical Institutions

PURPOSE: To compare depression treatment preference among untreated depressed persons from different ethnic groups and to relate these preferences to attitudes toward depression treatment.

METHODS: The Centers for the Epidemiological Studies of Depression (CES-D) scale was adapted as an interactive screening test on a popular internet health site. Persons with a positive screening test (CES-D > 22) were instructed to seek a medical opinion and asked to complete a survey of treatment history, attitudes and preferences that could be printed to take a physician. CES-D and survey responses were anonymously compiled over eight months. Our main outcome was preference for counseling or medications to treat depression.

RESULTS: The CES-D identified 14,185 persons with depression, 6,641 (47%) of whom had no past history of treatment for depression. Untreated depressed persons were 72% female, 15% minorities (4% Asian, 6% African American, 5% Latino), and 47% were ≤ 29 years old. Some untreated depressed persons ($N = 1,565$) indicated no specific preference for counseling or medications. Among persons who indicated a preference for treatment ($N = 4,894$), a higher percentage of untreated depressed African Americans (77%), Asians (68%), and Latinos (67%) preferred counseling over antidepressant medications, compared with 53% of Whites ($p < 0.001$). After adjustment for age and gender, African-Americans, Asians, and Latinos were still more likely to have a preference for counseling with relative odds (RO) of 2.8 (CI 2.13-3.68), 1.7 (1.24-2.34), and 1.7 (1.33-2.27), respectively, compared with whites ($p < 0.01$).

Asians had the strongest concerns about stigma of depression with 46% agreeing that their family would be disappointed if they had depression compared with 31% of whites ($p < 0.001$). A higher percentage of African Americans (39%), Asians (22%) and Latinos (22%) strongly agreed that prayer was helpful in healing depression, compared with 14% of whites ($p < 0.01$). Minorities were more likely to strongly agree that medications were addictive with RO of 2.6 (CI 2.1-3.1) compared with whites. After adjustment for these attitudes, the strength of association of preference for counseling and ethnic group was attenuated, with African Americans, Asians, and Latinos having RO 2.46 (CI 1.8-3.3), 1.44 (CI 1.03-2.0) 1.36 (CI 1.02-1.8), respectively, compared with whites. After receiving feedback on their depressive symptoms, 45% felt that they needed treatment for depression.

CONCLUSION: Ethnic minorities with untreated depression indicated a preference for counseling over medication that was partially explained by their reported differences in attitudes. When identifying untreated depressed individuals, physicians should discuss patient preferences and concerns about treatment, and consider counseling as an initial treatment for depression, especially among minorities.

PATIENTS' COMPREHENSION OF IMPRECISE INFORMATION. J Hsu, P Ubel, DA Asch, Division of Research, Kaiser Permanente, Oakland, CA; Division of General Internal Medicine, University of Pennsylvania, Philadelphia, PA

PURPOSE: Recent studies have suggested that patients have difficulty comprehending medical risk information. One explanation is that these patients have difficulty interpreting numbers. Most persons, however, use numbers in other uncertain contexts, e.g., when deciding on whether to carry an umbrella to work in the morning or on purchasing services that might vary in price. This study seeks to determine whether subjects are able to use imprecise numerical information when making decisions in non-medical and in medical contexts.

METHODS: 145 prospective jurors received a written, self-administered questionnaire involving scenarios with imprecise probability information on the chance of rain, on the cost of a car repair, and on the chance of developing a hypothetical disease. Within each scenario, subjects received a probability estimate with three levels of precision, e.g., a 30% probability of rain $\pm 5\%$, $\pm 10\%$, and $\pm 20\%$. In both the weather and the medical test scenarios, we varied the probability estimate (from 50% to 30%); and in the weather scenario, we also varied the ranges of precision (from ± 1 , 5, 10% to ± 5 , 10, 20%) around the probability of rain. The questionnaires were otherwise identical. The study assessed subject's ability to compare the precision provided by the three options within each scenario.

RESULTS: Within this convenience sample of potential jurors, the mean age was 40 years; 70% were female; 66% received greater than 12 years of education; 44% were Caucasian; and 76% were employed. Subjects correctly identified the choice with the most precise information in 81%, 88%, and 69% of the responses in the weather forecast, in the car repair, and in the medical test scenarios respectively ($p < 0.001$). Furthermore, when the risk estimate varied from 30% to 50%, there was no significant difference in patients' identification of the most precise weather forecast (89% v. 84%, $p > 0.5$) or in their identification of the most precise medical test (67% v. 70%, $p > 0.7$). When the width of the confidence intervals varied between $\pm 10\%$ to $\pm 20\%$, there also was no statistically significant difference: 69% v. 84% ($p > 0.09$).

CONCLUSION: In conclusion, the subjects' ability to distinguish choices on the basis of information precision varied by context, but not by base rate or degree of precision around that base rate. These data suggest that contextual factors may play a larger role in subjects' ability to comprehend imprecise risk information than issues of numeracy, and that presenting uncertainty in medical settings faces particular challenges even to patients who can understand uncertainty in non medical settings.

THE MAGNITUDE AND TIMING OF EFFECT OF PREVENTIVE STRATEGIES IN TYPE 2 DIABETES MELLITUS. ES Huang, JB Meigs, DE Singer, General Internal Medicine, Massachusetts General Hospital, Boston, MA

PURPOSE: Generalists face the challenge of managing multiple risk factors when caring for patients with type 2 diabetes. We carried out a meta-analysis of hyperglycemia, hypercholesterolemia, and hypertension treatment trials (RCTs) to quantify the magnitude and timing of effect of these strategies.

METHODS: Using Medline and review articles, we systematically searched for RCTs focusing on type 2 diabetes or diabetic subgroups. We included RCTs measuring pre-specified macrovascular and microvascular outcomes and abstracted characteristics of the studies, patients, and outcomes. We calculated treatment effect by odds ratio(OR) and patient-years of treatment needed to prevent one event(PYT). ORs were pooled using the DerSimonian-Laird model (RevMan 4.0). We estimated the time to treatment effect (TTE) from the separation of Kaplan-Meier curves.

RESULTS: Four glycemic, 7 cholesterol, and 13 hypertension studies were used. For macrovascular events(cardiovascular mortality, myocardial infarction (MI), and stroke), hypercholesterolemia and hypertension control had larger treatment effects than glycemic control (see TABLE):

	Cardiovascular Mortality	OR [95% CI]	PYT (patient-years, range)	TTE (years, range)
Glycemic Control		0.98 [0.78, 1.24]	909-no estimate from studies with no benefit	12-no estimate
Cholesterol Control		0.70 [0.57, 0.86]	63-170	1-4.5
Blood Pressure Control		0.56 [0.41, 0.75]	51-250	0.7-5.5

In MI, cholesterol lowering had the largest magnitude of effect (OR 0.43 [0.18, 1.02] & PYT 30-135), whereas in stroke, blood pressure control had the largest treatment effect (OR 0.59 [0.43, 0.81] & PYT 55-370). For renal failure, BP lowering had a larger effect (OR 0.48 [0.19, 1.23] & PYT 9-1111) than glycemic control (0.74 [0.33, 1.68] & PYT 5000), though endpoints were few. Microvascular events, overall, were prevented at 3 years for BP control and 9 years for glycemic control.

CONCLUSION: Hyperglycemia is the characteristic focus of type 2 diabetes management. However, macrovascular disease poses the greatest threat to such patients. BP and cholesterol treatment leads to larger and more rapid effects on macrovascular complications than glucose control. Even for microvascular nephropathy(renal failure and macroalbuminuria), hypertension appears to be as effective as hypoglycemic therapy. Glycemic control is important in type 2 DM. Our analyses indicate that cholesterol and BP management is even more important.

CARING FOR ADULTS WITH CEREBRAL PALSY: EMPLOYMENT RAMIFICATIONS FOR FAMILY MEMBERS. *N. Hupert, EP McCarthy, RB Davis, U Iezzoni, Beth Israel Deaconess Medical Center, Harvard Medical School, Boston, MA*

PURPOSE: People with cerebral palsy (CP) and other potentially serious congenital and childhood onset medical conditions are increasingly surviving into adulthood. In small studies, parents of children with these conditions express concern about how to provide adequate care as their children grow into adulthood. We identified care providers for adults with CP in a national sample and assessed the impact of providing such care on employment of family members.

METHODS: We used the 1994-95 National Health Interview Survey (NHIS, a nationally representative sampling of non-institutionalized, civilian U.S. residents) and NHIS Disability Supplement (NHIS-D, a detailed follow-up survey for persons with selected conditions including CP). We used SUDAAN to calculate national population estimates and descriptive statistics.

RESULTS: 129 persons age 18 or older reported cerebral palsy (representing an estimated 174,000 civilian, community-dwelling adults), of whom 76% completed the NHIS-D. The mean age of the sample completing both surveys was 36.4y (SE 1.6). 67% needed at least one regular helper for activities of daily living (ADLs, such as bathing) and/or instrumental ADLs (such as telephoning). A parent was the primary helper for 38%, spouse for 15%, paid employee for 13%, and other relative or friend for 34%. 34% of the total population either lived alone or were married. Of those who lived with someone else but were not married, 62% (or 41% of the total population) said that they did so because of their health or physical problem. 73% of these people lived with parent(s), 12% with a sibling, and 15% with another relative or non-relative; 47% of this group were 40 years old or older. 16% of all persons with CP (and 26% of those who lived with someone else because of health problems) said that their medical problem caused a family member to quit a job, reduce working hours, or change jobs. Family members affected included parents (64%), spouses (19%), and siblings (17%).

CONCLUSION: Parents, spouses, and other relatives provide the majority of personal assistance for non-institutionalized adults with CP living in the U.S. Approximately 1/3 of this population continue to live with parents well into middle age because of their medical condition. A substantial minority report adverse impact of their medical condition on family members' employment. These results suggest that greater financial and/or personnel assistance for the medical care of adults with CP may be warranted.

INCOME LEVELS AND ALCOHOL-RELATED HOSPITAL ADMISSION RATES. *SW Hwang, RH Glazier, F.K. Morrow Inner City Health Research Unit, St. Michael's Hospital, Toronto, ON, Canada*

PURPOSE: To determine the relationship between income levels and alcohol-related hospital admission rates within a large urban area.

METHODS: We examined hospitalizations among the approximately 635,000 adult residents of the downtown core of Toronto, Ontario, a socioeconomically diverse city with geographic clusters of high, middle, and low income neighborhoods. We used a comprehensive database of hospital admissions that obtains mandatory reports from every acute care facility in Ontario. Alcohol-related admissions during the period April 1996 through March 1997 were identified by ICD-9 codes for the following conditions: alcohol psychoses, alcohol dependence syndrome or alcohol abuse, alcoholic fatty liver, acute alcoholic hepatitis, alcoholic cirrhosis or other alcohol-related liver damage, toxic effects of alcohol, or accidental poisoning by alcohol. If a patient was admitted more than once, only the first admission was counted. Each patient's place of residence within Toronto was established using postal codes.

The study area consisted of 1,408 census enumeration areas. We obtained 1996 census data on population composition and average household income in each enumeration area. The enumeration areas were grouped into 5 income quintiles, each with approximately equal population. In each quintile, age- and sex-adjusted admission rates for alcohol-related conditions were calculated using direct standardization.

RESULTS: Alcohol-Related Hospital Admission Rates per 10,000 Population in Downtown Toronto

Income Quintile	Total (age- and sex-adjusted)	Men (age-adjusted)	Women (age-adjusted)
1 (Poorest)	46.7	71.9	23.4
2	30.5	49.2	13.2
3	26.7	39.0	15.4
4	24.9	35.7	14.9
5 (Richest)	17.8	24.9	11.2

CONCLUSION: Men and women living in the lowest income quintile areas of Toronto have particularly high admission rates for alcohol-related conditions. Among men, admission rates decrease steadily with increasing income, but admission rates among women are similar in all income quintiles except the lowest one. These findings suggest significant gender differences in the correlation between socioeconomic status and health-related consequences of alcohol use.

RISK OF DEATH DURING PERIODS OF HOMELESS SHELTER USE. *SW Hwang, F.K. Morrow Inner City Health Research Unit, St. Michael's Hospital, Toronto, ON, Canada*

PURPOSE: Homeless persons have high mortality rates, but it is unknown whether the homeless state itself affects the risk of death. The purpose of this study was to determine whether the risk of death increases during periods of homeless shelter use.

METHODS: We studied a cohort of 8769 men 18 to 64 years old who used homeless shelters in Toronto, Ontario, at least once during 1995. The cohort was followed until the end of 1997. All shelter admissions were tracked using a citywide database. Deaths were ascertained using the Ontario death registry. Survival analyses were performed using Cox regression. For each month of follow-up, a time-dependent covariate modelled the hazard associated with homeless shelter use during that month compared to no homeless shelter use. In a multivariate model, we adjusted for age and for pattern of shelter use during the one year preceding entry into the cohort. Entry into the cohort was defined as the first shelter admission in 1995. Pattern of shelter use was classified as transitional, episodic, or chronic using a previously described cluster analysis technique.

RESULTS:

Variable	Adjusted Relative Risk (95% CI)
Age (per additional decade)	1.7 (1.5, 1.9)
Shelter use during each month of follow-up (time-dependent)	1.8 (1.3, 2.7)
Pattern of shelter use during 1 year preceding entry into cohort	
— No shelter use (reference group)	1.0
— Transitional (1-2 brief stays)	1.1 (0.8, 1.7)
— Episodic (multiple stays of varying lengths)	1.5 (0.7, 3.1)
— Chronic (1-2 very long stays)	1.1 (0.3, 4.6)

The hazard associated with months during which shelters were used was consistent across age groups, and persisted when the analysis was restricted to men with little or no history of homelessness prior to 1995. For death due to homicide or suicide, the adjusted relative risk during shelter use months compared to months in which shelters were not used was 5.5 (95% CI, 1.8 to 16.5). An episodic pattern of homelessness prior to cohort entry was associated with a significantly elevated risk of death in univariate but not multivariate analyses.

CONCLUSION: Among men who use homeless shelters in Toronto, the risk of death almost doubles during months of shelter use. This association could be due to factors such as increased substance abuse, exposure to violence or the elements, or homelessness-related exacerbation of pre-existing medical problems. Alternatively, some persons may enter the shelter system because of deteriorating health.

DO OLDER AFRICAN-AMERICAN AND WHITE PATIENTS DIFFER WITH RESPECT TO THEIR ATTITUDES TOWARD JOINT REPLACEMENT? *SA Ibrahim, CJ Burant, LA Siminoff, K Kwoh, Medicine, Louis Stokes Dept. of VAMC; University Hospitals of Cleveland; Case Western Reserve University, Cleveland, OH*

PURPOSE: Although ethnic disparities in the utilization of joint replacement for osteoarthritis have been documented, the reasons behind these disparities are not clear. Since differences in patient attitudes about joint replacement may play a role in their utilization of joint replacement, older male AA and white veterans with chronic knee and/or hip pain were surveyed with respect to their attitudes toward joint replacement.

METHODS: A cohort of 285 patients aged 50 or older (45% AA and 55% white) with moderate to severe knee and/or hip pain of more than six months duration were surveyed. Patients who had had joint replacement were excluded. Patients were asked whether they would consider surgery if their knee or hip pain became severe and the procedure was recommended by their physician. We also assessed their perception of the efficacy of joint replacement using a component from the Efficacy of Specific Treatments for arthritis (Bill-Harvey). Other measures we assessed included the severity of their arthritis (Lequesne Scale), arthritis-specific health status (WOMAC), comorbidity (Charlson), Fatalism (Powe) and educational level.

RESULTS: AA and white patients were comparable with respect to age (67 ± 9 vs. 68 ± 9 , $p = 0.53$); severity of arthritis (mean score 11 vs. 12, $p = 0.09$); arthritis-specific health status (mean score 17 vs. 18, $P = 0.25$); comorbidity (mean score 2.6 ± 2.1 vs. 2.6 ± 2.1 , $p = 0.99$); Fatalism (mean score 2.1 ± 0.5 vs. 2.0 ± 0.6 , $p = 0.63$). AAs were less likely than whites to be high school graduates (54% vs. 66%, respectively, $p = 0.03$). AAs were significantly less likely than whites to consider joint replacement as an option (crude odds ratio = 0.42, 95% CI 0.21 to 0.83). This difference persisted after adjusting for age, severity, health status, comorbidity, fatalism, educational level and other demographic, clinical, and psychosocial covariates (adjusted odds ratio = 0.48 95% CI 0.23 to 0.99). In addition, AAs were more likely than whites to perceive joint replacement as "Not at all helpful" (14.6% vs 7%, $P = 0.006$).

CONCLUSION: In this cohort of AA and white male veterans with moderate or severe chronic knee and/or hip pain, AA patients were less likely than whites to consider joint replacement as an option for future therapy and they were also less likely to perceive joint replacement as being helpful for arthritis. These ethnic differences in patient attitudes may help to explain differences in the utilization of joint replacement for arthritis.

CLINICAL PREDICTORS OF MENTAL DISORDERS IN PRIMARY CARE, REPLICATION OF THE S4 MODEL. JL Jackson, S Hanling, J Houston, K Terhaar, JS Yun, Medicine, Uniformed Services University, Bethesda, MD

PURPOSE: Mental disorders are common in primary care. Unfortunately, primary care providers frequently miss such disorders. We previously described 4 clinical markers that predicted the presence of mental disorders: recent Stress, poor functional Status, greater number of physical Symptoms and increased symptom Severity (S4 model). The purpose of this study was to validate this model in another primary care cohort.

METHODS: Two hundred fifty patients presenting to a primary care walk-in clinic with a physical symptom were surveyed. Surveys assessed patient demographics, symptom characteristics, serious illness worry, recent stress, functional status (MOS SF6), and mental disorders (PRIME-MD).

RESULTS: A mental disorder was present in 26% of patients. Eighteen percent had a depressive disorder (5%-Major Depression), 11% anxiety (2%-Panic), and 8% a somatoform disorder. On univariate screen, each of the previously described clinical predictors were associated with the presence of a mental disorder: Stress (79% v 36%, $p < 0.0001$); poor functioning (32% v 15%, $p < 0.003$); more than 5 other physical symptoms (53% v 14%, $p < 0.0001$); and greater symptom severity ($p = 0.03$). On multivariate (logistic regression) analysis, recent Stress (OR: 6.2, 95% CI: 3.0-12.1), more than 5 other physical symptoms (OR: 4.5, 95% CI: 2.3-8.8) or poor overall functioning (OR: 8.1, 95% CI: 1.1-35.0) were significant predictors of the presence of a mental disorder, while symptom severity was not (OR: 1.2, 95% CI: 0.6-2.5). The 3 or 4 variable regression models both reasonably fit the data (Hosmer-Lemeshow $\chi^2(7) = 4.23$, $p = 0.75$) with an area under the ROC of 0.82. There was a stepwise relationship between the number of predictors present and the likelihood of a mental disorder with 8%, 28%, 66% and 100% having underlying mental disorders among patients with one, two, three or all four of the clinical predictors present, respectively.

CONCLUSION: In this cohort, 3 (Stress, number of Symptoms, poor functional Status) of the original 4 variables were found to predict the presence of a mental disorder. Symptom severity did not add to the 3 variable regression model. Since these cues often emerge during "normal" clinical encounters, they may help clinicians select patients for more time-consuming mental interviews. There is a stepwise relationship between the number of variables presence and case-finding for mental disorders is particularly fruitful when at least 3 of these variables are present.

THE IMPACT OF UNMET EXPECTATIONS AMONG ADULTS PRESENTING WITH PHYSICAL SYMPTOMS. JL Jackson, K Kroenke, Medicine, Uniformed Services University, Bethesda, MD; Medicine, Regenstrief Institute, Indianapolis, IN

PURPOSE: To assess the prevalence and impact of unmet expectations on patients presenting with physical symptoms.

METHODS: Design Prospective cohort

Setting Primary care walk-in clinic, most patients seeing physicians new to them.

Patients 750 adults with the principal reason for visit being a physical symptom.

Measurements Previsit surveys assessed symptom characteristics, expectations, functional status (Medical Outcome Study Short Form-6), serious illness worry and mental disorders (PRIME-MD). Surveys immediately after the visit and at 2 weeks assessed patient satisfaction, residual serious illness worry and unmet expectations. Two-week assessment also included symptom outcome and functional status. Health utilization for 3 months was assessed using the DOD computerized health data system.

RESULTS: Nearly all patients endorsed at least one pre-visit expectation. Eighty-one percent hoped for a diagnosis, 63% prognostic information, 60% a prescription, 54% a diagnostic test, 45% a subspecialty referral and 7% an excuse from work. Immediately after the visit, 33% had an unmet expectation for an explanation of the symptom's cause and 51% an unmet desire for an estimate of how long to expect the symptom to last. Eleven percent had a variety of other unmet expectations, including desires for diagnostic tests, subspecialty referrals, prescriptions, or sick slips. Patients who reported receiving diagnostic or prognostic information were more likely to be fully satisfied with their care ($p < 0.001$) were less likely to be worried their symptom was due to something serious ($p < 0.001$) experienced greater symptom ($p < 0.01$) and functional status improvement ($p < 0.003$) at 2 weeks and had lower 3 month utilization rates. ($p = 0.002$) Residual expectations for a subspecialty referral or diagnostic testing were also associated with lower levels of satisfaction ($p < 0.00001$) and with continued serious illness worry ($p < 0.001$).

CONCLUSION: Receiving diagnostic and prognostic information from physicians during encounters with symptomatic patients is associated with greater satisfaction, reduced serious illness worry, and greater symptom and functional status improvement by two weeks.

DELIVERY OF PREVENTIVE HEALTH SERVICES TO A HISPANIC HMO POPULATION. EA Jacobs, DS Lauderdale, Collaborative Research Unit, Cook County Hospital; Department of Health Studies, University of Chicago, Chicago, IL

PURPOSE: Population-based studies have shown that the US Hispanic population receives fewer preventive health services than non-minority comparison groups. Limited access to care as measured by insurance, income and continuity of care and cultural factors including acculturation, limited English-proficiency, and specific health beliefs, have been associated with reduced preventive health service

use in this population. We explored whether Hispanic patients who have enhanced access to care receive preventive health services at the same rate as a non-Hispanic comparison group.

METHODS: We conducted a cross-sectional study of two groups of adult HMO members who had been continuously enrolled for two years: (1) Hispanic members identified through a surname matching program [HG] and (2) a comparison group [CG] of a 10% random sample of all other adult members. The validity of surname matching was verified using Hispanic census tract information. Demographic data, Spanish language indicators and preventive service use were abstracted from an administrative database.

RESULTS: The two groups were similar in age and sex distribution and total years of enrolment, but differed significantly in census median income (HG = \$36,465 vs. CG = \$40,772; $p < 0.01$). Rates of mammography, breast exams, pap smears, fecal occult blood testing and flu immunizations were calculated per person-year for the age and sex appropriate members of each group. The rates were not significantly different between the two groups (see table).

Preventive Service	CG (n = 4053)	HG (n = 1703)
Mammograms	0.65	0.59
Breast Exams	0.55	0.53
Pap Smears	0.73	0.72
FOB	0.83	0.78
Rectal Exams	0.52	0.45
Flu Immunizations	0.73	0.63

Multiple logistic regression analysis of each of the rates controlling for age, sex, total years of HMO enrolment, and language produced similar results. The only significant associations were pap smears and Spanish-language (OR 0.75; 95% CI 0.59-0.96), fecal occult blood testing and female gender (OR 0.67; 95% CI 0.54-0.87) and years of enrolment (OR 0.90; 95% CI 0.82-0.98) and flu immunizations and years of enrolment (OR 0.82; 95% CI 0.70-0.96).

CONCLUSION: These findings suggest that Hispanic disparities in receipt of preventive health care may be reduced by enhancing their access to health care through insurance and a continued relationship with a single health care system.

SURVEY OF VIOLENCE AGAINST WOMEN IN AN INTERNAL MEDICINE RESIDENTS' CLINIC. MJ Jelley, EM Mutzig, DZ Akberzie, JC Matthews, Department of Internal Medicine, University of Oklahoma, Tulsa, OK

PURPOSE: Domestic violence is a serious public health issue but has only recently become part of the curriculum in residency. Several studies have shown high incidence of violence against women among acute and primary care patients. The purpose of this study is to determine current incidence of violence among female clinic patients and gather information on related health problems in order to improve training in violence intervention for residents.

METHODS: A self-administered 30-question survey was given to consecutive female patients at the internal medicine residents' clinic during a five day period. The survey was titled "Women's Health Survey" and questions on violence followed several demographic and general health questions. The women completed the survey in the waiting room or in the exam room.

RESULTS: Of 264 patients approached, 169 complete surveys were returned (64%). When women were asked if they had ever been physically abused as a child, 28% responded "yes". When asked if they had ever been sexually abused as a child, 35% responded "yes". Physical or sexual abuse in adulthood was present in 47% of women. Physical abuse by a partner within the last year was present in 13% of women. Women with a history of abuse as an adult were significantly more likely to have had a miscarriage than those without an abuse history (55% vs. 32%). They were also more likely to have attempted suicide and visited an emergency room recently. There were no significant differences in age, education or income between those with a history of abuse and those without.

CONCLUSION: The rate of current and past abuse among these primary care female patients is alarmingly high. The abuse may be contributing to increased use of health care services. Recognition of this problem is the first step to decreasing the incidence of abuse. Primary care residents should have ample opportunity to become well trained in recognition and initial intervention in violence against women.

PATIENTS TALK ABOUT SPIRITUALITY AND THERAPEUTIC RELATIONSHIPS WITH PHYSICIANS. MW Jenckes, R Hebert, D O'Connor, DE Ford, L Cooper-Patrick, Medicine, Johns Hopkins Medical Institutions; Oncology, Johns Hopkins Medical Institutions, Baltimore, MD

PURPOSE: To define good spiritual care and provide insight and guidance to physicians caring for seriously ill patients through analysis of recovering patients' comments on care they received.

METHODS: Two focus groups totaling fourteen participants (mean age 53; eight women, four African Americans). Participants were recruited through their physicians; all had experienced a life-threatening illness in the preceding year. Discussions addressed facilitators and barriers to talking about spiritual beliefs with physicians, physician characteristics, and recommendations for physicians. Discussions were audio taped, transcribed verbatim, and reviewed independently by two investigators who grouped distinct comments into categories with specific themes. Differences were adjudicated by a third investigator. Categories and comments within categories were then checked independently for relevance and con-

sistency by a health services researcher and a clinical social worker and further adjudicated as necessary.

RESULTS: Four hundred thirty one distinct comments on spirituality were generated with 95% falling into 8 major categories (see Hebert et al). The category of physician-patient communication was particularly interesting to us ($n = 104$). All group participants joined in describing a need for empathic communication from their physicians. The most common expression was the need for comfort. Patients mentioned prayer with their physicians, physical contact, eye contact, body language, and words that gave messages of comfort. Personality, trustworthiness, ability to give hope, genuine caring, and interest in patients' personal and family life were also used as descriptors. Patients with a strong sense of spirituality spoke of physicians who were able to give them comfort as *spiritual*. Patients demonstrated a yearning for this level of communication with their physicians. Several patients indicated willingness to break off care and move to a physician who could meet this standard, even in the middle of a serious illness, thereby disrupting their care. When physicians met their standard of spirituality, patients reported improved ability to cope with the burden of illness (medicines, tests, etc.) and to accept the outcome of their illness with peace.

CONCLUSION: Our study further supports the therapeutic effects of empathic care giving by physicians. Patients describe empathic physicians as spiritual, indicating that this skill set is sacred to them. Furthermore, this particular aspect of interpersonal care defined through patients' eyes is important to patients' ability to cope with illness and remain in a therapeutic relationship with their physician.

PREDICTORS OF SMEAR NEGATIVE PULMONARY TUBERCULOSIS: A UNIQUE SCORING SYSTEM. AM Kanaya, DV Glidden, HF Chambers, Medicine; Biostatistics, University of California, San Francisco, CA

PURPOSE: Clinicians face a dilemma when managing patients who are suspected to have tuberculosis (TB) but have negative sputum smears. Mycobacterial culture results may take weeks which delays the initiation of treatment and may allow further transmission of TB. We aimed to identify the clinical, demographic and radiographic characteristics that identify the smear-negative patient who has tuberculosis.

METHODS: A retrospective review of microbiology and medical records was conducted at San Francisco General Hospital between 1993 and 1998. All patients were hospitalized with a suspicion of pulmonary TB and were sputum smear-negative. Culture results were used as the gold standard to determine if TB was present. Patients with a past history TB or those currently being treated for TB were excluded.

RESULTS: 47 cases had positive TB cultures and 141 controls were culture-negative. Univariate analyses of demographic, clinical and radiographic characteristics were performed with the chi-square test. Odds of a positive TB culture were decreased among whites (Odds Ratio (OR) 0.46, 95% confidence interval (CI) 0.23-0.94), persons born in the U.S. (OR 0.36, 95% CI 0.17-0.75), and among those with any cough (OR 0.17 (0.08-0.37)), expectoration (OR 0.25 (0.12-0.50)), fever $>38.5^{\circ}\text{C}$ (OR 0.45 (0.23-0.88)) or an atypical chest xray infiltrate (OR 0.19 (0.09-0.39)). Odds of a positive TB culture were increased among those born in Central America (OR 3.11 (1.34-7.23)), with a history of incarceration (OR 2.72 (1.25-5.95)), a positive Purified Protein Derivative (PPD) (OR 4.27 (1.67-10.84)), and mediastinal lymphadenopathy on chest xray (OR 3.54 (1.40-8.96)). In a multivariate logistic model, a positive PPD (OR 4.55 (1.83-11.29)), mediastinal lymphadenopathy on chest radiograph (OR 3.56 (1.12-11.48)), expectoration (OR 0.29 (0.13-0.65)), atypical infiltrate (OR 0.38 (0.17-0.86)) and foreign-born status (OR 2.16 (0.90-5.17)) were the most significant predictors of TB. A score was developed which summed the risk factors for each patient by adding +1 for each positive predictor and -1 for each negative predictor. Having a score of zero or higher had a sensitivity of 85% for predicting TB.

CONCLUSION: A positive PPD, mediastinal lymphadenopathy, and foreign-born status are positively associated with TB in smear-negative patients while the presence of expectoration or an atypical chest xray infiltrate are negatively associated. The score of zero or more helps identify patients who are more likely to have TB and those who should be treated while awaiting culture results.

COLON CANCER SCREENING AS A WOMEN'S HEALTH ISSUE: FACTORS THAT INFLUENCE PATIENT COMPLIANCE WITH RECOMMENDED SCREENING. CL Karmen, S Warshafsky, S Hussain, W-N Lee, A Sandberg, K Sang, C Carosella, SJ Peterson, Medicine, New York Medical College, Valhalla, NY

PURPOSE: To identify the factors that influence a woman's compliance with recommendations for colonoscopy and sigmoidoscopy.

METHODS: Cross sectional survey. Women over the age of 50 presenting to a general internal medicine faculty practice over a five-month period were interviewed with regard to the dates of their most recent cancer screening tests. Colon cancer screening was defined as a sigmoidoscopy or colonoscopy in the past five years. A logistic regression model was used to assess predictor variables including age, tobacco use, vitamin use, personal or family history of breast and gynecologic cancer, and personal opinions regarding colon cancer and cancer screening tests.

RESULTS: 248 patients were studied. The mean age of the study population was 65 ± 10 yrs. 57% of patients had either a sigmoidoscopy or a colonoscopy. 85% of the women had fecal occult blood testing. Predictor variables found to significantly influence the likelihood of receiving either a sigmoidoscopy or colonoscopy were family history of colon cancer (OR 2.03; 95% CI: 1.00 to 4.10, $p = 0.048$), perception that colon cancer is an important health issue (OR 2.09; 95% CI: 1.10 to

3.96, $p = 0.024$), and age ($p < 0.001$). Other predictor variables including tobacco use, vitamin use, and personal or family history of breast and/or gynecologic cancer were not significant. In addition, breast cancer or cervical cancer screening did not significantly influence the likelihood of having either sigmoidoscopy or colonoscopy. 43% of the women learned about colon cancer from the media and 18% from health care professionals. None of the patients attributed insurance restrictions as a reason for not having the screening tests done. Only 2 patients could identify a female celebrity who had colon cancer.

CONCLUSION: In our study, women who agree to breast and cervical cancer screening are no more likely to have colon cancer screening than those who refuse. Despite an increased incidence of colon cancer in women with a personal history of breast and some gynecological cancers, women with these risk factors are no more likely to have colon cancer screening tests. Women with a family history of colon cancer and women who believe that colon cancer is an important health issue, however, are more likely to agree to colon cancer screening. We suggest that the media and health care professionals further stress the importance of colon cancer screening in women over the age of 50.

CAUSES AND PROGNOSIS OF SEVERE HYPONATREMIA IN HOSPITALIZED PATIENTS. J Kato, Department of Medicine, Okinawa Naha Hospital, Naha, Okinawa, Japan

PURPOSE: Hospitalized patients in various disease states can develop severe hyponatremia during their course of illnesses and some of them may succumb to death during hospitalizations. This study was designed to evaluate the etiology and prognosis of severe hyponatremia and determine the predictive factors for mortality.

METHODS: Adult cases (>19 years old) of severe symptomatic hyponatremia (serum sodium $<116\text{mmol/L}$) hospitalized in a community hospital were collected from medical records from October 1998 to November 1999. Etiology for hyponatremia was determined by documentation of each medical record. In-hospital mortality and their potential predictive factors were evaluated. Stepwise logistic regression analysis was involved for determination of risk factors for short term mortality by analyzing age, underlying disease, clinical severity, and some laboratory data.

RESULTS: There were 67 cases of severe symptomatic hyponatremia (35 men, 32 women) with mean age of 61 (27 to 94). Mean nadir of serum sodium was 104 mmol/L (92 to 115). Major symptoms included vomiting, consciousness disturbances, and seizure. SIADH (16 cases), iatrogenic dilution (15) and congestive heart failure (11) were frequently attributed to causes of hyponatremia. In-hospital mortality was 48% and its significant risk factors were hyponatremia less than 3.0g/dl (odds ratio 2.40 95% confidence interval 1.26-4.56) and the longer duration from the admission date to the day of nadir of serum sodium (odds ratio 1.06 95% CI 1.11-1.79 for each incremental day).

CONCLUSION: Mortality rate of hospitalized patients who were complicated by severe hyponatremia remained high, and its predictive factors included low serum albumin and longer hospitalization period to develop hyponatremia.

DETERMINANTS OF SURGICAL TREATMENT CHOICE FOR WOMEN WITH NON-INVASIVE AND INVASIVE BREAST CANCER. SJ Katz, PM Lantz, JK Zemencuk, Medicine and Health Management and Policy; Consortium for Health Outcomes, Innovation and Cost-Effectiveness Studies (CHOICES), University of Michigan; Center for Practice Management and Outcomes Research, Ann Arbor V.A., Ann Arbor, MI

PURPOSE: The increased use of mammography screening has led to an enormous increase in the incidence of non-invasive breast cancer, ductal carcinoma in-situ (DCIS). Because modified radical mastectomy remains a frequent surgical treatment for DCIS there is widespread concern among some patients, advocacy groups and clinicians that breast conserving surgery is heavily underutilized among women with non-invasive disease. However, there have been no population-based studies to assess determinants of surgical treatment type, the process of decision-making or outcomes such as patient satisfaction for patients with DCIS.

METHODS: We performed a population-based pilot survey of 183 women from the metropolitan Detroit SEER cancer registry catchment area diagnosed with DCIS and invasive breast cancer during a 5 month period in 1998.

RESULTS: The response rate was 71.2% with 80% of women completing the survey within 4 months of diagnosis. Half the women had DCIS. Overall, 51.9% of women received a mastectomy (54.3% of women with DCIS vs 49.4% of women with invasive disease, $p = 0.34$). Attending surgeons appeared to have played a dominant role in decision-making: One third of women did not perceive that they were given a choice between treatment options; an additional 38% of women received a surgeon recommendation (92% of whom received the recommended treatment). Patient attitudinal factors such as concerns about the clinical benefits and risk of surgical treatment options were important determinants of treatment choice and favored receipt of mastectomy. Knowledge about differences in clinical benefits and risks between surgery options was low: less than one third of women knew that survival was the same for the two treatments. Women who did not perceive choice would have "liked to have been more involved" (38% vs 9% of those who perceived choice, $p < 0.1$) and "did not have as much say as I wanted" (35% vs 11% of those who perceived choice, $p < 0.1$). These results were the same across stage of disease.

CONCLUSION: Determinants of surgical treatment type appeared to be the same for women with DCIS and invasive breast cancer, with surgeons playing a dominant role in the decision-making process. Low patient knowledge of differences in

clinical benefits and risks between treatment suggest that there is need to improve communication between surgeons and patients. Lower satisfaction among women who perceived less control suggests that the process of decision-making may be more important than the actual treatment chosen.

IMPACT OF CASE MANAGEMENT FOR HIV-INFECTED PERSONS. *MH Katz, WE Cunningham, JA Fleishman, RM Andersen, T Kellogg, SA Bozzette, MF Shapiro,* Department of Public Health, San Francisco, CA; *UCLA School of Public Health, Los Angeles, CA;* Agency for Healthcare Research and Quality, Bethesda, MD; *Veteran's Affairs San Diego Healthcare System, University of California, San Diego, RAND, San Diego, CA; University of California, Los Angeles, RAND, Santa Monica, CA*

PURPOSE: Although case management has been advocated as a method for improving the care of HIV-infected persons, its effectiveness is poorly understood. We evaluated the impact of case managers on unmet need for supportive services, medical service utilization, and medication utilization among a population based sample of 2,445 HIV-infected adults receiving medical care in the United States (HCSUS study).

METHODS: The HCSUS study employed a multistage design in which geographical areas, medical providers, and patients were chosen. In this analysis, case management and potential confounders (gender, age, ethnicity, HIV risk group, education, region, drug dependence, income, insurance status, housing status, living alone, sum of needs, and CD4 count) were assessed at baseline and outcomes were measured at the first follow-up visit (median duration between interviews was 8 months).

RESULTS: At baseline, 57% of the sample had contact with a case manager in the prior six months. In multiple logistic regression analysis, adjusting for potential confounders, having contact with a case manager at baseline was associated with decreased unmet need for income assistance (OR = 0.45; 95% CI = 0.31–0.65), health insurance (OR = 0.60; 95% CI = 0.36–0.99), home care (OR = 0.30; 95% CI = 0.16–0.58), emotional counseling (OR = 0.63; 95% CI = 0.41–0.96) and any unmet need (OR = 0.46; 95% CI = 0.33–0.66) at follow-up. Contact with case managers was not associated with utilization of ambulatory care (OR = 0.87; 95% CI = 0.65–1.15), emergency care (OR = 1.27; 95% CI = 0.97–1.68), or hospitalization (OR = 1.09; 95% CI = 0.81–1.47). Contact with case managers was associated with higher utilization of two (OR = 1.58; 95% CI = 1.23–2.03) and three drug antiretroviral treatment (OR = 1.34; 95% CI = 1.00–1.78) as well as treatment with protease inhibitors and non-nucleoside reverse transcriptase inhibitors (OR = 1.26; 95% CI = 1.01–1.58) at follow-up. Contact with case managers was not associated with receipt of prophylaxis against *Pneumocystis carinii* pneumonia and toxoplasmosis (OR = 1.15; 95% CI = 0.83–1.60).

CONCLUSION: Providing case management services to persons with HIV infection appears to decrease their unmet needs for supportive services and improve their medical care.

TAKING CARE OF MOTHER: A NATIONAL STUDY OF GENDER DISPARITIES IN RECEIPT OF HOME CARE FOR THE DISABLED ELDERLY. *SJ Katz, MU Kaboto, KM Langa,* Division of General Medicine; Consortium for Health Outcomes, Innovation and Cost-effectiveness Studies (CHOICES), University of Michigan; *VA Center for Practice Management and Outcomes Research, Ann Arbor, MI*

PURPOSE: The demand for home care for the disabled elderly has increased markedly in industrialized nations because age-specific disability has increased, populations are aging and more disabled elderly are living alone. This is of particular concern to women who represent a large proportion of the elderly disabled in these countries and may be vulnerable to unmet need because they live alone or they may be caregivers themselves. Yet, we have little information about gender differences in receipt of informal home care in the United States.

METHODS: To illuminate this issue we used the first wave of a large nationally representative longitudinal survey of the elderly (AHEAD, 1993, N = 7,443) to assess gender differences in receipt of unpaid (informal) home care to the disabled aged 70 and older. We used a two-part regression model to quantify the number of hours per week of informal home care after adjusting for age, level of disability and living arrangements.

RESULTS: Forty two percent reported one or more current ADL or IADL impairments (N = 3109, mean number of impairments = 2.9 out of 11) of whom 65.6% were women. The Table shows the number of informal care hours per week by gender and living arrangement adjusted for age and disability. Disabled women received much less informal home care than their male counterparts because they were much less likely to be married and married women received many fewer hours of care than married men. Indeed, disabled married women received only slightly more care than disabled women living alone; while disabled married men received three times more care than disabled men living alone.

Living Arrangement	Females Percent	Mean Hours (95% CI)	Males Percent	Mean Hours (95% CI)
Unmarried, living alone	45	10.9 (10.1–11.7)	16	8.2 (7.6, 8.8)
Unmarried, living with others	27	19.7 (18.4–21.0)	10	18.7 (17.4, 19.9)
Married	28	13.7 (12.6–14.8)	74	25.8 (24.0, 27.5)
All	100	14.5 (13.5–15.6)	100	24.3 (22.7, 25.8)

CONCLUSION: These results suggest that women may be particularly vulnerable to the consequences of disability even if they live with a spouse. Programs providing home care support to the elderly need to consider these enormous gender disparities when developing targeting strategies in the community.

TARGETING T-PA THERAPY: VALIDATING AN INDEPENDENTLY-DERIVED PREDICTIVE MODEL ON THE GUSTO TRIAL POPULATION. *DM Kent, JL Griffith, RA Hayward, S Vijan, JR Beshansky, RM Califf, HP Selker,* Medicine, New England Medical Center/Tufts University School of Medicine, Boston, MA; *Medicine, University of Michigan, Ann Arbor, MI; Medicine, Duke University Medical Center, Durham, NC*

PURPOSE: Previously, we developed a model, based on the Thrombolytic Predictive Instrument (TPI), to select patients with acute myocardial infarction (AMI) most likely to benefit from tissue-plasminogen activator (t-PA) over streptokinase (SK). Our model suggested that, when patients are stratified, 85% of the lives saved with population-wide use of t-PA could be obtained by treating only 50% of patients. Also, when the incremental risk of intracranial hemorrhage (ICH) was included, our model suggested that patients in the lowest benefit quartile would actually be more likely to be harmed by t-PA relative to SK, so that treating half of patients with t-PA would be essentially "quality-neutral" to treating all, if patients are properly selected. We now attempt to validate this model directly on the population in the GUSTO trial, the landmark study that demonstrated that t-PA was, on average, superior to SK in reducing mortality in AMI.

METHODS: Because the GUSTO database did not contain all the variables required by the TPI, we first derived a "Modified (GUSTO-compatible) TPI" on the original TPI database of 4,911 patients, and ensured that the new model's predictions had a very high correlation to the predictions of the original TPI (pearson correlation = 0.98). "Modified-TPI" predictions were then obtained on 24,146 patients in the GUSTO trial database. Patients were categorized into quartiles of predicted mortality benefit and quartiles of predicted composite (mortality and intracranial hemorrhage [ICH]) benefit.

RESULTS: Mortality and composite outcomes for patients treated with t-PA and SK are shown in the Table below, divided, respectively, into quartiles of expected mortality benefit (mort) and quartiles of expected composite benefit (comp). In terms of lives saved, 85% of the benefit of t-PA seen in the GUSTO trial was captured in the first two quartiles of mortality benefit, exactly as predicted. Also, when the risk of ICH was considered, a trend toward harm was observed in the lowest benefit quartile and, as predicted, treatment of all patients in the GUSTO

Quartile (mort)	I	II	III	IV
SK	14.2	7.5	4.0	1.9
t-PA	12.1	6.5	3.8	1.6
Quartile (comp)	I	II	III	IV
SK	13.9	7.6	4.1	2.7
t-PA	12.1	6.3	3.6	3.0

trial would be essentially no better than limiting treatment to the 50% of patients with the highest predicted benefit.

CONCLUSION: Using multivariate risk/benefit models within research protocols can help uncover patient subgroups likely or unlikely to benefit. If such models are made clinically available, clinicians could better target therapies with significant costs or risks.

RISK STRATIFICATION FOR VASCULAR COMPLICATIONS OF PERCUTANEOUS TRANSLUMINAL CORONARY ANGIOPLASTY (PTCA). *MKB Kesani, C Monsen, D Nayak, SJ Peterson,* NYMC

PURPOSE: To identify risk factors for vascular complications at the site of arterial puncture in patients undergoing PTCA.

METHODS: Retrospective study design. All cases of PTCA from Jan 1999 to Aug 1999. End point is development of psuedoaneurysm requiring operative intervention.

RESULTS: Retrospective chart review of 914 cases of PTCA from Jan 1999 to Aug 1999 showed 38 psuedoaneurysms (2%). Out of 914 total cases 51% were on heparin (466) and 49% (448) were on heparin plus IIb/IIIa inhibitors. Patients from same PTCA group who did not develop psuedoaneurysms were used as controls. Out of 38 cases of psuedoaneurysms 15 were excluded because of insufficient information and delayed onset of complication. Out of remaining 23 cases 19 were on heparin and 4 patients on both heparin and IIb/IIIa.

HEPARIN	HEPERIN+IIb/IIIa		
PSEUDOANEURYSM+	19	4	P < 0.01
PSEUDOANEURYSM(–)	3	20	P < 0.01

CONCLUSION: It appears that heparin is the major contributing factor for vascular complications after PTCA. Addition of IIb/IIIa inhibitors does not seem to increase the risk. In fact it will allow use of lower dose heparin, ultimately decreasing the incidence of vascular complications. The cost of IIb/IIIa inhibitors remains a limiting factor for their routine use.

SYMPTOMS IN PRIMARY CARE: ETIOLOGY AND OUTCOME. *AA Khan, A Khan, K Kroenke,* North Vernon Internal Medicine, The Care Group, Inc.; *Regenstrief Institute; Department of Medicine, Indiana University, Indianapolis, IN*

PURPOSE: While symptoms account for over half of all outpatient visits, or more than 400 million clinic visits in the U.S. alone each year, the etiology and outcome of common symptoms remains poorly understood.

METHODS: In a chart review study of 289 patients from March to December 1997 we recorded a total of 432 symptoms. Patients were randomly selected from 4 different months to avoid seasonal bias. We used explicit criteria to assign symptom etiology (physical, psychiatric, or idiopathic) and a 30% sample of the charts were co-audited to minimize investigator bias. Interrater agreement was high ($\kappa = 0.75$ for etiologic category, and 0.84 for outcome). Symptom type, visit type, demographics, provider relationship (new vs established), treatments, tests, referrals, and final diagnosis were examined. Follow-up notes for one year after the index visit were reviewed to determine chart-documented outcomes.

RESULTS: Patients were 73% women, 59% African-American, with a mean age of 56. Various joint and URI complaints were the most common followed by the following specific symptoms: back pain ($n = 42$), headache (27), dyspnea (26), abdominal pain (19), and chest pain (19). In two-thirds of the cases, symptoms noted in the chart were the principal reason for the visit. Prescribing a new (39%) or continuing an old (32%) medication was the predominant symptom-specific physician action, while ordering tests (19%), referrals (11%), or nonpharmacological treatments (10%) were less common. Using explicit criteria, 53% of the symptoms were classified as physical in etiology, 10% as psychiatric, and 37% as idiopathic. Back pain and headache were the most likely to be idiopathic (71% and 48%). A psychiatric cause was somewhat more common in younger, white patients. Investigators had the highest diagnostic certainty for physical causes and least with idiopathic at both initial and final visits.

At the index visit, nearly half (45%) of the symptoms were chronic, recurrent, or persistent since a prior visit. In follow-up notes, symptom outcome was documented as improved in 19%, the same or worse in 25%, and not mentioned in 56%. Poor outcome (i.e., same/worse) was associated with longer symptom duration at presentation, younger age, and psychiatric etiology. Outcome was unrelated to specific provider actions. Serious diagnoses (e.g., cancer) not suspected at the index visit did not emerge during follow-up.

CONCLUSION: Nearly half of physical symptoms documented in primary care lack a clear-cut physical etiology. While serious physical causes not suspected initially seldom become manifest at follow-up, symptom persistence is an important concern in at least one out of four patients. Better management strategies for such patients is a priority.

OUT-OF-HOSPITAL CARDIAC ARREST IN OCTOGENARIANS AND NONAGENARIANS. C. Kim, L. Becker, MS Eisenberg, Robert Wood Johnson Clinical Scholars Program, University of Washington; Emergency Medical Services Division, King County Department of Health; Department of Medicine, Division of Emergency Medicine, University of Washington, Seattle, WA

PURPOSE: To compare the survival after out-of-hospital cardiac arrest of octogenarians, nonagenarians, and younger patients. To determine the influence of age on survival after adjusting for factors known to influence out-of-hospital cardiac arrest outcome.

METHODS: Design: Retrospective cohort study.

Setting: Suburban King County, Washington.

Patients: 5882 patients with out-of-hospital cardiac arrest from presumed cardiovascular disease between January 1, 1987 through December 31, 1998, who received cardiopulmonary resuscitation by bystanders and/or emergency medical technicians and advanced cardiac resuscitation by paramedics.

Measurements: Rates of hospital admission and survival to hospital discharge were recorded for octogenarians, nonagenarians, and younger patients. Factors thought to influence outcome such as initial cardiac rhythm, witnessed arrest, time from collapse to emergency medical service care, bystander cardiopulmonary resuscitation (CPR), location of arrest, gender, as well as age, were compared by χ^2 for discrete variables and t-tests or F-tests for continuous variables. Multiple logistic regression identified independent correlates of hospital admission and survival to hospital discharge.

RESULTS: In patients with out-of-hospital cardiac arrest due to a cardiac etiology, younger patients had higher discharge rates than octogenarians, who in turn had higher discharge rates than nonagenarians (19% vs. 9.4% vs. 4.4%, $p < 0.0001$). However, survival to hospital discharge improved significantly for younger patients, octogenarians, and nonagenarians, who had ventricular fibrillation or pulseless ventricular tachycardia (VF/VT) (35% vs. 24% vs. 17%, $p < 0.0001$). After multiple logistic regression controlling for other factors, increased age was weakly associated with decreased survival to hospital discharge (OR 0.99, 95% CI [0.98, 1.00]).

CONCLUSION: Octogenarians and nonagenarians have lower survival to hospital discharge than younger patients, but age is a much weaker predictor of survival than other factors such as initial cardiac rhythm. Decisions regarding resuscitation should not be based on age alone.

OUT-OF-HOSPITAL CARDIAC ARREST IN MEN AND WOMEN. C. Kim, CE Fahrenbruch, LA Cobb, MK Copass, MS Eisenberg, Robert Wood Johnson Clinical Scholars Program; Division of Cardiology, Department of Medicine; Department of Neurology; Division of Emergency Medicine, Department of Medicine, University of Washington, Seattle, WA

PURPOSE: To determine the influence of gender on survival to hospital admission and hospital discharge after out-of-hospital cardiac arrest. To determine if men

and women have similar predictors of survival after out-of-hospital cardiac arrest, particularly initial cardiac rhythm, witnessed arrest, bystander cardiopulmonary resuscitation (CPR), and public location.

METHODS: We conducted retrospective analyses of 4,334 patients in suburban King County, Washington and 3656 patients in Seattle, Washington who experienced cardiac arrest with a cardiac etiology between 1990 and 1999. By gender, we compared favorable predictors of survival such as rates of ventricular fibrillation/pulseless ventricular tachycardia (VF/VT) as opposed to bradycardia or asystole, witnessed arrest, bystander CPR, public location, and response times, and also survival to hospital discharge and hospital admission. We performed multivariate logistic regression to determine if gender independently influenced survival apart from other predictors.

RESULTS: In both datasets, women were significantly older than men, had less frequent rates of witnessed arrest, VF/VT, and public location of arrest; women had poorer unadjusted survival to hospital discharge than men (18% in men and 13% in women, $p < 0.0001$ in suburban King County, and 15% in men and 9% in women, $p < 0.0001$ in Seattle). Multivariate logistic regression analysis showed that gender did not independently influence survival to hospital discharge in either dataset (OR 0.69 [0.37, 1.31] in suburban King County and OR 1.01 [0.78, 1.31] in Seattle). Female gender seemed to predict hospital admission in Seattle (OR 5.48 [2.17, 13.8]), particularly for younger patients, but not in suburban King County (1.31 [0.44, 3.89]); in suburban King County, the interaction between gender and age was inconsistent.

CONCLUSION: Gender does not independently predict survival to hospital discharge, and female gender is associated with poorer predictors of survival such as older age. Gender does not seem to consistently predict hospital admission apart from local emergency medical system factors.

THE RELATIONSHIP BETWEEN SUBCLINICAL CORONARY ATHEROSCLEROSIS AND CORONARY HEART DISEASE RISK. B. Kinosian, R. Cato, H. Glick, M. Wolfe, D. Rader, Medicine, University of Pennsylvania, Philadelphia, PA

PURPOSE: To define the relationship of a well-calibrated coronary heart disease (CHD) risk rule to subclinical coronary atherosclerosis (SCA) measured as coronary artery calcification (CAC) using electron beam CT (EBCT).

METHODS: Clinical population of 1324 without known CHD or lipid lowering medication (57% male, mean age 53 ± 7.3) scanned by EBCT, with concurrent measurement of other CHD risk factors. The current, discriminating (ROC areas for CHD events of .74-.77) and well-calibrated Framingham Heart Study risk rule (termed Framingham Risk Score [FRS]) was calculated for each participant using both the LDL and total cholesterol (TC) scoring tables. FRSs were categorized separately for men and women so that high risk (HR) equaled an equivalent annual CHD risk ($>1\%$ /yr). Two-stage (probit/OLS) regression was used to predict SCA (any CAC) and extent of clinically significant SCA ($CAC > 80$). ROC areas (reported as the c-statistic from logistic regression) were used to determine the ability of EBCT to discriminate HR determined by calibrated FRS, and the ability of FRS to identify clinically significant SCA.

RESULTS: Similar groups were classified at HR by FRS (22%) and EBCT (19.25%). Among men, 10% of those at low risk (LR; $<5\%$ /yr, $n = 240$) by FRS had clinically significant SCA by EBCT; 29.7% of HR men ($n = 202$) had no SCA as did 15% of HR men aged >60 ($n = 64$). Among FRS HR women ($n = 92$), 47% were without SCA, as were 20% aged >65 ($n = 20$). Over 15% of both men and women with CHD risk $>2\%$ /yr ($n = 60$) were without SCA. Using probit regression, LDL, HDL, age, gender, diabetes, and hypertension predicted the presence of SCA, smoking did not (pseudo $R^2 = .13$). Using OLS for those with any SCA, age, gender, diabetes, smoking and hypertension predicted the extent of SCA; lipids did not ($r^2 = .145$). To identify CHD risk $>10\%$ /10-yr, EBCT had moderate discrimination (ROC areas = .70 men, .66 women). As a measure of SCA, FRS had moderate discrimination (ROC = .68 [LDL]-.71 [TC]), but poorer discrimination among patients aged >60 yrs (ROC areas = .57 [LDL]-.59 [TC]).

CONCLUSION: FRS can discriminate among people with and without significant SCA. Serum lipids are significantly related to the presence of SCA. The traditional CHD risk factors predict SCA, but only modestly predict CAC quantitatively. A significant minority of FRS low-risk men have clinically significant SCA, which is not identified by the FRS risk factors. A substantial proportion of older FRS high risk patients have no detectable SCA (by EBCT). We confirm the generally close relationship between CHD risk and subclinical coronary atherosclerosis, but identify significant subpopulations where a clinical risk rule alone may be inadequate.

EFFECTIVENESS OF EXERCISE DEMONSTRATION FOR ARTHRITIS PATIENTS. KM Kolbusz, IA Elcier, J. Kaur, EB Mak, D. Collins, Edward Hines Jr VA Hospital, Hines; LUMC, Maywood, IL

PURPOSE: Most arthritis patients are unaware of the benefits of exercise, and chronic arthritis is commonly known to cause pain and limit daily activity. Exercise demonstrations and interactive participation will heighten patient awareness of the benefits of participating in an exercise program. This session will provide a practical way for the patient to learn methods to alleviate pain and discomfort caused by arthritis. Hence, improving their quality of life through life-style modifications.

METHODS: In a period of two years (1997-1999), 94 male veteran patients with a variety of arthritis conditions (46 osteoarthritis, 23 rheumatoid arthritis, 7 gout, 4 spondylolithesis, 2 hemoarthrosis, 3 osteopenia, 9 low back pain) were asked to voluntarily attend a 20-minute interactive exercise demonstration conducted by a

licensed physical therapist in a clinic waiting area. Subjects, age range 40–80 years, were recruited from the arthritis clinic while waiting to be seen by a physician. The importance of movement and flexibility were explained to the participants. The exercise sessions consisted of thera-band resistive strengthening exercises, range of motion exercises combined with Tai Chi movements, other general conditioning and energy conservation techniques. After each demonstration, the subjects filled out a questionnaire regarding the effectiveness of the demonstration. One-week later phone surveys were conducted to assess the frequency of use and benefits of the demonstration.

RESULTS: The results are as follows:

% Answered (N = 94)	Survey Responses
97.9	I have learned a lot of new information from the presentation
98.6	The information will be useful in performing my daily activities
94.5	I am definitely going to use the exercises I have learned
95.9	I would recommend this presentation to a friend with arthritis
% Answered (N = 55)	
Phone Survey Statements (only 55 were available for interview)	
78.2	Still using the exercises at home one week later
69.1	Reported feeling better as a result of exercising
52.7	Reported sharing information from exercise session with a friend
58.2	Would return for another exercise session

CONCLUSION: These interactive sessions demonstrated a convenient and practical form of exercise to the subjects. This could boost subjects' self-confidence and knowledge to exercise on their own. Telephone surveys indicated that there is dissemination of knowledge about exercise and arthritis among the participants and their peers. This interactive, user friendly, economical exercise technique is an effective and worthwhile approach for educating individuals about arthritis and pain management.

A NEW MEASURE OF DEPRESSION SEVERITY: THE PHQ-9. *K Kroenke*, RL Spitzer, JB Williams, Regenstrief Institute, Indianapolis, IN; Columbia University, New York, NY

PURPOSE: While considerable attention has focused on diagnosing depression in primary care, it increasingly has been noted that monitoring treatment response and adjusting therapy is equally important. Therefore, we have developed and validated a simple depression outcome measure in two large prospective studies.

METHODS: The Patient Health Questionnaire is a recently validated, self-administered version of the PRIME-MD diagnostic instrument for mental disorders in primary care. Its depression measure includes the nine DSM-IV criteria for mood disorders, each item scored from 0 ("not at all") to 3 ("nearly every day"). The PHQ was administered to 6000 patients in 7 general internal medicine and family practice clinics (n = 3000), and 7 obstetrics-gynecology clinics (n = 3000). Internal reliability was good in both samples (Cronbach's alpha = .90 and .86). Functional status was assessed with the SF-20, and sick days and clinic visits during the preceding 3 months by patient self-report. A sample of 580 patients had a telephone-administered structured psychiatric interview by a mental health professional who was blinded to the results of the PHQ.

RESULTS: Data for the 3000 general internal medicine/family practice patients is shown in the table. As PHQ-9 depression severity increased, there was a substantial, stepwise decrement in functional status for all six SF-20 subscales. Also, sick days and health care utilization increased. Using the MHP reinterview as the criterion standard, a PHQ-9 score of 10 or greater had a sensitivity of 88% and specificity of 88% for major depression. Analysis by type of depressive disorder—major (10% of sample), other (6%), and none (84%)—revealed that PHQ-9 thresholds of 5, 10, 15, and 20 represent minor, moderate, major, and severe depression cutpoints. Findings were essentially the same for the Ob-Gyn sample, thus providing external validity.

PHQ-9 score	SF-20 scale scores Mental health	Social function	Role function	Overall health	Past 3 months Sick days	Clinic visits
0–4	81	92	86	70	2.4	1.0
5–9	65	77	63	50	6.7	1.8
10–14	51	65	53	40	11.4	2.0
15–19	43	55	42	33	16.6	2.4
20–27	29	40	27	27	28.1	3.7

CONCLUSION: A validated diagnostic tool, the PHQ-9 also has considerable promise as an outcome measure in the treatment of depression. It is currently being used in several large treatment trials which will be important to demonstrate its sensitivity to change.

SYMPTOMS AT THE END OF LIFE: VARIATION BY DIAGNOSIS. *JS Kutner*, C Kassner, D Nowels, DGIIM: Family Medicine, UCHSC, Denver, CO

PURPOSE: Most previously-published data about symptoms at the end of life focus exclusively on end-stage cancer. The purpose of this study was to describe

symptom variation by diagnosis among hospice patients, from the perspective of hospice staff.

METHODS: Cross-sectional study conducted in the Population-based Palliative Care Research Network (PoPCRN), a community-based hospice research network. Hospice staff collected data during interdisciplinary team meetings, including demographic information (age, sex, date of admission, diagnoses, treatment setting, date of data collection) and a modification of the Memorial Symptom Assessment Scale. Chi squares determined associations between diagnoses and presence of severe (severe or very severe) and frequent (frequently or almost constantly present) symptoms.

RESULTS: Sixteen hospices returned data on 348 patients. The patients, of whom 55% were female, had a median age of 78 years. Most were receiving care at home (57.8%) or in a nursing home (24.7%). The most common diagnoses were cancer (55.2%), cardiac disease (13.8%), and neurologic disorders (12.1%). Cancer patients differed from non-cancer patients in presence of severe and frequent symptoms (see table).

	Cancer (%)	Non-cancer (%)	p value
Severe difficulty concentrating	23.8	42.6	0.005
Severe nervousness	8.2	27.4	0.003
Severe shortness of breath	17.2	37.3	0.004
Severe constipation	13.5	27.7	0.042
Frequent nervousness	16.7	32.8	0.023
Frequent shortness of breath	42.1	59.5	0.030
Frequent difficulty swallowing	40.8	60.4	0.048

Patients with non-cancer diagnoses had symptom constellations consistent with the biologic bases of the underlying diseases. For example, demented patients were more likely to have frequent (92.0% vs. 53.3%, p = 0.001) and severe (75.0% vs. 26.9%, p = 0.001) difficulty concentrating and to have frequent (76.5% vs. 50%, p = 0.036) and severe (50.5% vs. 19.8%, p = 0.010) drowsiness. Patients with global geriatric decline were more likely to have severe agitation (42.9% vs. 15.7%, p = 0.022). Patients with COPD were more likely to have frequent (81.8% vs. 42.5%, p = 0.001) and severe (57.6% vs. 18.6%, p = 0.001) shortness of breath.

CONCLUSION: These data demonstrate variation by diagnosis in symptom frequency and severity among hospice patients. Cancer patients are not representative of all hospice patients. Thus, those who care for terminally ill individuals must be facile in assessment and management of a wide range of symptoms, and must be aware of potential disease-specific symptom constellations.

PHARMACOLOGIC STRESS TESTING FOR CORONARY ARTERY DISEASE DIAGNOSIS. *YS Kwok*, C Kim, P Heagerty, RF Redberg, Division of General Internal Medicine; Robert Wood Johnson Clinical Scholars Program; Department of Biostatistics, University of Washington, Seattle, WA; Division of Cardiology, Department of Medicine, University of California, San Francisco, CA

PURPOSE: To obtain estimates of the sensitivity and specificity of adenosine, dipyridamole, and dobutamine when combined with echocardiography (echo), or single photon emission computed tomography (SPECT) for the diagnosis of coronary artery disease (CAD).

METHODS: Data Sources: Articles published in MEDLINE (1975–1999) regarding pharmacologic stress tests and their listed references were retrieved using the search strategy "coronary disease/diagnosis AND (pharmacologic stress);" experts were consulted to further identify studies.

Study Selection: Studies were included if all subjects underwent at least one echo or SPECT pharmacologic stress test with adenosine, dipyridamole, or dobutamine for diagnosis of CAD; all subjects underwent coronary angiography; and data was presented in a manner that allowed the calculation of sensitivity and specificity of the tests.

Data Extraction: Two independent reviewers abstracted population characteristics, technical factors, methodological factors, and results. Disagreements were resolved by conference.

RESULTS: Ninety-seven studies met inclusion criteria. Adenosine echo and dipyridamole echo had similar sensitivities, 72% (95% confidence interval [CI], 68%–77%) and 71% (95% CI, 68%–73%); and specificities, 92% (95% CI, 86%–95%) and 93% (95% CI, 90%–95%). Likewise, adenosine SPECT and dipyridamole SPECT studies had similar sensitivities, 90% (95% CI, 88%–92%) and 87% (95% CI, 85%–89%); and specificities, 76% (95% CI, 71%–82%) and 70% (95% CI, 66%–73%). The dobutamine echo and SPECT studies had similar sensitivities, 80% (95% CI, 79%–82%) and 82% (95% CI, 79%–85%); but dobutamine echo had a higher specificity, 83% (95% CI, 81%–85%), than dobutamine SPECT, 74% (95% CI, 70%–79%).

CONCLUSION: Adenosine and dipyridamole have similar sensitivities and specificities as pharmacologic stressors for the diagnosis of CAD. The use of SPECT with adenosine or dipyridamole offers higher sensitivity but lower specificity than the use of echo. In dobutamine studies, SPECT and echo offer similar sensitivities, but the specificity of echo is higher.

FACTORS PREDICTING FUTURE ABSTINENCE IN AN OUTPATIENT STOP SMOKING PROGRAM. *D Lakich*, TD MacKenzie, Community Health Services, Denver Health Authority, Denver, CO; General Internal Medicine, University of Colorado Health Sciences Center, Denver, CO

PURPOSE: To determine the long term smoking cessation rate and the predictors of cessation after outpatient group counseling in an urban public health care organization.

METHODS: Smokers ($n = 156$) who attended the first of a series of four free outpatient stop smoking group classes at an urban public hospital or at either of two associated community health clinics between August 1996 and July 1997 were surveyed using a written questionnaire. A follow-up survey by phone was attempted for all individuals (64% response rate, average length of follow-up = 27 months). Results from these surveys along with class attendance and information from a pharmacy database were used to determine the self-reported abstinence rate and to identify factors predictive of abstinence.

RESULTS: Features of the initial cohort include 62% women, 40% disabled, 50% minority, average age 48 years, and an average habit of 25 cigarettes per day. Participants attended an average of 3.4 of the 4 classes. 77% filled a prescription for transdermal nicotine at an agency pharmacy. However, only 22% returned for all the patch refills. Of the 101 patients who were contacted, 17 reported no current smoking. Factors associated with self reported abstinence at 2 years using multiple logistic regression include high self efficacy to quit at 6 months (OR 10.7, $p = .05$) and age greater than 65 years (OR 8.2, $p = .06$). None of the following factors correlated with abstinence: stage of change, nicotine dependence, gender, race, use of nicotine patches, class location, cigarettes per day, employment status, marital status, or insurance status.

CONCLUSION: The long-term self-reported abstinence rate among our low-income urban participants in group counseling compares favorably with results from the published literature. Similar to our inpatient follow-up program, a simple 10-point linear scale of self-confidence to quit at 6 months was highly predictive of abstinence.

THE QUANTITY AND COST OF INFORMAL CAREGIVING FOR THE ELDERLY WITH DEMENTIA: ESTIMATES FROM A NATIONALLY REPRESENTATIVE SAMPLE. KM Langa, MU Kabeto, A Herzog, ME Chernew, M Ofstedal, RJ Willis, RB Wallace, L Mucha, W Straus, A Fendrick, University of Michigan, MI; University of Iowa, IA; Merck and Company

PURPOSE: As the US population ages, the increasing prevalence of dementia will create a substantial societal burden. Prior studies suggest that the cost associated with family caregiving—not direct medical costs—accounts for most of the total cost of dementia. However, existing estimates of caregiving costs derived from small, non-representative samples are likely biased. Accordingly, our objective was to obtain better estimates of the cost of family caregiving by using a nationally representative sample.

METHODS: We used data from the 1993 Asset and Health Dynamics Study (AHEAD), a national longitudinal survey of people aged 70 or older ($N = 7,443$). Using the validated AHEAD cognitive status measures, all respondents were classified as having: Normal Cognition; Mild Dementia; Moderate Dementia; or Severe Dementia. A two-part regression model was used to estimate the additional weekly hours of family caregiving attributable to dementia, after controlling for sociodemographics (age, gender, race, net worth), health status (chronic health conditions), and level of social support (unmarried living alone, unmarried living with others, married).

RESULTS: About 10% of respondents showed evidence of cognitive impairment consistent with dementia. There was a strong positive association between dementia severity and weekly hours of family caregiving ($p < .01$) (Table).

Cognitive Status	Hours/Week	\$/Year
Normal Cognition	4.6	\$2,000
Mild Dementia	13.4	\$5,800
Moderate Dementia	19.2	\$8,300
Severe Dementia	40.9	\$17,700

Using the average hourly wage of a home health aide in 1997 (\$8.30) as the value of a family caregiver's time, the expected yearly caregiving cost per case ranged from \$5,800 for Mild Dementia to \$17,700 for Severe Dementia. This represents a total yearly cost in the US of about \$17 billion for dementia-related informal caregiving.

CONCLUSION: Family caregiving time and costs for those with dementia are substantial and increase sharply with the severity of cognitive impairment. Accurate estimates of caregiving costs are, therefore, essential to the validity of future evaluations of interventions that decrease the incidence or slow the progression of dementia.

SMOKING AND MENTAL ILLNESS: A POPULATION-BASED PREVALENCE STUDY. KE Lasser, JW Boyd, SJ Woolhandler, DU Himmelstein, D McCormick, DH Bor, Medicine; Psychiatry, The Cambridge Hospital/Harvard Medical School, Cambridge, MA

PURPOSE: To assess smoking rates in adults with mental illness, a group that may be particularly vulnerable to tobacco marketing efforts.

METHODS: We analyzed data from the National Comorbidity Survey, a Congressionally-mandated study of the prevalence of psychiatric disorders in the United States. Half of respondents were surveyed regarding tobacco use. We analyzed rates of current smoking (within the past month) and lifetime smoking (daily smoking for at least one month at any time in the past) in persons with and without mental illness in their lifetime, persons with mental illness in the past month, and according to individual DSM III-R diagnosis. We used chi-square tests to

compare smoking rates in those with and without mental illness, and multivariate logistic regression to control for age, sex, and region of the country.

RESULTS: 4411 respondents answered questions about both mental illness and smoking. The overall prevalence of current smoking was 28.5%, and the lifetime prevalence was 47.1%. Respondents with mental illness had consistently higher rates of smoking. Only 22.5% of those without mental illness reported smoking fairly regularly in the past month, vs. 34.8% of those with any history of mental illness, and 41.0% of persons with recent (past month) mental illness ($p < .0001$ for all comparisons). Conversely, 40.6% of all U.S. smokers have experienced mental illness within the past month. Lifetime prevalence of smoking was 39.1% in respondents without a history of mental illness, 55.3% in those with a lifetime history of mental illness, and 59.0% in those with mental illness in the past month ($p < .0001$). Smoking rates among people with a lifetime history of more severe DSM III-R diagnoses were even higher; after excluding persons with phobias, 36.2% were current smokers and 57.7% had smoked in the past. For every DSM III-R diagnosis, the lifetime prevalence of smoking was greater than the current prevalence of smoking, consistent with a quit rate of about 1/3. In logistic regression analyses, the adjusted odds ratios for current and lifetime smoking in respondents with a mental illness in the past month compared to respondents without mental illness were 2.6 and 2.8, respectively. All comparisons were statistically significant at $p < .0001$.

CONCLUSION: Smoking rates are consistently higher in persons with psychiatric illness, especially among those with more severe illness. Therapeutic nihilism about smoking cessation interventions in this population is not appropriate, given the substantial quit rates. Physicians should support public policies such as higher tobacco taxes and restrictions on industry marketing to protect this and other vulnerable groups.

MEDICAL COMPLICATIONS AFTER HIP FRACTURE REPAIR. VA Lawrence, SG Hilsenbeck, H Noveck, RM Poses, JL Carson, Medicine, Univ of TX Health Science Ctr and VA Med Ctr at San Antonio, San Antonio, TX; Medicine, Brown University, Providence, RI; Medicine, University Medicine and Dentistry, RWJ Medical School, New Brunswick, NJ

PURPOSE: The epidemiology of postoperative (postop) medical complications (COMP) has typically been reported by individual, unique COMP, with little focus on patients (pts) and their overall outcomes.

METHODS: We performed a retrospective cohort study of pts ≥ 60 years old, with no acute preoperative medical COMP, undergoing hip fracture repair at 20 hospitals in 1982–1993. We reviewed medical records for COMP (using explicit criteria) until death, discharge, or 30-days, whichever was first. The IndexCOMP was the first complication plus any others within the next 24 hours.

RESULTS: Of 8930 pts, postop COMP occurred in 1737 (19%); 1531 (88%) had one, 182 (10%) had two, and 24 (1%) had ≥ 3 IndexCOMP. Cardiac (720 pts, 41%, 30 day mortality [30DM] = 78 pts, 11%, 1 year mortality [1YM] = 221 pts, 31%) and pulmonary (344 pts, 20%, 30DM = 50pts, 14%, 1 YM = 141 pts, 41%) were the most frequent IndexCOMP and were associated with high mortality. GI bleeding, combined cardiac and pulmonary (card-pul) complications, deep venous thrombosis or pulmonary embolism, transient ischemic attack or cerebrovascular accident were less frequent IndexCOMP. Most cardiac IndexCOMP were classified as nonserious, and more pts had serious pulmonary (13%) than serious cardiac (10%) IndexCOMP.

Pts with combined cardpul IndexCOMP had significantly higher 30-day and one-year mortality than pts with either IndexCOMP alone ($p < 0.001$). Highest mortality rates were for pts with combined card-pul IndexCOMP (30DM = 33%, 1 YM = 58%), renal failure (30DM = 38%, 1YM = 62%), and multiple IndexCOMP (30DM = 31%, 1YM = 48%). IndexCOMP and subsequent death occurred significantly earlier for cardiac compared to pulmonary COMP (1 vs 4 days, 2 vs 8 days, $p < 0.0001$ for both).

CONCLUSION: Cardiac and pulmonary remain the most frequent postop COMP. Further: 1) serious pulmonary are as or more common than serious cardiac COMP; both are associated with high short and long term mortality; 2) pts with multiple COMP have the highest mortality; and 3) differences in length of stay are in part due to earlier cardiac COMP and associated death compared to pulmonary COMP and death.

A SYSTEMATIC REVIEW OF THE ASSOCIATION BETWEEN WHITE COAT HYPERTENSION AND HYPERTENSIVE END-ORGAN DAMAGE. FV Lefevre, D Mark, N Aronson, General Internal Medicine, Northwestern University Medical School; Technology Evaluation Center, Blue Cross/Blue Shield Association, Chicago, IL

PURPOSE: The decision to treat patients with white coat hypertension (WC HTN) is controversial. The degree to which WC HTN is associated with end-organ damage may be a crucial factor in this decision; this projects seek to further define this relationship.

METHODS: Literature search from 1966-present was performed to identify all studies of patients with WC HTN, defined by an elevated office blood pressure (BP) and a 'normal' BP on 24 hour ambulatory BP monitoring. Inclusion criteria for this review were: study identified and compared three groups of patients: normotensive, WC HTN and true hypertensive individuals, and included at least one outcome measure of hypertensive end-organ damage (cardiac, renal or ophthalmologic). Quantitative synthesis of outcome measures was performed by the Confidence Profile Method.

RESULTS: A total of 16 studies met the inclusion criteria. The most common measure of cardiovascular risk in these studies was left ventricular mass (LVM) ($n = 14$). The results for LVM were highly variable, however, in all cases measured LVM for patients with WC HTN was intermediate between normotensive patients and true hypertensive patients. Combined analysis estimated that there was an approximately 10% increase in LVM for patients with WC HTN, as compared to normotensive patients. While it is difficult to estimate the clinical significance of this difference, it may be roughly equivalent to the risk of a 10mm increase in SBP. For other endpoints (diastolic function, proteinuria, retinopathy), similar results were obtained from a smaller number of studies. The main sources of variability in these data are likely to arise from the definition of WC HTN, which differed considerably among studies, and in the severity of hypertension present in the true HTN group.

CONCLUSION: Patients with WC HTN have a degree of hypertensive end-organ damage that differs from normotensive patients. Although the ultimate impact on cardiovascular morbidity and mortality remains ill-defined, this analysis does not support the strategy of withholding antihypertensive medications in this group of patients.

THE LAST SIX MONTHS OF LIFE WITH CONGESTIVE HEART FAILURE. JW Levenson, EP McCarthy, J Lynn, RB Davis, RS Phillips, Division of General Medicine and Primary Care, Beth Israel Deaconess Medical Center, Boston, MA; Center to Improve Care of the Dying, George Washington University, Washington, DC

PURPOSE: To characterize the experiences of patients with congestive heart failure (CHF) during their last 6 months of life.

METHODS: Retrospective analysis of data from 539 patients with CHF who died within 1 year of enrollment in the Study to Understand Prognoses and Preferences for Outcomes and Risks of Treatments (SUPPORT), a prospective cohort study. Patients or their surrogates were interviewed and charts abstracted at several time points in SUPPORT. We constructed 4 observational windows backward in time, beginning with patients' dates of death and ending with their date of enrollment in SUPPORT or 6 months prior to death, whichever came first. For each outcome, patients contributed information to all windows for which they had data collected. We present frequency distributions for each outcome over time and report tests for trend. Outcomes included: estimates of six-month survival; functional status; occurrence of severe physical and emotional symptoms; patients' preferences for care; and the financial impact of patients' illnesses on their families.

RESULTS: As death approached, patients' prognosis became poorer and illness more severe. Median Acute Physiology Scores for hospitalized patients rose from 33 in the interval 6 months to 3 months prior to death, to 44 within 3 days of death. However, the median model-based estimate of six-month survival was 54% even within 3 days of death. Functional impairment, measures of depression and the percent of patients reporting severe pain or dyspnea increased as death approached. Perceived quality of life did not change appreciably, with 29% to 58% of patients reporting good to excellent quality of life in all intervals prior to death. As death approached, the percent of patients preferring not to be resuscitated (DNR) rose from 33% 6 months to 3 months prior to death, to 47% 1 month to 3 days before death ($p < .05$), and the frequency with which DNR orders were written for hospitalized patients also increased. The patients' illnesses had marked financial impact on their families, with 23% of patients' families reporting the loss of most or all of family savings at the time of the patient's death.

CONCLUSION: During the last six months of life in CHF, illness becomes more severe, disability and severe symptoms more prevalent, and patient preference not to be resuscitated more common. However, there is no significant decrement in quality of life as death approaches. The unpredictable course of CHF is reflected in the finding that, during their last 30 days of life, many patients retain good median model-based six-month prognoses and enjoy good to excellent quality of life.

PATIENTS' AND SURROGATES' EXPECTED BENEFITS FROM GASTRIC FEEDING TUBES. CL Lewis, LC Hanson, N Phifer, G Hunter, J Darter, TS Carey, Sheps Center for Health Services Research, University of North Carolina at Chapel Hill, NC

PURPOSE: The advent of non-surgical means for placement of gastric feeding tubes resulted in a significant reduction in the risk of this procedure and a steady increase in utilization; however, mortality for patients receiving these feeding tubes remains high. The purpose of this study is to determine the expected benefits of gastric feeding tube placement from the patient and surrogate perspectives.

METHODS: We interviewed patients and/or surrogates who had an initial gastric feeding tube placed to determine their perceptions of the benefits of feeding tube placement and of the decision-making process surrounding the placement. Patients older than 21 were identified by daily procedural logs of interventional radiology and the gastroenterology suites at a major teaching hospital and a community hospital. Patients were interviewed by telephone or in person within a month of the feeding tube placement. Surrogates were interviewed if patients were unable to be interviewed because of cognitive impairment or intubation. Patients with diagnoses of trauma or non-head and neck malignancy were excluded.

RESULTS: To date 159 patients were eligible and 24 patients and 68 surrogates have been interviewed (57%). The mean age was 66 years, 66% were white, and 49% were women. Major diagnoses were: 28% stroke, 19% neurodegenerative disorders, 16% head and neck cancer, and 37% other serious medical illness. Most patients/surrogates thought that the gastric feeding tube would improve their nutrition (99%), and make them more comfortable (72%). A majority also thought that it would allow them not to feel hungry or thirsty (68%) and to have less pain

(60%). Many thought the gastrostomy would provide greater independence (54%), and almost all thought it would prolong life (99%). Although most reported that it would improve quality of life (91%), there were no differences when respondents rated the quality of life on a 1–10 scale before the gastric feeding tube was placed (4.9) and after it was placed (5.0). Respondents felt that they had gotten adequate information to make the decision whether to get the feeding tube (98%) and had no regrets about the decision (98%).

CONCLUSION: Respondents reported that they had been adequately informed about the feeding tube. Patients and surrogates had similarly high expectations regarding the benefits to be derived from gastric feeding tubes, many of which may not be realized. Follow up interviews are in progress to determine if these expectations change with time and experience with the gastric feeding tube.

ANTIBIOTIC USE IN SMOKERS WITH ACUTE BRONCHITIS: A SYSTEMATIC REVIEW OF THE LITERATURE. JA Linder, I Sim, Division of General Internal Medicine, University of California, San Francisco

PURPOSE: Ninety percent of community physicians in the United States report prescribing an antibiotic to smokers with acute bronchitis. We performed a systematic review of the literature to determine the efficacy of antibiotic use in smokers with acute bronchitis without underlying chronic lung disease.

METHODS: We searched MEDLINE using the subject headings "bronchitis," "cough," and "antibiotics" to identify English language trials, reviews, letters, and editorials published between January 1966 and October 1999. Titles and abstracts were screened for suitability and those deemed appropriate were retrieved and reviewed. Reference lists of articles were used to identify additional appropriate sources.

RESULTS: One hundred and six articles were initially retrieved and reviewed, yielding 10 randomized placebo-controlled trials and 19 randomized active-controlled trials. None of the studies specifically addressed antibiotic use in smokers with acute bronchitis. The 10 randomized, placebo-controlled trials tested various antibiotics—demethylchlortetracycline (1 trial), trimethoprim/sulfamethoxazole (1 trial), doxycycline (4 trials), and erythromycin (4 trials)—in both smokers and non-smokers with acute bronchitis. Six of the ten trials, representing over 310 smokers, reported no benefit to antibiotics versus placebo in smokers. In one of these six trials, a benefit was found for non-smokers. Three trials, representing 160 smokers, reported decreased duration of cough, feeling of illness, congestion, or days off work for antibiotic-treated patients. One trial gave no information about smoking status. Meta-analysis of these results was not possible due to (1) data not uniformly reported or available by smoking status, and (2) different outcomes among trials, including duration of cough, sputum production, fever, days off work, general feeling of well-being, and resumption of normal activities. Of the 19 trials comparing two or more antibiotics only three reported baseline information about smoking habits, and none reported outcomes stratified by smoking status.

CONCLUSION: Although antibiotic prescription for smokers with acute bronchitis is common, this practice has not been rigorously evaluated and the evidence from placebo-controlled trials suggests it may be ineffective.

PATIENT DESIRE FOR DECISION MAKING IN COLORECTAL CANCER SCREENING. BS Ling, PC Schroy, MA Moskowitz, Department of Medicine, University of Pittsburgh, Pittsburgh, PA; Department of Medicine, Boston University, Boston, MA

PURPOSE: Five acceptable screening alternatives for colorectal cancer have been endorsed recently by an expert panel established by the Agency for Healthcare Research and Quality. This panel recommended that patients participate with their physician in selecting one of these to be performed. The purpose of this study is to determine patients' desire to participate in this decision making process.

METHODS: The study was performed at a hospital-based general medicine practice in a cross-sectional survey design. Inclusion criterion was age 40 to 75. Using a previously validated instrument (Autonomy Preference Index-API) which measures the extent patients prefer to participate in decision making for general health care issues, a modified version specific for colorectal cancer screening was developed (modified API). The modified API consisted of five items with a five-point Likert scale used for responses. A total score was tabulated for each participant and then linearly adjusted from 0 to 100 with larger numbers representing a higher desire for patient participation. Psychometric testing showed a Cronbach's alpha coefficient 0.72. Student's t-test and generalized linear regression models were used for the analysis of the data.

RESULTS: The survey was administered to 199 patients (70% response rate). The mean score for patient desire to participate in the decision making for colorectal cancer screening was 42.9 $\pm$ 18.6 (0 = desire to have the physician decide alone, 50 = desire to have the patient and physician decide together equally, and 100 = desire to have the patient decide alone). Females (mean 46.9 $\pm$ 17.3) prefer to share in the decision making process more than males (mean 37.3 $\pm$ 18.8) ($p < .001$). Patients less than age 60 (mean 45 $\pm$ 18.2) show a greater desire to participate in the decision making than those age 60 and greater (mean 38.8 $\pm$ 18.7) ($p = .02$). No differences were seen across race or education level. Patients who have ever discussed colorectal cancer screening with their physician exhibit less desire than those who have not (mean 39.4 vs. 48.1 respectively, $p = .001$). Multivariate analyses reveal that younger participants ($p = .001$), females ($p < .001$), and previously not having discussed colorectal cancer screening ($p = .01$) predicted higher scores on the modified API.

CONCLUSION: This study reveals that patients prefer physicians to assume more of the decision making power for colorectal cancer screening though patients want some role in this process. Differences exist between certain patient groups for which physicians should be specifically sensitive. Physicians should assess each of their patients' desire to participate in this decision making process and act accordingly. This will allow for the accommodation of each individual patients' preferences and needs.

RELIGIOSITY, SPIRITUALITY, AND HEALTH STATUS: FINDINGS FROM A NATIONAL SURVEY OF PATIENTS WITH HIV INFECTION. KA Lorenz, NS Wenger, RD Hays, PD Cleary, H Liu, MF Shapiro, General Internal Medicine, VA Greater Angeles Healthcare System; General Internal Medicine, UCLA, Los Angeles; RAND, Santa Monica, CA; Harvard University, Cambridge, MA

PURPOSE: Religious belief and practices may be important resources for HIV-infected individuals. We characterized the religious belief and practice of a nationally representative sample of HIV-infected patients and examined the association between these variables and clinical status.

METHODS: Participants in the HIV Cost and Services Utilization Study ($n = 2267$) were surveyed in late 1997 about religious involvement and practices. Those questions were used to create a 5-item measure of religiosity ($\alpha = 0.84$) and a 4-item measure of spirituality ($\alpha = 0.84$). We examined associations of religiosity and spirituality with demographics, acute stressors, and circumstances of HIV testing and care. Significant demographic variables were entered with religiosity and spirituality in multivariate models of recent viral load, CD4 count, symptom and disease incidence.

RESULTS: Many patients indicated that religion (42%) and spirituality (59%) were very important in their lives. In bivariate analyses, we noted significant positive relationships between religiosity (REL), spirituality (SPI) and need for income assistance ($pREL = 0.002$, $pSPI = 0.004$), housing ($pREL = 0.002$, $pSPI = 0.001$), financial aid with rent ($pREL = 0.008$, $pSPI = 0.005$), and need for home delivered meals or food bank assistance ($pREL = 0.036$, $pSPI = 0.005$). Circumstances of initial HIV testing and time to care did not differ by religiosity or spirituality, but individuals most anxious about revealing their HIV diagnosis to family ($pREL = 0.002$, $pSPI = 0.001$) or friends ($pREL = 0.000$, $pSPI = 0.038$) demonstrated higher religiosity and spirituality as did nonwhite race ($pREL = 0.000$, $pSPI = 0.000$), residents of regions other than the Western United States ($pREL = 0.000$, $pSPI = 0.000$), and non-homosexuals ($pREL = 0.000$, $pSPI = 0.000$). Age, gender, race/ethnicity, region, sexual orientation, and exposure to HAART were entered into multivariate models regressing religiosity and spirituality on clinical variables. A small significant relationship between higher spirituality and increased symptom incidence ($\beta = 0.0067$, $p = 0.009$) and disease incidence ($\beta = 0.002$, $p = 0.000$) was noted.

CONCLUSION: Religious belief and practices are common among patients with HIV infection. Cross sectional comparison demonstrated no consistent relationships between religiosity, spirituality, and clinical status. A small significant relationship was noted between spirituality and increased symptom and disease incidence; relationships between acute stressors, religiosity and spirituality also suggest that religiosity and spirituality are important resources for people facing HIV infection.

SEQUENTIAL EVALUATION OF SYNCOPE: A PROSPECTIVE POPULATION-BASED STUDY. M Louis-Simonet, A Rajeswaran, D Carballo, PF Unger, AF Junod, FP Sarasin, Dpt of Internal Medicine, Geneva University Hospital, Geneva, Switzerland

PURPOSE: To measure prospectively in primary-care patients the diagnostic yield of a standardized sequential workup of patients with syncope.

METHODS: All consecutive patients presenting in the emergency department with syncope as a chief complaint were included in the study. Stepwise workup included: 1) a complete history, physical examination and 12-lead ECG in all patients; 2) targeted test(s) when a specific entity was suspected by suggesting signs and/or symptoms; and 3) extensive testing (i.e., 24 hrs Holter, ambulatory loop recorder, echocardiography, upright tilt test and electrophysiologic studies in selected cases) for patients in whom the cause of syncope remained undetermined after the initial steps. Diagnostic criteria were based on existing literature.

RESULTS: Among all patients ($n = 650$) included over a two-year period, the cause of syncope was established after the initial evaluation in 452 (69.5%): vasovagal 37% ($n = 240$), hypotension (drug or hypovolemia-related, postprandial and idiopathic) 24% ($n = 158$), arrhythmias 3.5% ($n = 23$), other cardiac 1% ($n = 7$) and others 3.5% ($n = 24$). In 47 patients (7%) out of 66, targeted test(s) confirmed symptoms-based suspected diagnosis: aortic stenosis ($n = 7$), pulmonary embolism ($n = 7$), seizures ($n = 22$), stroke ($n = 8$), and others ($n = 3$). Extensive workup was completed in 110 of the 151 remaining patients. A cause for syncope was established only in 27 of them: cardiac 2.5% ($n = 16$), neurocardiogenic 1.5% ($n = 11$), the remaining patients ($n = 83$, 14%) being considered as having syncope of unknown etiology.

CONCLUSION: In unselected patients, the diagnostic yield of a noninvasive workup for syncope, including targeted testing for patients with suggesting signs and/or symptoms, was higher (77%) than currently reported.

SOCIAL SUPPORT DURING ELECTIVE JOINT REPLACEMENT ASSOCIATED WITH LOWER COST AND IMPROVED OUTCOMES. RM Lubitz, JL Ramsey, VL Dacey, RL Robinson, Department of Medicine, St. Vincent Hospitals and Health Services; Eli Lilly and Co., Indianapolis, IN

PURPOSE: Over 500,000 elective joint replacements are performed annually, making it one of the most common surgeries in the United States. Hip and knee replacement improves mobility, decreases pain and improves quality of life for patients with severe large-joint arthritis. Previous studies demonstrate that a patients' tangible and intangible supports ("social support") play an important role in rehabilitation and recovery. The relationship of social support to other health-related outcomes is poorly described. The purpose of this study was to examine the association between social support and hospital length of stay (LOS), cost of care, and health-related functional status.

METHODS: We surveyed 883 consecutive patients undergoing elective joint replacement at two suburban Indianapolis hospitals between January 1, 1998 and March 31, 1999. Patients were interviewed by telephone prior to surgery and 6 months post-surgery. The survey included reliable and valid measures of sociodemographic factors, comorbid conditions, total social support at time of surgery (MOS-20), general physical health and emotional function (SF-12 PCS and MCS), and lower extremity joint-specific pain and function (WOMAC). Hospital LOS and actual costs were obtained through the hospital's administrative database.

RESULTS: Mean age of the cohort was 67.4 years (range 21–100), 62.8% were women, and 90.8% were Caucasian. Elective knee replacement was performed on 62.4% and hip replacement on 37.6%. The average total social support was 88.7% (range 0–100). Subjects with higher social support were more likely to be undergoing knee replacement, a non-smoker, reside in a non-urban area, and discharged to home following surgery. Median hospital LOS was 4 days, and median hospital costs were \$6,418. Using linear regression, controlling for demographic and behavioral variables, procedure, comorbidities and baseline function, increasing social support was associated with a shorter LOS and lower hospital costs (both $p < 0.05$), and greater improvement in PCS score ($p = 0.07$). Social support was not associated with joint-specific outcomes.

CONCLUSION: Patients undergoing elective joint replacement have high levels of social support, and greater social support is associated with improved outcomes and lower hospital costs. Whether interventions targeted to patients with lower pre- and post-operative social support will improve outcomes requires further study.

BARRIERS TO SIGMOIDOSCOPY MAY BE INFLUENCED BY HEALTH BELIEFS AND ETHNICITY. PM Lukoschek, AB Moadel, Department of Epidemiology, Albert Einstein College of Medicine, Bronx, NY

PURPOSE: Colon cancer screenings are lower among ethnic minorities contributing to inequalities in cancer mortality. Barriers to cancer screenings need to be explored to reduce health disparities among ethnic groups.

METHODS: Participants ($n = 172$) in an urban, ethnic diverse community were recruited after local church and community events and invited to participate in telephone interviews. The questionnaire focused on knowledge of cancer screenings and various health beliefs (misconceptions, risk perceptions, cancer-related anxiety) based on medical literature and beliefs elicited from the community.

RESULTS: Participants mean age was 54.7, with 58% ($n = 110$) above the age of 50 (mean age: 63, range: 50–84), of which 48% were African-American, 76% were female, 50% had less than a college education, 49% were employed (43% retired, 2% disabled), and 97% had medical insurance. The reported sigmoidoscopy screening rate (37%) was higher when compared to the national average, although only 27% had the appropriate colon cancer screening within the last 5 years. Univariate analyses showed that gender, age and education was not associated with sigmoidoscopy use, however African-Americans and employed participants were less likely to have had a sigmoidoscopy when compared to Whites (71% vs. 51%, $p = 0.034$) and non-employed participants (72% vs. 50%, $p = 0.023$) respectively. Knowledge about cancer screenings and anxiety of cancer showed no association with reported sigmoidoscopy use. However, participants who believed that their personal risk of cancer was lower than that of others (81% vs. 54%, $p = 0.014$) and those who held the misconception that surgery spreads cancer were less likely to have had a sigmoidoscopy (71% vs. 49%, $p = 0.034$) compared to those who did not hold these beliefs. Multivariate logistic regression analysis of all significant univariate variables revealed that ethnicity (OR = 2.5, CI:1.03–6.18), personal risk perception (OR = 4.6, CI:1.48–14.4) and misconception about surgery (OR = 2.5, CI:1.01–6.11) were significant predictors of self-reported prior sigmoidoscopy use.

CONCLUSION: This research suggests that educational interventions could benefit from focusing on cancer risk perceptions and misconceptions in an ethnically appropriate manner to increase sigmoidoscopy screenings.

PATIENTS UNREALISTICALLY EXPECTING TO BE CURED OF ASTHMA HAVE WORSE FUNCTIONAL OUTCOMES AND HIGHER RESOURCE UTILIZATION. CA Mancuso, M Rincon, ME Charlson, Internal Medicine, Cornell University, New York, NY

PURPOSE: The goal of this study was to compare short-term outcomes in patients expecting a cure from asthma treatment versus patients not expecting a cure. Outcomes were change in function and resource utilization.

METHODS: Patients were eligible if they had moderate asthma and were 18–62 years old. Patients were interviewed in-person during routine office visits. On enrollment, patients completed the Asthma Quality of Life Questionnaire (AQLQ) measuring function with an overall and four domain scores, and other asthma questions, including measures of asthma knowledge and asthma self-efficacy. Patients were also asked the open-ended question "What are your expectations of asthma treatment?". Their responses were categorized with qualitative techniques

and corroborated by an independent investigator. Patients again completed the AQLQ during in-person interviews during interval follow-ups. Asthma resource utilization was measured by patient report every three months by telephone.

RESULTS: A total of 212 patients was enrolled, mean age 41 ± 11 , 83% women, mean follow-up 10.0 months. Surprisingly, 35% of patients expected to be cured of asthma. In multivariate analyses, patients expecting a cure were more likely to be currently using inhaled corticosteroids ($p = .004$), to have never required oral corticosteroids ($p = .02$), and to have lower asthma self-efficacy ($p = .008$).

Outcomes: In follow-up analyses of within-patient change in AQLQ scores, patients expecting a cure did not improve by a clinically important difference in any domain, while patients not expecting a cure improved by a clinically important difference in all domains. Also, more patients expecting a cure declined in various AQLQ domains and in the overall score ($p = .03$), compared to patients not expecting a cure. In multivariate analyses, only older age ($p = .02$) and expecting a cure ($p = .03$) remained predictors of worsened function. Patients expecting a cure also had more hospitalizations, emergency room visits and non-routine office visits for asthma compared to patients not expecting a cure (1.8 ± 5.0 versus 1.3 ± 2.8). In addition, patients expecting a cure had a stepwise increase in requiring at least some urgent resource utilization for asthma (52% versus 39%) compared to patients not expecting a cure.

CONCLUSION: Surprisingly, 35% of patients had the unrealistic expectation of being cured from asthma treatment. Expecting a cure predicted functional decline and more frequent asthma resource utilization. Addressing unrealistic expectations that are potentially modifiable may help to improve short-term asthma outcomes.

VIROLOGIC RESPONSE TO INDINAVIR (IDV) AND NELFINAVIR (NFV) COMBINATION REGIMENS IN CLINICAL PRACTICE. LE Markson, JB Greene, PT Duong, PB Landsman, SG Sajjan, JF Murray, Outcomes Research & Management, Merck & Co., Inc., West Point, PA; Target Health, New York, NY

PURPOSE: To evaluate factors associated with viral load suppression in patients treated with IDV or NFV regimens in clinical practice.

METHODS: Data from 2 clinics were retrospectively reviewed for 298 patients who have received either IDV or NFV (excluding regimens of four or more drugs) during January 1995 through April 1999. All patients had detectable viral RNA (vRNA > 400 copies/mL, Roche Amplicor) prior to regimen start. Regimens were classified as triple therapy (IDV or NFV plus 2 NRTI) or Other regimen. Multivariate logistic regression analysis was used to assess odds of achieving suppressed vRNA (vRNA < 400 copies/mL in at least 1 measure after regimen start), adjusting for age, race, gender, baseline CD4 and vRNA, PI experience, injection drug use (IDU), simultaneous start (antiretrovirals (ART) started within 14 days of PI start date), triple regimen and propensity scores. Propensity scores from logistic regression model were used to adjust for baseline differences between IDV and NFV groups including age, race, gender, baseline CD4 and vRNA, PI experience, and IDU.

RESULTS: Median baseline CD4 and vRNA in IDV group ($n = 217$) were 188/mm³ and 63,030/mL, respectively. Median baseline CD4 and vRNA in NFV group ($n = 81$) were 243/mm³ and 46,952/mL, respectively. From the propensity score model, a lower odds of receiving IDV compared to NFV was observed for the following characteristics: African-Americans (OR 0.35; CI 0.17-0.7; $p = 0.003$), baseline CD4 > 200/mm³ (OR 0.4; CI 0.2-0.8; $p = 0.009$) or reported IDU (OR 0.46; CI 0.25-0.85; $p = 0.013$). The odds of achieving suppressed vRNA, adjusting for propensity scores, was significantly higher for the following variables: PI naive (OR 2.6; CI 1.3-5.4; $p = 0.007$), triple therapy (OR 3.4; CI 1.6-6.9; $p = 0.001$) and simultaneous starts at regimen start (OR 2.49; CI 1.3-4.6; $p = 0.004$); the odds of vRNA suppression approached significance for IDV compared to NFV (OR 1.97, CI 0.98-3.97, $p = 0.057$). The study power for detecting significance ($p < 0.05$) between IDV and NFV groups was limited ($b = 0.70$) with a study sample of $n = 298$.

CONCLUSION: In routine care, this analysis supports the strategy of starting triple ART simultaneously in PI naive patients to achieve suppressed vRNA response. Assessment in a larger observational cohort is needed to determine if IDV and NFV have differential response rates in clinical practice.

IMPROVING HEALTH CARE IN RUSSIAN-SPEAKING PATIENTS. UG Mason, I Pines, K Doll, D Main, PS Mehler, Denver Health Medical Center, Denver, CO

INTRODUCTION: Ethnic inequalities in health care may be attributed to cultural and communication barriers between patients and health care providers. We postulate that a Russian patient's inability to speak English may interfere with the patient's ability to follow instructions, and/or cause distrust of the caregiver, thereby contributing to poor glycemic control.

PURPOSE: We assessed glycemic control in Russian-speaking only patients from the same health care system before and after the employment of a bilingual, Russian-born physician who was medically trained in both countries.

METHODS: We conducted a retrospective chart review (1997-1999) to compare the values of hemoglobin A1c, as an indicator of glycemic control, in our monolingual, Russian-speaking patients who, prior to October 1997 were seen by English-speaking physicians at Denver Health, with the values after they began to receive treatment from a bilingual, Russian physician. All of the patients were treated on the main campus of Denver Health Medical Center, and the chart reviews were performed by the physician primary and co-primary investigators.

RESULTS: Twenty-two Russian-speaking diabetics were evaluated for glycemic control. Their average age was 67.0 (11 males and 11 females). Initial mean A1c measurements were 9.23 ± 1.44 with a minimum value of 6.7 and a maximum of

12.5. The second A1c measurements, obtained approximately eight to twenty-four months after the initial values, were a mean of 7.9 ± 0.86 with a minimum value of 6.9 and maximum of 10.0. The mean decline in A1c was -1.3 ± 1.0 with the greatest decline in A1c values of -3.2 (-3.2 to $+0.7$). A p -value for the t -test of the mean change was significant at $p < 0.0001$.

CONCLUSION: The significant decline in HgbA1c from the first to second measurement indicates improvement in glycemic control of the Russian-speaking population likely due, in part, to language and cultural concordance between these new emigres and their primary care provider. Further evaluation will be necessary to define the optimal means of caring for new immigrant patients.

PREDICTORS OF OBJECTIVELY MEASURED PHYSICAL ACTIVITY LEVELS IN MEN AND WOMEN WITH PERIPHERAL ARTERIAL DISEASE. MM McDermott, A Priyanath, P Greenland, K Liu, GJ Martin, W Pearce, Medicine, Preventive Medicine, and Vascular Surgery, Northwestern University Medical School, Chicago, IL

PURPOSE: To identify characteristics associated with higher physical activity (PA) levels in patients with lower extremity peripheral arterial disease (PAD).

METHODS: PAD participants age 55 and older were identified from a non-invasive vascular laboratory. Non-PAD participants age 55 and older were identified from the non-invasive vascular laboratory and from a general medicine practice at the study institution. Participants underwent a 2.5 hour study visit, during which PAD severity was assessed with the ankle brachial index (ABI), a detailed medical history was obtained, and objective measures of functioning were administered. To objectively measure PA, participants were outfitted with a Caltrac accelerometer that was worn continuously for seven days. The Caltrac accelerometer calculates kcals expended based upon measured vertical movement at the hip. At the end of seven days, participants were contacted by telephone and asked to read the number of PA kcals displayed on their Caltrac accelerometer. Participants then mailed the Caltrac to the study team.

RESULTS: A total of 312 participants (228 with PAD) wore the Caltrac monitor for seven days. In age-adjusted analysis, PAD participants had significantly lower PA than non-PAD participants (802 vs. 1128 kcals, $p < 0.001$). Results according to physical activity tertile among PAD patients are shown in the table.

	1st PA tertile	2nd PA tertile	3rd PA tertile	Trend p value
Age	74.4	75.0	70.8	<0.001
ABI	0.65	0.67	0.72	0.02
Exertional leg pain	64%	78%	82%	0.05
4 meter walking velocity (m/sec)	0.76	0.87	0.95	<0.001
Distance achieved in six minute walk (feet)	904	1121	1297	<0.01

In multivariable regression analyses among PAD patients only, ABI (regression coefficient = 608 kcals/1 unit ABI, $p < 0.01$) and walking velocity over four meters (regression coefficient = 996 kcals/1 m/sec, $p < 0.01$) were associated independently with PA, adjusting for age, sex, race, body mass index, exertional leg symptoms, and comorbid diseases.

CONCLUSION: Higher physical activity levels are associated with a higher prevalence of exertional leg pain in PAD. Both PAD severity, as measured by ABI, and walking velocity over four meters are associated independently with objectively measured PA in PAD patients. Further study is needed to determine whether slower walking velocity precedes decline in PA or whether slower walking is a consequence of reduced PA.

QUALITY OF LIFE AND 5-YEAR MORTALITY AMONG VETERANS WITH CORONARY ARTERY DISEASE: RESULTS FROM THE AMBULATORY CARE QUALITY IMPROVEMENT PROJECT (ACQUIP) PILOT STUDY. MB McDonnell, JA Spertus, SD Finn, HSR&D Center of Excellence, VA Puget Sound Health Care System; Medicine, University of Washington, Seattle, WA; Medicine, University of Missouri-Kansas City, Kansas City, MO

PURPOSE: While Quality of Life (QOL) measures may be useful to identify "at risk" patients, few studies have examined the relationship between disease-specific health status and long-term mortality. This study examined the relationship between QOL and all-cause 5-year mortality among VA outpatients with coronary artery disease (CAD).

METHODS: Three VA General Internal Medicine Clinics participated in the study as pilot sites for the Ambulatory Care Quality Improvement Project (ACQUIP). This prospective cohort study examined 1679 patients with CAD who responded to QOL surveys mailed between November 1993 and April 1994. Quality of life was measured using 4 scales of the Seattle Angina Questionnaire (SAQ): physical function, disease perception, angina frequency and disease stability. All scales were scored 0 (worst) to 100 (best). Comorbid conditions and demographic information were obtained from patients at entry into ACQUIP. Death information was obtained from local and national VA information systems. Mortality from all causes was included. Data were examined using survival analysis and Cox Proportional Hazards regression.

RESULTS: The cohort was 98% male and 92% Caucasian with a mean age of 67. Overall, 469 (28%) of the 1679 died during the five year follow-up period. The cumulative proportion surviving at five years was .69. Patients were stratified into 2 groups for all scales: those scoring < 50 and those scoring > 50 . The cumulative

proportion surviving at 5 years was .63 for patients scoring = <50 on the physical scale compared to .79 for those scoring >50 ($p = .000$). For the frequency scale, the cumulative proportion surviving was .63 for patients scoring = <50 and .70 for those scoring >50 ($p = .005$). These differences remained significant when stratified by age and co-morbid conditions. Differences in survival were not significant for the disease perception and stability scales. In a multivariate model including physical function, age, and comorbid conditions, the relative risk of death was 2.0 (95% CI = 1.62–2.46) for patients scoring = <50 compared to those scoring >50 ($p = .000$). In a model including angina frequency, age, and comorbid conditions, the relative risk of death risk was 1.4 (95% CI = 1.13–1.76) for those scoring = <50 ($p = .003$).

CONCLUSION: Lower quality of life, as measured by the SAQ physical function and frequency scales, may be a significant risk factor for all-cause mortality in veterans with CAD. QOL as measured by the disease perception and symptom stability scales did not significantly relate to risk of death.

TRANSGENERATIONAL RECRUITMENT OF MINORITY WOMEN TO BREAST AND CERVICAL CANCER SCREENING. SM McGraw, H Giff, L Costa, KW Smith, D Cohen, K Bullock, New England Research Institutes, Watertown, MA; Yale Primary Care, Family Health Center, St. Mary's Hospital, Waterbury, CT

PURPOSE: We report on the results of recruitment to a two-armed randomized trial to test an innovative intervention to increase rates of breast and cervical cancer screening among older African-American and Latina women. The intervention relied on younger women, who typically use preventive services more often than do older minority women, to encourage the older women to seek screening.

METHODS: Both study recruitment and the intervention were conducted by outreach workers (OW) familiar with study neighborhoods. Younger women (ages 30–45) were asked to nominate and work with at least one older African-American or Latina woman (50 years and older). Recruitment was conducted at sessions randomized in advance to treatment or control conditions. OW were blinded to the randomization until just prior to a session. In treatment sessions, OW explained the study and taught the younger women how to intervene with the older women. In control sessions, the OW explained the study. One week after recruitment, the nominators (younger women) in both study conditions were contacted to determine the older woman's willingness to participate. Also, in the intervention condition, nominators were contacted six weeks later to reinforce their work encouraging the older nominee to get an exam.

RESULTS: A total of 151 recruitment sessions were held over a 22-month period. They were conducted at health fairs, in employment training, housing or energy assistance offices, at churches, hair salons, and health clinic waiting rooms and door-to-door. A total of 549 nominators were willing to contact an older woman to participate in the study, but only half were able to obtain agreement from their nominees. Fifty-eight percent of the older women nominated by the younger women in the control condition agreed to participate, compared to 48% in the intervention condition. After 6 weeks, 75% of the intervention group nominators reported talking with the older women about the screening exams.

CONCLUSION: This unique intervention represents an effort to create a community-based approach to contact women who can be difficult to reach for preventive screening exams. Recruitment to the study required considerable effort with outreach in a number of different recruitment sites. Recruitment rates were slightly higher in the control condition which demanded less of the participants.

TEMPORAL TRENDS IN THE CONTROL OF HYPERLIPIDEMIA IN TYPE 2 DIABETIC PATIENTS. PS Mehler, S Biggerstaff, Denver Health Medical Center, Denver, CO

INTRODUCTION: There is a markedly increased risk of cardiovascular mortality in type 2 diabetic patients. Inadequate achievement of published guidelines for the control of hyperlipidemia may be contributing to this result.

PURPOSE: The purpose of this study is to assess the degree of lipid control, longitudinally over five years, in a cohort of type 2 diabetic patients.

METHODS: Nine hundred fifty (950) type 2 diabetic patients enrolled in the Appropriate Blood Pressure Control in Diabetes (ABCD) trial had fasting blood samples obtained to determine low-density lipoprotein (LDL) cholesterol, total cholesterol, high-density lipoprotein (HDL) and triglycerides levels. Temporal trends in the degree of lipid control were analyzed from five years of the study period. Subgroup analysis in patients with concurrent coronary disease was also performed to determine the degree of lipid control in this population. Multivariate logistic regression analysis was used to evaluate for potential factors associated with different degrees of lipid control.

RESULTS: The mean baseline total cholesterol levels in these type 2 diabetic patients at the initiation of the study was 218 mg/dL and the LDL level was 130 mg/dL. Only 18.15% of these patients had a LDL level less than 100 mg/dL at baseline. After five years of being enrolled in the study this number remained unchanged (18.44%). The percentage of patients achieving a LDL level less than 130 mg/dL also did not improve from baseline through the end of the study (51.56% vs. 53.01%). Of the patients in the study, 18.40% had a baseline LDL which exceeded 160 mg/dL, and 14.36% of the patients continued to have a level greater than 160 mg/dL at the completion of the study. There were 214 patients with a history of coronary disease at the onset of the study; only 17.3% had a LDL less than 100 mg/dL and 23.8% had a LDL greater than 160 mg/dL. In multivariate logistic regression analysis none of the variables (gender, age, drug assignment, race/ethnicity, insurance [yes/no]) retained a statistically significant association with attainment of published guidelines for lipid control in diabetics.

CONCLUSION: Despite a number of studies which have proven the benefit of aggressive lipid lowering in diabetic patients, the attainment of adequate cholesterol control is markedly suboptimal. Major changes in treatment strategies will likely be required to ensure better cardiovascular health for diabetic patients.

TUBE FEEDING IN END STAGE DEMENTIA: PREDICTORS OF GASTROSTOMY TUBE PLACEMENT AND IMPACT ON SURVIVAL. DE Meier, RS Morrison, SA Baskin, J Morris, JC Ahronheim, Hertzberg Palliative Care Institute, Mount Sinai School of Medicine; Department of Medicine, St. Vincent's Hospital, New York, NY

PURPOSE: To identify risk factors for tube feeding in hospitalized end-stage dementia patients and examine the impact of tube feeding on survival.

METHODS: Design: Prospective cohort study.

Setting: Hospital.

Subjects: Fifty seven end stage dementia patients who did not have a feeding tube hospitalized for an acute illness at a large New York City Hospital. Patients hospitalized for placement of a feeding tube were excluded.

Main Outcome Measures: Placement of a gastrostomy tube, re-hospitalization rates, mortality.

RESULTS: Twenty two of 57 subjects (39%) had a feeding tube placed during their hospitalization. Risk factors for feeding tube placement by logistic regression included poor oral intake while hospitalized (odds ratio of 8.92, 95% confidence interval 2.2–40.0) and being African American or Latino (odds ratio of 5.34, 95% confidence interval 2.1 to 37.0). Previous hospitalizations for pneumonia, a history of weight loss, dementia stage, presence of a pressure ulcer, absence of an advance directive, pre-hospitalization residence (nursing home vs home), or dementia stage did not predict tube feeding. Median survival following hospital admission in subjects receiving a feeding tube was 195 days (range 21 to 1405 days) as compared to 189 days in subject who did not receive a feeding tube (range 4 to 1502 days). Tube feeding was not associated with survival in a Cox proportional hazard model ($p = 0.9$)

CONCLUSION: End stage dementia patients hospitalized with an acute medical illness are at high risk for placement of a gastrostomy tube. Risk factors for tube placement include poor oral intake while in the hospital and being non white. End stage dementia is associated with a median survival of 6 months following hospitalization for an acute illness. Tube feeding does not appear to appreciably prolong survival.

THE PREVALENCE AND IMPLICATIONS OF INCIDENTAL FINDINGS ON EMERGENCY DEPARTMENT ABDOMINAL CT SCANS. WA Messersmith, DFM Brown, MJ Barry, Internal Medicine; Emergency Medicine, Massachusetts General Hospital, Boston, MA

PURPOSE: As CT scanning replaces other more focused tests (such as IVP) for common urgent problems such as renal colic, incidental findings are more common. We examined the prevalence and implications of these "incidentalomas," defined as findings unrelated to the original purpose of the scan which may either go unnoticed, or raise concern and generate further diagnostic work-up.

METHODS: We reviewed the reports from 321 consecutive emergency department (ED) noncontrast, helical "renal stone" abdominal CT scans obtained between 9/96–6/97 for the presence of incidental findings. These findings were then subdivided by two independent reviewers into 2 groups based on the level of concern they raised for serious pathology. The hospital records of these patients were then examined to see whether these findings were followed up in any way in the ensuing 2–3 years.

RESULTS: The mean age of the patients was 43 (range 16–81); 36% were female, and 67% had evidence of nephrolithiasis confirmed by CT. Of the 321 patients, 14 (4%) were excluded due a clear diagnosis other than nephrolithiasis being made on CT scan. 145 (45%) of the remaining 307 scan reports described incidental findings, of which 61 (45%) were of "moderate" or "serious" concern to both reviewers. Follow-up information was obtained on 80% of the cases with incidental findings, either from PCP notes (29%), specialty clinic notes (47%), or the hospital's computerized laboratory results. Not surprisingly, only 1 of 118 scans where both reviewers had "little or no" concern about the findings had subsequent follow-up at our hospital. However, only 11 (18%) of the 61 cases deemed of "moderate/severe" concern by both reviewers had any evidence of follow-up despite data on 51 (83%) of them. In only 27% of the total cases, and the same 27% of those with "moderate/severe" incidental findings, were they mentioned in the ED record.

CONCLUSION: Incidental findings on ED abdominal CT scans are common and sometimes entail further diagnostic testing. Most are never noted or followed up, however, which raises a quality issue, emphasizing the need for effective communication between physicians.

THE RELATIONSHIP BETWEEN PATIENT SATISFACTION AND HEALTH RELATED BELIEFS. MA Micek, CH Braddock, KA Bradley, D Martin, MB McDonnell, SD Fihn, HSR&D, Center of Excellence, VA Puget Sound Health Care System; Department of Medicine, University of Washington, Seattle, WA

PURPOSE: To examine the relationship between patient satisfaction and patient beliefs relating to their health and the medical system.

METHODS: Participants for this study included patients followed at 3 VA general internal medicine clinics participating in the VA Ambulatory Care Quality Improvement Project (ACQUIP), a multicenter randomized controlled trial to test the

effectiveness of an information feedback system on improving health status and satisfaction outcomes. A random subsample of these patients received the Health Beliefs Survey, which contained items adapted from previously published health beliefs scales. The survey included items asking about general faith in doctors and medical science, the harmful effects of medical treatment, and the value of treatments not based on medical science. An additional item asked patients if they felt their beliefs about health were similar to their physicians' beliefs. Response options ranged from 1 (strongly disagree) to 5 (strongly agree). Patient satisfaction was measured with the 12-item humanistic component of the Seattle Outpatient Satisfaction Questionnaire (SOSQ), which is scored from 0 (low) to 100 (high). Additional data gathered included patient age, ethnicity, level of education, and income. We used simple and partial correlation techniques to evaluate the association between patients' reports of their health beliefs and patient satisfaction scores, while adjusting for patient demographic characteristics.

RESULTS: We analyzed the survey responses of 127 patients. Patient satisfaction scores had a mean of 75 and were skewed towards high ratings. No significant associations were found between patient satisfaction and items asking about faith and skepticism in doctors and medical science. However, the degree to which patients felt their beliefs about health were similar to their provider's beliefs was positively associated with patient satisfaction (simple correlation coefficient = .25, $p = .015$), and this association persisted after adjustment for patient demographic characteristics (partial correlation coefficient = .25, $p = .02$).

CONCLUSION: Patient beliefs relating to faith and skepticism in doctors and medical science were not significantly associated with patient satisfaction. However, the degree to which patients felt their beliefs about health were similar to those of their provider was significantly related to patient satisfaction. Further studies are needed to clarify the relationship between health beliefs and patient satisfaction, and to evaluate whether patient-provider discussions of health beliefs could improve patient satisfaction.

THE EFFECTIVENESS OF INTRAVENOUS PAMIDRONATE IN TREATING OSTEOPOROTIC PATIENTS. RG Miller, KK Chang, MJ Klag, LA Mead, MA Levine, Dept. of Medicine; Dept. of Pediatrics, Johns Hopkins School of Medicine, Baltimore, MD

PURPOSE: The use of oral bisphosphonates for osteoporosis is highly effective but unfortunately limited by gastrointestinal side effects. We examined the effectiveness of pamidronate, an intravenous bisphosphonate, as an alternative therapy.

METHODS: We performed a nonconcurrent, prospective study of 78 patients with osteoporosis of varying cause treated in an osteoporosis clinic at an urban tertiary care hospital; 26 were treated with pamidronate, and 52 received other treatments including estrogen, calcitonin, and/or oral bisphosphonates (standard care). Osteoporosis was defined as a T-score of ≤ 2.5 at the L-spine, femoral neck, or trochanter on dual energy X-ray absorptiometry (DEXA) scan. Standard care patients were matched 2:1 to the pamidronate group by age, race, and sex. Baseline was defined as the start of pamidronate therapy or the first clinic visit. Pamidronate was administered as a 30 mg infusion over 4 hours every 3 months. Bone mineral density (BMD) was measured at baseline and approximately annually thereafter using DEXA scan. Response to therapy, defined as either stabilization or increase in BMD at one or more years of follow-up, was the primary outcome variable.

RESULTS: Mean age was 58 and 54 years and body mass index was 23.5 and 23.8 kg/m² for the pamidronate and standard care groups, respectively. Mean follow-up was 1.3 years for pamidronate patients and 1.8 years for standard care patients. There was no statistically significant difference in percent responders between the pamidronate and standard care groups at any anatomical site (see table). Among responders, mean percent increase in BMD at the lumbar spine and femoral neck was comparable in pamidronate and standard care patients.

Response to Therapy

	% Responders (95% CI)	% Change per year in BMD Standard Care	Pamidronate	Standard Care	p-value
L-spine	64% (45-83)	64% (51-77)	4.0%	5.2%	0.33
Femoral Neck	65% (47-83)	57% (43-71)	4.3%	6.2%	0.29
Trochanter	67% (48-86)	52% (38-66)	2.1%	7.8%	0.001

CONCLUSION: In this observational trial, the response rate of osteoporotic patients to intermittent intravenous pamidronate is similar to the response rate to standard treatment. On average, pamidronate patients gained 2.1-4.3% in BMD the first year. Thus, intermittent infusion of intravenous pamidronate provides an acceptable and convenient alternative treatment for patients with osteoporosis who cannot tolerate oral bisphosphonates or other therapies.

THE EFFECT OF MONASCUS PURPUREUS ON LIPID PROFILES IN HYPERCHOLESTEROLEMIC INDIVIDUALS. RJ Millev, SA Wasco, AG Myrdal, AH Kirkpatrick, TN Bilodeau, GM Kline, PS Freedson, DP O'Brien, JM Rippe, Research Laboratory, Center for Clinical and Lifestyle Research, Shrewsbury, MA

PURPOSE: The purpose of this study was to investigate the effects of *Monascus purpureus* supplementation, incorporated into a snack bar, on blood lipid levels.

METHODS: Fifty men and thirty-six women, with a mean total cholesterol (TC) level of 246.9 mg/dL (SD $\pm$ 23.21 mg/dL) and a mean low density lipoprotein cholesterol (LDL-C) level of 165.1 mg/dL (SD $\pm$ 21.34 mg/dL), were randomly assigned to a treatment or control group. Subjects were administered either a pla-

cebo bar or the treatment bar (2.4g/day of *Monascus purpureus*) and were counseled throughout the study (12 weeks) by a registered dietitian on how to adhere to the American Heart Association (AHA) Step One Diet.

RESULTS: A two-factor repeated measures ANOVA revealed a statistically significant group by time interaction for the treatment group where TC decreased by 14% (34.7 mg/dL $p < 0.0001$) and LDL-C decreased by 22% (35.9 mg/dL $p < 0.0001$). In contrast, the control group demonstrated a 1.4% increase in TC and no change in LDL-C levels. There were no statistically significant changes in triglyceride levels, high density lipoprotein cholesterol levels, liver function tests, or body weight for the treatment group. Also, there were no statistically significant group by time interactions for caloric intake, dietary cholesterol intake, percent calories from fat, or percent calories from saturated fat.

CONCLUSION: This study demonstrates that supplementation with a traditional Chinese food, *Monascus purpureus*, incorporated into a snack bar, is effective in reducing TC and LDL-C levels beyond the level of reduction obtained through the use of the AHA Step One Diet. In addition to its effectiveness in patients with hypercholesterolemia, *Monascus purpureus* is well tolerated with relatively few reported side effects.

SHORT TERM EFFECTS OF SHARED DECISION MAKING ON PATIENT ADHERENCE, LEVELS OF HEMOGLOBIN A1C, AND CHOLESTEROL IN DIABETES MELLITUS. ES Moore, M Agarwal, GL Barbour, Georgetown University Medical Center, Washington, DC; VA Medical Center Washington DC

Background: Shared medical decision making (SMDM) is believed to improve adherence in chronic disease management. This has been shown in diabetic populations for Hemoglobin A1C but not for cholesterol levels.

Methods: We studied the effect of SMDM on common parameters in diabetes (HbA1c, lipids, visit patterns to Primary Care, Eye and Podiatry clinics and medication usage). Patients in Group A (N = 75) were provided with active SMDM (self-help texts and RN initiated telephone counseling) for three months and then followed routinely; Group B patients (N = 73) were matched controls given only routine care. Fourteen months after beginning the intervention we compared the two groups ("post" values) and evaluated the same parameters in the 14 month period prior to the institution of SMDM in both groups ("pre" values).

Results:

HbA1c (%) ($\pm$ SD)	Chol (mg/dl) ($\pm$ SD)	Prim. Care visits	Eye Clinic Visits	Pod. Clinic Visits
Gp A pre	8.42 (2.15)	193.0 (43.0)	549	175
Gp A post	8.76 (1.93)	189.0 (40.6)	692	175
Gp B pre	8.07 (2.59)	220.0 (102)	663	156
Gp B post	8.17 (1.40)	197.0 (50.0)	777	155

There was no significant difference between groups for any of the parameters studied. There was virtually no change over time for the blood levels; the change in clinic visit pattern appeared in both the study and control group.

Conclusion: Over the time frame of 14 months (with only three months of active intervention) there was no significant impact of SMDM shown on HbA1c, cholesterol or clinic usage pattern in Primary Care of diabetes. The intervention perhaps occurred over too short a time to produce change. It is possible that factors such as more intense reinforcement, a different type of patient involvement, or a dual intervention directed at both patients and providers may be needed to see an effect.

PAIN AND DELIRIUM IN ELDERLY HIP FRACTURE PATIENTS. R Morrison, J Magaziner, M Gilbert, AL Siu, Geriatrics, Mount Sinai School of Medicine, New York, NY; Gerontology, University of Maryland, Baltimore, MD; Orthopedics, Mount Sinai School of Medicine, New York, NY; Medicine

PURPOSE: To identify risk factors for post-operative delirium in elderly hip fracture patients and to examine the relationship between pain and opioids and the development of delirium.

METHODS: 650 patients over age 55 with femoral neck or inter-trochanteric fractures admitted to 4 New York hospitals from 8/87 through 7/98 were eligible for inclusion. 571 (88%) consented to participate and were prospectively interviewed on a daily basis, five days a week, with the Confusion Assessment Method (CAM) for the presence of delirium. Pain was assessed through daily interviews using a 5 point numeric rating scale. Subjects were included if they were not delirious on admission and if they did not develop delirium pre-operatively. Medical record review was used to supplement CAM observations for the presence of delirium. Risk factors for the development of delirium were identified through review of the literature and categorized (e.g., demographic, operative, biomedical). Forward stepwise multiple logistic regression was used to examine the association between risk factors and development of delirium in all patients and in cognitively intact and dementia patients. Included in the model were all variables of at least borderline significance ($p < .15$) by univariate analyses. If a risk category did not contain a variable of at least borderline significance, the most clinically relevant variable was entered. Mean daily opioid doses in morphine sulfate equivalents were forced into the models.

RESULTS: 490 of 571 subjects were not delirious prior to surgery. Of the 490 eligible subjects, 57 (11.6%) developed delirium post operatively. Risk factors associated with delirium included a diagnosis of dementia (OR 16.9, 95% confidence interval 3.72-76.41) and more than 3 hours spent in the emergency department (OR 2.0 95% confidence interval 1.07-3.78). Risk factors for delirium in the cognitively

intact subjects included: poor baseline functional status (OR 1.45 95% confidence interval 1.08–1.97) and severe pain (OR 4.3 95% confidence interval 1.183–15.36). Risk factors for delirium in dementia subjects included: good baseline functional status (OR 1.3 95% confidence interval 1.03–1.63). Opioid analgesics were not associated with the development of delirium in any of the models.

CONCLUSION: Inadequately treated post-operative pain is a strong risk factor for post-operative delirium in cognitively intact elderly hip fracture patients. Patients with a diagnosis of dementia are also at very high risk.

PREDICTORS OF PNEUMONIA-RELATED AND NON-PNEUMONIA-RELATED MORTALITY IN COMMUNITY-ACQUIRED PNEUMONIA. EM Mortensen, DS Obrosky, CM Coley, DE Singer, TJ Marie, WN Kapoor, MJ Fine, University of Pittsburgh, Pittsburgh, PA; Massachusetts General Hospital, Boston, MA; University of Alberta, Edmonton, AB, Canada

PURPOSE: All previous studies of prognosis assessed predictors of all-cause mortality (MORT) for patients with community-acquired pneumonia (CAP). The aim of this study was to compare the predictors of CAP-related (CAP-R) and CAP-unrelated (CAP-U) 90-day MORT for patients (pts) with CAP.

METHODS: This analysis was conducted as part of the Pneumonia Patient Outcomes Research Team (PORT) prospective, multicenter cohort study. Baseline demographic and clinical data, including comorbid illnesses, laboratory and radiology results, were obtained for all pts. For all deaths, a synopsis of all pre-death clinical information was independently reviewed by a 5 member review panel to determine by consensus whether CAP was an underlying or immediate cause of death, or if CAP played a major role in causing death (i.e., CAP-R). Baseline pt data were used as independent variables in 3 Cox proportional hazards models, using 3 MORT outcomes (CAP-R, CAP-U, and all-cause) as the respective dependent measures. Variables that were statistically significant in any of the 3 Cox models were used in each final model to predict CAP-R, CAP-U, and all-cause MORT.

RESULTS: Of the 2287 patients, 208 (9.1%) died within 90-days. MORT was CAP-R in 69% and CAP-U in 31%. Significant predictors of CAP-R MORT were hypothermia (hazard ratio 1.9 with 95% confidence interval 1.0–3.5), age per decade (1.6, 1.4–1.9), altered mental status (2.3, 1.5–3.4), liver disease (3.9, 1.2–12.7), leukopenia (3.0, 1.1–8.0), aspiration (3.1, 1.9–5.0), and arterial hypoxemia (2.0, 1.3–3.0). Significant predictors of CAP-U MORT were age per decade (1.3, 1.1–1.5), systolic hypotension (2.6, 1.2–5.8), cancer (2.8, 1.7–4.5), dementia (2.8, 1.6–5.0), immunosuppression (2.0, 1.1–3.6), aspiration (3.6, 2.0–6.5), and multilobar x-ray infiltrates (1.6, 1.1–2.5). Significant predictors of all-cause MORT were age per decade (1.5, 1.3–1.6), male gender (1.5, 1.1–2.0), tachycardia (1.6, 1.1–2.5), systolic hypotension (1.8, 1.0–3.0), hypothermia (1.8, 1.1–2.9), altered mental status (1.9, 1.4–2.6), cancer (1.9, 1.4–2.7), immunosuppression (1.7, 1.1–2.6), dementia (1.9, 1.3–2.8), anemia (1.7, 1.1–2.4), elevated blood urea nitrogen (1.8, 1.3–2.4), arterial hypoxemia (1.7, 1.2–2.2), high-risk etiology (1.8, 1.2–2.6), and aspiration (3.4, 2.3–4.9).

CONCLUSION: In conclusion, this study demonstrates differences in the predictors of CAP-R and CAP-U MORT for patients with CAP. Given prior work demonstrating significant differences in the timing of CAP-R and CAP-U MORT, future studies of CAP outcomes should distinguish CAP-R and CAP-U MORT in addition to reporting all-cause MORT.

A COMPARISON OF ATTITUDES BETWEEN GENERAL INTERNAL MEDICINE PATIENTS AND ONCOLOGY PATIENTS REGARDING PHYSICIAN INQUIRY INTO SPIRITUAL AND RELIGIOUS BELIEFS. AK Moyer, C Puchalski, Internal Medicine, The George Washington University Medical Center, Washington, DC

PURPOSE: Different studies of Internal Medicine patients have shown that many patients think spiritual issues influence their health and that physicians should ask them about their religious and spiritual beliefs. National surveys and other studies have also shown that 65–75% of the general population would like their spiritual issues to be addressed by their physicians. Other studies have shown that spiritual beliefs have aided Oncology patients in coping with their illness and in creating a better quality of life. These types of results have suggested that spirituality may be an important dimension of patient centered care that needs to be addressed. However, there have been no studies that directly compare the attitudes of general Internal Medicine patients with a group of patients with more chronic illness such as cancer. This survey compares Internal Medicine patients with Oncology patients at an academic practice.

METHODS: Self-administered questionnaires were handed to patients arriving for appointments at an Oncology and an Internal Medicine outpatient clinic at a University teaching facility. The questionnaire included questions on demographics, self-rated health status, attitudes about the impact of spirituality/religion on health, whether or not doctors should ask patients about their spiritual/religious beliefs, a Duke Religiosity Index, and a Facit Spirituality Scale.

RESULTS: 106 Internal Medicine patients and 76 Oncology patients completed the questionnaires. 65% of the Oncology patients and 46% of the Internal Medicine patients agreed that physicians should ask patients about their religious and spiritual beliefs, a statistically significant difference between the two patient groups ($p = 0.02$). Factors associated with patients' agreement that doctors should ask patients about their spiritual/religious beliefs were better self-rated health status, higher spirituality scores, higher religiosity scores, and the belief that spiritual/religious faith and practice can have a positive impact on one's

mental health and physical health. No variable was found to explain the difference in agreement between the Internal Medicine patients and the Oncology patients.

CONCLUSION: This study shows that patients, particularly those with chronic illness such as cancer, would like their spiritual issues to be addressed by physicians. This is an interesting finding that this issue is more important for oncology patients than internal medicine patients. More research is needed to better understand why. This data supports the need to recognize patients spirituality in patient centered care.

AGE MODIFIES THE EFFECT OF SEX ON LONG-TERM SURVIVAL FOLLOWING ACUTE MYOCARDIAL INFARCTION. KJ Mukamal, MA Mittelman, Medicine, Beth Israel Deaconess Medical Center, Boston, MA

PURPOSE: Whether sex affects the long-term prognosis of acute myocardial infarction (AMI) is controversial. No previous study has formally examined age-related differences in the effect of sex on long-term prognosis. We therefore studied the associations of age and sex with long-term survival in early survivors of AMI enrolled in the Determinants of Myocardial Infarction Onset Study.

METHODS: We studied 1935 patients with AMI interviewed between August 1989 and September 1994 a median of four days following AMI. Trained interviewers performed standardized interviews and chart reviews to obtain demographic and clinical information. We searched the National Death Index for patient deaths through December 31, 1995. We used Cox proportional-hazards regression to estimate hazard ratios (HRs) for total mortality, adjusting for previous AMI, diabetes, hypertension, current smoking, former smoking, obesity, usual frequency of exertion, alcohol abstinence, use of thrombolytic therapy, use of cardiac medications, and congestive heart failure or ventricular tachycardia during hospitalization. We categorized age into three groups (<50, 50–69, 70+) and analyzed survival with dummy variables, within individual strata of age, and with an interaction term between age and sex.

RESULTS: The table shows adjusted HRs for total mortality with their 95% confidence intervals (CI) according to age and sex. Men under 50 represented the reference group for these comparisons. Women had higher mortality among subjects under 50, but men had higher mortality among subjects aged 70 and older. The interaction between age and sex was significant ($p = 0.04$). In adjusted stratified analyses, the HRs and 95% CI for total mortality for women relative to men were 1.4 (0.5–3.9) among patients under 50, 1.1 (0.7–1.7) among patients aged 50–69, and 0.6 (0.4–0.8) among patients aged 70 and older. Exclusion of patients who died within 30 days of AMI ($n = 44$) did not change our results.

CONCLUSION: In this post-AMI population, the effect of sex on long-term survival depended upon patient age. After multivariate adjustment, women seemed to have worse long-term survival than men. At older ages, however, women had better long-term survival than men. Further research is needed to clarify the mechanisms responsible for this interaction.

Age (years)	Less than 50	50 to 69	70 and older
Men	1.0 (reference)	1.4 (0.8–2.4)	3.7 (2.1–6.5)
Women	1.6 (0.7–4.0)	1.3 (0.7–2.5)	2.4 (1.3–4.4)

ALCOHOL CONSUMPTION AND SURVIVAL FOLLOWING ACUTE MYOCARDIAL INFARCTION. KJ Mukamal, MA Mittelman, Medicine, Beth Israel Deaconess Medical Center, Boston, MA

PURPOSE: Moderate drinkers have a lower risk of acute myocardial infarction (AMI) than abstainers. Whether moderate drinkers have a different long-term prognosis following AMI than abstainers is unknown. We therefore assessed the association of alcohol consumption and long-term survival among early survivors of AMI enrolled in the Determinants of Myocardial Infarction Onset Study.

METHODS: We studied 1935 subjects with AMI interviewed between August 1989 and September 1994 a median of four days following AMI. Trained interviewers assessed usual alcohol consumption with a standardized questionnaire. We searched the National Death Index for patient deaths through December 31, 1995. We used Cox proportional-hazards regression to estimate hazard ratios for total mortality, adjusting for age, sex, current smoking, former smoking, educational attainment, usual frequency of exertion, previous AMI, diabetes, use of thrombolytic therapy, and congestive heart failure during admission. We confirmed the proportionality of hazards with time-varying covariates.

RESULTS: Of the 1935 patients, 47% reported no alcohol consumption, 36% consumed less than 7 drinks weekly, and 17% consumed 7 or more drinks weekly. The unadjusted risks of death in these groups were 22%, 13%, and 9%, respectively. The table shows adjusted hazard ratios and 95% confidence intervals according to usual alcohol consumption. After adjustment, moderate alcohol consumption was associated with lower mortality following AMI. The effect of alcohol was similar in both men and women. Exclusion of subjects who died within 30 days of AMI ($N = 44$) did not change our results.

CONCLUSION: Moderate drinkers have lower long-term mortality following AMI than abstainers, regardless of sex.

Drinks per week	None	Less than 7	7 or more
All Subjects	1.0 (ref)	0.7 (0.5–0.9)	0.6 (0.4–0.9)
Men	1.0	0.7 (0.5–1.0)	0.6 (0.4–1.0)
Women	1.0	0.6 (0.3–0.9)	0.2 (0.0–1.2)

REGULAR MAMMOGRAPHY USE IN WOMEN IN THE FRAMINGHAM OFFSPRING STUDY. JM Murabito, MG Larson, JC Evans, KM Freund, BE Kregar, GL Splansky, PWF Wilson, MA Moskowitz, NHLBI's Framingham Heart Study and Section of General Internal Medicine, Boston Medical Center, Boston, MA

PURPOSE: To describe regular mammography use in a population-based sample of women.

METHODS: 683 women participating in the Framingham Offspring Study completed a breast health questionnaire in 1996-97. Two measures of regular mammography use were created: 1) meeting age-appropriate guidelines, that is, having mammography use since age 40 years equal to or exceed the recommended number of mammograms for one's age; and 2) meeting guidelines for age-appropriate use since the first mammogram (if after age 40 years). Multiple imputation was used to assign values for those women with incomplete data for age at first mammogram or total number of mammograms. Five hundred replicates were simulated for each incomplete observation, using normal random deviates, to estimate compliance distributions.

RESULTS: Almost all women had at least one mammogram, however, recent (≤ 2 years) mammography use was lower than ever use. Compliance with both measures of regular use was substantially less than ever or recent use and dropped off with age. Even after women undergo their first mammogram, compliance with regular mammography use is low, particularly in older women.

Age group	N	Ever	Recent ≤ 2 yrs	Compliance Since Age 40	Compliance Since First
40-49	93	95%	72%	46%	57%
50-59	268	96%	88%	26%	44%
60-69	222	96%	88%	11%	26%
70-79	100	94%	79%	3%	14%

CONCLUSION: Women are not undergoing mammography according to screening guidelines even after their first mammogram. Efforts to improve regular mammography use are needed.

EFFECT OF FOR-PROFIT STATUS ON LENGTH OF STAY IN HOME CARE. RL Murkofsky, EP McCarthy, RB Davis, RS Phillips, Division of General Medicine and Primary Care, Beth Israel Deaconess Medical Center, Harvard Medical School, Boston, MA

PURPOSE: Prior to the 1997 Balanced Budget Act, home health agencies (HHAs) were reimbursed primarily on a fee-for-service basis. We examined whether the for-profit status of HHAs affected length of stay (LOS) in home care in 1996.

METHODS: We performed a secondary data analysis of current and discharged home care patients reported in the 1996 National Home and Hospice Care Survey. Data were collected on a sample of current and discharged patients from representative home health and hospice agencies. We analyzed patient demographics, diagnoses, dependency in activities of daily living (ADLs) and instrumental activities of daily living (IADLs), and the for-profit status of agencies. We developed a Cox Proportional Hazards model for LOS (days) in home care, using SUDAAN to adjust for the complex sampling design.

RESULTS: In 1996, an estimated 10.2 million patients received home care in the US (64% female, 61% white, 62% Medicare, 68% ≥ 65 years). Of these patients, 12% had CHF, 11% COPD, 15% diabetes, and 11% cancer; 40% had one or more ADL dependencies, 34% had one or more IADL dependencies, and 34% received care from for-profit HHAs. The median calculated Charlson-Deyo Comorbidity Index score was 1. The mean LOS for current home care patients was 332 days, while the mean LOS for discharged patients was 98 days. Among those discharged, 68% met their goals and 4% died. The remainder were admitted to a hospital (11%) or nursing home (4%), moved or changed HHAs (6%), or had another reason for discharge (7%).

After adjusting for sex, race, living alone, referral source, Charlson-Deyo Comorbidity Index, COPD, insurance type and census region, we found the following factors to be significantly associated with increased LOS in home care: For-profit HHA (Hazard Ratio for HHA discharge 0.77 [95% CI: 0.61, 0.97]); age < 18 years (0.52 [0.32, 0.86]), 45-64 years (0.75 [0.57, 0.99]), 65-74 years (0.71 [0.54, 0.93]), 75-84 years (0.65 [0.50, 0.84]), and 85 years or older (0.60 [0.44, 0.82]) (all compared to age group 18-44 years); non-metropolitan residence (0.74 [0.61, 0.91]); diabetes (0.79 [0.68, 0.93]); dependency in IADLs (0.89 [0.80, 1.00] per dependency) and ADLs (0.93 [0.90, 0.97] per dependency). The following factors were associated with decreased LOS in home care: cancer (1.63 [1.28, 2.08]), and referral by a hospital (1.30 [1.04, 1.62]).

CONCLUSION: After adjusting for patient age, diagnoses, function, and area of residency, receiving care from a for-profit home health agency in 1996 was associated with longer length of stay in home care. These data suggest that the projected reductions in home care funding imposed by the 1997 Balanced Budget Act will differentially affect length of stay in for-profit home health agencies.

THE PREVALENCE OF HEPATITIS C IN AN URBAN PRIMARY CARE PRACTICE. B Murphy, J Andrilli, P Bushkuhl, General Medicine, Saint Vincents Hospital of New York, New York, NY

PURPOSE: To assess the prevalence of hepatitis C antibody among a randomly selected group of primary care patients in New York City and to gauge associated risk factors for acquiring the disease.

METHODS: Patients were selected from 3 primary care practices in New York City. Patients represented all 5 boroughs of the City and 200 were randomly selected using a random number generator. Sample size was computed for an $\alpha = 0.05$ and $\beta = 0.20$. Chi-square analysis was performed to assess statistical significance. A $p < 0.05$ was deemed statistically significant.

RESULTS: Of 200 patients screened, 16 (8%; $p < 0.001$) were positive for exposure to hepatitis C based on ELISA screening and RIBA confirmatory testing. Only one of the 16 knew they were positive prior to testing. Of those testing positive, 75% were male and the average age was 47yo. All patients testing positive for hepatitis C tested positive for prior exposure to hepatitis A ($p < 0.01$) and 13 of 16 had antibody to hepatitis B ($p < 0.01$). There were no chronic hepatitis B carriers among those positive for hepatitis C. Two patients were co-infected with HIV. Half of the hepatitis C patients were positive for antiphospholipid antibody. The average hepatitis C viral load was 75,000 IU/cc. Only 1 of the 16 hepatitis C patients (6%; $p < 0.01$) had an abnormally elevated ALT. None of the hepatitis C patients ever received a blood transfusion; 38% reported a prior history of illicit drug use; 25% reported a possible exposure to body fluids infected with hepatitis C; 37% reported no known risk factors for acquiring hepatitis C.

CONCLUSION: The prevalence of hepatitis C in large urban areas may far exceed that of the national average of 1.8%. Physicians should be aware that many patients may not have biological markers suggesting disease and thus follow-up of historical clues relating to prior drug use, multiple sexual partners, recurrent thrombotic events, or exposure to other viral diseases should prompt testing for hepatitis C exposure.

CORRELATION BETWEEN HEALTH-RELATED QUALITY OF LIFE MEASUREMENTS AND CLINICAL PARAMETERS IN HEMODIALYSIS PATIENTS. M Mya, M Klein, C Yuscak, W Lee, Internal Medicine; Nephrology, Westchester Medical Center; Social Worker, Bradhurst Dialysis Center, Valhalla, NY

PURPOSE: Past studies have demonstrated a powerful correlation between morbidity, mortality and patient-assessed health-related quality of life in hemodialysis patients as measured by medical outcomes study short form-36, also called SF-36. The aim of this study was to identify which of the clinical parameters routinely measured in ESRD patients on hemodialysis correlate with patient-assessed health-related quality of life.

METHODS: A retrospective chart review was done on 74 patients dialyzed at a single dialysis center who had completed a SF-36 between June, 98-March, 99. 13 charts were excluded due to incomplete response. Personnel instructed on the use and purpose of the SF-36 (in this study the social worker administered the survey to new patients and to other patients every 6 months. We scored the surveys by computer and filed the results in the medical records. Only results of the first survey of each patient completed in the study window were used for this analysis. The average hemoglobin, albumin levels, total hours of hemodialysis, urea reduction ratio [URR] of each patient measured during a month prior to the date of SF-36 survey were obtained. We used multivariate regression method to control for all covariates and compare with physical and mental summary scores. Of 61 patients, 24 were women, 37 were men. 39% had diabetes, 72% had hypertension and 21% had coronary diseases.

RESULTS: Mean age of the patients was 59 years. Mean hemoglobin was 11.3 g/dl. Mean albumin level was 4.3. Average time of dialysis in a month was 43.4 hours. Mean URR was 68.3%. Mean physical summary score was 35.4 $\pm$ 11.5, mean mental summary score was 50.4 $\pm$ 11.7. In this multivariate regression analysis, there were no statistically significant correlation between mental summary score and albumin, URR, hours of hemodialysis per month. No statistically significant correlation was found between physical summary score and any of the variables. Age and hemoglobin showed positive correlation with mental summary score, with P value of 0.003 and 0.01 respectively. An increase in hemoglobin 1 g/dl correlated with increase in mental summary score of 3.2 with P value 0.01. Every increase in age by one year correlated with increase in mental summary score of 0.36 with P value of 0.003.

CONCLUSION: Association between hemoglobin and mental summary scores suggested that sense of mental wellbeing as measured by SF-36 is higher in patients with higher hemoglobin levels. Whether improving hemoglobin with therapy affects the mental wellbeing needs to be assessed with larger prospective study needs to be assessed. A small sample size and effect of other comorbid conditions are our limitations.

THE POTENTIAL CLINICAL AND ECONOMIC EFFECTS OF HOMOCYSTEINE LOWERING: A DECISION ANALYSIS. BK Nallamothu, AM Fendrick, M Rubenfire, S Saint, RR Bandekar, GS Omenn, Department of Internal Medicine; Department of Biostatistics, University of Michigan, MI

PURPOSE: Elevated total homocysteine (tHcy) levels have been identified as a risk factor for coronary heart disease. Because clinical trials have just recently begun, the benefits of lowering tHcy levels with folic acid and vitamin B₁₂ will not be known for several years. We therefore used decision analytic modeling to estimate the likely clinical and economic effects of tHcy lowering with folic acid and vitamin B₁₂.

METHODS: We modeled 3 strategies: 1) No Intervention-no screening, no treatment; 2) Treat All-no screening, daily folic acid (400mcg) and vitamin B₁₂ (500mcg) for all; 3) Screen & Treat-screening, followed by daily folic acid and vitamin B₁₂ for individuals with elevated tHcy levels (≥ 11 μ mol/L). Each strategy was assessed in 2 simulated cohorts: 40 year-old men and 50 year-old women. First, a decision

tree using clinical inputs and cost estimates specific to each strategy was used to determine 1) the likelihood of identifying elevated tHcy levels (if screening occurred); 2) the effectiveness of folic acid at lowering tHcy levels; and 3) initial costs. Next, Markov simulations incorporating age, gender and CHD-specific mortality rates were generated to estimate the effect of tHcy-lowering on survival and ongoing costs. The published literature was reviewed to determine the best estimates for the Base-Case Analysis, while sensitivity analysis was used to evaluate uncertain estimates across a broad range of potential values. Cycles during the simulation were one year in length with subjects followed until age 85 or death. The outcomes we assessed were overall life expectancy in discounted life-years and total costs in discounted 1998 dollars.

RESULTS: If lowering tHcy with folic acid and vitamin B₁₂ reduced excess coronary heart disease risk by 40% (Base-Case Analysis), the two intervention strategies cost between \$13,000 and \$77,000 per life-year saved when compared to No Intervention. Although the Treat All strategy was slightly more effective overall, the Screen & Treat strategy resulted in a much lower cost per life-year saved (\$13,700 in men; \$27,500 in women) when compared to No Intervention. Incremental cost-effectiveness ratios for the Treat All strategy when compared to the Screen & Treat strategy were over \$500,000 per life-year saved in both cohorts. The Screen & Treat strategy remained the most cost-effective intervention under multiple scenarios evaluated during sensitivity analysis.

CONCLUSION: Lowering tHcy with folic acid and vitamin B₁₂ could result in substantial clinical benefits at reasonable costs. If tHcy lowering is considered, a Screen & Treat strategy is likely to be more cost-effective than universal folic acid supplementation.

ELECTRON-BEAM COMPUTED TOMOGRAPHY IN THE DIAGNOSIS OF CORONARY ARTERY DISEASE: A META-ANALYSIS. BK Nallamothu, S Saint, LF Bielak, M Rubenfire, PA Peyser, SS Sonnad, AM Fendrick, Department of Internal Medicine; Department of Epidemiology; Department of Surgery, University of Michigan, MI

PURPOSE: Electron-beam computed tomography (EBCT) is a new, non-invasive method of obtaining cross-sectional images of the heart that is increasingly being advocated as a method to diagnose coronary artery disease. Before clinical use of EBCT can be justified, however, its overall diagnostic accuracy must be clearly defined. We thus undertook a formal meta-analysis to estimate the accuracy of EBCT in the diagnosis of significant coronary artery disease.

METHODS: English-language studies published from 1979 through 1999 were retrieved using MEDLINE, Current Contents, bibliographies, and consultation with experts. We included a study if it: 1) used EBCT as a diagnostic test for significant coronary artery disease ($\geq 50\%$ stenosis); 2) reported results in absolute numbers of true-positive, false-positive, true-negative, and false-negative cases; and 3) used coronary angiography as the reference standard for determining the presence of significant coronary artery disease. Data from the included reports was combined using summary receiver operating characteristic (ROC) analysis.

Results from 14 EBCT reports, with a total of 1662 subjects (71% men), were combined. Study groups in all instances consisted of individuals undergoing coronary angiography for evaluation of coronary artery disease. The mean number of subjects per report was 119. Reported sensitivity ranged from 81% to 99% and specificity ranged from 21% to 83%. From its summary ROC curve, the maximum joint sensitivity and specificity rate for EBCT was found to be 75%. Overall discriminatory power for EBCT, summarized as a diagnostic odds ratio, was 9.2 (95% confidence interval, 3.9 to 21.4). The use of different thresholds for a positive test in the included studies had only a small and non-significant association with the test's overall discriminatory power. For a threshold value that results in a sensitivity of 90%, specificity can be expected to be 54%; if sensitivity is 80%, specificity will likely be 71%. We found no evidence that the accuracy of EBCT was different in specific age or gender groups.

CONCLUSION: Based on our meta-analysis, EBCT appears to be sufficiently accurate at identifying coronary artery disease in individuals, with sensitivity and specificity rates that are comparable to traditional exercise stress testing. Further studies are needed to determine the precise role of EBCT among the current armamentarium of noninvasive tests for coronary artery disease. Until then, our study provides clinicians with estimates of the overall accuracy of EBCT, allowing them to better interpret results from this rapidly diffusing diagnostic innovation.

INDEPENDENT EFFECTS OF CARDIOVASCULAR DISEASE AND DIABETES ON ACUTE HEALTH CARE UTILIZATION AND HEALTH STATUS. S Natarajan, WD Bradford, TV Kleckley, MD Silverstein, Center for Health Care Research; Division of General Internal Medicine, Medical University of South Carolina; Ralph H. Johnson VAMC, Charleston, SC

PURPOSE: Current guidelines from the American Diabetes Association recommend treating patients with diabetes as intensely as patients with established cardiovascular disease. However the comparative risk for adverse events requiring acute health care in patients with the two conditions is not clear. The aim of this study was to compare the independent effect of diabetes and cardiovascular disease on acute health care utilization and health status. Our hypothesis was that patients with cardiovascular disease would have greater health care utilization and lower perceived health status than patients with diabetes.

METHODS: Data were analyzed from the 1996 Medical Expenditure Panel Study (MEPS), a nationally representative survey of medical care use and costs among more than 23,000 Americans. The MEPS allows evaluation of the impact of patient and family characteristics on medical care use and spending. It includes 841

diabetics and 463 patients with cardiovascular disease. Our indicator for cardiovascular disease included patients with coronary heart disease, cerebrovascular disease and peripheral vascular disease. We examined the independent effect of diabetes and cardiovascular disease on health care utilization and perceived health status while adjusting for age, sex, race, education, and occupation. Multivariate logistic regression analyses were performed using SAS which does not take into account the complex sampling frame. Results are expressed as odds ratios (OR) with 95% confidence intervals (CI)

RESULTS: In 1996, emergency room visits were more likely in patients with cardiovascular disease (OR 3.5, 95% CI 2.7-4.5) than in patients with diabetes (OR 1.7, 95% CI 1.4-2.1). Similarly the likelihood of hospitalization was also much greater in patients with cardiovascular disease (OR 5.9, 95% CI 4.6-7.5) than in persons with diabetes (OR 2.1, 95% CI 1.7-2.6). However when perceived health status was examined, the odds of patients with cardiovascular disease reporting superior health status was greater (OR 0.33, 95% CI 0.26-0.43) than in patients with diabetes (OR 0.22, 95% CI 0.19-0.28).

CONCLUSION: Patients with cardiovascular disease had significantly greater acute health care utilization than patients with diabetes. Current guidelines may need to be refined to match the intensity of management to risk of adverse outcomes needing acute care. Though patients with diabetes had lower health care utilization than patients with cardiovascular disease they had worse perceived health status.

IMPACT OF MULTIPLE CARDIOVASCULAR RISK FACTORS ON ACUTE HEALTH CARE UTILIZATION AND HEALTH STATUS. S Natarajan, PJ Nietert, TV Kleckley, MD Silverstein, Center for Health Care Research; Division of General Internal Medicine, Medical University of South Carolina; Ralph H. Johnson VAMC, Charleston, SC

PURPOSE: To examine the impact of multiple cardiovascular disease risk factors such as hypertension (HTN), diabetes (DM) and high cholesterol (HC) on acute health care utilization and health status. We hypothesized that patients with multiple risk factors would have greater health care utilization and lower perceived health status than patients with a single risk factor.

METHODS: Data were analyzed from the 1996 Medical Expenditure Panel Study (MEPS), a nationally representative survey of medical care use and costs among more than 23,000 non-institutionalized Americans. It includes 2,199 hypertensives, 841 diabetics, and 625 persons with high cholesterol. We examined the independent effect of HTN, DM, HC and combinations of these on health care utilization and perceived health status while adjusting for age, sex, race, education, occupation, and veteran status. Multivariate logistic regression was performed using SUDAAN to account for the complex sampling design. Comparisons were made to persons without cardiovascular risk factors and reported as odds ratios (OR) with 95% confidence intervals (CI).

RESULTS: Among the risk factors, DM had the greatest impact with significantly greater emergency room visits (OR 2.1, 95% CI 1.6-2.8), hospitalizations (OR 2.6, 95% CI 1.9-3.5) and lower odds of reporting superior health status (OR 0.23, 95% CI 0.17-0.30). HC had the least impact while HTN was intermediate. Among persons with multiple risk factors, emergency room visits were more likely for patients with DM and HTN (OR 1.8, 95% CI 1.3-2.4), DM and HC (OR 2.0, 95% CI 0.8-5.2), HTN and HC (OR 1.4, 95% CI 0.9-2.1) and in patients with DM, HTN and HC (OR 2.1, 95% CI 1.1-4.0). Hospitalization was also greater in patients with DM and HTN (OR 2.6, 95% CI 1.9-3.6), DM and HC (OR 2.0, 95% CI 0.8-4.9) HTN and HC (OR 1.4, 95% CI 0.9-2.3) and in patients with DM, HTN and HC (OR 2.6, 95% CI 1.3-5.3). The odds of reporting superior health status were significantly lower in patients with DM and HTN (OR 0.13, 95% CI 0.09-0.18), DM and HC (OR 0.3, 95% CI 0.13-0.68), HTN and HC (OR 0.44, 95% CI 0.31-0.62) and in patients with DM, HTN and HC (OR 0.07, 95% CI 0.03-0.2).

CONCLUSION: Among the cardiovascular risk factors we examined, diabetes is the driver of increased health care utilization and decreased perceived health status. Multiple risk factors were not synergistic in that we found no significant increase in health care utilization nor significantly decreased health status in patients with multiple risk factors compared to patients with a single risk factor.

OUTCOMES ASSOCIATED WITH OMISSION OF RADIOTHERAPY AFTER BREAST-CONSERVING SURGERY, AMONG OLDER WOMEN WITH EARLY STAGE BREAST CANCER. AB Nattlinger, RG Hoffmann, RT Kneusel, MM Schapira, General Internal Medicine; Biostatistics, Medical College of Wisconsin, Milwaukee, WI

PURPOSE: Previous studies have shown that substantial numbers of older women who undergo Breast-Conserving Surgery (BCS) for early stage breast cancer do not receive the recommended radiotherapy (RT). We evaluated whether omission of RT adversely affected outcomes among older women who underwent BCS.

METHODS: We studied a population-based cohort of 2,781 women aged 65 or older included in the Surveillance, Epidemiology, and End Results (SEER) registry, who underwent mastectomy or BCS treatment in 1986-87 for local or regional breast cancer, and for whom Medicare part A and B claims were available for at least 6 years following diagnosis (or until death). Since SEER does not collect information on disease recurrence, disease recurrence was defined as Medicare claims for mastectomy ≥ 6 months after diagnosis, RT ≥ 8 months after diagnosis, or chemotherapy ≥ 15 months after diagnosis. A proportional hazards model was used, and analyses were adjusted for age, stage, race, and socioeconomic status.

RESULTS: Initial treatment was mastectomy in 77% subjects, and BCS in 13% subjects. Of those treated with BCS, 55% underwent RT. Stratifying by stage, women who underwent BCS without RT had a significantly higher risk of recur-

rent disease (OR = 1.38, $p = 0.037$) compared to women who underwent mastectomy. Women who underwent BCS with RT had a recurrence rate that was similar (OR = 0.91, $p = 0.45$) to women undergoing mastectomy. As expected, there was a higher probability of recurrence among regional stage patients. However, the odds ratios for recurrence for women who underwent BCS without RT vs mastectomy were consistent within each stage (OR = 1.36 for local stage, OR = 1.32 for regional stage.) To help evaluate the face validity of our claims-based methodology for determining disease recurrence, we studied disease recurrence as a predictor of death. In the cohort as a whole, subjects who developed recurrent disease had a significantly greater risk of dying. For all cause mortality, the OR = 2.33 ($p = 0.0001$) for women with recurrent disease. For breast cancer specific mortality, the OR = 7.84 ($p = 0.0001$).

CONCLUSION: As has been found in the randomized trials of younger women, this study found omission of RT with BCS to be associated with an elevated risk of disease recurrence. This population-based study suggests poorer outcomes among older women who undergo BCS without RT.

DOES FOOD INSUFFICIENCY AFFECT HEALTH STATUS AND HEALTH CARE UTILIZATION AMONG DIABETICS? DATA FROM NHANES III. K Nelson, W Cunningham, R Andersen, G Harrison, L Gelberg, Division of General Internal Medicine and Health Services Research, Department of Medicine, School of Medicine; Department of Health Services, School of Public Health; Department of Family Medicine; Department of Community Health Sciences, School of Public Health, UCLA, Los Angeles, CA

PURPOSE: Preliminary clinical studies have shown that food insufficiency has adverse health consequences, including hypoglycemic episodes and increased need for health care services, among adult diabetics. The purpose of this study is to determine the prevalence of food insufficiency and describe the association of food insufficiency with health status and health care utilization in a national sample of diabetics.

METHODS: We analyzed data from diabetics ($n = 1503$) interviewed in the Third National Health and Nutrition Examination Survey (NHANES III), a cross-sectional representative sample of the civilian non-institutionalized population living in households. Individuals were classified as food insufficient if a family respondent reported that the family sometimes or often did not get enough to eat or that adults in the family had to cut down on the size of their meals because of financial constraints. Both bivariate and multivariate analysis were used to examine the impact of food insufficiency on self-reported health status, hospitalizations, and physician utilization.

RESULTS: Six percent of diabetics reported food insufficiency, representing over 568,600 diabetics nationally (95% CI 368,400; 768,800). Diabetics who were food insufficient were more likely to report fair or poor health status than those who were not food insufficient (63% vs. 43%, OR 2.2, $p = .05$). In a multivariate analysis, fair or poor health status was independently associated with non-white race, poverty, low educational achievement, and number of chronic diseases, but not with food insufficiency. Diabetics who were food insufficient reported more physician encounters, either in clinic or by phone, than those who were food secure (12 vs. 7, $p < .05$). In a multivariate linear regression, food insufficiency remained independently associated with increased physician utilization among diabetics. There was no significant association between food insufficiency and rates of hospitalization.

CONCLUSION: Food insufficiency is relatively common in this national sample of diabetics and increased physician utilization and reduced health status. Prospective studies are needed to understand the impact of food insufficiency on the health status of and health care utilization by diabetics.

FAILURE TO DIAGNOSE AND TREAT PRIMARY CARE PATIENTS WITH OSTEOPOROTIC VERTEBRAL COMPRESSION FRACTURES. JM Neuner, JK Zimmer, MB Hamel, Division of General Internal Medicine, Beth Israel Deaconess Medical Center, Boston, MA

PURPOSE: Many experts define a non-traumatic vertebral compression fracture as an osteoporotic fracture. As treatments are available to reduce the risk of recurrent osteoporotic fractures, we sought to determine whether osteoporosis was diagnosed or treated in primary care patients (pts) with non-traumatic compression fractures.

METHODS: To identify pts with vertebral compression fractures cared for in a hospital-based primary care internal medicine practice, we performed a computerized search for the terms vertebral "compression," "wedging," or "loss of height" from all chest, thoracic and lumbosacral spine X-rays performed between 1/1/97 and 1/1/99 as part of usual care. We excluded pts with X-rays or medical records indicating trauma (defined as fall from more than standing height) or metastatic cancer. We reviewed pt's outpatient medical records for demographics, risk factors for osteoporotic fractures, comorbid illnesses, and diagnosis or treatment of osteoporosis. To confirm non-prescription therapy, we conducted telephone surveys in 12/99.

RESULTS: Of 171 pts (median age 76), 66% were female, 12% had taken corticosteroids, 32% had an X-ray ordered for back pain or a fall, and 56% had multiple compression fractures. Primary care physicians (PCPs) documented fractures for only 32% of pts. As some pts were diagnosed with osteoporosis without documentation of compression fractures, 39% of pts were diagnosed with osteoporosis overall. Bone densitometry was performed for 21% of pts between 1992-99. A minority of pts was treated for osteoporosis: 42% were taking calcium, 35% were taking a multivitamin/vitamin D, and 37% were offered ≥ 1 prescription drug (20% alendronate, 12% calcitonin, 23% estrogen). Rates of prescription treatment

were higher when osteoporosis was diagnosed (76% when diagnosis made vs 12% when diagnosis not made). In multivariable analysis adjusted for age and using the generalized estimating equation to account for clustering by provider, pt characteristics independently associated with receiving prescription treatment were: female sex (AOR 2.2, 95% CI 1.0, 4.6), steroid use (7.9 [2.9, 21.0]), back pain or fall as indication for X-ray (5.7 [2.4, 13.6]), ≥ 2 X-rays with fracture (2.9 [1.3, 6.9]), lower weight (0.9 [0.8, 1.0] per ten pound increase) and fewer comorbidities (0.7 [0.5, 0.9] per unit increase in Charlson comorbidity score).

CONCLUSION: Most pts with non-traumatic vertebral compression fractures were not treated for osteoporosis. This suboptimal care is explained partially by PCP's failure to diagnose osteoporosis after a vertebral compression fracture was described on an X-ray report. Education of PCPs and radiologists about the diagnosis and treatment of osteoporosis and better communication between physicians may improve care for pts with osteoporosis.

EFFECTIVENESS OF TOBACCO INTERVENTIONS IN MINORITY COMMUNITIES. Q Ngo-Metzger, R Phillips, R Davis, J Reede, Division of General Medicine and Primary Care, Beth Israel Deaconess Medical Center; Faculty Development and Diversity, Harvard Medical School, Boston, MA

PURPOSE: Tobacco use is a leading cause of death in the US, especially in minority populations. Increasing the unit price of tobacco (IUP) and community-wide interventions (CWI) are two methods to deter tobacco use. We conducted a systematic review of the published literature to determine whether evidence exists supporting the use of these interventions in minority communities.

METHODS: We searched the literature for the years 1976-1998, using 3 electronic databases (Medline, EconLit, and CDC's Office of Smoking and Health Database). Inclusion criteria were: 1) English language publications; 2) population-level interventions; and 3) outcomes measures for tobacco use. A second search was done using the key terms: Asian, Black, Hispanic, Minority. Only 1 of 62 IUP papers and 18 of 174 CWI papers focused on minorities. We reviewed the articles using a standardized study abstraction form developed by the Task Force for the Guide to Community Preventive Services. Abstracted data included components of the intervention, characteristics of the study populations, and exposure and outcome measurements. Quality of study design and execution were evaluated by two independent reviewers. Discrepancies were resolved by consensus.

RESULTS: IUP was found to be an effective intervention for decreasing cigarette sales to Hispanic (H) and African American (AA) smokers. After controlling for income and education, price elasticity of demand (% change in demand resulting from 1% change in price) was -1.89 for H and -0.32 for AA smokers, compared with -0.14 for white smokers. This study is consistent with 44 other studies done in non-minority populations. The table below describes CWI in the 6 studies representing the best evidence subset (Good or Fair Quality Rating Score):

Author	Study Design	Target Population	Numbers	Odds of Quitting
Resnicow, 1997	Randomized Trial	African Americans	1,154	OR = 1.4 (0.9, 2.1)
Jason, 1988	Randomized Trial	African Americans	137	OR = 2.7 (1.0, 7.2)
McAlister, 1992	Non-Randomized Trial	Hispanic Americans	295	OR = 2.5 (1.2, 5.4)
Marin, 1994	Time Series	Hispanic Americans	7,667	OR = 1.1 (0.6, 1.9)
McPhee, 1995	Non-Randomized Trial	Vietnamese Americans	5,376	OR = 1.1 (0.7, 1.7)
Jenkins, 1997	Non-Randomized Trial	Vietnamese Americans	4,873	OR = 1.7 (1.3, 2.2)

CONCLUSION: Increasing the unit price of tobacco is an effective intervention to deter smoking among Hispanic and African American smokers. Community-wide interventions may increase smoking cessation, although only 2 of 6 studies achieved statistical significance. These data suggest the need to increase the unit price of tobacco and to further study community-wide interventions in minority communities.

USEFULNESS OF PRE-BIOPSY CLINICAL DIAGNOSIS FOR PATIENTS WITH RENAL DISEASES. H Nomura, I Koni, E Furui, J Koizumi, Department of General Medicine; 2nd Department of Internal Medicine, Kanazawa University Hospital, Kanazawa, Ishikawa, Japan

PURPOSE: To determine the usefulness of making a pre-biopsy clinical diagnosis in patients with renal diseases.

METHODS: Clinical information and diagnoses were recorded by the physicians and submitted with the 1,591 renal biopsy specimens to the renal pathology lab during 1992-1999. Physicians were encouraged to select one of the following clinical diagnoses: Acute Nephritic Syndrome (ANS), Acute Renal Failure (ARF), Chronic Nephritic Syndrome (CNS), Chronic Renal Failure (CRF), Diabetic Nephropathy (DN), Nephrotic Syndrome (NS), Persistent Hematuria (PH), Rapidly Progressive Glomerulonephritis (RPGN), and Lupus Nephritis (LN). Clinical diagnoses were assumed to be made for predicting combinations of pathological diagnoses with similar therapeutic preferences detailed in the table. Positive post-test probabilities (Prob.) were calculated based on the frequencies of each pathological diagnosis cluster appearing in the database.

RESULTS:

Clinical Dx	Pathological Dx	N	Spec.	Sens.	LR(+)	Prob.
ANS	Acute Glomerulonephritis	40	.982	.524	29.1	.286
ARF	ATN, Acute Cortical Necrosis	63	.971	.947	32.7	.286
CNS	Mes PGN, Nephrosclerosis	814	.675	.738	2.3	.655
CRF	End-Stage Kidney	21	.987	.100	7.7	.000
DN	Diabetic Glomerulosclerosis	11	.999	.173	173	.855
NS	MCNS, MN, MPGN, FSGS	298	.918	.658	8.0	.643
PH	Mild Mes PGN, Minor Abnormality	226	.911	.500	5.6	.444
RPGN	Crescentic Glomerulonephritis	55	.984	.408	25.5	.565
LN	Lupus Nephritis	58	.995	.785	157	.872

Spec., Specificity; Sens., Sensitivity; LR(+), Positive Likelihood Ratio; ATN, Acute Tubular Necrosis; Mes PGN, Mesangium-proliferative Glomerulonephritis; MCNS, Minimal Change Nephrotic Syndrome; MN, Membranous Nephropathy; MPGN, Membranoproliferative Glomerulonephritis; FSGS, Focal Segmental Glomerulosclerosis

CONCLUSION: Post-test probabilities were not sufficiently high for any clinical diagnosis entities, suggesting that making a pre-biopsy clinical diagnosis, at least with the classification used in this study, is not useful in predicting pathological diagnosis clusters with similar therapeutic preferences. Caution should be used in interpreting these results since in some clinical diagnosis entities such as DN and PH, renal biopsies might have been avoided assuming the clinical diagnoses were accurate.

STREET DEATHS AMONG BOSTON'S HOMELESS PERSONS: JULY 1998 THROUGH JANUARY 1999. JJ O'Connell, SE Swain, Boston Health Care for the Homeless Program, Boston, MA

PURPOSE: In February and March 1999, the Boston Health Care for the Homeless Program (BHCHP) conducted a review of the circumstances surrounding the deaths of 12 homeless men and 1 homeless woman (a second woman remains unidentified in the city morgue) who died on the streets of Boston between July 1998 and January 1999.

METHODS: Medical charts of these individuals at BHCHP's two hospital based primary care clinics as well as records from a detoxification program where the primary author serves as medical director were reviewed. Of note this study has not yet been able to review the utilization of the mental health system by this group.

RESULTS: The review revealed that not only did these individuals suffer from complex medical conditions but that 100% suffered from chronic substance abuse (ETOH was the primary substance of choice in 12 of the 13) and two-thirds suffered from severe mental illness. In addition the review showed multiple emergency room visits and hospitalizations particularly in the six months prior to death. Strikingly, 9 of the 13 (69%) had been seen in the emergency room or admitted to the hospital within three weeks of the day of death. Many had been seen multiple times, with several seen within 72 hours of death. Half of those who died had also been admitted to the Barbara McInnis House (BHCHP's medical respite facility) with acute medical problems during the six months prior to their deaths. This high frequency of utilization of the health care system was paralleled by a dramatic number of admissions to publicly-funded detoxification programs. Virtually all of the decedents had multiple admissions during the year prior to death. Ten of the 13 (77%) had received detoxification from alcohol and other drugs within 6 weeks of death, with 3 (23%) having died on the streets within one week of discharge from a detoxification unit.

CONCLUSION: The homeless persons who died on the streets suffered from an unusually high "tri-morbidity" of acute and chronic medical illnesses, persistent mental illness, and relapsing addictions to alcohol and other substances. Contrary to our original hypothesis that these individuals had fallen through the safety net, this group was not only well-known to homeless providers on the streets, but also had extraordinarily high utilization patterns of the medical care system (especially emergency rooms, acute care hospitals, and respite care) as well as the substance abuse system (primarily the acute detoxification units). Additional study is needed to understand the utilization of the mental health system by these individuals.

PSYCHOSOCIAL VARIABLES AND SUBCLINICAL CORONARY ARTERY DISEASE IN A SCREENING POPULATION. PG O'Malley, AJ Taylor, DL Jones, General Internal Medicine and Cardiology, Walter Reed Army Medical Center, Washington, DC

PURPOSE: The relationship between depression, anxiety, and hostility has been well studied in clinical coronary artery disease (CAD), but not subclinical CAD. We studied the relationship between multiple psychosocial variables and subclinical CAD, as represented by coronary calcification

METHODS: Cross-sectional study of a prospective, consecutive sample of 630 consenting, active duty U.S. Army personnel (39-45 y.o.) without known CAD, who were undergoing a routine physical as required by regulations. Each participant was surveyed with validated instruments to assess for depression, anxiety, somatization, stress, and hostility level. Predicted risk was assessed using the Framingham risk index, based on measured conventional CAD risk predictors. Subclinical CAD was determined using EBCT to quantify coronary artery calcifica-

tion (CAC; Agatston method). Assessments of psychosocial factors and CAC were blinded to each other.

RESULTS: Of 700 personnel screened, 630 agreed to participate. The mean age was 42; 83% were male, 72% white, 83% college graduates, and 76% officers. The 10 year predicted risk of clinical CAD was 4.5%. The prevalence of CAC was 19% (mean CAC = 10.3). There was a modest relationship between predicted risk and CAC (ROC area 0.61, $p = .004$). The prevalence of prior or current criteria-based DSM-IV psychiatric disorders was 14.6% (depression: 6.3%, anxiety: 1.4%, somatoform disorder: 9.5%, with some overlap). As a group, 90% of participants had hostility scores lower than the median normative scores. There was no correlation between CAC score and depressive ($r = -0.05$, $p = 0.25$), anxiety ($r = 0.01$, $p = 0.8$), somatization ($r = -0.08$, $p = 0.14$), stress ($r = -.04$, $p = 0.45$) or hostility ($r = -0.06$, $p = 0.24$) scores. Similarly, there was no correlation between any of these psychosocial variables and predicted risk. Those without psychiatric disorders had higher CAC than those with prior or current psychiatric disorders (11.6 vs 1.3; $p = 0.0001$).

CONCLUSION: In a relatively young screening population, psychosocial factors were not related to the presence or extent of early, subclinical atherosclerosis. These data suggest that any association between psychosocial factors and coronary events would be through mechanisms other than the development of atherosclerosis, such as through interactions with vulnerable plaque.

COLORECTAL CANCER SCREENING: ARE WE SCREENING APPROPRIATELY? Y Oppenheim, J Oh, A Eichorn, A Jotkowitz, Division of General Internal Medicine, Long Island Jewish Medical Center, New Hyde Park, NY

PURPOSE: Colorectal cancer is the second leading cause of cancer related death in the United States and data suggest that the risk of mortality from colorectal cancer can be reduced by screening. Previous studies indicate that physicians are neither knowledgeable nor compliant with screening recommendations. The purpose of this study is to determine whether internists practice according to current guidelines for colorectal cancer screening.

METHODS: We conducted a survey of all hospital and community based medical attending physicians at a tertiary care hospital to determine whether physicians screen in concordance with current guidelines. The survey included questions about physician gender, specialty, practice setting and patterns regarding colorectal cancer screening. Endpoints of interest were based on current recommendations and consisted of appropriate age (50 years old) and methods of screening of asymptomatic individuals. Appropriate methods of screening were defined as the use of home based fecal occult blood testing, flexible sigmoidoscopy, colonoscopy or barium enema but not digital rectal exam or office based guaiac testing after digital rectal exam. Data was analyzed by logistic regression to determine independent predictors.

RESULTS: Surveys were sent to 491 physicians. Valid responses were received from 215 physicians, of whom 85% were community based. From these responders, 38.7% were general internists, 14.6% were gastroenterologists and 46.7% were other subspecialists. Fifty-three percent of physicians started screening at the appropriate age and 82% of physicians did not have an upper age limit for screening. Ninety-two percent of physicians utilized inappropriate initial screening tests in addition to or in place of the recommended screening tests. Working in a hospital based practice was the only independent predictor of screening at the appropriate age (OR = 2.47, CI = 1.08 to 5.68) and using appropriate screening methods (OR = 7.86, CI = 2.67 to 23.14).

CONCLUSION: Many physicians initiated screening at inappropriate ages or utilized inappropriate screening methods in addition to or in place of recommended colorectal cancer screening methods. Hospital based physicians were more likely to start screening at the appropriate age and screen in accordance with current screening recommendations for colorectal cancer. This may have been due to their positions as role models and educators and their need to remain updated on current issues. Our data suggests that increased efforts are needed to encourage and educate physicians to practice in accordance with these guidelines. Additional studies are necessary to determine whether these early findings are reproducible and whether they apply to other screening recommendations.

HEALTH CARE UTILIZATION, HIV/AIDS PROGRESSION AND USE OF HIGHLY ACTIVE ANTIRETROVIRAL THERAPY (HAART) IN THE WOMEN'S INTERAGENCY HIV STUDY (WIHS) COHORT. H Palacios, A Munoz, K Fox, M Young, M Cohen, J Richardson, T Wilson, H Sacks, University of California, San Francisco, San Francisco, CA; John Hopkins University, Baltimore, MD; Georgetown University Medical Center, Washington, DC; Cook County Hospital, Chicago, IL; University of Southern California, Los Angeles, CA; SUNY Health Science Center, Brooklyn; Mount Sinai School of Medicine, New York, NY

PURPOSE: To characterize health care utilization in women of comparable HIV/AIDS progression in calendar periods with varying prevalence of HAART use.

METHODS: Of cohort enrolled in 1994, 496 (77%) with AIDS at baseline (B-AIDS), 1283 (91%) AIDS-free and 475 (85%) HIV- had ≥ 1 semiannual visit between 10/95 and 9/98, the study period. The initially AIDS-free women who developed AIDS were time-dependent categorized as incident AIDS (I-AIDS) and the complement were analyzed as the "HIV+" group. Outcomes were the proportions with >1 hospitalization (HOSP) and >1 emergency room (ER) visit over the preceding 6 months. Calendar was partitioned as 10/95-9/96, 10/96-3/97 and 4/97-9/98 corresponding to no to low HAART use (low-era), transition-era, and high use (high-

era). HOSP and ER associations with the era of HAART, stratifying by HIV/AIDS group, were assessed by logistic regression.

RESULTS: Overall, HOSP and ER were similar in B-AIDS and I-AIDS, but higher than in HIV+ and HIV- which were similar to each other. HOSP and ER were significantly ($p < 0.05$) higher in the low-era than in the transitional era for B-AIDS women (OR = 1.66 & 1.51) as well as HIV+ women (OR = 1.35 & 1.33). In the 3 HIV infected groups, HOSP and ER in the high-era were slightly higher than in the transition-era but were consistently lower than those in the low-era. Furthermore, in the high-era, the HOSP and ER use of the HIV+ were no different than those in the HIV- whose HOSP and ER remained stable over time.

CONCLUSION: At the population level, widespread use of HAART has resulted in substantial decrease in the utilization of high-cost health care services. The slowing of progression to AIDS by HAART resulted in health care utilization that was similar by HIV+ and HIV- women.

STaT: A THREE QUESTION SCREEN FOR INTIMATE PARTNER VIOLENCE. A Paranjape, LM Sullivan, JM Liebschutz, Section of General Internal Medicine, Boston Medical Center/Bedford VA; Department of Statistics, Boston University, Boston, MA

PURPOSE: An important barrier to universal screening for intimate partner violence (IPV) is lack of a well validated, easy to use screening tool. We conducted a study to develop a simple, validated screening tool for the detection of a lifetime history of IPV in women.

METHODS: We interviewed 75 women in the emergency room of an urban teaching hospital. Consecutive English-speaking female patients waiting for non-urgent medical problems, between the ages of 18 and 65, were eligible. The initial interview consisted of demographic questions and potential screening items; the items encompassed all aspects of IPV, were drawn from published literature and clinical experience, and had dichotomous response options. A semi-structured clinical interview followed; we obtained details of all previous and current relationships, using probes for specific violent acts. We defined lifetime IPV as any one or more of the following acts committed by an intimate partner: sexual abuse as any sexual act that was coerced or committed under threat of violence, physical abuse as repetitive, intentional violent acts directed at the subject, emotional abuse as threats of serious injury to subject or her children, deliberate denial of access to money, car or house keys. Using receiver-operator curves (ROCs), we compared the estimated sensitivity and specificity for each item with respect to IPV and determined the best 4 items. We compared the estimated sensitivity and specificity for all response options for all combinations of the best 4 items using ROCs and calculated 95% confidence intervals (CIs) of the sensitivity and specificity of the final screen.

RESULTS: The prevalence of lifetime IPV was 63%; of those abused 23% reported IPV in the past 1 year. The mean age was 36 years (SD:10 years); 75% of subjects had an annual income $< \$20,000$. The three items with the best operating characteristics were: "Have you been in a relationship where your partner has pushed or slapped you?", "Have you ever been in a relationship where your partner threatened you with violence?" and "Have you ever been in a relationship where your partner has thrown, broken and punched things?". The sensitivities of these items were 87%, 78% and 83% respectively. For a positive response to any 1, any 2, or all 3 questions, the sensitivity (with 95% CIs) of the three-question screen STaT to detect IPV was 96% (90%, 100%), 89% (81%, 98%) and 64% (50%, 78%) respectively. The corresponding specificity of each response combination was 75%, 100% and 100%.

CONCLUSION: A single positive response to a simple three-question screen STaT has a sensitivity of 96% and specificity of 75% to detect lifetime IPV. This screen may improve physician efforts at universal detection.

VALIDITY OF SURROGATE RATINGS OF SYMPTOMS, SATISFACTION, AND QUALITY OF CARE FOR TERMINALLY ILL PATIENTS. PM Pasley, M Rahn, DP Sulmasy, Depts. of Medicine and Ethics, Saint Vincents Hospital, New York; The Bioethics Institute, New York Medical College, Valhalla, NY

PURPOSE: While it is widely thought that the quality of care rendered to dying patients can be improved, one obstacle is that many dying patients are too sick to discuss the quality of care they have received. While previous studies of the quality of end-of-life care have often interviewed surrogates, such as family members, the Institute of Medicine and others have noted that there is scant evidence regarding the validity of surrogate's assessments of patient's experiences. We therefore designed a study to determine the validity of surrogate assessments of the hospital experience of dying inpatients regarding symptoms, satisfaction, and quality of care.

METHODS: Twenty-six patients and their surrogates at two major northeastern teaching hospitals received a detailed interview regarding three aspects of the hospitalization: a 4-point scale of symptoms developed by Lynn, et al., and scales of satisfaction and the quality of the interpersonal aspects of care, adapted from instruments developed by Feinstein and Matthews. The patients ranged in age from 36 to 94 years, with an equal distribution of the sexes. They were predominantly Caucasian (89%) and most had underlying diagnoses of either malignancy (39%), cardiovascular disease (34%), or HIV/AIDS (15%). Sixteen had Do Not Resuscitate orders.

RESULTS: Spearman rank correlations for symptom ratings by patients and surrogates were: dyspnea ($p = 0.58$; $P < .01$), restlessness (.47; $P = .02$), nausea (.54; $P < .01$), constipation (.40; $P = .05$), feverishness (.63; $P < .01$), fecal incontinence

(.57, $P < .01$), and dry mouth (.58; $P < .01$). Symptoms of pain ($p = .27$; $P = .21$) and urinary incontinence (.13; $P = .55$) did not meet statistical significance.

Pearson correlations between patient and surrogate assessments of care were: satisfaction with nursing care ($r = .06$; $P = .79$), satisfaction with physician care (.16; $P = .93$), quality of nursing care (.17; $P = .43$), and quality of physician care (.48; $P = .02$). McNemar testing revealed no systematic tendency for surrogates to assess care either more positively or more negatively.

CONCLUSION: We conclude that the use of surrogates to assess the symptoms of terminally ill patients is reasonably accurate, except for pain and urinary incontinence. However, with the possible exception of the quality of physician care, surrogates' ratings of satisfaction and the quality of care do not reflect patients' views. While our sample size is small and further study is needed, these results suggest that surrogate ratings of many symptoms can be used as valid outcome measures in interventions to improve end-of-life care. However, for the important symptom of pain and for levels of satisfaction and quality, other instruments or strategies will need to be developed.

COMPARISON OF FOUR STRATEGIES FOR DIAGNOSING DEEP VEIN THROMBOSIS: A COST-EFFECTIVENESS ANALYSIS. N Perone, H Bounameaux, A Perrier, Department of Internal Medicine, Geneva University Hospital, Geneva, Switzerland

PURPOSE: Four diagnostic strategies for clinically suspected deep vein thrombosis (DVT) have been recently validated in large scale outcome studies, resting on various combinations of clinical probability assessment, serial lower limb venous compression ultrasonography (US) and plasma D-dimer measurement (DD). We compared the cost-effectiveness of these diagnostic algorithms.

METHODS: Formal cost-effectiveness analysis by a decision analysis model. Outcomes considered were costs per patient, 3-month mortality, number of lives saved per 1000 patients and incremental costs per life and per year of life saved.

RESULTS: Under baseline conditions (prevalence of DVT, 24%), the effectiveness of all strategies was similar (4.2 to 4.4 lives saved per 1000 patients managed). The most expensive strategy was serial US (second US on day 7 in all patients with a normal first US). Performing a second US only in patients with an elevated DD level reduced costs. Taking clinical probability into account (serial US only in patients with an intermediate clinical probability of DVT) yielded further savings. DD as a first test, followed by a single US only in case of an abnormal DD level, and by phlebography only in patients with a normal US and a high clinical probability of DVT, was the cheapest, and, therefore, the most cost-effective option. This strategy allowed a 15% reduction in incremental costs compared to the most expensive algorithm.

CONCLUSION: Combining clinical probability and DD to a single US in the diagnostic workup of DVT is highly cost-effective, allowing to reduce costs without any increase in mortality. Serial US appears less cost-effective.

FEMALE PATIENTS' PREDICTORS OF COMPLIANCE WITH MAMMOGRAPHY. NW Phifer, S Cykert, AF Walker, G Kissling, Internal Medicine Program, Moses Cone Hospital, Greensboro; Division of General Internal Medicine, The University of North Carolina School of Medicine, Chapel Hill; Department of Sociology, UNC Greensboro; Department of Math, UNCG Greensboro, Greensboro, NC

PURPOSE: To identify predictors that determine which women will comply with mammography, using a model that incorporates demographic data as well as attitudinal scales of race, religiosity, and trust.

METHODS: We recruited a stratified random sample of 101 women aged 50 to 75 yrs. from church groups, health fairs, and medical clinics in urban areas of North Carolina and Florida. We administered a structured questionnaire to obtain demographic information, to assess the above attitudes, and to ascertain compliance with mammography.

RESULTS: Sixty-six percent of the patients interviewed received mammograms within the last two years. 41% of those interviewed were African-American and the remainder were white. In the logistic regression the predictors of whether or not a patient received a mammogram were income, race, and self-reported health status. Religiosity, trust, and perceptions of racism did not predict whether or not a woman received a mammogram. The odds ratio for income less than \$20,000 was .199 (Wald confidence limits .082 to .480), for African-American race .237 (Wald confidence limits .08 to .70), and for average or poor self-reported health .287 (Wald confidence limits .097 to .852).

CONCLUSION: African-American race remains an independent predictor of decreased breast cancer screening. The specific reason for reluctance to obtain mammography remains unclear.

DOES EPIDURAL SCARRING CAUSE PAIN AFTER LUMBAR SURGERY? A CRITICAL REVIEW. R Piefrobob, SS Rathore, WJ Richardson, TS Carey, Division of General Internal Medicine and Clinical Epidemiology, University of North Carolina School of Medicine, Chapel Hill, NC; Division of Orthopaedic Surgery, Duke University Medical Center, Durham, NC

PURPOSE: Epidural scarring resulting from lumbar surgery has been implicated as the cause of post-procedural low back pain and sciatica, though the evidence surrounding this association is unclear. We performed a critical literature review to assess the relationship between scar formation and post-procedural pain.

METHODS: A Medline search was performed for all English language articles concerning human subjects published between 1966-1999 first using the key words

scar, fibrosis, and cicatrix. In the second stage, identified articles were searched using key words failed back, sciatica, back pain, and adcon (surgical device used in lumbar surgery). Articles were evaluated for data concerning the causal relation between post-procedural scarring and back pain or sciatica. Five criteria were utilized in the evaluation of causation: consistency (a scar, when present, is associated with pain); specificity (a scar is associated with specific type of pain); strength (larger size scars lead to more pain); temporality (scar formation precedes pain); and coherence (a decrease in scar size leads to a decrease in pain). Articles were abstracted by two separate reviewers (RP, SR).

RESULTS: Of 262 articles initially identified, only 15 reported data relevant to the causal relation between scarring and post-procedural sciatica and back pain. Articles included 10 case series (8 prospective, 2 retrospective), 3 case-control studies, and 2 randomized clinical trials with a mean of 81 patients per study. Data sufficient to evaluate consistency was reported in 11 studies; scarring did not meet this causal criteria in any of the 11 studies. Of 6 studies providing data to evaluate specificity, only 1 study reported findings sufficient to support causal specificity. Of 6 articles reporting data concerning temporality, none reported that scarring preceded low back pain or sciatica. Only one article reported data concerning coherence and it did not meet the requirement of causality. Of 10 studies reporting data concerning strength, 8 found no association between size of scar and post-procedural pain. Of interest, the two studies reporting a positive strength association between scar size and post-procedural pain involved two randomized trials of a surgical device intended to decrease scar size.

CONCLUSION: While there is suggestive evidence of an association between scar size and extent of pain as reported in the surgical device trials, there is little evidence in the literature to support a causal relationship between scar formation and post-procedural low back pain or sciatica after lumbar surgery.

EVALUATION OF A NOVEL COMORBIDITY INSTRUMENT IN COPD INPATIENTS. **RG Pinckney**, RP O'Brien, JF Piccirillo, B Littenberg, University of Vermont, Burlington; Washington University, St. Louis, MO

The Charlson Comorbidity Index (CCI) has been previously validated in many populations and is used widely to describe and adjust for burden of comorbidity in clinical research. A previous attempt to validate its use in patients with Chronic Obstructive Pulmonary Disease (COPD) failed. The Adult Comorbidity Evaluation (ACE-27), a modification of the Kaplan-Feinstein index, is designed to improve on prior comorbidity indexes by including information on an increased number of comorbid conditions as well as their severity. Using medical record review, the ACE-27 records the severity of 27 conditions. Each is graded absent (0) or mild to severe (1-3) with specific clinical criteria for each grade. The index condition (in this case, COPD) is not counted as a comorbid condition. We evaluated this new instrument and compared its performance to the CCI in hospitalized COPD patients.

Index	Univariate HR	P	Multivariate HR	P
Charlson Comorbidity Index	1.14	.136	1.11	.208
Continuous ACE-27	1.09	.111	1.09	.135
Prognostic comorbidity present	1.82	.059	1.78	.072

We identified a cohort of 74 inpatients with COPD exacerbations in Burlington, Vermont in 1995. Comorbidity scores and covariates were obtained by chart review, and information on mortality was retrieved from charts and the Social Security Death Index. Two methods of scoring the ACE-27 were evaluated: a continuous scale of 0-81 points and a summary "prognostic comorbidity" which is positive if any one grade 3 (severe) or any two grade 2 (moderate) comorbidities were present. Analyses were performed using Cox proportional hazards models and chi-square analysis.

The mean age of the 74 inpatients was 69.5 years, 53% were male, 19% were in the intensive care unit and two died in the hospital. Over five years of follow-up, 58% died with average time of observation of 2.5 years. The CCI failed to significantly predict survival in this population. The continuous ACE-27 score performed similarly. However, the presence of prognostic comorbidity was a powerful predictor. The 5-year death rate was 51% in patients without prognostic comorbidity and 76% in patients with prognostic comorbidity ($p = 0.047$ by chi-square analysis). Cox proportional hazards analysis showed similar results. Correction for age, gender and acute illness in multivariate models had little effect on the hazard ratios (HR).

The CCI was not a significant predictor of mortality in patients with COPD. The ACE-27, especially when used to indicate prognostic comorbidity, performed better. Limitations of the ACE-27 include lack of validation in other populations and the necessity for skilled chart review. Future studies should attempt to further validate the ACE-27 in other patient populations.

A MULTICENTER STUDY OF ELECTROCARDIOGRAM (ECG) LEFT VENTRICULAR HYPERTROPHY (LVH) AND BUNDLE BRANCH BLOCK (BBB) IN THE TRIAGE AND OUTCOME OF EMERGENCY DEPARTMENT (ED) PATIENTS WITH SUSPECTED ACUTE CARDIAC ISCHEMIA. **JH Pope**, R Ruthazer, JR Beshansky, JL Griffith, HP Selker, Division of Clinical Care Research, New England Medical Center, Boston, MA; Baystate Medical Center, Springfield, MA

PURPOSE: To determine the triage and outcome significance of ECG LVH and BBB in ED pts with acute cardiac ischemia.

METHODS: Analysis of data from a multicenter prospective clinical trial in EDs of 6 hospitals of varying types across the US including pts presenting with symptoms consistent with acute cardiac ischemia. Confirmed diagnosis determined by initial and follow-up ECGs and cardiac enzymes (99% follow-up).

RESULTS: Clinical and ECG data from the 5,324 pts:

ECG ST-T Abnormality:	LVH	LBBB	RBBB	Primary	Variant	None
N	182	158	184	2,314	349	2,137
	(3.4%)	(3.0%)	(3.5%)	(43.5%)	(6.6%)	(40.1%)
Mean Age (years)	68**	70**	70**	59**	56**	53
Chief Complaint-Chest Pain (%)	59**	57**	54**	67**	68	72
-Shortness of Breath (%)	24**	32**	21**	18**	15	12
First Systolic Blood Pressure (mm Hg)	152**	142	140	143*	152**	141
History-Hypertension (%)	79**	67**	62**	57**	63**	42
-Diabetes (%)	29**	34**	33**	24**	19	16
-Angina (%)	54**	60**	55**	41**	31	27
-Myocardial Infarction (%)	42**	46**	46**	31**	18*	14
Confirmed Diagnosis-Hypertension (%)	3**	1	0	2	3*	1
-CHF (%)	19**	28**	21**	11**	5*	3
-UAP(%)	15**	13**	11*	11**	8	6
-AMI (%)	7**	13**	8*	12**	3	3
-ACI (%)	23**	26**	19**	23**	11	9
Triaged-Admit, No ACI (%)	77**	84**	70**	61**	49	47
With ACI (%)	100**	93	97	98**	95	90
30-day Mortality (%)	5**	3*	4**	3**	0	1

* $p < 0.05$, ** $p < 0.001$ compared to the None group (ACI-acute cardiac ischemia, AMI-acute myocardial infarction, CHF-congestive heart failure, LBBB-left bundle branch block, LVH-left ventricular hypertrophy, MI-myocardial infarction, None-no ST-T wave abnormalities, Primary-primary ST-T wave abnormalities, RBBB-right bundle branch block, UAP-unstable angina pectoris, Variant-other secondary ST-T wave abnormalities)

CONCLUSION: Of patients with symptoms suggestive of acute cardiac ischemia, the 10% with ECG LVH or BBB were older and as likely to have acute cardiac ischemia as patients with primary ST-T wave abnormalities. Of note to the clinician, compared to patients without ST-T wave abnormalities, these patients had more false positive admissions, yet more confirmed diagnoses of ACI or CHF, and had higher 30-day mortality rates possibly related to greater age and rates of CHF.

MEASUREMENT OF PHYSICAL ACTIVITY LEVELS IN PATIENTS WITH PERIPHERAL ARTERIAL DISEASE: COMPARISON OF A VALIDATED ACCELEROMETER TO THE CARDIOVASCULAR HEALTH STUDY QUESTIONNAIRE. **A Priyanothi**, E O'Brien, GJ Martin, MM McDermott, Division of General Internal Medicine, Northwestern University Medical School, Chicago, IL

PURPOSE: Patients with Peripheral Arterial Disease (PAD) have an increased rate of cardiovascular mortality which may be due to reduced physical activity levels.

OBJECTIVES: First to compare physical activity levels using a validated accelerometer between men and women, with and without PAD. Second, to use a commonly used physical activity questionnaire to measure physical activity.

METHODS: 302 participants from a prospective study of lower extremity functioning in PAD completed the Cardiovascular Health Study (CHS) physical activity questionnaire. 75 patients with PAD and 48 patients without PAD wore an accelerometer continuously for seven days to measure physical activity. The accelerometer measures physical activity levels based on vertical movement at the hip joint.

RESULTS: PAD participants had lower physical activity levels than non-PAD participants when measured by the accelerometer (820 kcal/week $\pm$ 397 vs 1045 kcal/week $\pm$ 483, $p = 0.006$). Physical activity levels in PAD participants when measured by the CHS questionnaire were 767 kcal/week $\pm$ 990 compared to 878 kcal/week $\pm$ 890 in non-PAD participants ($p = 0.319$). In linear regression analysis, ABI was an independent predictor of accelerometer-measured physical activity (regression coefficient = 325 kcal/one unit ABI, $p = 0.0237$) after adjustment for age. In multivariable analyses, ABI was independently associated with physical activity levels measured by the accelerometer after controlling for age, sex, race, and comorbid illness (regression coefficient 288.4 kcal/one unit ABI, $p = 0.05$).

CONCLUSION: Men and women with PAD have significantly lower physical activity levels than patients without PAD when measured with an accelerometer. A commonly used questionnaire for measuring physical activity levels was not a sensitive measure of physical activity when compared to the accelerometer. Future studies should be directed towards improving physical activity in patients with PAD in order to prevent cardiovascular mortality and functional decline.

CHELATION THERAPY USE IN PATIENTS UNDERGOING CORONARY ANGIOGRAPHY. **H Quan**, WA Ghali, MJ Verhoef, CM Norris, ML Knudtson, Community Health Sciences; Medicine, University of Calgary, Calgary; Public Health Sciences, University of Alberta, Edmonton, AB, Canada

PURPOSE: Chelation therapy is a treatment for atherosclerosis that involves the intravenous administration of ethylene diamine tetraacetic acid (EDTA). The mechanism of action of EDTA is unknown but clinical trials consistently reveal

that chelation therapy has little or no beneficial effects. Given the fact that patients are nonetheless using the treatment, we conducted a survey to examine sociodemographic and clinical variables related to chelation therapy use in patients undergoing coronary angiography.

METHODS: The Alberta Provincial Program for Outcome Assessment in Coronary Heart Disease (APPROACH) is a cohort study of all patients undergoing coronary angiography in the province of Alberta, Canada. The APPROACH database captures information on sociodemographic, clinical, and treatment variables. Data on quality of life (EuroQoL) and chelation therapy use were ascertained through a survey mailed to patients one year after angiography. For this analysis, we included patients undergoing angiography in 1995–96 and surviving at least one year after angiography. Two logistic models were fit to identify variables associated with chelation therapy use: one for use of chelation therapy at some time (users') and another for use at the time of the survey (current users').

RESULTS: Among 5933 patients responding to the one year survey (69% response rate), 466 (7.8%) reported use of chelation therapy at some time and 212 (3.6%) were current users. The variables significantly associated with being a user were: abnormal coronary anatomy (e.g., odds ratio 3.2 for 3 vessel disease and 2.9 for left main relative to those with normal anatomy), coronary artery bypass surgery (CABG) treatment prior to angiography (1.4), score of 1.0 on EuroQoL scale (0.8), angioplasty (PTCA, 0.7) and CABG (0.6) in the first year after angiography. Current use was associated with abnormal coronary anatomy, diabetes (0.6), PTCA (0.7) and CABG (0.4) in the first year after angiography.

CONCLUSION: Chelation therapy users appear to have more extensive coronary artery disease and poorer quality of life than do non-users and are less likely to have been treated with revascularization after coronary angiography compared with nonusers. This suggests that chelation therapy users may be turning to the treatment as a 'last resort' after being judged not suitable for revascularization. Alternatively, some patients may be declining revascularization in favor of this less invasive, but unproven, treatment option.

THE PREVALENCE OF DISTRESSING SYMPTOMS AMONG SELECTED OUTPATIENTS AT THE BEGINNING OF THE END OF LIFE. *MW Rabow*, Division of General Internal Medicine, UCSF, San Francisco, CA

PURPOSE: Much attention has focused on the high prevalence of distressing symptomatology, especially pain, in hospitalized patients at the end of life. We assessed the prevalence and severity of physical symptoms and the level of psychosocial well-being among seriously ill outpatients with cancer, congestive heart failure (CHF), and chronic obstructive pulmonary disease (COPD) in a general medicine clinic.

METHODS: During a 6-month period in 1999, patients in an academic general medicine clinic were identified by their Primary Care Physician (PCP) as having severe COPD, severe CHF, or cancer. Those patients willing to participate in a controlled trial of interdisciplinary care were recruited for participation in an ongoing research project entitled "The Comprehensive Care Team." For administrative purposes, within the clinic, PCPs and their patients are divided into two equivalent practices. Patients in one practice were assigned to the intervention group; patients from the other practice received usual care. To assess patients' baseline level of symptomatology prior to any intervention, all patients completed a self-administered questionnaire regarding their pain, shortness of breath, depressive symptoms, and quality of life. The questionnaire included previously validated instruments, including the Brief Pain Inventory, the UCSD Dyspnea Index, the CES-D, and the Padilla Quality of Life Assessment.

RESULTS: 48 patients were enrolled during the 6-month period (31 in the intervention group (64.6%) and 17 in the control group (35.4%)). The average age for all patients was 66 years (standard deviation (SD) = 14.3), with an average education level of 13.6 years (SD = 3.4). 73% of patients were female, 52% white, and 23% African-American. 33.3% of the sample had COPD, 25% CHF, 22.9% COPD and CHF, 14.6% cancer, and 4.2% cancer and CHF. On a 0–10 scale, patients reported an average pain intensity of 4.0 (SD = 2.5) with an average maximum pain during the last 24 hours of 4.7 (SD = 3.2). 88% of patients reported dyspnea. A CES-D score greater than 16 suggests the need for anti-depressant medication. Among patients surveyed here, the mean CES-D score was 18.7 (SD = 13.3). 10% of the sample had a CES-D score of greater than 40. On a 0–10 scale, the average patient quality of life was 7.2 (SD = 2.6), with 23.4% of the sample rating their quality of life as 5 or less. At baseline, there were no statistically significant differences between intervention and control patients.

CONCLUSION: As is true among hospitalized patients at the end of life, significant pain, dyspnea, and depression are common among outpatients with serious CHF, COPD, and cancer in a general medicine clinic. Primary care physicians appear to be undertreating physical symptoms and depression, even among patients they identify as having serious illness.

ECONOMIC EVALUATION OF BCG REVACCINATION AMONG THE SCHOOL CHILDREN IN JAPAN. *M. Rahman*, M Sekimoto, T Fukui, General Medicine and Clinical Epidemiology, Kyoto University Hospital, Kyoto, Japan

PURPOSE: Bacillus Calmette Guerin (BCG) revaccination has been implemented in Japan among the tuberculin-negative 1st grade primary and junior high school students for decades. Controversies regarding the effectiveness of BCG revaccination and low incidence of tuberculosis (TB) among the children were the impelling factors to conduct this study. The main objective of this study is to compare the costs and benefits of BCG revaccination in Japan.

METHODS: Cost-benefit analysis was conducted for cohort of school children who underwent revaccination during 1996 from the perspective of the payer for health care. The study population comprised of estimated 1.4 million 1st grade primary school and 1.5 million 1st grade junior high school students enrolled in 1996 in all over Japan. Assuming 50% vaccine efficacy for revaccination, 10 years of duration of protection, and 5% annual discount rate, calculation was made on the total hypothetical number of TB averted by revaccination. Both direct and indirect costs were included in the analysis.

RESULTS: The revaccination program for 1996 school children cohort would prevent 294 TB cases over a 10-year period at a cost of US \$115,147 per case averted. The direct and indirect costs of revaccination were estimated at US \$31.4 and US \$2.5 million, respectively. The direct benefits, the treatment costs saved for future cases averted by revaccination, were estimated to be US \$2.3 million. The major indirect benefit, the averted loss of work output attributable to TB-related morbidity, was estimated at US \$1.6 million. The benefit-cost ratio remained at 0.12 with baseline assumptions and ranged from 0.04 to 0.28 for one-way sensitivity analysis. The costs of revaccination were found to exceed benefits even after inputting assumptions which tend to favor revaccination.

CONCLUSION: From the economic point of view, BCG revaccination policy is not favorable in the present Japanese situation.

RETROSPECTIVE CLASSIFICATION OF PROSTATE-SPECIFIC ANTIGEN TESTS: DIFFERENTIATING SCREENING FROM DIAGNOSTIC CLINICAL ENCOUNTERS. *SS Rothore*, AV Kshirsagar, MJ Schell, A Amamoo, J Freeman, RP Harris, PA Godley, Department of Medicine and Lineberger Comprehensive Cancer Center; Department of Biostatistics; Department of Surgery, University of North Carolina, Chapel Hill, NC

PURPOSE: Accurate retrospective classification of prostate specific antigen (PSA) tests is crucial for observational evaluations of PSA testing's effectiveness in preventing death from prostate cancer. We sought to evaluate the validity of a directed retrospective medical chart review as one such classification method.

METHODS: We evaluated all PSA tests ordered at University of North Carolina Hospital, Chapel Hill, NC in May 1998, excluding patients with prior diagnoses of prostate cancer. Of the 163 PSA tests identified, medical charts were obtained for 120. Each chart was reviewed by 4 physicians: a medicine resident (RES), oncologist (ONC), urologist (UR), and medicine attending (GM) using predefined standardized criteria in order to classify the PSA test as a screening or diagnostic service. The attending physician (ATTEND) who ordered the PSA test completed a similar questionnaire in which they classified their intent in ordering the PSA test as either screening or diagnostic using identical criteria. Concordance rates were calculated between each rater and ATTEND's classification. Valid responses were received for 95 tests (79%), which constitute the study cohort.

RESULTS: Raters demonstrated a high level of agreement with ATTEND's classification as a majority of tests generated complete agreement between all 4 raters and ATTEND (n = 51,53%). Concordance rates by individual rater were 0.75 (GM), 0.77 (ONC), 0.78 (UR), 0.80 (RES) for an average concordance of 0.78. Of note, in 7 tests all 4 raters provided identical classifications, yet disagreed with ATTEND's classification. Follow-up discussions with ATTEND indicated ATTEND misclassified all 7 tests as screening services though the tests clearly met prespecified diagnostic criteria. Correcting the 7 tests inappropriately classified by ATTEND increased concordance rates to 0.84 (ONC), 0.87 (RES), 0.88 (UR), 0.91 (GM), for overall corrected average concordance of 0.88.

CONCLUSION: Standardized criteria applied to medical chart data provide a valid method of retrospectively classifying PSA tests as screening or diagnostic. While concordance rates vary based on rater specialty, variances are small and likely not meaningful. Caution, however, should be taken when evaluating results dependent on the retrospective evaluation of PSA ordering intent. Our data provide validation for retrospective evaluation of screening status, as part of case-control studies of PSA screening efficacy currently under way.

A COMPARISON OF QUANTITY-FREQUENCY QUESTIONS, CAGE, & AUDIT QUESTIONNAIRES IN OLDER ADULTS. *MC Reid*, ME Tinetti, PG O'Connor, J Concato, VA Connecticut, and Yale University

PURPOSE: Although several methods are available to measure alcohol use, the most useful method for older adults remains uncertain. We sought to determine the levels of agreement and concordance between 3 commonly used measures in a study of older (≥ 65 years) community-living adults. We conducted surveys of (1) veterans enrolled in the West Haven, VA primary care clinic (N = 303), and (2) Medicare beneficiaries (N = 511) in the greater New Haven, CT, region. We administered standard quantity-frequency (QF) questions, the CAGE, and the Alcohol Use Disorders Identification Test (AUDIT) to all subjects who reported use of alcohol in the past 12 months: 59% of VA, and 51% of Medicare participants. To assess agreement, we determined the percentage of subjects who screened positive (+) by each measure defined as: QF + = ≥ 14 drinks per week; CAGE + = score ≥ 2 ; and AUDIT + = score ≥ 8 . We assessed concordance by calculating kappa values for each of the pairwise comparisons.

Participation rates were 71% and 81% for the VA and Medicare samples, respectively, and the mean ages were 73 and 76 years. The proportions of current drinkers who screened positive in the VA and Medicare samples, respectively, were: QF + = (14% and 4%); CAGE + = (19% and 4%); and AUDIT + = (16% and 5%). Lev-

els of agreement (%) and concordance (kappas) were highest for the QF vs. AUDIT comparisons, but were marginal for the other comparisons as shown below

	VA Sample (N = 179)			Medicare sample (N = 261)		
	QF+	CAGE+	AUDIT+	QF+	CAGE+	AUDIT+
QF+	—	35% (.29)	72% (.76)	—	18% (.16)	50% (.56)
CAGE+	48% (.29)	—	41% (.25)	20% (.16)	—	29% (.29)
AUDIT+	84% (.76)	35% (.25)	—	70% (.56)	36% (.29)	—

Our results show that commonly used alcohol measures have limited agreement and concordance in older adults, with the possible exception of the QF vs. AUDIT comparisons. The use of any single measure may fail to detect a substantial number of older individuals with potentially important alcohol exposures. Research is needed to determine the optimal screening strategies for detecting alcohol disorders among older populations.

WHEN SHOULD WE STOP SCREENING? S Rich, H Welch, VA Outcomes Group, White River Junction, VT

PURPOSE: Although the age at which to start screening has been the subject of considerable debate, the question of when to stop has received little attention. To help address this question, we calculated the days of life given up by stopping screening at various ages.

METHODS: For each of three cancers (breast, cervical and colon), we used life tables to calculate life expectancy at various ages. The days of life given up by stopping screening at a specific age is the difference in life expectancy between two life tables: one for stopping screening at that age and the other for continuing screening. Based on the randomized trial data, we used a 30% reduction in cancer-specific mortality for breast and colon cancer screening. Because there are no comparable data for cervical cancer, we assumed a 30% reduction in the "best guess" analysis and a 70% reduction in the "best case" analysis. We also assumed there was no harm with screening.

DATA: All-cause and cancer-specific mortality were obtained from the National Center for Health Statistics and the Surveillance Epidemiology and End Results Survey (SEER).

RESULTS: Assuming a starting age of 50, screening throughout life has a maximum benefit in life expectancy of 42.7 days for breast cancer and 28.1 days for colon cancer. Assuming a starting age of 20, screening throughout life for cervical cancer has a maximum benefit of 7.1 days and 46.6 days in the "best guess" and "best case" analyses, respectively.

Days of life given up by stopping screening (% of total benefit)

Stop screening at age:	Breast Cancer	Colon Cancer	Cervical Cancer
	best guess	best guess	best case
65	20.5 (52%)	17.6 (37%)	1.4 (80%)
75	8.5 (80%)	8.9 (68%)	0.5 (93%)
80	4.6 (89%)	4.8 (83%)	0.2 (97%)
85	1.8 (96%)	1.7 (94%)	0.1 (99%)

CONCLUSION: Even assuming the mortality reduction with screening persists in the elderly, 80% of the benefit is achieved before age 75 in breast cancer, age 80 in colon cancer, and age 65 in cervical cancer. The small benefit of screening in the elderly may be outweighed by harm, e.g., anxiety, additional testing, and unnecessary treatment.

SMOKERS TAKING BUPROPION GAIN LESS WEIGHT AFTER SMOKING CESSATION: A GENDER-RELATED EFFECT. NA Rigotti, AN Thorndike, MJ Durcan, JD White, JA Johnston, R Niaura, D Gonzales, DPL Sachs, JT Hayes, RD Hurt, General Medicine Division, Massachusetts General Hospital, Boston, MA; Glaxo Wellcome R&D, Research Triangle Park, NC; Brown University, Providence, RI; Oregon Health Sciences University, Portland, OR; Palo Alto Center for Pulmonary Disease Prevention, Palo Alto, CA; Mayo Clinic, Rochester, MN

PURPOSE: Weight gain, actual or feared, is a major barrier to smoking cessation, especially for women. Effective ways to avoid or treat post-cessation weight gain are lacking. Bupropion sustained-release (SR) is an FDA-approved smoking cessation aid. A 7-week course of the drug temporarily blunts post-cessation weight gain. We hypothesized that a longer course of therapy might have a more permanent effect on post-cessation weight gain.

METHODS: We analyzed data from a multicenter relapse prevention trial in which 784 adult smokers took open-label bupropion SR 300 mg/d for 7 weeks. At week 7, the 432 smokers who had quit (no cigarettes for the past 7 days) were randomly assigned to a double-blind trial comparing an additional 45 weeks of bupropion vs placebo. Weight was assessed at end of drug therapy (52 weeks) and at 13, 15, 18, and 24 months. Analysis was limited to smokers who were continuously abstinent for 24 months (n = 97).

RESULTS: At the end of drug treatment, patients who were assigned to receive bupropion long-term gained less weight than those assigned placebo (difference, -4.1 kg; 95% CI: -6.1, -2.1). The effect persisted through 24 months (difference -2.2 kg), remained statistically significant (p < .05) through 18 months (difference, -2.5 kg; 95% CI: -4.7, -0.2) and was greater in women than men. In women, the drug-placebo difference in weight gain was -6.3 kg at 52 weeks (95% CI: -8.8, -3.7) and -3.7 kg at 18 months (95% CI: -7.0, -0.4). In men, differ-

ences were smaller and not statistically significant (-1.8 kg at 52 weeks, -1.3 kg at 18 months).

CONCLUSION: Long-term treatment with bupropion is associated with a clinically significant reduction in weight gain after smoking cessation, especially in women. The effect lasts beyond the end of drug treatment. Bupropion may be a good choice for smokers who are concerned about gaining weight after smoking cessation.

CIGAR SMOKING BY U.S. COLLEGE STUDENTS. NA Rigotti, H Wechsler, Massachusetts General Hospital, Harvard Medical School; Harvard School of Public Health, Boston, MA

PURPOSE: Cigar sales have increased dramatically since 1993 following a tobacco industry marketing campaign. Recent national data on cigar smoking prevalence are limited. The data available suggest that cigar use rates are rising fastest in young adults. If so, cigar use rates may be high in college students.

METHODS: We analyzed tobacco use data from the 1999 College Alcohol Survey, a random sample of 14,138 students drawn from a nationally-representative sample of 119 4-year U.S. colleges. Correlates of cigar use were identified with multiple logistic regression.

RESULTS: Among college students, 37% (56% of males, 25% of females) have ever smoked a cigar and 9% (16% of males, 4% of females) are current (past 30 day) cigar users, but <1% smoke daily. Cigar use is more common in males (OR 4.6, 95% CI 4.0, 5.4) and blacks (OR 3.2, 95% CI 2.1, 4.6) and less common in hispanics (OR 0.6, 95% CI 0.4, 0.9) and Asians (OR 0.7, 95% CI 0.6, 1.0). Current cigarette smokers (OR 3.7, 95% CI 3.1, 4.2), marijuana smokers (1.4, 95% CI 1.2, 1.7), and binge drinkers (OR 2.0, 95% CI 1.7, 2.4) are more likely to smoke cigars, but <10% of students who smoke both cigars and cigarettes started smoking cigars first. Cigar use is more common in students who are dissatisfied with their education (OR 1.3, 95% CI 1.0, 1.7) and who rated as important participation in fraternities/sororities (OR 1.4, 95% CI 1.1, 1.9), parties (OR 1.2, 95% CI 1.1, 1.4), and sporting events (OR 1.2, 95% CI 1.0, 1.4).

CONCLUSION: Cigar use is common in college students, especially white and black males, cigarette smokers, users of other substances, and students whose priorities are more social (parties, fraternities, attending sporting events) than educational. We find no evidence to support concerns that cigars may serve as a "gateway" to cigarette smoking in college populations. However, cigar use is not harmless and ongoing efforts to reduce tobacco, alcohol, and substance use in college should also address cigars.

DO ABNORMAL LIVER FUNCTION TESTS PREDICT THE YIELD OF IMAGING STUDIES IN HOSPITALIZED PATIENTS? JM Rothschild, DW Bates, JM Fiskio, R Hanson, R Khorasani, Internal Medicine, Brigham and Women's Hospital; Information Systems; Radiology Information Systems, Partners HealthCare System; Radiology, Brigham and Women's Hospital, Boston, MA

PURPOSE: To determine whether the severity and quantity of liver function test (LFT) abnormalities can predict the yield of abdominal imaging exams performed on inpatients.

METHODS: The study was performed in a 720 bed tertiary teaching hospital. All inpatient-imaging exams are requested on-line using physician order entry with indications chosen from predefined menus. During the 27 month study period (Dec. 1995-March 1998), there were 89,450 non-obstetrical admissions. In all, 856 exams were performed to evaluate abnormal (abn) LFTs, including 540 US (63%) and 316 CT (37%) exams. Altogether, 759 image reports were successfully matched to lab results. Reports were coded as: Positive (pos-abn and related to the diagnostic indication), Normal (NI) and Other. A total of 10,613 LFTs [AST, ALT, total (TBR) and direct (DBR) bilirubin, and alkaline phosphatase (AP)] completed within 3 days of the exam orders were analyzed. LFT results were classified by severity as follows: nl, mild abn (1-2xnl), moderate (mod) abn (2-10xnl), and severely abn ($\geq 10xnl$). The 5 labs were collapsed into 3 categories: AST/ALT, AP, and TBR/DBR. For multiple testing, the most abnormal LFT was selected. Severity of lab category results were defined by the most abn lab test within each category (ie if mild abn AST and mod abn ALT/mod abn ranking). To determine the number of "abn lab categories" per patient, lab categories $>2xnl$ (only mod or severe abn) were considered. Image yield of abn lab categories is the rate of pos exams/all exams.

RESULTS: The 759 imaging exams were coded as follows: 196 positive (26%), 79 nl (10%) and 484 other (64%). The yield of positive exams was stratified by level of lab category abn: ALT/AST-nl-16% (34 pos exams/209 nl labs), mild abn-28% (70 pos exams/244 mild labs), mod abn-30% (89 pos exams/293 mod labs) and severe abn-27% (3 pos exams/11 sev labs); AP-nl-17% (23/139), mild-21% (33/159), mod-31% (113/360), and severe-27% (27/99); DBR/TBR-nl-23% (49/202), mild-28% (31/108), mod-26% (68/266) and severe-27% (48/180). The imaging yield of pos exams/all exams by frequency of "abn lab categories" per patient is: 0 abn labs-21% (19 pos exams/92 exams), 1 abn lab-21% (56/268), 2 abn labs-28% (71/256) and all 3 abn labs-35% (50/143).

CONCLUSION: While over a quarter of inpatient abd imaging exams can explain abn LFTs, both the severity and frequency of abnormal LFTs correlate poorly with the diagnostic yield. Not uncommonly, the imaging indication of "abnormal LFTs" were associated with normal or near normal labs, and the diagnostic yield was comparably good.

PATIENT RATINGS OF HEMODIALYSIS AND PERITONEAL DIALYSIS IN THE CHOICE STUDY. *HR Rubin*, L Zhou, J Kerman, M Jenckes, N Fink, N Powe, Medicine; Health Policy and Management; Epidemiology, The Johns Hopkins University, Baltimore, MD

PURPOSE: To compare how similar patients evaluate hemodialysis (HD) and peritoneal dialysis (PD) a few weeks after beginning dialysis.

METHODS: Patients were participants in the CHOICE study, a national study comparing hemodialysis and peritoneal dialysis outcomes. Because patients were sampled within centers, this analysis was limited to centers offering both modalities. 593 of 644 eligible patients completed baseline rating surveys after a few weeks on dialysis which included 3 overall ratings and 20 items rating specific aspects of dialysis care as poor, fair, good, very good, or excellent. Survey items have been published and were developed through exploratory in-depth interviews and importance ratings by 100 dialysis patients of over 100 candidate items.

RESULTS: Patients on peritoneal dialysis were younger (22% PD vs. 13% HD < 40), more educated (19% PD vs. 33% HD < high school), more likely to be Caucasian (80% PD, 60% HD), working (26% PD vs. 10% HD), and married (66% PD vs. 51% HD). PD patients were more likely than HD patients to live >30 miles away from the dialysis center (39% PD, 9% HD). PD patients had fewer comorbidities (ICED 0-1 for 45% of PD and 30% of HD patients). There were no differences among PD and HD patients in sex, or SF-36 physical and mental component health status self-ratings (mean PCS 32 and MCS 45 for both groups).

Patients on peritoneal dialysis gave higher ratings of dialysis care overall (PD 82% vs. HD 55% excellent) and of all specific aspects of care rated. Items with the greatest differences were in the domain of information provided (average difference of 35 in % respondents rating excellent) and technical quality, such as the amount of fluid removed (difference 36% excellent). Smaller differences were seen in ratings of coordination among nephrologists and other doctors, availability of nephrologists, and center staff's helpfulness and concern. Adjustment for patient age, race, education, health status, marital status, employment status, distance from the center, and time since starting dialysis did not reduce, and for most items increased, the difference between the groups.

CONCLUSION: Both with and without adjusting for patient characteristics, after a few weeks on dialysis patients on peritoneal dialysis rate their care higher than those on hemodialysis. Primary care practitioners may wish to recommend this modality preferentially to end-stage renal disease patients who are eligible for both. Future CHOICE analyses will determine if these results are sustained over a longer term of dialysis, and will compare patient ratings of care to clinical outcomes.

PHYSICAL ACTIVITY AND MORTALITY IN THE PHYSICIANS HEALTH STUDY ENROLLMENT COHORT. *M Sabolsi*, I-M Lee, JE Buring, JM Gaziano, Division of Preventive Medicine, Brigham and Women's Hospital, Boston; Department of Medicine, The Cambridge Hospital, Cambridge; MAVERIC, VA Boston, Boston, MA

BACKGROUND: Physical activity is associated with lower total mortality rates among individuals without a history of heart disease. However, it is unclear whether mortality rates continue to decrease commensurately with higher levels of physical activity.

METHODS: We examined the relationship between frequency of vigorous physical activity and mortality in a prospective cohort of 86,738 US male physicians from the enrollment cohort of the Physicians' Health Study. Study participants were 40-84 years old without prior history of myocardial infarction (MI), stroke, cancer, or liver disease at baseline. On the baseline questionnaire, men reported how often they exercised vigorously enough to work up a sweat. Outcome measures were total and disease-specific mortality.

RESULTS: There were 3161 deaths during the mean follow-up period of 5.5 years. We observed an L-shaped relationship between the frequency of vigorous physical activity and total mortality. After adjustment for other predictors of mortality, compared to the men who reported the lowest (rarely/never) frequency of vigorous physical activity, more active individuals had highly significant reductions in the risk of death during follow-up. Those who reported exercising vigorously: one to three times per month, once per week, two to four times per week, five to six times per week, and daily, had relative risks for total mortality of 0.71, 0.62, 0.59, 0.59, and 0.66, respectively (p for linear trend 0.0001). The inverse association between frequency of vigorous physical activity and total mortality was statistically significant among men aged 40-59, 60-69, and 70-84. Lower mortality rates among men who exercised vigorously was primarily due to lower CVD mortality rates. Men who exercise vigorously also experienced lower mortality rates from total cancer and other (non-CVD, non-cancer) causes. Much of the inverse association between frequency of vigorous physical activity and cancer mortality appeared to be explained by a significant decrease in lung cancer mortality (p for linear trend 0.01). The remaining four most prevalent causes of cancer death in this cohort—colon, prostate, pancreatic and hematologic—were not associated with frequency of vigorous physical activity.

CONCLUSION: These data support a L-shaped relationship between frequency of vigorous physical activity and total mortality, even among older men aged 70-84. The shape of the curve appears to be similar for mortality from cardiovascular disease, cancer, and other (non-CVD, non-cancer) causes of mortality. The inverse relationship between frequency of vigorous physical activity and cancer mortality appears to be largely due to lung cancer mortality. No statistically significant associations between frequency of vigorous physical activity and mortality were seen for other cancers.

HORMONE REPLACEMENT THERAPY (HRT) AS PREVENTIVE MEDICINE FOR POSTMENOPAUSAL WOMEN (PMW): A SHARED DECISION-MAKING MODEL. *HS Sacks*, DN Rose, JM Clark, J Lau, Mount Sinai School of Medicine, New York, NY; Long Island Jewish Medical Center, New Hyde Park, NY; Johns Hopkins University, Baltimore, MD; New England Medical Center, Boston, MA

PURPOSE: Previous decision analyses showed the effect of individuals' risk factors for disease on the decision to take HRT for primary prevention. The purpose of this study was to determine the effect of current data on risks and benefits, as well as individuals' preferences for the various outcomes, on the decision.

METHODS: We used a Markov decision analysis model for 50-year-old PMW considering HRT for preventive medicine. The outcome measure was the extension of quality-adjusted life months (QALMs) as a result of taking HRT. We used the best available data on risks of breast cancer (BC), coronary heart disease (CHD), and osteoporosis-associated hip fracture (HF) and the effect of HRT on these risks, and the median utility values chosen by 52 women who evaluated the pertinent health states. For the sensitivity analysis, we varied the utilities from the 25th to the 75th percentile of these values. We assumed the effect found in a trial of HRT for secondary prevention of CHD is the same for primary prevention.

RESULTS: HRT increases BC incidence and decreases HF. For CHD (from the HERS Study) there is an initial increase in incidence, followed by a decline. PMW at average risk gain 2.2 QALMs by taking HRT. PMW who value CHD and HF as having the worst impact on health and BC as having the mildest impact gain the most from HRT, 7.9 QALMs. PMW who value CHD and HF as having the mildest impact on health and BC as having the worst impact lose 0.2 QALMs from HRT. Depending on the utility values chosen and on individuals' risk factors, HRT results in a gain of up to 22.5 QALMs for PMW with low risk of BC and high risk for CHD and HF) or a loss of up to 14.1 QALMs for PMW with high risk of BC and low risk for CHD and HF). HRT results in smaller gains for black PMW than for white PMW, and smaller gains for older PMW than for younger PMW.

CONCLUSION: Unless future studies show a larger benefit on CHD mortality or other health states, HRT decisions for PMW should include careful consideration of individual preferences for all of the potential outcomes. The model can readily incorporate data on new treatments and other outcomes as they become available.

HORMONE REPLACEMENT THERAPY (HRT) AS PREVENTIVE MEDICINE FOR BREAST CANCER SURVIVORS (BCS): A DECISION ANALYSIS. *HS Sacks*, DN Rose, JM Clark, J Lau, Mount Sinai School of Medicine, New York, NY; Long Island Jewish Medical Center, New Hyde Park, NY; Johns Hopkins University, Baltimore, MD; New England Medical Center, Boston, MA

PURPOSE: A previous decision analysis suggested that, depending on individual risk factors, HRT might increase life expectancy for many postmenopausal women, but the risks and benefits of HRT for BCS have not been studied.

METHODS: We used a Markov decision analysis model for 50-year-old BCS considering HRT for preventive medicine. The outcome measure was the extension of quality-adjusted life months (QALMs) as a result of taking HRT. We used the best available information on risks of breast cancer recurrence, coronary heart disease (CHD), and osteoporosis-associated hip fractures, and the effect of HRT on these risks. We assumed the effect of HRT found in a randomized trial of HRT for secondary prevention of CHD (the HERS study) is the same for primary prevention: an initial increase in incidence, followed by a decline. For the base case analysis, we used the median of the utility values chosen by 52 women who evaluated the pertinent health states. For the sensitivity analysis, we varied the utilities from the 25th to the 75th percentile of these values.

RESULTS: BCS at average risk for CHD and hip fracture lose 6.6 QALMs by taking HRT. BCS who value the CHD and hip fracture states as having the worst impact on health and the breast cancer state as having the mildest impact on health lose the least from HRT, 3.0 QALMs. BCS who value the CHD and hip fracture states as having the mildest impact on health and the breast cancer state as having the worst impact on health lose the most from HRT, 9.3 QALMs. Depending on the utility values chosen for health states and on individuals' risk factors for disease, HRT results in a gain of up to 5.4 QALMs (for BCS with high risk for CHD and hip fracture) or a loss of up to 10.4 QALMs (for BCS with low risk for CHD and hip fracture).

CONCLUSION: Unless future studies show a larger benefit on CHD mortality or other health states, HRT decisions for BCS should include careful consideration of individual preferences for all of the potential outcomes. The model can readily incorporate data on new treatments and other outcomes as they become available.

SCREENING FOR BREAST AND CERVICAL CANCER VARIES BY PROVIDER SPECIALTY: RESULTS FOR THE MEDICAL EXPENDITURE PANEL STUDY. *R Sadurski*, MD Silverstein, Center for Health Care Research, Medical University of South Carolina, Charleston, SC

PURPOSE: Compliance with screening mammography and Pap smear testing may be affected by nature of usual source of care. We compared breast and cervical cancer screening in women age 50 and older by usual source of care and specialty of provider to identify differences in utilization.

METHODS: The 1996 U.S. Medical Expenditure Panel Survey, a nationally representative survey of health care use for the civilian population, was studied. The dependent variables were proportion of women age 50 and older who had Pap smear or mammography testing within the two years prior to survey. Results are stratified by age.

RESULTS: Information from 2678 women, over 50 years old (range 50–90 years), was analyzed. The tables summarize percentage of women that reported testing in the last two years.

Pap Smear

Age category (years)	No Provider	Gen/Family Practice	Internal Medicine
50–64	52.0	75.8	78.7
65–74	34.4	60.3	68.8
>75	30.8	39.9	46.6

Mammography

Age category (years)	No Provider	Gen/Family Practice	Internal Medicine
50–64	50.7	73.5	87.1
65–74	35.9	67.4	91.3
>75	29.3	50.0	62.8

CONCLUSION: The proportions of women 50 years and older who reported breast or cervical cancer screening differed on the source of care. The proportions were lowest for women who reported having no usual source of care. Patients with an internist as provider were more likely to be screened than patients with a general practitioner/family practice physician as provider.

DECREASED ADHERENCE TO ANTIRETROVIRAL MEDICATIONS IN HIV-INFECTED PATIENTS WITH ALCOHOL PROBLEMS: IS IT THE ALCOHOL? JH Samet, LM Sullivan, JB Savetsky, T Tripp, KA Freedberg, General Internal Medicine, Boston University School of Medicine, Boston; DM-STAT, Medford, MA

PURPOSE: Because alcohol has been identified as one risk factor for decreased adherence to antiretroviral therapy (ART), we sought to identify all the factors associated with decreased adherence to ART in a cohort of HIV-infected patients with a history of alcohol problems.

METHODS: We measured quantity of alcohol consumed in the past month and adherence to ART in 157 HIV-infected patients with a history of alcohol problems (identified by either ≥ 2 positive responses to the CAGE questionnaire or by clinical assessment). Adherence was assessed by self-report using a modification of the AIDS Clinical Trials Group (ACTG) adherence instrument. This information was summarized in a continuous variable defined as the percentage of prescribed doses of ART taken over the prior 3 days. We evaluated other variables previously demonstrated to be associated with poor adherence to ART: social support; cocaine use; heroin use; depressive symptoms; and number of ART doses per day. We first examined the crude unadjusted relationship between alcohol and adherence using correlation analysis and then examined the relationship between alcohol and adherence using multiple linear regression adjusting for variables determined to be significant in bivariate analyses.

RESULTS: Subjects had the following characteristics: 19% women; 39% black, 21% Latino, 35% white, 5% other; mean age was 41 years; mean CD4+ count was 406/ μ L. Sixty-four percent (100/157) of patients reported 100% adherence in the prior 3 days (mean number of drinks 33); 36% (57/157) reported $<100\%$ adherence (mean number of drinks 80). The correlation between the number of drinks in the past month and percentage 3-day adherence was -0.29 ($p = 0.0002$). In the multivariate analysis controlling for cocaine and heroin use, social support, ART doses prescribed, and depressive symptoms, we found that number of drinks in the past month was inversely related to and the only variable significantly associated with 3-day adherence to ART ($p = 0.02$).

CONCLUSION: In HIV-infected patients with a history of alcohol problems, the number of drinks consumed was the only significant variable associated with decreased prior 3-day adherence to ART. Based on this, brief interventions for alcohol problems, which have been demonstrated to decrease alcohol consumption, should be part of a comprehensive effort to improve adherence to ART.

EFFECTIVENESS AND COST EFFECTIVENESS OF INFLUENZA IMMUNIZATION AMONG WORKING ADULTS (IN ARGENTINA). N Santoro, JT Insua, Occupational Medicine, Du Pont, Berazategui; Health Care Policy and Management, Austral University, Pilar, Buenos Aires, Argentina

PURPOSE: There is no consensus about the need to immunize healthy working adults, no data are available in Argentina about this issue. We evaluated the effectiveness and cost-effectiveness of an influenza immunization program in a working population.

METHODS: A prospective non-randomized cohort study comparing immunized patients from 1994–1998 versus controls was performed in Du Pont Occupational Medicine Clinic, Berazategui (Argentina). Also, a retrospective pre-post immunization program effectiveness study comparing 1991–1993 vs. 1994–1998 period was conducted. A cost-minimization study of the program was also performed. 811–785 workers per year were studied, on average 51% were immunized, with the recommended vaccine. Influenza like illness (ILI) episodes and days lost to work were measured in immunized (I) and controls (C), all new cases were recorded monthly since 1991. Annual risk difference (RD) for I and C episodes and days lost was computed for period 1994–1998, and for 1991–1993 preimmunization vs. 1994–1998 post immunization period. Costs of the program and of sick leave were obtained. **RESULTS:** Monthly incidence curves of new ILI episodes and for days lost to work varied for each year. Consistently more protection was obtained in the I vs. Cs. ILI episode RD between immunized and C was -12 episodes/100 person years (95% CI -10.90 , -13.09 , $p < 0.001$, vaccine effectiveness was 67%. Difference of days lost due to ILI was -39.80 days/100 person years (95% CI -37.18 , -42.42), $p < 0.001$.

Both results were more evident during influenza seasons. Comparing pre and post immunization periods, RD was -28.45 episodes/100 person years (95% CI -26.4 , -30.5), $p < 0.001$, vaccine effectiveness was 78%. Days lost to work were -112 days/100 person years (95% CI -109.5 , -114.4), $p < 0.001$. Savings were 39.83–147.8 U \$ per vaccine dose administered.

CONCLUSION: Influenza vaccine among the working population of the Du Pont plant in Argentina, is effective and cost effective.

IMPACT OF SEXUAL AND PHYSICAL ABUSE ON SUBSTANCE ABUSE-RELATED PROBLEMS. JB Savetsky, R Saitz, JM Liebschutz, LM Sullivan, C Lloyd-Travaglini, L Weinstein, JH Samet, Section of General Internal Medicine, Boston University School of Medicine; Biostatistics and Epidemiology; Data Coordinating Center, Boston University School of Public Health, Boston, MA

PURPOSE: Past interpersonal trauma is associated with worse physical and mental health; however its relationship to addiction severity is unclear. We hypothesize that a history of sexual and/or physical abuse (SPA) is associated with increased substance-related problems in a drug and alcohol-dependent population.

METHODS: We interviewed consecutive subjects at an inpatient detoxification unit between 6/97 and 4/99. Trained interviewers collected data on physical and sexual abuse history, demographics, physical and mental health, and they administered the Inventory of Drug Use Consequences (INDUC), a fifty-question, validated instrument measuring lifetime substance abuse consequences. All analyses were stratified by gender. We examined the relationship between SPA and INDUC score (range 1–45). Variables associated with the INDUC at $p < 0.20$ in bivariate analysis or deemed clinically relevant were entered into multiple linear regression models. In multivariable analysis we adjusted for age, race, depressive symptoms, physical functioning, drug of choice, polysubstance use, homelessness, and having a partner/spouse.

RESULTS: The 470 subjects were 24% female, 46% African American, 15% Hispanic, and had a mean age of 36 years. A majority (69%) graduated from high school and 46% were homeless. Drugs of choice were heroin (27%), cocaine (32%), and alcohol (39%). Among women 81% (90/111), and among men 69% (247/356), disclosed SPA. While women experienced predominantly both sexual and physical abuse, males reported mostly physical abuse. The median age of reported first abuse was 11 years of age. In bivariate analysis, INDUC scores were significantly higher in both men (means 37.7 vs. 34.7, $p < .0001$) and women (means 34.2 vs. 25.1, $p = .0008$) with SPA. In the multivariate model, SPA remained significantly associated with increased substance abuse consequences (male adjusted means 36.3 vs. 34.0, $p = .001$; female adjusted means 33.1 vs. 27.8, $p = .006$).

CONCLUSION: A history of sexual and physical abuse is common among persons admitted for detoxification, and this trauma is associated with greater substance abuse severity in both men and women. These findings support efforts to develop and evaluate substance abuse treatment strategies that simultaneously address SPA and addiction.

IMPROVING THE ACCURACY OF BREAST CANCER RISK PERCEPTIONS WITH A COMPUTER BASED TAILORED RISK INFORMATION INTERVENTION. MM Schapira, AB Nattlinger, TA McAuliffe, LJ Eparvier, Internal Medicine; Biostatistics, Medical College of Wisconsin, Milwaukee, WI

PURPOSE: It is well known that many women overestimate their personal risk of breast cancer. The objectives of this research are 1) to compare the accuracy of 5 year and lifetime breast cancer risk perceptions, and 2) to test the effect of presenting tailored risk information on the accuracy of breast cancer risk perceptions among a primary care population.

METHODS: A randomized controlled trial of tailored breast cancer risk information versus a control intervention was conducted. Women 40 years and older were randomly recruited from two primary care practices. Women in the intervention group were presented with tailored risk information derived from the computer based National Cancer Institute Breast Cancer Risk Tool. The tailored risk information was presented to the patient in an illustrated pamphlet. Control subjects received an American Cancer Society pamphlet on breast cancer prevention. One month after the intervention follow-up risk perceptions were assessed. Five year and lifetime risk estimate ratios were defined as the subjects' five year and lifetime risk estimates divided by the computer based calculated risk estimates. Risk ratio differences were defined as the five year and lifetime baseline risk ratios minus the five year and lifetime follow-up risk ratios, respectively. A Mann-Whitney test was used to compare the five year and lifetime risk ratio differences between intervention and control groups.

RESULTS: Seventy-seven women have been enrolled in the study to date. Forty-two percent (42%) of subjects had a high school level education or less, 93% were Caucasian, and 16% were less than 50 years of age. At baseline women overestimated their five year risk of breast cancer by a factor of 15 and their lifetime risk by a factor of 4. The table below presents risk estimates for the intervention and control groups.

	Baseline 5-Year Risk Ratio	Follow-Up 5-Year Risk Ratio	5-Yr Risk Ratio Difference	Baseline Lifetime Risk Ratio	Follow-Up Lifetime Risk Ratio	Lifetime Risk Ratio Difference*
Intervention (n = 46)	15.5 (18.2)	11.1 (20.1)	4.4 (16.8)	4.5 (3.9)	3.4 (4.6)	1.07 (4.5)
Control (n = 31)	13.6 (13.5)	17.1 (22.5)	-3.5 (22.1)	4.4 (4.3)	4.2 (5.0)	0.16 (3.0)

* $p < 0.05$

As seen in the table above, women in the intervention group significantly improved their 5 year breast cancer risk estimates compared to women in the control group.

CONCLUSION: We report for the first time that women overestimate short term breast cancer risk to a greater degree than lifetime breast cancer risk. However, an intervention that presents tailored risk information to patients in a quantitative and graphic format can improve the accuracy of risk estimates. These findings can be applied to patient education efforts in the primary care setting.

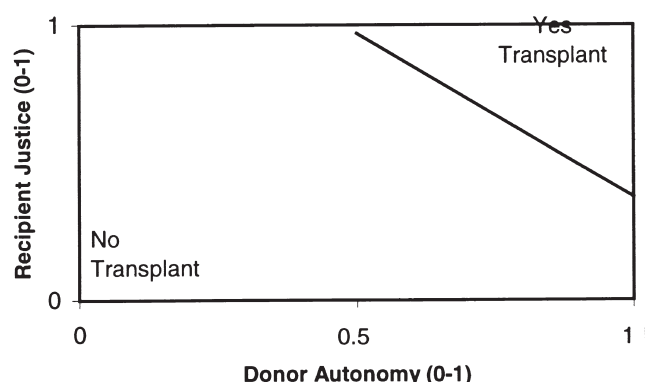
A GENERIC MODEL TO EXAMINE ETHICAL ISSUES SURROUNDING KIDNEY TRANSPLANTATION. J Stahl, R Kolarik, B Cohen, University of Pittsburgh, Pittsburgh, Pennsylvania

PURPOSE: Decision analysis is used to examine the risks and benefits of different decision strategies and has not traditionally been applied to ethical problems. UNOS recommends that Justice, Autonomy and Utility should all be considered when making an organ allocation decision. The risks and benefits of kidney transplantation traditionally have been examined from the perspective of either the patient or society. Another potentially useful perspective is that of the dyad of donor and recipient as both are part of the transplant process and both incur risks and benefits.

METHODS: A generic decision analytic model was constructed to examine two strategies: 1) Donor allocates organ to recipient [Strat1], 2) Donor does not allocate organ to recipient [Strat2]. The Utility component of the model was constructed as a semi-Markov process with 7 health states: Well, HTN, CRI, on Dialysis, Transplanted, post-graft failure on dialysis, and Dead. The concept of Justice was operationalized as a function of Need, Urgency and Outcome following Kamm. Need and Urgency were modeled as quality adjustment factors for outcome [range: 0-1]. Autonomy was defined, as per Faden and Beauchamp, as a function of Intent, Understanding and Controlling Influence (compulsion or restraint). These also were modeled quality adjustment factors [range: 0-1]. The model was designed to allow for the weighting of Justice, Autonomy and Utility within and across individuals thereby allowing the examination of a range of philosophical perspectives. In our base case, both Justice and Autonomy were considered as independent within and across individuals. The model generates an overall Ethically Adjusted Life Year (EALY). Finally, the EALY for each individual in the dyad was weighted by a factor, which allowed for differential weighting of each individual, and through sensitivity analysis, can represent a range of philosophical views from ranging from utilitarianism toward Rawls concept of absolute priority to the worst off.

RESULTS: In the base case analysis, the net EALY for the Strat1 outweighed Strat2. But, for example, if Donor Autonomy was compromised, i.e., fell below .47, then the transplant should not be performed. Donor Autonomy must exceed this level before any level of Recipient Justice would imply a net benefit to Strat1.

Recipient Justice v. Donor Autonomy



CONCLUSION: If Justice and Autonomy are important considerations in the allocation of organs then developing and using tools to assess the donors and recipients perceived level of justice and autonomy prior to the allocation of organs is necessary.

SEPSIS SYNDROME: VARIATION IN RESOURCE USE AMONG ACADEMIC CENTERS. DI Yu, R Platt, ER Black, KE Sands, JS Schwartz, PL Hibberd, PS Graman, PN Lanken, KL Kahn, DR Snyderman, J Parsonnet, R Moore, DW Bates, for the AMCC Sepsis Project Working Group.

PURPOSE: Treatment of sepsis syndrome is expensive, often encompassing a number of discretionary modalities such as pulmonary artery catheterization and albumin. The goal of study was to assess intercenter variation in resource use among patients with sepsis syndrome.

METHODS: We conducted a prospective cohort study of 1028 adults admitted to 8 academic medical centers between January 1993 and April 1994. This sample represented a weighted random sample of patients who were in an ICU or had a

positive blood culture, and all patients who received a novel therapeutic agent or who died in an emergency department or ICU. Information on all other patients admitted to the 8 centers during the study period ($n = 248,761$) was also obtained. The main outcomes were differences in total hospital charges (CHARGES) and length of stay (LOS) for patients with sepsis syndrome compared to the remainder of patients in each center. Rates of treatment with varying procedures used after sepsis onset were also measured. All calculations used appropriate case weights.

RESULTS: The difference of CHARGES between patients with sepsis syndrome and all other patients varied from \$53,985 per case to \$212,096 per case across sites. The difference of LOS varied from 16.1 days per case to 28.1 days per case. Treatments used frequently after the first onset of sepsis among patients with sepsis syndrome were pulmonary artery catheter (19.4%), ventilator support (21.8%), pressor support (45.8%) and albumin (14.4%). Pulmonary artery catheter, ventilator support and albumin had moderate variation profiles, varying 3.17- to 4.85-fold, while the rate of pressor support varied only 1.92-fold across centers.

CONCLUSION: These data demonstrate significant variation in resource use among academic centers, and in addition variation in the use of technologies in treatment of sepsis syndrome. Further analyses will focus on patient outcomes and their relation to variation in resource use in this study population.

RESOURCE UTILIZATION IN PATIENTS WITH SEPSIS SYNDROME. D Yu, R Platt, E Black, K Sands, J Schwartz, PL Hibberd, PS Graman, PN Lanken, K Kahn, D Snyderman, J Parsonnet, R Moore, D Bates, for the AMCC Sepsis Project Working Group.

PURPOSE: Sepsis syndrome is a common and serious condition associated with substantial mortality, and these critically ill patients often receive expensive and intensive therapy. The goal of study was to assess resource utilization associated with sepsis syndrome in 8 academic medical centers.

METHODS: We performed a prospective cohort study in which we evaluated 1028 adult admissions between January 1993 and April 1994, representing a weighted random sample of patients who were in an ICU or had a positive blood culture, and all patients who received a novel therapeutic agent or who died in an emergency department or ICU. The main outcomes were length of stay (LOS) and total hospital charges (CHARGES). We compared these outcomes for cases with those of all other patients admitted to the 8 centers during the study period ($n = 248,761$). In the sepsis group, we also determined correlates of post-onset LOS. All means and their differences were calculated using appropriate case weights.

RESULTS: In linear regression models adjusted for age, gender, insurance status, Charlson comorbidity score, ICU stay, discharge DRG weight and study site, the mean LOS for patients with sepsis syndrome was 18.2 days, which was 11.0 days longer than for all other patients ($p < 0.0001$; $r^2 = 0.19$), while the mean difference in CHARGES was \$43,942 ($p < 0.0001$; $r^2 = 0.43$). Eight independent correlates of increased post-onset LOS at or prior to onset of sepsis included pre-onset LOS; stay in an ICU; steroid therapy; surgery; burn injury, multiple trauma, bowel perforation or severe pancreatitis; Hickman catheter; transplant of liver, heart or lung; and cardiogenic shock.

CONCLUSION: The large total charge and length of stay of patients with sepsis syndrome compared with those without emphasize the need to assess the proportion of this utilization attributable to sepsis syndrome. Correlates of resource use in this population included a number of factors associated with tertiary care.

PROVIDER/ORGANIZATION CENTERED RESEARCH

COMMUNITY RESIDENTS PERCEPTIONS OF SERVICES PROVIDED BY COMMUNITY HEALTH WORKERS. KE Aaron, LR Bone, DM Levine, HR Rubin, Health Policy and Management, Johns Hopkins School of Hygiene and Public Health; General Internal Medicine, Johns Hopkins School of Medicine, Baltimore, MD

PURPOSE: Community Health Workers (CHWs) are used increasingly to supplement health care providers, particularly to educate and motivate patients with chronic disease to follow medical recommendations. Because CHW services have rarely been evaluated systematically, we sought to develop an instrument for clients to evaluate CHW care.

METHODS: To identify salient aspects of care, we content-analyzed 18 in-depth interviews with clients receiving home visits from CHWs in a program to improve blood pressure control in a low-income urban community. Then, we sampled 192 clients from 3 CHW programs addressing cardiovascular diseases to rate the importance of 65 specific aspects of CHW care (from 1, not at all, to 4, extremely important (EI)). Within each program, we calculated the proportion of respondents rating each item EI, and the proportion rating each domain EI by averaging across items. We averaged item-level and domain-level results for the three programs.

RESULTS: The five domains of CHW care were CHW attributes, CHW services, benefits from CHWs, visit arrangement, and role of CHWs in improving health care access. 84 clients completed the importance ratings, 35 from Program 1, 32 from Program 2, and 17 from Program 3 (response rates 37-46%). The highest rated domain was the role of the CHWs in improving access to care, e.g., care for the elderly, access to unavailable services, and supplementing doctors' care. Over-

all 60% of respondent rated that role extremely important. The second most important domain (rated extremely important by 42% of respondents) CHW attributes, e.g., worker knowledge, communication and interpersonal skills. The next most important domain, rated by an average of 40% of respondents, was benefits from CHWs, e.g., social support and lowered blood pressure. extremely important. Domains rated fourth and fifth in importance were CHW services and visit arrangements respectively. CHW services, e.g., checking blood pressure and providing health information was rated extremely important by 33% of respondents. Visit Arrangement, e.g., call prior to home visit and the location of the visit in the home, was rated extremely important by 27% of respondents.

CONCLUSION: Clients note diverse important aspects of CHW care, and most highly value the role that CHWs play in extending medical care in underserved communities. Evaluation instruments obtaining client rating of CHW services should include aspects of care salient to service recipients and should be used to assess care.

ANALYSIS OF USER-FEEDBACK AS A TOOL FOR IMPROVING SOFTWARE QUALITY. SA Abokire, MT Martin, JM Telch, GJ Kuperman, DW Bates, Information Systems, Partners HealthCare System, Boston, MA

PURPOSE: Clinical software is having a growing impact on patient care, with an influence ranging from passive information to active monitoring and therapeutic intervention. Since patient outcomes are increasingly affected by clinical software, a process to monitor and improve the quality of clinical software is warranted. One key source of information about problems and their clinical impact are the end-users. However, user-feedback is typically analyzed in a piecemeal fashion, and is not usually analyzed in an ongoing way. The purpose of this study was to develop a meaningful classification structure of clinical software user-feedback. Further, our goal was to analyze our findings to evaluate potential impact of problems, to evaluate rates of problem reporting, and to extract lessons for developers of new clinical applications.

METHODS: We evaluated all user feedback from a newly implemented computerized physician order entry system at Massachusetts General Hospital, a tertiary hospital in Boston, Massachusetts. From the start of the application in October 1998 through December 1999, all of the 888 feedback comments were analyzed. Comments that related to operation, policy, or were general remarks were removed, leaving 785 user comments. From these, 18 categories were developed to delineate the software function being performed. This scheme appears reasonably complete (all user comments can be readily assigned), and includes functions such as alerts, co-signature, authentication, order activation, screen formatting, medication start or stop, order status, and results viewing. Next, we rated the impact of the problem as either one of cost, safety, or efficiency. In each category, a range of one to four was applied to indicate severity of impact.

RESULTS: In all, 634 comments had an impact on efficiency (70%), 233 comments had a potential impact on patient safety (26%), and comments with a cost impact totaled 36 (4%). (The total number is greater than 785 since some comments had impact on more than one category). The rate of feedback declined from a peak of 231 comments per month initially to a steady average rate of 20 per month, suggesting either responsiveness to early suggestions or user fatigue. Several fundamental problems of importance for future developers emerged, including inadvertent order activation, insufficient alerting, potentially confusing templates, and necessity of safety measures for drug ordering boundaries.

CONCLUSION: A clear-cut mechanism for assembling and learning from user-feedback is possible, and is a beneficial cornerstone to developing correct and safe clinical software. When user feedback is classified and analyzed useful lessons can be learned and applied to future software.

TELEMEDICINE FOR DELIVERY OF PULMONARY OUTPATIENT CARE TO A RURAL SITE: A COST ANALYSIS. Z Agha, RM Schapira, B Varkey, JA Frahm, BE Dunn, Internal Medicine, Medical College of Wisconsin and Zablocki VAMC; Pathology, Medical College of Wisconsin, Milwaukee, WI; Iron Mountain Veterans Affairs Medical Center, Iron Mountain, MI

PURPOSE: The Milwaukee Veterans Affairs Medical Center (MVAMC) is the referral center for Iron Mt. VAMC, a rural hospital in the MI upper peninsula. In the past, patients from the Iron Mt. made a 450 mile round-trip to MVAMC for pulmonary consultation which necessitated lodging at the MVAMC. In 1998, a pulmonary telemedicine clinic was started between MVAMC and Iron Mt. to provide pulmonary services to Iron Mt. patients. We have conducted a cost analysis to determine if the telemedicine program is cost effective.

METHODS: We conducted a cost minimization analysis of telemedicine by building a model to compare telemedicine with face-to-face care. For cost minimization analysis, comparisons of competing options are analyzed and the one that costs least is favored. Analysis was conducted from both societal and health care provider (Veterans Affairs) perspective. We have captured both direct medical costs and lost productivity costs for societal analysis. However, intangible costs were not included. The time frame of the project was set at 5 years. Standard cost analysis procedures were used and annual cost of managing a patient in each program was calculated. All cost data were standardized to 1999 dollars, future costs were discounted at 3% rate. For base-case analysis, personnel and capital cost data were used from the ongoing telemedicine program and face-to-face care. Sensitivity and threshold analyses were conducted for all variables.

RESULTS: 74 patients (116 visits) have been seen in the telemedicine clinic. 9 (12%) of these patients subsequently made a face-to-face visit to MVAMC for a

procedure. Results of base-case analysis showed an annual societal cost of \$441 per patient managed by telemedicine versus \$540 per patient managed by face-to-face care. From a health care provider's perspective the base-case cost of telemedicine (\$253/patient/year) was higher than that of face-to-face care (\$216/patient/year). Sensitivity analysis demonstrated that if the number of patients seen by telemedicine was doubled, or the duration of telemedicine visit was reduced from 45 (base-case) to 30 minutes, or the cost allocated to lodging for face-to-face care was increased from \$30/night (base-case) to \$58/night (national average), then telemedicine becomes the favored option.

CONCLUSION: Pulmonary telemedicine clinic is a cost effective way to deliver outpatient care to patients residing in an area of limited access to pulmonologists. Telemedicine would be even more cost effective if the analysis included the monetary value of intangible benefits of telemedicine, such as improved access and decreased demands related to patient travel.

IMPACT OF HOUSESTAFF TEACHING ON SMOKING CESSATION ADVICE DOCUMENTATION. M Alam, A Hussain, S Chhabra, LT Clark, Medicine, SUNY HSC at Brooklyn, NY

PURPOSE: Smoking is one of the primary modifiable risk factors for coronary artery disease (CAD). Smoking multiplies the effect of other CAD risk factors. Smoking cessation is one of the most effective and certainly the least expensive approaches for preventing disease progression in native coronary vessels and bypass grafts.

Approximately 70% of current smokers report a desire to stop smoking completely. However, data from the 1991 National Health Interview Survey of the Center for Disease Control (CDC) indicate that only a little more than half of the smokers who had at least one outpatient visit with a physician or other health-care professional during one year period were advised to quit.

In a retrospective chart review of 32 consecutive patients with ICD 9CM code: 410 (Acute Myocardial Infarction) discharged from our hospital's coronary care unit (CCU), documentation of smoking cessation advice was present in 11 patients' charts (34%). This data prompted us to do a prospective analysis of this important quality of care indicator.

METHODS: The housestaff on monthly rotation in the the CCU were given a comprehensive overview of risk factors for CAD and the pervasive problem of the lack of documentation of smoking cessation advice was emphasized in a briefing at the beginning of the month for two consecutive months. The charts were reviewed by an independent observer for the documentation of smoking cessation advice prior to the discharge.

RESULTS: During this two month study period, 18 patients with the diagnosis of acute myocardial infarction at discharge from the CCU were identified as current smokers at the admission. Thirteen (72%) patients were male and 5 (28%) were female. Smoking cessation advice was documented in 11 patients' charts (61%). When interviewed by the independent observer at the time of discharge, 5 patients (28%) were determined to quit smoking by themselves, 7 patients (39%) expressed to seek help for smoking cessation, and remaining 6 patients (33%) were not sure of quitting.

CONCLUSION: Our data show that taking a simple measure like housestaff briefing, the documentation of smoking cessation advice improved from baseline of 34% to 61%. This suggests that housestaff motivation can indeed improve the outcome related with quality of care issues. A significant improvement in many other aspects of patient care may be achieved by conscious efforts in their training.

WHAT THEY CARRIED: A DESCRIPTIVE STUDY OF PATIENT LISTS IN PRIMARY CARE PRACTICE. G Albertson, J Kutner, L Schilling, CT Lin, E Cyran, L Ware, J Steiner, RJ Anderson, University of Colorado, Denver, CO

PURPOSE: To describe a patient population presenting to a primary care clinic with a written list. We sought how often patients bring a pre-written list, the demographics of this population, their health status and the reason for the visit.

METHODS: 507 consecutive patients enrolled in an academic internal medicine clinic presenting for primary care were studied. Patients were asked to complete a pre-visit questionnaire asking about demographic data, reason for today's visit, health status, length of relationship with their PCP, whether they usually bring a pre-written list and the reason for the list. Providers were surveyed with a post-visit questionnaire and asked to rate the medical/social complexity of the patient, to list known or suspected mental health diagnoses and to rate their overall visit satisfaction.

RESULTS: 111/507 patients (20%) presented during the study period with a pre-written list. Patient gender, level of education, annual income and ethnicity did not differ significantly between those with or without lists. Patients with a list self-rated their overall health lower than patients without a list ($p = 0.029$). Patients with a list tended to be older (age >45 years) compared to patients without a list ($p = 0.08$). Duration of the patient-provider relationship did not influence whether patients presented with or without a list ($p = 0.56$), but patients were more likely to bring a list when their appointment was with a provider other than their PCP. Lists were more common in encounters scheduled for medication refills (31% vs 15%, $p = 0.001$), follow up on an existing concern (61% vs 49%, $p = 0.028$), to discuss health screening (11% vs 5%, $p = 0.046$) or to request a test (17% vs 7%, $p = 0.002$). Patients who brought a list to the study encounter were more likely to bring a list when seeing any doctor compared to patients who did not bring a list to the study encounter (69% vs 14%, $p = 0.001$). Patients cited help with recall (63%), improvement in visit efficiency (46%), and ease of explanation of symp-

toms/concerns (26%) as the most common reasons for bringing a list. Providers rated social/medical complexity higher in patients with a list compared to those without ($p = 0.029$). Known or suspected psychiatric co-morbidity rated by the provider did not differ significantly between the two groups. Providers found lists to be helpful in 75% of patients who brought a list and PCPs felt a list would have been helpful in 10% of patients who presented without a list. PCP rating of overall visit satisfaction was comparable in patients with and without a list.

CONCLUSION: Patients commonly bring pre-written lists to encounters with their PCPs. Few demographic data characterize this group of patients. The old adage that once a list writer, always a list writer seems to in this medical setting. Patients use pre-written lists to facilitate communication of symptoms with their doctor and improve visit efficiency. Providers view patient lists as helpful and no association is noted between list-bringing behavior and an underlying mental diagnosis.

WEB-BASED ELECTROCARDIOGRAM RETRIEVAL: DOES IT REDUCE ORDERS? J Algozy, K Traber, L Bellini, S Schwartz, D Asch, Robert Wood Johnson Clinical Scholars Program, University of California Los Angeles, Los Angeles, CA; Department of Anesthesia; Department of Internal Medicine, University of Pennsylvania Health System; Department of Internal Medicine, Philadelphia Veterans Administration Medical Center, Philadelphia, PA

PURPOSE: The purpose of our study was to evaluate the installation of a web-based electrocardiogram (EKG) archive retrieval system at a major tertiary care academic hospital. Few studies have been conducted to investigate the benefits of these new data technologies and their impact on provider practice.

METHODS: In a before and after trial, all consecutive patients seen in the Admissions Evaluation Center at the Hospital of the University of Pennsylvania between February and April 1998 were enrolled as pre-intervention subjects prior to the installation of a web-based EKG archive retrieval system in November 1998. All patients in the AEC between February and April 1999 were enrolled as post-intervention subjects after the installation of the system. We used logistic regression to study EKG orders and adjusted for comorbidities, age, sex, and whether the patient was a walk-in or pre-scheduled visit.

RESULTS: A total of 1543 patients were enrolled in the pre-intervention group and 1005 patients in the post-intervention group. Adjusting for age, sex, comorbidities, and scheduling status, 71.0% of patients had an EKG ordered after computer installation compared to 75.3% prior to installation ($p < 0.043$).

CONCLUSION: Health care institutions continue to install expensive medical information systems. This study showed a statistically significant, although clinically modest reduction in EKGs ordered after installing an expensive new computer technology. Health care institutions must continue to evaluate whether medical information systems are worth the significant expense incurred during their installation.

THE ACHIEVABLE BENCHMARK OF CARE: AN EFFECTIVE AND INEXPENSIVE ENHANCEMENT TO PROVIDER FEEDBACK. JJ Allison, CI Kiefe, NW Weissman, S Person, General Internal Medicine; Preventive Medicine; School of Health Related Professions; Center for Outcomes and Effectiveness Research and Education, University of Alabama at Birmingham, Birmingham, AL

PURPOSE: Provider feedback, when effective, generally produces modest behavior changes. We sought to determine if enhancement of feedback with an Achievable Benchmark of Care (ABC), a data-driven, peer-based, measure of excellence, would increase feedback effectiveness.

METHODS: Medical records were reviewed for 1360 diabetic patients from the practices of 70 Alabama primary care physicians. Physician performance on diabetes-related indicators (measurement of HgA1c, cholesterol and triglycerides, influenza immunization, and foot examination) was calculated. Indicator performance rate was the percentage of patients who received the procedure at least once during the preceding 18 months. We calculated an ABC for each indicator based on a subset of top-performing physicians. Physicians were divided randomly into a group that received feedback of personal performance and mean group performance and a group that received similar feedback enhanced with an ABC. Baseline measurement was from 1/94-6/95 and follow up from 1/97-6/98.

RESULTS: The table shows baseline performance and absolute improvement for physicians receiving ABCs compared to physicians receiving usual feedback for three indicators:

		Baseline (%)	Improvement (%)	Odds Ratio	p-value
HgBA1c	ABC	31	38		
	Usual	29	33	1.33	<0.001
Influenza	ABC	38	18		
	Usual	39	9	1.57	<0.001
Foot exam	ABC	45	13		
	Usual	31	11	1.33	<0.02

Physicians receiving ABC enhanced feedback improved more than physicians receiving usual feedback (absolute incremental improvement: 5% for HgBA1c, 9% for influenza vaccination, and 2% foot exam, $p < 0.05$ for all). Similar results (not shown) were achieved for measurement of cholesterol and triglycerides. Multivariable analyses adjusted for clustering of patients within physicians without loss of significance, and revealed that fewer years in practice correlated positively with performance improvement.

CONCLUSION: The ABC offers a low-cost, effective approach to enhancing provider feedback. With high face validity, the ABC has many advantages over subjectively defined benchmarks.

UNDERSTANDING THE REASONS FOR COST SAVINGS ASSOCIATED WITH A HOSPITALIST SERVICE. N Armistead, H Palmer, S Manivannan, M Elnicki, K Evans, West Virginia University, Morgantown, WV

PURPOSE: Many hospitalist programs have demonstrated significant cost reductions but the reasons for these cost reductions are not clear. The goal of this study is to define specific patient parameters that may contribute to the savings. We hypothesized that hospitalists are more effective in implementing savings in patients with more common diagnoses, those admitted via the ED (as opposed to ICU) and those who are able to get part of their therapy at home.

METHODS: During the 98-99 academic year, WVU implemented a hospitalist service (H) which was compared with the two other medicine services, one staffed by a General Internist (GIM) and the other by a subspecialist (SS). A total of 2577 patients were randomly assigned to the teams based on housestaff call schedule. The discharged patients from the three teams were retrospectively analyzed, specifically looking at the following subgroups: 1) Top 10 percentile cost patients 2) Patients admitted via ED vs "other" source (outside referrals, clinics, direct admits) 3) patients who spent any time in the intensive care unit 3) The top 5 DRGs 4) Patients discharged with home health.

RESULTS: The table below lists number of patients and mean costs for some of the subgroups examined. The H group demonstrated significant cost reductions in both patients admitted from the ED and outside referrals compared with SS. Similarly the H group also demonstrated savings in the top 5 DRG and in patients discharged with home health (n/mean cost H 81/6452, GIM 70/8908, 66/13784*). No cost differences were demonstrated in ICU patients and in top 10 percentile patients for the total group (mean costs H 38769, GIM 34510, SS 32693).

Total (SD)	ED admits	ICU transfer	other admits	top 5 DRGs		
H	859 5716 (12734)	503 3113	90	26229 266	3699	179 3195
GIM	796 6297 (13522)	449 3913	69	29480 278	4395	197 4143
SS	922 7906* (14635)	532 4879*	111	27565 279	5856*	204 5662*

* $p < 0.01$ v. H; there was no difference in patient mix and outcome measures

CONCLUSION: Hospitalists are more likely to decrease costs in patients who do not spend time in an ICU, patients whose diagnoses are one of those more commonly seen and patients discharged with home health, perhaps reflecting quicker mobilization of resources. Source of admission (ED versus other) does not impact on the savings. They have a lesser impact on the "most-expensive" patients.

VARIATION AMONG VA HOSPITALS IN RISK-ADJUSTED POSTOPERATIVE PULMONARY COMPLICATION RATES. AM Arozullah, W Henderson, S Khuri, J Daley, University of Illinois, Chicago; Hines VA Hospital, Hines, IL; Boston VA Healthcare, West Roxbury; Partners Health Care, Boston, MA

PURPOSE: To determine if variation exists in postoperative pneumonia (POP) and respiratory failure (RF) rates following noncardiac surgery among VA hospitals after adjustment for patient-specific risk factors.

METHODS: Cases were selected from prospective observational data collected through the National VA Surgical Quality Improvement Program. POP was defined as positive sputum culture with antibiotic treatment or an infiltrate on chest x-ray diagnosed as pneumonia or pneumonitis following surgery. RF was defined as mechanical ventilation greater than 48 hours postoperatively and/or reintubation subsequent to postoperative extubation. Logistic regression models using patient-specific risk factors to predict POP and RF were developed using Phase I cases (10/91-12/93, 44 VA hospitals) and validated using Phase II cases (1/94-8/95, 123 VA hospitals). The models were used to estimate the probability of POP and RF for each case with the sum within each hospital designated as the expected number of complications. Observed to expected (O:E) ratios and 95% confidence intervals (CI) were computed for each hospital.

RESULTS: Phase I POP rates ranged from 0.8% to 8.5%. After risk adjustment, 7 hospitals (16%) were high outliers (CI > 1), 12 hospitals (27%) were low outliers (CI < 1), and 9 hospitals (20%) changed rank by more than 5. Phase I RF rates ranged from 1.0% to 6.0%. After risk adjustment, 5 hospitals (11%) were high outliers, 7 hospitals (16%) were low outliers, and 21 hospitals (48%) changed rank by more than 5. No high outlier hospital for RF was a high outlier for POP. Four of the 7 low outlier hospitals for RF were also low outliers for POP. Phase II POP rates ranged from 0% to 10.5%. After risk adjustment, 26 hospitals (21%) were high outliers, 21 hospitals (17%) were low outliers, and 38 hospitals (31%) changed rank by more than 10. Phase II RF rates ranged from 0% to 6.0%. After risk adjustment, 15 hospitals (12%) were high outliers, 23 hospitals (19%) were low outliers, and 44 hospitals (36%) changed rank by more than 10. Nine of the 15 high outlier hospital for RF were high outliers for POP. Eleven of the 21 low outliers for POP were low outliers for RF.

CONCLUSION: Significant variation exists in risk-adjusted rates of POP and RF among VA hospitals. There is only moderate correlation between the POP and RF rate within each hospital. Hospital POP and RF rates are potential outcome-based measures that could be used in addition to mortality rates to evaluate surgical quality of care. Future studies focusing on the impact of hospital structure and

process of care measures on POP and RF rates in high and low outlier hospitals may provide insight into how to decrease POP and RF rates.

VARIATION IN PROCESSES OF CARE FOR PATIENTS WITH COMMUNITY-ACQUIRED PNEUMONIA AT U.S. ACADEMIC HOSPITALS. *SJ Atlas, JJ Dedier, DE Singer, General Medicine Division, Massachusetts General Hospital, Boston, MA*

PURPOSE: Processes of care associated with decreased mortality have been reported for patients hospitalized with community-acquired pneumonia (CAP). Widespread collection and reporting of such quality indicators have been advocated as a means to compare health care organizations. We assessed the variation in such processes of care for CAP among 38 U.S. academic hospitals participating in the University HealthSystem Consortium pneumonia benchmarking project.

METHODS: Adults with a discharge diagnosis of CAP and no exclusion criteria (e.g., HIV, tuberculosis, cystic fibrosis, immunosuppressed, hospitalized in past 10 days) were eligible. Retrospective hospital records were reviewed for up to 40 consecutive eligible patients discharged between December 1, 1997 and February 28, 1998. Process measures assessed included: initial administration of antibiotics (ABX) within 8 hours of arrival, collection of blood cultures (BCX) prior to antibiotic administration, and blood and sputum culture (SPX) collection and oxygen saturation (O2sat) measurement within 24 hours. We report the median percentage of patients achieving each process measure by hospital and their range. Differences across hospitals were assessed controlling for the severity of CAP at admission using the Pneumonia Severity Index.

RESULTS: The population included 1100 eligible patients admitted to 38 hospitals. The median percentage of patients achieving each process measure by hospital and the range across hospitals is depicted.

Process measure	% of Pts Achieving by Hospital, Median	% of Pts Achieving by Hospital, Range
ABX within 8 hours	77.5	53.8–100
BCX before antibiotics	74.8	11.1–100
BCX within 24 hours	82.0	53.6–100
SPX within 24 hours	50.2	12.0–87.1
O2sat within 24 hours	91.0	50.0–100

After controlling for pneumonia severity on admission, the percentage of patients achieving these each of process measures varied significantly across hospitals (all $p < 0.005$).

CONCLUSION: In conclusion, more than 2-fold variation exists in achieving markers of quality care among patients admitted with CAP across U.S. academic institutions.

PHYSICIAN ATTITUDES TOWARDS THE HOSPITALIST MODEL OF CARE: RESULTS OF A NATIONAL SURVEY. *AD Auerbach, PK Lindenauer, B Nelson, SZ Panfilat, P Katz, RM Wachter, Department of Medicine, UCSF, San Francisco, CA; Baystate Med Ctr, Tufts U. School of Medicine, Springfield, MA; ACP-ASIM, Philadelphia, PA; Inst. of Health Policy and Research, UCSF, San Francisco, CA*

PURPOSE: To determine availability of and physician attitudes toward the hospitalist model in a national sample of internists.

METHODS: We performed a telephone survey of American College of Physicians—American Society of Internal Medicine members who identified themselves as generalists. Physician reports of inpatient care arrangements and responses to questions regarding attitudes toward the hospitalist model were primary outcome measures. A summary score of physician attitude was derived from highly correlated survey questions ($\alpha = 0.84$). Multivariable models were employed to determine factors associated with attitude summary score.

RESULTS: Four hundred physicians (51%) agreed to participate. The median age of respondents was 45 years; 81% were male, 50% practiced in independent or small group settings, and most admitted patients to private, non-profit teaching hospitals. Most physicians reported they cared for 2 or fewer hospitalized patients each day. Ninety-four percent were familiar with the term hospitalist, "65% reported that the model was available in their community, and 25% used hospitalist services for all or some of their inpatients. The model was more commonly available in Western states ($p < 0.05$). Few (2%) physicians reported the presence of the mandatory" hospitalist model in their community. Seventy-three percent felt that the hospitalist model would reduce continuity of care. Only 28% of physicians felt that their patients would prefer care from an inpatient care specialist, but 51% felt that patients might get better care in such a system. In multivariable models, physicians who were in solo practice had a less favorable attitude towards the hospitalist model ($p < 0.01$). Physicians who had training in specialties with more inpatient practice had a more negative attitude ($p = 0.047$), as did physicians who had 15 or more patients hospitalized each month ($p = 0.011$). However, physicians who had hospitalists in their community had a more positive impression of the hospitalist model ($p < 0.001$). Factors such as physician age and gender, or hospital characteristics were not associated with differences in attitude score.

CONCLUSION: In this national sample, hospitalist services were widely available; mandatory hospitalist services were rare. Despite feeling that quality of care might be improved, physicians were concerned about the hospitalist model's effects upon patient-doctor relationships and patient satisfaction. Investigations are needed to determine true effects of the hospitalist model upon patient satisfaction and the doctor-patient relationship.

THE EFFECT OF MANAGED CARE ON PREVENTABLE HOSPITALIZATION RATES. *LI Backus, M Moron, P Bacchetti, AB Bindman, Medicine; Epidemiology & Biostatistics, UCSF, San Francisco, CA*

PURPOSE: Preventable hospitalization rates are a previously validated measure of primary care effectiveness. To assess managed care's impact on primary care effectiveness, we examined whether an increase in managed care over time was associated with a decrease in preventable hospitalization rates.

METHODS: Hospitalization rates and managed care penetration were calculated on an annual basis for adults ages 18–64 years in each of the 394 zip-code defined primary care service areas across California. Annual statewide hospital discharge data provided information on every hospitalization and whether it was for a patient insured by managed care. Preventable hospitalizations were identified by ICD-9 codes as those with a primary diagnosis of asthma, CHF, COPD, diabetes or HTN. Hospitalization rates were calculated using census estimates of the 18–64 year old population for each area. To diminish the impact of inaccurate population estimates, we created "standardized" preventable hospitalization ratios by dividing the combined rates of these 5 conditions by the admission rate for appendicitis and GI obstruction in each area, as these two conditions were assumed to be stable over time. Managed care penetration in each area was determined on an annual basis as the percentage of hospitalizations for that area that were insured by managed care. Change in managed care penetration was calculated as the average of the 1995 and 1997 values minus the average of the 1990 and 1992 values. Changes in standardized preventable hospitalization ratios were log transformed and then calculated similarly. Because these were not normally distributed, rank correlations (r) were calculated, both overall and within each quartile of initial standardized admission ratios.

RESULTS: From 1990 to 1997, managed care penetration increased from 24.3% to 42.2% and overall admission rates declined from 956 per 10,000 to 785 per 10,000. The change in managed care was not associated with a change in standardized preventable hospitalizations overall ($r = -0.04$; 95% CI $-0.14, 0.06$; $p = 0.42$). However, the negative association between increasing managed care penetration and changes in the standardized preventable hospitalization ratios increased progressively when moving from areas in the lowest quartile of baseline preventable hospitalization ratios towards those in the highest quartile. In the 98 areas with the highest initial ratios, increases in managed care penetration were significantly associated with a decline in the standardized preventable hospitalization ratios ($r = -0.3$; 95% CI $-0.47, -0.11$; $p = 0.003$).

CONCLUSION: An increase in managed care, particularly in areas with initially high preventable hospitalization rates, was associated with a decrease in these rates over time suggesting an improvement in primary care effectiveness.

OBSERVATIONS ON HOW CLINICIANS USE SUMSEARCH. *RG Badgett, CD Mulrow, LS Levy, J Arterburn, Medicine, Univ. of TX Health Science Center at San Antonio and the Audie L. Murphy Memorial Veterans Hospital, San Antonio, TX*

PURPOSE: SUMSearch (<http://SUMSearch.uthscsa.edu>) is a new method to find medical evidence that uses meta-searching and contingency searching. As SGIM is considering endorsing this product, we summarize its recent usage.

METHODS: We report the results of the last 502 consecutive searches. We optimistically define a successful search as retrieving at least one article from MEDLINE.

RESULTS: 421 (84%) of searches were successful. Previously, we reported that 22% of searches used more than two words within any one search term ("long terms") and these searches were significantly more likely to fail. Prior to the current series, SUMSearch was programmed to warn clinicians when they were about to submit long terms. In the current series, 12% of searches contained long terms and the presence of long terms was no longer a predictor of failed searches. SUMSearch now records when the MESH browser is used to select search terms. Searches used the MeSH browser in sixteen percent of searches. These searches were significantly less likely to fail (8% versus 16%).

In 50% of the successful searches, clinicians clicked at least one link to a document. The median time was 155 seconds to finish selecting a median of two documents. The shortest time needed was 10 seconds to click a single document. The longest time occurred when a search that was executed, then three hours later 12 documents were clicked over 20 minutes.

CONCLUSION: 84% of searches retrieved at least one document. Based on the current and a previous analysis, searches that use the MeSH browser and avoid long search terms are more likely to succeed. Clinicians click documents from only 50% of successful searches. Possible explanations for not selecting documents include: perceived lack or genuine lack of germane citations, overlooking germane articles, and insufficient time to review list of documents. These results for clinicians using SUMSearch are consistent with prior research that showed 50% of literature searches done by medical librarians are able to answer clinical questions. We plan to redesign the list of citations returned by SUMSearch to facilitate review of the many citations.

DIABETES MANAGEMENT BEFORE AND AFTER INITIATING MEDICAID MANAGED CARE. *JE Bailey, RJ Womeodu, JE Wan, DL Van Brunt, GW Somes, Medicine and Preventive Medicine, University of Tennessee, Memphis, TN*

PURPOSE: Managed care may increase emphasis on outpatient management of chronic diseases such as diabetes, but little firm data are available. This retrospective cohort study sought to determine the impact of managed care on effec-

tiveness of diabetes management in Tennessee, where a statewide experimental program (TennCare) delivers services to all Medicaid enrollees through capitated managed care organizations (MCOs).

METHODS: A cohort of diabetic enrollees in the state's largest academic Medicaid MCO was assessed longitudinally to compare their exposure to diabetic services before and after TennCare. A total of 88 cohort members met the following inclusion criteria: 1) age 18–64 throughout the study period (92–96), 2) continuous enrollment for at least 320 days in two years of TennCare (95 and 96), and in the Medicaid program in two years before TennCare (92 and 93), 3) at least one diabetes diagnosis or medication prescription in each year of the study, and 4) no Medicare or other third party insurance. Claims data were used to assess baseline characteristics and chart review data were used to assess health services utilization for 82% of cohort members ($n = 72$) for whom complete medical records were available. Data for 94, the year of transition to the TennCare program, was excluded from the analysis. The paired t -test was used to compare exposures to clinical services and outcomes before (92–93) and after TennCare (95–96).

RESULTS: For the cohort participants, average age was 48.5, 76.4% were female, 90.3% were black, 2.8% white and for 6.9% race was unknown, 86.4% were eligible on the basis of disability, 10.2% were eligible on the basis of AFDC, 3.4% were eligible on the basis of blindness. A large percentage were identified to have complications of diabetes during the study period, including nephropathy (14.9%), retinopathy (22.0%), neuropathy (14.9%) and osteomyelitis (2.8%). Subjects had an average of 6.0 vs. 7.9 outpatient clinic visits per year ($p = .003$), 0.5 vs. 0.7 diabetic eye exams ($p = .08$), 0.2 vs. 0.5 foot exams ($p = .1239$), 0.4 vs. 0.6 cholesterol assessments ($p = .0042$), and 0.4 vs. 0.9 glycosylated hemoglobin assessments annually ($p = .0011$). No other significant differences were found in rates of cholesterol assessments, urine protein assessments, mammography, average blood pressure, average total and LDL cholesterol. However, average glycosylated hemoglobin decreased from 9.6 to 8.3 ($p = .0165$).

CONCLUSION: Diabetic enrollees experienced increases in utilization of many recommended health services, and despite disease progression, experienced improved glucose control following the initiation of Medicaid managed care. These increases may reflect historical trends or, alternatively, improved chronic disease care in a primary care gatekeeper system.

MEDICAID PATIENTS REPORT BARRIERS TO ASTHMA-RELATED CARE. YA Bair, JA Garcia, RL Kravitz, Center for Health Services Research in Primary Care; Division of General Medicine, University of California Davis, Sacramento, CA

PURPOSE: It is unclear whether Medicaid managed care programs, including the newer capitated plans, affect health care access. This study examines differences in self-reported access among 478 adult, non-elderly, asthmatics followed in a large Northern California medical group.

METHODS: This survey-based study of access to care, service utilization and satisfaction with care, was administered at outpatient clinic sites in an academically affiliated group practice consisting of 120 primary care providers. There were no significant differences in terms of age, gender or ethnicity between responders (75%) and non-responders. The current study focuses on the 478 (12% of 3,990) patients who answered "yes" to the question "Do you have asthma?".

Categorical variables were analyzed using Fisher's exact test for comparison of proportions. In comparisons of means for continuous variables, 2-sided t -tests were used. Results with a two-tailed p -value less than or equal to 0.05 were considered statistically significant. The independent association of insurance status with perceived access problems and emergency department use was assessed using logistic regression analysis.

RESULTS: Compared to patients with other forms of insurance in the same managed care network, patients covered by the state's Medicaid program were more likely to report access problems for asthma-related care, including difficulties reaching a health care provider by telephone and obtaining clinic appointments and asthma medications. Adjusting for relevant clinical and sociodemographic variables, Medicaid patients were significantly more likely to report at least one access problem compared to non-Medicaid patients (AOR = 3.3, $p < .0005$). Patients reporting at least one access problem were also more likely to have made at least one asthma-related emergency department visit within the past year (AOR = 2.4, $p < .0001$). Reported barriers to care did not translate into reduced patient satisfaction.

CONCLUSION: These findings suggest that health insurance alone does not guarantee health care access. Comparison of Medicaid patients to other enrollees within the same managed care organization provides a reliable assessment of perceived barriers to care by controlling for variation in the system of care delivery. Additional study is needed to determine why some Medicaid patients, who have financial access to services and are documented users of primary care services, nonetheless report barriers to obtaining care. The existence of multiple barriers to the use of services in an insured population has important implications for patients, purchasers and community health planners, particularly in light of ongoing efforts to produce the highest quality, most efficient system of care.

HEALTH RISKS IN CORRECTIONAL HEALTH CARE WORKERS IN RHODE ISLAND. RA Ballard, CM Salas, AC Spaulding, JD Rich, JB Nemhausen, D Trout, RRM Gershon, Rhode Island Hospital; The Miriam Hospital, Providence, RI; The Johns Hopkins University, Baltimore, MD; CDC, Cincinnati, OH

PURPOSE: To assess Rhode Island correctional health care worker's (CHCWs) risk of exposure to tuberculosis and hepatitis B and C.

DESIGN: Cross-sectional pilot study, part of a larger three-state study.

Setting: The adult state correctional facility in Rhode Island.

Subjects: 102 CHCWs including physicians, nurses, psychologists, dentists and hygienists.

Method: An assessment was conducted as follows: 1) a 13-page (175 item) self-administered questionnaire was completed, which included questions on demographics, adherence to universal precautions (UP), work stress, and risk factors for tuberculosis (TB) and hepatitis; 2) a 5 unit PPD tuberculin skin test (TST) was placed, and 3) one tube of blood was drawn for hepatitis B and C serologies.

RESULTS: Of 102 eligible CHCWs in RI, 81 agreed to participate (79%), and 59 (73%) completed all three parts. Out of 62 TSTs, two positive reactions ($>10\text{mm}$) were found (3.2%), and one borderline conversion at 9mm (1.69%). The two positive TSTs included one who had been treated for TB in the past, and one who noted it was his usual BCG reaction. In comparison, Rhode Island hospital tested 3357 employees between January and September of 1999, reporting a conversion rate ($>10\text{mm}$) of 0.42%. Six CHCWs (7.5%) were positive for hepatitis B, all past infections (hepatitis B surface antigen negative), but two were previously unknown. Of the two unknowns, one had indicated non-occupational risk factors for hepatitis on the questionnaire. The other CHCW had only core antibody (IgG) positive, no surface antibody, suggesting remote infection. CHCWs immunity to hepatitis B due to vaccination was also reported. Five CHCWs (6.8%) other than those positive for hepatitis B, did not receive the entire series of immunizations. Of those who did, 19 (28%) had no hepatitis B surface antibody (no evidence of immunity). Three CHCWs (3.8%) were positive for hepatitis C antibody, two previously unknown infections. One CHCW scored the second lowest on adherence to UP. Although the other CHCW was in the top 25% of UP adherence, previous employment in a high risk job was reported.

CONCLUSION: This is the first study to examine the occupational risk of tuberculosis and bloodborne infections in CHCWs. The data suggest that CHCWs may be at increased risk than health care workers (HCWs) for tuberculosis. The prevalence for hepatitis B is lower than that reported for HCWs in the medical literature (10%–20%), which may be due to the observed rate of hepatitis B vaccination. The prevalence for hepatitis C in CHCWs was slightly higher than reported values for HCWs (1%–2%). Ongoing research will determine if these trends are observed in other correctional facilities.

IMPACT OF CARDIOVASCULAR PROCEDURES USE ON HOSPITAL DEATH AMONG ISCHEMIC HEART DISEASE PATIENTS IN NEW YORK CITY. JM Barnhart, J Fang, MH Alderman, Epidemiology & Social Medicine, Albert Einstein College of Medicine, Bronx, NY

PURPOSE: The use of diagnostic and therapeutic procedures for coronary disease has been shown to differ by race and gender. We sought to determine the impact of coronary angiography and revascularization (PTCA/CABG) use on hospital death rates among non-Hispanic Black and White patients with ischemic heart disease (IHD) in New York City (NYC).

METHODS: We analyzed the New York State Department of Health's Statewide Planning and Research Cooperative System (SPARCS) database for patients admitted to NYC hospitals in 1996 and had a principal discharge diagnosis of IHD (ICD-9 codes 410–414). Frequencies of procedure use and crude death rates were calculated (per 100 IHD patients). Race and gender specific adjusted odds ratios and 95% confidence intervals (C.I.) for hospital mortality were calculated, controlling for potential confounders, using multiple logistic regression analyses. Odds ratios (OR) compared death rates among those who had the specified procedure with those who did not.

RESULTS: There were 36,683 IHD patients. Their mean age was 65.8 years (S.E. = 12.6). Sixty-one percent of the Whites were men; 55% of the Blacks were women. Frequencies for procedure use, crude hospital death, and the adjusted OR (95% C.I.) for hospital death, respectively are listed below.

	Angio %	PTCA %	CABG %	Crude Death %	Angio	PTCA	CABG
Black Women	29.1	9.4	7.7	3.4	.48 (.30–.77)	.23 (.07–.71)	2.5 (1.5–4.0)
White Women	37.2	17.0	13.3	6.0	.35 (.28–.42)	.17 (.11–.26)	.82 (.65–1.1)
Black Men	33.1	11.9	9.1	2.7	.35 (.19–.65)	.40 (.15–1.1)	1.6 (.83–3.0)
White Men	47.4	24.6	21.6	3.6	.30 (.25–.36)	.19 (.13–.27)	.63 (.50–.78)

Whites underwent angiography (angio), PTCA, and CABG more often than Blacks. However, the crude hospital death rate was highest among White women and lowest for Black men. Using hospital death as the outcome variable and controlling for age, insurance status, and co-morbidities (hypertension, diabetes, heart failure, and hypercholesterolemia), the race and gender specific OR (95% C.I.) revealed that Black women who underwent CABG were more likely to die, compared to Black women who did not have CABG (OR = 2.5; 95% C.I. (1.5–4.0)). A similar trend was observed among Black men, while CABG was associated with lower death rates among Whites.

CONCLUSION: These data suggest that CABG was associated with higher in-hospital mortality among Blacks with IHD. Referral bias or limited access to coronary angiography and PTCA/CABG may preclude Blacks from obtaining the maximum benefit of these procedures. Prospective studies on diverse populations with documented IHD are needed to shed insight into patient outcomes, following PTCA/CABG.

MEDICAL CARE UTILIZATION OF PATIENTS WITH SOMATIZATION AND HEALTH-RELATED ANXIETY. AJ Barsky, SL Ettner, J Horsky, DW Bates, Psychiatry, Harvard Medical School, Boston, MA; Medicine, University of California at Los Angeles, Los Angeles, CA; Medicine, Brigham & Women's Hospital; Medicine, Harvard Medical School, Boston, MA

PURPOSE: The primary care of patients whose symptoms are medically unexplained (est. 25%–50% of ambulatory visits) often frustrates providers and results in patient dissatisfaction. There are few empirical, quantitative studies that assess the costs of ineffective and maladaptive utilization of medical care by somatizers.

METHODS: Consecutive patients presenting for an office visit in a primary care clinic at a large urban hospital were asked to complete a 27-question screening survey. Items adapted from the Somatic Symptom Inventory and the Whiteley Index of hypochondriasis evaluated bodily complaints, health anxiety, and bodily preoccupation. Patients with scores above ($N = 212$) and below ($N = 664$) the 86th percentile were compared. This threshold was selected in order to identify the estimated 14% of primary care patients with a diagnosable somatoform disorder. Demographic characteristics along with morbidity, resource utilization, insurance status and service costs were obtained from hospital records for 12 months preceding the screen.

RESULTS: Of the 3344 patients invited to participate, 788 (24%) were ineligible, 720 (28%) declined, 224 (7%) did not complete the survey, and 114 (3%) were excluded for incomplete records. Patients in the below-threshold group were randomly selected from each score tertile. After adjusting for comorbidity (Charlson Index), and sociodemographic characteristics, the top-scoring group had higher total, physician, lab and procedure costs, and more primary care and specialist visits. Emergency ward visits and hospitalization rates were not significantly different.

Adjusted differences in selected outcome measures by whether above or below somatization score threshold ($N = 876$, SE in parentheses)

Outcome Measure	Scores Below Threshold	Scores Above Threshold	P-Value
Total Costs	\$953 (40)	\$1312 (84)	<0.001
Lab/Procedure Costs	\$335 (23)	\$518 (57)	0.002
N° of Visits to All Providers	6.3 (0.2)	9.2 (0.5)	<0.001
N° of Primary Care Visits	2.7 (0.1)	3.4 (0.2)	0.001
N° of Specialist Visits	3.6 (0.2)	5.9 (0.5)	<0.001
N° of Specialists Seen	1.9 (0.1)	2.7 (0.2)	<0.001

CONCLUSION: Somatizers are disproportionately high utilizers of medical services. Even after controlling for serious morbidity, they accrued costs 38% higher than other patients, made about 3 more office visits, and saw one additional specialist per year. Early detection and intervention may prevent the development of an enduring pattern of maladaptive and costly utilization.

THE ROLE OF PRIMARY CARE PHYSICIANS IN CANCER PREVENTION TRIALS. TA Battaglia, AM Harrington, A Ash, MN Prout, KM Freund, Section of General Internal Medicine, Boston University School of Medicine, Boston, MA

PURPOSE: While cancer treatment trials historically involved oncologists as investigators and recruiters of study subjects, the successful recruitment to cancer prevention trials, like the Breast Cancer Prevention Trial (BCPT), relies on the active participation of primary care physicians (PCPs). We surveyed PCPs to investigate which factors influence their decisions to encourage enrollment of patients into such trials.

METHODS: A self-administered survey was sent to a random sample of general internists and family practitioners in 3 regions. We described a medical scenario of a woman at high risk for breast cancer and asked PCPs if they would encourage her enrollment into a BCPT (yes/no). All clinical features of the case were held constant, while 5 patient characteristics: age, race, socioeconomic status, physical mobility, and health status varied dichotomously (e.g., 80 v 65 years of age). Each survey included 1 of 16 possible scenario versions. We also developed scales assessing the PCPs' perception of patient trust in them after trial enrollment (0–100 scale); perception of control in their patients' care after enrollment (3 point scale); knowledge of the BCPT (6 point scale); and attitudes toward the BCPT and tamoxifen (0–100 scale). Chi-square and t-statistics were used.

RESULTS: Based on interim results: 179 physicians (32%) completed the survey; 47% were male; 42% were general internists and 58% were family physicians. Fifty-two percent of PCPs encouraged enrollment of the woman in the scenario into a BCPT. Four factors were associated with their decision. 1) Trust: PCPs who anticipate a decrease in trust by their patients after trial enrollment are less likely to encourage enrollment (mean trust score 45 v 58, $p = 0.0001$). 2) Control: PCPs who perceive a decrease in control of their patient care are less likely to encourage enrollment (mean control score 1.8 v 2.0, $p = 0.02$). 3) Patient age: PCPs are less likely to encourage enrollment of 80 year olds than 65 year olds (40% v 60%, $p = 0.03$). 4) Knowledge: PCPs with lower knowledge scores of the BCPT are less likely to encourage enrollment (mean 2.9 v 3.3, $p = 0.08$). No other patient characteristic (race, socioeconomic status, physical mobility, health status) nor physician attitudes toward a BCPT or tamoxifen had an association with PCPs' encouragement to enroll.

CONCLUSION: PCPs are unlikely to encourage enrollment of older women to a BCPT. Additionally, PCPs who perceive a loss of patient trust or a loss of control in patient care subsequent to trial enrollment, or are less knowledgeable about the BCPT are less likely to enroll patients in a BCPT. Therefore, trial investigators

must not only educate PCPs about cancer prevention trials, but work to ensure a continued, strong PCP-patient relationship if successful recruitment is to be achieved.

PATIENT RESPONSE TO OFFICE VISITS IN WHICH PHYSICIAN SELF-DISCLOSURE OCCURS. MC Beach, D Roter, R Frankel, W Levinson, D Ford, Johns Hopkins University, Baltimore, MD; University of Rochester, Rochester, NY; University of Chicago, Chicago, IL

PURPOSE: Physician self-disclosure can be defined as statements made by a physician that describe his/her personal experience in areas that have medical and/or emotional relevance for a patient. Although the use of physician self-disclosure has been regarded as either a boundary violation or as a means of building trust and rapport with patients, little is known about its prevalence or impact on the physician-patient relationship.

METHODS: We utilized an existing database of physician-patient encounters that consists of approximately 10 audiotaped office visits for each of 124 physicians (59 primary care physicians and 65 surgeons). Patients in the study were mostly Caucasian (85%) and female (55%). The mean age was 51 years. A trained coder listened to each audiotape and coded for physician self-disclosure as defined above using the Roter Interaction Analysis System. After the encounter, each patient completed a survey that assessed sociodemographics, health status and various satisfaction measures. The differences between patient response to visits with and without physician self-disclosure were examined using ANOVA and linear regression.

RESULTS: Self-disclosure occurred with approximate equal frequency in primary care visits (102/589, 17%) and in surgical visits (93/676, 14%). The relationship between physician self-disclosure and patient-reported outcomes differed by physician specialty. For surgeons, self-disclosure was associated with higher patient ratings in nonverbal communication ($p = .000$) and in the American Board of Internal Medicine communication behavior scale ($p = .003$). These patients were significantly more likely to report greater feelings of warmth/friendliness ($p = .007$) as well as overall satisfaction ($p = .001$). For primary-care physicians, self-disclosure was consistently associated with poorer ratings of these same measures. In fact, following visits with primary care physicians who made statements of self-disclosure, patients were significantly less likely to report that they would recommend their doctor to a friend ($p = .05$). There was a statistically significant interaction for physician specialty and self-disclosure on all of the above measures. These relationships were not changed when controlling for patient age, gender, race, education, self-rated physical and emotional status or the number of prior visits with that physician.

CONCLUSION: Physician self-disclosure appears to be predictive of greater patient satisfaction with surgical but not primary care visits. These surprising findings suggest that the communication between patients and physicians across specialties may carry subtle differences in dynamics and consequences that define the patient experience in ways that merit further investigation.

SURROGATES' DESIRE FOR PARTICIPATION IN END-OF LIFE DECISIONS IN THE INTENSIVE CARE SETTING. MC Beach, M Jenckes, EB Bass, HR Rubin, K Lane, N McBee, N Kass, MA Williams, Johns Hopkins University, Baltimore, MD

PURPOSE: Although many have investigated the degree to which patients prefer to participate in their own medical decisions, the purpose of this study was to measure surrogates' desire for participation in end-of-life decisions in the intensive care unit (ICU) setting and to compare that desire with their impression of how decisions were actually made.

METHODS: All patients who died in any of 3 ICUs (surgical, medical, or cardiac) at a large inner-city academic medical center between April and October 1999 were eligible for inclusion in the study. Trained interviewers conducted telephone or in-person interviews with one closely involved family member or other surrogate within 2 months after each patient's death. Each surrogate was asked about his/her desire to participate in decisions for the patient and then to describe how decisions were actually made. Responses were categorized into 3 domains: decisions made mainly by surrogate, mainly by physician, or jointly by surrogate and physician.

RESULTS: Of 136 deaths over the 6-month period, we identified a surrogate for 104 patients. 49 of 104 surrogates responded to the survey and 35 of 49 respondents indicated that there were decisions to be made for the patient. 43% of respondents were African-American, 52% were >65 years old, 52% had at least some college and 60% were a spouse, 24% were a child, 10% were a parent, and 6% were a sibling of the patient. 51% of surrogates reported a desire to make joint decisions with the physicians, although 37% reported a desire to make decisions themselves and 12% preferred that physicians make all the decisions. When comparing surrogates' desire for participation with their actual self-reported level of participation, there was only 57% observed agreement with a weighted Kappa statistic of 0.33 (95% confidence interval, 0.06 to 0.59). Of 15 surrogates whose desire did not match their actual involvement in decisions, 12 (80%) were given more responsibility in decision making than they preferred to have. Most of the mismatch occurred when the surrogate desired a joint decision. Of 18 surrogates who reported a desire for joint decisions, only 7 (39%) reported that the decision was made jointly.

CONCLUSION: Most families or other surrogates want primary or joint responsibility for decision-making for terminally-ill patients in the ICU setting but a substantial number of surrogates report that they are given more responsibility than

they actually prefer to have. These results suggest that physicians need to communicate more effectively about the decision-making roles of surrogates of terminally-ill patients in the ICU setting.

THE CARE OF CANADIAN INPATIENTS ON WEEKENDS. CM Bell, DA Redelmeier, Department of Medicine; Institute of Medical Science, University of Toronto; Clinical Epidemiology Unit, Sunnybrook and Women's Health Science Centre, Toronto, ON, Canada

PURPOSE: Health care staffing in hospitals tends to be less intense on weekends than on weekdays, despite the relatively consistent day-to-day burden of disease. We tested whether mortality rates differed for patients with serious emergencies who were admitted to hospital on a weekend rather than on a weekday.

METHODS: All acute care hospitalizations resulting from an emergency department visit in Ontario, Canada were analyzed between 1988 and 1997 ($n = 3,805,788$). Hospital mortality for patients admitted on weekends was compared to hospital mortality for patients admitted on weekdays using a chi-square analysis and logistic regression techniques.

RESULTS: We found no difference in aggregate hospital mortality rates for patients admitted on a weekend rather than a weekday (5.9% vs 5.9%, $p > 0.20$). Patients with ruptured abdominal aortic aneurysms ($n = 6,661$), however, were significantly more likely to die if admitted on a weekend rather than a weekday (38.2% vs 32.7%, $p < 0.001$). Similarly, patients with unstable diabetes ($n = 102,857$) were more likely to die if admitted on a weekend (4.6% vs 4.2%, $p = 0.007$), and patients with pulmonary embolism/infarction ($n = 14,244$) were more likely to die if admitted on a weekend (14.3% vs 12.0%, $p = 0.001$). Adjustments for age and gender revealed similar results for all three diseases ($p < 0.015$ for all). Patients with acute myocardial infarction ($n = 172,969$) were only marginally more likely to die if admitted on a weekend (15.7% vs 15.4%, $p = 0.098$) and patients with intracerebral hemorrhages ($n = 12,082$), were not more likely to die if admitted on a weekend (43.4% vs 43.3%, $p > 0.20$). We found no medical disease with significantly higher survival for those admitted on a weekend rather than a weekday (25 other well-specified conditions examined so far).

CONCLUSION: Patients with diseases that have a significant short-term mortality and significant effective acute medical treatments are more likely to die if they seek medical care on a weekend rather than a weekday. This shortfall, however, is not universal and may spare those patients primarily treated in a critical care unit.

PROSTATE CANCER SCREENING COUNSELING—GETTING PRACTICAL. NA Benton, DA Nardone, Primary Care, VHA Medical Center, Portland, OR

PURPOSE: Improve performance for prostate cancer screening counseling.

Background: In FY 1997 the Department of Veterans Affairs mandated primary care providers (PCPs) counsel men between 50–69 years of age regarding risks and benefits of prostate cancer screening. In FY 1998 our performance was 25% based on number of progress notes indicating risks and benefits of prostate cancer screening discussed (numerator) and total number of progress notes from a male population ages 50–69 years (denominator) as conducted by the External Peer Review Program (EPRP) random chart review. In FY 1999, prostate cancer screening counseling became an institutional priority. PCPs felt EPRP data were not reliable, as sample sizes were too small.

METHODS: A multidisciplinary Health Promotion Interest Group (HPRO) identified strategies for improvement—approving patient information handout for screening, performing internal record reviews independent of External Peer Review Program (EPRP), instructing nurses to provide patients information regarding prostate cancer screening, offering men the opportunity to ask questions of primary care providers (PCPs), purchasing and modifying an electronic record clinical-note-template to assure quick and uniform documentation of interventions, educating nurses to document interventions electronically (as completed) for men who had no questions, and providing feedback to nurses and PCPs regarding performance.

RESULTS: Baseline EPRP performance for FY 98 was 25%. EPRP performance for Jul-Aug-Sep and Oct-Nov 1999 was 45% ($n = 76$) and 70% ($n = 54$) respectively. Random internal data collection demonstrated performance in Aug, Sep, Oct, and Nov to be 75% ($n = 73$), 68% ($n = 94$), 81% ($n = 75$), and 90% ($n = 128$). There was an average of 26 charts per month reviewed by EPRP and 92 per month internally.

CONCLUSION: PCPs and nurses accepted internal review performance data as credible since sample sizes were higher. Better performance with internal chart review could be due to preferential selection of records from full-time versus part-time PCPs. Success in performance can be attributed to nurse-driven, information-sharing, and uniform electronic-data-entry systems. It is unrealistic and inappropriate clinically to mandate that PCPs counsel all men regarding prostate cancer screening. Providing information and responding to questions meet the spirit and letter of the performance measure. Let's get practical!

ARE MANUSCRIPT SUBMISSIONS TO U.S. GENERAL MEDICAL JOURNALS DECLINING IN AN ERA OF FISCAL CONSTRAINT? AN ANALYSIS OF TRENDS, 1994–1998. M Berkwitz, WB Bilker, FF Davidoff, Division of General Internal Medicine; Department of Biostatistics and Epidemiology, University of Pennsylvania, Philadelphia, PA

PURPOSE: To determine if manuscript submissions to U.S. general medical journals have declined relative to those at specialty and international journals since 1994, and to determine trends in representation of U.S. and international contributors.

METHODS: We solicited submissions information from 55 medical journals, including 6 general and 25 specialty journals in the U.S., and 5 general and 19 specialty journals internationally. Editors provided data on total manuscript, research, and non-research submissions, and on authors' countries of origin between 1994 and 1998. Trends for individual journals were modeled using linear regression. Trends by journal category and comparisons between general and specialty and U.S. and international journals were analyzed using generalized estimating equation regression for absolute and relative change.

RESULTS: All 6 U.S. general medical and 20 other journals responded, for a response rate of 47%. Results are reported as (absolute; percent) change in manuscripts per year. Submissions rose and fell for U.S. general journals individually but no significant change was detected in aggregate for total (20.3 [−37.4, 77.9]; 3.5% [−1.4, 8.4]), research (11.4 [−39.8, 62.6]; 2.0% [−2.0, 6.0]), and non-research submissions (1.7 [−10.8, 14.1]; 2.2% [0.17, 4.1]), or for numbers of U.S. (−11.0 [−59.2, 37.1]; 2.7% [−4.0, 9.4]) and international contributors (25.98 [−10.3, 62.2]; 6.5% [−2.0, 14.9]) (all p values $> .10$ except $p = .03$ for relative change in non-research submissions). By contrast, U.S. specialty journals saw an aggregate increase in total (16.0 [41.4, 104.0]; 10% [4.4, 14.9]), total research (62.9 [36.7, 89.1]; 11% [4.7, 17.0]) and clinical research (32.2 [13.4, 51.0]; 12% [5.4, 18.9]) submissions (all $p < .001$). Numbers of U.S. contributors to these journals held steady but contributions from international authors increased significantly (64.7 [38.5, 90.9], 10.4% [7.5, 13.4], $p < .001$). Despite these changes, no significant differences were detected when general and specialty journals were compared, though the difference in total research submissions to U.S. general and specialty journals approached significance (−55.6 [−112.5, 1.4], $p = .056$; −8.7% [−16.1, 1.4], $p = .02$). There were insufficient data at the time of the analysis to make meaningful comparisons between U.S. and international general medical journals.

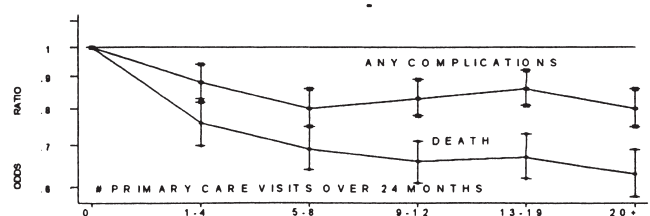
CONCLUSION: In a period when fiscal constraint threatened academic productivity there were no apparent declines in manuscript submissions to U.S. general medical journals, and no changes in numbers of U.S. contributors. Possible differences in research submissions to general and specialty journals may be attributable in part to increased contributions by international authors

PRIMARY CARE AND THE RISK OF COMPLICATIONS AND DEATH IN A NATIONAL SAMPLE OF U.S. ELDERLY WITH DIABETES. AG Bertoni, JS Krop, GF Anderson, F Brancati, General Internal Medicine, Johns Hopkins Medical Institutions, Baltimore, MD

PURPOSE: Although the efficacy of specific elements of primary care (e.g., glucose control) is well established, the overall effectiveness of primary care for adults with diabetes is less certain, particularly at older ages. We therefore conducted a prospective study of 131,595 Medicare beneficiaries (aged 65+) from a 5% national sample, with diabetes (ICD9 250.xx) who were alive on 1/1/96 and free of serious diabetes complications.

METHODS: Primary care visits were defined as outpatient visits in 1994 and 1995 to internists and other primary care providers identified as the usual source of care (most frequent billing contact). Main outcomes, ascertained from claims data, were death and serious complications including amputation, gangrene, metabolic disturbances (diabetic ketoacidosis, hyperosmolar syndrome, hypoglycemia), blindness, osteomyelitis, and pyelonephritis that occurred between 1/1/96 and 12/31/96. Covariates included demographic variables, comorbidities and average household income by county from 1990 Census data.

RESULTS: During 12 months of follow-up, 8539 elders died and 14470 developed complications including 8771 with amputation or gangrene, 4807 with metabolic disturbances, and 2582 with other complications. Compared to their counterparts who made 1 or more primary care visits, those with no visits were at greater risk for any serious complication (Odds Ratio (OR) 1.24; 95% confidence interval [CI] 1.2, 1.3) and for death (OR 1.7; 95% CI 1.6–1.8). After adjusting for gender, race, Charlson comorbidity index, Medicaid status, hospitalization, region, household income, diabetes comorbidities, and visits to ophthalmologists and podiatrists, there was a graded relationship between primary care visits and reduced risk of complications and death.



This effect appeared specific: there was no relationship between primary care visits and brain or lung cancer.

CONCLUSION: Primary care visits are independently associated with a reduced risk of death and serious diabetic complications in the Medicare population. Improvements in access to or use of primary care may reduce morbidity and mortality in US elders with diabetes.

COORDINATION OF EARLY-STAGE BREAST CANCER CARE. NA Bickell, GJ Young, LR Sternberg, Mount Sinai School of Medicine, NY, NY

PURPOSE: Effective coordination of care in inpatient surgical wards and intensive care units has been found to be associated with lower patient morbidity and mortality. Little is known about the way care is coordinated in an outpatient setting. Adjuvant local and systemic treatments of early stage breast cancer improve health outcomes and are delivered mostly in the outpatient setting, often by different specialists. To assess approaches important in coordinating care for women with early-stage breast cancer, we performed a case study, interviewing health care providers involved in delivery of breast cancer care.

Semi-structured 1 hour interviews were developed by the research team including an expert in coordination of care. Fifty-nine interviews, including 30 physicians, 13 nurses, 5 senior clinical or quality directors, 7 administrative assistants, 2 patient navigators, 1 social worker and 1 tumor registrar at 4 hospitals in the downstate NY metropolitan and 1 in upstate NY regions were conducted. Downstate hospitals had participated in a breast cancer quality improvement project, the upstate hospital is a designated community cancer center. An average of 12 individuals per site participated. Interviews were taped and notes taken during the interviews. Descriptions of categories of coordination mechanisms and providers' perceived value of each mechanism were recorded.

Coordination approaches perceived as being valuable include: 1) Real time multidisciplinary breast cancer patient management meetings; 2) Systematic use of patient educators and navigators; 3) Routine assessments of performance using national guidelines and the hospital tumor registry. Physicians at all sites use follow-up appointments as a way to keep track of patients' test and consult results. Few use computerized ticklers to trigger future appointments. No one had an organized system to trigger referrals or track their results but this was not viewed as a problem.

We identified coordination approaches deemed valuable by providers of breast cancer care. The effectiveness of these approaches to improve rates of receipt of adjuvant local and systemic therapies remains unknown.

REASONS FOR UNDERUSE: THE CASE OF BREAST CANCER. NA Bickell, MD McEvoy, MR Chassin, Mount Sinai School of Medicine and Medical Center

PURPOSE: Despite numerous randomized trials showing the health benefits of radiotherapy (RT) following breast conserving surgery and systemic treatments for stage 1B & 2 breast cancer, underuse of these therapies remains a significant quality problem. Understanding the root causes of underuse is necessary to design effective interventions to improve care.

In a prior study, using evidence-based guidelines created by local breast cancer experts, we found that 1 in 6 (44/275) women did not receive beneficial adjuvant treatment in accordance with the guidelines; 22 had RT and 22 had systemic therapy omitted. To understand the reasons their physicians had for treatment omission, we conducted semi-structured interviews and audio-taped all 13 surgeons at a tertiary academic medical center whose early-stage breast cancer patients did not receive local or systemic treatments in accordance with the guidelines. One investigator reviewed all interview transcripts, a second reviewed 40 transcripts. Based upon content, categories were created and reasons for underuse coded with excellent inter-rater reliability ($\kappa = 1.0$).

Reasons for underuse were classified into categories. The physician thought: 1) the patient should have been treated and could not explain why they weren't (34%); 2) the patient should not have been treated due to an extenuating common clinical circumstance not addressed by the guideline, e.g., age (30%), or an unusual clinical circumstance, e.g., concurrent second cancer (11%); 3) the patient should have been treated but they refused treatment (16%); 4) the guideline did not apply to a patient because they disagreed with the clinical content of the guideline (9%).

Categories of root causes of underuse of breast cancer adjuvant treatments were identified. In no instance was the physician unaware of guideline-recommended treatments. Interventions targeted at root causes of underuse such as a tracking system or patient navigator to prompt and follow-up on referrals, and discussion at multidisciplinary rounds of patients with extenuating clinical factors could address the majority of reasons for underuse of beneficial breast cancer treatments.

BARRIERS TO DOMESTIC VIOLENCE SCREENING IN URGENT CARE AND EMERGENCY ROOM SETTINGS. LM Birkovich, S Berg, PY Takahashi, Internal Medicine; Social Services, Mayo Clinic, Rochester, MN

PURPOSE: The Abuse and Violence Prevention and Education Committee was interested in determining the prevalence of domestic violence (DV) in the local population, educating staff and patients about DV, learning about barriers to screening, and intervening in the cycle of violence.

METHODS: Nursing staff of the Urgent Care Center (UCC) underwent a one hour training session by a physician/social worker team. Nursing staff of the emergency room (ER) received a four hour interactive session with an outside trainer/expert and women's shelter advocate. During a six month period, all male and female patients 18 years and older presenting to the UCC or the minor trauma/medical corridors of the ER were eligible for screening. Patients were asked by the nursing staff three questions re: the experience of physical or emotional abuse within the past year and whether the patient would like more information. Data was collected from cards assigned to each eligible patient and totaled. Near the completion of the 6 month pilot project, questionnaires were sent to the nursing staff regarding their attitudes and experience in asking about DV.

RESULTS: In the UCC, of 17,441 adult patients seen, 5949 (34%) were screened with 263 positive responses (incidence 4.4%). In the ER only 2152 patients were screened (less than 20%) with 263 positive responses (incidence 6.7%). In the UCC, 944 cards (16%) and in the ER 810 cards (38%) were returned marked "unable to screen".

CONCLUSION: The prevalence of recent abuse in patients successfully screened at the UCC was 4.4% and in the ER 6.7% (overall 5%). This prevalence is consistent with published data considering our population included both men and women. Our concern is that only 34% patients in the UCC and less than 20% in the ER were successfully screened. Common barriers to screening include 1) patient not alone, 2) mental or physical status of patient, 3) language barrier, and 4) caregiver's lack of comfort in asking the question. Physician support of routine screening is important to compliance with screening. Despite an increased length of the training session and an outside trainer/expert, the ER was less successful at screening. In spite of the barriers, both practice areas are planning to continue universal screening.

ARE PHYSICIANS EQUIPPED TO ADDRESS THE OBESITY EPIDEMIC? JP Block, W Fisher, KB Desalvo, Internal Medicine, Tulane University School of Medicine; Biometry and Genetics, Louisiana State University School of Medicine, New Orleans, LA

PURPOSE: Despite the lack of formal training in the treatment of obesity, internists have been called upon to serve as front line educators for the growing numbers of clinically obese patients. To analyze whether physicians are currently suited to fill that role, we surveyed internal medicine residents regarding their attitudes and knowledge about obesity.

METHODS: Trained researchers surveyed in person a convenience sample of 58 house officers from an inner-city internal medicine training program. These physicians direct the care of chronically-ill patients of which 80% are clinically obese. Physicians were asked to complete a 28 question survey and self-report their own BMI, height and weight. The instrument demonstrated reliability of the knowledge and attitudes subscales of 0.98. The subscales were fitted with probalistic conjoint models, and linear logit scores were used for summary knowledge and attitudes values.

RESULTS: The majority of respondents were male (55%), and most (93%) reported no prior special training in obesity. The average self-reported BMI was 20 (range 1-90) compared with an average actual BMI of 23 ($p = .06$). Responses did not vary significantly by year of training or physician gender. Despite having reasonable understanding of the medical consequences of obesity, basic obesity knowledge questions were the most difficult for respondents as assessed by probabilistic conjoint models. 66% did not know the threshold BMI for obesity, and 69% did not know that waist circumference is a reasonable measure of obesity. 41% failed to report their own BMI within 10% of its actual value; inability to accurately estimate one's own BMI was correlated with decreased confidence in the ability to treat obesity ($p = 0.04$). Although 95% of physicians agreed that treating obesity was important, attitudes towards treatment were generally negative. Respondents overwhelmingly reported a lack of treatment success (70%), 59% felt unqualified to treat obese patients, and nearly 30% reported treatment to be futile. 66% indicated confidence in their skills, and confidence was associated with an increased self-perception of qualification ($r = .317$, $p = .004$) and success in treating obesity ($r = .575$, $p < .001$). Summary knowledge and attitude scores were not correlated ($r = 0.07$).

CONCLUSION: Despite a good knowledge of the medical outcomes of obesity, the respondents had a poor grasp of the basic tools necessary to identify obesity. Many negative attitudes toward the treatment of obesity were also expressed. If physicians are to be the major change agents for obese patients, better physician educational efforts must be designed to address both knowledge deficits and induce attitude change.

CAN PATIENTS WITH HYPERTENSION ACURATELY REPORT THE QUALITY OF THEIR CARE? LE Boulware, LA Mead, MJ Klag, LH Rohlf, DM Ford, Department of Medicine, Johns Hopkins Medical Institutions, Baltimore, MD; Hoechst Marion Roussel

PURPOSE: Although clear guidelines have been set for providers in terms of blood pressure (BP) control, little is known regarding patients' abilities to appropriately assess the adequacy of the care they receive for high blood pressure (BP). Measuring quality of care for hypertension would be more efficient if patients can accurately rate the specific care they receive for their condition.

METHODS: As a part of a feasibility study for a large prospective study, a three-page questionnaire was mailed to a 2.5% random sample of U.S. recipients of a health promotion newsletter, stratified by geographic region, 77% of whom had hypertension. Patients reported the self-perceived quality of their care, average number of times BP was taken by a health professional per visit, average last recorded blood pressure, and their clinical characteristics.

RESULTS: Of 2132 hypertensive participants (28% response rate), 51% were male, 91% were white, 85% graduated from high school, 80% were older than 60 years of age, 77% reported taking a calcium channel blocker (CCB), 98% had been on anti-hypertensive medications for longer than 1 year, 12% reported their BP as being controlled to <130/85mmHg, and 21% rated the quality of their care for hypertension as excellent. The percent rating their care as excellent increased in a stepwise fashion in relation to the number of times BP was measured: 24% for persons with BP taken 0-1 times, 30% for BP taken 2 times, and 41% for BP taken >3 times ($p < 0.001$). Likewise, the percent rating their care as excellent increased in a stepwise fashion in relation to levels of BP control: 20% for persons

with SBP >160mmHg or DBP >100mmHg, 26% for SBP 140–159mmHg or DBP 90–99mmHg, and 42% for BP <130/85mmHg ($p < 0.001$). In a logistic regression model adjusting for age, sex, and race, people reporting excellent quality of care were more likely to have better BP control (OR 3.0 [95% CI 2.1–4.1, $p < 0.001$]) for BP <130/85 vs. SBP >160 or DBP >100, have their BP taken more frequently (OR 1.5 [95% CI 1.20–1.8, $p < 0.001$]) for BP taken >2 times) and to be taking CCBs (OR 1.3 [95% CI 1.1–1.7, $p < 0.001$]). In analyses stratified by presence of a college education, results were similar. The sensitivity of reporting excellent care in predicting BP <130/85mmHg and BP taken >2 times, however, was only 42% and 32% respectively; specificity was 75% and 76% respectively.

CONCLUSION: Although these measures have substantial misclassification, patients' perceptions of the quality of hypertension care are directly associated with objective measures of adequacy of hypertension care, such as levels of BP control and provider vigilance in monitoring BP.

WHAT FACTORS PREDICT PHYSICIAN INVOLVEMENT OF PATIENTS IN CLINICAL DECISIONS? CH Braddock III, KA Edwards, M Micek, W Levinson, Department of Medicine; Department of Medical History & Ethics, University of Washington, Seattle, WA; Section of General Internal Medicine, University of Chicago, Chicago, IL; Center of Excellence in Health Services Research and Development, VA Puget Sound, Seattle, WA

PURPOSE: To identify factors associated with physician performance of informed decision making (IDM).

METHODS: We performed a cross-sectional study of IDM in private primary care physicians' ($n = 59$) and surgeons' ($n = 65$) offices. Completeness of IDM was measured by content analysis of audiotaped physician-patient discussions of clinical decisions, using criteria representing the minimum discussion for basic IDM: discussion of the nature of the decision and either the patient's role in decision making or the patient's preferences. We used the percentage of all decisions for each physician that were complete as an aggregate measure of physician performance of IDM (IDM score). Additional data included physician and patient demographics, number of years in practice and hours per week in practice for physicians, physician perceptions of the how well they knew the patient, attitudes and beliefs towards malpractice risk, and beliefs about the importance of attention to psychosocial concerns. We used multivariate linear regression with generalizing estimating equations to ascertain associations between these characteristics and IDM score.

RESULTS: We analyzed 1057 audiotaped office visits, with approximately ten patients for each physician. Physicians averaged 20% complete decisions by minimum criteria, with a range of 0–50%. In multivariate analysis of all physicians, fewer years and fewer hours per week in clinical practice were both independent predictors of higher IDM score ($p = 0.005$, 0.028 respectively). The physician's perception that they knew the patient "well" had a negative association with IDM score, ($p = 0.037$). Surgeons had higher IDM scores than primary care physicians, though not statistically significant. (22 vs. 19, $p = 0.098$) No other physician or patient characteristics, nor visit length, showed significant association with IDM score.

CONCLUSION: We found that few physicians meet minimum criteria for IDM in routine office practice. Fewer years in practice and fewer hours per week predicted better IDM, while believing to know the patient well predicted poor performance. Physicians in practice fewer years may have less paternalistic notions of decision making than their senior colleagues. Those working fewer hours per week may perceive less time pressure, despite lack of independent association between visit length and IDM score. Physicians who believe they know the patient well may assume that patients no longer desire involvement, though it is unclear whether their patients would concur with that assumption.

THE INFLUENCE OF SOCIODEMOGRAPHIC CHARACTERISTICS ON QUALITY OF CARE FOR MEDICARE BENEFICIARIES WITH DIABETES IN MANAGED CARE. AF Brown, RS Starr, PG Gutierrez, J Adams, RH Brook, MF Shapiro, CM Mangione, Department of Medicine, UCLA, Los Angeles; RAND, Santa Monica, CA

PURPOSE: To evaluate the relationship between sociodemographic characteristics and quality of care for Medicare beneficiaries with diabetes enrolled in managed care.

METHODS: Telephone interviews and clinical examinations were performed on a random sample of subjects with diabetes cared for in a Medicare managed care plan that contracts with 17 provider groups in Los Angeles. Participants were questioned about diabetes-specific service use in the past year. Blood pressure and glycosylated hemoglobin (HbA_{1c}) were measured during a standardized clinical examination. The six performance measures evaluated were based on Diabetes Quality Improvement Project (DQIP) indicators. These include HbA_{1c} measurement, cholesterol check, dilated eye exam, and foot exam in the past year and evidence of controlled blood pressure (mean arterial pressure [MAP] <100) and HbA_{1c} <9.5. The influence of age, gender, race/ethnicity, income, Medicaid, and education on each quality of care indicator was tested with a series of logistic regression models, adjusting for health status using the SF-12 and for comorbid illness.

RESULTS: Interviews were completed by 484 persons (response rate 66%). Mean age was 75 ± 8 years; 51% were women; 47% were white, 24% Latino, 16% African American, and 8% Asian Pacific Islander; 57% earned under \$20,000 per year; 7% received Medicaid; and 31% had not graduated from high school. In the past year, 92% reported HbA_{1c} testing, 79% cholesterol testing, 81% had a dilated

eye exam and 67% a foot exam performed. Blood pressure was controlled in 67% and 73% had HbA_{1c} <9.5.

Odds Ratios (p) for Significant Correlates ($p < .05$) of Markers of Better Quality of Care (— = NS)

	HbA _{1c} tested	Cholesterol tested	Dilated eye exam	MAP≤100	HbA _{1c} <9.5
Age (inc. by 5 yrs)	—	0.6 (.005)	0.4 (.003)	—	—
Female	0.4 (.02)	—	—	—	—
Medicaid	—	—	0.4 (.02)	0.4 (.04)	—
Income <\$20k	—	—	—	—	0.5 (.02)

Performance of foot examinations was not affected by sociodemographic factors.

CONCLUSION: It is reassuring that race and education were not associated with differences in quality of care. However, although all subjects in this study were insured and had identical health benefits, older persons, women, and those with lower income were at greatest risk for not receiving needed care for their diabetes.

PRE-OPERATIVE MEDICAL CONSULTATION AND PATIENT OUTCOMES. JM Bugar, WA Ghali, JB Lemaire, H Quan, Medicine; Community Health Sciences, University of Calgary, Calgary, AB, Canada

PURPOSE: Specialized pre-operative assessment clinics (PACs) are now widely used in North America, but a positive effect on patient outcomes has never been proven. We compared outcomes for patients seen by various consultants in a tertiary care centre's PAC.

METHODS: We used hospital discharge data to identify admissions for selected surgical divisions (general surgery, cardiovascular-thoracic, neurosurgery, oral, orthopedics, plastics, urology, gynecology, and otolaryngology) from July 1, 1996 to March 31, 1998. Pre-operative clinic booking sheets identified patients seen in the PAC. We compared mortality, complications, cardiac complications, length of stay (LOS), and hospital costs of patients seen and not seen by specific consultants in the PAC (nursing, general internal medicine [GIM], and anesthesia). We used logistic regression to control for age, sex, co-morbidity, and major vs. minor procedure.

RESULTS: Of 9603 surgical admissions, 58.8% were seen in the PAC by nursing, 16.4% by GIM, and 23.2% by anesthesia. 56.9% of patients were female, 23.7% had hypertension, 15% had neoplastic disease, 12.4% had chronic obstructive pulmonary disease, and 7.4% had history of myocardial infarction. Adjusted outcomes by various consultants are shown in the following table:

Consultant	Seen	Mortality	Compl.	Card. compl.	LOS (days)	Hospital cost
Nursing	yes	0.42%	26.4%	5.2%	4.8***	\$5006***
	no	0.58%	27.1%	5.5%	5.3	\$5930
GIM	yes	0.21%*	28.6%*	3.1%***	5.9***	\$5549
	no	0.57%	26.3%	5.9%	4.8	\$5355
Anesthesia	yes	0.63%	28.1%**	7.4%***	4.8**	\$5411
	no	0.41%	26.2%	4.4%	5.1	\$5379

* $p = 0.02$, ** $p = 0.04$, *** $p < 0.001$

Despite a higher burden of co-morbidity, patients seen by GIM had significantly lower rates of both mortality and cardiac complications. As well, patients seen by nursing showed statistically significant decreases in LOS and cost. Although the adjusted rates of complications and cardiac complications were higher in patients seen by anesthesia, this may reflect a tendency for anesthesia to see most high risk cases.

CONCLUSION: We found that an association exists between pre-operative GIM consultation and reduced cardiac complications and mortality, even though overall complications, LOS, and cost are not improved. Further work is now needed to explore the mechanisms for this encouraging association.

IMPROVING THE CARE FOR PATIENTS WITH COMMUNITY ACQUIRED PNEUMONIA: A MULTI-INSTITUTION STUDY. MA Callahan, DS Battelman, Public Health, Cornell University Medical College, New York, NY

PURPOSE: Community Acquired Pneumonia (CAP) is one of the most common reasons for hospitalization in the US. However, variations in appropriateness of care exist. We sought to improve the care for patients across an academic hospital network through a program of practice profiling and feedback, academic detailing, and implementation of standardized order sets.

METHODS: 100 cases of CAP were randomly chosen from each of 7 hospitals for the baseline year. A standard data extraction form was used. Cases were excluded if the patient was immunocompromised, not admitted through the ER, was under the age of 18, was suspected of having TB, or if the admitting diagnosis was not pneumonia. Analysis and profiling of year 1 data was accompanied by academic detailing and development of standardized order sets for use by emergency room physicians. Following a 6 month implementation phase, 80 cases per hospital were sampled for the followup year to look for changes in the target outcomes.

RESULTS: 609 cases were included in the baseline year analysis. 600 cases were included in the followup year. Patient demographics were similar between the two samples. Baseline data showed a strong correlation between initiating antibiotics in the ER and shorter LOS. However, in the baseline year only 63% of patients received their initial dose of antibiotics in the ER (range 51% to 92% by hospital).

The baseline year also demonstrated that only 52% of patients received appropriate antibiotics (range 34% to 72%) based on an expert panel using the 1998 IDSA guidelines for hospitalized CAP. During the implementation phase, improvement efforts focused on two recommendations: initiating antibiotics in the ER and choosing appropriate antibiotics for treatment. Followup data after implementation showed significant changes from the baseline year. Overall delivery of the first dose of antibiotics in the ER increased from 63% to 89% (OR 4.8, 95% CI 2.3–9.7). Appropriateness of antibiotic choice increased from 52% appropriate to 86% appropriate (OR 5.7, 95% CI 2.9–10.9). All hospitals in the study improved on both measures.

CONCLUSION: A program of clinical profiling, academic detailing, and targeted areas for improvement is associated with improvement in the ER care of patients with community acquired pneumonia. Future programs designed to improve care and reduce medical errors may benefit from similar approaches.

CRITICALLY APPRAISED TOPICS PROVIDE EVIDENCE BASED REVIEWS OF MEDICAL LITERATURE FOR WEB USERS. *JD Cappuccio*, W Polashenski, B Robbins, Medicine, University of Rochester, Rochester, NY

PURPOSE: Evidence based medicine strives to match clinically derived questions to available medical research. The use of Critically Appraised Topics (CATs) is one tool used to efficiently bring evidence-based medicine into clinical practice. We wished to determine whether medical care personnel would reference CATs if they were readily available via the Internet.

METHODS: We created a Web site at the University of Rochester School of Medicine and Dentistry. CATs were predominately submitted by residents and faculty members associated with the University of Rochester's Combined Medicine and Pediatric Program, and the Primary Care Program in Internal Medicine. CATs were reviewed by 2 faculty experienced in evidence based medicine, and scored in a standardized method. After review, their original authors revised the CATs to reflect improvements suggested by reviewers. CATs were published on the Internet at <http://www.urmc.rochester.edu/medicine/res/CATS/Cathome.html>. Statistics looking at amount of activity at the site, location of activity, and degree of repetitive use were measured.

RESULTS: In November 1999 there were 4660 hits to the University of Rochester CAT Bank representing 877 sessions in the month or 29 uses per day. 53% came from the United States, 21% came from international sites and 26% came from unknown sites. The average session length was 6:54. 107 users visited the site more than once. Sites within the University of Rochester or its affiliated hospitals accounted for 4 of the top 10 individual users. Within the United States, 94% sessions originated in Virginia and 76% sessions started in Reston Va. Top referring sites included the University of Rochester, the Center for Evidence Based Medicine in Oxford England, popular search engines and several evidence based medicine sites.

CONCLUSION: We conclude that evidence based reviews made available as Critically Appraised Topics have generated a large amount of national and international interest. Further investigation is needed to determine by whom and how the CATs are being utilized. Clearly, the information is being viewed and recognized by several important computer domains in the United States and abroad.

NO CARE FOR THE CAREGIVERS: DECLINING HEALTH INSURANCE COVERAGE FOR HEALTH CARE PERSONNEL, 1990–1998. *BGS Case*, DU Himmelstein, S Woolhandler, Harvard Medical School, Boston; Department of Medicine, The Cambridge Hospital/Harvard Medical School, Cambridge, MA

PURPOSE: The proportion of Americans lacking health insurance increased from 13.9% in 1990 to 16.3% in 1998. We examined trends in health insurance coverage of health care workers and their children between 1990 and 1998.

METHODS: We analyzed data from the annual March supplements of the Current Population Survey (CPS) for 1991 through 1999, a Census Bureau survey that collects information about health insurance during the previous calendar year from a nationally representative sample of non-institutionalized US residents. We used Standard Industrial Classification (SIC) codes to identify health care personnel.

RESULTS: The number of uninsured health care personnel under the age of 65 years increased 63.5% between 1990 and 1998, from 834,000 to 1.36 million; the proportion uninsured rose from 9.0% to 12.2%. For workers in other industries, the proportion uninsured increased less dramatically over this period, from 15.1% to 17.9%, but remained greater than the proportion among health personnel. In 1998, 20.0% of nursing home personnel were uninsured vs. 8.2% of hospital workers and 8.7% of those employed in medical offices. The proportion of registered nurses in nursing homes who were uninsured rose from 0.5% in 1990 to 12.2% in 1998. Among occupational groups, health aides experienced the highest uninsurance rate, 23.8%. RNs suffered the greatest rise in uninsurance since 1990, increasing 118% from 2.3% to 5.0%. LPNs experienced an increase from 7.0% to 14.5%, and uninsurance among physicians rose from 2.9% to 5.4%. African-American women health workers, who constitute 12.9% of the health care work force, had an uninsurance rate of 21.2% vs. 10.9% for other personnel. Workers earning <\$25,000, who constituted almost half of all personnel, were three times as likely as higher-income workers to lack coverage. Among public sector health workers 7.8% were uninsured in 1998; union members were half as likely to lack health insurance as other personnel. In 1998, 1.12 million uninsured children lived in a household with a health care worker, accounting for 10.1% of all uninsured children in the U.S.

CONCLUSION: Increasingly, the health care sector is consigning its own workers and their children to the ranks of the uninsured.

IMPACT OF MEDICAID MANAGED CARE IN A COMMUNITY: FOCUS ON SAFETY NET PHYSICIANS. *RD Cebul*, J Sudano, R Jean-Baptiste, D Einstadter, Center for Health Care Research & Policy, Case Western Reserve University at MetroHealth Medical Center, Cleveland, OH

OBJECTIVE: To describe the effects of mandatory Medicaid managed care (MMC) on physicians and their patients in a large urban community.

DESIGN, SUBJECTS: Mailed survey to physicians (MDs) in Ohio's largest county. **MAIN MEASUREMENTS:** Department of Human Services data were used to describe the entrance and exit of MMC companies in the county from the beginning of mandatory MMC (7/96) to 5/99. MD's were classified as Safety Net or Lower Volume MDs by self-reported Medicaid penetration in their practices. Main outcomes were responses to items on: importance of caring for Medicaid patients to MDs practice identity/social commitment; MD severing of MMC contracts; impact of MMC on practice revenues and financial well-being; and MDs opinions re: effect on their patients' quality of care and access to needed services.

MAIN FINDINGS: 6 companies exited over the study period, covering 41,552 enrollees and 35.5% market share. 50% of the MDs declined or severed contracts during the study period. 42% of the 319 responding MDs were classified as Safety Net MDs and were much more likely than Lower Volume MDs to report that caring for Medicaid patients was "very important" to their practice identity/social commitment (63% vs. 14%, respectively, $p < 0.001$). Safety Net MDs also were more likely to report reduced practice revenues (78% vs. 53%, $p = 0.01$), worsened financial well-being of their practices (67% vs. 46%, $p = 0.08$), and reduced quality (42% vs. 27%, $p = 0.04$) and access (56% vs. 33%, $p = 0.001$) for their patients. Primary Care MDs, somewhat overrepresented in the sample, were generally less negative on all measures than non-Primary Care MDs.

CONCLUSIONS: In this community, Safety Net MDs and their patients are more vulnerable to the adverse financial and quality effects of MMC. Loss of these MDs and their practices may threaten the health care safety net.

EARLY ADOPTION OF TROPONIN AS A PROGNOSTIC TEST IS ASSOCIATED WITH APPROPRIATE REDUCTIONS IN CARDIAC CATHETERIZATION RATES FOR MEDICARE PATIENTS HOSPITALIZED WITH UNSTABLE ANGINA. *RM Centor*, JJ Allison, NW Weissman, T Shaneyfelt, G Heudebert, CI Kiefe, Internal Medicine, University of Alabama at Birmingham School of Medicine, Birmingham, AL

PURPOSE: In 1997, Alabama hospitals began using serum troponin as a prognostic test for patients admitted with unstable angina. We sought to determine if early use of troponin, taken as a marker for early adoption of new technology in general, was associated with changes in use of an older technology—cardiac catheterization.

METHODS: From 16 cardiac catheterization capable hospitals in Alabama, we used ICD-9 codes to identify 1187 Medicare patients admitted with unstable angina in 1995 and 1031 in 1997. Centrally trained abstractors reviewed the complete medical records. We confirmed cases using a guideline-based rigorous clinical algorithm of abstracted variables. We defined early adopters as hospitals who measured troponin in 1997 and late adopters as those who did not. Then, we compared changes in cardiac catheterization rates for early versus late adopters. We also examined the relationship between high troponin values and use of cardiac catheterization at the patient level. We used multivariable regression to adjust for clustering of patients within hospitals and for other covariates.

RESULTS: In 1995 there was no use of troponin at any of the hospitals. In 1997, 9 of the 16 hospitals (early adopters) started using troponin. In these 9 hospitals, 273 out of 601 patients (45%) had their troponin measured on admission. Overall, the cardiac catheterization rate in the 16 hospitals declined slightly from 46% in 1995 to 44% in 1997. The table shows percentages of patients undergoing catheterization each year, by adopter of troponin status for the hospital:

	Had cath 1995	Had cath 1997	Two-year change
Early adopters	55%	43%	-12%*
Late adopters	38%	45%	+7%*

* $p < 0.05$ for two-year change in early adopters.

Early adopters reduced their cardiac catheterization rates over two years, whereas late adopters increased their rates. In addition, for those patients in whom troponin was measured, those with abnormal troponin values had a higher probability of cardiac catheterization than patients with normal values (65% versus 49%, $RR = 1.67$, 95% CI: 1.08–2.62). The regression model determined that early adoption of troponin was the most important predictor of cardiac catheterization.

CONCLUSION: Early adoption of a new non-invasive diagnostic technology (troponin) was associated with a decreased use of an old invasive technology (cardiac catheterization). Furthermore, our study shows a relationship between the results of the troponin test and the use of cardiac catheterization.

KNOWLEDGE AND PERFORMANCE OF PREVENTIVE CARE AMONG PRIMARY CARE PHYSICIANS. COMPLIANCE WITH THE RECOMMENDED GUIDELINES. *A Ciapponi*, FA Rubinstein, Division of Family and Preventive Medicine, Hospital Italiano, Buenos Aires, Argentina

PURPOSE: To evaluate the knowledge and level of compliance with different Preventive Practices (PP) among primary care physicians (PCP) in Argentina

METHODS: We used a questionnaire to assess PCP's knowledge (universal, selective or not recommended) and frequency of recommendation (5 point likert scale (never to always) to assess self reported performance) of 64 practices. Sample: non-probabilistic national sample of 247 PCP selected to participate in a multi-center study of PP. Outcome measures: Adequate level of knowledge (AK = >60% of recommended (AB) and non-recommended (DE) PP correctly identified in the survey. Performance was evaluated separately for AB and DE PP. Perf AB (>60% of all AB PP recommended when applicable), Perf DE (>60% of DE PP not recommended) and Adequate Preventive Performance (APP = >50% of both AB and DE PP).

RESULTS: We obtained 192 surveys for analysis (77% response rate). Overall, 43% of respondents achieved AK, PerfAB = 31%, PerfDE = 42% and APP = 19%. Compliance with selected sentinel PP such as diet and physical activity (79 and 69%), mammogram and pap smear (68 and 76%), cholesterol (78%), PSA (17%) and bone densitometry (13%) was deemed satisfactory. Sigmoidoscopy and FOB test, although correctly identified as selectively recommended (66%), was rarely indicated (15-20%). Variables included in the analysis were: age, sex, years since graduation, resident status, specialty (generalist vs internist), patients per week, orientation to PP, knowledge of guidelines, academic center and location in the capital of the country (capital). In the univariate analysis AK was significantly associated with age, guidelines, academic center and capital. In the multivariate model, only academic center remained significant (OR 4.7 [1.8-12.2] $p < 0.02$). In the univariate analysis the three preventive performance measures were significantly associated with specialty, orientation to PP, guidelines and academic centers. In addition, Perf DE was also significantly associated with age and AK and APP added sex, years since graduation and capital. In the multivariate model the variables that remained significant were: Perf AB: generalist (OR 3.7 [1.5-9] $p < 0.005$) and orientation to PP (OR 4.8 [1-19] $p < 0.052$); Perf DE: AK (OR 6.5 [2.8-15.3] $p < 0.0001$) and generalist (OR 2.8 [1.2-6.5] $p < 0.02$); and APP: academic centers (OR 6.9 [2.2-18.4] $p < 0.001$) and capital (OR 4.1 [1.2-14.4] $p < 0.029$).

CONCLUSION: Among highly motivated physicians, compliance with selected PP is acceptable but the overall performance with PP is low. We need to develop a strategy to improve the provision of PP by better integrating the recommended guidelines into clinical practice, especially for non generalists and those practicing in non academic centers outside the capital

PHYSICIANS' IMPACT ON THE USE OF HORMONE REPLACEMENT THERAPY (HRT). JM Clark, EB Bass, KM Bass, TL Bush, The Johns Hopkins University School of Medicine; University of Maryland School of Medicine, Baltimore, MD

BACKGROUND: Use of HRT is related to patient demographics, but little is known about physicians' role in women's choices. We investigated the use of HRT in postmenopausal women in Maryland to determine the role of general internists, family practitioners, and gynecologists in prescribing HRT, and the associated reasons for starting HRT.

METHODS: In 1999, we mailed questionnaires to a population-based sample of 1000 randomly selected women ages 50 and over living in the 5 largest counties in Maryland. We asked about demographics, source of information about HRT, use of HRT, characteristics of their primary care physician (PCP) and HRT prescriber, and reasons for starting or not starting HRT.

RESULTS: 29 surveys were undeliverable; 603 surveys were returned, of which 408 (42% overall) were completed and 195 were blank. The respondents had a mean age of 63.6 years (SD 9.5), 75% were white, 18% African-American, 65% married, and 35% had a household income less than \$40,000. 46% were currently taking HRT, 15% took it in the past, and 39% never took it. Overall, 64% of women cited a physician as their primary source of information about HRT, 9% the news media and 7% a woman's magazine; 4% had never heard of it. Those who received the information from a physician were 7.5 times more likely to take HRT than those who didn't, after controlling for age, race, income and education ($p < .001$). Although 60% of respondents' PCP's were general internists, 27% family practitioners and 7% gynecologists, 76% of current users were prescribed HRT by gynecologists and only 14% by general internists and 4% by family practitioners. Depending on the specialty of the prescriber, current users had different reasons overall for starting HRT ($p = .03$): women prescribed HRT by gynecologists as compared to internists were less likely to begin HRT for osteoporosis prevention (18% vs. 36%), or cardiovascular prevention (23% vs. 32%) and more likely for control of menopausal symptoms (22% vs. 18%). Among women who never took HRT, 40% never discussed it with their PCP. Of those who discussed it, 40% chose not to take it because their PCP did not recommend it, and only 8% because of fear of breast cancer.

CONCLUSION: In this population-based sample, nearly half the respondents were currently taking HRT. Physicians are the primary source of information about HRT for most women and appear to have an enormous impact on a woman's choice to take HRT.

CRITICAL PATHWAYS: HOW STRONG IS THE LITERATURE SUPPORTING THEIR USE? DS Conner, WD Rifkin, AS Silver, BE Flansbaum, DN Rose, Dept. of Medicine, Long Island Jewish Medical Center, New Hyde Park; Dept. of Medicine, Nassau County Medical Center, East Meadows, NY

PURPOSE: Across the nation, hospitals and other health care institutions are developing critical pathways to improve health care and control costs. Many studies

have been published supporting the use of critical pathways, but the quality of this literature has not been formally examined. We systematically reviewed the critical pathway literature to evaluate the quality of such work.

METHODS: We searched Medline databases from 1990 through 1998 for published reports of studies comparing conventional care to care integrating a critical pathway. We developed a tool to evaluate four aspects of the quality of these studies: study design and choice of control group, statistical treatment of the results, patient selection and evaluation of the intervention's use. Two trained readers evaluated each article; differences were resolved by consensus.

RESULTS: Our search identified 588 articles pertaining to critical pathways. We excluded 493 because they did not have a control group and 19 because they did not meet our definition of a critical pathway, leaving 76 articles for analysis. Most critical pathways involved surgical (67%) or medical (13%) diseases and the vast majority (92%) were in hospital settings. **STUDY DESIGN:** A randomized controlled trial design was used in one article (1.3%). An observational design with historical controls was used in 95% of the trials and with concurrent controls in 11% (some trials had more than one control group). **TREATMENT OF RESULTS:** Costs were reported in 71% of the trials and 91% of these showed a benefit (lower costs). These benefits were supported statistically in only 36%. Hospital length of stay (LOS) was reported in 90% of the trials and 84% of these showed a benefit (shorter LOS). These benefits were supported statistically in only 49%. Accounting for potential differences in case-mix between groups occurred in only 38% of trials. **PATIENT SELECTION:** Patient selection most commonly used procedure codes (63%) or billing codes (35%). Clinical criteria to define eligibility were reported in 33% of trials. Seventy-five percent had specific enrollment dates for all groups in the study. **EVALUATION OF USE:** Only 37% of the articles reported some analysis to assure the reader that the intervention (the critical pathway) was used correctly. The most common method reported was training of users (75%) and less often reported was actual frequency of use (39%).

CONCLUSION: The preponderance of literature supporting the effectiveness of critical pathways is based upon non-randomized observational studies using historical controls. Most studies do not attempt to assess patient case-mix. Before continuing widespread implementation, we should design trials that address previous methodological pitfalls and validly prove or disprove the effectiveness of critical pathways.

PATIENT-PHYSICIAN RACE CONCORDANCE AND COMMUNICATION IN PRIMARY CARE. L Cooper-Patrick, DE Ford, HT Vu, NR Powe, DM Steinwachs, DL Roter, Johns Hopkins University School of Medicine; Johns Hopkins University School of Hygiene and Public Health, Baltimore, MD

PURPOSE: To compare patient ratings of partnership with physicians and physician and patient communication behaviors in race-concordant and race-discordant relationships.

METHODS: Cross-sectional post-visit surveys and audiotape analysis of 283 patients seeing 31 physicians in 15 urban primary care practices who identified themselves as being either African American or white. Main outcomes were patient ratings of physicians' participatory decision-making (PDM) style and communication behaviors. Analysis of variance and multiple linear regression were used to measure the associations between race-concordance and the outcomes.

RESULTS: Patients, aged 18-88 years, were 58% African American, 68% female, and 17% college graduates. Physicians were 58% African American and 55% female. African American and female physicians had higher PDM scores than their counterparts, $p = 0.05$ and $p = 0.01$, respectively. Patients in race-concordant relationships rated their physicians' PDM styles as more participatory ($p = 0.07$). The length of visits ranged from 13 minutes for African American patients with white doctors to 25 minutes for white patients with African American doctors ($p = 0.001$). Differences in communication behaviors included: psychosocial, positive, and emotional talk, and patient-centered interviewing by physicians, and psychosocial talk by patients. White physicians were the least patient-centered regardless of patient race, African American physicians with African American patients were intermediate, and African American physicians with white patients were the most patient-centered. There were no differences in physician verbal dominance, physician use of facilitators, or physician and patient biomedical talk by race-concordant status.

Race-concordant status and physician communication behaviors [mean (SE)]

MD	Patient	Psychosocial	Emotional	Positive	Patient-centered
race	race	Talk,	Talk,	Talk,	Interviewing
		utterances	utterances	utterances	Score
white	white	6.3 (4.0)	3.4 (1.1)	4.1 (1.4)	8.2 (19.3)
white	AA	9.1 (4.8)	2.1 (1.3)	3.4 (1.6)	18.4 (23.0)
AA	AA	12.2 (3.6)	5.3 (0.9)*	7.3 (1.2)*	29.9 (17.2)
AA	white	20.1 (6.1)*	9.7 (1.6)*	8.6 (2.1)*	80.3 (29.1)*

Analyses adjusted for length of visit. * $p \leq 0.05$, ** $p \leq 0.01$.

MD = physician, AA = African American.

Reference group is white MD/white patient.

CONCLUSION: Patient ratings of partnership with physicians are higher in race-concordant relationships. Audiotapes confirm differences in several patient-centered communication behaviors by race-concordant status. Future analyses based on underlying theories of partnership between physicians and patients may clarify the nature and context of these differences.

LOSS OF ADL FUNCTION IN OLDER HOSPITALIZED PATIENTS: INCREASED VULNERABILITY WITH AGE. KE Covinsky, RH Fortinsky, M-M Chren, RM Palmer, SR Counsell, L Sands, C Landefeld, University of California, San Francisco, CA; University of Connecticut, Farmington, CT; Cleveland Clinic, Cleveland, OH; Indiana University, Indianapolis, IN

PURPOSE: Older patients hospitalized with medical illnesses are frequently discharged with worse ADL function than their pre-illness baseline. Our goal was to describe the course of functional loss in these patients and to examine the impact of increasing age on functional loss.

METHODS: On hospital admission (ADM), we asked 2309 patients (mean age 79) or their surrogates about the patient's independence in 5 ADL (bathing, dressing, eating, transferring, toileting) at baseline two weeks before ADM, and at ADM. We interviewed them again at DC. We defined functional decline as the loss of independence in at least one ADL.

RESULTS: Overall, 799 (35%) patients declined in function between baseline and DC. Of these, 402 patients declined before ADM and 397 patients declined between ADM and DC. Of the 1510 (65%) patients who did not decline between baseline and DC, 455 patients declined before ADM, but recovered before DC, while 1055 patients did not decline either before or after ADM.

As shown in the table, the oldest patients were the most likely to decline between baseline and DC, both in bivariate analyses, and in multivariate analyses controlling for APACHE scores, Charlson comorbidity scores, presence of dementia, and baseline ADL and IADL function.

Age	Proportion Declining	Multivariate Odds Ratio
70-79	25%	1.0
80-84	38%	1.5 (1.2-1.9)
85-89	49%	2.2 (1.7-2.9)
90 or more	61%	3.0 (2.2-4.1)

CONCLUSION: We conclude that functional decline is common in older hospitalized patients, often occurs after hospital ADM, and the risk markedly increases with advancing age.

INSURANCE PREFERENCES OF THE UNINSURED. M. Danis, AK Biddle, SD Gool, National Institutes of Health, Bethesda, MD; University of North Carolina, Chapel Hill, NC; University of Michigan, Ann Arbor, MI

PURPOSE: Over 40 million Americans are uninsured. Among the major obstacles to universal insurance are a lack of consensus about a basic benefit package and concern about cost. To address these barriers to universal coverage a project was designed to provide uninsured patients the opportunity to define a benefit package within cost constraints.

METHODS: Groups of uninsured individuals recruited from community settings and ambulatory care sites located in central North Carolina were convened to solicit their insurance choices using a simulation exercise, *Choosing Healthplans All Together*. In this group exercise, participants design a managed care package by selecting from 15 categories of services, trading between breadth of services and flexibility of restrictions (e.g., formulary, co-payments, utilization review) within the constraints of a fixed monthly premium. The premium is comparable to the typical per member/per month premium paid by U.S. employers for managed care plans and permits coverage of 60% of the services offered in the exercise.

RESULTS: Fifty-seven participants in 5 groups, out of a planned total of 240 participants in 20 groups, participated in the exercise. Participants were predominantly male (86%), minority (88%), and socio-economically disadvantaged (90%) had an annual income <\$15,000. Participants picked the following services in descending order of frequency: hospitalization (98%), pharmacy (91%), dental (91%), vision care (81%), tests (79%), primary care (74%), long term care (70%), home health (67%), mental health (61%), specialty care (58%), uninsured coverage (56%), last chance (transplantation and clinical trials, 53%), allied health and other services (51%), complementary medicine (49%), and infertility (23%). Those services that were more frequently included in participants' plans were also more likely to have the more flexible option chosen. Participants chose to include an average of 10 services with intermediate flexibility (score of 1.8) on a scale from 1 (tightly managed care) to 3 (more flexible care) during their first selection, and slightly broader coverage (10.25 services) with less flexibility (1.75) during their final selection following role-playing illness episodes.

CONCLUSION: These data suggest that socio-economically disadvantaged, uninsured individuals tend on average to pick a package that balances breadth and flexibility of health services when offered the opportunity to design their own health insurance coverage within the constraints currently experienced by many U.S. employees.

NATIONAL PATTERNS OF AMBULATORY MEDICAL CARE IN THE SERIOUSLY MENTALLY ILL. GL Daumit, RM Crum, NR Powe, DE Ford, Johns Hopkins University School of Medicine, Baltimore, Maryland

OBJECTIVE: Seriously mentally ill (SMI) individuals have higher mortality rates than the general population that are only partially accounted for by their mental illness. This population is at risk for not receiving adequate general medical and preventive care, but little is known about their use of outpatient medical services. **METHODS:** We conducted a cross-sectional study using National Ambulatory Medical Care Survey (NAMCS) data from 1993-97. The NAMCS is an ongoing an-

nual survey of random samples of U.S. office-based physicians that records information on individual patient visits. We defined SMI as having an ICD-9 diagnosis of schizophrenia, bipolar disorder, or recurrent psychotic depression, or taking an anti-psychotic drug or lithium. We combined data from 1993-97 and included visits for individuals aged 18-65. For visits to general medical providers, we examined reason for and duration of visit, occurrence of risk factors and preventive counseling, and disposition after visit. We calculated national estimates and compared SMI visits to the general population (GP) aged 18-65.

RESULTS: SMI individuals made 1939 visits to 547 physicians, representing 1.5% of all physician visits nationally during this time period. Visits to general medical providers (internists, family practitioners, or ob/gyns), accounted for 22% percent of SMI visits in contrast to 64% of visits in the GP. The following results are reported for general medical provider visits. SMI individuals had a mean age of 45 years, were 59% female, and 83% white. One-third of SMI were covered by Medicaid, 23% by Medicare, and 23% were in an HMO. The SMI made visits to a new physician only 8% of the time. In the SMI, 29% stated a mental health reason for the visit, compared to 13% of the GP ($p < .001$). The mean visit duration was 19.2 minutes (SE 0.77) in the SMI patients compared to 16.8 minutes (SE 0.06) in the GP ($p < .002$). Blood pressure checks were not different in SMI (78%) and GP visits (75%). The SMI were more than twice as likely to be reported as obese as the GP (25% SMI vs. 12% GP, $p < .001$); for visits with reported obesity, the SMI were just as likely to receive counseling on diet (47% SMI vs. 46% GP) and exercise (34% SMI vs. 32% GP). The SMI were twice as likely to report smoking (32% SMI vs. 16% GP, $p < .001$), but were somewhat less likely to receive smoking-related counseling (25% SMI vs. 30% GP, $p < .001$). SMI visits were more likely to result in a specific time for follow-up (74% SMI vs. 60% GP, $p < .001$).

CONCLUSION: The SMI are at high risk for obesity and smoking. Consistent with their complex needs, they have longer visits and are more often scheduled for follow-up visits. The quality of care for these seriously mentally ill individuals receiving general medical care appears comparable to the general population.

PRIMARY CARE PHYSICIANS' RATINGS OF QUALITY INDICATORS IN OFFICE PRACTICE.

GL Daumit, JA Flynn, NR Powe, DE Ford, Johns Hopkins University School of Medicine, Baltimore, MD

OBJECTIVE: Many quality improvement programs target specific diseases, but little work has been done to identify structural components of primary care offices related to quality of care such as number and type of staff, or office procedures. In the process of developing a self-assessment tool for primary care practices, we aimed to identify important characteristics of office practice associated with primary care quality from the physician's perspective.

METHODS: We recruited internists and family practitioners for two-hour focus groups. The sessions were audiotaped and transcribed, and two physicians (GLD, JAF) reviewed transcripts to identify all office characteristics possibly related to quality of care; a third adjudicated disagreements (DEF). We assembled these 60 characteristics in a questionnaire and distributed them to focus group participants. Participants ranked the characteristics on two scales from 1 to 10 based on the importance to quality and the degree to which the characteristic was under control of the physician. We calculated mean scores for importance to quality and control.

RESULTS: Twenty-five clinicians from different practices participated in focus groups and 23²⁵ returned ranking questionnaires. Sixteen of the 23 were internists, 10 were in solo practice, 11 had more than 1/2 of patients capitated, 5 were women, and 6 were non-white. The table shows the mean (SD) for the five highest and five lowest ranked of the 60 practice characteristics.

Characteristic	Importance	Control
1. Patients believe MD cares	9.1 (1.3)	9.0 (1.5)
2. Courteous office staff	9.1 (1.2)	7.7 (2.7)
3. Patients know they can go to PCP first with problem	9.0 (0.9)	7.7 (2.5)
4. Legible medical records	8.9 (1.2)	9.0 (1.9)
5. Practice educates patients about managing health	8.8 (1.8)	8.7 (1.7)
56. Disease management programs	4.9 (2.8)	4.1 (3.1)
57. First visit needs full history and physical	4.7 (2.7)	7.1 (3.1)
58. In-office lab	4.5 (2.6)	5.0 (2.9)
59. Helping patients purchase equipment	4.5 (3.1)	6.5 (3.0)
60. Patient contact outside clinical setting	3.6 (2.5)	4.4 (3.1)

We also asked how MDs would spend an extra \$50,000 to improve quality. Of the 47 comments, 20 mentioned increasing salary or hiring new staff, 10 mentioned computers, and 7 mentioned new office equipment.

CONCLUSION: Physicians rated interpersonal aspects of care and staff at the top of the list for impacting quality. Rankings for specific office procedures were more disparate. These physicians appear not to embrace disease management programs, an evidence-based intervention, for improving quality of office-based primary care.

CAN A SINGLE ASSESSMENT OF VITAL SIGNS OBTAINED DURING ROUTINE PRIMARY CARE VISITS PREDICT ADVERSE VASCULAR EVENTS? S. Dove, M Brunt, J Kesterson, XH Zhou, WM Tierney, Indiana University, Regenstrief Institute, Roudebush VA Medical Center, Indianapolis, IN

INTRODUCTION: Hypertension is among the most common treatable conditions in primary care and is a risk factor for morbid vascular events. We studied whether a single outpatient blood pressure (BP) reading can predict such events among hypertensive inner-city primary care patients.

METHODS: We identified primary care patients with hypertension listed among their outpatient problems during their first scheduled visit in 1993 to the primary care practice affiliated with an urban public hospital. Data were extracted from a 25-year old electronic medical record system and included the systolic BP (SBP), diastolic BP (DBP), and heart rate (HR) routinely recorded during their index visit, prior comorbid illnesses and results of laboratory and imaging tests recorded on or before the index visit in 1993, and hypertensive drugs taken at the index visit. We then searched their next 5 years' records for inpatient and outpatient diagnoses and laboratory tests indicating the first occurrence of myocardial infarction (MI), stroke (CVA), heart failure (HF), ischemic heart disease (IHD), chronic renal insufficiency (CRF), and all-cause death. For each adverse outcome, independent (predictor) variables were assessed via Cox regression. Those variables found to be univariately significant predictors were entered in final multivariable Cox models.

RESULTS: We identified 5825 hypertensive primary care patients who kept scheduled visits in 1993; 69% were women and 63% black. Their mean age was 57 years. In the next 5 years, 433 (7%) had an MI, 1017 (17%) a CVA, 1385 (24%) IHD, 1309 (22%) HF, 718 (12%) CRF, and 766 (13%) died. Controlling for clinical factors and prior history, the SBP (but not DBP) recorded at the index visit was found to be a significant multivariable predictor of CVA ($p = 0.001$), combined CVA/MI ($p = 0.002$), IHD ($p = 0.005$), and CRF ($p < 0.001$). HR predicted death ($p = 0.003$), and no vital sign predicted MI alone or HF. A 10 mmHg higher SBP was associated with an increase of 13% (95% CI 6–21%) in the risk of CRF, 9% (3–15%) for CAD, 7% (3–11%) for CVA, and 6% (2–9%) for combined CVA or MI. A 10 beat/min increase in HR was associated with a 16% (5–26%) increase in the risk of death. No significant association with any of the target outcomes was found for mean BP, pulse pressure, the HRBP index, or the ratio of SBP to DBP.

CONCLUSION: Vital signs recorded routinely during primary care visits have surprisingly significant prognostic value. Specifically, a single systolic blood pressure reading predicted the subsequent occurrence of CVA, combined CVA/MI, IHD, and CRF while heart rate predicted death. These are thus true vital signs which should be attended to during primary care visits.

APPROACHES TO ADHERENCE IN RANDOMIZED CONTROLLED TRIALS. MM Davis, P Rathouz, Robert Wood Johnson Clinical Scholars Program; Department of Health Studies, University of Chicago, Chicago, IL

PURPOSE: Adherence to medication is a major determinant of therapeutic effectiveness. Recent advances have led to significantly improved methods of monitoring and promoting adherence to therapy, and several statistical approaches have been proposed for analyzing adherence data. Yet, classic intent-to-treat (ITT) analyses of randomized controlled trial (RCT) data intentionally ignore subjects' adherence; this approach may leave clinical decision-makers with more limited information than they would desire. The purpose of this study was to examine recent, clinically influential RCTs, to assess how the latest studies measure, report, and interpret data regarding adherence.

METHODS: High-impact RCTs within the fields of general internal medicine and medical subspecialties were identified through a review of all articles presented in the 1997–98 and 1998–99 "Update" series in the *Annals of Internal Medicine*. Cited articles were excluded if they described: non-drug treatment, non-self-administered study drug(s); therapy related to perioperative care or outpatient procedures; or a comparison of inpatient vs. outpatient therapy. 65 articles cited in the "Update" series and published in 1997 or 1998 qualified for analysis. Assessment included: size and site of study; duration of active therapy; measures of daily adherence to therapy; and report and analysis of adherence data.

RESULTS: Among recent influential RCTs, there was a broad range of study size (range = 20–18,790 subjects; median = 450) and duration of active therapies (range = <1 week–10 years; median = 48 weeks). Only 35 articles (54%) included at least one measure of daily adherence to study therapy; 20 of these utilized frequently biased measures such as pill counts or patients' self-report. Measures of adherence data were more commonly employed in smaller (<450 subjects) vs. larger trials (odds ratio (OR) 2.59 [95% CI 0.96, 7.00]), and in shorter (<48 weeks) vs. longer studies (OR 2.30 [0.85, 6.19]). Only 9 articles in the sample described any effort to promote adherence to therapy among study participants. Of the 35 articles that monitored adherence, only 13 presented both ITT and supplemental analyses that reflected the impact of subjects' adherence patterns.

CONCLUSION: Recently published, clinically influential RCTs commonly offer limited information regarding daily adherence to medical therapy. Moreover, methods most often utilized to monitor, promote, and interpret adherence do not reflect the current state of the art. Clinical investigators should consider implementing more reliable methods regarding adherence, in order to offer health care providers a more detailed sense of how to apply RCT findings to clinical decision-making.

BURDEN OF ILLNESS IN ADDICTED PERSONS WITHOUT PRIMARY MEDICAL CARE. J DeAlba, JH Samet, R Saitz, Section of General Internal Medicine, Boston Medical Center-Boston University School of Medicine, Boston, MA

PURPOSE: Persons with alcohol and drug addictions are known to be at risk for a variety of medical illnesses. Little is known about the frequency of illness in persons with addictions who lack primary medical care. Our objective was to assess

the burden of medical illness in this population and to identify substances of abuse and other patient characteristics associated with worse health.

METHODS: Patients without primary medical care who were admitted for alcohol, heroin or cocaine detoxification were enrolled in the Health Evaluation and Linkage to Primary Care (HELP) study. Subjects completed standardized interviews assessing demographics, social support, substance use and medical problems. Burden of illness was assessed by self-report of physicians' diagnoses, and by the Short Form Health Survey (SF-36). To characterize subjects with the most illness, we assessed the relationship between patient characteristics and the SF-36 Physical Component Summary (PCS) score in bivariate and multivariable analyses.

RESULTS: A total of 470 persons were enrolled. The mean age was 35.7 (SD 7.8) years, 76% were male and 46% were African-American; 57% had an annual income of less than \$20,000 and 38% were unemployed. Almost half of all patients (45%) reported being diagnosed with a chronic illness, 80% had prior medical hospitalizations and 21% had been prescribed medications for a chronic physical illness. The most common diagnosed chronic conditions were asthma/chronic obstructive lung disease (20%), hypertension (16%) and chronic liver disease (12%). Only 2.6% reported being HIV positive. Many (45%) reported acute medical illness (i.e., pneumonia), serious injury or a sexually transmitted disease in the past six months. The mean PCS score for the group was 48 (SD 10.7), which is similar to norms for the U.S. population. Cocaine and alcohol abuse were not significantly associated with lower scores. In a multivariate analysis adjusting for race (model R-square 0.16), older age (–2.64 PCS score points per decade, $p = 0.0001$), female gender (–3.46, $p = 0.0001$), unemployment (–3.27, $p = 0.001$), living alone (–3.42, $p = 0.01$), and problem heroin use (–3.08, $p = 0.002$) and hallucinogen use (–2.56, $p = 0.04$) were associated with lower PCS scores.

CONCLUSION: Despite youth's protective effect, alcohol and drug dependent persons lacking primary care had a substantial burden of serious medical illness. While all such persons with addictions might benefit from linkage to primary medical care, factors associated with worse health in our analysis help to identify those with the greatest need.

THE RELATIONSHIP BETWEEN QUALITY CARE PROCESS MARKERS, SEVERITY OF ILLNESS, AND OUTCOMES IN PATIENTS HOSPITALIZED WITH COMMUNITY-ACQUIRED PNEUMONIA AT U.S. ACADEMIC INSTITUTIONS. JJ Dedier, SJ Atlas, DE Singer, General Internal Medicine, Massachusetts General Hospital, Boston, MA

PURPOSE: Timely blood cultures (BCx), measurement of oxygenation (O2Sat) and prompt administration of antibiotics (ABX) are generally considered markers of high quality care in the treatment of community-acquired pneumonia (CAP). A large study of Medicare patients reported lower 30-day mortality in those receiving ABX within 8 hrs of arrival (odds ratio [OR] 0.85, $p < 0.05$) or BCx within 24 hrs of arrival (OR 0.90, $p = 0.07$).

The objective of this study was to determine whether early achievement of quality process markers in CAP is associated with i) higher proportions reaching clinical stability within 48 hrs of admission by the criteria of Halm et. al., ii) decreased hospital length of stay, or iii) fewer inpatient deaths.

METHODS: Retrospective chart review identified 1100 eligible patients discharged with a primary or secondary diagnosis of CAP between 12/1/97 and 2/28/98 from among the 38 U.S. hospitals in the University Health System Consortium pneumonia benchmarking network. Logistic regression was used to assess the univariable and severity-adjusted relationship between each process measure and the 3 clinical outcomes listed above. Test of trend assessed the relationship between process markers and admission pneumonia severity (PSI).

RESULTS: All 3 process markers were achieved more frequently the higher the PSI class ($p < 0.05$). In univariable and PSI-adjusted analyses, process measures were not significantly associated with better outcomes (Table). However, the PSI-adjusted ORs for early ABX and BCx were compatible with shorter lengths of stay and fewer inpatient deaths, respectively.

MARKER	OR FOR INSTABILITY AT 48 HRS		OR FOR LOS >4 DAYS (median)		OR FOR INPT. DEATH	
	Crude	Crude PSI-adj[95% CI]	Crude	Crude PSI-adj[95% CI]	Crude	Crude PSI-adj[95% CI]
ABX≤8h	1.22	1.15[.85, 1.56]	0.94	0.90[.67, 1.22]	1.68	1.53[.77, 3.05]
BCX≤24h	1.69	1.54[1.12, 2.14]	1.37	1.26[.90, 1.76]	1.27	0.81[.39, 1.70]
O2Sat≤24h	1.12	0.99[.60, 1.64]	1.70	1.55[.93, 2.59]	1.54	1.37[.40, 4.75]

CONCLUSION: In our analysis, process markers were not significantly associated with improved outcomes. Our results may be subject to residual confounding by pneumonia severity, and their interpretation may be limited by insufficient sample size. However, they suggest these process markers are not strong or uniform determinants of outcome in the context of care academic hospitals provide.

SELF-REPORTED QUALITY OF CARE AND PREVENTIVE SERVICES AMONG HISPANIC ADULT MANAGED CARE ENROLLEES. DE DeLaet, O Carrasquillo, Division of General Medicine, Columbia University College of Physicians and Surgeons, New York, NY

PURPOSE: While many studies have compared managed care plans (MCPs) to fee-for-service systems (FFS), few have examined if ethnic minorities enrolled in MCPs receive better quality of care than those in FFS.

METHODS: We used the data from the first three rounds of the 1996 Medical Expenditure Panel Survey. We analyzed self-reported satisfaction with care and use of preventive services among the 7,328 Hispanic and Non-Hispanic White adults (NHWs) aged 18–64 who reported having private health insurance. Persons in

MCPs were those enrolled in a Health Maintenance Organization or a plan with a gatekeeper system. Otherwise they were considered as having FFS. We used SUDAAN to adjust for survey design and to calculate nationally representative estimates. Logistic regression was used to adjust for differences in educational level, age, gender, and self-perceived health status.

RESULTS: As compared to NHWs, Hispanic adults were much less likely to have private insurance (47% vs 78%, $P < .001$), and among those privately insured much more likely to belong to MCPs (67% vs 52%, $p < .001$). Hispanics in MCPs were of similar age, education, and had similar self-perceived health status as those in FFS, but slightly more females were in MCPs. Hispanics in MCPs tended to report lower levels of satisfaction as compared to those in FFS including lower overall quality of care (81% vs 74%, adjusted odds ratio [OR] 0.63, 95% confidence intervals [CI] 0.37–1.08), lower satisfaction with office staff (77% vs 70%, OR 0.70, CI 0.42–1.15), were less likely to be asked about other treatments (88% vs 78%, OR 0.45, CI 0.25–0.80) and were less confident in their provider's ability (99% vs 93%, $P < .01$). In general these trends were similar to those noted for NHWs.

However, Hispanics in MCPs were more likely to report having received preventive services than those in FFS, including having had their blood pressure measured (92% vs 83%, OR 2.36, CI 1.46–3.81), among adults age >35 having had a cholesterol test (76% vs 71%, OR 1.29, CI 0.81–2.06), among females having had a PAP smear (85% vs 75%, OR 1.82, CI 1.08–3.10), and among females age >40 having had a mammogram (74 vs 66%, OR 1.71, CI 0.80–3.67) and having had a breast examination (85% vs 73%, OR 2.24, CI 1.00–5.01). Differences in preventive service delivery between MCPs and FFS were of greater magnitude among Hispanics than among NHWs.

CONCLUSION: Employing commonly used "report card" measures of quality that focus on preventive services, privately-insured Hispanics in MCPs appear to do better than those in FFS, even more so than NHWs. Yet Hispanics were as dissatisfied as NHWs with the quality of their care under MCPs when compared to FFS.

THE LACK OF PHYSICIAN CONCORDANCE WITH GUIDELINES IN THE PERIOPERATIVE USE OF BETA-BLOCKERS. *HR Delbecq, A Karim, DS Conner, DN Rose, Dept. of Medicine, Long Island Jewish Medical Center, New Hyde Park, NY*

PURPOSE: The American College of Physicians guidelines for perioperative care recommend the use of beta-blockers for certain patients to improve outcomes after surgery. Studies of physician behavior, with respect to guidelines and recommended practices, have shown that beta-blockers have been underutilized after myocardial infarction. We evaluated physicians' concordance with the perioperative use of beta-blockers.

METHODS: We reviewed the medical charts of adult patients having a cholecystectomy at a tertiary care medical center from 1997 through 1998. Two trained chart abstractors reviewed the medical records for demographic and clinical information and determined whether a cardiologist or non-cardiology medical specialist performed a preoperative evaluation. Patients met criteria for perioperative beta-blocker use if they had a history of coronary artery disease or if they had the presence of two or more risk factors: age greater than 65 years, history of hypertension, diabetes mellitus, hypercholesterolemia, or current smoking. We also noted emergency operations and the presence of contraindications to beta-blocker use.

RESULTS: Among the first 87 cases of cholecystectomy we reviewed, criteria for beta-blocker use were met in 42 (48%) of patients. Among these, 4 (9%) had emergency operations and 2 (5%) had contraindications to beta-blocker use. For the remaining 36 patients, the average age was 68 years, 55% were female, 26% had coronary artery disease, 50% had a documented preoperative medical evaluation in the chart by a non-cardiologist, 22% had an evaluation by a cardiologist, and 28% had no documented evaluation. Of the 9 patients with documented coronary artery disease, 6 were evaluated by non-cardiologists and 3 by cardiologists. There were 5 (14%) patients on beta-blockers prior to admission and 31 patients not on beta-blockers. Patients evaluated by cardiologists were more likely to be on beta-blockers (38%) than those evaluated by non-cardiologists (7%, $p = 0.06$). Of the 31 patients not on beta-blockers at admission but meeting criteria for their use, none were started on beta-blockers preoperatively. The only postoperative cardiovascular complication was one case of congestive heart failure.

CONCLUSION: We found that perioperative beta-blockers are underutilized in patients with risk factors for coronary artery disease, despite evidence that their use in appropriate individuals may be life-saving. Our findings are consistent with previous surveys of physician practice on beta-blocker use. If larger studies confirm these findings, quality improvement initiatives are indicated to reduce perioperative and postoperative complications of coronary artery disease.

EAT SMART! EFFICACY OF A LOW INTENSITY CLINICIAN CENTERED DIETARY INTERVENTION. *HK Delichatsios, R Lobb, MK Hunt, K Emmons, MW Gillman, Harvard Medical School/Harvard Pilgrim Health Care; Dana-Farber Cancer Institute, Boston, MA*

PURPOSE: Poor diet is a leading cause of chronic disease. Patients value nutrition advice given by their primary care providers (PCPs), who seldom provide it due to lack of time, expertise, or confidence. To help patients improve their eating habits, we evaluated a multifaceted intervention that included PCP endorsement.

METHODS: We randomized 6 Harvard Pilgrim Health Care HMO centers to an intervention or control group. Eligible patients were >18 years old and had scheduled routine appointments with their PCPs. All subjects completed a survey at baseline and at 3 months. Both surveys included a short food frequency question-

naire and a health habits survey. The intervention comprised 1) mailed dietary recommendations focusing on fruits and vegetables, meats, whole fat dairy, and low fat dairy, tailored to patients' baseline intake and stage of change; 2) verbal endorsement by the PCP of the benefits of these recommendations; 3) two telephone motivational interviews with study staff to set dietary goals; and if needed, 4) a telephone consultation with the study nutritionist. To account for cluster randomization, we used a mixed model analysis of variance.

RESULTS: We recruited 230 subjects in the intervention and 274 in the control group. Demographics for the intervention and the control groups were, respectively: mean age, 50 and 57 years ($p < 0.01$); proportion female, 77% and 63% ($p < 0.01$); race/ethnicity, 78% white, 13% African American and 95% white, 1% African American ($p < 0.01$). 89% of the participants completed the final survey. The baseline mean (SD) intake of fruits and vegetables was 3.0 (1.7) servings/day in the intervention group and 3.3 (1.6) servings/day in the control group. At 3 months, intake of fruits and vegetables had increased by 1.1 servings/day in the intervention and by 0.3 servings/day in the control group. Using an intent-to-treat analysis and adjusting for age, sex, race, and baseline intake of fruits and vegetables, we estimated that the change in fruit and vegetable intake in the intervention group was 0.56 (95% CI 0.31, 0.81) servings/day higher than in the control group. The change in consumption of red and processed meats, whole dairy, and low-fat dairy, did not differ between the control and intervention groups. Process data suggested that most subjects focused on their intake of fruit and vegetables rather than the other 3 food groups. The intervention was well received by both participants and PCPs.

CONCLUSION: This low intensity nutrition intervention was effective in helping patients increase their fruit and vegetable intake. Clinical interventions that combine practice supports and brief physician endorsement may be effective and efficient in producing behavior change among patients.

AGREEMENT AMONG RESIDENT PHYSICIANS IN CHOOSING FOLLOW-UP INTERVALS. *JP Deluca, D Lefrancios, C Cunningham, D Kapell, G Paccione, Department of Medicine; Residency Program in Social Medicine; Department of Family Medicine, Montefiore Medical Center, Bronx, NY*

PURPOSE: To investigate the degree of agreement between resident physicians on the assignment of follow-up interval and the identification of new diagnostic questions in ambulatory patient visits.

METHODS: Medical residents completed a standardized patient data collection form for the "first" and the "most interesting" patients in a given three hour clinic session. For each case, the residents designated the follow-up interval and identified new diagnostic questions. The next day, the group of residents (average 4–5) met with an attending for "follow-up interval" rounds at which randomly selected cases were presented. For each case presented, each resident in the group independently chose a follow-up interval and identified new diagnostic questions.

Measurements: To quantitate the residents' responses regarding the assignment of follow-up interval and the identification of new diagnostic questions, we defined the variable, "agreement." For each case, the residents were considered to have "agreement" in follow-up interval if: (1) the same follow-up interval was chosen by $\geq 75\%$ of the respondents, or (2) all respondents chose a follow-up interval that was within $\pm 50\%$ of the mean. "Agreement" in identification of new diagnostic questions was defined as at least two-thirds of the respondents identifying the same diagnostic question in a given case.

RESULTS: A total of 82 cases were presented. Agreement in the assignment of follow-up interval was seen in 46% (38/82) of the cases, with the average length of follow-up interval of 4.2 weeks. Agreement in the identification of new diagnostic questions was seen in 82% (67/82) of the cases. In cases where residents agreed on the identification of new diagnostic questions, the agreement on follow-up interval was 52%; however, in those cases where residents did not agree on the identification of new diagnostic questions, there was only a 20% agreement on follow-up interval.

CONCLUSION: When presented with the same clinical data there is poor agreement (46%) between resident physicians in the assignment of follow-up interval. One factor which appears to be associated with the level of agreement in follow-up interval is the level of agreement in identifying new diagnostic questions. However, even at high levels of agreement (82%) in the identification of new diagnostic questions, there is poor agreement in the assignment of follow-up. This suggests that there are other factors which influence residents' decision making regarding follow-up interval.

PREDICTING PHYSICIAN PRACTICE PATTERNS USING CLINICAL VIGNETTES. *KB Desalvo, WW Merrill, Internal Medicine, Tulane University School of Medicine, New Orleans, LA*

PURPOSE: Monitoring physician practice patterns by chart abstraction is expensive and subject to error. Some evidence suggests that clinical vignettes may be a reasonable tool to predict physician behavior in actual practice.

METHODS: We prospectively studied the practice patterns of Internal Medicine residents in the assignment of revisit intervals (RVI) for their patients with chronic disease. 222 physician-patient encounters were studied. Physicians ($n = 59$) completed a questionnaire at the end of the visit that collected information on the predictors of interest. Data gathered included demographic information on the provider, physician perceptions of the patient (disease stability and compliance) and management plans (ordering diagnostic tests and changing therapy for the main diagnosis). Physicians assigned an open-ended RVI for each encounter. Practice

patterns of these physicians in assigning RVI had previously been assessed using clinical vignettes of patients with chronic disease (Desalvo, KB et al; Arch Int Med; In Press). Physicians were ranked according to the mean RVI from the clinical vignette study and this ranking was used as a predictor for actual practice patterns. Provider specific predictors were sequentially entered into a mathematical model to determine their role in explaining the variance in revisit intervals.

RESULTS: The characteristics that contributed to the shortening of the revisit interval included physician perceptions of patient stability ($p < .001$) and patient compliance ($p = .012$). Physician decision to order diagnostic tests ($p = .15$) and to change the treatment of the main diagnosis ($p < .001$) also resulted in shorter RVI. When studied in actual practice, these physician characteristics accounted for 29% of the variance in the assignment of the revisit interval ($r = .53$). When the provider's rank from the clinical vignette data was added into the model, these five variables accounted for 88% of the variance in assignment of revisit intervals ($r = .93$).

CONCLUSION: These data suggest that clinical vignettes assessing physician behavior can strongly predict physician practice patterns. Combined with other physician level data, information from properly constructed vignettes gives excellent correlation with observed physician practice variation. This model needs to be validated in other physician populations and expanded to other practice parameters.

PHYSICIAN GENDER INFLUENCES THE ASSIGNMENT OF REVISIT INTERVALS FOR CHRONIC DISEASE. KB Desalvo, WW Merrill, Internal Medicine, Tulane University School of Medicine, New Orleans, LA

PURPOSE: Understanding the contributors to physician practice variation in the assignment of revisit intervals (RVI) may lead to strategies to decrease the inappropriate use of clinic visits.

METHODS: We prospectively studied practice patterns of Internal Medicine residents caring for indigent patients in inner city clinics in New Orleans, Louisiana. Consecutive clinic encounters ($n = 228$) for hypertension (HTN) and diabetes mellitus (DM2) were selected for inclusion. Physicians ($n = 59$) completed a questionnaire at the end of the visit that collected the information of interest including: data on the providers, their perceptions of the patient, plans for therapy and the open-ended RVI for the next encounter. Mean RVI between groups was compared using appropriate tests. Analysis of variance was used to evaluate the effects of independent variables on the return visit interval.

RESULTS: For 68% of encounters, HTN was the primary diagnosis and DM2 was the primary diagnosis in 32%. In 27% of encounters, patients had both. The mean RVI was 13 weeks for HTN and 11 weeks for DM2 with a range of 1–42 weeks. Level of training had no effect on RVI. Physiologic markers of disease severity contributed to physician perceptions of the patient's stability but did not significantly contribute to the assignment of the RVI. Several physician specific characteristics were associated with shortening of the RVI. Patients perceived to be non-compliant were assigned a shorter revisit interval (10 v. 13 weeks, $p < 0.012$) as were those considered unstable (8 weeks v. 15 weeks ($p < .001$)). Ordering diagnostic tests shortened the RVI from 14 to 12 weeks. When therapy for the primary diagnosis was changed, the RVI was significantly shortened (15 v. 10 weeks, $p < .001$). 29% of the variance in assignment of RVI was accounted for by physician perceptions about patient stability and compliance and practice parameters ($r = .53$). 71% of the encounters studied were with male physicians. Disease severity for patients of male and female residents did not differ. Female physicians were more likely to overbook themselves to see their patients ($p = .016$). The sex of the physician was a significant predictor of RVI. Females assigned a shorter RVI (10 v. 13 weeks, $p = .013$) irrespective of level of training, perceptions of the visit, or other practice parameters monitored.

CONCLUSION: Among physicians trained at the same institution, variation exists in the assignment of the RVI that is not explained by patient characteristics. The timing of the RVI was also predicted by physician gender. This may reflect the female physicians inclination to provide more health education and preventive care.

BARRIERS TO HEALTH CARE AMONG PRIMARY CARE PATIENTS IN A LARGE URBAN PUBLIC HEALTH SYSTEM. AL Diamant, R Hays, L Morales, W Ford, D Calmes, S Asch, N Duan, E Fielder, S Kim, J Fielding, M Shapiro, D Hayes-Bautista, L Gelberg, Department of Health Services, UCLA, Los Angeles and The Los Angeles County

PURPOSE: Problems with access to health care in the public health system have been previously reported. This is the first study to examine barriers to health care associated with patient and provider characteristics in a representative sample of patients in a large urban health care system.

We surveyed adult patients and the guardians of pediatric patients, among a stratified cross-sectional probability sample ($n = 3,193$) of all patients receiving primary care services through the Los Angeles (County-Department of Health Services) who had made a visit within 12 months of their current visit. Interviews were performed at all four types of medical facilities (Comprehensive Health Centers, Health Centers, Hospital Outpatient Clinics and Public/Private Partnership Clinics) distributed throughout Los Angeles County, either operated by or contracted with the LAC-DHS. The response rate was 80% ($n = 2,564$). The study sample was racially and ethnically diverse (Latino/Hispanic 61%, African American 21%, non-Hispanic white 15%, Asian/Pacific Islander 2% and Other 1%); 66% were female; the median age was 37 years (range from 1 to 87 years) the median income was between \$5,001 and \$10,000; and 68% of respondents were uninsured for health care.

A total of 32% reported having postponed needed medical care during the twelve months preceding the interview. Overall 12% had postponed needed care because they could not take time off work, 12% did not have any way to get to a medical appointment, 11% had responsibilities for taking care of others, and 10% were too sick. Moreover, 21% of the sample had gone without needed health care during the past 12 months because they had to spend the money on food, clothing, or rent. Additionally, 45% had gone without needed medical care at a County facility during the past 12 months because they thought it would have been too expensive. Women, adults, Latino/Hispanics, patients without health insurance, and patients at contracted community clinics were the most likely to report having gone without care because of competing needs.

Barriers to needed health care continue to exist, even for patients who rely on Safety Net Providers to fulfill their medical needs. Competing needs for other basic necessities are an important factor contributing to delays in care. Identification of these barriers is important for eradicating disparities in access to and receipt of medical care.

PATIENTS' USE OF THE INTERNET FOR MEDICAL INFORMATION. JA Diaz, PD Friedmann, AW Moulton, Division of General Internal Medicine, Rhode Island Hospital; Brown University School of Medicine, Providence, RI

PURPOSE: To identify patients' reasons for seeking health information on the Internet, type of information sought and perceptions of the quality of information available; and to establish whether medical information obtained is shared with medical providers.

METHODS: We mailed a confidential survey to 1000 patients, age 18–65, randomly selected from billing files of a general internal medicine practice. The survey consisted of demographics, questions about patient use of the Internet for medical information, frequency of use, reasons for use, and respondents' perception of quality of information. The survey also assessed whether medical information acquired on-line is shared with medical providers.

RESULTS: The response rate after the first mailing was 27%. Respondents' mean age was 47.6 (SD 12.1) and 62% were female; 74% indicated that they had used the Internet for medical information. Common reasons for using the Internet were to obtain information on: nutrition/diet (64%), side effects and/or complications of medical therapy (61%), alternative/complementary medicine (44%), second opinions about medical conditions (40%), cancer (17%), diabetes (9%) and women's health (9%). Of those using the Internet, 59% felt that the quality of information was "excellent" or "very good", and 41% assessed the quality of information as "good" or "fair". Of Internet users, 65% rated information on the Internet as the same or better quality than that from their doctor. Most (60%) indicated that they do not discuss this information with their physicians, and 12% indicated that they went "on-line" for health information instead of seeing or speaking with their physician.

CONCLUSION: Patients use the Internet to seek information on a broad range of medical questions, and a majority are confident that the information is as good or better than the information that they receive from their physician. Physicians need to understand the benefits and limitations of this extensive medical resource available to patients and to recognize the possibility that patients may use the Internet as an alternative to consulting their physician.

PATIENT AND PHYSICIAN SATISFACTION WITH A TELEPHONE-BASED ANTICOAGULATION SERVICE. AL Doggette, GA Bonet, P Milligan, A Frazier, E Verzino, B Walton, BF Gage, General Medical Sciences, Washington University School of Medicine, St. Louis, MO; Health Management Partners, St. Louis, MO

PURPOSE: How disease-state management programs affect patient and physician satisfaction is largely unknown. We compared the satisfaction and knowledge of 2 cohorts of patients: (1) patients randomized to have access to a multi-disciplinary telephone-based anticoagulation service (AS), and (2) patients receiving their anticoagulation care from their primary-care physician. We also assessed physician satisfaction with the AS.

METHODS: We surveyed 300 patients (mean age = 73) about their satisfaction with their anticoagulation care and their knowledge about warfarin therapy. We also surveyed the 17 referring physicians about their experience with the AS.

RESULTS: In an intent-to-treat analysis, patients randomized to the AS sites were more satisfied with the timeliness of getting their blood test results ($P = .02$), were more likely to know what a safe international normalized ratio (INR) value was ($P = .001$), and felt safer taking warfarin ($P = .04$) as compared with patients at the control sites. In an on-treatment analysis of patients enrolled in the AS versus control site patients, AS enrollees were more satisfied with the timeliness of getting their blood test results ($P = .001$), with the availability of the staff ($P = .03$), and with the overall quality of their anticoagulation care ($P = .02$). AS enrollees also were more likely to know what a safe INR value was ($P = .001$), what a low blood test or INR meant ($P = .01$), and felt safer taking warfarin ($P = .02$). Physicians reported that using the AS saved, on average, 4 minutes of their time and 13 minutes of their staff's time, per INR. All physicians were willing to recommend the use of the AS to a colleague.

CONCLUSION: Patients and physicians are satisfied with anticoagulation care provided by a multi-disciplinary, telephone-based AS. These findings suggest that a disease-state management program can be endorsed by primary-care physicians and improve patients' satisfaction with and knowledge about their anti-thrombotic therapy.

QUALITY OF LIFE AND TREATMENT DIFFERENCES BETWEEN PRIMARY CARE AND OTOLARYNGOLOGY IN RHINOSINUSITIS. RJ Dolor, DL Witsell, DL Simel, Health Services Research, Durham VA Medical Center, Durham, NC

PURPOSE: To compare patient outcomes with rhinosinusitis managed by primary care providers (PC) versus otolaryngologists (ENT) with respect to disease severity, comorbidities, general and disease-specific quality of life (QOL), patient satisfaction, and utilization of VA services.

METHODS: On 165 consecutive patients treated for rhinosinusitis from the PC and ENT clinics we collected demographic and comorbidity information. At enrollment, 6 weeks, and 12 weeks later we recorded responses to surveys on general QOL (Short Form-12, SF-12), rhinosinusitis QOL (Sinonasal Outcome Test-20, SNOT-20), disease severity and duration (Chronic Sinusitis Survey-Duration-Based, CSS-D), and patient satisfaction (Visit Specific Questionnaire, VSQ). Prescriptions, radiology, and laboratory data, and number of visits for rhinosinusitis were also collected at the same intervals. Group differences were compared using Wilcoxon analysis and t-tests as appropriate.

RESULTS: Baseline demographic variables between groups (86 PC and 79 ENT) were equal except for a higher proportion of patients with a history of nasal polyposis in the ENT group (41% ENT vs. 20% PC, $p = 0.004$). At baseline, ENT patients had lower physical health scores (PCS-12, $p = 0.02$), but PC patients had lower mental health scores (MCS-12, $p = 0.03$). There was no difference in SNOT-20 scores, or symptom severity scores (CSS) between the groups. However ENT patients had a longer duration of nasal congestion ($p = 0.006$), previous antibiotic use ($p = 0.006$) and nasal spray use ($p < 0.001$). Overall patient satisfaction was higher in patients treated by ENT ($p = 0.01$) due to differences in time spent with the provider ($p = 0.02$) and waiting time ($p < 0.001$). At 6- and 12-week follow-up, there were no differences in the proportions of patients who were cured, improved, the same, or worse. At 6- and 12-weeks, SF-12 scores were unchanged. In both groups, SNOT-20 scores improved from baseline to 6 weeks ($p = 0.007$ PC, $p = 0.03$ ENT), but did not change between 6 to 12 weeks. Symptom severity scores decreased in both groups at 6 weeks, however ENT patients had continued decreases in severity scores at 12 weeks. ENT patients had higher resource utilization (more visits, labs, radiological tests, and surgical procedures) during the 12-week follow-up period ($p = 0.001$).

CONCLUSION: We conclude that outcomes of patients treated by PC or ENT improve by 6 weeks and remain stable at 12 weeks as measured by disease-specific QOL and symptom severity surveys. Variations in antibiotics prescribed, follow-up visits, labs, radiological tests, and surgical procedures between PC and ENT are due to differences in type of sinusitis. PC subjects are more likely to have acute or recurrent sinusitis that responds to initial treatment. ENT subjects are more likely to have chronic symptoms or are treatment failures.

CENTRAL GEORGIA DISTRIBUTION OF HIV INFECTION BY ZIP CODE. HL Durham, HP Katner, FC Dane, N Patel, Internal Medicine, Mercer University School of Medicine, Macon, GA

PURPOSE: To examine the geographic change of HIV cases in a 13 county mixed small urban/rural population in central Georgia over an 11 year period

METHODS: The Internal Medicine Department of the Mercer University School of Medicine in Macon, Georgia compiled a computerized 104 item per patient Microsoft Access database of the approximately 600 active patients in the Bibb County health district HIV clinic. In the data is the patient's residential zip code at initial clinic presentation. For the period 1989 to 1999 yearly counts and percentages of new patients for each zip code were assembled using the SAS statistical package release 6.12. The study has approval from the Mercer University IRB and the Bibb County Health Department.

RESULTS: The table shows the results for the three most frequently encountered zip codes (31201, 31204, 31206) during the study period. The only other zip code with notable percentages of new cases was 31030 which had 7% of the new patients in 1996 and 11% in 1998. Zip code 31030 is located thirty miles from the core zip codes. The median number of zip codes in a year was 11 with a minimum of 3 in 1989 and a maximum of 28 in 1997. Generally over the study period the number of new patients increased yearly and the number of zip codes also increased; but the three core zip codes always contributed at least 50% of the new cases.

CONCLUSION: The three core zip codes, located in the largest population concentration of the region (Macon, Georgia) and representing 45% of the thirteen county population of 250,500, contributed at least 50% of the new cases in any year. One possible confounding factor explaining this could be the location of the clinic in one of the core zip codes. Because of transportation or other potential issues, HIV patients in outlying zip codes may be presenting to clinics outside of the region. Alternatively before diagnosis patients may be relocating to the core zip codes to access improved health and social services.

Zip	1989	1990	1991	1992	1993
31201	1 (33%)	1 (13%)	5 (39%)	3 (12%)	7 (23%)
31204	1 (33%)	2 (25%)	3 (23%)	10 (39%)	7 (23%)
31206	0 (0%)	1 (13%)	2 (15%)	2 (8%)	4 (13%)
TOTALS	3 (66%)	4 (51%)	10 (77%)	15 (59%)	18 (59%)
# Zip Codes	3	7	5	9	10

Zip	1994	1995	1996	1997	1998	1999-June
31201	16 (27%)	20 (26%)	19 (19%)	18 (17%)	21 (18%)	5 (16%)
31204	8 (14%)	14 (19%)	19 (19%)	16 (16%)	20 (17%)	5 (16%)
31206	5 (9%)	13 (18%)	12 (12%)	17 (17%)	21 (18%)	7 (22%)
TOTALS	29 (50%)	47 (63%)	50 (50%)	51 (50%)	62 (53%)	17 (51%)
# Zip Codes	20	15	25	28	25	11

BARRIERS TO CARE IN PATIENTS ADMITTED WITH SEVERE, UNCONTROLLED HYPERTENSION. SM Dy, JH Young, K Roberts, M Lawrence, A Jacobsen, D Brotman, S Putman, MJ Klag, Division of General Internal Medicine, Johns Hopkins, Baltimore, MD

PURPOSE: Admissions for severe, uncontrolled hypertension (HTN) are common at our urban teaching hospital, often resulting in unnecessary morbidity. To evaluate how best to prevent these admissions, we assessed correctable risk factors for poor control in these patients, including lack of knowledge, access to care, and medication-related issues.

METHODS: We prospectively screened all African-American urban residents admitted to our internal medicine services between 8/99 and 1/00 for the presence of severe HTN. Patients were eligible if the average of 2 blood pressures in the emergency room was at least $180/110$. Patients were excluded if a secondary cause (e.g., pregnancy) or other medical condition (e.g., pulmonary edema) explained the blood pressure elevation. Physicians abstracted medical information from charts and patients completed a detailed interview.

RESULTS: Of 62 eligible patients, 57(92%) agreed to participate. The mean age was 51 years (range, 28 to 81 years), 45% were female, 44% had finished high school, and 52% were employed. Half were smokers, and 21% reported active cocaine use. The mean blood pressure was $209/130$, and the mean duration of HTN was 10 years. Half had been previously admitted for HTN within the last five years, and 80% rated their health as fair or poor.

Only 13% had not been on antihypertensives and only 10% had not seen a doctor within the past year. Two-thirds reported having a primary care physician; of these, 80% were able to give his or her name. All but one patient was able to correctly name at least one adverse consequence of uncontrolled HTN, and 80% could name two or three. Two-thirds of patients had medical insurance, but only half of these had full pharmacy coverage.

Forty-six percent had difficulty paying for medications, and 79% of these said that they sometimes missed medications because of financial problems. However, 65% of patients said that they took their medication every day when they had it. Seventy percent of patients were not actively taking blood pressure medication at the time of admission, and half of the rest were poorly controlled at baseline. Difficulty with cost was the most common reason why patients were not taking their medication. Eighty-six percent of patients with cost problems were not taking their medications on admission, as compared to 47% of those without cost problems.

CONCLUSION: Most of these severely affected hypertensive patients had a regular physician and knew about the consequences of HTN. Most were trying to take medications, but many had difficulty because of the expense. Interventions to reduce hospitalizations for severe HTN should include improving access to antihypertensive medications.

SELF-REPORTED PATTERNS OF ALTERNATIVE CARE USE IN PERSONS WITH DIABETES 1996. LE Egede, X Ye, MD Silverstein, General Internal Medicine; Center for Health Care Research, MUSC, Charleston, SC

PURPOSE: An estimated 16 million people in the United States (US) have diabetes mellitus. Diabetes cost the US \$98 billion in 1997. 42% of the US population was estimated to use alternative care in 1997 at a cost of \$21 billion, and alternative care was most commonly used for chronic medical conditions. We evaluated the prevalence and pattern of alternative care use in persons with diabetes in 1996.

METHODS: The Medical Expenditure Panel Survey (MEPS) is a survey of the United States civilian and non-institutionalized population, drawn from the National Health Interview Survey database, and it provides a nationally representative estimate of health care utilization, expenditure, and insurance coverage. We did a cross-sectional study using 883 MEPS diabetic respondents that reported using any form of alternative care treatment in 1996 to determine patient characteristics of alternative care users compared to non-users and patterns of use.

RESULTS: The prevalence of alternative care use in diabetics was 7.8% compared to a prevalence of 4.3% in the general MEPS sample in 1996. Among all diabetics, alternative care users differed from non-users. Users were more likely to be between the ages of 45-64 years (57% Vs 38%), have >12th grade education (54% Vs 24%), and have employment other than self-employment (52% Vs 32%). Users of alternative care were more likely to receive nutritional advice (41%), to receive alternative care from a nurse provider (14.5%), and were more likely to have discussed it with their primary care provider (55%).

CONCLUSION: In this sample of diabetic persons, the prevalence of alternative care use is lower than estimated in the literature (7.8% vs. 42%). Diabetic alternative care users were more likely to be older, have >12th grade education, and be employed when compared to diabetic non-alternative care users, and pattern of use involved nutritional therapy which is consistent with the disease manifestations.

MENTAL DISORDERS IN A NEUROLOGY CLINIC SETTING. J Ekstrand, P O'Malley, R Labutta, J Jackson, Dept of Medicine; Dept of Neurology, Walter Reed Army Medical Center, Washington, DC; Dept of Medicine, EDP, Uniformed Services University of the Health Sciences, Bethesda, MD

PURPOSE: The purpose of this study was to determine the prevalence of common psychiatric disorders among patients during their first visit to a neurology clinic, and the likelihood of establishing a neurologic diagnosis if a psychiatric disorder is present.

METHODS: We administered a prospective diagnostic survey to a consecutive sample ($n = 234$) of new referrals to a hospital-based neurology clinic. Before their visit, all patients completed the PRIME-MD. After the visit, the study neurologist, blinded to the PRIME-MD results, completed a questionnaire regarding their diagnostic assessment, which included the presence or absence of psychiatric disorders.

RESULTS: Psychiatric disorders were detected in 39% of patients. The most common diagnoses were somatoform disorder (24%) and depression (23%) (some patients had multiple disorders). A neurologic diagnosis was made in 85% of cases, 79% of which were felt to be primarily organic, 5% psychiatric and 16% both organic and psychiatric. The presence of any psychiatric disorder was associated with the lack of a neurologic diagnosis (OR 0.5, 95% CI 0.3–0.9). Subgroup analysis found specifically the diagnosis of somatoform disorder (OR 0.5, 95% CI 0.3–0.98) or anxiety disorder (OR 0.3, 95% CI 0.1–0.8) associated with the lack of a neurologic diagnosis.

CONCLUSION: Thirty nine percent of new patients seen in a general neurology clinic had a psychiatric disorder, higher than the rate found in primary care (20–29%). The presence of any psychiatric disorder, somatoform disorder, or anxiety disorder reduced the likelihood that an organic cause would be found for the patients' symptoms. Prospective studies are needed to determine if screening for psychiatric disease prior to referring patients with unexplained neurologic complaints would reduce costs or improve recognition of potentially treatable psychiatric disorders.

DIRECT OBSERVATION OF COUNSELING ON COLORECTAL CANCER IN PRIMARY CARE PRACTICE. EF Ellerbeck, KK Engelman, J Gladden, MC Mosler, GS Raju, JS Ahluwalia, University of Kansas Medical Center, Kansas City, KS

PURPOSE: The purpose of this study was to determine how often doctors, in actual clinical practice, discuss colorectal cancer with their patients.

METHODS: Trained medical students directly observed doctor-patient encounters in 38 primary care physician's practices. They recorded each time discussions occurred on colorectal cancer screening and identified who initiated the discussion. Telephone follow-up was performed on a subset of patients 1–2 days following the office visit. Medical students also conducted assessments of office systems specific to colorectal cancer in each physician practice.

RESULTS: In 2,480 visits with patients over 50 years of age, colorectal cancer was discussed in 344 visits (13.9%). The frequency of colorectal cancer discussions varied among the practices from 0 to 41.3% (with a median of 12.8%) of office visits. Colorectal cancer discussions were initiated by the physician in 86.9% of the cases when it was discussed. These discussions were more common on new patient visits (odds ratio [OR] = 2.8; 95% confidence interval [CI] 1.6–4.8; $P < .01$), and for patients <75 years of age (OR = 1.4; 95% CI 1.1–1.8; $P < .01$), but these patient-specific factors explained little of the variance in rates of colorectal cancer discussions. Flowsheets that included colorectal screening were used in 23.7% of offices while colorectal cancer patient education material was present in 60.5% of the offices. In visits where flowsheets were present, doctor-initiated discussion of colorectal cancer screening took place 16.1% of the time versus 10.4% of the time when flowsheets were not present (OR = 1.65; 95% CI 1.27–2.15; $P < .001$). Colorectal discussions occurred during 15.8% of visits in offices with patient education materials, compared to 10.9% in offices without educational materials (OR = 1.53; 95% CI 1.20–1.97; $P < .001$). In telephone follow-up with 125 patients over 50 years of age, 58 (46.4%) patients said that they had discussed either FOBT, sigmoidoscopy, or colonoscopy with their physician at some time in the past.

CONCLUSION: Although discussions of colorectal cancer occur commonly in clinical practice, the frequency of these discussions varies. Many patients over the age of 50 report that they have never discussed colorectal cancer screening with their doctor. Although flowsheets and other reminder systems have been shown to be effective, flowsheets were used in a minority of practices. Flowsheets and patient education materials, however, were associated with increased discussion of colorectal cancer screening.

TREATMENT OF ANXIETY DISORDERS IN GENERAL MEDICAL PRACTICES. JKL Esler, RT Dolan, MJ Fagan, RB Weisberg, L Culpeper, MB Keller, Psychiatry and Human Behavior; General Internal Medicine, Brown University School of Medicine, Providence, RI; Family Medicine, Boston University Medical Center, Boston, MA

PURPOSE: Research on the phenomenology of mental disorders has recently broadened beyond descriptions of symptoms, diagnoses, and global impairment. The purpose of the present study, the Primary Care Anxiety Project is to document the psychopathology, clinical course, psychosocial functioning, treatment received and economic burden of anxiety disorders in subjects presenting to a general medical setting for treatment of a non-psychiatric medical illness. This is a pro-

spective, naturalistic (i.e., observational), longitudinal follow-up study and subjects will be followed for a minimum of two to three years.

METHODS: We report on the intake data from 282 subjects who screened positive for anxiety and who met DSM-IV criteria for Panic Disorder, Generalized Anxiety Disorder, PTSD, Social Phobia, or Mixed Anxiety Depression Disorder on a structured clinical interview. The screening measure was an adaptation of DSM-IV criteria into self-report questions. This is the first prospective follow-up study of anxiety disorders in a general medical setting. This paper reports findings on the mental health and medical treatment utilization of PCAP subjects.

RESULTS: Forty percent of all subjects with anxiety disorders also had a comorbid major depressive episode (MDE). While the presence of MDE did not increase the likelihood of psychosocial treatment (e.g., psychotherapy), the presence of depression clearly increased the likelihood of receiving any psychotropic medication (Chisq = 24.6, $p < .001$). For example, 43% of anxiety-disordered subjects with MDE versus 14% of anxiety-disordered subjects without MDE received an SSRI (Chisq = 29.6, $p < .001$) and 23% of subjects with comorbid MDE received an anxiolytic compared to 14% without an MDE (Chisq = 3.21, $p < .07$). Multiple regressions indicated that, even while controlling for the number of medical illnesses and other predictors, the presence of Post-Traumatic Stress Disorder was strongly related to high numbers of general medical visits (OR:1.93, $p = .04$) and hospitalizations (OR:2.12, $p = .03$).

CONCLUSION: These data indicate that anxiety disorders present a major "cost" to the individual and society in the sphere of treatment utilization. Patients with anxiety disorders in primary care settings, particularly those without a comorbid MDE, receive psychotropic medication at relatively low rates. Primary care physicians may benefit from further education regarding the assessment and treatment of anxiety disorders.

PHYSICIANS' PARTICIPATION IN CAPITAL PUNISHMENT. NJ Farber, BM Aboff, EB Davis, J Weiner, EG Boyer, PA Ubel, Dept of Medicine, Christiana Care Health System, Wilmington, DE; Dept of Management & Information Systems, St. Joseph's University; Management Dept, Drexel University; Dept of Medicine, U of Pennsylvania, Philadelphia, PA

PURPOSE: In a previous study, we found that physicians generally condone colleague involvement in capital punishment. However, there are no data about physicians' own involvement in cases of lethal injection for the purpose of capital punishment. We therefore surveyed physicians about whether they would be involved in such cases, and what factors influenced their decisions about involvement.

METHODS: A survey instrument was developed and pretested which asked how willing physicians would be to perform 8 American Medical Association (AMA) proscribed actions and 2 allowed actions involving lethal injection, based on a 4 point Likert scale. Questions assessing reasons for participation in capital punishment, and attitudes toward the death penalty and assisted suicide were included. The impact of attitudinal and demographic variables on the number of proscribed actions respondents would be willing to perform were analyzed via Analysis of Variance (ANOVA) and multiple logistic regression analyses.

RESULTS: Of the 962 surveys which were received by subjects, 413 (43%) were completed and returned. Forty-one percent of respondents indicated they would be willing to perform at least one proscribed action, while 25% were willing to perform 5 or more actions, and 14% were willing to perform all 8 proscribed actions. Perceived duty to society ($p < 0.001$), and approval of the death penalty ($p < 0.001$) and of assisted suicide ($p = 0.015$) all correlated with an increased number of actions willing to be performed by respondents. Previous membership in the AMA also correlated with an increased willingness to be involved in lethal injection actions ($p = 0.031$). Only 3% of respondents knew of any guidelines on this issue.

CONCLUSION: It is disturbing that many physicians would be willing to be involved in the execution of adults against their will, despite clear opinions against such involvement in cases of capital punishment by the medical profession. However, these opinions have not been well disseminated among physicians; medical societies and ethicists must therefore better inform the profession about these issues. The issue of physician involvement in the process of capital punishment should also be addressed at a societal level, given the impact that societal norms have on physicians' decisions in these cases.

VARIATION IN PHYSICIANS' MANAGEMENT OF CELLULITIS IN A TEACHING HOSPITAL. DRE Farquhar, J Kim, Division of General Internal Medicine, Queen's University, Kingston, ON, Canada

PURPOSE: Cellulitis is a common soft tissue infection which frequently requires hospitalization. We wished to describe current practice in managing cellulitis at our institution, with particular reference to variations in care related to the specialty of the admitting physician.

METHODS: We conducted an audit of care provided to all patients admitted to our teaching hospital with a diagnosis of cellulitis from 1994–1997. Using a standardized data extraction form, information was obtained about patient demographics, clinical and laboratory features on admission, attending physician base specialty, investigations, consultations, choice of antibiotics, duration of therapy, surgical procedures, length of stay, and follow-up arrangements. Patients admitted to surgical (SURG) and medical (MED) services were compared in both univariate and multivariate analyses.

RESULTS: During the four-year period of observation, 103 patients were admitted with a diagnosis of cellulitis. Of these, 36 (35%) were admitted to SURG, and 67

(65%) to MED. Comparative features of these two populations of patients are highlighted in the table below.

Variable	SURG (N = 36)	MED (N = 67)	p-value
Age (yrs)	45.8	58.5	0.0009
% Febrile	8.6	26.9	0.005
% Treated with cloxacillin as first-line therapy	61.1	31.3	0.003
% Treated with cephalosporin as first-line therapy	55.6	77.6	0.02
% Seen in follow-up by attending physician	80.0	19.1	<0.0001
Median length of stay (days)	4.5	5.0	0.021

No differences between the two groups were seen in the presence of leukocytosis, the frequency of comorbidities (diabetes, peripheral vascular disease, immunosuppression), prior antibiotic use and consultation requests, the duration of antibiotic therapy, or readmission rate. In multivariate analysis, patient age and the presence of fever retained a highly statistically significant association with physician base specialty. Linear regression showed that duration of fever, but not physician base specialty, was independently associated with length of stay.

CONCLUSION: At our institution patients admitted to medical services tend to be older and are more likely to be febrile than patients admitted to surgical services. Surgeons tend to use cloxacillin more commonly as first-line therapy, whereas internists prefer cephalosporins. Length of stay is comparable on surgical and medical services, and is significantly associated with duration of fever. Surgeons are far more likely to see patients in follow-up than internists. Our data show considerable practice variation, and suggest that the development of more standardized approaches to management would be desirable.

NEW ORGANIZATIONAL STRATEGY TO IMPROVE VACCINATION COVERAGE AGAINST INFLUENZA IN ELDERLY PATIENTS. B Favrat, M Rickenbacher, A Pecoud, B Genton, Medical Outpatient Clinic, University of Lausanne, Switzerland

PURPOSE: Influenza immunization rate in elderly patients remains dramatically low, even with several educational interventions based on physicians. We hypothesized that physicians' omission to propose vaccination to the patients during the consultation was a major reason for low influenza vaccination coverage. Objective: To evaluate the impact of a new strategy for immunization consisting of the intervention of a medical student to propose the influenza vaccination systematically to all patients prior to the physician's visit.

METHODS: We conducted a study comparing immunization rates between 1998 and 1999 in patients aged 65 years or older. The intervention took place in an academic primary care center during the last trimester of 1999. Patients were routinely offered influenza vaccine by a medical student prior to the physician's visit, provided there were no contraindications. A questionnaire was filled in including vaccination status in the previous year, potential contraindications and reasons for refusal, if any. Standard procedures in 1998 and 1999 were training workshops for physicians, leaflets and reminder letters for patients.

RESULTS: In 1998, 229 eligible patients were seen. In this group, the vaccination rate was 39.3% (90/229). In 1999, 289 patients were eligible. Vaccination coverage reached 84.4% (244/289); 3.9% of patients wished to get vaccination elsewhere, 0.3% had medical contraindications and only 11.4% refused.

CONCLUSION: A strategy based on the intervention of a medical person prior to the physician visit led to a substantial increase in vaccination coverage. The number of refusals was low which suggests that physicians' omission was the major cause of low coverage. Intervention to increase vaccination coverage should not only focus on educational program but also on the organizational change.

WILLINGNESS TO RETURN OR REFER: INFLUENCE OF PHYSICIAN BEHAVIOR AND PROCESS OF CARE IN THE PRIMARY CARE SETTING. AD Federman, EF Cook, RS Phillips, AL Puopolo, JS Haas, TA Brennan, HR Burstin, Division of General Medicine, Brigham and Women's Hospital; Division of General Medicine, Beth Israel-Deaconess Medical Center, Boston, MA; Division of General Medicine, San Francisco General Hospital, San Francisco, CA

PURPOSE: In an era of managed care, patients are increasingly concerned about the physician-patient relationship and access to needed services. In this study, we assessed the impact of physician (MD) behavior and process of care on patients' intention to leave their usual site of care and unwillingness to recommend that site to family or friends.

METHODS: All patients aged 20-75 with at least one visit in the prior year to a staff physician at 11 ambulatory care sites were randomly selected for telephone survey and chart review. Chi-square and t-tests were used to examine associations between the outcomes and demographics, health status, reports and ratings of care, duration and site of care, number of visits and type of provider. Potentially modifiable aspects of MD behavior and process of health care delivery were analyzed with logistic regression, adjusting for other univariate predictors of unwillingness to return or recommend.

RESULTS: Of 2669 respondents (response rate 55%), 5.8% were not willing to return and 6.9% would not recommend their site of care. In multivariate analysis, patients unwilling to return were more likely to be dissatisfied with the amount of time spent with the MD (OR 2.0, 1.3-3.3), report that the MD did not listen to what they had to say (OR 12.5, 5.0-31.2), and to report not receiving all needed services in the past year (OR 3.1, 1.7-5.4). Similarly, patients unwilling to recom-

mend were more likely to be dissatisfied with the amount of time spent with the MD (OR 1.7, 1.0-2.8), to report that the MD did not listen (OR 6.4, 2.5-16.3), and that they did not receive needed services (OR 7.6, 4.5-12.7). Furthermore, these patients were more likely to be dissatisfied with the in-office waiting time (OR 4.3, 2.5-7.2), and to report that the MD did not address the reason for their visit (OR 4.0, 2.1-7.7). In a subgroup analysis of patients who had clinical tests, failure to explain their purpose also predicted unwillingness to recommend (OR 9.9, 1.9-52.4).

CONCLUSION: Our findings suggest that the extent to which MDs inform and involve patients may influence their intention to return or refer others to their usual source of care. Perception of the amount of time spent with the MD was predictive of both outcomes. As rationing of time and services under managed care increases, inattention to these potentially modifiable aspects of patient care could result in worse doctor-patient communication and increased attrition in primary care practices.

RANDOMIZED TRIAL OF NURSE PRACTITIONER-COMMUNITY HEALTH WORKER INTERVENTION: IMPACT ON YOUNG BLACK MEN'S SATISFACTION WITH HIGH BLOOD PRESSURE CARE. K Felix-Aaron, MN Hill, HR Rubin, Health Policy and Management, Johns Hopkins School of Hygiene and Public Health; Medicine, Johns Hopkins School of Medicine; Johns Hopkins School of Nursing, Baltimore, MD

PURPOSE: An intensive nurse practitioner-community health worker (NP-CHW) intervention improved blood pressure modestly among young urban black hypertensive men. We investigated the impact of the intervention on patient satisfaction with medical care for high blood pressure (HBP).

METHODS: 309 men were randomly assigned to receive a less intensive (LI) intervention (one-time HBP education and referral) or a more intensive (MI) intervention, including HBP education and HBP management by a NP-CHW team. At 24 month follow-up, participants received a satisfaction questionnaire containing rating of overall satisfaction (1, not at all, to 5, extremely satisfied) and 37 ratings of quality of 10 aspects of care (1, poor, to 5, excellent (E)).

RESULTS: 82% (N = 255) of the men completed the questionnaire. MI and LI groups were similar at baseline in age (mean MI 42, LI 41 years), education (mean MI 11, LI 12 years), and the proportions using illicit drugs (51% MI, 48% LI), reporting excellent health status (8% MI, 4% LI), and without health insurance (58% MI, 48% LI). Both groups rated items about provider technical quality (35-37% E), information (33-37% E) and interpersonal treatment (35-37% E) better than access issues (20-26% E). More than twice as many MI patients were extremely satisfied with the way their MD or NP took care of them (55% vs. 27%), and with "medical care for HBP" (55% vs. 26%). Domains with the largest differences between groups in E ratings were information (24-28% difference, e.g., 49% vs. 21% E rating of helpfulness of advice on HBP and how to keep healthy, difference 28%, 95% CI (17%, 39%)); technical quality (27-28% difference, e.g., 46% vs. 19% E rating of the MD or NP's ability to figure out what's wrong with you, difference 27%, 95% CI (17, 37); and interpersonal treatment (25-27% difference, e.g., helpfulness, friendliness, and caring of the MD or NP, 46% vs. 20% E rating, difference 26%, 95% CI (16, 36)). There was less marked but still substantial differences between the MI and LI patients in ratings of access issues (e.g., ease of getting medications, 30% vs. 14% E rating, difference 16%, 95% CI (6, 26). Groups were similar in a few items.

CONCLUSION: An intensive NP-CHW intervention that modestly lowered blood pressure also dramatically improved several dimensions of patient satisfaction with medical care for HBP in young black urban men, a group with low ratings of care at baseline and at high risk for treatment dropout. Future research should explore whether these better ratings predict adherence to medical treatment and lower risk of preventable cardiovascular disease.

THE INFLUENCE OF GENERALIST PHYSICIANS' PERSONAL CHARACTERISTICS, ATTITUDES, AND PRACTICE DEMOGRAPHICS ON PROPENSITY TO REFER TO SUBSPECIALISTS. M Filipek, MT Donohoe, RL Kravitz, T Shuell

PURPOSE: To determine the effects of generalist physicians' personal characteristics, attitudes, and practice demographics on referral propensity.

METHODS: A convenience sample of 400 practicing primary care physicians in Oregon was surveyed regarding their personal demographics and practice characteristics; career satisfaction; attitudes towards uncertainty [Gerrity 1983] and cost effectiveness [Goold 1994]; patients' requests for referrals on their Lickert-Scale ranked propensity to refer a series of 37 vignettes, from which a referral propensity score (RPS) was generated. The 37 previously pilot-tested vignettes ranged from the simple [well within the purview of generalist care, as defined by the knowledge and skills graduating residents are expected to acquire according to the ACP and the AAFP, eg. uncomplicated HTN, DM, and asthma] to the complex [clearly requiring subspecialty consultation for testing or treatment, eg. post-MI angina requiring cardiac catheterization and presumptive temporal arteritis requiring diagnostic biopsy].

RESULTS: 269 physicians responded [172 general internists (GI), 84 family physicians (FP), and 13 general practitioners (GP): 84 were residents, 185 were non-resident practicing clinicians]. On factor analysis [STATA], alpha = 0.85 for reliability of RPS. On univariate analysis, generalists practicing in both small and large single specialty practices were less likely to refer than those in multi-specialty group practices [p = 0.001]. There was no effect of generalist type, training status, or years in practice on RPS. Lower comfort with uncertainty was associated with higher RPS (p < 10⁻⁵). While more cost-conscious residents had lower

RPSs [$P = 0.004$], more cost-conscious practicing clinicians did not. Generalist with greater career satisfaction had higher RPSs [$P = 0.027$]. The percentage of capitated patients in a generalists' practice had no bearing on physicians' comfort with uncertainty, cost-consciousness, career satisfaction, or RPS.

CONCLUSION: GIs, FPs, and GPs do not differ in their RP. However, generalists with less comfort with uncertainty and higher career satisfaction are more likely to refer, and housestaff with greater cost-consciousness are less likely to refer. While the percentage of capitated patients in generalists' practices does not seem to affect RP, comfort with uncertainty and career satisfaction do.

RELATIONSHIP OF GENERALIST PHYSICIANS' PRACTICE DEMOGRAPHICS TO COST-CONSCIOUSNESS, COMFORT WITH UNCERTAINTY, WILLINGNESS TO ACCEDE TO PATIENT'S REQUESTS FOR REFERRAL, AND CAREER SATISFACTION. M Filipek, MT Donohoe, RL Kravitz, T Shuell.

PURPOSE: To determine the relationship of generalists physicians' practice demographics to cost-consciousness, comfort with uncertainty, and career satisfaction. **METHODS:** A convenience sample of 400 practicing generalist physicians in Oregon was surveyed regarding personal characteristics; practice demographics, including setting, patient population, and method of reimbursement; willingness to accede to patients' request for referrals; and comfort with uncertainty [Gerrity 1983], cost-consciousness [Goold, 1994], and career satisfaction.

RESULTS: 269 [67%] physicians responded [172 generalist internists, 84 family physicians, and 13 general practitioners; 84 were residents and 185 were non-resident practicing clinicians]. On univariate analysis there was no significant difference between physician specialty, level of training and practice size or setting, AND sub-scale scores or willingness to accede to patient request for referral, except that attending generalists working in HMO settings had lower career satisfaction scores than those working in non-HMO settings ($p = 0.04$). Generalists with greater levels of cost-consciousness were more willing to accede to patient request for referral ($p = 0.01$). In response to a request for written comments, 62 of 269 respondents [23%] complained of malaise and burnout, excessive patient loads, inadequate visit length, poor compensation from insurance companies, and loss of autonomy.

CONCLUSION: Other than the lower career satisfaction of non-resident generalists in HMO settings and the increased willingness of cost-conscious generalists to accede to patient request for referral, there were no significant differences between generalists' field and practice demographics AND cost-consciousness, comfort with uncertainty, career satisfaction, and willingness to accede to patients' request for referrals. Even so, written comments by almost $\frac{1}{4}$ of respondents revealed an undercurrent of dissatisfaction with the practice of primary care today. Further study of generalist physicians' demographics and attitudes, complemented by analysis of practice patterns, is warranted to determine the effects of the contemporary health care system and of medical training on the provision of primary care in the U.S.

RANDOMIZED CLINICAL TRIAL OF A PRACTICE GUIDELINE TO REDUCE LENGTH OF STAY FOR PATIENTS WITH PNEUMONIA. MJ Fine, RA Stone, JR Love, DS Obrosky, LJ Hough, WN Kapoor.

PURPOSE: To assess the effectiveness and safety of a practice guideline (PG) in reducing the duration of intravenous (IV) antibiotic therapy and LOS for patients (pts) hospitalized with community-acquired pneumonia (CAP).

METHODS: A randomized trial of PG implementation was conducted at 1 university, 3 community teaching, and 3 community non-teaching hospitals in western PA. The PG recommended conversion from IV to oral antibiotics and/or hospital discharge (DC) when pts met specified clinical criteria. Randomization to the intervention arm (I-A) or control/usual care arm (C-A) was at the physician (MD) group level for internists, family MDs, and pulmonary or infectious disease MDs in each hospital (709 MDs in 173 groups). Pts admitted by these MDs were identified prospectively using standard eligibility criteria for CAP. When pts met all criteria for IV antibiotic conversion and/or DC, only physicians in the I-A received the PG intervention, including (1) placing a detail sheet with PG recommendations in the medical record, and (2) nurse-mediated physician reminders and concurrent feedback. Primary outcomes were LOS and duration of IV antibiotic therapy. Secondary outcomes were 30-day mortality, readmission, and return to usual activities. Multivariate analyses of outcomes were performed using a clustered discrete proportional hazards model, adjusting for hospital site and CAP severity risk classes I-V at baseline.

RESULTS: Overall, 608 eligible pts were enrolled (median age 71, 46% male, 83% white) from 116 physician groups. No baseline differences existed for I-A and C-A pts. Hazard ratios (HR) and 95% confidence intervals (CI) for the primary outcomes in I-A and C-A are tabled below (larger HR = shorter LOS or duration IV).

Outcome	HR (95% CI)	I-A Mean Days $\pm$ SD	C-A Mean Days $\pm$ SD
LOS	1.12 (0.94-1.34)	5.89 (± 4.21)	6.04 (± 4.08)
IV Duration	1.12 (0.91-1.38)	5.18 (± 3.02)	5.25 (± 2.45)

There was a significant interaction ($p = 0.01$) between hospital site and intervention effectiveness for LOS, with HR by site ranging from 0.69 (CI = 0.40, 1.17) to 2.24 (CI = 1.54, 3.25). HRs were larger in subgroup analyses that excluded 61 (10%) pts admitted from nursing homes and 96 (16%) pts in risk class V (HR for LOS = 1.18, CI = 0.97, 1.42; HR for IV antibiotic duration = 1.22, CI = 0.97,

1.52). No significant differences in any secondary outcomes existed among I-A and C-A pts.

CONCLUSION: Dissemination of a PG did not compromise pt outcomes and resulted in small yet statistically significant site-specific reductions in LOS. The larger reductions in duration of IV antibiotic therapy and LOS in less severely ill non-nursing home pts suggest that future PG implementation should focus on this more homogeneous low-risk subgroup of pts with CAP.

RANDOMIZED TRIAL OF A PRACTICE GUIDELINE TO REDUCE LENGTH OF STAY FOR PATIENTS WITH PNEUMONIA. MJ Fine, RA Stone, JR Love, DS Obrosky, LJ Hough, WN Kapoor.

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GUIDELINES TO ASSIST PATIENTS IN THE RECOGNITION OF MYOCARDIAL INFARCTION MAY DISCOURAGE THEM FROM SEEKING CARE. HSF Fraser, L Ohno-Machado, N Shapur, WJ Long, J Griffiths, HP Selker, Endocrinology, Childrens Hospital; Decision Systems Group, Brigham and Womens Hospital; Department of Medicine, New England Medical Center, Boston; Lab for Computer Science, MIT, Cambridge, MA

PURPOSE: The majority of deaths from acute myocardial infarction (AMI) occur out of hospital, but many could be prevented by prompt treatment. Patients delay a median of three hours before seeking treatment. We studied the performance of guidelines designed to help patients recognize the symptoms of AMI and encourage them to seek treatment appropriately.

METHODS: Data on symptoms and cardiac risk factors (but not ECG) of 5773 patients, presenting to the emergency department with possible AMI, came from the 6 hospital Acute Cardiac Ischemia Predictive Instrument trial [NEJM 1984; 310: 1273-8]. Patients met the Imminent Myocardial Infarction Rotterdam criteria, shown to capture more than 90% of patients who have AMI in the community. Patient guidelines came from leaflets, books or the Web (AHA, AMA and others). Symptoms were combined using AND/OR to form logical rules that were tested on the clinical database to measure Sensitivity (SN) and specificity (SP). Logistic regression was also employed to find the ideal prediction variables for AMI in this database.

RESULTS: Most patient guidelines included chest or arm pain plus some combination of nausea/vomiting, dyspnea, dizziness/syncope and sweating. One guideline implied all symptoms must be present, giving a low SN of 40.6% and SP 66.0%. Other guidelines suggested any one symptom was enough, giving SN 99.5% but SP 0.4%. A more effective guideline (AHA, <http://www.amheart.org>) required: (chest OR arm pain) AND (nausea OR vomiting OR dyspnea OR dizziness OR syncope OR sweating), giving SN of 64.2% and SP 37.8% (sweating was not in the database).

Logistic regression indicated the most predictive features (Odds ratios, 95% CI) were: age >55 (1.85, 1.60-2.17), arm pain (1.50, 1.39-1.61), chest pain (1.35, 1.20-1.50), severe pain (1.30, 1.19-1.41), & male sex (1.23, 1.14-1.32). Negative predictors were

stabbing pain (0.63, 0.53–0.73), palpitations (0.68, 0.62–0.74), pain over left nipple (0.76, 0.66–0.86), and dizziness (0.80, 0.72–0.88), all $P < 0.0001$.

CONCLUSION: Many guidelines tested had low sensitivity due to requirements for classic combinations of symptoms, or ambiguous wording. The absence of a clear broad description of symptoms may discourage patients from seeking assistance. Multi-variate models such as logistic regression, can reveal important predictors not previously considered and simplify the use of negative evidence. Future guidelines should be developed and evaluated with patient data, preferably collected in a community setting to reflect the population at risk.

PAST MEDICATION COMPLIANCE AND ADHERENCE TO CURRENT TUBERCULOSIS TREATMENT BY HISPANIC TEENAGERS. L Friedman, J Berg, C Sipan, M Hovell, A Catanzaro, K Moser, N Kelley, Departments of Pediatrics and Medicine, University of California; Center for Behavioral Epidemiology, San Diego State University; County Department of Health Services, Tuberculosis Control Unit, San Diego, CA

OBJECTIVE: Literature shows that many patients taking long term medication regimens have difficulty with compliance. One of the important predictors for medication adherence is past adherence behavior.

DESIGN: As part of a larger study to increase medication adherence in Latino adolescents with latent TB (tuberculosis), 130 PPD+ teens were surveyed concerning past medication-taking behavior.

RESULTS: Baseline data indicate that 54% reported that they did not finish taking some type of medicine in the past. Ninety-three percent used compliance aids to help them take their medicine. Popular aids included: calendars ($n = 34$); taking medication at a particular time ($n = 71$); having someone remind them ($n = 71$); putting their medication where they can see it ($n = 84$); and putting their medication near something they use daily ($n = 71$). Barriers to missing doses of medication were also identified by 75%. Barriers related to time issues, such as oversleeping and coming home late as well as running out of medication. Preliminary findings from monthly self-reports on INH adherence indicate that 75% of subjects are not adherent with medication-taking.

CONCLUSION: These findings underscore the need to develop adherence behaviors and to decrease barriers which affect INH therapy.

MANAGED CARE AND ACCESS TO OUTPATIENT DRUG ABUSE TREATMENT ON DEMAND. PD Friedmann, S Lemon, MD Stein, TA D'Aunno, Brown University & University of Chicago

PURPOSE: To examine the relationship between outpatient drug abuse treatment (ODAT) program's managed care (MC) involvement and potential patients' access to 'treatment on demand' (TOD).

METHODS: As part of a panel study, we surveyed a nationally representative sample of 618 ODAT programs in 1995. The dependent variable, TOD, was defined as a director's report of an average wait of zero days for treatment. They reported MC involvement as the percent of HMO-/PPO-insured patients, of patients requiring prior authorization, and of patients whose ongoing treatment is subject to concurrent review; these MC variables were categorized *a priori* into none (referent), 1 to 25%, and >25%. Logistic regression models evaluated the association between MC involvement and TOD.

RESULTS: Of 589 programs with wait time data, 66% reported TOD. Controlling for profit status, concurrent review for >25% of patients (OR, 0.43; 95% CI, 0.20–0.92), and prior authorization for 1 to 25% (OR, 2.0; 95% CI, 1.0–4.0) or >25% (OR, 3.2; 95% CI, 1.3–7.7) of patients were related to TOD. HMO-/PPO-insurance of 1–25% of patients may also influence the availability of TOD (OR, 0.59; 95% CI, 0.34–1.0). Private for-profit programs (87%) were also more likely to report no wait than not-for-profit (63%) or public (60%) programs ($P = 0.001$). Ongoing analyses will further evaluate the effect of managed care involvement on access from 1988 to 1995.

CONCLUSION: Preliminary results suggest a stepwise relationship between prior authorization and greater access to treatment on demand. Conversely, concurrent review and patients' HMO/PPO involvement were associated with decreased access to immediate treatment. Private, for-profit programs also appear to have more readily available treatment capacity than public or not-for-profit programs.

LINKAGE TO MEDICAL SERVICES IN THE DRUG ABUSE TREATMENT OUTCOMES STUDY. PD Friedmann, MD Stein, S Lemon, TA D'Aunno, Brown University and University of Chicago

PURPOSE: To examine whether on-site service delivery, referral mechanisms, case management and transportation promote patients' medical service utilization in the context of drug abuse treatment.

METHODS: We analyzed the Drug Abuse Treatment Outcomes Study (DATOS), a 12-month longitudinal study of 6148 patients enrolled in a purposive national sample of 71 drug abuse treatment programs from 1991–1993. Director's surveys provided information about medical services delivery. Analyses evaluated the effect of on-site location of services within the drug treatment program, external referral (formal and informal), the program's emphasis on case management, and the availability of transportation on the log-transformed number of medical visits reported by patients at a three-month in-treatment personal interview. Multivariate, three-level hierarchical linear models addressed patient clustering within programs, and programs within the four treatment modalities, methadone maintenance (MM), long-term residential (LTR), short-term inpatient (STI), and outpatient drug-free (ODF), as well as patient-level case-mix variables from the intake interview.

RESULTS: On-site delivery increased medical service utilization in programs with on-site care only (parameter estimate [β], .154; standard error [SE], .057), but lessened utilization in programs with a mix of on-site and formal (β , -.123; SE, .043) or informal (β , -.230; SE, .053) referral, compared with programs with informal referral only. Transportation services increased utilization (β , .070; SE, .027) but a greater emphasis on case management decreased utilization (β , -.278; SE, .043). Other variables that increased medical utilization included program-level factors (treatment modality and the extent to which referrals are monitored), and baseline patient-level factors (worse self-reported health, more medical comorbidity, greater treatment motivation, greater medical utilization in the 12 months prior to addiction treatment, and black race).

CONCLUSION: Patients in addiction treatment programs that focus on centralized delivery mechanisms (on-site delivery only) most reliably utilize medical services, while those in programs with a mix of on-site and referral receive medical services least reliably. In the context of addiction treatment, transportation services are a more important facilitator of medical service delivery than is case management. These findings suggest that initiatives, grants and organizational policies to promote linkage of addiction treatment clients to medical services should emphasize focused, integrated delivery mechanisms, and transportation.

SCREENING & INTERVENTION FOR ILLICIT DRUGS BY PRIMARY CARE PHYSICIANS AND PSYCHIATRISTS. PD Friedmann, D McCullough, R Saitz, Brown University, University of Chicago, & Boston University

PURPOSE: To describe adult primary care physicians' and psychiatrists' approach to drug abuse screening and intervention, and to identify correlates of more optimal practices.

METHODS: We performed a mail survey of a national sample of 500 general internists, 500 family physicians, 500 obstetrician-gynecologists (OBG), and 500 psychiatrists (PSY) from the AMA Masterfile. Likert-type scales assessed reported frequency of asking new outpatients about drug use, treatment recommendations in cases of diagnosed illicit drug abuse, and barriers to these practices.

RESULTS: Among the 853 respondent physicians (adjusted response rate, 57%), 68% usually or always ask new outpatients about illicit drug use. For patients with diagnosed illicit drug abuse, 55% usually or always offer referral to a formal treatment program, but 15% do not intervene at all. In multivariate logistic models, correlates of usually/always asking about drug use included OBG (adj. OR, 2.5; 95% CI, 1.5–4.2), and PSY (OR, 2.9; CI, 1.3–6.2) specialty, confidence in drug history-taking (OR, 2.1; CI, 1.7–2.6), and optimism about the effect of treatment (OR, 2.5; CI, 1.1–5.7). Barriers to screening included greater concern that patients do not want to be asked about substance use (OR, .72; CI, .59–.86), and perceived time constraints (OR, .83; CI, .70–.99). Psychiatrists (OR, 2.8; CI, 1.0–7.6) were most likely, and OBG least likely (OR, .50; CI, .29–.84) to intervene.

CONCLUSION: Most primary care physicians and psychiatrists ask about illicit drug use. In cases of diagnosed drug abuse, about half of physicians routinely offer referral to formal treatment, but many fail to intervene. Initiatives to promote physician involvement with drug abuse should improve self-confidence in this area, dispel pessimism about treatment, disseminate evidence that patients do not mind being asked about drug use, and address perceived time constraints.

ATTITUDES TOWARDS OUTPATIENT DEEP VEIN THROMBOSIS TREATMENT: A SURVEY OF GENERAL INTERNISTS AND FAMILY PHYSICIANS. RM Gaddis, SR Deitcher, R Uppal, AK Jaffer, Internal Medicine Residency Program; Head, Section of Clinical Thrombosis, Department of Vascular Medicine, The Cleveland Clinic Foundation, Cleveland; Medical Student, The Ohio State University College of Medicine, Columbus; Staff Physician, Department of General Internal Medicine, The Cleveland Clinic Foundation, Cleveland, OH

PURPOSE: Low molecular weight heparins (LMWH) have been shown to be safe and effective for the outpatient treatment of acute deep venous thrombosis (DVT). We conducted a survey to study the attitudes of primary care physicians towards the use of LMWH for outpatient DVT management and to identify common and potentially correctable barriers to the implementation of this form of treatment.

METHODS: Cleveland Clinic Health System primary care physicians ($n = 344$), including general internists ($n = 229$) and family physicians ($n = 115$), were sent confidential surveys between September and November 1999. The survey consisted of 18 questions. Physician training background, practice type, attitudes towards LMWH and barriers to the outpatient treatment of DVT with LMWH were assessed. Barriers were assessed using a 5-point Likert scale (1 = strongly disagree; 5 = strongly agree). Results were tabulated and intergroup comparisons performed using appropriate statistical tests.

RESULTS: A total of 151 (44%) physicians responded to the survey. Thirty-eight physicians (26%) who responded had used LMWH for the outpatient treatment of acute DVT. Physicians who had used LMWH in this setting felt significantly more knowledgeable about and comfortable with LMWH ($p < 0.001$) than physicians who had not used LMWH in the treatment of DVT. There was a trend towards more LMWH use by general internists ($p = 0.06$) than family physicians. General internists also felt more knowledgeable about LMWH and its use compared to family physicians ($p = 0.014$). Each proposed barrier to outpatient LMWH use for DVT treatment received a mean score greater than 4 (4 = agree). Barriers to treatment included an ability to get insurance approval to cover the cost of LMWH, a lack of ready access to duplex ultrasound, a lack of a nurse for patient education, a perception of patient non-compliance and availability of LMWH at a local pharmacy.

CONCLUSION: According to our study, LMWH has not become widely used by pri-

many care physicians for the outpatient treatment of DVT despite its potential for cost savings and reduced hospital length of stay. Further educational efforts need to be undertaken to expand the use of LMWH in the outpatient setting because of its many benefits over unfractionated heparin.

GROUP PRACTICE STRATEGIES TO MANAGE HMO NETWORK PHARMACEUTICAL COSTS. KA Galt, EC Rich, PD Turner, JE Krolewski, R Feldman, B Dowd, TS Bernhardt, Center for Practice Improvement and Outcomes Research; Department of Medicine, Creighton University, Omaha, NE; University of Minnesota; Blue Cross Blue Shield of Minnesota, Minneapolis, MN

PURPOSE: The purpose of this study was to evaluate the relationship of pharmaceutical costs to various administrative features of group practice relevant to pharmaceutical care (e.g., pharmaceutical detailing policies, care management techniques, office medication records).

METHODS: Sources of data were a survey of primary care group practice organizations (103 respondents) affiliated with Blue Cross Blue Shield of Minnesota (BCBSM) and corresponding BCBSM claims data from members continuously enrolled in the responding practices from 1/1-12/31/95 (n = 76,389). Bivariate analysis revealed that larger group practices were significantly more likely to have primary care gatekeepers (p < .001) and outpatient case managers (p < .001). Multivariate linear regression analysis was performed to evaluate the independent effects of these group practice strategies on the natural log of patient pharmacy expenditures PMPY. This analysis controlled for patient and group practice characteristics.

RESULTS: The following strategies were independently associated with lower pharmaceutical expenditures PMPY: employment of outpatient nurse case managers (p < 0.008), primary care physician gatekeeping (p < 0.001), policies to control pharmaceutical detailing (p < 0.000), and medication lists and outpatient charts (p < 0.000). Log linear conversion of drug charges indicates 8.3% lower costs related to office and medication records, 5.6% lower for pharmaceutical detailing, 5.2% lower for pharmaceutical detailing policies, 5.2% lower for outpatient nurse case managers, and 3.8% lower for primary care physician gatekeeping. These data reveal substantial variations in prevalence of administrative techniques to managed pharmaceutical costs within primary care group practices in an HMO network.

CONCLUSION: Groups which use these strategies have significantly lower pharmaceutical costs PMPY. Over half of the group practices surveyed had some policy in their clinic regarding physician interaction with pharmaceutical sales representatives (PSRs). Most policies related to managing the PSR visit times or limiting visits to groups rather than to individual physicians. Very few clinics reported more stringent policies, such as, not accepting sample medications or prohibition of gifts. Nonetheless, the presence of these modest policies was associated with lower prescription drug costs.

MEDICATION ERRORS AND POTENTIAL ADVERSE DRUG EVENTS AMONG OUTPATIENTS. JK Gandhi, SN Weingart, LL Leape, DL Seger, JM Rothschild, J Borus, J Fleming, DW Bates, Division of General Medicine, Brigham and Women's Hospital, and CareGroup Healthcare Systems, Boston, MA

PURPOSE: Medication errors are common and preventable in the hospital setting and can lead to adverse drug events (ADEs). However, comparatively little is known about these events in the outpatient setting. We sought to examine the frequency and type of medication errors and potential ADEs (medication errors with potential for patient harm) that occur among outpatients.

We prospectively collected copies of prescriptions given to patients by 13 primary care providers at two general medicine practice sites in Boston, each over 4 weeks. Site 1 was an academic hospital practice with electronic prescription writing (but no error checking). Site 2 was a community-based practice with handwritten prescriptions. Prescription copies were screened by a pharmacist for errors.

Among 384 prescriptions, there were 68 medication errors (rate, 18%), and of these, 14 were potential ADEs. The most common errors were dose (60%) and frequency (16%) errors. The rate of medication errors at Site 1 (electronic) was significantly lower than at Site 2 (handwritten) (6% vs 34%, p < .0001).

At Site 2, 5/54 (94%) of medication errors could have been prevented by electronic prescribing linked to the medical record. However, the frequency of potential ADEs at the 2 sites was similar. In addition, there were significantly more medication errors and potential ADEs related to new prescriptions than to refills.

We conclude that medication errors are common in the outpatient setting and that new prescriptions more often contain medication errors and potential ADEs. Outpatient computer systems that include electronic prescribing may be an effective way to reduce the number of medication errors, although error checking may be necessary to reduce serious error rates.

	Electronic Site 1	Handwritten Site 2	Total
No. of prescriptions screened	224	160	384
Medication errors*	14 (6%)	54 (34%)	68 (18%)
- Potential ADEs	6 (3%)	8 (5%)	14 (4%)
New Prescription (n = 180)		Refill (n = 204)	
Medication errors*	46 (26%)	22 (11%)	
- Potential ADEs**	12 (7%)	2 (1%)	

*p < .0001 **p < .005

LINKING DOCTOR-PATIENT RELATIONSHIP QUALITY TO OUTCOMES. D Gelb Safran, A Murray, H Chang, J Montgomery, J Murphy, WH Rogers, The Health Institute, Division of Clinical Care Research, New England Medical Center, Boston, MA

PURPOSE: To evaluate the role of doctor-patient relationship quality in determining patients' adherence to medical advice and voluntary disenrollment from a doctors' practice.

METHODS: Data derive from a longitudinal observational study of insured adults (1996-1999). Study participants completed a self-administered questionnaire at baseline (68.5% response rate) and follow-up (69.4% response rate). The baseline questionnaire measured 4 structural features of care (access, continuity, integration, duration of the primary care relationship) and 4 interpersonal features of care (communication quality, interpersonal treatment, doctor's knowledge of the patient, and patient trust). Multivariable logistic regression was used to examine each scale's role in predicting adherence to the primary physician's advice, and voluntary disenrollment from the physician's practice. There were 2 indicators of adherence—one measuring attempts to change any of 5 behavioral risk factors (smoking, alcohol use, obesity, sedentary life-style, nonuse of seatbelts) based on the physician's advice ("process" adherence), and the other measuring actual change in the risk factor ("outcome" adherence). Voluntary disenrollment was determined using patient-provided information linked to a statewide physician registry. Regressions controlled for patients' sociodemographic profile, health status, and ambulatory utilization.

RESULTS: Trust was the leading predictor of voluntary disenrollment from a primary care physician's practice (OR = 1.58, p < .001), but was followed closely by the other 3 indicators of relationship quality (OR = 1.49 to 1.53, p < .001). Patients with baseline trust scores in the 5th percentile were 3.3-times more likely to disenroll than those with 95th percentile trust scores. "Process" adherence (attempted behavior change) was best predicted by 2 measures of relationship quality: doctor's knowledge of patient, and patient trust (p < .001). "Outcome" adherence (successful behavior modification) was predicted only by patient trust (p = .05).

CONCLUSION: The quality of doctor-patient relationships, particularly patients' trust in their primary physician, is strongly predictive of adherence to medical advice, and of loyalty to a physician's practice. In an era marked by increasing pressure on physicians to attend to such factors as market share, productivity, and efficiency, these findings remind us that the very essence of medical care delivery remains the interaction of one human being with another, and suggest that medical practices cannot afford to ignore this in the race to the bottom line.

MORTALITY AFTER NON-CARDIAC SURGERY: PREDICTION FROM ADMINISTRATIVE VERSUS CLINICAL DATA. JM Geraci, ML Johnson, HS Gordon, NJ Petersen, J Daley, K Hur, WG Henderson, SF Khuri, NP Wray, Houston VA Medical Center and Baylor College of Medicine, Houston, TX; Institute for Health Policy, MGH/Partners HealthCare System, Boston, MA; VA Cooperative Studies Program, Hines VA Medical Center, Hines, IL; Surgical Service, West Roxbury VA Medical Center, West Roxbury, MA

PURPOSE: To determine whether the administrative data in the VA Patient Treatment File (PTF) can adequately adjust for patient severity and identify hospital mortality outliers, when compared to predictive models developed using clinical data from a primary data collection, the National VA Surgical Quality Improvement Program (NSQIP).

METHODS: We developed logistic regression (LR) models to predict operative mortality following non-cardiac surgery for 5 VA patient groups: 17,768 general surgery (GEN), 17,448 orthopedic surgery (ORTHO), 4,843 thoracic surgery (THOR) and 9,811 vascular surgery (VASC) and all noncardiac surgery (ALL) patients. These patients underwent surgery in 44 VA hospitals during fiscal years 1991-1995. PTF LR models included demographic, principal diagnosis and comorbidity information (ICD-9-CM codes handled per the Modified Wray-Petersen risk adjustment approach). Clinical LR models used information from the 60 variables collected according to the NSQIP protocol. Model performance was assessed by the c-index, or area under the ROC curve. Outlier VA hospitals for mortality were calculated using the ratio of observed-to-expected 30-day post-operative mortality, and 90% confidence intervals.

RESULTS: PTF models had slightly less predictive power than NSQIP models (PTF model c-indices 0.69-0.87; NSQIP model c-indices 0.77-0.93). The PTF and NSQIP ORTHO models identified the same, isolated high mortality outlier; PTF GEN identified 4 of 4, PTF THOR 1 of 2, PTF VASC 3 of 4 and PTF ALL 5 of 5 NSQIP high mortality outliers. These PTF models also identified 1 or more additional hospitals as high mortality outliers. PTF models identified fewer of the low outliers found by the NSQIP models: PTF GEN 2 of 5, PTF THOR 0 of 2, PTF VASC 1 of 1 and PTF ALL 6 of 7 low mortality outliers identified by NSQIP. There were no low outlier hospitals for ORTHO by either PTF or NSQIP data.

CONCLUSION: For most surgical populations, LR models developed from administrative data identified most, though not all, the same operative mortality outliers as those identified by LR models constructed using clinical data. LR models developed from administrative data might be sufficient to select hospital surgical services for subsequent review of their patient care practices. Further studies are needed to determine the validity of both administrative and clinical data used for outlier assessment.

PREScription OF OPIATE MEDICATIONS IN OUTPATIENT CARE. AL Gifford, C Quach, Medicine, University of California; Health Services Research, VA San Diego Healthcare, San Diego, CA

PURPOSE: Pain control in primary care is a high clinical priority, and opiate medications are an important and controversial part of the analgesic pharmacopeia. Yet little is known about how often opiates are prescribed, physician opiate management workload compared with workload for other drug classes, or what proportion of opiate prescribing occurs in primary care settings.

METHODS: Prescription data for fiscal 1999 were obtained for all opiates (codeine or hydrocodone±acetaminophen [DEA Schedule III]; oxycodone+acetaminophen, hydromorphone, methadone, morphine, fentanyl [Schedule II]) dispensed from all outpatient pharmacies of an urban, academically-affiliated VA medical center, with active outpatient specialty practices and a large multi-site primary care continuity practice. Comparison data on all antilipid drugs were obtained. Drugs were quantified as prescriptions filled, pills, and total drug-days (pills dispensed divided by maximum pills/day).

RESULTS: In fiscal 1999, opiates were prescribed by 545 MDs, of whom 28 were Primary Care MDs (PCMDs); board-certified general internists on staff, responsible for longitudinal patient panels. Opiates were 3.7% of all outpatient prescriptions. Quantity of outpatient opiate use is shown in the table, compared with antilipid drug data. Of all opiate users, 1003 patients (21%) were heavy users (≥ 60 drug-days/year), using 230,461 (83%) drug-days. The 28 PCMDs prescribed nearly half (44%) of all opiate drug-days in the medical center, and wrote or refilled a median of 1.5 opiate prescriptions per day of primary care (interquartile range [IQR], 1.2–3.3), compared with 1.9 antilipid drugs prescribed or refilled per day of primary care (IQR, 1.3–2.5). PCMDs were 2.9 times more likely than other providers to provide care for the heavy opiate users (95% CI 2.5–3.4). Schedule II drugs were 50% of opiate drug-days prescribed, and were used in similar proportions by PCMDs (46%) and non-PCMDs (53%).

	Opiate Drugs	Antilipid Drugs
Prescriptions	16,602	13,845
Total Pills	1,408,431	1,500,506
Drug-days	278,806	1,095,995
Patients Treated	4883	4007

CONCLUSION: In spite of regulatory hurdles to writing and dispensing, and concerns about the appropriateness of opiates, use of these drugs is common. Managing patients on opiates may be a substantial part of outpatient care. If these results generalize to other care settings, they suggest the need to recognize the importance of chronic pain and opiate management, and develop programs accordingly to optimize and rationalize pain care.

DETERMINANTS OF USE OF BREAST-CONSERVING SURGERY: CHANGE OVER TIME. M Gilligan, RG Hoffmann, RT Kneusel, AB Nattinger, Medicine, Medical College of Wisconsin; Medicine, Zablocki VAMC; Biostatistics, Medical College of Wisconsin, Milwaukee, WI

PURPOSE: Previous studies have shown that age, socioeconomic status (SES), and urban versus rural residence are determinants of use of breast conserving surgery (BCS). We evaluated whether differences in use by these factors have diminished with increased adoption of BCS use over time.

METHODS: The Surveillance, Epidemiology, and End Results (SEER) database was used to select a cohort of 158,496 women ≥ 30 years diagnosed with local or regional stage breast cancer from 1983 to 1996 who received surgical treatment. Income and education were estimated using census data. Logistic regression models were developed to predict BCS use by age, education, income and population density, controlling for other factors. Models for each of the factors were tested for changes over time in ordering of groups, spread between highest and lowest use groups, and changes in slope. Differences in ordering were determined by group-wise comparisons of the average standard error for each group at four points in time. Change over time in spread was assessed by year-wise comparisons of the difference of the adjusted means of the highest and lowest use groups.

RESULTS: The use of BCS increased overall by 344% from 1983 to 1996, and increased among all groups examined. Women ≥ 80 yrs. had highest BCS use compared to other age groups in 1983 ($p < 0.001$), yet had the lowest use by 1996 ($p < 0.001$). Although BCS use increased in all age groups, the increase in use over time was significantly lower for the oldest age group. For local stage disease, the spread between the highest and lowest use age groups decreased over time ($p = 0.01$). For regional stage disease, the spread increased ($p = 0.01$). During the period of study, there was a persistently higher use of BCS among higher income groups. The increase in use of BCS was slowest among women in the poorest quartile leading to a widening of the spread over time between the highest and lowest income groups ($p < 0.00001$). Results by educational status were similar to the results by income status. Women residing in more urban areas had persistently higher BCS use over time. There was a significant increase in the spread between the most urban and least urban groups from 1983 to 1996 ($p = 0.00007$).

CONCLUSION: Despite an overall increase in the use of BCS over time, there remain significant differences in the use of BCS among the groups examined. Factors associated with lower use of BCS continue to include lower income, lower education and more rural residence. Older women, who previously had the greatest use, now have the lowest use. This analysis suggests that differences in BCS use by age, SES, and population density will persist, or even increase, for the immediate future.

HYPERTENSION: LEVEL OF BLOOD PRESSURE OR RISK OF THE PATIENT? NE Gimpel, V Schaj, D Terceiro, M Boccardo, V Finkelsztain, A Rubinstein, Division of Family and Preventive Medicine, Hospital Italiano, Buenos Aires, Argentina

PURPOSE: The risk of cardiovascular disease in patients with hypertension is determined by the level of blood pressure and also by the presence or absence of target organ damage (TOD) or other risk factors (RF). The JNC VI provides a practical classification by level of blood pressure (stage 1, 2, 3) and presence of TOD or RF (risk group: A, B, C). This report makes emphasis on absolute risk and benefit and uses risk stratification as a guidance to the treatment strategy.

We aimed to evaluate whether the processes and outcomes of care were associated to the baseline risk group in addition to the initial level of blood pressure.

METHODS: We examined the medical records of 922 randomly selected patients with hypertension who received their regular care in our primary care clinic with at least one year of follow up. Based on the JNC VI we obtained the information to assign each patient to his/her corresponding stage and risk group. We recorded information on demographics, treatment, and level of control ($< 140/90$ mmHg). We also defined intensity of therapy (increasing the dosage or adding a new drug) as a marker of the process and the mean decrease of systolic blood pressure (SBP) and diastolic (DBP) as a marker of outcome of care.

RESULTS: The mean (SD) age was 59.5 (15.6) years and 43.2% were men. The mean follow up was 42.8 $\pm$ 22.7 months with 3.42 visits per year. Overall, 53.5% of the patients had their blood pressure (BP) under control. Stratifying by level, stage 1: 62.2%, 2: 53.1% and 3: 39.1% ($p = 0.001$). The degree of control was similar in the three risk groups after adjusting for stage using Mantel Haenszel χ^2 ($p = 0.25$).

The mean SBP at the initial visit was 155 $\pm$ 20.0 mmHg and the mean DBP was 98.9 $\pm$ 40.6 mmHg. The mean decrease in SBP and DBP between initial and last visit was 5.6, 15.8 and 30.9 ($p = 0.0001$) and 5.2, 11.3 and 20.2 ($p = 0.0001$) for stage 1, 2 and 3 respectively.

Adjusting for stage, the mean decrease in SBP and DBP was 15.6, 14.1 and 16.7 ($p = NS$); and 10.5, 9.9 and 11.4 ($p = NS$) in group A, B and C respectively.

The intensity of therapy by stage was 53%, 66% and 81% for stage 1, 2 and 3 respectively ($p = 0.09$).

Adjusting for stage, we did not observe differences across groups ($p = NS$).

CONCLUSION: Although BP was reduced as a consequence of more intensive treatment according to initial stage, it appears that physicians are not considering the patient's global risk profile when making decisions about the management of hypertension. Doctor's need not only to incorporate level of BP but also overall risk to improve the effectiveness of their interventions.

THE EFFECT OF A NARCOTICS MANAGEMENT SYSTEM ON NARCOTICS PRESCRIBING AND UNSCHEDULED VISITS IN A LARGE VETERANS PRIMARY CARE CLINIC. KC Goldberg, EZ Oddone, DL Simel, Ambulatory Care, Durham VA Medical Center, Durham, NC

PURPOSE: National guidelines for the management of chronic pain with narcotics recommend establishing a written pain contract, limiting the number of providers, and providing consistent, scheduled supplies of narcotics. It is not known how adoption of these interventions affects provider or patient behavior. This study measured the effect of an explicit pain management program on unscheduled patient visits, narcotics prescribing behavior, and oxycodone use in a large ambulatory care clinic at a university-affiliated VA hospital.

METHODS: In 1997, the Durham VA Ambulatory Care Clinic initiated a comprehensive pain management program to help primary care providers manage patients whose treatment involved chronic narcotics. This program involved a specific pain contract signed by the provider and the patient and a clerical mechanism that assisted with scheduled prescribing of controlled substances. All patients with evidence of such a pain contract were identified retrospectively. A matched control group of patients followed in the ambulatory care clinics during the same period that received comparable amounts of narcotics, but did not have a pain management plan on their chart, was also identified. Demographic information, all outpatient visits, and narcotics prescriptions were abstracted and summarized for both groups. Visits and narcotics prescriptions from other nearby VA hospitals were collected for intervention patients. Emergency department visits, the number of separate providers issuing narcotics prescriptions, the number of unique narcotics prescriptions, and the amount of oxycodone consumed were summarized by calendar quarter for intervention patients before and after the execution of their pain management contract, and for control patients throughout the study period.

RESULTS: 91 patients with a pain management contract were identified. Compared to the time period prior to the contract, visits to the emergency department, the number of providers issuing prescriptions, the number of separate prescriptions for narcotics, and the number of dispensed oxycodone tablets all decreased significantly ($p < 0.001$ for each measure) after implementation of the contract. Visits to other nearby VA hospitals did not increase. In a matched group of 224 patients receiving narcotics, emergency department visits decreased during the observation period, but to a lesser degree. The number of separate providers issuing narcotics and the number of unique prescriptions did not change over time, although the number of oxycodone tablets consumed increased steadily for this control group.

CONCLUSION: A pain management program made explicit by a provider-patient contract can result in fewer unscheduled visits, a consolidation of narcotics prescribing, and lower overall use of narcotics.

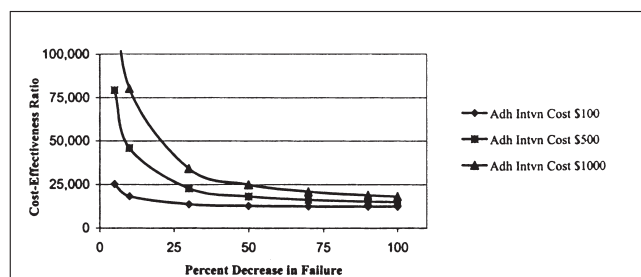
COST-EFFECTIVENESS OF INTERVENTIONS TO IMPROVE ADHERENCE TO HIGHLY-ACTIVE ANTIRETROVIRAL THERAPY (HAART) IN HIV-INFECTED PATIENTS. SJ Goldie, AD Paltiel, MC Weinstein, E Losina, GR Seage, CJ Cohen, AD Kimmel, H Zhang, KA Freedberg, Harvard School of Public Health, Boston, MA

BACKGROUND: Highly-active antiretroviral therapy (HAART) has reduced HIV-related morbidity and mortality, but strict adherence is required to prevent the development of drug resistance.

PURPOSE: To determine the clinical impact and cost-effectiveness of various adherence interventions as a function of virologic suppression and cost.

DESIGN: A state transition model of the natural history of HIV infection was implemented as a Monte Carlo simulation to estimate lifetime costs and quality-adjusted life months (QALMs) for several hypothetical adherence interventions. Intervention effectiveness was expressed as a percentage reduction in the failure to achieve HIV-RNA suppression on HAART. Data were derived from clinical trials, cohort studies, national databases, and published literature. Sensitivity analyses were conducted to assess the stability of the results to uncertainty regarding each intervention effectiveness, cost, time of initiation, and duration of benefit.

RESULTS: HAART with three drugs increased quality-adjusted life expectancy from 27.44 to 69.42 QALMs and lifetime costs from \$52,160 to \$110,250, compared with no therapy. An adherence intervention which decreased virologic failure rates by 5%, 10%, and 25%, provided a gain of 1.2, 2.7, and 8.7 QALMs, respectively. The relationship between the intervention's monthly cost, effectiveness, and cost-effectiveness is shown below.



CONCLUSIONS: Adherence interventions which decrease failure rates by only 10 to 25% will be cost-effective, even if expensive. Studies evaluating such interventions should report results in terms of both change in failure rate and cost, in order to allow rapid assessment of cost-effectiveness for development of HIV health policy.

REDUCING ANTIBIOTIC USE IN AMBULATORY PRACTICE: IMPACT ON PATIENT-CENTERED OUTCOMES. R Gonzales, JF Steiner, JH Maselli, A Lum, PH Barrett, Jr, Medicine, University of Colorado Health Sciences Center; Kaiser Permanente of Colorado, Denver, CO

PURPOSE: In the first year of a patient and clinician educational intervention, we decreased antibiotic treatment of adults with uncomplicated acute bronchitis from 80% to 45%.

The purpose of this study was to evaluate the impact of reduced antibiotic use for acute bronchitis on duration of illness and patient satisfaction.

METHODS: We conducted a telephone survey of adults diagnosed with acute bronchitis at a clinic where antibiotic treatment of acute bronchitis had been decreased, and a usual care clinic of similar size and demography, belonging to a group-model HMO in the Denver metropolitan area. Patients were contacted 2-3 weeks following their office visit. Major outcomes were time to illness resolution, and satisfaction with the episode of care. A proportional-hazards model was used to evaluate duration of illness (relative risk (RR) and 95% confidence intervals (CI) reflect probability of symptom resolution). Satisfaction between groups was compared using the Wilcoxon rank sum test.

RESULTS: Antibiotics were prescribed to 63% and 84% of respondents from the intervention and usual care clinics, respectively ($p < 0.001$). Using multivariate analysis, increased age was associated with longer duration of illness (age > 65 : RR = 0.57 95% CI = 0.34-0.95), but not gender, comorbidity, duration of illness prior to the visit, or antibiotic treatment. After adjusting for these factors, there was no difference in duration of illness between intervention and usual care clinics. Satisfaction with the visit did not differ between patients from intervention and control clinics, nor between patients receiving and not receiving antibiotics.

CONCLUSION: Reducing the rate of antibiotic treatment of adults with uncomplicated acute bronchitis does not prolong illness duration or produce dissatisfaction with care.

SUSTAINABILITY OF AN INTERVENTION STRATEGY FOR REDUCING ANTIBIOTIC USE FOR UNCOMPLICATED ACUTE BRONCHITIS. R Gonzales, JF Steiner, JH Maselli, R Miller, A Lum, PH Barrett, Jr, Medicine, University of Colorado Health Sciences Center; Kaiser Permanente of Colorado, Denver, CO

PURPOSE: In the first year of a patient and clinician educational intervention, we decreased antibiotic treatment of adults with uncomplicated acute bronchitis from 80% to 45%.

The purpose of this study was to evaluate the sustainability of this intervention in the subsequent year.

METHODS: We calculated visit and antibiotic prescription rates for incident, uncomplicated acute bronchitis from administrative claims data from a group-model HMO. We compared utilization between sites from the initial post-intervention

study period (11/97-11/98) to the follow-up period (11/98-2/99). The positive results of the initial intervention were publicized system-wide in the fall of 1998, and during a single staff meeting at the full intervention site. No other patient or staff education took place in the follow-up period. Bronchitis visit rates and antibiotic prescription rates were compared between sites using a mixed-effects logistic regression model, adjusted for patient and provider characteristics.

RESULTS: Antibiotic prescription rates remained high at the control sites, and remained low at the full intervention site during the follow-up period (78%, 80% and 53%, respectively; $p = 0.001$). 30-day return visits for bronchitis or pneumonia also remained constant during the follow-up period. The likelihood that a patient with an office visit for acute bronchitis during the first intervention period sought care again during the follow-up period was lower at the full intervention site compared to control sites (OR = 0.4, 95% CI = 0.2, 0.8).

CONCLUSION: During the subsequent winter, a successful intervention to reduce antibiotic treatment of uncomplicated acute bronchitis was sustained with minimal reinforcement. This intervention also reduced subsequent visits for uncomplicated acute bronchitis among patients who previously sought care for this illness.

HERBAL AND NUTRITIONAL SUPPLEMENT USE IN GENERAL MEDICINE CLINICS: FREQUENT USE, AND FREQUENT POTENTIAL FOR DRUG INTERACTIONS. CB Good, P Glassman, L Trilli, B Leong, Section of General Internal Medicine and Pharmacy, Veterans Affairs Medical Center, Pittsburgh and West Los Angeles; University of Pittsburgh, Pittsburgh, PA

PURPOSE: To assess use of herbal and nutritional supplements (herb/nutr supps), and evaluate for potential interactions with prescription medications in 2 outpatient clinics.

METHODS: A convenience sample of patients waiting for clinic appointments was surveyed in 1999 for use of herb/nutr supps. Patients were asked about current and past use of herb/nutr supps, what they were taking, where purchased, monthly costs, sources of information for these medications, and whether the patient has discussed use of these herb/nutr supps with their medical provider (PCP). Medication profiles for all patients taking herb/nutr supps were reviewed for potential drug interactions.

RESULTS: 458 patients were surveyed, 260 from the Pittsburgh VAMC (Pgh), and 198 from West Los Angeles VAMC (LA). Overall, 43% of patients were currently taking herb/nutr supps (38% Pgh, 49% LA), and an additional 18% had previously taken these medications. Most commonly taken herb/nutr supps are listed in table. Patients took an average of 7 prescription oral medications in Pgh. Potential drug interactions between prescription drugs and herb/nutr supps were common (table). However only 2% were felt to be potentially serious interactions. 70% of Pgh and 40% of LA patients had informed their PCP of their herb/nutr use; of those who did not discuss, 59% of Pgh and 88% of LA patients state their PCP had never asked about herb/nutr supps use. Monthly costs for herb/nutr supps were modest (table). Patients rarely identified their PCP as a source of information about herb/nutr supps (16% for both sites).

	Currently Taking Herb/Nutr Supps	
	Pgh N = 98	LA N = 97
Vitamins/Minerals	65%	86%
Garlic	37%	27%
Ginkgo Biloba	22%	24%
Saw Palmetto	15%	13%
Ginseng	12%	13%
Potential Drug Interaction	48%	42%
Cost . \$25/month	17%	24%
PCP Aware	70%	40%

CONCLUSION: Use of herb/nutr supps is very common in general medicine patients who take prescription medications. Potential for interaction between herb/nutr supps is common, although most potential interactions are minor. PCPs need to inquire about use of herb/nutr supps in their patients, and consider potential adverse interactions with prescription medications.

DEVELOPMENT OF AN INSTRUMENT TO MEASURE TRUST IN HEALTH INSURERS. SD Gould, D Fessler, Internal Medicine, University of Michigan, Ann Arbor, MI

PURPOSE: Patient-centered assessments of healthcare are increasingly important, and have included measures of quality, satisfaction, access and views of the doctor-patient relationship. There are no measures of perceived fairness or trust in organizational decision making, despite work that has shown their importance to patients and healthplan enrollees. This study aimed to develop a reliable measure of trust in health insurers.

METHODS: Insured residents of SE Michigan were interviewed by telephone using a structured instrument. The survey included established measures of doctor-patient trust, satisfaction with care, health status, healthcare use, and demographics. New items were based on interviews with healthplan enrollees and the literature on institutional trust, and included content areas of beneficence, advocacy, competence, honesty and openness, and fairness. Principal components factor analyses using Varimax rotation, retaining factors with eigenvalues > 1 were used to identify potential items for scales. Scales constructed to measure insurer trust were tested for internal consistency using Cronbach's alpha. Univariate analyses for statistical associations used Pearson's r or student's t-tests.

RESULTS: Of 984 contacted households, 400 adults completed interviews (RR = 42%). Conversions were not pursued as the intent was instrument development, not generalizability.

Respondent Characteristics	Household Characteristics	
Age (Mean, range)	Income <35k (%)	28
Women (%)	Employer source of insurance (%)	80
Minority race (%)	>3 years in healthplan (%)	69
Health Status (%fair/poor)	Chronic/serious illness (%)	34

One predominant factor accounted for 40–68% of the variance and had an eigenvalue >20. It included items about competence ("The management at my health insurance company knows what it's doing"), advocacy, beneficence ("My health insurance company has my interests at heart"), fairness, honesty and openness, and one global item ("I trust my insurance company's decisions"). Two scales were developed that include items representing all these domains of institutional trust (Long: 15 items, $\alpha = 0.95$, Short: 8 items, $\alpha = 0.89$). Both instruments correlated with existing measures of Trust in Doctors (Pearson $r = .49$ Long, $.46$ Short) and Satisfaction with care ($r = .70$ Long, $.66$ Short), and less strongly with an item assessing overall worry about health insurance ($r = .37$ Long, $.35$ Short). Respondents who would change healthplans if they could had lower scores than those who did not ($p < .001$).

CONCLUSION: Using original statements from managed care enrollees and a theory of institutional trust, we developed a comprehensive, reliable and reasonably brief measure of trust in health insurance plans.

GUIDELINE CHAOS: COMPETING RECOMMENDATIONS FOR PREOPERATIVE CARDIAC ASSESSMENT. *AJ Gordon*, DS Macpherson, VA Pittsburgh Center for Health Services Research, Pittsburgh, PA

PURPOSE: Two widely disseminated guidelines for preoperative cardiac assessment for non-cardiac surgery (American College of Cardiology/American Heart Association (AHA), American College of Physicians (ACP)) use different approaches to assess risk of surgery. We sought to learn if these two guidelines lead to important differences in recommendations for a group of patients who attended a Medical Pre-operative Evaluation Clinic (MPEC).

METHODS: We reviewed computerized records of all patients evaluated at MPEC in a Veterans Hospital between January 1 and April 1, 1998. Using data abstracted from these notes, we determined what testing should have been done if each guideline was followed and what testing was actually ordered. Each note was reviewed twice separated in time. During each review, only one guideline was followed. On the second review, the result of the first review was not known. Inter-rater reliability regarding key variables was determined by a second author on 10% of patients. We simplified guideline recommendations to 1) operation without testing (OR), 2) non-invasive stress testing (NIT), 3) cardiac catheterization (CATH), and 4) OTHER (cancel or delay surgery).

RESULTS: 138 charts were reviewed. Reliability testing revealed 100% agreement among 30 key variables. The sample was older (median age 69), male (97%), and had histories of current angina (26%), bypass surgery (15%), and shortness of breath (32%). Electrocardiograms were abnormal in 51%. Patients were undergoing primarily orthopedic (36%), head and neck (20%), neurological (15%), and abdominal (11%) surgeries. Recommendations were discordant between guidelines for 17% of patients (bold numbers, Kappa = 0.382, McNemar's $p = 0.093$). Guidelines never agreed on the need for NIT. Extreme differences in recommendations (i.e.: one recommends OR, the other CATH) occurred in 9 (7%) patients. MPEC physicians recommended 102 patients go to the OR, 3 for CATH, and 6 for OTHER. They ordered NIT more often ($n = 27$) than either guideline. In this subgroup of patients, the two guidelines also differed (Kappa = 0.263, McNemar's $p = 0.065$) and reversible ischemia occurred in 31%.

	ACP			
	OR	NIT	CATH	TOTAL
OR	106	3	4	113
NIT	10	0	0	10
CATH	5	1	9	15
TOTAL	121	4	13	138

CONCLUSION: When applied to a cohort of real patients being evaluated for surgery, AHA and ACP guidelines significantly differ in recommendations for further testing prior to surgery. Results have clinical implications for implementation, management, and practitioner adherence to published guidelines.

QUALITY OF CARE BY RACE FOR DIABETES, CONGESTIVE HEART FAILURE, AND CHRONIC OBSTRUCTIVE LUNG DISEASE. *HS Gordon*, ML Johnson, CM Ashton, Department of Medicine, Houston VAMC, Houston, TX

PURPOSE: Racial variations in the utilization and outcomes of health care are well documented; however, few studies have compared quality of care in Black and White patients. The purpose of this study is to examine whether process of care (quality) differs in Black and White United States veterans.

METHODS: Data were from the medical records of patients discharged from one of 12 southern Veterans Affairs Medical Centers (VA) with diabetes (DM; N = 678), congestive heart failure (CHF; N = 850), or chronic obstructive lung disease

(COPD; N = 1292). Quality of care was assessed with disease-specific explicit criteria for the process of hospital care. Criteria were developed for the admission work-up, evaluation and treatment, and readiness for discharge for each disease by panels of expert physicians. Adherence scores (percent of applicable criteria met) were produced for admission (ADM), evaluation and treatment (TRT), and discharge (DC) for each patient. Bivariate associations of race with quality (adherence score) were examined with the t-test. After controlling for severity of illness (APACHE II) and other covariates, the independent association of race with quality was determined for ADM and TRT with linear regression and for DC with logistic regression, because DC scores were not normally distributed.

RESULTS: Mean age and mean APACHE score were similar ($P > .05$) in Black and White patients for DM, CHF, and COPD. Mean ADM scores were higher (82.0 vs. 79.4%; $P < .001$; CHF and 71.7 vs. 68.7%; $P < .001$; COPD) in Blacks and were also higher ($P < .001$; CHF and COPD) in teaching hospitals. Mean TRT scores were higher ($P = .05$; COPD) in Blacks and were higher ($P < .001$; CHF and COPD) in teaching hospitals. Mean DC scores did not vary by race or teaching hospital. Blacks were more likely ($P < .001$) to be admitted to teaching hospitals. After controlling for age, APACHE score, comorbidities, and other covariates Blacks had higher ADM scores than Whites with CHF ($\beta = 2.15$; $P = .006$), COPD ($\beta = 2.89$; $P < .001$), but not with DM ($\beta = 1.25$; $P = .20$). Adjusted TRT score was higher ($\beta = 3.64$; $P = .03$) in Blacks with COPD. Blacks had lower (OR 0.67, 95% CI 0.47–0.95) adjusted DC scores for CHF. However, when we controlled for teaching hospitals ADM and TRT scores did not differ by race; only the DC score for CHF was significant (0.67, 95% CI 0.47–0.95) in Blacks.

CONCLUSION: We found that adjusted quality was significantly higher in Blacks at ADM (CHF, COPD) and at TRT (COPD) and was significantly lower in Blacks at DC (CHF). After adjusting for teaching hospitals, quality was similar by race at ADM and TRT for each diagnosis, but remained lower in Blacks for one of three diagnoses (CHF). Differences in quality by race may be influenced by higher quality at teaching hospitals and a higher likelihood that Blacks were treated in teaching hospitals.

EFFECTIVE TECHNIQUES FOR REDUCING HOSPITAL EMPLOYEE WORKER'S COMPENSATION COSTS. *J Green-McKenzie*, University of Pennsylvania Medical Center, Philadelphia, PA

In order to assess the effect of various cost control and managed care tools on Workers' Compensation costs at a self insured University Hospital, a study was conducted on six cohorts of injured workers spanning calendar years 1993–1998. Cost control techniques were started in 1993 and included institution of an early return to work program, internal administration of legal cases, institution of an injury prevention program, and utilization of modified duty assignments in the workplace. Managed care techniques were instituted in 1997. Such strategies included aggressive case management and tailoring of the provider panel.

Each successive intervention resulted in a reduction in total expenditures, both medical and indemnity. In the early years (1993–96) a reduction in expenditure of more than 50% was realized for both medical and indemnity payments. This trend continued when managed care techniques were later introduced (1997) as indemnity payments were reduced almost 40% and lost time cases were reduced over 50% from calendar year 1996 to 1997. In addition, the percentage of patients who chose to remain in the system increased from 50% to over 90% from 1993 to 1998. The high quality of the provider panel, measured by provider reputation, training and board certification status, was maintained.

Sequential application of cost control and managed care techniques can result in significant reductions in Worker's Compensation costs, both medical and indemnity, in this setting without a decrease in patient satisfaction or a compromise in the quality of care.

THE EFFECT OF TIME TO SURGERY ON MORTALITY IN HIP FRACTURE PATIENTS. *JP Grimes*, H Noveck, P Gregory, M Butler, JL Carson, General Internal Medicine, Family Medicine, Orthopedics, UMDNJ—Robert Wood Johnson Medical School, New Brunswick, NJ

PURPOSE: There is a perception that the standard of care is to surgically repair hip fracture within 24 hours of hospitalization. Prior studies found that outcome improved with rapid surgery, but may not have controlled adequately for underlying illness that delayed surgery. We evaluated the association between time to surgery and 30-day mortality.

METHODS: We performed a retrospective cohort study in consecutive hip fracture patients, age 60 or older, who underwent surgical repair at 20 US hospitals between 1983 and 1993. Patients were initially excluded with metastatic cancer, trauma resulting in multiple injuries requiring surgery, or if declined blood transfusion for religious reasons. For this analysis, we also excluded patients with a fracture occurring >48 hours before admission to the hospital. In the 8,383 patients, we identified those who had surgery delayed for "medical" reasons (N = 2550), which was defined as dehydration, electrolyte imbalance, serious infection, congestive heart failure, diabetes mellitus out of control, acid base disturbances, exacerbation of COPD/asthma, acute hepatitis, uncontrolled blood pressure, myocardial infarction, and arrhythmia. The primary outcome was 30-day mortality.

RESULTS: In the overall population, the median time to surgery was 23 hours (interquartile range 17–38 hrs) and 30-day mortality was 4.7% (95% confidence interval (CI) 4.3%–5.2%).

Thirty-day mortality rates were 3.9% (95% CI 2.4%–4.4%) in those without medical problems and 6.7% (95% CI 5.7%–7.7%) in those with medical problems ($p <$

0.01). Furthermore, in patients without a medical reason for delay the median time to surgery was 20 hours (interquartile range 15–24 hrs) while those with medical reason for delay the median time to surgery was 45 hours (interquartile range 34–70 hrs) ($p < 0.01$). Time to surgery was associated with 30-day mortality after controlling for age, gender and co-morbidity ($p = 0.008$). However, when we added the variable “surgery delayed for medical reasons” to the logistic regression model, time to surgery was no longer significant ($p = 0.3$).

CONCLUSION: Time to surgery in hip fracture patients was not associated with 30-day mortality. Our results imply that time to surgery is a proxy for the burden of medical illness and that the delay in surgery in prior studies was a marker for active medical problems. These results suggest that it may not be as important to rush patients to surgery as previously recommended.

A COST-EFFECTIVENESS COMPARISON OF THE CYCLOOXYGENASE-1-SPARING AGENTS CELECOXIB AND ROFECOXIB WITH THREE OTHER STRATEGIES FOR HIGH-DOSE NONSTEROIDAL ANTI-INFLAMMATORY DRUG THERAPY. PW Groeneveld, Department of Medicine, University of California, San Francisco; Division of General Internal Medicine, San Francisco VA Medical Center, San Francisco, CA

PURPOSE: Since the recent U.S. Food and Drug Administration approval of celecoxib and rofecoxib, these cyclooxygenase-1 (COX-1)-sparing nonsteroidal anti-inflammatory drugs (NSAIDs) have been widely used as replacements for cheaper, non-selective NSAIDs in the treatment of inflammation and pain. Recent studies have verified the clinical benefit of COX-1-sparing agents compared with non-selective NSAIDs in reducing peptic ulcer incidence, yet the higher relative cost of the newer agents has raised questions of whether their use is cost-effective.

METHODS: In this study, a Markov decision model was designed to compare the projected costs and effectiveness of four different therapeutic strategies for high-dose NSAID use. These were: 1) ibuprofen alone; 2) ibuprofen plus misoprostol; 3) ibuprofen plus omeprazole; 4) celecoxib or rofecoxib. Similar clinical efficacy of each regimen at the respective maximum dose for the treatment of inflammation and pain was assumed. The analysis was conducted from a payer perspective and the time horizon was ten years. Each strategy's ability to prevent NSAID-induced ulcers was modeled from published evidence in the medical literature. Utility values for the quality of life effects of peptic ulcer were estimated from previous cost-utility studies in ulcer disease. Cost data on ulcers were obtained from a recently published study of the economics of ulcer disease in a managed care setting. Drug costs were obtained from the 1999 Drug Topics Red Book. All cost and utility estimates were maximally “biased” to be economically favorable to the COX-1-sparing agents.

RESULTS: For the base-case analysis (assuming an ulcer incidence of 0.5% per year), the marginal cost-effectiveness ratio (CER) for COX-1-sparing agents was \$1,100,000 per quality-adjusted life year (QALY) gained compared to the use of ibuprofen alone. Ibuprofen combined with omeprazole was similar in cost-effectiveness to celecoxib or rofecoxib. The misoprostol/ibuprofen strategy was dominated unless the annual ulcer rate exceeded 3%. Sensitivity analyses indicated that the marginal CER for COX-1-sparing agents fell below \$100,000 per QALY when the annual risk of peptic ulcer was greater than 5% per year. The marginal CER was less than \$50,000 per QALY only when the annual ulcer risk was 8% or greater.

CONCLUSION: This study suggests that even in a model that maximally favored the clinical and economic performance of COX-1-sparing NSAIDs, such drugs are only cost-effective in patients who are at substantially increased risk of developing NSAID-induced peptic ulcers and their complications.

DETERMINANTS OF PHYSICIAN ADOPTION OF A NEW CANCER PREVENTION STRATEGY. CP Gross, M Cruz-Correa, MI Cantor, T Valente, CJ McNeil-Solis, NR Powe, Yale School of Medicine, New Haven, CT; Johns Hopkins Medical Institutions, Baltimore, MD

PURPOSE: To examine whether physician beliefs, colleagues' practices, and other personal and practice characteristics were associated with the use of a new cancer prevention therapy for patients at high risk for esophageal carcinoma.

METHODS: We mailed a self-administered questionnaire to a national random sample of gastroenterologists. The survey included case scenarios involving patients with Barrett's esophagus. The dependent variable was the use of columnar epithelium ablation therapy (AT) in these scenarios. The independent variable was the belief that close colleagues (defined as the three physicians with whom clinical issues are most frequently discussed) would use AT for the same scenarios. We asked multiple choice questions about sources of information, personal and practice characteristics, and Likert-type questions about the efficacy, safety, costs, and medical literature pertaining to AT. Logistic regression was used to identify physician characteristics and beliefs that were associated with use of AT in at least one of the case scenarios.

RESULTS: We received 279 completed surveys (50% response rate; 93% male, 52% academic). About 36% of respondents reported that they would use AT; 56% thought their colleagues would use it. The most commonly cited sources of information about AT were published literature (52%) and medical conferences (44%); only 16% cited their colleagues. The use of AT was strongly associated with the belief that AT was: used by their colleagues (OR: 9.51; 95% CI: 4.60–19.67), supported by the literature (OR: 5.77; 95% CI: 2.61–12.72), effective (OR: 3.83; 95% CI: 1.95–7.54), and cost saving (OR: 5.07; 95% CI: 1.89–13.61). Respondents who had completed their training more than 7 years earlier were also more likely to recommend AT (OR: 4.37; 95% CI: 2.15–8.87). There was no association between

use of AT and gender, academic appointment, caseload, mechanism of reimbursement, or the belief that ablation is unsafe. After adjusting for physician characteristics and beliefs in the multivariate model, the perceived use of AT by colleagues remained strongly associated with use of AT (OR: 11.86; 95% CI: 4.75–29.58). Further, in the 135 respondents who did believe that AT was effective, cost-saving, or supported by the literature, perceived AT use by colleagues was still strongly associated with AT use (OR: 12.21; 95% CI: 3.98–37.42).

CONCLUSION: Although few physicians indicated that their colleagues were a major source of information about AT, adoption of this new cancer prevention strategy appears to be influenced by colleagues' practices, independent of beliefs about the pertinent medical evidence, risks, and benefits.

DETERMINANTS OF HOSPITAL TRANSFER IN THE MANAGEMENT OF ACUTE MYOCARDIAL INFARCTION. J Gurwitz, J Gore, R Goldberg, H Barron, A Tiefenbrunn, P Frederick, J Malmgren, Meyers Primary Care Institute, University of Massachusetts Medical School and Fallon Healthcare System, Worcester, MA; University of California and Genentech, Inc., San Francisco, CA; Washington University School of Medicine, Saint Louis, MO; University of Washington, Seattle, WA

PURPOSE: Many factors precipitate the transfer of hospitalized acute myocardial infarction patients including the patient's clinical status and the need for diagnostic testing and therapeutic interventions that are not available at the admitting hospital. In this study, we examined non-clinical determinants of hospital transfer of acute myocardial infarction patients.

METHODS: We employed a large national registry of patients hospitalized for acute myocardial infarction in the United States from June 1994 to March 1998 (National Registry of Myocardial Infarction-2). The study population included 537,283 patients who survived the index hospitalization to either transfer or discharge. We utilized a forward stepwise multivariable logistic regression model with non-transfer as the dependent variable to examine the independent effect of patient age, gender, race, and insurance status, adjusting for various clinical and hospital characteristics. Clinical characteristics included prior medical history, pre-hospital delay, Killip class on presentation, and therapeutic interventions. Hospital characteristics included bed size, presence of on-site facilities for cardiac procedures, and volume of acute myocardial infarction cases.

RESULTS: 152,310 patients (28.3%) were transferred to another hospital. Of these patients, 34,190 (22.4%) were transferred in <24 hours, 27,162 (17.8%) in 24–48 hours, and 90,958 (59.7%) in >48 hours after admission. In a multivariable model, we found age greater than or equal to 75 (versus age <65, adjusted odds ratio 2.32, 95% CI 2.26–2.38), female sex (adjusted odds ratio 1.19, 95% CI 1.17–1.21), and African American race (versus white, adjusted odds ratio 1.40, 95% CI 1.35–1.45) to be significantly associated with non-transfer. Relative to patients with commercial insurance, self-pay or Medicaid patients (adjusted odds ratio 1.47, 95% CI 1.43–1.52) and Medicare patients (adjusted odds ratio 1.38, 95% CI 1.35–1.42) had a greater odds of non-transfer. HMO-covered patients had a higher odds of transfer relative to patients covered by commercial insurance (adjusted odds ratio 1.18, 95% CI 1.14–1.20).

CONCLUSION: Non-clinical factors including age, gender, race, and insurance status are associated with whether or not acute myocardial infarction patients are transferred to another hospital.

ADVERSE OUTCOMES OF INSTABILITY ON HOSPITAL DISCHARGE IN PATIENTS WITH PNEUMONIA. EA Halm, MJ Fine, DE Singer, WN Kapoor, TJ Marrie, AL Siu, Health Policy, Mount Sinai Hospital, New York, NY; Medicine, Univ. of Pittsburgh Medical Center, Pittsburgh, PA; Medicine, Massachusetts General Hospital, Boston, MA; Medicine, Victoria General Hospital, Halifax, NS, Canada

PURPOSE: Pressures to reduce the cost of hospital care have resulted in rapidly declining length of stay (LOS). As LOS falls, the risk of patients (Pts) being discharged from the hospital prior to becoming clinically stable increases. We sought to measure rates of instability on discharge (DC) in Pts hospitalized with community-acquired pneumonia (CAP) and their associated clinical outcomes.

METHODS: Daily clinical data were collected on a subset of 680 Pts hospitalized with CAP as part of the Pneumonia PORT multicenter prospective cohort study. Deaths, readmissions, and return to usual activities (RTUA) within 30 days of DC were ascertained by telephone interview. Logistic regression and survival analyses were used to examine associations between instability on DC and post-DC outcomes. Instability on DC was defined as DC prior to being stable for 24 hrs according to a previously validated definition (Temp ≤ 100 F, RR ≤ 24 , HR ≤ 100 , SBP ≥ 90 , and O₂ sat $\geq 90\%$, and baseline mental and eating status).

RESULTS: Pts mean age was 58, and 75% had ≥ 1 comorbidity. By the Pneumonia Severity Index, 70% of Pts were low risk, 21% moderate risk and 8% high risk. The median LOS was 6 days. Of the 680 Pts, 135 (19.8%) had ≥ 1 instability on DC (O₂ sat (6.6%), RR (3.8%), HR (3.5%), Temp (3.4%), abnormal mental status (1.6%), abnormal eating status (1.9%), and SBP (1%). Overall, 122 Pts had 1 instability on DC, 12 Pts had 2, and 1 Pt had 3. Within 30 days of DC, 23 (3.4%) Pts died, 67 (9.8%) were readmitted, 80 (11.7%) died or were readmitted (major events), and 223 (34.8%) did not RTUA. Pts discharged with ≥ 1 instability on DC had 2.7-fold increased odds of death (CI, 1.1–6.4) and 1.6 times increase in major events (CI, 0.96–2.8). Pts with ≥ 2 instabilities on DC had 22.5 times increased risk of death (CI, 6.7–76) and 6.9-fold increase in major events (CI, 2.2–21). Pts discharged with ≥ 1 instability were more likely not to RTUA (OR = 1.7, CI: 1.2–2.6). Time to death, readmission, major events and RTUA were all associated with the number of instabilities on DC (log rank tests $p < .04$ for all). The number of instabilities on DC

remained a significant predictor of all post-DC outcomes even after controlling for Pt age, sex, pneumonia severity, comorbidities, admission source, DC location, and DNR status.

CONCLUSION: Instability on DC is associated with poor post-DC outcomes even after controlling for other important risk factors. Pneumonia guidelines and pathways to reduce LOS should include objective criteria for judging appropriateness for DC to safeguard against Pts being sent home 'sicker and quicker'.

DIFFICULT END OF LIFE TREATMENT DECISIONS: DO OTHER FACTORS TRUMP ADVANCE DIRECTIVES? SB Hardin, YA Yusufaly, Internal Medicine, VA Medical Center; Internal Medicine, LLU School of Medicine, Loma Linda, CA

PURPOSE: While advance directives (AD) have been widely promoted as a means to ensure that patients' treatment preferences are followed, there is extensive evidence for their apparent failure. We conducted a survey of physicians to examine how they would respond to difficult clinical situations with an explicit AD and the reasons for their treatment decisions.

METHODS: We mailed surveys to all internal medicine faculty and resident physicians at an academic medical center. The survey consisted of 6 hypothetical clinical scenarios. Each described a seriously ill patient with an explicit AD. The scenarios contained a potential conflict between the AD treatment preference, clinical prognosis, and/or family/friends. Three scenarios described a fair prognosis with an AD declining treatment; the remaining 3 described a poor prognosis with AD preferences for aggressive treatment. Respondents were asked to make a treatment decision and then indicate the rationale for their decision.

RESULTS: A total of 117/250 (47%) analyzable surveys were returned. Overall, decisions by faculty and residents were NOT consistent with the AD in 65% of cases (see Table.) Physicians cited a variety of reasons for their treatment decisions including the AD (37%), family/friends (when present in scenario, 15%), likelihood of survival (18%), and quality of life (34%). When physicians made decisions inconsistent with the AD, they were more likely to list other reasons for their decisions (89%, $p < 0.001$.) Similarly, of those making decisions consistent with the AD, a majority of faculty and residents cited the AD as their rationale in those cases (86%, $p < 0.001$.)

Scenarios	Prognosis	AD Preference	Family/Friends	Treatment decisions NOT consistent with AD
1.	Poor	Treat	Decline involvement	53%
2.	Fair	Don't treat	Treat	58%
3.	Poor	Treat	Not involved	53%
4.	Poor	Treat	Don't treat	66%
5.	Fair	Don't treat	Unsure	82%
6.	Fair	Don't treat	None	79%

CONCLUSION: Despite the presence of an explicit AD, internists frequently make treatment decisions that are not consistent with the AD. In difficult clinical situations, internists appear to consider other factors such as prognosis, perceived quality of life, and family/friends wishes as more determinative than the AD. Given the potential for conflict between ADs and clinical/social factors, future work needs to elucidate how strictly patients might want their AD preferences followed.

TESTING FOR AND TREATING INFLUENZA IN PATIENTS WITH INFLUENZA-LIKE ILLNESS. S He, DN Rose, Department of Medicine, Long Island Jewish Medical Center, New Hyde Park; Department of Medicine, Albert Einstein College of Medicine, New York, NY

PURPOSE: Clinical decision making for patients with influenza-like illness has become more complex with the recent introduction of rapid office tests for the identification of influenza infection and the recent approval of a new class of antiviral drugs, the neuraminidase inhibitors. These drugs are active against both influenza type A and type B viruses, in contrast to the older antiviral drugs which are active only against influenza type A. The purpose of this study was to determine the utility of rapid testing for influenza followed by antiviral therapy and the utility of empirical antiviral therapy.

METHODS: We performed a decision analysis and cost-effectiveness analysis, using results of clinical trials of rimantidine, an older antiviral drug, and zanamivir, a neuraminidase inhibitor, for the treatment of average-risk (for complications of influenza) and high-risk patients with influenza-like illness. Four strategies were compared to symptomatic treatment alone: empirical rimantidine treatment, empirical zanamivir treatment, test for influenza A and treat positives with rimantidine, and test for influenza A and B and treat positives with zanamivir. Outcome measures were reductions in the number of days of fever and influenza symptoms and the cost per quality-adjusted life year (QALY) saved.

RESULTS: Compared to symptomatic treatment, the empirical rimantidine treatment strategy results in the greatest reduction in the number of days of fever, 1.4 for average-risk persons and 2.0 for high-risk persons, relative to symptomatic treatment alone. The strategy of testing for influenza A and B and treating positives with zanamivir results in the smallest reductions, 0.7 and 0.9 days, relative to symptomatic treatment alone. Testing for influenza adds cost but does not improve outcome. Empirical zanamivir treatment is the optimal strategy if influenza type B causes more than half the documented influenza cases in the general population, an uncommon occurrence. The cost of the four strategies ranges from \$37 to \$63 per person, including the cost of treating symptoms and bacterial complica-

tions if they occur. The cost per QALY saved ranges from \$11,800 and \$8,600 for average-risk and high-risk patients with the empirical rimantidine treatment strategy to \$42,100 and \$32,900 for average-risk and high-risk patients with the strategy of testing for influenza A and B and treating positives with zanamivir.

CONCLUSION: Antiviral therapy results in modest reductions in symptoms of patients with influenza-like illness. The cost for these benefits is also modest. Testing for influenza adds cost without additional benefit. The most effective and cost-effective strategy is empirical treatment with rimantidine, an older antiviral drug.

EFFECT OF A PRIMARY CARE DELIVERY MODEL ON SATISFACTION WITH MEDICAL CARE AMONG YOUNG ADULT PATIENTS. DH Hickam, SK Joos, HSR&D Program, Portland VA Medical Center; Oregon Health Sciences U., Portland, OR

PURPOSE: Delivery of medical care via a primary care provider is a well accepted model for patients with chronic diseases. Younger adult patients often do not have chronic conditions and require medical care only on an intermittent basis. These patients may not necessarily perceive that they need a primary care provider. This study's objective was to determine whether young adults who receive care within a primary care system have different perceptions of the adequacy of their health care than those who do not receive primary care services.

METHODS: The study was conducted among patients who receive care at an urban VA medical center with an integrated primary care system. Subjects included all individuals who were discharged from military service after August, 1990 and had received medical care between October 1997 and September 1998. In January 1999, a questionnaire was mailed to 1113 patients who met these criteria. The questionnaire included demographic items, the SF-36V health status instrument and the VA Customer Satisfaction Survey (developed in collaboration with the Picker Commonwealth Foundation). Respondents were classified by whether they had received care from an assigned primary care provider. Comparisons of satisfaction items between patient groups were adjusted for age, gender, and health status by using analysis of covariance and logistic regression.

RESULTS: Of 467 respondents, 15 were excluded because their age was greater than 60 years. Median age of the remaining subjects was 38 years, 86% were male, 84% were Caucasian, and 69% had an assigned primary care provider (PCP). Patients having a PCP had higher satisfaction with their most recent clinic visit ($p = .05$), but they were not more satisfied with overall care. Patients having a PCP did not consistently have more favorable perceptions of specific provider behaviors. Among patients with a PCP, 18% reported that they were not involved in decisions about their care as much as they desired, 23% reported not having complete confidence in their provider, and 23% reported not receiving as much information about their condition as they desired. These proportions did not differ significantly from those reported by patients without a PCP. However, more of the patients with a PCP (75%) than of those without a PCP (55%) reported that their providers were always familiar with their recent medical history ($p < .001$).

CONCLUSION: Among this young adult population, a primary care system appeared to affect some perceptions of medical care but not overall satisfaction. For some physician communication behaviors, primary care providers do not perform better than other providers. Primary care providers may benefit from initiatives to improve communication behaviors.

VARIATION AT PROVIDER, TEAM AND CLINIC SITE LEVEL FOR PROFILE MEASURES RELATED TO DIABETES CARE. TP Hofer, S Krein, RA Hayward, HSR&D, VA; Internal Medicine, UM, Ann Arbor, MI

PURPOSE: We used computerized hospital databases to define a set of indicators representing important processes of care related to the management of diabetes. We asked whether the variation in the indicator performance occurred more at the provider, primary care team or clinic site level within a group of 20 primary care clinical sites across eight hospitals comprising one Veterans Administration Service Network (VISN).

METHODS: Data files were obtained from each facility containing patient demographic information, laboratory data, pharmacy data, and encounter data (dates and diagnoses). Separate databases contain patient assignments to primary care providers and teams.

Inclusion criteria required that a patient have at least one outpatient visit during FY98, and a prescription for insulin or an oral hypoglycemic medication or home glucose monitoring supplies during FY 98. We included 18723 patients and 265 providers in 50 teams.

The indicators were taken from the Veterans Health Administration (VHA) Diabetes Clinical Guidelines, and included receipt of eye care, and indicators related to obtaining a measurement, the mean level and proportion of patients above a threshold level of HgbA1c and LDL.

Multilevel random effects models were used to estimate the variance explained by clinic site, team and provider for each of the indicators.

RESULTS: Glycemic control at sites (HgbA1c 7.3%-8.8%) is similar or better than most patient populations reported in the literature although these reports predate the publication of recent trials showing the benefits of tighter control. However, some sites still have many patients with poor glycemic control (clinic rate of patients with a last HgbA1c >9.5% ranges from 8%-31%).

Monitoring and treatment of lipids is an area that can be improved at all facilities. Many patients are not being monitored and high LDLs are relatively common in those patients who have measures.

The site explained the most variance in mean HgbA1c and LDL level at 8.7-9.1% while PCP explained from 0-2% across the facilities. The largest amount of varia-

tion at the provider level was found for obtaining a HgbA1c or LDL measurement in the past year and varied substantially across facilities (percent of variation due to provider ranging from 2–20%). Eye exam rates varied most at the clinic site level (9% of the variance in rates) and little or none at the physician level.

CONCLUSION: Clinic site level profiles for important processes of care related to diabetes can be feasibly developed and will be reasonably reliable given the number of patients and the magnitude of facility differences. While physician level profiles will usually have too low reliability to be useful, for some indicators at some sites investigating provider level differences in performance may be possible.

HAVE DEATHS DUE TO MEDICAL ERRORS BEEN OVERESTIMATED? *TP Hofer*, R Hayward, HSR&D, VA; Internal Medicine, UM, Ann Arbor, MI

PURPOSE: Physician implicit review has been used to assess whether hospital deaths are "preventable." Studies using such reviews have suggested that the number of deaths in U.S. hospitals that are due to medical errors is shockingly high. We sought to uncover what physicians mean when they describe a death as "preventable" in terms of the probability of a patient surviving had care been optimal.

METHODS: We conducted 344 reviews of 102 different eligible hospital deaths, at seven Department of Veterans Affairs Medical Centers. Cases with a terminal diagnosis who were admitted to the hospital predominantly for comfort care were excluded. A total of 15 physician reviewers were trained in the use of the implicit review instrument for the study. Each was a board certified internist with extensive experience in inpatient medicine. A previously reported instrument was used. The question on preventable death was rated on a 5 point scale (definitely yes, probably yes, uncertain, probably not, definitely not). In this study we added to the review instrument several questions asking the probability of certain events, such as the likelihood that the death would have been prevented if care had been optimal, rated from 0%–100%.

The unit of analysis was the review although we accounted for the clustering of reviews by patient and reviewer using multilevel statistical models.

RESULTS: 7% of the reviews rated overall care as being substandard and 14% reported care as of borderline quality or worse. Similar to previous studies, in 8% of the reviews death was rated as probably or definitely preventable by optimal care, and in 17% of the reviews the preventability of death was rated as uncertain. In addition, the inter-rater reliability of ratings of the preventability of death was similar to previous studies (ICC = 0.22 for two reviewers).

If a death was rated as probably or definitely preventable, the probability of preventing death with optimal care was estimated to be 57% (95% CI 47%, 66%), however, the likelihood of surviving three months with good cognitive and physical function was estimated at only 11% [4%, 19%]. Amongst all deaths reviewed, the physicians estimated that, had care been optimal, the average probability of living 1 year or more with good cognitive and physical function was 0.9% [0.08%, 0.17%].

CONCLUSION: The alarming statistics about preventable deaths in US hospitals should be tempered by an understanding that the deaths are identified, by physician review, in very sick patients whose chances of surviving to leave the hospital would be uncertain, even with optimal care. A further implication relevant to this finding is that hospital mortality rates can not capture much of the difference in preventable death rates since many such patients will die before discharge even with optimal care.

TRENDS IN FLUOROQUINOLONE USE AND RESISTANCE AT AN ACADEMIC MEDICAL CENTER FROM 1993–1999. *ES Huang*, DE Singer, DC Hooper, General Internal Medicine; Infectious Diseases, Massachusetts General Hospital, Boston, MA

PURPOSE: The expanded use of the newest generation of fluoroquinolones has raised concerns about inducing resistance among gram-negative rods (GNR). We describe trends in antibiotic use and resistance among GNR isolates.

METHODS: We collected data on inpatient antibiotic (fluoroquinolones, third-generation cephalosporins, and aminoglycosides) prescribing from our hospital pharmacy records. Volume of prescribing was expressed as defined daily doses (DDD) per year. Data on the most commonly prescribed outpatient antibiotics from 1997–1999 were obtained for patients belonging to a large local insurance plan. The annual proportions of resistance among major GNRs were extracted from microbiology reports. We used simple linear regression (SLR) and chi-square for trend to detect significant changes over time.

RESULTS: Inpatient fluoroquinolone prescribing rose from 11,140 to 43,384 DDD per year during 1993–1999. The increase in use was greatest from 1998–1999 (25,871 to 43,384). Cephalosporin use increased from 15,845 to 26,258, and aminoglycoside use dropped from 18,249 to 11,319 (all $p < .01$ by SLR). Among outpatient antibiotic prescriptions, ciprofloxacin was the 8th and 10th most prescribed drug for 1998 and 1999; levofloxacin emerged as the 6th leading expense in 1999. GNRs had significant increases for inpatient quinolone resistance:

TABLE OF %RESISTANT ISOLATES

Organism	Annual # of isolates (range)	1993	1994	1995	1996	1997	1998	1999
<i>E. coli</i>	989–1453	0.6	0.5	0.5	0.6	1.7	2.5	3.0
<i>E. cloacae</i>	218–365	2.5	1.9	2.2	3.2	6.4	6.5	6.9
<i>K. pneumoniae</i>	373–671	3.9	2.1	6.5	5.7	6.7	8.0	5.4
<i>K. pneumoniae</i> (oxytoca)	97–177	1.1	2.3	3.1	3.7	5.0	8.0	11.3
<i>P. aeruginosa</i>	396–669	7.8	8.4	11.7	9.2	14.7	15.4	16.9

$P \leq .002$ for all row trends. In contrast, the only significant trend for the other two drug classes was a decline in aminoglycoside resistance for *K. pneumoniae* (10.9–3.2 all isolates).

CONCLUSION: Inpatient fluoroquinolone use at our institution has risen markedly since the introduction of levofloxacin. There was a concomitant increase in inpatient GNR resistance to fluoroquinolones, notably for *Pseudomonas aeruginosa*, with little change in resistance for the other 2 antibiotic classes. As fluoroquinolone use and resistance are increasing, strategies are needed to define and promote the optimal use of these new broad spectrum antibiotics.

PROMOTING INFLUENZA VACCINATION OF PATIENTS AGED BELOW 65 IN PRIMARY CARE. *J-P Humair*, T Martinez Guzman, H Stalder, Department of Community Medicine, Geneva University Hospital, Geneva, Switzerland

PURPOSE: Influenza vaccination is recommended for elderly people and chronic patients of any age but is also available for healthy adults aged under 65. During a community campaign, we implemented a multifaceted intervention in a university-based primary care centre to promote influenza vaccination of high-risk people. Study aims were: (1) to measure influenza immunisation rates among ambulatory patients younger than 65 with or without a chronic disease; (2) to assess the impact of this intervention on vaccination coverage.

METHODS: In this pre-/post-intervention study, we analysed data from a random sample of patients aged below 65 who attended our academic primary care clinic during autumn trimesters of 1995 and 1996. Respectively 405 and 445 patients were included before and during intervention trimesters. The intervention comprised patient information by leaflets and posters, a walk-in vaccination clinic, a physician training workshop, reminders on medical records and periodic feedback to physicians on vaccination performance. Data were collected from the computerised billing file and the review of medical records.

RESULTS: At baseline, immunisation coverage of outpatients younger than 65 was 5.2% with higher rates among those aged 50–64 (11%) or with a chronic illness (19.2%). Vaccine uptake was lower among diabetics (11.5%) than in patients with a pulmonary (30%), cardiac (20%) or another chronic disease (e.g., renal failure, immunosuppression) (28.6%). During intervention, global vaccination coverage rose to 10.6% (odds ratio (OR) = 2.2 [95% CI: 1.3–3.7]). Immunisation rates especially increased among patients with diabetes (45.5%; OR = 6.4 [95% CI: 1.5–27.7]) or any chronic illness (42.6%; OR = 3.1 [95% CI: 1.3–7.6]). In a logistic regression model, the strongest predictors of influenza vaccination were the presence of a pulmonary disease (adjusted OR (AOR) = 5.4 [95% CI: 2.4–19.2]) or of another chronic condition (AOR = 7.1 [95% CI: 1.6–30.9]). Exposure to intervention (AOR = 2.6 [95% CI: 0.9–7.5]) and diabetes (AOR = 3.2 [95% CI: 0.9–11.4]) approached statistical significance to be independent predictors of influenza immunisation.

CONCLUSION: At baseline, few ambulatory patients younger than 65 received the influenza vaccine, even those at high risk. Vaccine uptake doubled after an intervention targeting high-risk people and combining strategies directed at patients, physicians, clinic organisation and the community. The largest effect was observed among chronic patients for whom influenza immunisation is recommended. However, as their immunisation coverage is not optimal, we must develop new strategies and intensify our efforts to promote influenza vaccination of high-risk patients aged below 65.

MEDICAL CARE OF ADULTS WITH CEREBRAL PALSY: LACK OF COORDINATION OF CARE UNDER MEDICAID. *N Hupert*, E McCarthy, RB Davis, L Iezzoni, Beth Israel Deaconess Medical Center, Harvard Medical School, Boston, MA

PURPOSE: People with serious congenital and childhood onset medical conditions are increasingly living into adulthood, but little is known about their health care as adults. Using cerebral palsy (CP) as a paradigm for these conditions, we examined insurance status and coordination of medical care for adults with CP in the United States.

METHODS: We used the 1994–95 National Health Interview Survey (NHIS, a nationally representative sampling of non-institutionalized, civilian U.S. residents) and NHIS Disability Supplement (NHIS-D, a detailed follow-up survey for persons with selected conditions including CP). We used SUDAAN to calculate national population estimates and descriptive statistics.

RESULTS: 129 persons age 18 or older reported cerebral palsy (representing an estimated 174,000 civilian, community-dwelling adults), of whom 76% completed the NHIS-D. The mean age of the sample completing both surveys was 36.4y (SE 1.6). 73% had impairments of activities of daily living (ADLs) and/or instrumental ADLs. 78% were limited or unable to work or retired due to health problems. 70% had a regular physician (34% family practitioner, 24% general practitioner, 10% internist); 89% with a regular physician had seen a doctor in the last year. 18% said they coordinated their own medical care. Of those unable to coordinate their own medical care, 37% reported that no physician, member of a physician's office (MPO), or other person provided this service. 25% said care was coordinated by a physician or MPO, 11% by a person outside a physician's office, and 9% by both. 32% had Medicaid +/- private insurance coverage ("Medicaid"), 26% had private insurance only, 17% had Medicare +/- private ("Medicare"), 18% had both Medicaid and Medicare ("dual eligibles"), and 8% had no private or public insurance. Among persons who identified a regular physician and did not coordinate their own medical care, 50% on Medicaid reported no coordination of medical care, which was greater than the proportion under private insurance (24%), Medicare (21%), or dual eligibility (21%). Among persons with no regular doctor, 100% of those covered under Medicaid vs. 83% with private insurance, 40% with Medicare, and 75% dual eligibles reported no coordination of care. Among those who did not

coordinate their own care, only the uninsured had a higher overall proportion of lack of coordination of care (63%) than those covered under Medicaid (60%).

CONCLUSION: Adults with CP comprise a large community-living population with substantial health care needs. Many of those covered under Medicaid lack coordination of medical care, regardless of presence of a regular physician. Future studies should examine the association between type of health insurance, provision of medical care, and health outcomes for this population.

REFERRAL PATTERNS FOR RENAL INSUFFICIENCY. M Hyder, G Manchester, P-Y Fan, Department of Internal Medicine, University of Massachusetts Medical Center, Worcester, MA

PURPOSE: We reviewed our primary care database to determine the level of renal insufficiency and other factors associated with referral for nephrology evaluation. **METHODS:** We defined the median referral SCr as the SCr at which 50% of patients had nephrologic evaluation. We screened lab results for nearly 15,000 patients in the primary care panel for index years 1996 and 1997. Records of patients with SCr ≥ 2.0 were reviewed for referral within 12 months of the index year, age, sex, hypertension (HTN) and diabetes (DM). To assess rate of progression of renal disease, we calculated the Δ SCr within the index year. Exclusion criteria included acute renal failure with subsequent SCr < 2.0 , follow-up < 1 year, and maintenance dialysis or kidney transplant prior to the index year.

RESULTS: Of 155 patients with SCr ≥ 2.0 , 72 (46%) were referred. For SCr ranges 2.0–2.3, 2.4–2.7, 2.8–3.1, and ≥ 3.2 , the percentages of patients referred to nephrology were 23%, 49%, 69%, and 87%, respectively. Due to the small number of patients at each specific SCr, the median referral SCr could not be precisely determined, but was estimated to be 2.6–2.8.

	Referred	Nonreferred	Men- Ref	Women- Ref	Men- Nonref	Women- Nonref
Mean SCr	3.5	2.3	3.6	3.4	2.4	2.2
Mean Δ SCr	0.8	0.3	0.9	0.6	0.2	0.5
Mean Age	66	74	66	67	74	77

As shown in the table, referred patients were younger ($p < 0.005$), had higher SCr ($p < 0.0001$), and higher Δ SCr ($p < 0.005$) than those who were not referred. Only 39% (41/104) of men compared with 61% (31/51) of women were referred ($p < 0.02$). This finding was not due to gender-related differences in age, SCr, Δ SCr, HTN, or DM. Referred men were younger ($p < 0.03$), and had higher Δ SCr ($p < 0.01$) and were more likely to have HTN and DM ($p < 0.02$) than nonreferred men. While referred women were younger ($p < 0.02$), there was no significant differences in Δ SCr, HTN, or DM between referred and nonreferred women.

CONCLUSION: The median referral SCr was 2.6–2.8 at our tertiary medical center. A greater percentage of women were referred than men. Factors associated with referral of men included SCr, age, Δ SCr, HTN and DM referral. However only SCr and age correlated with referral of women. Further studies are needed to confirm our findings and delineate the factors responsible for gender differences in referral patterns.

PERSONAL USE AND PRESCRIBING PATTERNS OF ALTERNATIVE MEDICINE AMONG UNIVERSITY PHYSICIANS. BP Jacobs, M Lee, M Patterson, A Avins, Osher Center for Integrative Medicine, University of California-San Francisco; General Internal Medicine, San Francisco Veterans Affairs Medical Center; Department of Epidemiology, University of California-San Francisco, San Francisco, CA

PURPOSE: Little information is available describing physicians personal use and prescribing patterns for CAM. The aim of this study is to assess the prevalence and predictors of personal use and prescribing patterns of CAM therapies among university physicians.

METHODS: A mail survey of physicians at four university-affiliated hospitals was performed. Eligibility was limited to the departments of internal medicine, pediatrics, obstetrics and gynecology and family medicine.

RESULTS: The response rate was 84% (383/456). 52% (95% C.I. 47%–57%) of physicians had used at least one type of CAM therapy to address their own health care needs. 51% (95% C.I. 46%–56%) of physicians had actively suggested at least one type of CAM therapy to their patients. Among physicians using these therapies, 43% had informed their own primary care provider. 32% (95% C.I. 27%–36%) of physicians reported feeling competent to prescribe CAM therapies. 32% (95% C.I. 27%–37%) of physicians reported a desire to gain expertise in at least one CAM modality.

Multi-variate analysis revealed several independent ($p < 0.05$) predictors of personal use and prescribing patterns of CAM. Personal use was associated with non-subspecialty training (OR = 1.8), and current family member use (OR = 2.3). Active suggestion of CAM therapies to patients was associated with younger age (OR = 1.03, per one year incremental decrease), non-subspecialty training (OR = 2.7), percent time in clinical practice (OR = 2.4), and family member use (OR = 2.2). A trend for significance ($p < 0.10$) was identified for family member use during childhood and female gender. Reported competence to prescribe CAM was associated with female gender (OR = 1.8), percent time in clinical practice (OR = 2.2), and non-subspecialty training (OR = 1.9). A trend for significance was identified for family member use (OR = 1.7). A desire to gain expertise in a particular CAM therapy was associated with younger age (OR = 1.07), percent time in clinical practice (OR = 2.9) and family member use during childhood (OR = 2.0). Gen-

eralists were no more likely than subspecialists to desire to gain expertise in a particular CAM therapy.

CONCLUSION: Personal use and recommendation of CAM therapies to patients is common among university physicians at these teaching hospitals. Similar to general population surveys, physicians do not inform their primary care providers that they use such therapies. This survey identifies characteristics of physicians likely to use CAM in treating their own health problems or prescribe CAM in clinical practice.

PERSONAL USE AND PRESCRIBING PATTERNS OF COMPLEMENTARY AND ALTERNATIVE MEDICINE AMONG UNIVERSITY PHYSICIANS. BP Jacobs, M Lee, M Patterson, A Avins, General Internal Medicine Section, San Francisco Veterans Affairs Medical Center; Department of Epidemiology; Osher Center for Integrative Medicine, University of California-San Francisco, San Francisco, CA

PURPOSE: The aim of this study is to identify predictors associated with physicians who use complementary and alternative medicine (CAM) to treat their own medical conditions and who prescribe CAM in their clinical practice.

METHODS: A mail survey to physicians at four university-affiliated hospitals was performed. Eligibility was limited to the departments of internal medicine, pediatrics, obstetrics and gynecology and family medicine.

RESULTS: The response rate was 84% (383/456). 52% (95% C.I. 47%–57%) of physicians had used at least one type of CAM therapy to treat their own medical conditions. Among these users, 43% had informed their own primary care provider. 51% (95% C.I. 46%–56%) of physicians had recommended at least one type of CAM therapy to their patients. 32% (95% C.I. 27%–36%) of physicians reported feeling competent to prescribe a CAM therapy. 32% (95% C.I. 27%–37%) of physicians reported a desire to gain expertise in a CAM therapy.

Multi-variate analysis revealed several independent ($P < 0.05$) predictors of physician personal use and prescribing patterns of CAM. Personal use was associated with non-subspecialty training (OR = 1.8) and family member use (OR = 2.3). Recommendation of CAM therapies to patients was associated with younger age (OR = 1.03, per one year incremental decrease), non-subspecialty training (OR = 2.7), percent time in clinical practice (OR = 2.4), and family member use (OR = 2.2). A trend for significance was identified for family member use during childhood (OR = 1.8) and female gender (OR = 1.7). Reported competence to prescribe a CAM therapy was associated with female gender (OR = 1.8), non-subspecialty training (OR = 1.9), and percent time in clinical practice (OR = 2.2). A trend for significance was identified for family member use (OR = 1.7). A desire to gain expertise in a CAM therapy was associated with younger age (OR = 1.07), percent time in clinical practice (OR = 2.9), and family member use during childhood (OR = 2.0). Generalists were no more likely than specialists to desire to gain expertise in a particular CAM therapy.

CONCLUSION: Personal use and recommendation of CAM therapies to patients is common among university physicians at these teaching hospitals. Similar to general population surveys, physicians do not inform their primary care providers that they use these therapies. This survey identifies characteristics of physicians who use CAM to treat their own medical conditions and who prescribe CAM in their clinical practice.

TO DETERMINE PHYSICIANS' UNDERSTANDING OF AND ADHERENCE TO FAMILY MEMBERS' PREFERENCES FOR ICU CARE. MW Jenckes, M Beach, N Kass, MA Williams, HR Rubin, Medicine; Neurology; Health Policy & Management, Johns Hopkins Medicine, Baltimore, MD

PURPOSE: To determine how closely care delivered matches families' stated desires in an ICU setting.

METHODS: Following each death in 3 ICUs between Apr-Oct '99, we identified one family member and one MD who reported close interaction with the patient prior to death and interviewed each, families by telephone and MDs self administered surveys. Here we focus on agreement between 3 items: the level of care the family wanted for the patient; the level of care the MD reported the family wanted for the patient; the level of care the MD reported that the patient received. The response set was: 1 To do everything no matter what to keep my family member alive 2 To go a little longer and see if my family member improved; if they did not then to stop 3 To limit care underway at that point.

RESULTS: Of 136 ICU deaths identified, we contacted 113 family members (83%) and 103 MDs (76%). 49 (43%) of families agreed to participate and completed. 43% of respondents were Black, 52% were > 65 years old and 52% had at least some college. 60% were spouses, 24% children, 10% a parent and 6% a sibling. For 18 patients, both MDs and family members completed the items of interest. 3 of these (18%) desired to "do everything" for their family member and 9 (56%) desired to "go a little longer". In contrast, MDs wanted to "do everything" for 1 patient (5%) and to "go a little longer" for 6 (33%). Agreement between the family member's wishes and the MD's perceptions of those wishes was 50% (8/16, kappa = .22). Of the 3 patients whose family members desired to "do everything", 1 received "everything", 1 had mid-level care and 1 had less aggressive care. Families stated wish and MDs's report of what was done agreed for 6 patients (22%), kappa = .03. In contrast, MD's statements about what they wanted for patients and what was done agreed for 15 of 17 patients (88%), kappa = .76.

CONCLUSION: In a small sample from one AMC, family members were more likely than physicians to desire aggressive care for their dying relative in an ICU. MDs often misperceived family members' wishes, and what was done for patients was much closer to what MDs wanted than to what family members wanted. Ethical

analyses may support ignoring family members' wishes for more aggressive care in some cases, e.g., when MDs believe patients wish less aggressive care, or when MDs believe that patients/family members wish aggressive care due to misunderstanding, and that aggressive care would not achieve desired outcomes. If others confirm these results, to develop care standards for reducing while still providing ethical patient care, and to find ways to help family members manage any necessary remaining gaps between their wishes and what is actually done.

DO ICU MDs DO WHAT THEY THINK PATIENTS WANT OR WHAT THEY WANT TO DO FOR PATIENTS? MW Jenckes, M Beach, MA Williams, HR Rubin, Medicine; Neurology, Johns Hopkins Medical Institutions, Baltimore, MD

PURPOSE: To determine whether care provided in Academic Medical Center (AMC) ICUs reflects what they believe patients want or what they want for patients.

METHODS: Following each of 113 eligible deaths Apr–Oct 1999 in 3 ICUs in a tertiary AMC, MDs received self administered surveys on the deceased patient's care. Survey items measured the MD's report of aggressiveness of care desired by patient and by the MD and that actually delivered. Possible responses were: 1 To do everything no matter what to keep my family member alive 2 To go a little longer and see if my family member improved; if they did not then to stop 3 To limit care underway at that point. Agreement, kappa, and positive and negative predictive values were calculated comparing the care delivered with 1) MD's perceptions of patients' wishes for aggressive care with and 2) MD's wishes for aggressive care. Using 95% CIs, we determined the statistical significance of differences between 1) and 2) above for agreement, kappa, and predictive values.

RESULTS: Thirty-four (33%) of 104 surveys had eligible responses to the three items. MDs reported that 3 patients (10%) had everything done, that 13 (40%) went a little longer, and that 17 (50%) had limited care. On patient wishes, MDs said that 4 (12%) wanted everything, that 14 (41%) wanted to go a little longer, and that 16 (47%) of patients wanted to limit care. MDs reported that they themselves wanted to do everything for 3 (9%) of patients, go a little longer for 13 (38%), and limit care for 18 (53%). MDs' own wishes for the patient concurred with their perception of the patient's wishes for 30 (88%) patients. For 3 (9%) of patients, physicians' wishes were for less aggressive care and for 1 (3%) case, for more aggressive care than they thought patients wanted. Care delivered reflected what physicians said they wanted for the patient for 27 (79%) of patients (kappa = .64, 95% CI .41, .86). The predictive value of physician wanting to limit care was .83 (95% CI .75, .92), and of physician wanting to do everything .33 (95% CI .07, .59). Care reflected MD's reports of what patients wanted for 27 (79%) of patients (kappa = .65, 95% CI .42, .88). The predictive value of the MD reporting that the patient wanted to limit care was .88 (95% CI .80, .95), and of the MD reporting that the patient wanted to do everything was .50 (95% CI .26, .74).

CONCLUSION: In one tertiary AMC ICU, aggressiveness of care MDs delivered agreed as closely with what they believed patients wanted as with their own wishes. Care was more likely to reflect MDs perceptions of patients wishes when patients wanted limited care. Further research should examine these findings in larger samples and if confirmed examine reasons for discrepancies between care and patients' wishes for aggressive care.

THE EFFECT OF RACE ON MORTALITY IN MEN HOSPITALIZED IN THE VETERANS AFFAIRS HEALTH CARE SYSTEM. AK Jha, MG Shlipak, W Hosmer, CD Francis, WS Browner, Division of General Medicine, Veterans Affairs Medical Center, Department of Medicine, University of California, San Francisco, San Francisco, CA

PURPOSE: Race affects the delivery of healthcare but the effect of race on health outcomes has been inconsistent. Racial disparities are at least partly due to differences in access to healthcare. The VA operates a nation-wide, government-funded healthcare system with few financial barriers and may allow for improved access to care for black patients. Therefore, we studied the effect of race on mortality for patients admitted to Veterans Affairs (VA) hospitals, a healthcare system that offers potentially equal access to care.

METHODS: We studied 39,190 patients admitted to one of 147 VA hospitals during the fiscal year 1996 for one of six common medical conditions: heart failure, angina, diabetes mellitus, pneumonia, chronic obstructive pulmonary disease, or chronic renal failure. Due to relatively small numbers, we excluded women and other racial groups. For patients with more than one hospitalization, a single admission was randomly selected. We used logistic regression and proportional hazard models to adjust for differences in patient demographic characteristics, comorbidity score (based upon number of concurrent illnesses), length of hospital stay, and hospital characteristics. The primary outcomes were unadjusted and adjusted 30-day mortality rates for blacks and whites; we also compared unadjusted and adjusted inpatient and 6-month mortality rates. Finally, we compared adjusted mortality rates after stratification by variables that we hypothesized might interact with race (age, financial status, comorbidity score, length of stay, coronary bypass surgery availability, urban location, region, and disease.)

RESULTS: Overall mortality at 30 days was 4.5% in black patients and 5.8% in white patients (relative risk, 0.77; 95% confidence interval, 0.69 to 0.87; $P = 0.001$). Mortality was lower among blacks for each of the six medical diagnoses. Multivariate adjustment had only a modest effect (relative risk, 0.84; 95 percent confidence interval, 0.74 to 0.94; $P = 0.004$). Black patients also had lower adjusted inpatient mortality (relative risk, 0.86; 95 percent confidence interval, 0.75 to 0.99; $P = 0.03$) and 6-month mortality (relative risk, 0.87; 95 percent confidence interval, 0.79 to 0.97; $P = 0.02$). These effects were consistent across sub-

groups based on age, income, urban and rural locations, and region of the country.

CONCLUSION: Black patients admitted to VA hospitals with common medical conditions have lower mortality rates than white patients. Though the survival advantage of black patients is not readily explained, the favorable outcomes for black patients at the VA are likely due to equal access to healthcare and quality inpatient treatment.

A NATIONAL SURVEY OF INTERNAL MEDICINE-PEDIATRIC SUBSPECIALISTS: PROCEDURAL PRACTICE PATTERNS. M Johannesson, M Ciccarelli, K Kroenke, D Litzelman, Department of Medicine, University of Rochester School of Medicine and Dentistry, Rochester, NY; Department of Medicine, Indiana University—Regenstein Institute, Indianapolis, IN

PURPOSE: To describe the demographic characteristics and procedural practice patterns of combined Internal Medicine-Pediatric (Med-Peds) graduates who sought subspecialty training after residency, and to compare the practices of these subspecialists (SS) with primary care (PC) Med-Peds graduates.

METHODS: In a national survey of 1481 Med-Peds graduates, three questions were asked about 58 procedures: 1) How often do you perform the procedure? 2) Would you have liked more, less or the same amount of training in residency? and 3) Do you wish to perform the procedure more frequently?

RESULTS: An overall response rate of 60% was obtained which included 219 SS and 889 PC respondents. The most common subspecialties are: infectious disease (18%), emergency medicine (9%), rheumatology (9%), cardiology (8%), critical care (7%), gastroenterology (7%), pulmonary (6%) and hematology/oncology (6%). Almost three-fourths indicated board certification in their subspecialty with one-third certified for both internal medicine and pediatrics in their subspecialty. Two-thirds are women, and most practice in cities greater than 50,000 people (85%) in a group setting (94%). Almost half of SS graduates spend greater than 80% of their time in direct patient care and one-third indicated a university based practice. Of the 58 procedures, only 13 are performed by more than half of SS graduates, vs 31 for PC graduates. Five of the top 10 procedures performed by the largest proportion of Med-Peds graduates were similar for SS and PC: chest xray interpretation, EKG interpretation, venipuncture, cerumen removal, and peripheral intravenous catheter placement. For no procedure did more than half of SS graduates desire additional training, in contrast to 16 procedures identified by more than half of PC graduates. Although SS graduates perform fewer procedures, they desired more training during residency in procedures that PC graduates also indicated were important. Eight of the top 10 procedures which both SS and PC graduates desired more training are the same and are shown.

Percent of Med-Peds Graduates Who Desire Additional Procedural Training

Procedure	Lamp	Splint-	Sigmoid-	Nasal	Punch	Joint	Tread-	Remove
Subspecialty	48	41	40	40	37	37	36	ingrown
Primary Care	62	68	65	58	64	62	57	toenail

CONCLUSION: Med-Peds subspecialists appear to perform procedures less frequently than primary care Med-Peds physicians unless the procedure is in their area of expertise. However, physicians who pursue subspecialty careers still indicate a desire to learn primary care procedural skills in residency training.

REGIONAL VARIATION OF DIAGNOSES SEEN BY INTERNISTS IN AMBULATORY CARE SETTING. AB Joffe, JH Oh, AM Eichorn, SA Wartman, Division of General Internal Medicine, Long Island Jewish Medical Center, New Hyde Park, NY

PURPOSE: In the last 20 years concerns with cost-effectiveness and the rise in the penetration of managed care have produced changes in the type of diagnosis that are treated as outpatients. However, these changes may have been disproportional between metropolitan and non-metropolitan areas. The purpose of this study is to determine the regional differences in the most common diagnosis seen by internists.

METHODS: The National Ambulatory Medical Care Survey (NAMCS) is an annual probability sample survey conducted by Centers for Disease Control on patients' office visits. The 1996 NAMCS dataset was used to generate lists of the most common diagnoses seen by internists in outpatient setting in both metropolitan and non-metropolitan areas. Metropolitan and non-metropolitan areas were defined per NAMCS protocol. Sampling weights were applied to generate national estimates of total number of visits. Diagnostic clusters (version 4.1 1998) developed by Schneeweiss et al were used to group diagnostic codes representing similar conditions. Clinically significant difference was defined as a relative difference of at least 50% between metropolitan and non-metropolitan areas.

RESULTS: The estimated number of patient visits to internists was 100,977,712 in 1996. We evaluated the top 12 most common diagnostic clusters, which constituted more than 50% of the reason for the outpatient visits in both metropolitan and rural areas. Six diagnostic clusters were common to both rural and metropolitan areas: hypertension (13.7%), diabetes mellitus (5.1%), acute upper respiratory infection (4.6%), acute lower respiratory infection (4.4%), ischemic heart disease (3.2%), and asthma (2.4%). In non-metropolitan areas chronic diseases were also common: COPD (6.0%), chronic rhinitis (3.1%), lipid disorders (2.3%), and degenerative joint diseases (2.2%). Diagnostic clusters common to the metropolitan areas were general medical exam (5.0%), malignancies (3.5%), acute sprain or

strains (3.0%), sinusitis (2.9%), peptic disease (2.7%), and depression/anxiety/neuroses (2.6%).

CONCLUSION: Our results show a difference between metropolitan and rural areas in terms of the most common diagnosis seen. General medical exams and minor injuries were more commonly seen in the metropolitan areas and could be related to an increased penetration of managed care in these areas. The impact of these differences on health care delivery is unknown and needs to be further studied.

OFFICE EVALUATION OF WOMEN WITH CHEST PAIN—FOCUS ON PREVENTION. DR Judelson, Cardiology, Women's Health Institute, Cardiovascular Med. Group of Southern California, Beverly Hills, CA

PURPOSE: Opportunities for appropriate screening, early and accurate detection and management of coronary heart disease (CHD) in women are frequently missed because risk factor evaluation and modification, diagnosis, and treatment protocols have been based on the traditional non-gender specific model of disease.

This is of particular concern to physicians who routinely care for populations of women at risk for cardiovascular disease.

Participants will receive vital information on identification and treatment of risk factors, the role of diet therapy for both prevention and treatment, symptoms and outpatient diagnostic testing. This seminar will enable physicians to educate their patients about their personal risk factors, understand how to intensify their prevention methods for their patients, and to improve the critical evaluation of potential cardiac symptoms of CHD in women.

Participants will receive a comprehensive packet that includes information on key points discussed for providing complete patient education: presentation summary; references; copies of the slides; case studies; and a patient education brochure, "Guide to Heart Healthy Eating."

LEARNING OBJECTIVES: This program is flexible as it is divided into four segments—risk factors, diet, symptom issues, and diagnostic testing—each with their own learning objectives. Using different modules, the presentation can touch on one or two aspects or provide a comprehensive understanding of CHD in women. Outline of program content:

I. Welcome, introduction of presenter, workshop rationale, review of learning objectives, description of comprehensive participant handout, and appropriate disclosures.

II. Slide lecture:

A. Risk factors and their modification

B. Diet therapy as prevention and treatment

C. Symptoms and treatment issues

D. Diagnostic testing

III. Summary of physician, testing, and patient-related issues

IV. Question and answer session

RESULTS: Our objective is to provide current information on the prevention, detection, and management of CHD, focusing on gender specific guidelines for health care professionals. Participants are expected to share this information with their colleagues and female patients, thereby decreasing the incidence of and mortality from CHD among a large population of women.

CONCLUSION: Participants will be provided with a complete guide they may use in presenting this information to students and physicians.

The material included in the participant handout will be expanded upon by the presenter.

Equipment needs:

Preferred: LCD projector, PC compatible computer (if available), slide projector, 3 carousels, screen, podium with speaker light, lavalier microphone, and laser pointer.

SEVERITY-ADJUSTED OUTCOMES OF INTENSIVE CARE IN A VA AND PRIVATE SECTOR HOSPITALS. PJ Kaboli, MJ Barnett, SM Fuehrer, GE Rosenthal, Univ of Iowa and Iowa City VAMC, Iowa City, IA; Cleveland VAMC, Cleveland, OH

PURPOSE: Compare intensive care unit (ICU) mortality, length of stay (LOS), and treatment intensity in a VA hospital and in private sector hospitals serving a large metropolitan region.

METHODS: We conducted a retrospective study of 1246 consecutive patients admitted to the medical (n = 571) and surgical (n = 675) ICUs of a university-affiliated VA hospital during 11/93–12/96. Clinical and demographic data were abstracted from patients' medical records. Admission severity of illness was measured using the Acute Physiology and Chronic Health Evaluation (APACHE) III methodology. ICU treatment intensity was measured using a modified version of the Therapeutic Intervention Scoring System (TISS), which considered the use of 30 diagnostic and therapeutic interventions in the first 24 hours of ICU care. Observed mortality, LOS, and TISS scores were compared to values that were predicted on the basis of validated multivariable models that were developed in patients in 27 hospitals serving the same metropolitan region in 1993–96. The models accounted for age, gender, comorbidity, APACHE III Acute Physiology scores, admission diagnosis, and admission source.

RESULTS: The mean age of VA patients was 62.6 years, 97.7% were men, and 65% were white. Observed VA mortality was lower than predicted ($p < .001$) in all patients (Table). In stratified analyses, VA mortality was lower for surgical ICU patients, but not medical ICU patients. Mean observed VA ICU LOS was similar in all patients, and in medical and surgical patients when examined separately. Mean

TISS scores were higher in VA patients, suggesting that VA patients received more active treatments during the first day of ICU care.

		All Pts [95% CI]	Medical	Surgical
Mortality	Observed	14.0%*	23.1%	6.4%*
	Predicted	18.1% [16.4–19.9]	25.6% [22.7–28.3]	11.9% [9.8–14.0]
Mean LOS	Observed	4.3	4.7	4.0
	Predicted	4.4 [4.1–4.6]	4.8 [4.4–5.2]	4.0 [3.7–4.3]
Mean TISS	Observed	2.42*	2.69*	2.21*
	Predicted	1.77 [1.60–1.94]	1.92 [1.65–2.20]	1.75 [1.55–1.95]

* $P < .001$

CONCLUSION: Severity-adjusted ICU mortality and LOS at a VA hospital were similar or lower than in private sector hospitals serving the same metropolitan region, although VA patients received more intensive ICU treatment. If generalizable to other VA hospitals, these data suggest that outcomes of ICU care may be comparable in VA and private sector hospitals. The degree to which the lower mortality in VA patients can be attributed to more intensive treatment merits further study.

PHYSICIAN CHARACTERISTICS ASSOCIATED WITH JOB STRESS, PRACTICE OVERLOAD AND COMPROMISED INTERPERSONAL CARE. SH Kaplan, S Greenfield, IB Wilson, S Supran, Family Medicine and Community Health, Tufts University School of Medicine, Boston, MA; Medicine, New England Medical Center, Boston, MA

PURPOSE: In other industries it has long been observed that when the morale of the workforce is compromised the quality of the product suffers. In a 1998 national survey of 6,536 physicians in 20 U.S. cities, we compared physician job stress, morale, time pressures, practice volume and sense of autonomy over practice to quality of physician communication with patients.

METHODS: Practicing physicians were sampled from the national roster maintained by the publishers of the Physicians' Desk Reference. The mean physician age was 51.7; 14.7% were female. We used previously validated multi-item measures of physician job stress ($\alpha = .80$), morale ($\alpha = .87$), time pressures ($\alpha = .85$), autonomy over practice (single item), and quality of patient communication ($\alpha = .89$). Practice overload was measured as the difference between the number of patients seen per hour and the number reported as comfortable to see per hour.

RESULTS: Study physicians reported substantial job stress (mean = 55.5, SD [20.5]) and low morale (mean = 47.1, SD [18.7]). Physicians in the highest quartile of physician-patient communication scored 9.91 points lower on the job stress scale than those in the lowest quartile (effect size $> .50$). In multi-variate models relating quality of communication with patients to job stress, low morale, time pressures, practice autonomy and practice volume overload, all independent variables were statistically significant ($p < .001$). Two physician characteristics, age and gender, increased the explained variance of the model by 46.6%. Women reported similar levels of job stress, morale, time pressures and overload compared with men, but these variables were not significantly associated with quality of patient communication for women physicians. Younger physicians (age < 45) reported significantly more job stress (57.3 vs. 46.0, $p < .001$) and practice overload (.36 vs. .04, $p < .001$) than older physicians (> 60). These differences were related to differences in quality of patient communication.

CONCLUSION: We conclude that, as for other industries, job stress, compromised morale, time pressures and lower autonomy or control over the work environment, are related to lower quality of performance in health care, at least as measured by interpersonal care. Further, the quality of interpersonal care of certain physicians, particularly young men, may be most adversely affected by pervasive changes in the clinical environment, such as increased practice volume and decreased control over practice circumstances.

WHY DON'T PHYSICIANS FOLLOW COMPUTERIZED OUTPATIENT REMINDERS? AS Karson, GJ Kuperman, DG Fairchild, J Fiskio, A Mekala, DW Bates, General Medicine, Brigham & Women's Hospital, Boston, MA

PURPOSE: Guidelines have been developed to assist physician decision making, however abundant data demonstrate that even generally accepted guidelines are often not followed. To improve physician adherence to guidelines we developed a patient-specific computerized outpatient reminder system. Although many reminders were followed, many were not. Structured review of the medical record was used to classify potential reasons why reminders were not followed.

METHODS: Reminders were in 4 categories: 1. diabetes care (eye exams, HbA1C and cholesterol studies, nursing education visits); 2. therapeutic recommendations (ACE-inhibitor use in diabetic hypertensives, and ASA, beta-blocker, and statin use in coronary artery disease(CAD) patients); 3. expensive medication use (NSAIDS, H2-blockers, statins, ACE-inhibitors), and 4. health maintenance (Pap, mammogram, cholesterol, Pneumovax). During a 1 year period reminders were delivered to 197 PCPs affiliated with a tertiary care institution. All reminders were printed on the ambulatory face-sheets that PCPs receive at the time of patient office visits. We assessed how often patients received the recommended care items after reminders were delivered. When recommended care items did not occur, we performed chart reviews on a random sample of records to identify potential reasons why reminders were not followed. We evaluated charts spanning 5 reminder types: HbA1C, diabetic eye exams, CAD and no ASA, CAD and no Beta-Blocker, and Expensive ACE-Inhibitor.

RESULTS: Including all 16 types of reminders, 55,215 reminders were delivered and in 17,918 instances the recommended action occurred shortly after the visit; in 37,297 instances the action did not. For the 5 reminder types we reviewed, 5,930 reminders were delivered; 1,557 subsequent actions were taken. Among the 4373 times in which actions did not occur; a stratified random sample of these instances (650 records) was reviewed; findings fell into several categories. For CAD and no ASA reminders: 46% of the time there was a potential contraindication to ASA (30% of time patient on warfarin); 25% of time no reason could be found for not adhering to guideline, 13% of time patient was actually on ASA, 8% of time patient ASA allergic, 6% of time the patient did not have CAD. For diabetic eye exam reminders: 47% of the time no reason could be found for not adhering to guideline, 22% of time exams were planned but not done, 19% of time exam done >2 months after reminder, 2% of time patient did not have DM.

CONCLUSION: The reasons for failure to adhere varied dramatically by guideline. This analysis demonstrates a number of opportunities for improvement including: better reminder logic, improved capture of coded data, and development of systems to ensure that planned tests actually get accomplished.

CORRELATES OF GOOD GLYCEMIC CONTROL IN PRIMARY CARE. AS Karson, EF Cook, AL Puopolo, JS Hass, TA Brennan, HR Burstin, General Medicine, Brigham & Women's Hospital, Boston, MA; General Medicine, UC San Francisco, San Francisco, CA

PURPOSE: While studies have increasingly demonstrated the benefits of good glycemic control for Type II diabetes mellitus (DM), little is known about glycemic control in primary care. In this study, we examined rates and correlates of glycemic control.

METHODS: All patients aged 20–75 with at least one visit in the prior year to a staff physician at 11 ambulatory care sites were randomly selected for chart review and survey. Chi-square and T-tests were used to identify associations between good glycemic control and sociodemographic factors, health status, continuity of care, non-PCP management of DM, patient satisfaction, and compliance with DM process measures (eye exams, foot care, cholesterol and urine protein studies). Multivariable analyses were used to assess independent correlates of good glycemic control.

RESULTS: Of the 2858 respondents (69% response rate), 374 had Type II DM. Overall, 44.6% patients achieved good glycemic control (glycosylated hemoglobin < 8.0). After adjusting for patient age, sex, site of care, and insulin or metformin use, patients with <high school education were less likely to have good glycemic control OR 0.21 (0.07, 0.67). Compliance with DM process measures was not correlated with glycemic control or a provider other than the PCP managing the patient's diabetes care.

CONCLUSION: Patients in selected primary care practices were not consistently achieving acceptable levels of glycemic control. Compliance with the DM process measures often used as proxies of quality diabetes care was not correlated with good control. Targeted initiatives for poorly educated diabetics in primary care may improve glycemic control.

PATIENT-SPECIFIC COMPUTERIZED OUTPATIENT REMINDERS TO IMPROVE PHYSICIAN COMPLIANCE WITH CLINICAL GUIDELINES. AS Karson, GJ Kuperman, J Horsky, DG Fairchild, JM Fiskio, DW Bates, General Medicine, Brigham & Women's Hospital, Boston, MA

PURPOSE: Clinical guidelines have been developed to assist physician decision making, although even generally accepted guidelines are often not followed. In order to improve adherence to guidelines we developed, implemented, and assessed the impact of a patient-specific computerized outpatient reminder system.

METHODS: A multidisciplinary team created reminders in 4 categories: 1. diabetes care (eye exams, HbA1C and cholesterol studies, nursing education visits); 2. therapeutic recommendations (ACE-inhibitor use in diabetic hypertensives; aspirin, beta-blocker, and statin use in coronary artery disease (CAD); 3. expensive medication use (for patients on NSAIDs, H2-blockers, statins, ACE-inhibitors); 4. health maintenance (Pap, mammogram, cholesterol, Pneumovax). For reminder categories 1–3 the 197 PCPs affiliated with a tertiary care institution were randomized into control and intervention groups; control groups did not get reminders. For category 4 all PCPs received reminders. All reminders were printed on face-sheets that PCPs receive at the time of patient visits. The intervention lasted for 1 year. For reminders in categories 1–3, we compared, using generalizing estimating equations, how often patients received the recommended care items after office visits with intervention versus control PCPs. For reminders in category 4 we compared, using T-tests, the percentage of patients in compliance with the recommended item at the beginning versus the end of the study.

RESULTS: Diabetics seeing intervention PCPs more often received overdue HbA1C studies (60 vs. 46%, $p < 0.0001$), cholesterol studies (39 vs 20%, $p < 0.0001$), eye exams (17 vs 10%, $p < 0.04$), and nursing visits (3.1 vs 1.4%, $p < 0.64$). Diabetic hypertensive patients seeing intervention PCPs were started on ACE-inhibitors more frequently (14 vs. 8%, $P < 0.01$); CAD reminders did not lead to statistically significant differences between groups. Reminders led to switching expensive medications to lower cost alternatives in the ACE-inhibitor (4.9 vs 2.7%, $p < 0.04$), H2-blocker (5.4 vs. 0.8%, $p < 0.001$) and statin (7.3 vs 4.4% $p = 0.15$) classes. Expensive NSAIDs were not switched to inexpensive NSAIDs at a significantly increased rate however they were discontinued more often in the intervention group (10.9 vs 6.0%, $p < 0.02$). Health maintenance reminders led to increased compliance rates in all areas studied (all $p < 0.001$).

CONCLUSION: Computerized patient-specific outpatient reminder systems significantly improve physician compliance with many recommended guidelines. Patterns of adherence varied by guideline and reminders for lab tests and ophthalmology referrals were followed more often than those related to medication prescribing.

IMPLEMENTATION OF SMOKING CESSATION GUIDELINES IN PRIMARY CARE: THE ROLE OF CLINIC NURSES. DA Katz, DR Muehlenbruch, J Martin-McKay, N Zillmer, MC Fiore, TB Baker, Dept of Medicine; Center for Tobacco Research and Intervention, U. Wisconsin, Madison, WI

PURPOSE: Based on a systematic literature review, the AHCPR Smoking Cessation Guideline describes counseling and treatment approaches that have been shown to be effective in randomized controlled trials. One of the key findings of the guideline is that counseling by nurses (and the involvement of multiple types of providers) is associated with significantly higher quit rates. This study describes 1) the participation of nurses and medical assistants (MAs) in the routine identification and counseling of smokers and spit tobacco users at each clinic visit, and 2) overall concordance with AHCPR guideline recommendations approximately 3 years following their dissemination.

METHODS: We conducted exit interviews of 4071 consecutive adult patients who presented for routine, non-emergency care at 8 community-based primary care practices (7FP, 1IM) during the pre-intervention phase of a controlled guideline implementation trial.

RESULTS: Overall concordance with guideline recommendations at the time of the clinic visit is shown in the table (% concordance by nurses/MAs in parentheses):

AHCPR Guideline-recommended Intervention	Cigarette smokers (N = 784) % Concordance	Spit tobacco users (N = 64) % Concordance
Ask about tobacco use*	58 (35)	31 (16)
Ask about willingness to quit*	23 (5)	8 (0)
Advise pt to quit*	28 (5)	16 (3)
Help pt to set quit date	2 (0)	2 (1)
Discuss pharmacotherapy*	11 (<1)	3 (0)
Arrange follow-up	3 (0)	2 (0)

*p ≤ 0.05 for comparison between smokers and spit tobacco users

CONCLUSION: The majority of smokers who present for routine primary care in our region are asked about their smoking status; nurses (and MAs) are integrally involved in this assessment. Of those patients who receive any counseling, the primary care clinician is typically the only team provider to offer advice to quit. Spit tobacco use is generally not assessed routinely and few users receive any counseling. Increased involvement of nurses in the implementation of smoking (and spit tobacco) cessation guidelines requires clinic leadership and support, identification of cessation as a high priority preventive care objective, and training of nurses in effective brief intervention techniques.

THE GAP BETWEEN EVIDENCE AND PRACTICE: FACTORS ASSOCIATED WITH THE EMERGENCY DEPARTMENT TRIAGE OF PATIENTS WITH POSSIBLE ACUTE CARDIAC ISCHEMIA. DA Katz, JR Beshansky, JL Griffith, PS Rahko, HP Selker, Dept of Medicine, University of Wisconsin, Madison, WI; Division of Clinical Care Research, Tufts-New England Medical Center, Boston, MA

PURPOSE: To improve the quality and cost-effectiveness of medical care, the AHCPR Unstable Angina (UA) Guideline stratifies patients into 3 risk groups and recommends outpatient management for patients in the low risk group. To gain further insight into emergency department (ED) decision making, we identified patient characteristics that predicted concordance of actual triage with guideline-recommended triage.

METHODS: Guideline criteria for ED triage decisions were retrospectively applied to a patient sample derived from a prospective clinical trial. We included 2245 patients who presented to the ED of 10 hospitals with symptoms suggestive of acute cardiac ischemia (ACI) in 1993. We abstracted medical records and electrocardiograms (ECGs) to classify patients based on the guideline. Within each risk group, we used logistic regression models to predict concordance of actual triage decisions with the guideline.

RESULTS: Application of the guideline showed that $142/302$ (47%) of low-risk patients were triaged to home and $163/1943$ (84%) of intermediate-high risk patients were admitted to telemetry or CCU, as recommended. Among low-risk pts, increased age, male sex, white race, prior myocardial infarction, and private insurance predicted hospital admission; women and blacks were more likely to be triaged to home (OR = 2.75, $p = 0.001$; OR = 1.86, $p = 0.04$, resp). Among intermediate-high risk pts, increased age, known coronary artery disease, hypercholesterolemia, and ST-segment shifts ≥ 1mm predicted admission to a telemetry/CCU bed, as recommended; blacks were less likely to be triaged in agreement with the guideline (OR = 0.76, $p = 0.04$).

CONCLUSION: There are significant discrepancies between observed and recommended ED triage decisions in pts with symptoms suggestive of ACI. Whether these discrepancies are attributable to deficiencies in clinical practice, deficiencies in the guideline, or other factors remains to be understood. However, certain important risk factors known to be associated with adverse short-term outcomes (diabetes, congestive heart failure) did not appear to impact upon triage decisions,

whereas other factors without prognostic importance (gender, race, insurance) were incorporated in these decisions. Cognitive feedback to contrast how ED physicians use certain cues with how well these cues actually predict ACI-related short term outcome may improve ED decision making.

TREATMENT OF EARLY STAGE BREAST CANCER—DOES SEEING A MEDICAL ONCOLOGIST PRIOR TO SURGERY INFLUENCE TYPE OF SURGERY AND SATISFACTION?

NL Keating, JC Weeks, MB Landrum, E Guadagnoli, Dana-Farber Cancer Institute, Brigham and Women's Hospital; Department of Health Care Policy, Harvard Medical School, Boston, MA

Previous studies suggest that medical oncologists are more likely than surgeons to favor breast conserving surgery (BCS) over mastectomy for treatment of early stage breast cancer. We examined whether discussing surgical options with a medical oncologist was related to type of surgery and satisfaction with treatment decision-making in two states where we have demonstrated differences in rates of BCS.

We identified women with stage 1 or 2 invasive breast cancer at 17 Massachusetts hospitals and 30 Minnesota hospitals. Rates of BCS among women eligible for BCS were 74% in MA and 48% in MN. We abstracted type of surgery and other clinical characteristics from medical records. We surveyed 792 women in MA (61%) and 1633 women in MN (83%) two months after surgery and asked about the number of physicians with whom they discussed treatments, the specialty of the physicians, and, for the 3 physicians they considered most important, if surgical treatments were discussed prior to final decisions about surgery. We also asked about satisfaction with treatment decision-making. We assessed the effect of discussing surgical options with a medical oncologist on both type of surgery and satisfaction using propensity score analysis to adjust for patient, hospital, and surgeon characteristics.

Women in MA saw a mean of 3.5 physicians compared to 2.8 in MN ($P < 0.001$). 58% of women in MA and 28% in MN discussed surgical options with a medical oncologist ($P < 0.001$). In MA, discussing surgical options with a medical oncologist was not related to type of surgery but was associated with greater satisfaction with treatment decision-making (Table). In MN, where the rate of BCS was much lower, discussing surgical options with a medical oncologist was associated with BCS but was not associated with greater satisfaction.

The effect of discussing surgical options with a medical oncologist on treatment and satisfaction is complex. When BCS is standard therapy in a community (MA), seeing a medical oncologist does not influence treatment but is associated with greater satisfaction, possibly due to providing women with more information. When BCS is not standard care, seeing a medical oncologist is associated with greater use of BCS but does not affect satisfaction, possibly because surgeons' and medical oncologists' recommendations conflict.

Outcome	Massachusetts OR (95% CI)	Minnesota OR (95% CI)
Adjusted odds of BCS	1.2 (0.8 to 1.8)	1.7 (1.1 to 2.6)
Adjusted odds of reporting that very satisfied with treatment decision-making	1.6 (1.1 to 2.3)	0.8 (0.6 to 1.1)

HOW ARE PATIENTS' EXPERIENCES DURING THE CLINICAL ENCOUNTER RELATED TO TRUST AND CONSIDERATIONS OF CHANGING PHYSICIANS?

NL Keating, DC Green, AC Kao, JA Gazmararian, YV Wu, PD Cleary, Division of General Medicine, Brigham and Women's Hospital; Department of Health Care Policy, Harvard Medical School, Boston, MA; USQA Center for Health Care Research™, Atlanta, GA; the American Medical Association, Chicago, IL

To better understand the effect of specific experiences during patient-physician interactions, we asked patients about ambulatory care experiences and assessed the relationships between problem experiences and both patients' trust in their physicians and whether they had considered changing physicians.

We surveyed 2086 patients (response rate 61%) insured by a large national health insurer who had at least one visit with a primary care provider in 1995. To assess problem experiences, we asked if their physician always (1) gives them enough time to explain the reason for the visit (2) gives answers to questions that are understandable (3) takes enough time to answer questions (4) asks about how their family or living situation affects their health (5) gives as much medical information as they want, and (6) includes them in decisions as much as they want; and we identified a problem experience if their physician did not. We also asked about trust in their physicians (using a 10-item scale) and if they had considered changing their doctor due to concerns about their care. We used linear regression to assess the effect of problem experiences on trust and logistic regression to assess the effect of problem experiences on considering changing physicians. In both models, the independent variables of interest were the 6 problem experiences, and control variables included patient, physician and plan characteristics.

Most (78%) patients reported at least one of the six problem experiences noted above and 49% reported at least two. Patients with more problems had lower trust scores ($P < 0.001$) and more often reported having considered changing physicians ($P < 0.001$). In the linear regression model, each of the six problems remained independently associated with lower levels of trust (all $P < 0.005$). In the logistic regression model, three of the problem experiences were independently related to considering changing physicians. Patients whose physicians did not always give enough time to explain the reason for the visit (OR 1.9; 95% CI 1.3 to

2.9), did not always take enough time to answer questions (OR 3.2; 95% CI 2.1 to 5.0), and did not always give enough medical information (OR 4.0; 95% CI 2.4 to 6.9) more often considered changing physicians.

A majority of patients report problem experiences in the ambulatory setting. These experiences are strongly related to lower levels of trust in their physicians and several are also strongly associated with considering changing doctors. Efforts to improve patients' experiences may promote more trusting relationships and greater continuity and therefore should be a priority for physicians and health care organizations.

MEASURING ANTIDEPRESSANT PRESCRIBING PRACTICE IN A HEALTH CARE SYSTEM: IMPLICATIONS FOR QUALITY MEASUREMENT.

EA Kerr, EA McGlynn, KA Van Vorst, SL Wickstrom, VA Center for Practice Management and Outcomes Research, Ann Arbor VA Healthcare System; Department of Medicine, University of Michigan, Ann Arbor, MI; Center for Research on Quality in Health Care, RAND Health, Santa Monica, CA; Ingenix; Center for Health Care Policy and Evaluation, UnitedHealth Group, Minneapolis, MN

PURPOSE: To determine how different specifications in the construction of quality of care measures from administrative data regarding depression treatment influence conclusions about the adequacy of antidepressant prescribing practices.

METHODS: We selected all adult members of two United Healthcare plans who had at least one outpatient or inpatient claim with a diagnosis of depression during the years 1993-1995 and were continuously enrolled for 12 months. Pharmacy claims data were used to construct measures of duration of treatment, dose, and type of antidepressant. We examined the effects of two different definitions of a new episode of depression (using a 4-month versus 9-month clean period prior to the index depression visit) on the likelihood of being treated with antidepressants at therapeutic doses. Further, we compared two different ways of identifying an episode of depression (requiring 1-visit versus 2-visits with a code for depression) with respect to their influence on antidepressant prescribing practices (dose and duration). We also examined whether antidepressant type was related to the likelihood that antidepressants were prescribed at therapeutic doses.

RESULTS: The duration of the clean period was not related to conclusions about antidepressant prescribing practices. Patients with two or more visits with depression diagnosis codes were significantly more likely to receive antidepressants than those with only one visit and were more likely to receive therapeutic doses at each time period (86% vs 72% at 1 month of treatment and 23 vs 15% at 5 months, $p \leq 0.0001$). Among persons receiving antidepressants, those receiving selective serotonin reuptake inhibitors (SSRIs) were more likely to receive therapeutic doses and to continue treatment for at least five months than those prescribed other classes of antidepressants. In multivariate analysis, being prescribed an SSRI versus another class of antidepressants was significantly associated with receiving both one month (OR = 7.3 [5.7-9.3]) and five months (OR = 2.0 [1.6-2.5]) of therapeutic treatment.

CONCLUSION: Our results indicate that CONCLUSION regarding appropriateness of antidepressant prescribing can vary markedly depending on how the quality measure is specified. Given that administrative data are and will continue to be used for both monitoring and quality improvement purposes in the short run, it is critical that we understand how variations in measurement specifications influence the CONCLUSION that are drawn about treatment of depression in health plans.

ASSOCIATION OF ELEVATED SERUM PHOSPHORUS AND PROGRESSIVE HEART VALVE CALCIFICATION IN CHRONIC HEMODIALYSIS PATIENTS.

S Khan, S Joseph, R Garrick, P Visintainer, Medicine; Nephrology; Basic Sciences, WMC, Valhalla, NY

PURPOSE: Cardiovascular disease is the major cause of mortality in ESRD >45% (USRDS 97). Ischemic and calcific valvular heart diseases are implicated.

METHODS: A retrospective case control study was done to evaluate the role of serum Calcium [Ca], phosphorous [Po4], Ca-Po4 product, inorganic parathyroid hormone [iPTH] and age in progressive heart valve calcification in chronic Hemodialysis [HD] patients. Fifty-five (55) patient's charts and echos were reviewed. The study group (SG-n = 37) was defined as patients on HD with progressive heart valve calcification, and control group (CG-n = 18) was defined as patients on HD with no progression of heart valve calcification. Inter echo interval was 6-72 months for both the groups. Baseline values for serum Ca, Po4, iPTH were obtained by averaging the values for the 3 months period prior to the first echo. In the SG, 43% were female, with 39% female in the CG ($p = 0.78$). The mean age in the SG was 69(+/-12.8) yrs and 59.6 (+/-11.9) yrs in CG ($p = 0.011$). Mean base line serum calcium was 9.5 (+/-0.7)mg/dl in SG, compared to 9.3 (+/-1.2)mg/dl in CG ($p = 0.42$). Mean base line serum Po4 was 6.6 (+/-0.2) mg/dl in SG, compared to 4.5 (+/-0.3) mg/dl in CG (p less than 0.001). Mean Ca-Po4 product was 62.6 (+/-10.3) in SG, compared to 41.0 (+/- 10.2) in CG (p less than 0.001). Median iPTH was 139 pg/ml in SG and 97 pg/ml in CG ($p = .41$). Median duration on HD prior to the 1st echo was 10 months in SG and 1.5 months in CG ($p = 0.41$).

RESULTS: Statistical analysis was done using T-test, chi-square, and receiver operating characteristic [ROC] curves and logistic regression models (adjusted for age and duration on HD). Using ROC curves, AUC (area under the curve) was 0.59 for serum calcium, 0.90 for serum Po4, 0.96 for Ca-Po4 product and 0.73 for age. With logistic regression models odds ratio for serum Po4 was 9.62 (p less than 0.001), odds ratio for serum Ca was 1.07 (p less than 0.85) and odds ratio for Ca Po4 product was 1.30 (p less than 0.001).

CONCLUSION: These data demonstrates that an elevated serum Po4 at the time of base line echo is a strong independent predictor of progression of heart valve calcification in chronic HD patients. Ca-Po4 product derives its influence from serum Po4. Age is an independent predictor of progressive heart valve calcification but serum Ca, iPTH and duration on HD are not.

IMPROVED CARE FOR PATIENTS WITH UNSTABLE ANGINA AFTER NATIONAL GUIDELINES RELEASE: RELATIONSHIP WITH SOCIOECONOMIC CHARACTERISTICS IN A MULTIHOSPITAL LONGITUDINAL STUDY. Cl Kiefe, NW Weissman, JJ Allison, E Funkhouser, J Canto, R Centor, Medicine; Epidemiology, University of Alabama at Birmingham, Birmingham, AL

PURPOSE: While clinical practice guidelines have flourished recently, their effectiveness in improving care remains unproven. In 1994, the Agency for Health Care Policy and Research released evidence-based guidelines on the diagnosis and management of unstable angina. We studied adherence to these guidelines prior to release, and then again twice after release, for the same hospitals. In particular, we investigated possible differences in longitudinal change by patient ethnicity, sex, and insurance status.

METHODS: For 22 hospitals in one state, we used automated ICD-9 code criteria and then a guideline-based rigorous clinical algorithm to identify random samples of Medicare patients admitted with unstable angina in '93 (preguideline release in '94), '95, and '97. Centrally trained abstractors reviewed the complete medical records. We used guideline-based criteria for patients to receive, if indicated, aspirin (ASA) within 24 hours of admission, beta blockers, and heparin.

RESULTS: Patients had a mean age (SD) of 72 (10) years, 11% were African-American (AA), 55% female, and 16% were also on Medicaid (used as a marker for low income). The table shows performance on the indicators (% adherence), by year, overall and for Medicaid patients only:

YEAR	N	Heparin All pts*	McAid pts*	Beta block All pts*	McAid pts	ASA 24hr All pts*	McAid pts*
1993	605	34.6	16.7	33.7	34.1	52.3	36.2
1995	1334	38.3	30.0	41.7	48.1	58.3	47.2
1997	1113	42.7	32.7	55.6	50.0	71.7	57.9

*P < .05, * < .005 for time trend across years.

Performance for Medicaid patients tended to start out lower and remain lower. Similar patterns were noted for AAs versus whites and, to a lesser extent, for women versus men. For example, for AA women, adherence to beta blockers went from 32% to 38% to 46%. Multivariable adjustment for hospital characteristics, and for patient clustering within hospitals, did not change these contrasts.

CONCLUSION: Care improved from the year before to the year after release of guidelines, and was even more improved two years later. This was also true for low-income, minority, and female patients, although differences for disadvantaged populations persisted. The extent to which the improvements are attributable to guideline release is the object of further study. Nevertheless, care is evolving towards guideline adherence.

ADHERENCE TO NATIONAL CHOLESTEROL EDUCATION PROGRAM GUIDELINES IN A GENERAL INTERNAL MEDICINE PRACTICE. BD King, B Tafari, TW Gress, LJ Goebel, Internal Medicine, Marshall University, Huntington, WV

PURPOSE: Because of previous reports in the literature of poor physician compliance with National Cholesterol Education Program (NCEP) guidelines, we designed a retrospective cohort study to assess patients' achievement of their target low density lipoprotein cholesterol (LDL-C) and physician compliance with NCEP guidelines in our practice.

METHODS: Investigators randomly selected 70 patients with a diagnosis of hyperlipidemia seen in our academic Internal Medicine practice in 1996 and followed them for 2 years. We collected data including demographics, past medical history, medication use, and lipid profiles by chart review. We assessed compliance with NCEP guidelines by 1) identifying those patients that reached their target LDL-C based on risk profile and 2) determining the number of cholesterol management changes made during the study period.

RESULTS: Of 70 patients reviewed thus far, 57 met eligibility criteria for the study. At baseline, the cohort had the following characteristics: 94% white; 47% male; average body mass index 31.7 kg/m² (sd = 5.7 kg/m²); average cholesterol 219 mg/dl (sd = 27 mg/dl); 95% had health insurance; 26% had coronary heart disease; and 61% had two or more cardiac risk factors. Lipid analysis was performed at least once during the study on 68% of patients. Of these, 54% attained their target LDL-C. Patients without coronary heart disease were more likely to be at their target LDL-C compared to those with coronary heart disease, but the difference was not significant (56 vs. 33%; p = 0.17). Physicians had 35 opportunities during the study period to make adjustments in dietary or medication management for those patients not at their target LDL-C. Eighteen management changes (51%) were made, with 60% and 48% taking place in those with and without coronary heart disease, respectively.

CONCLUSION: We found that only half of our patients with a diagnosis of hyperlipidemia were at their target LDL-C. Additionally, physicians missed half of the opportunities to adjust cholesterol management. Further study of an intervention to improve cholesterol management in our practice, such as feedback of the results of this study to our physicians, is needed.

LOW RATE OF INFLUENZA VACCINATIONS AMONG HEALTH CARE PERSONNEL: A NATIONAL STUDY. WD King, SJ Woolhandler, DU Himmelstein, DH Bor, Department of Medicine, The Cambridge Hospital/Harvard Medical School, Cambridge, MA

PURPOSE: Randomized controlled trial and other evidence indicates that flu shots for health care workers can decrease nosocomial infection and inpatient mortality; reduce absenteeism and provide model behavior for patients to be immunized. Previous studies have described low vaccination rates of staff at individual health facilities. We present the first national analysis of flu shot receipt by health care workers.

METHODS: We utilized the 1997 National Health Interview Survey (NHIS), a nationally representative sample of the civilian noninstitutionalized US population. We analyzed receipt of flu shots within the past year among persons currently employed by hospitals or other health care settings. We used weights supplied by NHIS to allow extrapolation to the US population.

RESULTS: NHIS surveyed 2204 health care workers. After weighting, this represented 5.09 million hospital workers and 6.86 million workers in other health care settings. Only 35% of all health care workers had received a flu shot within the past year (as compared to 26% of all employed persons). Hospital workers were more likely than workers in non-hospital health care settings to have received a flu shot 38.0% versus 32.8%.

Receipt of flu shots increased with age: 30.5% of persons less than 45 years of age received a flu shot as compared to 43.6% of persons age 45-64 and 54.8% of persons age 65 and older. Men were more likely than women to have received a flu shot, 38.3% versus 34.1%. Persons with a family income of \$20,000 or more were more likely to have had a flu shot than persons with lower family incomes, 36.0% versus 29.8%.

Among occupational groups, health diagnosing professionals (mostly physicians) were most likely to receive a flu shot 46.1%. At 36.6%, flu shot rates for health assessing and treating professionals (mostly registered nurses) were little better than for health aides (33.1%), technicians/technologists (35.5%) and cleaning/maintenance personnel (35.5%).

Receipt of flu shots seemed to correlate with the employee's personal access to health care. Individuals who had visited a physician within the past year were more likely to have had a flu shot than those with no physician visit, 37.1% versus 25.8%. 67.6% of individuals who had ever received pneumonia vaccine had received a recent flu shot.

CONCLUSION: The national flu shot rate for health care workers is only 35%. Health care workers who are poorer and have less access to care have even lower rates. This is not a site-specific problem but a national one. Exhortation has not caused a majority of health care workers to comply with recommendations. The focus must change to education programs, stringent accreditation and regulatory standards, and improving low-income health workers' personal access to care.

PRIMARY CARE PHYSICIANS' ATTITUDES AND BELIEFS REGARDING THE DETECTION OF DEPRESSION AND DISTRESS IN PRIMARY CARE PATIENTS. JC Klapow, RM Centor, LA Renkl, MA Stafford, LL Casebeer, University of Alabama School of Medicine, Birmingham, AL

PURPOSE: Understanding primary care physicians' attitudes and beliefs concerning psychosocial issues should inform content development in medical education. The present study characterized physicians' attitudes and beliefs about the detection of depression and distress across 4 domains: Assessment Strategies, Assessment Confidence, System Barriers, and Comfort with Uncertainty.

METHODS: The subjects were 49 primary care physicians from the southeastern United States who volunteered to participate in a medical education program focused on the detection of depression and distress. We administered a 56 item survey (The Physician Belief Scale, Physician Reaction to Uncertainty Scale, and eleven items assessing attitudes towards health care system barriers). We recorded responses on a Likert scale (1 = disagreement and 5 = agreement).

RESULTS: Survey respondents included 30 males and 19 females. 22 were trained in Internal Medicine, 25 in Family Practice and 2 in other fields. Subjects had an average age of 45 years (30-72) and were in practice for an average of 16.7 years (sd = 9.6). Responses to selected items from each scale illustrate the results. Responses varied greatly within each of the four domains: Assessment: (#4) importance of ruling out organic disease prior to psychosocial assessment; (#12) burden of evaluating and psychosocial problems; (#17) scarcity of time to evaluate psychosocial problems. Confidence in assessment: (#34) confidence preparation to assess. Health Care System Barriers: (#42) perceptions of inadequate reimbursement for mental health services; (#45) lack of confidence in mental health care quality. Uncertainty: (#51) comfort with medical uncertainty.

Question #	4	12	17	34	42	45	51
1 = Disagree	18%	31%	27%	8%	43%	16%	4%
2	18%	27%	22%	27%	28%	33%	33%
3	31%	12%	18%	10%	4%	27%	22%
4	23%	24%	25%	43%	8%	18%	29%
5 = Agree	10%	6%	8%	12%	17%	6%	12%

CONCLUSION: While primary care physicians are often viewed as a monolithic group, this survey indicates considerable diversity in attitudes and beliefs across the 4 domains. These differences occurred in a primary care physician sample at risk for selection bias because of a stated interest in depression and distress. Characterizing primary care physicians as a homogeneous group may not accurately identify specific continuing medical education needs regarding the assessment and management of psychosocial problems.

IS QUALITY OF IN HOSPITAL CARE OF PATIENTS ADMITTED FOR CONGESTIVE HEART FAILURE INFLUENCED BY AGE ?

MP Kossovsky, FP Sarasin, TV Perneger, P Chopard, J-M Gaspoz, Internal Medicine; Quality of Care Unit; Groupe de Recherche et d'Analyse en Services et Soins Hospitaliers (GRASSH), Hopitaux Universitaires de Geneve, Geneva, Switzerland

PURPOSE: To examine if suboptimal care of patients admitted for congestive heart failure (CHF) is linked with age.

DESIGN: Retrospective chart review. Population: a random sample of patients hospitalized with a principal diagnosis of CHF, discharged alive between 1996 and 1998. Dependent variable: explicit criteria assessing the quality of processes of care (Ashton et al., Med. Care 1994;32:755-70), grouped into 3 scores: admission work-up; evaluation and treatment during the stay; readiness for discharge. The association between age, clinical data, length of hospital stay and quality of care was analyzed in linear regression models.

RESULTS: 371 patients were included in the study. There was no relationship between age and the scores of quality of the admission work-up and of readiness for discharge. However, the proportion of criteria met for evaluation and treatment during the stay decreased with age:

Age	Proportion of criteria met		95% CI	p
	Mean	Decrease Reference		
<70	71.3%	—	—	—
70-79	68.7%	-2.6%	+1.9 to -7.1	0.26
80-89	62.6%	-8.7%	-4.3 to -13.0	<0.001
≥90	52.2%	-19.0%	-11.5 to -26.6	<0.001

In the multivariate model, admission to an intensive care unit and length of hospital stay were positively related to the proportion of criteria for evaluation and treatment during the stay that were met; however, this did not significantly modify the negative relationship with age.

CONCLUSION: Greater age was significantly associated with less adequate in-hospital management of patients with CHF. This result is disturbing because the incidence of CHF increases with age and CHF is among the leading causes of hospitalization among the elderly.

ASSOCIATION BETWEEN LENGTH OF STAY AND READINESS FOR DISCHARGE IN PATIENTS WITH CONGESTIVE HEART FAILURE.

MP Kossovsky, TV Perneger, FP Sarasin, M Louis-Simonet, J-M Gaspoz, Department of Internal Medicine; Quality of Care Unit; Groupe de Recherche et d'Analyse en Services et Soins Hospitaliers (GRASSH), Hopitaux Universitaires de Geneve, Geneva, Switzerland

PURPOSE: To examine if readiness for discharge of patients admitted for congestive heart failure (CHF) is linked with length of stay (LOS).

METHODS: Design: Retrospective chart review. Population: a random sample of patients hospitalized at the Department of internal medicine of the Geneva University Hospitals, Switzerland, with a principal diagnosis of CHF, discharged alive between 1996 and 1998. Dependent variable: explicit quality criteria assessing readiness for discharge (Ashton et al., Med. Care 1994;32:755-70). Adjustments for age and clinical data was made by means of a linear regression model.

RESULTS: 371 patients were included in the study. Among them, 137 patients stayed less than 10 days, 234 stayed 10 days or more. The proportion of quality criteria assessing readiness for discharge steadily increased during each hospital day until day 10 and then reached a plateau.

Length of stay	Proportion of criteria met		95% CI Change for each day	p
	Constant			
Less than 10 days	52.7 %	+ 3.20 %	1.78 to 4.62	<0.001
10 - 29 days	82.7 %	- 0.20 %	-0.67 to 0.28	0.42

In the multivariate model, living alone at home was positively related to the proportion of quality criteria met assessing readiness for discharge, while non sinus cardiac rhythm on admission affected this proportion negatively; however, both these relationships did not significantly modify the association with LOS.

CONCLUSION: CHF patients with LOS <10 days are less likely to meet explicit quality criteria of readiness for discharge than patients staying longer. Therefore, because insufficient readiness for discharge may have a negative impact on patient outcomes, reductions in LOS in these patients should be accompanied by measures assuring that readiness for discharge is reached faster.

IMPACT OF A NURSE-MANAGED ANTICOAGULATION CLINIC IN A STAFF-MODEL HMO.

DS Kountz, S Blumenthal, T Miller, D Smith, Robert Wood Johnson Medical School, New Brunswick, NJ

PURPOSE: To determine the impact of a nurse-managed anticoagulation clinic (AC) vs usual care (UC) on the management of patients on oral anticoagulation therapy in a staff-model HMO.

METHODS: Retrospective chart review of 108 patients (70 in AC, 38 in UC) over 12 month period from 1997-1998 using a standardized worksheet.

RESULTS: Charts were reviewed of patients who were seen at a staff-model HMO with approximately 20,000 members in central New Jersey. Enrollment of patients in the AC was voluntary at the discretion of the attending physician. All patients had an indication for warfarin based on recommendations of the American College of Chest Physicians Consensus Conference on Antithrombotic Therapy. A

total of 1296 patient months were reviewed. The mean age of patients in both groups was 70. There was no statistical difference in mean prothrombin time (PT) or international normalized ratio (INR) between groups. In addition, there were no differences in indications for anticoagulation between groups. Although there was no obvious indication of complications associated with under- or excessive-coagulation in the charts, data related to INR monitoring was either missing or uninterpretable in the UC patient charts twice as likely as in the AC charts. One of the differences in the charts was the utilization of an anticoagulation flowsheet developed by the medical group. It was present in all of the AC charts but either not present or not utilized by physicians in the UC group.

CONCLUSION: Nurse or pharmacist-directed anticoagulation clinics have been shown to facilitate close monitoring of patients on oral anticoagulation therapy in various clinical settings. While there was a trend toward more consistent monitoring in the AC group, it was not statistically significant. Despite the development of an anticoagulation flowsheet patient information related to therapy was often missing. Other health-care providers would have difficulty managing anticoagulation for a patient if that physician were unavailable, increasing the likelihood of complications. The presence of a nurse-directed anticoagulation clinic is one strategy to address this problem.

APPROPRIATENESS OF EARLY CORONARY ANGIOGRAPHY AFTER ACUTE MYOCARDIAL INFARCTION: COMPARISON OF EXPERT PANEL RATINGS AND A DECISION-ANALYTIC MODEL.

KM Kuntz, J Tsevat, MC Weinstein, L Goldman, Department of Health Policy and Management, Harvard School of Public Health, Boston, MA; Department of Medicine, University of Cincinnati, Cincinnati, OH; Department of Medicine, University of California, San Francisco, CA

PURPOSE: We previously compared the appropriateness of coronary angiography as judged by expert panel ratings vs. decision-analytic models for the post-discharge period for patients after acute myocardial infarction. The purpose of this study was to compare expert panel ratings vs. decision-analytic models for angiography prior to hospital discharge.

METHODS: We constructed a decision-analytic model based on the clinical variables utilized by an expert panel to develop appropriateness ratings for the use of pre-discharge coronary angiography after myocardial infarction. Variables considered were age, use of thrombolytic therapy, and whether or not the infarction was complicated by persistent chest pain, mitral regurgitation or ventricular septal defect, cardiogenic shock, persistent pulmonary edema, severe left ventricular dysfunction, stress-induced myocardial ischemia, or recurrent ventricular tachycardia or fibrillation at least 24 hours after myocardial infarction. We used the decision model to simulate the 32 clinical scenarios evaluated by the panel to calculate the gain in quality-adjusted life years (QALYs) and incremental cost-effectiveness ratio of pre-discharge coronary angiography compared with no angiography, and compared these outcomes with the appropriateness score and appropriateness classification estimated from the expert panel ratings.

RESULTS: The Spearman correlations were 0.29 and -0.62 for appropriateness score vs. QALY gain and incremental cost-effectiveness ratio, respectively. Of the clinical scenarios that were categorized as "appropriate," the average gain in QALYs was 0.62 years and the average incremental cost-effectiveness ratio was \$27,000 per QALY gained, where averaging was based on the likelihood of a patient being in each clinical scenario. Of the clinical scenarios that were categorized as "equivocal," the average gain in QALYs was 0.28 years and the average incremental cost-effectiveness ratio was \$42,000 per QALY gained. None of the clinical scenarios was categorized as "inappropriate" by the expert panel. Cardiogenic shock was the most influential variable for both the decision-analysis outcomes; age was the most influential variable in determining the appropriateness score.

CONCLUSION: There was good agreement between the appropriateness score and the incremental cost-effectiveness ratio of pre-discharge coronary angiography compared with no angiography after acute myocardial infarction. The agreement between the appropriateness score and the gain in QALYs was much weaker.

SYMPTOM BURDEN AT THE END OF LIFE: UNMET NEEDS.

JS Kutner, C Kassner, D Nowels, DGIM; Family Medicine, UCHSC, Denver, CO

PURPOSE: To describe symptom prevalence, severity and frequency among hospice patients, from the perspective of hospice staff.

METHODS: Cross-sectional study conducted in the Population-based Palliative Care Research Network (PoPCRN), a community-based hospice research network, over 6 weeks. Hospice staff recorded patient data during interdisciplinary team meetings, including demographic information (age, sex, date of admission, diagnoses, treatment setting, date of data collection) and a modification of the Memorial Symptom Assessment Scale. Descriptive statistics were calculated for each of the variables. "Unmet needs" were defined as symptoms that were frequent (frequently or almost constantly present) or severe (severe or very severe).

RESULTS: Sixteen hospices returned data on 348 patients. The patients, of whom 55% were female, had a median age of 78 years. Most were receiving care at home (57.8%) or in a nursing home (24.7%). The most common diagnoses were cancer (55.2%), cardiac disease (13.8%), and neurologic disorders (12.1%). The most common symptoms were lack of energy (83.0%), pain (75.6%), lack of appetite (62.6%), difficulty concentrating (59.8%), and feeling sad (50.9%). These patients had a number of unmet needs, with frequent and severe occurrences of common symptoms (see table). Hospice staff were unable to rate a number of potentially important symptoms, including: problems with sexual interest or activity (63.1%), change in self image (30.4%), and worrying (26.2%). There were no significant as-

sociations between length of time of receipt of hospice care and staff ability to rate a particular symptom.

Symptom	Present, n (%)	Frequent, when symptom present, n (%)	Severe, when symptom present, n (%)
Lack of energy	289 (83.0)	216 (74.7)	132 (45.7)
Pain	263 (75.6)	90 (34.2)	38 (14.4)
Lack of appetite	218 (62.6)	119 (54.6)	67 (30.7)
Feeling drowsy	213 (61.2)	111 (52.1)	45 (21.1)
Difficulty concentrating	208 (59.8)	120 (57.7)	65 (31.3)
Feeling sad	177 (50.9)	56 (31.6)	33 (18.6)
Shortness of breath	167 (48.0)	84 (50.3)	43 (25.7)
Agitation	166 (47.8)	50 (30.1)	28 (16.9)
Worrying	151 (43.4)	51 (33.8)	23 (15.2)
Cough	145 (41.7)	47 (32.4)	16 (11.0)
Feeling nervous	145 (41.7)	35 (24.1)	23 (15.9)

CONCLUSION: Rapid, quality data collection is feasible in a community-based hospice research network. This study identifies a significant unmet symptom burden among hospice patients. Primary care physicians, who often care for patients with life-threatening illnesses, must be prepared to assess and manage these symptoms.

PATTERNS OF OUTPATIENT CARE AND HOSPITAL USE FOR PERSONS WHO ABUSE DRUGS. C Laine, J Cocroft, J Rothman, WW Hauck, BJ Turner, Department of Medicine, Jefferson Medical College, Philadelphia, PA; AIDS Institute, New York State Department of Health, Albany, NY

PURPOSE: To explore the relationship between patterns of outpatient care and subsequent hospital use by persons who abuse drugs.

METHODS: Retrospective cohort study of 44,327 HIV- and 14,363 HIV+ drug users enrolled (>9 months of each year) in NYS Medicaid during 1996-97. Using Medicaid claims data, we defined patterns of outpatient care in 1996: regular substance abuse care only (>5 mos with one provider), regular medical care only (>35% of care from one provider), both, or neither. For HIV+ patients, we also determined whether the medical care provider was an HIV specialist. We then identified any hospitalization in 1997 and, if hospitalized, the percentage of follow up time spent hospitalized. Separate multivariable models for HIV- and HIV+ patients adjusted for patient characteristics.

RESULTS: The rates of any hospital stay were 36% for HIV- and 56% for HIV+ drug users. Among those hospitalized, HIV- and HIV+ patients spent, respectively, 6.5% and 8.0% (mean) of follow-up time hospitalized. HIV- patients with regular substance abuse care alone or in conjunction with regular medical care had the lowest adjusted odds of any hospital stay (OR .64 [95% CI .57, .66] and OR .64 [.59, .69], respectively), followed by only regular medical care (OR .89 [.85, .93]) compared to those with neither type of care. Among HIV+ patients, the adjusted odds of any hospitalization were lowest among those with both regular substance abuse care and non-HIV specialty care (OR .58 [.49, .68]) or regular substance abuse plus HIV specialty care (OR .67 [.59, .76]), followed by only regular HIV specialty care (OR .74 [.66, .82]), only regular substance abuse care (OR .76 [.68, .84]), and only regular non-HIV specialty medical care (OR .81 [.72, .91]). Similar adjusted associations appeared between care types and percentage of follow-up spent in the hospital for HIV- patients. Among HIV+ patients, those with regular substance abuse care with regular non-HIV specialty care fared the best, having a 23% reduction in the average proportion of days spent in the hospital (95% CI .13%, .32%). HIV+ patients with only regular medical care (HIV specialty or not) did not differ significantly from those without regular substance abuse or medical care with respect to the proportion of time spent in the hospital.

CONCLUSION: Drug users' patterns of outpatient care are associated with subsequent hospital use. In both HIV- and HIV+ drug users, those with a regular source of substance abuse care appear to use the hospital less. A regular source of medical care also appears to play a role in reducing inpatient care of drug users.

WHY WE DO THE THINGS WE DO: UNDERSTANDING PHYSICIAN TEST ORDERING BEHAVIOR. W Land, D Litaker, J Isaacson, N Mehta, C Asher, Internal Medicine and Cardiology, The Cleveland Clinic Foundation, Cleveland, OH

PURPOSE: Clinical practice guidelines are tools designed to improve the test-ordering system. However, not all guidelines are effective in meeting this objective because they commonly address *systems* rather than the *behaviors* underlying these systems. The Theory of Planned Behavior (TPB) has been used to identify factors motivating people to perform certain behaviors and may be helpful in better understanding factors that influence test-ordering behavior. The primary behavioral concepts of this theory include *behavioral beliefs* (an action will have a positive or negative outcome), *normative beliefs* (identifying people who influence an action), and *control beliefs* (factors that promote or deter an action). The objective of this study was to determine the applicability of the TPB to physician decision making with regard to test ordering.

METHODS: Echocardiograms (ECHO) are a commonly ordered and perhaps over utilized test; therefore, we applied the TPB to identify behaviors influencing physicians to order them. Using a modified focus group approach, eight major behavioral beliefs, five normative beliefs, and six control beliefs that influenced ECHO

ordering among physicians were identified. A survey based on these belief domains was developed and published on the World Wide Web. 1327 members of the SGIM were invited to participate through an electronic mail message. The survey required each respondent to rate responses on each belief using a seven point Likert Scale (1 = very strongly disagree, 7 = very strongly agree). Descriptive statistics without subgroup analysis were reported as part of this exploratory study. **RESULTS:** 320 Physicians completed the survey of which 282 (88%) were staff physicians, 24 (8%) were Fellows, and 14 (4%) were residents. Behavioral Beliefs most influential in the process of ordering an ECHO included clarification of equivocal physical examination findings and affirmation of diagnostic decisions. Cardiologists were identified as the group most likely to influence the ordering of an ECHO reflecting a Normative Belief. Reminders of recently done ECHOs were most likely to prevent a respondent from ordering another reflecting a Control Belief.

CONCLUSION: This study showed that the TPB might be relevant in understanding physician decisions with regard to test ordering. An approach addressing behavioral beliefs bolstered by recommendations of specialists (those influencing normative beliefs), and incorporating reminders of duplicate testing may be more effective in modifying test ordering behavior than guidelines alone. Incorporation of the TPB may be helpful in designing future strategies to modify physician behavior.

FREQUENCY, CONTENT, AND PERCEIVED EFFECTIVENESS OF PHYSICIAN LOBBYING OF MEMBERS OF CONGRESS. SH Landers, AR Sehgal, MetroHealth Medical Center, Case Western Reserve University, Cleveland, OH

PURPOSE: Physicians are uniquely qualified to educate legislators about health care issues, but little is known about how physicians lobby members of Congress. We sought to determine the frequency, content, and effectiveness of physician lobbying of Congress.

METHODS: From the staff of 84 randomly selected members of Congress (49 Senators, 35 Representatives), we interviewed the legislative assistants who work on health care legislation and meet with constituents and lobbyists on behalf of their Senator or Representative. We asked about (1) the frequency of meetings between legislative assistants and physicians (2) the issues discussed, and (3) the perceived effectiveness of lobbying.

RESULTS: Senate and House legislative assistants met with an average of 10.0 and 4.0 physicians per month, respectively. This suggests that approximately 29,000 such meetings occur annually. The most common issues discussed were Medicare reimbursement (mentioned by 80% of subjects), managed care reform (75%), and funding for medical research (25%). Other issues such as access to care for the uninsured, tobacco control, abortion rights, and violence prevention were rarely discussed. Most legislative assistants rated physicians as very effective (44%) or somewhat effective (46%) as lobbyists. The most common suggestion for improving physician lobbying was to focus less on reimbursement and to address a broader range of health care issues.

CONCLUSION: Physicians are frequent and effective lobbyists on reimbursement, managed care, and research issues. Policy makers appear receptive to increased physician input on a broader range of health care issues. Including these issues in lobbying efforts has the potential to shape health policy in a way that improves both patient care and public health.

HEALTH PLAN PREDICTORS OF PERFORMANCE ON THE CONSUMER ASSESSMENTS OF HEALTH PLANS SURVEY (CAHPS). BE Landon, AM Zaslavsky, ND Beaulieu, J Shaul, PD Cleary, Department of Health Care Policy, Harvard Medical School; Division of General Medicine and Primary Care, Beth Israel Deaconess Medical Center; Harvard Business School, Boston, MA

PURPOSE: With the growth of managed care, many purchasers and consumers of health care have become concerned about the quality of care being delivered within managed care. In response, measures of technical quality of care and patient experiences with care including the Consumer Assessment of Health Plans Survey (CAHPS) have been developed for the purpose of monitoring health plan performance. Little is known, however, about the health plan characteristics associated with better performance.

METHODS: Cross-sectional surveys of Medicare beneficiaries in managed care plans that had contracted to provide care to Medicare beneficiaries in 1997 and associated information about the health plans obtained from the Interstudy Competitive Edge Database. The main outcome measures were 4 ratings (of the health plan, care, primary physician, and specialist physician) and 4 composites of questions about care (relating to access, advice, care delivery, and customer service) derived from the Medicare CAHPS survey.

RESULTS: In multivariate analyses, regional effects were strong predictors of performance on the CAHPS measures. Health plan type, size, and age of plan were related to health plan performance. For-profit (FP) health plans had lower adjusted performance on the customer service and access composites and the plan ratings while health plans with national affiliations had lower adjusted scores in both the customer service and delivery composites as well as in ratings of care, specialists, and the plan. Controlling for other variables, the plan ratings and customer service composite scores of national plans, regardless of tax status, were more than one standard deviation lower than those of local not-for-profit (NFP) plans and the ratings of local FP plans were about one-half standard deviation lower. In addition, the scores on the customer service composite were over one standard deviation worse for both national FPs and national NFPs and the scores on the access

composite were about one-half standard deviation lower for both national and local FP health plans. Twelve percent of national FP health plans, however, still received ratings that were above the median rating for local NFP plans. NCQA accreditation and Federal qualification were not related to performance on most measures. For the access composite, health plans with full NCQA accreditation performed better.

CONCLUSION: Among the health plan characteristics we examined, for-profit status and national affiliation had strong negative associations with health plan performance. NCQA accreditation was not associated with performance.

SPECIALTY VERSUS SPECIALIZATION: THE CASE OF HIV CARE. BE London, IB Wilson, NS Wenger, MF Shapiro, SA Bozzette, PD Cleary, Harvard Medical School; New England Med. Ctr., Boston, MA; UCLA, Los Angeles, CA; Beth Israel Deaconess Med. Ctr., Boston, MA; UCSD and the VA Health System, San Diego, CA

PURPOSE: There is debate about whether generalists or specialists provide better quality HIV care. To assess this, we examined the relationships of formal specialty training and expertise (reflected in HIV caseload) to knowledge levels and referral practices in a national sample of HIV providers.

METHODS: Participants were physicians caring for patients in HCSUS, a study of a probability sample of 2864 HIV-infected individuals in the U.S. The survey included questions about specialty training, case-load, CME activities, self-assessed expertise, and referral practices. A 10-item assessment of HIV-related knowledge was developed. First the relationship of specialty training, case-load, and CME activities with self-assessed expertise was described. Next, the relationships of specialty training and case-load with knowledge and referral practices were examined. For multivariable models the knowledge scale was dichotomized at a score of 75%.

RESULTS: The analytic sample included 378 physicians (response rate = 72%). Of the 378, 152 had infectious disease (ID) training, and 226 were generalists; 4% of ID-trained MDs and 37% of generalist MDs did not consider themselves HIV experts. The median current caseloads were 150 and 200 for ID and generalist experts respectively. In contrast, the caseload for non-expert generalists was 5. Experts attended more local and national CME meetings than non-experts (9.2 v. 2.6, $p < .01$ and 2.3 v. .3, $P < .01$, respectively). In bivariate analyses, both ID training and caseload were associated with HIV knowledge and referral practices ($p < .01$). Mean scores on the knowledge scale were similar for ID and generalist experts (9.2 vs. 8.8, $p = .01$) though ID experts were less likely to ever refer than generalist experts (14% v. 27%, $p < .01$). In multivariable models that considered specialty training and caseload jointly, MDs with caseloads of 20-49 and >50 were more likely to have a high knowledge score (OR = 2.48 $p = 0.11$ and OR = 4.18 $p < .01$, respectively), and specialty was no longer significant. In the models predicting referral practices, both experience (OR = .22, $p < .01$ and OR = 13, $p < .01$ for caseloads of 20-49 and >50 respectively) and specialty (OR = .18, $p < .01$ and OR = .07, $p < .01$ for generalist and ID experts respectively) were significant. **CONCLUSION:** In this analysis, HIV-specific knowledge was more strongly associated with experience with HIV than specialty training. In addition, though referral practices were related to both experience and specialty, generalist experts and ID physicians reported similar behaviors. This suggests that generalist physicians, through clinical experience and self-education, can develop specialized knowledge in HIV care.

ARE MINORITIES SYSTEMATICALLY UNDERTREATED FOR MENTAL ILLNESS? KE Lasser, DU Himmelstein, SJ Woolhandler, D McCormick, DH Bor, Medicine, The Cambridge Hospital/Harvard Medical School, Cambridge, MA

PURPOSE: The prevalence of psychiatric illness in minority and white patients has been shown to be similar. We sought to compare their rates of outpatient mental health treatment.

METHODS: We analyzed data from the 1997 National Ambulatory Medical Care Survey (NAMCS) and the 1997 National Hospital Ambulatory Medical Care Survey (NHAMCS), nationally representative samples of office-based and hospital-based outpatient visits, respectively. For visits to Primary Care Providers (PCPs), defined as General and Family Practice, Internal Medicine, Geriatric Medicine, Adolescent Medicine or General Pediatrics, and to Psychiatric providers, we analyzed total visit rates and visit rates for the following mental health services: mental health counseling, psychotherapy, and "drug therapy"(DT), defined as treatment with anti-depressant, anti-anxiety, anti-psychotic or sedative-hypnotic medications. We used population weights and the Census Bureau's 1997 population estimates to calculate yearly visit rates per thousand population according to race and ethnicity as designated by the provider.

RESULTS: In the combined NAMCS and NHAMCS samples, total visit rates to PCPs (per 1000 in population per year) were 1350, 1340, and 1250 for whites, Hispanics, and blacks, respectively. Visit rates for psychotherapy and mental health counseling in Primary Care were higher for whites (10.3 and 43.9) than for Hispanics (9.1 and 19.2) and blacks (5.5 and 19.7). Similarly, whites had a higher visit rate for DT (109.0) than Hispanics (48.3) or blacks (73.7). However, blacks received sedative-hypnotic prescriptions at a much higher rate (16.2) than whites (9.0) or Hispanics (6.6). Overall visit rates to Psychiatric providers by Hispanic (48.2) and black (37.8) patients were markedly lower than the visit rate by whites (106.0). Visit rates to psychiatrists for psychotherapy and mental health counsel-

ing were lower for Hispanics (20.9 and 24.8) and blacks (21.0 and 23.6) than for whites (67.7 and 44.1). Whites had a visit rate for psychotherapy that was more than 3 times that of minority patients, a greater difference than that observed for mental health counseling. DT visit rates at psychiatric appointments were also disparate at 29.1, 38.3, and 71.8 for blacks, Hispanics, and whites, respectively. **CONCLUSION:** Minorities receive outpatient psychiatric care at approximately half the rate of whites. This difference may be due to poor minority access to care and real or perceived cultural barriers between patients and providers. Undertreatment of mentally ill minority patients may cause undue suffering. Moreover, it may be an underestimated factor in the persistence of racial inequalities, such as disproportionately high rates of homelessness and incarceration in minority communities.

AN ASSESSMENT OF THE CONSORT STATEMENT'S IMPACT ON PEER REVIEW. DL Lebeau, MA Winker, MD Schwartz, PB Fontanrosa, Medicine, New York University, New York, NY; JAMA, Chicago, IL

PURPOSE: The CONSORT statement was designed to assist "peer reviewers in evaluating reports of trials submitted for publication". We conducted a pilot study to evaluate CONSORT's utility for improving the methodological aspects of reviews.

METHODS: A single-blind randomized control trial (RCT) was conducted by enrolling consecutive RCT manuscripts submitted to JAMA during a six-week period. Up to ten content reviewers per manuscript were recruited and randomized to receive either a "standard" or "CONSORT" packet. A study instrument measured main outcomes. Section one contained the Feurer format and section two used the review quality instrument (RQI). Section three adapted the Chalmers and Jadad RCT manuscripts scales to grade reviewers' reviews for methodological completeness (number of 22 methodological features mentioned), and accuracy (percent of manuscripts' methodologic flaws identified).

RESULTS: Five of six possible manuscripts were enrolled. Forty of 46 randomized reviewers returned reviews. CONSORT reviewers did not differ from standard reviewers on the Feurer scale or RQI (Table). Non-significant trends suggested CONSORT reviewers had improved methodological completeness and accuracy. Four reviewers reporting epidemiology or statistics backgrounds were more methodologically complete (background 7.26, no background 3.06; p -value = 0.008) and accurate (background 25.0%, no background 10.5%; p -value=0.08). Editors' rated more favorable reviews that were more methodologically complete and accurate (p -value < 0.001 for both trends). Post-hoc calculations found that our sample size had the power to detect a difference in means for methodological accuracy of 15 percent, and would have required a sample size of 330 to detect a difference of ten percent. The study instrument had an inter-rater correlation coefficient of 80 percent.

CONCLUSION: We created and evaluated a unique study instrument which may have utility in measuring the frequency of which methodological concerns are addressed by peer reviewers. Our study instrument or sample size may be unable to measure existing differences. Reviewers may have not used the CONSORT statement because its inclusion was under emphasized to avoid the Hawthorne effect. Low methodologic accuracy and completeness scores suggest reviewers do not focus on methodology or lack sufficient background to comment on it. Editors may favor more comprehensive methodologic commentary.

	Median	CONSORT packet	Standard packet	2-tailed t-test
Feurer (98)	61.1	62.0	60.0	$p = 0.635$
RQI (14)	6.17	6.68	5.58	$p = 0.191$
Methodology: completeness (22)	3.59	4.23	2.84	$p = 0.160$
Methodology: accuracy (100%)	12.6	14.9	10.0	$p = 0.334$

USE OF TRADITIONAL CHINESE MEDICINE VS. WESTERN MEDICINE IN A CHINESE-AMERICAN PATIENT POPULATION. BB Lee, BG Fincke, XS Ren, Section of General Internal Medicine, Boston Medical Center; Section of General Internal Medicine, Boston VAMC, Boston, MA; HSR&D, Bedford VAMC

PURPOSE: To examine use of Traditional Chinese Medicine (TCM) and Western Medicine (WM) in a Chinese-American patient population attending a conventional urban community health clinic.

METHODS: We developed and administered a Chinese language survey which included a 12-item scale of physical and mental health status, two 5-item acculturation scales, demographic determinants, and queries of frequency and type of TCM and WM used. Trained multilingual interviewers administered the survey in one of two Chinese dialects. The subjects were a convenience sample of adult primary care patients 18 years or older, who self-identified as being of Chinese descent.

RESULTS: We administered the survey to 97 subjects (67% response rate). The respondent's average age was 46 years, 70% were female, and 65% had an annual household income under \$10,000. Subjects showed low acculturation with an average score of 1.4 out of 5 on both scales. Eighty-six percent of respondents used some form of TCM in their lifetime with an average frequency of once every eight months. Nine percent of respondents used TCM more often than WM. When asked

about preference for TCM or WM for various conditions, the subjects responded as follows:

Condition	Use TCM 1st	Use WM 1st	Use Together 1st	Use Neither
Health Maintenance (n = 97)	51%	4%	31%	14%
Arthritis (n = 97)	27%	60%	12%	1%
URI (n = 97)	15%	67%	8%	10%
Depression (n = 95)	14%	58%	7%	21%
Cancer (n = 95)	3%	83%	14%	0%
Heart Attack (n = 97)	2%	91%	6%	1%

Of those subjects who chose to use WM first, depending on the condition, 73–79% would switch to TCM if WM failed to work. Regression analysis showed a correlation between frequency of use of TCM and worse physical health status ($p < .05$). Twenty-eight percent of subjects said they do not tell their WM health care provider about TCM use.

CONCLUSION: Despite their low acculturation, our subjects were strongly predisposed to use WM, though this varied according to type of medical condition. They favored TCM to maintain health, but the vast majority preferred WM for life-threatening conditions. A number of subjects would choose to use TCM and WM concurrently, and a majority would switch to TCM if they perceived WM to be ineffective. TCM was commonly used, but often not divulged. It is important for WM providers to speak with these patients about TCM in order to recognize the effect it may have on treatment.

PHYSICIANS' PERCEPTIONS OF BENEFITS FROM GASTRIC FEEDING TUBES. CL Lewis, LC Hanson, G Hunter, J Darter, N Phifer, TS Carey, Sheps Center for Health Services Research, University of North Carolina at Chapel Hill, Chapel Hill, NC

PURPOSE: Placement of gastric feeding tubes has increased despite 50% mortality at six months in those who receive them. This suggests that physicians are unaware of the poor prognosis of patients with feeding tubes, perceive other benefits besides survival, or place them for other reasons.

METHODS: We interviewed the attending physicians caring for patients who had recently had a feeding tube placed to determine their perceptions of the benefits of gastrostomy placement. Patients older than 21 who had a new gastric feeding tube placed were identified by daily procedural logs of interventional radiology and the gastroenterology suites at a major teaching hospital and a community hospital. Physicians responsible for ordering the placement of these gastric feeding tubes were interviewed by telephone or were sent a written interview. Patients with diagnoses of trauma or non-head and neck malignancy were excluded. Physicians were interviewed independent of patient participation in the study.

RESULTS: We contacted 157 physicians and interviewed 140 (89%). Patients' major diagnoses were 28% stroke, 19% neurodegenerative disorders, 16% head and neck cancer, and 37% other serious medical illness. The mean age of the patients was 66 years, 65% were white, and 49% were women. Most (54%) of the physicians had cared for the patient for two weeks or less. Physicians' expected benefits of gastrostomy placement included: improvement in nutritional status (94%), improvement in hydration (67%), prolongation of life (66%), administration of medications (61%), and prevention of aspiration (55%). A minority reported (25%) that it would facilitate placement of the patient, and 15% reported conflict with family members over the decision. Physicians perceived the mean quality of life to be 33 before the gastrostomy was placed and 45 afterward ($p < .05$, 100 = perfect health; 0 = death). They estimated that 11% of the patients would be alive in one year without the feeding tube, but 72% would survive more than one year with the feeding tube and 20% would survive 10 years or more.

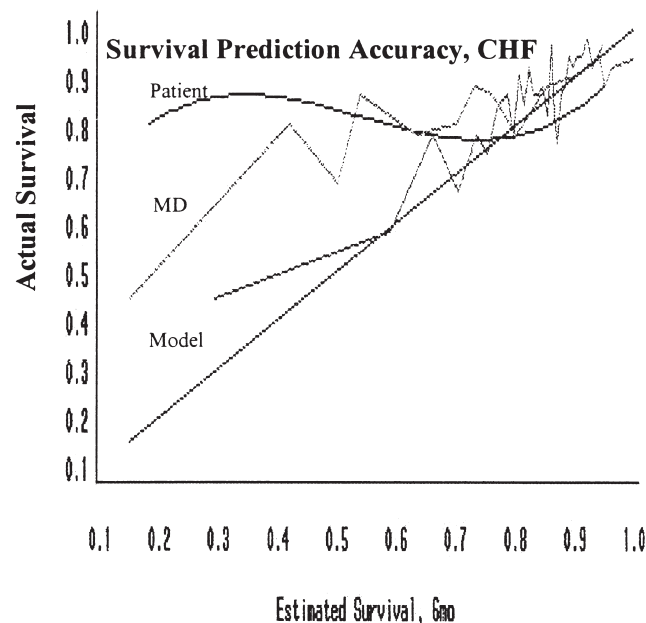
CONCLUSION: Physicians expected improvement in nutrition and hydration in patients who received gastric feeding tubes. A minority reported conflict over the decision with family members or pressure from long-term care facilities. Physicians perceived that placing the gastrostomy would substantially prolong their patients' lives, which is not supported by the current empirical literature. This misperception by physicians may contribute to the increase in utilization of this procedure.

PATIENT, PHYSICIAN, AND PREDICTIVE MODEL CALIBRATION: DIFFERENCES BY DISEASE TYPE IN PATIENTS WITH SERIOUS ILLNESS. J Lynn, HR Arkes, Z Zhong, Center to Improve Care of the Dying, The George Washington Univ., Washington, DC; Department of Psychology, Ohio University, Athens, OH

PURPOSE: To assess calibration and discrimination of prognostic estimates by patients/surrogates, physicians, and predictive model, and especially to learn the variation in prognostic accuracy by diagnosis. Diagnoses include: severe chronic obstructive pulmonary disease with exacerbation (COPD), severe chronic heart failure with exacerbation (CHF), cirrhosis, metastatic colon cancer, and metastatic lung cancer.

METHODS: The SUPPORT Project enrolled 9105 seriously-ill hospitalized patients; developed a prognostication model with good overall performance characteristics; interviewed patients, family surrogates, and physicians about expectations of likely survival; and followed patients for six months to six years to assess survival. Here, we stratify by main diagnosis and test calibration and discrimination of the statistical model, the physicians' estimates, and the patients' estimates (with surrogate substitution).

RESULTS: In all diseases, the model and the physicians' estimates exhibited good calibration and discrimination. However, patients' estimates exhibited poor calibration and discrimination, the latter index reflecting few differences between the most severely ill and those with more reserve.



CONCLUSION: The SUPPORT prognostic model and physicians calibrate and discriminate well in each of these diseases. However, patients do not recognize the difference between those with very short expected lifespans and those who are expected to live somewhat longer. This is especially true of lung cancer, CHF, and COPD patients. Perhaps patients with these conditions so often survive multiple life-threatening exacerbations, and thus these patients have few cues to inform them when the risk of death is increasing. Perhaps also, the time course of colon cancer and cirrhosis give more reliable cues, such as known limited aggregate survival, weight loss, and discovery of metastases. The poor calibration and discrimination for patients/surrogates might direct improvement efforts at sharing prognosis better between patients and physicians or at ensuring that appropriate advance care plans are made for all with serious disease.

AN INNOVATIVE ELECTIVE TRAINING IN MANAGED CARE COMPETENCIES. TK Malik, LG Smith, Mount Sinai School of Medicine, New York, NY

OBJECTIVE: To implement an innovative educational experience in managed care for senior internal medicine residents. The specific focus at this institution was to initiate a pilot program that tried to impart competency in managed care for trainee residents. The larger objective was fostering of mutual understanding and working relationships between academic health centers (AHC) and managed care (MC).

DESIGN: An administrative elective was created from one of two yearly outpatient blocks of PGY3 residents. It comprised:

a) Rotation at the corporate headquarters of a for-profit-MC organization, b) experience at a full risk office practice, c) a practical introduction on setting up and managing a medical office, and d) membership in a committee at the AHC corresponding to that of the MCO, designed to help understand health-care delivery models from different perspectives.

Results of pre and post-training evaluation among PQE residents and comparison group (CG) were collated from a total of 56 grantee programs.

RESULTS:

(A). When perceived mastery of managed care competency in 18 topics was compared PQE residents had better 'well prepared to perform' scores in following 13 asterisked (*) topic areas versus CG group ($p < 0.01$):

1. Patient Care. (Evidence Based Medicine*, Epidemiology, Cost-effective Decisions*, Population-Based*, Disease management, Prevention, Practice Guidelines, Patient Satisfaction*, Time Management).

2. Performance Monitoring. (Utilization management, Profiling*, Continuous Quality Improvement*).

3. Team work and Coordination of Care. (Collaboration, Referral*, Case Management*).

4. Organizational Issues. (MC and Ethics*, MC and Health Delivery*, MC and Economics*).

(B). When attitudes about managed care were compared:

PQE residents had better 'strongly agree' scores of positive impacts of MC versus CG group ($p < 0.01$) in following 6 areas: Emphasize Preventive Care, Improve Continuity, Decrease Unnecessary Utilization, Increase Primary Care Physician's Role, Improve Quality of Care, Equal Access. Decrease in Health Care Cost impact was not statistically significant.

CG group had more 'strongly agree' scores of negative impacts of MC versus PQE residents ($p < 0.01$) in following 4 areas: Restrict Referrals, Threaten Patient Relationship ($p < 0.05$), Causes Poorer Outcomes for Sicker Patients, and Omission of Beneficial Tests.

CONCLUSION: This program was funded by Partnerships for Quality Education (PQE), an \$8.3 million collaborative grant initiative from Pew Charitable Trusts and Commonwealth Fund. Such an innovative program with academic health centers, managed care organizations and funding agencies learning together can be a major instrument of change.

USE OF VISION SERVICES AND SATISFACTION WITH CARE AMONG MEDICARE BENEFICIARIES WITH FEE-FOR-SERVICE VERSUS MANAGED MEDICARE. *CM Mangione, EB Keeler, JL Adams, PP Lee, AL Coleman, MF Shapiro, Dept. of Medicine, UCLA School of Medicine, Los Angeles; Health Program, RAND, Santa Monica, CA*

PURPOSE: To determine whether there are differences in use of vision services, quality of vision care, and overall satisfaction with care among Medicare patients in fee-for-service (FFS) versus managed care (MC).

METHODS: Cross-sectional survey of 1061 randomly sampled MC patients from 27 physician groups who contract with one large for-profit HMO in Los Angeles and 236 FFS patients matched to the MC sample by zip code, comparing use of vision services, visual disability, presence of chronic eye and medical problems, SF-12 health status, satisfaction as measured by the CAHPS 1.0, and demographic characteristics. Multivariate analyses compare rates of annual dilated eye exams and cataract surgery, selected CAHPS scales, and severity of visual disability by FFS versus MC. Also survey of physician groups (response rate 96%) to determine whether features of MC physician groups influence rates of dilated eye exams and cataract surgery, selected CAHPS scales and visual disability. Domains measured included: linkage of PCP compensation to specialty referral volume; capitation experience; and availability of specialty care within the group. All regression analyses were adjusted for independent effects of age, gender, income, race, Medicaid, health status, medical and eye conditions.

RESULTS: Response rate was 65%. Demographics were mean age of 75 years, 59% female, 18% Latino, 17% African American, and 7% Medicaid. All demographic characteristics were similar for FFS versus MC except for Latino ethnicity (20% in MC sample vs 10% in FFS, $p = .001$). Prevalence of medical and eye diseases and mean SF-12 scores did not vary by FFS versus MC. Multivariate analyses indicate that FFS patients were more likely to have had dilated eye exams during the past year at 55% vs 48% for MC ($p = .05$), had similar rates of cataract surgery at 11% vs 12% for MC, and reported less visual disability ($p = .01$). Those in FFS rated quality of eye care and health care overall significantly higher than persons in MC ($p < .05$). CAHPS rating of getting needed referral care was high and similar in both settings. Persons cared for in groups where PCP compensation was linked to referral volume had a trend toward fewer dilated eye exams ($p = .06$), and toward reporting more trouble getting needed care ($p = .07$). Patients cared for in groups less experienced with capitation had higher rates of cataract surgery ($p = .03$).

CONCLUSION: Persons in FFS had significantly more eye exams, better visual functioning, and better satisfaction with care when compared to those with managed Medicare. Linkage of PCP compensation to volume of referral care within managed care may lead to less specialty service use and lower perceived access to care.

VARIATION BY REGION IN THE FINANCIAL AND ORGANIZATIONAL STRUCTURE OF PHYSICIAN GROUPS IN CALIFORNIA AND THE PACIFIC NORTHWEST. *CM Mangione, C Damberg, M Horst, A Castles, PR Gutierrez, AF Brown, M Spar, D Carlisle, KL Kahn, Dept of Medicine, UCLA School of Medicine, Los Angeles; Pacific Business Group on Health, San Francisco, CA*

PURPOSE: To describe variation in the financial and organizational structure (FOS) of physician groups by region.

METHODS: Developed and administered a 45 minute telephone survey to measure structure in the following areas: 1) immutable characteristics of provider groups such as size and profit status; 2) compensation arrangements for physicians; and 3) the process of monitoring referral care and resource use.

RESULTS: Medical Directors from 54 of 57 physician groups who are participants in the 1998 Physician Value Check Survey project completed the FOS (96% response rate). These physician groups which provide the majority of managed care in these markets are geographically dispersed with 48% in Southern California, 39% in Northern California, and 13% in the Pacific Northwest. The percent of revenue derived from capitated contracts, where groups are compensated at a fixed amount per member per month, was highest in Northern California at a mean $\pm$ SD of 86 $\pm$ 24% and lowest in the Pacific Northwest at 33 $\pm$ 14% ($p = .0001$). The dominant form of primary care physician (PCP) compensation also varied by region:

	Pac NW (%)	N. Cal (%)	S. Cal (%)	p
Salary	0	29	42	.001
FFS	100	19	15	
Capitation	0	48	38	
Mixed mode	0%	4%	5%	
Bonus/Withold	29	62	46	.85

With full risk contracting, physician groups are motivated to monitor the appropriateness and volume of referral care. The most frequent management strategies used in all regions at similar proportions include: mandatory pre-authorization from the group for specialist visits (43%) or procedures (76%); pre-authorization by the PCP for specialist visits (72%); PCP-level profiling with feedback (74%); and implementation of formal guidelines that provide criteria with regard to appropriate use of referral care (74%).

CONCLUSION: Financial and organizational structure varies widely for the groups which provide the majority of care to the managed care populations in these regions. Linkage of these structural characteristics to patient level data will be critical to determine whether physician group characteristics are associated with better outcomes of care.

WHICH PRIMARY CARE PATIENTS WITH ALCOHOL DISORDERS RECEIVE TREATMENT? *D Mansell, A Spiro III, A Lee, L Kazis, General Internal Medicine, University of Alabama at Birmingham, Birmingham, AL; CHQOER, Bedford VAMC, Bedford, MA*

PURPOSE: Little is known about possible differences between primary care patients who have and have not been treated for alcohol disorders. We examined the health status and clinical characteristics of ambulatory patients with alcohol disorders (AD) who had and had not received treatment.

METHODS: Cross-sectional sample and survey of 6829 males who use VA ambulatory services in the greater Boston area. Patients were defined as having AD if CAGE was greater than or equal to 2 and current (CAD) if they had had a drink in past year, else abstinent (AAD). According to self-report of treatment, patients were further classified as treated (CADT or AADT) or never treated (CADNT or AADNT); patients with a CAGE = 0 and no history of treatment were classified as NOAD. The total number of physical (PHYS) and psychiatric (PSYCH) comorbidities was based on a medical history interview. Health status was measured with the SF-36 from the Medical Outcomes Study using two summary scores, PCS (physical component summary scale) and MCS (mental component summary scale). Higher scores denote better function. Patients with current AD completed the alcohol section of the QDIS, a diagnostic interview for DSM-III-R. ANCOVA adjusted for age was used to compare PHYS, PSYCH, PCS, and MCS scores. Tukey's test was used for multiple between group comparisons; all p values are < 0.05 two-tailed.

Results: 2425 of 4236 (57%) eligible patients completed the survey.

	CADNT	CADT	AADNT	AADT	NOAD
N	170	141	69	124	1458
Age	56.4 ^a	55.5 ^a	63.3	58.0 ^a	63.9
Married	45.5 ^a	39.7 ^a	60.3	41.5 ^a	64.5
PHYS	5.1	6.4 ^a	5.2	5.6	4.9
PSYCH	0.8	1.5 ^a	0.9	1.4 ^a	0.6
PCS	38.7 ^b	33.4 ^a	35.2	37.2	37.0 ^b
MCS	45.7 ^{a,b}	39.7 ^a	47.6 ^b	44.1 ^a	49.8 ^b

^adiffers from NOAD, ^bdiffers from CADT

CONCLUSION: Patients who achieved abstinence without treatment (more than 10% of all those with AD) have health and health status comparable to those without alcohol disorders. Our results suggest that patients with the greatest burden of illness are those who receive treatment and that abstinence improves mental health status.

CAUSES OF ANTIBIOTIC RESISTANCE: ATTITUDES AND PERCEPTIONS OF RESIDENTS COMPARED TO ATTENDING. *E Martinez, WC Wester, L Durairaj, DN Schwartz, S Husain, AT Evans, Department of Medicine, Cook County Hospital, Chicago, IL*

PURPOSE: Antibiotic resistance is a growing health problem. Little is known about the attitudes and perceptions of physicians regarding its causes, and no study has ever been published comparing internal medicine residents to attendings. It is perceived that residents don't have an adequate appreciation of antibiotic usage and its ramifications.

METHODS: A survey containing a 19 item self-administered questionnaire was sent to 488 internal medicine physicians in 4 Chicago hospitals, of which 424 (87%) were returned completed. The survey asked physicians to rank potential causes of antibiotic resistance, ranging from "unimportant" to "extremely important." The sample included 243 internal medicine residents and 181 internal medicine attendings. The residents included 143 from a public hospital and 100 from a university hospital, and the attendings included 114 general internists, 21 ID specialists, and 46 other specialists.

RESULTS: Resident physicians consistently ranked all the potential causes of antibiotic resistance as high or higher (in terms of potential importance in causing the problem) than attending physicians. Specifically, residents ranked 6 potential causes at a statistically significant higher level compared to attendings. These included: 1. Prescribing antibiotics for self-limited bacterial infections. 2. Prescribing antibiotics for self-limited non-bacterial infections. 3. Prescribing antibiotics for shorter than recommended duration. 4. Prescribing antibiotics empirically without a definite diagnosis of bacterial infection. 5. Prescribing broad spectrum antibiotics when equally effective narrower spectrum antibiotics are available. 6. Having poor access to good information on local antibiotic resistance patterns. There were 12 potential causes ranked at similar levels by residents and attendings. Only one cause was ranked at a statistically significant lower level by resi-

dents compared to attendings (on the influence of pharmaceutical company advertising and drug promotion). There were no significant differences between residents training in a public hospital (predominantly from international medical schools) and those training in a private university hospital (predominantly from US medical schools).

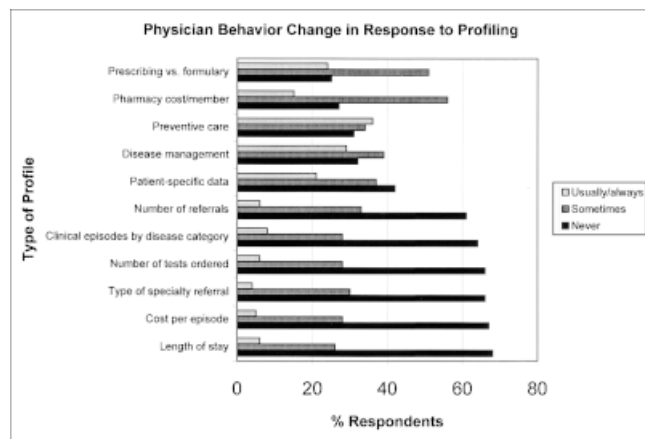
CONCLUSION: Residents in internal medicine programs surveyed in this study attribute as much or more importance to the various potential causes of antibiotic resistance as attending physicians. Whether this implies that current residents in training are more aware of this problem than attendings, or that perceptions may change with time, remains a question for further study.

PHYSICIAN-REPORTED BEHAVIOR CHANGE IN RESPONSE TO PROFILING. KW Mayer, JS Kutner, GIM, UCHSC, Denver, CO

PURPOSE: The objective was to describe physician behavior changes in response to physician profiling and to explore possible reasons why physicians change their behavior.

METHODS: Primary care physicians (internal medicine and family medicine) who are members of the Colorado Medical Society (n = 1740). A survey regarding physician profiling was mailed to identified physicians.

RESULTS: 337 of 1651 eligible physicians completed the survey (34% internal medicine, 62% family medicine, 4% other), for a 20.5% response rate. 99 retired physicians were excluded. 93% of respondents work in an outpatient setting. Practice location and types varied (31% urban, 46% suburban, 22% rural, 60% single-specialty, 17% multi-specialty, 18% solo practice, and 4% academic/administrative). The most common type of patient insurance was HMO, followed by PPO, medicare and medicaid. 81% of respondents currently receive profiling data. The figure describes how often physicians reported changing their behavior based on the type of profile received. Seventy-nine percent reported that data from different sources was contradictory, 71% found the data confusing and difficult to interpret, 68% did not believe the data were statistically valid, and 50% did not believe the data were collected correctly. Fifty-three percent agreed that they received too much data to process meaningfully, and 62% did not believe that data helped them practice medicine better. Only 38% of respondents found the data useful in improving patient care, 40% found the data useful in controlling costs at their practice, and 42% used the data to identify areas to expand their knowledge base. Forty-nine percent reported that they wanted their profiles to resemble those of their colleagues, but 65% reported changing their behavior to keep their managed care contracts.



CONCLUSION: Despite the proliferation of profiling data, physicians report infrequent behavior change based on profiles received. Physicians appear to be skeptical of the data quality and application to clinical practice. Pressures related to managed care contracts seem to be the primary motivator for behavior change based on profiling data. Current physician profiling practices do not appear to change physician behavior.

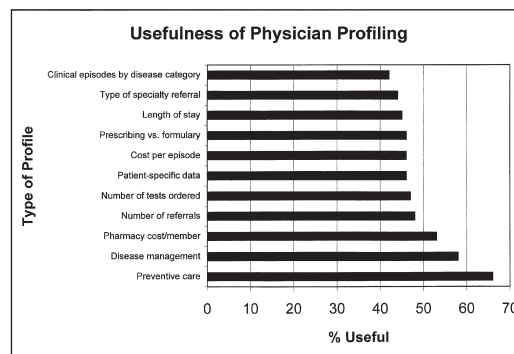
PHYSICIAN PROFILING PATTERNS AMONG PRIMARY CARE PHYSICIANS. KW Mayer, JS Kutner, GIM, UCHSC, Denver, CO

PURPOSE: To describe physician profiling patterns and the perceived usefulness of profiling among primary care physicians.

METHODS: Primary care physicians (internal medicine and family medicine) who are members of the Colorado Medical Society (n = 1740). A survey regarding physician profiling was mailed to identified physicians.

RESULTS: 337 of 1651 eligible physicians completed the survey (34% internal medicine, 62% family medicine, 4% other), for a 20.5% response rate. 99 retired physicians were excluded. 93% of respondents work in an outpatient setting. Practice location and types varied (31% urban, 46% suburban, 22% rural, 60% single-specialty, 17% multi-specialty, 18% solo practice, and 4% academic/administrative). 81% of respondents currently receive profiling data. Types of profiling information received included: pharmacy/cost per member (66%), pharmacy versus formulary (62%), preventive care (57%), number of referrals (44%), types of

referrals (42%), disease management (42%), patient-specific data (42%), number of tests ordered (27%), cost per episode (27%), length of stay (24%), and clinical episode by diagnostic category (15%). Format of profiling data received included: physicians compared with self (51%), physicians compared with group and individual names given (31%), physician groups compared with each other (25%), physicians compared with group and no individual names given (13%), and hospitals compared with each other (10%). The figure describes how useful physicians found each type of profile. Physicians who received profiles were more likely than those who did not receive them to find data on numbers of referrals (59% vs. 45%, $p = 0.022$), tests ordered (68% vs. 46%, $p = 0.001$), and disease management (52% vs. 32%, $p = 0.001$) NOT useful. Those receiving profiles, compared with those who did not, found data comparing the physician to self (55% vs. 39%, $p = 0.008$) NOT useful, while physicians compared with group-names given-WAS useful (51% vs. 36%, $p = 0.018$).



CONCLUSION: Physician profiling is common, but not necessarily perceived as useful by those who are being profiled. Overall, preventive care profiles received the most favorable response from physicians. Profiles were more useful in theory than in practice; for certain types of profiles, those who received them were less likely than those who did not to find them useful.

AN EVALUATION OF A COMPUTERIZED TELEPHONE REMINDER SYSTEM IMPLEMENTED TO IMPROVE APPOINTMENT SHOW RATES IN INNER CITY, GENERAL INTERNAL MEDICINE MANAGED CARE PRACTICES. J McCauley, MW Jenckes, M Atwood, Y Corley, S Svec, R Pilgrim.

BACKGROUND/AIMS: Missed appointments result in disruption of care and decreased office productivity. Published average "show rates" (SR) are 58% and are lower in younger and lower income patients. "I forgot" is the most common reason for a no show. Computerized telephone reminder systems have not been studied in primary care settings with high percentages of capitated Medicaid patients (CMA). We studied the effectiveness of such a system, Smart Talka (ST), in 2 inner city practices to determine: 1) what percentage of our patients could be reached to confirm or reschedule and 2) the effect of ST on show rates.

METHODS: We implemented ST at 2 inner city, general internal medicine practices (GIM) serving 14,000 capitated patients with 81% African Americans and 70% CMA insurance. We programmed ST to call each patient up to 6 times between 10 AM and 8:30 PM on Mondays, Wednesdays, and/or Fridays. ST recorded the number of homes reached with or without confirmation (confirming or rescheduling via touch-tone) and those homes not reached. Reasons for not reaching homes include those with no phone, wrong number, disconnected (no number), and called but could not complete the number of programmed tries before the appointment, the latter usually because the phone was busy or not answered (past date). We collected SR results via our patient scheduling system. Pamphlets on ST were available to patients at the practices.

RESULTS: In a 1-month period, ST attempted calls to 2,728 GIM patient numbers and reached 1,762 (64%). The average show rate for all patients was 54%. Sixty nine per cent of CMA patients v 72% of capitated commercial patients confirmed and showed (ns).

	Reached Confirmed	Re-scheduled	Not Reached No Confirm	No Number	Past Date
Number (%)	993 (36%)	77 (3%)	662 (24%)	167 (6%)	829 (30%)
Show Rate	70%	16%	47%	41%	46%

CONCLUSION: Computerized telephone reminders can be effective in contacting inner city, primary care patients and improving show rates. This results in improved continuity of care and office productivity. Reports can help practices improve by identifying problem areas, e.g., no confirmation, wrong phone number. Based on initial program success, we now call patients five days a week and plan more intensive education to improve our confirmation rate. We also feed back wrong numbers to the practices, asking them to find alternative contact numbers.

HOW EFFECTIVE IS A VIDEO MEDICAL EDUCATION UPDATE ON FAMILY ABUSE IN A MANAGED CARE PRACTICE? J McCauley, MW Jenckes, D Morse, C Gibbons, B Friedman, M McGuire, Medicine, Johns Hopkins Medical Institutions, Baltimore, MD

PURPOSE: Clinicians in busy medical practices often do not have access to updates in treatment of abused patients; lack of time and access to educational materials are major barriers. We developed a video for clinicians on screening and treatment of patients for child, domestic or sexual, or elder abuse. We used pre- and post-viewing tests to evaluate the effectiveness on improving knowledge and changing attitudes.

METHODS: Physicians, social workers, researchers, and legal staff worked with a professional filmmaker to develop a 35 minute movie which included epidemiology, typical patient presentations, and recommended treatments (including legal reporting requirements) for child, domestic and sexual, and elder abuse. Experts associated with the host institution (Johns Hopkins Medical Institutions) gave interviews and role-played typical patient-physician visits for each abuse type. The scripts included information required by JCAHO. Our pre- and post-test included 24 knowledge (11 circle correct answer and 14 T/F) and 24 attitude (5-point scale, strongly agree to strongly disagree) items. Knowledge items were balanced between the abuse areas. Attitude items tested clinician comfort with, and understanding of, abuse issues and clinician reaction to common myths about abuse. We included a 7 item, 5-point Likert scale evaluation of the video on the post-test. **RESULTS:** In our first month we received 102 surveys from eight sites in two academic medical institutions: ages 20-65+; 40% graduated since 1990; 57% MD, 28% RNs, 10% other. The MDs were 60% GIM or FP, and 15% PEDS or OB (25% other). 64% had received some training in the area of abuse, 40% 1-2 lectures and 11% a 2-3 day short course. The number giving the correct answer increased for 23 of 24 knowledge items, 12 (50%) of them significantly. The proportion of "agree" or "strongly agree" responses increased on all but two items, 19 of 25 (76%) significantly. We found a impressive 41% positive attitude shift in both "likelihood to consider abuse when faced with mental decline in an elder" ($p < .0001$) and in "avoiding discussing abuse with patients so as not to offend" ($p < .0001$). On the evaluation section, 32% rated the video "excellent", and 55% indicated they were very likely to recommend the video to a colleague.

CONCLUSION: This videotape was an effective means of standardizing education and rapidly disseminating it within and across medical institutions. We found clinicians had adopted a number of myths and other incorrect information. Clinicians showed significant improvement in knowledge and positive attitude shifts after viewing clinical role plays and reliable informant presentations. They also rated the video highly. We recommend videos for clinical updates for busy medical practices.

DO GENERALISTS IN ARGENTINA BELIEVE THEY KNOW ENOUGH ABOUT DOMESTIC VIOLENCE? R Mejia, M Aleman, A Fernandez, EJ Perez-Stable, Programa de Medicina Interna General, Universidad de Buenos Aires, Buenos Aires, Argentina; Division of General Internal Medicine, Department of Medicine, University of California, San Francisco; Division of General Internal Medicine, Department of Medicine, San Francisco General Hospital, San Francisco, CA

PURPOSE: Although 50% of Argentine women are estimated to experience domestic violence (DV) at some point in their lives, clinicians have historically not been formally trained in detecting and evaluating women at risk for DV. We surveyed practicing generalists to assess their level of knowledge and practice styles regarding DV and their perception of need for training in this area.

METHODS: Surveys were distributed to 291 generalists from 11 medical centers in 4 regions of the country. We used a 10-point scale to assess their knowledge of DV. We asked them about the frequency with which they discussed DV with their patients, and had them rate need for training.

RESULTS: 175 surveys were completed (60% response). 66% were men, mean age was 46 y, the average number of women patients seen per month was 143. Mean knowledge score was 4.6 (± 1.8). On average, the respondents had discussed DV during the last month in 1.2 (± 0.22) opportunities and DV was diagnosed 0.8 (± 0.18) times during the previous month or in 0.6% of women seen. 78% of physicians reported no previous training about DV; 65% considered themselves "not properly equipped to diagnose or treat" DV victims. 47% showed an interest in undergoing some type of formal training. 70% wanted written materials.

CONCLUSION: Practicing generalist clinicians in Argentina score poorly on an assessment of knowledge of domestic violence and perceive a need for additional training in this area.

PATIENT PREFERENCES AND WILLINGNESS TO PAY FOR INPATIENT CARE BY PRIMARY CARE PHYSICIANS VERSUS HOSPITALISTS. DO Meltzer, JM Herlko, Department of Medicine, University of Chicago, Chicago, IL

PURPOSE: Hospitalists are increasingly being used in place of primary care physicians to care for hospitalized patients. However, patients may prefer to be cared for by their primary care physician, with whom they may already be familiar. The objective of this study is to determine patients' preferences and willingness to pay to be cared for by their primary care physician versus a hospitalist during a hospitalization for a general medicine condition.

METHODS: A willingness to pay survey was administered by telephone to a nationally representative sample of 308 persons in July and August 1999. The main outcome measures were preferences and willingness to pay for care by the patients' primary care physicians versus a hospitalist over the length of a hospitalization for a general medical condition.

RESULTS: 87% of patients currently have a primary care physician. Among patients with a current primary care physician, 70% prefer care by the primary care physician, 23% state no preference, and 7% prefer care by a hospitalist. Preference for

and willingness to pay for care by the primary care physician increased with the duration of and patients' satisfaction with the relationship with the primary care physician. Across all patients, mean willingness to pay for care by a primary care physician versus a hospitalist was \$200, but 8% of patients reported willingness to pay of \$1000 or more. Eliminating those patients for whom willingness to pay was \$1000 or more reduced the mean willingness to pay to \$95. Eliminating the 16% of patients for whom willingness to pay was \$500 or more reduced the mean willingness to pay to \$51. Eliminating the 20% of patients for whom willingness to pay was \$250 or more reduced the mean willingness to pay to \$34.

CONCLUSION: Most patients prefer to be cared for during a hospitalization by their primary care physician, but willingness to pay is modest compared to the \$300-\$700 per admission cost savings found in randomized trials of hospitalists. Some patients, especially with strong relationships with their primary care physician, report much larger willingness to pay. The design of hospitalist programs should be sensitive to these concerns. The option to have care by one's primary care physician at a modest copayment might substantially mitigate the adverse effects of hospitalists for the fraction of patients who highly value care by their primary care physician while decreasing costs for the majority of patients. For example, with a \$250 copayment, the 20% of patients valuing care by their primary care physician at \$250 or more would pay this amount to have care by their usual doctor. The remaining 80% of patients would choose care by hospitalists, potentially reducing costs by hundreds of dollars while on average valuing the loss of care by their usual doctor at only \$34.

COUNSELING FOR DEPRESSION BY PRIMARY CARE PROVIDERS. LS Meredith, RM Mazel.

PURPOSE: Primary care providers (PCPs) deliver a significant amount of depression care, yet little is known about the content of their clinical encounters with depressed patients. We describe the extent to which PCPs' encounters with depressed and non-depressed patients involve psychotherapeutic counseling during primary care visits.

DESIGN: Cross-sectional descriptive evaluation of audiotaped office visits between October 1997 and September 1998. **Setting and Participants:** 154 encounters of 27 PCPs at 3 Veterans Health Administration clinics in California. Between 2 and 26 visits were taped for 19 providers; 8 had only 1 visit taped. **Main Outcome Measures:** Analysis of audiotaped patient-provider verbal interaction using standard talk categories from the Roter Interaction Analysis System and specific measures of depression counseling coded for sequences of depression talk. **Analysis:** Analysis of variance and covariance to evaluate differences in depression type adjusted for encounter length and patient characteristics (demographics and casemix).

RESULTS: PCP encounters with depressed patients who had a structured diagnosis of major depression were shorter in overall duration but duration of talk focused on depression was longer by 1-2 minutes, and included significantly more of this depression-focused talk (76.9 utterances, $p = .0219$) and less biomedical talk ($p = .0028$) compared with patients who had only subthreshold depression (53.8 utterances) or no depression (27.2 utterances).

CONCLUSION: PCPs delivered more psychotherapeutic depression counseling to their patients with major depressive disorder compared with patients who had no depression or symptoms but no disorder. Findings suggest that PCPs are addressing the needs of their depressed patients and practice structures that support more counseling time may increase the likelihood that patients receive appropriate care for depression.

HOUSE STAFF INTERVENTION TRIAL TO IMPROVE PATIENT SATISFACTION. CL Meyer, D Bor, Department of Medicine, Cambridge Hospital, Cambridge, MA

PURPOSE: Patient satisfaction has much to do with areas not formally addressed during house staff rounding and in the traditional daily "SOAP" progress note. A pilot survey done at our hospital found that patient concerns were addressed infrequently. House staff have the opportunity to engage patients regularly around their fears, worries or uncertainties. We set out to examine the hypothesis that a simple intervention of the house staff asking for and documenting patients daily concerns would improve patient satisfaction.

METHODS: Half of the interns on ward service (one team of 3 interns) were assigned to amend the traditional daily SOAP note to a daily SCOAP note: Subjective, Concerns (open-ended inquiry into major concerns for the day), Objective, Assessment, Plan. The other team of 3 interns continued with the traditional daily SOAP note. The setting was an adult inpatient ward in a community hospital and the trial was conducted over two months (two consecutive pairs of intern teams). Daily concerns of the intervention arm were tabulated and all patients were asked to complete a patient satisfaction survey at the time of discharge.

RESULTS: Data were collected from 41 patients (24 from the intervention arm and 17 from the control arm). Patients whose concerns and worries were elicited by house officers during rounds and documented in the chart (the SCOAP note arm) were more likely to be satisfied with their care (92% vs. 71%) and were more likely to feel their day-to-day concerns were met (96% vs. 59%) than the control patients. Overall, 48% of the patients had concerns about discharge planning and 48% had concerns of pain and discomfort, followed by food and diagnosis (each 32%), sleep (24%), and concerns of upcoming tests/procedures/surgeries (20%).

CONCLUSION: The traditional SOAP progress note should be changed to a SCOAP note (where daily concerns are inquired about and documented) to improve patient satisfaction. In addition, every effort should be made to keep patients abreast of any and all planning around their discharge.

A COLLABORATIVE HEALTH CARE IMPROVEMENT PROJECT BETWEEN A PRIMARY CARE PRACTICE AND A HEALTH INSURER. CE Milich, JM Edmunson, MH Eckman, EJ Barnhart, HP Selker, New England Medical Center; Blue Cross Blue Shield of Massachusetts, MA; University of Cincinnati Medical Center

PURPOSE: Primary care physicians and health insurers have been the focus of efforts to improve the quality and delivery of preventive health care such as smoking cessation counseling. We describe a collaborative effort between a primary care practice and an insurer to address the mutual goal of helping patients stop smoking.

METHODS: The goal of the project was to increase patient smoking cessation rates and documentation of smoking cessation status and counseling by instituting systematic methods to identify smokers.

We designed and implemented a three-armed trial that compared two smoking cessation interventions and usual care. Both interventions met the Agency for Healthcare Research and Quality's Clinical Practice Guidelines for instituting a systematic method to identify all patients who smoke. The interventions differed in their primary purposes: one intervention was aimed at only identifying and documenting the smoking status of patients; the other intervention also reminded clinicians to provide smoking cessation counseling and counseling documentation.

RESULTS: The design phase of the project involved: 1) obtaining agreement by the health insurer and clinical practice leadership on clinical objectives, standards for documentation, and payment for smoking cessation counseling; 2) jointly developing a research protocol and implementation plan; 3) obtaining the endorsement of clinicians on the intervention teams; and 4) training clinicians and their staff to use the interventions.

During the implementation phase, the insurer provided: clinician education seminars on effective and efficient smoking cessation counseling, patient information materials, and all intervention materials.

CONCLUSION: This project illustrates a collaborative approach between a primary care practice and a health insurer that leads practices to institute clinical practice guidelines. It also demonstrates how a practice can implement improvements in patient care that meet mutually shared goals between a health insurer and a primary care practice (improving identification of patients who smoke and reducing patient smoking rates), as well as goals that are more important to one party, such as improved counseling documentation rates (insurer), and time-efficient smoking cessation counseling techniques (clinicians).

ROLE OF AXILLARY NODE DISSECTION IN PATIENTS WITH T1a AND T1b BREAST CANCER. BA Mincey, T Bommer, ES Atkinson, EA Perez.

PURPOSE: Axillary lymph node status remains the best prognostic indicator for patients with invasive breast cancer. Recently, the role of axillary lymph node dissection in patients with T1a or T1b tumors has been debated. Some authors have suggested that the incidence of node positivity is so low in these patients that routine axillary dissection should be eliminated; others maintain that it should continue to be the standard of care because of its role in the determination of prognosis and systemic therapy decisions.

The purpose of this study was to establish an institutional database of axillary lymph node status in patients with early breast cancer, to determine whether incidence of nodal positivity justifies axillary dissection in all patients regardless of tumor size.

METHODS: All patients with T1a (less than or equal to 5 mm) and T1b (6–10mm) breast cancer who underwent both primary breast surgery and axillary node dissection at Mayo Clinic Jacksonville from 1992 through 1998 were identified through the tumor registry. The following data were collected for each patient: tumor size, biological grade, estrogen/progesterone receptor status, number of nodes harvested, and number of positive nodes.

RESULTS: Of 163 patients evaluated, 39 had T1a and 124 had T1b tumors. Node positivity was 0% for T1a and 11.3% for T1b tumors ($p = 0.028$). There was no relationship between lymph node involvement and estrogen receptor status ($p = 0.29$). However, there was a significantly higher risk of lymph node positivity for progesterone receptor negative ($p = 0.012$) and for estrogen receptor negative/progesterone receptor negative lesions when compared to those with hormonally receptor positive tumors ($p = 0.040$). Further, we found a significantly higher risk for lymph node positivity as tumor size increased ($p = 0.002$). Finally, higher tumor grade conferred a higher risk for lymph node involvement.

CONCLUSION: Our data demonstrate that, while T1a lesions have a minimal risk of nodal positivity and may not necessitate routine axillary dissection in the future, T1b lesions should be managed with routine analysis of axillary lymph node status. Whether sentinel node mapping can change this standard awaits further study.

EXAMINATION OF ANTIHYPERTENSIVE POLYPHARMACY PREDICTORS FROM CLAIMS DATA. B Misra, RS Stafford, Institute for Health Policy, Massachusetts General Hospital, Boston, MA; Howard University College of Medicine, Washington, DC

PURPOSE: To use administrative claims data to examine and evaluate clinical and nonclinical predictors of antihypertensive polypharmacy.

METHODS: We utilized 1998 medical and prescription claims data from an employer-based capitated insurance plan for an integrated health delivery system covering 126,000 members in Eastern Massachusetts. We evaluated potential predictors of the concurrent use of multiple antihypertensives. We identified and analyzed 80,692 antihypertensive claims made by 11,383 members. Using pre-

scription date and pill quantity combined with an assumed defined daily dose for each antihypertensive, we calculated the start and end dates for each prescription. We then ascertained whether 3 or more (3+) antihypertensives were used concurrently. This information was merged with medical claims to determine if selected co-morbidities were present during the year. We also assessed other factors, including age, gender and whether the patient had been to three or more providers. Logistic regression was used to evaluate the independent impact of these predictors on the likelihood of receiving multiple antihypertensives.

RESULTS: Of all members receiving antihypertensive medications, 35% concurrently received 3+ antihypertensives at some point during the year. Cardiac comorbidities significantly increased the likelihood of antihypertensive polypharmacy: heart failure (OR:3.1, 95% CI:2.4–4.1), coronary artery disease (1.9, 1.7–2.1), atrial fibrillation (1.3, 1.0–1.8), and hypertension (2.1, 1.9–2.3). Diabetes also predicted the use of multiple antihypertensives (1.7, 1.5–1.9). Increasing patient age made polypharmacy more likely (OR 2.2, 1.9–2.7 for Age ≥ 65 vs. 18–44). Female patients also were more likely to be on multiple antihypertensives (1.2, 1.1–1.3). The strongest independent predictor of antihypertensive polypharmacy was visits to more than three providers (3.9, 2.8–5.5).

CONCLUSION: Antihypertensive polypharmacy may be legitimate, even necessary in patients with complicated cardiac diagnoses. In addition to clinical predictors, nonclinical factors such as increasing age and female gender independently increased the likelihood of multiple antihypertensives. Our study may suggest that better coordination is needed between multiple physicians caring for the same patient. In addition, consolidation of therapeutic regimens may be possible. While administrative claims data have several weaknesses, these data can identify predictors of polypharmacy. In addition, such information can provide physicians in capitated plans with useful feedback on polypharmacy. While such feedback may identify legitimate polypharmacy, it also may identify potentially dangerous, but unnecessary drug combinations.

PATTERNS AND VARIATIONS OF HYPERTENSION TREATMENT IN A CAPITATED INSURANCE PLAN. B Misra, RS Stafford, Institute for Health Policy, Massachusetts General Hospital, Boston, MA; Howard University College of Medicine, Washington, DC

PURPOSE: To examine patterns and quality of hypertension treatment from administrative claims data from a health care delivery system in comparison with national prescribing practices.

METHODS: We analyzed 1998 prescription claims data from a commercial plan of an integrated delivery system that covers 126,000 members in 11 geographic regions in Eastern Massachusetts. From over 600,000 prescription claims, we identified 80,692 for antihypertensive drugs. We categorized these prescriptions by drug class: Angiotensin Converting Enzyme Inhibitors (ACEI, including A2 Receptor blockers), Calcium Channel Blockers (CCB), Beta Blockers (BB), Diuretics (DIUR), and others. Percent totals of each class were calculated for individual regions and in total. Costs per antihypertensive prescription also were computed and then compared to both national treatment guidelines and national data on patterns of hypertension treatment.

RESULTS: Among all regions, 29% of all antihypertensive prescriptions were ACEI, 31% for BB, 19% for DIUR, 16% for CCBs, and 6% for others. This pattern was substantially different from published national patterns: 33% for ACEI, 11% for BBs, 8% for DIUR, 38% for CCB, and 9% for others. Within the delivery system there were wide variations in prescribing by region. CCBs ranged from 9.5% of prescriptions in one region to 22% in another. ACEIs ranged from 24% to 34% between regions, while the combination of BBs and DIURs ranged from 39% to 57%. The average cost of antihypertensives per prescription was \$32.90, but varied between \$24.60 to \$33.80 by region. Difference in costs between drug classes, ranging from \$58.35 for CCBs to \$6.63 for diuretics, were a major contributor to these regional differences.

CONCLUSION: Overall antihypertensive prescription patterns from this capitated plan appear are more consistent with JNC VI national guidelines than published national statistics. Consistent with these guidelines, the use of BBs and DIURs predominates and results in lower treatment costs in comparison to national patterns. The wide regional variations between antihypertensive class and average prescription cost indicate room for further change. ACEIs and CCBs remain a large share of prescriptions and significantly increase the cost of therapy. For example, an estimated \$300,000 in pharmacy costs could be averted within this delivery system by reducing CCB use to the low level (10%) found in one region. The magnitude of health care costs involved suggests the value of interventions that result in prescribing that is more consistent with national guidelines.

WHAT DO PATIENTS WANT IN A PRIMARY CARE PHYSICIAN? AL Moes, RA Murden, Internal Medicine, The Ohio State University, Columbus, OH

PURPOSE: In the managed care era, insurance companies often dictate primary care practice characteristics and allowable practice divisions between primary care and subspecialty providers. The purpose of this study is to seek patient preferences in this area.

METHODS: A total of 700 patients of a multispecialty (primary and subspecialty care) academic internal medicine practice were randomly surveyed. The survey was divided into two sections. The first asked for patient preferences in terms of primary care versus subspecialty management for 26 different conditions and procedures. The second section asked for patient preferences concerning individual primary care physician characteristics and primary care practice characteris-

tics. Usable surveys were returned from 652 patients, including 443 females and 209 males ranging in age between 18 and 84. A total of 597 patients reported having a regular primary care physician with 555 of these patients visiting a primary care office more than once in the last year.

RESULTS: Of the conditions/procedures that were listed, primary care management was strongly favored in 12 (pneumonia, arthritis, asthma, back pain, stomach ulcer, fatigue, hyperlipidemia, fracture, suturing, joint injection, lancing a boil, rectal exam). Subspecialty management was strongly favored in 10 (Alzheimer's disease, angina, depression, personal problems, mole removal, sigmoidoscopy, cardiac testing, diaphragm fitting, pelvic exam, prostate biopsy). Four items (impotence, vaginal infections, diabetes, and chest pain) received equal preferences for primary and subspecialty care.

The primary care physician characteristics that were identified as most important were compassion, accessibility, and spending time during the visit. The least important were liking the patient personally and being the same sex as the patient. The most preferred practice characteristics were answering all questions, spending time during the visit, sending letters about results, same day accessibility for acute problems, and always seeing the same physician. Answering phone calls within an hour and high med school class rank were viewed as least important. In a final dichotomous question, 66% of respondents preferred a primary care physician who ranked in the bottom 50% of his/her med school class but was compassionate, whereas 64% preferred a subspecialist who lacked compassion and a good bedside manner but who finished in the top 50% of his/her med school class.

CONCLUSION: Patients have surprising and specific preferences for their primary care providers and their practices. These preferences should be routinely assessed and be considered, along with physician preferences, by insurers when forming policies.

CARING FOR REFUGEES AND SURVIVORS OF TORTURE. A. Moreno, M Grodin, Section of General Internal Medicine; Department of Psychiatry; Boston Medical Center, Boston University School of Medicine; Health Law Department, BU School of Public Health, Boston, MA

PURPOSE: Despite the growing numbers of refugees and asylum seekers and their complex medical, mental health, social, and legal needs, little descriptive data exists about this patient population. We describe the patients enrolled through primary care in a refugee/survivor of torture program.

METHODS: We reviewed the charts of all patients seen between 12/98 and 12/99.

RESULTS: 50 patients were enrolled of which 33 (66%) were males. The mean age was 41 years for women and 40 years for men; age ranges were 30-70 years and 18-65 years, respectively. 47% of the patients were from Central America and the Caribbean, 34% from Africa and the Middle East, 10% from South America, and 8% from Europe. The main reasons for uprooting and torture were civil war (34%), political activities (14%), ethnic persecution (14%), relative of a victim (10%), humanitarian activities (6%), refusal to cooperate (6%), and religious activities (6%). Patients were subjected to threats (86%), harassment (40%), beatings (38%), battlefield conditions (22%), destruction of property (22%), sexual torture (16%), and electric shocks (12%). After uprooting and torture, 32% of the patients continued to experience significant psychological trauma. Common psychiatric conditions were post-traumatic stress disorder (54% met all Diagnostic and Statistical Manual (DSM) IV criteria and 10% had some DSM IV criteria), depression (38% met all DSM IV criteria and 4% had some DSM IV criteria) and anxiety (40%). 28% percent of the patients had scars and another 8% showed evidence of organ dysfunction or limb deformity. 30% of the patients were active or former smokers and 22% had a CAGE score equal or greater than 3. No patient had recreational drug abuse. 36% of the patients were enrolled in mental health care and 16% declined. 6% of the patients declined to continue their primary care with the program. 15 affidavits were written and 3 court testimonies were made on behalf of these patients. 21 patients had active legal needs; attorneys working with the Center took 38% of these cases. 20% of the patients were referred for the first time to a social worker. 33 patients had previously seen a healthcare provider of which 82% were never asked if they had suffered uprooting or torture. The mean time between arrival to the country and first contact with the program was 4 years, a range of 3 months-15 years.

CONCLUSION: Chronic physical and mental sequelae are common among this patient population, as the continuation of psychological trauma after uprooting and torture. Another important problem is the non-recognition by healthcare providers of this vulnerable population, which in part explains the long delay between arrival to the country and access to specialized care.

RESIDENT PHYSICIANS' PERCEPTIONS OF THE SAFETY OF PERFORMING INVASIVE PROCEDURES IN PATIENTS WITH DISORDERED HEMOSTASIS. KA Myers, G Hanson, P Parks, DRE Farquhar, Division of General Internal Medicine, Queen's University, Kingston, ON, Canada

PURPOSE: Resident physicians are often required to perform invasive procedures at the bedside on patients with disordered hemostasis. We wished to determine the threshold levels of International Normalized Ratio (INR) and platelet count at which resident physicians felt comfortable proceeding with three common procedures without prior correction of the hemostatic derangement.

METHODS: As part of a seminar on quality assurance, 19 residents in internal medicine (PGY1 = 7, PGY2 = 7, PGY3 = 5) were presented with three clinical scenarios, each of which portrayed a patient in urgent need of one of the following procedures: paracentesis(PC), thoracentesis (TC), or lumbar puncture (LP). In the

first iteration of each scenario, respondents were asked to indicate the threshold level of INR above which they would not perform the procedure without first infusing fresh frozen plasma (FFP). In the second iteration, they were asked to record the threshold platelet count below which they would not proceed without platelet transfusion.

RESULTS: Threshold values (see table for median values and ranges) showed wide variation in each iteration of all three scenarios. For example, while four respondents stated they would perform a paracentesis even if the INR were above 3.5, almost a third would give fresh frozen plasma prior to the procedure if the INR exceeded 1.5. Although the data show that residents tend to exercise greater caution with LP than PC or TC, four residents indicated that they would perform LP without platelet transfusion even in the face of severe thrombocytopenia (<20). With increasing resident seniority (PGY1 and PGY2 vs PGY3), a trend toward a more cautious approach to LP and a more liberal approach to TC, was also evident (data not shown).

Procedure	Median Threshold INR (Range)	Median Threshold Platelet Count (Range)
PC	2.5 (1.5-3.5+)	40 (<20-75)
TC	2.0 (1.5-3.5+)	50 (<20-100)
LP	1.5 (1.5-3.5+)	50 (<20-100)

CONCLUSION: There is wide variation in resident physicians' perceptions of the indications for correction of hemostatic derangements prior to performing invasive procedures. Potential adverse consequences of inadequate knowledge in this area include the unnecessary delay of procedures, or inappropriate utilization of blood products. Although this pilot-study is small, it strongly suggests that there is a need for the dissemination (and development, where necessary) of guidelines for the transfusion of blood products prior to performing these common procedures.

BREAST SELF-EXAMINATION: WHO TEACHES IT AND WHO IS TAUGHT? L Nekhyudov, MB Barton, JG Elmore, SW Fletcher, Department of Ambulatory Care and Prevention, Harvard Medical School/Harvard Pilgrim Health Care, Boston, MA; Department of Medicine, University of Washington School of Medicine, Seattle, WA

PURPOSE: Breast self examination (BSE) is a patient-centered, convenient, non-invasive and inexpensive screening modality for breast cancer. Having been taught BSE by a medical provider predicts whether it is performed. The purpose of this study was to determine which patient and provider characteristics are associated with the teaching of BSE.

METHODS: Medical records were reviewed for documentation of whether BSE was taught to 2245 randomly selected women, ages 40 to 69, who were enrolled continuously for 10 years in an HMO, had at least one screening clinical breast exam and had no history of breast cancer. Using univariate, logistic and generalized estimating equations multiple regression analyses, patient and provider variables were tested for their association with BSE teaching.

RESULTS: Over 10 years, 68% (n = 1534) of the women had BSE teaching documented at least once (median 1.6, range 0-13). Documentation of BSE teaching was more likely among women under age 65 than those over (70% vs. 64%, p = 0.005), women of income greater than \$35,000 than those under (69% vs. 63%, p = 0.025) and women who had been prescribed estrogen replacement therapy than those not (72% vs. 66%, p = 0.004). White women were taught more commonly than black women (70% vs. 62%, p = 0.012) and were also more likely to be taught at least twice (45% vs. 29%, p = 0.001). Body mass index, marital status and having a relative with breast cancer were not associated with documentation of BSE teaching. BSE teaching was directly related to the number of clinical breast exams (p = 0.001) and screening mammograms performed (p = 0.001). Multivariable GEE logistic regression analysis found patient age, frequency of clinical breast exams and frequency of screening mammograms to be independent correlates of BSE teaching. Of the primary care providers who saw at least 3 patients in the study sample (n = 248), 88% of nurses, nurse practitioners and physician assistants, 75% of internists and only 17% of the obstetrician-gynecologists documented BSE teaching at least once (p = 0.001). Female providers taught more than did male providers (79% vs. 64%, p = 0.01). These associations remained significant in multivariable logistic regression.

CONCLUSION: Two-thirds of female patients in this HMO setting were taught BSE at least once during the 10-year period. Patient age, general breast cancer screening frequency as well as provider type and gender were associated with the documentation of BSE teaching.

FACTORS ASSOCIATED WITH ADHERENCE TO NATIONAL CHOLESTEROL EDUCATION PROGRAM GUIDELINES FOR PRIMARY AND SECONDARY CAD PREVENTION. JM Neuner, RS Phillips, EF Cook, A Puopolo, JS Haas, TA Brennan, HR Burstlin, Div of General Internal Medicine, Beth Israel Deaconess Med Center; Div of General Medicine, Brigham and Women's Hospital, Boston, MA; Div of General Medicine, San Francisco Gen Hospital, San Francisco, CA

PURPOSE: To examine factors related to adherence to National Cholesterol Education Project (NCEP) guidelines for treatment of elevated cholesterol in primary care patients (pts) with or without coronary artery disease (CAD).

METHODS: We performed a chart review and patient survey (response rate = 69%) of 4856 pts aged 20-75 seen by a staff physician in the previous year at 11 academically-affiliated practices in Boston. Pts with CAD were adherent with NCEP guidelines if: 1) cholesterol (CHOL) or LDL was measured and 2) diet/drug

therapy offered for LDL not done or ≥ 100 mg/dL. Pts without CAD were adherent with NCEP if: 1) CHOL measured, and 2) LDL measured for CHOL ≥ 200 mg/dL or HDL < 35 mg/dL, and 3) diet/drug therapy offered for LDL ≥ 130 mg/dL (≥ 2 risk factors for CAD) or LDL ≥ 160 mg/dL (< 2 risk factors). Full adherence to NCEP guidelines was analyzed using the generalized estimating equation to account for clustering of treatment by provider. Potential correlates of guideline adherence included pt age, gender, education, income, race, smoking, diabetes, hypertension, other co-morbid conditions, family history of CAD, use of cardiac medications or procedures, a single-item self-rated health measure (from poor to excellent), # of primary care physician (PCP) visits in prior year, and out-of-pocket payments for medications.

RESULTS: For CAD pts ($n = 278$, median age 62.5, 52% male, 23% non-white), overall adherence with NCEP guidelines was 67%. After adjustment for age, gender, and co-morbid illnesses, patients with a CABG or coronary angioplasty (AOR 2.32, 95%CI 1.13, 4.74), > 3 PCP visits/year (1.95 [1.13, 3.32]), or aspirin treatment (2.41 [1.35, 4.32]) were more likely to be adherent with NCEP. For pts without CAD ($n = 4578$, median age 41, 39% male, 27% non-white), overall adherence with NCEP was 56%. After adjustment for age, gender, and co-morbid illnesses, pts with hypertension (1.27 [1.04, 1.55]), > 3 PCP visits/yr (1.53 [1.24, 1.87]), or better health (1.11 [1.03, 1.20] per unit increase) were more likely to be adherent with NCEP, while current smokers (0.81 [0.65, 1.00]) and patients with a high school education or less (0.79 [0.65, 0.95]) were less likely to be adherent with NCEP guidelines.

CONCLUSION: While adherence to NCEP guidelines for cholesterol screening and treatment was higher for patients with CAD, overall adherence was low. Several factors, including frequent contact with PCP and CAD risk, were significantly associated with adherence. These data suggest the need for targeted efforts to increase adherence with NCEP guidelines, particularly for at-risk patients.

RISK ASSESSMENT IN HEMODIALYSIS PATIENTS IN A DISEASE MANAGEMENT PROGRAM: PREDICTING RESOURCE UTILIZATION. AR Nissenson, T Steinman, J Dickmeyer, W Mattern, T Parker, T Litchfield, UCLA Medical Center, Los Angeles, CA; Beth Israel Deaconess Medical Center, Boston, MA; RMS Disease Management, McGaw Park, IL; UNC School of Medicine, Chapel Hill, NC

PURPOSE: Disease management (DM) utilizes coordination of care across the disease spectrum and the settings where care takes place, generally is based on a capitated payment and risk-taking, and has been successful in the management of patients with a variety of chronic diseases. Since DM is a form of proactive patient care, it is essential to be able to predict which patients are likely to have adverse outcomes and to target care to prevent complications where possible. Unfortunately, however, there are no generally accepted, validated risk assessment tools for patients undergoing dialysis.

METHODS: Renal Management Strategies (RMS), an affiliate of Baxter Healthcare, has been providing DM for over 1200 patients enrolled in a large national managed care organization in 16 cities. We developed a unique risk assessment tool with 10 items based on a variety of patient characteristics and previous history of resource utilization, and also used the ICED index to evaluate patient risk.

RESULTS: The following correlations have been found between elements of these two instruments and patient outcomes:

	ALBUMINAL RISK ASSESSMENT	ALBUMINAL ICED	ADMITS RISK ASSESSMENT
Pearson Correlation	-.0124*	-.0137**	0.127**
Significance	.022	.002	.009
N	266	430	341

	ADMITS ICED	DAYS RISK ASSESSMENT	DAYS ICED
Pearson Correlation	0.057	0.119*	0.05
Significance	.079	.014	.100
N	617	341	617

*Correlations significant to the .01 level (1-tailed)

**Correlations significant to the .05 level (1-tailed)

CONCLUSION: Although additional refinement of the risk assessment tools is needed, particularly further simplification, such instruments are of considerable value as more dialysis patients are cared for using a DM approach. The multi-faceted risk tool which includes utilization data is more predictive of risk than the ICED, which includes only co-morbid and physical conditions.

PRIMARY CARE CLINICIANS' CHALLENGES WITH END OF LIFE CARE. MA Norman, MW Rabow, S Dibble, Internal Medicine; Institute of Health and Aging, University of California, San Francisco, San Francisco, CA

PURPOSE: Traditionally, primary care clinicians provide care to patients near the end of life. The quality of patient care often depends upon the ease of access and the willingness of primary clinicians to refer to ancillary services such as pharmacists, home care, hospice, psychologists, spiritual leaders and social workers. The purpose of this study was to investigate referral patterns and challenges of primary care clinicians for patients near the end of life.

METHODS: Using a written questionnaire with 29 items, we surveyed primary care clinicians in a university-based general internal medicine practice regarding

their referral patterns for seriously ill outpatients with chronic obstructive pulmonary disease, congestive heart failure or cancer. Five-item Likert scales were used to quantify the ease of providing care (1 = difficult, 5 = easy), number of referrals made (1 = none, 5 = > 3 referrals) and areas in which the primary clinicians needed help (1 = no help, 5 = a lot of help). We calculated mean scores and compared them between residents and faculty using the t-test statistic.

RESULTS: Of the 81 clinicians surveyed, 64 responded (79%). The clinicians were predominantly female (56%), Caucasian (59%) residents (63%) and had a mean age of 33.3 (S.D. 7). Overall, primary care clinicians reported difficulty in arranging homecare (mean score = 2.2), providing psychosocial care to patients (mean score = 2.4) and families (mean score = 2.2), and providing adequate access by phone for patients (mean score = 2.2). Residents indicated greater difficulty than faculty with conducting advanced directive discussions (mean score 2.1 vs. 3.1, $p < 0.01$) and with providing psychosocial care to patients (mean score 2.0 vs. 3.0, $p < 0.01$). Compared with faculty, residents indicated more help was needed with hospice referrals (4.0 vs. 3.0, $p < 0.02$).

CONCLUSION: To maximize the quality of care for patients near the end of life, faculty and residents in the university setting need increased access to home care services and psychosocial supports. This assistance may most easily be provided through an integrated multidisciplinary team designed to provide end of life care.

TRENDS IN DIAGNOSIS SEEN BY INTERNISTS IN AMBULATORY CARE SETTING IN THE LAST 20 YEARS. JH Oh, A Jotkowitz, AM Eichorn, SA Wartman, Division of General Internal Medicine, Long Island Jewish Medical Center, New Hyde Park, NY

PURPOSE: In the last 20 years many changes have occurred in health care delivery. Concerns with cost-effectiveness, rise in the penetration of managed care systems, and an ever-aging population has most likely produced changes in the type of diagnosis that are treated as outpatients. The purpose of this study is to determine the changes in the most common diagnoses seen by internists in the past 20 years.

METHODS: The National Ambulatory Medical Care Survey (NAMCS) is a probability sample survey conducted by Centers for Disease Control on patients' office visits since 1973. The 1977 and 1996 NAMCS datasets were used to generate lists of the most common diagnoses seen by internists in outpatient setting. Sampling weights were applied to generate national estimates of total number of visits. Diagnostic clusters (version 4.1 1998) developed by Schneeweiss et al were used to group diagnostic codes representing similar conditions. Clinically significant change was defined as a relative change of at least 50% over the last 20 years.

RESULTS: The estimated number of patient visits to internists has increased 63% from 61,918,658 in 1977 to 100,977,712 in 1996. Five diagnostic clusters remained common over the last 20 years: Hypertension (12.4% to 11.2%), Diabetes Mellitus (5.0% to 4.4%), Acute Upper Respiratory Infection (4.7% to 4.0%), Malignancies (2.3% to 2.7%), and Peptic Disease (2.0 to 2.1%). Common clusters with significant increases in frequency were General Medical Exam (2.2% to 3.8%), Acute Lower Respiratory Infection (2.4% to 3.8%), Sinusitis (1.1% to 2.4%), Acute Sprains or Strains (1.5% to 2.3%), and Asthma (0.5% to 2.1%). There was also a significant decrease in the frequency of the following common diagnostic clusters: Ischemic Heart Disease (9.1% to 2.8%), Degenerative Joint Disease (3.7% to 1.8%), Depression/Anxiety/Neurosis (3.2% to 2.1%), Medical Surgical Aftercare (1.5% to 0.5%), and Obesity (1.5% to 0.7%).

CONCLUSION: Our analysis shows an increase in acute illnesses seen in the outpatient setting. This result could be attributed to the recent shift of acute care from inpatient and emergency departments to the outpatient setting due to the increased penetration of managed care. The increase in general medical exams may be due to the need to establish a primary care physician in many health plans or the increased emphasis on preventive medicine. The reasons for the three-fold decrease in the frequency of ischemic heart disease needs to be investigated further. In view of these results it may be beneficial for internal medicine residency and CME programs to ensure that proper attention is being given to the diagnosis and management of acute illnesses.

COMPARISON OF TRIAGE DECISIONS IN PATIENTS WITH COMMUNITY-ACQUIRED PNEUMONIA WITH THE PORT STUDY PREDICTION RULE. E Ortiz, CH Quach, LA Lenert, VA San Diego Healthcare System, San Diego, CA; University of California, San Diego

PURPOSE: The Pneumonia Patient Outcomes Research Team (PORT) developed a prediction rule to assist clinicians with hospitalization decisions in patients with community-acquired pneumonia (CAP). This study was undertaken to assess adherence to the prediction rule and evaluate its potential impact as a triage tool.

METHODS: We performed chart reviews on all patients with CAP who were seen in the Urgent Care or General Medicine clinics at the VA San Diego Healthcare System during 1998 and met the inclusion criteria. Demographics, treatment setting, comorbid conditions, signs/symptoms of CAP, laboratory and x-ray results, antibiotic use, resolution of CAP, mortality, resource use, and adherence to the prediction rule were analyzed.

RESULTS: There were 161 patients in the study, including 85 who were initially managed as outpatients and 76 who were initially hospitalized. Of the 85 outpatients, 20 (24%) were in risk class III (intermediate risk), where either a brief hospitalization or outpatient treatment is appropriate, and 3 (4%) were in risk class IV (high risk), where hospitalization is recommended. Five (25%) class III patients and 1 (33%) class IV patient required subsequent hospitalization. No deaths occurred in any of these patients. Of the 76 hospitalized patients, 20 (26%) were in

risk classes I-II (low risk) and should have been managed as outpatients. These patients had more comorbidities (1.6 vs 0.7, $p < 0.001$) and signs/symptoms (3.6 vs 2.5, $p < 0.001$) than the 62 class I-II patients who were managed as outpatients. No deaths or rehospitalizations occurred in these patients compared with no deaths and a 3.2% rehospitalization rate for the class I-II patients who were managed as outpatients. No deaths occurred in any patients in risk classes I-III, regardless of treatment setting. Mortality rate was 14% in risk class IV and 33% in risk class V.

CONCLUSION: Agreement between observed clinical practice and the prediction rule was 86%. Deviations from the prediction rule were skewed toward over-hospitalization, as 26% of our low-risk patients were managed as inpatients, while 4% of our high-risk patients were managed as outpatients. Some of this deviation may be attributed to factors such as compliance or social issues, patient preferences, etc. that are not addressed by the prediction rule and may not be captured in the medical record. We believe most of it is attributable to clinicians overestimating the severity of illness in low-risk patients. Prospective use of the prediction rule can help identify these low-risk patients and triage them to outpatient management, which should result in cost savings without adversely affecting outcomes.

APPLES AND ORANGES?: TESTING RESPONSE RELIABILITY TO COMPUTERIZED AND PAPER-BASED QUESTIONNAIRES. PA Ott, DG Litaker, L Muscatello, The IH Page Center for Health Outcomes Research; Department of General Internal Medicine, The Cleveland Clinic Foundation, Cleveland, OH

PURPOSE: Internet surveys are an increasingly common way to access large populations at minimal costs in obtaining research data. Although many factors make this approach attractive, the reliability of responses obtained is unclear, even if a paper-based questionnaire is placed on the Internet unaltered. The purpose of this study was to establish reliability in administering a questionnaire by both traditional and computerized interfaces.

METHODS: A questionnaire consisting of the Rhinoconjunctivitis Quality of Life Questionnaire (RQLQ) and the allergy-specific Work Productivity and Activity Impairment Index was developed for three different interfaces: paper form (P), touch screen (TS), and web-based (WB) form. Volunteers answered this questionnaire in two of the three formats, assigned randomly and in random order. No supplemental instructions were provided by the research assistant, who administered all questionnaires sequentially in an office setting. Responses were transformed to a summary score for the RQLQ, compared for each individual, and differences by survey format were explored using one way ANOVA. Test-retest responses for each individual were compared on an item-by-item basis using Pearson's correlation and kappa coefficients. Completion time for each test was monitored and compared for each format pair using Student's *t* test. Respondent acceptance was ascertained using an open-ended question upon completion of the second questionnaire.

RESULTS: 75 volunteers completed 150 questionnaires in one of three possible format pairs (P-TS, P-WB, and TS-WB). 54 (72%) of respondents were female and the median age group for respondents was 40-49 years. Responses for most items were highly correlated for all formats ($p < 0.001$) although correlation coefficients ranged from .575 to .960. Kappa coefficients suggested excellent concordance in responses obtained by each format pair. Interestingly, the RQLQ score was modestly, although significantly higher for the paper-based form, while similar for both TS and WB formats. Completion time did not differ by questionnaire format and acceptance of each testing mode by respondents was similar.

CONCLUSION: Although concordance and correlation of responses to a questionnaire administered using different interfaces is similar, a systematic bias may have resulted in higher scores for respondents using the paper-based form. This study suggests alternate methods of data acquisition may have significant value in obtaining research data without affecting the reliability of results. Nonetheless, careful testing prior to implementation is important to identify other factors that may lead to small, but significant systematic differences.

ASSOCIATION OF SILDENAFIL USE TO IMPROVEMENTS IN EMOTIONAL WELL-BEING AND RELATIONSHIP WITH SEXUAL PARTNER. NM Paige, RD Hays, MS Litwin, J Rajfer, MF Shapiro, Divisions of General Internal Medicine and; Urology, University of California, Los Angeles, Los Angeles, CA

PURPOSE: We evaluated the impact of sildenafil use on erectile function, relationship with sexual partner, health-related quality of life (HRQOL), and emotional well-being among individuals with erectile dysfunction.

METHODS: Patients were recruited from a university hospital urology clinic, a university hospital internal medicine clinic, and from university-affiliated community primary care clinics. Letters were sent to 236 eligible patients by the 25 primary care providers and 2 urologists in the study asking their patients if they would participate in the study. Respondents ($n = 140$) who returned a postcard were sent the survey materials. 124 (or 52.5% of the eligible sample) completed and returned a survey; the 85 patients who reported current use of sildenafil were included in this analysis. Sexual and erectile function were assessed using the International Index of Erectile Function (IIEF). Items from the marital interaction scale from the Cancer Rehabilitation Evaluation System Short Form (Cares-SF) were used to assess relationship with sexual partner. The survey also included the SF-12 and the emotional well-being scale from the SF-36. Respondents were asked to report about their status for both before and after treatment with sildenafil.

RESULTS: The mean age of the respondents was 60.5 years (range 36-80). 53% were married; 95% were white; 62% had a 4-year college degree; mean household income was \$75,000. Physical and mental health for the sample at baseline was similar to the U.S. population; the average SF-12 physical health summary score was 52.9; the average mental health summary score was 51.4. Users of sildenafil reported a 54% increase in overall IIEF scores, 88% increase in erectile function scores, 60% increase in overall sexual satisfaction, and 36% increase in intercourse satisfaction (all $p < 0.01$). 38% of respondents indicated that using sildenafil had definitely improved the quality of their life. Likewise, 29% of respondents indicated that using sildenafil had definitely improved their relationship with their partner. With the use of sildenafil, there was a statistically significant improvement in scores for erectile function ($p < 0.01$), relationship with sexual partner ($p = 0.01$), and emotional well-being ($p < 0.01$). In a multivariate model, improvements in erectile function and relationship with sexual partner were each significantly associated with improvement in emotional well-being ($R^2 = 0.19$, $p < 0.01$).

CONCLUSION: Users of sildenafil reported dramatic improvements in erectile and sexual functioning. These improvements were associated with improvements in their scores on emotional well-being and relationship interaction measures. Prospective studies are needed to further investigate these potentially important associations.

EXPLORATION OF ECONOMIC IMPLICATIONS OF ROFECOBIX VERSUS NSAIDS: A DECISION-ANALYTIC APPROACH. JM Pellissier, WL Straus, DJ Watson, SX Kong, Merck & Co., Inc., West Point, PA

PURPOSE: Phase III clinical trial results of rofecoxib (VIOXX), a specific inhibitor of Cox-2, showed osteoarthritis (OA) patients treated with VIOXX had significantly fewer clinically significant gastrointestinal (GI) adverse events than those who received non-selective NSAIDs. We estimated the economic implications of these results.

METHODS: A model-based decision analytic approach was used, focusing on events related to GI problems that imply healthcare resource use (gastrointestinal perforations, symptomatic gastrointestinal ulcers or upper GI bleeding (PUBs) and minor GI problems, e.g., dyspepsia). Event probabilities came from clinical trials of VIOXX. To extrapolate trial results to practice, resource utilization profiles for GI events were developed using the literature, updated by chart review studies and costed using current published sources. Average wholesale prices were used for base case drug costs, weighted by 1999 market share for generic and branded products. Efficacy of GI co-medications (co-meds) and mortality risk associated with major GI problems were inferred from the literature. A 75% reduction in future prophylactic GI co-med prescription rates with VIOXX was assumed for the base case. Base case 1 year analyses were done with the PUB data obtained from a prespecified pooled analysis of the VIOXX clinical trials. Analyses were also performed using pooled results of two 12 week endoscopic surveillance trials, with adjustments for silent ulcers of 40% and 85%. Sensitivity analyses explored the effects of varying costs, hospitalization and surgery rates for PUBs, efficacy of GI co-meds and rates of GI co-med use. We calculated iatrogenic cost factors, which express a drug's economic burden as the ratio of the total expected cost of drug treatment (drug cost plus costs of secondary drug effects) to drug cost alone.

RESULTS: The iatrogenic cost factors implied by the base case were 1.86 for NSAIDs and 1.18 for VIOXX. The NSAID iatrogenic results are consistent with previously published results. Under base case conditions, the expected cost savings in GI problems and co-meds averted with VIOXX versus NSAIDs was \$0.81 per day, representing a 95% offset of the difference in drug price. In analyses based on endoscopic data, therapy with VIOXX was less expensive than therapy with NSAIDs regardless of silent ulcer adjustment. Results were most sensitive to prophylactic GI co-med rates, and robust over a range of model assumptions and costs.

CONCLUSION: In this analysis based on differences in clinically significant GI events and dyspepsia for OA patients, VIOXX had a markedly lower iatrogenic cost compared to NSAIDs due to cost savings in GI problems and co-meds averted. When endoscopic data alone were considered, VIOXX was cost-saving across all assumptions about silent ulcer rates.

A SURVEY OF HMO PATIENTS' ATTITUDES TOWARD FINANCIAL INCENTIVES. AG Pemberton, SD Pearson, Department of Ambulatory Care and Prevention, Harvard Medical School and Harvard Pilgrim Health Care, Boston, MA

PURPOSE: To assess patients' knowledge of and attitudes toward physicians' financial incentives in a large mixed-model HMO.

METHODS: A written survey was sent to 2,000 randomly selected patients in an HMO in New England. The survey included descriptions of three basic compensation models: salary with <10% withhold (SALARY); fee-for-service with < 10% withhold (FFS); and group capitation (>10 physicians) with stop-loss protection (CAPITATION). Patients were asked what their level of comfort would be if their physicians were paid by each of the three models.

RESULTS: Among patients surveyed, 1125 (56%) responded. Respondents had a mean age of 56 years (sd 16.5) and 61% were women. Only 2% of patients had ever had a discussion with their PCP about how they were paid, but 62% desired more information about this. Of those who wanted more information, 60% preferred it to come from the HMO, 17% from their physician, 5% from both, and 19% were not sure. The percentage of patients who said they would be "uncomfortable" or "very uncomfortable" with each payment model differed significantly:

with SALARY 16% [95% CI 14%, 18%]; with FFS 25% [22%, 28%]; and with CAPITATION 53% [50%, 56%]. Discomfort with CAPITATION was more common among patients with the following characteristics: under age 50 (62% vs. 47%, $p < 0.001$), white race (55% vs. 36%, $p < 0.001$) annual household income $> \$20,000$ (61% vs. 18%, $p < 0.0001$), college education (69% vs. 36%, $p < 0.001$), commercial insurance (65% vs. 33%, $p = 0.001$), and excellent or good health (55% vs. 36%, $p = 0.001$). With the exception of race, discomfort with FFS was also significantly related to all of these characteristics. As shown in the table, multiple logistic regression models found education, insurance type, and income to be independent correlates of discomfort with both FFS and CAPITATION:

	FFS, adjusted OR	CAPITATION, adjusted OR
Income $> \$20,000$	2.2 (1.15, 4.17)	3.8 (2.31, 6.17)
College degree	1.7 (1.20, 2.43)	2.0 (1.47, 2.73)
Commercial insurance	1.6 (0.97, 2.56)	2.2 (1.45, 3.32)

CONCLUSION: A majority of patients wanted more information about how their PCP was paid. Discomfort with fee-for-service and capitation were evident compared to salaried compensation. Patient concerns about financial incentives are important to consider in the design and disclosure of physician compensation.

PROCESS OF CARE AND OUTCOME AFTER ACUTE MYOCARDIAL INFARCTION IN VA COMPARED WITH MEDICARE. LA Petersen, S-LT Normand, M Volya, C Brown, BJ McNeil, HSR&D, VAMC, Houston, TX; Harvard Medical School, Boston, MA

PURPOSE: To compare process of care and outcome after acute myocardial infarction (AMI) in VA and non-VA hospitals, we used clinical data from 2 groups of patients with confirmed AMI in 1994–95.

METHODS: The first group was a stratified national random sample of 2486 male veterans age 65 and over cared for in 81 VAs. The second was a random sample of 29,457 males age 65 and over cared for in 1530 non-VAs under fee-for-service Medicare financing in 7 states. Clinical data and use of aspirin (ASA), beta-blockers (BB), angiotensin converting enzyme inhibitors (ACEI), and thrombolytics were collected on both samples using the Cooperative Cardiovascular Project data collection instrument. Ideal candidates for medications were identified using published criteria. To reduce confounding, we matched each VA patient to a Medicare patient using caliper matching based upon propensity scores. Propensity scores were estimated using a logistic regression model that included patient and hospital correlates of VA use.

RESULTS: The mean age of VA patients was 73.4 years (+5.7) and of Medicare patients 75.5 years (+7.0) ($p < 0.001$). Ideal VA and Medicare candidates were equally likely to undergo thrombolysis (51.2% vs. 47.0%, respectively; $p = 0.19$). Of ideal candidates for ASA, 77.4% in VA and 67.8% in Medicare were prescribed ASA on discharge ($p < 0.001$). Of ideal candidates for BB, 64.2% and 54.7% respectively were prescribed them on discharge ($p < 0.001$). In the subgroup of ideal candidates with recurrent chest pain, prior MI, or positive stress test, only 42.7% of VA patients and 36.0% of Medicare patients were prescribed BB on discharge ($p < 0.001$). Of ideal candidates for ACEI, 62.2% and 53.4% were prescribed them on discharge ($p = 0.015$). In the subgroup of ideal candidates with poor left ventricular function and diabetes, 63.0% in VA and 44.4% in Medicare were prescribed ACEI on discharge ($p < 0.001$). In the propensity score approach, 92% of the VA patients were matched to Medicare patients, and the analysis was restricted to the matched pairs. Of matched ideal candidates, VA patients were more likely than Medicare patients to receive thrombolysis (50.6% vs 40.4%; $p = 0.013$), or to be discharged on ASA (77.3% vs 68.6%; $p < 0.001$) but equally likely to be discharged on BB or ACEI. There was no significant difference in age-adjusted risk of 30-day mortality in VA vs. Medicare (Cochran-Mantel-Haenszel relative risk [95% CI] 0.9 [0.9–1.0]).

CONCLUSION: Process of care and outcome for AMI is superior or comparable in VA as compared with Medicare. For both VA and Medicare, significant opportunity for improving medication use in subgroups of high-risk patients exists. Further studies should assess long-term use of these medications in appropriate candidates.

FEMALE PATIENTS' PREFERENCES REGARDING MASTECTOMY COMPARED TO BREAST-CONSERVING SURGERY AS THERAPY FOR EARLY STAGE BREAST CANCER. NW Phifer, S Cykert, AF Walker, Internal Medicine, Moses Cone Hospital, Greensboro; Division of Internal Medicine and Clinical Epidemiology, The University of North Carolina School of Medicine, Chapel Hill; Department of Sociology, The University of North Carolina at Greensboro, Greensboro, NC

PURPOSE: Despite convincing evidence that breast-conserving surgery (BCS) followed by local irradiation produces equivalent cancer cure rates compared to mastectomy in early stage breast cancer, the utilization of BCS for patients who meet clinical criteria remains low.

METHODS: Using a structured questionnaire, we interviewed 101 women aged 50 to 75 yrs. to assess their strength of preference for BCS and mastectomy when presented with scenarios offering these therapies for their newly diagnosed breast cancer. The scenarios were presented in a standard gamble format and preferences are reported as utility scores (1 = perfect health, 0 = death). Bivariate analyses to identify predictors of utility scores and to compare same subject utility scores for BCS vs mastectomy were performed using non-parametric techniques.

RESULTS: For the new diagnosis of early stage breast cancer followed by BCS

then irradiation the mean utility score was .83 (95% CI .77, .89) compared to .82 (95% CI .76, .88) for this diagnosis followed by mastectomy. The .01 mean utility score difference was not statistically significant. In bivariate analyses, "good" self-reported health predicted higher utility scores for both procedures while income greater than \$50,000 per year predicted a higher score for mastectomy. Age, race, education, health insurance status, and marital status did not predict any significant differences in utility scores for either BCS or mastectomy.

CONCLUSION: We conclude that the very limited use of BCS is driven by factors other than patients' preferences.

PATTERNS IN THE UTILIZATION OF CONSULTANTS AT A UNIVERSITY HOSPITAL. RL Powers, P Hortsman, West Virginia University, Morgantown, WV

PURPOSE: The goal of this study is to examine the timeliness and utility of hospital consultative practices on non-pregnant adult medical and surgical patients.

METHODS: The initial time and date for all consults over a three month period from four medical and surgical nursing stations were recorded. The consult was tracked by retrospective chart review to determine the date of its completion and the potential for impact on patient care through delay in performance, change in medication (med), change in therapy (tx), new diagnoses (dx), recommendation to undergo additional tests (new test), recommendation to undergo a procedure (new proc), and length of stay (LOS). Analysis was performed on the following groups of consultants: General Internal Medicine (GIM), Medical Sub-specialty (MSS), General Surgery (GS), Surgical Sub-specialty (SSS) or Cardiovascular Surgery (CVS). **RESULTS:** Ninety-nine percent of the 847 consults were seen within 24 hours, prohibiting assessment of the impact of delay in consult performance. Eighty-one percent of the patients undergoing consults had greater than the average hospital LOS (4 days). The individual services receiving the highest number of consults were gastroenterology (95), GIM (67) and cardiology (63). Analysis of consultant recommendations identified trends among specialties, with internists (especially GIM) serving more of a management and evaluative role, sub-specialists more of a diagnostic role and surgeons and gastroenterologists (GI) more of a procedural role. Chi-square for a gradient in the proportions tests were used to test the significance of the associations between ordered physician groups and their recommendations.

Consults and the Recommendations Recorded by Specialty Groups

CONCLUSION: Chart review found that hospitalized patients receiving consults have a longer than average LOS despite rapid response by the consultants. The type of recommendations differed by specialty. Internists (especially GIM) provide more management and evaluative guidance, sub-specialists offer more diagnostic suggestions and surgeons and gastroenterologists propose more procedures. Determination of the impact of these differences on patient outcome and the reasons for choosing a particular consultant requires further study.

	n	Δmed	Δtx	Δdx	new text	new proc
GIM	67	64%	31%	16%	76%	6%
MSS	379	46%	27%	31%	54%	34% (GI 74%)
GS	68	29%	25%	19%	56%	62%
SSS	135	29%	23%	24%	36%	31%
CVS	48	15%	19%	10%	33%	58%
p for trend		<.0001	.008	.0020	<.0001	<.0001

GENDER BIAS IN CLINICAL TRIALS ENROLLMENT: DO DOUBLE STANDARDS STILL APPLY? K Ramasubbuj, HS Gurm, DG Litaker, Internal Medicine Residency Program, The Cleveland Clinic Foundation, Cleveland, OH

PURPOSE: Differential enrollment into clinical trials by gender has been previously described. Because study conclusions arising from a predominantly male population must be extrapolated to all patients, a concern for generalizability has been raised. In 1993, recommendations were made by the Food and Drug Administration specifically encouraging the inclusion of women to address this potential issue. The purpose of this study was to review clinical trial (CT) enrollment among studies published in a major medical journal to reassess changes in this practice. **METHODS:** We conducted a systematic search of all articles published in the Original Articles section of the New England Journal of Medicine from 1994–1999. All randomized CTs in which the primary end point was either total mortality or included mortality in a composite endpoint were considered for abstraction. We excluded all studies concerning gender-specific diseases, subset analyses of other trials, and metaanalyses. Trials were characterized by primary specialty and the primary factor of interest was percentage of women enrolled. Data were analyzed using Student's t test and ANOVA.

RESULTS: 1322 original articles were published in the period 1994–1999 including 423 CTs. Although 121 met initial inclusion criteria, we focused attention on the four specialties responsible for publishing the majority of these reports. 94 CTs were identified for abstraction and represented publications related to cardiovascular disease in 50 (54.3%) articles, treatment of cancer in 21 (22.3%) articles, infectious diseases in 12 (12.8%) articles, and the gastrointestinal system in 10 (10.6%) articles. Considering all study results together, only 11 (11.7%) studies reported the primary outcome mortality by gender. The percentage of women enrolled ranged from 0 to 62.5% with an average of 25.9% across all studies. Analyzing percentage of women enrolled by specialty of study, funding source, and primary country of study origin demonstrated no significant differences between

groups. Interestingly, of 13 studies funded by the NIH, the percentage of women enrolled was lowest (20.7%).

CONCLUSION: At a time when evidence-based medicine is emphasized in residency training and medical practice, there are surprisingly few gender-specific data available to guide the treatment of a substantial segment of the population. Despite the recommendations of a federal agency, enrollment of women into CTs published in a major medical journal is disproportionately low. These results suggest an ongoing need for a conscious effort at enrollment of women in future research.

POLYPHARMACY IN A HOSPITAL BASED AMBULATORY PSYCHOTIC DISORDER CLINIC POPULATION. SS Rathore, E Rothman, D Perkins, Department of Psychiatry, University of North Carolina Hospital, Chapel Hill, NC

PURPOSE: The benefit of multiple psychotropic medication use (polypharmacy) among patients with schizophrenia is uncertain. However, clinicians may attempt to manage psychotic patients' residual symptoms with additional psychotropic agents. Despite this practice, the prevalence of polypharmacy, and characteristics of patients receiving multiple psychotropic agents in clinical practice, is unknown. **METHODS:** We evaluated patients treated at a hospital-based ambulatory psychotic disorder clinic (Schizophrenia and Treatment Evaluation Program) at the University of North Carolina Hospital, Chapel Hill, NC to determine the incidence and characteristics of patients receiving psychotropic polypharmacy. Clinic patients' physicians were contacted and asked to provide data including patients' demographic characteristics, baseline psychiatric diagnoses, aspects of medical history, and current psychotropic medication use. Patients receiving polypharmacy were identified and polypharmacy was evaluated for associations with patient characteristics using t-test and chi-square analysis. Of the 130 patients enrolled in the clinic, we obtained data on 106 patients (82%) whose results are reported here.

RESULTS: Patients were 34 ± 12 years of age, predominantly male (63%) and Caucasian (79%), though the clinic contained a substantial population of African-American patients (19%). Primary patient DSM IV axis I diagnoses were schizophrenia (56%), schizoaffective disorder (35%), non-organic source psychosis (7%) and bipolar disorder (6%); few patients (4%) presented with axis II disorders. Patients had received psychiatric treatment for an average of 10.5 ± 9.1 years and most (80%) were treated with an atypical antipsychotic agent. In addition to antipsychotic agents, patients were treated with antidepressants (12%), mood stabilizers (5%), antianxiety agents (4%). More than a third (34%) were treated with an antipsychotic agent and two or more psychotropic agents, while 8% were treated with two antipsychotic agents. Psychotropic polypharmacy was noted in 62% of patients, with an average number of 2.3 medications. Patients prescribed multiple medications were older (mean: 33.6 vs 26.7 , $p < 0.01$), had been in psychiatric treatment longer (mean: 10.0 vs 4.5 , $p < 0.02$), and were more likely to be white (86.2% vs 70.6%, $p < 0.05$); gender and a usual source of medical care were not associated with polypharmacy.

CONCLUSION: A significant portion of patients undergoing treatment at a hospital based ambulatory psychotic disorder clinic receive psychotropic polypharmacy. Further research is needed to elucidate the appropriateness of this polypharmacy.

RACIAL VARIATION IN THE INCIDENCE OF INTRAPARTUM HEMORRHAGE. SS Rathore, MJ McMahon, AF Olshan, Department of Obstetrics and Gynecology; Department of Epidemiology, University of North Carolina School of Public Health, Chapel Hill, NC

PURPOSE: Epidemiologic surveillance studies have reported a higher incidence of intrapartum hemorrhage—placental abruption, placenta previa, and other unspecified hemorrhage—among babies born to black mothers. It is unknown, however, to what extent racial variation in hemorrhage incidence may be explained by other clinical risk factors.

METHODS: We conducted a retrospective evaluation of births in North Carolina in 1997 to determine the incidence of intrapartum hemorrhage and its association with maternal race. Live singleton births ($n = 105,950$) with complete data regarding maternal race and the incidence of intrapartum hemorrhage were evaluated for univariate association with maternal race by chi-square analysis. Logistic regression models evaluated the adjusted odds of occurrence of intrapartum hemorrhage and three constituent etiologies—placental abruption, placenta previa, unspecified hemorrhage—among each racial group. Models adjusted for established hemorrhage risk factors included maternal age, education, gravidity, parity, prior termination, gestational age, adequacy of prenatal care, marital status, tobacco or alcohol use during pregnancy, pregnancy weight gain and pregnancy complications including anemia, cardiac disease, acute/chronic lung disease, gestational diabetes, hemoglobinopathy, hypertension, renal disease, eclampsia, uterine bleeding early in pregnancy, and premature rupture of membranes.

RESULTS: Black mothers had similar rates of intrapartum hemorrhage compared to white mothers (1.42% vs 1.41%, $p = 0.93$). Black mothers were more likely to experience intrapartum placental abruption (0.82% vs 0.65%, $p = 0.003$) than white mothers and were less likely to experience other unspecified hemorrhage (0.34% vs 0.48%, $p = 0.02$); race was not associated with the occurrence of placenta previa (0.33% black vs 0.33% white, $p = 0.95$). Maternal race remained associated with intrapartum hemorrhage after multivariate analysis, though the direction of the association was reversed. Black mothers were less likely to experience intrapartum hemorrhage (OR 0.76, 95% CI 0.65, 0.88) and other un-

specified hemorrhage (OR 0.63, 95% CI 0.48, 0.83) than white mothers (OR 1.00) and were statistically comparable for risk of placental abruption (OR 0.83, 95% CI 0.68, 1.03) and placenta previa (OR 0.83, 95% CI 0.62, 1.12).

CONCLUSION: While black mothers may have higher rates of intrapartum placental abruption, the increased incidence can be attributed to a disproportionately higher burden of other hemorrhage risk factors compared to white mothers. The decreased risk of other unspecified hemorrhage among black mothers requires further examination.

EXPERIENCE MAKES A DIFFERENCE IN PHYSICIANS' GRIEF REACTIONS TO THEIR PATIENT DEATHS. EM Redinbaugh, SD Block, DL Seltzer, N Gadmer, A Mitchell, R Arnold, Medicine, University of Pittsburgh Medical Center, Pittsburgh, PA; Psychiatry, Dana-Farber Cancer Institute, Boston, MA

PURPOSE: Caring for dying patients has been linked with emotional exhaustion and burnout among oncologists. To examine the relationship of experience (e.g., intern, resident, attending) with physicians' emotional reactions to dying patients this study assessed grief reactions, burnout, and coping behavior in a sample of physicians who recently experienced a patient death on a general medicine unit in an academic hospital.

METHODS: Of the 61 physicians eligible for the study, 89% ($n = 11$) agreed to participate. Participants included 18 interns, 18 residents, and 19 attendings the majority of who were male (64%), married (63%), and caucasian (76%). Each week two patients who died on the medical service of two large teaching hospitals were randomly chosen, and the attending, resident, and intern who cared for the patient at death were selected for study inclusion. The identified physicians were contacted by phone and asked to participate in the study which involved completing self-administered questionnaires as well as participating in an interview about their experience in caring for the patient who died. Interviews were transcribed and read for common themes. The Maslach Burnout Scale, the Grief Reaction Inventory, and 6 items from the Brief Ways of Coping Scale assessed burnout, grief, and coping, respectively. Questionnaire data were analyzed with SPSS-9.0 software. One-way ANOVAs tested the relationship between level of training and reactions to patient deaths, and hierarchical multiple regression tested the effect of experience (in years) on the relationship between grief reactions and burnout.

RESULTS: Attending physicians reported significantly fewer grief reactions ($F(2,52) = 4.65$, $p < .02$) and burnout symptoms ($F(2,52) = 3.40$, $p < .05$) than interns and residents; however, groups did not differ in the number of coping behaviors used to manage grief reactions ($F(2,52) = 1.9$, $p = .16$). Regression analyses indicated that number of years experience mediated the relationship between grief reactions and burnout ($\beta = -.36$, $p < .01$); as physicians gained experience, grief was less likely to be associated with burnout symptoms. Interview data suggested that interns were more affected by patient deaths because they were more intimately involved with patients and their families. Furthermore, interns indicated that they were less likely to receive the support they needed when compared to the reports of residents and attendings.

CONCLUSION: A greater sense of emotional closeness to patients and families as well as a lack of support from colleagues appeared to make interns more vulnerable to the effects of patient deaths when compared to more experienced physicians.

VARIATION AND INACCURACY IN PHYSICIANS' ASSESSMENT OF CHEST PAIN PATIENTS IN THE EMERGENCY DEPARTMENT. B Reilly, J Schaidler, A Evans, Y Wang, R Rydman, Medicine; Emergency Medicine, Cook County Hospital; Rush Medical College, Chicago, IL

PURPOSE: Prior to implementing a standardized decision aid for chest pain patients in our Emergency Department (ED), we sought to understand how our physicians diagnosed and triaged such patients without it.

METHODS: We presented 20 written cases to 147 physicians specializing in internal medicine, emergency medicine and cardiology at one institution. Each case contained clinical data needed to estimate the patient's probability of coronary artery disease (pCAD), acute myocardial infarction (pMI), and major complications (pComp) as reported in large published cohort studies. Respondents were asked to estimate these probabilities, make a provisional diagnosis (MI, unstable angina, other) and a triage decision (CCU, non-CCU telemetry, other). Accuracy of probability estimates was measured in two ways: 1) as proportions within 5 percentage points of the best point estimates reported in the literature, and 2) as proportions within clinically relevant threshold ranges (e.g., pMI $< 5\%$, 5–25%, $> 25\%$). Agreement was measured as interquartile ranges (IQR) and reliability coefficients (R) for continuous variables, and as kappa statistics (k) for categorical variables.

RESULTS: Overall, agreement among physicians was moderate for pCAD ($R = 0.49$) and pMI ($R = 0.45$) but only fair for pComp ($R = 0.30$), diagnosis ($k = 0.31$), and triage decisions ($k = 0.36$). Wide variation persisted after controlling for physicians' specialty and experience. Accuracy of estimates of pCAD, pMI, and pComp compared with the literature ($\pm 5\%$) were only 18%, 26%, and 97%, respectively. Median probability estimates exceeded clinically relevant threshold ranges in all cases except pCAD when pCAD was high. Among the lowest risk cases, physicians overestimated pCAD, pMI, and pComp in 81%, 62%, and 67%, respectively.

CONCLUSION: Our physicians varied widely in their diagnoses and triage decisions in simulated cases of ED patients with chest pain. They consistently overestimated pCAD, pMI, and pComp. Achieving consensus in decision-making will require physician education and, perhaps, standardized decision aids.

TRIAGE OF EMERGENCY DEPARTMENT PATIENTS WITH CHEST PAIN: FROM DIAGNOSIS-BASED DECISION MAKING TO OUTCOMES-BASED DECISION MAKING. B. Reilly, J. Schaidler, Y. Wang, A. Evans, Medicine, Emergency Medicine, Cook County Hospital; Rush Medical College, Chicago, IL

PURPOSE: Before implementing an outcomes-based triage aid for chest pain patients in our Emergency Department (ED) (derived from Goldman et al's prediction rule; NEJM 1996), we compared the triage aid with our physicians' decision-making in simulated cases.

METHODS: In 20 simulated cases, we compared trichotomous triage decisions (CCU, non-CCU telemetry, other) by 147 physicians at one institution with decisions recommended by the decision aid. Using recursive partitioning, we modeled the multivariate predictors of physicians' triage decisions: clinical variables of the cases (eg, electrocardiogram (ECG) changes, congestive heart failure (CHF), hypotension, prior myocardial infarction (MI) or revascularization); physicians' probability estimates (for coronary artery disease (CAD), MI, and cardiac complications); physicians' diagnoses (MI, unstable angina), specialty (Medicine, Emergency Medicine, Cardiology), level of training (resident, attending), and experience. Using the decision aid as the gold standard model, we measured the accuracy of physicians' decisions. We also examined the impact of physicians' diagnoses and probability estimation on the accuracy of both the gold standard model and the recursive partitioning model.

RESULTS: In contrast to the decision aid (which is based on probability of complications), the physicians' decision model involved 3 sequential branch points: 1) diagnosis of MI or unstable angina; 2) presence of CHF; and, 3) abnormality of ECG. Compared with the decision aid, physician's triage decisions were inaccurate in 42% of cases. Perfecting physician's diagnoses (in their model) reduced inaccuracy slightly, to 40%. Conversely, using physicians' probability estimates in the gold standard model actually increased inaccuracy, to 56%.

CONCLUSION: Our physicians use a diagnosis-based triage strategy for ED patients with chest pain. Adoption of an outcomes-based strategy will require physician's acceptance of both its logic and its empirically derived probability estimates.

ARE CALIFORNIA MANAGED CARE ORGANIZATIONS USING HOSPITAL OUTCOME STUDIES AND DATA? PS Romano, JA Rainwater, Division of General Medicine, UC Davis School of Medicine, Sacramento, CA

PURPOSE: Managed care plans have a major role in selecting health care providers, but there is little evidence that quality plays a role in this process. We surveyed health plan executives to assess whether and how they use publicly available or internally generated hospital outcomes data.

METHODS: The CA Department of Corporations provided a list of managed care plans with active Knox Keene licenses as of June 1998. We attempted to contact all 47 HMOs that contracted with hospitals to pay for acute inpatient care. We mailed a questionnaire to a key respondent at each HMO, and followed up with telephone calls. Thirty HMOs provided a usable questionnaire or interview (63.8%). Several HMO characteristics were examined, including size, profit status, model type, and accreditation.

RESULTS: Health plan executives identified the following factors as most important in contracting with hospitals (in descending order): JCAHO accreditation, geographic location, and negotiated price. Respondents from for-profit HMOs assigned greater importance to price than respondents from nonprofit HMOs ($p < .02$). Other factors deemed very or extremely important by at least 80% of respondents were disciplinary actions against the hospital, the hospital's reputation, and its commitment to quality improvement. By contrast, specific outcome measures (e.g., mortality, complication, or readmission rates; incidence of potentially over-used procedures; transplant success rates) were deemed less important. Respondents from staff model HMOs viewed these outcome measures as more important than those from other HMOs ($p < .03$). Although most respondents (70%) had reviewed at least one source of publicly available information on hospital outcomes, the usefulness of each source (except www.healthscope.org) was rated "poor" or "fair" by at least 40% of those who had used it. Ten respondents, including 7 of the 9 accredited HMOs, reported conducting internal studies of hospital performance. Health plan executives saw collecting and disseminating hospital quality information as primarily the responsibility of government agencies and accrediting organizations (e.g., JCAHO). About 73% of respondents said that HMO's should collect and analyze their own information on hospital quality. The majority of our respondents agreed that hospital outcome studies would ultimately lead to improved quality of care (87%) and less unnecessary and inappropriate care (70%).

CONCLUSION: Managed care leaders in CA base hospital contracting decisions primarily on accreditation, geographic location, and price. Objective data on hospital quality were not viewed as extremely important in this process. However, respondents expressed optimism that such information may become more useful and important in the future.

HYPERGLYCEMIA IN ACUTE ISCHEMIC STROKES: PREVALENCE, SEVERITY, TREATMENT, AND EFFECTS ON OUTCOMES. J. Rotich, LS Williams, A. Bruno, SE Fineberg, WM Tierney, Regenstrief Institute, Indiana University Center for Healthcare Improvement, Roudebush VAMC, Indianapolis, IN

PURPOSE: Preliminary data from animal and human studies suggest that hyperglycemia during acute ischemic strokes is associated with higher morbidity and worse short-term sequelae. If true, this might identify a potential intervention to

improve stroke outcomes. Using data from a comprehensive electronic medical record in an inner-city public hospital, we sought to assess (1) the proportion of patients with acute stroke who are hyperglycemic when admitted (2) the severity and treatment of the hyperglycemia, and (3) its association with stroke outcomes. **METHODS:** Each patient's first hospitalization for ischemic stroke (i.e., primary discharge ICD9 code 434 or 436) during the 5-year period from 7/93 through 6/98 was identified. From patients' electronic records we extracted data on demographics, prior diagnoses, all blood glucose (BG) results performed by the inpatient lab (finger-stick BG results were not available), hypoglycemic drug treatment, inpatient duration and charges, and mortality within 30 days of discharge. Stroke patients with admission hyperglycemia (first BG on admission ≥ 130 mg/dL) were compared with euglycemic stroke patients.

RESULTS: We identified 671 patients with acute stroke. Their mean age was 62 years, 56% were women, 55% were African-American, and 51% had prior diabetes. Of the 656 (98%) stroke patients with inpatient BG results, 269 (41%) were hyperglycemic on admission, and 153 (23%) had an admission BG ≥ 180 . Patients who were hyperglycemic on admission were more often women (64% vs 50%, $p < .001$) and diabetic (80% vs 30%, $p < .001$). An average of 8 additional BG tests were obtained for 183 (68%) of the patients with admission hyperglycemia, with a mean value of 202. Inpatient drug treatment of patients with admission hyperglycemia was suboptimal: 44% received no diabetes drugs, 10% received only oral hypoglycemic drugs, and 13% received only sliding scale insulin. Patients with admission hyperglycemia had 20% longer inpatient stays (7.1 vs. 6.0 days, $p = .02$), 24% higher total inpatient charges (\$10,661 vs \$8,581, $p = .03$), and twice the risk of dying within 30 days of discharge (11% vs 5.7%, $p = .02$). Controlling for age, sex, race, and prior diabetes, the odds ratio for 30-day mortality among patients with admission hyperglycemia was 1.9 (95% CI 1.0-3.7, $p = .049$). The highest BG obtained during hospitalization also predicted 30-day mortality: a 50 mg/dl increase in the highest BG increased the odds of death by 20% (95% CI 10-25%, $p < .001$).

CONCLUSION: Admission hyperglycemia is common, costly, and morbid among inner-city patients with acute ischemic strokes. It is also inadequately treated. These data support the implementation of clinical trials of intensive management of hyperglycemia in patients with acute ischemic strokes.

DIFFERENCES BETWEEN GENERALISTS AND INTERNISTS IN THE MANAGEMENT OF HYPERTENSION IN A MANAGED CARE SETTING. AL Rubinstein, V Schoj, D Terceiro, M Boccardo, V Finkelsztain, N Gimpel, Division of Family and Preventive Medicine, Hospital Italiano, Buenos Aires, Argentina

PURPOSE: Hypertension (HT) remains a serious public health problem and a major risk factor for coronary heart disease (CHD) and stroke. Despite the availability of evidence-based guidelines to help improve the consistency and quality of interventions, there is still much variability among different primary care doctors (PCP). Our aim was to evaluate differences in HT care between generalists (GEN: family doctors and general internists), as compared to categorical internists (INT), from a university-affiliated HMO.

METHODS: We randomized 20-25 medical records of patients (pt) with HT per PCP (30 GEN and 10 INT) for a total of 925. Several indicators of **process**: adequacy of work-up (W), % recording in notes of life-style changes (LSC), % of pt with non-pharmacologic treatment (NPT), use of drugs, % of pt on diuretics and/or β blockers (DBB), intensity of drug treatment (IDT), etc; and **outcomes**: % of pt with HBP control ($\leq 140/90$), and mean change in SBP and DBP, were evaluated. We also analyzed HT care in different subgroups: % of pt with ACE inhibitors (ACE) in type 2 DM or CHF, and % of pt on NPT in JNC VI stage 1-group A and group C. Univariate and multivariate analysis were performed to explore the association of GEN/INT with each indicator, adjusted for potential confounders.

RESULTS: As compared to GEN, pt seen by INT were older (65.4 vs. 57.9 ys, $p < .001$) and more likely to be female (65.2% vs. 54.4%, $p > .01$). No differences were shown regarding pt distribution according to JNC VI stage (1-3) or group (A, B, C). We found no differences between GEN and INT with respect to % of pt with HT under control (52.7% vs. 55.8%, $p = 0.4$), mean change in SBP (14.7 vs. 17.1 mmHg, $p = 0.1$), or DBP (11.1 vs. 10.5 mmHg, $p = 0.5$). GEN and INT differed in the process of care: W: 96.7% vs. 79.1%, adjusted OR = 6.6 [5.8, 7.5]; LSC: 80.3% vs. 48.1%, OR = 2.8 [2.3, 3.2]; NPT: 24.4% vs. 6.3%, OR = 3.2 [2.6, 3.8]; DBB: 70.6% vs. 41.9%, OR = 2.7 [2.1, 3.2]; and IDT: 66.6% vs. 54.8%, OR = 2.1 [1.6, 2.6]. Although INT as compared to GEN used more ACE (47.1% vs. 36.4%, $p < .001$), this difference was no longer significant when we analyzed only pt with DM or CHF (57.1% vs. 54.9%, $p = 0.8$). GEN indicated less calcium-channel blockers than INT (16.3% vs. 42.2%, OR = 0.26 [0.23, 0.5]). On the other hand, 16.3% of group C pt seen by GEN, were under NPT, as compared to only 3.3% of pt seen by INT.

CONCLUSION: Although both groups of PCP's achieved similar results on HT control, the type of care delivered was different. GEN were more conservative in the work-up, more prone to NPT, more likely to initiate drug treatment with diuretics and β blockers and less likely to use newer and more expensive drugs. This variation may have important consequences both in quality and cost-effectiveness.

CRITICAL PATHWAYS: SEPARATING THE WHEAT FROM THE CHAFF. S. Saint, TP Hofer, J Rose, LF McMahon.

PURPOSE: Most US hospitals use critical pathways for at least some of their patients. A goal of pathways is to maximize efficiency of care by reducing patient length-of-stay and resource utilization; however, pathway effectiveness in improving efficiency has been inadequately evaluated. We therefore assessed whether or

not critical pathways as a clinical management technique have been successful in reducing patient length-of-stay and resource utilization at our academic medical center.

METHODS: We evaluated all critical pathways begun in our medical center between 1993 and 1996 in which at least 50 adult patients could be evaluated in the year preceding and in the year succeeding pathway implementation. Each pathway was developed separately, based on the recommendations of a multidisciplinary team. Using a before-and-after design, we evaluated each pathway's effect on patient length-of-stay and resource utilization (as determined by relative value units). We constructed separate models for each pathway using each outcome and included patient data one year before and one year after pathway implementation. We accounted for both secular trends and DRG case-mix changes within pathway groups in our analyses. Statistical significance was based on a two-sided *P* value of less than 0.05.

RESULTS: 13 pathways satisfied inclusion criteria. The diagnoses (or procedures) covered by these pathways were: acute myocardial infarction (AMI), acute pancreatitis, asthma, breast surgery, caesarian section, community-acquired pneumonia (CAP), hip arthroplasty, kidney transplantation, knee arthroplasty, liver transplantation, lung lobectomy, percutaneous transluminal coronary angioplasty (PTCA), and radical nephrectomy. 4 of the 13 pathways significantly decreased patient length-of-stay more than what would be expected based on the secular trend: the pathways for AMI (by 10%), caesarian section (by 7%), kidney transplant (by 12%), and PTCA (by 11%). 4 pathways also significantly reduced ancillary resource utilization more than expected: the pathways for caesarian section (by 26%), CAP (by 10%), kidney transplant (by 14%), and PTCA (by 12%). 3 pathways—caesarian section, kidney transplant, and PTCA—were successful in significantly reducing both patient length-of-stay and ancillary resource utilization. The majority of pathways did not significantly decrease either patient length-of-stay or resource utilization.

CONCLUSION: Vast amounts of resources are currently expended on the development and implementation of pathways; to justify this expenditure, pathway efficacy should be demonstrated. Our evaluation reveals that though some pathways significantly reduced length-of-stay and resource utilization, the majority did not. Thus, efforts should be placed on critically assessing the effectiveness of pathways and understanding the reasons behind successful and unsuccessful ones.

LACK OF PHYSICIAN AWARENESS OF URINARY CATHETERS IN THEIR PATIENTS: IGNORANCE IS NOT BLISS. S. Saint, J. Weise, J. Amory, M. Bernstein, U. Patel, J. K. Zemencuk, S. J. Bernstein, B. A. Lipsky, T. P. Hofer, Department of Internal Medicine, University of Michigan, Ann Arbor, MI; Department of Medicine, UCSF, San Francisco, CA; Department of Medicine, University of Washington, Seattle, WA

PURPOSE: One-quarter of hospitalized patients have an indwelling urethral catheter placed at some time during their stay. Urinary tract infection (UTI) is the most common nosocomial infection, with catheter-related infection accounting for the vast majority of nosocomial UTIs. Catheter-related UTI leads to increased morbidity and healthcare costs. In addition, the majority of patients given indwelling catheters find the catheters uncomfortable and restrictive of their daily activities. Indwelling catheterization is inappropriate in about one-third of catheterized patients; however, the reasons behind inappropriate catheter use are unclear. We hypothesized that catheters are commonly used inappropriately because physicians are often unaware of their presence. We sought to assess: 1) how often physicians and medical students know if their own hospitalized patients have a catheter; and 2) whether catheter awareness depends on catheter appropriateness.

METHODS: The physicians and students responsible for patients admitted to the medical wards at four U.S. hospitals were the study subjects. Each provider was given a list with the names of the patients on their service and for each patient was asked: "As of yesterday afternoon, did this patient have an indwelling urethral catheter?" Respondent's answers were compared to the actual patient situation. Chart review assessed catheter appropriateness.

RESULTS: Of 469 patients, 117 (25%) had indwelling catheters. Catheterization was considered appropriate in 70% of these patients. Among all catheterized patients, 18% of medical students, 22% of interns, 28% of residents, and 35% of attending physicians were unaware that their patients were catheterized. Among inappropriately catheterized patients, 24% of medical students, 31% of interns, 41% of residents, and 51% of attending physicians were unaware that their patients were catheterized. After adjusting for patient age and gender, hospital, and level of respondent training, respondents were significantly more likely to be aware of catheters that were appropriate (OR = 3.6; 95% CI, 1.6 to 7.8).

CONCLUSION: Physicians are commonly unaware that their own patients are catheterized. Inappropriate catheters are more often "forgotten" than appropriate ones. Since the best method of limiting catheter-related UTI is to limit inappropriate urethral catheterization, system-wide interventions aimed at altering the process of catheterization are needed. We plan to initiate innovations limiting urethral catheterization, thereby improving the safety of hospitalized patients.

CAN PROVIDING PHYSICIANS WITH PATIENT-SPECIFIC INFORMATION INCREASE THE LIKELIHOOD OF ALCOHOL COUNSELING? R. Saitz, LM Sullivan, MA Moskowitz, JH Samet, Section of General Internal Medicine, Boston University School of Medicine, Boston Medical Center; Boston University School of Public Health, Boston, MA

PURPOSE: To test whether providing primary care physicians (PCPs) with patients' alcohol screening results and individualized recommendations increases the likelihood of alcohol counseling.

METHODS: PCPs in an urban academic practice were randomized to an intervention or control group. During office visits, intervention PCPs received patient's alcohol screening results (CAGE and consumption), a readiness to change measure, and action recommendations based on problem severity; control physicians received no such information. Eligible patients visiting these PCPs were explicitly defined current hazardous or harmful drinkers. Patient demographics, comorbidity, alcohol consumption, and problems were assessed in standardized interviews using validated instruments. PCP counseling was assessed by patient report immediately after the visit, and was defined as specific advice (regarding safe drinking, to quit, or cut down) or referral (to alcohol treatment or a mutual help group). Power was 80% to detect a 50% increase in counseling rates between PCP groups. **RESULTS:** Of 41 PCPs, 20 were in the intervention group, 48% were residents, 55% male, and 65% white. PCPs were similar by randomized group, including demographics, training, and attitudes towards patients with alcohol problems. The 312 patients were 64% male, 56% black, 19% white, 16% Latino, 63% high school graduates, 40% unemployed. Mean age was 44. Median annual income was \$7500. For most (66%), routine care was the reason for the visit, though 67% had 2 or more concerns to discuss. Patients drank a mean of 6 drinks per drinking day, and 36% were in Precontemplation, 32% Contemplation, and 32% the Action stage of readiness to change. A consistent trend favored intervention group PCPs: they were more likely to provide alcohol counseling (mean physician counseling rate 49% vs. 40%, *P* = 0.32); to give patients any advice about drinking (56% vs. 44%, *P* = 0.19); to have a discussion with the patient about drinking (68% vs. 57%, *P* = 0.23); and to be the initiators of such a discussion when one occurred (64% vs. 52%, *P* = 0.25). Intervention group PCPs were significantly more likely to give safe drinking limit advice (19% vs. 6%, *P* = 0.04). Adjusted analyses considering the hierarchical nature of the data, and physician, patient, and visit characteristics, yielded similar results.

CONCLUSION: A simple intervention, providing screening results and recommendations for action, was associated with modest effects: a trend towards a small increase in physician counseling for alcohol problems and a greater likelihood of safe drinking advice. To have greater impact on alcohol counseling by PCPs, more substantial interventions are needed.

FAILURE OF LINKAGE WITH PRIMARY MEDICAL CARE: A PROSPECTIVE COHORT OF PERSONS WITH ADDICTIONS. R. Saitz, M. Larson, L. Jacobson, M. Winter, LM Sullivan, JH Samet, Section of General Internal Medicine, Boston University School of Medicine and Boston Medical Center, Boston; New England Research Institutes, Watertown; Boston University School of Public Health, Boston, MA

PURPOSE: To identify patient characteristics and health care experiences associated with a subsequent failure to link with primary care.

METHODS: We studied a prospective cohort of subjects in a randomized trial of an intervention to engage adults with addiction in primary care, the Health Evaluation and Linkage to Primary care (HELP) study. Eligibility criteria were: admission to a detoxification unit; alcohol, heroin, or cocaine as drug(s) of choice; and no primary medical care. Exclusions were: inability to provide 3 contacts to facilitate follow-up; pregnancy; plans to leave the area; dementia; and an inability to speak English or Spanish. Subjects were evaluated by interview using standardized measures. Linkage, defined as seeing a primary care clinician at least once, was assessed at follow-up by interview. We included clinically and statistically important predisposing, enabling and illness variables in multivariable logistic regression models.

RESULTS: Most subjects were male (76%); the mean age was 36; 46% were black, 37% white, 11% Hispanic; 38% were unemployed; 60% uninsured; 54% recently incarcerated; 47% homeless; and 47% had a chronic medical illness. Most (81%) reported health care utilization in the past 6 months, including addictions, mental health and episodic medical care. Most (87%) had polysubstance abuse problems; 86% had alcohol problems, 75% cocaine, 69% marijuana, and 38% heroin problems. Of 470 subjects, 348 (74%) completed at least 1 follow-up interview over a 2-year period; 142/348 (41%) did not link with primary medical care. In a multivariable model adjusting for age and randomization assignment, the following factors identified at enrollment were independently associated with a failure to link with primary care: male (Odds Ratio 2.6, 95% Confidence Interval 1.5–4.5), white (OR 1.7, CI 1.1–2.8), friends and family that do not support abstinence (OR 2.1, CI 1.3–3.5), no recent medical visits (OR 1.7, CI 1.0–2.9), and absence of chronic illness (OR 1.8, CI 1.1–2.8). Health insurance, employment, homelessness, past incarceration, addiction severity, psychiatric illness, and recent addictions and mental health utilization were not significant predictors.

CONCLUSION: This cohort of young addicted patients without primary care had very high utilization of health care services but many failed to link with primary medical care. Characteristics and experiences identified in this study suggest that linkage with primary medical care will require aggressive strategies that make use of episodic health care encounters, and address patients' perceptions of need for primary care.

PATIENT PERCEPTIONS ON THE CAPABILITY OF INTERNISTS: A MULTI-CENTER SURVEY. SM Salerno, FJ Landry, PJ Kaboli, Department of Medicine, Walter Reed Army Medical Center, Washington, DC; Department of Medicine, University of Vermont School of Medicine, Burlington, VT; Department of Medicine, University of Iowa School of Medicine and Iowa City VA Medical Center, Iowa City, IA

PURPOSE: Misconceptions exist in patient understanding of internists' role in health care. The American College of Physicians-American Society of Internal

Medicine has attempted to raise public awareness of the capabilities of internists through their "Doctors for Adults" campaign. We wished to determine whether patients presenting to general internal medicine clinics understand the capabilities of internists and whether they could discern important differences between internists and other primary care providers. We also wanted to see if patients felt most confident in internist treatment of specific diseases, symptoms, or preventive medicine skills.

METHODS: Adults over age 17 enrolled in internal medicine continuity clinics in Georgia, Iowa, and Vermont were given a written survey to complete before seeing their physician. Patients were asked eleven questions using a 5 point Likert scale on perceived internist characteristics. The subjects also selected which of 24 diseases, symptoms, or preventive medicine skills they felt were within an internist's scope of care. Patient age, gender, and education level were also collected.

RESULTS: Of 911 surveys distributed, 601 (66%) were completed. Very few (19%) respondents confused an internist with an intern. Nearly half of patients (45%) confused internists with a general or family practitioner and 39% thought internists could treat children. Only 50% of respondents thought internists were trained in women's health and only 38% felt they could perform a gynecologic exam. Patients had more confidence ($p < 0.05$) in an internist's ability to treat symptoms than their ability to treat specific chronic diseases or perform specific preventive medicine skills. College educated patients displayed significantly fewer ($p < 0.05$) misconceptions about internists in most categories than those with less education.

CONCLUSION: Patients lack understanding of the capabilities of internal medicine physicians, especially on how they differ from other primary care specialties. Although patient confidence in internists' ability to treat symptoms is high, confidence in treatment of chronic diseases and preventive medicine skills could be improved. Continued effort is required in public education designed to promote better understanding of the role of the internist as a specialist in adult medicine.

THE MANAGEMENT OF WARFARIN THERAPY BY ANTICOAGULATION CLINICS. C Sam, JL Speckman, MA Moskowitz, JE Ansell, Evans Department of Medicine, Boston University School of Medicine, Boston, MA; Division of General Medicine, Emory University School of Medicine, Atlanta, GA

PURPOSE: There is increasing use of anticoagulation clinics (ACC) to care for patients on warfarin. This study assessed the manner of health care delivery in ACCs and the correlation of these characteristics with patient anticoagulation control.

METHODS: A survey was mailed in 1999 to 581 anticoagulation clinics that were members of a professional ACC association. The survey asked about clinic characteristics including age and size, staffing and sources of referrals; clinical practices including initiation of therapy, frequency of INR testing, use of *Point-of-Care* monitoring and home monitoring; and patient outcomes. Two measurements of control were the proportion of INRs in range of those performed on a single designated day (cross sectional), as well as over a 6 month period.

RESULTS: 233 centers (40%) responded, the majority (66%) had been in operation for less than 5 years and 34% monitored more than 400 patients. While 64% had a physician director, over half reported being staffed by nurses (63% full-time), and two-thirds by pharmacists (48% full-time). Half were affiliated with either teaching or community hospitals, and 28% were affiliated with group practices. General internists and cardiologists accounted for a two-thirds of the referrals, 44% of which were for atrial fibrillation, 15% for heart valve replacements and 11% for deep vein thrombosis. Among patients under 60 years old, 61% of ACCs initiate warfarin therapy with a 5mg dose, 34% with larger doses, and 6% with smaller. Among patients over 60 years old, 40% of clinics start with 5mg, 20% with larger doses and 40% with smaller. Half of the clinics were utilizing *Point-of-Care* testing in the clinic, and 18% had patients using instruments at home, but only 3% of patients were managing their own therapy. The 2 measures of control were highly correlated; 62% in range over a 6-month period and 68% in range on a single day ($p < 0.001$). There was no relationship in the percent of INRs in range and the frequency of ordering INRs or the size of the ACC. The major bleeding complications in a 1-year period were gastrointestinal bleeds (0.6%), genitourinary bleeds (0.3%), soft tissue bleeds (0.2%) and cerebral bleeds (0.1%). There were 1.5% thrombotic events in the same period. Clinics affiliated with the Veteran's Administration (VA) had more INRs in range, over a 6-month period, compared to non-VA affiliated centers (74% vs. 62%, $p < 0.02$).

CONCLUSION: ACCs monitor warfarin therapy for patients often cared for by primary care providers. These clinics are large and staffed predominantly by nurses and pharmacists. Control rates are high and complication rates are low. The use of these clinics needs to be considered by primary care providers.

LINKAGE TO PRIMARY MEDICAL CARE: A RANDOMIZED CONTROLLED TRIAL OF A MULTIDISCIPLINARY HEALTH EVALUATION IN A DETOXIFICATION UNIT. JH Samet, MJ Larson, JB Savetsky, M Winter, L Sullivan, R Saitz, Section of General Internal Medicine, Boston University School of Medicine, Boston, MA; New England Research Institutes, Watertown, MA; Data Coordinating Center; Biostatistics and Epidemiology, Boston University School of Public Health, Boston, MA

PURPOSE: To develop and assess the effectiveness of a novel multidisciplinary medical clinic for linking patients from a residential detoxification program to primary care.

METHODS: We enrolled 470 subjects undergoing inpatient detoxification from alcohol, heroin or cocaine in a randomized controlled trial, the Health Evaluation

and Linkage to Primary Care (HELP) Project. The intervention consisted of a medical evaluation at the detoxification unit by a nurse, social worker and physician trained in motivational interviewing and a specific referral to primary care. Prior to randomization all subjects were interviewed by a research associate concerning demographics, substance abuse, medical problems, social support, utilization, and HIV risk behaviors. The primary outcome of interest was a visit with a primary care clinician after discharge from the detoxification unit.

RESULTS: Of the 470 subjects enrolled, 235 were randomized to the intervention. Baseline characteristics included the following: 76% male; 46% black, 37% white, 15% Hispanic; mean age 36 years. Alcohol was identified as the primary or secondary drug of choice by $298/470$ (63%). Heroin or cocaine were identified as primary or secondary drug of choice by $354/470$ (75%). Comparison of control and intervention subjects revealed no differences in terms of demographic, substance use and medical severity. Among all subjects with follow-up within two years of enrollment (348/470, 74%), linkage was achieved in 65% of the intervention group and 52% of the control group ($p = 0.01$). Results among those with follow-up at 6 months, found 61% and 36% linkage in the intervention and control groups respectively ($p < 0.001$). The findings of increased linkage were consistent in subgroups of subjects who identified alcohol, cocaine or heroin as their drug of choice.

CONCLUSION: Linkage of addicted patients to primary medical care occurs among patients in a detoxification unit and can be enhanced by a multidisciplinary medical clinic. Effective linkage of substance-dependent patients to primary care is the first step in engaging medical providers to address these individuals' health and addiction issues.

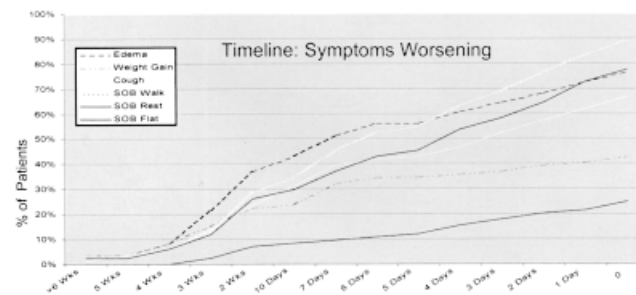
NATURAL HISTORY & CAUSES OF CONGESTIVE HEART FAILURE DECOMPENSATION AT A PUBLIC HOSPITAL. GD Schiff, RA McNutt, S Fung, HC Aggarwal, T Speroff, Medicine, Cook County Hospital, Chicago, IL; Vanderbilt University Medical Center, Nashville, TN

PURPOSE: Because decompensated congestive heart failure is a leading cause of medical hospitalizations, developing a better understanding of its causes and natural history is a priority for prevention efforts. Previous literature and institutional experience suggest there are multiple opportunities for earlier recognition, improving medication usage, and more timely intervention to help patients whose CHF is worsening. We sought to study underlying causes and their time course leading to hospitalization.

METHODS: Using a structured questionnaire, a research nurse interviewed patients admitted for CHF within 24 hours of admission. Data was obtained and analyzed regarding presenting symptoms, patient response to worsening, factors contributing to decompensation, and their temporal relationships.

RESULTS: Of 104 patients 87 (82.9%) presented with dyspnea, 53 (50.5%) edema, 13 (12.4%) chestpain (Decompensation timeline below). 32 Patients (30.5%) ran out of medications, 45 (43.2%) skipped meds of whom $29/42$ (90.6%); $29/45$ (64.4%) respectively stated medication lapse was a major factor in their decompensation. 24.4% of patients also identified physical stresses and 49.9% identified emotional/social stresses as factors. Twice as many pts increased water intake in response to worsening as self-increased diuretic (22 vs. 12).

CONCLUSION: Significant opportunities exist to prevent admission-requiring exacerbation of CHF. Most patients demonstrated a pattern of worsening over several weeks with a paucity of self directed effective measures taken to reverse the worsening. Medication lapses, physical and social stresses, care access barriers, and missed appointment were identified at major potentially ameliorable factors. Distinct patterns of worsening were identified that we are now applying to a prevention/early treatment strategy, as part of National IHI Collaborative.



SECONDARY PREVENTION OF CORONARY ARTERY DISEASE AT A LARGE TEACHING HOSPITAL. JL Schnipper, RS Stafford, General Medicine Division, Massachusetts General Hospital, Boston, MA

PURPOSE: To determine the extent to which patients with coronary artery disease (CAD) receive adequate secondary prevention at outpatient practices associated with a large teaching hospital.

METHODS: As part of a Primary Care Operations Improvement (PCOI) project at Massachusetts General Hospital (MGH), we identified 3,933 patients with a diagnosis of CAD (ICD-9 codes 410.0-414.99) who had been seen in one of ten MGH outpatient practices between 10/1/96 to 9/30/97, were younger than 85, living at home, and were free of metastatic cancer. Outcomes of interest for this cohort included aspirin use, beta-blocker use, blood pressure control, LDL testing, and

LDL control. Information on lipid testing and results were collected using electronic laboratory data. For the 1,799 patients in Internal Medicine Associates (IMA), we abstracted information from the CoSTAR electronic database on aspirin and beta-blocker use, related medications, medical problems and allergies. On a randomly selected subset of 163 patients, we performed a manual chart review to determine average blood pressures during the two-year study period.

RESULTS: Of the 3,933 patients included in the study, 808 (21%) had not received an LDL test during the two-year study period. Among the patients who had been tested, 2016 (65%) had a last LDL level above 100 mg/dL, and 854 (27%) had an LDL above 130 mg/dL. Among a sample of patients without an LDL test or an LDL >130 mg/dL, 52% were not on a HMG-CoA reductase inhibitor ("statin"), and only 2% were on a maximum dose of a statin. The vast majority of patients not on a statin were nevertheless on aspirin or a beta-blocker.

Among 1630 IMA patients with at least 3 recorded medications, 444 (27%) were not taking aspirin and 479 (29%) were not on a beta-blocker. Among the 129 patients with CAD from the manual chart review, 40 (31%) had a recorded systolic blood pressure of 140 mm Hg or greater.

Age over 75 had no effect on LDL testing, but did predict decreased statin use. Beta-blocker use was also less common in the elderly, but was not affected by the presence of relative contra-indications to their use.

CONCLUSION: Our findings suggest that the biggest deficiency in the secondary prevention of CAD among outpatients at MGH is in the management of hyperlipidemia. Both the lack of lipid testing and the limited aggressiveness of controlling LDL cholesterol to nationally recommended levels may contribute to sub-optimal patient outcomes. On the other hand, aspirin use, beta-blocker use and blood pressure control, while not optimal, seem to be significantly better than that shown from previous national and international studies. Efforts aimed at improving the quality of care in patients with CAD, such as the PCOI project at MGH, should consider lipid management a vitally important area for potential improvement.

INTENDED AND UNINTENDED IMPACTS OF DIFFERENTIAL COST SHARING FOR ACE INHIBITORS ON ANTI-HYPERTENSIVE THERAPY. S Schneeweiss, SB Soumerai, RJ Glynn, M MacLure, C Dormuth, AM Walker, Harvard School of Public Health, Boston, MA; Harvard Medical School, Boston, MA; Ministry of Health, Victoria, BC, Canada

PURPOSE: Increased copayments for prescription medication with higher prices are frequently required by health plans without conclusive data about unintended effects. We analyzed the impact of differential cost-sharing for ACE inhibitor on drug utilization, savings and potential substitutions by other medications.

METHODS: We analyzed 36 months of claims data in British Columbia, which implemented reference pricing in January, 1997. The study patients (119,074) were all non-institutionalized Pharmacare beneficiaries 65 years of age or older who used ACE inhibitors. Time series analyses were conducted to determine the effect of the policy on drug utilization and expenditures. Logistic regression was used to identify predictors for switching or stopping medications.

RESULTS: We observed a steep 29% decline in use of cost-sharing ACE inhibitors (ACEI) immediately following initiation of the policy ($p < 0.0001$). A longitudinal analysis of medication switching dynamics showed that 80% of all policy related switching to no-cost medications took place within the first three months. During a transition period of three months following the policy implementation overall utilization of ACEI was reduced by 18% compared to the predicted pre-policy trend. Afterwards the overall ACEI utilization rate was still 11% lower than the extrapolated pre-policy trend but increasing at the same rate ($p = 0.14$). Utilization rates of other anti-hypertensives were unchanged, suggesting no net substitution by alternative medications. The policy saved Can \$7.2 million in pharmaceutical expenditures during 12 months. Patients with low income status, heart failure or diabetes were among the most likely to stop all anti-hypertensive therapy ($n = 1463$) compared to those switching to the reference drugs (OR = 1.6, 1.3, 1.4, respectively).

CONCLUSION: The implementation of reference pricing in British Columbia is effective in terms of a sustained reduction in drug expenditures. However, differential cost-sharing affects anti-hypertensive drug therapy disproportionately, particularly in those with low income status, who appear to stop therapy in some cases. However, these impacts raise questions about ways of implementing such policies that avoid inequities in access to essential medications. Further research is needed on the effects of such policies on health outcomes and total expenditures.

AUTOLOGOUS BLOOD TRANSFUSION—WHO IS NOT GETTING THEIR SHARE? JB Segal, JS Kerman, NE Fink, NR Powe, Division of General Internal Medicine, Johns Hopkins University School of Medicine; Department of Epidemiology, Johns Hopkins University School of Hygiene and Public Health, Baltimore, MD

PURPOSE: The use of autologous blood transfusion, which lessens the risk of blood-borne infectious diseases and immunological reactions, has increased over the last decade. Little is known about the patients who receive autologous blood compared to those who receive packed red blood cells from anonymous donors.

METHODS: We conducted a cross-sectional study of patients with hospital admissions in 1996. We used the Nationwide Inpatient Sample, which is a stratified probability sample of hospitals from 19 states that includes information on 6.5 million admissions. From among the patients who received autologous blood transfusion (ICD-9 procedure code 9902), we identified the five most frequent diagnoses. Within each diagnostic category, we compared patients who received autologous transfusion to patients who received packed red cells but not autologous

transfusion. Chi-square tests and multivariate logistic regression were used for analysis.

RESULTS: The five most frequent diagnoses for which patients received autologous blood were osteoarthritis, prostate cancer, complications of a device or implant, spondylitis, and bone disease. For each diagnosis except device complications, age was inversely associated with the use of autologous blood ($p < .0001$). Only for osteoarthritis were women less likely to use autologous blood than were men, when controlled for age and race, with an OR of 0.89 [95% confidence interval (C.I.) 0.81–0.97]. In all five diagnostic categories, blacks were less likely than white patients to use autologous blood, with odds ratios from 0.23 [95% C.I. 0.15–0.33] for device complications to 0.51 [95% C.I. 0.32–0.81] for spondylitis. Similarly, Hispanic patients were significantly less likely to use autologous blood in 3 of 5 diagnostic categories. In all five diagnostic categories, patients with Medicaid were less likely to receive autologous blood than those with private insurance or Medicare. OR's for receiving autologous blood ranged from 0.27 [95% C.I. 0.16–0.46] to 0.58 [95% C.I. 0.42–0.81] when Medicaid recipients were compared to privately insured patients, controlling for sex, age, and race.

CONCLUSION: Minority patients and patients with Medicaid are considerably less likely to receive autologous blood transfusion than are white patients and those with private insurance or Medicare. These discrepancies could be due to differences in patient preparation prior to hospitalization, patient preference, or to differences in patient-provider discussions regarding treatment options.

THE PATTERN OF INFLUENZA AND PNEUMOCOCCAL VACCINATION IN LONG-TERM CARE FACILITIES IN KANSAS. M Sharma, MC Mosier, GF Solman, DR Colkins, Medicine-Pediatrics, University of Kansas Medical Center; Preventive-Medicine; Medicine, KUMC, Kansas City, KS; Medicine, Harvard Medical School, Boston, MA

PURPOSE: Immunization trials in reasonably healthy institutionalized elderly (>50 years) have documented reductions in morbidity and mortality rates due to pneumonia. There is no study to document the current profile of standard policy and procedures in place or to evaluate the impact of educational intervention on the immunization rate in long term care facilities (LTCFs). This study was designed to evaluate (i) the current pattern and variables of immunization rates; (ii) the effect of educational intervention on immunization rates in LTCFs.

METHODS: Surveys were designed with the focus on variables including standard policy and procedure, standing order for immunization in place, standard policy for healthcare workers and the medical records of the annual assessment of all the residents. The survey was sent to 415 LTCFs in the State of Kansas. Written material informing about immunization was then sent to the directors of the nursing (DON) of all 415 LTCFs and post intervention survey was requested from all LTCFs. We also randomly selected 55 of the LTCFs for review of their medical records to assess flu and pneumonia IR. The study was conducted over a period of 14 months.

RESULTS: Only 105 out of 415 LTCF responded to the survey. For influenza, 87% and for pneumonia, 34% of the responding LTCFs had a Policy/Procedure in place. Of these LTCFs, standing orders for immunization for influenza and pneumonia were present in 79% and 74% respectively. Policies for influenza immunization of healthcare workers was present in 78.1% of LTCFs. We received matching immunization rates and DON survey data from 45 LTCFs. Post-intervention immunization rates were available from only 26 LTCF. Pre and post intervention immunization rates for influenza were 74% and 79% respectively. However pre and post intervention immunization rates for pneumococcal vaccination were 10% and 75% respectively. Educational intervention had minimal effect on influenza immunization rates but a significant increase in pneumococcal immunization rates ($p < 0.001$).

CONCLUSION: We conclude that a tremendous variability in the policy and procedure profile for influenza and pneumococcal vaccination exists and educational intervention can improve the rate of immunization for pneumonia in the LTC facilities in the State of Kansas.

Limitations: There was poor response rate as only 25.9% DON filled out the survey. Only 47.27% of randomly selected LTCFs responded to post-intervention survey. We could send video tapes to 75 of the 405 LTCFs due to limited resources. To keep the uniformity of intervention (written material only), we used data from only 45 of the 55 randomly selected LTCFs.

PHYSICAL RESTRAINTS AND THE RISK OF FALLS IN HOSPITALIZED PATIENTS. R Shorr, K Guillen, K Walker, L Rosenblatt, C Caudle, S Kritchevsky, Preventive Medicine, University of Tennessee; Methodist Hospitals of Memphis, Memphis, TN

PURPOSE: There is considerable controversy surrounding the use of physical restraints in hospital settings. Whether restraints protect patients from falling is not known.

METHODS: Case-control study of inpatients at a 528 bed urban acute care hospital. Cases were prospectively identified using a "fall evaluator"—trained hospital externs or residents, who provided full time coverage during the study. Controls were matched to cases by nursing unit and length of stay. Restraint use was ascertained for both cases and controls through blinded evaluation of medical records. Multivariate analysis was performed using conditional logistic regression controlling for demographic factors, use of psychotropic drugs, cardiovascular drugs and other medications.

RESULTS: During 194 days of observation, 226 hospital patients fell (mean age: 63.7 ± 16.2 years; 50% female). Overall, the rate of falls was 3.88/1000 patient-days. Unit rates varied from 0.97/1000 patient-days, on a general medical unit, to

10.45/1000 patient-days, on a psychiatric unit. At the time of the fall, physical restraints were in use by 9.6% of cases and 2.6% of controls. Compared to patients not in restraints, the multivariate relative risk of falling among restrained patients was 7.8 (95% CI, 1.7 to 36.6).

CONCLUSION: The use of physical restraints is associated with a high risk of falls in hospitalized patients. Because of potential "confounding by indication" a causal path cannot be established. However, these data suggest that restraints should be used with extreme caution in hospitalized patients.

WHAT SIGNIFICANCE DOES INSURANCE STATUS HAVE UPON AVOIDABLE ADMISSION RATES TO ACUTE CARE HOSPITALS? SK Sigworth, JB Perlin, WR Smith, General Internal Medicine, Medical College of Virginia of VCU, Richmond, VA

PURPOSE: To determine the effect that insurance status plays on avoidable admission rates at a large urban inner city hospital

METHODS: A retrospective, cross sectional analysis of medical records, during the 4th quarter of 1998, was performed on all patients admitted to the Internal Medicine service of a large university medical center, whose length of stay was less than 48 hours. Exclusionary criteria included elective admissions and any admission to the Intensive Care Unit. Payor status was identified for each patient, categorizing them into Medicare, Medicaid, Commercially Insured, or Uninsured Indigent. Subsequently, these admissions were classified as avoidable or unavoidable. Avoidable admissions were determined based on published criteria for appropriate use of acute inpatient services (InterQual®, 1997; Milliman & Robertson, 1998).

RESULTS: 2427 patients were admitted to the Medicine service over the study period with 518 (21.3%) discharged within 48 hours. Of these, 279 met the inclusionary criteria. Analysis of insurance status revealed 88 (33.7%) with Medicare, 36 (12.9%) with Medicaid, 61 (21.9%) with commercial insurance and 94 (33.7%) without insurance. Chi-square analysis revealed no overall statistical significance ($p = 0.433$) among avoidable admission rates by insurance type. When compared with the avoidable admission rate for patients with any form of insurance 25.4% (47 of 185), the uninsured rate 30.9% (29 of 94) was also not of statistical significance ($p = 0.334$).

CONCLUSION: In our study, no statistical significance was seen in avoidable admission rates by insurance status. These so-called "soft admissions" for ambulatory sensitive conditions may be more thoroughly scrutinized, especially for the uninsured patients due to their potential non-payor status. We found no evidence to support this increased scrutiny. Of note, a higher avoidable admission rate was evident in the uninsured indigent group, proving that payment mechanism was not an important factor in the decision making process of a potentially "soft" admission.

VETERANS' ACCESS, UTILIZATION, AND SATISFACTION WITH CARE: RESULTS FOR THE MEDICAL EXPENDITURE PANEL STUDY. MD Silverstein, X Ye, PJ Nietert, Center for Health Care Research, Medical University of South Carolina, Charleston, SC

PURPOSE: Many veterans have access to the Veterans Health Administration facilities for ambulatory, emergency, and hospital care in addition to the civilian (non-VA) health care system. We hypothesized that veterans would therefore have improved access, increased utilization, and increased satisfaction with health-care compared to non-veterans.

METHODS: The 1996 U.S. Medical Expenditure Panel Survey population was studied. This is a random sample of U.S. non-institutionalized civilian population. The primary independent variable was veteran status; covariates included age, sex, race/ethnicity, marital status, education, employment, health insurance (private insurance or Medicaid), and self-reported physical and mental health. The dependent variables were access to health care (usual source of care, physician or other provider as usual source of care, office based provider as a source of care, and difficulty obtaining access to care), utilization of health services (number of physician visits, emergency department visits, and hospital discharges), and satisfaction with care. Logistic regression and linear regression analyses were performed using SUDAAN to account for the clustered sample design.

RESULTS: Information was available from approximately 16,000 adults age 18 or over in the 1996 MEPS panel. After adjusting for covariates, veterans were less likely to identify a person as a provider of their usual source of health care (odds ratio 0.78, 95% CI 0.67-0.90), or an office-based physician as a usual source of care (odds ratio 0.57, 95% CI 0.46-0.70). Overall veterans were equally likely to have a usual source of care (odds ratio 1.12, 95% CI 0.96-1.31) and did not report less difficulty in access to care (odds ratio 0.94, 95% CI 0.77-1.15). Veterans reported significantly more physician visits (mean 2.06, 95% CI 1.91-2.22), than non-veterans (mean 1.64, 95% CI 1.60-1.68). Veterans were equally likely to have an emergency department visit (odds ratio 1.04, 95% CI 0.86-1.26), but were more likely to be admitted to hospital (odds ratio 1.89, 95% CI 1.51-2.36). Veterans were equally satisfied with care (odds ratio 1.10, 95% CI 0.93-1.30).

CONCLUSION: In 1996 U.S. veterans were less likely to report having a specific provider or office based provider as a usual source of care but did not otherwise have problems with access to care. Veterans had more physician office visits and were more likely to be admitted to hospitals. Veterans were equally satisfied with their health care compared to non-veterans.

TO HELP OR HINDER—WHICH IS MORE IMPORTANT IN INFLUENCING PHYSICIAN RATINGS OF HEALTH PLAN QUALITY? MA Smith, AB Bindman, MK Davis, MD Finch, Population Health Program, Department of Preventive Medicine, University of Wisconsin-Madison Medical School, Madison, WI; University of California San Francisco, San Francisco, CA; University of Minnesota School of Public Health; Center for Health Care Policy and Evaluation, United Health Group, Minneapolis, MN

PURPOSE: To determine how health plan practices contribute to physicians' overall ratings of health plan quality.

METHODS: The Physician's Evaluation of Health Plans (PEHP) Project surveyed by telephone a probability sample of 1757 generalist physicians in 16 health plans in 5 metropolitan areas nationwide. Physicians reported only on the plan from which they were sampled. 70% responded. Three multi-item scales (internal reliabilities: 0.75 to 0.88) assessed physicians' perceptions of facilitators and barriers to high quality care in the plans and the clinical capabilities of plan physicians. Facilitator items included the presence of guidelines, patient education and flagging patients for preventive care. Barrier items included authorization procedures, cost and time constraints, and access to specialists. A single item ("I would recommend this plan to a friend or family member") rated overall health plan quality. All items used five-point scales ("excellent" to "poor," or "strongly agree" to "strongly disagree"). Our analytic goal was to explain generalist physicians' overall rating of health plan quality using the three multi-item scales and their demographics (age and gender). We compared the explanatory power of each variable using R-squared.

RESULTS: Among univariate models, the facilitators scale explained the most variation in physicians' overall ratings of health plan quality (R-squared = 0.44). In the multivariate model, the listed variables explained 55% of the variation in generalist physicians' ratings of overall health plan quality. The explanatory power of the multivariate model decreased the most when the facilitators scale was removed (R-squared loss = 0.09), followed by the barriers and clinical capabilities scales.

Predicting Generalist Physicians' Ratings of Overall Health Plan Quality

Variable	R-squared for univariate model	R-squared loss if removed from multivariate model
Facilitators	0.44*	0.09*
Barriers	0.35*	0.04*
Clinical Capabilities	0.14*	0.01*
Physician Gender	0.01*	0.00
Physician Age	0.00	0.00

* $p < 0.001$ Multivariate model included indicator variable for each health plan

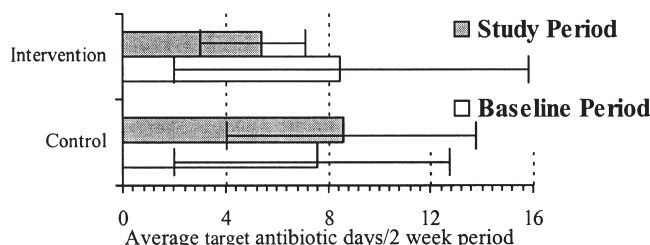
CONCLUSION: Generalist physician ratings of health plan quality are more highly related to what health plans do to facilitate care than the barriers created by plans in managing care.

ACADEMIC DETAILING TO IMPROVE USE OF BROAD-SPECTRUM ANTIBIOTICS AT AN ACADEMIC MEDICAL CENTER. DH Solomon, L Van Houten, RJ Glynn, L Baden, KA Curtis, H Schrag, J Avorn, Pharmacoepidemiology, Brigham and Women's Hospital; Infectious Diseases, Brigham and Women's Hospital, Boston, MA

PURPOSE: Excessive reliance on broad-spectrum antibiotics is costly and can exacerbate anti-microbial resistance, yet over-use of such antibiotics is common. The goal of this research was to test a targeted one-on-one educational program ("academic detailing") designed to improve the appropriateness of broad-spectrum antibiotic use.

METHODS: Using block randomization, we assigned 17 general medical, oncology, and cardiology teams in a large urban teaching hospital to intervention or control status. During an 18-week study period from January 1999 through May 1999, an infectious diseases consultant or clinical pharmacist discussed rational antibiotic use with house officers who prescribed either levofloxacin or ceftazidime to patients on intervention teams for possibly inappropriate indications. Alternative antibiotic choices were explained. The outcome of interest was the number of days of levofloxacin or ceftazidime administered on each team during the study period, adjusted for baseline prescribing.

RESULTS: Prior to the trial, intervention and control teams had similar prescribing patterns for the target antibiotics, and used the drugs for similar indications during the study period (all p -values > 0.1). During the intervention, days of levofloxacin and ceftazidime prescribed for patients on the intervention teams were 58.8% lower than for patients on the control teams (adjusted p -value = 0.0003). Length of stay, intensive care unit transfers, re-admission rates, and in-hospital deaths were similar in both groups (all p -values > 0.1). Hospital-wide use of these agents was stable during the study period.



CONCLUSION: A targeted one-on-one educational intervention can be a practical, effective, and safe method for reducing excessive broad-spectrum antibiotic use. Wider use of such programs may help slow the growth of nosocomial anti-microbial resistance.

BARRIERS TO APPROPRIATE ANTIRETROVIRAL THERAPY AMONG A STATE-WIDE COHORT OF HIV-INFECTED PATIENTS. JM Sosman, LM Bogart, SL Catz, University of Wisconsin, Madison, WI; Kent State University; CAIR, Medical College of Wisconsin

PURPOSE: Highly active antiretroviral therapy (HAART) consisting of combinations of antiretroviral medications has improved the survival of HIV-infected patients. The Wisconsin AIDS/HIV Drug Reimbursement Program (WADRP) provides Wisconsin residents access to all FDA approved anti-HIV medication. The WADRP does not impose expenditure caps or have waiting lists, all physicians can participate and patients can receive total reimbursement for medications. The purpose of this study was to identify patient and physician characteristics associated with substandard antiretroviral prescriptions provided to a state-wide cohort of HIV-infected patients.

METHODS: WADRP patient data (birth date, gender, risk factor for HIV, ethnicity, income, antiretroviral medications) were linked to physician prescriber data obtained from the AMA Master File. Antiretroviral regimens were classified according to the DHHS guidelines as "optimal" (HAART) or "suboptimal". Chi-square analyses were conducted on the classification of prescribed anti-HIV medication for each patient variable and the likelihood of prescribing "optimal" (HAART) regimens by physician characteristics.

RESULTS: Of 369 new WADRP enrollees from 1997 to mid-1999, 116 were prescribed "suboptimal" and 200 were prescribed "optimal" therapy. Patients who were not White were less likely to receive any antiretroviral treatment and less likely to receive "optimal" regimens, $\chi^2 (2) = 8.40, p < .05$. Patients with incomes below the federal poverty level were more likely to receive "suboptimal" regimens, $\chi^2 = 6.20, p = .05$. Ninety six physicians wrote antiretroviral prescriptions for the WADRP patient cohort. Chi-square analyses indicated that physicians who had more experience (6 or more WADRP patients) were more likely to prescribe "optimal" regimens, $\chi^2 (3) = 16.77, p < .01$. Physicians who practiced in sparsely populated areas (<250,000) were less likely to prescribe "suboptimal" regimens ($\chi^2 (1) = 5.73, p < .05$), and physicians whose practices were office-based were less likely to prescribe "suboptimal" regimens, $\chi^2 (1) = 4.55, p < .01$. Physician age, ethnicity and specialty were not significant predictors.

CONCLUSION: Disparities exist for HIV-infected patients enrolled in a state-wide program which provides antiretroviral medications to all enrollees. Individuals who were poor and not White were more likely to receive either "no" or "suboptimal" antiretroviral therapy. Physicians with less prescribing experience, practicing in more urban and non-office-based professional settings were more likely to prescribe "suboptimal" regimens. Additional methods to limit these disparities are needed.

INAPPROPRIATE USE OF THROMBOLYTICS AND EXCESS MORTALITY IN THE ELDERLY. SE Soumerai, TJ McLaughlin, D Ross-Degnan, JH Gurwitz, Harvard Medical School/Harvard Pilgrim Health Care, Boston; The Meyers Primary Care Institute, Worcester, MA

PURPOSE: National guideline implementation programs have substantially increased use of thrombolytic therapy for elderly patients with acute myocardial infarction (AMI). However, evidence of benefit to support use of thrombolytic therapy in the old-old (age 75 and older) is lacking. We studied a large community population of elderly AMI patients in order to answer the following questions: (1) What proportion of eligible and ineligible elderly AMI patients receive thrombolytic therapy? (2) Controlling for baseline risk status and demographics, what is the association between thrombolytic use and in-hospital mortality by age, and among patients with or without medical contraindications to thrombolytic treatment?

METHODS: We abstracted the medical records of 2,659 elderly patients admitted with AMI at 37 Minnesota community hospitals for the periods 1992-93 and 1995-96. We measured in-hospital mortality, thrombolytic use among eligible and ineligible patients, and demographic, comorbidity and severity-of-illness variables at presentation. We conducted multiple logistic regression analyses to determine whether age and presence of medical contraindications modified the effect of thrombolytics on mortality, controlling for and stratifying by propensity scores (based on all demographic, comorbidity and severity-of-illness variables predicting thrombolytic use).

RESULTS: Sixty-two percent of eligible patients (n = 684) and 16% of ineligible patients (n = 1972) received thrombolytic therapy. Thus, 42% of thrombolytic recipients were ineligible for treatment, mostly due to medical contraindications. Among the old-old with absolute medical contraindications (n = 371), 25% of thrombolytic recipients died as compared with 12% of non-recipients (p < 0.001). In the total sample, there was no significant association between thrombolytic use and in-hospital mortality (O.R. = 0.90, p = 0.70), controlling for propensity to receive thrombolytics. However, for every one year increase in age, the adjusted odds of dying for thrombolytic recipients vs. non-recipients increased significantly by 1.04 (p = 0.03). Among patients with absolute medical contraindications (n = 457), the adjusted odds of death among thrombolytic recipients was 2.5 as compared with non-thrombolized patients (p = 0.008).

CONCLUSION: Overall, there was a significant association between the age of elderly AMI patients and in-hospital mortality following thrombolytic therapy. Most of this association was explained by use of thrombolytic therapy in older patients with absolute medical contraindications. Our findings suggest the need for more care in the selection of elderly patients for thrombolytic therapy.

ORGANIZATION OF CLINICAL PRACTICE. M Spar, D Carlisle, C Damberg, C Mangione, A Brown, KL Kahn, University of California, Los Angeles; RAND, Santa Monica, CA

PURPOSE: We used systematic literature review, clinical judgment, and administrative experience in the clinical and management arena to develop a conceptual framework to understand how the clinical structure of organizations influence patient care and outcomes. Donabedian's traditional model of health care quality describes structure, process, and outcomes as essential elements. The last decade has resulted in the centralization of many aspects of clinical care so that medical organizations are implementing plans to use resources to systematically effect the care of cohorts of patients. Whereas the process of medical care pertains to a one-on-one interaction between patients and providers, the structure of clinical care pertains to ways in which resources are devoted to the care of cohorts of patients. We propose that a conceptual model that articulates ways in which medical organizations can influence the care of patients will allow us to better predict patient care and outcomes.

METHODS: In the development of this conceptual frame-work, five domains emerged as unique types of structure that might influence patient care and outcomes: patient support, provider support, continuity of care, locus of control, and provider-patient match. These five domains are noted to effect specific components of the clinical structure of care including structure aimed at optimizing: a) patient care (including patient education, use of physician extenders, training of clinical staff); b) the use of evidence based medicine (including use of patient and provider reminders, use of clinical practice guidelines); c) the patient-provider encounter (including use of multidisciplinary teams, medical record availability); d) data exchange (including access to lab data and measures of communication); and e) involvement of the central medical organization (including the extent of their support for assessing and giving provider feedback). We used the conceptual model to generate a focused survey, the "Clinical Structural Survey" or CSS of the corporate clinical leader of medical organizations. Challenges in the development of the survey included: a) type of organization to survey regarding structure; b) type of respondent to interview c) definition of "policy" pertinent to structure, d) discrepancies between reported "policies" and actual practice and e) changes in corporate policy with time.

RESULTS: Our survey has been conducted with the leaders of 56 medical organizations. They were able to complete the survey (average time 45 minutes), and repeatedly noted that the survey captured important aspects of the structure of care that their organization considered or implemented. The survey worked equally well in the interview of directors from medical groups and Independent Practice Associations (IPAs), but IPAs had more "not applicable" responses consistent with our conceptual model that medical groups had more centralized policy pertinent to structure than did IPAs, on average.

CONCLUSION: Our clinical structure model helps to frame how medical organizations allocate resources to optimize the care of cohorts of patients. We have articulated five key domains that can be analyzed to understand which domains of structure actually do impact patient care and outcomes in the real world of managed care.

PREDICTORS OF SELF CARE BEHAVIORS OF DIABETES TYPE I ADULTS IN THE OREGON HEALTH PLAN. M Srinivasan, A McNeill, S Garfield, N Swigonski, Indiana University School of Medicine, Indianapolis, IN; Research Triangle Institute, NC

PURPOSE: Improved self-care behaviors in patients with DM Type I (DMI) have been linked to improved clinical outcomes. However, self-care behaviors among primarily Medicaid adults with DMI have not been previously reported. The purpose of this paper was to determine predictors of self-care (SC) behaviors recommended by the American Diabetes Association in Oregon Health Plan (OHP) adults based on an adapted Glasgow personal beliefs and barriers to self care model.

METHODS: 97% of all OHP pts with a DMI ICD9 code were initially included. After application of exclusion criteria, 94 non-pregnant DMI pts 18-45 yo continuously enrolled for 10 or 12 months from June 1997-May 1998 were included. From a 1998 telephone survey, we constructed a two level SC index of the frequency of 5 behaviors: adjustment of insulin, following a meal plan, checking blood sugar, checking feet, smoking (worse = 1-3, better = 4-5). DMI related quality of life (DMQOL) was assessed with a 5 point, 5 item scale, examined by patient quartiles. Barriers/enablers to SC were determined through unmet medical needs, number of visits (0, 1-2, 3-6, >7), receipt of standard medical care (0-3, for education, referrals, physical examination), & health status (PCS12, 0-100 [high]). Three patient demographics, age, gender, and ethnicity, were based on self-report. **RESULTS:** Our pt population was 47% female, 79% white, aged 32 ± 6 yo. Pts averaged 3.8 ± 4.7 visits over 3 months, and averaged an MCS of 42 ± 11, and PCS of 44 ± 13. Pt's self-reported quality of life averaged 16 ± 4. 32% patients reported

regularly adjusting insulin, 44% following a meal plan qd, 5% checking blood sugar qd, 62% checking their feet. 47% were non smokers. 26 patients (28%) reported better SC. Using logistic regression in SUDAN 7.5.2, patients were more likely to report better SC, at $p < .05$, if they had >7 visits in the past 3 months (OR 92.6) or were older (OR 1.2). Patients were more likely to report worse SC if they had a DMQOL of 5-13 or 14-17 (OR .13, .15) or were male (OR .2).

CONCLUSION: Diabetes is inherently a disease of self management, in which complications can be prevented by judicious practice of self-care behaviors. However, these behaviors involve lifestyle changes that are difficult to practice consistently, even in the most motivated patient group. Our data suggest that DMI patients in low income populations who report a lower quality of life, who are male, or who are younger may need special attention to enhance self care behaviors. Patients with more encounters with their provider report markedly better SC. These data provide some evidence to suggest areas in which patient education and reinforcement of self-care behaviors can have significant impact on future morbidity from DMI.

MEDICAL CARE FOR ACUTE LOW BACK PAIN IN THE OREGON HEALTH PLAN. PREDICTORS OF OVER-UTILIZATION. M Srinivasan, M Przybylski, N Oldridge, R Lubitz, N Swigonski, Indiana University School of Medicine, Indianapolis, IN

PURPOSE: Acute Low Back Pain (LBP) is generally a self-limited disease process. Recent recommendations from the Agency for Health Policy Care & Research (AHCPR) emphasize the conservative nature of LBP standard medical care (SC). In 1994, the Oregon Health Plan (OHP) expanded insurance coverage to residents under the federal poverty limit with a prioritized benefits program, in which LBP was not reimbursed. The purpose of this study was to determine SC predictors for acute LBP management, utilizing AHCPR recommendations & their framework for evaluating clinical guidelines.

METHODS: Using 1995 chart review & Medicaid encounter data, we identified 175 non-pregnant patients (pts) with a LBP ICD9 who were 18-65 yo, with $\geq$ one episode of new/acute LBP, & without cancer, urinary incontinence or spinal cord disease. Based on chart review, pts were considered to have received SC during their first acute LBP visit if three AHCPR recommended management decisions occurred: the provider ordered medications, no imaging studies were ordered, & no referrals made to MD/DO/chiropractor.

RESULTS: Our pt population was 65% female, aged 38 ± 10 yo, 88% white, 3% African American (AA), 5% Hispanic; 19% were ≥ 225 lbs, 6% were noted to be sedentary, 7% had poor muscle tone, & 12% were non-smokers. In 1995, these pts had 2.8 ± 2 LBP visits. Medications were ordered for 92%, no imaging tests ordered for 81%, & no referrals made for 75%. SC was received by 99 pts (56%). From logistic regression, pts were more likely to receive SC (indicating medications were ordered, but not referrals or imaging studies) if they were if they were AA (OR 2.4) or sedentary (OR 7.3) ($p < .05$), and if ≥ 225 lbs (OR 2.5) ($p < .10$). Pts were less likely to receive SC if they had poor muscle tone (OR 0.08), lived in a more metropolitan county (OR 0.42) or had more frequent LBP visits (OR 0.68) ($p < .05$), and if enrolled in a salaried health plan (OR 0.39) ($p < .10$). Factors not correlated with SC include age, gender, documentation of abnormal physical exam findings, non-smoking status, & patient non-compliance.

CONCLUSION: Paradoxically, "less" care indicates better adherence to AHCPR quality of care standards for low back pain. Based on our data, pts who are sedentary, AA, with less frequent visits, & living in less urban areas are more likely to receive SC. These pts are less likely to have referrals/imaging studies, and more likely to have medications ordered. Therefore, our SC index provides an interesting look into the provision of OHP non-reimbursed services, with potential over-utilization of resources in select pt groups with LBP. This data suggests that paradigms for SC should include tracers for both potential over- and under-utilization of medical services.

OPTIMIZING PRECEPTING TIME AND PATIENT THROUGHPUT: WHAT IS THE BEST RATIO OF RESIDENTS TO PRECEPTORS? JE Stahl, G d. la Perez, SL Cohen, University of Pittsburgh School of Medicine; University of Pittsburgh Graduate School of Engineering, Pittsburgh, PA

PURPOSE: To determine an optimal ratio of residents to preceptors that both minimizes the wait for the preceptor and maximizes patient flow.

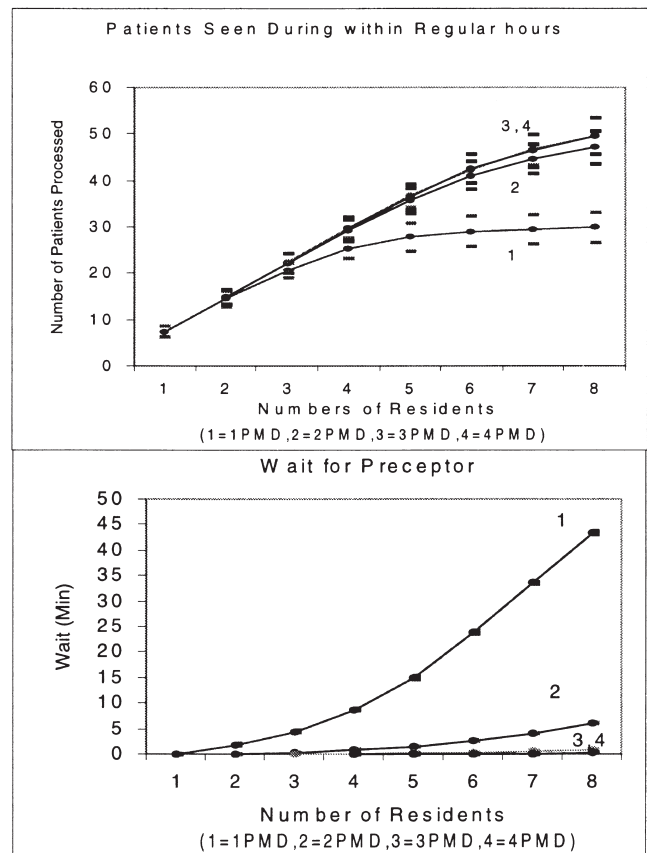
SETTING: Teaching Ambulatory Care clinic

MEASUREMENTS: Patient flow time through clinic, Patient waiting time for physician, Resident waiting time for Preceptor, Resident time spent with preceptor, Clinic overtime, Number of patients seen per day.

METHODS: A discrete event simulation model of a teaching ambulatory clinic was built where the number of residents and preceptors could be varied. The number of residents was varied between 1 to 8 and the number of preceptors between 1 to 4 to yield 26 different combinations where the number of residents was equal to or greater than the number of preceptors. Two rooms were made available to each resident and one nurse for each pair of rooms. The resident's patients arrived at a rate of 1 every 30 minutes with a SD of 15 minutes. A 10% no show rate was assumed. Each scenario was run for the equivalent of 247, 8-hour days to simulate a working year. Data was derived from physician estimates and scheduling data. The outcome measures were compared with ANOVA and paired t-tests.

RESULTS: Wait time for a preceptor ranged from less than 1 minute for Preceptor:Resident (P:R) ratios of 1:1 to 43.5 (+/-19.1) min for a ratio of 1:8. Time spent precepting ranged from 10.9 (+/-6.2) min across all permutations once precepting was initiated. The total number of patients processed during regular working hours increases on average at ~3.1 patients/additional resident with one precep-

tor, at 5.5 with 2 preceptors, at 5.5 with 3 preceptors and at 4.5 with 4 preceptors. The baseline number of patients processed rises with the number of preceptors.



CONCLUSIONS: As expected in a model that allows queuing for resources, waiting time and delays rise rapidly when processing time becomes greater than the arrival rate. When the P:R ratio falls below 1:4 there is a resulting rapid rise in time spent waiting for precepting and a decreased proportion of total registered patients seen within regular working hours. Scenarios with 2 or 3 preceptors benefit most from additional residents. Therefore, the optimal Preceptor:Resident ratio should not fall below 1 to 4.

BRIEF ALCOHOL INTERVENTION FOR NEEDLE EXCHANGE (BRAINE) CLIENTS: EARLY FINDINGS. MD Stein, A Gogineni, A Charuvastra, M Sobota, R Swift, R Longabaugh, Medicine, Brown University School of Medicine, Providence, RI

PURPOSE: To test if a brief multidimensional intervention using motivational enhancement therapy (MET) reduces needle sharing among injection drug users (IDUs) who have concomitant hazardous alcohol use.

METHODS: A randomized, controlled trial of MET vs. assessment only. The goal of MET is for an interventionist to mobilize a patient's resources to change behaviors. The MET arm consists of two, 90-minute sessions, one month apart, focusing on alcohol and drug use reduction. Population: AUDIT-positive (≥ 8) IDUs recruited from Needle Exchange (1998-1999). Outcome: Reduction or maintenance of HIV risk days (needle sharing) at 6 months, measured by 30-day Time Line Followback.

RESULTS: Of 140 persons recruited, 89% were white, 37% female, and the mean age 36.5. The six-month follow-up rate was 99% ($n = 139$). At baseline, the mean AUDIT was 21.4. Subjects had mean needle sharing days of 7.1 in the last month. 65 subjects (47%) had no needle sharing days in the month prior to study entry; at six months, 60 of these persons (92%) maintained their zero risk. 74 persons had ≥ 1 risk days at baseline; at six months 65 persons (88%) lowered their risk days in the last month. MET subjects were more likely to lower their risk by ≥ 1 day compared to controls (93 v. 82%; $p = .30$).

CONCLUSION: Preliminary analyses indicated that brief MET lowers needle sharing, but not significantly more than an assessment only. Subgroup analysis on the full sample may reveal IDUs who benefit most from MET.

IMPACT OF DISEASE MANAGEMENT(DM)ON HOSPITALIZATION RATES IN PATIENTS(PTS) WITH END STAGE RENAL DISEASE(ESRD). I Steinman, AR Nissen, J Dickmeyer, W Maftren, T Parker, J Nielsen, T Litchfield, Beth Israel Deaconess Medical Center, Boston, MA; UCLA Medical Center, Los Angeles, CA; RMS Disease Management, McGow Park, IL; UNC School of Medicine, Chapel Hill, NC

PURPOSE: Understanding the causes for hospitalization and developing a program to decrease this morbidity in pts with ESRD achieves a positive cost-benefit ratio.

METHODS: Renal Management Strategies (RMS), an affiliate of Baxter Healthcare, has implemented a nationwide renal DM program and has accumulated over 15,000 pt months experience over the past 2 years.

RESULTS: Complications of vascular access (e.g., thrombosis, poor blood flow, infection) are the most common causes for hospital admissions, followed in order by cardiovascular (CV) diseases, congestive heart failure (CHF), sepsis (usually related to the vascular access), and pneumonia. Introduction of targeted RMS DM programs utilizing the expertise of Health Service Coordinators (HSCs) to collect data and implement the CGI process, has resulted in a progressive decrease in hospitalization rates:

Hospitalization reasons (number of primary reasons/pts in month)

Months	Access	Cardiac	CHF	Sepsis	Pneumonia
1 99	.031	.022	.023	.021	.007
2 99	.015	.016	.010	.010	.009
3 99	.030	.026	.012	.019	.008
4 99	.022	.015	.013	.015	.004
5 99	.034	.016	.007	.020	.003
6 99	.041	.020	.011	.010	.006
7 99	.034	.015	.011	.020	.004
8 99	.028	.022	.012	.015	.004
9 99	.020	.028	.015	.015	.005
Pearson Correlation	.124	.234	-.308	-.131	-.640*
Significance	.375	.272	.210	.369	.032

*Correlation is significant at the .05 level (1-tailed).

The RMS vascular access initiative used an educational program for local nephrologists and dialysis units based on NKF-DOQI guidelines; CV admissions decreased in conjunction with a comprehensive anemia management program; CHF admissions fell coincident with a comprehensive educational program focused on achieving true dry weight for dialysis pts.

CONCLUSION: DM for ESRD improves the quality of care by decreasing hospital admissions for the 5 most common causes for hospitalization using CGI processes.

CHANGING RESIDENT PHYSICIANS' PRESCRIBING IN THE HOSPITAL SETTING: A RANDOMIZED CLINICAL TRIAL. SA Sternberg, JA Berlin, BL Strom, Department of Medicine, University of Chicago, Chicago, IL; Department of Biostatistics and Clinical Epidemiology, University of Pennsylvania, Philadelphia, PA

PURPOSE: Academic detailing is an effective method to change physicians' prescribing practices in the ambulatory setting. However, few controlled trials have studied methods to change prescribing in the hospital setting. The objective of this three armed study was to compare the effects of academic detailing versus a mandatory drug order form versus control on prescribing of histamine 2 receptor antagonists (H2) in the hospital setting.

METHODS: A RCT was implemented at an academic medical center between September 1993 and April 1994. Subjects were resident physicians in all departments who wrote prescriptions. Clinical services were randomly assigned to one of three arms: 1) face to face education by pharmacist detailers, 2) a required drug order form or 3) a control intervention of printed material about H2. An expert panel devised appropriateness criteria for the inpatient use of H2. The primary outcome measure was the mean proportion of appropriate prescriptions per physician before and after the one month intervention. A secondary outcome measure was the change in the total number of prescriptions.

RESULTS: Of 62 physicians who completed the trial, 15 did not prescribe H2 in either the pre or post intervention period leaving 47 physicians with complete data. Six of 17 physicians in the academic detailing group did not complete the intervention due to time constraints, despite an average of 26 minutes spent by the detailers trying to contact and accommodate the physicians. The mean proportion of appropriate prescriptions per physician did not change in the academic detailing ($p = 0.06$) or the printed material control group ($p = 0.39$) and decreased after the intervention in the order form group ($p = 0.003$). There were no differences between groups in either the mean proportion of appropriate prescriptions or total number of prescriptions per physician after adjusting for baseline differences.

CONCLUSION: This RCT comparing academic detailing, a required drug order form, and a control intervention of printed material was unable to demonstrate an improvement in prescribing of H2 by resident physicians in the hospital setting. The unique difficulties of performing this kind of study in the hospital rather than ambulatory setting are highlighted. These difficulties include the challenge of identifying heavy prescribers, the limited time for follow-up due to one month clinical rotations, and the lack of receptivity of resident physicians to participate. Well designed trials are needed to determine the most effective means of improving physicians' prescribing in the hospital setting.

ERROR, STRESS, AND TEAMWORK IN MEDICINE AND AVIATION: A MULTINATIONAL STUDY. EJ Thomas, B Sexton, RL Helmreich, Department of Medicine, University of Texas - Houston Medical School, Houston; Department of Psychology, University of Texas Austin, Austin, TX

PURPOSE: The Institute of Medicine recently concluded that preventable errors were the 8th most common cause of death in the United States and that promoting effective team functioning, possibly using an aviation model, is one way to prevent errors. We surveyed operating room and intensive care unit (ICU) personnel about attitudes concerning error, stress, and teamwork, and compared these attitudes to those of airline cockpit crew members.

METHODS: Cross-sectional surveys were administered to medical personnel (physicians, fellows, residents) from urban teaching and non-teaching hospitals in the United States, Israel, Germany, Switzerland, and Italy; and cockpit crew members (captains, first officers, second officers) from several major United States airlines. Common items from four different surveys administered between 1992 and 1999 are reported here.

RESULTS: Surveys from 1,033 medical personnel and over 30,000 cockpit crew members were analysed. Typical response rates from cockpit crews were 45%; for operating room personnel ($n = 851$) 40% to 100%; and for intensive care unit (ICU) personnel ($n = 182$)

the response rate was 59%. 70% of surgical attendings agreed with the statement "even when fatigued I perform effectively during critical times" compared to 46.6% of anesthesia attendings and 22% of airline captains, $p < .001$. 55% of surgical attendings disagreed with the statement "Junior team members should not question decisions by senior team members" compared to 92% of ICU attendings and 92% of captains $p < .001$. In addition, surgeons perceive teamwork and communication in the OR team to be of a higher quality than the rest of the team. Finally, although 94% of ICU personnel believed all errors were important, only one-third reported that medical errors were handled appropriately in their ICU and over 60% agreed that errors were difficult to discuss because of each of the following factors: threat of malpractice suits, threat to job security, personal reputation, expectations of the patients' family/society, and possible disciplinary actions by licensing boards.

CONCLUSION: Medical personnel reported that error is important, but difficult to discuss and not handled well in their hospital. These barriers are even more significant given that there appears to be a culture in medicine that denies the impact of stress on performance. Further barriers include differing perceptions of teamwork among various team members, and for operating room teams, support of hierarchies that do not accept input by junior team members.

THE ABILITY OF PEAK FLOW AND QUALITY OF LIFE MEASUREMENTS TO PREDICT EXACERBATIONS OF ASTHMA. WM Tierney, LA Mamlin, R Seshadri, MD Murray, M Weinberger, Indiana University School of Medicine, Regenstrief Institute; Roudebush VA Medical Center, Indianapolis, IN

Current asthma guidelines recommend regular assessment of peak expiratory flow rates, symptoms, and quality of life (QoL). We assessed the clinical utility of these recommendations among patients who filled prescriptions for inhaled or oral breathing medications at 36 CVS pharmacies. Patients were eligible if they admitted having "a breathing problem," were taking at least one inhaled or oral asthma drug, and were receiving $\geq 70\%$ of their asthma drugs at a single study pharmacy. Patients with asthma (rather than COPD as defined by American Thoracic Society criteria that use age and smoking history) completed the McMaster Asthma Quality of Life (AQoL) Questionnaire. Peak flow was measured at enrollment and during monthly follow-up calls; patients were also asked about recent exacerbations (ER or hospital visits for asthma or lower respiratory infections). We then assessed (1) the correlations between baseline AQoL scores and peak flows and (2) the ability of these baseline measures and monthly follow-up peak flows to predict exacerbations.

We enrolled 954 (71%) of 1352 eligible patients. This report is limited to the 562 patients (59%) with asthma. Their mean age was 53 years, and 75% were women. Their mean baseline peak flow rate was 351 L/min (71% of predicted maximum). Mean AQoL scores on a scale from 1 (worst) to 7 (best) were: 4.5 for activity, 4.4 for symptoms, 4.3 for emotion, 4.3 for environment, and 4.4 for overall QoL. Baseline AQoL scores were correlated with baseline peak flows: highest for the scales assessing symptoms ($r = 0.24$, $p < 0.0001$), activity ($r = 0.19$, $p < 0.0001$), and overall QoL ($r = 0.21$, $p < 0.001$), but lower for emotion ($r = 0.15$, $p = 0.0004$), and environment ($r = 0.11$, $p = 0.009$). Over a mean of 6 months of follow-up, there were 52 asthma exacerbations. Univariate Cox regression showed that having an exacerbation was significantly predicted by each baseline AQoL scale (hazard ratios [HR] = 0.62 to 0.76, $p = 0.0002$ to 0.005) but marginally by baseline peak flow ($p = 0.05$). By multivariable Cox regression, exacerbations were predicted by black race (HR = 0.38, $p = 0.001$) and overall AQoL (HR = 0.69, $p = 0.004$), but not baseline peak flow ($p = 0.36$). Patients had a mean of 5 monthly follow-up peak flows (range 1-13). Controlling for baseline peak flow and AQoL scores, follow-up peak flows did not predict asthma exacerbations within the following 30 days ($p = 0.12$).

Baseline peak flow was correlated with baseline asthma symptoms and QoL measures, yet only overall QoL predicted subsequent asthma exacerbations. Baseline peak flow did not, nor did monthly assessments of peak flow. These results do support the NIH guideline's recommendations for the routine assessment of QoL in patients with asthma, but not the routine assessment of peak flow.

EVALUATION OF THE DO-NOT-RESUSCITATE ORDERS AND END-OF-LIFE CARE IN TERMINAL CANCER PATIENTS AT A TEACHING HOSPITAL IN JAPAN. Y Tokuda, Department of Medicine, Okinawa Chubu Hospital, Gushikawa, Okinawa, Japan

PURPOSE: This study was designed to analyze the pattern of end-of-life decision making for terminal cancer patients at a teaching hospital in Japan.

METHODS: Retrospective descriptive study was conducted for the review of the course of 124 patients who died of terminal cancer during two twelve-month periods, in 1989 and 1999, at a teaching hospital in Okinawa, a tropical island in Japan. Basic demographic data, notification to patient of having cancer, patient involvement at the time of writing a do-not-resuscitate (DNR) order, and various medical interventions, which were performed within a month before the time of end-of-life, were evaluated. Fisher's exact test was employed for comparison of the data between 1989 and 1999.

RESULTS: There were 79 men and 45 women cancer deaths with mean age of 69 (63 in 1989 and 61 in 1999). Only 6 patients were notified to have cancer. 113 (91%) of patients had a DNR order written, with none of patients but 95% of families being involved. Medical interventions near the time of end-of-life, included antibiotic administration in 55 patients, total parenteral nutrition in 14 patients, and intravenous albumin infusion in 9 patients. There were no difference in pattern of the above practice between 1989 and 1999 (Table, $p > 0.05$ Fisher's exact test).

year	1989	1999
total	63	61
men:women	43:20	36:25
mean age	67	70
number of informed as having cancer	2	4
number of DNR orders written	56	57
number of DNR orders with patient consulted	0	0
number of patients given intravenous albumin	4	5
number of patients given antibiotics	25	30
number of uses of total parenteral nutrition	6	8

CONCLUSION: Almost all dying cancer patients were not notified of having cancer and seldom involved for DNR decision making in a teaching hospital in accordance to Japanese tradition and custom. Potentially futile interventions were frequently performed near the time of end of life. Those pattern of practice had not changed in recent 10 year interval.

MORTALITY IN A PROSPECTIVE STUDY OF PHYSICIANS. DM Torre, N-Y Wang, LA Mead, DE Ford, JH Young, T Houston, A Kouzis, MJ Klag. Medicine, The Johns Hopkins Medical Institutions, Baltimore, MD

PURPOSE: Socioeconomic status is strongly related to most health outcomes. Physicians enjoy high socioeconomic status including high levels of access to care and health knowledge but the magnitude of health benefits in this group is unclear.

METHODS: We compared the mortality experience of a cohort of 1,213 white male medical students, graduating classes 1948-64, to the U.S. white male population using age-race-sex and calendar time specific death rates from U.S. vital statistics. Mortality through 1998 was assessed by annual questionnaires, contacting family members and co-workers, scanning obituaries and National Death Index searches. Vital status was known for >99% of the cohort.

RESULTS: The average year at entry was 26 years of age, average follow-up 40 years and mean age at death 58 years. Prevalence of smoking decreased from 51% in medical school to 5.6% at the end of follow-up. During 48,589 person-years of follow-up, 195 deaths occurred. The risk of all-cause mortality was 57% lower than expected (standardized mortality ratio [SMR] 0.43, 95% confidence interval 0.37-0.49), compared to the general population. All cause mortality did not differ by specialty. SMR was similar before (0.53, 95% CI 0.40-0.69) and after (0.38, 95% CI 0.32-0.44) 1973. For specific causes of death, the risk was lowest for cardiovascular diseases 0.26 (95% CI 0.20-0.34), coronary artery disease 0.15 (95% CI 0.10-0.23), and stroke 0.21 (95% CI 0.6-0.54). Risk of dying from cancer was also less than expected, 0.46 (95% CI 0.35, 0.60). SMR's were markedly less for diseases strongly linked to smoking, e.g., respiratory diseases 0.27 (95% CI 0.11-0.52), and lung cancer 0.31 (95% CI 0.17-0.51). Suicide was the only cause of death where risk was greater than the general population (SMR 1.76; 95% CI 1.09-2.69). The risk of dying from suicide was lowest among surgeons (SMR 0.26, 95% CI .00-1.43) and highest among psychiatrists (SMR 4.86, CI .98-14.1). Internists (SMR 1.11, 95% CI 0.3-2.84) and pediatricians (SMR 3.42, 95% CI 0.69-9.95) had intermediate death rates from suicide compared to the general population.

CONCLUSION: These results demonstrate the substantial health benefits associated with being a physician, compared to the general population.

EFFECTIVENESS OF PATIENT LETTER REMINDERS ON CERVICAL CANCER SCREENING: A META-ANALYSIS. DS Tseng, E Cox, M Plane, K Hla, General Internal Medicine; Pediatrics; Family Medicine, University of Wisconsin, Madison, WI

PURPOSE: To investigate the effectiveness of patient letter reminders on improving health care visits for cervical cancer screening using pap smears by performing a meta-analysis on existing randomized controlled trials.

METHODS: A literature search was conducted for all relevant articles between the years of 1966 and 1999. The bibliography of all articles obtained was also reviewed for pertinent studies. A search for unpublished studies was pursued by making inquiries of authors located by the literature search. Eligibility criteria included randomized controlled studies that looked at populations requiring pap smears for screening purposes. The intervention studied was in the form of a re-

minder letter only and there were no restrictions as to country of origin. The studies retained were scored for quality by an adaptation of the method reported by Chalmers. The data were extracted from eligible articles in order to compare the proportion of women screened between the intervention and the control groups during the study period. Two authors independently extracted the data. The Mantel-Haenszel method was used to measure the summary effect size of the intervention. A test for homogeneity using the Mantel-Haenszel method was also performed to test the hypothesis that all of the studies were measuring the same effect (null hypothesis).

RESULTS: Of the initial 421 articles found on literature search, 23 articles that dealt with the primary question were retained. Four additional articles were obtained by bibliography review. Fourteen of the studies were randomized controlled trials. The inquiry for unpublished studies yielded two studies. Nine articles were left after excluding those articles that did not fit all of the inclusion criteria, including one unpublished study. Most of the exclusions were due to using combined interventions such as letter and physician reminders. Individual odds ratios and confidence intervals were calculated. The test for homogeneity showed a non-significant chi-square value of 5.78 with 8 degrees of freedom. The hypothesis of homogeneity was accepted. The pooled odds ratio showed that patients who received the intervention were significantly more likely to have had cervical cancer screening than those who did not (OR 1.86, 95%CI 1.56-2.22).

CONCLUSION: The use of patient reminders in the form of mailed letters is an effective method of increasing the rate of cervical cancer screening.

HEALTH CARE PREDICTORS OF QUALITY OF ANTIRETROVIRAL THERAPY (ART) IN HIV+ WOMEN. BJ Turner, C Laine, D Zhang, RJ Pomerantz, L Cosler, WW Hauck, Department of Medicine, Jefferson Medical College, Philadelphia, PA; New York State (NYS) Department of Health, Albany, NY

PURPOSE: Because access to appropriate ART represents a key aspect of HIV quality of care, we explored the impact on type of prescribed ART of HIV specialty care from an HIV specialist (i.e., infectious diseases or heme/oncology or receiving an enhanced Medicaid fee for HIV care) as well as the effect of methadone treatment for drug users.

METHODS: We determined the type of ART prescribed in four 6 mo. intervals from 1997-98 from claims for HIV+ women on NYS Medicaid who met the following criteria in the interval of interest: prescribed >30d ART, currently not pregnant, and fully eligible (N by interval = 340, 403, 384, and 387). These women delivered previously in 1993-96. We linked vital statistics for their last delivery to claims files from 1992-98. We estimated logistic regressions for two outcomes (with variances corrected for clustering) and adjusted for demographics and clinical factors: 'good' ART in an interval (i.e., no monotherapy or contraindicated combination) and, among persons on 'good' ART in an interval, 'best' ART (ie, highly active ART [HAART]).

RESULTS: Depending on the time interval, no HIV specialty care was received by 23-35% of women, 1-2 visits by 11-17%, 3+ visits by 52-66% while methadone was received by 11-13%. The proportion with 'bad' ART diminished over the time from 26 to 15% while, of women on 'good' or 'best' ART, the proportion with the latter rose from 26 to 65%. The adjusted odds ratio (AOR) of 'good' ART was increased ($P < 0.05$): two-fold for later time periods, NYC residents vs. rest of state (AOR = 1.46, 95% CI 1.01, 2.13), women treated in sites offering ART trials (AOR = 1.43, 95% CI 1.06, 1.94), women with 1-2 HIV specialty visits (AOR = 1.71, 95% CI 1.11, 2.63) or 3+ visits (AOR = 2.10, 95% CI 1.55, 2.85) vs. none, and methadone-treated women (AOR = 2.06, 95% CI 1.01, 4.21) vs current drug users. Among women on 'good' ART, AOR of 'best' ART were increased ($P < 0.05$) for: later time periods, NYC residence (AOR = 1.55, 95% CI 1.06, 2.25), high school graduates (AOR = 1.50, 95% CI 1.17, 1.92), and methadone treated women (AOR = 2.81, 95% CI 1.41, 5.60). Only 3+ fee/specialty visits had a borderline ($P = 0.06$) association with best ART (AOR = 1.32, 95% CI 0.99, 1.76) vs no such visits.

CONCLUSION: Among these previously pregnant HIV+ women, we conclude that specialty care for HIV and methadone treatment for drug users diminish the likelihood of patients being prescribed types of ART that are not recommended. HIV specialty care showed less of an effect on 'best' ART (HAART), perhaps because 'good' ART is adequate for some women.

COST OF CARE IN ACADEMIC HEALTH CENTERS: THE JULY PHENOMENON REVISITED. PD Turner, JD Bramble, EC Rich, Center for Practice Improvement and Outcomes Research; School of Pharmacy and Allied Health Professions; Department of Medicine, Creighton University, Omaha, NE

PURPOSE: Every July, academic health centers prepare for the influx of new physicians recently graduating from medical schools. While the evidence is inconclusive that new interns have an adverse affect on the cost and quality of care, the colloquial term "July Phenomenon" has been used to describe the trend in which the efficiency of inpatient care decreases each July, and gradually increases over the subsequent academic year. This study evaluated changes over the academic year in the total adjusted charges and length of stay of inpatient care in a national sample of teaching and nonteaching hospitals to investigate the effect of house-staff experience related to the cost of care.

METHODS: A retrospective cross-sectional research design was used to conduct a secondary analysis of a 10% subsample of the Healthcare Cost and Utilization Project (HCUP) 1996 Nationwide Inpatient Sample (NIS). This database approximates a 20% sample of U.S. hospitals. The top 40 DRGs were identified and categorized into either medical, surgical, or Ob/Gyn. Patients admitted during 1996

with these preselected DRGs to urban, nonfederal, nonpediatric acute care hospitals were examined. This resulted in 169,168 patients in 422 hospitals in 13 states. The level of housestaff experience was measured as the number of months from July in which the patient was admitted. Multiple regression analyses were used to evaluate the different effects of experience on patient care in teaching hospitals compared with nonteaching hospitals, controlling for patient, hospital, and market characteristics. Our variable of interest was the interaction term, "Teaching/Nonteaching" $\times$ "Experience", and when significant, indicates that the passage of one month since July is associated with a different change in outcome (i.e., total adjusted charges or LOS) in teaching hospitals than nonteaching hospitals.

RESULTS: The "July Phenomenon" was observed for total charges with the subset of patients with medical diagnoses—i.e., over the course of the academic year, these patients at teaching hospitals experienced a significant decrease in total costs relative to similar patients at nonteaching hospitals ($b = -2.20E-3$, $p = .003$). There were no meaningful effects of housestaff experience on total charges for the surgical and Ob/Gyn diagnoses, and length of stay for all three groups of diagnoses.

CONCLUSION: Housestaff training is significantly related to resource utilization with patients admitted with medical diagnoses. The training of inexperienced physicians may represent an important source of fiscal inefficiency for teaching hospitals in a competitive market.

COST-EFFECTIVENESS INFORMATION AND SCREENING RECOMMENDATIONS: ARE PHYSICIANS SET IN THEIR WAYS? PA Ubel, C Jepson, DA Asch.

PURPOSE: Physicians are increasingly being faced with situations in which one screening option is both more effective and more expensive than another. Economists say that the proper way to make such a choice is to look at the incremental cost (IC) of the more costly option over the less costly one. IC is essentially the extra cost of the given option divided by the extra return—that is, the cost of getting each unit of return that would not be achieved by the less costly option. IC is often counterintuitively high. For example, the IC of annual Pap smears, compared to Pap smears every three years, is nearly \$1 million per year of life saved. The counterintuitive expense of some screening tests raises questions about whether physicians' screening recommendations would be influenced by the type and amount of cost information they receive.

METHODS: We created a set of three scenarios, each asking the respondent to make a screening recommendation for a hypothetical patient; the scenarios involved decisions about breast, cervical, and colorectal cancer screening respectively. Three versions of each scenario were created, containing either no cost-effectiveness information, average cost-effectiveness information (e.g., the average cost-effectiveness of yearly Pap smears is \$39,000 per life year), or incremental cost-effectiveness information (incremental cost-effectiveness of yearly Pap smears is \$1 million). In addition, for each scenario, a parallel scenario was created in which the cost information was the same, but the nature of the screening was unfamiliar (gastric cancer, Pap tests using a new procedure to increase yield, and an unlabeled cancer and tests). Average and incremental cost versions of the unfamiliar scenarios were created. We randomized 1,500 U.S. primary care physicians to receive one of the resulting 15 scenarios.

RESULTS: A total of 950 physicians responded (62%). Overall, subjects were less likely to recommend the most expensive screening test in unfamiliar scenarios than in familiar ones ($p < 0.001$). Their willingness to recommend expensive tests in unfamiliar scenarios was significantly reduced by receiving incremental cost information ($p < 0.001$). However, the type of cost information provided had no effect on recommendations in familiar scenarios ($p > 0.2$).

CONCLUSION: Subjects' screening recommendations are influenced by cost information in unfamiliar screening scenarios but not in familiar ones.

RISK FACTORS FOR DISCHARGES AGAINST MEDICAL ADVICE IN NEW YORK STATE IN 1998. UE Uwagoma. Division of Sociomedical Sciences, Joseph Mailman School of Public Health, Columbia University, New York, NY; Department of Internal Medicine, Harlem Hospital/Columbia University College of Physicians and Surgeons, New York, NY

PURPOSE: To determine the social, clinical demographic characteristics, and risk factors of patients discharged against medical advice [AMA] in New York State.

METHODS: An analysis of discharges from non-federal acute hospitals in New York State in 1998 from the Statewide Planning and Research Cooperative Systems (SPARCS) data set. Patient-specific data including age, sex, race, zip code, and various clinical information were abstracted along with the principal diagnosis, discharge status and disposition. A subset of the SPARCS data set for the year 1998 containing about 1.2 million of the 2 million discharges in New York state in 1998 were utilized for this study. Patients with a discharge disposition of leaving against medical advice were identified. In this study the ICD9 code for the principal diagnosis was translated into major diagnostic groupings. The other covariates are included in the study included self-pay patients identified as uninsured, as well as those residing in Harlem zip codes 10029, 10037, 10039. The statistical analysis was conducted on SAS software version 6.12. The initial baseline characteristics of the whole sample and those discharged AMA were obtained. A logistic regression analysis was run with AMA discharges as the dependent variable and the other covariates were the independent variables.

RESULTS: Of the sample of 1.16 million discharges there were 22,542 AMA discharges [2%]. The largest age group were those between 31 and 40. They con-

sisted of 34% of AMA discharges versus 14% of the sample population. Most of the AMA discharges were male (64%), emergency admissions (69%), mental disorders (39%) and Medicaid patients (59%). The odds ratio for those aged 30–40 was 2.68 [$p = 0.0001$], emergency admissions 1.30 [$p = 0.0001$], males 1.77 [$p = 0.0001$], Medicare patients 1.93 [$p = 0.0001$], Medicaid patients 3.44 [$p = 0.0001$] uninsured 3.83 [$p = 0.0001$] mental disorders 4.88 [$p = 0.0001$] whites 0.81 [$p = 0.0001$] and Harlem residents 1.22 [$p = 0.0001$].

CONCLUSION: AMA discharges are a significant problem in our healthcare system. Those at higher risk for AMA discharges are primarily young males, people with mental disorders, emergency admissions, minorities, Medicaid/Medicare patients, and the uninsured. AMA discharges should be evaluated as an important healthcare outcome measure.

VALIDITY OF CLAIMS DATA FOR HIV/AIDS SURVEILLANCE IN TENNESSEE'S MEDICAID PROGRAM. DL Van Brunt, JE Bailey, R Phillippi. Preventive Medicine, University of Tennessee, Memphis; Tennessee Department of Health, Knoxville, TN

PURPOSE: This study assessed the validity of a method for using Medicaid claims data for HIV surveillance in Tennessee for a period covering both traditional Medicaid and Medicaid managed care, and examined the time interval between identification in the state's HIV/AIDS Reporting System (HARS) and detection using a claims casefinding method.

METHODS: Tennessee's Medicaid claims data for the years 1992–7 were compared to the state Health Department's mandatory HARS records. All subjects who had at least one health care claim containing the ICD code for either HIV or asymptomatic HIV, or who had at least one claim for an antiretroviral medication were considered positive for HIV for study purposes. The Kappa statistic was used to assess agreement in identification of enrollees with HIV or AIDS. A time interval was computed between the HARS date of identification and the first qualifying claim from the administrative data, with correspondence rates for 60 day, 1 year, and two year intervals described.

All direct and indirect identifiers from the research database were replaced with a randomly generated case identifiers for each record to prevent identification of individual enrollees.

RESULTS: Overall, 1,415,991 people were enrolled in TennCare during the study period. 4,395 individuals were identified as HIV+ by claims and HARS, 788 by claims only, and 2,051 by HARS only. Inter-rater (HARS v. Claims) agreement yielded Kappa = .76 ($p < .001$). Using the HARS as criterion, this makes claims data sensitivity = 68% and specificity = 99% for HIV surveillance. The ability to identify HIV cases was limited in regards to time, however, with 5.4% of cases being identified within 30 days before or after the HARS date, 16.8% within 6 months in either direction (one year interval), and 25% within 1 year before or after HARS identification (2 year interval). Of the 788 cases identified by claims and not by HARS, a limited chart review confirmed that 187 of these cases were true positives.

CONCLUSION: Claims casefinding methods appear to be useful for HIV surveillance during the period 1992–7 in the Tennessee Medicaid population. However, the ability to assign time of disease onset is less accurate, making this method more appropriate for period prevalence estimates over longer time spans than for incidence or period prevalences over short time spans. For this study, it appears that longer exposure to the claims collection system increases the likelihood of identification. Research that depends on assessment of time at-risk should not rely solely on administrative casefinding algorithms, but should attempt to validate data from other outside sources such as chart or laboratory data. Claims data may be a useful adjunct to other surveillance systems.

HIGH SERUM VITAMIN B12 LEVELS: CLINICAL ASSOCIATIONS AND POTENTIAL MECHANISMS. H Vasireddy, I Aurangzeb, B Bhushute, K George, R Carmel, HEM & ONC, NYM Hospital, Brooklyn, NY

PURPOSE: Much is known about low B12 levels, but elevated levels has not been systematically investigated even though they are common. So we prospectively compared the clinical associations of elevated B12 levels with those of normal B12 levels. We also explored the mechanisms by studying the distribution of B12 in the blood among its transport proteins or transcobalamins (TC).

METHODS: The clinical laboratory provided all 94 sera that had B12 level $>900\text{ng/L}$, out of 670 consecutive assays (14% incidence) over 75 days. Results were confirmed in follow-up blood samples, after excluding duplicate submissions and patients without confirmatory specimens or with normal repeat B12 levels. Clinical and laboratory data from 60 patients (group I) were compared with those of 75 consecutive controls that had normal B12 levels (group II). The distribution of B12 between TCI and TCII was determined by the QUISO method; normally $>75\%$ of serum B12 is carried by TCI (holoTCI), which has only a storage function, and $<25\%$ by TCII (holoTCII), which rapidly delivers B12 to cells.

RESULTS: (1) Among all the clinical conditions, only renal failure ($p = .01$) and diabetes mellitus ($p = .04$) were significantly more frequent in group I than II. Multivariate linear regression (MLR) analysis showed only kidney disease to be independently associated with a high B12 levels. None of the patients with high B12 levels, had myeloproliferative disease or leukemoid reaction, the best known causes of high B12 levels, and only 3 had recent B12 injections as possible explanations. (2) Among all laboratory data, only high creatinine ($p = .0001$), hypoalbuminemia ($p = .0001$), anemia ($p = .001$) and high alkaline phosphatase ($p = .01$) were significantly more common in group I than group II. MLR showed only high creatinine level and hypoalbuminemia to be independently associated with a high

B12 level. Serum B12 levels didn't correlate with WBC counts. (3) Holo TCI was inversely and holoTCII directly correlated with serum B12 levels ($p = .0001$) and mean % holo TCI was significantly lower in group I than II ($69.6\% \pm 19.5$ vs $87.5\% \pm 10.0$, $p = .001$).

CONCLUSION: Elevated B12 levels are common in clinical practice (14% prevalence). Our data shows renal failure to be the only disorder independently associated with high B12 levels, while better known causes are actually rare. Ours is the first demonstration that B12 seems to be abnormally distributed in the patients with high B12 levels, with less of it being carried by TCI and more by TCII. This suggests that the probable mechanism involves diminished delivery of B12 to cells by TCII. For example, many high B12 levels may be due in part to the failure of normal uptake of TCII-bound B12 by TCII receptors in the kidney in renal failure.

SHOULD WE SCREEN FOR DEPRESSION IN PRIMARY CARE? A COST-UTILITY ANALYSIS. S Vijan, M Valenstein, J Zeber, VA HSR&D, Ann Arbor VAMC; Department of Internal Medicine; Department of Psychiatry, University of Michigan, Ann Arbor, MI

PURPOSE: Depression is a common condition that is associated with substantial disability and health care costs. Although depression is very common in primary care, it often remains undiagnosed. A number of policy-makers have recommended depression screening programs as one strategy for increasing detection; however, studies of screening have had mixed results. To further inform the debate, we examined the cost-utility of screening for depression versus usual care. **METHODS:** We developed a Markov model to examine the lifetime costs and benefits of periodic (annual or opportunistic) or one-time screening for depression in typical primary care settings. Parameter estimates were drawn from the published literature. We simulated hypothetical cohorts of 20, 40, and 60 year old primary care patients. In each cohort, lifetime costs and quality adjusted life years (QALY) were tabulated with and without screening for depression. One, two, and three-way sensitivity analyses were conducted in order to determine the robustness of the model to variations in the estimates and to determine the variables that were most influential in defining the cost-utility of screening for depression.

RESULTS: In the base case, annual depression screening had an incremental cost-utility (C/U) ratio of about \$225,800 per QALY gained when compared to usual care. Opportunistic screening had an even higher C/U ratio. However, one-time screening (at entry to the health care system) had a C/U ratio of \$48,900 per QALY gained. Estimated C/U ratios were most sensitive to the utilities assigned to major depression, prevalence of depression, costs of screening, treatment initiation once depression is diagnosed, remission rates with treatment, and rates of diagnosis in usual care. Sensitivity analyses indicated that annual screening would achieve a C/U ratio below \$50,000 per QALY gained only under very specific multiple conditions. A population with a high prevalence of depression ($>13\%$) and high treatment success rates ($>70\%$) were both required, along with any one of the following: high treatment rates for those diagnosed with depression ($>70\%$); very low utility of depression (<0.54); or low cost of screening ($<\$3.00$). In contrast, a change in nearly any single parameter increased the C/U ratio of one-time screening above \$50,000 per QALY gained.

CONCLUSION: Annual or periodic screening for depression is unlikely to be a cost-effective intervention in primary care unless substantial improvements are also made in the quality and effectiveness of depression treatment. However, one-time screening for depression may be reasonably cost-effective even with current practice patterns.

DEVELOPMENT OF A NEW WEB-BASED EDUCATIONAL RESOURCE IN EVIDENCE-BASED MEDICINE FOR PRIMARY CARE PHYSICIANS: THE PRIMARY CARE HEALTH INFORMATION PROJECT. EW Vogel, RC Maulitz, K McCain, A Smolen, MCP Hahnemann University, Philadelphia, PA

PURPOSE: To develop an innovative, evidence-based information source for primary care physicians to use in their outpatient practice, which is accessed from the World Wide Web.

METHODS: The goal of the Primary Care Health Information Project (P-CHIP), supported by a grant from The National Institute of Standards & Technology, was to develop a Web-based resource of high-quality, concise, dynamically interactive & evidence-rated information that primary care physicians could quickly access to answer patient care questions. A team of information scientists and primary care physicians designed a database of medical information from important original journal research, and a user interface for physicians to search the database for information.

RESULTS: The database consists of modules on common topics of importance for primary care physicians (e.g., diabetes or asthma), of which seven have been developed so far. For each topic, information from one hundred or more original journal articles was entered in the database. For each article, the database includes the abstract of the article, study type, clinical topic, medical subject heading (MeSH) terms assigned to the article, and a clinical "pearl" developed by a generalist physician domain expert. The "pearl" summarizes the main points of interest for primary care physicians needing to apply the information to patient care. Physicians access the database through the search interface, allowing retrieval of articles meeting criteria requested by the physician, including study type, evidence quality, and subtopic (e.g., epidemiology or diagnosis). The database is dynamic, as new high-quality journal articles can be added to the database as they become available. The search interface also accesses information on the specified topic from a complementary database, which contains audio-visual segments from recorded lectures given by experts on the topic of interest. The

user interface and database is being tested during the year 2000 with a group of practicing primary care physicians for the accuracy of search results and the applicability of the medical information in the database to patient care.

CONCLUSION: The Primary Care Health Information Project has developed a new Web-based resource of high-quality information for primary care physicians. Further testing is under way to optimize the value of this database in applying the information to patient care.

HMO PATIENT SATISFACTION FOLLOWING BRIEF ALCOHOL ADVICE. KA Vom Eigen, D Dauser, JC Higgins-Biddle, TF Babor, General Internal Medicine; Community Medicine, UConn School of Medicine, Farmington, CT

PURPOSE: We designed a multi-site randomized trial to examine the impact of brief advice for risky drinkers on behavioral, health, economic and satisfaction outcomes in managed care settings.

METHODS: Three primary care clinics in each of three HMOs (9 sites) were randomized to one of three conditions: brief advice by their provider (P), brief advice by another trained staff member (S), or usual care (C). Patients in each clinic completed a self-report questionnaire. Those screening positive as risky drinkers in the P and S sites received brief advice. A two-month follow-up telephone interview included the HEDIS 3.0 Member Satisfaction Survey, the PCAS Interpersonal Treatment Scale, and global satisfaction items. Satisfaction scores were compared using Wilcoxon Rank Sum tests and linear regression analysis.

RESULTS: Over 900 patients (544 Males, 375 Females; P 248, S 242, C 429) were enrolled from the nine sites, beginning in March 1998. In univariate comparisons, satisfaction scores were slightly lower for participants in intervention (P, S) vs. control (C) sites. In linear regression analysis, adjusting for age, gender and overall health status, participation in intervention sites (P, S) was associated with slightly lower satisfaction. However, when we adjusted for enrollment from one of the P sites where specific local management problems had occurred, there were no significant satisfaction differences between intervention and control sites. There were no significant differences in satisfaction scores of participants in S vs. P sites, in univariate or multivariate comparisons.

CONCLUSION: Although intervention (P, S) participants had slightly lower satisfaction scores than controls, our preliminary data suggest that these differences are due primarily to specific local management problems at one of the P sites. Controlling for this one site, it appears that screening and brief advice does not negatively affect patient satisfaction. Further data collection and analyses will provide more information on patient satisfaction with these interventions, and on how satisfaction measures relate to health and behavioral outcomes.

PHYSICIAN ATTITUDES TOWARD PSA TESTING: CHANGES FROM 1993 TO 1998. JD Voss, JM Schechtman, Internal Medicine, University of Virginia, Charlottesville, VA; Health Care Sciences, George Washington University Medical Center, Washington, DC

PURPOSE: To evaluate attitudinal changes over time regarding PSA test use in prostate cancer (PC) screening, a sample of primary care physicians was surveyed in 1993 and 1998.

METHODS: The study population was comprised of general internists and family physicians participating in a 90,000 member mixed HMO affiliated with a major academic medical center serving metropolitan Washington, D.C. One hundred seventy six physicians completed the 1993 survey (response rate of 76%). After adjustment for 35 physicians who had retired or could not be located, 108 1998 responses were received (response rate of 76%). Physicians were asked whether they agreed, disagreed or were uncertain regarding statements about PSA test use in PC screening. Query statements included acceptance of PSA tests as a standard, malpractice liability for not ordering, PSA screening test characteristic acceptability and statements regarding the impact of screening and treatment on outcomes. Responses in 1998 were compared to 1993 results with a two-tailed $p < 0.05$ as criteria for statistical significance.

RESULTS: Significant increases in agreement with attitudes favoring PSA testing were observed from 1993 to 1998. The percentage of physicians expressing favorable attitudes toward PSA testing showed an absolute increase of 11% to 17% (relative increase of 21% to 52%) across the questions repeated over time. In both periods, fewer physicians indicated agreement with statements related to improved clinical patient outcome from early intervention (efficacy of radical prostatectomy, benefits of treatment outweigh risks) than with non-patient related statements. Community-based physicians reported higher levels of agreement with attitudes favoring PSA testing than academic-based physicians in 1993 and 1998.

Attitudes and Beliefs	Percent Agreement		p value
	1993	1998	
Yearly PSA tests for asymptomatic men 50 or greater are a standard of care	53	64	0.006
MDs face malpractice liability if PSA not tested and PC detected later	36	53	0.003
Radical prostatectomy for early PC reduces PC specific mortality	29.3	44.6	0.013
PSA has acceptable sensitivity, specificity and PPV	46.9	59.2	0.036
Morbidity and mortality of treating PC may outweigh the benefits	Percent Disagreement		
	1993	1998	
	35.4	46.5	0.071

CONCLUSION: From 1993 to 1998, a significant number of physicians reported increasing agreement with attitudes favoring PSA test use for PC screening. However, a substantial proportion remains unconvinced of its merits as a screening tool, consistent with the evidence based medical literature in this area.

PSA TESTING TRENDS FROM 1993 TO 1998: REPORT OF A PHYSICIAN SURVEY. JD Voss, JM Schectman, Internal Medicine, University of Virginia, Charlottesville, VA; Health Care Sciences, George Washington University Medical Center, Washington, DC

PURPOSE: We report the results of a longitudinal study that surveyed a sample of primary care physicians in 1993 and 1998 regarding their PSA testing practices. **METHODS:** The study population was comprised of general internists and family physicians participating in a 90,000 member mixed model HMO affiliated with a major academic medical center serving metropolitan Washington, D.C. One hundred seventy six physicians completed the survey in 1993 (response rate of 76%). Respondents were resurveyed in 1998. After adjustment for 35 physicians who could not be located or who had retired, 108 responses were received (response rate 76%). Physicians were asked what percentage of time they ordered PSA tests for asymptomatic men aged 50 or greater presenting for general physical examinations. Physicians were categorized as low test orderers (0–30%), moderate test orderers (40–70%) or high test orderers (80–100%) based on examination of the 1993 self reported PSA ordering frequency distribution. Responses in 1998 were compared to 1993 results with a two-tailed $p < 0.05$ as criteria for statistical significance.

RESULTS: In 1993, 26% of physicians reported low, 13% medium and 65% high PSA testing frequencies. From 1993 to 1998, physicians reported a rise in mean percentage of time they ordered PSA tests from 72.8% to 80.5%, an absolute increase of 7.7% (CI 2.6–12.9%, $p = 0.004$), and a relative increase of 10.6%. Between 1993 and 1998, 43.3% of physicians reported no change in test ordering, 43.3% reported increased PSA test ordering and 13.4% reported decreased PSA test ordering ($p = 0.001$). Among high PSA testers in 1993, 87% continued to report high test ordering in 1998. Among moderate testers in 1993, 83% changed to high test ordering in 1998, 17% remained moderate testers and none changed to low testers. Among low testers in 1993, 18% reported high test ordering, 32% reported moderate testing and 50% continued to report low test ordering (McNemar $p = 0.024$). In contrast, physicians reported performing digital rectal examinations in 94% and 96% of screening general examinations in 1993 and 1998 ($p = 0.110$).

CONCLUSION: Significant changes in self reported PSA testing practices were seen over this recent five year period. Substantial increases in PSA use were noted among initial moderate and low test utilizers despite a lack of published scientific evidence supporting its use. Physician behavior is likely being driven by factors other than the published medical literature.

APPLICATION OF DISEASE MANAGEMENT PRINCIPLES TO ACHIEVE NATIONAL GUIDELINE-DEFINED GOALS FOR PATIENTS WITH CORONARY HEART DISEASE. MN Walsh, GJ Wan, LC Kuo, DA Eisenberg, RJ Simpson, TA Pearson, CM Alexander, School of Medicine, Indiana University, Indianapolis, IN

PURPOSE: To determine whether a site-defined practice improvement program would increase the proportion of patients achieving AHA/ACC guideline-defined goals for coronary heart disease.

METHODS: Nine site-defined quality improvement programs used a variety of provider and patient-based disease management tools to improve care. Cardiovascular risk factors, laboratory values, and medication use were collected prospectively (August 1997–June 1998) from these sites (2 internal medicine, 2 academic, 2 managed care, 3 cardiology; $N = 688$ patients).

RESULTS: Compared to baseline, at 6 months total cholesterol <200 mg/dL increased from 57% to 74% ($P < 0.001$), LDL-C <100 mg/dL increased from 33% to 74% ($P < 0.001$), HDL-C >35 mg/dL decreased from 27% to 19% ($P < 0.001$), rigorous exercise of 30 minutes, 3 times a week increased from 37% to 51% ($P < 0.001$), lipid medication use increased from 72% to 79% ($P < 0.01$), and aspirin use increased from 84% to 90% ($P < 0.001$). No differences in LDL-C goal achievement were seen in patients treated with the most commonly used statins (i.e., 82% = simvastatin; 77% = atorvastatin; $P = NS$) at 6 months. Despite some improvement in TG levels and blood pressure control, differences at 6 months compared to baseline were not statistically significant. There were no significant changes in diabetes management, smoking cessation, body mass index, and beta-blocker use at 6 months. Successful implementation of this program was related to 1) the total number of patients enrolled and 2) the use of a nurse study coordinator. No differences in LDL-C goal attainment were found between patient and provider interventions.

CONCLUSION: These data appear to demonstrate an opportunity to improve patient care by the use of disease management principles using a variety of interventions and measures to promote best practices in the treatment of CHD patients. The amount of improvement demonstrated correlated more with the dedication of each site to quality improvement (evidenced by the use of a nurse study coordinator and total enrollment achieved) than any particular patient or provider centered intervention. There is a need for additional studies to further evaluate optimal interventions to improve care of patients with CHD.

THE IMPACT OF THE "LAZY SUNDAY" ON LENGTH OF STAY AND CHARGES FOR CARE. MG Weiner, A Cohen, Division of General Internal Medicine, University of Pennsylvania School of Medicine, Philadelphia, PA

PURPOSE: The quality and intensity of inpatient hospital care should be the same regardless of the day of the week the care is being provided. However, weekend days, particularly Sundays, seem less active in terms of clinical interventions and discharges. This study examines impact of this "Lazy Sunday" effect on length of stay (LOS) and hospital charges and shows how the phenomenon has changed between 1988 and 1995 in the face of increasing cost-containment pressure. We hypothesized that admissions on Monday would be more likely to have a LOS <5 , enabling a pre-weekend discharge. Patients admitted on Thursday would be less likely to be discharged in <5 days since the typical peak discharge day would fall on the "Lazy Sunday."

METHODS: We utilized data from the Nationwide Inpatient Sample (NIS) collected as part of the Healthcare Cost and Utilization Project. To ensure comparisons of patients with similar characteristics, we only selected admissions among patients over 65 from Subsample 1 of the NIS from 1988 and 1995 having the 4 most common DRG's. The NIS also names the weekday of each admission and the corresponding LOS and charges. Since the "Lazy Sunday" effect is hypothesized to impact typical admissions, outlier admissions (LOS >25 days, charge $>\$25,000$) were excluded. The significance of differences in average LOS and average charges was assessed with the Student t-test. The Chi square test was used to assess variation in the cumulative percent discharge by day 5.

RESULTS: In 1988 and 1995, the 4 most common DRG's among patients over 65 (CVA, pneumonia, CHF and angina) accounted for 23159 and 33592 admissions respectively. The table shows data for average LOS, the cumulative percent discharge by day 5 (CP5) and the average charge (AC) for admissions on Monday (M) and Thursday (Th).

Year	LOS-M	LOS-Th	CP5-M	CP5-Th	AC-M	AC-Th
1988	6.8 $\pm$ 4.7**	7.2 $\pm$ 4.7**	42**	30**	5360 $\pm$ 3868*	5587 $\pm$ 4092*
1995	5.4 $\pm$ 3.7**	5.9 $\pm$ 3.8**	55**	40**	7487 $\pm$ 4829*	7888 $\pm$ 4932*

* $p < 0.01$ ** $p < 0.001$ for corresponding Monday-Thursday differences

CONCLUSION: Although the "Lazy Sunday" effect occurs whenever an admission spans a Sunday, the analysis shows that, consistent with our hypothesis, the effect is greater when the admission occurs later in the week, resulting in a 0.5 day increase LOS and increased charges of about \$400. While the average LOS has decreased and charges increased over the period between 1988 and 1995, the impact of the "Lazy Sunday" as measured by the absolute difference in LOS on Monday and Thursday has not changed. Given the Monday-Thursday charge difference in 1995, this study suggests that more aggressive discharge planning for the nationwide set of patients over 65 with common DRG's could save at least \$90 million in charges annually.

HEALTH EXPENDITURE AND UTILIZATION PATTERNS FOR MEDICARE BENEFICIARIES WITH ALZHEIMER'S DISEASE. M Weiner, WE Weller, TJ Shaffer, NR Powe, GF Anderson, Indiana Univ. Center for Aging Research, Regenstrief Institute, Indianapolis, IN; Johns Hopkins Medical Institutions, Johns Hopkins Univ., Baltimore, MD; Dept. of Medicine, Indiana Univ., Indianapolis, IN

PURPOSE: Risk adjustment is beginning to include factors based on specific clinical conditions. Although Alzheimer's disease (AD) is a major cause of morbidity and death among older adults, we know little about how expenditures and utilization for persons with AD change over time. We sought to compare 1994–1996 patterns of resource utilization for Medicare beneficiaries with AD to those for beneficiaries without AD and to test the hypothesis that AD is associated with increased expenditures independent of demographic factors.

METHODS: We examined a 5% nationally random sample of paid fee-for-service Medicare claims for services rendered to beneficiaries between 1994 and 1996, using diagnostic coding for AD to define cases. We identified a 1994 cohort and then assembled data reflecting the cohort's 1994, 1995, and 1996 claims. We assessed usual sources of care, comorbidity, and Medicare expenditures by service and physician specialty.

RESULTS: Of 1,138,445 beneficiaries in 1994, 1% had a 1994 diagnostic code for AD. In 1994, per capita expenditures for beneficiaries with AD were \$8,676, or 2.1 times those for beneficiaries without AD. Expenditures for those with AD declined after 1994 (\$6,744 in 1995 and \$7,163 in 1996), whereas expenditures rose slightly and steadily for the non-AD population (\$4,617 in 1995 and \$5,250 in 1996). In 1994, 48% of beneficiaries with AD were hospitalized, compared to 23% of beneficiaries without AD; hospitalization was 1.5 times more common with AD in 1995, and it was 1.3 times more common with AD in 1996. Ratios of per capita expenditures for Medicare beneficiaries with AD to per capita expenditures for Medicare beneficiaries without AD were high for services provided by psychiatrists, geriatricians, or neurologists, relative to those provided by internists. Absence of comorbidity is much less common among beneficiaries with AD (9%), compared to those without AD (27%). Compared to beneficiaries without AD, a smaller proportion of beneficiaries with AD had usual sources of care, and this difference increased over the 3-year period. In multivariate regression analysis, age, race, gender, time enrolled in Medicare, comorbidity, and AD diagnosis were independently associated with total 1996 expenditures.

CONCLUSION: AD was independently associated with greater Medicare expenditures from 1994 to 1996, and for AD, expenditures in psychiatry, geriatrics, and neurology are relatively higher than in internal medicine. Medicare beneficiaries with AD may also lack continuity of care. These findings suggest that the role of cognitive function in risk adjustment requires further investigation.

DO SUBSPECIALISTS WORKING OUTSIDE OF THEIR SUBSPECIALTY PROVIDE LESS EFFICIENT INPATIENT CARE THAN PRIMARY CARE PHYSICIANS? *SR Weingarten, LE Lloyd, GD Braunstein, Cedars-Sinai Health System, Los Angeles, CA; Cleveland Health Quality Choice*

Studies have demonstrated that subspecialists (SUBS) provide better quality care when working within their subspecialty as compared with primary care physicians (PCPs). However, many SUBS care for patients outside of their subspecialty. There are no data demonstrating that SUBS treating patients outside of specialty provide worse care than PCPs or SUBS practicing within specialty. We studied 5,112 patients hospitalized with pneumonia (CAP), acute myocardial infarction (AMI), congestive heart failure (CHF), and upper gastrointestinal hemorrhage (UGIB) at 6 hospitals in Cleveland. There were statistically significant increases in severity-adjusted length of stay (LOS) comparing SUBS practicing outside of specialty with those practicing within specialty for patients with CHF (23% increase, 95% CI 15%, 32%), UGIB (22% increase, 95% CI 7%, 39%) and CAP (14% increase, 95% CI 4%, 24%). Severity-adjusted mortality (MORT) was higher for all patients combined when comparing SUBS outside of specialty with those within specialty (mortality odds ratio 1.46, $p < 0.05$). When comparing PCPs with SUBS practicing outside of specialty, LOS was longer for SUBS practicing outside of specialty for CHF (16% increase, 95% CI 8%, 26%), UGIB (15% increase, 95% CI 2%, 30%), CAP (18% increase, 95% CI 9%, 28%). Our data suggest that SUBS caring for patients outside of subspecialty may have longer LOS than SUBS practicing within specialty or PCPs, and may have higher MORT. Our findings may have implications for health care systems where SUBS care for patients outside of their chosen subspecialty.

WHY DON'T PATIENTS WITH ATRIAL FIBRILLATION RECEIVE WARFARIN? *S Weisbord, J Whittle, R Brooks, Medicine, University of Pittsburgh Medical Center; Medicine, VA Pittsburgh Healthcare System, Pittsburgh, PA*

PURPOSE: Though there is widespread agreement that warfarin use decreases the risk of stroke in patients with atrial fibrillation (AF), studies suggest that many such patients do not receive it. Since studies to date have not determined the reasons for apparent underuse, we conducted a study of the reasons primary providers do not use warfarin in patients with AF.

METHODS: We used computerized records to identify all patients with a hospital discharge diagnosis or clinic visit diagnosis of AF between 10/1/95 and 5/31/98 within the Pittsburgh VA Healthcare System (PVAHS), a tertiary care referral VA. We eliminated patients who had not returned since 7/1/97, had received warfarin since 9/30/95, had no electronic medical records (EMR), or died prior to 1/1/99. A medical resident (SW) reviewed all pertinent EMR for remaining patients to: 1) confirm the diagnosis of AF and that warfarin was not being used; and 2) identify definite contraindications or stated reasons for non-anticoagulation. A board-certified internist (JW or RB) reviewed each remaining case (paper and EMR). For all such cases, if any of the three reviewers thought the reason for non-use was not clear, the primary care provider (PCP) was asked to complete a confidential survey regarding warfarin use in that patient.

RESULTS: We identified 1289 AF patients who had been seen since 7/1/97, 844 (65%) of whom, had a warfarin prescription. Of 445 remaining, 19 had died, 43 had inadequate EMR and 43 were on warfarin from elsewhere, leaving 340. Of these, 206 no documented AF (though they may have had a distant history), and 49 had transient AF (absent for greater than 6 months). Of the remaining 85 patients, there were documented reasons for warfarin non-use in 53 (62%). These were divided among patient refusal (15), fall risk (15), bleeding risk or history (15), alcohol abuse (10), or compliance (3). PCP surveys provided reasons for 11 more persons, leaving 21 with no or unusual reasons (e.g., asymptomatic status, history of ulcers without bleeding). Thus, $21/445$ (4.7%) of apparent non-use may be inappropriate.

CONCLUSION: In contrast to studies suggesting that warfarin is underused in AF, we found high rates of use. Moreover, persons not on warfarin had credible reasons for non-use. In the majority of cases either chart review or PCPs justified non-use. These findings should be considered in light of the setting—a university affiliated tertiary referral center. Nonetheless, we believe that the primary reason for our more favorable findings is our use of physician review of the entire chart, and direct communication with PCPs, which allowed for judgement regarding such cases as multiple very brief episodes of AF, lack of reliable follow up or documented patient refusal.

CHANGING CLINICIAN BEHAVIOR: CHARACTERISTICS ASSOCIATED WITH DECREASING ANTIBIOTIC PRESCRIBING IN AMBULATORY PRACTICE. *K White, PH Barrett, Jr, D Price, JH Maselli, R Gonzales, Medicine, University of Colorado Health Sciences Center; Kaiser Permanente of Colorado, Denver, CO*

PURPOSE: In the first year of a patient and clinician educational intervention to decrease antibiotic use for adults with uncomplicated acute bronchitis, antibiotic prescribing decreased from 80% to 45% of incident visits at the full intervention clinic, and this effect was sustained during the subsequent year with minimal reinforcement. The purpose of this study was to evaluate clinician characteristics associated with decreased antibiotic prescribing behavior.

METHODS: We calculated antibiotic prescription rates for uncomplicated acute bronchitis, based on administrative claims data, for individual clinicians at the control, limited and full intervention sites belonging to a group-model HMO. The primary outcome measure was percent change in antibiotic prescription rate,

from baseline winter to year 1 and year 2 winters, for each clinician. Only clinicians with 10 or more visits represented in consecutive years were included in the analysis. We compared antibiotic prescription rates between different clinician types—physicians (MD), nurse practitioners (NP); and different specialties (Family Medicine (FM), Internal Medicine (IM)) during each period. Statistical significance of differences between clinician type and specialties were assessed using Student's t-test.

RESULTS: Sixteen, 14, and 11 clinicians from the control, limited and full intervention sites, respectively, were eligible for analysis. The proportion of clinicians with >10% decrease in antibiotic prescription rates in year 1 were 25%, 42%, and 73%. At the full intervention site in year 1, antibiotic prescription rates decreased a greater amount among NPs ($n = 6$) than MDs ($n = 5$) (−51% vs. −5%, $p = 0.02$). However, by year 2, the percent change in antibiotic prescription rates were similar between NPs (−48%) and MDs (−36%) ($p = 0.46$). During either period, there was no difference between clinician specialty in percent change in antibiotic prescription rates.

CONCLUSION: Physicians may be slower to adopt changes in prescribing practices in response to quality improvement efforts than mid-level providers. Evaluating the impact of such efforts should include a sufficient time interval to detect changes by all providers.

DOCTOR/PATIENT AGREEMENT REGARDING BENEFITS OF CORONARY REVASCULARIZATION. *J Whittle, J Conigliaro, CB Good, KC Goldberg, P Sankar, M Skanderson, M Kelley, Pittsburgh VA Healthcare System; Center for Research on Health Care, University of Pittsburgh*

OBJECTIVE: Little is understood about how patients receive information about the potential benefits and risks of medical procedures. We compared physician and patient assessments of the benefits of coronary revascularization.

METHODS: We prospectively enrolled patients who were offered or received percutaneous angioplasty (PTCA), coronary artery bypass grafting (CABG) or minimally invasive coronary artery bypass (midCAB) at a VAMC or its university affiliate. Patients were asked if they thought the procedure would improve survival, symptoms (e.g., pain control), and functional status. They were also asked what "the doctor" had told them. A cardiologist or cardiology fellow involved in decision making indicated whether the procedure was expected to increase survival, prevent myocardial infarction, or decrease symptoms.

RESULTS: Surveys were completed by 348 patients. In 230 cases, physicians also provided useable surveys. Patients almost always believed the procedure would improve survival, symptoms in general, pain and functional status. (96%, 93.3%, 97.7% and 93.4%, respectively), though they less often reported that the physician told them that (47.3%, 48.6%, 53.5% 60.4%). This did not vary by procedure. Patients were slightly less likely to believe there would be a survival benefit if they had not heard their physician say this than if they had (93.3% vs 98.5%, $p = 0.04$). For symptoms and functional status, the difference was greater, though the vast majority of patients who had not heard their physician say there would be a benefit still expected one (85% vs 99%, $p < 0.001$). Physicians listed improved survival as a benefit for 12% of PTCA patients, 83% of CABG patients and 67% of midCAB patients. The proportion of physicians who said there was a survival benefit was similar among patients who did and did not expect such a benefit for PTCA (12.5% vs 10.6%), CABG (79.7% vs 89.1%) or midCAB (66.7% vs 66.7%), all $p > 0.2$. Physicians more frequently predicted a symptomatic benefit (PTCA 90%, CABG 69.6%, midCAB 89%), but there was still no agreement between physician response and patient report of what the doctor had said.

CONCLUSION: Most persons considering coronary revascularization believe it will increase survival and improve symptoms, though many acknowledge their doctor did not tell them this. Moreover, even when the patients did recall the doctor telling them there was a survival benefit, this did not correspond to what doctor reported. More accurate communication regarding benefits of medical therapies may require assessment of baseline patient understanding. We conjecture that this may help patients to more clearly understand the risks and benefits of these procedures.

RANDOMIZED TRIAL OF CASE-FINDING FOR DEPRESSION IN ELDERLY PATIENTS. *M Whooley, B Stone, K Soghikian, VA Medical Center and University of California, San Francisco; Kaiser Permanente Medical Care Program, Oakland, CA*

PURPOSE: To determine the effect of case-finding for depression on frequency of depression diagnoses, prescriptions for antidepressants, prevalence of depression, and health care utilization in elderly patients.

METHODS: We randomly assigned 13 primary care clinics at Kaiser Permanente Medical Center in Oakland, CA, to intervention (7 clinics) or control (6 clinics) conditions. A total of 2346 patients aged 65 or older who were attending appointments at these clinics completed the 15-item Geriatric Depression Scale (GDS). Primary care physicians in the intervention clinics were notified of their patients' GDS scores. We suggested that participants with severe depressive symptoms (GDS score ≥ 11) be referred to the Psychiatry Department and participants with mild to moderate depressive symptoms (GDS score 6–10) be evaluated and treated by the primary care physician. Depressed (GDS ≥ 6) intervention group participants were also offered a series of organized educational group sessions on coping with depression led by a psychiatric nurse. Primary care physicians in the control clinics were not notified of their patients' GDS scores (usual care). Physician diagnosis of depression, prescriptions for antidepressant medications, and health care utilization during two years of follow-up were determined by review of

administrative databases. Prevalence of depression at two-year follow-up was measured using the GDS.

RESULTS: A total of 331 participants (14%) had depression (GDS ≥ 6) at baseline, including 162 in the intervention group and 169 in the control group. During the two-year follow-up, 56 (35%) intervention participants and 58 (34%) control participants received a physician diagnosis of depression [odds ratio (OR) 1.0, 95% CI, 0.6–1.6; $p = 0.96$]. Prescriptions for antidepressants were received by 59 (36%) intervention participants and 72 (43%) control participants (OR 0.8, 95% CI, 0.5–1.2; $p = 0.25$). Two-year follow-up GDS scores were available for 206 participants (69% of survivors): at that time, 42% (41/97) of the intervention participants and 50% (54/109) of the control participants had depression (OR 0.7, 95% CI, 0.4–1.3; $p = 0.30$). Comparing participants in the intervention and control groups, there were no significant differences in mean $\pm$ SD GDS change scores (-2.4 ± 3.7 vs. -2.1 ± 3.6 ; $p = 0.50$) at the two-year follow-up, nor were there significant differences in mean number of clinic visits (1.8 ± 3.1 vs. 1.6 ± 2.8 ; $p = 0.5$) or mean number of hospitalizations (1.1 ± 1.6 vs. 1.0 ± 1.4 ; $p = 0.8$).

CONCLUSION: Case-finding for depression did not affect patient outcomes at two years. Perhaps case-finding followed by more intensive interventions than those applied in our study will improve the outcomes of patients with depression.

EVALUATION OF EMPLOYEE ATTITUDES ON BUDGET-DRIVEN MEDICAL SERVICE REDUCTIONS. LM Winterbottom, GL Noel, JA Jackson, C Borrego, LS Snodgrass, DC Smith, Portland VA Medical Center, Portland, OR

PURPOSE: To assess whether groups of employees with substantially different clinical roles differ in their perceptions of appropriate reductions of medical services.

METHODS: To meet an expected 10 million dollar budget deficit, the members of each clinical department of a large tertiary VA medical center were presented with detailed data about their costs; they prioritized 25% of these as least essential to good patient care. Two hundred and ninety eight items for potential cost reductions were proposed. All 2000 employees of the medical center were invited to provide feedback on the items.

RESULTS: Twenty eight percent ($n = 565/2000$) of the employees participated in the poster feedback session and indicated support or opposition for items. Nineteen percent of responders were "practitioners" (MDs, PhDs, NPs and PAs), 55% were employees who provide clinical support (RN, secretaries, therapists and other staff in direct contact with patients) and 26% were non-clinical support staff. Employees supported, defined as more than 50% endorsement by all groups, 80% of the proposed cost reduction measures. There was universal support for more than 87% of items within the categories of drug restrictions, improved efficiency, revenue generation, test restriction or new services proposed to save money. One quarter of items of service restriction (items proposing restriction of a disease specific therapy) or drug substitutions were opposed. Opposed items included elimination of palliative services for terminal or chronic conditions, treatment or diagnosis of politically popular diseases, items lacking rigorous evidence based standards, and limited access to users outside our immediate geographic area. The clinical support staff tended to endorse cuts less often ($p < .05$) than practitioners in the categories listed above, as well as for items shifting workload, those implying medicolegal risks, those proposing decreased length of stay and items proposed solely to satisfy the quota of 25% of departmental costs. Service restriction items were opposed more often than all other types of items (28% vs. 15%, $p < .01$).

CONCLUSION: Using financial information, clinicians can identify inefficient practices and non-obligatory services they will sacrifice to maintain adequate funding for essential care, and there is employee consensus for most proposed reductions. Employee consensus might be improved through increased participation in item identification and education on evidence based and community standards.

FREQUENCY OF VISITS TO ALTERNATIVE PROVIDERS. PM Wolsko, RS Phillips, RB Davis, DM Eisenberg, Center for Alternative Medicine Research and Education, Beth Israel Deaconess Medical Center, Boston, MA

PURPOSE: Previous studies have estimated that in the United States in 1997, there were approximately 1.6 times as many visits to alternative practitioners as there were to primary care physicians. Few data are available to understand characteristics of high frequency users of AM providers. Understanding important correlates of high frequency AM provider use is important, particularly as insurers feel increasing pressure to offer new or expanded AM benefits.

METHODS: To determine correlates of frequent visits to AM providers in a random national sample of AM users, we utilized data collected in a nationally representative telephone survey of adults conducted between November 1997 and February 1998. The data were weighted to adjust demographic features for aggregate discrepancies between sample distributions and population distributions provided by the US Census Bureau. Respondents were asked about a list of common medical conditions, their use of conventional medical services, and their use and insurance coverage of 20 AM therapies. We defined high frequency AM use as 5 or more visits annually to a specific type of AM provider and examined use in those who gave information on visits to AM providers in the last year. We used logistic regression to identify factors independently associated with high frequency AM provider use.

RESULTS: Of those eligible, sixty percent completed the interview. Of 2055 respondents, 19.1% gave information regarding visits to AM providers. The median and mean number of visits to a given type of AM provider in the last year were 4 and 9.9, respectively. Mean number of visits to an AM provider varied by the degree of insurance coverage as follows: 12.6 for full, 9.9 for partial, and 8.3 for no coverage. Adjusted odds ratios (AOR) for the likelihood of having made five or more visits to a specific type of AM provider are shown below.

Patient characteristic associated with increased AM provider use	AOR (95% CI)
Full insurance coverage of AM provider	2.61 (1.47, 4.64)
Partial insurance coverage of AM provider	2.23 (1.35, 3.67)
Used AM provider for wellness/prevention	2.42 (1.57, 3.72)
Saw provider for back or neck problems	1.65 (1.07, 2.56)

The effect of full insurance coverage was not significantly different from partial coverage when compared in the model. Having visited a conventional medical doctor 5 or more times in the last 12 months had borderline significance as a correlate of high frequency AM provider use, OR (1.61 (0.99, 2.64)).

CONCLUSION: Lack of insurance coverage appears to be a limiting factor in determining the number of visits made to an AM provider. As insurers extend coverage of AM services, the number of visits made to AM providers might be expected to increase.

CHOICE OF INSURANCE PLANS OFFERING ALTERNATIVE MEDICINE BENEFITS. PM Wolsko, RS Phillips, RB Davis, D Eisenberg, Center for Alternative Medicine Research and Education, Beth Israel Deaconess Medical Center, Boston, MA

PURPOSE: Due to widespread interest in use of alternative medicine (AM), insurers have begun to provide coverage for AM services. However, few data are available describing patient demand for AM insurance benefits.

METHODS: To describe demand for AM insurance benefits in a random sample of AM users, we utilized data collected in a nationally representative telephone survey of adults conducted between November 1997 and February 1998. The data were weighted to adjust demographic features for discrepancies between sample and population distributions provided by the US Census Bureau. Respondents were asked about a list of common medical conditions, use of conventional medical services, and use of 20 AM therapies. A random subset of respondents who had used at least one AM modality or self-prayer in the last year were also asked whether, if presented with a choice of two insurance plans that were otherwise equivalent, they would choose the insurance plan offering AM benefits. We used logistic regression to determine factors associated with being more likely to choose the insurance plan offering AM benefits.

RESULTS: Of those eligible, sixty percent completed the interview. Of 2055 respondents, 54.6% had used at least one AM modality or self-prayer in the last year. Of those, half were surveyed about their likelihood of choosing an otherwise equivalent insurance plan offering AM benefits: 68.9% reported they would be more likely to sign up with that plan, 24.7% were indifferent, and 6.5% would be less likely to sign up with that plan. Adjusted odds ratios (AOR) for being more likely to choose a plan with AM benefits are shown below.

Factor associated with preferring a plan with AM benefits	AOR (95% CI)
Age <65	6.13 (3.21, 11.7)
Live in the West vs. the rest of the country	3.05 (1.62, 5.77)
Used AM for wellness/prevention in last 12 months	2.22 (1.42, 3.48)
Have health insurance vs. no health insurance	2.18 (1.16, 4.11)
Less than very confident in their medical doctors	2.09 (1.25, 3.52)
Education level college or greater	1.82 (1.05, 3.16)
Saw AM provider in the last year	1.64 (1.03, 2.62)

Having a greater number of medical conditions or a greater number of visits to conventional medical doctors in the last year were not associated with choosing an insurance plan with AM benefits.

CONCLUSION: Insurance benefits for AM were desired by the majority of respondents who had used any type of AM in the past year. Extrapolation of these results to the US population conservatively estimates that there are at least 75 million Americans for whom the presence or absence of AM insurance benefits may impact the selection of their insurer.

EXPERIENCES OF DIABETIC PATIENTS UNDER MEDICAID MANAGED CARE. RJ Womeody, JE Bailey, M Graney, DL Van Brunt, Internal Medicine, Preventive Medicine, University of Tennessee, Memphis, TN

PURPOSE: This descriptive study of diabetic enrollees in an academic Medicaid managed care organization (MCO) sought to determine satisfaction with accessibility and quality of care and exposure to recommended health services for diabetes before and after program initiation.

METHODS: The target population included individuals who were Medicaid eligible prior to 1994, when all patients were required to enroll in a Medicaid managed care system (TennCare). Eighty-eight cohort members met inclusion criteria, and 62 enrollees completed a telephone survey conducted by trained interviewers. Administrative claims data and chart review data were collected for the same enrollees.

RESULTS: The respondents were predominately African American (90%), females (75%), non-smokers (86%), with BMIs of 33.5 ± 9.4 , 55.7 ± 8.9 years of age, completed 8.9 ± 3.0 years of education, permanently disabled (63%), with annual incomes of less than 10k dollars (58%). Their mean age when diagnosed with diabetes was 39.8 ± 11.3 years, and most (65%) were currently prescribed insulin. During the implementation of TennCare most patients choose their MCO, and kept the same physician (63%). Twenty-eight percent rated their current healthcare under TennCare as excellent, 10% very good, 40% good, 14% fair, and 5% rated their current healthcare as poor. Thirty-eight percent felt the quality of their healthcare before TennCare was the same; 22% felt it was better or much better; and 40% felt it was worse or much worse. Respondents reported that their blood sugars were controlled most of the time now compared to before TennCare (45% vs. 29%, $p = .05$), and that they were more likely to check finger stick blood sugars now than before TennCare (74% vs. 43%, $p = .05$). They were less likely to check urine dipsticks (10% vs. 19%, $p = .05$). Patients also reported out of pocket expenses for transportation and medications prior to TennCare to be the same (35%); better or much better (10%); or worse or much worse (55%). More reported that they did not have a test performed because of lack of approval under TennCare (63% vs. 23%, $p = .06$), and seeing a diabetes specialist at least once a year (45% vs. 33%, $p = .05$), but there were no significant differences in the percentage of patients who reported receiving detailed information about diabetes (92% vs. 75%), or receiving dietary information (93% vs. 78%).

CONCLUSION: After the initiation of Medicaid managed care patients were more likely to report home glucose monitoring, and control of their blood sugars most of the time. The majority was satisfied with their MCO, and reported fewer out of pocket expenses. From the patient perspective enrollment in Medicaid managed care had several positive outcomes, and appeared to result in improved access to appropriate diabetic care.

PRESCRIBING PROTEASE INHIBITORS FOR HIV: DOES PATIENT ADHERENCE MATTER?

MD Wong, MF Shopiro, PD Cleary, IB Wilson, BE Landon, WE Cunningham, RM Andersen, SA Bozzette, H Liu, NS Wenger, The HIV Cost and Services Utilization Study (HCSUS) Consortium, UCLA, Los Angeles, CA; Harvard Medical School, Boston, MA; RAND, Santa Monica; UCSD and the VA Health System, San Diego, CA

PURPOSE: Rates of receipt of protease inhibitors in HIV disease vary substantially by race, gender and other patient characteristics. There is concern that physicians may withhold therapy based on their (not necessarily accurate) assessment of patients' likely adherence. We surveyed physicians to assess whether prescribing protease inhibitors to patients is affected by perceived patient ability to adhere to treatment regimens.

METHODS: Health care providers of a nationally representative sample of 2864 HIV-infected individuals in the U.S. were surveyed in 1998. Using a 5-point Likert scale, they were asked whether they agreed that a patient's likelihood of adherence is a very important factor in their decision to prescribe protease inhibitors and whether they prescribe protease inhibitors only if a patient is likely to be adherent with the regimen. We examined the association between providers' attitude toward prescribing protease inhibitors and their knowledge of HIV treatment and personal characteristics. The response rate from the provider survey was 72%.

RESULTS: Among 411 providers, 89% agreed that adherence is an important factor when deciding to prescribe protease inhibitors and 69% percent agreed that they prescribe protease inhibitors only if a patient is likely to be adherent. In a stepwise logistic regression analysis, providers with better knowledge of HIV treatment and general internists (as compared with infectious disease specialists) were more likely to consider their prospective assessments of patient adherence in their decision to prescribe protease inhibitors (OR = 4.18 $p = 0.04$ and OR = 1.56 $p = 0.06$, respectively). Factors not associated with consideration of patient adherence were physician's gender, ethnicity, years in practice, sexual preference, preference not to treat intravenous drug users, and the belief that many of their patients cannot adhere.

CONCLUSION: Among physicians caring for a nationally representative sample of HIV-infected individuals in the U.S., most said they would prescribe protease inhibitors only if the patient is likely to be adherent. This finding raises concern because existing evidence suggests that physicians are poor judges of patients' adherence. Efforts to increase delivery of protease inhibitors to undertreated groups need to address physician perceptions of patients' adherence.

DO PHYSICIANS DECIDE NOT TO OFFER THEIR PATIENTS USEFUL SERVICES THAT ARE NOT COVERED BY HEALTH PLANS? A NATIONAL SURVEY. **MK Wynia, DS Cummins, JB VanGeest, IB Wilson, Institute for Ethics, American Medical Association, Chicago, IL; Department of Clinical Care Research, New England Medical Center, Boston, MA**

PURPOSE: Little is known about how often physicians (MDs) are faced with the circumstance of a patient's insurance not covering a useful service, or how MDs respond to this dilemma. We examined the frequency of, and some factors related to, one potential response—not offering such care to patients.

METHODS: The sample included practicing MDs selected randomly from the AMA Masterfile and surveyed by mail in 1998 (64% response rate). Participants were asked how frequently in the last year each had "decided not to offer a useful service to a patient because of health plan coverage rules." For analyses, ordinal responses were dichotomized (never or rarely vs. sometimes, often, or very often). Multiple logistic regression was used to determine correlates of not offering useful services. Independent variables included MD's demographic and practice characteristics, attitudes about utilization review, satisfaction with practice, and MDs'

reports of how often patients asked that they mislead payors to get coverage for uncovered services.

RESULTS: Respondents were 36% primary care physicians, 13% internal medicine subspecialists and 23% surgeons. They had been in practice for a mean of 17 years, 20% were women, and 40% were foreign born. Overall, 31% of physicians reported not offering useful but uncovered services to patients at least sometimes over the last year. Those who reported this were less likely to be satisfied with medical practice (OR 0.58, CI 0.38–0.86) and more likely to report recent patient requests to mislead their insurers to obtain coverage for uncovered services (OR 2.26, CI 1.52–3.37). Other weaker, but statistically significant, correlates were having more than 25% of one's patients under Medicaid (OR 1.7, CI 1.07–2.89) and being American born (OR 1.5, CI 1.02–2.34). Non-significant variables included age, gender, specialty, and practice characteristics such as practice setting, primary method of payment, number of managed care contracts, and proportion of income from managed care. Of those reporting not offering useful services, 76% reported doing so more often in 1998 than 5 years previously.

CONCLUSION: While not offering useful services to patients is an undesirable response to the dilemma of such services not being covered, it is an increasingly common one. Offering such services may prompt patients to request that MDs manipulate reimbursement rules, which in turn may motivate MDs not to discuss useful but uncovered services in the first place. In a cross-sectional analysis the direction of causation cannot be known, but such "no-win" scenarios for MDs may contribute to reduced MD satisfaction.

PATIENT TRUST OF PHYSICIANS AT OUTPATIENT CLINICS. **S Yamashiro, S Emura, M Shinozuka, M Yamada, Y Oda, H Onishi, K Yamamoto, N Miyoshi, K Ishii, T Mouri, S Koizumi, General Medicine, Saga Medical School, Saga; Internal Medicine, Yomise Clinic, Tokyo, Japan**

PURPOSE: To investigate the correlation between patient satisfaction with care and patient trust of physicians at two sites: A university hospital outpatient (general internal medicine) clinic and a private primary-care clinic.

METHODS: A pilot study of patient trust was conducted in November and December of 1999. Adult patients who were cared for by the same physicians for more than 3 months and who had at least 4 visits during this time, were selected for this study. The patient trust of physician ratings were obtained by self-administered questionnaires, the Trust in Physician Scale (modified by the Stanford Trust Study Group). In addition to patient trust, we obtained patients' characteristics, the lengths of the patient-physician relationships, the reasons for the patients' choice of physicians, the satisfaction with care, and a general trust scale (Yamagishi, Japan). 275 patients used the questionnaire to rate 11 different physicians during their visits.

RESULTS: The Trust in Physician scale (5 Likert scales: 1 = totally disagree to 5 = totally agree) were transformed to a 0 to 100 scale, and showed high internal consistency (Cronbach's $f_c = .82$) for Japanese patients. Mean scores of male and female were 86.3 (SD: 10.3) and 85.9 (SD: 10.5) respectively. The trust scores were not statistically associated with gender, age, education levels and the lengths of the relationships. However, the trust scores were strongly associated with the reasons for the patients' choices, the general trust scores, and the satisfaction with care after adjusted by patient characteristics.

CONCLUSION: The patient trust was a useful concept to assess the relationship between patient and physician, especially the satisfaction with care. The patient trust, one of the health care seeking behaviors, was also associated with the societal trust in general in our study. However, we need further investigations such as the relationship between patient trust and compliance and consequence of medical care.

FLOW SHEET AND FEEDBACK: AN INTERVENTION TO IMPROVE COMPLIANCE WITH AMERICAN DIABETES ASSOCIATION GUIDELINES. **N Yaqub, TW Gress, A Habte-Cheno, LJ Goebel, Internal Medicine, Marshall University; Internal Medicine, Veterans Administration Hospital, Huntington, WV**

PURPOSE: We hypothesize that the addition of a diabetic flow sheet along with physician feedback on compliance with American Diabetes Association (ADA) guidelines improves the quality of care for diabetic patients in an ambulatory care clinic.

METHODS: We conducted two cross-sectional retrospective chart reviews of patients with type 2 diabetes mellitus. Group 1 comprised 77 randomly selected patients from 1995, and Group 2 comprised 151 randomly selected patients from 1999. In 1996, we collated information from Group 1 and presented a written summary of results to physicians regarding compliance with ADA guidelines. One year prior to review of Group 2, we introduced a flow sheet of ADA guidelines in diabetic patients' charts.

RESULTS: Groups 1 and 2 were similar in age (61.8 vs. 61.4; $p = 0.84$), sex (57 vs. 45% male; $p = 0.08$), years since diagnosis of diabetes (median category 5–10 years both groups; $p = 0.59$), and presence of hypertension (57 vs. 65%; $p = 0.27$). In comparative analysis, we found Group 2 to have significant increases in the use of angiotensin converting enzyme inhibitors (31 vs. 58%; $p = 0.001$), documentation of glucose self-monitoring (63 vs. 80%; $p = 0.007$), documentation of physician goals of tight blood sugar control (79 vs. 96%; $p = 0.001$), podiatry referrals (10 vs. 21%, $p = 0.03$), and foot exams (40 vs 68%; $p = 0.001$). We found decreases in the mean values for glycohemoglobin (10.0 vs 9.2%; $p = 0.03$), fasting serum glucose (176 vs 134 mg/dl; $p = 0.001$), random nonfasting glucose (210 vs 187 mg/dl; $p = 0.03$), total cholesterol (260 vs 209 mg/dl; $p = 0.001$), and

low density lipoprotein cholesterol (140 vs 120 mg/dl; $p = 0.02$). Although urinalysis testing decreased (66 vs 38%; $p = 0.001$), evaluation for microalbuminuria increased (9 vs 30%; $p = 0.001$), as did 24-hour urine collection (4 vs 13%; $p = 0.02$).

CONCLUSION: A relatively simple intervention of a flow sheet and written feedback on compliance with ADA guidelines is associated with an increased level of implementation of these guidelines and improvement in several measures of diabetic care in an ambulatory care clinic.

CONSUMER SATISFACTION OF CARE IN ELDERLY PATIENTS IN CAPITATED MANAGED CARE SETTINGS. M Yu, AF Brown, PR Gutierrez, EB Keeler, MF Shapiro, CM Mangione, Dept of Medicine, UCLA School of Medicine, Los Angeles, CA; Health Program, RAND, Santa Monica, CA

PURPOSE: To determine whether there are differences in the satisfaction of care in an older, sicker subset of patients enrolled in Medicare managed care (MC) plans.

METHODS: We studied the relationship of demographic and clinical characteristics to satisfaction with care for 1061 randomly sampled Medicare beneficiaries cared for by 27 medical groups who contract with one large for-profit HMO in Los Angeles. Data collection included the Consumer Assessment of Health Plans Study (CAHPS™) 1.0 questionnaire, the SF-12, medical conditions, and demographic characteristics. The CAHPS 1.0 measures the following domains of satisfaction: overall ratings of the health plan, specialists, personal doctor, and quality of care; the ease of getting needed care, finding a primary provider, seeing a specialist; and communication with the health providers. We compared a subset of patients, 75 or older, with 2 or more chronic medical conditions to the remainder of the sample for each of the above 8 CAHPS sub-scales. Multivariate analysis compared selected CAHPS sub-scales. We also surveyed physician groups (response rate 96%) to adjust for the independent effect of characteristics of MC physician groups that may influence selected CAHPS scales. Independent variables from the physician groups adjusted for in multivariate models included: linkage of the PCP's compensation to volume of specialty care; use of a bonus system in the PCP's compensation; experience with capitation; amount of specialty care available within the group; and whether the dominant form of PCP compensation was capitation. All regression analyses were also adjusted for the independent effects of gender, race, and Medicaid status.

RESULTS: In the older/sicker subset ($N = 261$), the mean age was 80 ± 4 years and the mean number of comorbidities was 2.9 ± 1.2 , compared to 73 ± 5 years and 1.4 ± 1.2 comorbidities for the remainder of the sample. After adjustment for demographic and provider group characteristics, we found that older patients with more chronic medical conditions did not have significantly different CAHPS sub-scales scores from a younger and healthier subset with one exception: older, sicker patients reported more difficulty finding a personal doctor ($OR = 0.6$, $p = 0.03$).

CONCLUSION: Despite the current economic and administrative constraints of Medicare managed care plans, older patients with multiple chronic illnesses appear to have similar levels of satisfaction as younger, healthier Medicare managed care beneficiaries.

THE INFLUENCE OF HOSPITAL OWNERSHIP ON USE OF INSTITUTIONAL POST-ACUTE CARE AND OUTCOME AFTER STROKE. DS Zingmond, SL Eftner, DM Carlisle, Division of General Internal Medicine, UCLA School of Medicine, Los Angeles, CA

PURPOSE: Patients hospitalized for acute stroke have prolonged care needs that conflict with organizational incentives promoting shorter lengths of stay and shifting of care into subacute care settings. The purpose of this study is to identify whether hospital ownership and profit-status influences choice of post-acute care and patient outcome as measured by hospital readmission and total mortality.

METHODS: Longitudinal analysis of Medicare patients aged 65 years and older with principle diagnosis of stroke (DRG 14) discharged from non-Federal acute care hospitals in California between 1995 and 1996 with identification of deaths through linkage to state death registry. Logistic and Cox proportional hazard models were used to assess influence of hospital ownership-profit status on patient discharge location, readmission, and mortality. Linear regression models assessed acute length of stay. Models controlled for patient demographics, previous admissions, disease severity, stroke type, and other hospital factors. Analyses were clustered by hospital to account for intra-hospital correlation.

RESULTS: A total of 55,103 discharges met the selection criteria with 73% patients cared for in private not-for-profit hospitals (NFP), 15% in private for-profit hospitals (FP), and 11% in public hospitals. Most patients were discharged to skilled nursing (SNF) (39%) or rehab (13%) care. Ten percent of patients (5,598) died in the hospital. Overall, 33% (17,949) died and 48% (26,494) were readmitted during the study period. Average acute length of stay was 4.86 days. In multivariate logistic models, patients in FP hospitals had longer acute length of stay ($b = 0.26$ days, $p = 0.009$) were more likely discharged to rehab care (Odds Ratio (OR): 1.53, 95% Confidence Interval (CI): 1.10 to 2.12) and less likely discharged to SNF (OR: 0.72, 95% CI: 0.59 to 0.88) or home health care (OR: 0.70, 95% CI 0.51 to 0.95) as compared to NFP hospitals. In multivariate Cox models stratified to post-acute care discharge, patients discharged from FP hospitals were more likely to be readmitted (Hazard Ratio (HR) 1.12, 95% CI: 1.05 to 1.19) or to be readmitted or die outside of the hospital (HR 1.08, 95% CI 1.02 to 1.14) versus NFP hospitals. Total mortality was not different in the stratified model.

CONCLUSION: Medicare patients discharged to post-acute care from FP hospitals have longer acute length of stay, are more likely to use rehab care, and are more likely to be readmitted after discharge. Alternative measures of outcome indicate discharges from FP hospitals to post-acute care appear sicker, suggesting potentially inappropriate overuse of rehab services.

MEDICAL HUMANITIES/CONCEPTUAL ETHICS

HUMANITIES AND MEDICINE: A CURRICULUM FOR BIO-ETHICS AND PROFESSIONALISM. J Basmajian, J Boal, D Muller, Department of Medicine, Mount Sinai Medical Center, New York, NY

PURPOSE: It is generally accepted that an understanding of bio-ethics is crucial to delivering good patient care. Despite its importance in medical education, bio-ethics is either not taught at all or not taught well in many residency training programs. Reasons include the lack of clinical applicability of some teaching methodologies and the fact that ethical concerns or disputes are often viewed as having no discrete answer or solution. Literature, film and theater create an atmosphere that provides an intimate, fascinating and safe glimpse of situations that would otherwise be impossible to replicate. They allow residents to become deeply invested in the characters and outcomes without confusing the issues with medical jargon and the defensive posturing of case presentations. The goal of our project is to create a curriculum in bio-ethics and professionalism by using the arts as a means for small group discussions of complex and controversial topics.

METHODS: The framework consists of didactic lectures, small group discussions and theater/film events that highlight various ethical issues encountered in the practice of medicine. Each discussion or media event is designed to enhance the educational development of physicians in training by augmenting the individual's knowledge, skills and attitudes in medical ethics and professionalism. For example, various short stories, poems or plays are chosen to evoke reflection on topics such as death/dying, patient suffering, patient-doctor relationships and altruism. The unique feature of the program consists of monthly events for residents to experience plays, films or public lectures that also raise awareness and knowledge that is applied to ethics and humanism. Each event is followed by a discussion with faculty and a Likert-type scale survey to assess the educational value of the experience. The housestaff were asked four questions about the event to evaluate the specific learning objectives, the influence on one's attitude toward doctoring, the sense of humanism and the relevance to clinical training.

RESULTS: Surveys from three events were obtained for a total of 25 responses. Responders assigned a weight of one point to strongly disagree up to a weight of five points to strongly agree to each question. The average response for each question was as follows: learning potential 4.32, career reflection 3.77, humanism 4.0 and relevance to training 4.75.

CONCLUSION: Our survey results support that residents view a curriculum in bio-ethics and professionalism as a highly relevant component of their training. With the help of the arts, educators can further augment post graduate medical education by enabling residents to attain skills and attitudes in bio-ethics and professionalism.

"TURFING" PROFESSIONALISM: THE PATIENT'S PERSPECTIVE. CV Caldicott, RM Frankel, Internal Medicine, SUNY Upstate Medical University, Syracuse, NY; Primary Care, Strong Health/Highland Hospital, Rochester, NY

PURPOSE: Medicine residents who receive "turfs" are known to feel abused and overpowered by other doctors who elect not to care for these patients. This pilot study examines whether the patients caught in such transactions of social triage and rejection can discern that they are perceived as "turfs." We further wondered if the unprofessional conduct of their physicians adversely affects their medical care.

METHODS: Study subjects included 1) transfers to medicine teams in University Hospital (UH) either from an outside institution or from another specialty service within UH, and 2) patients evaluated in the Emergency Department by a specialty service but ultimately admitted to medicine. Audiotaped interviews were conducted to elicit patient experiences and perspectives on care. Major themes were identified by reviewing the tapes. In brief audiotaped interviews, medicine house-staff caring for these patients explored feelings and experiences regarding appropriateness of transfer or triage. Based on their comments, and applying previously described criteria, the patient interviews were assigned either "appropriate" or "inappropriate" (i.e., "turf") status. Where the housestaff were ambivalent in their feelings about the patient, an "equivocal" status was assigned.

RESULTS: Ten of 18 patients were considered appropriate by their housestaff, five inappropriate, and three equivocal. The most common themes among the appropriate patients were a general desire for more communication between doctors and patients, satisfaction with their medical care and hospital experience at UH, and complaints about their care at the outside facility. Among the "turfed" patients, communication was also a theme, but it was expressed as a complaint about communication both between doctors and patients and among doctors themselves: "I got a different story from every doctor I talked to. There was no

communication between doctors." In another theme, "turfs" detected "turf battles" between doctors of different specialties involved in their care even though the patients were not openly informed of such: "There was an obvious dispute between two groups of doctors, a turf battle. [It was] very negative." In the "turfed" patients' last theme, they questioned doctors' motivations, specifically financial remuneration and insurance status: "[not having insurance] was probably half the reason she didn't get taken care of like she should."

CONCLUSIONS: "Turfed" patients understood their unfortunate status and perceived unprofessional motivations and conflicts among their doctors. Although less satisfied with their care than appropriately transferred controls, actual consequences of "turfling" on outcomes of care should be investigated in future studies.

BARRIERS TO THE INTRODUCTION OF CURRICULUM OF LITERATURE AND MEDICINE FOR RESIDENTS. PA Cohen, CU Viscoli, AH Fortin VI, Department of Medicine, The Cambridge Hospital, Cambridge, MA; Department of Internal Medicine, Yale School of Medicine, New Haven, CT

PURPOSE: To assess the barriers on the inpatient wards to introducing a structured, case-based literature and medicine curriculum focused on the humanistic aspects of patient care in a primary care internal medicine residency program.

METHODS: Seven attending physicians, twenty-six residents and five medical students at two community hospitals completed anonymous questionnaires before and after participating in the literature and medicine curriculum.

RESULTS: Before and after questionnaires were collected from all seven attending physicians and 28 (90%) of the 31 residents and students. In open-ended responses all attending physicians described the lack of time on the inpatient wards as the major barrier to the curriculum. Lack of time on the wards was also the most common barrier cited by residents and students (36% of questionnaires). The second most common barrier described by attending physicians, residents and students was lack of participation and/or negative attitudes of a minority of residents. Attending physicians had a high level of interest in teaching the humanities and medicine curricula both before and after participation (mean 4.0 [SD 1.1] on a Likert scale of 0 [no interest] to 5 [very interested]). Residents and students had high levels of interest in learning more about the humanities and medicine (mean 3.7 [SD 1.1] on the same scale). However, most participants both before and after were unsure of the ability of humanities and medicine curricula to improve patient care.

CONCLUSION: Given the high level of interest in the humanities and medicine among attending physicians, residents and medical students, educational resources should be utilized to overcome the identified barriers including lack of time, negative attitudes of a minority of residents and uncertainty regarding clinical relevance. Suggestions for overcoming the barriers include: (1) development of brief literature and medicine curricula for focused use in both the inpatient and outpatient settings; (2) further research of attitudes of residents towards the introduction of humanities and medicine curricula during residency training; and (3) further research of the ability of humanities and medicine curricula to develop humanistic qualities and attitudes during residency training.

PLACEBO DIAGNOSIS: HISTORICAL ANALYSIS OF ITS USE IN ONE MEDICAL PRACTICE, 1900-1920. CW Crenner, History and Philosophy of Medicine; Department of Medicine, University of Kansas School of Medicine, Kansas City, KS

PURPOSE: Diagnostic procedures have several potential secondary uses in medical practice beyond the identification of disease. They may, for example, provide reassurance to a patient or physician. An extreme form of such secondary use is placebo diagnosis for reassurance. An influential physician, Dr. Richard C. Cabot, in the early twentieth century explored the use of placebo diagnostic procedures. METHODS: Random number sampling of patient records from the private practice of Richard C. Cabot, 1900 to 1920 demonstrated instances of placebo diagnosis and treatment. A follow-up examination of individual records, including letters from Cabot's patients, revealed detailed information about these practices. Historical context was provided by review of early twentieth-century medical literature on the use of diagnosis and placebo, and by review of Cabot's own published writings on the subjects of diagnosis, placebo, and relationships with patients.

RESULTS: Cabot experimented with the use of a diagnostic procedure as a placebo. For example, he collected sputum for staining for acid-fast bacilli, and then reassured a patient about the absence of tuberculosis without examining the sputum. Cabot's use of placebo diagnosis coincided with his early experimentation with the self-conscious use of placebo treatments for his patients. These practices were consistent with Cabot's stated concern about the relative over-valuation of laboratory diagnosis and specific therapeutics, and about the influence of these biases on professional identity and autonomy. Cabot's published views on placebos, however, soon hardened in response to a different set of concerns about professional obligations and image. If the public recognized the wide use of placebos, Cabot soon warned, they would denounce the practice. If they did not recognize it, they would increasingly come to believe inappropriately that "every symptom needs a drug." Cabot subsequently abandoned his own use of placebo treatments and diagnostic procedures, consistent with his changing views.

CONCLUSION: For a thoughtful and innovative practitioner like Richard Cabot there was flexibility in the secondary use of novel diagnostic procedures flowing from the laboratory in the early twentieth century. Cabot's exploration of placebo diagnosis, for example, waned seemingly in response to changing concerns about appropriate ideals for professional behavior. Concepts of ideal professional obliga-

tions, identity and autonomy seem to have deeply influenced his use of new diagnostic procedures.

THE HISTORY OF EVIDENCE-BASED MEDICINE IN AMERICAN MEDICAL EDUCATION. ML Green, Internal Medicine, Yale University School of Medicine, New Haven, CT

PURPOSE: To trace the evolution of EBM in the history of medical education.

METHODS: Historical analysis of primary and secondary documents.

RESULTS: EBM educators can trace their ancestry to a reform of the American system in the late 19th century, which saw the rise of university-based medical schools, raising of admissions standards, expansion of curricula, and the embrace of "progressive education." The latter recognized the importance of experiential and life-long learning and the stifling effects of deference to authority. These notions found expression in the use of laboratories and clerkships to supplement lectures. One of the many forces driving this reform was the influence of Europe. German scientists pioneered experimental laboratory science, while the French systematically documented clinical observations. In 1836, Dr. Pierre Louis's observational study challenged the accepted efficacy of blood-letting. His empirical approach established clinical research as a pathway to the understanding of disease. From this point on, medical knowledge would evolve and expand. American educators recognized physicians' need to continually update their knowledge. Medical students must graduate with the thirst, critical skills, and "informatics" for lifelong learning. Introduced by Osler in 1875, the journal club became an important venue for information exchange. Its primary purpose was to help physicians keep abreast of the emerging world's literature. In the early 20th century, therapies for dramatic acute infections were developed in the laboratory without the need for the persuasion of a p-value. Later, extending Dr. Ernest Codman's "end result" idea, investigators turned to patient-based research to study the subtler statistical outcomes of chronic diseases. Practitioners needed a framework to appraise and apply this new literature. Critical appraisal eventually found its way into medical education curricula, often in journal clubs. Now students in this forum not only "kept up" with articles, but also critically scrutinized them. Although educators also intended journal clubs to impact on clinical practice, patients were rarely discussed. In the last 10 years, educators folded critical appraisal into the larger process of evidence-based medicine, which begins and ends with an individual patient. This required new technologies to serve practitioners on the front lines of patient care. In addition to informing clinical decisions, EBM represented an alternative approach to life-long learning in which the patient encounter cues the acquisition of new knowledge.

CONCLUSION: EBM's experiential, self-directed educational approach extends the century-old "progressive education" mandate to include recent advances in clinical epidemiology, medical informatics, and decision science. Historical analyses may inform the current EBM debate.

RACIAL INTEGRATION AND AN OASIS OF BLACK HOSPITALS: LINCOLN HOSPITAL OF DURHAM, NC, 1901-1976. P Reynolds, Medicine, Johns Hopkins University, Baltimore, MD

PURPOSE: W. Montague Cobb, renowned anthropologist and anatomist, surveyed the landscape of black hospitals after World War II publishing his findings in two landmark articles in 1947 and 1948. In his judgement the quality of patient care services and teaching faculty in predominantly all-black medical schools did not equal that found in all-white institutions. His analysis of Lincoln Hospital, built in 1901, however, prompted him to label it "an oasis of black hospitals." This paper seeks to explore the basis of Cobb's praise and the extent to which policies of race influenced this vocal and prolific physician-scholar.

METHODS: Historical sources include archival records of Lincoln Hospital, archival and personal papers of leading individuals, records of the Durham Regional Health Planning Council, and oral history interviews with physicians, nurses, and Durham citizens.

RESULTS: Lincoln Hospital opened in 1901 with funds provided by Washington Duke, founder of the Duke Tobacco Company. It expanded in 1925 again with resources provided by the Duke family with a matching 1:1 gift of \$75,000. That same year Lincoln Hospital was accredited as one of 14 black hospitals in the country approved for internship training. Close examination of racial policies are instructive. The medical staff was composed of African-American and white physicians from Durham with both groups operating on and caring for patients, and both involved in teaching interns and residents, nurses and allied health professionals. The chiefs of service for nearly every area were white because of the requirement of board certification for accreditation, and thus, they participated in monthly staff meetings and continuing education programs held at Lincoln Hospital. In 1949 the By-Laws of Lincoln Hospital were changed to reflect a policy of open integration for patients and the medical staff. In addition, involvement of Duke Hospital surgery residents and faculty in the teaching program at Lincoln Hospital beginning in the 1930s, and the active engagement of Duke faculty and family members as trustees of the hospital helped strengthen the facility and its teaching programs for the next four decades. The policy and practice of racial integration of the Lincoln Hospital medical staff continued until the hospital closed in 1976.

CONCLUSION: Cobb's assessment of Lincoln Hospital as an oasis results on its modern facilities when compared to other predominantly all-black hospitals, its residency, nursing and allied health training programs, the high quality of African-American physicians and surgeons at Lincoln Hospital, and its teaching fac-

ulty and clinicians who included many of the preeminent white physicians and academic scholars in Durham and at Duke Hospital.

QUALITATIVE RESEARCH

AN EVIDENCE-BASED REVIEW OF THE INDEPENDENT CONTRIBUTION OF BEHAVIORAL INTERVENTIONS FOR HYPERTENSION: DOES ADDING MORE INTERVENTIONS IMPROVE HYPERTENSION CONTROL? L Boulware, GL Daumit, KD Frick, C Minkovitz, RS Lawrence, NR Powe, Johns Hopkins School of Public Health, Baltimore, MD

PURPOSE: Behavioral interventions such as counseling, training courses, and self-monitoring of blood pressure (BP) are commonly used as alternative or adjunct measures in the treatment of hypertension. Whether employing additional types of behavioral interventions contributes to improved outcomes is unknown. **METHODS:** We systematically reviewed the literature to assess the independent and additive effects of behavioral counseling, self-monitoring of BP, and structured training courses on BP control. We searched the medical and psychology literature from 1970-99 for peer-reviewed articles reporting on interventions for hypertension management, excluding studies reporting solely on efficacy of medical therapy. Data were abstracted on content and quality of studies. Our analysis considered studies containing only the following interventions: counseling, self-monitoring of BP, training courses, counseling and training courses, counseling and self-monitoring of BP, or counseling, self-monitoring of BP, and training courses. We calculated pooled estimates across studies of mean change in systolic blood pressure (SBP), mean change in diastolic blood pressure (DBP), and proportion of subjects with blood pressure control (BPC, ranging in studies from 140-160/90-95 mmHg).

RESULTS: Of 232 articles identified, 104 met inclusion criteria. Of these, only 12 articles contained interventions focusing exclusively on counseling, self-monitoring of BP, and/or training courses. For articles reporting on counseling interventions, 5 studies ($n = 509$) reported change in DBP (-4.99 mmHg [95% CI -5.93 , -4.05]), 4 studies ($n = 418$) reported change in SBP (-5.72 mmHg [95% CI -7.47 , -3.97]), and 4 studies ($n = 351$) reported proportion of subjects with BPC (0.53 [95% CI 0.37, 0.57]). One article with monitoring alone ($n = 34$) reported on DBP (-8.9 mmHg [95% CI -12.56 , -5.24]). One article with training alone ($n = 333$) reported on DBP (-0.8 mmHg [95% CI -1.78 , 0.18]) and SBP (-0.8 mmHg [95% CI -2.37 , 0.77]). One article with counseling plus training ($n = 63$) reported on BPC (0.95 [95% CI 0.87, 0.99]). One article with counseling plus training plus monitoring ($n = 86$) reported on DBP (-6.0 mmHg [95% CI -8.36 , -3.64]) and SBP (-8.0 mmHg [95% CI -11.64 , -4.36]).

CONCLUSION: Existing evidence over 3 decades suggests that combining counseling and training courses has a significantly larger effect on BP control than counseling alone. Although pooled point estimates suggest that the combination of self-monitoring of BP, training and counseling may have a larger effect on lowering DBP and SBP than counseling alone, overlapping confidence intervals suggest no consistent improvement in BP measures when adding behavioral therapies.

A JOURNEY INTO LIGHT AND DARKNESS: USING REFLECTION TO ENHANCE SELF-AWARENESS DURING RESIDENCY TRAINING. DW Brady, G Corbie-Smith, WT Branch, Jr, Division of General Medicine, Emory University, Atlanta, GA

PURPOSE: We reviewed narratives written over three years of residency training by our initial cohort of primary care residents to assess them for common themes and evidence of self-reflection.

METHODS: Four times, twice as interns and then once each of the next two years, the residents met with 2 faculty members to write narratives and share their stories, voluntarily, with the group. The residents were instructed at the beginning of each session to spend thirty minutes writing about any event in their lives that they viewed as important and that had influenced who and where they are today. Using the constant comparative method, all three authors reviewed the narratives—first as a whole, then each one was reviewed and compared to prior and subsequent narratives—to identify emerging themes and concepts. A series of team meetings were used to refine the meaning of each content area (early PGY1, late PGY1, middle PGY2, late PGY3) discuss alternative interpretations, and reach consensus on common themes.

RESULTS: Early PGY1 narratives focused on the interns' experiences with patients and/or their illnesses and shared a common theme—a search for professional identity and values. By the end of the PGY1 year, the stories took a more varied path. Some continued clarifying their professional identities and retained a connection to the idealism with which they began their training. Others, however, began to describe a darker side of residency training, full of fear and hopelessness. PGY2 trainees described a sense of despair and detachment and seemed lost in the depths of very agonizing experiences; the idealism of the internship was gone. The PGY3 narratives reflected on the experience of residency and how it influenced the trainees' perspectives, with half of the papers centered around patient encounters and the other half reflecting on the overall training experience; from the PGY3 stories emerged common themes of hope and reconciliation.

CONCLUSION: The residents were able to come to know each other at a level not often achieved by casual discourse. Residency was a journey that began with an idealistic search for professional identity and values, often fell into a darkness where these came into question, and then, for most, returned "into the light" where their values and identities were affirmed by their experiences. Residency is a difficult time given its temporal, mental, and emotional demands. Written narratives may be one method to encourage residents to process their experiences in order to make them better physicians, more compassionate caregivers, and more fully developed human beings.

ELDERLY DRIVERS: MAKING THE MOST OF MEDICAL EVALUATIONS FOLLOWING MOTOR VEHICLE ACCIDENTS. CK Brands.

LEARNING OBJECTIVES: 1) Highlight the medical literature on elderly drivers and motor vehicle accidents. 2) Review the medical evaluation of elderly drivers involved in motor vehicle accidents.

CASE PRESENTATION: A sixty six year old white male is brought to the office by his wife who is concerned about his poor concentration. The patient has also noted difficulty with concentration during the last two weeks. Twenty days ago he was found by the local police to have left the scene of an accident he caused and was ordered to appear before the municipal judge. The judge ordered a complete medical evaluation prior to consideration of reinstatement of driving privileges. Paperwork for a physician statement from the board of motor vehicles has been presented by the couple at this visit. The patient is unable to recall the date of the accident. He recounts that he was having difficulty concentrating at the wheel of his vehicle and collided with another vehicle on a straight stretch of road. He recognizes that he did not stop at the scene but does not know why he did not stop. He sought no medical attention following the accident. His wife, with him in the office for the first time, reports a change in personality over the last five months. She attributes this change to his difficulty concentrating which has been progressive during this time. She is most concerned about what she identifies as imbalance which she has noted over the last three weeks. She has prevented him from falling several times in the last week when his gait was unsteady. His past medical history is remarkable for resection of Dukes A colon cancer two years ago. He had coronary revascularization ten years ago and makes regular office visits for blood pressure control. He continues to smoke cigarettes. Medications include: atorvastatin 10 mg po qd, aspirin 325 mg po qd, HCTZ 25 mg po qd, atenolol 50 mg po bid. Review of systems is remarkable for a productive cough for two weeks. He denies dyspnea, hemoptysis, orthopnea, and chest pain. He reports seeing television in shades of gray, despite watching a color monitor. He has had anorexia and nausea for two weeks, but no vomiting. Physical examination reveals an elderly man with difficulty registering questions from the examiner. Vitals: 96.7 F, HR 80, RR 20, BP 160/64. HEENT: unremarkable. Disc margins are sharp. Thyroid: normal. Lungs: clear. Cardiovascular exam: apical II/VI systolic ejection murmur as identified on previous visits. Abdomen: well healed, midline scar. Nontender, without masses, and no organomegaly. Prostate without masses. Mental status testing: fully alert and oriented. Patient recalls one of three items at five minutes. He cannot count backwards from 100 by ones. Strength is $\frac{1}{2}$ in left upper and lower extremities. DTRs are 2+ symmetrically. He cannot tandem gait. There is slight dysmetria in the upper extremities bilaterally. There is no adenopathy.

PLANNING FOR DEATH BUT NOT THE END-OF-LIFE: VIEWS OF HOMEBOUND ELDERLY PATIENTS. JA Carrese, JL Mullaney, RR Faden, TE Finucane, Johns Hopkins University, Baltimore, MD

PURPOSE: Patients worry about the care they will receive when they are sick and dying, yet efforts to promote advance planning for end-of-life care have been largely unsuccessful. This study sought to examine and understand how homebound elderly patients think about and approach future illness and the end-of-life.

METHODS: In-depth semi-structured interviews lasting an hour were conducted with 20 randomly selected patients over the age of 74 who receive their medical care from a university affiliated elder house call program. Interview transcripts were independently coded by two readers and compared for agreement. Content analysis identified several major categories of themes. Relationships between themes were examined and organized conceptually. Findings were reviewed and verified with a subset of patients during follow-up interviews.

RESULTS: Five dominant themes (i.e., views expressed by at least half of the informants) are presented. (1) Most informants do not think about or even acknowledge the future, and (2) most informants do not plan for the future. Consistent with these views (3) most informants are resistant to planning for future illness and the end-of-life. Explanations for resistance to end-of-life planning include the view that it is upsetting, associated with uncertainty, or, if considered to be God's domain, unnecessary. Accordingly (4) patients adopt a "day-at-a-time", "what is to be will be" approach to life. However, seemingly paradoxically (5) most informants have made "final arrangements" for their death (e.g., estate wills, funeral plans, cemetery plots). Similarly, significant percentages of patients have completed some form of advance directive (health care agent, 35%; living will, 55%), but these directives are poorly understood, and intended for use only in limited situations: "dire" states when the patient is pre-terminal, the problem irreversible, and the condition intolerable.

CONCLUSION: These patients do plan for certain aspects of the future: they consistently make final arrangements for their death. To a lesser extent they are also willing to plan, through the completion of advance directives, for dire states. But

they are reluctant, or frankly opposed, to planning for the end-of-life if they find it upsetting, if they consider it to be God's domain, or if it involves discussing scenarios of serious future illness associated with substantial uncertainty. These findings raise questions about the usefulness of the advance care planning model for addressing end-of-life issues with patients like our informants.

PERCEPTION AND REGULATION OF QUALITY IN THE ORGANIZATION OF HEALTH CARE: A QUALITATIVE STUDY. P. *Chopard*, J-L Denis, Quality Group, University Hospitals, Geneva, Switzerland; Health Services Administration, Montreal University, Montreal, PQ, Canada

PURPOSE: To highlight perceptions and regulations of quality in an organization of care.

METHOD: Qualitative analysis of semi-structured individual interviews, audio recording.

Participants: 12 participants chosen because of their function and their hierarchical positions (medical (2), nurses-paramedical (5), administrative-logistic staff (3), board of directors (2)).

RESULTS: Several meanings of quality cohabit in an organization and are controlled in different ways. For the professionals, the "historical" meaning is that of excellence, but a new meaning has appeared, that of values to be negotiated. For the administrative staff or that of logistic functions the main meaning of quality is that of standards to be respected. Methods of regulation differ according to the meaning given to quality. For people in the medical or paramedical professions, they are passing from implicit regulation based on the excellence of training to scientific regulation, which is based on guidelines or on evidence based medicine. For the non-medical managers, regulation is based on the same principles as those used to regulate quality in the industrial world.

CONCLUSION: Quality has different meanings and regulations, which are not univocal. In addition, there exists a true dynamic of quality with the emergence of new meanings and new regulations. Our study suggests that the different regulations of quality extend its "determinants" largely beyond the care itself. Thus, managers but also other health professionals should pay more attention to the evaluation of the relations which link quality of care to management modes and organizational structures.

PERCEPTION OF DRUG ERRORS BY HEALTH CARE PROFESSIONALS: A QUALITATIVE STUDY. P. *Chopard*, E Perrin, V Dupont, P Vernet, V Malegue.

PURPOSE: To investigate how perception of drug errors influences management of errors.

METHODS: Qualitative analysis of semi-structured individual interviews. Audio recording. Coding and analyzes were made separately by two analysts. Any discrepancies were resolved by discussion. Software Atlas-ti was used.

Participants: 15 professionals selected according to their hierarchical position and to their function (Nurses: 8, Nursing Management staff: 5, Doctors: 2).

RESULTS: Frequency of errors is underestimated because only errors, which present a serious risk to the patient, are taken into account. Drug errors are perceived as being consecutive to individual failure (lack of information, inattention). Means to prevent errors are formulated in the same terms (better training, better information, more vigilance, sanctions). Thus, discussions about errors are rare, informal, limited to some individuals and aim at pointing out the lack of attention.

CONCLUSION: Perception of drug errors concerns pre-industrial logic, which is associated with an individual learning pattern based on three principles: training, vigilance and sanctions. This learning pattern runs counter to the logic of the drug dispensing process, which is done according to industrial logic. Therefore, the mechanisms of control devised for individual learning cannot apply with success to the industrial method of dispensing drugs used in most modern hospitals.

"MY PERIOD IS TOO LONG": A PRESENTATION OF DOMESTIC ABUSE. SR *Colbert Threats*.

LEARNING OBJECTIVES: (1) Diagnose Domestic Abuse when it presents as somatic complaints. (2) Recognize that mood disorders may be complicated by domestic abuse. (3) Recognize culturally biased wording on medical questionnaires.

CASE PRESENTATION: A previously healthy 31 year old African-American woman, Gravida 3, Para 3, presented for an initial office visit with a chief complaint of irregular menses. The patient had regular monthly menses every 28 days, with an average duration of 7 days, until two months ago when her menses lasted for 16 days. The following month her menses lasted ten days. The genitourinary review of systems was unremarkable except for pelvic discomfort with menses, chronic dyspareunia and chronic sexual disinterest. The patient had a normal pelvic exam four months ago. Her history was otherwise unremarkable. Social history was notable for the following: She was a Nigerian born US citizen. She had been married for nine years and her spouse, an Air Force captain, was also a native Nigerian. For the past five years, the patient has been a labor and delivery nurse. Six weeks, prior to this visit the patient had suddenly quit her job. She denied smoking, alcohol use or illicit drug use. A general health questionnaire revealed multiple complaints in every organ system, with positive answers to every cardiac, abdominal and genitourinary question. Psychiatric review of system was notable for anxiety and depression. She wrote: "I'm stressed out, typical working mother things, maybe I need some Prozac". On the question regarding domestic

abuse: *Are you now or have you ever been a victim of violence?* The patient checked no.

Her physical exam, including a pelvic, was unremarkable. When told that she had a normal exam, the patient became tearful and said, "Then I must be just crazy, I'm falling apart." The patient admitted that for six months she has had a disrupted sleep pattern, poor appetite, a ten pound weight loss, poor concentration, anhedonia, feelings of shame, guilt, irritability, poor self-worth, hopelessness, and despair. The patient denied suicidal or homicidal ideation. The examination then progressed to the following:

PHYSICIAN: "And your husband, how's that relationship?"

PATIENT: "I should kill him. Just joking. We've been fighting alot. He spends his check on the boat(gambling). We have a lot of bills and I pay them all. I tried to talk to him, but he just calls me stupid, dumb, unsupportive."

The patient also admitted to years of forced sexual contact and hitting which she was taught was normal. Recently, the hitting had become more violent. The patient admitted to quitting her job because a coworker had seen her bruises.

DISCUSSION: The above case illustrates that underlying domestic abuse needs to be considered in the differential diagnosis of multiple somatic and mood complaints. Brief questionnaires may be inadequate to distinguish the presence of domestic abuse, especially in certain cultures.

HOW PATIENTS PERCEIVE PHYSICIAN COMMUNICATION REGARDING CARDIAC TESTING. TC *Collins*, LA Petersen, N Kressin, J Clark, Health Services Research, Houston Center for Quality of Care and Utilization Studies, Baylor College of Medicine, Houston, TX; Health Services Research, Edith Nourse Rogers Memorial Veterans Hospital, Center for Health Quality, Outcomes, and Economic Research, Bedford, MA

PURPOSE: We investigated patients' perceptions of their communication with providers regarding cardiac testing.

METHODS: We convened 4 focus groups with 13 patients who had undergone cardiac stress testing with positive results, stratified by race (white vs. black) and whether or not they had undergone coronary angioplasty or bypass. Verbatim transcripts were analyzed qualitatively by a team of behavioral scientists and general internists in order to identify significant dimensions of communication and patient-provider relationships.

RESULTS: We identified three principal domains of communication regarding cardiac testing: the substance of the information communicated, the meaning of that information for the patient, and the quality of the patient's relationship with the physician. In reviewing the substance of the communication, patients revealed ambiguity in the information received. One patient stated "They said the heart was fine but there was some blockage in the artery." One patient who was advised to have a heart catheterization described the communication as "They just said that they felt like that it needed to be done. I asked questions that morning when I was there. I wasn't satisfied, but I had to get this done. It was real vague." Focusing on the meaning of the information, patients felt that the terms used to describe their heart disease lacked clarity. A patient discussed being told that he should have a catheterization "because there was something going on, he described it then as a leaky heart. I'm not aware just what a leaky heart is or why its called a leaky." While there was no apparent variation by race in patients' perception of the information received, black patients repeatedly expressed a preference for building a relationship with physicians before agreeing to an invasive cardiac procedure. Statements made by black patients were "to make a decision on a major operation, even to angioplasty for me, I maybe would have agreed to that if I had known the physician doing the treatment," "They didn't know too much about me and my health status."

CONCLUSION: We found problematic aspects of the patients' experiences regarding communication of cardiac testing. Our findings suggest that although patients desire clarity from physicians, they are often confused regarding the information received. Both a lack of substance and vagueness of the information received may be linked to feelings of mistrust toward physicians when considering further diagnostic testing. Black patients appeared more concerned with building a relationship with physicians before agreeing to invasive testing.

MEDICAL RECOMMENDATIONS AS JOINT SOCIAL PRACTICE. BA *Costello*, FD Roberts, Division of General Internal Medicine, Mayo Clinic, Rochester, MN; Department of Communication, Purdue University, West Lafayette, IN

PURPOSE: To understand how patients' responses to medical recommendations influence the course of the medical interview.

METHODS: The data for this study comprise 16 videotaped consultations from an Internal Medicine clinic and 21 audiotaped consultations from an Oncology clinic, both based at large university medical centers. The visits ranged in length from 3 to 58 minutes. Combining the data, fourteen physicians and thirty-seven patients were recorded. Each clinic visit was transcribed in detail. Analysis was undertaken using the qualitative methodology of conversation analysis which allows understanding of the sequential and substantive organization of talk and interaction.

RESULTS: We find that patients verbally respond to physician recommendations in three different ways. First, patients may respond to clinical recommendations regarding their care with unconditional agreement such as "Okay. Fine". Patients thereby ratify the physician's suggestion and let it stand as the plan to be followed. Second, patients may use weak displays of resistance, such as silence, after a recommendation to signal that something is problematic about the doctor's

suggestion. Physicians then commonly reformulate the recommendation to make it more acceptable to the patient. Third, patients may offer counterproposals which show disagreement with the physician's recommendation. These counterproposals may be accepted by the physician or may prompt him/her to justify the original recommendation.

CONCLUSION: We demonstrate that on the level of individual medical encounters, patient agency is both apparent and operative, and that physicians are not unilaterally determining treatment plans. Physicians may be contributing to differential treatment approaches by recommendations which are based on their knowledge of standard therapies combined with their perceptions of patients, but patient preferences also affect selection of disease management. What emerges from our data is a demonstration of how treatment plans are negotiated during the medical encounter, how physicians are sensitive to subtle interactional cues, and thus how patients contribute to the final formulation of recommendations. Hence, the outcome or product of the medical encounter (the treatment plan) can be associated with patient as well as physician actions. In clinical practice, physicians attuned to these different patient responses will be able to benefit by understanding how to more actively engage patients in negotiating final treatment plans.

EXPERIENCES OF BLACKS DURING MEDICAL TRAINING: A REFLECTION OF OUR SOCIETY. *GO Darko*, M Bharel, JD Orlander, JM Liebschutz, Section of General Internal Medicine, Boston Medical Center, Boston, MA

PURPOSE: Existing studies describe the prevalence but not the impact of discrimination against minorities during medical training. Little is known about the unique experiences of Black physicians. This qualitative research study explored the experiences of Blacks and the subsequent impact during their medical training.

METHODS: Black residents from two teaching hospitals in Boston were recruited. Open-ended individual 60-90 minute interviews were audio-taped, transcribed, then analyzed and coded through repeated close readings.

RESULTS: The 17 participants (9 men, 8 women), had a mean of 3 years of post-graduate training. Eleven participants were primary care residents and 6 were surgical or subspecialty residents. Five themes emerged from the analyses. 1) All participants were continually aware of the low numbers of Black physicians. As a result, participants felt highly visible and closely watched, which led some to feel "uptight" and "self conscious." In addition, 13/17 participants reported feeling socially isolated in their training programs, which led to depression for some and for others, a heightened focus on family and community for support, rather than their professional colleagues. Also, participants reported incidents of mistaken identity (e.g., nurse or orderly). Some reacted by ensuring they could easily be identified as physicians. 2) Fourteen participants believed that if they performed poorly, they would be assessed unfairly or treated more harshly than their majority colleagues. Fearing harsh treatment, participants worked harder than expected and carefully avoided making mistakes. 3) Fourteen participants perceived themselves as representing all Blacks to non-Blacks, which motivated them to set higher standards for themselves to pave the way for future Black trainees. Our participants were viewed with pride by Black patients and received special treatment by the Black support staff, which reinforced their belief that they represent their race. 4) The 15 participants who had mentors found them to positively influence their self-esteem and career choices. Black mentors' familiarity with Black experiences were particularly helpful. 5) Lastly, 13 participants described specific unpleasant interactions with their majority colleagues which they perceived as discriminatory yet they doubted their perceptions.

CONCLUSION: Black trainees' unique experiences draw from their small numbers, their visibility, a sense of expectation from majority physicians as well as a drive to exemplify their race. These parallel experiences and perceptions of Blacks in American society. Residency program directors need to be aware of these unique pressures in order to improve medical training for Black physicians.

STRUCTURED IMPLICIT HOSPITAL CHART REVIEW: WHAT ARE THE REVIEWERS THINKING?

MH Farrell, TP Hofer, RA Hayward, Internal Medicine and Pediatrics, Yale University School of Medicine, New Haven, CT; Internal Medicine, University of Michigan, Ann Arbor, MI

PURPOSE: Peer review is a key component of quality assessment in hospital care. Suggestions to improve the reliability and validity of chart review include training and sharing multiple viewpoints to increase mutual understanding of a case. "Structured implicit" methods direct an experienced reviewer's attention and judgement to specific processes of care. It is not clear if these added structures eliminate decision-making heuristics based on anecdote or prejudice.

METHODS: As part of a larger study across 8 VA hospitals, we taped and transcribed eight reviewers as they discussed initial quality ratings in 95 patient charts. All reviewers were board certified/board eligible in internal medicine and were trained in structured implicit methods for a study of adverse events related to inpatient laboratory abnormalities (hypokalemia, hyperkalemia, renal failure, hyponatremia and digoxin toxicity). We used qualitative grounded theory methods to build a coding structure for systematic analysis using Atlas software.

RESULTS: The reliability of the ratings of quality of care and preventability of complications was between 0.2 and 0.4, which is comparable to other published studies. As demonstrated by quotes, several themes emerged among the reviewers' discussions about quality. Reviewers were averse to extreme ratings (e.g., "I don't like to say definitely to anything," or "nothing is 100% preventable"). A reviewer often claimed special expertise for a question, and their partner usually de-

ferred to their opinions. The reviewers compared how they would have done in the situation depicted in the chart, and were reluctant to "second-guess" the clinicians. They commonly cited unrelated favorable events in the chart to excuse a specific instance of poor quality, such as globally good care during an episode of hypokalemia. They cited positive and negative prejudices based on identifying information in the chart, such as a certain hospital, physician, or specialty service (e.g., surgery).

CONCLUSION: Our implementation of structured implicit review was comparable in design and achieved reliability to the published literature using this method. Despite the specific training to objectively focus reviewers on particular processes of care, reviews still displayed biased heuristics or emphasize irrelevant or prejudicial information when grading quality. Implicit review is a valuable tool for research, but there appears to be room for further improvement in the methods used to reduce bias and improve reliability.

STARVATION AND ACETAMINOPHEN: A TOXIC COMBINATION. *S Fein*, EF Yee.

LEARNING OBJECTIVES: 1. Recognize starvation as a risk for potentiating acetaminophen-induced hepatotoxicity.

2. Distinguish acetaminophen-induced hepatotoxicity from other causes of acute hepatitis.

CASE PRESENTATION: A thirty-six year old woman was admitted with a four day history of fatigue, weakness, anorexia, fever, diarrhea, nausea, jaundice and glossitis. Two days prior to admission the patient developed dysuria and abdominal pain and was prescribed acetaminophen with codeine (Tylenol #3) and TMP/SMZ (Bactrim) by her primary physician. Her anorexia was compounded by both her glossitis, which made eating painful, and her habit of starving herself to control weight. She had a history of alcohol abuse but reported thirty days of sobriety. In the Emergency Department the patient was febrile at 39.2c with a pulse of 111. She was alert, extremely thin, with icteric sclera, right upper quadrant tenderness, a mildly enlarged liver, and cervical motion tenderness on pelvic exam. No ascites, petechiae or spider angiomas were noted. Labs included AST 2434, ALT 1986, alkaline phosphatase 199, total bilirubin 3.5, direct bilirubin 3.2, PT 15.0, PTT 39.5, and INR 1.3. The patient was admitted with a presumptive diagnosis of acute hepatitis. She was monitored, hepatotoxins withheld, and antibiotics started for treatment of presumed concomitant PID. Results of hepatitis panels, HIV and P24 tests, GC and Chlamydia cultures were all negative. Upon further questioning, the patient revealed that she took four Tylenol #3 each day (1.2 grams of acetaminophen) for two days prior to admission, but had also been taking four hydrocodone/acetaminophen (Vicodin) a day (2 grams of acetaminophen) for two weeks since a dental procedure. An acetaminophen level was then obtained. Thirty-six hours after admission the level was two. Acetylcysteine (Mucomyst) use was discussed but not given since the patients' transaminases were normalizing with supportive care. She was discharged 10 days after admission.

DISCUSSION: 1. Starvation is an important cofactor that increases a patient's susceptibility to acetaminophen-induced hepatotoxicity. Ten grams of acetaminophen in 24 hours is toxic. As little as 4 grams in 24 hours is toxic in alcoholics. Deliberate (dieting) and unintentional (pain with glossitis) starvation predisposed this patient to succumb to low dose acetaminophen-induced hepatitis. The physiologic effect of starvation on acetaminophen-induced hepatotoxicity will be discussed in further detail.

2. Even without an elevated acetaminophen level, the high AST (seen in toxic acetaminophen ingestion and viral, but not alcoholic, hepatitis) was a clue to the final diagnosis. Careful histories can decrease time-to-diagnosis and insure appropriate treatment with acetylcysteine. A review of the evidence for the treatment of unintentional acetaminophen toxicity with acetylcysteine will be presented.

GRIEF, GRATITUDE, AND CHAOS: STRANGERS AT THE DEATH BED. *NM Gadmer*, PG

Ruopp, RM Arnold, EM Redinbaugh, MC McDonald, DL Selzer, MDV Good, SD Block, Adult Psychosocial Oncology, Dana-Farber Cancer Institute; School of Education, Harvard University, Boston, MA; University of Pittsburgh, Pittsburgh, PA; Department of Social Medicine, Harvard Medical School, Boston, MA

PURPOSE: Some of the underlying barriers to competent palliative care by physicians are thought to be attitudinal. However there are few empirical data that address this issue. To investigate this question, we explored physicians' attitudes and experiences in providing care to patients who died.

METHODS: Two patients a week were randomly selected from among all deaths on the medical service of two large teaching hospitals. The attending, resident, and intern who cared for the patient at the time of death were selected for study inclusion and were contacted within 2 weeks for a semi-structured interview that lasted approximately 80 minutes. The response rate was 88 percent. The interview incorporated open-ended questions and quantitative measures to assess the physician's experience in caring for the dying patient. The following dimensions were explored: physician-patient relationship, relationship with the family, response to the patient's death, and perceived quality of care. Interview transcripts were analyzed using a grounded theory approach. Members of the research team independently read each transcript and identified themes which were discussed as they emerged. The themes were then compiled and used in coding. Atlas/ti, a qualitative analysis software, was used to assist in the coding and analysis of the transcripts.

RESULTS: Of 153 physicians, 46 interns, 45 residents, 50 attendings, and 12 other physicians cared for 49 patients. Physicians' relationships with patients were brief and not considered close. Physicians reported stronger ties with fami-

lies. Physicians were most likely to characterize the dying process as positive when death was anticipated and/or viewed as timely; when patients and family were regarded as rational; when patients and family goals were concordant with those of the medical team; when error or adverse events were absent; and when physicians felt connected to families or experienced their gratitude. Deaths were regarded as difficult when unexpected and untimely; when family and medical team were in conflict about decisions; when death was hastened through active interventions; and when physicians either identified with or disliked the patient or family. Deaths that were perceived as both good and bad elicited distress and self-doubt about the quality of care delivered.

CONCLUSION: Physicians have clear ideas about what constitutes "good" and "bad" dying. Experiences of discontinuity in care, lack of closeness in relationships with patients and family, and concerns about error when caring for dying patients are barriers to competent palliative care.

PANCOAST TUMOR PRESENTING AS CHRONIC NECK PAIN AND C5-6 RADICULOPATHY. KM Greifer.

LEARNING OBJECTIVES: To recognize the presentation of Pancoast tumors and to review the diagnostic work-up of cervical radiculopathy

CASE PRESENTATION: A 52 year old previously healthy woman presented to her primary physician with right sided neck pain radiating to her right upper extremity and hand. Physical exam was unremarkable including neurologic exam. Cervical spine XRAYs demonstrated neuroforaminal narrowing at C5-C6. Anti-inflammatory medication was prescribed. A few months later the patient was seen again with similar complaints although she stated her pain had not improved, and had in fact worsened. Her physical exam again was normal. She was referred to orthopaedics and magnetic resonance imaging (MRI) of the cervical spine demonstrated mild disc disease with minimal encroachment at the C5-C6 level. She was sent for physical therapy. Her pain continued to worsen and she was subsequently sent to a pain specialist for cervical epidural injections. By then, the patient was on narcotic analgesics for the pain. The injections failed to relieve her pain and she began to complain of paresthesias and weakness in her right upper extremity. A neurosurgical consult was obtained but it was felt that the disease seen on her MRI was not severe enough to warrant surgery. Subsequently an electromyogram/nerve conduction study was done. This study raised the possibility of a brachial plexopathy rather than a cervical radiculopathy. At this point the patient had overt weakness in her grip strength. An MRI was done of the brachial plexus which revealed a large tumor encasing the brachial plexus and subclavian artery arising from the apex of the right lung consistent with a Pancoast tumor. Chest/abdomen computed tomography showed only the large tumor and no evidence of other disease or metastasis. Biopsy subsequently showed non-small cell lung cancer.

DISCUSSION: Pancoast tumors can present with neurologic findings—more commonly with a unilateral Horner's syndrome. This patient demonstrated another presentation. It is important to recognize that Pancoast tumors often do not present with cough or hemoptysis or shortness of breath as do other lung tumors. In addition, this patient had pain out of proportion to her radiographic findings. Nerve conduction studies should be considered early in the diagnostic work-up where there are questions as to the correlation between radiography and symptoms.

CAN STATES ENSURE QUALITY FOR MEDICAID BENEFICIARIES ENROLLED IN MANAGED CARE?: THE CONNECTICUT EXPERIENCE. AR Gupta. HM Krumholz, Department of Medicine, Yale School of Medicine, New Haven, CT

PURPOSE: Using preventive health measures for women enrolled in Medicaid managed care in Connecticut, we reviewed state mandated performance rates and analyzed health plan organizational characteristics to identify high performers and assess the state's commitment to ensuring quality of care for Medicaid beneficiaries.

METHODS: A survey on plan characteristics, quality structures, and improvement efforts was conducted with all managed care organizations (MCOs) with contracts to enroll Medicaid beneficiaries from 1996 to 1998. Personnel at the six plans that had participated in state mandated quality reporting were interviewed via telephone by a single individual.

RESULTS: Based on annual preventive data reports submitted by MCOs to the state, the number of Medicaid beneficiaries 18 years and older who received a Pap smear was 6723 (34%) in 1996, 12720 (43%) in 1997, and 13219 (42%) in 1998. The number of Medicaid beneficiaries 40 years and older who received a mammogram was 787 (27%) in 1996, 1001 (26%) in 1997, and 1216 (26%) in 1998. Interviews with MCOs with Medicaid contracts revealed extreme heterogeneity in terms of organizational characteristics, quality measurement structures, guidelines for mammograms and Pap smears, and quality improvement initiatives. Of the six MCOs, 4 offered another product line (i.e., commercial plan or Medicare plan) in addition to their Medicaid line. MCOs with more than one line of business were more likely to have NCQA accreditation and to be involved in managed care longer than plans enrolling only Medicaid patients. MCOs with multiple product lines tended to use the medical record for data collection whereas Medicaid only plans relied heavily on encounter forms with little correlation to the medical record. Guidelines regarding the use of mammograms and Pap smears for screening also varied among the plans and were not consistent with the measurement specifications and screening requirements of the state. In response to poor performance rates from 1996 to 1998, only one MCO initiated an intervention to improve the

rates of women screened. MCOs attributed their low performance rates to the small percentage of women in the Medicaid population as well as the absence of target goals and lack of sanctions for poor performance by the state.

CONCLUSION: Marked differences in plan characteristics and approaches to quality measurement and improvement of managed care organizations with Medicaid enrollees do not account for low performance rates on women's preventive health measures. State mandated public reporting of these low rates in and of itself did not encourage plans to initiate quality improvement programs. States must employ other strategies to ensure high quality care for Medicaid beneficiaries enrolled in managed care.

HOSPICE PATIENT ATTITUDES REGARDING SPIRITUAL DISCUSSIONS WITH THEIR DOCTORS. A Hart, Jr. J Kohlwe, L Rhodes, R Deyo, HSRΔ, Center of Excellence, VA Puget Sound Health Care System; Department of Medicine, University of Washington, Seattle, WA; Department of Medicine, UCSF, San Francisco, CA; Department of Anthropology, University of Washington, Seattle, WA

PURPOSE: To assess hospice patient attitudes regarding physician discussion of religious or spiritual issues in the medical setting.

METHODS: We conducted fifteen semi-structured interviews of hospice patients in Seattle, aged 63-86. The interviews focused on patients' living with their illnesses, physician-patient relationships, and spirituality and religion with open-ended questions. The interviews were audio-taped, transcribed and analyzed for dominant themes. Trustworthiness, an ethnographic concept akin to reliability and validity, was assured by transcript review with two independent readers.

RESULTS: Four dominant themes were identified. First, patients expressed a desire to be treated as "whole persons," which includes discussion of non-medical issues. Secondly, patients wanted doctors to be more sensitive to their social and spiritual needs. Thirdly, patients were generally in favor of doctors asking about their religious or spiritual beliefs. Finally, although a majority of patients wanted to be asked about their spiritual beliefs, they did not want to be preached to! No dominant themes emerged about specific spiritual issues patients wanted their doctors to discuss. Although not a dominant theme, we found it interesting that several patients believed doctors are an "extension of God" in healing.

CONCLUSION: Hospice patients in our study viewed the acknowledgment of a patient's religion or spirituality as an aspect of treating the "whole person." However, patients do not expect physicians to be their primary spiritual advisor, and doctors should let patients set the agenda if patients indicate that they want to discuss spiritual issues. Physicians may need to develop special training opportunities to feel empowered to engage patients in religious or spiritual discussions. Ultimately, more training in relationship building, in addition to good interviewing technique, may enhance discussion of these important issues.

TRUST ME, I'M A DOCTOR: MANAGED CARE AND THE PATIENT-PHYSICIAN RELATIONSHIP. JM Hauser. R Gorawara-Bhat, W Levinson, Medicine, University of Chicago, Chicago, IL

PURPOSE: To examine physicians' own views of the effects of managed care on their relationships with patients by exploring (1) the types of problems that arise in patient encounters (2) how physicians address these, and (3) the ways that they experience "trust" in the patient-physician relationship.

METHODS: Qualitative methodology consisting of semistructured interviews with a convenience sample of 22 primary care physicians (19 internal medicine, 2 family practice, 1 pediatrics) in two cities. Interviews were audiotaped, transcribed and were analyzed for significant themes using grounded theory methods.

RESULTS: The sample reported an average of 11.3 (sd 7.1) years in practice and saw an average of 86.2 (sd 31.8) patients per week, of whom 52.5% (sd 11.7) were in capitated managed care plans. All of the physicians reported that managed care had changed their relationships with patients in ways that ranged from increased "hassle" on non-medical tasks to a more profound mistrust that they perceive from their patients. The specific areas that physicians identified as problematic included formulary restrictions, specialty referrals, diagnostic testing, and decisions around hospitalizations. While some physicians viewed disclosure of financial arrangements as a remedy to address these conflicts, they varied in their opinions about how and when to disclose such information to patients. Almost all believed that the managed care plans themselves should be clearer in their disclosures. Physicians' in this sample largely defined "trust" as the assumption that they act on their individual patients' best interests and worried that in the current environment, the "feeling that your physician is your advocate . . . is not nearly as solid as it once was."

CONCLUSION: Managed care raises troubling dilemmas for this sample of physicians in their relationships with patients in areas ranging from administrative hassles to diagnostic decisions. Many physicians see previously fulfilling relationships being replaced by mistrust in their roles, in the words of one subject, "holding both the purse strings and the prescription pad." Although disclosure of financial arrangements represents one potential remedy, further research is needed to determine whether or how these discussions should take place in the setting of the patient-physician relationship.

PATIENT PREFERENCES FOR DISCUSSING SPIRITUALITY WITH PHYSICIANS. RS Hebert. MW Jenckes, DR O'Connor, DE Ford, L Cooper-Patrick, Department of Medicine; Department of Oncology, Johns Hopkins University, Baltimore, MD

PURPOSE: To identify patients' interests and concerns about discussing spiritual and religious beliefs with physicians.

METHODS: Two focus groups totaling fourteen participants (mean age 53; eight women, four African Americans). Participants were recruited through their physicians; all had experienced a life-threatening illness in the preceding year. Discussions addressed facilitators and barriers to talking about spiritual beliefs with doctors, physician characteristics, and recommendations for physicians. Discussions were audiotaped, transcribed verbatim, and reviewed independently by two investigators to group distinct comments into categories with specific themes. Differences were adjudicated by a third investigator. Comments within categories were then checked for relevance and consistency by a health services researcher and clinical social worker.

RESULTS: Four hundred thirty one distinct comments were generated with 95% falling into one of eight broad categories: 1) Patients' description of their spirituality/religiosity 2) Church attendance 3) Mention of God/Holy Spirit 4) Prayer 5) Physician-patient communication 6) Patient preferences of whom they would prefer to discuss spirituality with 7) Context for discussion and 8) Patient recommendations to physicians. About two-thirds of patients described themselves as spiritual, half described themselves as religious, and a quarter as neither. God and prayer were often mentioned as a source of comfort, guidance, or healing. Twenty percent of comments pertained to physician-patient communication, especially psychosocial and emotional aspects such as respect for a patient's individuality and whether a physician engendered trust, comfort, and hope. Physicians with strong interpersonal skills were viewed as spiritual. Half the participants felt that physicians should talk about spiritual beliefs; this dialogue was believed to be a valuable part of the healing process. The other half felt physician-initiated conversation was not expected, but would be welcomed. Patients had different views about the context for these discussions. Some felt the appropriate setting was during serious illness; others felt discussion at this time would imply a poor prognosis. They believed the topic would be better broached during routine care. Participants recommended that physicians ask patients about their use of spirituality and sources of support, refer them to spiritual counselors, or pray with them.

CONCLUSION: Patients are interested in discussing spiritual and/or religious issues with their physicians. This can be accomplished through an open and enhanced doctor-patient relationship. Future work should explore patient characteristics associated with preferences for spiritual discussion.

AVAILABILITY OF A NEW MENTORING SURVEY INSTRUMENT. VA Jackson, A Palepu, RC Barnett, C Caswell, S Regan, TS Inui, PL Carr, Harvard Medical School, Boston, MA; University of British Columbia, Vancouver, BC; Harvard University, Boston, MA

PURPOSE: Mentoring relationships are often cited as key to the development of productive academic careers in medicine, yet the characteristics of these experiences, particularly how they fail, are poorly understood. Description of mentoring quality and variation has been hampered by the lack of survey instruments applicable to academic medical faculty. We have completed preliminary qualitative research to develop a comprehensive survey instrument.

METHODS: The instrument was developed after review of the mentoring literature, existing surveys, and the analysis of a focus group. Focus group participants, chosen for spectrum of mentoring experiences, were recruited from among prior respondents to the Faculty Advancement Study Questionnaire (1995) and were asked to define and describe optimal and sub-optimal mentoring. The focus group was comprised of 7 academic faculty—4 men and 3 women, 4 clinical and 3 basic science faculty. Six participants had more than ten years' experience at their current institution. Focus group proceedings were audiotaped and analyzed by five readers who identified key words, phrases, and topics, grouping them by consensus into major themes.

RESULTS: The 6 major themes were: 1) Critical mentoring situations—dealing with failure, promotion, negotiation, and publishing. 2) Critical mentoring functions—assisting the mentee in balancing professional and personal lives, aiding in networking, providing feedback, and fostering growth as an academic professional. 3) Critical mentor roles—working as a career strategist, helping the mentee develop an independent academic identity, avoiding exploitation of the mentee, holding the mentee to a high achievement standard, acting as an advocate. 4) Dynamics of the mentoring relationship—mentor learning from the mentee, producing career change for both, taking risks with traditional boundaries. 5) Effects of gender and race—including gender bias, difficulties of cross-gender mentoring, importance of minority mentors when preferred. 6) Resource recommendations—building visible, formal, mandatory, systems for mentoring, formally rewarding mentoring effort.

CONCLUSION: Better understanding the elements of successful mentoring may assist in promotion of these relationships where they exist and provide resources to foster them where they do not prevail. We have built, pre-tested, and are fielding a comprehensive survey instrument to study the academic mentoring experience which we will make available to other investigators.

ACUTE CYSTITIS PATHWAY. A PHONE TRIAGE SUCCESS. K Kerwin, WA Harb, Medical Education, Oakwood Hospital, Dearborn, MI

PURPOSE: Applying triage pathways in the ambulatory setting can result in substantial healthcare savings.

METHODS: Great variability exists in practice pattern amongst physicians in the ambulatory clinics. Patients with an acute problem may wait for an appointment 1–2 days to be seen, thus delaying therapy and potentially complicate outcome.

Some physicians treat simple problems through phone triage, others do not. Variability also exists in the choice of antibiotic and in the length of therapy.

A task force was empowered to set up an acute cystitis pathway. A triage nurse phone interviews patients with suspected UTI. The goal is to establish the diagnosis and to exclude any systemic or complicating factors that might prevent treatment over the phone. The nurse discusses results with physician, then calls prescriptions to the pharmacy and instructions to the patient.

Results: 43 patients were triaged, 39 were treated over the phone. Three were asked to be seen and one had no UTI. None of the patients enrolled had any complications. Time lapse from complaint to prescription called in was one half hour to 4 hours. Total savings in this trial was \$5088. Since we see an average of 25 UTI's per month, the potential savings is \$42,000 per year.

RESULTS: When the acute cystitis pathway is applied to low risk patients, with proper screening tools and back up, there is a very significant savings to our health care system. It also results in improved patient satisfaction, speed of therapy and lack of time off from work along with good outcome. It allows physicians to concentrate on patients with higher acuity of illness. We recommend using this model in all ambulatory settings.

HOW DOES A SCREENING MAMMOGRAM IMPACT A WOMAN'S PERCEPTION OF HER OWN HEALTH? GC Lamb, M Kupst, AB Nattinger, Medical College of Wisconsin, Milwaukee, WI

PURPOSE: To explore the effect of mammograms on the perceived emotional and physical well being of healthy women.

METHODS: Four focus groups were conducted in September, 1999. Potential participants were identified by random selection from a list of all women who received mammograms or breast biopsies at our institution during December, 1998. Patients were excluded if they were diagnosed with cancer, unable to give consent, or did not speak English. The groups included 1) women with normal mammograms, 2) women with abnormal mammograms requiring 6 month follow-up, 3) women requiring biopsies, and 4) women of lower socioeconomic status as defined by insurance status (general assistance/Title 19) at the time of the mammogram. All focus groups were run by a female professional facilitator with experience in breast cancer research. Discussions addressed anticipation of the mammogram, waiting for results, and health related behavior following the mammogram. Transcripts of the sessions were independently coded and analyzed by two of the investigators.

RESULTS: A total of 45 women participated in the 4 focus groups. A wide range of opinions were expressed. Prior to having the mammogram, several women admitted to being "scared", or "concerned" about finding cancer. Worry about pain from the test was also expressed. Others described feeling "good" about having finally scheduled the procedure. Waiting for the results, many women "think about it a lot" and are quite anxious. After news of a normal mammogram, some women are more serious about self exams and health maintenance while others feel "covered" and become "lazier" about self exam. Women with abnormal results requiring 6 month follow-up had reactions ranging from "never thought about it" to substantial concerns that affected how they responded to minor irritations, how they treated those around them and their desire for knowledge about breast cancer. The most striking responses were seen in women who were advised to undergo a biopsy. These women had intense experiences that had an impact on their physical health, emotions, work, family interactions and sexual relations. The nature of an individual's response seemed to be related to both patient factors and system issues. Patient factors included faith in doctors, technology or religion; previous health experiences and other stressors at home. System issues included the doctor's attitude, nature of explanations, counseling, provider consistency and time spent waiting for results.

CONCLUSION: These focus groups identified a variety of ways in which women perceive that mammograms impact their emotional and physical well being. While these findings cannot be generalized to the population at large, they can be used to guide future quantitative research into the impact of screening mammography and develop approaches to mitigate adverse effects.

UNILATERAL BREAST MASS AS A MANIFESTATION OF ANTIRETROVIRAL TREATMENT-RELATED LIPODYSTROPHY. LM Losater, W Burman.

LEARNING OBJECTIVES: 1. Recognize gynecomastia as a possible complication of antiretroviral therapy

2. Adopt a conservative approach to the diagnostic evaluation of a breast mass found in a male taking antiretrovirals for the treatment of HIV infection

CASE PRESENTATION: A 48 year old male with HIV infection presented with a breast mass first noted 3 weeks earlier. He denied any history of trauma, nipple discharge, weight loss, or the use of non-prescription medications or recreational drugs in the past 3 years. He had been treated with zidovudine and didanosine for 2 years, but was switched to nevirapine, stavudine, and lamivudine 3 months ago. He had a good immunologic and virologic response to the new regimen, achieving a viral load <25 copies/mL. His only other medications included fluoxetine and trimethoprim-sulfamethoxazole. On examination, he had a hard, fixed, tender mass superior and lateral to the right areola, but no adenopathy. He also had truncal obesity, wasting of the cheeks, and atrophy of the extremities. Testicles were normal in size. Because his presentation was unusual for gynecomastia and concerning for malignancy, he was evaluated by a multidisciplinary team which included an oncologist, radiologist and surgeon and subsequently underwent a right total mastectomy, without an antecedent biopsy or mammogram. The histo-

pathology of the 3 × 1 cm mass removed was consistent with gynecomastia. He developed a similar mass in his left breast approximately 5 weeks after the first but did not report it for 1 year. At that time, serum concentrations of free testosterone, human chorionic gonadotropin, estrone, estradiol, androstenedione, prolactin, and thyroid stimulating hormone levels were within normal limits.

DISCUSSION: The syndrome of lipodystrophy is being increasingly recognized as a complication of protease inhibitors. In men, the most common manifestations include truncal obesity, buttock, extremity and facial wasting, and a dorsocervical fat pad. Breast hypertrophy, although a common complication among women, has been reported infrequently in men. This was a case of gynecomastia presenting as a unilateral, eccentric mass which was most likely due to antiretroviral treatment and resulted in a major operation. Only 5 cases of gynecomastia secondary to HIV antiretroviral treatment have been documented in men. Gynecomastia has not previously been described with the use of nelfinavir, nor does it usually present unilaterally. HIV infection alone can cause gynecomastia, however the prevalence of gynecomastia in HIV infected men has not been reported. Medical providers should identify antiretrovirals as a possible cause of a breast mass in males and choose a more conservative approach, such as needle biopsy, for the initial diagnostic evaluation.

TESTING POSITIVE FOR HIV AMONG DRUG ADDICTS: PROVIDING A FUTURE. J Liebschutz, C Kerner, J Samet, Section of General Internal Medicine, Boston University School of Medicine, Boston, MA

PURPOSE: The drug use consequences of testing positive for HIV in drug addicts have not been well studied. We conducted a qualitative research study to understand the impact of testing positive for HIV on individuals with a history of drug dependence.

METHODS: HIV-infected people with prior heroin or cocaine use were recruited with a particular focus on those who tested recently. Open-ended individual interviews were audio-taped and transcribed. The transcriptions were analyzed via multiple readings, coded using NUD*ST software and evaluated by a group of HIV-infected drug addicts, substance use counselors and healthcare providers. Grounded Theory methods were used to develop the theory and conclusions.

RESULTS: Of the 26 subjects, 10 were women and 16 men, with a mean age 42 years. Other characteristics included 68% prior or current homelessness; 77% minority ethnicity; 13 mean years of illicit drug use; drug of choice: heroin 50%, cocaine 50%; 3 mean years since initial HIV test (range 0.1–11). At the time of HIV testing 19/26 were using illicit drugs daily as compared to 5/26 at the time of the interview. All had some periods in recovery since testing positive. For subjects in recovery at the interview, testing positive for HIV was one of a number of 'wake-up calls' that shifted their awareness only of the next high to a sense of past, present and foreshortened future. In some cases this 'wake-up call' was precipitated by developing symptoms from HIV, long hospital confinement, or incarceration. Some saw a premature death as a motivation to stop using drugs because they did not want to die an addict. All participants held the belief that using drugs would worsen their HIV status, either directly through illicit drug effects on the immune system or indirectly through poor self-care. This belief was an important motivation to decrease drug use. They were proud of their efforts to get responsible including getting off the streets, eating regularly, and keeping appointments. The longevity promised by antiretroviral therapy further motivated many to develop a stable lifestyle in order to benefit from medication. However, suicidal ideation and depression in 13/26 participants dampened this motivation. Subjects reported access to HIV priority housing and residential treatment programs as important resources for recovery. Subjects viewed health care providers as key members of their support system, in part, because they were frequently isolated from family. **CONCLUSION:** Testing positive for HIV among drug addicts appeared to interrupt a long-standing lifestyle and, paradoxically, to provide potential for a better future. Exchanging addiction for the promise of health, in combination with available social resources, helped decrease subsequent drug use and increase stable lifestyles.

WHICH PATIENTS GET AN "A" FOR ADHERENCE? SE Malcolm, JJ Ng, RK Rosen, MD Stein, VE Stone, Brown University, Providence, RI

BACKGROUND: Adherence has been a key issue determining the success of treatment in patients with HIV/AIDS. Studies examining adherence to highly active antiretroviral therapy have focused on predicting adherence rates within demographically defined groups of patients. However, few have attempted to identify and study patients who demonstrate excellent adherence. Understanding the factors that have enabled these patients to achieve superior adherence rates may help to identify new strategies for enhancing adherence to HAART.

OBJECTIVE: To examine the attitudes and behaviors of HIV/AIDS patients with excellent adherence and how they differ from those with suboptimal adherence.

METHODS: A diverse group of patients with HIV/AIDS on HAART were interviewed, using a semi-structured qualitative format. We compared patients with excellent adherence to those with suboptimal adherence in terms of motivators, patient-provider relationship, substance abuse, social support and mental health. Patients categorized as having excellent adherence were those who reported taking their medication consistently 90–100% of the time or who hadn't missed a dose within 3 months of the interview. Patients who reported taking their medication less than 75% of the time were categorized as having suboptimal adherence.

RESULTS: There were 43 patients: 63% were males, 37% nonwhites, 44% former substance abusers, 40% active users and 16% nonusers. 25 patients were excel-

lent adherers and 18 patients were suboptimal adherers. Patients with excellent adherence rates voiced the following themes when compared to less adherent patients: 1) They believed that adherence rates needed to be 90–100% for the medication to be effective; 2) The primary care provider was regarded with a great degree of respect and trust; 3) They continued to follow their medication regimens even when actively using alcohol and/or drugs; 4) Patients were open with family and friends about their HIV status. They received substantial support from their families and attended support groups; 5) Patients cited staying healthy as their key motivator. Those with excellent adherence were not actively depressed; all had normal CESD scores. They expressed a strong desire to live and actively took steps to manage feelings of depression. In contrast 31% of suboptimal adherers were actively depressed. Of the excellent adherers, 16 were former substance abusers, 4 were current users, and 5 nonusers. Of the suboptimal adherers, 13 were current substance abusers, 3 were former users and 2 nonusers.

CONCLUSION: Those with HIV/AIDS who demonstrate excellent adherence to HAART differ from their less adherent counterparts in terms of key health-related attitudes and behaviors. Former substance abuse does not appear to be an obstacle to achieving excellent adherence. Identifying and studying excellent adherers appears to provide new insights and strategies for enhancing adherence to HAART.

TRANSITION POINTS: CHANGING NEEDS FOR INFORMATION AND SUPPORT DURING BREAST CANCER DIAGNOSIS, TREATMENT, AND SURVIVORSHIP. ED McKinley, S Saleh-Jones, KC Stange, Center for Health Care Research and Policy, MetroHealth Medical Center, Cleveland, Ohio

PURPOSE: Little is known about how breast cancer patients' information and psychosocial needs change during the diverse phases of diagnosis, treatment and survivorship. We hypothesized that there would be identifiable transition points during these phases where patients' needs change in predictable ways.

METHODS: We recruited forty-two breast cancer survivors in Cleveland who were at least one year from last treatment to participate in 8 focus groups. Women were introduced to the idea of a "process diagram" outlining their steps through the process of breast cancer diagnosis, treatment, and survivorship. We explored if, where, and why women experienced problems with information, support, or communication. All women completed an exit survey, and all transcripts were read and coded by the PI, a research assistant, and a qualitative analyst.

RESULTS: The mean age of participants was 56, 85% were white, 50% had incomes over \$50,000, 49% underwent mastectomy, 50% received chemotherapy, and 31% had breast reconstruction. We identified three transition points where women said their needs were not being met: 1) the transition from wellness to sickness, 2) the transition from being uninformed to making informed decisions, and 3) the transition from acute treatment to follow-up care. The transition from wellness to sickness occurred at diagnosis. Women repeatedly said that they received too much information and not enough emotional support here. The second transition occurred gradually over weeks to months as women had to make many complex decisions about surgery and adjuvant treatments. Most women described an "information learning curve" here. The third transition occurred when active treatment ended. Here, all women reported being eager for information but receiving very little. No woman reported being "prepared for survivorship".

CONCLUSION: We identified three transition points in the process of breast cancer care in which women's needs for information and support changed in predictable ways and were not being met. These findings and participant feelings indicate that care could be improved with relatively simple interventions that involve tailoring the delivery of information and support to the patient's stage along the diagnosis/treatment/survivorship continuum.

A RARE MYOPATHY IS UNMASKED BY AN EXERCISE PROGRAM IN A YOUNG ADULT. H Moturi, K Barnard, N Busis.

LEARNING OBJECTIVES: Objective: (1) Consider metabolic myopathy in the differential diagnosis of a patient with rhabdomyolysis. (2) History of exercise intolerance and exercise-induced cramping is highly suggestive of an underlying myopathy.

CASE PRESENTATION: Patient is a 23-year-old, healthy, male graduate student from Thailand, who presented with generalized muscle aches. One week prior to the onset of his symptoms he initiated an intense exercise program-situps, running, and lifting weights 2 hrs per day for 3 days. Lab studies showed a total creatine kinase (CK) of 6000. He was managed conservatively with oral hydration. Ten days later he developed left arm swelling, difficulty in rising from a chair, walking upstairs, and combing his hair. He was found to be in rhabdomyolysis with total CK increased to 21,600. On exam the patient was found to have weakness, generalized muscle tenderness and edematous extremities. A doppler ultrasound study of the left brachial vein revealed a deep vein thrombosis. He was admitted to the hospital where intravenous hydration and rest were initiated. Muscle injury was evident with high CK, transaminases, Aldolase, and Myoglobinuria. Further questioning revealed exercise intolerance during sports requiring endurance and speed and longer recovery from muscle cramps. He never noticed dark urine. His father and paternal uncles had similar complaints of longer duration of muscle cramps and decreased endurance. His father had proximal hip girdle weakness elicited on exam but with normal CK. There is no documented case of myopathy in the family.

The patient denied any history of drug abuse. In laboratory data-TSH, T3, T4 were normal and ANA, ANCA, Ds-DNA, and monospot were negative. A workup for hy-

percoagulable state was negative, and the DVT was attributed to exertion. Eight weeks later CPK still fluctuated around 1000. Muscle biopsy was then performed which showed muscle necrosis. Biochemical analysis of his muscle biopsy revealed Phosphofructokinase (PFK) deficiency. In 4 months CPK normalized.

Discussion: In this patient, starting a new exercise program precipitated an acute rhabdomyolysis. This, in conjunction with a prior history of exercise intolerance strongly suggested an underlying myopathy, leading to muscle biopsy confirming the diagnosis of PFK deficiency. PFK deficiency is a rare autosomal recessive disease, which inhibits muscle capacity to regenerate ATP through glycolysis. Hence during increased metabolic demand muscle uses fatty acids and ketones as alternative energy source. Exercise tolerance can be enhanced by a slow warm-up or brief periods of rest allowing for the start of second wind phenomenon (switching to utilization of fatty acids). A high fat diet should theoretically enhance the muscle energy, however the clinical effectiveness is only anecdotal.

ADHERENCE TO HIGHLY ACTIVE ANTIRETROVIRAL THERAPY IN SUBSTANCE ABUSERS WITH HIV/AIDS. JJ Ng, RK Rosen, SE Malcolm, MD Stein, VE Stone, Brown University School of Medicine, Providence, RI

BACKGROUND: Adherence to Highly Active Antiretroviral Therapy (HAART) in substance abusers with HIV/AIDS is generally considered to be inadequate, and detailed studies are lacking. Since adherence to medical regimens is a complex behavior, insight into factors that influence adherence in this population may be gained through qualitative research methods.

OBJECTIVES: To investigate factors and identify themes that may influence adherence among substance abusers with HIV/AIDS.

METHODS: We interviewed 44 persons with HIV/AIDS using a semi-structured instrument. The sample was chosen to represent a broad spectrum of substance abuse patterns. Of the 44 patients, 42 were receiving HAART and 2 were receiving other antiretroviral regimens. The interview was partially based on the *Health Belief Model* of patient behavior. The interviews were taped, transcribed, and analyzed using ATLAS text indexing software. Transcripts were analyzed by two of three researchers.

RESULTS: Four major patient beliefs emerged regarding substance abuse and adherence to HAART: 1) alcohol and drugs interfered with the actions of antiretrovirals, 2) alcohol and drugs were so detrimental to health that it was futile to take medications concurrently, 3) the side effects of HAART worsened hangovers and withdrawal symptoms, and 4) current drug or alcohol use made medication adherence a low priority. However, independent of substance abuse status, many subjects reported that their adherence improved over time. Reasons given for this included: 1) incorporating medications into their daily routine took time, 2) major life changes renewed interest in their health and led to increased adherence.

CONCLUSIONS: This study generates several hypotheses regarding adherence to HAART. Substance abusers with HIV/AIDS may have specific misconceptions about HAART and drugs of abuse which, when addressed, may enhance adherence. Clinicians and researchers assessing adherence need to take into account that adherence may change over time.

QUALITY OF CARE FROM PERSPECTIVE OF ASIAN AMERICAN PATIENTS. Q Ngo-Metzger, M Massagli, M Mannoia, B Claridge, R Phillips, Division of General Medicine and Primary Care, Beth Israel Deaconess Medical Center; Picker Institute; Center for Survey Research, University of Massachusetts, Boston, MA

PURPOSE: Asian Americans (AA) are the fastest growing minority group in the United States, with a 400% increase in the last three decades. Despite this rapid growth, little is known about AA's perspectives on the components of high-quality care, especially among those who are Limited-English-Proficient (LEP). We used patient group interviews to evaluate quality of care from the view of LEP-AA patients (pts).

METHODS: LEP-AA pts of Vietnamese origins were recruited from an urban community health center. We developed a moderator guide and trained bilingual group moderators to conduct interviews. We assessed quality of care using the domains of the Picker Instrument for Outpatient Care, with additional domains assessing the quality of interpretation services and patient-provider communication about the use of non-Western medicine. We interviewed 22 pts (mean age 46, 50% female, 100% foreign born, 95% LEP, 58% completed high school) in 2 gender-specific groups. Video recordings and transcriptions were prepared for each group interview. One investigator reviewed these recordings and identified recurring themes about the attributes of quality health care.

RESULTS: Quality of interpretive services was considered to be an important aspect of care. LEP-Vietnamese-American (VA) pts preferred using trained interpreters rather than family members. For older VA pts, gender concordance of interpreters was reported to be very important. LEP-VA pts valued providers and interpreters who show "respect" to the pts, demonstrated by having a welcoming and courteous manner. They also valued providers who show knowledge of non-Western medical practices, such as "coining" or "cupping." Many feared disclosing the use of these practices because of negative responses they have previously received from their providers, but felt that the culturally competent provider would understand, if not condone, these traditional practices. They valued providers who are conscientious in caring for pts' social and psychological needs as well as medical needs, and wanted specific treatment recommendations at the conclusion of visits. VA pts also rated highly having adequate access to their providers, adequate time with their providers, and having good continuity of care.

CONCLUSION: LEP-VA pts regard having access to professional medical interpreters as an important indicator of quality care. These pts define a culturally competent provider as one who is knowledgeable about non-Western medical practices and is not condemnatory toward the pts for using these practices. LEP-VA pts also value having adequate access to provider and good continuity of care as two important quality domains.

DO MEDICAL STUDENTS' CLINICAL ABILITIES MEASURED BY OBJECTIVE STRUCTURED CLINICAL EXAMINATION REFLECT ACTUAL PATIENT SATISFACTION IN OUTPATIENT CLINIC? Y Oda, S Yamashiro, H Ohnishi, S Emura, N Miyoshi, K Ishii, M Yamada, S Koizumi, General Medicine, Saga Medical School, Saga, Japan

PURPOSE: To determine whether Objective Structured Clinical Examination (OSCE) scores of medical students reflect actual patient satisfaction in a university hospital outpatient clinic.

METHODS: In 1999, an OSCE which consisted of interview, general physical examination and neurological examination stations was administered to 81 senior year medical students. 281 patients seen in our outpatient general medical clinic rated these medical students by using six items from the American Board of Internal Medicine Patient Satisfaction Questionnaire (PSQ). Students were divided in quartiles according to OSCE scores and relationships to patients' ratings were analyzed.

RESULTS: OSCE items showed a good internal consistency with Cronbach's alpha (f_c) of 0.89 as well as the items of PSQ ($f_c = 0.83$), but no overall correlation between these two scores were seen. Medical students in lowest quartile, however, tended to be rated lower by patients as well ($p = 0.12$).

CONCLUSION: Medical students' clinical abilities measured by 4 station OSCE did not correlate well with patients' satisfaction as a whole, although lowermost scores might be used to identify under-performing students who require special attention.

PATIENT-IDENTIFIED BARRIERS TO TAKING ANTI-HYPERTENSIVE MEDICATIONS IN AFRICAN-AMERICANS. G Ogedegbe, J Allegrante, C Mancuso, M Charlson, Medicine, Weill Medical College of Cornell University, New York, NY

PURPOSE: Hypertension is a highly prevalent disease in African-Americans. Identifying issues hindering patients from taking their medications as prescribed is essential in order to address the issue of poorly controlled blood pressure in this patient population.

The purpose of this study was to identify the barriers faced by a group of hypertensive African-American patients in taking their prescribed anti-hypertensive medications.

METHODS: One hundred African-American patients with hypertension followed at an internal medicine primary care practice participated. Computerized medical records were reviewed to identify eligible patients and to obtain demographic data, comorbidity data, and blood pressure. In-depth interviews were conducted with each patient either at the time of ambulatory clinic visits or by telephone. During the interviews, patients' experiences with anti-hypertensive medications and their perceptions about the challenges they face in taking their medications were explored. Some of the open-ended questions asked included: 1) What difficulties do you think people with high blood pressure have in taking their medications? 2) What has been your experience with taking high blood pressure medications? 3) Are there things or situations that make it difficult or easier for you to take your blood pressure medications? and 4) What do you think are the potential risks and barriers of taking your high blood pressure medications? All responses were recorded verbatim during each interview and then sorted and categorized using qualitative analysis.

RESULTS: Of the 100 patients interviewed, mean age was 55 years (range 28-80), mean years of education was 13, and 60% were women. The mean systolic blood pressure was 140 mmHg, and the mean diastolic was 85 mmHg. The most important issues identified by patients that hindered them from taking their medications as prescribed were: forgetfulness (28%), side effects (21%), dislike for pills (19%), and inconvenience (13%). Others included: fear of dependence (10%), swallowing big pills (10%), absence of symptoms (7%), reluctance to take their medications in social situations (6%), cost (4%), preference for folk remedies (4%), unable to refill prescription in a timely manner (4%), and too busy to take medications (3%).

CONCLUSION: Patients cited multiple, varied issues that hindered them from taking their anti-hypertensive medications. Identifying these issues will guide the development of specific interventions aimed at helping patients overcome them and thus lead to better control of their hypertension with improved long-term outcomes.

COMMUNICATION WITH PATIENTS ABOUT SEXUAL RISK-TAKING: IT'S THE RELATIONSHIP! B Ogur, D Stone, D Hatem, Internal Medicine, Cambridge Health Alliance, Cambridge; Internal Medicine, University of Massachusetts, Worcester, MA

PURPOSE: The purpose of the study was to better understand the cognitive, affective, situational, and motivational aspects of the practice of safe and unsafe sex among those at risk for HIV infection through in-depth interview, in order to provide insight into possible implications for provider-patient communication about risk reduction.

METHODS: 20 in-depth, semi-structured, open-ended interviews were conducted with patients with evidence of high-risk sexual behavior. Subjects were selected from patients from the investigators' clinical practices with self-report of an STD, multiple sexual partners, or unwanted pregnancy in the past 5 years, or with patient expressed concerns about consequences of recent unsafe sexual activity. Interviews were audio-taped, transcribed and analyzed independently by each coinvestigator. Major themes were identified by iterative consensus coding.

RESULTS: In almost every instance, subjects described both safe and unsafe sexual experiences contextually in terms of their relationship with the sexual partner. Subjects did not spontaneously focus on specific sexual behaviors, nor on condom use, not even with specific questioning. Subjects frequently articulated a hierarchy of relationships, with condom use having different meanings and requiring different styles of negotiation in different types of relationships. In relationships "just for sex", talking about safer sex was perceived as more interpersonally intimate than actually having intercourse, which often occurred without verbal communication. In these situations, patients who were successful at talking to their partners described the use of rule-based ("I always use protection"), or, alternatively, more playful forms ("No party-hat, no party!") of communication, avoiding the need for intimate communication and the potential for causing offense to the partner. In more committed relationships, the notion of "trusting" the partner frequently took the place of careful risk assessment, resulting in unsafe behavior. Paradoxically, in these situations, patients described the communication about safer sex as potentially destructive to intimacy.

CONCLUSION: Counseling which is focused only on behavioral aspects of sexual risk-taking may ignore the major significance of the meaning of the interpersonal interaction in which sexual behavior occurs. Helping patients to understand their own framework of interpersonal meaning and how they construct condom use within relationships, and assisting them in anticipatory communication appropriate to the type of sexual relationship in which they are engaged may be a more productive way to help patients reduce their unsafe practices.

MINIMIZING AND MEASURING IDIOSYNCRATIC INTERPRETATIONS OF QUALITATIVE DATA. HS Perkins, AA Meny, HP Hazuda, Medicine, University of Texas Health Science Center, San Antonio, TX; English, Harvard College, Cambridge, MA

PURPOSE: Because qualitative data raise concerns about idiosyncratic interpretation, techniques to minimize and measure it must be developed. We describe one here.

METHODS: In analyzing patient interviews about dying, we wanted responses to speak for themselves, as undistorted by coder bias as possible. So we designed a two-stage content analysis to minimize interpretive bias and measured residual idiosyncrasy in the results. Seeking coders likely to bring different perspectives to the analysis, we chose four coders with different genders, ethnicities, and professional disciplines. We tried to promote both a balanced perspective and an interpretive synergy by requiring at each stage two of three coders to read responses blindly, independently, and randomly and then to reach consensus about interpretation. The fourth coder decided persistent disagreements.

RESULTS: In the first stage of analysis, coders A and B read all interviews and devised a thematic coding scheme from subjects' own words. The scheme was conservative: Themes suggested by one coder made the scheme only if agreed to by the other. The final saturated scheme had 81 themes. A agreed to 55% of themes suggested by B, and B agreed to 50% suggested by A. These modest agreement rates show the dependence of theme selection on individual coder perspective and, thus, the need for multiple coders and iterations in devising a coding scheme. In the second stage, coders A and C applied the coding scheme to each interview. Coder A ensured consistency of theme concept from the first stage; coder C, a fresh perspective on the interviews and theme meanings. Definitive determination of theme presence required agreement by two coders. A and C agreed on 97% of final determinations of theme presence. The other 3% required Coder D to side with either A or C. Final determinations identified any given theme in 3 to 325 responses to questions. These determinations agreed with 86% of A's initial codings and with 79% of C's, indicating substantial interpretive consistency between coders.

CONCLUSION: Thus, qualitative studies can build in steps to minimize and measure idiosyncratic interpretation, and reporting such steps should become standard for these studies.

CULTURAL DIVERSITY AND ADVANCE CARE PLANNING. HS Perkins, A Gonzales, CMA Geppert, JD Cortez, HP Hazuda, Medicine, University of Texas Health Science Center; Administration, Bexar County Hospital District, San Antonio, TX; Psychiatry, University of New Mexico, Albuquerque, NM; Intercultural Development Research Association, San Antonio, TX

PURPOSE: A patient's satisfaction with critical care depends on implementing the patient's wishes. But those wishes vary widely and seem unpredictable so far. We suspect culture underlies many wishes about critical care and can serve as a guide for advance care planning. Thus, we explored patients' culture-based wishes about critical care and dying.

METHODS: We interviewed 26 Mexican Americans (MAs), 18 Euroamericans (EAs), and 14 African Americans (AAs).

RESULTS: Content analysis revealed some themes that were common to all groups and likely reflected core American culture. For example, majorities of all groups (69% of MAs, 78% of EAs, and 50% of AAs) believed patients deserve a role in determining their care. Majorities (92% of MAs, 83% of EAs, and 79% of AAs) also liked advance directives as one form of advance care planning. But other themes, likely reflecting specific ethnic subcultures, distinguished the groups. For

example, most MAs (58%) believed the medical system controls treatment but trusted the system to serve them well (62%) and wanted no life support when terminal (50%). Most EAs (78%) believed patients—not the system—control treatment but trusted the system to serve them well (72%) and had defined for themselves unacceptable outcomes (50%). Most AAs (64%) believed the system controls treatment, few trusted the system to serve them well (36%), and most believed a patient should not express treatment wishes until very sick (50%).

CONCLUSION: Thus, advance care planning must take account of both core American cultural values and specific ethnic cultural values. Core values support advance care planning in general, but such planning should be tailored to specific ethnic views about the medical system, its trustworthiness, and terminal care.

A BEDSIDE DIAGNOSIS OF PULMONARY ACTINOMYCOSIS. C Pietrantonj, M Budev, T Mekhail, W Tomford.

LEARNING OBJECTIVES: 1. The bedside diagnosis of Actinomycosis.

2. Recognize the indolent progression of this infection.

3. A low incidence of clinical recognition.

CASE PRESENTATION: A 67 year-old alcoholic presented with a rapidly expanding chest "lump" sustained after a contusion 2 weeks prior. Examination revealed a disheveled male with numerous dental caries and a circumferentially raised 4-cm, non-pulsatile, indurated, and erythematous, para-sternal lump. CT scan of the chest revealed a left upper lobe consolidation with extension of a similarly attenuating inflammatory mass into the soft tissues of the anterior mediastinum and chest wall, without osseous involvement. A prompt bedside needle aspirate of the lesion and microscopic examination by the house-staff yielded the presence of pathognomonic "sulfur granules". Eventually, cultures confirmed *A. israelii* with a few anaerobic gram-negative and gram-positive bacilli. Further surgical drainage was initially considered, however serial aspirations resulted in partial resolution of the "lump". Four weeks of metronidazole and high dose intravenous penicillin-G were given, followed by 12 months of oral penicillin. One year later, the lesion completely resolved without complications.

DISCUSSION: *A. israelii*, a non-acid fast, gram-positive, filamentous anaerobic bacteria, is the most common agent of pulmonary actinomycosis. It should be suspected in chronic, infiltrative, cavitary, and tumor-like mass lesions of the pulmonary parenchyma. Fundamental bedside techniques can assist to establish an immediate diagnosis, without awaiting culture assays. Chest wall lesions consist mainly of granulomatous tissue with extensive fibrosis and small abscesses, originating as a direct invasion from a subpleural lesion. Since once the organism is established locally, infection spreads contiguously in a progressive manner without regard for anatomical barriers. Sinus tracts will often extend from the abscess to the skin or adjacent organs. Overlying skin may suppurate, forming tracts that spontaneously close and reform. Cervicofacial infection is most common, followed by abdominopelvic infection, while pulmonary actinomycosis only contributes to 15 percent of cases. Our patient manifested two pathognomonic features of thoracic actinomycosis, extension to the chest wall and conglomerations of filamentous rods in mineralized tissue, called "sulfur granules". An animal model of infection has not been established with Actinomycosis spp. alone, infection by co-pathogens supports the concept that additional microbes are important for the initiation of infection. Most mild cases are eradicated by the common use of antibiotics, thereby contributing to the overall rarity and indolent course of this infection. The diagnosis of Actinomycosis is truly the mark of a vigilant clinician.

AN ETHNOGRAPHIC STUDY OF PATIENTS' CAUTIONARY VIEWS ABOUT INTEGRATION OF SPIRITUALITY INTO THEIR MEDICAL CARE. J Ramirez, D Castro, LK Loo, LE Skoretz, DL Stottlemeyer, Faculty of Religion, Loma Linda University, Loma Linda, CA; Medicine, Loma Linda University, Loma Linda, CA

PURPOSE: Spirituality and prayer are important to most people. Two-thirds of the American public believe physicians should talk to their patients about spiritual issues and pray with those who request it. Through the use of ethnographic, in depth interviews, patients were invited to discuss their perspectives on the integration of spirituality into their current medical care.

METHODS: This study was the qualitative component of a larger survey research project addressing the same purpose. 600 General Internal Medicine patients from University, Veterans, and County hospitals agreed to complete the survey. From this group, a sample of convenience of 15 inpatients agreed to in-depth interview.

RESULTS: A total of 71 transcribed pages produced 16 categories. From this matrix of 113 category entries emerged a prominent theme of "patients' cautionary words about integration of spirituality into medical care." 23 cautionary statements were organized into 7 areas of concern. First, patients perceived the need to find compatibility between the physician's meaning of prayer and their own belief system—"I don't know what kind of religion they are [MDs] . . . that would matter to me." Second, before prayer is offered the physician would be expected to asked for permission. Third, for spiritual care to be meaningful "it cannot be something that is going to be written in a textbook and you follow A, B, C because every one is different, both physician and patient." The fourth and fifth areas of concern are seemingly contradictory perspectives. Introducing prayer could be interpreted as a statement of "how bad" the patients' prognosis "really is." On the other hand, when the situation is desperate and there are seemingly no medical options, prayer is welcomed and accepted even by those who might reject it otherwise. The sixth caution was for physicians not to forget that they can use prayer as a vehicle for promoting hope and "positive feelings." The last cautionary statement was the

perceived need for a spiritual ambiance that includes cleanliness, religious paraphernalia (like crucifixes and paintings), and a welcoming spiritual milieu.

CONCLUSION: When inviting patients for prayer, physicians are expected to be personal and meaningful not generic or emasculated from a particular faith tradition. At the same time physicians need to carefully avoid specific doctrinal statements that might be offensive to patients. This process needs to begin by a systematic informed consent that forewarns the patient. The need to communicate hope is well served by prayer and spirituality; even those who would reject a physicians' prayer may welcome prayer from a different source.

PROMOTING EMERGENCY DEPARTMENT USE OF A TRIAGE DECISION AID FOR PATIENTS WITH CHEST PAIN: PERSUASION OR COERCION? B. Reilly, R. Roberts, J. Schaidler, A. Evans, L. Moran, K. Das, E. Martinez, J. Calvin, Medicine; Emergency Medicine, Cook County Hospital; Rush Medical College, Chicago, IL

PURPOSE: We describe the change process used to implement a new standardized triage decision aid for patients admitted from our Emergency Department (ED) with suspected acute cardiac ischemia (ACI).

METHODS: In an effort to reduce very low risk hospital admissions, an interdepartmental team (Emergency Medicine, Medicine and Cardiology) developed a triage decision aid based on a previously validated prediction rule (Goldman, NEJM 1996). Prior to beginning a 14-week prospective study of its safety and efficiency, we set as a minimum threshold 80% use of the decision aid by ED physicians. During a 17-week "phase in" period in 1999, we measured use of the decision aid for all patients meeting eligibility criteria for the study. We also measured changes in number of chest pain admissions to our Observation (OBS) Unit compared with similar weeks in 1998. In weeks 1-8, ED physicians received frequent feedback and encouragement about use of the aid. In weeks 5-8, non-MD ED staff received similar interventions. In weeks 9-13, ED staff were instructed that decision aid use was required before an inpatient or OBS bed could be assigned to admitted patients. In week 14, the Hospital Director personally delivered that same message.

RESULTS: The Table illustrates progressive increases in use of the triage aid after sequential interventions during the 17-week phase-in period and its sustained use during the study period.

Time Period (1999)	Eligible Patients N	Triage Aid Use mean weekly % (range)	Change in Number of ACI Patients Admitted to OBS Unit (compared to same weeks in 1998)
Week 1-4	325	50% (48-53)	+69%
Week 5-8	331	59% (49-68)	+12%
Week 9-13	381	73% (60-85)	+36%
Week 14-17	293	89% (83-94)	+125%
Week 18-31	1042	82% (58-93)	+79%

CONCLUSION: Major institutional changes in the process of clinical care may require both collaborative persuasion and constructive coercion. Impact on outcomes merits more study.

CONCEPTUALIZING SPIRITUAL WELL-BEING: A FOCUS GROUP STUDY OF OLDER ADULTS WITH CHRONIC ILLNESS. KE Rosenfeld, J. Steckart, R. Patterson, Medicine, VA Greater Los Angeles Healthcare System, Los Angeles, CA

PURPOSE: Spiritual well-being has recently been recognized to be important to chronically ill individuals health-related quality of life (HRQOL). This qualitative study sought to construct a conceptual model for spiritual well-being from the perspective of older adults with chronic illness.

METHODS: Eight focus groups of older individuals with chronic illness were conducted. A diversity of perspectives was sought through stratification by one or more of the following variables: age (65-74, >74 yo); race (Caucasian, African-American, Spanish-speaking Latino); chronic illness category (heart disease, lung disease, arthritis, visual impairment). Participants were asked about the meaning of spiritual health; the characteristics of spiritually healthy people; what they did to maintain spiritual health; and how aging and chronic illness impacted on spiritual health. Focus groups were audiotaped and transcribed verbatim (and translated for Spanish-speaking groups), then coded for thematic structure by two independent coders trained in content analysis. "Trustworthiness" (the qualitative equivalent of validity and reliability) was sought through checks on interrater reliability, respondent feedback to the findings, and concurrence by a panel of spirituality experts.

RESULTS: Participants expressed a wide range of perspectives on spiritual health. Many participants equated spirituality with formalized religious commitment, although a variety of non-religious elements were also identified. Analysis of the focus group transcripts yielded themes in five major areas, with substantial overlap between them: 1) spiritual orientation (e.g., religious or spiritual practices, connection to God); 2) existential state (e.g., meaning and purpose in life, attitude towards death); 3) interpersonal relationships (e.g., loving and being loved, helping others); 4) outlook (e.g., "glass half full") and 5) affect (e.g., inner peace, joy). Representative quotations for each of these theme areas will be presented.

CONCLUSION: Spiritual health is an important dimension of older, chronically ill individuals' HRQOL. Contrary to historical attempts to define spirituality in terms of its religious elements, our analysis suggested that a definition that includes its psycho/developmental elements may better represent patients' own models of spiritual health. Several theme areas identified in this study (e.g., affect, interpersonal relationships) correspond to previously identified HRQOL dimensions (emo-

tional well-being, social well-being), although other identified areas (spiritual orientation, existential, positive outlook) are not currently represented in HRQOL assessment tools. Expanding the conceptualization of HRQOL to include spiritual well-being may be appropriate in studies of HRQOL in aging and chronic illness.

A QUALITATIVE ASSESSMENT OF MEDICAL DECISION-MAKING IN AN INNER-CITY HOSPITAL. AM Torke, GM Corbie-Smith, WT Branch, Internal Medicine-Primary Care Residency; Division of General Medicine, Emory University School of Medicine, Atlanta, GA

PURPOSE: Although medical decisions are made daily between physicians and their patients, we know little about the effect of culture on this process. As a first step in understanding possible cultural influences on decision-making, we explored the opinions and attitudes of African American patients in an inner-city hospital as they considered undergoing a medical procedure.

METHODS: We conducted in-depth, semi-structured interviews with patients at an inner-city, hospital-based medicine clinic who were considering flexible sigmoidoscopy as a screening test for colon cancer.

RESULTS: We interviewed twenty-six African American subjects. Subjects named various reasons for undergoing the flexible sigmoidoscopy, including recommendations of their physicians, fear of cancer, and the desire to detect medical problems early. Most patients wanted detailed information about the test, especially information about pain and safety, the purpose of the test, and test results. Subjects overwhelmingly named their doctor as the most trusted source of health-care information. While full disclosure was extremely important to the subjects, most wanted the decision to be made in part or full by their physician. Trust in the physician was thought to be as important or more important than knowing risks and benefits of the procedure. Most subjects stated that the main purpose of the consent document was to protect the physician in case something went wrong. All subjects stated that they would feel comfortable refusing a medical test that they did not want.

CONCLUSION: In current ethical frameworks, autonomy is regarded as the primary principle of medical decision-making. Although patients in our study wanted extensive information about their medical care, most willingly deferred the decision-making to the physician. This voluntary surrender of the decision-making role raises important questions about the value placed on autonomy by African American patients in an inner-city clinic.

DIRECT-TO-CONSUMER PRESCRIPTION DRUG ADVERTISING: WHAT'S BEING SOLD? J. Tremmel, S. Woloshin, L. Schwartz, Department of Medicine, Dartmouth-Hitchcock Medical Center, Lebanon, NH; VA Outcomes Group, VA Medical Center, White River Junction, VT; Center for the Evaluative Clinical Sciences, Dartmouth Medical School, Hanover, NH

PURPOSE: The \$1.8 billion spent on direct-to-consumer advertisements by pharmaceutical companies now exceeds spending on advertising to physicians. We examined which products pharmaceutical companies are advertising to consumers and what messages are being communicated.

METHODS: We performed content analysis of advertisements (excluding the FDA required PDR-like "brief summary" page) appearing in 10 widely read magazines: 4 women's magazines (Family Circle, Ladies Home Journal, Better Homes & Gardens, Family Circle), 3 men's magazines (Sports Illustrated, Men's Health, Gentlemen's Quarterly) and 3 with general readership (Time, Newsweek, People). We examined the first issue of every other month during the year July 1998-July 1999.

RESULTS: Direct-to-consumer advertisements were about 3 times more common in magazines targeting women compared to those targeting men or a general readership (mean number of advertisements per issue: 5.1 (women's); 1.5 (men's) and 1.6 (general)). The products most commonly advertised were those designed to address symptoms (e.g., sneezing/runny nose, urinary incontinence, hair loss, impotence, menopausal symptoms) followed by products for treatment (e.g., Alzheimer's, HIV, diabetes, otitis media) or prevention (Lyme vaccine, bone density testing, breast cancer). Most advertisements described the benefit of a medication in vague, qualitative terms (e.g., "Zyrtec works fast") and made an emotional appeal to readers, most commonly to their desire to get back to normal (61%), to make their life better than it is (28%), or to avoid some feared outcome (21%). Half of the advertisements encouraged people to consider a medical cause for their experiences ("is it just forgetfulness, or is it Alzheimer's Disease?"). Only 14% of the advertisements described the benefit of the medication with data (e.g., "66% had visible hair growth"). In contrast to information on benefit, potential side-effects of medications were more often described with data (53%), for example, "side effects occurred in less than 2% of men". No advertisements mentioned cost.

CONCLUSION: Direct-to-consumer advertisements present little information to help patients make informed decisions but may stimulate demand for products by implying benefit, generating emotional responses, and encouraging consumers to medicalize ordinary (unpleasant) experiences. A substantial proportion of advertisements also invite patients to share in disease management decisions such as the choice of antibiotic or type of insulin. Whether this approach is in the best interest of the patient or the physician-patient relationship is unknown.

CONCEPTS OF HEALTH AND PERCEPTIONS OF BARRIERS TO HEALTH CARE AMONG AFRICAN AMERICAN MEN. EE Whitaker, WE Johnson, S. Avila, S. Thomas, J. Ravenell, T. Fisher, J. Ryu, LR Murray, RWJ Clinical Scholars Program; School of Social Services Administration, U. of Chicago; Trauma Department, Cook County Hospital; Woodlawn Adult Health Center, Chicago, IL

PURPOSE: African American men in the United States shoulder a disproportionate burden of disease and death compared to the rest of the population. Sadly, many of these men have ambulatory sensitive conditions such as hypertension, diabetes, and HIV/AIDS for which they often do not receive appropriate primary health care.

This study was conducted to 1) assess African American men's concepts of health and 2) perceived barriers to receiving health care.

METHODS: Eight focus groups, with 10 participants on average, were conducted with African American men in the Woodlawn community of Chicago. Men were divided into sub-groups including adolescents, elderly men, gay/bisexual men, and HIV positive men. Focus groups were audiotaped, transcribed and reviewed by two researchers to identify dominant themes.

RESULTS: Men's concepts of health included traditional health messages regarding diet and exercise but also a recognition of factors ranging from the social environment to unemployment that affect health. They perceived a strong link between fitness of mind and of physical health. Many men described spirituality as an important determinant of their health status. The men reported a number of barriers to care including 1) lack of respect displayed by physicians and other health facility staff, 2) the clinic setting which often includes women and children or long waiting times and 3) seeking care may lead to their being seen as weak or vulnerable.

CONCLUSION: This is one of the first studies to report African American men's own perceptions of barriers to care. Analysis of the participants' insights suggests that African American men recognize basic health issues and feel health effects from their social environment and unemployment. Additionally this study points to possible remedies to systematic barriers which have been proposed by the target population.

LISTENING TO EMPLOYEE BENEFITS DECISION-MAKERS DISCUSS HEALTH PLAN QUALITY: FOCUS GROUPS WITH POTENTIAL USERS OF PERFORMANCE MEASURES. MK Wynia, DS Cummins, GI Balch, EA Balch, Institute for Ethics, American Medical Association, Chicago; Balch Associates, Oak Park, IL

PURPOSE: Performance measures are often reported with individual consumers as the intended audience. Yet employers make most health plan choices when they select one, or a few, plans to offer their employees—making them important potential recipients of information on plan quality. This study explored employers' use of information on health plan quality.

METHODS: In December 1998 and January 1999 we conducted 6 national Computer Assisted Telephone focus groups, with 52 individuals responsible for selecting health care packages. There were 2 groups each from small (10-99 employees), medium (100-999 employees) and large (>1,000 employees) businesses. All participants were from areas with substantial competition among health plans and all offered their employees at least one plan. We asked what information on plan quality they used to select plans and where they obtained that information.

RESULTS: The first decision employers face each year is whether to alter the plan(s) offered. Making a change represents a substantial burden. Even among large employers, a very small group is responsible for making this decision and selecting plans to offer. Employee complaints and cost are the primary factors considered. Few actively seek employee input (this is "looking for trouble"), but most track customer service complaints carefully and assume acceptable quality in the absence of complaints. They rarely look at satisfaction or other traditional plan quality measures prior to making selections and they see report card information as incomparable across plans, confusing, or edited to favor a plan that presents it. Very few have heard of the most widely used quality reporting instrument, HEDIS. Experiences with employer purchasing groups are mixed. Almost all place great weight on the recommendations of an insurance broker both to suggest good plans and to "weed out" or "flag" worse quality plans. Larger employers sometimes use consultants rather than brokers for this purpose. Brokers often compile information and negotiate directly with plans, but usually also receive commissions from plans. Other trusted information sources are human resources industry peers and trade journals.

CONCLUSION: Most employers display little interest in conventional quality measures as they are now presented. Instead, they use employee complaints, their peers and insurance brokers or consultants to assess health plan quality. Insurance brokers and consultants may turn out to be the most important recipients of health plan quality report cards.

THE EAGLE AND THE MIRROR: THE ASSESSMENT OF METAPHORS FOR PATIENTS WITH HEART FAILURE, CANCER AND HEPATITIS C. SL Zickmund, DR Labrecque, R Oren, Internal Medicine, University of Iowa, Iowa City, IA

PURPOSE: Interactions between patients and physicians focus on information that may provide limited insight into the impact of the disease experience. To explore alternatives, we asked patients to metaphorically describe their experience.

METHODS: Patients with heart failure (CHF; n = 74; 55.0 ± 1.8 years; 33% women), cancer (C; n = 57; 50.5 ± 5.2 years; 63% women), and hepatitis C (HC; n = 74; 44.9 ± 1.3 years; 26% women) were randomly chosen to depict themselves as a living or inanimate object before and after their diagnosis. The answers were coded in five categories by a blinded investigator.

RESULTS: A thematic analysis revealed no gender differences. However, several distinctions emerged when comparing the groups. The majority of patients with heart failure (78%) or cancer (72%) used metaphors suggesting a worsening of their lives. For example, one man described himself as a soaring eagle before the

disease. Afterwards, he saw himself as an eagle caught in an oil slick. In contrast, half of the patients with hepatitis C (50%) depicted the disease as an enriching, positive influence. One woman described herself as a runaway train, out of control, before the diagnosis. She became a train on a life-saving track afterward. Metaphors related to a lack of energy or to slowing down predominated in the CHF patients (CHF:59%; C:16%; HC:22%). These included viewing oneself as a fast sports car only to become a scooter. About one third of the patients with hepatitis C depicted the disease as a chance to reflect or regain control of their lives, as expressed by the mirror which "makes me look at myself, change things I'd like to change" (CHF:0; C:4%; HC:30%). Cancer patients exhibited the least thematic consistency, articulating their experience as (1) being wounded such as an "injured dog helpless on side of road" (CHF:3%; C:19%; HC:6%) or (2) as revealing the deterioration of aesthetic dimensions, such as becoming a "tulip with no petals" (CHF:0; C:19%; HC:4%). Assessing the influence of age on the choice of metaphors demonstrated that individuals depicting a decrease in energy were older (54.8 ± 1.6 years) compared with those experiencing positive effects of the illness (45.7 ± 2.3 years). This was due to the younger age of patients with hepatitis C who chose metaphors indicating an increase of control as they started treatment protocols.

CONCLUSION: The metaphoric description of experiences provides a powerful tool, allowing patients to highlight the impact of illness on their lives. Despite differences in age and gender, patients with chronic diseases of heart and liver embrace similar thematic conceptualizations of their illnesses. While the answers of patients with malignant diseases did not reveal a predominant theme, the majority of individuals used images related to aesthetics or existential threat.

CLINICAL VIGNETTES

SARCIDOSIS PRESENTING AS CHRONIC COUGH. EM Aagaard, Internal Medicine, University of California, San Francisco, San Francisco, CA

LEARNING OBJECTIVES: 1) Evaluate a patient presenting with chronic cough. 2) Recognize the clinical manifestations of sarcoidosis, including associated medical illnesses.

CASE PRESENTATION: A 51 year old metal and wood shop teacher with a past medical history notable for hypertension, gout, nephrolithiasis, and a positive tuberculin skin test presented to the clinic with a chief complaint of cough of greater than 2 months duration. His cough was non-productive and associated with moderate fatigue, such that he was unable to participate in his favorite hobby, gardening. He denied any other recent upper respiratory symptoms, including nasal congestion, rhinorrhea and postnasal drip. He had occasional sour taste in his mouth and heartburn. He denied any fever, chills, night sweats, weight loss, chest pain, wheezing, or shortness of breath.

Past medical history was as noted above. He had recurrent nephrolithiasis, with calcium oxalate stones. He had mild renal insufficiency believed to be due to his nephrolithiasis. His positive tuberculin test was noted as a child. He did not receive therapy, but was followed with serial negative chest x-rays, the last of which was less than 1 year prior to presentation. His only medication was amlodipine. He had no significant family history. He was married, sexually monogamous, and without a history of tobacco, drug or alcohol use. He had no recent travel history. On physical examination, he was well-appearing with a frequent cough, but in no respiratory distress. He was afebrile, with a blood pressure of 160/94. His exam was otherwise normal. He was initially treated with a proton pump inhibitor for presumptive gastroesophageal reflux disease. However, given his history of a positive tuberculin skin test and possible toxin exposure through his work as a wood and metal shop teacher, a chest x-ray was ordered. In addition, because of his significant fatigue, a laboratory panel including complete blood count, renal function, liver function, thyroid function, calcium, and glucose were ordered. Blood tests were notable for a hematocrit of 40.2% and a creatinine to 3.3 mg/dL (up from 1.3 mg/dL 6 months prior). Other lab tests were normal. Chest x-ray revealed increased reticulonodular densities in the lower lobes. A renal ultrasound showed no evidence of hydronephrosis, but was notable for bilateral renal calculi, early medullary nephrocalcinosis, mild splenomegaly, and multiple enlarged peripancreatic and retroperitoneal lymph nodes. A thin section chest CT showed multiple nodules scattered diffusely throughout both lungs along the perivascular bundles with associated interlobular septal thickening. A bronchoscopy with transbronchial biopsy was performed revealing granulomas with multinucleated giant cells. Cultures and special stains were negative. The patient was diagnosed with sarcoidosis.

BILATERAL INTERNUCLEAR OPHTHALMOPLÉGIA IN THE PRIMARY ANTIPHOSPHOLIPID SYNDROME. JI Ameh, MT Choksi, A Munir, I Ashley, F Rosner, Medicine, Mount Sinai Services at Queens Hospital Center, Jamaica, NY

LEARNING OBJECTIVES: Recognize Primary Antiphospholipid Syndrome (APLS) presenting as bilateral internuclear ophthalmoplegia (BINO).

CASE: A fifty-two year old Haitian woman was hospitalized because of gait instability for one day. Past medical history included a spontaneous abortion 20 years

ago, a cerebrovascular accident (CVA) with central loss of vision in 1997, and a right lacunar infarct in the internal capsule in 1998 with residual left upper extremity paresis. She was on ASA 81mg daily since 1997. Her vitals were normal. The significant findings were gait instability, left upper extremity mild monoparesis, left pronator drift, left facial paresis and mild dysarthria. Computerized Tomographic (CT) scan of the brain showed a lacunar infarct in the anterior limb of the right internal capsule. Pertinent labs were aPTT 26.9 s, IgG anticardiolipin Ab 42.6 (normal <23 mcg/ml). The patient was anticoagulated with heparin. MRI on the following day showed an area of hypo-attenuation in the pons and medulla suggestive of an infarct. The next day, the patient developed bilateral Internuclear Ophthalmoplegia with worsening dysarthria and left facial palsy, followed within 24 hours by dysphagia and left hypoglossal palsy. On the fourth day of hospitalization, the patient developed acute respiratory distress and hypoxic respiratory failure. The patient was intubated and transferred to the MICU. A repeat CT scan of the brain did not reveal intracranial bleeding.

DISCUSSION: APLS manifests clinically as arterial or venous thrombosis, recurrent fetal loss, and/or thrombocytopenia in patients with persistent elevation of antiphospholipid antibody. Cerebral ischemia is the most common arterial thrombotic manifestation, which may be recurrent. Consideration of the combination of the pertinent clinical and laboratory findings make the diagnosis of APLS. CVA especially thrombotic in young patients, or recurrent arterial or venous thrombo-occlusive episodes in the absence of an obvious etiology should prompt an evaluation for the APLS. The tests that strongly suggest the presence of antiphospholipid antibodies—*a*) Lupus anticoagulants (LA) *b*) anticardiolipin (aCL) antibodies *c*) Beta-2 glycoprotein 1. Multiple tests need to be performed if the clinical index of suspicion is high since a single test will detect only 60–80% of cases. General aspects of treatment consist of anticoagulation with heparin followed by warfarin. All patients with antiphospholipid syndrome and a major thrombotic event should have lifelong anticoagulation with warfarin to maintain an INR of 3–4. Recognition of recurrent thrombotic events is an indication, which should lead a physician to think APLS in differential diagnosis. BINO is a usual manifestation of multiple sclerosis but rarely it could be of APLS.

SPONTANEOUS INTRAABDOMINAL VARICEAL BLEED—AN ATYPICAL PRESENTATION OF PORTAL VEIN THROMBOSIS. *A. Anand*, DC Wolf, Medicine, Westchester Medical Center, Valhalla, NY

LEARNING OBJECTIVES: Hypercoagulable state and its work-up, presentation and management.

CASE PRESENTATION: A 26 year old female with no significant past medical or surgical history, presented with a one week history of acute epigastric pain radiating to the back. The pain was continuous and associated with anorexia and night sweats. She denied nausea, vomiting or bowel complaints. On further questioning the patient noted a use of oral contraceptives for 2 years which she had stopped 2 months prior to presentation. A contrast CT scan of the abdomen was obtained, which revealed portal and superior mesenteric vein thrombosis, with patent hepatic and splenic veins. In addition, a mass was seen displacing the pancreas and associated vascular structures. An exploratory laparotomy was performed which revealed the mass to be a hematoma with reaction. Histopathology revealed fibroblastic proliferation. A work up to rule out hypercoagulable state including protein C, protein S, anti-thrombin III, factor V leiden, prothrombin 20210 and homocysteine was negative. Work-up for dysfibrinogenemia, polycythemia, antiphospholipid antibody syndrome, paroxysmal nocturnal hemoglobinuria was negative. The patient was anticoagulated with heparin and later with coumadin. A repeat CT scan revealed progression of the thrombosis to involve the splenic vein. A follow up CT scan a week later did not show any further progression of the thrombosis. An esophagoduodenoscopy showed grade II esophageal and gastric varices. The patient was discharged on nadolol and coumadin. On follow up the patient has been asymptomatic.

DISCUSSION: Portal vein thrombosis is idiopathic in 50% of cases. The most common presentation is gastrointestinal bleeding from gastric and esophageal varices. Acute portal vein thrombosis was described in 2 patients who presented with acute abdomen, but this is the first known case of portal vein thrombosis complicated by intraabdominal variceal bleed. Treatment is conservative with anticoagulation, although endovascular stenting and thrombolysis through a percutaneous transhepatic catheter have been reported in a few cases. Beta blockers are known to reduce incidence of variceal bleed and improve long term survival.

A PRIMARY CARDIAC LYMPHOMA AND A SUCCESSFUL USE OF A NOVEL THERAPEUTIC APPROACH. *N. Aghajanian*, AG Rosmarin, A Sadaniantz, Cardiology, The Miriam Hospital; Brown University School of Medicine, Providence, RI

LEARNING OBJECTIVES: 1) Primary cardiac lymphomas (PCL) involve the heart and pericardium and are often high grade B-cell lymphomas. PCL is mostly found in the right side of the heart with the majority of tumors involving the right atrium. 2) Diagnosis of PCL is still a challenge. Radiologic, histologic and cytologic modalities are used to diagnose PCL. 3) We report this unusual case and show a unique surgical method used to prevent catastrophic complication of right atrial wall perforation following initiation of chemotherapy.

CASE PRESENTATION: A 61 year old male with history of hypertension, presented with a three month history of cough, low grade fever, chills and shortness of breath. The patient had a twenty-year smoking history but quit seven years ago. He drank alcohol on occasions. On initial examination, the temperature was 101 F, blood pressure of 146/68 and pulse regular at 110/minute. Jugular venous pres-

sure was 10 cm. His lungs were clear. Auscultation showed normal heart sounds and no murmurs. Chest x-ray was non-contributory. EKG showed nonspecific inferior T-wave abnormalities. A transthoracic echocardiogram showed an unusual diastolic and systolic color Doppler jet in the vicinity of the tricuspid valve. A transesophageal echocardiogram showed a 4.0 × 4.5 cm right atrial mass which was obstructing flow from right atrium to right ventricle. Next day, at surgery a large tumor mass originating from the free wall of the right atrium above the inferior vena cava was found. Pathological analysis of the right atrial mass, atrial tissue surface and pericardium showed diffuse, large cell, high grade non-Hodgkin's B cell lymphoma. Five days later a large bovine pericardial patch was placed over the right atrium in order to prevent cardiac rupture from chemotherapy. One week later, the patient was admitted for chemotherapy with CHOP. A repeat transesophageal echocardiogram three months following his initial admission found no evidence of tumor in the right atrium. However, six months later he developed central nervous metastasis and ultimately died from its complication.

DISCUSSION: PCL is extremely rare with an incidence of 0.25% of reported tumors of the heart of collected autopsy series. Death from PCL is typically related to direct infiltration of the heart by the tumor and not from metastasis. Surgery, chemotherapy and radiation have been tried as treatment for PCL. We describe the successful application of bovine pericardium to the heart of a patient with PCL. This surgical intervention prevented the likelihood of catastrophic cardiac rupture at the time of chemotherapy treatment for PCL.

THE RAMIFICATIONS OF THE ASSOCIATION BETWEEN GUILLAIN-BARRE SYNDROME AND THE INFLUENZA VACCINE. *BH Ashar*, Medicine, Johns Hopkins University, Baltimore, MD

LEARNING OBJECTIVES: 1) Recognize the association between the influenza vaccine and Guillain-Barre Syndrome, and 2) Recognize the impediment that the association may pose to attempts at widespread influenza vaccination.

CASE PRESENTATION: A 62-year-old patient with a history of cervical stenosis presented with a three week history of worsening numbness in her hands, arms, feet, and legs. She also noted progressive weakness that caused her to fall numerous times. She had attributed the symptoms to worsening of her cervical stenosis and subsequently did not seek treatment until she was unable to walk. Her physical examination was notable for 1/5 strength of all muscles of the lower extremities. The wrist extensors also showed 1/5 strength. There was a loss of position sense and proprioception in her hands and feet. Deep tendon reflexes were unable to be elicited. Laboratory data revealed a normal CBC, chemistries, creatinine kinase, and sedimentation rate. Examination of the cerebrospinal fluid revealed 2–3 WBCs per high power field and a protein level of 249 mg/dl. The diagnosis of Guillain-Barre Syndrome was made. She was treated with intravenous immunoglobulin for five days and referred for rehabilitation. Of note, the patient denied any recent upper respiratory or gastrointestinal complaints. During her hospitalization, the patient's husband stated that he had searched the Internet and that he was convinced that the flu shot that his wife had received four weeks earlier was responsible for her illness. He also reported "warning" his friends of the potential dangers of the vaccine.

DISCUSSION: The causal association between the influenza vaccine and Guillain-Barre Syndrome is controversial. A relative risk of 1.7 has been reported. This translates to about one additional case of Guillain-Barre per million persons vaccinated against influenza. Despite this low risk and the potential benefits of vaccination, physicians' efforts at influenza prevention may be hindered by the dissemination of anecdotal cases. These anecdotes may become more powerful with further expansion of the Internet.

AN UNCOMMON ETIOLOGY OF BRONCHIECTASIS: COMMON VARIABLE IMMUNODEFICIENCY. *R. Awad*, M Hanley, UG Mason, Denver Health Medical Center, Denver, CO

LEARNING OBJECTIVES: 1) Review the differential diagnosis of bronchiectasis. 2) Recognize immunodeficiency as an etiology of bronchiectasis.

CASE: A 42 year old female presented with a history of 3 to 4 episodes of upper respiratory tract infections and sinusitis for 10 years. She underwent sinus surgery 3 years earlier with no improvement of the frequency or severity of sinusitis. For the past 2 years, she had recurrent prolonged episodes of sinusitis and bronchitis. Her past medical history and family history were unremarkable. A physical examination was remarkable for bilateral tenderness of maxillary sinuses, swollen nasal mucosa and turbinates with evidence of post nasal drip. Her lung examination was remarkable for diffuse rhonchi. A chest radiography revealed diffuse bilateral interstitial infiltrates and diffuse airway thickening which was unchanged from 5 months earlier. A CT scan of the sinuses revealed pansinusitis. A CT scan of the chest revealed bilateral symmetric cylindrical bronchiectasis with scattered areas of mucus plugging. HIV antibodies were undetected. Measurement of serum immunoglobulins revealed subnormal levels of IgG <33 mg/dL (505–1364), IgM <11 mg/dL (67–354) and IgA <7 mg/dL (67–288). Three months ago patient started receiving intravenous immunoglobulin on a monthly basis and she has had no further episodes of sinusitis or bronchitis.

DISCUSSION: Bronchiectasis is described as an abnormal irreversible dilation of the bronchial airways, which is usually a consequence of inflammation and destruction of the structural components of the bronchial wall. The etiology of bronchiectasis is diverse and can be divided into 2 groups. Focal bronchiectasis can be caused by endobronchial tumors, foreign body or extrinsic compression. Generalized bronchiectasis can result from viral, fungal, bacterial and mycobacterial in-

fections. Less common causes include: cystic fibrosis, ciliary dysfunction disorders and immunodeficiency disorders. Generalized impairment of pulmonary defense mechanism occurs with immunoglobulin deficiency and leads to recurrent infections and generalized bronchiectasis as highlighted in this case.

RECOGNITION AND MANAGEMENT OF FECAL IMPACTION IN THE ELDERLY. AN Babu, JL Chenoweth, St. Louis VA Medical Center, St. Louis, MO

LEARNING OBJECTIVES: 1) Recognize the protean manifestations of fecal impaction. 2) Review important concepts in prevention and management.

CASE: A 55 year-old male was admitted with fever and change in mental status of 2 days duration. His past medical history was significant for viral encephalitis after which the patient has been demented, non-verbal and confined to bed or wheelchair. On examination, the abdomen was mildly distended and non-tender. A large, firm, oval, supra-pubic mass seemed to be arising from the pelvis. Rectal examination showed soft, brown stool negative for occult blood. A Foley catheter that had been placed in the Emergency Room had drained about 500cc of urine. A CT scan of the abdomen demonstrated a diffusely dilated colon. There was severe fecal impaction in the rectum, pushing anteriorly the decompressed, non-distended urinary bladder. The pelvic mass became completely non-palpable after the patient was successfully disimpacted with digital evacuation and multiple enemas. The patient's mental status returned to baseline and he defervesed. Fecal impaction did not recur during the hospital stay after bowel prophylaxis was instituted.

DISCUSSION: Fecal impaction is a significant problem in chronically invalid patients and the elderly. Major etiologies include neurological problems such as spinal cord injury, stroke or multiple sclerosis, medications (especially narcotics and anti-hypertensives) and anorectal pathology. The physical examination can reveal altered mental status, tachypnea and fever. The abdomen may be distended with tympany to percussion. Digital rectal examination is usually diagnostic, revealing impacted stool. However, the absence of a palpable impaction does not exclude a more proximal one. Additionally, the presence of fecal incontinence should arouse suspicion for fecal impaction. Common complications of impaction include intestinal obstruction and perforation. Unusual sequelae encompass megacolon, volvulus, anorectal fistulae, stercoral ulceration and lower limb ischemia. Reported urological complications in males have included urinary retention, ureteral obstruction, hydronephrosis, renal failure and urinary infection. Three reports have presented the incidental detection during nuclear imaging of bladder displacement from fecal impaction. Our report is however the first to document a normal urinary bladder presenting clinically as a large, palpable pelvic mass due to fecal impaction, despite the presence of a functioning Foley catheter. Preventive measures against impaction should be used in high-risk patients, including a daily bowel regimen of stool softeners/laxatives and maintaining hydration. For patients with established impaction a thoughtful history and examination is usually diagnostic, supplemented as needed with imaging studies. Disimpaction, enemas and laxatives are the mainstays of treatment, with endoscopic/surgical approaches for refractory cases. Metabolic, neoplastic, neurological or inflammatory etiologies should be sought where indicated.

WHEN IRITIS IS REALLY SECONDARY TO SOMETHING ELSE. HA Batat, Division of General Internal Medicine, Denver Health Medical Center; Division of General Internal Medicine, University of Colorado Health Sciences Center, Denver, CO

LEARNING OBJECTIVES: 1) Diagnose acute iritis. 2) Recognize that acute iritis may be secondary to an underlying ophthalmologic infection.

CASE PRESENTATION: A 55 year-old female with a history of systemic lupus erythematosus on numerous immunosuppressive agents presented to the urgent care clinic complaining of five days of visual loss and pain in her left eye. She had no prior ophthalmologic disease or complaints. Interestingly, she had been seen by her primary care physician nine days prior to this visit and had complained of a left-sided headache for two days. At this visit her exam had demonstrated tenderness over her left forehead but was otherwise normal. Her headache was thought to be consistent with either a migraine or a viral syndrome. She had no complaints of eye pain or visual loss at this time. However, during her urgent care clinic evaluation she was noted to have a decrease in the visual acuity of her left eye compared with an ophthalmologic exam six months prior. In addition, her left pupil was non-reactive, and she complained of photophobia during the fundoscopic examination. Given these findings, the patient was felt to have iritis secondary to her underlying collagen vascular disease. However, further examination revealed a few crusted lesions on her left forehead. The patient admitted to having had a rash on the left side of her forehead beginning about five days prior to this clinic visit which was now resolving. Because of the concern for herpes keratitis given this finding, fluorescein staining of the cornea was performed, revealing a classic dendritic pattern. Emergent ophthalmologic consultation confirmed herpes keratitis with secondary iritis and she was admitted for intravenous acyclovir. Despite aggressive treatment, one month later she still had significant visual loss.

DISCUSSION: Iritis is often associated with underlying collagen vascular diseases. Classically, the patient with iritis presents with unilateral eye pain, visual loss, and photophobia. In advanced cases, the routine eye exam will show an unreactive pupil. Definitive diagnosis is made with slit lamp examination showing flare cells in the anterior chamber. In early disease, pain with pupillary constriction may be the only clinical feature distinguishing iritis from conjunctivitis. Herpes keratitis is a severe, rapidly progressive infection of the eye, usually occurring in patients who have trigeminal zoster. Untreated, both diseases can lead to perma-

nent visual loss. As illustrated with this case, iritis may be the primary eye pathology or a secondary reaction to an underlying infection.

THE CLINICAL DILEMMA OF HYPOXIA. AK Bedi, A Cheema, P Mehta, Medicine, New York Methodist Hospital, Brooklyn, NY

LEARNING OBJECTIVES: To diagnose bronchiolitis obliterans with organizing pneumonia.

To recognize the need for confirmation of diagnosis by lung biopsy for better management and prognosis of patients.

CASE: Acute hypoxemia, especially when due to alveolar hypoventilation, represents a medical emergency and has many potential causes. We present a case of a 49-year-old diabetic lady who came to our hospital with worsening shortness of breath, cough with yellowish sputum and chest tightness, which had become severe for one week before admission. She had 30 pack year smoking history, cocaine and marijuana use which she quit 5yrs ago and multiple sexual partners. On examination, she had a temperature of 98.6, BP was 144/70 mm Hg, respiratory rate of 44/min and oxygen saturation was 88% on room air. She had decreased air entry bilaterally with generalized wheezing on chest auscultation. Arterial blood gas was pH 7.44, PCO2 37 mmHg, PO2 56 mmHg, oxygen saturation 88.3%. Chest x-ray showed increased bronchial markings in the right base. A CT scan of the chest done to check for pulmonary embolism showed presence of multiple areas of ground-glass appearance occupying all zones in a non-diffuse, fan shaped manner and bilateral lymphadenopathy. The abnormal laboratory data included mildly elevated alanine and aspartate aminotransferases, alkaline phosphatase and lactate dehydrogenase. Hepatitis antibody profile, HIV testing, tests for mycoplasma and legionella infections were negative. The clinical diagnosis was atypical pneumonia versus Pneumocystis carinii infection and the patient was started on intravenous bactrim, azithromycin and corticosteroids. Bronchoscopy with transbronchial lung biopsy revealed chronic interstitial pneumonia with organizing alveolitis obliterans. Intravenous antibiotics were discontinued and the patient was continued on IV corticosteroids. Nine days after admission she was discharged home on oral corticosteroids.

DISCUSSION: Cryptogenic organizing pneumonitis or idiopathic BOOP (bronchiolitis obliterans with organizing pneumonia) is a relatively rare condition with an incidence of 6-7 per 100,000 hospitalizations in teaching institutes. It is characterized by excessive proliferation of granulation tissue within the small airways and alveolar ducts and associated inflammation in the surrounding alveoli. The clinical presentation often mimics community acquired pneumonia and is often preceded by flu like symptoms. Its presenting features include persistent non-productive cough, dyspnea with exertion and weight loss of more than 10 pounds. The overall prognosis is much better than conditions like idiopathic pulmonary fibrosis which is in the differential diagnosis. A high clinical suspicion and confirmation of the diagnosis of BOOP is very important because it can favorably change the management and prognosis of the patient.

RHABDOMYOLYSIS AFTER INADVERTENT STATIN INGESTION. IM Billingsley, K Myers, Queen's University School of Medicine, Kingston, ON, Canada

LEARNING OBJECTIVES: 1. Recognise the clinical presentation of drug-induced rhabdomyolysis. 2. Demonstrate the value of carefully scrutiny of all patient medications at presentation.

CASE PRESENTATION: Acute rhabdomyolysis is a clinical and laboratory syndrome that is the result of skeletal muscle injury with release of cellular contents in the plasma. The most common suspected aetiologies reported in the literature include alcohol and trauma, with increasing recognition of drug-induced rhabdomyolysis including that associated with use of lipid lowering statins. We report a case of rhabdomyolysis after mistaken ingestion of atorvastatin.

A 68 year old Caucasian male presented to the hospital emergency room with a two week history of myalgia, generalised weakness and fatigue. His past medical history included diabetes mellitus type 2. He admitted to the consumption of 5-6 units of alcohol per week and a past history of alcohol abuse. Six weeks previous to his presentation, the patient was attacked by a domestic cat (who subsequently attacked others and was later quarantined and destroyed) and had been started on a course of clavulanic acid.

On examination motor weakness, more proximally than distally, was noted. Admission laboratory investigations: Creatine Kinase (CK) 52 915, Urea 22.5, Creatinine 311, AST 2 549, ALT 501, LDH 2 009. Urine was noted to be smoky-grey in colour and positive for myoglobin. The patient's CK continued to rise and peaked at 185 810. His muscle weakness progressed such that he was unable to ambulate but had no respiratory difficulties. He developed oliguria and required haemodialysis. Further consideration of the aetiology of the rhabdomyolysis and review of his medication bottles brought to the hospital revealed that the patient had incorrectly been dispensed atorvastatin 40 mg instead of Accupril 40 mg approximately four weeks previous to his presentation. The patient's renal function and muscle weakness improved and dialysis was discontinued. The patient had a slow and complicated recovery.

DISCUSSION: Rhabdomyolysis is a well described adverse reaction associated with atorvastatin use. While patients are normally advised to discontinue the medication should myalgia or weakness occur, this patient received no such advice as he had not been prescribed atorvastatin and only received it inadvertently. This case illustrates an interesting presentation of a known adverse drug reaction and highlights the role for careful scrutiny of patient medications at presentation in the diagnosis of undifferentiated medical problems.

CLINICALLY DIAGNOSED HERPES ZOSTER ENCEPHALITIS (HZE) WITH CEREBRAL VASCULITIS IN AN IMMUNOCOMPETENT ELDERLY HOST. R Borra, T Bharatan, P Mehta, A Leong, T Evelyn, Internal Medicine, NY Methodist Hospital, Brooklyn, NY

LEARNING OBJECTIVES: Recognition, early diagnosis and treatment of HZE and associated cerebral vasculitis in the elderly to reduce mortality and morbidity.

CASE PRESENTATION: HZE is a rare condition with significant mortality and morbidity which can be reduced with prompt treatment. A previously healthy 80 year old woman became delirious and acutely encephalopathic several days after onset of disseminated cutaneous zoster with involvement over the right trigeminal nerve distribution. She had fever, headache, right ptosis, 6th nerve palsy, CSF lymphocytic pleocytosis with raised protein, increased signal density in the right temporal lobe on MRI suggestive of localized encephalitis but a normal EEG. Blood cultures, serum VDRL, rheumatoid factor, cryptococcal antigen and PPD testing were -ve and CD4, CD8 counts were normal. AFB smear, HZV DNA PCR, herpes simplex viral DNA 1 & 2 on CSF were -ve. Acyclovir, antibiotics and steroids for presumed vasculitis did not improve her condition. At autopsy, diffuse cerebral vasculitis, perivascular lymphocytic infiltration, trigeminal ganglionitis and neuritis was noted.

DISCUSSION: Encephalitis is a rare complication of HZ associated with disseminated cutaneous disease and cranial nerve lesions. HZE is a clinical diagnosis as the viral recovery from CSF or brain tissue has been generally unsuccessful. It should be suspected in patients who present in an encephalopathic state with a recent clinical HZ infection, an abnormal brain MRI and CSF examination. Cerebral vasculitis is a serious post-zoster CNS complication (especially with disseminated zoster and cranial nerve lesions) which can present as hemisphere strokes following the trigeminal distribution of zoster due to focal internal carotid and middle cerebral artery vasculitis. In rare occasions, diffuse cerebral vasculitis is described. Early administration of acyclovir produces complete resolution of encephalitis within 72 hours with normalization of EEG, and reduced mortality. However, delayed diagnosis and treatment reduces overall effect of acyclovir. Cases of cerebral vasculitis without preceding CNS infection have been treated successfully with immunosuppressive agents. However, treatment of cerebral vasculitis due to HZ is not clear and according to some reports, there is no treatment for Zoster associated cerebral vasculitis which may be responsible for this patients' poor response to therapy. Although acyclovir did not result in improvement in our patient, it could inhibit viral replication in the CNS or decrease the antigenic stimulus that drives the CNS immunological responses. Continued treatment of undiagnosed cerebral vasculitis associated with HZE with immunosuppressive agents may lead to neurological deterioration and treatment with acyclovir is potential cure.

RECOGNITION AND MANAGEMENT OF FATAL ANTICONVULSANT HYPERSENSITIVITY SYNDROME (AHS) DUE TO COMMONLY USED ANTICONVULSANTS, PLANNING SUBSEQUENT ANTICONVULSANT TREATMENT OPTIONS AND GENETIC ADVICE TO SIBLINGS FOR PREVENTION. R Borra, Internal Medicine, New York Methodist Hospital, Brooklyn, NY

LEARNING OBJECTIVES: Same as title.

CASE PRESENTATION: A 25 year old Hispanic male who was taking phenytoin for two months for post traumatic seizure prophylaxis presented with fever, cough, erythematous scaly macular rash of 5 days duration over whole body except palms and soles, facial pustules, fever (101.7 F), generalized lymphadenopathy. Labs : WBC: 15,200/L, 15.6% eosinophils. Serum chemistry and liver function tests were normal. Throat and blood cultures, viral culture for EBV, CMV, rubella, rubeola and tests for HIV and syphilis were -ve, ruling out most other causes of this clinical picture. A history of drug exposure and clinical findings led to the diagnosis of AHS due to phenytoin. He was treated with IV hydration and corticosteroids with resolution of rash, fever and lymphadenopathy within a week and uneventful recovery.

DISCUSSION: Considering high prevalence of epilepsy, clinicians must recognize potentially fatal AHS due to arene oxide-producing anticonvulsants (phenytoin, carbamazepine and phenobarbital) which presents as skin rashes, fever, lymphadenopathy, atypical lymphocytosis or eosinophilia, thrombocytopenia, hepatitis and nephritis. It occurs in 0.1%-0.01% of population, is more common in Afro-Americans, has diverse clinical features resulting in a delay of diagnosis and is fatal in 5-50% if hepatitis or toxic epidermonecrosis occurs. AHS is caused by an inherited deficiency of epoxide hydrolase, an enzyme required for metabolism of a toxic intermediate, arene oxide, produced during metabolism of above mentioned anticonvulsants by cytochrome P-450. Inheritance is autosomal co-dominant and siblings have a 25% risk of AHS and should be counseled. Arene oxides bind to cell macromolecules resulting in cell death (cytotoxicity), teratogenicity, mutation, tumor or can act as haptens and neoantigens resulting in hypersensitivity reactions. The lymphocyte cytotoxicity test for arene oxides is an in-vitro method for prediction and diagnosis of AHS but is not widely available. The diagnosis of AHS is made by most centers from the history of drug exposure and clinical findings. Prenatal prediction of risk of congenital malformations (Fetal Hydanotin Syndrome) is made by demonstrating low epoxide hydrolase activity in fetal amniocytes. Treatment: Discontinuation of the drug, supportive care with attention to fluids and electrolytes, admission to the burn unit depending on the skin exfoliation, and systemic steroids to reverse cutaneous eruptions. Internal manifestations do not seem to reverse with steroids. Benzodiazepines, gabapentine, lamotrigine are other anticonvulsant alternatives in these patients. Valproic acid, because of its hepatic metabolism should not be used during the acute phase.

WIDE QRS TACHYCARDIA REFRACTORY TO PHARMACOLOGIC INTERVENTIONS AND CARDIOVERSION. ES Brilakis, L Chen, PA Friedman, Internal Medicine; Cardiovascular Diseases, Mayo Clinic, Rochester, MN

LEARNING OBJECTIVES: 1. Recognize causes of pharmacologic or electrical treatment failure in patients with wide QRS tachycardia. 2. Identify alternative therapies to terminate a wide QRS tachycardia if all ACLS proposed actions fail.

CASE PRESENTATION: A 43-year-old white man with no history of cardiopulmonary disease presented with a 4-hour history of palpitations and fatigue. He denied chest pain, dyspnea or syncope. A prior electrocardiogram showed no pre-excitation or QTc prolongation. Blood pressure was 146/98 mmHg, heart rate was 181 beats/min and regular, and respiratory rate was 20 breaths/min. Cardiac examination revealed a regular tachycardia with normal S1 and S2; no S3, S4 or murmur was detected, and the lungs were clear to auscultation. Complete blood count, electrolytes (including calcium and magnesium), creatine kinase, and troponin I, all were within normal limits. A chest radiograph demonstrated a heart of normal size and clear lungs. A 12-lead electrocardiogram demonstrated a monomorphic regular wide QRS tachycardia with a ventricular rate of 188/min and right bundle branch block (RBBB) morphology. Valsalva maneuver, carotid sinus massage, administration of lidocaine, adenosine and procainamide did not affect the tachycardia. He received 6 synchronized shocks, 4 of which were at 400 Joules with no response. At that point a transvenous pacemaker was inserted in the right ventricle inducing a 5-beat run of ventricular tachycardia at a rate of 300 beats/min, followed by normal sinus rhythm. A transthoracic echocardiogram demonstrated normal left ventricular size; ejection fraction was 54%. An electrophysiologic study excluded the presence of an accessory pathway or dual AV node physiology and did not induce any arrhythmia. The patient was dismissed on oral verapamil, and remained asymptomatic 3 months later.

DISCUSSION: According to ACLS guidelines, hemodynamically stable patients with wide QRS tachycardia of uncertain type should be sequentially treated with lidocaine, adenosine, procainamide, and bretylium. If pharmacologic treatment fails, or if hemodynamic instability occurs, synchronized cardioversion is recommended. However, if all ACLS proposed interventions fail, as in our patient, overdrive pacing or internal cardioversion could be utilized.

WHY YOUR MOTHER TOLD YOU NOT TO TALK AND EAT AT THE SAME TIME: AN UNUSUAL DIAGNOSIS OF A CHRONIC PULMONARY SYNDROME. N Britwum, K Duru, JJ Shannon, Internal Medicine, Cook County Hospital, Chicago, IL

LEARNING OBJECTIVES: Recognize the clinical manifestations of occult foreign body aspiration.

CASE PRESENTATION: A 40 year old man without significant past medical history presented with right-sided chest and back pain, and a dry cough of six days' duration. The pain was non-pleuritic, but was aggravated with movement. The patient denied fever or chills. He had been treated for pneumonia six months ago for similar symptoms of right-sided chest pain. He completed a full course of antibiotics with complete resolution of symptoms. He denied alcohol or tobacco use. He had no history of seizures, and denied any recent dental procedures. The patient had a desk job and was not exposed to any dust or airborne fibers.

On exam, the patient did not appear acutely ill. He had a low-grade fever of 100 degrees. Other vital signs were unremarkable. Chest exam revealed dullness to percussion and diminished tactile fremitus over the right lung field. On auscultation, decreased breath sounds were heard in this region, but egophony was absent. Chest X-ray demonstrated an extensive right-sided pleural effusion suggestive of early loculation, and subsequent thoracentesis confirmed that the pleural fluid was straw-colored and turbid. The pH of 7.05 and fluid LDH of 1149 suggested the presence of a complicated parapneumonic effusion. Chest CT showed a loculated effusion with underlying right middle and lower lobe consolidation. A thoracic surgery consult was obtained, and the patient went for open thoracotomy and decortication. Intraoperative bronchoscopy revealed a 3 cm plug of aluminum foil occluding the right lower lobe bronchus. After removal of the foil decortication proceeded without incident and the patient's lung reexpanded. He did well after the surgery and was discharged home.

DISCUSSION: Foreign body aspiration is more common in children, but should be considered in adults who present with complicated focal airspace disease. The majority of cases involve the right lung, and particularly the right lower lobe, due to the relatively oblique orientation of the right mainstem bronchus. Recent reviews confirm that foreign body aspiration may occur without known risk factors for aspiration, and may be diagnosed several months to years after the initial event, particularly when the foreign body is radiolucent, as in this case (Prakash and Limper, Annals of Internal Medicine, 1990; Rodenstein et al. Chest, 1999). Such patients often present with subacute respiratory symptoms of chest pain, cough and wheezing, and may be diagnosed with recurrent pneumonias. Controversy exists whether rigid or flexible bronchoscopy is the most effective means of foreign body removal. Foreign body aspiration should be considered whenever physical exam or chest X-ray suggests the presence an occluded bronchus.

PE OR NOT PE: THAT IS THE QUESTION. ED Brownfield, Division of General Medicine, Emory University, Atlanta, GA

LEARNING OBJECTIVES: 1) Review the data from the PIOPE study. 2) Recognize the limitations of noninvasive testing for PE. 3) Discuss the difficulty of distinguishing between PE and pneumonia.

CASE PRESENTATION: A 49 year old African American female with a history of sarcoidosis, hypertension and end stage renal disease presented with shortness of

breath and right sided pleuritic chest pain. Patient had been in her usual state of health until ten days prior to admission, when she noted increasing left leg swelling after a long road trip. Five days prior to admission, patient developed sudden onset of right sided chest pain followed by a productive blood-tinged cough. She denied any fever, chills, night sweats. Her medications were notable for prednisone. Her exam was remarkable for a pulse of 101, respiratory rate of 22, temperature of 38.4, room air saturation of 93%. She had inspiratory crackles in the right middle lobe, her extremities were benign. Her WBC was 18.8, A-a gradient 50 mmHg, d-dimer 923. Her ECG revealed sinus tachycardia with LVH, her CXR showed a right middle low opacity. Lower extremity dopplers were negative and V/Q scan was intermediate probability. Patient was admitted with a working diagnosis of PE and started on heparin. AFB $\times$ 3 was negative. Patient improved over five days, her shortness of breath and chest pain resolved. She remained afebrile and her WBC slowly decreased. On hospital day #6, the patient acutely developed worsening shortness of breath with gross hemoptysis and suffered a respiratory arrest. Due to the unexpected clinical course an autopsy was requested. Autopsy revealed extensive bronchopneumonia secondary to *Serratia marcescens*.

DISCUSSION: The role of clinical suspicion and V/Q scanning for diagnosis of PE has been best evaluated in the PLOPED study. This case points out the limitations of PLOPED and other proposed tests used in the diagnosis of PE. It begs to ask if pulmonary arteriography is indicated in all cases of suspected PE. This case also brings up many clinical points, namely infection in the setting of steroids, incidence of PE in end stage renal patients. Finally, this case is ironic and humbling. The decision to treat for presumed PE was in some respect to not miss a life-threatening process. The question which still remains is should we have treated her for pneumonia from the beginning and would this have saved her life?

NOT A CLOT. J. Browning, K Szauter, L Tuft, P Leonard, Internal Medicine, The University of Texas Medical Branch, Galveston, TX

LEARNING OBJECTIVES: (1) Recognize the similarities between the presentation of a pulmonary embolism and that of pulmonary hypertension due to other causes.

(2) Recognize consequences of an uncorrected congenital cardiac defect in an adult.

CASE PRESENTATION: A 33 year old woman presented to the emergency room with left sided chest pain and dyspnea, bilateral lower extremity pain and mild edema of the right (R) lower leg. Her past medical history was notable for menorrhagia (controlled by oral contraceptives) alopecia, and hepatitis C. Surgical history included an open cholecystectomy 12 years earlier. The patient used tobacco and had previously abused alcohol and intravenous drugs. She had a sedentary lifestyle and her work in an institutional laundry facility necessitated long periods of sitting and standing. Physical examination was notable for an uncomfortable appearing young woman in mild respiratory distress. She had diffuse alopecia. Lung exam revealed moderate air movement throughout. Cardiovascular exam was notable for a paradoxically split S2 and a soft holosystolic murmur at the left lower sternal border. Abdominal examination was unremarkable. There was moderate edema of the lower extremities bilaterally. Initial laboratory studies including electrolytes and a CBC were normal. An arterial blood gas on room air revealed a pO₂ of 44. Her EKG was remarkable for an S wave in lead I and a Q wave and inverted T wave in lead 3. A ventilation-perfusion scan was read as low probability for a pulmonary embolism. However, because of the high clinical suspicion for a pulmonary embolism the patient was placed on oxygen therapy and started on heparin. The following morning the patient showed no clinical improvement. Lung examination revealed poor air movement diffusely. A chest radiograph showed a large heart and prominent pulmonary artery. Lower extremity doppler studies revealed no deep venous thrombosis. An echocardiogram showed (R) atrial enlargement (R) ventricular dilatation, and moderate tricuspid regurgitation. A rapid clinical decompensation necessitated her transfer to the intensive care unit. A pulmonary artery (PA) catheter was placed. Measurements demonstrated that the PA pressure exceeded systemic pressure. A second echocardiogram following the infusion of agitated saline showed significant right to left shunting at the atrial level consistent with an atrial septal defect (ASD). She was not a candidate for a heart-lung transplant and died within a few weeks of her presentation.

DISCUSSION: This patient presented with Eisenmenger syndrome due to an uncorrected congenital ASD. Despite a "textbook" presentation of a pulmonary embolism her limited response to initial therapy prompted a further evaluation resulting in an unexpected and unusual diagnosis. This case provided rich learning opportunities in both pathophysiology and clinical diagnosis.

LOW PULSE OXIMETRY LEADING TO THE DIAGNOSIS OF HEMOGLOBINOPATHY. CM Bruns, KM Hla, Department of Medicine, University of Wisconsin, Madison, WI

LEARNING OBJECTIVES: 1) Recognize hemoglobinopathy as a cause of low oxygen saturation.

CASE: A 53 year-old healthy Korean woman presented to the primary care clinic for follow up of screening flexible sigmoidoscopy. During the procedure, prior to sedation, arterial oxygen saturation (SaO₂) was measured at 83% on room air by pulse oximetry with a good tracing. She denied fever, cough, shortness of breath, chest pain, orthopnea, calf tenderness, tobacco use, environmental exposures or use of any medications. On physical exam, vital signs were normal, cardiopulmonary exam was normal, and there was no cyanosis or clubbing. Complete blood count, pulmonary function tests, and chest x-ray were normal. Pulse oximetry in the office was 83–84% on room air. Arterial blood gas obtained on room air re-

vealed a normal paO₂ of 94 mmHg while the SaO₂ was only 86%. Reduced hemoglobin was elevated at 12.9%. Methemoglobin and carboxyhemoglobin were normal. Administering 50% FIO₂ and 100% FIO₂ caused the paO₂ to rise appropriately, but the SaO₂ did not correct to expected levels and remained at 91% and 96%, respectively. Mild instability of hemoglobin was identified on isopropanol testing, and 40.3% of the patients' hemoglobin was abnormal by chromatography with the remainder being largely hemoglobin A. A DNA analysis was performed and showed that the patient is heterozygous for hemoglobin Rothschild, a beta chain point mutation substituting arginine for tryptophan in the hemoglobin molecule (beta37(C3)Trp→Arg). It is inherited as an autosomal co-dominant condition. Her two children also had low oxygen saturation when screened with pulse oximetry and were subsequently shown to have the same hemoglobin abnormality.

DISCUSSION: Pulse oximetry is a useful tool for noninvasive monitoring. However, its use is limited in the presence of dyshemoglobinemias, dyes and pigments (such as methylene blue), low perfusion states and anemia where SaO₂ is not reflective of paO₂. In our patient, the alteration in the beta chain of hemoglobin results in a decreased affinity of hemoglobin for oxygen. This leads to a shift to the right of the hemoglobin-oxygen dissociation curve and a subsequent low SaO₂ due to increased concentration of reduced hemoglobin. While the patient is largely asymptomatic from this benign hemoglobinopathy, there is the potential for confusion in the event of emergency management of an injury or during a procedure where pulse oximetry might be measured. The differential diagnosis of a low pulse oximetry reading should be expanded to include hemoglobinopathy.

AN UNUSUAL CAUSE OF OTITIS MEDIA. D. Buchan, S Nair, SUNY HSC at Syracuse, Syracuse, NY

LEARNING OBJECTIVES: 1. Recognize the clinical features of typhoid fever. 2. Distinguish usual from more rare causes of otitis media.

CASE: A previously healthy 24 year-old Indian accountant presented on 2/1/99 with a temperature of 103F, chills, productive cough, nausea, malaise of 4 days duration. He had returned one week prior from a three week stay in southern India. Nausea, malaise and a feverish feeling commenced on the plane trip back to the U.S. On exam he appeared ill, had a temperature of 101.9F, pulse of 68 and BP of 96/70. The left tympanic membrane was described as "fire red". He had frontal sinus tenderness and an erythematous pharynx without exudates. Lungs were clear. Neck was supple. He was diagnosed with otitis media and prescribed a 5 day course of azithromycin.

He presented 4 days later with unremitting fever, bifrontal headache, persistent cough and nausea. On exam he had a temperature of 102.1F, BP of 100/60, HR of 92 and RR of 16. He had a foul body odor but no rash. Neck was supple. Left tympanic membrane was dull and red. He had sinus tenderness on percussion. Lungs were clear. There was no hepatosplenomegaly. Chest x-ray was clear. He was switched to Amoxicillin/Clavulanate. Blood smears were negative for malaria. Stool culture was negative. CBC showed a WBC of 5.3 K/ul with 53% neutrophils, 13% monocytes, 3% eosinophils, 31% lymphocytes. Hb was 14.4 g/dl and platelet count was 225,000 K/ul. Blood cultures grew salmonella typhi and the diagnosis of typhoid fever was made. He was switched to ciprofloxacin and completed a 14 day course. He rapidly defervesced and began to feel well.

DISCUSSION: Typhoid fever should be suspected in travelers to endemic areas. Because this patient had a "fire red" tympanic membrane, the physician thought he had common otitis media/sinusitis. It was only when the patient failed to respond to azithromycin that an alternative diagnosis was sought. Acute otitis media is not a common manifestation of typhoid fever but it has been described. This patient had several features of typhoid fever including unremitting fever, headache, putrid body odor, travel to an endemic area, pulse-temperature deficit, and normal white blood cell count.

WHEN IS A TAN MORE THAN JUST SKIN DEEP? E. Cagola, DG Litaker, J Isaacson, General Internal Medicine, The Cleveland Clinic Foundation, Cleveland, OH

LEARNING OBJECTIVES: Recognize the clinical presentation of adrenal insufficiency and recall that an infectious cause of Addison's Disease can be extrapulmonary tuberculosis involving the adrenal glands.

CASE PRESENTATION: A previously healthy 71-year old Croatian born man presented with a two month history of persistent weakness, dizziness, anorexia, non-productive cough, fever, night sweats, and unintentional 15 pound weight loss. During a recent hospitalization for these symptoms, he was found to have bilateral upper lobe infiltrates on chest X-ray with a strongly positive PPD (20 mm induration). A bronchoscopy with transbronchial biopsy showed chronic inflammatory changes without caseating granulomata, bronchial washings were sterile and mycobacterial cultures were negative at 3 weeks. The patient had been discharged on prophylactic Isoniazid, but his symptoms failed to improve. He then presented to the outpatient clinic with worsening weakness, dizziness and persistent fevers. The patient resided in the US for over 30 years and worked as a metal grinder. He was a former smoker with 20 pack-year history and reported drinking 2–3 alcoholic beverages daily. Physical examination was notable for a cachectic appearance, orthostatic blood pressure changes and generalized hyperpigmentation, markedly accentuated at the palmar creases and at a vaccination scar on the left arm. Hyperpigmentation of the buccal mucosa was not present. The patient was admitted and placed in respiratory isolation. Laboratory studies showed serum sodium of 133 mmol/L, potassium 6.1 mmol/L, CO₂ 19 mmol/L, Glucose 89 mg/dL, BUN 64 mg/dL, and Creatinine 2.1 mg/dL. A Cosyntropin stimulation test

was abnormal along with a markedly elevated ACTH level. Abdominal CT scan demonstrated bilateral adrenal enlargement (L>R) without lymphadenopathy or other abnormalities. The patient was treated with corticosteroids, fludrocortisone and was placed on a 4 drug antituberculous regimen for a presumptive diagnosis of Tuberculous Addison's Disease. A CT-guided adrenal biopsy showed acute and chronic inflammation with necrosis, most consistent with an infectious process, but acid-fast bacilli and fungal stains were negative. The patient steadily improved on therapy and was discharged home. At 6 weeks, induced sputum cultures finally grew *M. tuberculosis*.

DISCUSSION: The clinical presentation of adrenal insufficiency can be varied, but a heightened index of suspicion is needed in patients who present with hyperpigmentation, weakness, weight loss and hypotension. Tuberculous Addison's disease, which once accounted for up to 90% of cases, is now an uncommon cause of primary adrenal insufficiency in the US. Nevertheless, it should be considered in immunocompromised patients and in immigrants from countries where TB is endemic.

WHEN STERNAL PAIN ISN'T COSTOCHONDRITIS. E. Caloia, J Isaacson, DG Lifter, General Internal Medicine, The Cleveland Clinic Foundation, Cleveland, OH

LEARNING OBJECTIVES: Recognize sternal metastases as a rare cause of sternal pain and an uncommon presentation of adenocarcinoma of unknown primary.

CASE PRESENTATION: A previously healthy 62-year old man presented with a 6-week history of sternal and upper back pain. Four weeks prior to presentation, he had been diagnosed with costochondritis and degenerative arthritis of the spine, which were treated with a 2-week course of a NSAID. His symptoms worsened on therapy and he began to complain of severe, constant, sharp sternal and upper back pain, exacerbated by arm movements, changes in position and deep inspirations. The patient had been previously in excellent physical condition and denied any headache, trauma, unusual physical activity, recent upper respiratory tract infection, cough, fever, nausea, vomiting, diaphoresis or weight loss. He was a former 40-pack year smoker, stopping eight years ago, and reported drinking 1/2 to 1 case of beer per week. Physical examination was notable for extreme tenderness just below the sternal angle, without associated overlying skin inflammation or any localized tenderness at the costochondral joints. There was tenderness to palpation of the upper thoracic spine, approximately 2-3 cm to the right of the T5 vertebral body. Rectal exam showed trace guaiac positive stool with a normal prostate. A complete blood count, metabolic panel and PSA test were within normal limits. A chest x-ray demonstrated a lytic bone lesion with an associated soft tissue mass involving the posterolateral aspect of the right fifth rib. Chest CT scan showed a 2 x 3-cm soft tissue mass involving the right posterolateral rib, two 5-8 mm non-calcified nodules in the periphery of the right and left lower lobes of the lungs, normal vascular structures and no adenopathy. A bone scan showed areas of increased tracer uptake in the sternum and the right posterior 5th rib. The patient underwent a sternal biopsy showing metastatic adenocarcinoma, possibly from a lung or prostate primary. Subsequent diagnostic tests including a CT scan of the abdomen/pelvis and colonoscopy failed to reveal a source of the primary cancer. The biopsy specimen did not stain for PSA, making metastatic prostatic adenocarcinoma much less likely. A pulmonary adenocarcinoma was considered the probable primary source.

DISCUSSION: Sternal pain associated with other areas of bone pain should be thoroughly investigated and not simply attributed to costochondritis. Costochondritis usually involves one of the costochondral joints and can be associated with sternal joint swelling (Tietze's Syndrome). Several cancers can metastasize to the ribs, chest wall and spine including: breast cancer, prostate cancer, sarcoma and plasma cell cytoma. Metastatic adenocarcinoma of unknown primary is a rare cause of sternal pain.

SHOULDER PAIN: WHEN A FOCUSED EXAM IS NOT ENOUGH. P. Cantey, D Brady, Division of General Medicine, Emory University, Atlanta, GA

LEARNING OBJECTIVES: 1) To recognize when to broaden the differential diagnosis of shoulder pain. 2) To demonstrate the need to treat the whole patient and not just the chief complaint.

CASE: The patient was a 43 year old male with a chief complaint of right shoulder pain. His job required a lot of heavy lifting, but he denied any injury or strain. The pain in his shoulder had persisted for several months, necessitating at least 3 prior visits to the urgent care center where he was prescribed non-steroidals and physical therapy. Despite treatment, he had not had any relief of his pain. He said that he could no longer raise his right arm. Focused examination of the right shoulder revealed markedly diminished passive range of motion with elevation, abduction, internal rotation, and external rotation. Strength around the right shoulder joint was 1/2 with the limitation being primarily secondary to pain. Distal strength and left shoulder strength were normal. Reflexes were normal. Upon further questioning, the patient revealed progressive dyspnea and weight loss over several months and denied any cardiac symptoms, melena, bleeding or bruising. He reported 60 pack years of cigarette smoking. Past medical history revealed admission to the hospital one year prior for a left upper lobe pneumonia. Further examination revealed bilateral temporal muscle wasting and prominent ribs. There were hard, fixed left anterior cervical lymph nodes, the largest being 1cm. Breath sounds were markedly diminished from the left base to the left apex with dullness to percussion throughout the left side. Chest radiograph revealed opacification of the left lung field with some leftward mediastinal shift and elevation of the left hemidiaphragm. Radiographs of the right shoulder revealed a large lytic lesion in the

humeral head, with some invasion into the humeral neck. There was no periosteal inflammation and no reactive bone formation. Chest computed tomography revealed a large left pleural effusion, truncation of the left mainstem bronchus about 1cm from the carina, encasement of the aorta, and poor visualization of the left pulmonary artery. The CT also confirmed the presence of a 6mm by 6mm soft tissue mass in the humeral head with bony erosion. Cytologic examination of the pleural fluid as well as bronchial biopsy confirmed the diagnosis of poorly differentiated non-small cell carcinoma of the lung.

DISCUSSION: Shoulder pain is a common complaint, usually benign, and mostly self-limited; even the occasional frozen shoulder has an excellent and self-limited prognosis. However, the physician must consider other diagnoses on the differential when there is no response to traditional treatment and there are other red flags in the patient's presentation. The physician should also be aware that the patient's primary complaint is not always the root of the problem.

QUADRUPLE AMPUTATION FROM CAPILLARY LEAK SYNDROME. MP Carson, Depts. of Medicine and Obstetrics, UMDNJ—Robert Wood Johnson Medical School, New Brunswick, NJ

LEARNING OBJECTIVES: 1) Recognize the differential diagnosis of capillary leak syndromes 2) Diagnose and treat Idiopathic Capillary Leak Syndrome

CASE PRESENTATION: A 36 year old female, G1P0 with a 16 week twin gestation, presented with syncope, vomiting and a loose bowel movement. Initial systolic blood pressure was 80 mmHg, but came up to 100 mmHg after IV fluid boluses. She only complained of swelling and pain of her hands and forearms. We were called in as medical consultants after 4 hours of anuria despite 3 liters of Lactated Ringers (LR) solution. Her HCT was 54, WBC count was 30,000 with 37% bands. Pulse oximetry was normal. Her exam revealed tense non-pitting edema of her hands and forearms, 1/2 radial pulses, clear lungs, and an inability to actively extend her fingers. Our first impression was sepsis syndrome so we started ceftriaxone and ampicillin to cover community acquired infections and *Listeria*. Infected pregnant women get noncardiogenic pulmonary edema, but we had never seen compartment syndrome. We called orthopedics to check forearm compartment pressures, which were only mildly elevated. When her lips began to swell we gave corticosteroids to treat possible angioedema. Anaphylaxis was also considered, and we gave her antihistamines. Epinephrine was withheld due to adequate blood pressure and no wheezing. She miscarried during the night. By the next morning she had received 10 liters of LR to maintain adequate urine output, her compartment pressures had increased and she was taken to the OR for fasciotomies of both forearms and calves. Serum creatinine never went above 1.0 mg%. Day #4: Spontaneous diuresis began. Day #7: She still had normal brain, lung, liver, and renal function and was taken to the OR for closure of the fasciotomies. They found liquefied muscle, and performed bilateral above knee and above elbow amputations. Final lab results included: negative maternal & fetal cultures/anti-Latex IgE /ANA; normal bone marrow biopsy/serum protein electrophoresis/complement levels/C1 esterase activity. After excluding sepsis, angioedema, vasculitis, and finding literature describing profound acute hypovolemia with compartment syndromes and central organ sparing, we diagnosed her with Idiopathic Systemic Capillary Leak Syndrome (SLCS).

DISCUSSION: Since 1960 fifty cases of SLCS have been reported. The hallmarks are hypovolemia, hemoconcentration and central organ sparing. Case reports describe leukocytosis, spontaneous diuresis on Day #4, compartment syndrome with fasciotomies, and death from airway edema, but never amputations. It has been associated with monoclonal gammopathy and can recur. Acute therapy is supportive. Terbutaline and theophylline may decrease capillary permeability and have been used as prophylaxis against recurrent attacks. This case was devastating to all involved.

NOT JUST ANOTHER TYPICAL HEADACHE. DH Chao, A Gomez, EF Yee, Medicine, UCLA/San Fernando Valley Program, Sepulveda, CA

LEARNING OBJECTIVES: 1. Recognize tuberculous meningitis as an unusual cause of headaches. 2. Review the diagnostic challenges for TB meningitis.

CASE PRESENTATION: A previously healthy 32 y.o. male from Mexico presented to a clinic with new-onset of a tense "squeezing-like" frontal/occipital headache and dull lower back pain radiating to the left hip. He had no history of fever, neck stiffness, visual changes, or weakness. Physical exam was negative for any neurological deficits or focal findings. Diclofenac was prescribed. Ten days later the patient returned with continued headache, worsening back pain, and a temperature of 38.4C. His history and physical were otherwise unchanged from the previous visit. A lumbar sacral spine x-ray was negative and he was prescribed ibuprofen and acetaminophen. Six days later, he was admitted with worsening headache, back pain, fever, and right sided weakness. The physical exam revealed nuchal rigidity, and inability to flex his back or hip due to pain. Motor strength was 1/2 in the right upper and lower extremities, 3/5 on the left. Lab showed Hb 11.8, Hct 37, WBC 10.6, 81% neutrophils, 8% lymphocytes, 12 monocytes; LDH 1035, HIV positive. A non-contrast MRI revealed no abnormalities. A lumbar tap showed an opening pressure of 24, glucose 27, protein 983, RBC 50, WBC 680, 83% neutrophils, 13% lymphocytes, 4% monocytes. CSF gram stain: 3+ wbc, no organisms, KOH stain and AFB smear negative. The patient was admitted with a presumptive diagnosis of bacterial meningitis and started on cefotaxime and ampicillin. Clinically, the patient deteriorated with worsening neurological findings, a CT scan showing left basal ganglion infarction, and a MRI showing diffuse lumbar spine enhancement. A repeat LP showed continued low glucose, elevated protein and

WBC values with neutrophilic predominance, and negative KOH and AFB smear. His skin tests were anergic. CSF culture for fungal, herpes, VDRL, and TB PCR were all negative. The patient was started on 4 anti-TB drugs empirically with improvement in his CSF values, but was still left with neurological deficits. It was not until one month after his hospitalization that his TB culture came back positive.

DISCUSSION: Headaches are a very common disorder that present to primary care providers. Early on, TB meningitis may present as a common, non serious headache because of subacute symptoms. Fever and neurological deficits may not present for 1–2 weeks. Even when considered, diagnosis of TB is challenging because CSF AFB are seen on smear only 20% of the time. CSF culture is positive in 80% of cases, but may take weeks to months to grow out. PCR is more rapid, but only 60% sensitive. One needs to have a high index of suspicion for TB when treating populations from endemic areas, or who are HIV positive. Empiric therapy should be initiated early on because of the serious sequelae of TB meningitis.

SUDDEN LOSS OF CONSCIOUSNESS—SYNCOPE OR SEIZURE? A CASE REPORT. LY Chen, ST Higano, Internal Medicine; Cardiovascular Diseases, Mayo Clinic, Rochester, MN

LEARNING OBJECTIVES: 1) Diagnose cardiac syncope from its clinical features 2) Recognize the value and limitations of electrophysiologic studies in the diagnosis of syncope

CASE PRESENTATION: A 76 year old man with a significant cardiac history presented with 2 episodes of sudden loss of consciousness. Past medical history is remarkable for coronary artery bypass graft surgery in 1993 and wide QRS complex tachycardia which was evaluated by electrophysiologic (EP) studies in 1998. The EP study was essentially normal with no inducible ventricular tachycardia (VT). He was started on sotalol with good results. He presented with 2 episodes of unconsciousness. The first episode occurred suddenly in a sitting position with no preceding aura or prodrome such as lightheadedness, nausea, epigastric distress or palpitations. There were no convulsions, incontinence, tongue biting or postictal confusion. The second event happened abruptly while he was driving with similarly no aura or prodrome. In hospital, his cardiovascular and neurologic examinations were unremarkable. A third spell happened in hospital and this time he had a premonition of presyncope just before losing consciousness in a sitting position. The ECG showed a sinus tachycardia of 150 beats/minute. There were no convulsions, incontinence or automatisms. Further continuous ECG monitoring only showed asymptomatic runs of supraventricular tachycardia. Tilt table testing with isoproterenol was normal. An EP study was essentially normal and did not demonstrate sustained monomorphic VT. Finally, an EEG showed potentially epileptiform abnormalities occurring independently over both temporal regions during wakefulness and sleep. The patient was started on phenytoin and will be followed up in a few months.

DISCUSSION: Cardiac syncope can often be diagnosed by its clinical presentation. Sudden loss of consciousness with no preceding aura or prodrome in an elderly patient with prior history of coronary artery disease and ventricular arrhythmias would strongly suggest a diagnosis of cardiac syncope. In such a patient, EP studies should be undertaken if noninvasive techniques fail to provide the diagnosis. A nondiagnostic study should prompt one to consider an alternate diagnosis. Temporal lobe epilepsy is known to mimic cardiac syncope. It is also associated with ictal tachycardia or bradycardia. With time, if phenytoin prevents further spells, a diagnosis of temporal lobe epilepsy is likely in this case.

MIGRATORY PEG (PERCUTANEOUS ENDOSCOPIC GASTROSTOMY). TS Choksi, P Mehta, S Sharma, Medicine, NY Methodist Hospital, Brooklyn, NY

LEARNING OBJECTIVES: *To create more awareness among general practitioners about complications of PEG procedure which is getting more common with passage of time.

CASE: A 79-year-old nursing home resident had recurrent diarrhea and fever. He had extensive evaluations, which were inconclusive. His past medical history was non-significant except for dementia. He had PEG placement 5 months previously, which was replaced 1 week later because it was nonfunctioning. His physical exam was significant for evidence of dementia, fever of 100.5 F, regular pulse of 104/min and peri-PEG site erythema with a foul smelling discharge through the PEG. The remarkable laboratory findings were WBC's 17,000/ul with a left shift in the differential count. He underwent a Gastrografin study via the PEG, which was diagnostic of gastrocolic fistula. The patient underwent laparotomy. Perioperatively, the PEG was found traversing transverse colon with obvious entry and exit sites before entering body of stomach. The gastrocolic fistula tract had surrounding granulation tissue. The PEG was removed and he had surgical repair of the gastrocolic fistula, with a feeding gastrostomy put in place. The subsequent course of events was unremarkable. His fever and diarrhea resolved spontaneously and he was discharged to a skilled nursing facility.

DISCUSSION: Gastrocolic fistula is a rare complication of PEG although its exact incidence is unknown. We suspect that physicians are going to be faced with increasing cases of this diagnosis because many patients are undergoing this procedure. An interesting feature of our patient is that both the fever and diarrhea were intermittent and resolved spontaneously at times. The two CAT scans showed the PEG tube in stomach. So it could be inferred that the PEG migrated through the fistulous tract and hence the symptoms occurred with the PEG in the colon but spontaneously resolved when the PEG moved back to the stomach. Thus a very

high index of suspicion and awareness about this complication of gastrocolic fistula is warranted in any patient who has undergone PEG.

SEVERE COPD: WHEN CIGARETTE SMOKING DOESN'T SEEM TO BE THE WHOLE STORY. CA Cole, Internal Medicine, Washington Hospital Center, Washington, DC

LEARNING OBJECTIVES: Recognize HIV disease as a cause or cofactor in the development of severe COPD resulting in cor pulmonale.

CASE PRESENTATION: A 44 year old black male presented to the emergency department with three to five days of dyspnea, paroxysmal nocturnal dyspnea, peripheral edema and abdominal and scrotal swelling. He reported being totally well prior to the development of these symptoms. He denied cough, chest pain, fever or chills. Past medical history was remarkable only for childhood seizures. There was no history of hypertension, asthma or surgical illnesses. Social history was remarkable for remote use of crack cocaine, no alcohol use, and a cigarette smoking history of 1/2 ppd for 25 years. Family history was notable only for hypertension and coronary artery disease. Review of systems was otherwise negative.

Physical examination:

Temp 98 Pulse 109 Respirate rate 28 BP 127/87

The patient was thin but healthy appearing, in no distress

Pertinent positives on exam:

Shotty cervical nodes

Jugular venous pressure 7 cm above sternal angle

Chest was barrel shaped

Lungs were clear to auscultation and percussion

Heart sounds were distant

Abdomen was slightly distended with the liver 11 cm by percussion

Extremity exam revealed +2–+3 edema, with edema of the scrotum also.

Admission labs:

WBC 2.2, Crit 47

ABG 7.346/67(PCO2)/41(PO2)

Chest X-ray: flattened hemidiaphragms consistent with COPD, cardiomegaly, left base atelectasis

EKG: right axis deviation, right atrial enlargement

echocardiogram: dilated right atrium and right ventricle, moderate tricuspid regurgitation, pulmonary artery systolic pressure of 55 mm HG

alpha one antitrypsin level: 137 (normal 110–200 mg/dl)

HIV: positive

Pulmonary Function Tests: FVC 1.60L (44% predicted)

FEV1 .68L (23% predicted)

TLC 6.87 L (128% predicted)

RV 4.98L (335% predicted)

FEF 25–75 .26 L/sec (8% predicted)

DISCUSSION: HIV infection has been shown to alter the natural history of many other diseases, among them tuberculosis, syphilis, and various malignancies. Added to this list should be chronic obstructive pulmonary disease. When the degree of COPD seems out of proportion to the amount of cigarette smoking, HIV disease should be included in the list of factors which can cause or exacerbate COPD.

A MASS OF TROUBLE: WHEN LUNG CANCER PRESENTS AS RIGHT VENTRICULAR OUTFLOW OBSTRUCTION. D Constantinescu, J Muneeyirci, B Flansbaum, J Oh, Dept. of Medicine, Long Island Jewish Medical Center, New Hyde Park, NY

LEARNING OBJECTIVES: 1) Recognize that symptoms suggestive of right-sided heart failure in the absence of a clear etiology may indicate a cardiac mass, 2) Learn that cardiac masses are commonly metastatic in origin, 3) Understand that lung cancer is a frequent cause of cardiac tumors.

CASE PRESENTATION: A 68 year old man with emphysema and an 80 pack-year history of smoking presented to his internist with sudden onset dyspnea on exertion followed by swelling of the lower extremities. He denied PND, orthopnea, dyspnea at rest, or chest pain. His physical exam revealed a change in his murmur and lower extremity pitting edema. A TTE demonstrated normal LV function, and the patient began oral diuretic therapy. The edema improved, but his dyspnea was unchanged. Over the following three weeks, the patient further noted increased lethargy, frequent chills, decreased appetite and weight loss. On re-evaluation, he was febrile and hypertensive with splinter hemorrhages. Blood cultures were positive for *E. coli*. At admission, the physical exam demonstrated JVD, bibasilar crackles, a 3/6 holosystolic murmur at the left second intercostal space, and trace lower extremity edema. The EKG revealed a new incomplete RBBB. The TEE reported a 7.2cm × 4.4cm homogeneous tissue density mass seen in the apex of the right ventricle causing RV outflow obstruction and severe right atrial enlargement. A biopsy obtained by right heart catheterization identified a non-small cell, poorly differentiated carcinoma suggestive of a lung primary. Spiral CT of the chest described a 1.8cm × 1.2cm spiculated nodule in the right upper lobe.

DISCUSSION: Physicians often overlook the possibility of an intracavitary tumor when evaluating symptoms suggestive of right ventricular outflow obstruction of uncertain etiology. However, cardiac tumors are known to present with this constellation of symptoms. Metastatic tumors occur 20–40 times more frequently than primary cardiac malignancies. In one series of 12,485 autopsies, secondary myocardial tumors were found in 55 (0.4%) patients. The most common etiologies were cancers of the esophagus (27%), lung (24%), lymphoma (16%), and liver (5%). Other studies also suggest breast cancer as a significant source of metastatic cardiac tumors. Patients with right intraventricular masses may present with dysp-

nea on exertion, syncope, lower extremity edema, murmur, and EKG changes (arrhythmias, right axis deviation, and RBBB). Echocardiography and MRI have been used to evaluate cardiac tumors with the latter providing greater anatomic detail. Our patient had right outflow obstruction as the initial presentation of a non-small cell lung cancer. This case reminds us that myocardial tumors should be included in the differential diagnosis of right-sided heart failure.

AUTOIMMUNE HEPATITIS PRESENTING AS PAINLESS JAUNDICE. S. Daud, R. Granieri, Internal Medicine, University of Pittsburgh, Pittsburgh, PA

LEARNING OBJECTIVES: 1) Recognize the clinical features and treatment of autoimmune hepatitis.

CASE PRESENTATION: A 35 year-old, previously healthy female was transferred from an outside institution for evaluation of jaundice. She noticed the jaundice five days prior, and noticed clay-colored stools and dark orange urine about three days prior to admission. She denied any recent abdominal pain, nausea, or vomiting. The patient had no other medical problems and was taking no medications other than a large amount of nutritional supplements. Two months before presentation, the patient took a trip to Africa and was on Mefloquine prophylaxis, two weeks before and four weeks after her journey. She has a 41 year old sister who was diagnosed with systemic lupus erythematosus at the age of sixteen. Her physical examination was remarkable for scleral icterus, a non-tender, smooth, palpable liver edge, three centimeters below the costal margin, and generalized jaundice. Her labs at the outside hospital revealed an aspartate aminotransferase (AST) of 1236 U/L, an alanine aminotransferase (ALT) of 989 U/L, a total bilirubin of 10 mg/dL and a gamma-globulin of 2.57 g/dL. Serologies for viral hepatitis were negative. A right upper quadrant ultrasound was negative, but a computerized tomography scan revealed periportal lymphadenopathy with the possibility of an infiltrative process. A liver biopsy was performed which showed severe acute hepatitis with areas of parenchymal collapse and necrosis, without atypical lymphocytes and was non-diagnostic, but suggestive of autoimmune hepatitis. Because of the family history of autoimmune disease and the increased gamma-globulin component, the diagnosis of autoimmune hepatitis was made. The patient was started on Prednisone with subsequent resolution of the jaundice, transaminitis, and hyperbilirubinemia.

DISCUSSION: Autoimmune hepatitis is an idiopathic hepatocellular inflammation, characterized by histological evidence of periportal necrosis, autoantibodies against hepatocyte surface antigens in the serum, and hypergammaglobulinemia. The disease is most predominant in young and middle-aged females. Symptoms includes those of fatigue, malaise, and anorexia with physical signs of jaundice and hepatomegaly, and the picture of acute hepatitis on laboratory evaluation. Although there are no pathognomonic features of autoimmune hepatitis, the diagnosis requires exclusion of other etiologies of hepatitis, including Wilson's disease, hemochromatosis, viral hepatitis, and cholestatic liver disease. Liver biopsy usually shows piecemeal necrosis. Treatment usually includes prednisone with or without azathioprine with a very small minority of patients requiring liver transplantation.

ABDOMINAL PAIN WITH A TWIST. JL Derby, Internal Medicine, Creighton University, Omaha, NE

LEARNING OBJECTIVES: Recognize that *Entamoeba histolytica* should be included in the differential diagnosis of abdominal pain and can closely mimic inflammatory bowel disease or appendicitis.

Recognize that immune status can affect invasiveness of amebiasis.

CASE PRESENTATION: Mr. M. is a 23 y.o. white male who presents with a long history of stomach problems. He reports that he has intermittent episodes of abdominal pain associated with vomiting since he was sixteen years old. He was hospitalized with his initial episode for a presumed appendicitis. Work-up was negative and no surgery was performed. Since that time, he has had intermittent episodes of severe abdominal cramping, vomiting and diarrhea, lasting 3-4 days. The vomiting is not usually associated with nausea, but is induced by the pain. When he vomits, he does not usually get any relief of his pain. The cramping is usually in his lower abdomen, bilaterally. It does not radiate to his back or periumbilical area. He will have diarrhea that is sometimes slightly bloody, but this did not start until a few months ago. He reports fevers, chills, and fatigue associated with these episodes. He feels well between these episodes, but this current episode has lasted several weeks and he has lost 20 pounds. He presents after seeing another physician two weeks ago, who did an upper GI with a small bowel follow through, which was negative. Lab done at that time including a CBC, comprehensive profile and amylase were all normal. He was treated with Prevacid which has given him no relief. He denies any family history of inflammatory bowel disease; he traveled to Colorado this summer, no travel out of the country except a trip to the Dominican Republic seven years ago. His PMH is otherwise negative, he has no HIV risk factors.

The only abnormalities noted on physical exam was a temperature of 99.6 and diffuse tenderness to deep palpation on abdominal exam. No rebound or guarding, hyperactive bowel sounds. A presumptive diagnosis of inflammatory bowel disease was made. Further lab showed an ESR of 5 and a lipase of 87. Stools were negative for WBC's, Giardiasis antigen. He was set up for a colonoscopy which was cancelled when his stool came back positive for *Entamoeba histolytica*. He responded to treatment with metronidazole and paromomycin, but the during the second week of treatment he came in complaining of SOB, DOE and weakness. A CBC

showed a HGB of 6.3, HCT 17.8 and a WBC of 7.1 which consisted of blasts with Auer rods.

DISCUSSION: Amebic dysentery can mimic other intestinal diseases. The typical presentation usually consists of sx that are of less than one month duration, although some cases will consist of a chronic nondysenteric infection in which there are episodic sx of diarrhea, abdominal pain and weight loss persisting for years. This is likely in this patient, who did not develop invasive disease until his immune system was presumably affected by his developing AML.

CURATIVE ACCESSORY SPLENECTOMY IN A PATIENT WITH RECURRENT IMMUNE THROMBOCYTOPENIA. EM Elamin, M Krishnamurthy, F Colella, Medicine, Division of Hematology/Oncology, NY Methodist Hospital, Brooklyn, NY

LEARNING OBJECTIVES: 1. Manage patients with recurrent immune thrombocytopenia

CASE: A 50-year-old woman underwent splenectomy for chronic immune thrombocytopenic purpura (ITP), at that time an accessory spleen was discovered and removed. After 9 years of complete remission, she developed symptomatic relapse. She responded well to medical treatment with steroids and danazol. However, she relapsed two years later with mucocutaneous bleeding and severe thrombocytopenia. Physical examination showed petechiae. The platelet count was 6,000/uL and antinuclear antibody was negative. Her peripheral blood smear showed thrombocytopenia and absence of Howell-Jolly bodies. CT scan of abdomen suggested a small accessory spleen. Radionuclide Tc 99m liver and spleen scan revealed physiologically active splenic tissue in the splenic bed. She was treated initially with steroids, danazol and Rh0 immune globulin with a good response. Subsequently she underwent exploratory laparotomy with successful removal of 4 accessory spleens measuring between 0.4 x 0.3 x 0.3 cm and 1.0 x 0.8 x 0.7 cm. Her platelet count abruptly increased to more than a million and stabilized around 600,000/mm³. She remains symptom free and in complete remission for more than two years.

DISCUSSION: Recurrent severe thrombocytopenia in chronic ITP after splenectomy is a common serious problem. An uncommon cause is presence of an accessory spleen, but the number of reported cases is increasing due to utilization of radionuclide splenic imaging studies. This is suggested by the absence of Howell-Jolly bodies in the peripheral blood smear and the diagnosis can be confirmed by radionuclide scanning. Literature review indicates that accessory splenectomy can be beneficial in the majority of patients but complete remissions are uncommon. Our patient achieved a sustained complete remission with a normal platelet count and no symptoms for more than two years after the second accessory splenectomy. This supports the benefit of accessory splenectomy and that it can be a definitive treatment for recurrent ITP. During the initial splenectomy, the surgeon should search thoroughly for accessory spleens, which if found should be removed. ITP patients with recurrence of thrombocytopenia even years after splenectomy should be investigated for accessory spleen/s.

SPLENIC COMPLICATIONS AFTER HEMATOPOIETIC GROWTH FACTORS, WITH LETHAL POTENTIAL AS WELL AS POSSIBILITY OF REVERSAL. EM Elamin, H Taha, H Dosik, Medicine, Hematology/Oncology Division, NY Methodist Hospital, Brooklyn, NY

LEARNING OBJECTIVES: 1. Monitor patients on combination of hematopoietic growth factors. 2. Diagnose complication of the commonly used growth factors.

CASE PRESENTATION: Patient #1: An 83-year-old woman had 5q-syndrome, marked by transfusion-dependent anemia (hematocrit (Hct) as low as 12%), neutropenia (absolute neutrophil count (ANC) was 800/uL) and mild thrombocytopenia (platelet count 103,000/uL). The spleen was slightly enlarged, increasing only from 2 cm to 4 cm below the left costal margin over 7 years. Because of progressive pancytopenia and increasing transfusion requirement, she was given subcutaneous Granulocyte colony-stimulating factor (G-CSF) 300mcg daily and erythropoietin (EPO) 70,000 units weekly. Her spleen enlarged within 5 weeks to 14 cm below the left costal margin. She was subsequently hospitalized with left upper quadrant pain and abdominal distension. Her Hct was 19%, ANC 10,180/uL and platelet count 9,000/uL. Her condition deteriorated and she died with acute respiratory distress syndrome. Autopsy showed multiple splenic infarcts and hemorrhagic bronchopneumonia.

Patient #2: A 54-year-old man with refractory anemia with excess blasts had a palpable spleen tip. His ANC was 900/uL, Hct 13% and platelets 81,000/uL. While awaiting a bone marrow transplant he received periodic blood transfusions. To reduce his transfusion requirement he was given G-CSF 200 mcg and EPO 20,000 Units daily. His spleen progressively enlarged to 10 cm below the left costal margin over 15 weeks on this treatment and his transfusion requirement increased. After both growth factors were discontinued his spleen size decreased to 7 cm below the left costal margin.

DISCUSSION: Splenomegaly is an uncommon complication of G-CSF and EPO and has been explained by extramedullary hematopoiesis. Splenic infarction and rupture are very rare. Both patients had progressive splenic enlargement. One patient developed splenic infarction and ultimately died. The second patient also developed splenomegaly but, with cessation of growth factors, the splenomegaly reversed to a large degree. Although EPO and G-CSF have each been reported to cause and aggravate splenomegaly, life threatening splenic complication is rare. We wish to alert clinicians to this uncommon complication particularly when G-CSF and EPO are used in combination in patients with splenomegaly. Patients receiving G-CSF and/or EPO, especially those with baseline splenomegaly, need fre-

quent follow-up and monitoring of spleen size. Discontinuing growth factors may reverse such splenomegaly and prevent lethal complications.

ADULT T CELL LEUKEMIA/LYMPHOMA PRESENTING AS OBSTRUCTIVE JAUNDICE WITH A MAJOR RESPONSE TO CHOP CHEMOTHERAPY. *EM Elamin*, P Mehta, M Krishnamurthy, Medicine, NY Methodist Hospital, Brooklyn, NY

LEARNING OBJECTIVES: 1. Recognize and treat an unusual presentation of Adult T cell Leukemia/Lymphoma (ATLL).

CASE: A 71-year-old Jamaican woman presented with hypercalcemia, itching and jaundice. Physical examination showed a maculopapular skin rash in the chest wall, moderate hepatomegaly and small bilateral axillary lymphadenopathy. Laboratory tests were remarkable for: White cell count of $19 \times 10^3/\mu\text{L}$ (70% lymphocytes), normal hematocrit and platelet count, ALT 82 U/L, AST 70 U/L, LDH 1071 U/L, calcium 12.4 mg/dl and progressive elevation of serum bilirubin to a peak of 23.9 mg/dl (>75% direct bilirubin), alkaline phosphatase (ALP) to a peak of 952 U/L and GGT to a peak of 898 U/L. The peripheral blood smear was remarkable for lymphocytosis with 10% atypical lymphocytes and nucleated red blood cells. Immunohistochemical studies in the bone marrow were compatible with T cell leukemia. Chromosomal analysis showed a normal karyotype. HTLV-1 antibody was positive. A bone scan showed a diffusely increased uptake. Ultrasound of the abdomen showed a moderate hepatomegaly without intra or extrahepatic biliary dilatation. A CT scan revealed a borderline enlarged spleen and mediastinal and retroperitoneal lymphadenopathy. Based on these findings she was diagnosed to have an acute type of ATLL and immediately started on CHOP (cyclophosphamide, adriamycin, vincristine, prednisone) chemotherapy. After 2 cycles of CHOP the jaundice resolved completely with normalization of the liver function tests. Although she remained in a clinical complete remission in the liver she relapsed with leukemic meningitis and died before the 4th cycle of CHOP.

DISCUSSION: The acute subtype of ATLL is characterized by a rapidly fatal course. Its treatment is still frustrating and the prognosis remains poor. Adriamycin based chemotherapy is the standard treatment with a median survival of 7-8 months and cures have not been reported. Our patient presented with an acute progressive obstructive jaundice most likely due to liver infiltration by the leukemic cells. This presentation is unusual as is the dramatic and complete resolution of obstructive jaundice upon combination chemotherapy. We would like to alert physicians to the mixed response of this fatal disease to combined chemotherapy.

INCREASING PRESENCE OF NEISSERIA GONORRHOEA ENDocardITIS IN THE ANTIBIOTIC ERA. *MD Elder*, D Preddie, P Gordon, DJ Tynes, Internal Medicine; Cardiology, Detroit Medical Center, Detroit, MI

CASE PRESENTATION: Neisseria gonorrhea infection has reached epidemic proportions in the United States. However, disseminated neisseria gonorrhoeae (DGI) accounts for only 3% of these cases in the antibiotic era. Our case is a 19 year old previously healthy male patient who presented with a two week history of nausea, vomiting and chest pain. Review of systems was positive for dark urine, fatigue and anorexia over the same two weeks. He denied arthralgias, dysuria, urethral discharge, or skin rash. On physical, the patient appeared acutely ill. Temperature was 102.0F, pulse 120 and respiratory rate in the 20s. Blood pressure was 100/70. The PMI was hyperdynamic and a loud holosystolic murmur was heard from the LSB to the apex. The chest was clear. Pertinent laboratory results included a white blood cell count of 24,400 with a predominance of PMNs and a hemoglobin of 8.8gm. With the exception of sinus tachycardia, the ECG was normal. Chest x-ray revealed cardiomegaly. A transthoracic echo revealed aortic and mitral vegetations and severe insufficiency of both valves. An emergent cardiac catheterization showed a central aortic pressure of 140/40 and normal coronary arteries. The patient was taken urgently to the operating room where transesophageal echo revealed additionally an aortic root abscess, with extension to the right ventricle through a ventricular septal defect and tricuspid vegetation. Mechanical aortic and mitral prostheses were placed, the VSD was repaired, and the abscess and tricuspid valve were debrided. Cultures of blood and of the vegetations grew N. gonorrhoeae. The patient was placed on ceftriaxone for 1 week and discharged on oral ceftriaxone for five weeks. Post-op laboratory evaluation revealed a terminal complement deficiency which is known to be a risk factor for DGI. Although DGI is rare, it should be considered in a sexually active young patient. It can present with tenosynovitis and urethral discharge, although the absence of these features as in this patient doesn't rule out the diagnosis. Gonococcal endocarditis can involve normal as well as diseased valves. Typically, gonorrhea endocarditis can involve any heart valve and often requires surgical repair secondary to valvular insufficiency. With the advent of penicillin-resistant gonococcal infection, third generation cephalosporins or fluoroquinolones are an alternative treatment pending sensitivities. Our patient did not present with typical genitourinary features of Disseminated Gonococcal Infections.

AN UNEXPECTED ANION GAP. *TB Ellerin*, EM Vargo, Medicine, Johns Hopkins Hospital; Medicine, Johns Hopkins Medical Services Corporation, Baltimore, MD

LEARNING OBJECTIVES: 1. Assess a patient with metabolic acidosis and an elevated anion gap.

2. Recognize anti-retroviral associated lactic acidosis.

CASE PRESENTATION: A 45-year old man presented with a two day history of epigastric pain radiating to his back associated with nausea and vomiting. He denied

alcohol use. His past medical history is pertinent for HIV infection, with the most recent CD4 count at 1,250 and an undetectable viral load, as well as hepatitis C infection. His medications were didanosine, stavudine, and nelfinavir. His physical examination was notable for mild hepatomegaly and moderate epigastric tenderness without rebound tenderness. Laboratory data revealed an elevated amylase and lipase and normal transaminases. A diagnosis of acute pancreatitis was made and his anti-retroviral medications were discontinued. Two days later, his abdominal pain had abated and his amylase and lipase returned to near normal values. He tolerated a regular diet. However, laboratory data revealed an anion gap of 20 (mEq/L) and a bicarbonate of 24 (mEq/L). His serum glucose was normal, there were no ketones in the urine, his serum and urine toxicology screen were negative, and there was no osmolar gap. His serum lactate was mildly elevated at 3 (mEq/L). The following day, he continued to improve but was found to have an increasing anion gap of 23. His serum bicarbonate was 20 and his serum lactate was 7. An abdominal CT scan demonstrated moderate pancreatitis and mild hepatomegaly with fatty changes. On the fifth hospital day, his anion gap had increased to 25 with a bicarbonate of 18. Liver tests revealed a mild transaminitis. An arterial blood gas revealed a pH of 7.30. A surgery consultation was obtained to evaluate for the possibility of mesenteric ischemia; no evidence was found to support this process. Over the next several days, the patient became progressively dyspneic and tachypneic, with increasingly diffuse abdominal pain, nausea, and vomiting. He was transferred to the medical intensive care unit where he was found to have a severe lactic acidosis, with a peak lactate of 40, and worsening transaminitis. He suffered cardiac arrest; arterial pH was 6.85 at that time. Attempts at resuscitation were unsuccessful.

DISCUSSION: This case illustrates two important points: the first is that an anion gap, especially when it is greater than 20 mEq/L, should be evaluated. Three common causes of anion gap metabolic acidosis are lactic acidosis, diabetic ketoacidosis, and uremia. The second point is the association between nucleoside analogues and lactic acidosis. The mechanism is not understood, but is thought to be related to mitochondrial toxicity. This case also serves to illustrate how well a patient can appear during the early stages of this syndrome. Treatment is supportive, while discontinuing the implicated antiretrovirals.

CARDIAC ARREST ASSOCIATED WITH HYPOKALEMIA IN A PATIENT RECEIVING ACETAZOLAMIDE(DIAMOX) AND TRIAMTERENE/HYDROCHLOROTHIAZIDE(MAXZIDE).

NK Ericsson, J Chang, Department of Medicine, Englewood Hospital and Medical Center, Englewood, NJ

LEARNING OBJECTIVES: Recognize that use of diamox with thiazide diuretics may result in malignant arrhythmias.

CASE: A 48-year-old woman with a 5-year history of hypertension, controlled with maxzide 750 mg p.o. daily and long-acting diltiazem 240 mg p.o. daily, was diagnosed with pseudotumor cerebri at another facility about one month prior to admission to this hospital. Her potassium level was 3.8 mEq/L. 5 days prior to admission, patient began taking diamox 500 mg p.o. B.I.D. On the day of admission, patient got out of bed to use the bathroom at about 4:30 a.m. About one hour later, patient was found unresponsive in bed. The paramedics arrived 40 minutes later and found patient in ventricular fibrillation. Patient developed a pulse after 13 minutes of resuscitative efforts. Patient was brought to the emergency department intubated and unresponsive. The patient did not use alcohol, cigarettes or illicit drugs. Physical examination revealed a moderately obese patient with blood pressure of 160/60, heart rate of 130 bpm, respiratory rate of 16 per minute and temperature of 97.6o. The patient exhibited decerebrate posturing to noxious stimuli, but with reactive pupils. Corneal and oculocephalic reflexes were intact. Babinski reflexes were negative. The general physical examination was otherwise unremarkable. Her potassium level was 2.4 mEq/L. Initial EKG showed sinus tachycardia at 132 bpm, rightward axis and non-specific ST-T abnormalities. C.T. scan of brain without contrast was normal. Chest x-ray showed no cardiopulmonary abnormality. The patient was admitted as a case of severe anoxic encephalopathy resulting from cardiac arrest most likely secondary to diuretic-induced hypokalemia. Patient's diuretics were held. Diltiazem was used for blood pressure control. A total of 910 mEq of potassium chloride was used to raise and stabilize the potassium level within normal limits. Subsequent echocardiography showed no structural heart abnormality. Electroencephalography showed severe non-specific cortical neuronal dysfunction. Patient remained in persistent vegetative state throughout her hospital stay. There was no recurrence of malignant arrhythmia.

DISCUSSION: Hypokalemia is prevalent in cardiac arrest. Some authors have suggested that hypokalemia may be the result rather than the cause of cardiac arrest. Ornato et al (1985) found that hypokalemia was associated with a 2.5-fold increase in relative risk of cardiac arrest. Although we would hesitate to draw conclusion from a single case, we feel that, in light of the temporal sequence of events in our patient, hypokalemia induced by the addition of diamox to maxzide is likely to have been responsible for our patient's cardiac arrest.

AN INFILTRATING DILEMMA. *S. Fazio*, J Sackey, Medicine, Beth Israel Deaconess Medical Center, Boston, MA

LEARNING OBJECTIVES: 1. Recognize restrictive cardiomyopathy in the differential diagnosis of dyspnea.

2. Diagnose amyloid deposition disease from its clinical and laboratory presentation.

CASE PRESENTATION: A previously healthy 52 year old lawyer presented with two months of exertional dyspnea. Review of systems was negative for other car-

diac or respiratory complaints. Past medical history was negative; the patient was on no medications. Initial exam was unremarkable with the exception of a mildly elevated diastolic blood pressure. EKG and chest x-ray were unrevealing. Pulmonary function testing was within normal limits. The patient's symptoms were felt to be consistent with angina and he subsequently underwent a stress test, where no significant EKG changes were noted. One month later, he presented in follow-up to his primary care physician. At that time, laboratory tests revealed a normal CBC and SMA-7. He was scheduled for an echocardiogram, which revealed moderate concentric left ventricular hypertrophy and an ejection fraction of 60%. He was referred to cardiology for further evaluation, where it was initially felt that his symptoms were related to diastolic dysfunction. He was begun on an ACE inhibitor. One month later, he noted pronounced dyspnea and fatigue with exertion, and given increasing concern for coronary artery disease underwent cardiac catheterization, which revealed normal vessels. An extensive laboratory evaluation revealed: mild anemia (Hct 35.5), creatinine of 1.5, BUN 25, ESR 5; normal electrolytes, liver and thyroid function, urinalysis and serum and urine protein electrophoresis. Eight months after his initial presentation, the patient returned with progressive symptoms, unable to walk the short distance between the train stop and his downtown office. He had developed new lower extremity edema on exam. A repeat echocardiogram revealed new bi-ventricular dysfunction. A right-sided catheterization was performed with endocardial biopsy, which stained Congo Red and Trichrome positive, with amyloid deposition in the interstitium, endocardium, and vascular walls. He was subsequently found to have amyloid involvement in his bone marrow and kidneys, negating candidacy for a heart transplant. The patient underwent placement of an implantable defibrillator and was begun on chemotherapy, but unfortunately succumbed to a malignant arrhythmia one year after his initial presentation.

DISCUSSION: Restrictive cardiomyopathies are the least common of the cardiac myopathic disorders (NEJM 1/23/97:267-274) and can be commonly overlooked in the differential diagnosis of a patient with subtle cardiac complaints. In any patient with suspected cardiomyopathy with anemia and renal insufficiency, amyloid must be considered. The absence of proteinuria and the negative serum and urine protein electrophoresis do not rule out the diagnosis.

"I HAVE SOME DIFFICULT NEWS . . .": LATE DIAGNOSIS OF HIV INFECTION IN A YOUNG WOMAN. MD Feldman, Medicine, UCSF, San Francisco, CA

LEARNING OBJECTIVES: 1) Recognize risk factors for HIV infection for women in the US. 2) Learn the sign/symptoms of HIV infection in women.

CASE PRESENTATION: The patient is a 33 year old administrative assistant followed for her primary care in an outpatient clinic affiliated with a University medical center. She had been well generally, but over the prior six months had been seen more frequently by several providers in the clinic for allergic rhinitis and 2 to 3 episodes of sinusitis. She was referred to ENT for evaluation; complete head and neck exam and sinus CT were consistent with the diagnosis of bacterial sinusitis. On follow up with her primary care provider, the patient reported that she had felt more fatigued recently and had noticed the gradual worsening of a rash, primarily on her face. She does not smoke, drinks rarely, and has never used intravenous drugs. She is monogamous with her husband of 8 years and has a healthy 4 year-old child. On physical exam, she is 5lbs below her normal weight, has a mild facial rash consistent with seborrheic dermatitis, no oral findings, and a normal chest, cardiac and abdominal exam. Laboratory data was obtained to evaluate her recurrent sinusitis as well as her fatigue and weight loss: CBC revealed a normocytic anemia, LFT's and TSH-normal, HIV antibody—positive. The patient's PCP had sent the HIV antibody test "just to be complete" and had not prepared himself, or the patient, for the possibility that it may come back positive. She had come to the appointment alone and was shocked by the result. It had never crossed her mind that she might be at risk for HIV. Repeat HIV antibody testing was also positive, with a viral load of 500,000 and CD4 count 75. She has had an excellent response to HAART and currently has an undetectable viral load.

DISCUSSION: HIV is the third leading cause of death among all women aged 25-44 in the United States, accounting for 11% of deaths. In spite of its prevalence, delayed diagnosis of women with HIV infection continues to be a major problem. Women, and clinicians, often underestimate the risk of infection with HIV, and several studies have shown that HIV screening that targets women who admit risk factors is a strategy that misses about half of infected women. It is critical that primary care physicians become familiar with these risk factors and recognize the unique manifestations of HIV infection in women. They must identify and treat women early in the course of the disease, and ideally counsel women in ways to reduce their risk of obtaining HIV infection. The primary care physician in this case was unaware of the epidemiology and clinical manifestations of HIV in women. As a consequence, the correct diagnosis was delayed, and the delivery of the bad news (the test result) was handled improperly.

"COLORS OF THE FALL"—A TRANSIENTLY ELEVATED PTT: THE ONLY INDICATION TO A DIAGNOSIS OF MASTOCYTOSIS. MC Flemmer, Internal Medicine, Eastern Virginia Medical School, Norfolk, VA

LEARNING OBJECTIVES: To recognize A) the manifestations of systemic mastocytosis. B) that transiently elevated PTT may be a biochemical abnormality aiding in diagnosis.

CASE PRESENTATION: 54 y.o. WM presented with a 6-yr h/o recurrent "spells". The spells consisted of diffuse flushing of the body, SOB, and lightheadedness leading to syncope. Some spells included chest tightness with diaphoresis. One

episode was associated with a grand mal seizure. Past medical history was non-contributory.

During severe spells, he typically presented hypotensive (systolic BP 40), cyanotic and with a violaceous rash. No other abnormalities were consistently noted. The patient usually was promptly resuscitated with fluids and pressors. Between spells, the physical exam was normal. Cardiac catheterization, CT scan of the head and abdomen, small bowel biopsy, EEG, tilt table, Holter monitoring, ECG and echocardiography were all normal. The CBC, electrolytes, LFTs, TSH, serum cortisol, urine metanephrines and 5HIAA were all normal. The only consistent abnormality was a prolonged partial thromboplastin time (PTT) during the spells, which resolved immediately thereafter. The serum tryptase level was slightly elevated at 1.2 ng/ml (N <1 ng/ml). BM biopsy confirmed the diagnosis of systemic mastocytosis. Systemic mastocytosis results from the overproduction of mast cells and their mediators within tissues. Heparin is a preformed secretory granule-associated proteoglycan released by mast cells. Clinically significant activity of heparin release in mastocytosis may include osteoporosis and inhibition of localized clotting. The release of heparin in mastocytosis may give rise to a transiently elevated PTT. Mastocytosis needs to be in the diagnostic repertoire of the general internist to streamline a usually invasive and unwieldy workup, as typified by this patient.

CARDIAC TOXICITY IN ACETAMINOPHEN OVERDOSE: EARLY RECOGNITION AND MANAGEMENT. D Gallagher, P Basaviah, L Lee, M Burns, Department of Medicine; Department of Emergency Medicine, Beth Israel Deaconess Medical Center, Boston, MA

LEARNING OBJECTIVES: 1. Recognize acetaminophen overdose can cause significant cardiac damage 2) Recognize importance of early cardiac monitoring in acetaminophen overdose

CASE: A 73 year old woman was brought to the Emergency Department by her family with vomiting and lethargy. Her vital signs were normal, and physical exam was remarkable only for decreased level of consciousness. A CT of the brain showed no evidence of bleeding or infarction. A toxin screen was positive for acetaminophen. Upon questioning, the patient admitted to ingesting 50 grams of acetaminophen (100 Extra Strength Tylenol) in a suicide attempt approximately 18 hours prior to presentation. The first acetaminophen level returned at 394 mcg/ml, which predicts a "high probability" for hepatic toxicity by the treatment nomogram. Additional relevant labs included AST 296, ALT 243, PT16.1/INR 2.1. EKG was notable for non-specific ST segment and T wave changes with a CK 500, CK MB index 6 and troponin greater than 50. N acetyl cysteine was administered at 22.5 hours post ingestion. Initial clinical impression was that the patient had early evidence of hepatotoxicity from her acetaminophen ingestion and had secondarily sustained a myocardial infarction. The patient was admitted to the MICU. On hospital day #3, AST and ALT peaked at 3000 each and INR at 6.0. Over the next 24 hours, the patients LFT's and CK's began to improve and by day #8, liver and cardiac enzymes and coagulation values returned to normal range. Despite aggressive electrolyte repletion, several cardiac dysrhythmias were noted including runs of ventricular tachycardia up to 30 beats and sinus node re entrant tachycardia from Day #3 to Day #12. Serial EKG's revealed deep T wave inversions in all leads. An echocardiogram revealed global hypokinesis with EF of 35%. A cardiac catheterization on Day #14 found no significant coronary artery disease. The EKG abnormalities and temporal rise and fall of CK and CK MB indicate that this patient sustained subendocardial necrosis from her acetaminophen ingestion, rarely reported previously.

DISCUSSION: Hepatic and renal toxicity following acetaminophen overdose are well recognized. However, case reports in the literature have also described cardiac toxicity. The exact mechanism of injury is unclear, but one hypothesis is that acetaminophen leads to the depletion of sulfhydryl groups and glutathione. In the myocardium, these SH groups are necessary for the activity of endothelin-derived relaxation factor, an endogenous vasodilator. Lack of EDRF produces coronary artery constriction and myocardial ischemia. As this case illustrates, the onset of this effect can precede serious liver toxicity. The mechanism leading to cardiac arrhythmias is unclear, though reports indicate multifactorial causes including electrolyte abnormalities, direct myocardial toxicity by acetaminophen by products, and altered serum free fatty acid levels. Accordingly, an EKG and early cardiac monitoring should be performed on all patients who present following massive acetaminophen ingestion. Though treatment is largely supportive, it is critical to recognize the potential for significant myocardial damage.

ACQUIRED CENTRAL DIABETIS INSIPIDUS. L Genao, General Medicine, Emory University, Atlanta, GA

LEARNING OBJECTIVES: 1. Review etiologies and clinical features of acquired diabetes insipidus (DI).

CASE PRESENTATION: A 64-y.o. male with a 12 pack-year smoking history presented with right shoulder pain for two weeks. He denied neurologic symptoms or history of trauma or heavy exertion. Active and passive range of motion (ROM) at shoulder joint was intact but difficult. Treatment with nonsteroidal anti-inflammatory drug (NSAID) yielded no improvement. Minimal relief was obtained with a new NSAID and acetaminophen. One month later, the patient had decreased visual acuity in right eye, followed two weeks later by polydipsia, polyuria and nocturia. He had no nausea, headache, fever, weakness, numbness, head trauma, seizure, weight loss or night sweats. On examination, he appeared fatigued. His pupils were equal, round and reactive to light with visual acuity of 20/60 in the right

eye and $20/25$ in the left. Fundoscopic examination was normal. Cranial nerves were normal as were motor, sensory and cerebellum testing. Pain localized to the proximal humerus limited ROM of right shoulder. No abnormal cardiopulmonary, abdominal findings or lymphadenopathy found. Glucose tolerance test and prostate ultrasound were within normal limits as were chemistries, CBC, and urine. Two days later, the patient lost vision in right eye. His visual acuity was $20/600$ in the right eye and unchanged on the left. Magnetic-resonance-imaging of the brain and orbits showed innumerable intracranial lesions, including one lesion in pituitary infundibulum. There were no lesions within the orbits. Right shoulder x-ray revealed lytic lesions over proximal humerus. Chest and abdominal computed tomography revealed an upper lobe mass in right lung, right hilar nodes and bilateral adrenal masses. Lung biopsy revealed squamous cell cancer. The patient was diagnosed with staged IV non-small cell cancer, DI due to pituitary metastasis and optic neuropathy due to meningeal compression. The patient received external beam radiation to the whole brain with no improvement. He refused chemotherapy. He was treated with desmopressin, opioids and anti-emetics. The patient died comfortably at home a few weeks later.

DISCUSSION: This case illustrates metastatic malignancy as an etiology of central DI. Central DI can be familial, due to head trauma, or iatrogenic in the setting of hypophysectomy or irradiation. Other causes include infectious, vascular, neoplastic, and infiltrative disorders. The onset of DI is variable and may occur at any age. Often the patient consumes large amounts of liquids and excretes large amounts of very diluted urine. The patient is usually awakened at night by thirst, which is generally not true in psychogenic polydipsia. As long as the patient is able to replace the urinary losses, the electrolytes remain normal.

ANALYZING THE DIAGNOSTIC PROCESS OF INTERNAL MEDICINE: PSEUDOVISCERAL SYNDROMES - WHEN HEURISTICS AREN'T ENOUGH. C Geophos, K Meyers, GW Rutecki, Internal Medicine, Evanston Northwestern Healthcare, Evanston, IL

LEARNING OBJECTIVES: 1. Recognize pseudovisceral syndromes as an etiology for abdominal pain. 2. Assess the diagnostic process of internal medicine vis-a-vis other specialties.

CASE PRESENTATION: A 48 y.o. M complained of RUQ abdominal pain (2 yrs.) with a recent increase in intensity. The pain was associated with bloating, constipation, & flatus. Physical exam was unremarkable, including a "symmetric" neurologic. He was hemocult negative. Sedimentation rate, liver function tests, amylase, CBC & abdominal ultrasound were also negative. He was treated with a proton pump inhibitor. Three months later, his pain increased with referral to his back. He also complained of abdominal distention & difficulty voiding. A T11, T12 level on sensory neurologic exam was noted with a positive Romberg to the right. MRI demonstrated a thoracic level schwannoma which was removed. The neurologic and abdominal findings disappeared following removal of the tumor.

DISCUSSION: A recent editorial in The Green Journal characterized the fundamental difference between internal medicine and family practice as one of heuristics vs. critical thinking. Essentially, heuristics or rule-based thinking can lead to the appropriate diagnosis most of the time, but the ability to think critically when the atypical presentation eventuates should be the internist's forte. Pseudovisceral syndromes (PsvS), or radicular pathology mimicking an intraabdominal process, are a true test of the critical diagnostic paradigm. Although rare, PsvS can mimic almost any abdominal disease & require appropriate diagnostic suspicion—particularly when initial workup for abdominal pain is negative. Unnecessary and expensive testing can be avoided and as such PsvS require critical thinking in the context of the history & physical exam. The words of Sir Zachary Cope may serve the internist well, "One often, if not always, learns more by analyzing the process of and detecting the fallacy in an incorrect diagnosis than by taking unction to oneself when the diagnosis proves correct." PsvS appear to serve as an appropriate paradigm for the diagnostic process of internal medicine.

THROMBOSIS AT AN UNUSUAL SITE DURING PREGNANCY. PS Gibson, K Rosene-Montella, Medicine, Women & Infants' Hospital of Rhode Island, Providence, RI

LEARNING OBJECTIVES: 1) To discuss unusual sites of venous thrombosis occurring in pregnancy. 2) To illustrate ovarian hyperstimulation syndrome as a risk factor for thrombosis during pregnancy.

CASE: A 36 year old woman presented to hospital complaining of left neck pain. She was eight weeks pregnant with twins, having undergone in vitro fertilization for ovulatory and severe male factor infertility. Her only past medical problem was thalassemia minor. She had been admitted six weeks prior with ovarian hyperstimulation syndrome, two weeks after her oocyte transfer. Her symptoms at that time had included nausea, vomiting, rapid weight gain and lower abdominal pain. An ultrasound had revealed ascites and bilateral enlarged multicystic ovaries consistent with ovarian hyperstimulation syndrome. She had improved with supportive care. Four days prior to presentation she noted the subacute onset of left neck pain, followed by mild dysphagia and then obvious swelling of the region. Examination revealed tender swelling of the left neck without erythema. A lateral neck x-ray was normal but ultrasound revealed left internal jugular vein thrombosis. She was hospitalized and treated with intravenous heparin. Testing for hypercoagulable states revealed her to be heterozygous for the Factor V Leiden mutation. She was switched to low molecular weight heparin in therapeutic doses. Her symptoms gradually resolved and the remainder of her pregnancy has been uneventful.

DISCUSSION: Pregnancy is a thrombophilic state, conferring a risk of thromboembolism that is five to six times that of the non-pregnant state. The overall risk of a thrombotic event is 0.5 to 3 per 1000 pregnancies and is even higher in

women with a prior thrombosis or thrombophilia. Thrombosis during pregnancy may occur at unusual sites including the pelvic veins, the ovarian veins and rarely the cerebral venous sinuses. Ovarian hyperstimulation syndrome (OHSS) results from ovulation induction with gonadotropins during the process of in vitro fertilization, with recruitment of multiple ovarian follicles leading to cystic enlargement of the ovaries. This is associated with a marked increase in vascular permeability with accumulation of ascites, pleural and pericardial fluid. Hypovolemia, hemoconcentration and hypotension may result. Venous thrombosis is common in this setting, occurring in up to ten to thirty percent of cases. The most frequent sites of thrombosis are the internal jugular and subclavian veins. Thrombosis at other unusual venous and arterial sites has also been reported. The etiology of OHSS and its complications are incompletely understood. To date there has been no systematic study of the frequency of thrombophilias in OHSS-associated thrombosis.

AN UNUSUAL PRESENTATION OF HIV IN PREGNANCY. PS Gibson, K Rosene-Montella, Medicine, Women & Infants' Hospital of Rhode Island, Providence, RI

LEARNING OBJECTIVES: 1) To review parotid disease in HIV-infected persons. 2) To discuss lymphoepithelial parotid cysts as a first manifestation of HIV infection. **CASE:** A 30 year old Liberian woman who was thirty weeks pregnant presented to hospital with a two day history of fever, chills and right jaw pain. Her past medical history included anemia and sickle cell trait. She had received a blood transfusion in Liberia five years prior after a surgical procedure. She reported three lifetime sexual partners with no history of sexually transmitted diseases or intravenous drug use. Her current pregnancy had been uneventful until 48 hours prior to presentation when she developed fever, chills and rigors. She also noted pain and tenderness at the angle of her jaw on the right side. On examination she appeared toxic with a temperature of 100.8 degrees and a blood pressure of $81/44$. She was noted to have a tender swollen lump in the region of her right parotid gland and bilateral soft, mobile cervical adenopathy. Her pharynx was unremarkable and the remainder of her examination was normal. WBC was 5.7 with frequent band forms. A chest x-ray was unremarkable. An HIV test was ordered. Blood and urine cultures were drawn and the patient was started on intravenous ampicillin/sulbactam and gentamicin. An ultrasound the next day revealed a 4 cm. heterogeneous hypochoic mass in the area of the right parotid gland, with no evidence of abscess. Cultures were negative but antibiotics were continued and her fever gradually resolved. An MRI study revealed multiple cystic masses scattered throughout the parotid glands bilaterally and extensive cervical lymphadenopathy, consistent with benign lymphoepithelial parotid cysts seen in HIV-infected persons. The HIV test was reported positive two days later. Her initial CD4 cell count was 249. She was discharged on antiretroviral therapy and antibiotics.

DISCUSSION: Parotid disease in HIV-infected persons has a lengthy list of potential causes including: lymphoma, Kaposi's sarcoma, viral or bacterial parotitis and mycobacterial lymphadenitis. The parotids are the only salivary gland with integrated lymphoid tissue. Lymphoepithelial parotid cysts may occur in HIV-infected persons due to hyperplasia of this lymphoid tissue leading to ductal obstruction and secondary cyst formation. The lymphoid tissue stains positive for p24 antigen. The cysts are multiple and bilateral, leading to painless parotid enlargement. This typically occurs early in HIV infection and may be the first clinical manifestation of HIV disease. Viral and bacterial superinfection of lymphoepithelial parotid cysts has been described among non-pregnant patients but this is the first such case reported as an initial manifestation of HIV disease during pregnancy.

A TICK BITE IN THE BRONX. P Gliatto, Medicine, Mount Sinai School of Medicine, New York, NY

LEARNING OBJECTIVES: 1. Recognize that Rocky Mountain Spotted Fever can occur in an urban setting. 2. Appreciate that Rocky Mountain Spotted Fever can have long term neurologic sequelae.

CASE PRESENTATION: A 58-year-old Hispanic native of the Bronx, NY presented to a tertiary teaching center with four days of fevers, myalgias, confusion and diffuse weakness after a tick bite. He had a history of type 2 diabetes mellitus, hypertension, myocardial infarction and a subcortical stroke with no neurologic sequelae. He was a binge drinker and had fallen asleep in a local park when intoxicated. Upon awakening he found a tick attached to his right shoulder. The following day he developed a headache, diffuse myalgias, and fever. Over the next three days his family noted that he became progressively weak and confused to the point that he could not get out of bed without assistance. On the day before admission a diffuse, non-pruritic erythematous rash appeared over his elbows and lower back. Upon arrival to the hospital he had a fever to 39.7 C, a HR of 128, RR of 24 and BP of $117/72$. Bibasilar rales we heard on lung exam. He was tachycardic but his heart sounds were normal and without murmurs. His abdominal exam was benign and no spleen tip could be appreciated. He had a diffuse, flat, blanching erythematous rash over his elbows, buttocks and back. His thighs and calves were tender. His neurologic exam was remarkable for lethargy and mild disorientation. Cranial nerves and cerebellar function were normal and muscle strength was $\frac{4}{5}$ in all extremities. WBC-10.1, Hb-15.4, plt-127, PT and PTT normal, SMA-7 normal except for Cr-1.5 and glc-199. AST-83 but all other LFTs were normal. ESR-55. CSF analysis: WBC-7, RBC-20, pro-106, glc-98, UA remarkable for protein and 5-10 RBCs/hpf. CXR showed faint bibasilar opacities. The patient was admitted and administered intravenous doxycycline and ceftriaxone. Blood and urine cultures were negative, as were Lyme serologies. Initial IgM for *R. rickettsii* was 1:16 and IgG was 1:32. Examination of the blood smear showed toxic granulations but no WBC inclusions. A biopsy of the rash revealed a leukocyto-

clastic vasculitis. The patient was persistently febrile for the first five days of admission, and then defervesced with a coincident clearing of his sensorium. The patient was discharged on the seventh hospital day with the presumptive diagnosis of Rocky Mountain Spotted Fever. Oral doxycycline was prescribed with the intent to treat the patient for a total of ten days. Upon follow-up in the internal medicine clinic three weeks later the patient complained of vertigo and had been walking with a cane. His exam at that time showed ataxia on finger-to-nose testing and a swaying to the right on gait testing. Repeat serologies were drawn. *R. rickettsii* IgG titer was 1:512. He was given meclizine for his symptoms, which gradually resolved over the next four months.

ACUTE, PROGRESSIVE, MUSCLE WEAKNESS IN A 54 YEAR OLD, ARTHRITIC MALE. [N Golechha](#), J Sunstrum, RD Hobbs, Internal Medicine, Oakwood Hospital, Dearborn, MI

LEARNING OBJECTIVES: To recognize drug interactions and patient compliance as a cause of serious illness.

CASE PRESENTATION: A 54 year old male was admitted to the hospital following a three week history of fever, nausea, vomiting, and increasingly severe weakness of his lower extremities. His past medical history was significant for rheumatoid arthritis, hypertension, hyperlipidemia and prostate surgery. There was no history of smoking, alcohol or illicit drug use. There was no recent travel. His chronic medications included an antihypertensive and a cholesterol lowering agent. He had been recently started on a new arthritis medication four weeks prior to admission. He did not recall the name of the medication but stated they were 12 "little white pills" and he was to take three per week. Because of his severe arthritis pain he completed his new prescription in four days.

Physical examination: Temp 101.2°F, BP 130/60, Pulse 92. He was alert and oriented x 3. His cranial nerves were normal. Strength in his upper extremities was 3/5 and 2/5 in the lower extremities. Deep tendon reflexes in the lower extremities were 2+. There were no tremors, fasciculations or sensory deficits. Plantar reflexes were normal. The remainder of the physical exam was unremarkable except for classic findings of rheumatoid arthritis.

Laboratory studies: WBC 16,200, Hgb 13.9g/dl, platelets 275,000, metamyelocytes 6%, bands 11%, PMNs 69%, lymphocytes 11%, monocytes 3%, total bilirubin 0.1 mg/dl, LDH 1,169 U/L, AST 1,426 U/L, ALT 700 U/L, Alkaline Phosphatase 37 U/L, GGTP 33 U/L. Hepatitis A, B, and C studies were negative. Urinalysis: protein >300 mg, hemoglobin - 4+, RBC - negative, coarse granular casts were seen. CPK 15,045 U/L (99% CK-MM), BUN 35 mg/dl, creatinine 1.4 mg/dl. A toxicological screen was negative. Acetylcholine receptor AB was normal.

DISCUSSION: The patient's "little white pills" were identified as methotrexate 2.5 mg. His other medications were cerivastatin 0.3 mg/day and lisinopril 5 mg/day. His progressive neurological deficits were a manifestation of methotrexate toxicity. Additionally, he developed both hepatitis and rhabdomyolysis. This case is a cautionary tale about drug interactions and patient compliance. Although the correct use of methotrexate was clearly listed on the prescription bottle and his physician had explained the purpose of the drug and its side effects, the patient self-medicated himself and accordingly suffered disastrous consequences. It also underscores the potential severity of drug interactions and should emphasize that when confronted with a diagnostic dilemma, the physician must always consider the patient's medications as a potential cause of complications and morbidity.

RECURRENT IDIOPATHIC INTRACRANIAL HYPERTENSION IN PREGNANCY. [DF Graham](#), L Larson, Division of Obstetric and Consultative Medicine, Department of Medicine, Women and Infants' Hospital, Brown University, Providence, RI

LEARNING OBJECTIVES: 1. Consider idiopathic intracranial hypertension in patients with headache and rapid weight gain. 2. Recognize the controversial association of idiopathic intracranial hypertension and pregnancy. 3. Manage idiopathic intracranial hypertension in pregnancy.

CASE PRESENTATION: A 22 year old gravida 3 para 1 woman presented at 17 weeks gestation with a 4 week history of intermittent pain behind both eyes associated with throbbing right sided headache radiating to the right occiput. The pain was worse at night time. She had gained 8 lb weight over the previous 4 weeks. Three years previously, in her first pregnancy she had been diagnosed with idiopathic intracranial hypertension after a similar presentation at 16 weeks gestation. She was treated in her first pregnancy with prednisone without complication, though had a premature delivery at 35 weeks. She had no other past medical history. She was taking acetaminophen with codeine as required and prenatal vitamins. She had no allergies. Her family history was of diabetes, hypertension and breast cancer. She had smoked 5-7 cigarettes daily for 8 years. She did not drink alcohol or take recreational drugs. She was recently unemployed. On examination she weighed 161 pounds, her pulse rate was 80/min, BP 118/70, her cardiovascular examination was normal. Her chest was clear and abdominal examination was unremarkable other than uterine enlargement consistent with dates. Neurological examination revealed a mild left ptosis and bilateral papilledema, but was otherwise normal. A presumptive diagnosis of idiopathic intracranial hypertension (IIH) was made and supported by normal cranial CT scan and elevated opening pressure of 50 mmHg on lumbar puncture with normal CSF. She was assessed by ophthalmology who agreed with the diagnosis and recommended acetazolamide, which was commenced at a dose of 250 mg bid. This resulted in symptomatic relief of her headaches, but resulted in side effects of nausea, mild metabolic acidosis and hypokalemia. Her acetazolamide was discontinued and she was com-

menced on prednisone, 20 mg bid which has been maintained throughout her pregnancy.

DISCUSSION: Idiopathic intracranial hypertension has been documented to have its onset in pregnancy, but it has recently been suggested that the association between IIH and pregnancy is coincidental due to the age and sex predilection of the disease. Onset of disease has been associated with rapid weight gain. This case illustrates onset of disease during pregnancy, with resolution postpartum, followed by recurrence in a subsequent pregnancy.

OBSTRUCTIVE SLEEP APNEA IN PREGNANCY. [DF Graham](#), R Powrie, T Lee, K Rosene-Montella, Division of Obstetric and Consultative Medicine, Department of Medicine, Department of Maternal-fetal Medicine, Women and Infants Hospital, Brown University, Providence, RI

LEARNING OBJECTIVES: 1. Assess patients with pulmonary hypertension in pregnancy. 2. Manage sleep apnea in pregnancy.

CASE PRESENTATION: A 41 year old gravida 1 woman presented at 16 weeks gestation with a few days gradually increasing peripheral edema and shortness of breath. She admitted to poor sleep and hypersomnolence. She had a 50 pack year smoking history and a family history of diabetes and premature coronary artery disease. Initial clinical assessment and chest X ray was suggestive of biventricular failure. A blood gas showed a pH of 7.40, pO₂ of 41, pCO₂ of 44 and total bicarbonate of 29. A V/Q scan was low probability. Swan-Ganz catheter measurements were consistent with secondary pulmonary hypertension with a pulmonary artery pressure of 41 systolic and pulmonary vascular resistance of 20 to 42. She was treated with diuretics and improved markedly. She was maintained on 100 mg furosemide bid. Physical examination at our medical center 2 weeks later revealed a morbidly obese woman with BP 136/60, pulse 98, respiratory rate 16. Her tongue was slightly large with a low lying soft palate. Her cardiac examination was unremarkable. There was 3+ pitting edema up to the level of her thighs. Her chest was clear. Abdominal examination showed morbid obesity. Pulmonary function tests revealed a restrictive pattern. Sleep study confirmed obstructive sleep apnea, with desaturations to 72% despite 2 liters of nasal oxygen. Echocardiogram showed normal left and right ventricular function. Doppler evaluation estimated a peak pulmonary systolic pressure of 62. She declined right heart catheterization. Her pulmonary hypertension was considered to be secondary to obstructive sleep apnea and restrictive lung disease imposed by her obesity. She was managed with bilevel positive airways pressure administration which prevented her oxygen desaturation during sleep. A multidisciplinary approach to her delivery involving maternal-fetal medicine, critical care and general medicine is planned with a slow early epidural and passive descent of the fetus.

DISCUSSION: Pulmonary hypertension in pregnancy is associated with substantial maternal morbidity and mortality. Presenting features, etiologies and management of this rare but critical medical complication of pregnancy will be discussed.

THE SEROTONIN SYNDROME: WHEN TWO COMMON MEDICATIONS DON'T MIX. [R Granieri](#), Internal Medicine, University of Pittsburgh, Pittsburgh, PA

LEARNING OBJECTIVES: 1. TO RECOGNIZE THE CLINICAL PRESENTATION OF SEROTONIN SYNDROME 2. TO RECOGNIZE A POTENTIAL ADVERSE DRUG REACTION WHEN TWO COMMON MEDICATIONS ARE USED SIMULTANEOUSLY
CASE PRESENTATION: A 61 year old female with hypertension and chronic depression presented with a one week history of nasal congestion, cough and maxillary sinus tenderness. She was diagnosed with a bacterial sinusitis and given trimethoprim-sulfamethoxazole and dextromethorphan. Five days later, she presented with nausea and vomiting. Her physical exam was unchanged. She was told to push fluids, continue the dextromethorphan but discontinue her antibiotic. Two days later she presented with dizziness and generalized anxiety. She was noted to be orthostatic, flushed, diaphoretic, tachycardic, hypertensive and delirious. She was admitted to the hospital.

In addition to her cough suppressant, medications included venlafaxine 37.5 mg 3 po bid and enalapril 10 mg po bid. On physical, she was a tremulous anxious appearing female with flushed facies. She was afebrile, pulse was 108 and her BP 190/66. On neurologic examination she was delirious with symmetric hyperreflexia. Muscle tone was normal. Her CBC, TSH, renal function and urinalysis were normal. CK was 606 with a RI 0.6. The patient's nausea, vomiting, flushing, diaphoresis, tachycardia, hypertension, delirium, and elevated CKs, in the absence of muscle rigidity and fever, were thought to be due to the SEROTONIN SYNDROME. Her venlafaxine and dextromethorphan were discontinued and she was treated with cyproheptadine with full resolution of symptoms. She was discharged to home on the fourth hospital day.

DISCUSSION: The serotonin syndrome is an important, albeit rare, complication of selective serotonin reuptake inhibitors (SSRIs). It can progress to cardiac arrest, seizures and multiorgan system failure. Suspicion for the serotonin syndrome should be raised in patients with exposure to a drug, or combination of drugs, that increase CNS serotonin activity. The classic presentation includes cognitive and behavioral changes, autonomic nervous system dysfunction and neuromuscular abnormalities. There are no specific laboratory abnormalities and an elevated CK is seen in 18% of cases, making it difficult to distinguish between neuroleptic malignant syndrome. In this case, precipitation of the Serotonin Syndrome was likely secondary to the use of dextromethorphan with venlafaxine. Dextromethorphan, a potent inhibitor of serotonin uptake, can lead to increase CNS serotonin activity. Treatment includes supportive care, discontinuation of serotonergic agents and antiserotonergic drugs (cyproheptadine).

Since SSRIs are commonly administered by general internists in the treatment of depression and since dextromethorphan is a common ingredient in prescription and over-the-counter cold remedies, knowledge of the serotonin syndrome is imperative.

EVALUATION OF UNINTENTIONAL WEIGHT LOSS. EH Green, R Granieri, Dept of Medicine, University of Pittsburgh Medical Center, Pittsburgh, PA

LEARNING OBJECTIVES: To discuss a rational approach to the evaluation of unintentional weight loss in the outpatient setting.

CASE: The patient is a 35 year old woman with a history of hypertension and Turner's syndrome who presented for a routine physical. She denied any complaints, including nausea, vomiting, early satiety, diarrhea, hematochezia, abdominal pain, oral ulcers, fevers, night sweats, chills, cough, dyspnea, hematuria, vaginal discharge, menorrhagia, or symptoms of depression. Medications included hydrochlorothiazide, loratidine, and oral contraceptives. Family history was negative for cancer, diabetes, or gastro-intestinal diseases. Social history included ½ pack per day of cigarette smoking without alcohol or illicit drug use. Her physical examination was notable for a 10 pound weight loss over three months, dropping from a stable weight of 124 pounds. The remainder of the exam, including rectal and pelvic exams, was normal. Laboratory data included a normal CBC, a normal TSH, normal liver injury and synthetic tests, and a random glucose of 172. A chest radiograph was within normal limits. Three months later the patient returned for further follow-up. She continued to be without complaints. Her weight had fallen another four pounds (to 110) and a fasting glucose was 87. Three months later her weight had stabilized at 110 pounds.

DISCUSSION: Asymptomatic weight loss is a common problem with potentially serious implications. It can be the first sign of an underlying malignancy, point towards the development of systemic disease, or expose underlying mental illness. Weight loss can be generally classified by four pathophysiologic mechanisms: decreased intake, impaired absorption, increased loss, and excess demand. There have been three well designed studies published in the last twenty years that provide a framework for a rational clinical evaluation. Each of them examined unintentional weight loss in a predominantly elderly age group. Although they differ in methodology and setting their results were concordant. Cancer was the most frequently diagnosed cause of involuntary weight loss (accounting for 16-36%). However, the vast majority of cases were not caused by an undiagnosed neoplasm. Depression, diabetes, thyroid problems, and gastrointestinal disorders all accounted for some cases. Approximately 25% of the cases never received a diagnosis. This cohort differs from others with asymptomatic weight loss in that their mortality and morbidity do not differ from age-matched controls. All three studies showed that when an etiology was discovered it was diagnosed by a careful history and physical exam coupled with basic blood work (CBC, LFTs, TSH), a chest radiograph, and age-appropriate cancer screening. In this case, no specific cause of weight loss was determined and the patient has had an uncomplicated course.

DOCTOR HEAL THYSELF: RECOGNIZING HAZARDS IN THE WORKPLACE. J Green-McKenzie, Occupational Medicine, University of Pennsylvania Medical Center, Philadelphia, PA

LEARNING OBJECTIVES: 1) Diagnose occupational asthma (occ asthma). 2) Recognize importance of workplace modification and manage return to work issues. **CASE:** A 46 yo clinical pathologist presents with complaints of increasing shortness of breath for the past 4 years. He was well until the beginning of his internal medicine residency 18 years ago when he noted an erythematous rash on his hands whenever he used latex gloves. This rash did not clear with self-prescribed steroid creams. At the beginning of his pathology residency 14 years ago he noted a non-productive cough after exposure to formalin. He remained in relatively good health except for occasional chest tightness, wheezing and coughing while at work where he was exposed to latex and formalin on a daily basis. These symptoms did not interfere with his daily activities which included vigorous exercise at least 3 times each week. Approximately 4 years ago he noted that his chest tightness, wheezing and cough became significantly and progressively worse after an exposure. He also developed progressive dyspnea on exertion such that, by the time of presentation, he was unable to walk more than a flight of stairs or a hundred yards without feeling markedly short of breath. His symptoms were worst at the end of the work week as was his self-monitored peak flow, while his symptoms lessened on weekends and during vacation and his exercise tolerance increased. He also noted heavy breathing when he "flipped" latex gloves off his hands and experienced similar symptoms when he ate fruit such as kiwi, avocado and banana. He was evaluated by an allergist a few months ago and was diagnosed with chronic sinusitis and allergy to formaldehyde and latex. He was treated with two courses of antibiotics without relief, and prescribed a steroid taper and a beta agonist inhaler to ameliorate his response to exposures at work.

PE: T98.7 P84 BP130/90 Gen: obese. Lungs: distant breath sounds, no wheezing. Cardiac: nl. Abd: obese. Skin: no rashes. Neuro: Aox3, nonfocal.

Labs: CXR: poor inspiration. Chest CT: mild bronchial wall thickening c/w mild airway dz/bronchitis. Spirometry: FEV1 = 77% predicted, FVC = 81% predicted, FEV1/FVC = 95% predicted. Cardiac ECHO: trace TR. Latex skin test:(+). Latex RAST:(-).

DISCUSSION: The prevalence of latex induced occ asthma is 2.5-6%. RAST is (-) in up to 30% of pts with latex allergy. Formaldehyde is an irritant, may exacerbate asthma and can lead to chronic bronchitis. This patient was removed from the workplace shortly after presentation as no reasonable accommodation could be

made for him and he was at risk for potentially serious and fatal anaphylaxis. Clinicians should suspect occupational asthma in patients with new onset asthma or asthma symptoms that worsen during or after work. Treatment for latex allergy is cessation of exposure.

AN UNUSUAL CAUSE FOR NEW ONSET OF FOCAL SEIZURES. A Gupta, General Medicine, SFGH-UCSF, San Francisco, CA

LEARNING OBJECTIVES: 1) Recognize neurocysticercosis as a cause of focal seizures. 2) Review diagnosis and management of neurocysticercosis.

CASE PRESENTATION: A 67 year-old homeless, white man presented to the clinic with spells of right facial twitching for two days. He denied associated loss of consciousness, headache, visual symptoms or other focal neurological symptoms. His past medical history was significant for remote tobacco and IVDU, diabetes mellitus, elevated PSA and recent negative HIV test. He denied any foreign travel. During the physical examination the patient was witnessed to have repetitive jerking movements of the right face lasting one minute. He was then noted to be dysarthric with a central 7th nerve palsy. His focal deficits resolved completely after ten minutes and the rest of his examination was normal. Work-up included normal CBC, electrolytes, RPR, urine tox and chest X-ray. A head CT with and without contrast revealed small, partially-calcified, ring-enhancing lesions. The largest in the left frontal lobe was associated with vasogenic edema. Neuroradiographically this was most consistent with neurocysticercosis but TB, metastases and toxoplasmosis could not be excluded. A PPD and serum cysticercosis immunoblot assay were negative. The patient refused lumbar puncture and refused treatment with anti-epileptics. A follow-up head MRI at 3 months revealed 5 partially calcified lesions with interval resolution of edema and no evidence of abnormal enhancement. The patient has remained seizure free for the past 8 months.

DISCUSSION: Neurocysticercosis is the most common parasitic central nervous system infection worldwide and is caused by *Taenia solium*, the pork intestinal tapeworm. It is an important cause of adult onset epilepsy particularly in patients from endemic areas namely, Mexico, Central and South America, India and sub-Saharan Africa. The incidence in the US is not known but estimated to be over 1000 cases per year. Transmission can be from either ingestion of undercooked pork or less commonly from ingestion of *T. solium* eggs in fecally contaminated food or direct contact with a tapeworm carrier. Head CT and MRI are the most effective means of diagnosis. The sensitivity of antibody testing is high for multiple active cysts (99%) but much lower for calcified stage of disease and therefore, a negative serology should not exclude diagnosis. Treatment with antihelminthic drugs is controversial as most disease is self-limited and in some cases treatment may precipitate complications. Stool O+P studies are insensitive but may be useful to screen close contacts of infected patients. This case is interesting in that it involves a patient who does not come from an endemic area but lives among many homeless people who do.

GEMFIBROZIL AND CERIVASTATIN: A POTENTIALLY DANGEROUS COMBINATION. AJ Haavisto, LJ Zakowski, Medicine, University of Wisconsin Medical School, Madison, WI

LEARNING OBJECTIVES: 1.) Recognize that the combination of gemfibrozil and cerivastatin can cause rhabdomyolysis. 2.) Identify the clinical spectrum of myopathy associated with cholesterol medications. 3.) Appreciate that certain patients are at greater risk for serious complications from this combination.

CASE: A 51 year old female presented to the emergency room with total body myalgias and lethargy. Four weeks earlier, the patient's primary physician discontinued simvastatin, because of complaints of arthralgias. At that time, the patient began taking both gemfibrozil and cerivastatin. There was no history of recent trauma, excessive exercise, infection, seizure or ethanol abuse. Past history was significant for an aortobifemoral bypass with a left nephrectomy in 1992 for peripheral vascular disease and renal artery stenosis. She also had a history of hypertension, hypercholesterolemia and coronary artery disease. In addition to gemfibrozil and cerivastatin, the patient was also taking metoprolol, amlodipine/benazepril, cyclobenzaprine and naproxen. Her baseline creatinine one month prior to this presentation was 1.6. On examination, the patient appeared acutely ill. Blood pressure was 170/70. The exam was otherwise unremarkable. Laboratories revealed potassium 8.0, creatinine 7.6 and creatinine kinase (CK) 60,000. EKG exhibited PR lengthening and peaked T waves. She was diagnosed with rhabdomyolysis and acute renal failure. She was given kayexalate, sodium bicarbonate, glucose, insulin, calcium chloride, IV fluids and mannitol. Following one course of dialysis, potassium was 3.9 and creatinine 4.9. Laboratory values returned toward normal over several days, and no further dialysis was necessary. One month later her creatinine was 1.3.

DISCUSSION: HMGCoA reductase (statin) medications, either alone or in combination with other medications, can cause myopathies including asymptomatic elevations of CPK, myalgias, myositis or rhabdomyolysis. Statins are metabolized through the cytochrome P450-3A4 enzyme; any medication also metabolized by this enzyme can cause an increase in statin activity and likelihood of myopathy. For both statins and gemfibrozil, the risk of myopathy increases with coexistent renal insufficiency. Approximately 5% of patients taking both gemfibrozil and a statin develop asymptomatic rises in CK that are usually reversible with discontinuation of the medications. Rhabdomyolysis is a very rare but serious complication of combination therapy, and only one case report has been published describing rhabdomyolysis in a patient taking both gemfibrozil and cerivastatin. All physicians prescribing statins should recognize that these medications, either

alone or with gemfibrozil, can cause serious myopathies, especially in patients with renal insufficiency.

THE LURE OF THE UNIFYING DIAGNOSIS. P. Haidet, R El Farra, Houston VAMC, Baylor College of Medicine, Houston, TX

LEARNING OBJECTIVES: 1) Recognize common manifestations of sarcoidosis and thyroiditis. 2) Consider the possibility of two simultaneous processes when constructing a differential diagnosis.

CASE PRESENTATION: A 52 year old African-American male presented to the Houston VAMC Emergency Department (ED) complaining of dyspnea. He related a two-week history of neck swelling and pain radiating into the right ear. One day prior to presentation, he noted fever and dyspnea. He denied cough, sputum, or pleuritic pain. His past medical history included type II diabetes mellitus, depression, and lower back pain, all longstanding. He denied recent travel. His diet reflected adequate iodine intake. He quit cigarette smoking several years prior. He previously worked as a welder. He denied any family history of cancer or endocrine/autoimmune diseases. In the ED, his exam revealed a temperature of 102F, pulse 130/minute, and 24 respirations/minute. Diffuse cervical and axillary lymphadenopathy were noted. Exophthalmos or lid lag were absent. Thyroid exam revealed a visible goiter, warm to the touch and tender to palpation. Auscultation of the chest was notable for bilateral basilar rales. Evaluation by flexible nasotracheal endoscopy revealed mild to moderate extrinsic compression of the trachea, and the patient was urgently intubated. After intubation, oxygen saturation was 96% on 60% FIO₂. Chest radiograph revealed hilar lymphadenopathy and a right posterior-basal infiltrate. Computed tomography revealed diffuse enlargement of the thyroid gland as well as cervical, axillary, hilar, and para-aortic lymphadenopathy. Fine needle thyroid aspirate showed nonspecific inflammation. Two separate cervical lymph node biopsies revealed non-caseating granulomas consistent with sarcoidosis, and bronchoscopy with biopsy also revealed non-caseating granulomas. TSH initially was normal, but became suppressed shortly after the CT scan (which included iodinated contrast).

DISCUSSION: In our attempt to identify a parsimonious diagnosis to explain all of the patient's findings, our list of possibilities included exceedingly rare entities such as sarcoidosis of the thyroid, lymphoma with thyroid involvement, and metastatic thyroid carcinoma. Upon review of incidence and prevalence data, we realized that the coexistence of two diagnoses (thyroiditis plus acute or chronic sarcoid) is more likely. Our search for a unifying diagnosis was based on the assumption that one event is more likely to occur than two simultaneous events. In this case, however, a single diagnosis necessitates the consideration of atypical presentations of rare entities. This is in contrast to the scenario of two typical presentations of more common entities. While the concept of parsimony is helpful in constructing a differential diagnosis, we found that strict adherence to the goal of a unifying diagnosis led us to overlook the more obvious choice.

YOU KNOW DOC, "IT'S LIKE I GOT ANTS IN MY PANTS." LK Halasyamani, IPC of Illinois, Hinsdale, IL

LEARNING OBJECTIVES: 1) Diagnose restless legs syndrome from its clinical manifestations. 2) Recognize that restless legs may represent a more serious underlying condition. 3) Treat the clinical syndrome of restless legs.

CASE PRESENTATION: JS is a 55-year-old school bus driver who presented to the emergency room complaining of "ants in his pants." He had increasing difficulty keeping seated while driving, and he could hardly sleep at night, pacing the floor of his bedroom. He also noted increased fatigue over the past few months and felt short of breath climbing two flights of stairs. JS had a PMHx of mild hypertension. He took no medications and had no known drug allergies. He smoked a ½ ppd and had 2-3 beers each weekend. He lived with his wife, who had also lost sleep from his midnight pacing. Family history was negative for mental illness and neurological conditions. On physical exam he appeared well, but had trouble sitting still. He repeatedly rubbed his legs and preferred to pace the floor. He had a BP of 145/95 and regular pulse of 104; he was not orthostatic. HEENT exam was notable for pale conjunctivae and non-icteric sclerae; he had no LAN; cardiac exam revealed a ½ systolic ejection murmur at the LUSB; lungs were clear; he had no hepatosplenomegaly; rectal exam revealed no masses and brown stool that was positive for occult blood; extremities showed trace edema; neurologic exam was unremarkable. Initial laboratory evaluation revealed a microcytic, hypochromic anemia, with a hematocrit of 23. The remainder of his CBC, electrolytes and liver function tests were within normal limits. His ECG revealed sinus tachycardia with evidence of LVH. His iron studies later revealed a ferritin of 8. JS was admitted to the hospital and received 2 units of PRBC. He subsequently underwent a colonoscopy which demonstrated a 3-cm malignant mass in his left colon. He underwent a partial bowel resection. His restless legs improved with iron repletion. JS has returned to driving his school bus, and both he and his wife are sleeping more soundly at night.

DISCUSSION: Restless legs syndrome can be diagnosed solely on the basis of a patient's history. The salient symptom is a creeping sensation primarily on the legs, prominent at rest and relieved by movement. In many cases, the syndrome of restless legs occurs as an idiopathic condition of many years' duration; however, in a small subset of patients, iron-deficiency anemia, malignancy, uremia, and pregnancy can predispose to restless legs, and treatment of the underlying condition can lead to marked improvement in symptoms. For patients with idiopathic restless legs, therapies are tailored to the severity of the syndrome. Mild cases can

be treated with benzodiazepines or opioids; moderate or severe symptoms are treated with carbidopa-levodopa, bromocriptine or pergolide.

EXCESSIVE ZINC INGESTION—A REVERSIBLE CAUSE OF ANEMIA AND NEUTROPENIA. WR Harper, Medicine, University of Chicago, Chicago, IL

LEARNING OBJECTIVES: 1) Learn how excessive zinc intake can lead to anemia and neutropenia. 2) Identify people at risk for this disorder. 3) Recognize the need to be thorough in assessing supplement use.

CASE PRESENTATION: The use of complementary medicine continues to increase. Zinc has been promoted for over 25 conditions, including the common cold. Excessive zinc intake impairs copper metabolism leading to reversible anemia and neutropenia. A case is presented of a 27 year old male without medical problems until 4 weeks prior to presentation when he noted progressive shortness of breath with exertion, ultimately with only one flight of stairs. He denied any fevers, chills, or weight loss. He was a non-smoker with no travel or known ill contacts. He did state that he had been taking 700-1000 mg of zinc per day for his acne. His physical exam was notable for orthostasis, pale conjunctivae, and normal pulmonary and cardiac exams. He had no lymphadenopathy. His evaluation included a normal CXR, EKG and Chem 17. His mononucleosis serologies were normal. His hemoglobin was 4.5, with a hematocrit of 14.7 and a MCV of 85.7. WBC count was 1.2 with 5% neutrophils, 70% lymphocytes and 25% monocytes. His reticulocyte count was zero, and his B12 and folate levels were normal. Iron was high at 168, his ferritin was 742. TIBC was normal at 269. His zinc level was 3.18 mcg/ml, normal being 0.66-1.10. Copper was 0.10 (0.75-1.45 mcg/ml). Withdrawal of zinc and administration of IV followed by oral copper normalized his zinc and copper levels. His anemia and neutropenia resolved. His diagnosis was zinc-induced hypocupremia with resultant anemia and neutropenia.

DISCUSSION: Zinc promotes the expression of its intestinal intracellular binding protein, metallothionein, which has higher avidity for copper. This leads to loss of copper as the intestinal cell is shed. As little as 2.5 mg over the RDA of 15 mg may lead to a negative copper balance. Hypocupremia is also seen in patients on TPN without trace elements, patients after bowel resection, and premature infants on formula. Copper is a key component of enzymes, including those necessary for iron release from stores and those of heme synthesis. There is a sideroblastic picture on bone marrow examination often with ring forms. The neutropenia arises from delayed maturation of WBC's caused by low levels of other copper enzymes. Traditional causes of drug-induced sideroblastic anemia have been isoniazid, chloramphenicol and alcohol. Zinc should be considered as well. The use of complementary medicine continues to rise. In one study, the use of 16 alternative therapies by a random population of individuals increased from 33% in 1990 to 42% in 1997. Accordingly, we must be extraordinarily vigilant in our history taking. We must also be increasingly knowledgeable about the potential complications of these supplements.

SACROILIAC PAIN AS A PRESENTING SYMPTOM OF BRUCELLOSIS. RH Harris, General Internal Medicine, University of Colorado Health Sciences Center, Denver, CO

LEARNING OBJECTIVES: 1) Understand that sacroiliac pain can be a presenting symptom of Brucellosis. 2) Recognize that dietary, travel and occupational histories are key in raising suspicion for Brucellosis in the correct clinical setting.

CASE PRESENTATION: A previously healthy 19 year-old hispanic male presented to an ambulatory clinic complaining of 2 weeks of progressively worsening left buttock pain. He denied trauma or inciting event. He denied fever, lower extremity numbness and bowel or bladder incontinence. He presented to clinic because of inability to ambulate secondary to left buttock pain and perceived left leg weakness. He denied intravenous drug use and reported that he had emigrated from Mexico 6 months prior to presentation. We later learned that he had illegally entered the United States, approximately one week prior to the onset of his buttock pain, by hiding in the trunk of a car. He reported giving himself an IM injection of penicillin in the left buttock 2 weeks prior to the onset of symptoms. On initial exam, he was afebrile and was noted to have severe left buttock and left sacroiliac tenderness. His lower extremity neurological exam was normal with only mildly diminished strength in the left hamstring. During the first day of hospitalization, spiking fevers to 39 Celsius were noted. The fevers occurred 3 to 4 times daily and were associated with profuse diaphoresis. A CT scan of the pelvis and abdomen revealed splenomegaly but no pelvic abnormalities. A whole body bone scan revealed clustered increased activity in the knees, ankles, shoulders, elbows and sacroiliac joints. The possibility of a bone marrow packing disorder or a systemic inflammatory arthropathy was entertained. Hematology/Oncology was consulted and a bone marrow biopsy was performed which showed a normocellular marrow with noncaseating granulomatous inflammation. The patient continued to have cyclic fevers and intense sacroiliac pain and tenderness. Further history taking revealed the patient's true travel history and a dietary history of goat milk injection. Blood cultures drawn on the fourth day of hospitalization grew a gram-negative coccobacillus organism which was later identified as *Brucella melitensis*. Patient's pain and fever responded quickly to intravenous gentamycin and oral doxycycline.

DISCUSSION: Brucellosis is a zoonosis which is transmitted to humans by contact with infected animals or animal products. The most common human form, *Brucella melitensis*, infects goats, sheep and camels. Unpasteurized milk product consumption, travel to high incidence areas and occupational exposure are important historical facts which may raise suspicion for brucellosis. Sacroiliitis can

occur either as a monoarticular septic arthritis or as part of a reactive asymmetric polyarthritis.

WHEN PATIENT AND FAMILY DISAGREE: DECISIONMAKING AND ADVANCE CARE PLANNING IN AN 85 YEAR OLD WITH ANEMIA. JM Hauser, Medicine, University of Chicago, Chicago, IL

LEARNING OBJECTIVES: 1) Recognize the difficulties of advance care planning in the absence of complete medical information. 2) Weigh family and patient perspectives in decisionmaking for an elderly parent.

CASE PRESENTATION: An 85-year-old woman with a history of anemia, weight loss and diabetes was admitted to the hospital with diffuse abdominal pain. She had been having intermittent pain for two weeks and recently developed a fever. A CT scan revealed extensive diverticulitis and abscesses. Her only laboratory abnormality was a carcinoembryonic antigen (CEA) of 26 (nl 0-5). A colonoscopy was recommended to assess her anemia and the possibility of a malignancy. As she had previously when her anemia was discussed, she declined to have the procedure. Her husband of 57 years, son, and daughter disagreed with her refusal and tried to persuade her to have the colonoscopy. But she insisted that she "did not want to know" whether she had cancer. After discharge, a hematocrit sent from home was 21.1, down from 33.9 on the previous day. Over the phone, her daughter reported that the patient felt tired and had had one "very dark" bowel movement but no other symptoms. She refused to come to the hospital. As it was not clear that she understood what was happening, she was seen at home that evening. Upon arrival, her family said: "we've decided she needs to go to the hospital and if she doesn't want to, we'd like you to call an ambulance and make her go. Is that OK?" On exam, she was drowsy but oriented. Her pulse was 92 and her abdomen was soft. She did not believe she had anemia. Although she understood that if she were bleeding, it might lead to her death, she refused to go to the hospital. At the same time, she "wanted to be around" for her daughter. Despite her protests, the decision was made to take her to the emergency room.

DISCUSSION: While models of decisionmaking and advance care planning strive for complete medical information and agreement between patient and family, these are often absent. In this case, a competent patient refused workup for a possible cancer, over the objections of her family. Subsequently, in an emergency when she was not competent, she was sent to the hospital over her objections. Could this have been foreseen? Should this patient, her family and her physician have previously come to a consensus about this type of situation? This case thus raises issues of the specificity of advance care planning, the situations when a patient's immediate wishes can be overridden and the balance of patient and family wishes in the face of urgent medical decisions.

EPHEDRINE NEPHROLITHIASIS ASSOCIATED WITH CHRONIC EPHEDRINE USE. S Hernandez, BA Costello, Dept of Internal Medicine, MAYO CLINIC, Rochester, MN

LEARNING OBJECTIVES: 1) Awareness of nephrolithiasis as a potential complication of ephedrine overuse. 2) Recognition of importance of asking patients about over-the-counter (OTC) supplements.

CASE: A 34 yo female was referred to our institution following a diagnosis of nephrolithiasis. Over the past 20 years, she reported 2-3 episodes of nephrolithiasis per year and was diagnosed with calcium oxalate stones. One day prior to admission, she developed severe left flank tenderness that radiated into the groin and was accompanied by nausea and chills. Prior to her transfer an IVP demonstrated a high-grade obstruction of the distal left ureter by a 5 mm stone. Upon admission, analgesia and hydration were provided; urine was strained for debris, 24 hour urine for citrate, oxalate, calcium, uric acid, and cystine was normal. She passed a stone that was analyzed and found to be composed of ephedrine. Initially she had reported only occasional use of albuterol for asthma, but upon further questioning, she reported long-term use of ephedrine and pseudoephedrine for nasal congestion up to 4 times daily. In the past year, she had also been using herbal supplements for weight loss. We advised her to discontinue all of these agents except albuterol. She has not had further episodes of nephrolithiasis after 6 months of follow-up.

DISCUSSION: Ephedrine nephrolithiasis has previously been reported in patients consuming ephedrine or ephedrine-containing herbal preparations (1, 2). Our case suggests that chronic use of ephedrine is sufficient to cause recurrent nephrolithiasis. Herbal supplements used for dietary purposes commonly contain ephedrine and we suspect our patient was likely receiving additional ephedrine through the supplement. Herbal supplements are not subject to the same review and approval process as are food products and medicines (3). When a supplement is on the market, lack of safety needs to be demonstrated before it can be restricted. In 1997, the FDA did call into question the safety of supplements containing ephedrine (4). However, it remains a readily available preparation and is often hidden in dietary supplements. Consumers do not usually recognize this fact. This case report add to previous findings of adverse effects related to ephedrine use.

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BRAIN DEATH !! OR ACUTE BARBITURATE INTOXICATION. D Hindka, T Rohr-Kirchgraber, Internal Medicine, Upstate Medical University, Syracuse, NY

LEARNING OBJECTIVES: Recognize & Manage Acute Barbiturate Intoxication.

CASE: 50 Year old woman, a nurse, was brought to the emergency room by the family in an unconscious state. Her husband stated that she had been sitting up conversing & suddenly rolled over & became unresponsive. Her family stated that the patient was a recovering narcotic & alcohol abuser & that she has been 'clean' for last 2 years.

They felt that she had been acting funny for 1-2 days prior to this incident. The patient denied to the family any drug use & a search of the home didn't reveal any drugs or drug paraphernalia. Patient was promptly intubated in emergency room because of respiratory depression & admitted to intensive care unit. Examination revealed a middle aged, obese woman, comatose & intubated, hypotensive & afebrile. Neurological-Pupils bilaterally constricted, sluggishly reactive, Doll's Eye response absent. Generalized motor flaccidity & absent deep tendon reflexes. Planters bilaterally absent. No spontaneous breathing activity. Chest, Cardiovascular & Abdominal exam were unremarkable.

Labs: Routine blood counts & Chemistries unremarkable. CAT Scan of brain was normal. **Diagnosis:** Brain Death !! **Treatment:** Trial of Flumazenil & Naloxone attempted without any change noted. But a few hours later the culprit showed. The urine toxicology returned positive for Barbiturate & Serum phenobarbital level was found to be in lethal range of 103mcg. She made a complete neurological recovery over the next several days. We hope that none of her patients suffered from seizures at the place she was performing her duties.

DISCUSSION: Acute Barbiturate Intoxication. Yes Barbiturate overdose can sometimes mimic brain death especially in late stages when pupillary dilation may occur. Incidence of barbiturate poisoning has declined markedly in recent years, largely due to decrease in their use as sedative-hypnotics. But still barbiturate poisoning remains a significant problem. Fatal Dose of Phenobarbitone is 6-10 Grams orally. Signs & Symptoms mainly consist of CNS & Cardiopulmonary depression. They may consist of unconsciousness, constricted pupils, depressed reflexes, hypothermia, hypotension, pulmonary edema, pneumonia & renal failure. During recovery hyperthermia can occur.

Management: 1. Gastric lavage even if ingestion is as late as 24hrs. because barbiturates slow the gastrointestinal motility & hence absorption is delayed. 2. Activated charcoal & cathartics. Charcoal is very good adsorbent. 3. Alkalinize the urine with intravenous sodium bicarbonate & diuresis to help excrete barbiturate in urine. 4. General supportive care. 5. Watch for pneumonia, hyperthermia & renal failure & treat as appropriate. 6. Hemodialysis is rarely necessary.

SINUS INFECTIONS AND ZEBRAS. RD Hobbs, Internal Medicine, Oakwood Hospital, Dearborn, MI

LEARNING OBJECTIVES: To recognize that other, more serious conditions can occasionally mimic routine problems such as a "simple" sinus infection.

CASE: A 76-year-old female presented to an outpatient clinic with a six-month history of sinus problems. She complained of stuffiness, a full sensation in the right maxillary and frontal sinus and some post-nasal drip. She denied any significant allergy history although she frequently had sinus problems during the fall and winter. She had treated herself with over the counter medications but nothing helped. Physical exam was unremarkable except for slight tenderness over the sinuses as well as post-nasal drip. Antibiotics and decongestants were prescribed. She returned to the clinic a week later and stated she felt no better. Her medications were changed and sinus films were ordered. The films showed opacification of the sinuses. She was lost to follow-up but reappeared a month later with persistent sinus discomfort, a very slight asymmetry of her face with slightly greater fullness of the right cheek, and the story that she had coughed up a piece of "cooked turkey meat." She had the sample in a plastic bag she had brought to the office. It contained what appeared to be a dry, bloodless piece of tissue. The specimen was placed in formalin, an emergency CT scan of the sinuses and a stat referral to ENT were ordered. Studies revealed an esthesioneuroblastoma—a highly malignant tumor that metastasizes early and widely. She went to surgery but despite chemotherapy and radiation therapy, died six months later.

DISCUSSION: This case illustrates that even routine clinical presentations such as sinusitis may occasionally mask or be mimicked by more serious and malignant conditions. Primary malignancies of the paranasal sinuses are rare conditions that few physicians will ever see in a primary care outpatient setting. Nonetheless, when considering the well known aphorism "When you hear hoof beats think horses and not zebras," the clinician while recognizing its wisdom, must never forget that the "zebras" are out there and that they can present at any time and sometimes in the most common of ways.

AN UNSEASONABLE TAN: A CASE OF ADDISON'S DISEASE. BD Hoerneman, CS Seibert, Medicine, University of Wisconsin Medical School, Madison, WI

LEARNING OBJECTIVES: 1) Recognize primary adrenocortical insufficiency. 2) Treat suspected adrenal crisis without obscuring diagnostic tests. 3) Differentiate primary from secondary adrenocortical insufficiency.

CASE: A 39 year-old woman presented with 5 days of increasingly severe fatigue, weakness, and lightheadedness. Two weeks before, she had several days of nausea, vomiting, and diarrhea, was diagnosed with gastroenteritis and treated with IV fluids. Her symptoms resolved except for fatigue, which later intensified. At this presentation, she denied fever, chills, anorexia, nausea, vomiting, and abdominal pain. Over the past 6 months, she reports a 40 lb weight loss. Three months ago, the patient moved to Wisconsin from Hawaii and noticed that her tan had not faded despite minimal sun exposure. She has had salt craving for at least two years, even salting ice cubes. On examination, the patient appeared acutely ill. Vitals revealed SBP of 60 mmHg, a pulse of 98/min, and temp of 96.3F. Sclera were anicteric. Skin was bronzed and tan lines were absent. Axillary and pubic hair were sparse. Oral mucosa was dry and gingiva were darkly pigmented. Heart, lungs, abdomen and neurologic exams were normal. Laboratories revealed Na 124, K 7.0, glucose 121, BUN 62 and creatinine 2.4. CBC and LFTs were normal. EKG demonstrated peaked T waves. Cosyntropin stimulation test could not be performed. Aggressive fluid resuscitation and IV hydrocortisone were given. 3 days after admission, with only minimal fatigue and normalized labs, the patient was discharged on a hydrocortisone taper and fludrocortisone. She was instructed to double her dose of hydrocortisone during illness and has done well.

DISCUSSION: Primary adrenocortical insufficiency (Addison's disease) is a potentially fatal condition resulting from deficiency of cortisol and aldosterone. Unlike secondary adrenocortical insufficiency where aldosterone production is preserved, patients with primary adrenocortical insufficiency can demonstrate marked volume depletion and hyperkalemia. Excess production of pro-opiomelanocortin in primary adrenocortical insufficiency results in the characteristic hyperpigmentation. Unfortunately, early diagnosis of adrenocortical insufficiency is often missed due to the nonspecific nature of symptoms. This can lead to patients presenting in adrenal crisis with severe hypotension and large electrolyte imbalances. High dose hydrocortisone therapy is commonly given, however dexamethasone may be used initially so diagnostic testing will not be distorted. Lifetime glucocorticoid therapy is required. Mineralocorticoid replacement is also likely to be necessary.

CHEST PAIN HERALDING A DEEPER ABNORMALITY. H Jasti, R Granieri, Medicine, University of Pittsburgh, Pittsburgh, PA

LEARNING OBJECTIVES: 1) Identify the clinical presentation of the antiphospholipid antibody syndrome (APS). 2) Review the diagnosis and management of this condition.

CASE PRESENTATION: An 18 year old white female with no significant past medical history presented with a subacute onset of right-sided pleuritic chest discomfort, with radiation to the right shoulder and breast, and slight shortness of breath. There was no associated nausea, vomiting, headache, visual changes, diaphoresis, vertigo, lightheadedness, or abdominal pain. Social history was remarkable for oral contraceptive and tobacco usage. Family history was significant for accounts of aunts having blood clots and strokes and a cousin with lupus. Physical exam revealed a pleasant young, white female in NAD. O₂ saturation was 94% on room air. The lung exam was remarkable for decreased breath sounds with dullness to percussion and decreased fremitus at the right lower base. Cardiac and abdominal exams were unremarkable. No rashes were noted. Lab work revealed a platelet count of 161,000, an INR of 0.9, and an increased PTT of 41. The ESR was 110. The RPR was positive with a negative FTA. The CXR revealed a moderate pleural effusion of the right lower lung and slight atelectasis in the left lower lung. Lower-extremity Dopplers and an ultrasound of her gallbladder were negative. A spiral CT revealed evidence of a PE in the left lower lobe. Subsequent lab work was positive for the lupus anticoagulant and anticardiolipin IgG & IgM.

DISCUSSION: APS is characterized by venous and arterial thromboses, pregnancy loss, and thrombocytopenia. The antiphospholipid antibodies constitute a heterogeneous group, including the lupus anticoagulant and the anticardiolipin antibodies, and are directed against the phospholipids involved in the coagulation process. Antiphospholipid antibodies are not specific for APS. For example, the prevalence of anticardiolipin antibodies in the general population ranges from 0-14%. The mechanism of the thrombosis is considered to be multi-factorial, including involvement of the B2-glycoprotein I pathway, enhanced platelet aggregation, inhibition of the protein C pathway, and altered endothelial cell function. Though arterial thrombi are less prevalent than their venous counterparts, stroke secondary to cerebral infarction may be a prominent manifestation. Pregnancy loss, especially in the second or third-trimester, is also of concern. The diagnosis is initially a clinical one, with further confirmation by laboratory studies, such as the modified Russell viper venom time and the sensitive PTT. Treatment after a thrombotic (venous or arterial) event includes long-term warfarin therapy. In addition, various factors such as oral contraceptives, obesity, hyperlipidemia, hypertension, and smoking can have an independent, additive effect. Thus management must also include attempts at reducing these risk factors.

A CASE OF ASCENDING PARALYSIS. H Jasti, R Granieri, Medicine, University of Pittsburgh, Pittsburgh, PA

LEARNING OBJECTIVES: 1) Identify the clinical presentation of Guillain-Barre syndrome. 2) Review the Guillain-Barre syndrome.

CASE PRESENTATION: A 77 year old white male presented with a five day history of progressive, bilateral lower extremity (LE) weakness and epigastric "burning" accompanied by nausea and emesis. He found himself to be unsteady upon walking and held on to furniture to keep from falling. He denied pain in his LE or lower back, or recent trauma. He also denied headache, lightheadedness, vertigo, visual

aura, dysarthria, blurriness of vision, CP or SOB. Physical exam revealed a weak and tired-appearing, but pleasant, elderly male in NAD. Cardiac, lung, and abdominal exams were unremarkable. Rectal exam revealed good sphincter tone and no loss of sensation around the perianal area. Neurological exam showed CN II-XII to be intact. Sensation was also intact throughout. The motor exam was significant for decreased strength in the LE in extension and flexion, both in the proximal and distal muscle groups. Deep tendon reflexes (DTRs) were absent. Cerebellar exam was normal. Labs, including a CBC, chem 19, TSH, CK, and a PSA were unremarkable. An LP demonstrated increased protein (174) and glucose (119) and no WBC. An EMG of the LE was consistent with the Guillain-Barre syndrome. The weakness quickly progressed to paralysis of his LE and subsequent respiratory distress. Plasmapheresis was initiated, with resultant marked improvement in his respiratory status and strength. He was discharged to home.

DISCUSSION: Guillain-Barre syndrome is the most frequently acquired acute inflammatory demyelinating peripheral polyneuropathy. In about 60-65% of the cases, the presentation may be preceded by an upper respiratory or gastrointestinal infection. Cardinal clinical features are symmetric and rapidly progressive weakness and decreased to absent DTRs. Definitive diagnosis includes an elevated protein without white blood cells on cerebrospinal fluid examination. Electrodiagnostic studies reveal a demyelinating neuropathy, with slow nerve-conduction velocities, prolonged distal latencies, and absent F-waves. Features casting doubt on the diagnosis include a marked, persistent asymmetry; bladder or bowel dysfunction at onset; and greater than 50 mononuclear leukocytes in the CSF. The disease reaches a plateau in about 2-4 weeks. Management necessitates close monitoring for potential respiratory and autonomic compromise. Treatment modalities include plasmapheresis or administration of IV IgG. Recovery is variable: 50-60% recover completely; 10-20% have residual deficits such as motor weakness; 10% have major deficits. Poor prognostic factors include age over 60 years, rapid disease progression to quadriplegia in less than 7 days, and the need for ventilator support. Mortality rate is approximately 5-8%. Intensive physical and occupational rehabilitation treatment is needed.

A CASE OF THE PUZZLED, STIFF MAN. H Jasti, R Granieri, Medicine, University of Pittsburgh, Pittsburgh, PA

LEARNING OBJECTIVES: 1) Identify the clinical presentation of the neuroleptic malignant syndrome (NMS).

2) Review the diagnosis of the NMS.

CASE: A 57 year-old white male presented with a two day history of disorientation and lethargy, punctuated by periods of restlessness. Past medical history was significant for hypertension, depression, and paranoid schizophrenia. Medications included atenolol, lithium, and thioridazine. Olanzapine had recently been added. Physical exam revealed a lethargic male who was uncooperative for most of the exam, though able to occasionally follow simple commands. Vital signs noted a BP of 170/90, pulse of 82, and a tympanic temperature of 100.3 F. Cardiac, lung, and abdominal exams were unremarkable. Neurological exam revealed him to be alert to person but confused to time and place. There was a generalized muscular rigidity with positive glabellar and palmar grasp responses. DTRs were 3+ in the upper and lower extremities. Plantar reflexes were upgoing. Lab work revealed an increased CK level (890) with negative MB fractions. An urinalysis was negative for myoglobin. Lithium levels were within normal levels. A head CT was negative. A LP revealed no WBC and had normal glucose and protein levels. The presentation seemed consistent with NMS. The patient was treated with aggressive supportive measures and Dantrolene. His mental status slowly improved over the course of a week and he was discharged to home.

DISCUSSION: NMS is characterized by fever, muscle rigidity, altered consciousness, and autonomic changes. However, patients presenting early in their course may not always have all the features. Lab work can reveal a leukocytosis and increased CK levels, the latter secondary to rhabdomyolysis resulting from muscle rigidity and breakdown. The syndrome occurs after the initiation of a neuroleptic drug or the addition of a second agent. The concomitant administration of lithium, tricyclic antidepressants, or benzodiazepines are particularly likely to evoke symptoms. Less commonly, the sudden withdrawal of levodopa, carbidopa, or amantadine has produced similar reactions. The mechanism involves the blockade of dopaminergic pathways in the basal ganglia and hypothalamus. The symptoms develop over 24-72 hours and may last as long as 5-10 days even after the medications are discontinued, given their long half-life. The overall mortality rate ranges from 10-20%, with the majority of fatalities resulting from cardiovascular collapse, renal failure, and respiratory inadequacy. Dantrolene is a muscle relaxant that reverses the muscle rigidity and has demonstrated some efficacy in reducing the duration of symptoms and associated mortality. Nevertheless, intensive supportive measures including drug withdrawal, vigorous cooling, and aggressive fluid replacement remain the cornerstones of therapy.

TUBERCULOSIS OF RIBS MASQUERADING AS GYNECOMASTIA: A CASE REPORT. H Jasti, N Afonso, G Alangaden, Internal Medicine, Detroit Medical Center, Detroit, MI

LEARNING OBJECTIVES: 1. True gynecomastia is a proliferation of the glandular component of the male breast tissue. Causes include altered estrogen androgen levels, aging, puberty, chronic disease, tumors and drugs. However breast enlargement can also occur as a result of underlying rib disease. 2. We are here reporting a case presented with bilateral gynecomastia. The patient had complete work up for major causes of gynecomastia. He was finally diagnosed with tubercu-

losis involving ribs. 3. Tuberculosis is an ancient infection. National surveillance data shows that foreign-born cases have accounted for about 60% of the increase in total TB cases.

CASE: A 30 y/o Asian-Indian male presented initially to the surgical service with complaints of bilateral breast enlargement. The breast masses had progressively increased in size over a period of two months and were associated with discomfort upon raising his arms over his head. He reported night sweats without fever or weight loss. The patient underwent evaluation for presumed gynecomastia including hormonal studies which were all normal. At this time he developed mildly painful swelling of his right ankle and the left metacarpal head of left middle finger. As part of the evaluation of the presumed gynecomastia he underwent bilateral breast biopsies. The initial Gram stain, acid-fast stain, and fungal stains were negative. The histopathology revealed necrotic bone with caseating granulomas. He was then referred to the medical service and was started on antituberculosis therapy with isoniazid, rifampin, pyrazinamide, and ethambutol. Four weeks later *Mycobacterium tuberculosis* susceptible to all antituberculosis agents was isolated from the mycobacterial cultures of the breast tissue. He denied any exposure to tuberculosis, however a PPD test revealed an induration of 12 mm. His chest radiograph was normal and his HIV antibody test was negative. Radiographic studies of his swollen ankle and hand were consistent with tuberculous osteomyelitis. He improved on antituberculosis therapy with resolution of his night sweats and improvement of his osteomyelitis. He remains on isoniazid and rifampin and is to complete at least 6 months of therapy.

DISCUSSION: This patient is an Indian, where the prevalence of TB is high. The unusual feature in this case was the gynecomastia secondary to involvement of the ribs. No other organ was involved except small joints of hand and ankle. This accounted for the delay in diagnosis even though the patient came from an area where tuberculosis is endemic. This observation demonstrates the difficult diagnostic situation presented by tuberculosis of the ribs. The key to diagnosis is to keep in mind that TB can present in multiple unusual sites, especially in immigrants from countries where TB is endemic. A positive PPD is helpful in the diagnosis.

ACUTE NECROTISING FASCIITIS. HS Kahlon, P Mehta, V Madhav, Medicine, New York Methodist Hospital, Brooklyn, NY

LEARNING OBJECTIVES: 1. NECROTISING FASCIITIS (NF) is one of the most serious infections and requires high index of suspicion and clinical awareness to be recognized. 2. Cornerstone of management in Acute Necrotising Fasciitis is early diagnosis and immediate surgical debridement alongwith antibiotics.

CASE PRESENTATION: A 47 year old male with known history of poorly controlled Diabetes Mellitus was admitted with complaints of high grade fever. Physical examination at the time of admission was normal. Patient's laboratory workup revealed White Blood Count (WBC) of 28000 with 90% neutrophils. Chest X-Ray and Urine analysis were normal. On second day of admission patient started complaining of excruciating pain in right thigh. Immediate X-Ray and CT Scan of the thigh were done which revealed gas in fascial planes of the thigh. Patient was diagnosed with Acute Necrotising Fasciitis and urgent surgical debridement was done alongwith immediate initiation of antibiotic coverage. Patient's blood cultures grew Group A beta hemolytic streptococcus.

DISCUSSION: Acute Necrotising Fasciitis is a rare soft tissue infection caused by toxin producing virulent bacteria, which is characterized by widespread fascial necrosis with relative sparing of the skin and underlying muscle. Polymicrobial necrotising fasciitis commonly involves aerobic and anaerobic gram negative enteric pathogens. Monomicrobial infection is commonly caused by Group A beta hemolytic streptococcus.

The condition is most commonly seen in Diabetics, immunosuppressed, iv drug abusers, alcoholics and patients with peripheral vascular disease. Common sites of infection are abdominal wall, scrotum, lower extremity, perineum and upper extremity. CT scan of the involved area shows presence and extent of gas tracking along the fascial planes-it characteristically does not involve muscle. Once Acute Necrotising Fasciitis is recognized prompt and aggressive therapy is essential, which includes: Resuscitation with IV fluids and hemodynamic stabilization, immediate institution of IV antibiotics with broad spectrum and anaerobic coverage, and early aggressive surgical debridement with sometimes repeat debridement within 24-48 hours. Untreated or delay in treatment of acute necrotising fasciitis can result in fatal outcome and high mortality rate. Failure to act vigorously in patients with suspected Acute Necrotising Fasciitis can lead to malpractice claims. Therefore being aware and early recognition of this clinical entity is imperative.

MYDRIASIS IN THE AMBULATORY SETTING: TO DRIVE OR NOT TO DRIVE. AR Kallianpur, KA Poehling, RS Diffus, Departments of Medicine and Pediatrics, Vanderbilt University; Quality Scholars Program and GRECC, Nashville VAMC, Nashville, TN

LEARNING OBJECTIVES: 1. To recognize that many patients experience real and distressing visual difficulty when driving after mydriasis. 2. To effectively counsel and manage patients for the prevention of complications from mydriasis-related visual impairment.

CASE PRESENTATION: The parents of a trauma surgeon, who live on Long Island, NY, recently had a frightening experience. Both in their late 70s, they stated that for years they have gone to the mall for their eyeglasses and never had a problem but were urged by a relative to see a "real eye doctor". They drove together to the recommended ophthalmologist's office early one fall afternoon, having received no prior information about what procedures would be done. Immediately after check-

in, they were both given mydriatic drops without either an explanation of the possible effects of the drops or inquiry into their plans for driving home. After an exam by the ophthalmologist, they mentioned to staff that they were anxious to get home because of a storm forecast and the darkening sky. They were not told to wait until the drops wore off and were not offered drops to reverse dilation. They left the clinic one hour after getting the mydriatic but discovered in the parking lot that neither of them could see well. They drove the 5 miles home at a crawl in cloudy, misting weather. Upon reaching home almost an hour later, they called their son, extremely upset at what had happened, adding that they had almost hit a parked car on the way. They chose not to tell the physician and resolved to go back to the mall in the future.

DISCUSSION: This case illustrates a common, dangerous, and underrecognized problem in the care of ambulatory patients. We heard similar anecdotes from other clinic patients. Prior studies have found distance visual acuity is not usually affected by mydriasis, but patients report difficulty with glare, reading road signs, judging distance, and seeing traffic lights on the way home. Extra caution may be needed in diabetics, six percent of whom fail to meet the minimum driving visual acuity requirement after mydriasis despite meeting it beforehand. Even unilocal dilation causes similar problems. Patient management strategies requiring further study include: the use of reversal agents, patient education about mydriasis, a reminder before the clinic visit to bring a driver and sunglasses, offering disposable sunglasses to patients without them, and scheduling patients who lack a driver early in the day to allow time for the mydriatic to wear off. We were unable to find documented guidelines that could assist physicians who use mydriatic drops in the ambulatory setting, and we urge the development of specific recommendations on this issue for improvement of quality of care and patient safety.

TREATMENT OF LOCALIZED PROSTATE CANCER: SUPPORTING PATIENTS IN MAKING INFORMED DECISIONS. B Kasravi, D Garcia, School of Medicine, UCLA, Los Angeles; Department of Medicine, Sepulveda Ambulatory Care Center, Sepulveda, CA

LEARNING OBJECTIVES: (1) Highlight the importance of a generalist in helping patients make informed decisions for the treatment of localized prostate cancer. (2) Review the evidence of different treatment options using current literature.

CASE PRESENTATION: A 70-year-old male with a history of severe chronic obstructive pulmonary disease (COPD), rheumatoid arthritis, and chronic bronchitis is found to have an elevated serum prostate-specific antigen (PSA) level of 13.5 ng/mL. Digital rectal examination reveals no palpable nodules. The patient is referred to a urologist and a biopsy is positive for adenocarcinoma with a Gleason score of 7. A bone scan and chest x-ray detect no metastasis. The urologist recommends either a prostatectomy or radiation therapy, but offers watchful waiting as an option. The patient and his wife return to their Primary Care Physician (PCP) with anxiety and fear over the new diagnosis. The PCP addresses each of the patients' concerns, reviews the risks and benefits for each treatment option, provides educational references, and coordinates the care plan by outlining steps that need to be taken. The patients' history of COPD, age, Gleason score, PSA level, and personal preferences are factored into the decision making. This patient eventually chooses radiation therapy as the treatment choice.

DISCUSSION: The clinical evidence for the best treatment option for localized prostate cancer is not clear. Current choices fall into three major categories including a radical prostatectomy, radiation therapy, and watchful waiting. Generally, patients with a PSA level of 10 or less ng/mL, a Gleason score of 7 or lower, and a tumor size of 1.5 mL or less have been shown to have a higher curability rate. Watchful waiting, has been shown to have a 66% survivability over 15 years in patients with confined disease. This option is mostly considered if the patient has various medical problems or is concerned about quality of life after treatment. Radical prostatectomy is also a favored option for younger patients with 47-70% of patients being free from recurrence in 10 years. Risks include death, impotence, urinary incontinence, and urethral stricture. Radiation therapy has also been shown to have a 10 year recurrence free rate of 64-86% in various studies. This option also carries risks such as death, impotence, anorectal complications, and acute GI or GU symptoms. This case illustrates how the PCP, urologist, and the patient can work together towards an informed decision for treating prostate cancer. The PCP's role is to help the patient in the decision process, aid the urologist on patient's health factors, support the patient throughout the treatment, and follow up with the patient after treatment.

THE NEED TO CONSIDER CYSTIC FIBROSIS IN ADULTS WITH UNEXPLAINED ELECTROLYTE ABNORMALITIES. SG Kassaye, DE Morrison, AS Dunn, Internal Medicine; Internal Medicine/ Pediatrics, The Mount Sinai Medical Center, New York, NY

LEARNING OBJECTIVES: 1) Recognize atypical presentations of cystic fibrosis, and 2) the differential diagnosis of hypokalemic metabolic alkalosis.

CASE PRESENTATION: J.T. is a 51 year old obese male who presented with one month of fatigue, weakness, and decreased appetite. He had no nausea, vomiting, diarrhea, cough, or shortness of breath. The patient worked as a cook on a tug-boat, and the onset of his symptoms coincided with a summer heat wave. The patient and his brother had been diagnosed in childhood with Bartter's syndrome and instructed to increase their sodium intake during times of excessive sweating. The patient was taking no medications and denied diuretic use. Physical exam was significant for a weight of 145 kg, blood pressure of 128/72, pulse of 75, respiratory rate of 16, oxygen saturation of 96% on room air, clear lung fields, trace pedal edema, and a normal neurologic exam. His laboratory findings included WBC 16.4, Hb 15.5, sodium 132, potassium 2.0, chloride 73, bicarbonate 35, BUN 15,

creatinine 0.7. Urine specific gravity was 1.030. Urine electrolytes were as follows: potassium 52.2, sodium 13.6, chloride 15. ABG revealed pH 7.53, pCO₂ 54, pO₂ 59. Chest x-ray was normal. ECG showed prolonged PR and QT intervals, U waves, diffusely flattened T waves, and ST segment depression in V2-V6, all of which were attributed to the profound hypokalemia. The patient was given IV fluid and potassium replacement. His electrolyte and ECG abnormalities rapidly corrected, and he was discharged on oral potassium supplements. After discharge, a sweat test was obtained and was abnormal. A presumptive diagnosis of cystic fibrosis (CF) was made and was confirmed by genotype analysis (mutation delta f508/R334W). Sputum culture grew *Pseudomonas aeruginosa* and *Stenotrophomonas maltophilia*, both organisms which typically colonize the airways of patients with CF.

DISCUSSION: This patient presented with dehydration, a hypokalemic metabolic alkalosis, hyponatremia, hypochloremia, high urinary potassium, and low urinary chloride. The differential diagnosis of hypokalemic metabolic alkalosis in a normotensive patient includes diuretic abuse, prolonged gastrointestinal fluid losses, Bartter's and Bartter's-like syndromes, and excessive chloride loss in sweat as occurs in CF. While the patient had previously been diagnosed with Bartter's syndrome, the low urinary chloride is inconsistent with this diagnosis. Cystic fibrosis usually presents in the pediatric age group with recurrent pulmonary infections, pancreatic insufficiency, and failure to thrive. The availability of sweat testing and genotype analysis allows clinicians to make the diagnosis in adults, and it should be considered when patients present with unexplained electrolyte abnormalities.

NO PLACE TO GO: THE DEMENTED HOMELESS. SG Kertesz, Dept. of Medicine, Boston University School of Medicine, Boston, MA

LEARNING OBJECTIVES: 1) Identify appropriate clinical and organizational response options when caring for demented homeless persons. 2) Recognize the impact of state and institutional regulations on the likely success of intervention.

CASE PRESENTATION: A 44-year-old homeless man with suspected dementia aroused clinical concern that he could die in the streets. Over several months health care providers had noted that the patient conversed fluently while often losing the thread of conversation. He was unable to recall information from previous visits. He often smelled of urine and/or alcohol. Once while sober the patient offered what little money he had to a clinic nurse without expectation of return. A staff member had observed the patient lying in the street being beaten without resisting his assailant. He was not thought to have a coexistent psychotic or major emotional disorder.

Past history was notable for chronic hepatitis C without cirrhosis, cerebral injury due to assault, occasional seizures attributed to that injury, and alcoholism. A prior CT scan had documented generalized brain atrophy as well as severe left frontotemporal encephalomalacia.

The patient was subjected to involuntary hospitalization in a locked medical-behavioral unit for short-term evaluation. Medical workup was negative and CT scan confirmed previous findings. Neuropsychological evaluation found the patient to be a danger to himself on the basis of dementia and recommended placement in a supervised nursing facility. Legal review showed that the patient could be detained if he was suicidal, homicidal or at imminent risk for harm to himself. No locked facility, however, had authority to hold this patient. Regulations precluded the Department of Mental Health from holding a patient suffering from an "organic" mental condition. In the absence of tuberculosis or a life-threatening medical condition, no other facility had authority to hold this patient. The patient was released to the street. The patient returned to clinic and threatened his doctor. He was rehospitalized involuntarily and a court found the patient incompetent to make health care decisions. A guardian was appointed. Because no facility had authority to hold the patient involuntarily, he was transferred to an unlocked floor, and he eloped. He continues to reside in the streets.

DISCUSSION: Clinicians should consider formal neuropsychological evaluation and legal consultation in the care of demented homeless persons. Most states permit involuntary hospitalization of persons shown to be a danger to themselves or others on the basis of a major mental illness. Some states do not provide facilities for persons at risk on the basis of dementia alone. Demented homeless persons may be at particular risk for death in the streets because of unique environmental exposures and the absence of facilities with legal authority to hold these patients.

BACK PAIN AND STROKE: STAPHYLOCOCCUS AUREUS ENDOCARDITIS WITH MULTIPLE SEPTIC EMBOLI. N. Khan, C Rangaswamy, S Schmitt, Department of Medicine, Cleveland Clinic Foundation, Cleveland, OH

LEARNING OBJECTIVES: Distinguish benign disc herniation from infective discitis. (2) Recognize sequelae of bacteremia and right and left sided endocarditis.

CASE: A 54-year-old man, with no significant past medical history, presented with back pain of acute onset while lifting objects at work. He denied history of direct trauma, or drug abuse, but had 50 pack years of smoking. Due to a normal neurological exam, his pain was treated symptomatically with NSAIDs. The following week his persistent pain was evaluated with an MRI which showed T12-L1 disc herniation. Since the neurological exam was unchanged, rest, corticosteroids and narcotic analgesics were recommended. Two weeks later, the family reported "recurrent disorientation and confusion", so the dosage of the narcotic analgesic was decreased. The following day, there was decreased mobility and strength in the right upper extremity. Admission exam also revealed a tender right mid-clavicular mass. A CT of the brain showed a left parietal lesion, consistent with acute ischemia, necessitating IV Heparin. Lab abnormalities on admission were platelets

= 41000 K/ul, creatinine 1.8 mg/dL (0.7-1.4 mg/dL), and normal WBC = 9.6 K/ul. A peripheral smear and bone marrow evaluation ruled out thrombotic thrombocytopenic purpura. Bone marrow and blood cultures grew methicillin susceptible *Staphylococcus aureus* (MSSA). An MRI of the spine showed discitis at T12-L1 levels. Given the sequence of events a transesophageal echocardiogram was performed, which revealed vegetations on the tricuspid and mitral valves. Subsequently, a CT revealed a right sternoclavicular abscess, splenic and right renal infarcts and bilateral pulmonary cavitations consistent with septic embolic episodes. MSSA bacteremia cleared with IV oxacillin. 10 weeks later, he remains with right hemiparesis, expressive aphasia, and respirator dependent.

DISCUSSION: Occupation related back injuries are common. When warranted by a thorough physical examination, appropriate radiographic studies can support a diagnosis of musculoskeletal back injury such as disc herniation. In the early stages, this is frequently difficult to distinguish clinically from infectious discitis. When symptoms fail to respond to conservative management with NSAIDs and rest, or when fever, leukocytosis, or elevated acute phase reactants are detected, further radiographic imaging is indicated to exclude infection or neoplasm. Endovascular infection with *S. aureus* is frequently complicated by seeding or embolization of infection to distant sites including solid organs and bones. As this case illustrates, patients may be afebrile and without leukocytosis. Also, use of narcotic analgesics may confound the significance of mental status changes. Patients with evidence of emboli should be thoroughly investigated for endocarditis.

DEEP VENOUS THROMBOSIS CAUSING RHABDOMYOLYSIS AND ACUTE RENAL FAILURE. J. Khan, RC Bussing, Medicine, Southern Illinois University School of Medicine, Springfield, IL

LEARNING OBJECTIVES: * Recognize that rhabdomyolysis can occur in the setting of extensive venous thrombosis and that renal failure can occur as a consequence of Rhabdomyolysis. * Remind physicians to check CK, renal function, and urine myoglobin in the setting of extensive venous thrombosis.

SUMMARY: Deep venous thrombosis is a very commonly diagnosed clinical entity. In the most severe forms of this condition, the integrity of vasculature to the involved extremity can be jeopardized. Compartment syndrome rarely has been described in association with venous thrombosis and can be limb threatening. Rhabdomyolysis is a well-described complication of compartment syndrome. Rhabdomyolysis can be associated with acute renal failure in less than 8% of patients. There have been no reported cases of deep venous thrombosis causing acute renal failure. We report a patient with extensive lower extremity venous thrombosis who developed acute renal failure and had evidence of rhabdomyolysis. Physicians should watch for evidence of rhabdomyolysis in patients with massive venous thrombosis.

CASE: A 51-year-old white male was admitted to the hospital with a subarachnoid hemorrhage. He had been previously in good health and was taking no regular medications. He had only a vague history of hypertension in the past. Surgical clipping of a bleeding anterior communicating artery aneurysm was done on the second hospital day. In the post-operative period, he remained unresponsive, was non-verbal, and did not follow commands. One month after the onset of illness he developed deep venous thrombosis of the left lower extremity. He was not anticoagulated because of the recent subarachnoid hemorrhage and a Greenfield filter was placed. The left leg swelling resolved gradually. Four weeks after the left leg deep venous thrombosis, an extensive venous thrombosis developed in the right lower extremity. The leg became extremely tense and swollen with bluish red discoloration. Color flow ultrasonography confirmed lack of spontaneous phasic flow and incompressibility of virtually all the deep veins of the right thigh and the right calf. The common iliac vein, as well as the inferior vena cava up to the level of the Greenfield filter, also showed evidence of thrombosis. The CPK rose to 10,439 U/L. The urine myoglobin rose to 18,740 NG/ML. The urine output dropped to 16cc/hour for 24 hours. The creatinine rose from 0.6 MG/DL to 3.4 MG/DL. The BUN rose from 11 MG/DL to 45 MG/DL. His treatment was with standard IV unfractionated heparin, leg elevation, and forced alkaline diuresis to keep the urine pH >7.5. In a week, his CPK normalized, the urine became free of myoglobin and the creatinine returned to normal. The patient returned to normal functioning with minimal leg swelling and normal renal function.

SUBACUTE THYROIDITIS: A COMMON CAUSE OF ANTERIOR NECK PAIN. N. Khan, S Reddy, A Hull, A Jaffer, Department of Internal Medicine, Cleveland Clinic Foundation, Cleveland, OH

LEARNING OBJECTIVES: (1) Recognize that subacute thyroiditis is a common cause of anterior neck pain. (2) Recognize the clinical features, course and treatment of subacute thyroiditis.

CASE PRESENTATION: A 70-year-old female presented to an outpatient clinic with a two week history of anterior neck pain and occasional dysphagia to solids and liquids. This began shortly after a bout of an upper respiratory illness. She reported poor appetite and weight loss of 10 lb in 7 weeks. She denied trauma, upper extremity weakness, fevers, chills, palpitations, irritability and tremors, but admitted to a history of chronic constipation. On physical exam she was tachycardic with tenderness in the neck muscles bilaterally, and a palpable symmetrical and tender goiter. There was no evidence of lymphadenopathy, proptosis or exophthalmus. The cardiovascular and pulmonary exams were normal. The DTR's were normal bilaterally. She was prescribed naproxen (500 mg/day) and heat compresses for symptomatic pain relief. Given her presentation, a screening TSH was obtained. The TSH = 0.02 (normal 0.4-5.5 uU/dL); subsequent thyroid indi-

ces revealed a Free Thyroid Index (FTI) = 19.3 (6–11 ug/dL), T4 = 13.9 (5–11 ug/dL), free T4 = 4.2 (0.7–1.8 ng/dL). A sedimentation rate (ESR) = 108. The patient's TSH-receptor antibody levels were within normal limits. One week later in follow-up, her neck pain was decreased and there was 'improvement' in her constipation. Her findings were consistent with hyperthyroidism from subacute thyroiditis. At the three week follow-up the patient denied neck pain, but complained of increasing fatigue, and had gained 11 lb. Neck palpation revealed a smaller nontender goiter. A repeat TSH was 5.83, FTI 3.2, and T4 3.3. This suggests a hypothyroid phase, which may need treatment with thyroxine if symptoms persist. Close follow-up is therefore needed.

DISCUSSION: Subacute thyroiditis is a common cause for anterior neck pain. It is a post viral inflammatory syndrome that causes release of thyroxine from follicular stores. Approximately 50% of patients present with signs and symptoms of hyperthyroidism. Symptoms may be subtle or overt with palpitations, tachycardia, diaphoresis or weight loss. Recovery may be followed by transient hypothyroidism which may require treatment. A high suspicion based on the patient's history and exam can enable an astute primary care provider to make this diagnosis on the initial visit which can later be confirmed using thyroid function tests and a sedimentation rate. An elevated ESR is so common that a normal value should question this diagnosis. Treatment involves using salicylates or NSAIDs for pain; steroids are reserved for patients with severe pain. Beta-blockers can be added for patients with symptomatic tachycardia and at risk for arrhythmias.

FOURNIER'S GANGRENE: EARLY DIAGNOSIS REMAINS A CHALLENGE. P. Khanna, K. Barnard, Medicine, UPMC-Shadyside Hospital, Pittsburgh, PA

LEARNING OBJECTIVES: 1) Recognize the importance of early surgical debridement in Fournier's gangrene. 2) Diagnose Fournier's gangrene when it presents with radicular leg pain. 3) Recognize the potential for necrotizing fasciitis in patients presenting with subtle features as pain or small induration or erythema.

CASE: Patient was a 48 year old African American gentleman with a history of chronic obstructive lung disease and alcohol abuse who presented with a painful cyst on the right buttock and was found to have perianal fistula. He was started on oral antibiotics and asked to return in 2 days for rectal evaluation under anesthesia or earlier if symptoms didn't improve. The patient returned a few hours later with severe left leg pain and shortness of breath. He was found to be afebrile, with pulse rate 100, blood pressure 110/70, respiratory rate 20 and pulse ox 88%. His lungs were clear, heart sounds were regular and abdomen benign. Patient was unable to dorsiflex his left foot due to pain. Right buttock examination revealed mild erythema without fluctuation. Diagnostic procedures focused on causes of left leg pain including doppler ultrasound to exclude acute arterial occlusion. DVT and pulmonary embolism were ruled out by negative doppler and ventilation perfusion scan. Patient was admitted and started on intravenous antibiotics and surgery consult was ordered. A few hours later the patient was found to be unresponsive and acidotic. The patient was immediately taken for debridement and was found to have extensive Fournier's gangrene extending into both thighs, full length of sacrum and base of scrotum. During the debridement the patient went into pulseless electrical activity, shock and disseminated intravascular coagulation and died a few hours later.

DISCUSSION: Fournier's gangrene is necrotizing fasciitis with extension into the perineal area and is caused by streptococcus pyogenes. It can present as unexplained fever or pain and early diagnosis is difficult. It can extend rapidly as in our patient where it extended from a small induration to extensive gangrene within a few hours leading to toxic shock syndrome.

MASSIVE SPONTANEOUS PARACENTESIS FROM AN UMBILICAL HERNIA. SJ Kolpak, General Internal Medicine, Denver Health, Denver, CO

LEARNING OBJECTIVES: 1) Recognize that umbilical hernia perforation is a serious, potentially fatal complication of ascites. 2) Treat ascites aggressively in the presence of umbilical hernia, especially when accompanied by overlying skin changes.

CASE PRESENTATION: A 46 year-old male with a history of alcoholism, hepatitis C, and hypertension presented to the clinic complaining of abdominal distention for two years. He was evaluated one year prior, and was found to have liver failure and ascites. He had been started on spironolactone and furosemide, which he had not taken for several months. Examination revealed a thin male with a distended, soft abdomen and hepatomegaly. An umbilical hernia nine centimeters in diameter was present, with brown discoloration and focal excoriation. There was no lower extremity edema. He was again started on diuretic therapy and told to return for a therapeutic paracentesis. The patient returned to clinic one month later with a flat abdomen and no lower extremity edema. He reported that two weeks prior, during a coughing paroxysm, his umbilical hernia had ruptured with the loss of his entire ascites volume within minutes, while leaning over the bathtub. His hernia had closed spontaneously, he had no recurrence of his ascites, and had no current complaints. Blood pressure was 100/62 and pulse was 80. He now had a flat, soft abdomen, with a seven centimeter flaccid umbilical hernia with a small healing eschar. There was no lower extremity edema, and he weighed 58 kg, a 12 kg loss from his previous visit. Electrolytes were normal and creatinine was 0.9, which was unchanged. Diuretic therapy was restarted, surgical consultation was obtained, and he underwent herniorrhaphy and peritoneovenous shunt placement six weeks later. He remained free of ascites three months later.

DISCUSSION: Umbilical hernia is common in patients with ascites, occurring in more than 20% of cases. However, spontaneous rupture of an umbilical hernia is

a rare and often fatal event. Since it was first described in 1901, approximately sixty cases have been reported in the English literature. Patients receiving only supportive care have had mortality rates as high as 60–90%, and even those patients undergoing urgent surgical repair experience a mortality of nearly 20%. The cause of death is usually peritonitis/sepsis, hepatic failure, renal failure, or gastrointestinal hemorrhage. In many cases, hernia rupture is heralded by skin changes in the hernia, most notably ulceration. Although this case had a favorable outcome, patients with ascites and umbilical hernia need aggressive reduction of fluid volume, usually followed by elective herniorrhaphy, to prevent the often mortal complications of spontaneous rupture.

DYSPHAGIA AND VOCAL CORD PARALYSIS DUE TO VARICELLA-ZOSTER VIRUS IN THE ABSENCE OF CUTANEOUS LESIONS. MJ Krasnoff, M-H Huang, Internal Medicine, State University of New York at Buffalo, Buffalo, NY

LEARNING OBJECTIVES: To diagnose varicella zoster virus (VZV) reactivation without the pathognomonic vesicles—zoster sine herpete.

CASE PRESENTATION: An 82-year-old man with mild hypertension noted a 5 day history of a sore throat, hoarseness, odonophagia, and dysphagia for solids and liquids. He also developed left-sided facial pain involving the jaw, cheek, and ear. He denied skin lesions, hearing loss, visual disturbance, headache, or shortness of breath. On examination, vital signs were normal. No oral lesions, facial rash, neck masses, or lymph nodes were detected, and his neurological exam was normal. A flexible laryngoscopic examination showed left true vocal cord paralysis. The left side of the epiglottis was erythematous with multiple ulcers. Treatment for acute epiglottitis with ceftriaxone was empirically begun. Four days later direct laryngoscopy showed that the vocal cord paralysis persisted, the epiglottitis appeared normal, and there was erythema of the left side of the arytenoid. Biopsy of this lesion showed an ulcerated epithelial surface with no multinucleated giant cells. A CT scan of the neck and chest revealed no mass. A barium swallow demonstrated a normal esophagus and no dysmotility. An MRI of the head was normal. CSF analysis showed 130 WBC/ μ L (92% lymphocytes), protein 61 mg/dL, and glucose 61 mg/dL. Gram's stain showed many WBCs and no organisms. Upon identification of the CSF lymphocytosis, empiric therapy with intravenous acyclovir was begun. The team ordered a VZV PCR on the CSF, but the lab ran a herpes simplex virus PCR (which was negative). The patient's dysphagia and hoarseness started to improve, and after 2 days he was switched to oral acyclovir to complete a 10 day course. One month later he still noted mild hoarseness and dysphagia.

DISCUSSION: The laryngoscopic findings of vocal cord paralysis explained his hoarseness, but the shallow lesion on the epiglottis did not seem sufficiently inflammatory to explain his severe dysphagia and odonophagia. Once extrinsic compression of the esophagus was ruled out, a polyneuritis was suspected. The diagnosis of a VZV infection causing a polyneuritis cranialis was supported by the CSF lymphocytic pleocytosis associated with slightly elevated protein and normal glucose. This patient manifested a painful reactivation of VZV in the absence of cutaneous lesions, known as zoster sine herpete. The CSF PCR can be utilized to rapidly diagnose VZV infection. IGM and IGG serology can be performed in the serum and CSF. The reactivation of VZV can affect cranial nerves 9 and 10 in addition to the more common syndromes involving cranial nerves 5 and 7.

UNCOMMON CELLULITIS: PRESENTATION AND PREVENTION OF LYME DISEASE. D Krause, Department of Medicine, Beth Israel Deaconess Medical Center, Boston, MA

LEARNING OBJECTIVES: 1) Recognize atypical presentations of Lyme Disease, 2) Reinforce the importance of early diagnosis and patient education.

CASE: A 69-year-old woman without significant past medical history presented with 2 days of low-grade fever, malaise, myalgia, and posterior knee pain. The patient lives in an urban suburb of Boston and is an avid gardener. She denied rash, trauma or recent travel. On physical exam she was afebrile and looked well. The posterior aspect of her left knee was diffusely erythematous and warm without scale or central clearing. No effusion, tenderness, or joint instability was found. Doppler ultrasound was significant for a 1.5 cm by 1.3 cm cystic structure in the left popliteal fossa without evidence of thrombus. She was treated with two successive courses of cephalexin 500 mg QID for 10 days, heat, and elevation for presumed cellulitis; each time she transiently improved, but worsened with cessation of antibiotics. One week after completing her second course of antibiotics she presented with a red, painful, and swollen left knee. Arthrocentesis revealed serousanguinous fluid with 475 WBC, 8225 RBC, no crystals or organisms seen on Gram stain, and subsequent cultures were negative. Lyme serology revealed positive IgG and IgM confirmed by Western blot analysis. She was treated with one month of Doxycycline 100 mg bid and clinically improved. Her Lyme studies remain positive.

DISCUSSION: Lyme disease is the most common tick borne illness in the United States with over 10,000 new cases reported annually. The probability of contracting the disease after a tick bite is dependent on tick stage of development, tick attachment time of greater than 48 hours, and prevalence of ticks infected with the spirochete *Borrelia burgdorferi* in a given area. Presentation is variable with fever, erythema migrans, malaise, and migratory arthralgias. Early recognition is critical as later sequelae including carditis with heartblock, neurologic complications, and musculoskeletal complaints can be avoided with prompt treatment. It is important to realize that only 60% of early infections present with the typical erythema migrans rash, and fewer than 30% of patients can recall the actual bite. Primary care physicians diagnose the majority of new cases. They are often faced

with requests for testing and empiric treatment. Although the chance of acquiring Lyme disease from a tick exposure is less than 10%, many physicians perform expensive serologic tests, and prescribe antibiotics regardless of actual risk. The estimated yearly economic burden of this disease approaches \$2.5 million dollars. Treatment, diagnosis and vaccination strategies are widely debated and can have a large impact on healthcare practice and economics. With serologic testing and vaccination readily available, we must continue to educate both physicians and the public regarding presentation, prevalence, interpretation serologic tests, and personal protection in approaching this disease.

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AN UNCOMMON COMMON COLD. L Krijger, A Nirula, E Yee, Medicine, UCLA/ San Fernando, Sepulveda, CA; Medicine, UCLA, Los Angeles, CA

LEARNING OBJECTIVES: 1. Recognize Common Variable Immunodeficiency as a cause of recurrent infections 2. Discuss the diagnosis and treatment of Common Variable Immunodeficiency.

CASE: A previously healthy 29 year old male presented with a 2 day history of fevers, chills, productive cough, and shortness of breath. He had two previous bouts of pneumonia in the last six months, one requiring hospitalization. In addition, he reported having 10 colds in the last 18 months and five sinus infections in the last 10 years. One year ago, he had an infection with *Giardia lamblia* following a camping trip in Australia. He worked as an ocean lifeguard, was training to work for the fire department, and had no history of tobacco, alcohol, or illicit drug use. He took ibuprofen for headaches as needed. On physical exam his temperature was 39.1°C, respiratory rate 22, pulse 114, BP 131/67, O₂ sat 95% on 3 liters. Pulmonary exam revealed crackles, egophony, fremitus and dullness in the left lower lobe. There was no lymphadenopathy or splenomegaly noted. A CBC showed WBC 22.9 (86% polys), Hb 14.6, Hct 42.4. A CXR showed a left lower lobe infiltration. The patient was admitted for pneumonia and treated with IV antibiotics. *S. pneumoniae* grew out on sputum cultures. During his hospitalization, he was evaluated for possible immunodeficiency because of his history of multiple infections. His HIV test was negative, but his immunoglobulin levels were profoundly low: IgG 36.6 (720-1680), IgM 10.1 (48-333), and an IgA 6.93 (73-455). The patient was diagnosed with common variable immunodeficiency disease (CVID) and referred for IV immunoglobulin therapy upon discharge. One month later he reported feeling better than he had ever felt in years. He has subsequently been hired by a fire department in another state.

DISCUSSION: Patients often present to primary care providers with a "common cold". Consider testing for CVID in an individual with a history of multiple respiratory infections. CVID is the most common symptomatic primary immunodeficiency syndrome and can be diagnosed with low immunoglobulin levels. It has an

estimated prevalence of 1/50,000-1/200,000 and affects men and women equally. Symptoms usually occur in the second to third decade of life, but may be seen at any age. Recurrent encapsulated bacterial infections involving the respiratory tract (sinusitis, bronchitis, otitis media, and pneumonia) are the most common manifestations. If untreated, CVID patients typically develop severe bronchiectasis, frequent complications, and oxygen dependency. CVID patients have a lowered life expectancy and higher rates of lymphoma (2-7%, a 30-200 fold increase) and gastric carcinomas (a 50 fold increase). Intravenous immunoglobulin therapy is a lifelong necessity. Patients with CVID should be counseled to be evaluated for any type of infection.

CALCIUM AND VITAMIN D: A BENIGN TREATMENT FOR OSTEOPOROSIS? A CASE OF VITAMIN D TOXICITY. MN Kulkarni, Internal Medicine, UTHSCSA, San Antonio, TX

LEARNING OBJECTIVES: 1) Diagnose secondary causes of osteoporosis. 2) Recognize potentially toxic side effects of vitamin D and calcium therapy if not appropriately monitored.

CASE PRESENTATION: Mr. G, an 87 year-old male was admitted to the ICU with a three week history of fatigue, generalized weakness, mild abdominal pain, polyuria, ten pound weight loss, and mental status changes, now requiring full help with activities of daily living. He was previously ambulatory, conversive, and able to perform small chores around the house. The patient had a past medical history of low back pain with a lumbar spine (L-1) compression fracture on xray diagnosed nine months prior, for which he was subsequently treated for osteoporosis with calcium citrate 950 mg twice daily, and ergocalciferol 50,000 units daily. He also had a remote history for colon cancer diagnosed twenty years ago, with normal colonoscopies and CEA since. Mr. G was difficult to arouse, disoriented, unable to follow commands, with gag and corneal reflexes intact, responding to pain only. Laboratory values revealed serum sodium 155, potassium 2.5, BUN/Cr 54/2.0, and corrected calcium 15.5. The vitamin D level was 402 ng/dl (normal 8.9-46.7 ng/dl), and PTHi was 13.5 mg/dl (normal 10-65 mg/dl), consistent with vitamin D toxicity. Initial workup for occult malignancy including CEA, PSA, skeletal survey, chest xray, upep/spep was normal. He was treated for hypercalcemia, with intravenous fluids, pamidronate, and methylprednisolone, with mildly improved mental status after two days.

DISCUSSION: Patients with osteoporosis must be evaluated prior to treatment for secondary causes such as hypogonadism, hypothyroidism, chronic steroid use, and hyperparathyroidism; and baseline calcium and vitamin D levels must be measured. In elderly patients, calcium levels must be closely monitored, and vitamin D therapy should be considered in those with vitamin D deficiency with the recommended dose of 800 IU daily or ergocalciferol 50,000 units weekly. Failure to seek secondary causes or to carefully monitor therapy may result in iatrogenic hypercalcemia with catastrophic outcomes as in this case.

NEUROSARCOIDOSIS IN A 25 YEARS OLD FEMALE. M Langiulli, R Garrick, S Peterson, Internal Medicine, Westchester Medical Center, Valhalla, NY

LEARNING OBJECTIVES: 1. The diagnosis of neurosarcoidosis in patients with no systemic manifestations of sarcoidosis. 2. To discuss the presentation of amenorrhea in a patient with neurosarcoidosis.

CASE PRESENTATION: The patient is a 25 years old female who presented to Westchester Medical Center (WMC) with a complaint of severe generalized headache with nausea and vomiting for three days. These headaches began six months prior to admission, were accompanied by right sided weakness, slurred speech and mental status change. She was admitted to another hospital where an MR of the head revealed leptomeningitis and a lumbar puncture found WBC 69, 97% lymphocytes, glucose 38 and a total protein of 213. Serology including ANA were negative. Viral meningitis was diagnosed and treated with steroids for three months with improvement. Upon presentation to WMC the patient had been off steroids for three months, she had no weakness and her speech was coherent. She also had amenorrhea. A CT scan of the head showed a low attenuation change in the forceps minor. A lumbar puncture showed WBC 32, 89% lymphocytes, glucose 34, total protein 229. CXR normal. ESR 52, ANA negative, RF <10, C3 185, C4 38, CH50 225, VDRL negative, ACE 20.8, all CSF cultures and stains for viruses, bacteria and AFB were negative. A prolactin level was 45.6, FSH 2.1, LH 0.7, Progesterone 0.9, Testosterone 1.3, Estradiol <20, TSH 1.3, Free T4 1.0, Total T4 6.9, T3 1.1, Cortisol 11.9, with a normal three day ACTH stimulation test. A Gallium Scan was done which showed increase isotope localization in both eyes, both hilar regions, right anterior mediastinal lymphadenopathy and increased renal uptake. Brain and spine MR with Gadolinium showed meningeal changes. Salivary gland, lung and bone marrow biopsy were done and were negative. Biopsy of the brain lesion found granulomatous disease with multiple caseating and noncaseating granulomas, consistent with neurosarcoidosis. The patient was discharged on oral steroids with resolution of symptoms.

DISCUSSION: Neurosarcoidosis occurs in only about 5% to 16% of patients with sarcoidosis, and most have systemic manifestations. This case is unusual because of its purely neurological presentation. While the hypothalamic-pituitary axis is often involved in neurosarcoidosis, the clinical presentation is usually diabetes insipidus. Amenorrhea, is extremely rare as a clinical manifestation of this disease.

TAKAYASU'S DISEASE IN A 22 YEARS OLD FEMALE. M Langiulli, S Pradhan, E Bueno, G Montouri, S Peterson, Internal Medicine, Westchester Medical Center, Valhalla, NY

LEARNING OBJECTIVES: 1. To discuss the presentation of a rare variant of Takayasu's Disease. 2. To discuss treatment of renovascular hypertension related to Takayasu's Disease. 3. To discuss the various complications of Takayasu's Disease.

CASE PRESENTATION: We present the case of a twenty-two year old Mexican woman who presented to the Westchester Medical Center ER with complaints of headache, blurred vision, chest pain radiating to her left arm and neck and epigastric pain. She had experienced bitemporal headaches, worse on the right side, increasing over the past eight months. Three weeks prior to admission, her abdominal pain worsened and her visual acuity diminished markedly. Her PMD found her to be severely hypertensive and started her on Nifedipine. The patient denied a history of IV drug use or significant alcohol use. She smokes less than ½ pack of cigarettes per month. Her mother is also hypertensive, but is well controlled. The patient was employed as a gardener and horse-groomer. On physical exam, blood pressure was noted to be elevated at $180/110$ in both arms. She had blurring of the optic disks, a systolic murmur at the apex, mild epigastric tenderness and bilateral abdominal bruits, L>R. Examination of the skin revealed several echymoses on the lower extremities. EKG showed left ventricular hypertrophy, but echocardiogram was normal. A renal duplex ultrasound examination was consistent with vasculitis of the abdominal aorta, with subsequent decreased flow through the superior mesenteric artery and the renal arteries, bilaterally. A CT scan of the abdomen demonstrated concentric thickening of the upper abdominal aorta from the celiac axis to the ostia of the renal arteries. Abdominal MRA confirmed these findings. She was diagnosed with bilateral retinal detachment, secondary to hypertensive disease or Takayasu's Disease. Friability of the inflamed vessel walls prevented biopsy. The patient was started on Prednisone 1mg/kg/day and discharged with plans for angioplasty.

DISCUSSION: This case is a rare variant of a rare disease. Pulses were normal and the aortic arch was normal. Retinopathy and chest pain are common in this illness. This case demonstrates that when a large vessel vasculitis is postulated, Takayasu's Disease must be high in the differential diagnosis.

OCCULT ATRIAL FIBRILLATION: AN UNDER-RECOGNIZED ENTITY. BP Large, NA Desbiens, Department of Medicine, University of Tennessee College of Medicine—Chattanooga Unit, Chattanooga, TN

LEARNING OBJECTIVES: 1. Recognize that atrial fibrillation may present occultly in patients with pacemakers.

CASE PRESENTATION: A seventy-two year old female presented with sudden onset of apparently altered mental status and weakness. She had a past medical history of coronary artery disease and sick sinus syndrome requiring implantation of a permanent pacemaker. Initial evaluation revealed an alert woman with receptive and expressive language defects consistent with Wernicke's aphasia. The examination was otherwise unremarkable, including a normal cardiac examination with regular heart rhythm and without murmurs or extra sounds. The initial electrocardiogram showed atrial and ventricular pacemaker spikes with regular QRS complexes. Echocardiogram showed atrial enlargement (diameter 5.6 cms.); no comment was made about atrial contractions. Inhibition of the pacemaker revealed underlying atrial fibrillation. The patient's aphasia improved and she was discharged on warfarin.

DISCUSSION: The prevalence of atrial fibrillation and the rate of pacemaker implantation are both increasing as the population of the U.S. ages. Atrial fibrillation can trigger rapid and irregular ventricular rhythm in patients with atrial synchronous pacemakers; however, in patients with permanent pacemakers atrial fibrillation can occur in the absence of its classic cardinal feature—an irregular rhythm. Our patient presented with new onset of CVA, a regular paced rhythm and absence of comment about atrial contractions on ECHO report. Without clinical suspicion and proper interrogation of the pacemaker, atrial fibrillation would have been missed. The possibility of occult atrial fibrillation should not be overlooked in paced patients and they should be intermittently checked for onset of atrial fibrillation. If it is discovered, anticoagulation should be strongly considered. The literature suggests that even in pacemaker clinics the diagnosis of occult atrial fibrillation is often missed, and an opportunity to prevent future strokes is lost.

SEVERE PREECLAMPSIA AND THROMBOPHILIA. LE Larson, RO Powrie, Medicine, Women and Infants' Hospital, Brown University, Providence, RI

LEARNING OBJECTIVES: 1. Recognize the association of severe early preeclampsia with the thrombophilias. 2. Illustrate the increased risk of thrombosis in pregnancy and the postpartum period. 3. Highlight the need to monitor women for postpartum depression.

CASE PRESENTATION: AS is a 32yo G2P0 (previous first trimester miscarriage) with a history of mild chronic hypertension and asthma who presented at 29 weeks gestation with headache, blurred vision and epigastric pain. Her physical exam revealed BP $169/107$, normal fundi, mild RUQ tenderness, 1+ edema, and normal neurologic exam with 3+DTR's. Labs were notable for 2+ proteinuria, plts 128K, hgb 12 g/dL, creat 0.6 mg/dL, AST 31U/L (12–30), and uric acid 3.8 mg/dL. The patient was presumed to have preeclampsia superimposed on chronic hypertension and was admitted for bedrest and monitoring. She was treated with steroids for fetal lung maturity. She was delivered by cesarean section for worsening preeclampsia after her platelet count dropped to 79K on hospital day 4. On postpartum day 1 she developed mild epistaxis and had plt 27K, hgb 8.4 g/dL, AST 99U/L, ALT 104U/L, Tbili 0.7 mg/dL, LDH 550U/L (85–200). She was felt to have HELLP syndrome and was treated with dexamethasone. On postpartum day

6 she developed worsening headache, mild mental status changes and low grade fevers. Her labs worsened with plt 12K, hgb 8.3 g/dL (decreased from 9.7), Tbili 1.3, LDH 6000 u/L and there were 15–20 schistocytes per hpf on her blood smear. Head CT was normal. The patient was transferred to the ICU for plasmapheresis for possible thrombotic thrombocytopenic purpura and given antibiotics for a left lower lobe infiltrate and possible endometritis. She had a prompt response with her platelets increasing to 135K and her LDH decreasing to 844 U/L after 3 treatments. Her ICU course was complicated by a right femoral hematoma secondary to line placement with a subsequent femoral neuropathy. She was discharged on a tapering dose of prednisone with resolution of the hemolysis but presented 2 weeks later with a left internal jugular thrombosis. A thrombophilia workup revealed decreased protein S levels and an elevated homocysteine level of 17 umol/L. The patient is now being treated for postpartum depression. The femoral neuropathy has resolved. Her son is doing well at home after 6 weeks in the NICU.

DISCUSSION: Women with thrombophilia are at increased risk for thrombotic events during pregnancy and the postpartum period. It is less well recognized by general internists that they are also at increased risk for obstetric complications such as severe early preeclampsia, intrauterine growth retardation and fetal loss. Women should be monitored for postpartum depression especially if they have the added stressor of a complicated pregnancy and postpartum course.

LEMIERRE'S SYNDROME: A RARE COMPLICATION OF A COMMON ILLNESS. S Levine, D Korenstein, Medicine, Mt. Sinai Medical Center, New York, NY

LEARNING OBJECTIVES: 1. To recognize Lemierre's Syndrome, a serious complication of pharyngitis. 2. To understand the indications for surgery in Lemierre's Syndrome.

CASE PRESENTATION: A previously healthy 34 year old black female presented to the ER with a complaint of sore throat, fevers, chills, neck and shoulder pain. The patient had been evaluated in the ER 10 days and 7 days prior to admission for sore throat, fever, and chills and was discharged home after negative rapid strep tests both times. On the third visit she had new left shoulder pain, cough, pleuritic chest pain, rigors, and vomiting. She appeared cachectic in moderate distress with mild scleral icterus. Vitals included a temperature of 37.9, p = 125, BP = $100/60$, RR = 18, and O2 sat of 100% on room air. She had an enlarged left tonsil with exudate, exquisite tenderness of the left neck with swelling, grade 2 systolic murmur at the left sternal border, and decreased breath sounds at the left lung base. She had an immobile left shoulder secondary to severe pain. Laboratory studies were significant for white blood cell count = 18,000 with 78% neutrophils and 4% bands, Hgb = 11, platelets = 81,000, albumin = 2.5 and total bilirubin = 2.3. Chest x-ray revealed infiltrate at the left base with a pleural effusion. A CT of the neck showed inflammation of the soft tissues on the left with an internal jugular vein thrombosis. MRI of the left shoulder revealed no fluid collection, and CT of the chest demonstrated multiple septic emboli throughout the lungs. After cultures were obtained, the patient was placed on clindamycin, ceftriaxone and heparin. Pus was aspirated from the neck under ultrasound guidance; gram stain showed many gram negative rods and gram positive cocci. The patient remained febrile to 40C and on hospital day #3 was taken to the operating room for ligation and excision of the left internal jugular vein. Culture of the neck aspirate and thrombus grew out *Streptococcus milleri*, although multiple blood cultures were negative. Repeat CT showed retained thrombus, however the patient's condition improved greatly after surgery. She was discharged after 11 days of hospitalization on amoxicillin clavulanate and warfarin.

DISCUSSION: (1) Lemierre's syndrome is a complication of pharyngitis with septic thrombophlebitis of the internal jugular vein and septic emboli, typically to the lungs and joints. (2) The most common organism is *Fusobacterium necrophorum* although other organisms such as *Streptococcus*, *Bacteroides*, and *Eikenella* can also be involved. (3) The mainstay of treatment is antibiotics directed at anaerobic microbes. The role of anticoagulant therapy is controversial. Patients with persistent sepsis and septic emboli despite medical therapy should be considered for surgical ligation and excision of the internal jugular vein.

AN ATYPICAL PRESENTATION OF BACTERIAL ENDOCARDITIS IN A WOMAN WITH A RARE CARDIAC ANOMALY. D-W Liao, N Yazbek, S Frost, General Internal Medicine, The Cleveland Clinic Foundation, Cleveland, OH

LEARNING OBJECTIVES: 1) Recognize the musculoskeletal manifestations of bacterial endocarditis. 2) Recognize the potential complications associated with a rare cardiac anomaly often discovered incidentally.

CASE PRESENTATION: A 64 yo woman with lumbar degenerative disc disease (ddd) was admitted with one week of worsening back pain exacerbated by movement. Narcotic analgesia prescribed two days earlier provided no relief. She denied recent trauma, neurological deficit, bowel or bladder incontinence, and fever or chills. Recent history was significant for ischemic colitis managed conservatively two months earlier. Remote history was remarkable for a heart murmur diagnosed in childhood. Physical exam demonstrated T = 36.5C, % systolic ejection murmur at the right upper sternal border, no peripheral or ocular stigmata of endocarditis, tenderness at L3/4, and a nonfocal neurologic exam. Lab demonstrated normal chemistry and urinalysis, WBC = 6.9, HCT = 35.1, ESR = 74, CRP = 14.8. EKG was normal. Lumbar MRI with gadolinium showed ddd without infection or inflammation. High dose patient controlled narcotic analgesia was necessary. A temperature of 38.1C was noted 29 hours after admission. Subsequent blood cultures were positive for enterococcus. Echocardiogram revealed a small,

mobile echodensity attached to the right coronary cusp of the aortic valve, and a dilated coronary sinus with high velocity flow (4m/s). Cardiac MRI demonstrated a large serpiginous fistulous tract from the origin of the right coronary artery to the coronary sinus with a high velocity jet emptying from the coronary sinus into a dilated right atrium. Ampicillin and streptomycin treatment resulted in resolution of fever and marked improvement in back pain over the following three days. **DISCUSSION:** Back pain presenting days to weeks prior to the development of fever is an uncommon but well reported singular presenting symptom of bacterial endocarditis. Evidence for spinal infection or inflammation is often lacking. Hypothesized etiologies include myalgia (due to vasculitis or arterial emboli), immune complex phenomenon, minimal osteomyelitis, reactive arthritis, and glomerulonephritis. Recognition of the musculoskeletal manifestations of bacterial endocarditis is important because of the potential to mimic other processes. Coronary arteriovenous fistula is an uncommon congenital anomaly often unrecognized until adulthood. Most adult patients are asymptomatic at the time of discovery, but complications including CHF, myocardial ischemia, pulmonary hypertension, and endocarditis can result in associated symptomatology. Surgical correction or percutaneous transcatheter coil embolization has been suggested in asymptomatic patients to prevent complications.

TRAPPED BY AN ABNORMAL TEST: PNEUMOCOCCAL PNEUMONIA MASQUERADING AS PULMONARY EMBOLISM. CH Lieu, Primary Care Internal Medicine, Waterbury Hospital Health Center, Waterbury, CT

LEARNING OBJECTIVES: 1) Recognize the importance of pretest suspicion of pulmonary embolism prior to ventilation/perfusion scanning. 2) Re-evaluate your patients' diagnosis during their hospitalization.

CASE PRESENTATION: A 55 yo previously healthy woman presented to the emergency department (ED) with three days of right-sided pleuritic chest pain which worsened with cough and deep inspiration. She denied any dyspnea and reported generalized malaise but no specific fever, chills, or productive cough. The patient had a distant history of nephrolithiasis and previously undiagnosed biliary colic. Pertinent social history included 50+ pack years of cigarette abuse. The patient was noted to be hypoxicemic on pulse oxymetry and a room air arterial blood gas revealed pH 7.44, pCO₂ 36 mm Hg, and pO₂ 54 mm Hg. A chest x-ray revealed only a patchy infiltrate at the right costophrenic angle. Concerned about pulmonary embolism (PE), the ED physician ordered a ventilation/perfusion (V/Q) scan which was interpreted as high likelihood for PE due to a matched defect at the right base and two areas of unmatched perfusion defects on the left. The ED record revealed no pretest suspicion for PE, and the patient was started on IV heparin and admitted to the housestaff who embarked on a hypercoagulable evaluation. After 24 hours of hospitalization, the patient's pulmonary examination changed from scant right basilar crackles to percussion dullness, egophony, and diminished breath sounds in her right lower lobe (RLL). Fever to 102.4° and a productive cough with blood-tinged sputum developed during hospital day (HD) #2. Worsening but transient hypoxemia occurred on HD #2. Antibiotics for community-acquired pneumonia were started approximately 36 hours after the initial presentation to the ED. Immediately prior to the patient's decompensation, a spiral CT of the chest was performed to rule out the presence of PE but its interpretation was not readily available. The housestaff caring for the patient overnight requested resumption of IV heparin, which had been discontinued by the attending physician due to low pretest suspicion of PE and changed physical examination. Interpretation of the spiral CT on HD #3 revealed consolidation of the entire RLL with an effusion and no evidence of PE. Blood cultures on the same day isolated *Streptococcus pneumoniae*. Despite the delay in diagnosis and treatment, the patient had an uneventful recovery with antimicrobial therapy and a diagnostic thoracentesis revealed an uncomplicated parapneumonic effusion.

DISCUSSION: The absence of level of clinical suspicion for PE limited the interpretation of the V/Q scan results for this patient. In the Prospective Investigation of Pulmonary Embolism Diagnosis trial, 2/3 patients with a low pretest clinical suspicion and a high probability V/Q can did not have a PE.

PERTUSSIS IN HOSPITALIZED ADULTS. JK Lim, Internal Medicine, Yale University School of Medicine, New Haven, CT

LEARNING OBJECTIVES: 1) Diagnose pertussis infection in adults. 2) Recognize/treat pertussis quickly to minimize spread. 3) Identify candidates for booster vaccinations.

CASE: 82 year-old female nursing home resident with hypertension, non-insulin dependent diabetes mellitus, and childhood whooping cough, presented to outside hospital in acute respiratory distress after experiencing dyspnea for 1 day. She noted fatigue and anorexia for 2 weeks without cough or chest pain. Exam notable for low-grade fever, bibasilar rales, and pedal edema. EKG revealed new LBBB. Chemistries unremarkable except for elevated creatine kinase (141), CK-MB (11.3), and troponin (8.3). CBC notable for elevated WBC (12.3) with PMN predominance. Patient emergently transferred to university hospital for cardiac catheterization, which revealed moderate non-ischemic cardiomyopathy with LVEF 35%. She was intubated in the coronary care unit for hypoxemic respiratory failure secondary to pulmonary edema and left lower lobe pneumonia, was started on intravenous ceftriaxone, and then switched to oral cefotetan after uneventful extubation. She experienced progressive clinical improvement except for persistent nonproductive cough since admission to university hospital. Sputum samples were normal, as were screens for adenovirus, influenza A/B, parainfluenza, and RSV. She was otherwise asymptomatic without fevers or leukocytosis. On transfer

to floor on day 12, paroxysmal nocturnal coughing episodes were noted. On day 14, respiratory isolation was initiated, nasopharyngeal samples were obtained, and erythromycin was started. Bordatella pertussis culture was negative; PCR analysis was positive. The coughing defervesced over 5 days, at which time she was uneventfully discharged to nursing facility.

DISCUSSION: Bordatella pertussis infection is a common yet underrecognized cause of persistent cough in adults. Readily identified by its characteristic high-pitched inspiratory "whoop" in children, it often passes undiagnosed by internists due to the absence of typical clinical findings (i.e., paroxysmal cough, marked lymphocytosis) and underestimation of its prevalence in adults. As illustrated, prior vaccination nor infection confers lasting immunity. Adults are the major reservoir of disease; therefore, timely recognition and treatment of infection is critical in minimizing epidemic spread. Diagnosis is limited by nonspecific symptoms and poorly sensitive lab tests. Cultures are universally negative in adults, and diagnostic PCR/DFA technology are often limited to research institutions. Nonetheless, pertussis should be suspected in adults with a persistent cough (over two weeks), particularly if associated with paroxysms, whoop, or posttussive emesis. Early administration of oral erythromycin may decrease infectivity and shorten illness duration. Improved acellular vaccines may soon lead to widespread immunization boosters in adults.

PERTUSSIS IN HOSPITALIZED ADULTS. JK Lim, Internal Medicine, Yale University School of Medicine, New Haven, CT

LEARNING OBJECTIVES: 1) Diagnose pertussis in adults. 2) Recognize and treat pertussis quickly to prevent epidemic spread. 3) Identify target candidates for booster vaccinations.

CASE PRESENTATION: 82 year-old white female nursing home resident with hypertension and non-insulin dependent diabetes mellitus presented to outside community hospital in acute respiratory distress after being found by facility staff to be diaphoretic and dyspneic at rest for one day. Patient had noted generalized fatigue, anorexia, and malaise for two weeks. Exam notable for low-grade fever, bibasilar rales, bilateral pedal edema. EKG notable for new LBBB. Chemistries unremarkable except for initial creatine kinase of 141, MB fraction of 11.3, and troponin of 8.3. Patient emergently transferred to university hospital for cardiac catheterization. Catheterization revealed depressed left ventricular systolic function of 35% with global hypokinesia, but otherwise minimal coronary disease and no valvular disease. Patient admitted to coronary care unit where patient emergently intubated for hypoxemic respiratory failure. Chest film revealed mild pulmonary edema and left lower lobe infiltrate. Patient started on empiric intravenous ceftriaxone which was switched to oral cefotetan after ten days, at which time patient was extubated. Hospital course marked by progressive clinical improvement except for persistent hacking poorly productive cough. Upon transfer to cardiology floor on day twelve, patient was noted to have paroxysmal nocturnal coughing episodes. On day fourteen, patient was placed in respiratory isolation, nasopharyngeal samples were obtained, and erythromycin therapy was started. Bordatella pertussis culture was negative; PCR analysis was positive. Previous six sputum cultures revealed normal flora. Screening for respiratory syncytial virus, influenza A and B, parainfluenza, and adenovirus was negative. Patient's coughing episodes gradually resolved over five days at which time patient uneventfully discharged to nursing facility.

CONJUNCTIVITIS AS A MANIFESTATION OF GONOCOCCAL INFECTION. HD Lu, P Haidet, Houston VAMC, Baylor College of Medicine, Houston, TX

LEARNING OBJECTIVES: 1) Review the differential diagnosis for conjunctivitis. 2) Recognize symptoms that necessitate urgent treatment. 3) Realize that conjunctivitis may arise secondary to other infections.

CASE PRESENTATION: A 31 year-old male with an unremarkable medical history presented with right eye redness, tearing and discharge for 3 days. He reported mild discomfort but no significant eye pain or photophobia. He denied any history of eye trauma, foreign object exposure, or ill contacts, and had not experienced conjunctivitis previously. System review revealed complaints of dysuria and a purulent urethral discharge that had started several days prior to the eye complaints. Social history was remarkable for recent episodes of unprotected sex with multiple partners. Physical exam revealed marked, diffuse erythema of the right conjunctiva with edema of surrounding orbital soft tissues. A thick and copious mucopurulent discharge was present with matting of the eyelashes. The cornea was normal in appearance. Palpation of the orbit revealed mild tenderness. Vision acuity was 20/20 by near-field Snellen. Lymph node examination revealed mild right preauricular lymphadenopathy. Examination of the genitalia revealed a small amount of purulent discharge at the urethral meatus. Gram stain and culture of this discharge revealed evidence of gonococcal infection. The patient was treated with intramuscular ceftriaxone and oral azithromycin, as well as ophthalmic erythromycin. He was counseled to advise his partners to seek medical attention.

DISCUSSION: Conjunctivitis is a common complaint encountered in the ambulatory setting. The differential diagnosis of conjunctivitis includes hyperacute, acute, and chronic bacterial conjunctivitis, as well as inclusion (chlamydial), viral, chemical, and allergic conjunctivitis. Most cases of conjunctivitis are self-limited and resolve without treatment. However, significant eye pain, copious mucopurulent discharge, photophobia, decreased visual acuity, corneal opacity or haziness, and injection of ciliary vessels are all causes for concern. This patient presented with a history and physical findings typical for hyperacute bacterial conjunctivitis. *Neisseria Gonorrhoeae* is the most common cause. Infection usually occurs

through direct autoinoculation from genitalia and may spread quickly to the other eye. Onset occurs within two to three days of inoculation. Hyperacute gonococcal conjunctivitis requires urgent treatment, since untreated cases often result in corneal ulceration, perforation and endophthalmitis. Diagnosis may be confirmed through gram-stain of discharge or conjunctival scraping. However, establishing a culture diagnosis should not delay treatment, which should include systemic as well as local antibiotics. Evaluation should also include identification and treatment of sexually transmitted diseases in the patient and partner(s).

HOME ASSESSMENT OF AN ELDERLY PATIENT. BM Matti, Medicine, St. Luke's/Roosevelt Hospital Center, New York, NY

LEARNING OBJECTIVES: 1. Acquire fundamental knowledge and skills to conduct a home visit effectively. 2. Recognize the value of home assessment in managing a patient's medical problem, such as asthma exacerbation.

CASE PRESENTATION: A 73 year-old, home-bound female was referred for increasing shortness of breath, dyspnea on exertion, and chest tightness of about one week's duration. The patient also experienced coughing but denied having fever or sore throat, and no orthopnea or leg swelling. Her symptoms were precipitated by exposure to dust from window-cleaning. For many years, the patient had mild and intermittent asthma, managed with as-needed Albuterol metered-dose inhaler. She did not smoke nor drink alcohol; she did have an underlying psychiatric disorder. The patient refused to go to the doctor's office or the emergency room for care. An emergent housecall was done with a visiting nurse. In addition to basic physician's instruments, a nebulizer machine, a peak flow meter, Albuterol respiratory solution and metered-dose inhaler were also brought. The patient lived alone on the 12th floor (accessible by elevator), in a one-bedroom apartment within a naturally-occurring retirement community. On examination, the patient was alert, coherent, oriented, and in no distress. She was mildly tachypneic (~18/min), with normal heart rate and blood pressure; her peak flow was about 300 ml/min. She tried to use her Albuterol inhaler which had expired. She was fairly nourished but poorly groomed and was able to ambulate slowly on her own. She did not use any accessory muscles of respiration. On auscultation, there was fair air entry with occasional expiratory wheezing. The rest of the examination was unremarkable. Inspection of the apartment revealed a wall-to-wall carpeting of dust and litter. The hallway and living room were poorly lit and difficult to walk through because of the stacks of magazines, newspapers, and various materials which the patient had collected through the years. The kitchen table had food scraps and bread crumbs. The sink had a stack of unwashed dishes which were several days old. There were also mice and cockroaches in the apartment. The patient was provided with a new Albuterol metered-dose inhaler as well as a prescription for Triamcinolone metered-dose inhaler 2 puffs twice daily. The nurse was instructed to do daily home visits for the next three days to assess patient's condition and possible need for oral steroids. A referral for psychiatric evaluation was also done. The building superintendent was later involved to assist in cleaning up the apartment.

DISCUSSION: There is an increasing need for physician-conducted home visits. "INHOMENESS" (standing for: Immobility, Nutrition, Housing, Others, Medication, Examination, Safety, Spirituality, Services) is an easy mnemonic for the evaluation of the patient's functional status and home environment.

PANCYTOPENIA IN THE SPRINGTIME. YS Meagh, LB Feuer, TL Simon, Medicine, Mount Sinai Medical Center, New York, NY

LEARNING OBJECTIVES: 1) Identify clinical manifestations of ehrlichiosis. 2) Utilize laboratory techniques to diagnose Ehrlichia infection. 3) Recognize the potential for coinfection with more than one tick-borne agent.

CASE PRESENTATION: A 74 year old woman with type 2 diabetes mellitus and benign hypertension presented in May with fever, malaise, myalgias, abdominal pain and confusion. She was in her usual state of good health until two weeks before admission when she experienced a tick bite on her abdomen while hiking in Westchester. Two days later she developed right upper quadrant pain, nausea and anorexia. In the subsequent week she noted aching of her upper and lower extremities and intermittent episodes of disorientation. On the day of admission the patient awoke with rigors, diaphoresis, confusion and an oral temperature of 103F. Physical exam revealed a lethargic, ill appearing woman. Temp-38.6C, HR-94 supine, 120 sitting, BP-79/36, O2 sat-92% room air. The sclera and skin were anicteric. Mucous membranes were dry. The neck was supple without adenopathy. Lungs were clear. Cardiac exam revealed a regular tachycardia and no murmurs. The abdomen was remarkable for diffuse tenderness most pronounced in the right upper quadrant, no rebound or guarding and no hepatosplenomegaly. Extremities were nonedematous and joints were without erythema or effusion. Skin exam only revealed a 0.5 cm erythematous macule in the right upper quadrant. Neurologic exam showed no focal deficits. WBC-2.5, Hgb-9.2, platelets-32,000, reticulocyte count-0.9%, fibrin split products and fibrinogen-wnl, Na-132, BUN-42, Creatinine-1.7. Serum transaminases and bilirubin were normal. CXR was unremarkable; EKG showed sinus tachycardia. The diagnosis of a tick-borne infection was suspected. The patient received intravenous fluids and oral doxycycline. Within 24 hours, the patient defervesced, symptoms subsided, renal function normalized and pancytopenia began to resolve. Examination of the buffy coat smear revealed intracytoplasmic granulocytic morulae. Lyme IgM and IgG titers were positive. PCR for ehrlichia and lyme were positive.

DISCUSSION: Ehrlichiosis should be considered in patients presenting with fever, pancytopenia and a history of a tick bite within the previous three weeks. The

clinical symptomatology is often nonspecific. A rash only occurs in one-third of patients. Severe complications may include multisystem organ dysfunction, disseminated intravascular coagulation and opportunistic infection. Although normal in this patient, elevation of hepatic enzymes is a hallmark of ehrlichia infection. In suspect patients, examination of peripheral and buffy coat smears may reveal intracytoplasmic morulae. Coinfection with the agents of Lyme disease, Rocky Mountain Spotted Fever and babesiosis should be considered in patients with ehrlichiosis.

REFRACTORY ABDOMINAL PAIN: A CASE OF A RUPTURED AORTIC ANEURYSM. AM Mehta, R Granieri, Medicine, University of Pittsburgh Medical Center, Pittsburgh, PA

LEARNING OBJECTIVES: 1. Discuss the presentation of RAAA. 2. Discuss indications for elective repair of AAA. 3. Discuss recommendations for monitoring of AAA.

CASE PRESENTATION: An 82 year old female with a history of a 4 cm AAA presented from a nursing home with a 3 week history of worsening lower abdominal pain. She denied fever, dysphagia, nausea, vomiting, diarrhea, melena, hematochezia, jaundice, or dysuria. She endorsed decreased appetite, decreased oral intake, and constipation. Multiple prior hemocults were negative. She had been started on narcotics for pain, without relief. PMH is significant for small bowel obstruction, umbilical hernia repair, hiatal hernia, recurrent constipation, rheumatoid arthritis, cavernous sinus aneurysm, hypothyroidism, and vertebral compression fractures. Medications included omeprazole, alendronate, and controlled release oxycodone. Physical exam revealed a thin female in no distress. T 36.8 P 84 R 22 BP 144/62. HEENT: Sclera anicteric. Neck, lungs, heart and extremities were normal. Abdomen was soft, nondistended, with bowel sounds, but with significant tenderness to palpation in lower quadrants bilaterally, without rebound or guarding. Rectal exam was heme negative. Na 127, K 2.6, Cl 87, CO2 30, BUN 13, Cr 1.0. LFTs, amylase, lipase, bilirubin, urinalysis all were normal. WBC 26.8 K with 90% polys and H/H was 8.9/25.2, which was decreased from baseline of 11/34. Abdominal x-ray showed cholelithiasis and calcified AAA without obstruction or free air. Abdominal ultrasound was significant for AAA measuring 5.5 x 5.3 cm involving the mid aorta, mild hydronephrosis, and cholelithiasis. Emergent abdominal CT scan showed a very large, contained, ruptured infrarenal AAA with extensive clot and no extravasation. The patient went for an emergent AAA repair, which she tolerated well, and left the hospital shortly thereafter.

DISCUSSION: RAAAs cause 1-2% of deaths in males over 65 years in western countries, with a rising incidence due to an aging population. Most RAAAs present with abdominal pain and tenderness, back or flank pain, and leukocytosis. Anemia and hypotension are uncommon. In one study of 23 patients, 61% were not diagnosed at presentation to internists. The mortality rate from RAAA exceeds 80%. For this reason, elective operative repair is recommended by many experts for RAAAs larger than 5 cm, or that grow by more than 1 cm per year. At this point, the risk of rupture outweighs the risk of surgery. The mortality risk for elective AAA repair is 4-8%. Monitoring is recommended every 6 months by ultrasound. Screening is recommended in individuals over age 50-55 with a family history of a AAA in a first degree relative.

ARCUS CORNEALIS IN A YOUNG PATIENT WITH HYPERCHOLESTEROLEMIA. RS Meisner, RD Hobbs, Internal Medicine, University of Michigan, Ann Arbor; Internal Medicine, Oakwood Hospital, Dearborn, MI

LEARNING OBJECTIVES: To recognize the earliest presentation, natural history and significance of arcus cornealis in a young patient.

CASE PRESENTATION: A 36-year-old Asian-American male with a history of hypertension presented for a health maintenance exam. His blood pressure was under good control on metoprolol. His main concerns were a family history of hyperlipidemia, diabetes, and coronary artery disease. He denied any symptoms of CAD. His past medical history was otherwise insignificant and he denied smoking or alcohol use. Physical exam revealed stable vital signs. Examination of the eyes revealed early arcus cornealis in the periphery of the cornea at the 12:00 o'clock position. It was only visible with elevation of the upper eyelids. No xanthomas were noted. The remainder of the exam was unremarkable. His lipid profile was elevated; LDL = 135, HDL = 38, triglycerides = 287, and total cholesterol = 230. He was informed of the significance of arcus cornealis at his young age as well as the implications of his family history. At this point he noted that other, younger family members also had the whitish discoloration of the cornea. No medications were prescribed since the patient wanted to first try lifestyle modifications. He was asked to follow-up for further cholesterol analysis.

DISCUSSION: This case illustrates several points. First, especially considering the high technology that is readily available to physicians, it emphasizes that a thorough physical exam still provides valuable diagnostic information. Although frequently discovered in the elderly and usually described as a benign finding, the presence of arcus cornealis, a lipid deposit in the peripheral cornea, is associated in a younger patient, with an increased risk of hyperlipidemia and cardiovascular disease. This may help the physician target these particular patients for more strict cardiovascular assessment beyond simple monitoring of cholesterol levels. Second, it points out the importance of elevating the eyelid when examining the eye for arcus cornealis. Since corneal arcus initially forms at the 12 o'clock sector of the periphery before spreading and encircling the cornea, many patients with early lesions and possibly significant hyperlipidemia may be missed if the entire cornea is not visualized.

AN UNUSUAL CAUSE OF RECURRENT DEEP VENOUS THROMBOSIS. RG Miller, V Reddy, Department of Medicine, Johns Hopkins University School of Medicine, Baltimore, MD

LEARNING OBJECTIVES: 1. Recognize the role of venography in the work-up of deep venous thrombosis (DVT). 2. Identify the May-Thurner syndrome as an underdiagnosed cause of DVT and recognize its management.

CASE: A healthy 28 year-old woman presented with progressive left thigh swelling and pain over 3 weeks. The pain was worse with weight-bearing. She denied trauma and was on no medications. Her past medical history was remarkable for three prior episodes of thigh swelling over 10 years, the first in 1989 during the last trimester of her only pregnancy. Each time, she was diagnosed with a DVT and treated with anticoagulation. There was no family history of clotting disorders.

Physical examination revealed moderate swelling of the left thigh with mild warmth but no erythema. The distal portion of the extremity was normal. Ultrasonography of the extremity was negative for clot. Due to a high clinical suspicion, venography was performed. This revealed compression of the left iliac vein by the right iliac artery with significant clot formation, consistent with the May-Thurner syndrome. Marked venous collateralization reinforced the chronicity of this process. The patient was treated with a 24 hour catheter-directed infusion of tissue plasminogen activator followed by angioplasty, and was discharged on anticoagulation. Her symptoms, however, recurred within two weeks, necessitating endovascular stent placement. She has since done well.

DISCUSSION: This case raises two important points. The first is that ultrasonography for the diagnosis of DVT, while excellent with a sensitivity of over 95% in symptomatic patients, is not the gold standard. When a clinician has an extremely high index of suspicion, venography should be pursued. The second point is that clinicians should remember that anatomical vascular anomalies may be a cause of DVT. As this case illustrates, the May-Thurner syndrome is a condition characterized by chronic compression of the iliac vein by the overlying iliac artery with resultant intimal hypertrophy and frequent clot formation. It is most common in young women and is often underdiagnosed, accounting for up to 2-5% of cases of lower extremity DVT. It is important to recognize vascular causes of DVT such as the May-Thurner syndrome, as stenting or surgical intervention may be necessary for optimal outcomes.

HAS CONTEMPORARY MEDICINE DULLED OCCAM'S RAZOR? A PATIENT FOR WHOM A SIMPLE EXPLANATION DID NOT SUFFICE. A Miller, S Hasabnis, S Moise, GW Rutecki, Medicine, Evanston Northwestern Healthcare, Evanston, IL

LEARNING OBJECTIVES: 1. Diagnose insulin resistance in the setting of hyperglycemia and mixed connective tissue disease. 2. Recognize the criteria for the diagnosis of Gitelman's syndrome. 3. Distinguish multiple, unusual diagnoses in complicated patients.

CASE PRESENTATION: We present a 74 y.o. F with type B insulin resistance, mixed connective tissue disease (MCTD) and GS. Background DM2 eventuated in her admission for hyperglycemia (572 mg/dl). 250 units of intravenous insulin/hour did not lower the serum glucose below 350 mg/dl. Insulin receptor antibodies were positive and the patient was treated with pulse steroids, cytoxan and plasmapheresis. Afterwards, 16 units of NPH insulin daily controlled her glucose. A positive RNP & Smith antibody were consistent with MCTD as the etiology for the insulin receptor antibodies. During the hospitalization, hypokalemic, metabolic alkalemia with normotension led to further workup. High urine chloride values with magnesium wasting and hypocalciuria were consistent with GS.

DISCUSSION: Extreme insulin resistance presenting in adulthood may be the result of the rare syndrome of type B insulin resistance. Measurable levels of insulin receptor antibodies are present and some of these patients have associated autoimmune diseases such as S.L.E. or Sjogrens. Also rare, and apparently unrelated to insulin resistance, is Gitelman's syndrome (GS). Previously subsumed under the umbrella of Bartter's, GS is comprised by normotensive, hypokalemic metabolic alkalemia with elevated urine chlorides, urine magnesium, renins and aldosterone, but with decreased urinary excretion of calcium. Although William of Occam's observation that simple explanations are usually the most reliable, contemporary medical sophistication may diagnose multiple zebras with today's complex hoofbeats.

MDMA-INDUCED ACUTE HEPATITIS IN A YOUNG PATIENT. MM Mohiuddin, I Tong, PD Friedmann, Rhode Island Hospital and Brown University School of Medicine

LEARNING OBJECTIVES: 1) Recognize that "Ecstasy" (methylenedioxymetamphetamine or MDMA) can present with a constellation of findings suggestive of hepatotoxicity with possible gallbladder involvement. 2) Alert physicians to consider drug use when confronted with unexplained symptoms and lab abnormalities indicative of acute hepatic disease.

CASE: A 27 year-old unmarried restaurant hostess presented with sudden and extreme abdominal pain. Pain was colicky, began in the epigastrium one day prior to admission, and gradually localized to the right upper quadrant. The patient described the pain as 10/10 without radiation. She had a history of GERD, but this pain was clearly different. She initially denied alcohol or other drug use. Her only medication was Triphasil. She denied acetaminophen use. Her restaurant served raw seafood, which she had neither ingested nor handled, and she had unprotected sex with a new partner for the previous 3 months. She reported no other risk factors for hepatitis, except, on further questioning she revealed that she had

used "Ecstasy" for the first time two days prior to admission. On physical exam, she was afebrile and anicteric with stable vital signs. Cardiac and lung exams were unremarkable. The abdomen was soft, mildly distended with pronounced percussive tenderness and guarding in the right upper quadrant. No organomegaly was present. Pelvic exam was unremarkable. Relevant labwork included: WBC = 5.4, Hgb = 15.5, AST = 943, ALT = 820, Alk Phos = 164, Total Bilirubin = 1.3, Albumin = 3.9, and INR = 0.9. Amylase and lipase were normal. Hepatitis serologies and toxicology were negative. Right upper quadrant ultrasound revealed minimal perihepatic and pericholecystic fluid with a thickened gallbladder wall. Abdominal CT showed pericholecystic fluid and intrahepatic biliary dilatation. The patient was placed on IV hydration and ampicillin/sulbactam with eventual resolution of pain. Laboratory abnormalities (peak AST = 1201 and ALT = 1066 on day 2) gradually declined and the patient was discharged without further complications.

DISCUSSION. Subacute idiosyncratic toxic hepatitis has been attributed to MDMA in nine previously documented cases. This form of hepatitis can occur after incidental or chronic ingestion, and the symptoms may develop a few days to a few weeks after ingestion. The spectrum of MDMA-associated liver disease is wide, including jaundice and anorexia, acute abdominal pain, and hepatocellular damage. Most reported cases regarding MDMA have noted severe colicky abdominal pain, which is a symptom generally not seen with hepatitis. Our patient had clinical, laboratory and radiologic findings to suggest both hepatotoxicity and acalculous cholecystitis. Clinicians should be aware that MDMA may cause a diverse constellation of hepatobiliary disorders.

A CHEST WALL MASS IN A YOUNG REFUGEE PREVIOUSLY TREATED WITH ISONIAZID. A Moreno, D Rishikof, J Bernardo, Section of General Internal Medicine; Section of Pulmonary, Critical Care, and Allergy; Boston Medical Center, Boston University School of Medicine, Boston, MA

LEARNING OBJECTIVES: 1. Recognize an unusual presentation of extra-pulmonary tuberculosis (TB). 2. Learn the differential diagnoses of a chest wall mass. 3. Recognize TB reactivation vs. re-infection and Isoniazid (INH) prophylaxis failure.

CASE PRESENTATION: The patient is a 29 year-old male refugee from Somalia, who presented on 09/02/99 with a one-month history of a left anterior chest wall mass and one-year of dull pain in the same area. The pain did not have any pleuritic features and was partially relieved with common analgesics. The patient denied any previous trauma to the affected area, drug abuse, weight loss, fevers, chills, or night sweats. The review of systems was negative. His past medical history was significant for malaria, intestinal Entamoeba Histolytica, and a 15 mm PPD test with a normal chest x-ray on 09/19/96. At that time, the patient was referred to the TB Clinic, where he was diagnosed with TB Class 2 (TB infection, no disease). For 6 months, he received INH and Vitamin B6 and claimed adherence to daily treatment. On physical exam, the left anterior chest wall, between the left para-sternal border and mid-clavicular line, over the 7th rib, revealed a poorly defined, tender, non-erythematous mass measuring approximately 5 cm in diameter. The lesion had minimal surrounding edema. No associated axillary or supra-clavicular lymph nodes were noted. The rest of the physical exam was normal. A CBC and a chest x-ray did not reveal any abnormalities. A CT scan of the chest revealed a 3 x 4 x 6 cm low attenuation soft tissue mass destroying adjacent rib and costochondral joint and several small prevascular lymph nodes. No pleural effusion or parenchymal lung involvement was noted. A fine needle aspiration biopsy revealed numerous neutrophils but no tumor cells. The aspirate's Gram and Kinyoun stain smears were negative, as well as the bacterial and fungal cultures. On 10/01/99, the laboratory reported a positive culture for a pan-sensitive Mycobacterium Tuberculosis. The patient is receiving treatment for TB Class 3 (TB infection, current disease) with Ethambutol, Pyrazinamide, Rifampin, and INH.

DISCUSSION: Although tuberculous osteomyelitis of a rib is an unusual presentation of TB, it should be considered along with other infectious or neoplastic processes in patients who present with chest wall masses. Other infectious etiologies include Staphylococcus, Pseudomonas, and Actinomyces. Sarcoma and metastatic disease are possible malignancies in this age group. Unless molecular epidemiological studies are done, it is clinically difficult to differentiate TB reactivation from re-infection. Non-compliance and drug resistance are common reasons for failure of INH prophylaxis; none of these reasons were present in this case.

ETHICS AND LAW: WHEN REFUGEES ARE PERPETRATORS AND VICTIMS. A Moreno, M Grodin, Section of General Internal Medicine; Department of Psychiatry; Boston Medical Center, Boston University School of Medicine; Health Law Department, BU School of Public Health, Boston, MA

LEARNING OBJECTIVES: 1. To recognize symptoms and signs of physical and psychological trauma. 2. To develop a strategy to approach situations when patients have committed crimes against humanity.

CASE PRESENTATION: The patient is a 57 year-old male from Guatemala. The patient came to the clinic seeking routine health care and was found to have symptoms of depression and alcohol abuse. The physician screened the patient for previous physical and psychological trauma because of his nationality and his mental health symptoms. The patient described how his alcoholic father abandoned him during his childhood and how he had to work at a very early age to help his mother support his younger siblings. During adulthood, he decided to become a police officer for two reasons: to gain power and a steady job. The patient denied receiving any special training at the police academy, such as interrogation techniques, but he was chosen to join an elite police unit responsible for investi-

gating political activists. He remembered one of his first assignments: clearing the city's downtown area of homeless people, most of whom had been displaced from rural areas by the war. It was his first opportunity to show his badge and authority. Soon, his superiors ordered him to guard political prisoners while senior officers interrogated them. He witnessed several individuals being tortured. The patient soon found in alcohol a relief for his increasing frustration. After the civil war in Guatemala, he was arrested and tortured on several of occasions for what initially appeared to be disciplinary incidents; however, he was questioned primarily about whether he knew who had ordered the arrest and torture of previous political prisoners. He went into hiding after many of his partners had disappeared or were assassinated. The patient escaped his country after receiving a phone call warning him about a plan to kill him. The review of symptoms was positive for symptoms suggestive of post-traumatic stress disorder (PTSD): flashbacks, intrusive memories, hyper-vigilance, and irritability. His past medical history was negative except for tobacco abuse. His physical exam was normal. The patient declined psychiatric and substance abuse counseling.

DISCUSSION: One study found that up to 10% of all foreign-born patients seen in urban medical centers are survivors of torture and other studies found that 5–35% of all refugees have been tortured. As this patient population continues to increase, practitioners should be prepared to screen for torture sequelae. A more complex problem is when a practitioner phases a patient who was a perpetrator and a victim, raising questions of neutrality during delivery of care, criminal complicity, and safety for other refugee patients.

ETHICS AND LAW: INFORMED CONSENT DURING THE MEDICAL CARE OF ASYLUM APPLICANTS. A. Moreno, M Grodin, Section of General Internal Medicine; Department of Psychiatry; Boston Medical Center, Boston University School of Medicine; Health Law Department, BU School of Public Health, Boston, MA

LEARNING OBJECTIVES: To recognize the ethical and legal dimensions of obtaining voluntary informed consent during the medical care of asylum applicants.

CASE PRESENTATION: The patient is a 28 year-old male asylum applicant from the Democratic Republic of Congo. Because a medico-psychological evaluation was needed as part of an asylum claim, his pro-bono attorney referred him to our refugee/survivor of torture program. The patient was a political activist, who was detained on 2 different occasions and tortured on multiple occasions during both detentions. Aside from symptoms consistent with post-traumatic stress disorder and depression, the patient complained of gingival bleeding for several months. The review of systems was negative. His past medical history was positive for a left forearm fracture and intestinal parasites. He denied any bleeding diathesis, as well as any tobacco, alcohol, or drug abuse. He reported having unprotected heterosexual intercourse and denied any sexually transmitted disease. On physical exam, the patient appeared well nourished. His gums revealed some degree of inflammation and bleeding with minimal blunt trauma. Few small bilateral axillary lymph nodes were noted. Skin and mucous membranes did not reveal any petechiae or ecchymosis. The rest of the physical exam was normal. The white blood cell count was 2.5. The platelet count was 120,000 per mm³ and the hematocrit was 40. While obtaining informed consent for an HIV test, the physician remembered that the diagnosis of some infectious diseases may have an adverse effect on immigration processes. It was decided to delay the test until a legal and an ethical consultation could be obtained. A moral dilemma became apparent as testing for HIV was in the patient's best interest, but positive test results could affect the patient's asylum claim, therefore, harming the patient. The consultants clarified that while some communicable diseases and welfare benefits bar individuals from gaining permanent residency, these restrictions do not apply to asylum claims. Nevertheless, if the patient is granted asylum and decides to apply for permanent residency in the future, he will need a waiver for this medical condition.

DISCUSSION: The elements and the goal of informed consent are uniform for all patient populations. However, the process of obtaining informed consent is a dynamic one that varies from individual to individual. When caring for asylum applicants, clinicians must pay special attention not only to the medical facts, but also to any legal and ethical ramifications. Healthcare providers, particularly social workers, should also pay special attention when this patient population is referred for welfare benefits.

NECROTIZING VASCULITIS WITH PREDOMINANT HEPATOSPLENIC INVOLVEMENT PRESENTING AS FEVER OF UNKNOWN ORIGIN (FUO) IN AN ADOLESCENT MALE. H. Moturi, AA Tulskey, R Erlich, A Selvaggio, N Rao, Internal Medicine, UPMC-Shadyside Hospital, Pittsburgh, PA

LEARNING OBJECTIVES: This is an example of systemic vasculitis presenting as FUO. Predominant hepatosplenic involvement, young age, lack of classic features of polyarteritis nodosa or microscopic polyangiitis with negative p-ANCA & c-ANCA, and focal granulomatous response, are unique features of this case.

CASE PRESENTATION: Patient is a 16 year old, Afro-American male with past history of seizures and without any history of recent travel, presented with recurrent fevers up to 104°F for five weeks. With a 15 lb. weight loss associated with myalgias, arthralgias generalized weakness, burning sensation in legs progressing to numbness and increasing frequency of seizures. Workup done two weeks after initiation of symptoms including cultures, ANA, RF, ACE level, sickle cell prep, monospot test and PPD was negative. Exam was remarkable for BP-150/95, temp -39.5 degree C, tachycardia, cervical micro-adenopathy, and absent deep tendon reflexes. No objective signs of neuropathy or organomegaly, and negative cardiopulmonary exam. Labs significant for normocytic-normochromic anemia, normal

WBC & platelets, no eosinophilia. Elevated GGT, mild elevated transaminases, hyponatremia, low complement levels, increased acute phase reactants-ESR, CRP, Ferritin; Cryoglobulins were positive, ANA—1:80. Microscopic hematuria and mild proteinuria were present. Bone marrow aspirate and biopsy were unremarkable. Imaging studies showed mild hepatosplenomegaly with splenic infarcts, and small wedge shaped defect in left kidney suggestive of infarcts. Echocardiography was negative for vegetations. Total body gallium scan was normal except for poor uptake in splenic bed. Patient continued to spike fevers with negative cultures. Splenectomy performed two weeks later revealed necrotizing vasculitis and focal granulomatous response in medium sized vessels with splenic infarcts. Liver biopsy showed prominent perivascular sclerosis, congestion, and chronic portal inflammation. Significant clinical improvement was noted after the patient was started on therapy with steroids and cyclophosphamide.

RENAL DYSFUNCTION AND UVEITIS IN A TRAVEL AGENT: WHAT ARE THE POSSIBILITIES? KA Myers, E Chan, AR Morton, Medicine, Queen's University, Kingston, ON, Canada

LEARNING OBJECTIVES: 1) To recognize the diverse causes of tubulointerstitial nephritis. 2) To demonstrate the utility of literature review in linking two uncommon medical conditions (tubulointerstitial nephritis and uveitis).

CASE PRESENTATION: A 46 year old travel agent developed nausea, anorexia and weight loss upon arrival in New Zealand for a holiday. By the time she returned home several weeks later, she had lost 20 pounds. In the year prior to the onset of her illness, she had traveled to Bonaire, Cuba and Jamaica. Her past medical history was remarkable only for a remote posthysterectomy deep vein thrombosis, and hypothyroidism. She was taking levothyroxine, but she did not take any other prescription or nonprescription medications. The patient's family physician referred her to a general internist for further assessment of her symptoms. Physical examination was normal. CBC showed a mild normocytic anemia with mild eosinophilia. The sedimentation rate was high (59 mm/hr). Creatinine was elevated at 2.2 mg/dL. Urine sediment contained granular but no cellular casts. Twenty-four hour urine collection showed 0.76 g/day protein. Renal ultrasonography was normal. The patient was admitted to hospital for a renal biopsy. The biopsy showed acute tubulointerstitial nephritis (ATIN), which prompted an extensive search for infectious and immunological causes of this condition. Chest films showed no signs of sarcoidosis. Studies for filaria, strongyloides, toxoplasmosis, leptospirosis, and hepatitis were negative. Antinuclear and antineutrophilic cytoplasmic antibodies were negative. While hospitalized for her renal biopsy, the patient complained of a painful, red left eye. Ophthalmological examination showed bilateral anterior uveitis. Review of the literature revealed a rare condition, called tubulointerstitial nephritis with uveitis (TINU) syndrome. Given the exclusion of other causes of ATIN, a diagnosis of TINU was made in our patient. Systemic steroids were started for treatment of ATIN, with topical steroids and atropine used to treat the uveitis. After several months of treatment, the patient's symptoms resolved. She has remained free of relapses since her initial treatment 5 years ago.

DISCUSSION: TINU syndrome was first described in adolescent girls. More recently, it has been diagnosed in the adult population, with the same female preponderance seen in the original pediatric case series. Although over 50 cases have been described in the literature, it is likely that it is underdiagnosed, because many physicians are unfamiliar with the syndrome. We were successful in making the diagnosis only after discovering TINU by linking uveitis and tubulointerstitial nephritis in the medical literature.

TICLOPIDINE INDUCED THROMBOTIC THROMBOCYTOPENIC PURPURA. N Naseer, A Aljaz, Y Agarwal, J Nelson, SJ Peterson, Medicine, Westchester Medical Center, Valhalla, NY

LEARNING OBJECTIVES: TTP is a known, although rare complication of Ticlopidine. We describe a case of TTP secondary to Ticlopidine occurring only two days after starting the drug. We feel great caution should be exercised before the drug is prescribed.

CASE PRESENTATION: A 72-year-old African American female with a past medical history of Hypertension, Coronary artery disease, Cerebrovascular disease was admitted to a community hospital with aphasia and right sided hemiparesis of a few hours duration. CT scan of the head showed left new occipitoparietal infarct. Her medications included Diltiazem, Enalapril, Atrovastatin and Aspirin. On admission her labs. were Hb 10 and Hct 30, Total WBC count 7.9, Platelets 157. Peripheral smear revealed normochromic, normocytic RBCs. Chemistry was essentially within normal limits except for LDH of 700 and CK of 423 attributable to her stroke. PT/PTT were also within normal limits. UA was unremarkable. At this point the patient was started on Ticlopidine. On the third day of admission patient's neurological status was unchanged. But her repeat labs. were remarkable: WBC 6.9, Hb 8.0, Hct 24, Platelet count 58 and LDH increased 1031. The PT/PTT were normal, the fibrinogen was elevated at 591. Peripheral smear showed anisocytosis, polychromasia, with schistocytes in almost all fields. Few platelets were noted with occasional large platelets. The next day her platelet count fell to 38 and her LDH further went up further to 3813. A diagnosis of Ticlopidine induced TTP was made and she was transferred to a tertiary care hospital for plasmapheresis. The platelet count normalized on plasmapheresis.

DISCUSSION: A number of case reports have been published which have linked the development of TTP/HUS with the use of Ticlopidine. Several characteristics have been described included an older age group, severe disease and a poor overall outcome, despite the use of plasmapheresis. The duration of Ticlopidine use had ranged from 2 weeks to 16 weeks before the onset of TTP/HUS. The above

case is unique because of the relatively brief duration of Ticlopidine usage. The mechanism by which Ticlopidine causes TTP remains unclear. Ticlopidine as an anti-platelet drug has been used increasingly in different clinical settings. We recommend more judicious use of the drug.

HEMOBILIA-INDUCED ACUTE PANCREATITIS OCCURRED 7 DAYS AFTER AN ULTRASOUND-GUIDED PERCUTANEOUS LIVER BIOPSY. Y-M Ning, Y-J Lee, M Abu-Mahfouz, RK Vyas, MN Ehrinpreis, Department of Internal Medicine, Wayne State University, Detroit, MI

LEARNING OBJECTIVES: 1) Diagnose hemobilia and hepatic pseudoaneurysm after percutaneous liver biopsy, and 2) recognize hemobilia-induced secondary disorders such as pancreatitis as a rare consequence of the procedure.

CASE PRESENTATION: A 62-year-old woman with a history of hepatitis C had an ultrasound-guided percutaneous liver biopsy for pathological evaluation of the disorder. Approximately 4 hours later, she developed transient hypotension with severe abdominal pain that resolved with fluid resuscitation and meperidine. She was discharged at the same day without further complications and was well until she presented with episodic sharp epigastric pain 7 days after the procedure. 24 hours later, she began to have hematochezia and hematemesis. Physical examination revealed epigastric tenderness, with stable vital signs, no icterus and no hepatosplenomegaly. The initial laboratory results showed strikingly elevated lipase 29,258 U/L and amylase 1,186 U/L. Liver function tests displayed AST 548 U/L, ALT 487 U/L, AP 199 U/L, Bilirubin T 1.7/D 0.9 mg/dl, Albumin 3.3 g/dl, PT 10.8 and PTT 22.9. Upper endoscopy revealed no active bleeding site. Abdominal ultrasound revealed hypochoic materials throughout the lumen of the gallbladder, some of which were in linear strands and others adherent to its wall. Abdominal CT suggested the presence of blood in the gallbladder. Hepatic angiography was therefore performed. A pseudoaneurysm was identified in a small branch of the right hepatic artery and transcatheter embolization occluded the forward flow from the pseudoaneurysm. The patient recovered without further symptoms. At discharge, her lipase and amylase returned to 452 U/L and 62 U/L, respectively, and liver function tests also improved significantly from maximal levels: AST 600 U/L to 504 U/L, ALT 690 U/L to 305 U/L, and Bilirubin T 4.9/D 4.8 mg/dl to T 2.8/D 2.0 mg/dl. In contrast, her Hgb dropped to 9.5 g/dl from the initial 13.2 g/dl.

DISCUSSION: Medline search yielded only one case of hemobilia-induced acute pancreatitis following percutaneous liver biopsy. It is important to be aware of such a potential complication of the procedure, although hemobilia itself has been well documented as an uncommon one. Angiography should be performed to confirm the diagnosis once indicated, especially if there is evidence of recurrent bleeding. These complications prompt further technical improvement to avoid such iatrogenic misfortunes.

MISDIAGNOSIS OF COEXISTANT GIARDIASIS AND COLITIS AS ANOREXIA NERVOSA IN A 17 YEAR OLD WITH WEIGHT LOSS. NG Peter, B Pawel, K Ginsburg, The Children's Hospital of Philadelphia, Philadelphia, Pennsylvania

Learning Objectives: 1. Distinguish organic diseases from anorexia nervosa.

A 17 year old female with a previous diagnosis of anorexia nervosa and *H. pylori* infection presented with nausea, vomiting and abdominal pain which began perimenstrually. She was admitted for dehydration. The patient had a history of being severely underweight. Despite the addition of caloric supplements to her diet she had failed to gain weight. She also had a history of severe perimenstrual abdominal pain and vomiting for 3 years, and had been admitted twice for dehydration. Treatment of *H. pylori* infection diagnosed by endoscopic biopsy had not improved her symptoms. The diagnosis of anorexia nervosa was made and she was assigned a counselor with whom to eat at school. The patient, however, felt strongly that she was too thin and wanted to gain weight. On admission she complained of epigastric and suprapubic abdominal pain which radiated to her back. She denied fever, dysuria, vaginal discharge, and diarrhea, but did note three to four formed stools per day. On physical examination she was cachectic and weak. Her heart rate was 96, temperature 38.2, and BP 129/90. She was orthostatic by blood pressure. She had mild epigastric and suprapubic tenderness. Stool was negative for occult blood. Laboratory studies revealed a WBC 8.5, Hgb 15.5 and ESR 5. UA showed ketones, trace blood and protein. Renal function, electrolytes, LFTs, albumin, amylase and lipase were normal. Corticotropin stimulation test and TFTs were normal. Endoscopic evaluation revealed *Giardia lamblia* as well as colitis consistent with possible inflammatory bowel disease. Anorexia nervosa is an eating disorder affecting primarily young women. It is characterized by a refusal to maintain body weight over minimal normal weight, an intense fear of weight gain although underweight, a disturbance in the way one's body is experienced and in most females amenorrhea. Patients who meet some, but not all criteria are often diagnosed with an atypical Eating Disorder. The patient presented here would have been very unusual in that she had neither a fear of gaining weight nor an abnormal perception of her body weight. Certain medical conditions, such as adrenal insufficiency and inflammatory bowel disease, may mimic anorexia nervosa by causing weight loss and fatigue in the absence of significant other symptoms. *Giardia lamblia* (GL) infection can mimic anorexia nervosa as well. GL is a flagellated protozoan classically thought to cause acute traveler's diarrhea, but also causes a more chronic syndrome of intermittent soft stools, abdominal cramping, weight loss, fatigue, and anorexia. Laboratory studies often reveal anemia or signs of malabsorption, but as in inflammatory bowel disease, may be normal. These patients may have unusual eating habits or signs of depression, but usually do not

exhibit all of the psychological features of anorexia nervosa. When evaluating a patient with weight loss believed to have an eating disorder, a high level of suspicion for an underlying organic cause is warranted even in the absence of significant symptoms or abnormal laboratory findings.

AN ATYPICAL PRESENTATION OF FITZ-HUGH-CURTIS SYNDROME. NG Peter, LR Clark, The Children's Hospital of Philadelphia, Philadelphia, Pennsylvania

Learning Objectives: 1. Recognize Fitz-Hugh-Curtis Syndrome when there is no evidence of salpingitis.

F.C. is a 17-year-old woman who presented with a history of right upper quadrant pain. The pain started one week prior to admission as diffuse abdominal pain accompanied by diarrhea. Two days prior to admission the pain localized in the right upper quadrant and right lower rib. It was pleuritic in nature and exacerbated by movement. The following day she developed a temperature of 101 F. Past medical history was significant for an appendectomy at age nine and a pregnancy terminated ten months ago. She took no medications except acetaminophen. Family history was significant for gallstones. She did not smoke, drink alcohol or use drugs. She reported sexual activity with two partners in her lifetime, but only one for the past year. Physical examination revealed a temperature of 37.9 centigrade, blood pressure 104/61, pulse 87, and respiratory rate 20. Pulse oximetry revealed an oxygen saturation of 100% on room air. Her lung examination revealed decreased breath sounds at the right base and mild crackles. Her cardiac exam revealed a 2/6 systolic murmur. She had tenderness over the right lowest rib anteriorly, the right upper quadrant, and the right back. Pelvic examination was unremarkable except for scant blood from the cervix. Laboratory studies revealed a WBC of 12.6 and an ESR of 73. Further evaluation included a normal right upper quadrant ultrasound, LFTs, amylase, lipase, bone scan, lung perfusion scan, and obstruction series. Blood, urine and stool cultures were negative. Chest radiography and chest computerized tomography revealed a trivial pleural effusion. The diagnosis of postviral pleurodynia was made and viral studies were sent which later were found to be negative. On the fifth hospital day her endocervical culture became positive for *Chlamydia trachomatis*. She was treated with doxycycline with rapid resolution of her symptoms. Fitz-Hugh-Curtis Syndrome (FHCS) is the syndrome of perihepatitis associated with pelvic inflammatory disease originally described by Stajano in 1919 and rediscovered by Fitz-Hugh and Curtis in the 1930s. *Neisseria gonorrhoeae* (GC) was initially thought to be the sole pathogen. Now it is thought that several organisms may play a role and that *Chlamydia trachomatis* is the most common. Patients with FHCS usually present with pleuritic right upper quadrant pain exacerbated by movement. It is often accompanied by fever. The ESR is almost invariably elevated. Most patients have evidence of current or past salpingitis. Occasionally, as in the patient described here, there is no evidence of salpingitis. In these cases establishing the diagnosis is very difficult. The diagnosis can be definitively made only via direct visualization at laparotomy or laparoscopy. In most cases the diagnosis is made clinically, by eliminating other etiologies and by detection of a causative organism. A response to antibiotics directed against GC or *Chlamydia trachomatis* may also support FHCS as the etiology.

DELAYED CONSEQUENCES OF CONGENITAL RUBELLA SYNDROME. ME Pickett, Medicine, Brigham & Women's Hospital, Boston, MA

LEARNING OBJECTIVES: 1) Recognize features of congenital rubella syndrome that may present in adulthood.

2) Review rubella disease incidence in recent years and recommendations for immunization

CASE: A 33 year old Puerto Rican woman presented with progressive throat pain for a week and vomiting and fatigue for two days. She had been diagnosed at age 31 with insulin dependent DM and had a history of ketoacidosis. She also had a history of an elevated TSH, for which she took a low dose of thyroxine. She reported the onset of fatigue with anterior throat pain that became so severe she was unable to swallow food. Despite her poor oral intake and good compliance with her insulin regimen, her glucose levels were increased. Past medical history was notable for bilateral sensorineural hearing loss which occurred at age 2, resulting in a significant speech disability. She had a history of a patent ductus arteriosus which required ligation. Family history was notable for a febrile illness in her mother occurring during this patient's gestation.

Physical examination was notable for an ill appearance and borderline tachycardia. She was slightly tachypneic. Skin and mucous membranes were dry. Her neck was tender anteriorly but had no lymphadenopathy. She had no pharyngeal erythema. Laboratories were notable for evidence of ketoacidosis with an anion gap of 23 and elevated β -hydroxybutyrate level. Glucose was 533. A TSH level was depressed at 0.41 despite having recently been normal with repletion therapy. Thyroid peroxidase antibodies were detected at a high concentration of 307 IU/ml. She was admitted for treatment of diabetic ketoacidosis and adjustment of her thyroid repletion therapy in the setting of a possible thyroiditis flare. Her throat pain subsided gradually over several weeks.

DISCUSSION: A review of this patient's gestational history confirmed that the in-utero infection was rubella. Congenital rubella syndrome (CRS) has been associated with the delayed onset of autoimmune polyendocrine failure, commonly resulting in type I diabetes mellitus and autoimmune thyroiditis. This pathology has been attributed to abnormal stimulation of the immune system during its early development. She exhibits many other features of CRS, including skeletal abnormalities, congenital heart disease, and hearing impairment which typically has its

onset at toddler age. The MMR vaccine has greatly reduced the incidence of rubella infection and CRS in the United States. Nonetheless, cases continue to occur and 122 cases of CRS were reported to a national registry between 1985 and 1996. Of these cases, it is notable that less than half could be explained by foreign exposure to rubella. It is also notable that among indigenous cases, a large majority had had previous pregnancies with missed postpartum opportunities for vaccination. 44% of CRS cases in the United States occurs in our Hispanic population.

MASSIVE PERICARDIAL EFFUSION IN A PATIENT WITH ANKYLOSING SPONDYLITIS. J Potter, S Frost, Internal Medicine, The Cleveland Clinic Foundation, Cleveland, OH

LEARNING OBJECTIVES: Recognize that pericardial disease may be associated with ankylosing spondylitis (AS).

CASE PRESENTATION: A 79 yo black man with AS was admitted with four days of progressive dyspnea on exertion and lower extremity edema. He denied cough, chest pain, orthopnea, paroxysmal nocturnal dyspnea, fever, chills, night sweats, weight loss, joint pain, pelvic pain, or change in his chronic lower back discomfort. Medical history was remarkable for pulmonary tuberculosis treated twenty years ago and stage I prostatic adenocarcinoma treated with radiation one year ago. He was not on any medications commonly associated with pericarditis. Physical exam demonstrated T = 36.3C, P = 140, BP = 90/62, RR = 26, room air pulse oximetry of 91%, pulsus paradoxus = 18, JVD, distant heart sounds with pericardial friction rub, egophony beneath the left scapula, scant bibasilar rales, 2+ lower extremity edema distal to the knees, limited cervical and lumbar range of motion, and no spinal or sacroiliac tenderness. Labs demonstrated normal chemistry, creatinine = 1.2, BUN = 25, AST = 59, alkaline phosphatase = 302, WBC = 8.6, HCT = 32.7, ESR = 105, TSH = 0.75, and PSA = 2.8. EKG showed atrial fibrillation with rapid ventricular response. CXR demonstrated markedly enlarged and globular appearing cardiac silhouette, widened upper mediastinum, and ankylosis of the thoracic spine. Echocardiogram revealed ejection fraction of 55%, large pericardial effusion, and significant inflow variation across the tricuspid and mitral valves suggestive of tamponade. After a failed pericardiocentesis, the patient underwent successful surgery for pericardial fluid drainage. Pericardial biopsy revealed only granular inflammation, and cultures were negative for bacteria, mycobacteria, and fungus. Pericardial fluid cultures were also negative, and cytology revealed no malignancy. Blood cultures for bacteria, mycobacteria, and fungus were unrevealing. Chest CT scan showed no mass, lymphadenopathy, or aortic pathology.

DISCUSSION: Although aortic valve and conduction system pathology is most commonly associated with AS, pericardial involvement has been reported. Clinical pericarditis is infrequent, but echocardiographic surveys have demonstrated asymptomatic pericardial effusions in up to 42% of patients with AS. Such effusions are rarely large, but cardiac tamponade has been reported. Other signs of active ankylosing spondylitis (i.e., sacroilitis, and uveitis) are often lacking. The inability to exclude all potential causes of pericarditis in this patient is acknowledged; however, the possibility that ankylosing spondylitis is associated is intriguing and underscores the importance of recognizing the association between cardiac pathology and connective tissue diseases.

AN UNUSUAL CAUSE OF ABDOMINAL PAIN. GA Prasad, FR Shaikh, NM Guda, MA Gennis, Internal Medicine, University of Wisconsin Medical School, Sinai Samaritan Medical Center, Milwaukee, WI

LEARNING OBJECTIVES: 1. Recognize that Sarcoidosis can present with isolated retroperitoneal lymphadenopathy without extra-abdominal disease. 2. Recognize Sarcoidosis as one of the unusual causes of abdominal pain.

CASE PRESENTATION: A 23-year-old African-American male presented with a seven day history of fever, abdominal pain and vomiting. On examination, he had a temperature of 102.3 F with normal pulse and blood pressure. There was no peripheral lymphadenopathy, or skin rash. The physical examination was normal except for tenderness in the right upper quadrant. The liver edge was felt 2 centimeters below the right costal margin. Initial laboratory tests included a normal CBC electrolytes and creatinine. Liver function tests revealed elevated serum alkaline phosphatase (382 U/L, normal <126 U/L) with normal transaminases. Blood and urine cultures were negative. A Chest X-Ray done on admission was normal. A CT scan of the abdomen showed significant retroperitoneal lymphadenopathy. Bone marrow aspirate and biopsy revealed normal cellularity and morphology. Chromosomal studies and lymphocyte flowcytometry did not reveal any evidence of a lymphoproliferative disorder. Serum alpha fetoprotein and beta human chorionic gonadotrophin levels were normal. The patient was anergic to PPD and control antigens. A serum ACE level was significantly elevated (244 U/L, normal <52 U/L). Laparotomy with biopsy of the liver and retroperitoneal lymph nodes revealed multiple non-caseating granulomas. Stains for acid-fast bacilli and fungal elements were negative. A diagnosis of Sarcoidosis was made. The patient was then placed on oral steroids which led to a prompt resolution of fever and other constitutional symptoms.

DISCUSSION: Sarcoidosis is a multisystem granulomatous disorder, usually affecting young adults and presenting with mediastinal/hilar lymphadenopathy, and a variety of pulmonary, cutaneous and ocular manifestations. Abdominal manifestations are less frequently diagnosed, as patients with sarcoidosis do not present with abdominal symptomatology. Intra-abdominal sarcoidosis is known to involve the liver, spleen, gastrointestinal tract, kidneys and lymph nodes. All reported cases of abdominal involvement in sarcoidosis are in patients with pulmonary manifestations. Isolated retroperitoneal lymph node involvement in sarcoido-

sis is rare, especially as a presenting manifestation. Our patient presented with fever and abdominal pain, with a normal chest X-ray. Review of the literature revealed only two reported cases with a similar presentation. Both patients however did have extraabdominal manifestations. Initial differentials in our patient included lymphoproliferative disorders, primary retroperitoneal malignancies and tuberculosis. This case is unique in that the disease was localized entirely to the abdomen.

PERIPARTUM CARDIOMYOPATHY IN A 25 YEAR OLD HISPANIC FEMALE. GD Price, D Messinger, J Lee, Medicine, Westchester Medical Center, Valhalla, NY; School of Medicine, New York Medical College, Valhalla, NY

LEARNING OBJECTIVES: 1. Recognize peripartum cardiomyopathy as a potential cause of heart failure in peripartum women. 2. Recognize that treatment of peripartum cardiomyopathy is similar to that of other etiologies of systolic left ventricular dysfunction.

CASE PRESENTATION: A 25 year old hispanic female was admitted to the hospital 10 weeks after a normal spontaneous vaginal delivery of her second child for an elective bilateral tubal ligation. Her two pregnancies had not been complicated in any way and the patient elected to have spinal anesthesia. In the operating room, after the induction of anesthesia, her blood pressure decreased to 100/40 with a pulse of 40. Ephedrine, 5 mg, was administered, producing ventricular bigeminy and the procedure was cancelled. An electrocardiogram revealed sinus tachycardia and ST-segment depression. Peak troponin-I was 21.4 at 0200 on 09/1/1999 (hospital day 2) which decreased to 10.4 by 1400 the same day. A cardiac catheterization revealed a diffusely hypokinetic left ventricle, and an ejection fraction of 30%. A MUGA scan quantitated the ejection fraction at 34%. Prior to admission the patient did not have any exercise intolerance of any kind and her only medical problem consisted of a microcytic anemia which was being treated with supplemental iron. The patient now had shortness of breath even with mild exertion. The patient was discharged home on hospital day 5 on vasotec 2.5 mg BID, lasix 20 mg QD, PRN, and digoxin 0.25 mg QD. Oral anticoagulation was not warranted. In subsequent follow up visits the patient's pharmacotherapy was advanced to include low dose beta blockade and vasotec 10 mg BID. A repeat echocardiogram in mid-November 1999 demonstrated normal left ventricular function with only minimal mitral regurgitation and her exercise intolerance has resolved.

DISCUSSION: Peripartum cardiomyopathy is a rare disorder that may present at any time during pregnancy and up to the fifth or sixth postpartum month. The diagnostic criteria are cardiac failure in the last month of pregnancy or within five months of delivery, absence of an identifiable cause, and absence of demonstrable heart disease before the last month of pregnancy. However the diagnosis should be considered in patients who develop systolic dysfunction of the left ventricle at any time during pregnancy. Once diagnosed, therapy is similar to that of other causes of systolic heart failure, with some modifications if the diagnosis is made during pregnancy. Recent studies indicate that 50% of patients have marked improvement and recovery usually occurs within six months of diagnosis. Those patients diagnosed more than one month postpartum may have a worse prognosis.

BILATERAL PULMONARY ARTERY EMBOLISM IN A 52 YEAR OLD MALE. GD Price, R Gunther, D Messinger, Medicine, Westchester Medical Center, Valhalla, NY

LEARNING OBJECTIVES: 1. Recognize clinical environment suggestive of pulmonary embolism.

CASE PRESENTATION: A 52 year old male was brought to the emergency room from a local correctional facility with a three to four day history of shortness of breath on exertion and severe intermittent substernal chest pain, diaphoresis, nausea, and tingling in the fingers of his left hand. The symptoms would last from a few minutes to as long as an hour. On the day of admission, his chest pain was relieved with one sublingual nitroglycerin tablet. His previous medical history was significant only for a repaired inguinal hernia. He had a 30+ pack year history of smoking and had been incarcerated for the past 5.5 months. His vital signs were temperature 98.1 F, pulse 128, blood pressure 128/60 and respirations 20. His physical exam revealed distant heart sounds, decreased dorsalis pedis pulses, and cool lower extremities. Laboratory examination was unremarkable. His initial electrocardiogram revealed sinus tachycardia at 119 beats per minute, left axis deviation, Q-waves in leads II and III and decreased ST segments in the anterior leads. His initial cardiac enzymes were CPK 53, CPK-MB 2.4, and Troponin I <0.3. A room air arterial blood gas (ABG) obtained later demonstrated 7.44/32/83/22/97% (A-a gradient of 27). On the second hospital day, the patient complained of chest pain and shortness of breath exacerbated by any movement. The patient was placed on oxygen by nasal cannula, given 325 mg of aspirin orally and 110 mg of enoxaparin SQ. A stat ABG showed 7.44/30/66/94% on 2 L/min. A spiral CT scan revealed a large amount of thrombus in both the right and left main pulmonary arteries extending into the lower branches of the pulmonary vasculature. The patient was started on intravenous heparin and transferred to the intensive care unit where tPA was administered secondary to hemodynamic instability. A hypercoagulable work up was negative, as were a duplex scan of the lower extremities and CT scans of the chest, abdomen, and pelvis. Transthoracic echocardiography revealed no thrombus or valvular lesion.

DISCUSSION: This case illustrates a life-threatening situation caused by bilateral pulmonary emboli. The admitting ABG was relatively unhelpful. High index of suspicion remains critical in the diagnosis of pulmonary emboli. This patient's risk factors include cigarette smoking and decreased activity due to incarceration.

NEUTROPENIA AS AN INITIAL PRESENTATION OF SLE IN A YOUNG HISPANIC MALE. II

Rafique, V.Dedhia, Montefiore Medical Center, Bronx, NY

LEARNING OBJECTIVES: 1) Ability to diagnose SLE in male patients with varying clinical manifestations.

2) Recognize differences in morbidity & mortality of SLE in male patients.

CASE PRESENTATION: Mr. A.R. is a 28 year old Hispanic male with PMH significant for NIDDM and HTN for two yrs, and Asthma who was in his USOH until one month ago when he developed persistent dry cough. A week PTA he also developed mild abdominal pain with associated with nausea and self-limiting diarrhea. One day PTA patient developed SOB, non-productive cough and fevers to 101. He was admitted to Montefiore for further evaluation of possible viral syndrome.

PSH: He is married with three children and monogamous. No tob, IV drug or ETOH use.

Meds: 1) Cozaar 50 mg po QD; 2) HCTZ 25 mg po QD; 3) Glucophage 500mg po QD; 4) Proventil MDI

NKDA

P.E. BP = 140/100 PR = 120bpm T = 101.6 RR = 28/min

Patient is an obese hispanic male, anxious yet in NAD

pertinent positives otherwise normal exam

Neck: 1cm B/L tender cervical nodes

Abdomen: obese, mild tenderness in upper quadrants

Labs:

WBC = 1.6 Hgb/HCT = 11/35 Plts = 86 Retic count = 0.4%

PT = 9.9 PTT = 26 BUN/Cr = 35/1.9 LDH = 371 Alb = 2.8

SGOT/SGPT = 11/68 D/T Bili = 0.2/0.4

Ferritin = 1350 ESR = 5

UA - ph5.0/1.022sg/3+prot/glu - nit - /mod occult blood/

wbc 40-50/rbc 3-5/bac some

Hepatitis serology negative, Parvovirus IgG negative,

Monospot negative,

Bone Marrow-Hypercellular Marrow with absolute granulocytic hyperplasia

no tumors, no granulomas, negative for AFB.

CXR-NAI CT Abdomen-normal

HIV Test was negative SPEP-neg

Patient was discharged with unexplained etiology for pancytopenia when his PMD noted the multi-systemic and inflammatory nature of clinical manifestation in a young male and decided to pursue an autoimmune process. In fact his ANA was 1:640 and Anti-dna >500, with hypocomplementemia. On further workup patient noted to be Nephrotic 10 gm/24 hours and Renal Biopsy confirmed Diffuse Proliferative Glomerulonephritis (DPGN) Stage IV. The patient has been doing well on treatment with Cytoxan and steroids. He remains physically active and enjoys his family life.

DISCUSSION: Despite gender predilection for women, young males with lupus are at increased risk of renal involvement and thromboembolic complications. They present less often with features such as arthritis, neuropsychiatric manifestations, skin lesions that are common in women. Therefore, the presentation may be obscure without clear defining criteria and often difficult to diagnose. However leukopenia are predictors of disease severity in males with Lupus. Most often, the primary renal manifestation is DPGN Type IV which portends higher degree of progression to ESRD. In addition thromboembolic complications were noted in higher frequency related to persistently elevated anti-cardiolipin. Therefore the ability to recognize sex-linked differences in clinical presentation and disease progression is paramount in males afflicted with Systemic Lupus.

RARE CAUSE OF ABDOMINAL WALL ABSCESS. A.Rajasekharaiah, R Hobbs, RJ Kiel, Internal Medicine, Oakwood Hospital, Dearborn, MI

LEARNING OBJECTIVES: To recognize that laparoscopic cholecystectomy for cholelithiasis does not eliminate gall stones as a cause for subsequent abdominal infection.

CASE PRESENTATION: Our patient was a 78 year old white female with a past medical history of hypertension, arthritis, diverticulosis, and peptic ulcer disease. She was admitted with a seven day history of swelling, pain and redness over the right flank. Her past surgeries included a laparoscopic cholecystectomy (LC) 5 months PTA for symptomatic cholelithiasis and appendectomy at age 17. On examination she had a temperature of 103°F. Local examination of the abdominal wall revealed an erythematous, hot, tender, indurated, non-mobile, non-fluctuant mass measuring 10 x 14 inches in the right lumbar area. Her WBC count was 14,100 cells/mm³, with a differential of 32% bands. A CT scan of the abdomen and pelvis revealed an 8 x 6 x 6 cm multi-loculated fluid collection with pockets of air in the lateral aspect of the abdominal wall at the level of the 12th rib. A large quantity of purulent material was drained from this area. During irrigation of the wound, a marble sized gall stone was removed. Wound cultures grew alpha-hemolytic streptococci and klebsiella. She was treated with clindamycin and levofloxacin and was later discharged home on the same antibiotics.

Review of her old records showed that at the time of the LC the gallbladder wall was accidentally punctured and several stones spilled. Despite irrigation, only one stone was retrieved at that time.

DISCUSSION: Cholelithiasis is a common disorder affecting 20 million individual in the US. LC has now become the preferred method of treatment of symptomatic patients.

The incidence of spillage of bile and stones is higher in LC. Removing all of the spilled stones can be difficult because of the restricted view and access. There are many anecdotal reports of complications secondary to spilled stones including ab-

dominal wall abscess 2-15 months post-operatively, adhesions, umbilical sinus tract formation, erosion into sigmoid colon, cholelithoptysis and others. A review of literature emphasizes the following points: (1) The time interval between laparoscopic intervention and clinical presentation of complications can be quite long; (2) Abscess may present at different locations unrelated to the gall bladder fossa; (3) The small percentage of intra abdominal abscess following spilled stones does not mandate conversion of LC to open cholecystectomy simply to retrieve gall stones; (4) Gall stones may not be seen on CT because of their small size and also because they may be isodense with the surrounding fluid. Ultrasound may be more useful for the demonstration of migrated stones especially when they are small and not calcified.

METRONIDAZOLE RESISTANT TRICHOMONAS VAGINALIS. S.Ramya, SA Gilroy, Department of Medicine, SUNY Health Science Center, Syracuse, NY

LEARNING OBJECTIVES: The prevalence of Trichomonas varies from 13 to 23% in women attending gynecologic clinics. It is sexually transmitted and can be found in the presence of other sexually transmitted diseases. The standard recommended treatment for Trichomonas has been Metronidazole. The cure rate is 90% with 2 gm single dose or 250 mg tid for 7 days. Treatment of male sexual partners is indicated. Estimates of the prevalence of infection due to metronidazole resistant Trichomonas range from 1 in 2000 to 3000 cases. 1. Physicians should be aware of the emergence of resistant strains of Trichomonas Vaginalis infection. 2. Use in-pouch TV culture methods for diagnosis and invitro metronidazole susceptibility testing in recalcitrant cases. 3. Treat the infection with available alternate regimens.

CASE PRESENTATION: A 20 yr. old Hispanic female presented to STD clinic with intermittent vaginal discharge since April 98. She had recurrent Trichomonas infections not amenable to treatment by metronidazole.

Treatments received-

a) Metronidazole 2 gm once, 500 mg tid x 7 days, 500 mg tid x 14 days, 500 mg tid x 28 days, metrogel for 1 week for Trichomonas

b) Boric acid capsules for a week for Trichomonas

c) Metronidazole for Bacterial vaginosis on 2 occasions

She was compliant in taking the medicines and she has been sexually abstinent from November 98. The infection was detected by the saline prep of the vaginal secretions.

During her August 99 visit to STD clinic she still had foul smelling vaginal discharge and itching. She has had one male partner, does not smoke and drinks alcohol occasionally. She is G2P1Ab1.

Significant findings-

General physical examination - normal, Vagina - red with thick white frothy discharge, Cervix - red with no overt cervical discharge, Bimanual examination - normal, Ph-5.0, whiff negative, few clue cells present, Saline prep showed motile Trichomonas, KOH prep - negative.

Lab results-

CBC - normal limits, Sed rate-1, Glucose-92, TFT- normal, PPD - negative, HIV and RPR tests - negative, Genprobe for Gonococcal and chlamydial - negative, In-pouch TV culture - Trichomonas, Sensitivity studies - Resistance to metronidazole. We treated her with paramomycin. Reculture done two weeks after treatment, was negative for Trichomonas.

DISCUSSION: Laboratory methods for identification of Trichomonas include—In-Pouch TV culture, Papanicolaou smear, DNA probe, PCR and traditional culture. Invitro metronidazole susceptibility test for Trichomonas was done using the In-Pouch TV culture. The InPouch TV culture system is reported to be very sensitive, cheap and practical.

Paramomycin is usually used to treat intestinal ameba, tapeworm infections and cryptosporidiosis. The cure rate is 85%, when administered topically as a 250 mg vaginal pessary for 14 days. Side effects of the treatment include erosions of the vulva and vagina. These are managed with saline soaks.

TUMORAL CALCINOSIS: A CLINICAL SCENARIO. PK.Rao, VV Madhav, A Bedi, PH Mehta, Department of Medicine, New York Methodist Hospital, Brooklyn, NY

LEARNING OBJECTIVES: 1. To recognize that tumoral calcinosis is an entity characterized by dystrophic soft tissue calcification of unknown etiology that occurs infrequently in hemodialysis patients. 2. Careful monitoring of serum phosphate levels and serum calcium-phosphorus product is essential in patients undergoing hemodialysis to prevent ectopic calcifications.

CASE PRESENTATION: A 61 year old white male on chronic hemodialysis for end stage renal disease, secondary to hypertensive nephrosclerosis, presented with a painful left hip mass of three weeks duration that progressively increased. There was no history of trauma to the hip. The patient was an active smoker and there was no significant family history. Pertinent clinical findings included a temperature of 101°F, bibasilar rales, a left hip mass about 14 x 16 cms which was warm, tender and non-pulsatile. Relevant lab data included a WBC count of 9,500/mm³, serum phosphate of 10.8 mg/dl, calcium of 8.5 mg/dl and albumin of 3.5 mg/dl. X-ray of the left hip revealed an extensive soft tissue calcification. Three phase total body bone and gallium scan showed abnormal intense uptake in the soft tissues of the left and right greater trochanter, proximal humeri, left ninth rib, symphysis menti and the right pubic bone indicating an active metabolic process (increased uptake of gallium) rather than an inactive calcified tumor or a mass. CT scan of the abdomen and pelvis showed dense calcifications involving muscle groups of buttocks, hips and thighs. Parathyroid scan was negative for an ade-

noma. Aspiration of the left hip mass revealed cheesy material resembling pus. A gram stain did not reveal any organisms or WBC. It was also negative for AFB and malignant cells. Crystalline, non-birefringent calcium deposits were noted. Excision biopsy of the mass showed prominent blood vessels and no evidence of malignancy. Correlation of imaging results and clinico-pathological findings reveal that this patient has tumoral calcinosis. A unique feature of this case is the first time description of dystrophic calcification using both gallium and bone scanning. **DISCUSSION:** Tumoral calcinosis is an ectopic calcification syndrome of uncertain etiology manifested clinically as slow growing periarticular soft tissue calcifying masses. Pathology is characterized by fibrous walled cystic spaces containing structureless calcific debris and variable inflammatory reaction. Supersaturation of calcium and phosphorus in the extracellular fluid is the most important factor and more than 60 (mg/dl)² in the calcium-phosphorus product is usually necessary for tissue calcification. Treatment includes discontinuation of calcium and vitamin D supplements, use of low calcium dialysate and treatment of established aluminium intoxication.

ARCANOBACTERIUM HEMOLYTICUM AND FUSOBACTERIUM NECROPHORUM: AN UNUSUAL CAUSE OF SEVERE SEPSIS AND ARDS. A. Rashid, K Chathargy, D Dugaescu, L Delorenzo, Internal Medicine, New York Medical College, Valhalla, NY

LEARNING OBJECTIVES: a. Recognize serious systemic manifestations of Arcanobacterium hemolyticum and Fusobacterium necrophorum. b. Recognize the importance of inhaled Nitric Oxide as treatment for ARDS (Adult Respiratory Distress Syndrome).

CASE PRESENTATION: A 19 year old female presented with sepsis manifested by fever, respiratory distress, acute renal failure and severe metabolic acidosis, one week after a sore throat. Initial laboratory work-up revealed anemia, thrombocytopenia, leukocytosis and abnormal liver function tests. The patient was started on broad spectrum antibiotics and placed on ventilatory support. Blood cultures grew Arcanobacterium hemolyticum and Fusobacterium necrophorum. Computed tomography of the neck was negative for septic thrombi. She developed Adult Respiratory Distress Syndrome with barotrauma resulting in multiple pneumothoraces. Despite the use of Pressure Control/Inverse Ratio ventilation gas exchange remained poor (PO₂/FiO₂ ratio 51). Therapy with inhaled Nitric Oxide was begun (20 ppm) with prompt improvement in oxygenation (PO₂/FiO₂ ratio 114). NO was discontinued after eight days. During her hospital course she developed several nosocomial infections including Vancomycin Resistant Enterococcus and Oxacillin Resistant Staph Aureus. Her condition progressively improved and she was discharged on day 53.

DISCUSSION: Arcanobacterium hemolyticum, a G positive aerobic rod, is a common cause of pharyngitis in the 10-30 year age group. It has rarely been associated with peri-tonsillar abscesses, sepsis, endocarditis and meningitis. Fusobacterium necrophorum, a G negative anaerobe, is a commensal of the alimentary canal which usually causes exudative pharyngitis. Cases of septicemia secondary to suppurative thrombophlebitis of internal jugular vein and complicated by multiple metastatic infection called Lemierre's disease have been reported with this organism. Clinical susceptibility testing is not routinely performed for these organisms. Published reports have shown Arcanobacterium to be sensitive to Erythromycin, Clindamycin, Chloramphenicol and Tetracycline; and Fusobacterium to Penicillin, Clindamycin and Ceftriaxone. In our case, inhaled Nitric Oxide, a selective pulmonary artery vasodilator improved oxygenation. We believe this was directly associated with the beginning of her overall clinical improvement.

DYSPHAGIA IN THE ELDERLY-NOT ALWAYS CANCER. A. Ravikanth, General Internal Medicine, University of Illinois at Chicago, Chicago, IL

LEARNING OBJECTIVES: 1) Distinguish achalasia from other causes of dysphagia. 2) Diagnosis and management options for achalasia.

CASE PRESENTATION: A 71 year old white female was admitted with increasing difficulty swallowing over the past five years. The dysphagia was worse for solids than for liquids. It was associated with postprandial pyrosis and lately with regurgitation of food into the mouth. Review of systems revealed a weight loss of 15-18 lbs in the last year.

Physical exam was unremarkable. Chest X-ray was done in the ER which revealed an upper mediastinal mass with an air-fluid level. A gastrografin esophagogram showed a dilated tortuous esophagus with multiple filling defects. There was no passage of dye into the stomach even after 45 minutes, consistent with achalasia. After nasogastric suction she underwent an Esophagogastroduodenoscopy (EGD) with balloon dilatation of the Gastroesophageal junction to 20 mm. Five days later she had another EGD with biopsy and injection of botulinum toxin. Biopsy was consistent with chronic inflammation. Her symptoms improved initially but she returned five months later with recurrence of dysphagia. She refused surgery on both occasions.

DISCUSSION: This case illustrates the following: 1) All elderly patients with dysphagia do not have cancer. The age specific incidence of achalasia climbs progressively with advancing years and about 15% of patients with achalasia present at >60 years age. 2) Dysphagia, regurgitation, weight loss are the commonest presenting symptoms. 3) Barium esophagogram is a simple and useful initial diagnostic test in a patient with dysphagia, before endoscopy.

THE FIVE LITER BLADDER: A CASE OF OBSTRUCTIVE UROPATHY CAUSING EDEMA. LE Regan, EF Yee, Medicine, UCLA/San Fernando Valley Program, Los Angeles, CA

LEARNING OBJECTIVES: 1. Recognize obstructive uropathy as an unusual cause of lower extremity edema 2. Discuss the workup of lower extremity edema.

CASE PRESENTATION: A 75 year old male presented to an outpatient clinic with bilateral lower extremity edema. Three years ago, he was diagnosed with lymphedema after a motor vehicle accident damaged his right leg resulting in swelling. Over the past year he experienced new onset left lower extremity edema as well as worsening right lower extremity edema. He had no chest pain, shortness of breath, orthopnea, or urinary complaints. His past medical history was significant for hypertension and osteoarthritis and total left hip replacement six years ago. Medications included hydrochlorothiazide/triamterene and vitamin supplements. Vital signs: BP 160/74, HR 77, RR 18, height 67 cm, weight 254.3 pounds. The patient was comfortable and appeared well-nourished. No JVD was present. Cardiopulmonary exam was unremarkable. Abdomen was obese, nontender, and nondistended. Prostate was symmetrical, nonenlarged, with no nodules. The lower extremities had pitting edema to the knees, 3+ on the left and 4+ on the right and his pulses were nonpalpable. No venous stasis changes, warmth, or erythema was noted on exam. The patient was given compression stockings. Over the course of four months, he underwent a workup for his edema including a bilateral venous doppler ultrasound (negative); electrolytes, BUN, Cr, TSH, and albumin (all normal); urinalysis (negative for protein); and echocardiogram (ejection fraction 69%). A CT scan of the abdomen and pelvis was obtained and showed a markedly distended bladder completely obliterating the inferior vena cava from the level of L4 to L5/S1. No evidence of hydronephrosis was seen. The patient was diagnosed with obstructive uropathy secondary to benign prostatic hypertrophy. A foley catheter was placed and five liters of urine was drained from the bladder.

DISCUSSION: Localized edema is usually indicative of venous or lymphatic obstruction. This patient developed bilateral lower extremity edema and was worked up for possible cardiac, renal, endocrine, hepatic, or nutritional disorders. His history, physical exam findings and labs did not support any of these diagnoses. While obstructive uropathy is a very uncommon cause of IVC compression and edema, benign prostatic hypertrophy is a very common clinical diagnosis. It is important to take a full history of urinary symptoms on older males. Though rectal exam is important for initial diagnosis of this condition, prostate size on digital rectal exam has not been shown to correlate with obstructive symptoms. This case is a rare example where physical compensation occurred so the patient experienced little or no urinary symptoms. It is important to maintain a high index of suspicion for obstructive uropathy in the older male population.

ACUTE SARCOIDOSIS PRESENTING AS BLURRY VISION, 7TH NERVE PALS, AND RASH. JJ Rencic, Internal Medicine, Hospital of University of Pennsylvania, Philadelphia, PA

LEARNING OBJECTIVES: 1) Recognize the clinical presentation of acute sarcoidosis. 2) Recognize and manage acute uveitis.

CASE: A 29 yo female presented to the emergency room with one month of blurry vision, a tender lower extremity rash, and new right facial droop. She had noted malaise without fevers and a ten-pound weight loss over the last month. ROS was otherwise negative. She had no significant medical or exposure history and was not taking any prescription or herbal medications. On physical exam the patient was afebrile with normal vital signs. Visual acuity of the right and left eye was 20/25 and 20/400 respectively, and the left eye was injected. Her right fundus was normal and the left fundus was not visualized. She had a right peripheral 7th nerve palsy. 1-2 cm erythematous, tender nodules were noted on her anterior shins. The remainder of her exam was normal. A chest x-ray showed large, bilateral hilar adenopathy. A panel 7, calcium, liver function tests, and CBC were within normal limits, and the RPR was negative. TB, syphilis, Behcet's, and lymphoma were felt to be unlikely given the history and other findings. The clinical diagnosis of acute sarcoidosis was made. A slit-lamp examination demonstrated bilateral anterior uveitis. The patient was discharged on oral prednisone and steroid eye drops.

DISCUSSION: Internists are familiar with the insidious form of sarcoidosis; however, 20-40% of patients present with subacute or acute manifestations. Two syndromes have been recognized in the acute presentation: Loeffgren's syndrome, which presents as erythema nodosum, arthritis, and bilateral hilar adenopathy, and Heerfordt-Waldenstrom syndrome, which presents as fever, parotid enlargement, anterior uveitis, and facial nerve palsy. The prognosis for acute presentations of sarcoidosis is very good, with only 15-20% of patients relapsing or progressing. Poor prognostic features are African-American race, age over 40, symptoms greater than 6 months, and lack of erythema nodosum. Sarcoidosis is considered a diagnosis of exclusion and virtually all patients require biopsy with historically appropriate lab testing to rule out other causes. This patient's finding of anterior uveitis is an ophthalmologic urgency. It presents as redness, pain, blurry vision, photophobia, with a small pupil, injection, and decreased visual acuity on exam. Visualization of inflammatory cells in the anterior chamber on slit-lamp examination confirms the diagnosis. Synechiae can form and lead to permanent visual impairment. The differential diagnosis includes sarcoidosis, Behcet's, Reiter's syndrome, IBD, psoriasis, and infections like herpes, syphilis, Lyme, and tuberculosis. Patients are treated with topical steroid therapy and pupillary dilation to decrease the risk of synechial formation. A careful search for an underlying cause is necessary.

JAUNDICE: FROM LITHIASIS TO CARCINOMA. CA Rettaly, Medicine, Louis A. Weiss Memorial Hospital, Chicago, IL

LEARNING OBJECTIVES: 1. Recognize features in the history of a jaundiced patient, indicative of hepatolithiasis. 2. Diagnose malignancies in patients with jaundice.

CASE PRESENTATION: A 71-year-old Russian woman presented with a chief complaint of itching for one week as well as right upper quadrant pain and icterus for 3 days. She had been having mild epigastric, post-prandial pain intermittently for almost two years thought to be related to gastroparesis and a paraesophageal hernia. The character of the pain had changed and appeared more localized to the right upper quadrant. She noted a 10-lbs weight loss over one month. She had a history of DM type 2, HTN, CVA, hypothyroidism, diverticulosis, NSAID associated gastritis and a paraesophageal hernia. In 1997 she had a laparoscopic cholecystectomy for cholecystitis complicated by a right subphrenic abscess that required prolonged antibiotics and percutaneous drainage. She was taking ASA, Glipizide, Metformin, Enalapril, Omeprazole and Acetaminophen with Codeine. On exam she was comfortable with a temperature of 97.8 F, blood pressure of $189/96$, heart rate of 76 and breathing 16 times per minute. The skin was icteric. The abdomen was soft, with guarding over the right side, no rebound, bowel sounds were present and rectal examination revealed no masses and the stools were negative for occult blood. An ultrasound examination showed dilated intrahepatic and extrahepatic bile ducts, surgical absence of the gallbladder and an area suspicious for a mass around the pancreatic head. A percutaneous transhepatic cholangiogram showed an obstruction of the extrahepatic common bile duct with proximal dilatation. A fine needle aspiration of the area of the pancreatic head was also performed while a stent was placed through the obstruction and into the duodenum obtaining good flow of bile. After 3 days of partial relief, she developed abdominal and back pain. The amylase rose to 1425 and a contrast enhanced CT of the abdomen revealed findings compatible with pancreatitis. No masses were seen around the head of the pancreas and a periaortic lymph node was thought to be enlarged. CA 19-9 was reported as 21000 and CEA as 18, while the aspirate from the head of the pancreas was reported negative for malignant cells. On day 6 her course was complicated by the development of a hospital-acquired pneumonia and on day 11 she had a sudden respiratory decompensation and ventricular arrhythmia refractory to ACLS protocols and expired. Autopsy findings: Acute pancreatitis with extensive fat necrosis; Well to moderately differentiated adenocarcinoma of the extrahepatic bile ducts with infiltration into the head of the pancreas, and metastasis to the regional lymph nodes, mesentery, peritoneum and pulmonary lymphatic-vascular invasion; Hepatolithiasis throughout the liver; Recent pulmonary emboli of small and medium vessels.

UNUSUAL CAUSES OF PULMONARY HYPERTENSION. M. Rihawi, M Gennis, Internal Medicine, Sinai Samaritan Medical Center, Milwaukee, WI

LEARNING OBJECTIVES: Recognize high output cardiac states as a cause of isolated severe pulmonary hypertension and right ventricular failure.

CASE 1: A 49 year-old healthy African American female presented with progressive massive edema over 3 weeks. Physical examination revealed thyromegaly, systolic murmur, loud S2, and anasarca. Routine labs and chest XR were normal. TSH was non-detectable. Free T4 was markedly elevated. Echo cardiogram showed right ventricular dilatation and failure with severe pulmonary hypertension. Left ventricular function was near normal. Evaluation for other causes of pulmonary hypertension included V/Q scan, ESR, ANA, arterial blood gases were negative. Right heart catheterization confirmed the echocardiographic findings. The patient responded well to treatment for Graves disease and within 6 months, had a complete resolution of her pulmonary hypertension with normal right ventricular function.

CASE 2: A 74 year-old Russian female with known severe congenital hemolytic anemia, presented with new dyspnea and progressive edema. Physical examination revealed scleral icterus, holosystolic murmur and edema. Lung exam was normal. Her initial labs were at her known baseline including Hgb 6 g/dl, reticulocyte count 8%, LDH >6000 IU/Liter, and total bilirubin 7 mg/dl. Echocardiogram showed severe pulmonary hypertension, right ventricular failure, and normal left ventricular function. Evaluation for other causes of pulmonary hypertension was negative.

DISCUSSION: High output states are known causes of left ventricular failure but not isolated pulmonary hypertension and right ventricular failure. There are rare case reports of beriberi, Graves Disease, and Pagets Disease with similar presentations. The etiology is unclear. It is possibly a combination of increased blood volume with more rapid venous return, along with organic changes in the pulmonary vasculature.

UNUSUAL PRESENTATION OF FULMINANT METASTATIC MELANOMA. RR Rohatsch, D Patel, B DiStante, SJ Peterson, Internal Medicine, Westchester Medical Center, Valhalla, NY

LEARNING OBJECTIVES: 1. Recognize that coagulopathy can complicate metastatic melanoma.

CASE PRESENTATION: A 45 year old previously healthy white male was brought to our emergency department after suffering severe bilateral lower extremity weakness followed by paraplegia. The patient remained incapacitated on the floor of his apartment consuming no food or water for almost two weeks before he was found by neighbors. On physical examination his vital signs were normal. His chest revealed a 15×14 cm rubbery mass adherent to the chest wall at the left superior anterior axillary line, and a 5×5 cm pedunculated friable mass in the left subscapular area. Neurologic exam was remarkable for total paresthesia and paraplegia below T-11, with areflexia, absent rectal tone and absent proprioception. Laboratory investigation showed a WBC-21, BUN/Cr-215/5.6, LDH-1805, CPK-256, amylase-244, lipase-581, and prothrombin time of 15 sec. Urine analy-

sis showed pH-5.5, 3+ blood, mass RBC, many bacteria and myoglobin of 3.4. CT scan revealed narrowing of the spinal cord at the T-10 to L-2 region with multiple pathologic fractures and lesions in the liver suggestive of metastatic disease. The patient was treated for urosepsis. Within 24 hours of admission he developed a profound coagulopathy and bleeding from various sites. His fibrinogen levels were extremely high, D-Dimer was positive and LDH was 3 times admission value. He soon developed pancytopenia. Biopsy of the back lesion was consistent with metastatic melanoma. Despite aggressive transfusion of blood products over the next 2 weeks along with fluids, antibiotics and vasopressors, the patient remained pancytopenic and febrile. He finally died 17 days after admission.

DISCUSSION: Metastatic Melanoma complicated by coagulopathy resembling Disseminated Intravascular Coagulopathy has been well described. However, this patient developed a remarkably high fibrinogen level, probably representing an acute phase reactant secondary to sepsis. Although sepsis was being treated with antibiotics, the coagulopathy and pancytopenia worsened, perhaps being driven by his extensive tumor burden. This patient's coagulopathy could be multifactorial. Hepatic dysfunction, sepsis, tumor driven hemolysis or a combination of all three may have implications in the resulting coagulopathy.

OSTEOPOROSIS IN A YOUNG WOMAN AFTER TAHBSO FOR OVARIAN CARCINOMA. IT'S NOT ALWAYS WHAT IT SEEMS: CELIAC SPRUE AS A DIAGNOSIS WITHOUT GI SYMPTOMS. TM Rohr-Kirchgraber, Medicine, Upstate Medical University, Syracuse, NY

LEARNING OBJECTIVES: 1. Learn to consider various etiologies when pursuing a diagnosis and 2. Learn how celiac sprue can cause osteoporosis.

CASE PRESENTATION: A 37 year old white female presents for further evaluation of osteoporosis. She had a TAH/BSO at age 15 for ovarian carcinoma (dysgerminoma) followed by XRT. She had been on Premarin 1.25 mg daily and had an adequate calcium intake, 4 extra strength Tums plus dietary calcium. A bone density study done at the patients request 7/96 demonstrated profound osteoporosis with substantial loss of bone compared with a previous study done five years earlier. She denied any fractures or loss in height.

PMH: as above

Social: married, lives with husband and two adopted daughters, denies tobacco, drug use, or ETOH use. No daily exercise, educated thru graduate school and drank <1 cup of coffee per day.

FHx: Mother, 68, good health; Father, 7, lung cancer; Brothers $\times 2$ alive and well. No h/o osteoporosis.

ROS: Menses, age 11, regular, last menses 6/75. Occasional fatigue, no recent wt changes. Denies ever taking steroids, thyroid hormone, or anticonvulsants. Denies kidney disease or stones.

PE: WDOWN, VS: BP $115/75$, HR 75, RR 14, Temp 36.5 C.

HEENT: wnl; Lungs: clear; Cardiac: RRR, S1S2, no murmurs; Abd: benign, liver span 10 cm, rectal brown guaiac negative; extremities and neuro exam WNL.

Data: Ca (mg/dl) 9.6, Mg (mg/dl) 1.7, PTH (10-65 pg/ml) 87, 25 OH Vit D (10-55 ng/ml) 24, 1,25 OH Vit D (21-48 pg/ml) 87, UrCa (75-250 mg/24 hr) 6, UrPhos (350-1000 mg/24 hr) 529. Blood tests for Na, K, CO2, Cl, Bun, Creat., Glu, T.Pro, alb, LFTs including Alk phos were normal. Bone mineral density (BMD) hip 0.688, % young nml 71; BMD L1-L4 0.686, % young nml 66.

Course: Pt started on 20 mcg of Calderol b.i.d. which was increased to t.i.d. Calcium intake was increased to 1800mg q day. The initial concern was that she had a marked intestinal resistance to the physiological action of 25-hydroxyvitamin D along with secondary hyperparathyroidism. 25-hydroxyvitamin D levels were increased to the point where urinary calcium excretion increased moderately and the secondary hyperparathyroidism was resolved. A marked improvement in bone mineral density resulted. A referral to GI for a colonoscopy was made which led to the pathological diagnosis of Celiac Sprue.

DISCUSSION: Celiac Sprue is a disorder in which gluten and related substances gliadin, are toxic and induce the binding of gluten to epithelial cells of the intestine with subsequent tissue damage. Usually characterized by malabsorption, abnormal small bowel structure, intolerance to gluten products, it can be found in a children and adults as well. Even in the absence of GI symptoms such as diarrhea and bloating, celiac sprue can be present and should be considered in the setting unexplained osteoporosis.

AN UNUSUAL CAUSE OF FLANK PAIN—AVASCULAR NECROSIS OF THE FEMORAL HEAD. LM Rucker, Medicine, Jacobi Hospital, Bronx, NY

LEARNING OBJECTIVES: 1) Diagnose Avascular Necrosis (AVN) of the femoral head. 2) Recognize the radiographic appearance of AVN.

CASE PRESENTATION: A 27 year old female presented to the walk-in clinic with one week of bilateral flank, back, and lower quadrant abdominal pain. The pain was severe and constant, worse with bowel movements and relieved by oxycodone and acetaminophen. She had completed a short course of nitrofurantoin for treatment of a bacterial cystitis one week prior to the onset of the pain. She had episodic diarrhea for two weeks, preceeding the antibiotic use. She had a past history of asthma, ectopic pregnancy, pelvic inflammatory disease, kidney stones, migraine headaches, hypertension, peptic ulcer, and bipolar disorder. Her meds included clonidine, diltiazem, oxycodone and acetaminophen, ranitidine, lithium, benztrapine, perphenazine, zolpidem, and antacids. She had taken prednisone off and on for years. She used inhaled flunisolide, ipratropium, and pirbuterol. Physical examination showed an obese woman in pain. Vital signs were normal. She had expiratory wheezes but good air movement. Abdominal exam was significant

for pink striae and tender lower quadrants without guarding or rebound. Pelvic exam showed right adnexal tenderness without mass and no cervical motion tenderness. Stool was guaiac negative. Subsequent exam was positive for flank pain and mild costovertebral angle tenderness bilaterally. Hip exam was significant for normal internal and external rotation. Flexion and extension were limited by, and reproduced, her flank pain. CBC, SMA, pregnancy test, abdominal and pelvic sonography, urine analysis, genprobes for gonorrhea and chlamydia, stool culture, C. difficile testing and a KUB (kidney, ureters, bladder) xray were all negative or non-diagnostic. Hip films with the note "Rule out AVN, long history of steroid use" showed increased density and flattening of the femoral heads consistent with AVN.

DISCUSSION: This patient had several possible causes of her pain. Because of her steroid history it was reasonable to consider AVN of the femoral head causing pain radiating to the flank. The physical examination of the hip was consistent with AVN, in that rotation was preserved. In osteoarthritis, limitation of internal and external rotation is one of the earliest findings. In either case pain localized to the groin is more common than flank pain. Onset of pain can be more rapid in AVN than the insidious pain characteristic of osteoarthritis. Findings of AVN on xray are easy to overlook (as they were in this patient's KUB), but are essential to search for. MRI is the most sensitive diagnostic test. The earlier the diagnosis of AVN is made, the better the likelihood the patient can have surgical decompression of the joint, delaying the need for total hip replacement.

BELLADONNA AS TREATMENT FOR GASTROINTESTINAL SPASM. DR. Saenger, D. Korenstein, Medicine, Mt. Sinai Medical Center, New York, NY

LEARNING OBJECTIVE(S): 1) Recognize the manifestations of gastrointestinal spasm, a functional disorder which can masquerade as abdominal mass. 2) Recognize the potential benefits of treatment of gastrointestinal spasm with belladonna.

CASE PRESENTATION: A 51 year old woman with history of hypertension, anxiety disorder and mild obesity presented with complaint of a palpable abdominal mass and early satiety. These symptoms were of recent onset and over the last several weeks had progressed to the point where she was nearly unable to eat solid food. The patient also complained of exacerbation of her anxiety disorder. On examination, the patient had a firm, approximately 5 cm. mass palpable just inferior to the xiphoid process. Compared with previous exam three months earlier, the patient had lost 10 pounds. Stool was hemoccult negative. The remainder of the physical examination was unrevealing. A therapeutic trial of clonazepam and omeprazole provided no relief of symptoms. Laboratory studies revealed iron deficiency anemia, with a hemoglobin of 10 mg/dL. CT scan of the abdomen and pelvis and upper endoscopy were both normal. The patient was diagnosed with gastrointestinal spasm and was treated with oral belladonna, which resolved the patient's symptoms completely. Following two months of belladonna treatment, the patient's oral intake had normalized, her weight had recovered, her anemia had resolved and her anxiety had improved.

DISCUSSION: Gastrointestinal spasm can be a challenging diagnosis. Although a functional disorder, it often manifests with physical symptoms and signs, even masquerading as a gastrointestinal malignancy. Belladonna, an anticholinergic drug, has both anxiolytic and smooth muscle relaxant properties. Studies in the 1960's and 1970's have demonstrated its efficacy in the treatment of psychosomatic gastrointestinal disorders.

STAPHYLOCOCCAL SEPSIS PRESENTING WITH MUSCLE PAIN. W. Sanchez, Internal Medicine, Mayo Clinic, Rochester, MN

LEARNING OBJECTIVES: (1) Recognize myonecrosis as a clinical manifestation of disseminated staphylococcal sepsis. (2) Identify pathologic features of widespread staphylococcal sepsis present on post-mortem examination.

CASE PRESENTATION: A 79 year-old man, who was in his usual state of good health, presents to his local emergency room after developing left-sided chest and shoulder pain one day after performing heavy lifting while gardening. Past medical history is significant only for diet-controlled Type II diabetes mellitus. After a negative evaluation for possible acute myocardial ischemia the patient is discharged to home. The patient continues to experience persistent shoulder pain he is subsequently seen by his local orthopedist who performs a steroid injection of the shoulder. Over the next several days the patient develops progressive pain, including bilateral thigh pain requiring hospitalization. On admission, the patient is found to have diffuse muscular pain, most prominently in his medial thighs and forearms. During his hospital course the patient continues to have progressive symptoms and develops darkening of his urine. Intravenous methylprednisolone and alkalinizing agents are administered for a presumptive diagnosis of polymyositis with rhabdomyolysis. The patient is then transferred to our institution for further evaluation.

On transfer the patient is noted to be lethargic but arousable. He is afebrile and other vital signs are stable. Physical examination reveals erythema and swelling of the tongue, some fullness of left supraclavicular lymph nodes, and diffuse muscular swelling with exquisite tenderness. Laboratory analysis are remarkable for a leukocyte count of $26.1 \times 1,000/\text{cu mm}$ with a prominent left shift, a creatine kinase of 1,825 U/L (reference range: 52-336 U/L) with an MB-fraction of 25.4 U/L (reference range: less than 4.3 ng/mL) and an elevated serum creatinine of 2.1 mg/dL. The patient had a fulminant hospital course, developing decreasing urinary output over several hours despite aggressive medical management. The patient's vital signs remained stable; however, shortly after admission he became

unresponsive. In accordance with the patient's advanced directives no resuscitative measures were taken and the patient expired.

DISCUSSION: This patient illustrates overwhelming staphylococcal sepsis masquerading as myositis, with steroid administration eliminating the febrile response. Blood and urine cultures obtained antemortem revealed the growth of *Staphylococcus aureus*. Postmortem examination revealed severe necrotizing myopathy of the extremities and myocardium resulting in elevated creatine kinase and MB-fraction levels. Further autopsy findings of widespread infection include microabscess formation in the lungs and prostate, tricuspid valve endocarditis, adrenal medullary necrosis and multifocal ischemic small and large bowel.

CALCIPHYLAXIS: METASTATIC CALCIFICATION COMPLICATING END-STAGE RENAL DISEASE. W. Sanchez, Internal Medicine, Mayo Clinic, Rochester, MN

LEARNING OBJECTIVES: (1) Identify the clinical features of calciphylaxis, a complication of end-stage renal disease. (2) Recognize the relationship of tertiary hyperparathyroidism and calciphylaxis.

CASE PRESENTATION: A 64 year-old woman presented to our institution for a second-opinion regarding a progressive, ulcerating wound of the left breast. She is a hemodialysis-dependant, type II diabetic who underwent a left breast biopsy as well as a left subclavian dialysis catheter placement approximately 8 weeks prior to admission. The wound from the biopsy became erythematous and subsequently began to ulcerate. She was treated with several courses of intravenous vancomycin and oral cephalexin and underwent two surgical debridements five and three weeks prior to admission.

The patient's past medical history is significant for a 25-year history of type II diabetes mellitus and has been insulin-requiring for the last 17 years. The patient has multiple complications of chronic diabetes, including end-stage renal disease (ESRD), secondary to diabetic nephropathy. She has been hemodialysis-dependant for a year prior to admission. At time of admission, the patient complains of pain and tenderness of the left breast, which is without any discharge. The patient also complains of multiple firm subcutaneous nodules which have become progressively more painful since she began dialysis. The patient is afebrile and denies any rigors or night sweats. Physical examination reveals a morbidly obese (body mass index 39.95 kg-square meter) female in a moderate amount of distress. There is a large necrotic ulceration of the inferior half of the left breast. The underlying soft tissue has a dry-gangrenous appearance and the lateral aspect of the breast is fluctuant. There are several firm, tender subcutaneous nodules scattered through her abdomen and lower extremities. One of these nodules located in her right groin is erythematous and has a macerated appearance. Admission laboratories are notable for an elevated serum creatinine (4.6 mg/dL), an elevated serum calcium (10.6 mg/dL), an elevated serum phosphorus (5.6 mg/dL). Laboratory analysis are further notable for a markedly elevated serum oxalate (24.5 mg/dL [reference range: 0.4-3.0]) and parathyroid hormone (42 mg/dL [reference range: 1.0-5.2]).

DISCUSSION: Calciphylaxis is a clinical syndrome that uncommonly occurs in patients with end-stage renal disease. The manifestations include calcium deposition in small to medium sized blood vessels which induces tissue ischemia, often causing painful non-healing ulcers. Metastatic calcification is induced by tertiary hyperparathyroidism and the severity of the disease can often be assessed by measuring the serum calcium-phosphorus product.

THE KING'S EVIL - AN INTERESTING CASE OF LYMPHADENOPATHY. MA. Schwartz, Internal Medicine, University of Chicago Medical Center, Chicago, IL

LEARNING OBJECTIVES: Learning Objectives: 1) Diagnose Scrofula—tubercular lymphadenitis—Appreciate the frequent absence of other clinical findings. 2) Recognize the need to perform a PPD in a patient with lymph node enlargement even without chest radiograph abnormalities. 3) Enjoy the fascinating history of the King's Evil as the standard paradigm of treatment for over 1000 years.

The evaluation of the adult with an enlarged, unexplained lymph node remains an important and challenging clinical scenario. The extensive differential diagnosis includes infectious, neoplastic, granulomatous, drug induced and rare or unusual causes. Frequently, the diagnosis is only made at the time of Fine Needle Aspirate (FNA) or by surgical lymph node biopsy. A case is presented of a 65 year old woman who was well and without medical problems or any clinical symptoms. She was referred by her employer who was concerned about a palpable 3 cm lymph node in the left posterior occipital chain. She had no recent travel but grew up in Belize and came to the United States 15 years before. Her physical examination was remarkable for a solitary enlarged cervical lymph node. Her evaluation included a normal CBC, Chest Xray, Chem 17 and negative serologies for syphilis, HIV and Infectious Mononucleosis. She underwent 2 nondiagnostic FNAs. A lymph node biopsy was ultimately performed and demonstrated classic caseating necrosis, multinucleated giant cells and had a smear positive for AFB. Her PPD was indurated at 20 mm. Her final diagnosis was isolated tubercular lymphadenitis-Scrofula. The final culture revealed *Mycobacterium tuberculosis* sensitive to all antibiotics. A review of the literature demonstrated that in countries where tuberculosis is no longer endemic, scrofula can present without other signs of active tuberculosis. It is also one of the most common sites of extrapulmonary tuberculosis. It can present as a single enlarged lymph node and frequently is associated with a normal chest Xray. As a result of these 2 findings, tuberculosis is commonly an afterthought and PPD placement is delayed. The presence of a positive PPD and an enlarged lymph node mandates histological or microbiological confirmation of tuberculosis as TB can coexist with other diseases that stimulate lymph

node enlargement. Historically, Scrofula was referred to as—the King's Evil because the kings of England and France were believed to have a divine gift to cure the disease by royal touch of an infected individual. For a thousand years, the Royal touch was the standard of practice to treat this once common disease. Though responsible for less than 5% of all causes of lymphadenopathy in the United States, the enormous prevalence of tuberculosis worldwide mandates that we continue to consider this etiology in all patients with an unexplained enlarged lymph node.

A WOMAN WITH ABDOMINAL PAIN AND A HISTORY OF COLON CANCER. Al Schwarz, SGIM, DHMC, Lebanon, NH

LEARNING OBJECTIVES: 1. Discuss the state of the art surveillance for recurrent colon cancer. 2. Using the Medical literature and emerging technology.

CASE PRESENTATION: Mrs A presents for her annual visit with a past medical history notable for having survived colon cancer resected 13 years ago. She had two isolated recurrences resected, one in the left upper and left lower lobes of the lungs. Her medical history also includes, osteoporosis, pericarditis, TIA's, SVT's and PUD. She takes digoxin, persantine and vitamins. She presented to the clinic with lower back pain. The pain was not activity related and was mild, dull and hard to localize. She had no nighttime symptoms, fevers or weight loss. On exam she had tender paraspinal muscles from T4-L2 on the left without spinal process tenderness or radicular neurologic findings. Over the course of several weeks the pain was alleviated by sitting up and leaning forward. Her exam was negative, a MRI of the spine, a bone scan and an Echocardiogram were ordered and all were normal. She asked for a second opinion and elected to then go the Mayo Clinic in Rochester Minnesota. There the physicians diagnosed her with a thoracic radiculopathy. Her oncologist released her from further follow up in the oncology clinic only to find that day her CEA rose into the abnormal range for the first time; going from 0.6 to 4.2 (JAMA 1993, 270:943-947). Subsequent exam, lab and CT scans were negative. One month later the patient again pleaded for more diagnostic and the new CEA monoclonal antibody nuclear imaging testing was discussed (J. Clin. Onc. 1999 16(5):1777-1787). The discussion of emerging technologies in medicine is a key lesson in this case. Few articles were available at the time of this case on radiolabeled monoclonal CEA imaging. Radiologist and Oncologist recommended a PET scan (positron emission topography) (J. Clin. Onc. 1999, 17:894-901 and Arch. Surg. 1999, 134:503-513). The scan was done and showed two areas of increased activity; one in the region of the pancreas and the other in the right inguinal area. The JAMA series on reading the medical literature will be reviewed in the context of interpreting studies examining a diagnostic test (JAMA 1994 271(9):703-707). On scheduled visits nine months since the onset of the pain syndrome her CEA rose to 11.5 and then 21.5. A CEA scan was performed locally and showed an increased area of activity in her right thigh. Plain films of the thigh showed an osteoblastic tumor, which was biopsied. The pathology showed well-differentiated adenocarcinoma. A repeat bone scan was ordered and showed multiple bony metastases and shortness of breath lead to the diagnosis of pulmonary lymphangitic spread. Her family was close by her side when she died 13 months after her pain syndrome started, 14 years after her initial diagnosis of colon cancer.

DISCUSSION: Recurrent colon cancer and use of emerging technologies.

A "PUFF OF SMOKE" IS A CLUE TO STROKE. JB Scott, Dept. of Medicine, Eastern Virginia Medical School, Norfolk, VA

LEARNING OBJECTIVES: Review common and uncommon causes of stroke in the young adult.

CASE PRESENTATION: A 31 year old female with a history of classic migraine with visual aura presented after a tonic clonic seizure. There was mild anterograde amnesia post-ictally. She denied visual change or focal paresthesias. She took no medications, smoked one pack of cigarettes daily, drank occasional alcohol, and used no illicit drugs.

Physical examination revealed a tongue laceration, subtle dysarthria, and a left palmomental reflex. The remainder of the examination, including detailed neurologic evaluation, was normal. CBC, chemistries, CXR and EKG were normal. Urine drug screen, blood cultures, and lumbar puncture results were negative. MRI of the brain revealed a lesion in the right frontal lobe, as well as multiple punctate lesions in the right and left centrum semiovale, consistent with ischemic infarcts. Evaluation for etiology of ischemic stroke, including carotid duplex examination, echocardiogram, and hypercoagulable states, was negative. Homocysteine and cholesterol levels were normal. Serologies for ANA, HIV, ANCA, VDRL, and antiphospholipid antibodies were negative. Cerebral angiogram revealed stenosis of the distal portions of the internal carotid arteries (ICA), and a network of dilated capillaries at the base of the skull. These findings are pathognomonic for Moyamoya disease, a rare disorder of unknown etiology in which progressive narrowing occurs in the terminal ICA and proximal portions of the anterior and middle cerebral arteries, followed by extensive collateralization of lenticulostriate arteries. "Moyamoya," a Japanese term for "hazy puff of smoke," describes the angiographic appearance of these vessels. Clinical manifestations result from subcortical cerebral ischemia, or from hemorrhage of dilated collateral arteries. Medical therapies include antiplatelet agents and steroids. Surgical revascularization, in which stenotic portions of the middle cerebral artery are bypassed, is an increasingly promising therapeutic option.

WHAT'S THE BIG DEAL? WHY NOT TO USE BLOOD CULTURES IN THE MANAGEMENT OF PYELONEPHRITIS. C Shang, D Brady, Division of General Medicine, Emory University School of Medicine, Atlanta, GA

LEARNING OBJECTIVES: 1. To discuss the low utility of blood cultures in the clinical management of pyelonephritis. 2. To discuss the negative financial impact of ordering blood cultures in pyelonephritis.

CASE PRESENTATION: A 32 year old female presented with a chief complaint of dysuria, malodorous urine, diffuse body aches, and subjective fever for 10 days. She also complained of nausea and decreased oral intake. She had been paraplegic since age 9 secondary to a vertebral fracture, had a chronic neurogenic bladder, and had had multiple previous urinary tract infections and episodes of pyelonephritis. 7 days prior to this visit, she was given levofloxacin for a presumed recurrent urinary tract infection and did not receive any relief. Her exam revealed temperature 36.6, pulse 101, respiratory rate 16, blood pressure 125/99, left costovertebral angle tenderness, and mild suprapubic tenderness. White blood cell count was 4.9 with 47% neutrophils. Her urinalysis showed no nitrite, 2+ leukoesterase, 10-25 WBC, 10-25 RBC, and numerous bacteria. She was admitted for a recurrent urinary tract infection recalcitrant to levofloxacin with a possibility of pyelonephritis. Urine and blood cultures were obtained, and her antibiotics were switched to ampicillin and gentamicin. Her symptoms resolved overnight, and the patient was discharged on the second hospital day. The urine culture grew more than three organisms, and the blood cultures were negative.

DISCUSSION: Multiple studies have shown that blood cultures rarely contribute to the management of pyelonephritis and, even when positive, seldom vary from the urine culture results. A recent study of 338 inpatients with pyelonephritis showed that, although 91% had blood cultures obtained, only 1 grew a pathogenic organism not found in the urine culture—with no impact on clinical management and a cost of about \$18,000. Two other studies showed that the blood cultures from over 200 patients with pyelonephritis had no impact on treatment. The yield of blood cultures in the face of potential pyelonephritis is very low, and the potential harm of false positives is significant. In addition to the potential side effects from antibiotics, studies have shown that treatment based on false positive blood cultures increased length of stay and cost more than \$3000 per patient treated.

WHEN THE FLU IS NOT THE FLU. IL Simon, Medicine, Mount Sinai Medical Center, New York, NY

LEARNING OBJECTIVES: 1) Recognize the clinical features of acute HIV infection which may mimic those of other common viral illnesses. 2) Utilize appropriate laboratory testing to diagnose acute HIV infection.

CASE PRESENTATION: The patient is a 28 year old financial analyst who was seen in early March for a general checkup. He had no acute complaints, nor any significant past medical or surgical history. He is a homosexual male who stated that he had a single partner with whom condoms were always used. The patient had been tested HIV negative approximately 1.5 years prior. Physical exam was entirely normal, and a repeat HIV test was also negative. The patient then returned later the same month with a two day history of fevers, joint and muscle aches, poor appetite, watery eyes, runny nose, mild chest tightness, and general malaise. Exam was again unremarkable except for a temperature of 37.8 celsius. There was no skin rash, no cervical adenopathy, no pharyngeal injection or exudates, and no hepatosplenomegaly. Influenza was diagnosed. The patient returned two days later with persistent symptoms. Now some abdominal pain and vomiting had developed, and a dry cough. Exam revealed a temperature of 102.2 F, and mild left upper quadrant tenderness, but was otherwise unchanged. CBC obtained at this visit revealed a WBC of 3.00, platelets of 72, and normal hemoglobin and hematocrit. Occasional large platelets were noted. Chemistries were significant for AST = 104, ALT = 86, and LDH = 467. Urinalysis was normal. Repeat labs two days later showed the same pattern with rising liver enzymes and LDH, and decreasing platelets. Heterophile antibody was negative, and hepatitis A,B, and C serologies were negative. The patient then went away on vacation for 10 days. Upon his return, he had lost eight pounds since his initial checkup. Symptoms were still present, though not as severe. Exam revealed temperature of 99.2 F, no rash, no icterus, normal pharynx, and normal abdominal exam. A few mobile left anterior cervical lymph nodes were palpated. Chest x-ray at this time was normal. Epstein-Barr antibody titers were negative for IgM, positive for IgG. Liver enzymes and WBC count had improved to almost normal. Repeat HIV testing was done, 19 days after the onset of the illness. HIV antibody screen (ELISA) was negative, but HIV DNA PCR was positive for detection of HIV-1 DNA. Subsequent viral load was 316.8 thousand copies/ml, T-helper count was 998, with a helper/suppressor ratio of 0.66.

DISCUSSION: Acute HIV infection may be easily confused with other viral entities such as influenza or infectious mononucleosis. Diagnosis requires a high index of suspicion and a thorough discussion of risk factors. ELISA may initially be negative. In this instance further testing, such as with HIV DNA PCR, is indicated. Early diagnosis and institution of treatment with highly active antiretroviral therapy may have an impact on the course of the disease.

QUICKSILVER, THE MAD HATTER, & MUNCHAUSEN'S REVISITED: ELEMENTAL MERCURY POISONING BY PARENTERAL ROUTE. LL Sohn, J Maks, C Francois, C Tuffon, Internal Medicine, Evanston Northwestern Healthcare, Evanston, IL

LEARNING OBJECTIVES: Diagnose elemental mercury poisoning; Recognize atypical Munchausen's syndrome

CASE PRESENTATION: Historically, the rims of felt hats were impregnated with mercury (Hg) and chronic cutaneous exposure to Hg could result in a progressive dementia—hence the phrase “Mad as a Hatter”. Hg, also known as “quicksilver,” has been injected by boxers as a perceived means of enhancing quickness. Hg injections for a “high” by drug users have also been described. We present an unusual case of parenteral, elemental Hg poisoning. A 34 yr old male presented to the emergency room with renal colic, chills, and a cough productive of green, blood-streaked sputum. He was admitted for fluids and pain control. Admission CT of the abdomen revealed a small nephrolith in the proximal left ureter and mild hydronephrosis. Admission CXR was negative. On hospital day two the patient complained of increasing shortness of breath and a repeat CXR was obtained. It revealed diffuse bilateral branching opacities greatest in the lower lobes thought to be consistent with a heavy metal exposure. CT revealed high density foci in multiple areas including the lungs, the anterior right ventricle, apex of the left ventricle, liver, and kidney. CT of the head was negative. Overwhelming evidence suggests that the route of Hg exposure was through intravenous injection. The patient denied knowledge of an intravenous injection. He, however, adamantly insisted that a broken sphygmomanometer sprayed Hg onto his head when the cuff was inflated. The preponderance of Hg in the pulmonary vasculature and the distinct lack of it in his upper airways and bronchial tree make inhalation exposure extremely unlikely. Most notably, the CXR had the appearance of a pulmonary angiogram. A piece of glass from the broken sphygmomanometer was found in the patient's drawer with a tourniquet. The X-ray of the patient's left hand at the site of an IV infiltration revealed small hyperdense foci consistent with traces of Hg in the skin. Initial blood level was 530 ug/dl. The first urine Hg level obtained after chelation had been implemented was 1193 ug/24 hours. The reason why this patient may have chosen to inject Hg may never be known, although features of Munchausen's syndrome are suggested. As Sherlock Holmes might have said, “It's elemental, dear Watson.”

COMPETENCE-DO ACTIONS SPEAK LOUDER THAN WORDS? JE Stahl, Medicine, Univ. of Pittsburgh Medical School, Pittsburgh, PA

LEARNING OBJECTIVES: Understand patient competence and who determines competence.

CASE PRESENTATION: A 48 y.o. man with a past medical history significant for HTN, poorly controlled DM with nephropathy, neuropathy, retinopathy, gastroparesis, impotence and resolved depression was first seen complaining of chest discomfort for 3 months. A stress test was arranged. When the patient arrived for his test his BP was $240/120$ accompanied by a headache. The patient was admitted for hypertensive urgency. After discharge the PCP and nurse educator engaged in intensive education on the management of HTN, diabetes and gastroparesis. When asked he could describe the implications of his disease and what the therapy did. Within a few weeks the patient began calling for frequent headaches accompanied by nausea and vomiting. Despite being encouraged to go to ED during these episodes he chose not. He stated the symptoms usually went away after the phone call. Within one month he was admitted for labile HTN, intractable nausea and vomiting. When his symptoms resolved, though his HTN was still labile, he demanded to be released from hospital. It was explained to him, that in his current condition his BP could lead to a stroke. He stated he understood and agreed to remain over night. This sequence repeated itself every day. Psychiatry determined that the patient did not appear to have any comprehension of his disease and that this behavior might be a form of a passive suicide attempt. He was determined incompetent and a restraining order was put in place. Shortly after, he walked out of hospital despite the restraining order. The police were contacted. The police chief informed the PCP that the patient was a known offender who had repeatedly gotten out of serving time by pleading his illness. After repeated calls, his wife called back to ask if his medications could be filled over the phone. She stated she did not know his location. At this time the VNA arrived at his home. She reported that she found him hypertensive, without any medicine and smoking marijuana. Shortly after the patient called asking if the restraining order could be rescinded since the police were after him. It was arranged for him to be seen at his local ED to be reevaluated. The psychiatrist there determined the patient to be competent. The patient did not return to clinic. Two months later he was admitted for a hypertensive emergency and died of a hemorrhagic stroke.

DISCUSSION: Competency has several components. The ability to communicate a choice, to have factual understanding of the situation, to appreciate the implications of the decision and be able to rationally manipulate the relevant information. In conversation, this patient appeared to be competent on each of these levels yet his actions were irrational with regard to his health. Competence does not necessarily imply rationality and actions can speak louder than words.

ALTERNATIVE THERAPIES FOR DEMENTIA IN THE ELDERLY. SA Sternberg, Department of Medicine, University of Chicago, Chicago, IL

LEARNING OBJECTIVES: Describe the risks and benefits of alternative therapies for dementia in the elderly.

CASE PRESENTATION: A 76 year old female recently diagnosed with probable Alzheimer's Disease presented to the geriatric clinic for follow up. The patient's daughter had recently taken over management of the patient's medications and finances. The patient needed reminders to bathe but was still quite functional. She did not have any behavioral or sleep problems, nor any falls or gastrointestinal complaints. The patient had a history of atrial fibrillation on warfarin and osteoarthritis on acetaminophen. Her physical examination was remarkable for an irreg-

ularly irregular heart rate of 80, marked short term memory deficits, word finding and naming difficulties, apraxia without parkinsonism or gait abnormalities or lateralizing neurological signs. The patient's daughter, after perusing the internet and visiting her local health food store, had purchased and started her mother on ginkgo biloba, lecithin, and coenzyme Q10. She wondered why her mother had not shown any improvement in cognition or physical functioning.

DISCUSSION: Alternative therapy use is increasing rapidly. Little is known about the use of alternative therapies by the elderly. The elderly are particularly at risk when using alternative therapies because they have limited incomes, many chronic conditions and medications, and caregivers who are demographically the highest users of alternative therapies. Ginkgo biloba is an extract of the maidenhair tree with antioxidant and anticoagulant properties. Three RCTs in the English language literature, evaluating 569 patients for an average of 29 weeks reported modest cognitive improvement. However, dropout rates were high often due to gastrointestinal side effects. Ginkgo biloba is a potent inhibitor of platelet-activating factor essential for platelet aggregation. Interactions with aspirin and warfarin resulted in a spontaneous hyphema and an intracerebral hemorrhage. Lecithin, a cholinergic precursor found in egg yolk, liver and soybeans was examined in at least five RCTs. No improvement in cognition or function was demonstrated. Inositol and choline, other cholinergic precursors, were each studied in one small RCT of short duration without benefit. Coenzyme Q10, a lipid soluble antioxidant, has not been studied in the elderly with dementia. Bovine phosphatidylserine, a membrane stabilizer, was studied in four RCT but cognitive improvement was not consistently demonstrated. Zinc, essential to antioxidant enzymes in the hippocampus, was not beneficial in one small study. The administration at high doses of vitamin E, an antioxidant, delayed death, nursing home placement, loss of function or worsening dementia but only after statistical data manipulation. Given the evidence and the patient's use of warfarin, she was advised to stop all the alternative therapies.

FEVER AND DELIRIUM IN A HOSPITALIZED MAN WITH DIFFUSE LEWY BODY DEMENTIA. PR Sutton, TJ Johnson, Department of Medicine, VA Puget Sound Health Care System; School of Medicine, University of Washington, Seattle, WA

LEARNING OBJECTIVES: 1) Diagnose neuroleptic malignant syndrome in the setting of inpatient medical consultation, and 2) Withhold therapy of uncertain benefit in a stable or improving clinical situation.

CASE PRESENTATION: A 75 year-old man with Diffuse Lewy Body Dementia (DLB) was transferred from the psychiatry service with a five-day history of fevers, confusion, and rigidity. He had been started on clozapine for psychotic symptoms and his dosage had been escalated 1 week prior to transfer. He was somnolent with an unsteady gait the following day. He developed intermittent fevers to 103°F. He was intermittently tachycardic and hypertensive. He was delirious with a coarse resting tremor. His neck was stiff. He had cogwheel rigidity on motor exam. His past history was notable for hypothyroidism. His medications included levothyroxine, valproic acid, and zolpidem. Laboratory evaluation was notable for a normal TSH, leukocytosis without left shift, mild prerenal azotemia, and an elevated creatinine kinase (CK, peak 9462 U/L). Troponin I was normal. Clozapine was discontinued. Blood cultures, chest X-ray, and a lumbar puncture were unrevealing. A urinalysis was positive for blood, but no red cells were present; there was no pyuria or bacteruria. A diagnosis of neuroleptic malignant syndrome (NMS) was made on the basis of (a) exclusion of other causes of fever, and (b) the clinical syndrome of delirium, rigidity, autonomic instability, and an elevated CK. Supportive care was administered: hydration and discontinuation of psychoactive medications. No pharmacotherapy specific for NMS was instituted. The patient's mental status, neurological examination, and laboratory values improved over the following five days.

DISCUSSION: DLB is the second most common cause of progressive senile dementia (7–30% of patients with dementia). Clinical diagnostic criteria emphasize fluctuating dementia, parkinsonism, and neuroleptic sensitivity in the diagnosis of DLB. In one small study, 29% of patients with DLB and 0% of patients with Alzheimer's Disease exhibited neuroleptic sensitivity. Frank NMS is less common, with an incidence of 0.02–3.23%. The incidence in patients with DLB is unknown. Risk factors for NMS include dehydration and neuroleptic dose. Atypical antipsychotics may cause NMS. Specific pharmacotherapy, including bromocriptine, dantrolene, and benzodiazepines, have been used in the treatment of NMS. None have been well-studied. Retrospective studies have suggested mortality benefits for dantrolene and bromocriptine, but included historical controls. Other studies have failed to demonstrate any benefit for these agents in hastening resolution of symptoms or reducing mortality. It is most reasonable to withhold therapy of uncertain benefit in a stable or improving clinical situation.

I'VE REVERSED MENOPAUSE! MD Swift, Medicine, Vanderbilt University, Nashville, TN

LEARNING OBJECTIVES: 1. Summarize current knowledge about the physiologic effects of estrogenic herbs, and patterns of herb usage in menopausal and perimenopausal women. 2. Identify herbs with estrogenic activity. 3. Effectively counsel women about estrogenic herbal supplements, especially women with a contraindication for unopposed exogenous estrogen.

CASE PRESENTATION: A 50 yo woman presented with a complaint of very heavy periods with some intermenstrual bleeding. She had experienced oligomenorrhea and hot flashes 6 months previously, but the hot flashes abated when she began taking a combination of several estrogenic herbs (dong quai, black cohosh, and goldenseal). Because of the loss of hot flashes and the onset of heavy vaginal

bleeding, she felt she had reversed her menopausal transition. Upon the request of her primary care physician, she stopped using the herbal estrogens. Her hot flashes returned. An endometrial biopsy revealed a proliferative endometrium, and a pelvic ultrasound revealed several fibroids. Her hot flashes were controlled with combination hormone replacement therapy, but she ultimately required a hysterectomy for heavily bleeding fibroids.

DISCUSSION: In the US, use of herbal remedies and supplements has risen 380% in the last decade, with an estimated 12.5% of American adults using herbal medicines in 1997. Women ages 35–49 use alternative medicines more than any other demographic group. In order to counsel their female patients, physicians must be knowledgeable about possible beneficial and harmful effects of those herbs with estrogenic effects. Although several herbs have estrogenic activity *in vitro*, clinically relevant estrogenic effects of herbs remain to be proven. At least one randomized trial of single herb treatment for hot flashes has shown no significant effect. However, many herbs are sold as combination tablets using several different herbs. Case reports such as presented here suggest that in high doses and in combination, estrogenic herbs may indeed exert a clinically relevant estrogenic effect. Patients taking such herbs may be taking active estrogen supplements, unaware of the potential risks and benefits of unopposed estrogen. Potential benefits include reduction of hot flashes and vaginal dryness, while potential harmful effects include stimulation of breast cancer cells, endometrial hyperplasia and endometrial carcinoma, hypercoagulability and hepatic injury. American women in their perimenopausal years are turning to herbal medicines in record numbers. The use of certain herbs, especially in combination, may indeed expose women to meaningful doses of exogenous estrogens. Therefore, physicians must be knowledgeable about these herbs and their potential actions in order to provide effective counsel to their patients. Further study is needed to ascertain the safety and effectiveness of higher doses and combination herbal therapies.

AN UNUSUAL PRESENTATION OF PERIPHERAL T-CELL LYMPHOMA. LK Tank, H Taha, P Mehta, B Bashey, L Krishnamurthy, Internal Medicine, New York Methodist, Brooklyn, NY

LEARNING OBJECTIVES: Recognize atypical clinical features of T-Cell Lymphoma.

CASE BACKGROUND: Peripheral T-cell Lymphoma is an aggressive type of Non Hodgkin's Lymphoma A subtype of T-cell angioimmunoblastic T-cell lymphoma occurs in older adults in whom it presents with generalized lymphadenopathy, rash and hepatosplenomegaly. It is locally aggressive and has a high recurrence rate after therapy. We report a case of angiocentric T/NK cell lymphoma presenting as a non-healing gangrenous foot ulcer, which is a very unusual presentation. **CASE:** An 86 year old woman presented with right leg cellulitis, deep vein thrombosis and femoral arterial stenosis. She reported right leg pain, swelling and redness for 2 weeks. The physical examination was unremarkable except for right leg cellulitis with erythema, edema of the second and fifth toes with laceration and drainage of serosanguinous discharge. CBC and serum chemistry results were within normal limits. The patient developed gangrene of the toes, secondary to peripheral vascular disease. A femoral popliteal bypass and transmetatarsal amputation of the toes was performed. Biopsy (transmetatarsal amputation) showed coagulative and liquefactive necrosis. The skin at the margin showed an atypical, predominantly perivascular lymphoid infiltrate also revealed blood vessels surrounded and infiltrated by a collar of atypical lymphoid cells associated with brisk mitotic activity. Small angulated cells with dark chromatin pattern and larger cells with open chromatin/prominent nucleolus predominated. Immunostains showed that the neoplasm had an unusual phenotype (CD2+, CD3-/+, CD5+, CD4+, CD8-, betaF1-, CD56-, TIA1-), best fitting a CD4+ Gamma/delta T-cell lymphoma. This type does not fit an NK cell neoplasm, which would be CD4-, CD8-, CF56+, TIA1+, nor the usual gamma/delta T cell lymphoma which would be CD4-, CD8-, CD5- and strongly TIA1+.

DISCUSSION: T Lymphocyte receptor has two heterodimer subunits, alpha/beta and gamma/delta. In the peripheral blood and lymph node gamma/delta receptor contribute 5% in the adult. Most gamma/delta T-cell lymphocytes are negative for CD4 and CD8. But a CD4+ and CD8-, gamma/delta T lymphocyte clone, which is a small population shows lytic activity against the Daudi lymphoma cell line and stimulates, Daudi cells to secrete more gamma interferon and GM-CSF when compared with CD4-, CD8-, gamma/delta lymphocyte. There has been only one previously reported case with this phenotype which presented with a cervical mass (Hum Patho 27:1370,1996.). Here we are reporting the second case of this disease which also had a unique presentation.

ENDOTIPSITIS: A NEW ERA OF HARDWARE INFECTIONS. ST Tekg, BD King, GE Oley, Internal Medicine, Marshall University, Huntington, WV

LEARNING OBJECTIVES: With the number of transjugular intrahepatic portosystemic shunt (TIPS) procedures increasing as a bridge to liver transplantation, recognition of infection of the TIPS and appropriate management is needed to prevent aborted transplantation and mortality.

CASE PRESENTATION: Two months after transjugular intrahepatic portosystemic shunt (TIPS) procedure and awaiting liver transplantation, a forty-five year old white male presented to the ambulatory clinic with fever, icterus and right upper quadrant pain. The patient complained of a five day history of fever, chills, myalgias, and cough. On physical exam, the patient was found to be lethargic, with scattered wheezes and no evidence of valvular heart disease. Laboratory findings demonstrated a cholestatic picture (alkaline phosphatase = 337 U/L, GGT = 576

U/L, and direct bilirubin = 6.6 U/L). New findings also included leukocytosis with a left shift. Chest x-ray and urinalysis were normal. Ultrasound of the abdomen did not show any biliary duct dilatation, ascites or abnormalities of the liver, pancreas or kidneys. Doppler of the TIPS revealed a patent shunt; however, there was evidence of a change in flow velocity from the proximal to distal end, consistent with stenosis and possible vegetation formation. Blood cultures were obtained on admission, which were positive for *Streptococcus viridans* and *salivarius*. The occurrence of fever, positive blood cultures and either a thrombus or vegetations present on the stent, or persistent bacteremia in patients with TIPS and no other detectable source of infection is diagnostic of endotipsitis. Our patient's presentation of fever, *Streptococcus viridans* and *salivarius* septicemia, and vegetation formation of the shunt is consistent with primary endotipsitis.

A CONFUSING PRESENTATION OF PRIMARY HYPERPARATHYROIDISM. SG Thompson, Medicine, Hospital of the University of Pennsylvania, Philadelphia, PA

LEARNING OBJECTIVES: 1. Recognize primary hyperparathyroidism in a patient with normal PTH. 2. Learn recent evidence about medical management of primary hyperparathyroidism.

CASE PRESENTATION: A 67 year old woman was brought to her physician by her sister for "confusion." Her history was notable for hypercholesterolemia, domestic violence, and depression. Medications were atorvastatin and aspirin. On physical exam, the patient was disheveled and tachycardic to 115, but the remainder of her exam was unremarkable. On mental status exam, she was alert, hypervigilant, and claimed to be "an African queen." Thought processes were tangential, and speech was pressured. Labs showed Ca 11.5 and albumin 3.8 with normal renal function. Urine toxicology was negative, and TSH was normal. Following psychiatric hospitalization and resolution of psychotic symptoms, the patient returned for followup. Review of her personal and family history failed to reveal nephrolithiasis or hypercalcemia. Repeat Ca was 11.2 and PTH was 7.2 pmol/L (1.3–7.6). 24 hr urinary Ca was 166 mg and creatinine clearance was 68 ml/min. DEXA scan is pending. The patient refuses a surgical procedure to correct her hypercalcemia.

DISCUSSION: We found mild hypercalcemia in this patient during evaluation for psychosis. We felt that the patient's elevated calcium warranted investigation but did not contribute to her psychosis. The profile of hypercalcemia and a normal, but inappropriately high, PTH in this patient is compatible with primary hyperparathyroidism or familial hypocalciuric hypercalcemia (FHH). Her 24 hr urinary Ca rules out FHH. Management of primary hyperparathyroidism is controversial. In 1990, an NIH Consensus Conference stated that all patients with primary hyperparathyroidism should be considered as surgical candidates, but that a subset of patients with asymptomatic primary hyperparathyroidism may be candidates for "surveillance." A recent observational study of individuals with primary hyperparathyroidism, most of whom were asymptomatic, demonstrated no change in serum Ca concentration, urinary Ca excretion, or bone mineral density after 10 years; 27% of patients developed symptomatic kidney stones. Observational data suggest calcium intake can and should be liberalized to 1,000 mg/d if the 1,25-dihydroxyvitamin D level is not elevated. Estrogen replacement therapy, demonstrated in a randomized, controlled trial to suppress bone turnover, reduce urinary Ca excretion, and increase bone mineral density in postmenopausal women with primary hyperparathyroidism, should be considered in appropriate candidates. The role of bisphosphonates in long term medical management of primary hyperparathyroidism remains to be studied.

RECOGNIZING EARLY WARNING SIGNS OF ANOREXIA AND BULIMIA IN A PRIMARY CARE PRACTICE. LW Tishler, Division of General Medicine, Brigham and Women's Hospital, Boston, MA

LEARNING OBJECTIVES: 1. Recognize early symptoms that may indicate an eating disorder. 2. Help patients to understand symptoms and behaviors that indicate anorexia or bulimia. 3. Facilitate referral and provide ongoing medical management as part of a team approach.

CASE PRESENTATION: A 30 year old woman presented to her primary care doctor in October of 1998 for routine care. She had numerous complaints, including "bowel problems" and infrequent stools. She was noted at that time to drink minimal water and eat little fiber. She was "thin but well" and weighed 117 lbs at 67" tall (BMI 18.3). On a return visit 7 months later, the patient complained of several months of amenorrhea. She was cold all the time; she developed sciatic pain while seated. "Eating issues" were noted and briefly discussed, but there was no scheduled follow up. When she presented to our clinic 4 months after the last primary care visit, she weighed 97 lbs (BMI 15.2). She met DSM IV criteria for anorexia nervosa, had osteoporosis of the lumbar spine, and required hospitalization for medical stabilization.

An 18 year old college student presented for routine primary care. She professed to be generally healthy, but was a reticent historian. Past medical history was most recently notable for an invasive GI workup in her home state. She had complained of constant nausea, mucous stools, and vomiting. There was no weight loss. She had a BMI of 24. There were no laboratory abnormalities. On further questioning, the patient revealed that she binged regularly, induced vomiting, abused laxatives and diet pills. She was also a binge drinker. No physician had previously asked about any of these behaviors with respect to her symptoms and she readily admitted that they might be related to her abdominal complaints.

DISCUSSION: Both of these patients met diagnostic criteria for eating disorders well before they were confronted with the diagnosis. They had presented numer-

ous times to primary care doctors and specialists with vague abdominal and constitutional symptoms. While eating disorders are often thought of as disease of children and adolescents, their incidence is increasing among young women in their 20's and 30's. Primary care doctors need to maintain a high index of suspicion about the presence of eating disorders in women of this age group. They should routinely question patients about eating behaviors and recognize the possibility of an eating disorder in women who present with certain complaints including amenorrhea, fatigue, weight loss, severe reflux, and bowel complaints with no other explanation. The primary care doctor can facilitate treatment by encouraging psychiatric referral, nutrition counselling, and seeing these patients frequently for monitoring and supportive care.

WHAT YOU DON'T KNOW CAN HURT YOU: SURREPTITIOUS USE OF CHINESE MEDICINES. J. Tung, General Internal Medicine, University of California, San Francisco, San Francisco, CA

LEARNING OBJECTIVES: 1) Obtain a thorough medication history, including nonprescription and herbal remedies. 2) Identify *Clostridium Difficile* infection in a patient with abdominal pain.

CASE PRESENTATION: A 72 year old woman with multiple medical problems including irritable bowel syndrome presented with worsening abdominal pain. She initially visited a community physician with complaints of increased abdominal cramping for two weeks. She also described occasional loose stools, nausea, and a loss of appetite. She denied vomiting, fevers, melena, or bloody stools. Her exam was reportedly normal, and she was sent home with loperamide. One week later she presented to our acute care clinic with no improvement in her symptoms. Stool studies were sent, including bacterial cultures, ova and parasites. No therapy was administered, and these studies subsequently returned normal. She followed up a few days later in the general medicine practice. Upon careful history taking, it was discovered that she had been taking a Chinese medication, a "stomach tonic", for over a month. Laboratory data at this time revealed a white cell count of 12,000 and normal liver function tests. Stool examination for *Clostridium Difficile* (C.Diff.) toxin was positive. She was asked to bring in the Chinese medicine, and one of the ingredients was identified as "SMZ" (sulfamethoxazole). The medication was discontinued, and she initiated a course of oral metronidazole. She returned to clinic two weeks later and reported resolution of her diarrhea and abdominal pain.

DISCUSSION: Patients with chronic abdominal pain are often difficult to evaluate. New symptoms or worsening of symptoms should prompt investigations into new medicines including nonprescription drugs and herbal remedies. Chinese medicines are frequently contaminated with antibiotics, steroids, analgesics, and other often-toxic substances. In this case the adulterant was declared in the packaging, however many herbal medicines come unlabelled. C. Diff. infection is associated with prolonged antibiotic use. While most inpatient cases of C. Diff. present fulminantly, outpatient infections may have a more indolent course, making recognition difficult. Symptoms may include chronic diarrhea, nonspecific abdominal pain, anorexia, and weight loss. Diagnosis is established with an assay for the toxin produced by the organism. Discontinuation of the offending antibiotic and administration of oral metronidazole is the treatment of choice.

FEVER AND ARTHRALGIAS IN A 19 YEAR OLD. K. Uomoto, A. Gomez, EF Yee, L. Mankin, Medicine, UCLA-San Fernando Valley Program, Sepulveda, CA

LEARNING OBJECTIVES: 1) To recognize systemic lupus erythematosus (SLE) in a young latino male. 2) To review the epidemiology (including the ethnic and socioeconomic predilection) of SLE.

CASE PRESENTATION: A previously healthy 19 year old Latino male presented with knee and lower back arthralgias, fever, and fatigue for 1 month. One week prior to admission the patient noted slowly progressive swelling in his lower extremities. He had no history of sexually transmitted diseases, IV drug use, recent travel or ill contacts. He was on no medications or herbal supplements. On physical exam the temperature was 37.3. The patient was alert, oriented and appropriate. He had no rash, alopecia or oral ulcers. The heart and lung exam were normal. The abdomen was diffusely tender to deep palpation but without rebound or guarding. There was trace pitting edema bilaterally. The ankles, knees, and lower back were tender to palpation but without erythema, increased warmth or edema. Admission laboratories were significant for a hemoglobin of 8.2 mg/dL, urinalysis with greater than 300 mg/dL of protein, 20 RBCs, 2 WBCs. The creatinine was 1.0 mg/dL. A 24 hour urine collection contained 7.4 grams of protein. The ESR was 75. The patient's course was complicated by worsening arthralgias, abdominal pain and renal function. A CT scan of the abdomen showed diffuse serositis. An ANA was positive with a 1:640 titer. A renal biopsy showed diffuse proliferative glomerulonephritis. The patient was diagnosed with Systemic Lupus Erythematosus (SLE) and treated with high dose steroids. The fever, abdominal pain and arthralgias resolved within 24 hours but the patient's creatinine continued to rise. The renal function eventually stabilized with the addition of cyclophosphamide. The post admission course was complicated by pulmonary embolism.

DISCUSSION: SLE affects women nine times more than men, except during childhood and over age 50 where the incidence is nearly equal. This case illustrates that SLE should be considered in young Latino men who present with arthralgias and renal impairment. SLE presents with higher frequencies among Latinos and other selected ethnic groups (African Americans and Asians) and earlier consideration of this diagnosis is warranted in these populations. This highlights the importance of ethnic consideration in the presentation of a not uncommon illness.

This patient had many features for SLE, but the diagnosis was not strongly considered early on because of the patient's age and gender. Early diagnosis is important because initiation of therapy is critical in preventing the disabling and life threatening complications of lupus. The patient's male gender is associated with worse prognosis. Prognosis may also be related to socioeconomic factors such as access to health care.

A PATIENT WITH SYSTEMIC LUPUS ERYTHEMATOSUS PRESENTING WITH AN "ATYPICAL" PNEUMONIA. EM Vargo, Medicine, Johns Hopkins Medical Services Corporation, Baltimore, MD

LEARNING OBJECTIVES: 1) Recognize the presentation of tuberculosis in an immunocompromised host; 2) Review the available data concerning the risk of active tuberculosis in lupus patients with and without immunosuppressive therapy; 3) Discuss the current recommendations for tuberculosis prevention in immunosuppressed patients.

CASE PRESENTATION: A 38-year old woman presented with a three week history of fever, shaking chills, night sweats and pleuritic chest pain. Her past medical history was significant for systemic lupus erythematosus (SLE) diagnosed four months previously, treated with prednisone 40 mg daily. On examination, she was a moderately-ill, cushingoid woman. Her temperature was 38.7 degrees Celsius, pulse 112/minute, respirations 24/minute, and blood pressure 160/94 mmHg. Lungs were noted to be clear to auscultation. The remainder of the examination was unremarkable. Her white blood cell count was 11.6/mm³. Chest x-ray revealed bilateral upper lobe infiltrates. Sputum obtained for Gram's stain, acid-fast bacillus stain, and silver stain showed no organisms. The patient was treated with intravenous cefuroxime and initially defervesced with resolution of many of her symptoms. However, she soon began to again manifest fever despite antimicrobial therapy. Suspecting atypical pneumonia, the inpatient physicians added high dose erythromycin to her regimen. She again defervesced after 24 hours. She was discharged to home on oral azithromycin. She was seen in the outpatient clinic one week later for follow up. At that time, the patient complained of continued fever and cough. Her temperature was 39 degrees Celsius, pulse was 120/minute, respirations 24/minute, and blood pressure was 175/92 mmHg. Examination notably revealed decreased breath sounds in the posterior upper lung fields and amphoric breath sounds in the apical regions. Review of the sputum culture data from the hospitalization revealed growth of *Mycobacterium tuberculosis* on day 12 of culture. Because of her respiratory distress and continued cough she was readmitted to the acute care hospital and placed in pulmonary isolation. She was started on a four-drug antituberculous regimen and gradually improved. On further questioning, the patient related a history of tuberculosis exposure as a child: her sister had died of pulmonary tuberculosis. Skin testing during childhood was reportedly negative and she did not receive prophylactic antituberculous therapy. Tuberculin skin testing was not performed prior to the initiation of corticosteroid therapy when the patient was diagnosed with SLE four months previously.

DISCUSSION: Patients with SLE are at increased risk for active tuberculosis, whether or not they are treated with immunosuppressive therapy. SLE patients tend to more often manifest with extrapulmonary tuberculosis or atypical presentations of pulmonary tuberculosis.

ATTEMPTED SUICIDE FOLLOWED BY FEVER AND RASH. EM Vargo, Medicine, Johns Hopkins Medical Services Corporation, Baltimore, MD

LEARNING OBJECTIVES: 1) Review the differential diagnosis of fever and rash in a patient with a history of intravenous drug use; 2) Recognize the rash associated with disseminated gonococcal infection; 3) Discuss the host and bacterial characteristics of disseminated gonococcal infection.

CASE: A 25-year old woman presented to the emergency department after a suicide attempt in which she lacerated both wrists. Her urine toxicology screen was positive for cocaine, opiates, cannabinoids, salicylates and acetaminophen. The wounds were sutured, and tetanus-diphtheria vaccination was given. She was admitted to the psychiatry service for detoxification and treatment of depression. Two days after admission, she became febrile with a temperature of 38.5 degrees Celsius, and painful, red and violaceous nodules were noted on her skin. She denied other constitutional symptoms, headache, chest pain, shortness of breath, abdominal pain, dysuria, or vaginal discharge. Her past medical history was significant for hepatitis C infection, and prior hepatitis B infection. She had been treated for gonorrhea and chlamydia cervicitis in the past, and was noted to be HIV negative one month prior to admission. Medications at the time of transfer were oxazepam and buprenorphine. Her social history was significant for the heavy use of both tobacco and alcohol, as well as intravenous heroin and cocaine use. She has supported herself through prostitution for the past five years. On examination, she was a thin, disheveled young woman. Temperature was 38.7 degrees Celsius, pulse was 90/minute, and blood pressure was 130/61 mmHg. Head and neck examination revealed poor dentition and a normal fundoscopic examination. She had no lymphadenopathy. Pulmonary, cardiac and abdominal examination were unremarkable. Examination of her extremities revealed sutured superficial skin wounds on both wrists, and nodules which appeared violaceous on an erythematous base without pustules or ulcers. There were nine lesions counted, which were scattered on the palms of her hands, arms, and legs. There were no warm, tender or swollen joints. Pelvic examination revealed no cervical motion tenderness or adnexal masses. On laboratory evaluation, the white blood cell count was 9.3/mm³, with normal hematocrit and platelet count. Serum chemistries were within normal limits. The urinalysis was normal, and a chest x-ray

was unremarkable. Blood cultures were obtained, and empiric antibiotics were administered. Serum rapid plasma reagin and HIV tests were obtained. A transthoracic echocardiogram was obtained and was normal. One day after transfer to the internal medicine service, the blood cultures became positive for gram-negative diplococci, subsequently identified as *Neisseria gonorrhoeae*. The antibiotic regimen was changed to cover for disseminated gonorrhea and the patient rapidly improved.

CALCIPHYLAXIS IN CHRONIC RENAL FAILURE. GN Varma, PG Jenkins, Department of Internal Medicine, Sinai Samaritan Medical Center, Milwaukee, WI

LEARNING OBJECTIVES: Recognize and diagnose the clinical condition of calciphylaxis in a patient with chronic renal failure.

CASE: A 46-year old African American female with end stage renal disease and obesity presented with tender subcutaneous nodule in the abdomen. She had been on dialysis for 12 years, and was on anticoagulants for recurrent fistula graft thrombosis. She was symptomatically treated initially in the out-patient clinic. In the next few weeks, she developed multiple painful nodules in the abdomen and breast. Lab values showed Calcium 10.5–11.5 mg/dL, Phosphorus 4.5–5.5 mg/dL, Intact PTH 433–651 pg/mL, Albumin 3–3.5 g/dL, BUN 40–50 mg/dL, and Creatinine 7–9 mg/dL. ANA and Anti-DNA were negative. Biopsy of nodule showed a medium sized artery with medial calcification and intimal thickening due to fibrosis. It also showed occlusion of an arterial branch by an organizing thrombus. This confirmed the diagnosis of calciphylaxis. The nodules were removed surgically. In the post-operative period, wound healing was poor and the patient developed secondary infections of the wound. This was treated with appropriate antibiotics. A subtotal parathyroidectomy was performed to control hyperparathyroidism. The PTH level decreased to 70 pg/mL post-operatively, and increased to 355 pg/mL in a few months. The patient has continued to develop multiple skin and subcutaneous lesions despite the treatment. Her condition is being managed with local wound care, analgesics and antibiotics, and surgery when necessary. Her phosphate, calcium, and PTH levels are monitored on a regular basis.

DISCUSSION: Calciphylaxis is a rare, painful, and debilitating disorder commonly seen in patients with end stage renal disease who are on dialysis or who have recently received a renal transplant. Early diagnosis and treatment may interrupt the progression of the disease process. Hyperparathyroidism, hyperphosphatemia, normal or increased plasma calcium concentration, and Vitamin D supplementation are some of the factors commonly seen in these patients. Epidemiological factors include white race, morbid obesity, administration of warfarin, and low serum albumin. The usual presentation is ischemic necrosis that affects the dermis, subcutaneous fat and, less often, muscle. The diagnosis is confirmed by skin biopsy which shows arterial occlusion and calcification of the arterial wall. Common complications include sepsis, gastro-intestinal hemorrhage, and coronary artery disease. Good control of calcium and phosphate levels, and regular wound care are important. Parathyroidectomy has shown to improve the condition in some patients.

EARLY DETECTION AND TREATMENT OF MYXEDEMA COMA. A Verna, WA Harb, S Gupta, Internal Medicine, Oakwood Hospital, Dearborn, MI

LEARNING OBJECTIVES: 1. Recognize Myxedema Coma in an unconscious patient. 2. Commence early treatment based on clinical suspicion.

CASE PRESENTATION: A 51 year old male with past medical history of Diabetes and Bipolar disorder was brought into ER in an unresponsive state with O₂ saturation of 80%. He was intubated because of respiratory failure and coma. On physical examination his Temp. was 97°F, BP 127/75 mm Hg., Pulse 60/min., Respiration 16/min. (on vent). Skin was cool, thick & doughy with dry, brittle hair. Pupils were equal & sluggishly reactive to light. Rhonchi were heard in both lung fields on chest auscultation. Cardiac examination revealed displaced apical impulse with muffled heart sounds, S4 was heard at the apex. Abdomen was obese, soft and non-tender, bowel sounds were present. There were some abrasions on the extremities with 1+ pulses felt bilaterally. Neurologically, he was unresponsive to all stimuli. He was spontaneously moving all the extremities and had intact brain stem reflexes. Lab data revealed WBC count of 15.3 with left shift. Na 137, K 3.8, Total CO₂:24, Blood Glucose 143, BUN 19 & Serum creatinine 1.0. Total TSH was 69.80 & Free T4 was 0.2. Cardiac enzymes were normal. Chest X-ray showed left lower lobe consolidation. EKG showed Sinus bradycardia (60), non-specific ST-T changes & QT prolongation. CK was 1494. CT of Head revealed mild atrophy, right maxillary & ethmoid sinusitis. Sputum culture grew streptococcus pneumoniae. 2-D Echo showed normal heart chambers, ejection fraction of 55% & mild pericardial effusion. CSF analysis was normal. Hypo & Hyperglycemia, drug overdose, metabolic derangements & myocardial infarction were excluded as the etiology of coma. Mechanical ventilation was started immediately upon presentation. Intravenous thyroxine 100 mg. Q8h. was started for the treatment of severe hypothyroidism, also IV Hydrocortisone was initiated to prevent adrenal crises. Levofloxacin & erythromycin were given for the treatment of pneumonia. Patient's condition remained critical despite all support. He was terminally weaned and given comfort measures as requested by the family.

DISCUSSION: Myxedema Coma is a very rare, but fatal condition. One should always be cognizant about its presence in an unresponsive patient, especially in conjunction with relative or absolute hypothermia. Prompt therapy should be instituted intravenously when clinical suspicion arises without awaiting for lab results.

ACUTE HIV-1 SYNDROME PRESENTING WITH A FALSE-POSITIVE MONOSPOT. JA Vidrih, RP Wolensky, KA Freedberg, General Internal Medicine, Boston Medical Center, Boston, MA; Infectious Disease, Massachusetts General Hospital, Boston, MA

LEARNING OBJECTIVES: 1. Assess the need to take an HIV risk history when a patient presents with a viral syndrome. 2. Recognize that a positive monospot does not exclude the diagnosis of Acute HIV-1 Syndrome.

CASE PRESENTATION: A 29 year-old male truck driver presented to the Urgent Care Clinic at Boston Medical Center with complaint of fevers, malaise and sore throat of two days duration. Physical exam was notable a temperature of 102°F, pharyngeal erythema without exudate and cervical adenopathy. Initially treated with Penicillin V for streptococcal pharyngitis, he returned 2 days later when he developed a rash on his face, neck and extremities. With no clinical improvement, a negative culture for Group A Strep and, a new rash, a sexual history was obtained which revealed unprotected sex with men. The patient underwent further testing which revealed a positive monospot (Epstein-Barr virus (EBV) heterophile), a negative HIV-1 ELISA and an HIV-1 RNA >500,000 copies/mL. Further testing revealed EBV serologies that were consistent with past infection or reactivation (high titer EBV Viral Capsid Antigen(VCA) IgG Ab), but not with recent infection (low titer EBV VCA IgM Ab). He was begun on stavudine, epivir and nelfinavir for acute HIV-1 syndrome 13 days after initial presentation.

Date	HIV-1 ELISA	HIV-1 RNA, copies/ml	CD4/uL	Monospot	EBV VCA IgM Ab	EBV VCA IgG Ab
5/20/99	Negative	>500,000		Positive		
6/2/99	Positive	26,000	512			
7/12/99					<1:10	>1:1280
10/28/99		<50	744			

DISCUSSION: Acute HIV-1 syndrome is an abrupt-onset, febrile illness defined by a high HIV-1 RNA and a negative HIV-1 antibody in the setting of clinical risk for HIV infection. Up to 90% of those newly infected with HIV will report an illness characterized by fevers, pharyngitis, lethargy, arthralgias and rash. Recognizing acute HIV-1 syndrome has important implications from both an individual patient and public health perspective. Early studies indicate that treatment of acute HIV-1 syndrome with combination antiretroviral therapy may preserve normal immune function, limit viral dissemination and potentially decrease transmission of the virus to others. However, the clinical similarity between acute HIV-1 syndrome and infectious mononucleosis makes the diagnosis difficult. This case suggests that clinicians considering the diagnosis of infectious mononucleosis should obtain a detailed sexual and drug use history and proceed to HIV RNA testing if clinically indicated. The finding of a positive monospot in a patient with a severe viral syndrome does not exclude the diagnosis of acute HIV-1 syndrome.

HELICOBACTER PYLORI AND GASTRIC MALT LYMPHOMA. DL Wahnner-Roedler, Internal Medicine, Mayo Clinic

LEARNING OBJECTIVES: To recognize that 1) gastric lymphoma of mucosa associated lymphoid tissue (MALT) almost always occurs in patients with underlying *Helicobacter pylori* (HP) infection; 2) eradication of HP is the first-line treatment of gastric MALT lymphoma; 3) regular endoscopic follow-up is mandatory to monitor treatment response.

CASE: A 79-year-old white male patient presented to his LMD in August of 1995 with weight loss and epigastric pain. A stomach x-ray was normal. Two weeks later the patient experienced a massive upper GI bleed.

Esophagogastroduodenoscopy (EGD) showed diffuse gastritis. Biopsies revealed a lymphoid infiltrate consistent with MALT lymphoma. Serology was positive for HP. CT scans of chest and abdomen were normal, bone marrow examination was negative for lymphoma. The patient was treated with Prilosec, Flagyl, tetracycline, and Pepto Bismol for 2 weeks with resolution of his symptoms. He was first seen at our institution in June 1996 with epigastric discomfort. EGD showed areas of erythema with edema of the mucosa scattered throughout the stomach. Multiple random biopsies revealed chronic gastritis, no HP, and minimal involvement with MALT lymphoma. In absence of any systemic disease and with his past history of treatment response he was again treated with tetracycline, Flagyl, Pepto Bismol, and Prilosec. Re-scoping in January 1997 revealed numerous areas of superficial ulcerations in the stomach and biopsies showed persistent MALT lymphoma. The patient was started on chlorambucil, 6 mg a day. Serial EGDs at six-month intervals with random biopsies were negative. Chlorambucil was discontinued after one-year treatment. There was no evidence of recurrent disease on EGD when the patient was last examined in August 1999.

DISCUSSION: HP has been implicated as a cause of gastritis, peptic ulcer disease, gastric carcinoma, and most recently in the development of MALT lymphoma. Although normal gastric mucosa is devoid of lymphoid tissue, chronic antigenic stimulation of HP may lead to the development of lymphoid follicles, and low-grade B-cell lymphoma has been postulated to arise in this mucosa associated lymphoid tissue. Patients often present with nonspecific upper intestinal discomfort or gastric bleeding. Endoscopic appearance may suggest benign gastritis and extensive biopsies may be required for diagnosis. Early HP eradication is essential for cure as untreated low-grade MALT lymphomas may progress to aggressive high-grade gastric lymphomas. Regular endoscopic follow-up with multiple biopsies is mandatory to monitor treatment response. If there is incomplete or no response to antibiotic therapy, chemotherapy should be undertaken or surgery if the lesion appears resectable.

AN UNCOMMON CAUSE OF CONGESTIVE HEART FAILURE. MD Wang, A Gomez, E Yee, Medicine, UCLA-Olive View Medical Center, Sylmar, CA

LEARNING OBJECTIVES: 1) To review the presentation of coarctation of the aorta. 2) To recognize socioeconomic factors that may delay the presentation of a congenital disorder into adulthood

CASE PRESENTATION: A 26 year old male from southern Mexico presented with a three month history of progressive dyspnea on exertion associated with right upper quadrant abdominal pain. He reported that 6 months ago he felt strong and could work endlessly. His exercise tolerance had since diminished progressively, and lately he could only walk 1-2 blocks. His RUQ pain was described as a pulsatile, pressure-like pain. Other symptoms included cough, two pillow orthopnea, PND, and intermittent nausea. He did report a doctor telling him when he was fourteen that he had some kind of heart problem. His sister and his aunt have heart problems. He was seen two months earlier for his symptoms, was found hypertensive and given an antihypertensive medication. On presentation, he was afebrile, tachycardic at 125, hypertensive $151/85$, and saturating at 95% on room air. His JVP 13 cm, carotids 2+, had a laterally displaced apical impulse, a left parasternal heave, a holosystolic blowing murmur at the apex that radiated to the axilla, and a similar murmur at the left lower sternal border but that enhanced with inspiration. He had rales half way up the lung fields. His abdominal exam revealed a 10 cm liver by percussion, a tender RUQ, but no signs of ascites. His radial pulses were 2+ bilaterally, but distal pedal and femoral pulses were unpalpable, though measurable by Doppler examination. He had 2+ ankle edema. His EKG revealed sinus tachycardia of 120, a right axis deviation of 140 degrees, left ventricular hypertrophy, left atrial enlargement, and nonspecific ST and T wave changes. His chest radiograph revealed a markedly enlarged heart with congestive heart failure. His laboratory results were significant for a creatinine of 0.9, AST 158, ALT 191, total bilirubin 1.6, direct bilirubin 0.1, albumin 2.9, and INR 1.45. His 2-D echocardiogram revealed severe TR and MR with an estimated pulmonary artery pressure of 90 mmHg, an estimated ejection fraction of 40%, with marked 4 chamber dilatation and global hypokinesis, and 2 flow jets through the tricuspid valve. A cardiac catheterization was performed. The diagnosis was coarctation of the aorta distal to the subclavian artery and possible 4 leaflet tricuspid valve. The patient was asymptomatic throughout his hospitalization and scheduled for surgery. **DISCUSSION:** This case illustrates that lack of routine healthcare in childhood can lead to a delay in presentation of an easily recognizable congenital cardiovascular disorder—coarctation of the aorta. It also illustrates the need for internists who care for immigrants or patients of lower socioeconomic status to consider congenital cardiovascular malformations as the etiology of symptoms in adults.

DOCTOR, I HURT EVERYWHERE. MD Wang, A Gomez, E Yee, Medicine, UCLA-Olive View Medical Center, Sylmar, CA

LEARNING OBJECTIVES: 1) To recognize that abuse, depression, and fibromyalgia are interrelated phenomena 2) To distinguish fibromyalgia from other disorders 3) To discuss the management of fibromyalgia

CASE: A 30 year old female presented with a ten year history of diffuse arthralgias and myalgias. She had constant pain of all muscles and joints, exhaustion, and dry eyes. She reported morning stiffness in her neck and upper back lasting for 60-90 minutes. There was no history of joint swelling, redness, or warmth. While her pain limited her activity, she had no actual weakness. Her past medical history was significant for depression, anorexia nervosa, and menometrorrhagia. During her childhood she had been severely verbally and emotionally, but not sexually or physically, abused. She had no history of thyroid disease, diabetes, seizures, renal dysfunction, photosensitivity, oral ulcers, or rash. Her medications included bupropion, oral contraceptives, acetaminophen, and vitamins. She denied the use of alcohol or illicit drugs. On physical examination, she appeared comfortable, but depressed and teary-eyed. Her skin revealed no rashes. None of her joints had any swelling, erythema or warmth, although her metacarpophalangeal and proximal interphalangeal joints were mildly tender. Thirteen of eighteen tender points for fibromyalgia were positive. Her neurologic examination was normal with no weakness in any muscle group. The Schirmer test was normal. The patient was presumptively diagnosed with fibromyalgia, but treatment withheld until her work up completed. She was scheduled for a follow up visit which she will have in the near future. Laboratory test results : CBC, electrolyte panel, and liver tests all normal. Calcium 9.8, creatine kinase 62, aldolase 3.7, glucose 86, ESR 9 mm/hr, rheumatoid factor negative, TSH normal. Radiographs of her chest, hands, feet, pelvis and cervical spine were not suggestive of any inflammatory or degenerative process.

DISCUSSION: Fibromyalgia is a diagnosis of exclusion supported by the absence of the following: history of myopathic drugs; objective weakness and arthritis on physical examination; evidence of systemic inflammatory processes or endocrinopathies on laboratory investigation. While most cases of fibromyalgia are idiopathic, the disorder can be associated with both connective tissue diseases and depression. Fibromyalgia is less commonly associated with a history of abuse than are somatoform disorders. This case, however, illustrates that emotional trauma, depression, and fibromyalgia seem to be a continuum of presentations that interplay with a history of depression, emotional trauma, and somatic symptoms consistent with fibromyalgia. The management of fibromyalgia will be discussed in more detail, but includes patient education, stress management, counseling, pharmacotherapy, exercise, and biofeedback.

THE USE OF THE PHYSICAL EXAMINATION IN THE EVALUATION OF A MEDICAL CRISIS. JG Wiese, Dept. of Internal Medicine, Univ. of California at San Francisco, San Francisco, CA

LEARNING OBJECTIVES: To incorporate physical exam findings in the diagnosis and treatment of patients in medical crisis.

CASE PRESENTATION: A 68-year-old man was admitted to the hospital for sub-sternal chest pain. The patient described progressive shortness of breath and a non-radiating, sub-sternal chest pain beginning twelve hours prior to admission. His past medical history was notable for COPD and long-standing tobacco use, but there was no history of exertional angina or coronary artery disease. With the exception of diffuse end-expiratory wheezes, the physical examination performed in the emergency department was normal. All laboratory values including a troponin, CXR, and EKG were normal. The patient was admitted to a telemetry floor to exclude myocardial ischemia. Twelve hours after admission, the patient had a syncopal episode during micturition, and a medical code was initiated. The patient had fully recovered by the time the physician had arrived. A repeat physical examination, however, revealed a persistently split S2 with a loud P2. A spiral CT revealed a large saddle pulmonary embolus. Thrombolytics were administered and the patient transferred to the ICU. Over the next two days, the shortness of breath and chest pain resolved. There were no additional syncopal episodes and no complications from thrombolytic therapy.

DISCUSSION: In many situations the physical exam can provide life-saving clues in the diagnosis and treatment of a medical crisis. Because the syncope occurred during micturition, it would have been reasonable to ascribe the event to neurogenic syncope. The repeat physical examination, however, provided evidence of an acute elevation in pulmonary artery pressure, prompting evaluation for pulmonary embolism. Aside from tobacco use, this patient had no risk factors for pulmonary embolism, and with the exception of the chest pain there was no historical data to suggest PE. The history of present illness often provides clues as to what the physical examination will show. This case, however, illustrates that while the history can be useful in guiding the exam, the physical exam remains an independent evaluative measure, and should be carefully performed. Fidelity to the diagnostic method of considering all possibilities and using the physical examination to assess each possibility was life-saving in this case.

MYOPATHY DUE TO A LOVASTATIN-INDINAVIR INTERACTION. JG Wiese, Internal Medicine, Univ. of California at San Francisco, San Francisco, CA

LEARNING OBJECTIVES: To recognize the importance of monitoring for drug toxicity from non-HIV medications upon institution of therapy for HIV.

CASE PRESENTATION: A 58-year-old HIV positive man presented with one week of progressive weakness. He had been seen in the clinic three days earlier, at which time normal CBC and electrolyte panels were obtained. There were no gastrointestinal, neurologic or cardiovascular complaints. The patient's past medical history was notable for hypertension and hypercholesterolemia for which he had been taking atenolol and lovastatin. There had been no recent alteration in the doses of atenolol and lovastatin. There was no history of opportunistic infections and a CD4 count one month prior was 275. A viral load at that time was 30,000 copies, prompting the addition of indinavir to his lamivudine-stavudine regimen. Physical examination revealed normal vital signs. There was objective proximal muscle weakness without atrophy or fasciculation. The remainder of the examination was normal. Repeat laboratory studies revealed an elevation in the creatinine (3.5 mg/dl), AST/SGOT (350 U/L), ALT/SGPT (180 U/L) and creatine kinase (CK >50,000 U/L). Rhabdomyolysis was diagnosed, and the patient was started on aggressive hydration with alkalinization of the urine. Lovastatin was discontinued. Over one week, the patient's laboratory abnormalities and physical examination returned to normal.

DISCUSSION: Rhabdomyolysis due to HMG-CoA reductase inhibitors (statins) is not uncommon, occurring in 0.7% of all patients receiving the therapy. This case illustrates that statin myopathy can occur due to elevated statin levels because of a drug interaction. The addition of the protease inhibitor decreased the clearance of lovastatin, presumably by inhibiting the hepatic P-450 CYP3A4 enzyme that is responsible for metabolism of both drugs. Similar drug interactions have been described with the combination of HMG-CoA reductase inhibitors with drugs that inhibit the CYP3A4 enzyme (gemfibrozil, cyclosporin, ketoconazole, nefazodone, and niacin). As the life expectancy of the HIV patient is extended, management of other medical disease such as hypercholesterolemia will become increasingly important. Because the medications used in HIV therapy commonly interact with the metabolism of other medications, physicians must exercise special vigilance in monitoring for side effects of all of the patient's medications. Where possible, medications that do not share the same method of metabolism and clearance as HIV medications should be chosen.

THAT DARN DOG! AN UNUSUAL PRESENTATION OF STAPHYLOCOCCAL ENDOCARDITIS. AL Wolfert, JZ Engel, Division of General Internal Medicine, Vanderbilt University, Nashville, TN

LEARNING OBJECTIVES: 1) Recognize the varied presentation of bacteremia in the elderly. 2) Distinguish possible versus definite endocarditis based on clinical criteria. 3) Recognize the high mortality associated with Staphylococcal Aureus endocarditis.

CASE: An 87 year old white female with history of hypertension and osteoporosis presented to the outpatient clinic with the chief complaint of "hurting all over". She had been well until ten days prior when she had tripped over her dog, landing on her backside. She sustained a small abrasion to her lower leg, had minimal pain, and resumed her normal activity. Three days later, her hip became extremely painful. Radiographs showed no fracture. She subsequently developed

progressive pain, diffuse myalgias, anorexia and dyspnea. She stayed in bed for the week prior to her office visit. She denied fevers or chills. Physical examination revealed a frail female with severe kyphosis, in mild distress. Her blood pressure was $88/50$, respiratory rate 32, pulse 90 and she was afebrile. Auscultation revealed no murmurs and clear lungs. Her right thigh was tender to the touch but there was no swelling or ecchymosis. There was a small abrasion on the right lower leg. Chest XRay revealed kyphosis, diaphragmatic eventration and a possible infiltrate. Oxygen saturation was 89% on room air. She was admitted to the hospital and ruled out for pulmonary embolism. She was started on treatment for community acquired pneumonia and placed on intravenous fluids for hypotension attributed to dehydration.

The first night in the hospital, she spiked a temperature to 101. Blood cultures subsequently grew *Staphylococcus Aureus* and remained persistently positive despite appropriate treatment. A trans-thoracic echocardiogram showed no vegetations. The patient's hospital course became increasingly complicated, with development of renal failure, bilateral pneumonia, and multiple right sided cerebral infarcts. Patient expired eight days after admission of multi-organ system failure. An autopsy revealed a 6 mm vegetation on the mitral valve, right temporal and occipital subarachnoid hemorrhage as well as multiple circular densities in the brain consistent with abscesses.

DISCUSSION: The diagnosis of acute bacterial endocarditis (ABE) should be entertained in any patient with persistent bacteremia without a source, especially in the presence of embolic phenomena. While nonspecific symptoms are common in all patients, the elderly are more likely to present with no fever and without a detectable murmur. Both Duke and Beth Israel have developed criteria for the diagnosis of ABE which include pathologic and clinical components. The mortality rate of *Staph Aureus* bacteremia (SAB) ranges from 15–40%. Patients over the age 65 have more than twice the mortality from SAB than their younger counterparts.

ERECTILE DYSFUNCTION: WHEN VIAGRA® IS NOT THE RIGHT ANSWER. W Wong, W Golanter, Section of General Internal Medicine, University of Illinois at Chicago Medical Center, Chicago, IL

LEARNING OBJECTIVES: 1) Recognize hyperprolactinemia as a rare cause of erectile dysfunction; and 2) Apply a cost effective screening strategy for diagnosing hyperprolactinemia in patients presenting with erectile dysfunction.

CASE: A 60 year old man with a history of diabetes mellitus presented to his primary care provider's office with the complaint of impotence. The patient had a history of normal erections until two months prior to presentation. He reported a moderate loss of libido, but denied depression or anxiety. The patient denied smoking or alcohol intake. Physical examination was unremarkable including normal genital, prostate, neurological, and peripheral vascular examinations. Serum glucose was 158 with a glycosylated hemoglobin of 9.9 mg/dl; electrolytes and renal function were normal. Cholesterol panel and thyroid function tests were normal. Serum testosterone was found to be 81 ng/dL (normal = 350–720 ng/dL). Additional laboratory evaluation revealed a prolactin level of 650 ng/mL (normal = 17–25 ng/mL). Magnetic resonance imaging of the head revealed an intrasellar pituitary mass lesion consistent with a pituitary macroadenoma. The patient was started on bromocriptine, however discontinued therapy due to intolerable gastrointestinal side effects, and was subsequently treated with cabergoline.

DISCUSSION: Erectile dysfunction is a common condition which affects an estimated 10 million men in the United States and is a common complaint encountered by primary care clinicians. Since the prevalence of hyperprolactinemia in patients with erectile dysfunction is low (1–5%), routine determination of prolactin is not justified. Screening for hyperprolactinemia should be limited to men with low sexual desire, gynecomastia, galactorrhea, or low testosterone. Applying this diagnostic strategy proves to be a cost effective means of identifying cases of erectile dysfunction due to hyperprolactinemia.

LACTATE DEHYDROGENASE IS SENSITIVE BUT NOT SPECIFIC IN DIAGNOSING PNEUMOCYSTIS CARINII PNEUMONIA. D C Yao, Division of General Internal Medicine, Johns Hopkins University School of Medicine, Baltimore, MD

LEARNING OBJECTIVES: (1) Recognize that serum lactate dehydrogenase (LDH) is a sensitive but not specific noninvasive test for diagnosing *Pneumocystis carinii* pneumonia (PCP). (2) Assess sensitivity, specificity, pre and post-test probabilities, and the likelihood ratio of disease. (3) Understand the diagnostic implications of these statistics.

CASE: A 45 year-old African-American man without significant past medical history presented with shortness of breath for two days. He also complained of fever, chills, night sweats, and productive cough with brown sputum. The patient had been using intravenous drugs for the past 15 years. An HIV test was negative 3 years ago. On physical exam, T = 99.1 F, HR = 100, BP = $146/84$, RR = 32, and O_2 Sat = 91% on room air. He was a thin man in moderate respiratory distress, using accessory muscles. The lung exam revealed crackles, rhonchi, and egophony bilaterally. Laboratory tests showed a white blood count of 1.8; LDH = 1540, AST = 170, ALT = 47. Chest x-ray revealed a bilateral diffuse infiltrate. Differential diagnosis included community acquired pneumonia versus PCP, tuberculosis, endocarditis, and possible HIV. Intravenous trimethoprim/sulfamethoxazole was started for presumed PCP. The hospital course included four out of four positive blood cultures for *Streptococcus pneumoniae*; induced sputa were negative for PCP by silver stain. Subsequent tests were positive for antibodies to hepatitis C virus and HIV.

DISCUSSION: The use of elevated LDH has been reported to correlate with the presence of PCP infection. The literature is reviewed in this case to delineate the utility of using LDH to diagnose PCP. Elevated LDH, hypothesized to originate from damaged lung tissue, can be found in PCP infections. However, elevated LDH can also be seen in other diseases, including bacterial pneumonia, tuberculosis, or lymphoma. The diagnostic test has a high sensitivity (83 to 100%), making it a good test to rule out PCP with a negative result (i.e., a high negative predictive value). However, with a specificity of 25 to 85%, it is a poor test to rule-in, or diagnose definitively, PCP even at a high level. Listed below is the likelihood ratio and post-test probabilities from one study, with a pre-test probability (prevalence of disease) of 57%, sensitivity of 92%, and specificity of 25%, showing the utility of LDH as a diagnostic test for PCP based on different levels.

LDH value (U/L)	Likelihood Ratio	Post-test Probability, %	Diagnostic Impact
>1,500	3.8	83	Rule in
1,000–1,499	3	80	
550–999	0.5	39	
<550	0.3	28	Rule out

AUTONOMIC DYSFUNCTION IN AN ELDERLY FEMALE. M Zulfiqar, M Bustamante, RD Hobbs, Internal Medicine, Oakwood Hospital, Dearborn, MI

LEARNING OBJECTIVES: To recognize the multifactorial nature and spectrum of illnesses that can lead to autonomic dysfunction in the elderly.

CASE PRESENTATION: A 76 year-old female with a past diagnosis of autonomic dysfunction and suspected Shy-Drager syndrome, presented to the hospital with orthostatic hypotension and difficulty swallowing both liquids and solids for the past six months. She had a recent history of dehydration, had lost 22 pounds in three months, and was profoundly depressed. She was bedridden. Her history was significant for hypertension, chronic insomnia, pulmonary embolus, myocardial infarction, possible Parkinson's disease and a 40-year history of depression that had been treated five times with ECT. Her chronic medications included nortriptyline 25 mg tid, fluoxetine 20 mg bid, lisinopril 40 mg qd, clonidine 0.2mg tid, zolpidem 10mg qhs, warfarin 2.5 mg qd, amlodipine 10 mg qd.

Physical exam revealed an elderly female with severe orthostatic hypotension, dehydration, difficulty swallowing, a distended bladder, and profound depression. There was no evidence of Parkinson's disease. She was rehydrated and nortriptyline was discontinued. She improved although orthostatic hypotension persisted to a lesser degree. Esophageal studies were normal. Bladder dysfunction disappeared. In the following days the family confided that she had recently been on 17 medications and had been diagnosed with complications related to polypharmacy. Her medications were a compromise between herself and her physicians. She refused to stop several medicines because she was convinced of their efficacy. She also had been stockpiling medications since 1962 and self-medicating. These medications included flurazepam and OTC sleep aids.

DISCUSSION: This case illustrates several points. First, it underscores the problem of drug interactions and polypharmacy. Her use of tricyclic antidepressants and antihypertensives all contributed to her orthostatic hypotension. Second, self-enforced bed rest and deconditioning magnified the problem. Third, since an asymptomatic orthostatic drop in blood pressure occurs in a large number of normal elders, age-related physiologic changes can be postulated as a contributing factor separate from any pathologic changes. Fourth, surreptitious use of medications, sedatives with long half-lives, and unknown over-the-counter medications all contributed to her problem and made diagnosis and treatment difficult. Lastly, this case underscores the multifactorial nature of illnesses that while less common in the young are so common in the elderly.

A PROSPECTIVE STUDY OF DIETARY LACTOSE AND OVARIAN CANCER. KM Fairfield, DJ Hunter, GA Colditz, CS Fuchs, FE Speizer, WC Willett, SE Hankinson, Dept of Med, Beth Israel Deaconess Medical Ctr; Depts of Nutrition & Epidemiology, Harvard Sch of Public Health; Dept of Med, Channing Laboratory, Brigham and Womens Hospital, Boston, MA

PURPOSE: The milk sugar lactose is a hypothesized risk factor for ovarian cancer because of direct effects of its metabolites on the ovary or by inducing hypergonadotropic hypogonadism. Women are encouraged to consume dairy products to prevent osteoporosis. We examined the relation of lactose consumption to ovarian cancer risk.

METHODS: We prospectively assessed consumption of lactose, milk, and milk products in relation to ovarian cancer risk among 80,326 women in the Nurses Health Study who had no history of cancer other than non-melanoma skin cancer. Food frequency questionnaires were included in 1980, 1984, 1986, and 1990. Women reported other known and suspected ovarian cancer risk factors on biennial mailed questionnaires from 1976–1996. Newly reported ovarian cancer diagnoses were documented by review of medical records. During 16 years of dietary follow-up (1980–1996), 301 cases of invasive epithelial ovarian cancer were confirmed, of which 174 were the serous subtype.

RESULTS: For all subtypes of invasive ovarian cancer, we observed a 44% greater risk for women with the highest consumption of lactose compared to the lowest (multivariate relative risk (RR) 1.44, 95% confidence interval (CI) 1.01 to 2.07; p for trend 0.07). We observed a two-fold increase in risk of the serous ovarian cancer subtype among women with the highest lactose consumption compared to the lowest (RR 2.05, 95% CI 1.25 to 3.35; p for trend 0.004). For each 11 gram per day

increase in lactose consumption (the approximate amount in one glass of milk), there was a 19% increase in risk of serous cancers (RR 1.19, 95% CI 1.03 to 1.38). Skim and low-fat milk were the largest contributors to dietary lactose. Women who consumed one or more servings of skim or low-fat milk daily had a 32% higher risk of any ovarian cancer (RR 1.32, 95% CI 0.96 to 1.80; *p* for trend 0.06) and a 66% higher risk of serous cancers (RR 1.66, 95% CI 1.10 to 2.51; *p* for trend 0.008) compared to women consuming 3 or fewer servings monthly. We observed an increased risk of serous cancers in women consuming 5 or more servings of yogurt per week (RR 2.35, 95% CI 1.09 to 5.07; *p* for trend 0.04), but no association with all tumor subtypes together. We found no evidence of confounding by fat or calcium intake and no effect modification by oral contraceptive use.

CONCLUSIONS: Our findings support the hypothesis that lactose intake from skim or low-fat milk increases risk of epithelial ovarian cancer. The excess risk was most apparent for the serous subtype of ovarian cancer, and was observed at consumption levels often recommended for women to reduce fracture risk.

DYKE—DAVIDOFF—MASSON SYNDROME AND CEREBRAL PALSY RELATED TO CONGENITAL CYTOMEGALOVIRUS INFECTION. *AA Karcic, E Karcic, M Hacena, Department of Internal Medicine, Nassau County Medical Center, East Meadow, NY*

LEARNING OBJECTIVE: (1). To recognize cerebral hemiatrophy (2). To understand its association with cerebral palsy

CASE: An 18-year-old male was seen after an episode of seizures. He had a history of congenital cytomegalovirus (CMV) infection, was deaf and legally blind, and severely mentally retarded. He had bilateral cryptorchidism, cerebral palsy (CP) and Dyke Davidoff Masson syndrome (DDMS). Past medical history included infantile spasms. At age 15, he developed left-sided focal motor seizures with secondary generalization. Exam revealed mild facial asymmetry, nondescended testes, spastic quadriplegia, more severe on the left, and bilateral planovalgus feet. A CT scan of the head showed right unilateral brain atrophy with ipsilateral calvarial hyperplasia and sinus enlargements, consistent with DDMS. After adjusting his antiepileptic medications the patient was discharged home

DISCUSSION: In 1933 Dyke, Davidoff and Masson described nine cases of a syndrome with cranial asymmetry, visible on plain film of the skull, associated with ipsilateral brain atrophy and contralateral hemiparesis. Often facial hemiatrophy, seizures and profound mental retardation are present. The syndrome is also known as cerebral hemiatrophy and is heavily underdiagnosed. DDMS and CP have often a significant etiologic and clinical overlap and may be different clinical and radiological manifestations of the same pathologic process. Their common etiology could be any type of insult to the growing brain (birth injury, trauma in early childhood, infection, compressive brain tumors etc). Our patient had a documented cerebral insult in his early life, that might have contributed to his clinical picture. The diagnostic modality of choice is CT scanning of the head.

GAVE: WHAT THE MIND DOES NOT KNOW THE EYES DON'T SEE. *P Khanna, A Shapiro, S Evans, J Ward, A Tulsky, UPMC Shadyside, Pittsburgh, PA*

LEARNING OBJECTIVE: Recognize gastric antral vascular ectasia as a differential diagnosis in recurrent upper gastrointestinal bleed.

CASE: A 65 year old female with history of recurrent gastrointestinal bleed requiring 26 units of blood transfusion over six months was admitted with anemia and extreme weakness. She had undergone multiple endoscopies and cauterizations of bleeding points but no specific diagnosis. On the fifth endoscopy, it was noticed that she had gastric vascular ectasia. She then had Roux-en-Y gastrojejunostomy. She was also found to have a cirrhotic liver nodule.

DISCUSSION: Gastric antral vascular ectasia or watermelon stomach is a clinicopathological condition characterized by severe gastrointestinal blood loss, iron deficiency anemia, endoscopic appearances of longitudinally prominent erythematous mucosal ridges converging on the pylorus and histological appearance of ectatic vascular channels, intravascular fibrin thrombi, and fibromuscular hyperplasia. Etiologic factors include hypergastrinemia, slower gastric emptying, increased levels of prostaglandin E2, and increased proliferation of local neuroendocrine cells containing 5 hydroxytryptamine and vasointestinal peptide. Diseases which gastric antral vascular ectasia is associated with include portal hypertensive states resulting from hepatic cirrhosis or venoocclusive diseases, cirrhosis without portal hypertension, autoimmune diseases, chemotherapy and growth factor use in bone marrow transplant patients. Diagnosis of gastric antral vascular ectasia is delayed for an average of 5–6 years as both endoscopic and histologic findings can often be missed or misinterpreted. This presentation focuses on such a case with pitfalls in diagnosis and treatment. Therapy consists of endoscopic laser ablation, oestrogens, progesterone or calcitonin. Antrectomy offers cure in patients who fail medical management. Awareness of gastric antral vascular ectasia can lead to early diagnosis by limiting the number of investigations, reducing morbidity associated with chronic blood loss and repeated blood transfusions. This review emphasizes the importance of being aware of the condition gastric antral vascular ectasia so that it can be diagnosed and appropriately treated.

POSSIBLE ASSOCIATION OF EPHEDRA/CAFFEINE SUPPLEMENTS AND DIABETIC KETOACIDOSIS. *V Rajasekaraiah, R Hobbs, MA Bustamante, Medical Education, Oakwood Hospital, Dearborn, MI*

PURPOSE: Recognize a possible association of Ephedra/Caffeine Supplements and Diabetic Ketoacidosis.

METHODS: A 25-year-old female with past medical history of IDDM for 2 years was admitted to our institution with a chief complaint of nausea and vomiting of one day's duration. The patient had no fever, chills, cough, sore throat, abdominal pain or chest pain. She reported daily monitoring of capillary blood glucose and no missed doses of Insulin. The blood sugar on the day prior to admission was 140–150 mg/dl. She denied any alcohol use, salicylate, methanol or ethylene glycol ingestion. She began using "Metabolife" to lose weight, 2 tablets twice a day, 4 days prior to admission. Examination revealed a moderately built female in no acute distress, breath smelling of ketone, otherwise normal exam. Labs showed blood glucose of 408 mg/dl, arterial pH of 7.03 with anion gap of 25. CBC, LFTs, TSH, Amylase, Lipase, Chest x-ray were normal. Blood cultures were negative. Lactate level was 1.37. Urine analysis showed ketones of 3+. She was diagnosed as diabetic ketoacidosis and treated with IV fluids and insulin.

RESULTS: Each tablet of "Metabolife" contains 12 mg of ephedrine and 40 mg of caffeine derived from the Chinese herb mahuang. This exceeds FDA recommendations for ephedrine. Ephedrine increases the metabolic rate, decreases appetite and increases insulin sensitivity. It acts as a sympathomimetic with agonistic effects on alpha and beta adrenergic receptors. It stimulates gluconeogenesis and blocks residual insulin release in patients with IDDM. The FDA has received 800 reports of adverse effects with these products since 1994 including headache, nervousness, insomnia and hepatotoxicity. A literature search and communication with the FDA office did not identify reports of diabetic ketoacidosis with "Metabolife." Based on the sequence of clinical events, an association between ephedra/caffeine use and diabetic ketoacidosis must be considered.

PULMONARY EDEMA AND RENAL ARTERY STENOSIS. *S Ramalakshmi, J Liput, C Jarmolowski, UPMC Shadyside, Pittsburgh, PA*

INTRODUCTION: Ischemic renal disease has been an emerging cause of renal failure in elderly. It is important to identify, as it is reversible if intervention is early. Among its various manifestations, it can present as flash pulmonary edema or acute renal failure. We had one such experience.

CASE: A 76 year old female patient with history of Coronary artery disease, diabetes mellitus II, hypertension, left renal artery stenosis with non functioning right kidney came in with complaints of shortness of breath and fatigue for two days. No other associated symptoms. Prior to admission a renal artery Doppler eight months back showed mild proximal left renal artery stenosis. Medications included Aspirin, amlodipine, quinapril, clopidogrel, atorvastatin, glipizide and omeprazole. On exam patient was tachypneic, tachycardic, blood pressure of 214/105, oliguric, with signs of pulmonary edema. Labs showed blood urea of 50, creatinine 2.9 with baseline 1.2. electrocardiogram had no changes. She was treated for the above and carefully monitored. With poor response to management and worsening symptoms, an emergency renal angiogram was done. It showed 95% stenosis of left renal artery and patient underwent angioplasty with stent placement. Post stent showed 0% residual (pictures available). Marked clinical improvement was noted within hours after the procedure. Blood pressure was normal and kidney functions returned to normal in two days. Also on cardiac evaluation patient had normal stress Cardioltite study with Ejection Fraction of 50%.

DISCUSSION: Timely recognition of acute progression of renal artery stenosis helped to restore the kidney functions. Delay in treatment could end in end stage renal disease. The postulated mechanism of pulmonary edema is due to direct effects of angiotensin II in myocardium, reduced pressure natriuresis and its resultant volume expansion. In our patient it was an acute renal failure rather than chronic and it was due to progression of stenosis itself rather than any precipitants. Non-invasive testing modalities include Doppler, Magnetic Resonance Imaging and spiral Computed Tomography. Treatment methods include angioplasty, stenting and surgical revascularization. Results of treatment for preservation of renal function have been encouraging.

OSETAMIVIR INDUCED RHABDOMYOLYSIS IN INFLUENZA A. *M Zulfiqar, R Hobbs, C Piko, Medical Education, Oakwood Hospital, Dearborn, MI*

PURPOSE: To recognize rhabdomyolysis as a possible complication of the new anti-influenza drug Tamiflu (Osetamivir).

METHODS: A 51 year old African American male presented to the hospital with a one day history of hematuria. It occurred five days after Tamiflu (Osetamivir), a new neuraminidase inhibitor, had been prescribed for a recent influenza infection. The patient's symptoms were improving at the time the hematuria began. There was no history of trauma, strenuous exercise, heavy alcohol intake, IV drug abuse, or other possible causes of rhabdomyolysis. He had continued taking the Osetamivir 75 mg twice a day even after the flu symptoms resolved. Muscle discomfort began as tenderness in the buttocks and progressed to the lower extremities three days after his flu symptoms had improved. Physical findings were positive for muscle tenderness in the gluteal areas and lower extremities. There was no rash. The patient's urinalysis was positive for hemoglobin although there were no red blood cells present on microscopic exam. His CPK was 109,817 and the myoglobin was found to be 2540. No electrolyte disturbances were found. The patient was treated aggressively with fluids and forced alkaline diuresis. He responded well. Renal function remained within normal limits.

RESULTS: Rhabdomyolysis is an established although less frequent complication of Influenza. It is most commonly seen in young children. The pathologic mechanism remains unknown but may be caused by a virus generated muscle-specific toxin. Influenza virus has also been isolated by direct hemagglutination and electron microscopy from the muscle specimen. In the current case the fact that the

muscle tenderness started after the improvement in the flu symptoms and in the absence of other demonstrable causes of rhabdomyolysis raises the possibility of a drug induced etiology. Oseltamivir, a new antiviral which inhibits the viral surface structure protein neuraminidase, has not been associated with rhabdomyolysis. Although a single case report is not enough to document a causal relationship, this patient's illness raises the distinct possibility that it may have been drug related or drug induced. Further postmarketing studies and further observations are in order.

CHEST PAIN, ELEVATED CARDIAC ENZYMES AND FOCAL EKG CHANGES: NOT ALWAYS A MYOCARDIAL INFARCTION. N Bettle, L So, K Barnard, UPMC Shadyside, Pittsburgh, PA

LEARNING OBJECTIVE: 1. Consider acute myocarditis in the differential diagnosis when young patients present with a clinical picture suggesting myocardial infarction.

CASE: A 34 year old male with a history of asthma and a recent upper respiratory tract infection was admitted for acute chest pain relieved by sublingual Nitroglycerin. The physical exam was unremarkable. An EKG revealed Imm ST-segment elevation and T wave inversion in leads V₁-V₃. Cardiac enzymes were mildly elevated. Echocardiogram showed antero-apical hypokinesis and an ejection fraction of 45%. A diagnosis of myocardial infarction was made. The patient was treated with Nitroglycerin and Heparin. Cardiac catheterization did not show significant coronary artery disease. The patient then developed 3rd degree AV-block leading to the first of four episodes of cardio-pulmonary arrest. At this point, global hypokinesis with an ejection fraction of 10% was noted on echocardiogram. The patient gradually improved with aggressive supportive measures (four pressor agents at maximal dose & intraaortic balloon pump) and empiric treatment with high-dose steroids and intravenous immunoglobulins. He was discharged with an almost normal ejection fraction and without neurologic deficits. The etiology of the myocarditis was not established.

DISCUSSION: Our patient presented with a picture strongly suggesting myocardial infarction. However, a sudden onset of focal cardiac abnormalities together with a history of a recent respiratory tract infection, is also the typical initial presentation of myocarditis in young men. This diagnosis was bolstered by findings of non-significant coronary artery disease and a rapid clinical deterioration, marked by arrhythmias, cardiac arrests and the development of global cardiac hypokinesis. The isolation of infective organisms from serology or cultures may give some clues as to the etiology but the isolated organisms may not be related to the myocarditis. Endomyocardial biopsy is considered the gold standard for diagnosis; however, it lacks sensitivity or specificity depending on the criteria used. Myocarditis is thus a clinical diagnosis. Treatment options beyond the management of heart failure and shock are controversial. This is probably due to a poor understanding of the disease process. For certain etiologies, specific therapy may treat the predisposing disease, but may not relieve the myocarditic process. The use of steroids, immunosuppressants and immunoglobulins have been studied with equivocal outcomes. What is known however, is that myocarditis is potentially self-limiting and that aggressive supportive measures must be accorded patients. If all measures fail, then cardiac transplantation can be a life-saving option.

AN UNEXPECTED CAUSE OF HYPERCALCEMIA IN A PATIENT WITH SUSPECTED RECURRENT MALIGNANCY. VV Joshi, G Tabas, UPMC Shadyside, Pittsburgh, PA

OBJECTIVES: (1) To recognize the work-up for hypercalcemia in the setting of previous malignancy. (2) To diagnose and manage hypercalcemia.

CASE: The patient was a 75 year old white female with a past medical history of non-small cell (squamous cell) lung cancer diagnosed one year ago, status post-right middle lobectomy. She presented to her physician's office with 6 weeks history of loss of appetite and constipation. The patient also had a history of frequent falls with the most recent fall causing an undisplaced radial fracture four days prior to admission. In the office before admission, serum calcium was 15 mg/dl and she was admitted for treatment of severe hypercalcemia. Physical examination revealed a thin frail lady Pulse-80, BP 130/80. Significant findings were dry oral mucosa, kyphosis and spinal tenderness from T₁₂ to L₁. Other lab data on admission showed phosphate 1.5 mg/dl, albumin 3.5 gm/dl and calcium oxalate crystals in urine. Chest x-ray showed cardiomegaly and right pleural thickening. Initial treatment included fluids and diuretics, which decreased Calcium to 14. Because of her past history of squamous cell cancer, the patient was worked up for recurrence of malignancy with CT scan of the chest/abdomen/brain, which were unrevealing. Initially the patient responded poorly to fluids and diuretics and required bisphosphonates. Because the patient was not responding to the above treatment, hypercalcemia secondary to primary hyperparathyroidism was considered, despite a normal calcium level one year ago. Her PTH levels and PTH-related protein levels were 924 (normal limit 12 to 72 pg/ml) and 0.5 gm/dl (normal limit 1.3 or less) respectively. Because of elevated PTH levels and calcium oxalate crystals in urine, she was considered a surgical candidate and underwent a thyroid Sestamibi scan which showed intense uptake in left parathyroid gland. Surgical exploration of her neck revealed a 3gm parathyroid adenoma. After removal she was discharged to home with a normal calcium level.

DISCUSSION: Hypercalcemia is a relatively frequent medical problem. The commonest causes are primary hyperparathyroidism and cancer. This patient with a history of lung cancer had severe hypercalcemia, which was thought to be caused by recurrent cancer, and much less likely by hyperparathyroidism. As compared to other types of lung cancers, squamous cell lung cancer has the highest associ-

ation with hypercalcemia. When determining the etiology for hypercalcemia in such group of patients one must not forget to consider hyperparathyroidism, as the diagnostic work up and management will be considerably different for these patients.

TUBERCULOUS MONOARTHRITIS IN A PREVIOUSLY ASYMPTOMATIC WOMAN. AA Karcic, V Maudar, E Karcic, T Mir, Department of Internal Medicine, Nassau County Medical Center, East Meadow, NY

LEARNING OBJECTIVE: 1). Diagnose Tuberculous Monoarthritis.

CASE: An 83-year old woman, presented with chronic, worsening, dull, right knee pain, worse with weight bearing. She reported intermittent discharge in the popliteal fossa. Physical exam revealed a flexed, swollen right knee with a small sinus draining serosanguineous fluid in the popliteal fossa. The erythrocyte sedimentation rate was 86 and C-reactive protein 10.3. The patient's chest radiograph revealed old right lower lung scarring. A plain radiograph of the knee showed joint space narrowing with metaphyseal and subchondral erosions and cysts. Knee arthrocentesis was unremarkable. A popliteal sinogram showed a branching sinus tract, opening into the popliteal fossa. The diagnosis was finally reached by arthroscopy with synovial biopsy that revealed multiple granulomas, multinucleated giant cells, and acid fast bacilli. Synovial tissue cultures grew mycobacterium tuberculosis; the sputum did not. The patient was tuberculin negative and aergic.

DISCUSSION: Tuberculous monoarthritis is a very rare condition in the developed world today, mostly because of availability of antituberculous therapy. It is seen in less than 1% of tuberculous infections. Untreated it may lead to complete joint destruction. The emergence of multidrug resistant (MDR) tuberculosis threatens to make tuberculous monoarthritis a more common diagnosis. Tuberculous arthritis should be considered particularly in tuberculin positive patients with chronic monoarthritis and abnormal chest and/or joint radiographs. Joints most frequently affected are the knee, hip, wrist and other small joints. Weight bearing joints are predisposed possibly because of daily microtrauma. Joint pain and swelling are the most common symptoms. Like other forms of extrapulmonary tuberculosis, tuberculous monoarthritis is usually due to reactivation of a hematogenously seeded dormant focus. Only a minority of patients will have active tuberculosis. Most others, but not all, will have evidence of exposure to tuberculosis, including abnormal chest radiographs and positive skin testing. The radiographical findings of joint space narrowing, metaphyseal and subchondral erosions and subchondral cysts are related to the dilatation, branching and looping of local capillaries, causing turbulence and predisposing to bacterial localization and growth. Sinuses or fistulas are rare. The gold standard of diagnosis remains synovial biopsy and tissue cultures. The diagnosis is frequently delayed. Treatment is the same as for other forms of tuberculosis.

CYCLOBENZAPRINE LEADING TO HYPOGLYCEMIA IN A PATIENT WITH DIABETIC CYSTOPATHY. E Karcic, AA Karcic, T Mir, Department of Medicine, Nassau County Medical Center, New York; Departments of Geriatrics at the St. Louis University School of Medicine; State University of New York at Stony Brook

LEARNING OBJECTIVES: 1) Be aware of diabetic cystopathy. 2) Know, before starting a medication, if it exacerbates cystopathy. 3) Assess the decrease in renal function and the decrease in insulin requirements.

CASE: A 66-year-old woman with insulin-dependent diabetes mellitus for 30 years (with diabetic retinopathy and sensory neuropathy), hypertension, chronic renal insufficiency, and fibromyalgia presented to the emergency department. At home, she became confused, dizzy, sweaty, and tremulous; then she lost consciousness. The finger stick glucose checked by her husband was 20 mg/dl. She regained consciousness after 50% dextrose was given intravenously by emergency services. Her medication list included: insulin, quinapril, amlodipine, furosemide, amitriptyline (for painful neuropathy), and cyclobenzaprine (started only recently for fibromyalgia). By the time she reached our hospital, her blood sugar level was 68 mg/dl; blood urea nitrogen was 46 mg/dl, creatinine was 1.9 mg/dl. The remaining work-up was unremarkable.

DISCUSSION: Prescribing multiple medications to patients with several medical problems is always a bit of a minefield. This problem is particularly acute in the elderly. We present a case of complex drug interactions that led to hypoglycemia. Case analysis suggested the following sequence of events: (1) cyclobenzaprine, together with amitriptyline, deteriorated an already existing baseline diabetic cystopathy (the patient had significant urinary retention in the past); (2) worsening urinary retention created an acute exacerbation of the already existing chronic renal failure; (3) declining renal function reduced insulin excretion and requirements, so that the unchanged dose of insulin became excessive and led to hypoglycemia.

CONCLUSION: A diabetic cystopathy has to be assumed in every patient with diabetic neuropathy, and should be considered when starting any new medications, to avoid potentially life-threatening complications of therapy.

ACUTE ABDOMEN CAUSED BY PANCREATITIS AS A RARE PRESENTATION OF POLYARTERITIS NODOSA. AA Karcic, M Shareeff, A Hamad, P Anand, Divisions of Internal Medicine and Rheumatology, Nassau County Medical Center, East Meadow, NY

LEARNING OBJECTIVES: (1) To be able to diagnose polyarteritis nodosa. (2) To consider polyarteritis nodosa in the differential diagnosis of pancreatitis.

CASE: A 41 year-old man with history of generalized weakness, intermittent fever, loss of weight, arthralgia, wrist drop (EMG showed axonal pattern neuropathy) and intermittent testicular pain (biopsy showed non-specific atrophy) presented with severe abrupt abdominal pain. Work up for polyarteritis nodosa (PAN) was done. Repeated mesenteric and celiac angiograms were normal. The patient was treated with low dose steroids, for presumptive diagnosis of PAN, any infective or other inflammatory etiology having been ruled out. CT scan of the abdomen revealed pancreatitis and a peripancreatic necrotic fluid collection. Laparotomy with pancreatico-bridement was performed. The biopsied tissue was consistent with PAN. The patient refused immunosuppressive therapy hence was maintained on steroids, analgetics, and antibiotics. He deteriorated gradually, developed polymicrobial sepsis and died few weeks later.

DISCUSSION: Polyarteritis Nodosa is a multisystem vasculitis that may affect any organ. Most frequently it presents with acute abdomen, occasionally due to pancreatitis. In order to be diagnosed The 1990 American College of Rheumatology requires 3 of the following 10 criteria to be present: (1) weight loss of more than 4 kg (2) livedo reticularis (3) testicular pains (4) mono- or polyneuropathy (5) myalgias (6) elevated blood urea nitrogen or creatinine (7) diastolic hypertension >90 mmHg (8) hepatitis B infection (9) visceral angiograms with aneurysms and occlusions (10) pathognomonic small or medium- sized artery biopsy specimen. (9) and/or (10) remain the gold standard of diagnosis. Polyarteritis survival at 5 years is about 55%, and is unaffected by adding cytotoxic agents to corticosteroid treatment.

SEVERE CORONARY ARTERY DISEASE IN A PREGNANT YOUNG WOMAN WITH SYSTEMIC LUPUS ERYTHEMATOSUS. AA Karcic, M Zihlif, A Conrad, Departments of Medicine and Cardiology, Nassau County Medical Center, East Meadow, New York

LEARNING OBJECTIVES: 1) To understand the role of Systemic Lupus Erythematosus (SLE) as a risk factor for coronary artery disease (CAD).

CASE: A thirty-one year old woman, mother of four, with a past medical history of SLE (diagnosed ten years ago, treated at that time with steroids), was admitted for new onset worsening angina with anterolateral ST depressions noted on the electrocardiogram. Two years ago she underwent a balloon angioplasty for a 100% blockage of the Left Anterior Descending coronary artery. Since then she remained asymptomatic. The patient had neither family history of CAD nor any other risk factors for CAD. Her home medications included aspirin and metoprolol. Incidentally she was found to be pregnant (8 weeks). The patient refused an angiogram because of the risk of radiation exposure to the child. She opted to be managed medically.

DISCUSSION: SLE is a significant risk factor for the development of CAD. The risk of myocardial infarction (MI) is increased up to 50-fold and the risk of fatal MI is increased three times over that in the general population. Atherosclerosis is the most common mechanism for CAD in SLE. Immune complex deposition causes the initial intimal damage, which is followed by accelerated development of atherosclerosis. Patients with advanced CAD also have a higher prevalence of pericarditis and valvular disease, which suggests that an immune factor is the cause of the CAD. The development of coronary atherosclerosis may be related to steroid use and prolongation of life (which allows more time for the development of atherosclerosis), or it could be related to the exacerbation by steroids of hypertension and hyperlipidemia. The treatment of CAD is the same regardless of the SLE status. If coronary bypass grafting is considered, arterial grafts rather than venous grafts should be used, because of a high thrombosis risk. After atherosclerosis, arteritis is the second most common type of CAD in patients with SLE. Pathologic examination is the only definitive method of distinguishing arteritis from atherosclerosis. Rapidly developing stenoses or restenoses suggest the diagnosis of arteritis. Clinically, the distinction is very important because to treat arteritis one has to increase the corticosteroid dose, while on the other hand increasing steroids might worsen risk factors in a patient with atherosclerotic disease. Finally, a high anticardiolipin antibody level, regardless if SLE is present or not, is an isolated risk factor for CAD.

THE TRIAD OF SARCOIDOSIS, RETROPERITONEAL FIBROSIS AND DIFFUSE PERIAORTITIS WITH VASCULAR INVOLVEMENT: A RARE ASSOCIATION WITH ADVANCED SARCOIDOSIS. AA Karcic, E Karcic, F Formato, M Hacena, Department of Internal Medicine, Nassau County Medical Center, East Meadow, NY

LEARNING OBJECTIVES: 1) To consider sarcoidosis in the differential diagnosis of ischemia. 2) Recognize the association between periaortitis, retroperitoneal fibrosis and sarcoidosis.

CASE: A 59-year-old woman with advanced sarcoidosis, retroperitoneal fibrosis, ureteral obstruction and bilateral ureteral stents was admitted to the urology service reporting difficult urinating. Ureterolysis was planned. Soon, she developed chest pain and pulmonary edema; she ruled out for myocardial infarction, but a nuclear stress test revealed reversible inferior wall ischemia. Furthermore, she complained of recurring abdominal pains, particularly after large meals. Her stools were positive for occult blood. There was a high clinical suspicion of mesenteric ischemia. A CT of the chest and abdomen, performed to evaluate the etiology of her pains, revealed extensive periaortic fibrosis (involving the entire aorta including aortic root and origins of coronary arteries, as well as the mesenteric arteries) and retroperitoneal fibrosis with ureteral obstruction and bilateral hydronephrosis. In addition there was lung fibrosis and hilar lymph node calcification

seen. The patient refused invasive work-up, and is currently managed conservatively.

DISCUSSION: Sarcoidosis is a multisystem granulomatous disease, most frequently affecting the lungs. The exact etiology is unknown. It is more common in young, black women and within families. This is the third reported case (Snow 1977, Godin 1980) describing the triad of sarcoidosis, retroperitoneal fibrosis and periaortitis. In this setting, vascular involvement and ischemia (cardiac, mesenteric, and peripheral) have been reported in several cases. In our patient with this triad, the cardiac and mesenteric ischemia were aggravated, if not caused, by extensive (peri)vascular fibrosis and compression of involved blood vessels. Sarcoidosis should therefore be considered as a rare cause of vascular insufficiency. Keeping in mind that both sarcoidosis and retroperitoneal fibrosis are relatively common, this "triad" is probably underreported.

FATAL OVERLAP SYNDROME OF LICHEN PLANOPILARIS AND SYSTEMIC LUPUS ERYTHEMATOSUS- AA Karcic, E Karcic, I Dominich, S Feffer, Nassau County Medical Center, East Meadow, NY

LEARNING OBJECTIVE: 1) To be aware of the existence of overlapping cases of lichen planus (LP) and systemic lupus erythematosus (SLE), two commonly seen conditions.

CASE: A 34-year-old man with schizophrenia and lichen planopilaris (skin biopsy), presented with fever of two days duration multiple pustules (with excoriations and crusts) over the entire head, oral thrush, a painless palatal ulcer and palmar and plantar erythema. Differential diagnosis included sepsis, endocarditis, syphilis, AIDS. He had mild anemia, leukopenia 2,600/mm³ and negative blood cultures. Facial pustule culture grew *S. aureus*; VDRL was nonreactive; CD4 count was 260, HIV negative; normal echocardiogram; palmar skin biopsy showed nonspecific lymphocytic vasculitis. Collagen vascular disease tests confirmed SLE; C₃ and C₄ complement levels were low. The patient deteriorated rapidly going into shock, acute renal, respiratory and marrow failure, bleeding diathesis, and hemolysis. Despite aggressive treatment (ventilation, dialysis, pressors, transfusions, plasmapheresis, antimicrobials, steroids, cytotoxics) the patient expired.

DISCUSSION: In practice it is often difficult to clearly distinguish between SLE and LP, or the coexistence of both disorders is suspected. SLE and LP have common clinical, histologic and immunohistologic features. Possibly, a common agent (viral, genetic, autoimmune-medications) could cause either disease or the overlap syndrome, depending on the genetic predisposition of the individual. The following are useful guidelines to distinguish between the two disorders: SLE is more likely to occur in those younger than 30, women and blacks, while LP usually occurs in older than 30 and shows no racial nor gender preference. SLE presents usually with an erythema (butterfly rash on the face) or with scaling reddish plaques (discoid lupus). LP typically presents with Polygonal, Purple, and Pruritic Papules (4 P's) seen on flexor sides of extremities and with Wickham's striae on the oral mucosa. Both disorders are systemic and have a strikingly similar appearance on histology and immunofluorescence. Serologic testing will confirm lupus. Unlike SLE, LP does not cause fatalities. A histochemical technique utilizing maleimide-derived fluorochrome is now available to distinguish between these two disorders.

INNOVATIONS IN MEDICAL EDUCATION

EVALUATING PHYSICAL DIAGNOSIS SKILLS USING A CARDIOLOGY PATIENT SIMULATOR: WE CAN DO BETTER! YK Agarwal, D Roth, SJ Peterson, CL Karmen, Internal Medicine, Westchester Medical Center, New York Medical College, Valhalla, NY; New York Medical College, Valhalla, NY

STATEMENT OF PROBLEM: Pacing changes in medical education with those happening in health care delivery has become a challenge. With less time spent at bedside, faster patient turnover and increasing demands on time; the ability to teach and evaluate bedside clinical and physical examination skills has become more challenging.

OBJECTIVES OF PROGRAM/INTERVENTION: We used "HARVEY" a Cardiology Patient Simulator to evaluate and compare the performance of third year medical students with those of second year medical residents

DESCRIPTION OF PROGRAM/INTERVENTION: "HARVEY" is a Cardiology Patient Simulator in the form of a lifelike mannequin capable of faithfully reproducing the arterial and venous pulsations, blood pressure changes, precordial impulses and auscultatory findings of a number of cardiovascular pathologies. Eleven third year medical students in the third and fourth week of their internal medicine clerkship at a University hospital were randomly presented with three Cardiovascular [CV] simulations. These simulations were Mitral Regurgitation [MR], Aortic Regurgitation [AR] and Aortic Stenosis [AS]. Their scores were compared to those of twelve medical residents in the early part of their second year of training at the same hospital. These residents had underwent a similar exercise in 1998 where we had used six randomly presented simulations. These being MR, AR, AS, Mitral Valve Prolapse [MVP], Atrial Septal Defect [ASD] and Pulmonic Stenosis [PS]. Only scores for the three common simulations were used.

FINDING TO DATE: Both residents and medical students were able to identify more than half the simulations correctly. Medical students performed just as well as the residents on these three not uncommon CV simulation exercises.

	% Correct Responses for Medical Students	% Correct Responses for Medical Residents
AS	69%	47%
AR	42%	71%
MR	60%	64%
TOTAL	61%	64%

KEY LESSONS LEARNED: 1. We need to improve physical examination skills at all levels of training. 2. Unlike the residents the medical students had spent some time with "HARVEY" prior to the test. This might have contributed to the lack of difference between the two groups and probably demonstrates the teaching effectiveness of simulation exercises. A formal teaching curriculum utilizing a CPS might enhance learning experience and improve upon our current abilities.

ACADEMIC INSTITUTIONS—SERVING ONE ANOTHER. Sy Anderson, BR Meyer, General Internal Medicine, Cornell University Medical Center, New York, NY

STATEMENT OF PROBLEM: Lack of on-site health service at a senior college.

OBJECTIVE OF PROGRAM/INTERVENTION: The Cornell Internal Medicine Associates (CIMA) and Hunter College entered into a relationship to create an on-site health service for the students of Hunter College. This relationship not only expands CIMA's community outreach but also exposes our ambulatory medicine residents to a population they are not frequently exposed to in their ambulatory medical practice.

DESCRIPTION OF PROGRAM/INTERVENTION: Both CIMA (part of the Cornell University Medical Center) and Hunter College (a senior college that is part of the City University of New York system) are located on the upper east side of Manhattan. Prior to the CIMA's involvement, Hunter College had no on-site medical care for a student body of approximately 20,000. The student body of Hunter College is very diverse. According to a comprehensive 1997 student profile survey, 50% of new freshman at the college were born outside the U.S. and 55% of the students had a language, other than English, as their primary language. Of the students (undergraduates and graduates) that responded to the survey in 1997, 42.8% identified themselves as White, 19.1% identified themselves as Black, 21.5% as Hispanic, and 15.5% as Asian/Pacific Islander. The average student is under 40 years of age (undergraduates and graduates). As with most academic medical practices that residents are exposed to, the patient population at CIMA is older and has multiple medical problems. In addition to providing much needed medical services to Hunter College, we also felt that this would be a unique opportunity for junior and senior medical residents (mostly primary care and a few categorical) to participate in the care of a younger and relatively healthy population. CIMA staffs the Hunter College Health Services two half sessions per week. Each session is precepted by an attending physician who supervises two residents. For continuity, the attending physician is consistent for each session and the same two residents attend each session on their ambulatory medicine block. In addition to usual medical exam, the residents and attending are strong advocates for preventative health care.

FINDING TO DATE: Since the Hunter College Health Services formally opened with medical services on February 17, 1999, we have seen nearly 2,000 students. Survey feedback from students has been extremely positive. Resident feedback for the experience has also been extremely positive.

KEY LESSONS LEARNED: As medical services at Hunter College expand, we will strive to continue to provide excellent health care to the student body. By exposing our residents to the college health service, we provide not only needed medical care, but also young positive role models for the students and a fulfilling experience for the medical residents.

AREAS OF CONCENTRATION: CURRICULAR VEHICLES FOR THE SENIOR YEAR OF MEDICAL RESIDENCY. P. Basaviah, S Fazio, M Aronson, C Hatem, C Bates, AOC Working Group, Medicine, Beth Israel Deaconess Medical Center, Boston, MA

OBJECTIVES OF PROGRAM/INTERVENTION: Many medical house officers at our institution have traditionally elected to do research during their senior year. Until recently, residents interested in medical education, healthcare policy, and specific areas of expertise in primary care have not had similar formal opportunities to develop these skills.

DESCRIPTION OF PROGRAM/INTERVENTION: We have created parallel tracks for elective time and self-directed education, called Areas of Concentration (AOC), which allow senior residents to gain expertise in specific arenas. These electives provide opportunities for senior residents to build unique skills in addition to those taught by the conventional curriculum.

FINDING TO DATE: We have designed longitudinal electives lasting 9–12 weeks in four Areas of Concentration: Medical Education, Healthcare Policy, Biomedical Research, and Primary Care. All residents in the program already have substantial primary care ambulatory block time. Those selecting the primary care AOC do advanced clinical training in Women's Health, HIV Medicine, Geriatrics, or Adolescent Medicine. All participating residents work closely with AOC directors and mentors to design scholarly works, including curricula, research papers, practice guidelines, educational workshops, and summary presentations. The residents also participate in longitudinal interactive didactic sessions relevant to their AOC.

KEY LESSONS LEARNED: Thirty-three of 55 (60%) senior residents chose the AOC elective in the first year of the program. Eight (24%) are in Medical Education, 5 (15%) in Primary Care, 3 (9%) in Healthcare Policy, and 17 (50%) in Biomedical Research. Medical Education AOC projects have included revised curricula for oncology training, evidence-based medicine, and obesity management, as well as a lecture series for third-year medical students doing their core rotation. Residents in the Primary Care AOC have focused on specific content areas for advanced clinical training including osteoporosis and menopause management; upcoming residents will do procedural training in women's health. Medical errors in the ambulatory setting and health care economics have been studied by Healthcare Policy AOC residents. Residents in the Biomedical Research AOC have investigated a wide variety of topics in both basic science and clinical research.

STATEMENT OF PROBLEM: The AOC program provides a vehicle for professional development and mentoring allowing senior residents to develop areas of expertise which enrich their traditional training. Future evaluation will assess the value of these programs for career planning.

THE MINI-CLINICAL EVALUATION EXERCISE (Mini-CEX) PROJECT—MAXIMIZING THE MEASURE. U. Blank, NL Grant, HL Lespoir, American Board of Internal Medicine, Philadelphia, PA

BACKGROUND: The mini-CEX is an efficient, effective evaluation method that promotes assessment of resident's clinical skills, attitudes and behaviors that are essential in providing high quality patient care. In conjunction with the ABIM's earlier work in developing the mini-CEX¹, the Board has recently expanded that activity toward the goal of national implementation among the 402 currently accredited internal medicine residency programs training over 23,000 residents. Twenty (20) residency programs located in the northeast and over 500 PGY-1s are now involved in an implementation pilot (July 1999–June 2000).

PURPOSE: The mini-CEX focuses on the core skills that residents demonstrate in patient encounters. It is a 15–20 minute snapshot of a resident/patient interaction that provides a valid, reliable measure of clinical performance based on multiple encounters (optimally four per year per resident) by different evaluators.

GOAL: The pilot project is designed to promote the use of the mini-CEX as a routine, seamless evaluation for residents during clinical rotations. It can be easily implemented by attending physicians in any setting.

FORMS AND RATING SCALE: Convenient pocket-size, duplicate forms were developed by participating program directors for use by attending physicians with the goal of immediate feedback to the resident and documentation for the program file. A nine-point scale is used; the rating of 4 is defined as "marginal" and conveys the expectation that with remediation the resident will meet the standards for Board certification.

CONCLUSIONS: The mini-CEX assesses residents in a much broader range of clinical situations than the traditional CEX, has better reproducibility, and offers residents greater opportunity for observation and feedback by more than one faculty member and with more than one patient. The ABIM encourages the use of this method in conjunction with or as an alternative to the traditional CEX.

1. Norcini JJ, Blank LL, Arnold GK, Kimball HR: The Mini-CEX (Clinical Evaluation Exercise): A Preliminary Investigation. *Annals of Internal Medicine*, 1995;123:795–799.

HEALTH CARE ON CONNECTICUT TOBACCO FIELDS: CONNECTING MEDICAL STUDENTS WITH MIGRANT FARMWORKERS. J. Botte, S Raczka, University of Connecticut School of Medicine, Farmington, CT

PURPOSE: In the spring of 1998, medical students from the University of Connecticut became involved with the Connecticut Valley Migrant Farmworker Network. This network is a coalition of agencies formed to provide health care to migrant and seasonal farmworkers in the Connecticut Valley. The purpose of this project was to develop a method of providing primary care to the workers via a mobile medical clinic.

METHODS: Establishing the clinic required the following: 1) Communication with UConn faculty and farm owners to allow the practice of medicine on the farmland, 2) Recruitment of students, physicians, and interpreters to staff the clinic, 3) Procurement of donations of medical supplies and medications, 4) Development of an intake form for record keeping and data collection. At the clinic patients were interviewed and completed the intake form. They also signed a waiver allowing information to be used for research purposes. Patients were then seen by a team of students, doctors, and interpreters. After appropriate care was administered and follow-up was scheduled, intake forms were collected and entered into a Microsoft Works database.

RESULTS: During the summer and fall of 1998, 211 men were seen. 67% were from Puerto Rico and Mexico, 21% were from Jamaica, 12% were from the United States. 7% were 15–19 years old, 15% were 20–24, 58% were 25–44, and 20% were 45–64. 17% were diagnosed with hypertension, 9% with a skin rash, and 8% with upper respiratory tract infection. In total, 60 UConn medical students and 25 physicians participated.

CONCLUSION: The UConn Migrant Farmworker Medical Clinic primarily served males between the ages of 25–44 who were of Jamaican, Puerto Rican, and Mexican origin. Their chief diagnoses were hypertension, skin rash, and upper respiratory infection. This differs considerably from the population studied in The National Ambulatory Medical Care Survey: 1993 Summary. Based on the above information three CONCLUSIONS can be made: 1) The population served by the UConn Migrant Farmworker Medical Clinic is a unique population with specific

needs. The information collected is critical for improving care and targeting preventive measures. 2) Based on the success and enthusiasm for the project, it can serve as a model for the development similar programs by other medical schools. 3) Through donations and volunteers, the cost of developing the entire clinic has been very low. With continued student and physician support, the UConn Migrant Farmworker Medical Clinic is expected to run throughout future growing seasons.

OPTIMIZING PATIENT SATISFACTION AND PAYOR MIX IN A RESIDENT CONTINUITY PRACTICE. TS Caudill, SA Haist, CH Griffith, Internal Medicine, University of Kentucky, Lexington, KY

STATEMENT OF PROBLEM: Traditional models of internal medicine resident continuity training often schedule faculty to see insured patients in non-teaching clinics to assure attending access for insured patients who choose not to be seen in a teaching setting. Residents are delegated the care of the poor and indigent in continuity teaching clinics.

OBJECTIVES OF PROGRAM/INTERVENTION: To fully integrate faculty and resident continuity care practices to attract a balanced payer mix optimizing resident continuity training, while maintaining patient satisfaction in the teaching setting. **DESCRIPTION OF PROGRAM/INTERVENTION:** We used managed care quality goals and enhanced patient education to market a new clinic model. The practice was renamed University Medical Associates to highlight the educational mission of the clinical practice. Groups of four residents (HCFA requirements) and clinic staff were assigned to an attending-led team scheduled together for all clinic sessions each week. Each resident schedule is tied electronically to their team attending in an electronic scheduling program, so patients requesting either practitioner will automatically be scheduled with both. The patients can also be scheduled with their attending when access to the resident for acute care visits is not available (night float, community-based rotations, etc.). New patients are informed of the educational mission and team structure by scheduling staff and sent a marketing brochure highlighting advantages of care in the academic setting. The patients are informed of being matched with a resident on their attending's team and every effort is made to ensure continuity at both the resident and faculty level for all ambulatory visits. One return appointment in each schedule was converted to a same-day visit slot to improve access for acute care needs.

FINDINGS TO DATE: The new model improved patient continuity and show rates at both the resident and attending levels (mean show rates improved from 65% to 82%). Payer mix also improved (31% insurance/managed care, 31% Medicare, 22% Medicaid, 8% patient pay, and 8% other). Productivity (14 RVU/half-day clinic), patient visits/faculty FTE/year (4875) and first available appointment (1 day) also improved. Patients from all payer types, including University professors, hospital administrators, medicaid and uninsured patients, report satisfaction in the amount of time their residents spend with them and increased satisfaction with the continuity of care with their attending doctor, despite annual resident attrition. Faculty and residents report improved continuity, as well as improved efficiency in patient management and resident supervision.

KEY LESSONS LEARNED: A fully integrated faculty/resident practice can retain an optimal payer mix, maintain satisfaction of insured patients and their providers, and increase patient continuity and efficiency of care in the continuity teaching setting.

A VERTICALLY INTEGRATED MEDICAL SCHOOL CURRICULUM IN EVIDENCE BASED MEDICINE. AS Cifu, D Altkorn, The University of Chicago, Chicago, IL

STATEMENT OF PROBLEM: Student exposure to an evidence based medicine curriculum at the University of Chicago Pritzker School of Medicine was not coordinated in any way. This situation risked repetition of some topics and premature introduction of others.

OBJECTIVES OF PROGRAM/INTERVENTION: Our objective was to introduce a vertically integrated curriculum in which each successive year builds on skills from the previous year. The curriculum would run through the preclinical and clinical years, involve required and elective work, and include both student and faculty development pieces.

DESCRIPTION OF PROGRAM/INTERVENTION: During the pre-clinical biostatistics course, students are introduced to skills in critically appraising clinical trials. The focus is on using statistics to assess the applicability of studies to individual patients.

During the 3rd year medicine clerkship, students are required to address a clinical question raised by a case seen during their ambulatory medicine rotation using a recent medical study. This study is then appraised using the questions from the "User's Guides to the Medical Literature" during a presentation to a faculty member and a small group of fellow students. Skills stressed are critical appraisal, application of research to clinical care and the dissemination of knowledge to colleagues.

During the 4th year, students may choose to take a one month selective, "Critical Appraisal of Influential Medical Literature." This is an intensive course aimed at improving a student's ability to critically read studies and apply them to clinical practice while emphasizing that current medical practice is grounded in the medical literature.

Associated with the above curriculum is a faculty development effort aimed at improving the skills of all medical and surgical faculty in teaching evidence based medicine.

FINDING TO DATE: The effort has been positively received by students. Students have exhibited an adequate knowledge base entering the fourth year course.

Those who complete the senior course excel on a post course knowledge assessment.

KEY LESSONS LEARNED: It is possible to institute a vertically integrated curriculum in the practice of evidence based medicine. Among the most challenging aspects is involving multiple specialties in the curriculum during the clinical years.

A RESIDENT-LED OSCE AT A PRIMARY CARE INTERNAL MEDICINE PROGRAM. G Coste, D Bor, Medicine, Cambridge Hospital, Cambridge, MA

STATEMENT OF PROBLEM: Observed Structured Clinical Exercises (OSCE) are commonly used in medical schools but rarely used in residency programs.

OBJECTIVE OF PROGRAM/INTERVENTION: Since 1988, our program has run an OSCE each fall. The objectives of the OSCE are 1) To provide interns a learning opportunity in a non threatening environment. 2) To provide PGY 2,3, and 4 residents teaching skills and a chance to practice them.

DESCRIPTION OF PROGRAM/INTERVENTION: The PGY 2,3, and 4 residents act as station managers. They are invited to a pre-OSCE workshop. This workshop offers an introduction to the principles of the "One Minute Manager." Station managers are asked to formulate educational goals for their station. A faculty member reviews the contents of the stations prior to its implementation. Examples of clinical stations include: Chest X ray reading, EKG interpretation, management of DKA, pain management, delivering "bad news." Two consecutive days are allotted to OSCE. This maximizes the number of interns who can participate. Each year 2/3 of the intern group participates. During a 90 minute session interns rotate from one station to another. Ten minutes is spent at each session. Immediately after the second OSCE session, all participants are invited to give verbal feedback to OSCE organizer at a luncheon conference. Each intern anonymously completes a survey to assess the quality of the teaching at each station and its clinical relevance. Results of the surveys are shared with each station manager.

FINDING TO DATE: We shall present quantitative and qualitative evaluation measures which indicate that OSCE is well received and found to be a valuable learning and teaching experience.

KEY LESSONS LEARNED: Residents would like to have two OSCE per year.

GENETICS IN PRIMARY CARE (GPC): A FACULTY DEVELOPMENT INITIATIVE. A Davis, W Burke, S Wartman, N Kahn, M Wilson, M Puryear, Society of Teachers of Family Medicine, Leawood, KS

STATEMENT OF PROBLEM: The rapidly changing health care delivery system brings to the forefront the role of the primary care physician to ensure appropriate care for patients in a cost-effective manner. Initiated in 1990, the Human Genome Project is expanding understanding of the genetic basis of disease, underscoring the need for primary care physicians to be cognizant of how new developments in this field impact on prevention, detection, and management of genetic conditions. *Genetics in Primary Care (GPC): A Faculty Development Initiative* is a three-year contract jointly funded (September, 1998-September, 2001) by the Maternal and Child Health Bureau and the Bureau of Health Professions of the Health Resources and Services Administration, the National Institutes of Health, and the Agency for Health Care Policy and Research to address the need for primary care faculty development in genetics.

OBJECTIVES AND DESCRIPTION: GPC's Scope of Work is to plan, implement, and evaluate outcomes of training programs in genetics which target family medicine, general internal medicine and general pediatrics faculty. An interdisciplinary Executive Committee (EC) manages the project. Advice and input are received from an interdisciplinary Advisory Committee (AC) with representation from 36 organizations. An eight-member Genetics Education Consultant Committee (GEEC) brings to the Executive and Advisory Committees expertise pertinent to development of the GPC training programs.

FINDINGS TO DATE: At the first meeting of the GPC Executive and Advisory Committees, held September 27-28, 1999, a course entitled *Genetics for Primary Care Providers* was conducted. Together the EC, AC and GEEC concluded that a case-based interactive approach to training primary care faculty in genetics may elucidate the contextual issues at the core of the interface between genetics and primary care.

NEXT STEPS: It is anticipated that a case-based interactive approach to training will be incorporated into the design of the GPC faculty development training program. By the time of the *Innovations in Medical Education* exhibit in May, more specific features and timing of the GPC training programs will be known and shared via the exhibit.

ASSESSING ESSENTIAL CLINICAL SKILLS OF MED-PEDS RESIDENTS. MB Duke, CH Griffith, SA Haist, JF Wilson, Internal Medicine; Behavioral Science, University of Kentucky, Lexington, KY

STATEMENT OF PROBLEM: In a recent survey, graduates of our combined med-peds residency indicated they had been well-prepared for practice in intensive care and inpatient situations, but believed they were underprepared in many ambulatory skills including counseling and in addressing the needs of patients with psychosocial problems. Since residents are rarely observed directly while interviewing or counseling patients, we have had limited ability to correct this perceived curricular deficiency, or to give specific individual feedback in areas of strength or weakness. This phenomenon appears to be common in university-based residency programs.

OBJECTIVES OF PROGRAM/INTERVENTION: To develop and implement a clinical performance exercise (CPE) for med-peds residents to assess psychosocial skills and to provide individual feedback.

DESCRIPTION OF PROGRAM/INTERVENTION: The CPE consisted of thirteen 20-minute stations including nine standardized patient (SP) stations. First and third year residents participated. The 9 SP stations included four pediatric/adolescent cases: 1) a child with chronic abdominal pain (school refusal); 2) telling parents bad news; 3) adolescent counseling; and 4) newborn hospital discharge instructions. There were five adult SP cases: 1) domestic violence; 2) cardiac risk factor reduction counseling; 3) HIV risk factor reduction counseling; 4) chest pain; and 5) adult survivor of childhood sexual abuse. The four non-SP stations included three EKGs, ten X-rays, twelve dermatological and ten laboratory slides. Criterion-based evaluation was used to determine excellent, adequate and unsatisfactory performances for each station. Two of the SP stations were videotaped and analyzed for specific verbal and nonverbal communication skills. Feedback was given to the resident individually and to the group for each station and for specific skills (interviewing, counseling, physical examination).

FINDING TO DATE: As a group, residents performed poorly on HIV counseling suggesting a need for curricular revision. Residents easily recognized depression in one SP but few inquired into the possible history of childhood sexual abuse. Many residents could not formulate a safety plan for the SP experiencing domestic violence.

KEY LESSONS LEARNED: To our knowledge, this is the first performance-based evaluation of residents in a combined med-peds residency. Our twenty minute stations are longer and address higher-order clinical skills than have been reported using the objective structured clinical examination (OSCE) format. The CPE has served as a useful tool for assessment of the individual residents and of the residency program.

A COLLABORATIVE APPROACH TO FACULTY DEVELOPMENT- HOW RESIDENTS CAN HELP US IMPROVE OUR TEACHING. E Eckstrom, JF Christensen, B Egener, Dept of Medicine, Legacy Portland Hospitals, Portland, OR

STATEMENT OF PROBLEM: As part of a new faculty development program, faculty and residents separately submitted a prioritized needs assessment for faculty development. Both groups identified a need to improve skills in providing excellent feedback to residents on inpatient rotations.

OBJECTIVES OF PROGRAM/INTERVENTION: We present an innovative, collaborative approach to improving faculty skills in providing feedback to residents on inpatient rotations.

DESCRIPTION OF PROGRAM/INTERVENTION: Three workshops were presented one month apart:

- 1) Skills Development—Using a local expert, we developed role play scenarios with interactive debriefing to model good and bad feedback. Faculty practiced individual skills and brainstormed effective techniques.
- 2) Feasibility planning with 6 resident attendees—Faculty and residents worked together to find a time, place, and format acceptable to all for providing feedback. Residents identified several issues of importance from their perspective and helped brainstorm suitable solutions.
- 3) Integration—Faculty reviewed skills, brainstormed recent feedback sessions to determine what could have gone better, and developed a formal process for providing feedback on inpatient rotations.

FINDING TO DATE: Six months after the workshops were given, residents completed a telephone survey (response rate 30%). Three resident representatives who had attended the feasibility session attended a 4th follow-up workshop, presented the results of the survey, and discussed ways to further improve.

Residents reported faculty to be slightly better to much better at providing excellent feedback. They noted: "feedback is more frequent, faculty is showing more interest, and mid-month evaluation is very good." When asked what was useful about the feedback they were given, residents replied: "feedback is useful when it is specific to a situation, when attendings give coping strategies and other process feedback instead of just content feedback, and when attendings take 5-10 minutes to discuss ways to improve." Faculty unanimously enjoyed the workshops, felt they had learned from them, and felt they had changed their practices in giving feedback.

KEY LESSONS LEARNED: Residents and faculty felt the collaborative workshops were a successful and valuable method to improve skills in giving excellent feedback on inpatient rotations. Both groups are now participating in a similar process to improve feedback in the outpatient setting.

WARD ETHICS: BUILDING SURVIVAL SKILLS FOR THIRD-YEAR STUDENTS. KA Edwards, CH Braddock, III, A Back, S Marshall, Medical History and Ethics, University of Washington School of Medicine; VA Puget Sound; Dean's Office, University of Washington School of Medicine, Seattle, WA

STATEMENT OF PROBLEM: Many institutions struggle with how to teach ethics and professionalism during the clinical years. Demands for student's time and attention are great, and yet, many formative lessons about being a medical professional are learned during clinical training. We created these "Ward Ethics" sessions in part based on student concerns that there are relatively few opportunities to meet and talk with colleagues about issues relating to professionalism and being a medical student on the wards.

OBJECTIVES OF PROGRAM/INTERVENTION: During these sessions, students will be able to:

- discuss with peers challenging experiences occurring in clinical rotations;
- outline how to address difficult issues with superiors and manage risks, mistakes, and failures on future rotations;
- identify strategies to maintain integrity throughout medical training and practice;
- improve awareness of personal values and beliefs, strengths and weaknesses, interests and aversions.

DISCUSSION OF PROGRAM/INTERVENTION: Ward Ethics sessions are primarily student-run, with an experienced clinician-teacher on hand to facilitate. Students are responsible for bringing the cases for discussion and reflection. The facilitator insures that the group works together to discuss alternative solutions, with a focus on building concrete skills in anticipation of future ethical or professional dilemmas.

The sessions occur during the first week of the Medicine Clerkship and repeat every 6 weeks with new students. By placing the discussions early in the Clerkship, we intend to focus on experiences from the prior clinical rotation and limit the pressures students often feel regarding evaluation with current faculty. As students have completed a variety of clinical rotations, the discussions are mostly anonymous and always confidential. Students and faculty-facilitator complete an evaluation form at the end of each session.

FINDING TO DATE: 1) Students reported overwhelmingly positive ratings for the sessions. 2) Students readily shared strategies for coping with difficult situations. They reported that it was helpful to hear that other students had similar experiences. 3) Faculty presence was appreciated by the students, particularly in creating the climate for students to openly discuss difficult situations and affirming their ideas for coping strategies.

KEY LESSONS LEARNED: 1) Facilitator training is essential. We have developed a tool-kit for facilitators and held brief workshops to equip them with teaching techniques as they shift roles from lecturer to facilitator. 2) Feedback from faculty and student evaluations drive modifications for future sessions. We plan to expand the program to include all required clerkships, with sessions occurring at multiple sites.

EVALUATION OF A FEEDBACK FORM, LESSONS LEARNED. C Evans-Moling, LJ Goebel, Internal Medicine, Marshall University, Huntington, WV

Providing feedback to residents in ambulatory care is an essential component of training programs, but it does not always occur in a frequent and timely manner. We hypothesized that a feedback form would encourage attendings to give feedback to residents, because it does not require excessive attending time. The investigators created a feedback form (check-off sheet) summarizing all written feedback given to Internal Medicine residents in the ambulatory clinic during the months of January and February 1997. The form contained three subheadings: major problems, minor problems, and improvement noted/good work. If a major problem was checked, the resident was required to respond. For major problems only, a copy of the form was placed in the resident's file and reviewed during ambulatory care grading sessions. We collected feedback forms used during January and February 1998 and compared them to written feedback given prior to introduction of the form. In May 1998, attendings and residents completed anonymous surveys regarding the feedback form. Prior to introduction of the form, feedback was given 137 times compared to only 32 times after the form. None of the feedback given during either time period was positive. Of attendings surveyed, 75% felt the form was a convenient, effective way to document problems and was helpful in filling out resident evaluations. However, only 50% used the form regularly. Reasons for non-use included the form not readily available, no time for the paperwork, and the perception that the residents viewed the form as punitive. Of residents surveyed, 80% felt the information given in the form was helpful. Because the feedback was usually negative, 56% of residents agreed or were unsure if the form made them angry. Sixty-seven percent of residents agreed that they would like to receive more feedback however, they were equally divided over their preference for verbal or written comments.

KEY LESSONS LEARNED: We conclude this feedback form was not associated with an increase in the frequency of feedback given to residents in our ambulatory clinic. Although we designed the form as a check-off sheet, many attendings still wrote notes on the form thereby not saving time. Furthermore, one attending expressed concern that those who used the form would have a disproportionate influence on the resident's evaluations. Additionally, although the residency program director appreciated having written documentation of resident behavior, the residents viewed this unfavorably. Most importantly, the form was never used to give positive feedback, strengthening the resident's negative reaction to the form. Future attempts at improving the frequency of feedback should focus on resolving these barriers and providing a better balance of positive and negative feedback.

LEARNER AS TEACHER INITIATIVE. S Fazio, P Basaviah, R Koffman, J Doweiko, Medicine, Beth Israel Deaconess Medical Center; Office of Educational Development, Harvard Medical School, Boston, MA

OBJECTIVES OF PROGRAM/INTERVENTION: It has been recognized for some time that residents play a central role in the education of interns and medical students, but receive little in the way of formal training to prepare them for this function.

DESCRIPTION OF PROGRAM/INTERVENTION: We sought to develop an educational module that would serve as an instructional paradigm for training medical residents to become more effective teachers. Our model utilized a multi-media ap-

proach with both didactic and role playing components, with the notion that the opportunity to practice new skills with immediate feedback improves learning and the acquisition of these skills.

FINDING TO DATE: Our program spanned a four week time period. Twelve senior residents participated, as well as ten students from the medical school who were rotating on the Internal Medicine service. The sessions were divided into three main modules, each focusing on an essential element of the teaching process: giving effective feedback, doctor-patient communication, and bedside teaching. Each session began with a brief didactic component, followed by an interactive set of video clips derived from popular culture as well as educational tapes. The second half of each meeting was devoted to role playing in smaller groups, utilizing patient-actors and pre-arranged scripts. In the final session, the students and residents were separated into three role-playing "stations" which required the application of skills learned from the prior lectures. Each resident was videotaped during the session, allowed to view the videotape alone, and returned to the group to receive feedback with faculty supervision.

KEY LESSONS LEARNED: Verbal and written evaluations revealed that the program was well-received by both residents and medical students, who agreed that it was a needed addition to the curriculum. Suggestions included having the sessions during the daytime rather than in the evening, and increasing the proportion of time devoted to role-playing exercises.

STATEMENT OF PROBLEM: The learner as teacher program is an effective tool to teach residents much-needed skills that are often overlooked in the formal curriculum. The use of an active participation model further strengthens the training exercise. The positive response from participants has prompted plans to incorporate similar modules into a longitudinal program to enhance the current residency curriculum at our institution.

A REQUIRED CLERKSHIP DESIGNED TO PROMOTE SELF-REFLECTIVE PRACTICE AND DEVELOP COMPETENCIES IN CLINICAL ETHICS AND END-OF-LIFE CARE. JJ Fins, CS Storey-Johnson, A Bernstein, R Payne, Weill Medical College of Cornell University, New York City

In response to the call for improved medical education about palliative care, the Weill Medical College of Cornell University designed Medicine, Patients, and Society III, a mandatory clerkship in the new curriculum to promote self-reflective practice and improve competencies in end-of-life care. This two-week course is notable for having students engage in what the anthropologist Renee Fox called, "participant observation." To help students develop their humanistic skills we have asked them to rotate on a clinical service at the Cornell campus of New York Presbyterian Hospital without having direct patient care responsibilities. They are required to keep a journal in which they note how care is given and received, observe contextual factors which influence the clinical dynamic, and reflect upon their own responses. Our goal in using this pedagogical strategy is to promote self-reflective practice. We believe that the cultivation of these observational and reflective skills is an essential ingredient in the promotion of professionalism and ethical practice. Course objectives include identifying psychosocial and contextual factors that influence care, ethical and legal issues at the end of life, and principles of pain and symptom management. Students are expected to demonstrate the ability to apply ethical norms to patient care, describe methods of pain and symptom management, communicate in an effective and humanistic manner, and articulate models of patient advocacy. Attitudinal goals include fostering professionalism in the care of patients, appreciating the importance of respecting cultural diversity, and assuming the responsibility for developing competency in these areas. To develop their technical skills in pain and symptom management, students participate in palliative care rounds at Memorial Sloan-Kettering Cancer Center, attend case management problem solving seminars in palliative care, and make a site visit to observe hospice care. Students attend additional seminar sessions to discuss their observations in patient care with a multidisciplinary faculty with expertise in palliative care and clinical ethics. These instructors are from the departments of medicine, neurology, psychiatry, pediatrics, and pastoral care. Students are evaluated on the basis of their seminar participation, written journal, and a 5-7 page paper in which they identify an ethical problem and formulate a solution which promotes patient advocacy. Preliminary evaluation of this clerkship shows that students report increased confidence in their palliative care skills and a heightened awareness of their ethical obligation to provide humane end-of-life care.

TEACHING PRIMARY CARE SKILLS. MA Fischer, EJ Malcolm, Internal Medicine, Stanford University, Stanford, CA

BACKGROUND: Increasingly medical practice is shifting from inpatient hospitalization to ambulatory treatment. Generalists are called upon to perform more procedures, and operate with greater speed. Though more training in outpatient sites is offered in some medical residency programs, categorical residents in many programs still spend the bulk of their training time on hospital based rotations. There is little time for exposure to, and proficiency in, skills required for success in outpatient primary care. Some primary care skills are procedure-based requiring interactive instruction.

GOAL: To develop primary care skills training sessions to address the educational needs of our residents and the diverse requirements of the subject matter, while integrating easily into our teaching program.

METHODS: The curriculum was developed through several methods: subject matter was generated from practicing clinicians in General Internal Medicine, Internal

Medicine subspecialties and other specialties; resident suggestion; published curricula. Final topics of two types were chosen: procedures or examination skills, and knowledge. Teachers were identified who could present the topics in small groups emphasizing hands on experience with models or volunteers, use of equipment, and open question-answer format. Sessions were scheduled for upper level residents (R2 and R3) once weekly. This limitation allowed smaller groups, and a narrower baseline of participant experience and skills.

RESULTS: Response to the program from both house staff and educators has been positive. We will discuss feasibility, attendance, teacher cooperation and selection, and program evaluation. Additionally we will offer details regarding the development of this curriculum that has become imbedded in our teaching program.

AN ONLINE PHYSICIAN SIGN-OUT PROGRAM. JO Gonzalez, N Hebela, L Bellini, Medicine, Hospital of the University of Pennsylvania, Philadelphia, PA

Traditional hand-written patient sign-out systems can be cumbersome and can lead to medical errors.

To assess improvement in physician satisfaction and perceptions of traditional and online sign-out systems as used by housestaff at a large academic teaching hospital.

A dynamic database-driven electronic alternative to the sign-out sheet used by housestaff throughout the hospital will be implemented. This electronic clinical reference serves as an on-line resource for physicians as they sign-out patients in their care. It is a fully functional, editable, on-line sign-out sheet. After receiving a user name and password, a physician may use the sign-out sheet by logging onto the site. They may then edit their patient list, change existing data, add, and delete information as necessary. Because the information can be accessed via the Internet, it is continuously available. Paper surveys were distributed to all housestaff in a university based training program in order to assess perceptions of the paper sign-out system. Survey questionnaires prior to the implementation of the online sign-out system were completed by 104 out of 173 (60%) department of medicine residents. These included 46 (68%) PGY 1; 33 (59%) PGY 2; and 25 (51%) PGY 3. Surveys showed that during their last clinical rotation: 1) 71% of all housestaff and only 58% of interns were satisfied with the quantity of information they received in a sign-out; 2) 60% of all housestaff and 57% of interns felt a poor sign-out affected patient comfort; 3) 38% of all housestaff and 30% of interns felt a poor sign-out affected patient safety.

KEY LESSONS LEARNED: Paper sign-out systems result in poor satisfaction with the quantity of data received and are perceived by housestaff to have a negative impact on patient comfort and safety.

INNOVATIONS IN CROSS-CULTURAL EDUCATION. AR Green, JR Betancourt, E Carrillo, Internal Medicine, New York Presbyterian Hospital; Internal Medicine, Weill Medical College of Cornell University, New York, NY

OBJECTIVES OF PROGRAM/INTERVENTION: Cross-cultural training is now recognized as a critical component of medical education, especially given the current attention to racial/ethnic disparities in health care. Initiatives in this area have not been widely implemented and have employed educational methods that are more theoretical than practical given the challenges of caring for diverse patient populations in busy clinical settings.

DESCRIPTION OF PROGRAM/INTERVENTION: We developed and implemented a cross-cultural curriculum for medicine residents with the overall goal of improving their ability to understand, communicate with, and care for patients from diverse backgrounds. Our specific objectives were to: Raise awareness of racial/ethnic disparities and sociocultural factors in health; Teach an individual patient based (rather than culture specific) approach to care; Emphasize practical, efficient application of skills in the clinical setting; Employ innovative, experiential teaching techniques; Use open-ended surveying methodology for continuous quality improvement.

FINDING TO DATE: The curriculum is composed of five modules (basic concepts, core cross-cultural issues, the meaning of the illness, the social context, and negotiation) taught in four weekly two-hour sessions. The following innovations facilitate discussion and feedback, and help residents put cross-cultural skills into practice: Cases for problem based learning are used to develop awareness of the issues; Videotapes of simulated cross-cultural doctor-patient encounters are shown to demonstrate the concepts and skills in action and to generate discussion; Actors for cross-cultural medical interviewing are employed to allow residents to put the skills to use and create a "living laboratory" for learning. Through these simulated cross-cultural encounters residents utilize the techniques by directly interviewing the actor/patient, by observing and critiquing each other, and through feedback from the actor and the instructors.

KEY LESSONS LEARNED: Qualitative data obtained from the residents through open-ended surveying techniques has been used for continuous quality improvement. This has led to the innovations described above and some of the key lessons learned below.

STATEMENT OF PROBLEM: Cross-cultural curricula can be feasibly integrated in standard medical training. Interactive case-based learning was greatly preferred over didactic presentation. Use of actors for simulated patient encounters was well-received and useful. Succinct introductory discussion of residents' own cultural backgrounds and attitudes towards health within their families was helpful. Medical literature and media coverage on racial/ethnic disparities in health was effective in increasing awareness of importance of "cultural competence."

INNOVATIVE TRAINING FOR THE TRANSITION FROM GENERAL INTERNAL MEDICINE (GIM) TO OCCUPATIONAL MEDICINE (OM). J. Green-McKenzie, E Emmett, Occupational Medicine, University of Pennsylvania Medical Center, Philadelphia, PA

STATEMENT OF PROBLEM: Many general internists move into the field of OM without formal training. This can limit their progression in the field. They may be excellent clinicians but lack in-depth training and experience in population based preventive medicine, risk assessment and communication, knowledge of workplace disability and work placement, and organizational management. In contrast, OM physicians with 1yr clinical residency training may lack the clinical expertise to evaluate complex causation and employment issues. Overall, there is a shortage of physicians with a combination of both in-depth clinical and OM training. Barriers that may deter otherwise motivated internists from specialty training in OM include the need to leave current employment, relocation and disruption of the family. Overcoming these barriers will allow better medical care for underserved workers and a better physician workforce in OM. Specialty training in OM consists of 1-3 clinical year(s) (as obtained in GIM), an academic year (off-campus MPH training is available), and a supervised practicum yr.

OBJECTIVES OF PROGRAM/INTERVENTION: To train up to 8 residents per year at the University of Pennsylvania (Penn) through an innovative high quality competency-based supervised practicum year without requiring them to leave their current employment.

DESCRIPTION OF PROGRAM/INTERVENTION: This external residency designed for physicians who have an MPH as well as have substantial clinical training, accepts trainees employed full-time in OM at a site with adequate supervision and appropriate training opportunities. Competencies to be achieved are based on Program Requirements for Residency Education in Preventive Medicine (ACGME) and those of the American College of Occupational and Environmental Medicine. Residents undertake a year long intensive 3-day per month didactic program in Philadelphia arranged around general requirements and 5, 2-month long competency modules: The Workplace-Hazard Recognition, Evaluation and Control; The Worker-Disability and Work Fitness; Organizational Management; Environmental Health and Risk Assessment and Population based OM. Residents complete an in-depth project in each module and may be required to take additional rotations at Penn. Four site visits are conducted to each resident's workplace by a faculty preceptor to monitor and augment training.

FINDING TO DATE: This program, now in its third year, is highly rated by the residents when compared with traditional residency experiences. Enrollees have numbered 4-8 per year. Graduates appreciate their enhanced ability to handle the myriad challenges facing OM physicians. All who have sat for board examinations have been successful.

KEY LESSONS LEARNED: Innovative solutions can provide excellent training for mid-career physicians seeking to alter their practice focus and specialty field.

A "NEWLYWED GAME" FORMAT TO PROMOTE DISCUSSION OF RESIDENT STRESS AND RELATIONSHIPS. C. Griffith, D Rudy, University of Kentucky College of Medicine, Lexington, KY

STATEMENT OF PROBLEM: The stresses of residency can be challenging for relationships. Anecdotally, we had observed that most discussions or seminars on resident stress and relationships were somber and tedious affairs, with limited participation from either residents or spouses, and what participation that occurred was so unrelentingly serious that we doubt any stress was alleviated.

OBJECTIVE OF PROGRAM: To remedy this, as part of two two-day resident retreats, we devised a "Newlywed Game" format to enliven discussion of the effects of resident stress on relationships.

DESCRIPTION: On the first day of the resident retreats, we recruited five resident couples to participate in the Newlywed Game, with the other approximately 30 residents as the audience. Although focused on married residents, it was stressed that the themes our game were exploring applied to all resident relationships, including family or any significant other (including same sex relationships). As in the Newlywed Game, initially husbands would leave the room while their wives would answer five questions, with the husbands returning and points received if the husband's answers matched their wives. Next, the wives would leave the room, with husbands answering questions and then wives returning to answer the same questions, to receive points for matched answers. Questions were devised to bring up major issues regarding resident stress in a relaxed, often humorous atmosphere, to engender and enliven discussion. Sample questions included: If you could add a medicine to your husband's food, what would it be? a) Prozac; b) Valium; c) Viagra; d) Ritalin; e) Insulin. Other questions asked, for example, what cartoon animal is your spouse most like when stressed? (most common answer was Eeyore); or what song best describes your love life? a) U Can't Touch This; b) You Shook Me All Night Long; c) Minute Waltz; d) All by Myself (most common answer was Minute Waltz). Full discussion of the issues (and conflicts) which may arise in relationships with resident stress occurred after the "game". The second day of the retreat focused on stress-reduction techniques, informed by the discussions.

FINDINGS: The Newlywed Game proved to be wildly popular with the residents, and engendered lively, interactive discussion of resident stress and relationships (including a great deal of discussion of the stresses on other relationships besides marriage). Spouses were much more involved in this format than in our usual small group discussions.

INTERPERSONAL ROUNDS: A NEW PATH TO RELATIONSHIP. BJ Guise, ZE Neuworth, Medicine, Lenox Hill Hospital, New York, NY

STATEMENT OF PROBLEM: There is a need to develop educational strategies, which will 1)improve a physician's understanding that the relationship between doctor and patient is integral to medical care, and 2)improve a physician's ability to create meaningful relationships with patients. Advances in communication skills training for physician's have provided necessary but not sufficient solutions to these problems.

OBJECTIVES OF PROGRAM/INTERVENTION: 1) To enable learners to increase their own sense of imperative to establish meaningful relationships with patients in order to succeed in providing excellent medical care. 2) To enable learners to improve their ability to establish meaningful relationships with patients. 3) To develop a laboratory for the study of empathy as an independent pathway to meaningful physician-patient relationship.

DESCRIPTION OF PROGRAM/INTERVENTION: Interpersonal Rounds consists of an hour-long learning experience with a medical service ward team, two faculty facilitators, and a patient as participants. Interpersonal Rounds occurs in three phases over the course of the hour. 1) The team is assembled on their inpatient unit in order to form the agenda for the rounds. Facilitators provide the general guideline of learners getting to know one of their patients better, as a person. The team selects a patient for whom this activity might clarify or address obstacles the team is facing. The team selects someone to lead a 15-minute bed-side encounter. The team then reassembles in the conference room to debrief. The debriefing is facilitated such that learners are most likely to reflect on their experience of exploring the patient as a person, and what new perspectives they may have acquired. Facilitators attempt to support the natural teaching relationships that form on the team in the dynamics for the discussion.

FINDING TO DATE: Thus far, our findings are based upon facilitator observations, and feedback from learners and chief residents (solicited and unsolicited). Learners enjoy the rounds and state that these conversations make a real difference for them and for patients. Learners report that they come to understand the patient's perspective on his/her life, experience of the illness, treatment, and hospitalization overall. Learners report that they want to and are more confident that they can have similar conversations with patients routinely. Participants reported that they had experienced making time-efficient, meaningful connections to patients.

KEY LESSONS LEARNED: Empathy can be observed and experienced in this practical and palatable educational format. Empathy holds promise as an accessible, independent pathway to meaningful physician-patient relationship. Learning experiences which occur in the context of patient care and permit learners to arrive at and address salient barriers to care are perceived as enjoyable, useful, and transformative by participants.

FACULTY DEVELOPMENT FOR PROBLEM-BASED LEARNING TUTORS THROUGH PEER OBSERVATION AND FEEDBACK. P. Haidet, S Pierrel, B Moran, L Yeoman, Houston Veterans Affairs Medical Center; Baylor College of Medicine, Houston, TX

STATEMENT OF PROBLEM: Problem-based learning (PBL) is a modality of teaching that is employed with increasing frequency in undergraduate and graduate education throughout North America. Primary care physicians make unusually good facilitators because of their tolerance for ambiguity and experience in physician-patient communication. A problem, however, is that most current faculty recruited to be facilitators have not experienced PBL during their own education. Further, extensive training in facilitation is often not feasible, given faculty time and practice commitments.

OBJECTIVES OF PROGRAM/INTERVENTION: Improve the quality of facilitation in PBL groups and enhance learning outcomes.

DESCRIPTION OF PROGRAM/INTERVENTION: We have developed an innovative process to provide facilitators with immediate, formative feedback. Our process, while based upon the experience of expert trainers, employs the facilitators themselves to observe one another and provide feedback. In this process, each facilitator participates in two sessions per semester, once to observe and once to be observed. These sessions provide an opportunity for facilitator's personal growth by focusing them on the *process* (rather than the *content*) of teaching/facilitating. Since facilitators may have varying levels of experience and sophistication in PBL, we developed a qualitative/quantitative instrument that guides the peer observer through a series of observational tasks focused on teaching behaviors. Using our instrument, we train facilitators in observation and feedback at the beginning of each semester.

FINDINGS TO DATE: Our initial experience in pilot testing this program has shown it to have two advantages over traditional forms of feedback. First, the feedback that is generated is immediate and is grounded in changeable teaching behaviors rather than in summative outcomes. Second, it requires a minimum of resources, since the facilitators themselves act as the observers.

KEY LESSONS LEARNED: This process has potential for faculty development beyond PBL, in that it can be adapted to a variety of small-group learning situations including attending rounds, journal clubs, and others. Our exhibit will feature descriptions of the development and use of our process. We will also have our observation instrument available and a training video that will allow participants to experience our guided observation process. We will be on hand to discuss our process and our plans for testing its effects on feedback, teaching behaviors, and learning outcomes.

PREPARING RESIDENTS TO SUCCEED IN MANAGED CARE: A FOUR-MODULE CURRICULUM. R Halpern, MY Lee, J Levin-Scherz, T Silverman, R Phillips, Tufts Managed Care Institute; Tufts University School of Medicine, Boston; Tufts Health Plan, Waltham, MA

STATEMENT OF PROBLEM: Residents need to be trained for effective practice in the managed care environment. However, programs have been slow to incorporate new curricula, in part because faculty themselves are unfamiliar with the content and lack easy-to-use instructional materials.

Objectives of Program/Intervention: As an original participant in the Partnerships for Quality Education initiative, Tufts Managed Care Institute (TMCI)—a non-profit educational collaboration of Tufts Medical School and Tufts Health Plan—proposed to develop faculty-friendly, modular courseware that would provide residents with quality instruction in managed care principles and practices. TMCI would conduct faculty development programs focused on the content and methods to facilitate implementation.

DESCRIPTION OF PROGRAM/INTERVENTION: TMCI developed courseware, *Preparing Residents to Succeed in Managed Care (PRS)*, comprised of four exportable modules in three formats: a self-paced CD ROM presenting the essentials of managed care; a rotation comprised of onsite learning experiences in community practices and managed care organizations; and structured classroom courses in patient-centered care and clinical practice guidelines. The CD ROM is an interactive learning tool that places the resident within a group practice; through a series of case-based scenarios the resident learns basic tenets, tools and techniques of managed care. The rotation module recommends flexible structured sessions with physicians, practice staff, hospital administrators, health plan managers, etc., designed to teach residents about the healthcare environment and keys for successful practice. The classroom modules incorporate didactic and role playing components including discussions of the ethics of managed care practice, negotiating treatment plans with patients, and how to find, assess and apply practice guidelines. TMCI conducts faculty development programs to train faculty to use these modules most effectively.

FINDING TO DATE: TMCI has trained faculty from 75 residency programs and distributed more than 500 copies of the CD ROM. Faculty have rated the faculty development programs highly (overall average score of 4.7 out of 5), and report that residents have responded well to the substance and instructional designs of the modules. Tests of resident attitudes towards managed care concepts before and after exposure to PRS modules are available with the courseware; seven programs have used the pre- and post-surveys, and results are being analyzed.

KEY LESSONS LEARNED: Flexible, pre-packaged teaching materials can help faculty more easily incorporate the principles and practices of managed care into residency training. Faculty themselves benefit from instruction in the content as well as in the use of these materials.

FACULTY DEVELOPMENT FOR HOUSESTAFF: THE ROLE FOR FORMAL EVALUATION SESSIONS DURING CLINICAL CLERKSHIPS. PA Hemmer, T Jamieson, LN Pangaro, Medicine, Uniformed Services University, Bethesda, MD

STATEMENT OF PROBLEM: In working with medical students, developing the evaluative and teaching skills of housestaff is a dynamic process. Faculty development initiatives should be tailored to the unique needs of housestaff, including the frequent change in services and their level of experience.

OBJECTIVES OF PROGRAM/INTERVENTION: The formal evaluation sessions held during the internal medicine clerkship at the Uniformed Services University (USU) provide a unifying forum of "real-time", "case-based" faculty development for housestaff at multiple teaching sites, while also yielding credible student evaluation and formative feedback.

DESCRIPTION OF PROGRAM/INTERVENTION: Formal evaluation sessions are planned meetings, held monthly at all 7 USU clerkship sites, at which the clerkship director, or on-site clerkship director, meets with all of the instructors who are working with students currently on the clerkship. Clerkship goal and expectations are reinforced (Reporter-Interpreter-Manager/Educator; Acad. Med. 1999;74:1203-7). Fifteen minutes is spent discussing each student. The clerkship director makes notes of the instructor's comments. Individual feedback is provided to the students the following day during meetings with the clerkship, or on-site, director. At the evaluation sessions, we model and teach the core components of the Stanford Faculty Development Program.

FINDING TO DATE: An intern will spend 8-10 hours in formal evaluation sessions each academic year; a resident will spend 4-5 hours. This allows us to focus our attention on the least experienced instructors. Through the use of formal evaluation sessions, a synthetic evaluation framework, as well as pre-clerkship orientation meetings and post-clerkship student feedback, we have achieved a reliability of evaluation similar to quantifiable examinations (Assoc. for Med Educ in Europe Conference, Sept. 1997). We have improved the identification of marginally performing students (Acad Med 1997; 72:641-3; Acad Med 2000; In Press), and identified students at risk of poor performance during internship (Acad Med 1998; 73:998-1002). This reflects, in part, the power of the evaluation sessions in helping housestaff mature in their role as evaluators by modeling and teaching effective skills for use in their interactions with students.

KEY LESSONS LEARNED: Making formal evaluation sessions work as a faculty development initiative entails: a realistic commitment of time and resources from the Department Chairperson; a clerkship (or program) director invested in the process; the identification and on-going training of onsite directors; avoiding schedule conflicts which might preclude housestaff from attending. Nevertheless, these sessions play a critical role in the development of the evaluative and teaching skills of some of our most heavily relied upon instructors—our housestaff.

TEACHING FUTURE TEACHERS TO TEACH. ST Hingle, PH Rockey, B Wang-Cheng, JS Lloyd, M Peterson, Medicine, Southern Illinois University School of Medicine, Springfield, IL; Medicine, Southern Illinois University School of Medicine, Springfield, IL; Medicine, Medical College of Wisconsin, Milwaukee, WI; Education, Loyola University Chicago, Maywood, IL

STATEMENT OF PROBLEM: Most all medical students enter residency training programs where one of their many responsibilities is teaching medical students. As they progress through the residency program, they also teach junior residents. Some residency training programs have developed Residents as Teachers programs. However, this is not yet universal. One purpose for getting senior medical students involved in teaching is to prepare them for residency. Another is to promote improved clinical skills. Also they serve as a resource for clinical faculty.

OBJECTIVE OF PROGRAM/INTERVENTION: The four presenters will share various experiences of their medical schools including successes and failures, barriers and pitfalls. The benefit of such training programs will also be explored.

DESCRIPTION OF PROGRAM/INTERVENTION: Each institution believes providing fourth year medical students with the skills to be better teachers is an important endeavor, but has approached it somewhat differently. We will discuss commonalities and differences in approach. Each panelist will take 10 minutes to describe his/her institution's approach. Specific examples of senior medical students demonstrating teaching techniques will also be displayed by videotape. The session will be continuously open for questions, answers, and sharing. A copy of all slides and summaries of the curriculum from each presenting institution will be provided to seminar attendees.

FINDING TO DATE: Survey data is currently being tabulated, to be summarized at a later date.

KEY LESSONS LEARNED: Utilization of MS 4's is an effective teaching tool. If properly taught, MS 4's are effectively utilized as teachers.

DEVELOPING A CONTEXTUAL CURRICULUM FOR UNDERGRADUATE MEDICAL EDUCATION. JC Hofmann, JE Swartzberg, WT Boyce, UCB-UCSF Joint Medical Program, UC Berkeley, Berkeley, CA

STATEMENT OF PROBLEM: As we educate physicians for the 21st century, a thorough understanding of the behavioral, psychosocial, and multicultural contexts of medical illness will be increasingly important in order to successfully manage chronic diseases of an aging population.

OBJECTIVES OF PROGRAM/INTERVENTION: To address these issues, the UC Berkeley-UCSF Joint Medical Program is developing an undergraduate medical curriculum in "contextual medicine" focused on the lives of individual patients in the contexts of their families, communities, and society at large. The objectives of this curriculum are: to create an educational setting in which medical students develop a greater understanding of the societal, ethical, and human "context" of medical care; to create a generalizable set of models for developing a contextually-based curriculum; and to study the effect of contextualization and evaluate outcomes.

DESCRIPTION OF PROGRAM/INTERVENTION: The curriculum in contextual medicine will feature a fully integrated, patient-centered curriculum. All core medical education courses will be focused upon a series of patient histories and the illumination of illness from behavioral, community, societal, and ethical perspectives integrated with those of human molecular biology and pathophysiology. Specific components of this curriculum are currently being developed by collaborative groups of research scientists, clinical teaching staff, practicing community physicians, and students. They include: integration of basic and clinical sciences courses around a core group of richly complex patient cases taught in cooperative, small group learning settings; an electronic curriculum database; a student-organized free clinic for the homeless; training projects and student research opportunities in community medicine and public health; year-long clinical experiences in which individual students follow and assist a single patient in the midst of a major life transition; and long-term student-faculty mentoring relationships.

FINDING TO DATE: The contextual curriculum is a work-in-progress. Faculty, administration, and students have given unanimous support for the development of this program and curricular changes are being well received. Our exhibit will describe the research design for the contextual curriculum, highlight faculty and curriculum development models, outline student evaluation techniques, and discuss applicability to other programs.

KEY LESSONS LEARNED: The process of increasing faculty awareness and knowledge of the entire curriculum, and engaging active faculty participation and cooperation in continuing curricular change is critical, as is "top-down" political support. We believe a contextually-based curriculum in medical education will better prepare physicians to deal more effectively with the behavioral and psychosocial aspects of medical care and improve the quality of patient care.

FACULTY DEVELOPMENT IN AMBULATORY TEACHING: INITIAL RESULTS FROM A NATIONAL CONFERENCE. I Houston, J Clark, D Kern, P Alguire, D Boulware, J Bowen, W Branch, R Esham, G Ferencik, R Kahn, R Horowitz, Johns Hopkins School of Medicine

Despite increasing emphasis on ambulatory teaching in medical education, faculty development in this area has been limited and fragmented.

To improve faculty development in ambulatory teaching on a national level. The General Internal Medicine Faculty Development Project: Generalist Faculty Teaching in Community-Based Ambulatory Settings, funded by HRSA, is a collaborative

effort of the major organizations in internal medicine. (Bowen et al. *AJM*: 107, 193-197) Teams from internal medicine teaching programs are selected to attend one of three national conferences based upon a submitted plan for faculty development addressing ambulatory teaching. Teams include an administrative team leader, a community-based teacher, and other key faculty members. Participants attend facilitated team meetings, plenary sessions, and 4 of 17 workshops related to educational skills and program implementation. Sixty-two teams (259 persons) were selected to attend the first conference in December, 1999. Response rate for the evaluation was 91%. Overall, 1% would not recommend the conference to a colleague, 9% would recommend as a satisfactory experience, 42% as a good experience, and 48% as an outstanding experience. Mean ratings for individual workshops ranged from satisfactory to outstanding. Workshops with the highest attendance were "Teaching in the outpatient setting: precepting skills" and "How Doctors Learn: Tips for Teaching Faculty to Teach." The "precepting skills" workshop received the highest rating (70% outstanding). Participants who identified themselves as being primarily community-based (CB) (N = 85) rated the conference similarly to hospital-based teachers (HB) (N = 130) (p = 0.4). On a scale listing prior training in 24 teaching skills, on average, CB teachers were trained in 4.5 fewer teaching skill areas, compared with the HB teachers (p < 0.001). CB teachers were less likely to have protected time for their faculty development project compared with HB teachers (odds ratio 0.49, 95% CI 0.24-0.98), but supervisor support and salary support rates were similar. Participants rated the likelihood that their group will accomplish the objectives of their plan with a mean = 4.2 (SD 0.73) on a likert scale from 1 = Not Likely to 5 = Extremely Likely.

KEY LESSONS LEARNED: This collaboratively implemented national faculty development project on ambulatory teaching succeeded in soliciting acceptable plans for faculty development from teams composed of hospital and community-based teachers from 62 teaching institutions. The first conference was highly rated. Information gathered will be used to improve subsequent conferences. We will assess the accomplishments of individual teams through prospective follow-up.

METACURRICULAR DEVELOPMENT: A CASE-BASED SESSION TO ENCOURAGE INNOVATION AND COLLABORATION IN CURRICULAR DESIGN. N. Hubert, SR Simon, Beth Israel Deaconess Medical Center, Harvard Medical School; Department of Ambulatory Care and Prevention, Harvard Medical School, Boston, MA

OBJECTIVES OF PROGRAM/INTERVENTION: General medicine fellows and junior faculty commonly are expected to develop educational curricula for housestaff and medical students, but rarely get formal teaching themselves about curriculum development. This may negatively impact the quality and effectiveness of their educational efforts.

DESCRIPTION OF PROGRAM/INTERVENTION: We developed a case-based, single-session exercise to help participants identify different approaches to curriculum development, improve their knowledge of the elements of curricular design, and practice skills of collaborative curriculum reform. Since this is a curriculum about developing curricula, we call our session "metacurricular development."

FINDING TO DATE: Participants are given a written case scenario and one-page assessment form that we created based on our prior experience in curriculum development and study of the literature on curriculum design (copies will be made available at SGIM). Participants are told they are newly hired faculty members at an academic medical institution and have one month to revise an existing curriculum for teaching Evidence Based Medicine (EBM). First, each participant writes down answers the following questions: Who is the target audience? Whom, if any-one, are they going to seek out for help? What concepts will guide their revision? What do they think a three-session EBM course should contain? How would they assess their curriculum? Participants then form small (3-4 person) groups and negotiate one "optimal" curriculum plan. Finally, the groups come together in a plenary session to present their thought processes and results. We will demonstrate the analysis grid we developed to compare these group data using concepts from literature on curriculum design.

KEY LESSONS LEARNED: This session was pilot tested among internal medicine and pediatrics fellows at Harvard Medical School in Fall 1999, with favorable informal feedback from most participants. Specifically, they reported that the session improved awareness of the potential for flexibility in curricular design and provided the opportunity to discuss the roles of experts and end-users (student learners) in the curriculum development process.

STATEMENT OF PROBLEM: This session on curriculum development using a realistic junior faculty scenario was well received by internal medicine and pediatrics fellows. In 1 1/2 hours, important concepts in curricular design and both intra-faculty and faculty-student collaboration can be addressed using this interactive case. We believe this session meets an important training need for future academic medical faculty.

A REMEDIATION PROGRAM IN CARDIOLOGY BEDSIDE SKILLS FOR INTERNAL MEDICINE RESIDENTS. S. Issenberg, MS Gordon, D Gordon, JW Mayer, JM Felner, Center for Research in Medical Education, University of Miami School of Medicine, Miami, FL; Office of Medical Education, Emory University School of Medicine, Atlanta, GA

STATEMENT OF PROBLEM: Essential bedside skills in cardiology are deficient among U.S. internal medicine residents. Reasons cited for these deficiencies include the lack of structured teaching and testing of these skills.

OBJECTIVES OF PROGRAM/INTERVENTION: We have developed a remediation program in cardiac bedside skills for second- and third-year internal medicine residents at the University of Miami School of Medicine.

DESCRIPTION OF PROGRAM/INTERVENTION: The program consists of physician led sessions and uses the UMedic multimedia computer curriculum that includes 10 patient-centered cardiology programs. The programs are linked to "Harvey," the Cardiology Patient Simulator, and include the history, bedside examination, laboratory data and treatment. Two sets of multimedia computer test items, equivalent in content and empirical performance, are used as pre- and posttests. (Reliability coefficients KR 20:0.81, 0.84, respectively.) Items assess auscultatory and non-auscultatory skills including the identification, interpretation and correlation of findings with cardiovascular diseases. The scope of bedside skills was framed by a multicenter consortium of cardiologists and educators.

Eight residents participate in five 2-hour sessions during a one-month ward teaching rotation that was created to provide structured learning time separate from their day-to-day patient responsibilities. Residents take the pretest at the beginning of the rotation and participate in instructor-taught sessions, using "Harvey" and UMedic, that focus on bedside skills. Residents take the posttest at the end of the rotation and complete an evaluation form designed to measure their satisfaction with the program and to quantify their patient care responsibilities during the rotation.

FINDING TO DATE: After the first 3 rotations, 25 out of a total of 90 second- and third-year residents completed pretests, 5 remediation sessions and posttests. Overall pre- and posttest means, standard deviations and ranges were 44.86, 14.86 and 20-64 and 70.43, 11.8 and 52-88, respectively (p < 0.001). Similar improvement was demonstrated in all areas of auscultatory and non-auscultatory findings. Residents rated all items regarding the program structure and content highly. Each house officer was responsible for an average of 36 inpatients and 18 outpatients during the 4-week rotation.

KEY LESSONS LEARNED: We have successfully developed a structured program in cardiac bedside skills using a simulator/computer-based standardized method to teach these skills to internal medicine residents that includes outcome measures. Residents enjoyed and appreciated the program and indicated that they wished to have similar experiences in other medical disciplines.

A SAFE PLACE TO GRIEVE-A GROUP FOR RESIDENTS TO DISCUSS SUFFERING AND DYING PATIENTS. VA Jackson, A Hollward, HG Worthen, RJ Pels, DH Bor, L Conant, Boston VA Healthcare System; The Cambridge Hospital, Harvard Medical School, Boston, MA

During residency, young physicians are frequently confronted with suffering and death. House officers report feeling increased isolation after the death of one of their patients (Ferris T et al., 1998). There is little time to engage in an in-depth discussion of these issues in traditional educational forums.

The goal of the group is to provide a safe place for medical residents to discuss experiences with suffering and dying. Our objectives are to improve patient care at the end of life as well as to foster the development of each resident by acknowledging and exploring the complicated interface of the personal and professional in our lives as physicians.

The group meets in the evenings for two hours twice monthly facilitated by two internists and one psychiatrist who share an interest in end of life issues and group process. The group begins with the sharing of a meal followed by a short silent writing exercise in transition to the time spent sharing thoughts and feelings about patients, family, and friends. The evening ends with a moment of silence, or a reading of poetry or prose. In the first year of the group, 66% of the medical house staff attended this voluntary evening group at least once. Twelve of 22 participants responded to a survey about the group. Mean values were calculated for the 5-point Likert scales. Residents felt the group provided a safe place to grieve (4.3) and that the group helped them explore their role as a doctor and as a fellow human being in caring for dying patients (4.5). Residents reported the group improved their care of patients at the end of life (4.0).

KEY LESSONS LEARNED: Residents will attend and value a group that provides a consistent respectful environment in which to discuss personal and professional experiences with suffering and death. Residents feel this type of group improves the care they give patients at the end of life. Facilitators face many challenges in offering an open-ended group with frequently changing busy participants.

OVERCOMING BARRIERS TO PHYSICIANS IN DELIVERY OF CARE TO PEOPLE WHO ARE HOMELESS. RJ Jahiel, PJ Ellis, Yale Primary Care Residency Program, New Haven and Waterbury, CT

PURPOSE: To understand and overcome barriers to physicians in the delivery of quality health care in homeless settings.

METHOD: Qualitative assessments of internal and external barriers were done before and/or during service of 30 general internal medicine residents at a homeless shelter clinic.

RESULTS: Internal barriers: 1) inability to comprehend the devastating effects of lack of a home; 2) spotty knowledge of homelessness and homeless populations; 3) stereotypes of and psychological distance from homeless people; 4) unrealistic view of physician roles in caring for homeless patients; 5) little preparation for work in low technology environments, 6) little advocacy training. External barriers: 1) time constraints; 2) incompatibility between circumstances of homelessness and medical management; 3) obstacles to follow up of patients in the medical care system; and 4) lack of needed resources.

CURRICULUM TO ADDRESS THESE BARRIERS: A) 90 minute seminar on: a) 10 functions of a home; b) diversity of homeless people and causes of homelessness; c) hazards of homeless life; d) psychological distancing, stereotyping, and unrealistic clinical expectations. B) 16-hour supervised clinical experience at the shelter

with a) 40 minutes per patient encounter; b) check list to elicit information on homelessness history and obstacles to a stable home; c) help in decisions on a medically optimal management versus a less-than-optimal one that is more feasible in homeless life; d) guided reflection on perceptions of homeless people; e) guidance to bypass obstacles to health care by adaptation of care, physician role modification, or team work. Outcome shows satisfaction of the medical residents. CONCLUSION: Efforts to improve the performance of physicians in the care of people who are homeless need to address internal and external barriers. It is feasible to package interventions targeting these barriers in a short time period in a community setting.

STANDARDIZED PATIENT WORKSHOPS TO TEACH RESIDENTS COMPLEX SKILLS IN WOMEN'S HEALTH AND PRIMARY CARE. D Kwolek, M Duke, Internal Medicine, University of Kentucky, Lexington, KY

STATEMENT OF PROBLEM: Optimal care of patients requires complex skills in interviewing techniques and nonverbal communication as well as medical knowledge. Direct faculty observation of residents to ensure competence in these skills is often limited by time constraints. Standardized patients have been used to assess these skills with medical students, but their use with residents has not been widespread. OBJECTIVES OF PROBLEM/INTERVENTION: In 1997 we developed clinical skills workshops on women's health and nonverbal communication that provide residents with focused instruction and feedback. The objectives of the workshops are to: 1. Evaluate residents' skills in interviewing, communication, and medical knowledge. 2. Integrate skills and knowledge relevant to the comprehensive care of primary care patients. 3. Refine residents' interviewing skills in difficult interviewing situations. 4. Provide residents with skills and knowledge relevant to the diagnosis of clinical problems in women's health, such as coronary artery disease, domestic violence, and sexual abuse.

DESCRIPTION OF PROBLEM/INTERVENTION: Using a modified Objective Structured Clinical Examination (OSCE) format, residents rotated through standardized patient stations to interview patients, including taking a psychosocial and sexual history. In other stations, residents evaluated risk of coronary artery disease and osteoporosis or dealt with issues of domestic violence and abuse. Some of the stations were videotaped. Feedback was provided to residents by standardized patients and faculty using predetermined checklists. Workshops were followed by an interactive, didactic session to reinforce key learning points. Videotapes were individually reviewed with each resident.

FINDING TO DATE: 1) Important deficits exist in resident evaluation of cardiac risk in women patients. 2) Many residents failed to uncover a history of childhood sexual abuse in a depressed patient. 3) Videotaping of selected workshops uncovered important problems in communication in some residents, which were previously undetected by faculty.

KEY LESSONS LEARNED: Clinical Skills Workshops are a valuable venue for resident teaching and may reveal deficiencies in resident skills that are not otherwise apparent. Limitations include cost, faculty time, and resident availability. Commitment by program directors to ensure attendance is important to maximize effectiveness of this intervention. Further evaluation is needed to determine if these workshops have a positive effect on patient outcomes.

HELPING RESIDENTS BECOME TEACHERS IN A COMMUNITY PROGRAM. HS Laird-Flick, K Taylor, Internal Medicine, St. Joseph Mercy Hospital, Ann Arbor, MI

OBJECTIVES OF PROGRAM/INTERVENTION: The transition from intern to senior resident requires management and teaching skills not explicitly taught within traditional medical curricula. Several of our residents expressed interest in such training.

DESCRIPTION OF PROGRAM/INTERVENTION: Our goal was to develop a brief, dynamic seminar to: 1) provide residents with a basic set of teaching skills; 2) improve teaching of interns and medical students; and 3) facilitate positive team interactions.

FINDING TO DATE: We conducted two, half-day workshops for senior residents, using a curriculum adapted from one previously described at Stanford. Topics included an introduction to educational frameworks, learning climate, communication of goals, understanding and retention, and feedback. Participants completed a post-seminar questionnaire (using a 5-point Likert scale) addressing overall usefulness of the program; its impact on attitudes; and a retrospective self-assessment of teaching performance.

		BEFORE	CURRENT
LEARNING CLIMATE	Stimulation	3.00 (0.69)	3.94 (0.42)
	Learner Involvement	3.00 (0.69)	4.17 (0.38)
	Respect/comfort	3.65 (0.93)	4.33 (0.69)
COMMUNICATION OF GOALS	Expression	2.78 (0.81)	3.94 (0.64)
	Negotiation	2.72 (0.83)	4.00 (0.51)
PROMOTING UNDERSTANDING AND RETENTION	Organization	2.85 (0.99)	3.80 (0.62)
	Clarity	3.05 (1.0)	3.85 (0.75)
	Emphasis	3.00 (0.65)	4.10 (0.55)
	Fostering Active Learning	2.65 (0.75)	3.95 (0.69)
FEEDBACK	Minimal	2.95 (0.89)	3.90 (0.72)
	Behavioral	2.85 (0.75)	4.05 (0.51)
	Interactive	2.63 (0.78)	3.95 (0.60)
CLINICAL TEACHING ABILITY	Inpatient setting	2.95 (0.83)	3.84 (0.50)
	As a lecturer	2.84 (0.76)	3.74 (0.65)

KEY LESSONS LEARNED: Overall, residents found the program useful (mean score 4.11, standard deviation (SD) 0.68), despite varying expectations beforehand (mean score 3.33, SD 1). They felt their teaching performance improved in all four topic areas (see table below). The program impacted their attitudes, knowledge, and teaching behaviors, and enhanced cooperative interactions among residents. They formulated a variety of personal goals, and identified barriers to their implementation, the most frequent of which was time.

STATEMENT OF PROBLEM: Senior residents in this community-based program derived benefit from a short, interactive program aimed at improving teaching skills and behaviors, despite variable expectations beforehand. Time constraints were felt to be the most important barrier to positive behaviors.

INTERNISTS' ATTITUDES TOWARD TEACHING RESIDENTS: PERSPECTIVES FROM A COMMUNITY HOSPITAL. HS Laird-Flick, M Yeh, Dept. of Internal Medicine, St. Joseph Mercy Hospital, Ann Arbor, MI

STATEMENT OF PROBLEM: Role modeling impacts the development of physicians-in-training. Medical educators have attempted to describe what makes a "good" role model, but little research exists regarding physician's decisions to teach.

OBJECTIVES OF PROGRAM/INTERVENTION: This study was designed to elucidate internist's attitudes toward working with residents, and factors influencing their decisions to teach, by surveying a spectrum of internists at a community-based hospital.

DESCRIPTION OF PROGRAM/INTERVENTION: A blinded, 29-item survey was distributed to 198 members of the Dept. of Internal Medicine at St. Joseph Mercy Hospital, Ann Arbor, Mich., which includes both teaching and non-teaching members. The survey addressed demographic features of respondents, attitudes toward teaching residents, and perceived benefits of or barriers to teaching. Univariate analyses on aggregate data were performed using SPSS software.

FINDING TO DATE: There was a 47% response rate. Most respondents indicated that they enjoyed working with residents (91%) and preferred settings which involved residents (88%). 63% agreed that they had made teaching a priority when selecting their current position. They identified five benefits of teaching: continuing medical education, keeping up-to-date with the current literature, personal and job satisfaction, and mentoring opportunities. They targeted four barriers: office time considerations, financial concerns, managed care, and administrative duties. Male and fellowship-trained physicians were more likely to have made teaching a priority (LR 7.41, $p = 0.006$ and LR 9.7, $p = 0.002$, respectively). Younger physicians were more likely to perceive faculty development and mentoring as benefits (LR 11, $p = 0.011$ and LR 10, $p = 0.018$). Physicians with more teaching experience were more likely to see different patient populations and compensation issues as barriers (LR 7.5, $p = 0.023$ and LR 5.9, $p = 0.05$), while those with less experience found faculty development to be of benefit (LR 7.5, $p = 0.02$). Those with more than 20 contact hours per week were more likely to identify time considerations as a barrier (LR 6.74, $p = 0.009$). Fellowship training increased the likelihood of identifying the four barriers of managed care (LR 7.5, $p = 0.006$), office time considerations (LR 3.9, $p = 0.047$), compensation issues (LR 14, $p = 0.001$), and administrative duties (LR 4.5, $p = 0.035$).

KEY LESSONS LEARNED: Community-based physicians in this study identified internally-oriented benefits and business-based barriers which did not preclude teaching. Several physician-specific characteristics influenced attitudes toward teaching. Understanding these influences may enable graduate medical programs to more effectively interact with community faculty members.

CONTINUOUS QUALITY CONTROL AND IMPROVEMENT OF AN INTERNAL MEDICINE CLERKSHIP. MY Lee, KM Collopy, PJ Wang, Office of Educational Affairs, Tufts University School of Medicine; Department of Cardiology, New England Medical Center, Boston, MA

STATEMENT OF PROBLEM: Quality control across multiple clinical teaching sites is a major concern for medical educators. Tuft's Internal Medicine clerkship presents major challenges due to 8 geographically dispersed hospital sites, each using multiple ambulatory sites to provide 33-50% ambulatory experience during the 12-week rotation.

OBJECTIVES OF PROGRAM/INTERVENTION: Tufts sought an evaluation system that would provide meaningful feedback for faculty to improve their clerkships, foster sharing of ideas among clerkship directors, allow comparisons across sites, adapt to needs of other core disciplines, and be administratively sustainable.

DESCRIPTION OF PROGRAM/INTERVENTION: Our standardized evaluation system for the Internal Medicine clerkship provides detailed feedback to every site. Web-based evaluation instruments administered to students yield a 97% response rate; general questions allow comparisons across all core disciplines, specific questions allow focused queries within a discipline. Quarterly reports of results (comparative bar graphs and extensive comments) are sent to clerkship directors, chairs, and directors of medical education, with established feedback loops among the dean's office, faculty, and sites. Every three years the Curriculum Committee performs in-depth evaluation of clerkships, with annual reports tracking the status of action plans. Targeted discussion of successes and problems occurs during regularly scheduled clerkship directors meetings. To maximize clerkship improvements, refinement of evaluation instruments and the feedback process is ongoing. An identical evaluation system is used for all 5 other core clerkships at over 100 departmental sites.

FINDING TO DATE: Student ratings of the Medicine clerkship have remained strong despite major increases in ambulatory sites. The system stimulates friendly competition among sites during discussions that also identify areas for deeper investigation through focus groups. Examples of impact include improving a problematic site (multiple systems problems were revealed, then corrected); providing focus for faculty development initiatives (workshops and discussions improved ambulatory teaching); and monitoring mid- and end-of-clerkship feedback to students (clerkship directors are improving compliance rates).

KEY LESSONS LEARNED: Tuft's Evaluation System provides a dynamic model for quality control and improvement across multiple clerkship sites. Capturing both student and faculty perspectives fosters critical discussion of content and teaching. Web-based evaluation instruments greatly increase administrative efficiency to convey essential results quickly to faculty and the Curriculum Committee.

DEVELOPMENT OF A BEHAVIORAL SCIENCE CURRICULUM IN A PRIMARY CARE INTERNAL MEDICINE RESIDENCY. BR Leslie, L Adams, L Cyran, S Kick, J Rifkin, Division of General Internal Medicine, University of Colorado Health Sciences Center, Denver, CO

PURPOSE: To describe the development and implementation of a behavioral science education program for primary care internal medicine residents.

METHODS: Psychosocial problems account for up to 45% of patient visits in a primary care setting. Physicians who can address these problems produce better patient outcomes and higher patient satisfaction. Recognizing the compelling need for behavioral science training, our primary care residency program decided to hire a full time psychologist in September 1999. Goals for this position included: 1) development and implementation of a three-year behavioral science curriculum; 2) provision of individual precepting for residents; 3) consultation with faculty and residents on specific patient problems; and 4) role modeling collaboration between physicians and behavior scientists in primary care settings. Needs assessment activities included individual meetings with faculty and residents, observation of residency clinic operations, and review of existing curriculum to identify opportunities for integrating behavioral sciences into traditional medical teaching. Existing curricula from family medicine programs and ABSAME (Association for the Behavioral Sciences and Medical Education) were reviewed.

RESULTS: The initiation of a more comprehensive behavioral science program has been well received. Feedback from faculty and residents has been positive and has helped prioritize the following program activities. 1) Curriculum development and implementation: The curriculum is delivered in multiple formats including longitudinal three year block sessions for complex topics (i.e., communication skills, behavioral change) and noon seminars on specific topics (i.e., somatic disorders, pain management). 2) Individual precepting: All residents have regularly scheduled one-hour behavioral science appointments. This time is used for case presentation or conjoint patient interviews. Videotapes of patient/physician interactions are also reviewed. 3) Precepting: Residents have continuity clinics at three sites. The behavioral scientist is regularly scheduled at these sites to consult with the residents and faculty around specific patient presentations. 4) Role modeling: As a result of consultation on specific patients, the behavioral scientist sees patients for assessment and psychotherapy. Care is delivered in collaboration with the resident or faculty member.

CONCLUSION: The need for behavioral science training in primary care residency is well documented. Implementation of a behavioral science curriculum involves identifying existing resources, designing a longitudinal curriculum, and responding to resident and faculty needs as they care for patients.

TEACHING MEDICAL ETHICS IN CYBERSPACE. AJ Lipman, RM Sade, MF Marshall, A Qlotzbach, C Lancaster, Medical University of South Carolina, Charleston, SC

PURPOSE: In medical training, computer-based methods of instruction offer the possibility of improvements in teaching clinical skills and professionalism. Without rigorous documentation of its pedagogic advantages, belief in the utility of internet-based teaching is not solidly grounded. We have carried out a prospective, randomized study of educational outcomes, comparing a traditional classroom course in clinical ethics with the same course supplemented by internet-based discussion.

METHODS: Introduction to Clinical Ethics is a sophomore medical school course that teaches a specific method for analyzing clinical ethical problems. Our sophomore class was randomly assigned to either classroom teaching alone (traditional group; n = 65) or classroom teaching supplemented with internet-based discussion of cases illustrating ethical issues (cyberethics group; n = 62). A final case analysis comprehensively evaluated student's understanding of the analytic method taught in the course. We compared the grades assigned by external reviewers to the case analyses of the two groups.

RESULTS: The student's understanding of ethical analysis, as measured by grades on the final paper, was significantly higher in the cyberethics than in the traditional group (3.0 ± 0.6 and 2.6 ± 0.7 , respectively; $p < 0.005$).

CONCLUSION: Our study provides objective data documenting the incremental value of internet-based teaching of clinical ethics to sophomore medical students.

A SYSTEMATIC APPROACH TO IMPROVING AN INTERDISCIPLINARY CURRICULUM. DC Lynch, AG Greer, K Springer, BS Harriet, LC Larsen, MC Clay, DM Cummings, East Carolina University School of Medicine, Greenville, NC

STATEMENT OF PROBLEM: Community-based interdisciplinary curricula comprise one approach to teaching medical students how to work with health care teams and access community resources to improve patient and community health. The relative novelty of this approach to medical education requires a systematic mechanism to (i) monitor curricular processes and outcomes, and (ii) facilitate continuous improvement of these curricula.

OBJECTIVES OF PROGRAM/INTERVENTION: The purpose of the community-based Interdisciplinary Rural Health Training Program (IRHTP) is to prepare students in medicine, health education, nursing, nurse practitioner programs, nutrition, pharmacy, physician assistant programs, and social work for interdisciplinary practice in rural settings and improve the health of the community.

DESCRIPTION OF PROGRAM/INTERVENTION: Consistent with the tenets of continuous improvement (CI), the Plan, Do, Study, and Act (PDSA) cycle was used to assist course planning, administration, and evaluation. The "Plan" phase occurred when an interdisciplinary faculty team revised IRHTP content and evaluation to enhance the experience for all students. The "Do" phase was characterized by implementation of planned revisions to the course. The "Study" phase involved examination of data obtained from students and faculty before, during, and at the end of, the revised course via questionnaires, journals, case conferences, web-based threaded discussions, and meeting notes. Further modifications were implemented during the "Act" phase.

FINDING TO DATE: The results indicated (i) there was a significant increase in student knowledge of the education and roles of other members of the health care team ($p < 0.005$) (ii) 83% of students could describe the inter-relatedness of social, political, legal, and economic factors in terms of their influence on community health, and (iii) 70% could list recommendations for improving community compliance with current standards of care. Other results, however, prompted several curricular changes. Structured opportunities for interactive discussion of interdisciplinary care among students and community members and for peer teaching were increased, time allocated toward lectures and independent reading was decreased, paperwork was reduced by integrating some assessments into required assignments and by adding banked cases to the case conference schedule, and the criteria for attaining outcome objectives were modified.

KEY LESSONS LEARNED: The PDSA cycle provided a systematic approach to improving an interdisciplinary rural health curriculum. Elements consistent with CI and central to the PDSA cycle included, 1) an intensive planning phase, 2) implementation of the revised curriculum and collection of relevant data, and 3) data review and utilization to guide further changes to curricular content and assessment.

INTERNET-BASED ROTATION AND ATTENDING EVALUATIONS BY INTERNAL MEDICINE RESIDENTS. ID Mackenzie, Community Health Services, Denver Health Authority; General Internal Medicine, University of Colorado Health Sciences Center, Denver, CO

STATEMENT OF PROBLEM: Prior to 1997, we were using multiple distinct forms for internal medicine residents to evaluate their experiences at 5 hospitals and 7 continuity clinic sites. Lack of standardization limited objective comparisons across sites. In addition, filing and distributing the forms was labor intensive.

OBJECTIVES OF PROGRAM/INTERVENTION: The primary objective of this project was to standardize resident's rotation and attending evaluations across five hospital sites and seven continuity clinic sites. Secondary objectives were to reduce the administrative expense of distributing, copying, and entering data from paper evaluations, to transition all 135 internal medicine residents to electronic mail (email) communication, to improve computer literacy of the residents, and to provide residents with greater assurances that their comments about rotations and attendings are confidential.

DESCRIPTION OF PROGRAM/INTERVENTION: In July, 1997, we required that all internal medicine residents in our program sign up for an email account. We then developed three standard evaluation forms that were approved by the participating University-affiliated institutions. These include a monthly rotation evaluation form, an attending evaluation form, and a continuity clinic site evaluation form. Using Microsoft FrontPage™, a web page development program, the author converted the evaluation forms into web pages accessible to anyone on the Internet. In 1998, it became clear that a more sophisticated server application was needed to maintain the complex relational database and to allow for easy updates to the system. In 1999, the forms and database were converted to Cold Fusion, a web server application with links to Microsoft Access. With this conversion, the completed evaluations are automatically sent by email to the program director, the chairman of Medicine, the attending of record, and the current Chief Residents. Also, residents and the supervising faculty are better able to track delinquent rotations evaluations.

FINDING TO DATE: Since the new pages went live in February, 1999 we have ostensibly required clinic evaluations every 6 months and rotation evaluations every month. We have received $103\%_{270}$ continuity clinic evaluations and $335\%_{1620}$ monthly rotation evaluations. We have also received 131 optional attending evaluations. The response rate is much less for senior residents than for interns. Despite the low response rate, we have been able to make objective comparisons across rotation sites for the first time. Feedback from faculty and residents is mixed.

KEY LESSONS LEARNED: In the eyes of those who look at evaluation forms, the program has been very successful. The challenges that face us are 1) acceptance by rotation directors so that paper forms can be discarded, and 2) building incentives for residents to complete the web pages.

A WEB SITE AS AN ADJUNCT TO A LONGITUDINAL AMBULATORY CLERKSHIP FOR MEDICAL STUDENTS. EJ Malcolm, SL McKenna, DE Merrill, VM Davis, RA Jensen, JR Stringer, R Steinhart, P Robertson, E Wolfe, Internal Medicine, Stanford University School of Medicine, Stanford, CA

In designing a nine-month longitudinal ambulatory care experience, we faced the difficulty of communicating with medical students and preceptors in diverse community and university settings.

- 1) Develop an interactive web site which allows students and preceptors in remote continuity clinic sites access to curricular materials and course information.
- 2) Insure frequent and timely feedback by making student evaluations available to preceptors on the web site.
- 3) Provide an on-line forum for students to discuss patient issues in between monthly didactic sessions.

Stanford Medical School has 4 years of experience providing comprehensive support for 41 medical school courses including: on-line searchable syllabi, streaming video, and interactive web based teaching programs. We have leveraged this experience to produce a web site for our longitudinal ambulatory clerkship for medical students that provides: (1) access to the web based clerkship syllabus; (2) online information about the clinic, searchable maps and didactic schedules; (3) electronic journals that can be accessed by students and preceptors in remote sites; (4) web-based feedback and evaluation forms; and (5) links to primary care resources on the web. Finally, in order to establish a base of institutional memory, we routinely update the web site with useful teaching points from the monthly student presentations, relevant communication from e-mail correspondence between students and preceptors, as well as frequently asked questions.

Students have found that the web site provides easy access to didactic materials and references. Students have spent an average of 7 minutes per page of didactic material. They have also found it useful for obtaining logistical clerkship information. The most commonly visited pages on the site (excluding lecture topics) included: "Goals and Objectives", "Course Structure and Policies", and "Purpose". We have not yet been able to evaluate preceptor use of the site as we have only recently acquired user access for the preceptors.

KEY LESSONS LEARNED: 1) The web site is a useful adjunct to traditional means of distributing course materials, particularly in a longitudinal ambulatory clerkship.

2) Preceptor use of the web site requires specific attention to education of the faculty regarding access and use of the web site.

3) The web site may provide a secondary incentive for community providers as it provides access to the medical library, librarian assisted searches and full text articles.

THE TEACHING OF CARDIAC AUSCULTATION DURING INTERNAL MEDICINE RESIDENCY: HAS ANYTHING CHANGED IN THE 1990s? S Mangione, DF Duffy, Department of Medicine and CRMEHC, Jefferson Medical College; American Board of Internal Medicine, Philadelphia, PA

STATEMENT OF PROBLEM: Increasing evidence gathered during the 1990s has indicated that errors in physical diagnosis may be quite common among physicians-in-training. Cardiac auscultation, a time-honored art and long the centerpiece of physical examination, seems to have particularly suffered. Whether this mounting evidence of trainee's deficits and errors might have caused a change in teaching practices and attitudes towards cardiac auscultation is unknown.

OBJECTIVES OF PROGRAM/INTERVENTION: 1) To assess the time and importance currently dedicated to the teaching of cardiac auscultation during internal medicine residency; 2) To compare 1999 data to those previously gathered by us in a similar survey conducted in 1990.

DESCRIPTION OF PROGRAM/INTERVENTION: We surveyed all accredited U.S. residencies in Internal Medicine. Data were collected by mailed questionnaire between January and June 1999. All non-respondents were sent a second copy of the survey one month after the initial letter. The survey instrument consisted of a one-page questionnaire identical to the one previously used in 1990. Time and importance dedicated to cardiac auscultation were then compared between 1999 and 1990 data.

FINDING TO DATE: There were $258/434$ responses in 1999 compared to $328/426$ in 1990. On a six-step scale, with six being the highest, 1999 respondents attributed significantly less clinical value than their 1990 counterparts to $7/11$ cardiac auscultatory findings. And yet, 1999 respondents gave significantly greater importance to cardiac auscultation ($5.6 \pm .6$ vs $5.3 \pm .8$, $P < .001$), and expressed greater desire for more auscultatory teaching during residency ($5.2 \pm .9$ vs 4.9 ± 1 , $P < .001$). This desire translated into a significant increase in the number of programs offering structured teaching of cardiac auscultation (48% compared to 27.1% for 1999 and 1990 respectively, $P < .001$). In contrast to the 1990 survey there was no inverse correlation between respondent's age and their wish for greater teaching of cardiac auscultation. There was, however, a persistently negative association between presence of structured auscultatory teaching and university affiliation of the residency program. This association was weak for 1990 (30.7% of non-university affiliated programs offered teaching compared to 21.9% of university-affiliated, $P = 0.07$) and stronger for 1999 (53.7% vs 40.2%, $P = .04$).

KEY LESSONS LEARNED: These data indicate a significant increase in the number of programs currently offering formal teaching of cardiac auscultation. This increase is significantly greater among non university-affiliated residencies. Whether this greater teaching emphasis will translate into greater clinical proficiency of primary care trainees needs to be demonstrated.

A QUALITATIVE AND QUANTITATIVE EVALUATION PROCESS FOR RESIDENT EDUCATION IN PRIVATE OFFICES. RM McGinn, M Smith, KD Ahern, Medicine, Staten Island University Hospital, Staten Island, NY

STATEMENT OF PROBLEM: The movement of resident ambulatory training from the traditional hospital based clinics to private office sites has necessitated the development of new educational strategies as well as evaluation tools and processes. Private office settings represent a significant educational challenge due to inherent differences in practice sites, patient populations and precepting styles. Therefore, a seamless evaluative process that specifically addressed resident education in the private office setting was developed to ensure a uniform high quality educational experience.

OBJECTIVES OF PROGRAM/INTERVENTION: Develop a Systematic Evaluation Plan; Develop Quantitative and Qualitative Evaluation Tools; Provide a Process for Formative and Summative Evaluation; Provide a Process for Resident and Mentor Feedback; Utilize Evaluation Outcome Data Effectively for Program Improvement.

DESCRIPTION OF PROGRAM/INTERVENTION: The evaluation program developed emphasized formative as well as summative evaluation. Components, tools, time lines, responsibility and outcome processes were included. Tools were developed to evaluate all components of the private office experience. These included evaluation of the experience by the residents and the physician preceptors, site evaluation, including resident patient panels, resident and preceptor performance and resident self assessment. The quantitative tools consisted of Likert Scale measures with open ended items and patient panel computerized tracking. Qualitative methodology included interviews and focus groups. Reliability and validity of the tools were established utilizing content and construct validity and for reliability internal consistency and concurrent review as appropriate.

FINDING TO DATE: The evaluation plan developed has yielded significant outcome data to date. The findings have supported that private office sites provide improved experiences in managed care, patient counseling and patient continuity without impacting on patient diversity. Important feedback from the physician preceptors indicated that mentoring did not interfere with patient relationships however areas of concern were staffing and productivity. Stipends for the mentors were also identified as important as was continued faculty development.

KEY LESSONS LEARNED: The findings from the evaluation have enabled refinement and future planning for a high quality educational experience for residents.

A COMPUTER-ASSISTED SYSTEM TO FACILITATE COMMUNICATION RESEARCH AND EDUCATION. KN McIntire, JA Tulskey, Internal Medicine, Durham VA Medical Center; Internal Medicine, Duke University Medical Center; School of Medicine, Duke University Medical Center, Durham, NC

STATEMENT OF PROBLEM: Recorded encounters with real and standardized patients are invaluable for teaching physicians, residents, and medical students to communicate with patients and for conducting research on the medical encounter. However, most researchers and educators use outdated audio technology to produce analog recordings that are cumbersome to work with and difficult to analyze.

OBJECTIVES OF PROGRAM/INTERVENTION: To develop a computer-assisted system that simplifies archiving and analyzing audiotape data, facilitates time measurement and improves sound quality.

DESCRIPTION OF PROGRAM/INTERVENTION: We adapted hardware and software initially developed for the music industry to meet our research and education needs. A PC computer equipped with a multi-media player and a 32-bit sound card converted our analog recordings to digital ".wav" (sound) files. Individual encounters were processed using Sound Forge XP filtering tools to enhance sound quality. The processed encounters were archived on compact discs (CD) as digital data files using Adaptec Easy CD Creator and a HP CD-R writer. Once archived, Sound Forge XP and Microsoft Access were used simultaneously on a split screen to analyze individual encounters. Sound Forge XP plays the audio file, provides a visual image of the sound frequencies, and enables researchers to time stamp significant events for easy retrieval and to time specified regions of talk to 0.001 seconds. While the recording plays, raters can complete the Access data-entry form that we programmed to automatically perform calculations and prepare the data for entry into SAS.

FINDINGS TO DATE: Digital file processing greatly enhances the sound quality of poorly audible analog recordings. Visualizing the sound frequencies enhances analysis. Events within encounters can be easily marked, timed and retrieved. Digital files may be sent via the internet, and data is easily incorporated into research and education presentations.

KEY LESSONS LEARNED: This user-friendly system makes audiotape analysis as easy as a mouse-click. Processing software makes noisy clinic backgrounds and background hiss disappear. Automated data processing minimizes errors, and digital archiving makes the data portable. We are currently using the system to evaluate an educational intervention designed to improve resident communication skills. We expect higher inter-rater reliability than previously possible. We will also use this technology to export data to distant collaborators for joint analysis and to share existing and future databases with other researchers. Furthermore, digital technology is a rapidly growing field, and we look forward to incorporating advances that will expand our capabilities and the system's efficiency including digital recorders and high-compression formats.

TEACHING RESIDENTS TO TEACH—THE RAT SURVEY. E Metallos, J Beddel, J Gross, R O'Keefe, J Arnstein, Montefiore, Bronx; Beth Israel, New York, NY

Departments of Medicine and Medical Schools rely on residents to teach interns and medical students. Most residents, however, have little formal training for this role. In 1996 Montefiore developed the Resident As Teacher (RAT) Project. RAT is a mandatory full day of interactive seminars designed to help PGY 2's become better teachers, evaluators, and feedback givers. This survey was conducted to evaluate the impact of RAT on resident's beliefs, skills, and attitudes towards teaching.

The instrument consisted of two surveys—a Pre-test and Post-test—administered and analyzed in a blinded manner. The Pre-test was given before RAT. The Post-test was given after the residents had completed at least one rotation where they were responsible for teaching interns/medical students, an interval of 2–4 months after the RAT intervention.

Thirty-six PGY 2's participated in the study, 30 returned Pre-tests (83%) and 34 returned Post-tests (94%). Before the intervention 93% of the participants reported they would like to become better teachers. However, prior to RAT, only 43% reported they knew how to achieve this goal. Prior to RAT, 43% described their previous teaching experience as limited (i.e.: "a topic presented on rounds") and 50% as moderate (i.e.: "math tutor in high school"). None described their prior experience as substantial. Of note, 90% reported they were unsure of their current role as teacher and reported little or no previous help developing their teaching style. After the RAT intervention, residents demonstrated improvement in three of the questions posed (on a Likert scale of 1–5, where 1 = strongly agree, 3 = neutral, 5 = strongly disagree): **1**) "I consider myself a teacher" ($p = 0.02$), **2**) "I feel I know how to become a better teacher" ($p = 0.04$) **3**) "I feel comfortable giving feedback and evaluation to medical students and interns" ($p = 0.05$). There was a trend towards improvement after RAT for five questions with $p < 0.1$, though they did not reach statistical significance: **1**) "The Department of Medicine relies on me to teach" **2**) "The Medical School relies on me to teach" **3**) "Giving feedback is an integral part of teaching" **4**) "I would like to be a better teacher" **5**) "I know how to achieve the characteristics which I feel make excellent teachers". After RAT 89% reported they had learned useful tools at the workshop to develop their teaching style, 73% reported making changes in their approach to teaching, and 73% reported feeling more prepared for their role as a teacher.

KEY LESSONS LEARNED: The RAT intervention has a positive impact on resident's beliefs, skills, and attitudes towards their role as a teacher.

RESIDENTS COST LESS THAN FACULTY: COST SAVINGS IN A STAFF MODEL CONTINUITY CLINIC/COMBINED FACULTY PRACTICE. S. Mitrovich, KJ Pituch, G Freed, S Clark, D Ollansky.

There is a general lack of understanding of what the costs are to train residents in community-based ambulatory settings. It is important to address this especially as patient care has shifted into ambulatory settings. In our general medicine clinic 6 faculty members see patients independently most of the week, but 4 faculty also spend one or more half days supervising 1–4 residents. Residents see both resident and faculty patients. Faculty supervise under HCFA primary care exception. This study compares practice income and expenditures when the faculty practice is compared with the resident practice.

To determine the added costs or savings attributable to a resident continuity practice in a combined faculty resident practice.

Fulltime equivalent providers were defined and salaries distributed according to percent effort in the faculty or resident portion of the practice. Faculty and resident visits were collected from the hospital billing system, as were billed clinic revenues and payor mix. Total practice overhead and support staff costs were taken from hospital cost reports. These costs were assigned proportionally by percentages of patients seen in each portion of the practice. Data was analyzed for a 4 month period. The monthly revenue or cost was determined for the faculty only practice and the resident practice. Costs to the resident practice included the portion of faculty salary used in supervision.

Faculty practice total patients seen per month were 1117 compared to 132 for the resident practice. The monthly revenue for the faculty practice is \$52,251 compared to \$25,736 for the resident practice. Faculty salary is \$51,826/mo at 3.98 Full time equivalents versus \$10,917 for the resident practice at 1.6 FTE. Total monthly costs including overhead and support staff are \$113,887 for the faculty practice and \$35,664 for the resident practice. The total monthly loss for the faculty practice is calculated to be \$15,487. The monthly loss for the resident practice is \$6,205.

KEY LESSONS LEARNED: In this community staff model clinic for resident training the resident portion of the practice was significantly less costly. The lower salary of residents with faculty supervision offset their lower patient volumes; also the support staff and overhead costs are significantly less for the resident portion of the practice. These variables contributed to the overall lower cost of the resident practice.

A MANAGED CARE CURRICULUM FOR INTERNAL MEDICINE RESIDENTS. M. Montes, Medicine, St Vincents Hospital and Medical Center, New York, NY

STATEMENT OF PROBLEM: As managed care becomes the predominant health care delivery system in the United States, there is a pressing need for physicians to be trained for practice in this new environment. This is especially true for many urban internal medicine programs where the majority of residents receive their training in traditional hospital based clinics. When New York State initiated a plan to implement mandatory managed care enrollment for all Medicaid recipients it became imperative that we appropriately prepare our residents for this new practice environment.

OBJECTIVES OF PROGRAM/INTERVENTION: In the summer of 1998 we developed and implemented a curriculum in "practice management skills for the managed care era" for our third year internal medicine residents. We wanted our residents to be familiar with the language of managed care, to understand the dynamics that led to its evolution, to recognize the various methods of assessing quality and controlling cost in a managed care environment, and to recognize ethical issues a physician might encounter in a managed care setting.

DISCUSSION OF PROGRAM/INTERVENTION: The curriculum consists of six units: 1) Health care financing and managed care, 2) Computers in clinical practice, 3) Cost effective clinical practice, 4) Quality and managed care, 5) Coding and billing and 6) Ethical issues and managed care. The didactic portion of the curriculum is taught during the ambulatory care block. The practical aspect involved a reorganization of our clinic practice site, specifically the development of attending and resident practice teams. Residents were given a pretest prior to the didactic sessions to assess their baseline knowledge. They were given the posttest an average of 6 months after completion of the didactic curriculum. Residents evaluated each of the units they completed.

FINDING TO DATE: Over the past year a total of 20 third year residents have participated in the didactic portion of the curriculum. General knowledge of managed care terminology and procedures improved from a mean score of 49% to a mean score of 62%. Response to the curriculum was overwhelmingly positive, with 60% of residents stating "strong agreement" and 40% stating "agreement" that the curriculum should be continued. The only suggestion for improving the curriculum was that it be offered earlier in their training and that more time be dedicated to it.

KEY LESSONS LEARNED: Internal medicine residents see managed care training as a much-needed part of their education. They are eager and receptive to interventions that will improve their knowledge of the key competencies needed for managed care practice. It is possible in a traditional hospital based clinic to implement a didactic curriculum that will improve resident's knowledge of managed care terminology and procedures.

TEACHING HUMAN RIGHTS WITH AN INTERNET-BASED COURSE. A. Moreno, RW Schadt, M Grodin, Section of General Internal Medicine; Department of Psychiatry; Boston Medical Center, Boston University School of Medicine, Boston, MA; Educational Technology; Health Law Department; BU School of Public Health, Boston, MA

STATEMENT OF PROBLEM: Despite the growing number of refugees, asylees, and survivors of torture, healthcare providers learn little during their training about caring for these individuals and other international human rights issues. In addition, providers working primarily in urban areas recognize they are ill prepared for this task.

OBJECTIVES OF PROGRAM/INTERVENTION: The project's primary objective is to devise an educational tool accessible to any healthcare provider or trainee who is interested in learning about victims of human rights violations. The project has 4 educational objectives: 1. To understand the scope and magnitude of human rights violations. 2. To recognize the symptoms and signs that victims of human rights violations present with. 3. To learn how to approach these patients. 4. To inform about other resources available for these individuals.

DESCRIPTION OF PROGRAM/INTERVENTION: We devised an introductory Internet-based course covering 8 areas: definitions, history, refugee status and asylum process, epidemiology, symptoms and signs, clinical approach, education and prevention, and resources. The project has a pilot phase during which the course's content and format will be tested and an implementation phase during which changes from the pilot phase will be incorporated into the project and the final course will be placed on the Internet.

FINDING TO DATE: The first phase of the project is complete and the first part of the second phase is underway. The course was introduced to faculty members, community-based healthcare providers, residents, and medical and public health students during a series of academic conferences. Overall the audiences agreed that such a course would fulfill knowledge needed in today's urban healthcare environment. Participants suggested expanding the following areas: education and prevention, definitions, and asylum process. Although designed to be an introductory course, we noted the importance of gaining feedback to further develop certain areas, such as epidemiology, education, and prevention for public health providers or signs, symptoms, and clinical approach for medical practitioners.

KEY LESSONS LEARNED: Teaching human rights through an Internet-based course demonstrates the flexibility and the power of this tool as an educational media. By facilitating access to course materials in human rights, levels of awareness and advocacy in healthcare providers will increasingly benefit a significant number of individuals whose human rights have been violated.

INCORPORATING LITERATURE INTO CORE BIBLIOGRAPHIES: GREAT BOOKS AND PROJECT PROFESSIONALISM. P.O'Brien, B Adelson, GW Rutecki, Medicine, Evanston Northwestern Healthcare, Evanston, IL

STATEMENT OF PROBLEM: In an article (JAMA 1987; 257:1629–1631), literary classics were applied to ethics. Well-known affinities exist between medicine and literature providing an alternative approach to traditional didactics, bedside discussions and forums. Yet, recent comprehensive bibliographies addressing areas wherein literature may lend to education—ethics (reports of the American Society for Bioethics and Humanities) and Project Professionalism (from the ABIM)—literature is absent.

OBJECTIVES OF PROGRAM/INTERVENTION: To introduce a literary context to graduate medical education with an emphasis on core education in professionalism.

DESCRIPTION OF PROGRAM/INTERVENTION: The "Great Books Seminar" offered by Evanston Northwestern Healthcare began in 1981. The course has evolved, but originally was intended to be a forum in which literature serves as a springboard for exploring issues in clinical medical ethics.

FINDING TO DATE: Currently core bibliographies present issues related to professionalism in isolation. For the resident, such an approach risks being didactic, presenting professionalism as another process to be mastered. Such an approach confounds the idea that professionalism is dynamic. Recent sample readings and the area they engage include: Face of Stone and The Use of Force by William Carlos Williams—portrayals of compassionate but overworked doctors confronting personal reactions to patients. The Student's Wife and Blackbird Pie concern academics, who while deeply committed to their work, ignore people close to them (written by Raymond Carver). Addressing the duty to treat during the AIDS epidemic, The Plague by Albert Camus and My Own Country by Abraham Verghese are foundations for discussion. Physician arrogance was discussed via the novel Frankenstein. Regarding physician impairment, the story entitled Old Doc Rivers by William Carlos Williams is contained in a larger volume entitled The Doctor's Stories. It is approximately 30 pages long and tells the story of a well-liked physician with a substance abuse problem. Although Doc Rivers got there—"anywhere, anytime, for anybody—no distinctions"—he began to take "the dope." His loyal clientel said, "he's a dope, but when he's right, you can't beat him." The bitter irony of his addiction was that "he became a legend and indulged himself the more."

KEY LESSONS LEARNED: As the "Great Books" seminar appears to be an appropriate venue for education in the core issues of professionalism.

PREPARING INTERNS TO BECOME RESIDENTS: HOW TO DEVELOP A SUCCESSFUL RETREAT THAT WILL BOOST MORALE AND TEACH PROFESSIONALISM TO YOUR HOUSESTAFF. JE O'Rourke, M Kulkarni, MC Henderson, Medicine, University of Texas, San Antonio, TX

STATEMENT OF PROBLEM: The job requirements of the resident differ vastly from those of the intern. Interns do not usually receive training for their role as a resident. A role that often requires leadership, teaching, efficiency and conflict resolution.

OBJECTIVES OF PROGRAM/INTERVENTION: The intern retreat teaches tools necessary to be an effective resident.

DESCRIPTION OF PROGRAM/INTERVENTION: Our Intern to Resident Retreat is held at a remote location on a weekday in June. The morning session consists of seven small groups discussing different aspects of the resident's job. The afternoon session is aimed at boosting morale, teaching professionalism and team building. Our exhibit will describe the goals, development and execution of a retreat. We will display the packet given to each participant, color photographs, and results of the written survey evaluating the educational impact of this exercise. We will also give concrete guidelines and instructions to those programs seeking to develop a similar retreat at their institution.

FINDING TO DATE: Improved intern to resident transition, decreased intern anxiety, and increased intern morale.

KEY LESSONS LEARNED: A retreat for interns prior to residency will smooth the transition from intern to resident. The faculty can teach upcoming residents what is expected of them and tools with which to accomplish it. The retreat is extremely well received by the interns increasing their confidence and decreasing their anxiety.

AN EVALUATION OF "INTRODUCTION TO PATIENT CARE"—A COURSE IN PATIENT ADVOCACY AND INTERDISCIPLINARY CARE FOR 1ST AND 2ND YEAR MEDICAL AND PHARMACY STUDENTS. JJ Peterson, K Schanche, MW Rabow, Division of General Internal Medicine, University of California, San Francisco, San Francisco, CA

STATEMENT OF PROBLEM: Health professionals generally receive little training in working with an interdisciplinary team or in patient advocacy.

OBJECTIVE OF PROGRAM/INTERVENTION: With the goal of improving patient quality of life and access to care, the "Introduction to Patient Care" elective course was developed to train 1st and 2nd year medical and pharmacy students to provide social support and advocacy for outpatients with end-stage CHF, COPD, or cancer in the General Medicine Practice at UCSF. Participating patients were those enrolled in a clinical research project entitled the Comprehensive Care Team (CCT)—an interdisciplinary team of professionals providing comprehensive health care and family caregiver support.

DISCUSSION OF PROGRAM/INTERVENTION: Groups of 2-3 students were assigned to serve as patient advocates to individual CCT patients. Students provided weekly patient contact in person or by telephone, primarily listening to patient stories and communicating needs back to the CCT. Students were required to attend weekly, one-hour classroom sessions. Sessions provided didactic instruction on the following topics: patient advocacy; working with an interdisciplinary team; the patient's experience of illness; caring for patients with CHF, COPD, and cancer; supporting family caregivers; dealing with death and dying; and advance care planning. Additionally, 50% of classroom time was reserved for students to share stories of their experiences with patients.

FINDING TO DATE: Of the 32 student participants, 8 (25%) were medical students and 24 (75%) were pharmacy students. At the end of the course, students completed anonymous, self-administered course evaluations (response rate = 63%).

Based on a 1-5 Likert scale (1 = strongly disagree; 5 = strongly agree), students agreed that the quality of the course was high (mean = 4.2) and that the course stimulated their interest in patient advocacy (mean = 4.5). Students felt comfortable speaking with their patients (mean = 4.0) and sharing their experiences with the class (mean = 4.2). While students felt that this course was more challenging than other electives (mean = 3.9), they felt that it was more valuable (mean = 4.5). All 20 respondents (100%) would recommend the course to a classmate.

CCT patients are surveyed semi-annually about their experiences with the CCT project. All 26 CCT patients (100%) working with student advocates reported that their contact with their advocate was one of the main benefits of the CCT intervention and one of the most satisfying aspects of their health care overall.

KEY LESSON LEARNED: A course promoting interdisciplinary care and patient advocacy for 1st and 2nd year medical and pharmacy students was well-received, considered important, and mutually satisfying to students and patients.

TRACKING RESIDENT CONTINUITY IN PRIMARY CARE SITES- MORE RESIDENTS HAVE HIGH CONTINUITY IN THE HOSPITAL CLINIC THAN IN COMMUNITY PRACTICES. KJ Pituch, S Mitrovich, KE Orsetti, SK Kesterson, DM Oliansky, PC Burghardt.

To follow patients over time is a major goal for continuity clinic training. Despite advantages to having residents train in community practices, residents risk losing continuity when they share patients with community preceptors.

1. To develop a tool to measure continuity simply, reliably, and completely. 2. To develop a resident activity report that includes continuity as a regularly reported variable 3. To compare patient continuity in resident clinics in the hospital vs. the community.

An electronic data base tracks resident-patient encounters using the billing/encounter form as the data collection tool. Patient demographics, visit type, diagnoses, procedures, and level of faculty supervision are recorded for each encounter. To determine continuity residents indicate with a check mark the number of times they have seen the patient previously. If the resident has seen the patient before, it is a familiar patient. Another check indicates the patient's primary care provider. The resident marks "Myself" to indicate a personal patient. These indicators of continuity were tallied for three months of weekly clinics for 50 PGY 2 and PGY 3 internal medicine residents.

Results are for the first three months of academic year 1999-2000. At the hospital clinic, 43 upper level residents were supervised by rotating faculty who did not see their own patients in this all resident practice. At two community sites, 7 upper level residents were supervised by faculty who had their own patients in those practices. The table summarizes the differences in 'familiar patients' and 'personal patients' at the three sites.

	Hospital Clinic	Community Clinics
'Familiar Patient' encounters/ total encounters	851/2023 = 42%	58/227 = 25
Range for individual residents:	10%-78%	5%-35%
'Personal Patient' encounters/ total encounters	1030/2023 = 51%	88/227 = 39%
Range for individual residents:	12%-80%	14%-59

KEY LESSONS LEARNED: 1. Residents at community sites saw significantly fewer familiar or personal patients in their primary care clinics compared to residents at the hospital clinic. 2. The range of continuity differed greatly from resident to resident, regardless of clinic setting. 3. The resident data base provides a simple tool for measuring continuity, as well as many other aspects of resident experience. 4. Program leadership hopes to use this information to optimize continuity in primary care training.

A SUCCESSFUL EVIDENCE BASED MEDICINE CURRICULUM FOR EVERYDAY USE IN RESIDENT TRAINING. W Polashenski, B Robbins, JD Cappuccio, Dept. of Medicine, University of Rochester, Rochester, NY

STATEMENT OF PROBLEM: Evidence based medicine is not always practiced. To be accepted and used, Evidence Based Medicine (EBM) must be taught and reinforced in a variety of clinical and educational settings.

OBJECTIVE OF PROGRAM/INTERVENTION: To develop and test a curriculum in which EBM provides a crucial structural component in residency education.

DESCRIPTION OF PROGRAM/INTERVENTION: Practicing EBM requires identification of a clinical question, literature review, critical appraisal of that literature, and application of the evidence to patient care. We have developed an EBM curriculum that involves all major aspects of our residency program. During the first year, residents are introduced to the concepts of EBM and learn the skills of asking four part questions encompassing the patient group, the clinical and comparison intervention and the outcomes of interest (PICO's questions). This is reinforced throughout the year in Intern Morning Report where questions are identified by the group and answered during the next session. In the second and third years of residency, EBM skills are sharpened in an advanced skills block and morning report. In these settings, residents complete Educational Prescriptions by developing PICO's questions, searching the medical literature, critically appraising the evidence, and considering the application to their patients. Educational Prescriptions are presented daily in morning report. The process is reinforced in the outpatient clinics where residents are encouraged to identify knowledge deficits in the form of a PICO's question and relate the answers in the form of an Educational Prescription. During their third year, residents are required to give

evidence-based conferences while on subspecialty rotations. For those residents who are interested in more in-depth training, a two-week, intensive EBM elective is offered for second and third year residents. To provide an easily accessible representation of resident work and to measure their acceptance of EBM, we created an Internet based bank of Critically Appraised Topics (CATs), a short formatted easily read style for educational prescriptions. Residents submit these one-page summaries of articles that are evaluated by two faculty members who provide specific structured feedback. Once revised, the CAT's are published on the World Wide Web. To promote interest in this, we have created a weekly contest for the best resident CAT and award the resident with the most weekly prizes with a trip to the McMaster EBM course.

FINDING TO DATE: In the first twelve months, we have received 105 CAT submissions from 23 residents. The program has been immensely popular with residents.

KEY LESSONS LEARNED: The use and acceptance of the CAT bank provides initial evidence that residents accept the application of EBM in daily clinical practice.

MAKING A CURRICULUM IN OBSTETRIC MEDICINE AVAILABLE TO OTHER EDUCATORS THROUGH THE INTERNET. RO Powrie, Brown University School of Medicine and Women & Infants Hospital of RI

STATEMENT OF PROBLEM: How to make a modular curriculum in obstetric medicine, designed for use in medical residency programs, readily available to educators across the country.

DESCRIPTION OF PROGRAM: Using an extensive review of medical consultations performed at a busy obstetrics and gynecology hospital, a brief practical curriculum in medical problems in pregnancy was developed for medical residents.

This curriculum included resident handouts, educator teaching scripts, bibliographies and cases for discussion for each of 13 specific topics. The curriculum was designed to be readily adaptable into a wide variety of educational forums within residency programs. The curriculum has previously been presented as a pre-course at the national meeting of the Society of General Internal Medicine in 1998 and has subsequently been sold to over 50 programs across the country.

Having completed the curriculum development project, we became concerned with how to more effectively disseminate the materials. Therefore, in association with the Society of Obstetric Medicine, we have placed our entire curriculum on the web. Our handouts, teaching scripts, cases and bibliographies can now be readily accessed and utilized by residency programs across the country. Slides can also be downloaded from the website and adapted to local needs. The website also provides access to an obstetric medicine examination which educators can employ to stimulate and document resident learning. We wish to present our website and make available to others the lessons we have learned thus far through its development and utilization.

THE MEDICAL REFERRAL: AN EDUCATIONAL TOOL FOR MEDICAL RESIDENCY TRAINING. S Reichman.

STATEMENT OF PROBLEM: The decision of a primary care physician to refer a patient to a specialist requires formulation of a question concerning diagnosis or management. In an era of managed care it is essential to create best practice strategies so that appropriate and cost effective referrals to medical specialties will be requested.

OBJECTIVES OF PROGRAM/INTERVENTION: In residency training it is important to have house staff (HS) recognize the value not only of framing a proper question for a consultant, but also of examining the pre-probability of benefit to the patient, and providing the appropriate history, physical findings and lab data expected of an internist.

DESCRIPTION OF PROGRAM/INTERVENTION: Four educational interventions were created. These varied from 1) a "consult" resident assignment to review requests of other HS to a specialist; 2) assignment of selected consults from HS to a resident for critique with other residents; 3) faculty presentation of incomplete consults; and 4) teaching a need to seek evidence to answer the posed question and determine benefit to the patient of the referral.

FINDING TO DATE: Review of consult requests by HS to five medical subspecialties revealed inappropriate and cost ineffective referrals. Preliminary examination of HS appreciation of the different interventions found general rejection of the first, but interest in selected consult review, faculty presentations and evidence gathering approaches. Fifty per cent of HS responding to an initial questionnaire agreed that phrasing a question, providing proper data and knowing established indications for recommending a procedure were all important in residency training. However, attempts to quantify measures of appropriateness of consults by attendings and specialists were hampered by lack of significant interrater agreement (kappa value <.6).

KEY LESSONS LEARNED: HS referrals to medical subspecialties arise from uncertainties in knowledge, diagnosis or management. Consult review provides a good base for examining educational needs and teaching appropriate communication to subspecialists, especially in an era of managed care restrictions.

STATEMENT OF PROBLEM: Primary care residents lack exposure to principles of geriatric medicine and are unfamiliar with care processes across the continuum of community-based sites available for frail older adults.

OBJECTIVES OF PROGRAM/INTERVENTION: To develop and evaluate a year-long program to deliver geriatric content to primary care residents using computer-based content linked to focused clinical experiences.

DESCRIPTION OF PROGRAM/INTERVENTION: 1) Six resident physicians (2 IM, 1 FP, 3 GYN) were pilot program participants. 2) The one-year program consisted of 10 sessions, held once monthly for 3 hours. 3) Independent review of geriatric content on one specified topic each month from: (a) new computer-based learning modules developed by the authors, and (b) selected learning modules on CD-ROM from The Computer-Based Self-Instruction Modules in Geriatric Medicine, vol. 1, Baylor College of Medicine; John A. Hartford Foundation Geriatric Consortium for Residency Training. 4) Group activities each month: faculty-lead, community-based clinical exposure focused on the monthly topic (e.g., wound care; geriatric assessment; gait/falls; care processes in long-term care, assisted living, and home care settings.) 5) Pre- and Post-testing of knowledge and attitudes using previously validated tools, and self-reported confidence about common geriatric syndromes. A satisfaction survey was conducted at program completion.

FINDING TO DATE: The completed one-year pilot project was well-received. Mean attendance was 60% of sessions. On a 5-point Likert scale (1 = poor, 5 = excellent), mean ratings were 4.3 for the overall program, 4.7 for the faculty-lead group clinical exposures, and 3.3 for computer-based materials. Residents felt this content was not redundant with other training. All six found the out-of-hospital clinical exposures valuable and felt they should be viewed as an integral component of the program. Applicability to future practice was rated more highly by IM and FP than by GYN residents. Confidence improved in all residents, and attitude improved in 5 of 6. On average, however, there was no demonstrated improvement in tested knowledge.

KEY LESSONS LEARNED: Residents appreciated the opportunity to learn geriatric content, and valued the faculty-lead, small group clinical experiences highly. Sources of dissatisfaction included computer-based learning methods, difficulty in finding time for independent study prior to the sessions, and logistical challenges in getting away from the hospital for outside experiences.

INTERNET PUBLISHING OF CRITICALLY APPRAISED TOPICS STIMULATES USE OF EVIDENCE-BASED MEDICINE. B Robbins, W Polashenski, JD Cappuccio, Dept. of Medicine, University of Rochester, Rochester, NY

STATEMENT OF PROBLEM: The entire process of evidence-based medicine (patient-based knowledge deficit, answerable clinical question, efficient search of the literature, critical appraisal, and application), and not only critical appraisal needs to be taught in residency training.

OBJECTIVES OF PROGRAM/INTERVENTION: To encourage the spontaneous use of evidence-based medicine (EBM) in residency training.

DESCRIPTION OF PROGRAM/INTERVENTION: Internal Medicine Residency programs have been increasingly emphasizing the teaching of evidence-based medicine (EBM) to their residents. The most common method currently employed is the traditional journal club where one or a few articles are critically appraised within the context of a group. While this has been shown to modestly increase resident's critical appraisal skills, it focuses almost solely on these skills alone, leaving the remainder of the EBM process not formally taught. For the past year we have been attempting to teach the whole process of EBM to our residents through formal feedback on their creation of critically appraised topics (CATs). In November 1998, we began a system within which residents were encouraged to answer their clinical questions using the process of EBM and formalize the answer in the form of a CAT. Their questions are drawn from the ambulatory clinic, subspecialty electives, intensive care unit, floor experiences, and our own EBM elective. The various attendings on the faculty would help them formulate their question and search, and they were directed toward a librarian facile in searching techniques as well. The evidence is summarized using the one-page CAT format and submitted to 2 faculty reviewers. The reviewers independently scored the CATs on explicitly stated criteria: how common, important, and useful is the information, how readable is their format, how well their comments help us apply the evidence, and how valid is the evidence their CAT is based on. We evolved this into a contest where the resident with the highest score on any CAT submitted that week won "CAT of the week." The resident with the most CATs of the Week in the academic year receives a trip to the EBM course at McMaster University.

FINDING TO DATE: In the first 12 months of the contest, we have 105 CATs spontaneously submitted by 23 different residents. We created a website (www.urmc.rochester.edu/Medicine/res/cats/index.html) where we publish these CATs for public use as well as E-mailing out the CATs to various clinicians around the world on a weekly basis. The website receives around 150 hits a day representing 30 different users.

KEY LESSONS LEARNED: By creating a process and contest whereby residents are encouraged to answer their own clinical questions and to report their findings in a standardized format, we have integrated the teaching of EBM into their daily clinical work.

GERIATRIC MEDICINE FOR PRIMARY CARE RESIDENTS: A PILOT PROGRAM OF COMPUTER-BASED LEARNING WITH TARGETED CLINICAL EXPERIENCES. SK Riegler, RD Laird, University of Kansas Medical Center, Kansas City, KS

INTRODUCING FIRST-YEAR MEDICAL STUDENTS TO PEER- AND SELF EVALUATION. DW Rudy, CH Griffith, III, JF Wilson, College of Medicine, University of Kentucky, Lexington, KY

STATEMENT OF PROBLEM: Peer- and self-assessment are important for continued quality improvement. Physicians-in-training are often resistant to peer-evaluation and may not have the skills necessary for self-assessment. One solution may be to address these issues throughout the medical curriculum. We describe a multifaceted approach to introduce peer- and self-evaluation to first-year medical students.

OBJECTIVES OF PROGRAM/INTERVENTION: First-year students will: 1. Understand the principles of giving and receiving feedback. 2. Practice giving verbal and written feedback to peers. 3. Assess one's own performance by discussion and by written evaluation form.

DESCRIPTION OF PROGRAM/INTERVENTION: The program takes place during our interviewing course. An introductory lecture is presented addressing the basic principles of giving and receiving feedback. To apply this information, students view a trigger tape of a medical interview and provide written feedback on a transcript of the interview. The class discusses student's responses to the trigger tape. Our interviewing course is conducted in a small group setting with individual students interviewing standardized patients in front of their group. After the interview, students are encouraged to critique themselves and their peers in a constructive and supportive fashion. Using the feedback skills practiced in the introductory lecture, the students comment upon their peer's skills. At mid-course, a student's interview with a patient is videotaped. After watching the videos, students complete self- and anonymous peer evaluation forms using rating scales of knowledge of the structure of the interview and interviewing style and techniques as well as providing open-ended comments. Evaluations are not used in grading. Students receive a written summary of their evaluations.

FINDING TO DATE: The written evaluations from the videotaped interviews have been evaluated. Peers generally rated their colleagues higher than the students did themselves or did the faculty, and provided very few negative comments. Peers scores on the numeric scales show little variation between categories (low interdimensional variance) or between students (low discriminability). Students were more critical on their own performances and were more likely to give themselves negative rather than positive comments.

KEY LESSONS LEARNED: First-year medical students were willing to provide feedback to their peers, but were reluctant to assign "grades" to their colleagues. Peer evaluations using an open-ended format for feedback may be more beneficial than quantitative assessment. Students had difficulties providing positive self-feedback. Interventions are needed to teach students how to have a more balanced perspective of their performance. In the future, students will be surveyed as to how useful they found conducting the peer and self-evaluation as well as receiving the written feedback from their classmates.

STIMULATING FIRST-YEAR STUDENT'S INTEREST IN PROFESSIONALISM AND PHYSICIAN COMPETENCIES. DW Rudy, CL Elam, Department of Medicine; Office of Academic Affairs, University of Kentucky, Lexington, KY

STATEMENT OF PROBLEM: Introducing professionalism into the pre-clinical curriculum is challenging in that many medical students immersed in studying basic sciences may not see the relevance of psychosocial issues. Beginning student's motivation for learning is external, i.e., passing examinations. Professional development occurs through internal motivation and life-long reflective learning. How might this discrepancy in student intellectual development and motivation and the goal to teach professionalism be resolved?

OBJECTIVES OF PROGRAM/INTERVENTION: The objective of our program is to encourage pre-clinical students to appreciate the importance of professionalism by allowing the students to become aware of their own perceptions, followed by observing role model physicians, self-reflection, and discussion.

DESCRIPTION OF PROGRAM/INTERVENTION: Our professional development curriculum begins during the application process. Applicants are asked to answer the following question on their secondary application "What are the competencies and qualities you feel a physician should possess for practice in the 21st century?" The composite results were presented to the students during a professionalism seminar in our Introduction to the Medical Profession Course. Students were challenged to strive to meet these expectations. During an early community-based clinical experience students were asked to observe the professional qualities of their physician preceptor and write an essay on how they plan to practice medicine in the future. Following the early clinical experience small groups of students met with a faculty member to discuss their experience.

FINDING TO DATE: The top 5 competencies were: technology (keeping up with technological advances) 63%; business (management, finance, cost) 62%; patient advocacy (helping, counseling, demonstrating concern) 53%; lifelong learning (keeping up with developments and research) 48%; communication (talking to patients and families, listening) 47 The top 5 qualities were: compassion (give aid or support) 49%, intelligence (ability to gain and apply knowledge, grasp of field) 45%; adaptability (flexible, open-minded, willing to change) 34%; honesty and integrity (sincerity, trustworthiness) 32%; dedication (dependable, hardworking, diligent) 29%. These qualities are consistent with expectations published in the literature.

KEY LESSONS LEARNED: Medical school matriculants are very insightful into understanding the professional attributes they will need to practice medicine in the 21st century. The above approach to introducing professionalism allows students to define necessary competencies and qualities of physicians and reflect on the importance of those attributes through observation and experience. Such experiences are designed to help students assimilate professional behaviors.

REAL TIME JOURNAL CLUB: A MULTI-PROFESSIONAL COLLABORATIVE MODEL. PA Schmidt, B Friedman, M Moore, Internal Medicine, Garden City Hospital, Mich. State Univ. COM/SCS, Garden City, MI; Faculty Development, Michigan State University, College of Osteopathic Medicine, Statewide Campus System, East Lansing, MI

STATEMENT OF PROBLEM: Lack of structured format to teach the principles and process of evidence based medicine, as described by Sackett et al. to community based Internal Medicine residents. Poor utilization of resources within the institution; ie medical librarian and clinical pharmacologist.

OBJECTIVES OF PROGRAM/INTERVENTION: 1. Present a description of an innovative format of internal medicine journal club. 2. Demonstrate the incorporation of Sackett's five step process of evidence based medicine; clinical query, search, appraise validity of evidence, apply and evaluate, within the Real Time Journal Club format. 3. Discuss ways to implement Real Time Journal Club into other institutions.

DESCRIPTION OF PROGRAM/INTERVENTION: Real Time Journal Club (RTJC) is a multi-professional, collaborative, learner-centered model emphasizing evidence relevant to patient care. It evolved from a facilitator-centered session, that emphasized critical appraisal of journal articles that didn't have direct relevance to patient care. A resident coordinator, general internist, medical librarian, and clinical pharmacologist conduct this once monthly one-hour session. The preceding month, the resident selects a clinical question of interest to him/her. Search techniques are pursued with the librarian. Abstracts and at most two articles are selected for the group to review in "real time" at the journal club session. Groups of three learners seek to answer the clinical question posed by the resident coordinator, and site the best evidence provided. Worksheets from the Information Mastery Working Group are utilized to enhance the critical appraisal process. Findings are summarized and implementation to patient care discussed.

FINDING TO DATE: Outcomes of RTJC are the following: Increased learner participation and enthusiasm for journal club with improved voluntary attendance at each session. Open dialogue, and improved communication between internal medicine residents and multi-professional staff on patient care issues. Observation of resident behaviors in the ambulatory internal medicine clinic, demonstrates self-directed generation of clinical queries, search for best evidence relevant to patient care, appraisal of what was found and discussion of the applicable literature with the supervising attending in the clinic. Preparation workload responsibility shared among participants decreases the amount of work for one person.

KEY LESSONS LEARNED: Changing the format from a facilitator-centered journal club, to a learner-centered, multi-professional collaborative model provides an environment for community based internal medicine residents to learn skills of evidence based medicine, which facilitate information management with application of relevant evidence to patient care.

RECRUITING AND REWARDING GENERALIST COMMUNITY PRECEPTORS. CS Seibert, LJ Zakowski, SS Van Eyck, Medicine, University of Wisconsin Medical School, Madison, WI

STATEMENT OF PROBLEM: With increasing emphasis on physician accountability and productivity, developing and maintaining a network of practicing physicians willing to serve as preceptors is one of the greatest challenges to successful ambulatory experiences.

OBJECTIVE OF PROGRAM/INTERVENTION: 1) Develop promotional material and incentives to recruit and retain generalist physician preceptors. 2) Share ideas and innovations with similar programs.

DESCRIPTION OF PROGRAM/INTERVENTION: For the past five years, the Generalist Partner Program (GPP) has been a vital and integral required component of the UW Medical School curriculum. This program, designed to give students early clinical experience, requires students spend at least one half day per month in a generalist's clinic during the first two years of medical school. Currently, there are 175 volunteer physician preceptors throughout Southern Wisconsin who devote countless hours and provide rich educational experiences for 288 first and second year students. We have attracted and retained a network of practicing community-based generalists to serve as preceptors using a variety of materials and methods. These include

A brochure briefly outlining the advantages of becoming a partner in the program. A handbook detailing the purpose of the program and responsibilities of preceptors and students.

A pocket guide that summarizes suggested clinic activities and tips on involving students in busy practices.

Free internet access including e-mail and electronic clinical resources.

Eligibility for volunteer clinical faculty appointment at the UW Medical School.

Use of instructional and recreational facilities including all libraries.

Recognition of program participants in the UW Medical School Alumni Magazine. UW Medical School GPP lapel pin.

University of Wisconsin bookstore gift certificate.

FINDING TO DATE: Our materials have served to 1) enhance visibility and recognition of our program, 2) increase physician interest in our program, and 3) improve retention of our generalist preceptors.

KEY LESSONS LEARNED: Recruiting and retaining a network of generalist physician volunteers is the key to the success of community based educational programs. A multitude of materials and rewards may help accomplish this goal. More studies are needed to determine the effectiveness of various materials and rewards.

I.M. QUIZITOR: TECHNOLOGY APPLIED TO INTERNAL MEDICINE RESIDENT EDUCATION. DL Shenson, SP Vogt, KK Stout, Internal Medicine, Oregon Health Science University, Portland, OR

STATEMENT OF PROBLEM: Teaching residents is challenging. Textbooks and didactic lectures have been traditionally used to teach housestaff. Existing computer technology, however, offers an opportunity to teach residents in a new format. Many internal medicine residency programs play a version of the hit television show Jeopardy® using medically based questions. We have created a computer based version that incorporates digital imagery, video, and audio media into an entertaining and educational conference.

OBJECTIVES OF PROGRAM/INTERVENTION: 1. Demonstrate how technology can be used to develop an interactive teaching tool for housestaff, students, and faculty. 2. Understand how the use of the Internet, digital images, and scanned pictures can be incorporated into teaching housestaff and students. 3. Demonstrate how I.M. Quizitor, a computerized version of the television show Jeopardy®, can be effectively used in medical education as an adaptable, interactive teaching tool.

DESCRIPTION OF PROGRAM/INTERVENTION: I.M. Quizitor is based on the hit television show Jeopardy®. We used PowerPoint® 2000 to create a template for answers to questions in 10 categories. Each month we create questions using current medical literature, digital images, radiographs, and auscultatory sounds obtained from the internet and our own patients. Internal medicine residents attend a monthly conference during which I.M. Quizitor is played. Teams are created based on housestaff rotation schedule for the month. Pairs of housestaff answer three consecutive questions. These pairs rotate so that all team members are able to participate. The final question is similar to final jeopardy and is a team effort.

FINDING TO DATE: Informal feedback has been quite positive. We are in the process of evaluating the educational value of this program as it relates to several important issues: use as a medical teaching tool, promotion of teamwork, interest in computers and the internet, as well as its entertainment value. Results of this evaluation will be available in early February.

KEY LESSONS LEARNED: The I.M. Quizitor conference is a well attended conference at the Oregon Health Sciences University. Novel conference formats should be included in residency curricula as they increase resident's interest in attending conferences and likely enhance learning.

SETTING HIGH STANDARDS IN RESIDENCY: THE USE OF AN INNOVATIVE INTERN RETREAT TO ADDRESS PROFESSIONALISM IN MEDICINE. PJ Sousa, Medicine, John A. Burns School of Medicine, University of Hawaii Internal Medicine Residency Program, Honolulu, HI

STATEMENT OF PROBLEM: Professionalism in medicine is facing serious challenges in this era of managed care, increasing cost containment, and medical economic turmoil, from both practice and medical education standpoints. Many post-graduate medical training institutions have found the task of developing a curriculum on professionalism that is both useful and meaningful equally challenging. In our own attempt to do this, we piloted a curriculum designed to address some of the many issues surrounding the teaching, learning, and instillation of professionalism in residency training.

OBJECTIVES OF PROGRAM/INTERVENTION: 1) Survey our interns about their beliefs and attitudes regarding professionalism in training; 2) Provide a realistic forum within which interns could define and identify different aspects of professionalism, share their experiences with professional and unprofessional behavior, and explore their responses to different clinical vignettes; and 3) Set the stage for collecting future data about the effectiveness of this approach.

DESCRIPTION OF PROGRAM/INTERVENTION: A mandatory, 8-hour retreat was held off campus for our interns only. They were relieved of all clinical duties that day and overnight. Pre- and Posttest questionnaires [modified 4-point Likert scale, Strongly Agree (SA) to Strongly Disagree (SD)] were administered at the beginning and end of the day, respectively. A brief didactic preceded small group breakouts which were then used to discuss 12 clinical vignettes, with summaries being presented to the large group, followed by a moderated group discussion. Lastly, the interns drafted a Code of Honor for their class, capturing the key elements of professionalism.

FINDING TO DATE: 100% A/SA that: professionalism was equally important as medical knowledge or clinical judgment/skills; they could improve their degree of professionalism. 95% A/SA that: professionalism could be taught and learned; their colleague's professionalism could be improved; they have a heightened sense of the behaviors which allow for a greater degree of professionalism; the retreat was useful/valuable. 80 to 85% A/SA that: professionalism is not stressed enough in medical education; they have a better understanding of altruism, excellence, duty, honor and integrity; they have a better appreciation of how power, arrogance, and greed can erode professionalism. 65% A/SA that more than just the retreat was needed to fully grasp the concepts discussed. 80% SD/D that learning about professionalism in other settings would be more valuable.

KEY LESSONS LEARNED: We felt this retreat was a success, and is a useful forum for our professionalism curriculum. Further retreats and other fora need to be developed, studied, and refined to improve this. Our residents (and faculty) enjoyed the retreat, and most are extremely willing to help plan and moderate future sessions.

GERIATRIC EDUCATION RESOURCES FOR RESIDENCY TRAINING. G Stratos, A Fabiny, W Hall, M Harper, R Luchi, D Reuben, G Sachs, G Sullivan, G Warsaw, Stanford University

In response to the need for enhanced residency training in geriatrics, eight academic medical centers (including Baylor College of Medicine; Harvard University; Johns Hopkins University; Stanford University; University of California, Los Angeles; University of Chicago; University of Connecticut; and University of Rochester) and the American Association of Family Physicians developed educational programs and implemented them locally during a 3-year initiative, funded by the John A. Hartford Foundation. In 1998, the sites formed a consortium with the purpose of creating exportable resources that would assist residency programs to strengthen the quality of geriatric education at their institutions. The resources include curriculum manuals, packaged methods for teaching geriatric skills, stand alone teaching aids, faculty development programs, and consultation programs. They are disseminated through the non-profit, Stanford University Geriatric Education Resource Center.

Between April 1998 and October 1999, the resources were disseminated to approximately 30% of internal medicine and family medicine residency programs. Evaluation of the resources by users has been very positive. Evaluation results are used to enhance the resources. Two key lessons have been learned: (1) a non-competitive, collaborative venture involving diverse academic medical centers is possible, and (2) funding is critical to bringing locally developed and tested resources to a national audience.

AN INNOVATIVE APPROACH TO A CONSISTENT AMBULATORY CURRICULUM FOR MEDICAL INTERNS. LW Tishler, AHM Chen, ME Pickett, JS Sillman, General Medicine, Brigham and Women's Hospital, Boston, MA

STATEMENT OF PROBLEM: 1. Very easy to develop a repeating, intern-centered primary care curriculum. 2. Adds confidence and quality to interns work in clinic as they become well versed in these topics. 3. Provides an excellent opportunity for senior residents to develop their teaching skills.

OBJECTIVES OF PROGRAM/INTERVENTION: As ambulatory education increases in residency training programs, it is common to have interns with various levels of ambulatory experience on block time through the year. It can be difficult to ensure that each intern, regardless of the time of year, receives basic instruction in essential primary care topics.

DESCRIPTION OF PROGRAM/INTERVENTION: The objective of our program is to ensure that each intern, during two weeks of block time in the first half of the year and again in the second half of the year, covers a variety of important topics encompassing both clinical practice and understanding of the medical literature.

FINDING TO DATE: We developed a repeating curriculum for our interns during their ambulatory time. We have new interns in clinic every 2 weeks. Each intern spends two weeks between July and December and two weeks between January and June on ambulatory block.

Each Thursday, the interns have didactic sessions. At noon, they have case-based conferences on telephone medicine and clinical time management taught by the general medicine faculty using materials developed by our curriculum committee. In the afternoon, they have conferences on primary care topics during the first half of the year and critical evaluation of the medical literature during the second half of the year.

A further exciting aspect of this program is the use of our primary care senior residents as teachers. The residents initially observe a faculty talk, then develop talks for the interns. During their talks, they are either videotaped or observed, so that their faculty mentor can provide feedback on their teaching skills.

KEY LESSONS LEARNED: 1. Improvement in intern confidence and independence in the most common problems and issues seen in our clinic. 2. Development of a standard primary care curriculum across the intern year. 3. Novel teaching opportunities for primary care senior residents in basic ambulatory topics with feedback from faculty and residents.

First Block	Second Block
Telephone Medicine	Critical Evaluation of the Medical Literature
Time Management	1. Reading Journals
Hypertension	2. Therapeutic Studies
Diabetes	3. Studies that determine risk
Health Maintenance	4. Overviews and Metaanalyses
Vaginitis	

PRIMARY CARE SURVIVAL SEMINARS. J Tjia, BW Martin, AH Jackson, Section of General Internal Medicine, Boston Medical Center, Boston, MA

STATEMENT OF PROBLEM: Graduates of hospital-based internal medicine residencies often lack essential skills to efficiently transition to office-based outpatient practice.

OBJECTIVES OF PROGRAM/INTERVENTION: To formally develop resident skills in outpatient medicine.

DESCRIPTION OF PROGRAM/INTERVENTION: The Primary Care Survival Seminars are intensive 3-hour skill-building sessions designed to equip residents with essential skills for outpatient practice. Sixteen (16) seminars are delivered over the course of 2 years in four-week blocks (one junior year and one senior year). These cover a wide variety of topics including: Medicolegal Aspects of Telephone Medicine, Current Problems in Managed Care I/II, Critical Evaluation of Literature I/II/III/IV, Common Problems in Rheumatology, Physical Therapy for Ambulatory Medicine, Arthrocentesis of the Knee and Shoulder in Primary Care, Nutrition for the Primary Care Provider, Motivational Interviewing Techniques for

Smoking Cessation, An Approach to the Management of Patients with Chronic Pain, Introduction to Alternative Therapies-Acupuncture, An Approach to Cross-Cultural Medicine, and Cost-Effectiveness in Ambulatory Medicine.

FINDINGS TO DATE: Residents find the seminars a valuable experience. The methods of case-based interactive discussions, role-plays, and emulations (e.g., simulating physical therapy exercises), along with minimization of lecture-based presentations, are key factors to motivate resident interest and learning.

KEY LESSONS LEARNED: Resident practice behaviors can change immediately as a result of this educational intervention. A major obstacle to continuing the program is providing adequate compensation to faculty presenters in the form of academic credit and financial support.

CAREER PATHWAYS: WHAT DO I DO AFTER TRAINING? JB Weiner, Medicine, Baylor College of Medicine, Houston, TX

STATEMENT OF PROBLEM: In 1995, the Baylor College of Medicine Department of Medicine surveyed current residents and recent graduates of the program and found that they felt there was a lack of career guidance. Residents and faculty joined together to develop a unique, resident-driven "Career Pathways" Program.

OBJECTIVE OF PROGRAM/INTERVENTION: The objective of this ongoing program is to facilitate the search and transition from training (residents and fellows) to career (fellowship or new practice).

DESCRIPTION OF PROGRAM/INTERVENTION: The program is organized by residents with the guidance of a faculty advisor. Though initiated at Baylor College of Medicine, the popularity of the program has led to its expansion regionally to include the participation of residents and faculty from the University of Texas at Houston and the University of Texas Medical Branch in Galveston. The program includes a year-long series of workshops and seminars including:

1) The Fellowship Forum: A dinner session with presentations by third year residents on how and where to apply for fellowship and discussion by faculty members from different subspecialties. 2) Interview Skills: An interactive presentation providing strategies on preparation for the time prior to, during, and after the interview. 3) Practice Choices: Which position is right for me?—A panel discussion of representatives from managed care, private, academic, and other types of practices. 4) Practice Management: A workshop on how to join or set up a practice. 5) Negotiating Your Contract: A seminar conducted by health care lawyers detailing specific issues concerning contracts and finding a lawyer. Materials developed for the program include the "Fellowship Guide for Dummies" and the Career Pathways Handbook "How to Get the Job that You Want."

FINDING TO DATE: Written evaluations are conducted after each session. Resident satisfaction with the programs has been high. Revisions to programs are made based on resident feedback. The evaluations have revealed that 1) the guidebooks have been found to be useful resources, 2) the panel discussion and session on legal issues are the most informative programs, and 3) location of venue and timing influence attendance.

KEY LESSONS LEARNED: Given faculty support, trainees can develop successful programs in areas where there is a perceived need. Both residents and faculty should share a role in assisting trainees in their career search. A "Career Pathways" Program with a series of workshops and seminars for trainees can lead to increased competence and comfort in making career decisions.

A CURRICULUM TO OBSERVE UNDERACHIEVING STUDENTS AND GIVE ASSISTED REMEDIATION (COUGAR). JG Wiese, Dept. of Internal Medicine, Univ. of California at San Francisco, San Francisco, CA

STATEMENT OF PROBLEM: Assisting the struggling student or intern is a difficult challenge. Success in clinical medicine requires proficiency in multiple interdependent skill areas, and a deficiency in one skill area can significantly impede a student's progress.

OBJECTIVES OF PROGRAM/INTERVENTION: This intervention diagnoses the struggling student's major skill deficiency using observational methods, written assessments and video techniques.

DESCRIPTION OF PROGRAM/INTERVENTION: Seven skill areas are addressed: attitude, communication, knowledge, organization, time management, physical examination, medical decision making, and interpersonal skills. A numerical score is given for each of the seven topic areas, and the area of greatest deficiency is addressed.

Students with low attitudinal scores undergo personal/professional goal setting and interest assessments. Interpersonal skills are augmented using an adaptation of Carnegie and Covey's techniques of conflict resolution. Using interactive video, students are shown situations requiring interpersonal skills and are asked to make decisions at critical moments in each scenario. Written communication is augmented using an interactive computer program. Oral communication is strengthened with digital subtraction video, in which student presentations are digitally edited and re-arranged to emphasize prioritization of data. Organization of thought is improved by showing students segments of a patient interview, and asking them to state their hypothesis at different parts of the presentation. Deductive reasoning exercises help students re-formulate their thought process at each point in the history/physical. The student's method of recording patient data is reviewed. Johnson's One-Minute Manager techniques are combined with exercises in prioritization to improve time management. Medical decision making is improved with exercises that teach the use of pre- and post-test probabilities. A pre-test is used to assess the student's knowledge base. Faculty provide a list of

reading/multi-media resources that are specific to the student's needs. The student's home library and reading strategy is reviewed.

FINDING TO DATE: This diagnostic strategy has assisted eight students and two interns. The areas of greatest deficiency have been communication skills (3 students/1 intern), interpersonal skills (2 students/1 intern), organization/time management (1 student), attitude (1 student) and knowledge (1 student). Every student successfully completed the medical clerkship, and both interns are currently re-engaged in a successful training program.

KEY LESSONS LEARNED: 1) Each struggling student is unique. 2) It is rare that clinical failure is due to global inadequacy. 3) A program tailored to the student's primary deficit can improve student progress. 4) Innovative technology can assist in timely correction of difficult problems.

IMPROVING RESIDENT TEACHING ABILITY WITH A VENUE-SPECIFIC TEACHING CURRICULUM. JG Wiese, DB White, Dept. of Internal Medicine, Univ. of California at San Francisco, San Francisco, CA

STATEMENT OF PROBLEM: Housestaff play a significant role in student education, but receive little formal education in teaching methods. In order to be effective educators, residents need practical teaching strategies and practice in developing these strategies.

OBJECTIVES OF PROGRAM/INTERVENTION: 1) Provide specific teaching strategies for three teaching venues: board teaching, bedside teaching, and teaching medical decision-making. 2) Illustrate teaching principles through exercises in which residents actively teach students. 3) Objectively evaluate this teaching curriculum with analysis of pre- and post-curriculum videotape of each resident's teaching.

DESCRIPTION OF PROGRAM/INTERVENTION: Before participating in the curriculum, each resident is videotaped teaching students. They are then given a packet of selected readings and asked to screen five selected movies that illustrate important principles of education. The teaching curriculum consists of 4 two-hour sessions in which residents learn education principles through active teaching of medical students. Didactic discussion is used only to consolidate lessons learned in the resident's teaching activities.

Session 1: Residents learn how to inspire learners, promote independent learning and teach methods of problem solving.

Session 2: Residents are videotaped while teaching students a 15-minute board presentation. Each resident's teaching is analyzed by the group and board teaching strategies are discussed.

Session 3: Each resident teaches the group a clinical topic at the patient's bedside. Through the bedside teaching sessions, residents learn to engage and educate the patient, to teach the physical examination while respecting the patient, and to involve the patient in clinical decision making.

Session 4: Each resident guides a student through the decision-making process of a complicated case. The group critiques each resident's performance, and additional methods of teaching medical thinking are reviewed.

At the conclusion of the curriculum, residents are videotaped teaching students. The pre- and post-curriculum videotapes are evaluated by a blinded, independent panel of observers.

FINDING TO DATE: 1) Resident teaching ability is improved by observed practice with immediate feedback. 2) Instructing residents how to teach at the bedside increases the frequency and quality of bedside instruction. 3) The prevalence of resident teaching increases as self-perceived teaching ability improves.

KEY LESSONS LEARNED: 1) Videotape is an effective way to review and critique resident teaching. 2) Venue-specific teaching strategies enable residents to use generic teaching principles in the context of their ward responsibilities. 3) Involving residents in active teaching exercises improves interest in the curriculum and refines teaching skills.

ACHIEVING EDUCATIONAL GOALS WITH THE HELP OF A CHECKLIST SYSTEM. LA Williams, T Tassava, Internal Medicine, St. Joseph Mercy Hospital, Ann Arbor, MI

STATEMENT OF PROBLEM: Ensuring that residents achieve the educational goals of a rotation is ideal, but difficult. It can be especially difficult in an ambulatory setting if there are multiple faculty and multiple residents with varied schedules. We faced such a situation in our general medicine and women's health portions of our two-month ambulatory block rotation.

OBJECTIVES OF PROGRAM/INTERVENTION: Our two main objectives were: (1) To standardize the content of the main educational topics in the rotation. (2) To develop a system to track resident's exposure to each educational topic in the curriculum for general medicine and women's health in the ambulatory block.

DESCRIPTION OF PROGRAM/INTERVENTION: We first developed a list of the core topics in the medicine and women's health curricula that we felt were most important to cover during this rotation. A set of reference articles and teaching materials were compiled for each topic. We then developed a combination orientation, evaluation and educational goals checklist. On a single sheet of paper for each resident we list the core topics with check boxes. Also included are empty lines for personal educational goals, mid and end-of-term evaluation feedback and future goals. On the same page is a checklist with all the administrative and orientation requirements of the rotation. This sheet is reviewed with each resident on the first day of the rotation, and used again at mid and end-of-term evaluations. Two other sheets are used as well. One lists the women's health and one lists the medicine core topics. Each list is in table form with each resident's name heading a column. These are taped to the wall of the charting rooms. As a faculty member

reviews a topic he or she initials it under each resident present. This serves as a visual reminder to the faculty to make sure that each resident receives teaching in every topic.

FINDING TO DATE: Since we began using this system we have noted several improvements. The orientation to the clinic is definitely more complete for each resident. The residents are more likely to receive mid and end-of-term evaluations. Use of the check sheets has allowed us to focus residents to certain types of patients and experiences so that their personal educational goals can be met. We feel more confident that each resident receives a more complete education and a more satisfying rotation.

KEY LESSONS LEARNED: Simple check lists in an ambulatory setting can help ensure that rotational goals are met.

A PRIMARY CARE APPROACH TO TEACHING WOMEN'S HEALTH. LA Williams, Internal Medicine, St. Joseph Mercy Hospital, Ann Arbor, MI

STATEMENT OF PROBLEM: The residents graduating from our Internal Medicine residency program did not feel prepared to handle gynecological and women's health problems with confidence. Our program formerly used two private ob/gyn practices for their gynecology experience, but the residents were not given adequate opportunity to perform exams and were in more of an observer role in that setting. Our other out patient teaching sites did not serve populations who presented commonly with problems that represented key aspects of the core curriculum in women's health recommended by the ABIM. Examples of lacking elements included sexually transmitted diseases, abnormal uterine bleeding, and contraceptive methods among others.

OBJECTIVES OF PROGRAM/INTERVENTION: The objectives we had were three-fold: (1) To teach residents the principles of women's health through a curriculum based upon the ABIM's core curriculum in women's health. (2) To teach in a setting that would provide the type of patients needed, and that would allow for residents to have hands-on experience in patient interviews, exams and procedures. (3) To use internists and other primary care providers in lieu of specialists to teach or residents women's health.

DESCRIPTION OF PROGRAM/INTERVENTION: Because we could not achieve our goals effectively within our current system of out patient teaching, we looked outside of it. In 1998 a collaboration was formed with a local university health center. The center is made up of a separate women's clinic in addition to a general medicine clinic. That spring, during a trial period, one faculty and one to two residents each month saw patients in the women's clinic two half days a week. By July of that year both the women's clinic and the general medicine clinic were incorporated into our two month ambulatory block rotation taken by seniors.

FINDING TO DATE: The response by the residents has been incredibly positive. The residents are the primary providers performing all initial interviews, exams and procedures. The population is very different from any of our other out patient sites. They are young, and generally healthy with no or few chronic medical problems. The residents get plenty of exposure to most aspects of the women's health curriculum. The pace is slow enough that a lot of case-based and didactic teaching can be done. The residents have voiced a much greater degree of confidence in practicing women's health independently. The general medicine clinic has also proven to be a unique experience as well in that it provides exposure to acute problems (such as URI's and injuries) that are not common in their continuity clinics.

KEY LESSONS LEARNED: Instead of struggling to change our current out patient structure to meet the educational needs of our residents we looked out side of it and formed a collaboration that not only met our original needs, but has become a resource for multiple educational programs.

QUALITATIVE STUDY OF THE EFFECTS OF A TEACHING SKILLS DEVELOPMENT CURRICULUM ON TEACHING PROVIDED BY RESIDENTS. GL Woods, GJ Lombard, I Meyer, Internal Medicine Residency Program; Division of Educational Services, University of Illinois College of Medicine, Peoria, IL

STATEMENT OF PROBLEM: The importance of the teaching role of residents is undisputed. However, much of the teaching that third and fourth year medical students receive is provided by residents who have had little or no instruction on how to teach.

OBJECTIVES OF PROGRAM/INTERVENTION: We described and cataloged the characteristics of the questions and statements that residents used in teaching medical students both before and after a course designed to develop their teaching skills. From this we characterized the teaching strategies that residents used and the influence that the course had on these strategies. We then looked at the strategies for common themes that could help us understand the conceptual models in use when residents teach, and if the course influenced these models.

DESCRIPTION OF PROGRAM/INTERVENTION: Thirty five first year residents elected to attend a longitudinal teaching skills development course. Before and again after the course, most of these residents were videotaped during three teaching interactions with standardized medical students. A sample of these tapes were reviewed. The characteristics of each question or statement made by a resident were identified and coded into categories using a coding system that has been previously developed and validated. The codes were tabulated and converted into frequencies. The independent samples T-test was used to measure the difference in the frequency with which each code appeared before and after the course. Codes which changed significantly in frequency were grouped and common

themes in these groups were sought. Independent coding of notes written by the residents after each videotaping will be used to triangulate these themes.

FINDING TO DATE: The preliminary review showed that first year residents frequently asked factual and convergent questions, gave direction on case presentation, and tended to analyze the case for the student. After attending a teaching skills development course, the same residents asked more process-focused questions, gave fewer directions on case presentation, and more frequently encouraged students to analyze the case themselves. Comparison with the coding of notes written by residents after the videotaping is pending. Coding of additional videotapes is underway.

KEY LESSONS LEARNED: A teaching skills development course appears to influence the teaching strategies that residents use from content-focused and teacher-centered toward process-focused and student-centered. Further study is underway to better understand these changes.

COMMUNITY-ORIENTED PRIMARY CARE CURRICULUM FOR INTERNAL MEDICINE RESIDENTS. DC Yao, BY Chen, NH Fiebach, Division of General Internal Medicine, Johns Hopkins Bayview Medical Center, Baltimore, MD

STATEMENT OF PROBLEM: Community-Oriented Primary Care (COPC) is an approach to primary care that integrates clinical medicine, epidemiology, social sciences, and health service research in a complementary fashion to develop programs to meet the health needs of a community. A COPC curriculum development team was formed at our institution, and based on targeted needs assessments from a piloted curriculum, a detailed curriculum was developed. The goal of this curriculum is to cultivate in general internal medicine residents the knowledge, attitudes and skills necessary to incorporate a COPC approach in their future practices.

OBJECTIVES OF PROGRAM/INTERVENTION: Specific learner objectives include: (1) Knowledge—Identify populations with which residents work; describe methods to assess the health needs of a community; describe a method to evaluate the impact of a COPC intervention. (2) Attitude—Identify own role as a primary care physician who addresses needs of community as well as individuals; rate as valuable the involvement of non-physicians and community representatives in COPC projects. (3) Skills—Define a community and its characteristics using databases; use results from community health assessments to identify and prioritize health problems; work in a group setting to design an intervention based on identified community health needs. Program level goals include incorporation of COPC in resident's future practices and community leader recognition of resident participation in community health issues.

DESCRIPTION OF PROGRAM/INTERVENTION: A longitudinal approach integrating didactic instruction with hands-on involvement in COPC projects was developed. The designed COPC curriculum is a two-year longitudinal course that consists of 8 sequential modules. Each module consists of one didactic and facilitated small group session and one flexibly scheduled project work session, incorporating selected readings and specific project assignments. Modules are led by GIM faculty, but include speakers from the community and other disciplines.

FINDING TO DATE: A pilot COPC curriculum was implemented in academic year 1998-1999. Residents undertook diverse projects such as preventing tobacco use through middle school student education; supporting caregivers of the home-bound elderly; and assisting medical clinic patients having difficulty obtaining prescription medications. Preliminary evaluation results show that residents have great enthusiasm for COPC, citing rekindled interest in volunteerism important to many of them prior to residency. Retained knowledge of the principles and practices of COPC was high.

KEY LESSONS LEARNED: (1) Residents value structured opportunities to learn about and participate in interventions in community health.

(2) A longitudinal, project-based learning approach appears to be a successful model for incorporating COPC training in a medical residency curriculum.

THE IRB AS VENUE FOR ETHICS EDUCATION: PROACTIVELY RESPONDING TO A CAVEAT AEGER. M Youtsey, B Adelson, GW Rutecki, Internal Medicine, Evanston Northwestern Healthcare, Evanston, IL

STATEMENT OF PROBLEM: May 14, 1999 marked the second anniversary of President Clinton's apology for the Tuskegee Study. The forty year U.S. Public Health Service Study allowed 399 African-American men to suffer the complications of syphilis and infect others without treatment or informed consent. During the same month, Duke University's license to conduct research was revoked for insufficient IRB oversight of 2,000 studies. Enrollment of research subjects at the University of Illinois, Chicago was recently halted, allegedly because participants were inadequately apprised of risks. Despite educational efforts as a result of the SUPPORT Study, protection of human subjects in research has been relatively neglected.

OBJECTIVES OF PROGRAM/INTERVENTION: To design an ethics rotation for Internal Medicine house-staff that provides engagement with the ethical dilemmas in the context of IRBs.

DESCRIPTION OF PROGRAM/INTERVENTION: The Institutional Review Board of a University-Affiliated Community Hospital (Evanston Northwestern Healthcare) will serve as the primary site for the rotation. The IRB convenes monthly and reviews approximately 10 new protocols with another 400 active protocols followed. The board is comprised of 16 physicians and researchers, as well as four members of the lay community.

FINDINGS TO DATE: Resident education will require attendance at four IRB meetings (3rd Friday every month). The HBO "made for TV movie" *Miss Evers Boys*—addressing the Tuskegee Study—serves as an introduction to the protection of human subjects. A 90-minute session follows the movie during which residents design a study with attention to informed consent, safety monitoring, the ethics of placebo-control and the reporting of adverse events. The first IRB meeting will consist of observation, during which time the resident will focus on the ethical scrutiny applied to protocols. In advance of the following three sessions, the resident will be provided with copies of two protocols to be reviewed at the meetings. The resident's role will be to highlight areas related to human subject protection. The resident would then attend the meeting at which the protocols are discussed, participate, and summarize the CONCLUSION. After each of the meetings, the resident and the IRB director (B.A.) review the resident's evaluative performance in the context of each protocol.

KEY LESSONS LEARNED: The IRB provides a fertile environment for IRB ethics education. It should become an integral portion of ethics and research training for internal medicine residencies.

INNOVATIONS IN PRACTICE MANAGEMENT

PREDICTION OF DAILY PATIENT VOLUME. HA Batat, JC Adams, PS Mehler, Denver Health Medical Center, Denver, CO

STATEMENT OF PROBLEM: In order to improve patient satisfaction and access to care a system allowing accurate staffing for expected patient volume was necessary.

OBJECTIVES OF PROGRAM/INTERVENTION: To develop a simple prediction equation to allow increased accuracy in predicting patient volume and staffing patterns.

DESCRIPTION OF PROGRAM/INTERVENTION: Daily patient volume from 2/98–1/99 was matched with calendar and weather variables. Weather data was taken from the National Weather Service and included total precipitation, snowfall, and temperature. Calendar variables included day and month of the year, season, and the day before, after, or of a major holiday. Stepwise linear regression analysis was performed with insignificant variables sequentially eliminated until all variables were significant at the 5% level. Randomly selected days from 2/99–7/99 were used as a validation set. The initial analysis resulted in a prediction equation of: daily patient volume = 57.0 + 12.0 January – 5.28 July – 5.57 August + 6.63 winter + 46.7 Monday + 37.2 Tuesday + 36.0 Wednesday + 29.4 Thursday + 24.1 Friday + 8.67 Saturday + 10.1 day after a holiday + 0.147 maximum temperature – 3.88 inches of snowfall. This equation accounted for 78.6% of the daily variation, with a p value of <0.01. However, this equation was felt to be cumbersome because of the reliance on weather forecasts. Thus, a second equation was developed utilizing only calendar variables. This equation was: daily patient volume = 66.2 + 11.1 January + 4.56 winter + 47.2 Monday + 37.3 Tuesday + 35.6 Wednesday + 28.2 Thursday + 24.2 Friday + 7.96 Saturday + 10.1 day after a holiday. This equation accounted for slightly less of the daily variation in patient volume, 75.2%, but also had a p value of <0.01. The validation set of data showed that the calculated patient volumes, using the equation including weather variables, averaged within 7.6% of actual. When the equation without the weather variables was used, the predictions were within 8.1% of actual patient volume. In 71% of cases the predicted number of patients seen was less than the actual number of patients seen. This is likely due to an 8–9% increase in patient volume since 2/99.

FINDING TO DATE: Using the prediction equation including only calendar variables, and adding a correction factor of 8% for the increase in volume, in 9/99 the clinic was accurately staffed on 70% of the days, overstaffed on 20% of the days, and understaffed on 10% of the days. Due to this, the number of patient complaints decreased by more than 50% and the number of patients who left without being seen, which had been increasing during the first half of 1999, decreased by 14% in September.

KEY LESSONS LEARNED: Accurate prediction of volume allows for improved staffing. Adequate staffing allows for a decrease in patient complaints and increased patient access.

PATIENT SATISFACTION IN THE RESIDENT CONTINUITY CLINIC. KD Baum, R Burgeson, S Johnson, Internal Medicine, University of Minnesota; Fairview Health Services, Minneapolis, MN

STATEMENT OF PROBLEM: Examine and compare patient satisfaction between a Resident Continuity Clinic (RCC) and a staff-run Primary Care Clinic (PCC) at the same site.

OBJECTIVES OF PROGRAM/INTERVENTION: The Resident Continuity Clinic is the cornerstone of the ambulatory care training most residents receive during Internal Medicine residency. While much effort is spent designing this portion of their curriculum to benefit the residents, few studies have sought to examine the impact these clinics have upon patient satisfaction. As such, we developed a

study to compare patient satisfaction at one of the RCC's in our program with that of the PCC at the same site run by the same staff members.

DESCRIPTION OF PROGRAM/INTERVENTION: A cross-section of patients seen in the previous 5 months at both the RCC and PCC received a mailed questionnaire covering 20 aspects of patient satisfaction in addition to overall clinic satisfaction and demographic data. Returned surveys were sent to an independent data entry firm for tabulation, and final analysis was done by our group using computerized statistical software packages. Pearson chi-square analysis was done for each question.

FINDING TO DATE: Of the 1028 surveys sent out to PCC patients, 335 were returned. 85 of the 350 RCC surveys were returned, for an overall response rate of 30.5%. This rate is consistent with most mailed surveys. Patients in the RCC group felt that routine appointments (p = 0.022) and phone call returns were not as convenient (p = 0.002), and it was more difficult for them to see the doctor of their choice (p = 0.015). In addition, RCC patient felt it was more difficult to have their questions answered during clinic hours (p = 0.024), and they were not as well informed of test results as those in the PCC clinic (p = 0.032). However, overall patient satisfaction did not statistically differ between the two groups (p = 0.268).

KEY LESSONS LEARNED: While it was heartening to find that patients in the RCC were just as satisfied with their medical care as those in the PCC, this study certainly identifies important areas of routine ambulatory patient care which would benefit from directed improvement in the RCC. Some of these areas speak to the decreased availability of residents (one half-day per week) compared to most faculty members, and as such are unlikely to be areas for possible improvement. However, there clearly is a disparity with regards to communication with patients, as seen by the lower ratings by RCC patients for questions such as phone call returns and communication of test results. These areas should be noted by the RCC directors for intervention development in order to improve the care and satisfaction received by patients who are generous enough to enroll in these clinics which help to further medical education.

A PERSONALIZED DECISION SUPPORT AID TO IMPROVE THE QUALITY OF CARE OF MENOPAUSAL WOMEN. NE Col, LK Hirota, AM O'Connor, JM Fortin, RJ Goldberg, Tufts-New England Medical Center, Boston, MA; University of Ottawa, Canada; University of Massachusetts Medical Center, Worcester, MA

STATEMENT OF PROBLEM: By 2020, one-third of North American women will be menopausal. The complex health needs of these women are poorly met by our current health care system.

OBJECTIVES OF PROGRAM/INTERVENTION: To improve the quality of care provided to menopausal women by developing a personalized decision support aid (DSA).

DESCRIPTION OF PROGRAM/INTERVENTION: We developed a DSA for menopausal women and their clinicians that consists of computer-generated personalized health reports. Women complete a brief written questionnaire about their life-style, family, and medical history. This information is entered into a software program that generates a comprehensive personalized report for the patient (35–45 pages) and a 2-page summary report for her clinician. Both reports include information about the woman's menopausal symptoms, her personal risks for common chronic diseases and the impact of hormone replacement therapy on these risks (based on published risk models), her utilization of recommended screening modalities, and general strategies for risk reduction and treatment. The DSA also screens women for depression, alcohol abuse, sexual dysfunction, and domestic violence, providing relevant information about diagnosis and treatment to clinicians and patients when needed. 10 clinicians and 13 patients have evaluated the DSA to date.

FINDINGS TO DATE: Most physicians rated the DSA very favorably (60% excellent, 30% good) and 70% would like to use it with their menopausal patients. None felt that it was scientifically inaccurate, and 90% reported that all of the topics they normally discuss with patients were included in the DSA. All felt that it increased their knowledge about their patient and their patient's knowledge about menopause. Most felt that it would improve their ability to counsel women about menopause, the efficiency of the consultation, and the quality of care. Half felt it would improve their interaction with the patient. Most patients felt the DSA would improve communication with their clinician (69%), help them identify questions to ask their clinician (77%), and improve the quality of care they receive (54%).

KEY LESSONS LEARNED: The menopausal DSA appears to be acceptable to clinicians and patients, may facilitate counseling, and may improve the quality of care provided to menopausal women.

Clinician Evaluation	Improve (%)	No Effect (%)	Worsen (%)
Interaction with pt	50	40	10
Quality of care	70	30	0
Ability to counsel pt	70	30	0
Efficiency of visit	60	10	30
Knowledge about pt	100	0	0

PHARMACEUTICAL COMPANY-SPONSORED PATIENT ASSISTANCE PROGRAMS: LOOKING A GIFT HORSE IN THE MOUTH. DM Eberwein, JR Kimberley, JL Wofford.

OBJECTIVES OF PROGRAM/INTERVENTION: Pharmacy industry sponsored patient assistance programs are increasingly available and hold the potential to offset drug costs for indigent patients. However, many providers are hesitant to use these programs due to perceived barriers.

DESCRIPTION OF PROGRAM/INTERVENTION: In order to evaluate the perceived usefulness and ease, we surveyed all providers at our urban community health center through a questionnaire given between June and August of 1999 (response rate 60%, $31/51$).

FINDING TO DATE: The questionnaire was used to help develop a systematic approach to enroll patients into the patient assistance programs. Data from the provider yielded information on the proportion of providers using the programs, the number of times the provider applied to a program, and the success rate for actually receiving medications. Data from the application yielded information on the method of enrollment, barriers specific to that medication or program, the amount of time it took to receive that medication and the overall satisfaction with the patient assistance program.

KEY LESSONS LEARNED: Fifty-eight percent (18/31) of responders had applied for medications through a patient assistance program at least once, and 55.5% (10/18) had applied four or more times. These applications represented 19 different medications. With most applications (14/39), providers became aware of the patient assistance programs through the clinic pharmacist. Written applications or letters to the pharmaceutical company were used to enroll patients most often (versus phone contact or sales representative). The most common barriers were receiving the medications and reapplying to the programs compared to the initial contact with the company. Medication was usually received in 1 to 6 weeks. The majority of providers (90%, $28/31$) felt that medications should be obtainable in <4 weeks and the best method of initial contact with the pharmaceutical company was through phone contact from the provider or provider representative. The providers (96%, $27/28$) felt the best reminder to reorder medication was a notice from the pharmaceutical company or pharmacist rather than a notice to the patient. Eighty-two percent (29/35) of providers were satisfied or very satisfied with the programs.

STATEMENT OF PROBLEM: The patient assistance programs were perceived as useful and fairly easy to use in this setting. We have designed multiple systems allowing providers easy access to program applications and follow up, which has allowed us to overcome barriers perceived by the providers. Our systematic approach towards enrollment resulted in providers using these programs more often. The response from the pharmaceutical companies towards the increased use of these programs may be a factor in how available these programs are in the future.

IMPROVING THE VALUE OF HEALTHCARE THROUGH AN INTERNAL SMALL GRANTS PROGRAM. BA Evanoff, J Garbutt, B Littenberg, G Medoff, WC Dunagan, Washington University School of Medicine, St. Louis, MO; University of Vermont, Burlington, VT

STATEMENT OF PROBLEM: There is insufficient data to choose between different options in health care delivery.

OBJECTIVES OF PROGRAM/INTERVENTION: To test innovative strategies for improving the value of healthcare by means of a small grants program.

DESCRIPTION OF PROGRAM/INTERVENTION: The Innovations in Health Care Program was started in 1994 as a partnership between medical school faculty and a regional integrated health system. Project funding is limited to \$25,000; proposals are solicited from all employees and medical staff. We use five simple guidelines to identify desirable projects: (1) clear-cut application to an important health care problem; (2) evidence that the innovation is likely to improve the value of healthcare; (3) high probability that benefits can be demonstrated in 18 months; (4) potential for applicability to multiple sites within the health system; and (5) rigorous evaluation of innovations (most funded studies have been randomized clinical trials).

FINDING TO DATE: We have funded 26 projects. The fifteen projects completed to date have demonstrated clear improvements in diverse outcomes including patient satisfaction, length of mechanical ventilation, and length of stay. Annual cost savings from completed projects (total funding of \$236,795) is estimated at \$2.2–3.1 million. Examples of projects which have demonstrated cost-beneficial clinical improvements include: the elimination of daily suction catheter changes in patients on mechanical ventilation; institution of protocol-driven weaning to reduce time on mechanical ventilation; substitution of tincture of iodine for povidone-iodine as a skin preparation to prevent contamination of blood cultures with skin flora; the substitution of normovolemic hemodilution for autologous blood donation in patients undergoing prostatectomy; and early ambulation for patients with community acquired pneumonia.

KEY LESSONS LEARNED: Most project proposals have come from within the academic medical center, while efforts to encourage research projects at community hospitals have met limited success. System-wide implementation of successful projects is difficult to achieve in a heterogeneous health care system. Despite these challenges, the Innovations in Health Care Program appears to be a viable model for encouraging changes in health care delivery.

PREDICTING MISSED OR CANCELLED APPOINTMENTS. RI Gianani, TD MacKenzie, General Internal Medicine, University of Colorado Health Sciences Center; Community Health Services, Denver Health Authority, Denver, CO

STATEMENT OF PROBLEM: The overall Westside Adult Clinic missed appointment rate is over 30%, which has resulted in long waits for new and follow-up appointments and uneven workloads for providers.

OBJECTIVES OF PROGRAM/INTERVENTION: To identify factors predicting missed or cancelled appointments in an urban adult medicine clinic.

DESCRIPTION OF OBJECTIVES/INTERVENTION: We analyzed 25122 clinic appointments made to the Internal Medicine clinic at Westside Family Health Center in Denver, Colorado during a 15 month period. The patient specific data include age, gender, race, language preference, primary care provider, and income. Provider-specific factors include training level and Spanish language ability. Visit-specific factors include prospective provider for the scheduled visit, whether or not prospective provider was the PCP, day of the week, time from appointment scheduling to visit, and type of visit.

FINDING TO DATE: Factors strongly associated with missing or cancelling an appointment are shown below.

	Odds ratio	CI
Not urgent care visit	2.24	(2.09–2.40)
PCP not at Westside	2.04	(1.94–2.15)
Not scheduled with PCP	1.71	(1.62–1.80)
Age <25 years	1.57	(1.44–1.70)
More than 30 days to appointment	1.38	(1.31–1.45)
Scheduled with NP	1.34	(1.27–1.42)

Factors that were strongly associated with missing or cancelling an appointment in the univariate analysis remain so using multiple logistic regression. Factors which were not strongly associated with missing or cancelling an appointment include language preference, language concordance between provider and patient, and income.

KEY LESSONS LEARNED: Patient-, provider-, and visit-specific factors strongly influence the likelihood of missing or cancelling an appointment.

ASSESSING ADVERSE SELECTION IN PRIMARY CARE CLINICS. JL Hamilton, E Rich, P Turner, Creighton University, Omaha, NE

Physicians practicing at a General Internal Medicine (GIM) clinic at a University teaching hospital were at risk of being terminated from a health plan due to consistently higher expenses than revenue. This project was undertaken to determine if the University GIM clinic had higher expenses than the other primary care clinics due to adverse selection.

Claims data for 1998 was obtained from 1 health plan. A case mix analysis was performed which utilized the first visit of the year for each patient. The top 30% of diagnoses were identified for each clinic and 2 physicians rated the diagnoses on a scale of 1 to 5 based on typical resource utilization expected over 1 year. The physicians chosen to rate the diagnoses included the Medical Director of the health plan mentioned as well as a practicing physician who is also the Utilization Management Director for the University. The number of visits per diagnosis were then multiplied by the average score for that diagnosis and an average was calculated for the clinic. Cohen's Kappa (Cohen, 1960) was used to assess physician agreement in rating the 21 diagnoses regarding resource utilization with an a priori Kappa of .40 set as acceptable based on Fleiss's (1981) criteria. The diagnoses were divided into tertiles of low, medium, and high resource utilization. Chi-square analyses with Bonferroni-adjusted planned comparisons were used to determine if there were significant differences among the clinics and their respective panel of patients based on resource utilization, and if so, whether the GIM clinic had a higher resource utilization patient panel than its 5 counterpart clinics. Case mix analysis demonstrated the GIM clinic to have the highest mean score indicating this clinic has patients with higher resource utilization. There was good agreement among the physician raters for the ratings of diagnoses for expected resource utilization (Kappa = .48). There was a significant difference among the six clinics regarding resource utilization expected in their panel of patients (Chi-Square = 112.58, $p < .001$). When compared to its counterparts, the GIM clinic was also seeing a different panel of patients (Chi-Square = 57.69, $p < .001$), specifically more patients with higher resource utilization and fewer with moderate or low resource utilization (Chi-Square = 40.31, $p < .001$; Chi-Square = 40.93, $p < .001$, respectively) compared to the other 5 clinics.

KEY LESSONS LEARNED: According to the case mix analysis, the University GIM clinic has patients who would be expected to have higher resource use than the patients in the other 5 clinics. These data were accepted by health plan administrators as an explanation for the higher utilization experienced by the GIM clinic. This methodology could be used in other teaching clinics where health plans do not have case mix and severity adjustment tools available to evaluate for adverse selection.

DEVELOPING AND IMPLEMENTING A PHYSICIAN-GENERATED, UNIFIED FORMULARY IN A MULTI-HOSPITAL, MULTI-PRACTICE NETWORK. JJ Heffernan, GW Smetana, N Minkoff, D Calabrese, Division of General Internal Medicine, Beth Israel Deaconess Medical Center, Boston; Deaconess-Waltham Hospital, Waltham; Provider Service Network, Caregroup, Boston, MA

STATEMENT OF PROBLEM: Physicians receive numerous, and often disparate, formularies from managed care organizations (MCOs). Although much overlap exists, and many of the included medications are appropriate, these formularies all differ and often promulgate the use of high cost, brand name pharmaceuticals to

maximize profit (or minimize loss) for the MCO by gaining access to corporate rebate programs, often to the detriment of provider units that, increasingly, bear full risk for pharmacy costs. As such, MCO formularies lack credibility among the provider community.

OBJECTIVES OF PROGRAM/INTERVENTION: To develop and implement a single-source, physician-generated unified formulary founded on the complementary principles of best efficacy and value to patients at least cost to provider groups.

DESCRIPTION OF PROGRAM/INTERVENTION: The Pharmacy and Therapeutics Committee of the Provider Services Network of Caregroup includes practicing network primary care clinicians and clinical pharmacists. This 22-member core committee, with input from 10 specialty subcommittees, has developed and implemented a unified formulary. Specific medication classes are reviewed on a monthly basis. The unified formulary has been distributed to the 2,800 physicians in this large, multi-hospital, multi-practice network in a number of formats: (1) multi-fold pocket cards (2) wall charts (3) binders (4) electronic versions on hospital intranets and the internet, and (5) downloadable electronic files for palm computing devices. Average wholesale prices of individual agents are featured prominently. The committee has counseled network physicians to use the unified formulary exclusively and to ignore formularies from individual MCOs. Pharmacy compliance is assessed against the PSN formulary alone.

FINDING TO DATE: Compliance with formulary recommendations has increased within every therapeutic class addressed in the document, yielding first year savings in excess of \$1.4 million. In certain drug categories, market share of PSN "preferred" drug products has risen by 30–40%. The unified formulary has become the cornerstone of other PSN pharmacy efforts within individual provider units, such as therapeutic interchange initiatives and educational efforts.

KEY LESSONS LEARNED: This physician-driven unified formulary has had tremendous support among network provider groups and enjoys substantial credibility as the work product of prescribing clinician's colleagues. It is a user friendly, single-source, easily accessible reference, and has become a widely used teaching tool, and the foundation of pharmacy interventions at both the network and local provider unit level.

INFORMATION MANAGEMENT PLANNING: EXPERIENCE OF AN ACADEMIC: HEALTH CENTER BUILDING IN ARGENTINA. *JT Insua, G Breitbart, R Martin, R Villalon, JC Di Lucca, The Austral University Hospital Planning and Development Group, Health Care Policy and Management Program, Austral University; Department of Internal Medicine, Austral University Hospital; Austral University Hospital Planning and Development Group, Pilar, Buenos Aires, Argentina*

STATEMENT OF PROBLEM: There is no experience in academic health center (AHC) development including information management planning (IMP) from the very first beginning of its operation in Argentina. It is important to mention that the market does not offer any support for health care information management planning.

OBJECTIVES OF PROGRAM/INTERVENTION: We conducted since 1998 an IMP process and designed the managerial and clinical information right into the "genetic code" of a new emerging AHC. The objectives were: a) to put the information management at the core of the future university hospital (UH) and integrated delivery system (IDS), b) to include paperless clinical information system along with administrative information management, through an electronic medical record (EMR) as a commanding system, c) develop the basis of a database for research, education, and health care provision and management.

DESCRIPTION OF PROGRAM/INTERVENTION: A program led by the medical and managerial groups was designed to: 1) describe an IDS process oriented planning as a template for the requirements to software and hardware potential providers of the hospital information management systems (HIS), 2) select a system available closest to the stated objectives stated, 3) use that providers system to develop the remaining objectives (such as the EMR, among others), 4) introduce system wide diagnostic and procedure coding, and 5) implementation of the previously described objectives.

FINDING TO DATE: 1) No similar experiences were found in Argentina up to date, 2) no EMR nor computerized medical record (CPR) were found in the market, 4) the best proposal of system provider met 60% of the requirements, 4) we worked with the provider in the development of the 40% of the remaining objectives not met and the EMR, 5) in order to use CPT coding and to be able to bill we had to develop an conversion table to the national practice coding system.

KEY LESSONS LEARNED: 1) Introducing IMP in the "genetic code" of an emerging AHC is in fact a significant advantage, 2) substantial effort in the planning phase is required, 3) there is organizational, medical and market immaturity in Argentina for the scope of the project, 4) organizational stress to adaptation of providers and physicians is widespread, 5) to upgrade a HIS into a clinical information system through an EMR was required, 6) there is a lack in procedure coding systems, 7) there is marketing and contractual risk of the coding decision performed, 8) difficult recruiting of trained human resources, 9) costs of the project.

IMPLEMENTATION OF A HOSPITALIST PROGRAM IN A LARGE PUBLIC TEACHING HOSPITAL. *JA Leviss, N Link, M Charap, Medicine, Bellevue Hospital/NYU School of Medicine, New York, NY*

The Bellevue Hospital Medical Service, 110 beds with 40,000 annual inpatient days, had a part-time faculty from the NYU School of Medicine (NYU) to supervise clinical care provided by house staff. Increasing oversight requirements and pro-

longed lengths of stay could not be addressed by the house staff-dominated system with only two hour daily rounds with attendings.

1) To create a full-time hospitalist attending program to supervise clinical care without disenfranchising the voluntary NYU Faculty nor ending its contribution to clinical care and teaching; 2) To reorganize the medical wards to: improve house staff oversight and attending accountability; meet Residency Review Commission requirements for distinct clinical and management attending rounds; improve discharge planning and outpatient followup; and decrease length of stay.

Since July 1, 1999, one hospitalist and two teaching attendings (voluntary) serve with two house staff teams per ward. The hospitalist oversees all patient care as the attending of record. Teaching attendings, with the hospitalist, conduct daily house staff teaching rounds. 16 hospitalists, chosen from the Primary Care Clinic for clinical and teaching skills, serve one month in every four. Five Physician Assistants (PA) were hired to follow hospitalist's clinic panels while on wards. PAs attend daily ward discharge planning rounds to facilitate outpatient followup. Hospitalists on wards still precept house staff continuity clinics.

1) Initial responses by attendings and house staff are positive as per multidisciplinary committees on Hospitalist and Residency Programs. End of year house staff and attending surveys will be compared to baseline data. 2) Surveys on all four wards two months in a row documented no house staff sense of decreased autonomy. 3) All hospital accreditation requirements were met. 4) Both Primary Care Clinic visits and medicine inpatient admissions increased 10% in July–December, 1999, compared to July–December, 1998. 5) Initial data indicates a 5% decrease in length of stay. 6) The program was funded at an incremental cost of less than \$400,000, including four PAs hired with a New York State Community Health Partnership grant. Hospitalist positions were created from existing primary care physician lines, with an 8% salary increase to compensate for increased duties.

KEY LESSONS LEARNED: A universally implemented, full-time hospitalist attending program, utilizing primary care physicians, can increase supervision of a house staff-centered medicine service without: decreasing house staff sense of autonomy; disenfranchising prior ward attendings; decreasing inpatient or outpatient productivity; or incurring great financial costs. Additionally, continuity of care to an outpatient setting can be facilitated.

FORECASTING THE IMPACT OF A CLINICAL PRACTICE GUIDELINE FOR PERI-OPERATIVE BETA BLOCKER USE. *PK Lindenauer, M Schmidt, JL Fitzgerald, EM Benjamin, Office of Clinical Practices Evaluation and Management and Department of Medicine, Baystate Medical Center and Tufts University School of Medicine, Springfield; Department of Epidemiology and Biostatistics, University of Massachusetts, Amherst; Office of Clinical Practices Evaluation and Management, Baystate Medical Center, Springfield, MA*

STATEMENT OF PROBLEM: Beta blockers have been shown to reduce morbidity and mortality when administered to high risk patients undergoing major non-cardiac surgery. Clinical practice guidelines (CPGs) are often used to enhance the adoption of effective treatment strategies but before implementing any CPG it is important to estimate its potential clinical and financial impact.

OBJECTIVES OF PROGRAM/INTERVENTION: We sought to determine how frequently peri-operative beta blockers were being used at our institution and to estimate the potential clinical and financial impact of a CPG designed to increase their use.

Description of Program/Intervention: We developed a CPG for identifying high risk patients undergoing major non-cardiac surgery and for facilitating and standardizing beta blocker usage in the peri-operative period. We conducted a retrospective analysis by chart review of patients undergoing major non-cardiac surgery at our institution during a 1 month period in 1999.

FINDING TO DATE: 1132 consecutive surgical patients were evaluated, 212 met inclusion criteria, and the charts of 158 patients were available for review. "Ideal candidates" were predefined as having established CAD or 2 or more major CAD risk factors and no initial contraindications to beta blocker use. $67/158$ (42%) patients appeared to be "ideal candidates" for treatment with peri-operative beta blockers, yet of these only $31/67$ (46%) received a beta blocker at any time during the course of their hospitalization and nearly all of these patients had been taking beta blockers chronically. Annual costs to the institution associated with additional drug and monitored bed use were estimated at \$32261 or \$60 per patient. Using the results of a large RCT to estimate the clinical benefit at our institution, an additional 60 lives might be saved each year at a cost of \$424 per life-year saved if all ideal candidates received beta blocker therapy.

KEY LESSONS LEARNED: A large opportunity exists to improve the quality of care at our institution at relatively modest cost. A clinical practice guideline may be one method to achieve this goal.

QUALITY IMPROVEMENT FOR ADVANCED HEART AND LUNG DISEASE. *J Lynn, MW Schall, KM Nolan, A Kabcenell, CC Milne, Center to Improve Care of the Dying, The George Washington University, Washington, DC; The Institute for Healthcare Improvement, Boston, MA; Associates in Process Improvement, Silver Spring, MD*

STATEMENT OF PROBLEM: Only a few generations past, most people died suddenly. Now, most will die after months or years of living with slowly progressive chronic illness. Care routinely mismatches needs, perhaps especially with advanced heart (CHF) and lung disease (COPD). Discontinuity and prognostic uncertainty make for aggressive medical treatment with inadequate advance planning, symptom management, and reliability.

OBJECTIVES OF PROGRAM/INTERVENTION: Working for seven months with 34 provider organizations, we aimed to learn how to improve care for patients with advanced CHF or COPD. Participating organizations included VA medical centers, hospices, integrated health systems, and one eldercare organization.

Description of Program/Intervention: The Breakthrough Series Collaboratives, a program of the Institute for Healthcare Improvement, recruits provider teams to learn methods to efficiently adapt improvements to their programs. In the Collaborative on Advanced Heart and Lung Disease, co-sponsored by the Center to Improve Care of the Dying and the Department of Veterans Affairs, 34 teams joined in three Learning Sessions, internet and telephone communication, site visits, monthly reports, and especially on-site work in quality improvement. Most teams were from the DVA or from hospices.

FINDING TO DATE: * Patients to target for advanced illness interventions are often best identified by asking their physician, "Is this patient sick enough that you would not be surprised if he/she died in the next few months?"

* Reliability over time is essential. The Collaborative developed a conception of good care that builds on making promises to patients: that medical treatment will be correct, that symptoms will never be overwhelming, that no gaps will affect care, that planning will encompass expectable emergencies, that care will be customized to patient/family preferences, that family issues will be important, and that caregivers will help to make every day as good as it can be.

* Good care systems turned out to need medication guidelines implementation, creative adaptation of medication schedules and costs, rapid response to serious symptoms, home medication kits, 24/7 on-call by caregiving team, patient/family self-management, and advance care planning.

* Care systems regularly cut emergency room and hospital use to about half of these patient's past utilization, and patients and families were much more confident and satisfied.

KEY LESSONS LEARNED: Clinical teams can substantially improve care for patients with advanced heart and lung disease, usually within a few months.

* Medicare financing will pose barriers to replicating this work in fee-for-service and many managed care settings. * Patients and family involvement in self-management is welcome and effective.

IMPROVING PROCESSES OF CARE IN PATIENTS WITH STROKE; THE ISCHEMIC STROKE INPATIENT PATHWAY (ISIP) STUDY. PJ Mahadevia, D Conner, N Pooran, U Gavarasana, P Jain, R Libman, R Kanner, Division of General Internal Medicine; Department of Neurology, Long Island Jewish Medical Center, New Hyde Park, NY

STATEMENT OF PROBLEM: Studies of critical pathways have focused more on resource utilization and length of stay than their effect on patient care.

OBJECTIVES OF PROGRAM/INTERVENTION: Our purpose was to evaluate the effectiveness of an ischemic stroke inpatient pathway (ISIP) on processes of patient care and outcomes.

DESCRIPTION OF PROGRAM/INTERVENTION: We performed a retrospective cohort study, comparing the care stroke patients received before and after implementation of a stroke pathway, ISIP. The ISIP consisted of admission pre-written order sheets, multidisciplinary rounding with a case manager and social worker, and a diagnostic algorithm based on local expert opinion by the Neurology department. Patients admitted by the neurology service for an ischemic stroke during September 1998 to February 1999 were in the pre-ISIP group. The ISIP was introduced March 1999. Post-ISIP patients were admitted from September 1999 to February 2000. Data was obtained from medical record review and a hospital database. We measured antiplatelet agent or anticoagulant use, deep vein thrombosis (DVT) prophylaxis, physiatry consultations, brain imaging tests, electrocardiograms (ECG), in-hospital mortality, the National Institutes of Health Stroke Scale (NIHSS), the Glasgow Outcomes Stroke (GOS) scores and hospital length of stay.

FINDING TO DATE: Seventy-three patients, 46 from the pre-ISIP and 27 patients from the post-ISIP cohorts, were evaluated. The groups had similar admission NIHSS (6.9 vs. 7.8) and GOS (2.2 vs. 1.9) scores. Among patients in the post-ISIP group there was a higher rate of either antiplatelet agent use or anticoagulation on admission (100% vs. 89%, $p < .002$), ordering physiatry consults on admission (100% vs. 48%, $p < .002$), and obtaining admission ECG (100% vs. 78%, $p < .02$). The rate of brain imaging tests (100% post vs. 98% pre) and DVT prophylaxis (75% post vs. 68% pre) was not significantly different between the groups. There were no differences in length of stay, in-hospital mortality, NIHSS or GOS scores between the groups at discharge.

KEY LESSONS LEARNED: Patients cared for using the ISIP had higher rates of therapies, tests and consultations that have been associated with better outcomes. Although the ISIP changed process of care we do not know its impact on outcomes. Since our study did not have the power to evaluate outcomes we suggest a larger prospective trial to evaluate the effect of a stroke critical pathway on patient care and outcomes.

ESTIMATION OF INFLUENZA VACCINATION RATES AMONG HIGH RISK PATIENTS USING AN ELECTRONIC MEDICAL RECORD. NS Mahaskar, FN Pelzman.

STATEMENT OF PROBLEM: Nearly 90% of influenza-related deaths occur in persons over the age of 65. Yearly immunizations can significantly decrease the morbidity and mortality related to influenza. While national recommendations call for immunization rates of 60% in these individuals, many medical centers fall short of this goal.

OBJECTIVES OF PROGRAM/INTERVENTION: We examined vaccination rates of patients over the age of 65 at a major university-affiliated general medicine prac-

tice. The results may be used to target our efforts to increase immunization of high risk populations.

DESCRIPTION OF PROGRAM/INTERVENTION: Our study dates were from August 1, 1998 to July 31, 1999. Using CLIMACS, a medical information and tracking system, we were able to identify all patients over the age of 65 who had an appointment made within two years of our study dates. We then generated a list of all patients who had received an influenza vaccine documented in CLIMACS's electronic flowsheet. By comparing these lists we were able to obtain an estimate of our overall vaccination rate. In addition, a second list of patients over the age of 65 who had an appointment between September, 1998 and January, 1999 was also obtained. A random chart review of 100 of these patients provided an estimate of vaccination rates among those who actually saw a physician during this period. We tabulated the number of vaccines given that were either documented in the flowsheet or recorded within the text of the progress note.

FINDING TO DATE: 5048 patients over the age of 65 had an appointment made between August, 1998 and July, 1999. 1443 of these were recorded on our CLIMACS system as having received a flu vaccine, corresponding to a 28.6% vaccination rate. 2530 patients had an appointment made from September, 1998 through January, 1999. Random chart review revealed that 91% kept their appointment. Of these, more than 2/3 (68%) received a flu vaccine. 27 patients did not, of which 14 refused, and 3 received a flu shot at another location.

KEY LESSONS LEARNED: Our overall vaccination rate was lower than national goals. However, of those patients who made and kept their appointments during the fall, a substantial proportion were vaccinated, exceeding national guidelines. These results imply the need to especially target those patients who are not seen by a physician in the months preceding the flu season. The differences in measured vaccination rates found by the two techniques of chart review reflect problems of compliance with flowsheet documentation within the electronic medical record. Potential strategies for improving compliance and record keeping include a computer generated letter reminder system, electronic chart reminders for physicians, and scheduling of annual exams during the fall months. The techniques outlined here can be applied to other areas of health care maintenance, such as pneumonia vaccination, to tailor efforts at improving preventive care to individual practices.

IMPACT OF PROACTIVE DIABETES MANAGEMENT BY CLINICAL PHARMACIST ON GLYCEMIC CONTROL AND COMPLICATION SCREENING PROCESS. S Mitrovich, H Choe, DA Dubay, Internal Medicine; Pharmacy, University of Michigan, Ann Arbor, MI

STATEMENT OF PROBLEM: 1. Case management and collaboration with a Pharm D in incorporated in a general medicine practice improves glycemic control. 2. Patient acceptance of Pharm D intervention is very high.

OBJECTIVES OF PROGRAM/INTERVENTION: Diabetes mellitus is a major health problem with the prevalence of type 2 diabetes rising. Studies have shown that good glycemic control improves outcomes. With standard diabetic management, patients with severe diabetes often attain suboptimal glycemic control and frequently do not meet the HEDIS standards.

DESCRIPTION OF PROGRAM/INTERVENTION: To compare proactive diabetes case management by a clinical pharmacist with usual care.

FINDING TO DATE: Fifty-four patients with type 2 diabetes with glycohemoglobin of $>10\%$ and 30-70 years old were identified. The group was blindly randomized to control versus intervention. Informed consent was obtained. The intervention group had the following services: comprehensive pt education, therapeutic medication management dosage adjustments per protocol, facilitate and follow diabetic complication screening exams, routine telephone follow-up and clinic visits and regular communication to the primary care provider via a patient summary report.

KEY LESSONS LEARNED: Data was analyzed for the first quarter. The average glycohemoglobin of the intervention group at baseline was 13.0% and 13.3% in the control group. Pharm D intervention at 3 months attained a 2.2 percent reduction in glycosylated hemoglobin. 93% of the patients randomized to the intervention group agreed to participate.

THE BOSTON CENTER FOR REFUGEE HEALTH & HUMAN RIGHTS. A Moreno, SM Akram, K Collins, P Gelfman, M Grodin, TM Keane, L Piowarczyk, Section of General Internal Medicine; Department of Psychiatry; Department of Pediatrics; Boston Medical Center, Boston, MA; Boston VAMC; Boston University School of Medicine; Health Law Department, BU School of Public Health; BU School of Law

STATEMENT OF PROBLEM: More than 3.5 million refugees, asylum seekers, and survivors of torture have resettled in the United States since 1975. Despite their growing number and their complex medical, mental health, legal, and social needs, only 14 health programs across the nation offer specialized services to these individuals. No program provides all services needed by this population, though most of their needs are related to uprooting, torture, and related trauma. Many of these individuals go without services.

OBJECTIVES OF PROGRAM/INTERVENTION: The Boston Center for Refugee Health and Human Rights' primary objective is to facilitate the re-integration of these individuals into society by delivering comprehensive culturally sensitive medical, mental health, and dental care for these individuals, coordinated with legal and social services. The Center has 3 other objectives: (A) Education and training for professionals and agencies serving these individuals. (B) Advocacy and pre-

vention. (C) Clinical, epidemiological, and legal research for the better understanding and improvement of healthcare and legal services.

DESCRIPTION OF PROGRAM/INTERVENTION: The Center brings together numerous healthcare providers and attorneys from different departments of BMC and schools of BU in a unique coordinated and collaborative effort for this purpose.

FINDINGS TO DATE: Since its inception in December 1998, the Center has cared for patients from all over the world, who suffer from a variety of conditions, including but not limited to post-traumatic stress disorder, depression, anxiety, parasites, and tuberculosis. The Center has provided the following services during the past year: adult primary care, mental health care, dental care, social work, legal aid, medical and psychological evaluations and affidavits for asylum cases, and consultations. Patients have ranked their needs as follows: 1. Stable job to sustain family. 2. Help with legal process. 3. Medical and mental health care. 4. English as a second language. Observations made during patient care suggest that patients who received integrated care suffer less psychological re-traumatization compared to patients who received segmented care. On average, patients have struggled for more than 4 years to complete their transition into society.

KEY LESSONS LEARNED: A comprehensive approach to the complex needs of these individuals is essential to facilitate their reintegration into society. Preliminary observations suggest that our model of care appears to decrease the incidence of psychological re-traumatization in this vulnerable population; however, studies are needed to confirm these anecdotal observations.

DEVELOPMENT OF A DECISION RULE FOR THE IMMUNIZATION AND TESTING OF HEPATITIS A AMONG PATIENTS IN A PRIMARY CARE NETWORK. B Murphy, General Medicine, Saint Vincent's Hospital of NY, New York, NY

STATEMENT OF PROBLEM: Thirty percent of the American population possesses antibody to hepatitis A. However it benefits health systems to devise a decision rule to avoid unnecessary screening or immunizations.

OBJECTIVES OF PROGRAM/INTERVENTION: To develop a decision rule to predict which patients will need hepatitis A antibody testing, hepatitis A immunization, or no intervention.

DESCRIPTION OF PROGRAM/INTERVENTION: Five hundred patients representing the 5 boroughs of New York City were randomly selected from a primary care practice and tested for antibody to hepatitis A, B, and C. In addition to obtaining demographic and past medical history information, patients were also screened for thyroid disease, abnormalities of liver function, evidence of blood dyscrasias, exposure to syphilis, and HIV status. Predictors were screened by univariate and multivariate analysis to identify risk factors associated with being hepatitis A antibody positive.

FINDING TO DATE: Patients were screened for evidence of antibody to hepatitis A (anti-HAV) with 321 (64.2%) being positive ($p < 0.01$). The overall average age was 49yo; the average age among anti-HAV (+) patients being 54yo; anti-HAV (-) being 41yo. Of those screened, 59% were male. Significant ($p < 0.10$) predictors on univariate screening included: age >48 yo, being an emigre from outside North America (USA and Canada), race other than Caucasian, and having a prior exposure to hepatitis B. However, multiple regression analysis revealed statistically significant predictors ($p < 0.05$) to be age greater than 48yo and being from outside North America. If the patient was from North America and less than 48yo, 83% were anti-HAV negative. However, if the patient was from outside North America and less than 48yo, only 22% were anti-HAV negative. Among patients older than 48yo and from North America, 40% were anti-HAV negative. Patients older than 48yo but from outside North America were 97.9% anti-HAV positive ($p < 0.001$).

KEY LESSONS LEARNED: If the patient is younger than 48yo and from North America, they should be vaccinated and not tested for prior exposure to hepatitis A. If patients are older than 48yo and from North America, they should be tested before vaccination. Patients older than 48yo and not from North America should neither be screened nor vaccinated. Patients not from North America and younger than 48yo are likely to have been exposed to hepatitis A but should probably be tested before ruling out vaccination. Non-North American patients from Asia, the Caribbean, Central and South America were more likely to have anti-HAV antibody than Western Europeans who they outnumbered 5:1. In screening patients from Western Europe, this model may not be as reliable as for other more represented groups.

SUCCESSFUL IMPLEMENTATION OF A DIABETES GUIDELINE. MA Nyman, ME Murphy, PG Schryver, JM Noessers, SA Smith, Internal Medicine; Health Sciences Research, Mayo Clinic, Rochester, MN

STATEMENT OF PROBLEM: Compliance with recommendations from the American Diabetes Association for the management of patients with diabetes has been shown to be suboptimal. Distribution of practice guidelines has had little impact on the performance of these recommendations. Changing physician practice patterns with provider focused interventions can be difficult.

OBJECTIVES OF PROGRAM/INTERVENTION: Through the use of multiple, provider focused interventions we sought to improve the performance of diabetic services and reduce overall Hemoglobin A1c (HgbA1c) levels for patients with diabetes.

DESCRIPTION OF PROGRAM/INTERVENTION: Continuous Improvement techniques were used throughout the implementation period. Multiple interventions were utilized and were mainly provider focused. Interventions included: small

group educational sessions led by opinion leaders, an electronic version of the guideline made available on all clinic computer workstations, chart audit with quarterly group feedback to the providers, and enhanced clinical orders support. **FINDING TO DATE:** Gradual, sustained, and statistically significant improvements in the three performance measures were demonstrated. Improvements included increases in the performance of annual urine protein testing from 24% to 66% ($p = 0.001$), annual dilated eye examinations from 63% to 84% ($p = 0.001$), and annual foot examinations from 86% to 97% ($p = 0.001$). Mean HgbA1c values for the continuing care diabetes population also improved from 7.8 (Standard Deviation 1.0) to 7.1 (SD 0.7) ($p < 0.001$) after implementation. In a paired analysis of unique patients with pre- and post-implementation data the overall mean HgbA1c improved from 8.02 (SD 1.06) to 6.97 (SD 0.82).

KEY LESSONS LEARNED: Continuous improvement efforts utilizing multiple interventions focused primarily on the providers resulted in improved delivery of care for patients with diabetes. Performance of diabetic services and overall HgbA1c values were improved. Interventions used to successfully implement this diabetes guideline included: 1) small group educational programs led by opinion leaders; 2) access to the guideline on computer workstations; 3) audit with quarterly feedback; 4) enhanced clinical orders support

A RESIDENT ENCOUNTER DATA BASE USING A MODIFIED PATIENT BILLING FORM AS THE DATA COLLECTION TOOL. KJ Pituch, S Mitrovich, PC Burghardt, DM Oliansky, Division of General Pediatrics; Division of General Internal Medicine; Information Technology Division, University of Michigan, Ann Arbor, MI

STATEMENT OF PROBLEM: Running a resident continuity clinic is a challenge for the program leadership and clinic management. Whether the clinic is in the hospital or in a community practice, managers need to make decisions regarding scheduling, faculty supervision, and clinic support staffing. Program leadership needs to monitor resident patient panels, patient continuity and quality of faculty supervision. At our institution, good data on resident experience was limited. Attempts to track resident experience and faculty involvement via questionnaire or log books had serious limitations.

OBJECTIVES OF PROGRAM/INTERVENTION: To develop and implement a simple but complete data base for use by clinic management and residency program leadership.

DESCRIPTION OF PROGRAM/INTERVENTION: The patient billing/encounter form was modified to include information on patient continuity and supervision level by the precepting faculty. Residents and faculty must complete the form prior to patient check out. Information from 16 data fields are entered daily into a Microsoft Access Database program. Reports can be generated for any date interval required and reviewed with the house officer, supervising faculty, or other authorized user. **FINDING TO DATE:** At 3 Internal Medicine clinics, 2 Pediatric clinics and 1 combined Med-Peds clinic, over 7000 patient encounters have been entered in 6 months. Three standard reports are being generated. A 'Resident Activity Report' summarizes for each individual resident the number of encounters, ages of patients, distribution of visit types, list of diagnoses (with frequencies), list of procedures, continuity indicators, and faculty supervision levels. The 'Resident Comparative Report' summarizes key indicators and compares the resident with his peers. The 'Faculty Report' summarizes the numbers of patients seen by each of his or her residents, the faculty involvement level, and the total RVU's earned by the resident. This report also monitors compliance with HCFA supervision requirements by flagging encounters which call for a level of supervision greater than that documented.

KEY LESSONS LEARNED: 1. Prompt feedback increases resident willingness to ensure that all items are filled in. 2. With feedback, some faculty pay more attention to charge entry and supervision requirements. 3. Program directors appreciate the comprehensive listing of diagnoses and procedures. 4. Several faculty members are requesting access to the data base to make inquiries about patient panels, continuity, and faculty supervision. 5. Since the hospital accounting system only assesses cost based on faculty billing, this data base will allow a more accurate analysis of the cost of resident training.

PHYSICIAN PRESCRIBING OF SYRINGES TO PREVENT HIV. JD Rich, M Mehrotra, TL Whitlock, ME Gaydos, M McKenzie, V Runarsoff, J Gornley, M Aboagye-Kumi, G Macalino, C Salas, Immunology Center, The Miriam Hospital; Brown University, Providence, RI

STATEMENT OF PROBLEM: Limiting access to clean syringes has driven the HIV/AIDS epidemic by increasing the reuse and sharing of syringes among injection drug users (IDUs). Until 1998, in Rhode Island, possession of a single syringe was a felony offense punishable by up to 5 years imprisonment. The penalty was reduced to a misdemeanor, but syringes are still illegal to purchase without a prescription. With the support of Rhode Island's Medical Society, Pharmacists Association, Health Department and others, physician prescription of syringes to prevent HIV was fully endorsed by the Director of the RI Department of Health in 1999.

OBJECTIVES OF PROGRAM/INTERVENTION: This is a pilot study to provide IDUs access to sterile syringes through syringe prescription by physicians. The project's goals are to reduce behaviors that put people at risk for the transmission of HIV and other blood borne pathogens, to provide access to primary health care services for IDUs, and to increase access to other services including substance abuse treatment, social and mental health services.

DESCRIPTION OF PROGRAM/INTERVENTION: Each participant receives free medical care from a physician, including a physical exam, free HIV and hepatitis

B & C screening, and, if deemed necessary by the physician, a prescription for sterile syringes. Follow-up appointments and referrals for substance abuse treatment, mental health and social services are scheduled as needed.

FINDINGS TO DATE: To date, we have enrolled 87 participants, and more data will be presented at the meeting. Of the first 35 participants, 57% are male, 74% are white, 57% homeless, and the average age is 41. On average, participants inject 4.4 times a day. The drug of choice is primarily heroin (83%). 57% have shared a syringe in the previous 30 days and, on average, participants reuse each syringe 16 times. Although 69% of participants use the Rhode Island syringe exchange program, of these, 63% were using it on a less than weekly basis. 51% of participants do not have a regular doctor or medical insurance. 86% have been in drug treatment. 17% reported being HIV positive and of the HIV negative participants, 54% believe that they are at risk for contracting HIV.

KEY LESSONS LEARNED: This is the first study to explore syringe prescription as a strategy to reduce the transmission of HIV and other blood borne pathogens in IDUs. Thus far, physician-patient interactions have gone well, illustrating that this is a feasible practice. In addition to providing legal access to sterile syringes, syringe prescription can increase access to medical care and substance abuse treatment services. Other physicians are encouraged to consider the prescription of syringes to IDUs with inadequate access to sterile syringes.

IMPLEMENTING "OPEN ACCESS" TO PRIMARY CARE IN A HOSPITAL-BASED, ACADEMIC PRACTICE. JA Sackey, JJ Heffernan, RA Parker, S O'Neill, D Greenberg, Division of General Internal Medicine; Ambulatory Services, Beth Israel Deaconess Medical Center, Boston, MA

STATEMENT OF PROBLEM: Patients perceive the quality of care to be high in academic-based practices, but they also report poor access to visits with providers of their choice. Traditional scheduling systems often defer current patient needs far into the future. With fully booked schedules, physicians are often unavailable to provide service to their own patients when they most need care for acute issues. Poor access may also drive patients to suboptimal care sites, such as urgent care centers and emergency rooms, and engender extra visits. Attraction and retention of new patients is correlated with how well a practice provides access.

OBJECTIVES OF PROGRAM/INTERVENTION: An "open access" system is predicated on the principle that a patient is "able to seek and receive care from a provider of choice at the time the patient chooses". Operationally, this implies that a practice must match, quantitatively, the demand for service with the capacity of each provider. Many non-academic practices have successfully implemented such systems of "doing today's work today". We have embarked on an ambitious plan to implement open access in a large academic-based primary care practice.

DESCRIPTION OF PROGRAM/INTERVENTION: Because many of our primary care faculty do not practice daily, we have modified the basic open access model: (1) establishing mini-care teams comprised of a faculty member, 4-5 comanaging residents and 1-3 nurse practitioners; and (2) expanding the time response to support the concept of "doing this week's work this week". Each team is responsible for servicing all its patient's needs within 7 days, with an expectation that most will be met on the day requested.

FINDING TO DATE: We have captured data establishing functional panel size and phone demand for visits with each faculty member. Those faculty who are implementing the new model have worked down backlogs of scheduled patients, through a number of interventions, such that the proportion of "open" schedule slots in the first and second months has risen from 0 and 6% to 34% and 58%, respectively. Eight to ten faculty members and their associated scheduling teams anticipate full implementation of open access by 5/00.

KEY LESSONS LEARNED: Implementation of open access requires continued monitoring of demand for visits, simplified schedule templates with a shift away from categorizing visits as "urgent" versus "routine", and aggressive management of available visit slots. Most importantly, open access necessitates a commitment to servicing the needs of patients as they arise. Change comes hard to physicians anxious about nominally unfilled schedules. Those patients and physicians who have already experienced improved access report improved satisfaction with no adverse effect on productivity.

OBSTACLES TO SUCCESSFUL LIPID DISORDER MANAGEMENT IN PRIMARY CARE. G Schectman, E Doerr, Z Agha, General Internal Medicine, Medical College of Wisconsin, Milwaukee, WI

STATEMENT OF PROBLEM: Although previous studies document poor physician compliance with practice guidelines for cholesterol management, the reasons that providers fail to achieve lipid goals are not well understood.

OBJECTIVES OF PROGRAM/INTERVENTION: We evaluated the relative importance of three potential areas of inadequacy facing the primary care provider: i) failure to screen, ii) failure to initiate therapy, and iii) failure to titrate therapy to optimal dose (lipid goal not achieved and patient receiving less than half of maximal statin dose).

DESCRIPTION OF PROGRAMS/INTERVENTION: Medical records from selected patients receiving primary care from 24 general internists and 5 nurse practitioners at an urban Veterans Affairs Medical Center were audited. Approximately 150 patients with at least one previous visit with the provider were selected randomly from each provider's panel over a nine month period beginning 1/99. The lipid goals were derived from National Cholesterol Education Program guidelines and defined as follows: CHD present-LDL <115, HDL >30, TG <250 mg/dl; CHD not present-LDL <160, TG <400.

FINDING TO DATE: During the nine months, 4757 patient records were reviewed from 29 providers.

Among all records audited, 82% (provider range 64-97%) showed evidence of timely lipid screening (yearly for patients with cardiovascular disease, otherwise every three years). Among those screened, 52% had abnormal lipids requiring drug therapy. Among these patients with a lipid disorder requiring drug therapy, 46% (provider range 25-70%) had not achieved their lipid goal. Among these patients, 42% (provider range 17-73%) did not receive lipid lowering therapy, and 40% (provider range 14-66%) received suboptimal doses of therapy. Therefore, optimal therapy could potentially have improved lipid responses in 82% of patients. Only 18% of patients who did not achieve lipid goals were receiving optimal therapy.

KEY LESSONS LEARNED: We conclude that failure of providers to initiate and optimize lipid lowering therapy were important contributors to most treatment failures. Teaching programs and/or reminder systems designed to overcome these deficiencies are likely to significantly improve provider performance in hypercholesterolemia management.

AN INNOVATIVE APPROACH TO HANDLING PATIENT PHONE CALLS. RJ Seballos, M Breudigam, CC Williams, E Pomiecko, Regional Medical Practice, The Cleveland Clinic, Strongsville, OH; Regional Health Affairs, The Cleveland Clinic, Cleveland, OH

STATEMENT OF PROBLEM: Cleveland Clinic Strongsville (CCS) is a satellite office of the Cleveland Clinic Foundation (CCF) that provides primary and surgical care. Eight primary care physicians (PCP) practice at this facility. Three primary care RNs answer phone calls that deal with prescription refills, referral authorizations, labs/x-ray notifications, and triage. Because of an increasing volume of phone calls and complexity of triage, patient complaints regarding the office's phone process are also increasing.

OBJECTIVES OF PROGRAM/INTERVENTION: CCS instituted an automated phone menu system on November 1, 1999. Its purpose is to improve the efficiency in managing patient phone calls by: 1.) reducing the length of time a patient was put on hold (hold time), 2.) decreasing the number of patients who hang up while waiting (abandonment rate), and 3.) facilitating patient access to same-day (urgent slots) or next-day appointments.

DESCRIPTION OF PROGRAM/INTERVENTION: The automated phone system allows patients to choose from the following selections: prescription refills; referral authorization; lab/x-ray results; non-urgent messages; or remain on hold if experiencing an urgent medical problem. Remaining on hold automatically connected the patients to more experienced triage nurses at CCF's Nurse On Call (NOC), a major component of CCF's call center. NOC has access to each PCP's schedule and can add patients to their urgent slots (typically 4/day per PCP). The three primary care RNs continue to handle all other calls.

FINDING TO DATE: Total calls to the primary care RNs decreased by approximately 50% (see Table below). Consequently, the average hold time and abandonment rate decreased, and continue to improve into the second month of the project. NOC triages all other phone calls and has been successful in facilitating same- or next-day appointments.

Phone Data	October, 1999	November, 1999	December, 1999
Total calls to IM and Peds	1,428	712	745
Mean hold time (min.)	4:00	1:44	1:33
Total calls abandoned (%)	656 (45.9)	166 (23.3)	157 (21.2)
Total calls abandoned after 30 sec. (%)	375 (26.3)	105 (14.7)	70 (9.4)
Symptom-based calls to NOC	0	299	153
Calls facilitated to same day appointments	0	42	26
Calls facilitated to next day appointments	0	58	29

KEY LESSONS LEARNED: Implementing an automated phone system menu and using existing resources (NOC) improved the office's efficiency in managing patient calls dealing with prescription refills, referral authorizations, lab/x-ray notifications, and triage.

A RANDOMIZED CONTROLLED TRIAL OF THE USE OF A NOVEL TEMPLATE WITHIN AN ELECTRONIC MEDICAL RECORD SYSTEM TO IMPROVE THE COMPLIANCE WITH EVIDENCE-BASED HEALTH PROMOTION INTERVENTIONS IN PATIENTS WITH DIABETES. J Shiao, S Wallenius, HN Lee, Internal Medicine, University of Toronto, Toronto; Family Medicine and Internal Medicine, Group Health Centre, Sault Ste Marie; Internal Medicine, McMaster University, Hamilton, ON, Canada

STATEMENT OF PROBLEM: Evidence-based screening interventions in patients with diabetes can improve clinical outcomes but compliance is suboptimal. It remains to be seen how an electronic medical record (EMR) system might facilitate physician and patient adherence to these health promotion interventions (HPI).

OBJECTIVES OF PROGRAM/INTERVENTION: [1] To determine if a simple reminder template introduced to the EMR of patients with diabetes might improve HPI; [2] To evaluate if a patient information card used in conjunction with the EMR template further improves HPI.

DESCRIPTION OF PROGRAM/INTERVENTION: A Randomized Controlled Trial of 21 Primary Care Providers (PCPs) allocated into three groups: Group 1 was the Usual Care Group, Group 2 had a diabetic template inserted into the EMR of pa-

tients with diabetes and Group 3 had the EMR template and an updated patient education card to give out if desired. The primary care providers were educated on how to update this template themselves. Documented compliance with five evidence-based HPIs (BP, Hb1Ac and foot exams every 4 months; annual dilated funduscopy and albuminuria screening) were used as the outcomes. These were measured at baseline, six months and 1 year. The main outcome for analysis was a composite of the five HPIs.

Group	Baseline	6 mo	1 yr
Usual	.38	.39	.49
Template	.54	.52	.47
Temp/card	.57	.72	.53

FINDING TO DATE: In October 1998, 304 consecutive patients of participating PCPs were randomly assigned to each group. PCPs were stratified by location within the building. There were 7 PCPs and 92 patients in Group 1, 7 PCPs and 118 patients in Group 2, and 7 PCPs and 94 patients in Group 3. One year follow up was over 99%. At baseline, Group 1 patients had significantly lower HPI adherence than Group 2 or 3 (38% vs 54% and 57%, respectively, see Table). At 6 months, only Group 3 patients had significantly better composite HPI than baseline, but this improvement was lost at one year. In fact, at study conclusion the only significant difference occurred in Group 1, possibly reflecting a Hawthorne effect or regression to the mean.

KEY LESSONS LEARNED: Although the combination of an electronic reminder template and a patient education card might transiently increase adherence to HPI guidelines in patients with diabetes, this improvement is not sustainable at one year. Ongoing EMR template updating and/or feedback might be necessary to achieve this.

OPEN ACCESS IN AN ACADEMIC MANAGED CARE ORGANIZATION. *DS Smith, C Russo, MB Wenneker*, Department of Internal Medicine, Yale Health Plan, New Haven, CT; The Bard Group, Newton, MA

STATEMENT OF PROBLEM: Yale Health Plan (YHP) is an academic, closed panel, staff model managed care organization that delivers primary care to 19,000 adults. After 28 years of operation using processes that had once served it well, YHP was no longer meeting the needs of its patients for timely access to their clinician of choice by appointment or telephone.

OBJECTIVES OF PROGRAM/INTERVENTION: Our goals were: 1) to improve significantly access to appointments of all types by creating a simplified and flexible scheduling system, 2) to increase face-to-face time with the clinician by maximizing efficiency of patient flow, 3) to improve the ability of patients to see their own clinician, and 4) to create a system of care which promotes population-based patient management.

DESCRIPTION OF PROGRAM/INTERVENTION: Based on a planning process involving every member of the department working in multidisciplinary teams supported by senior administration and an external consulting firm, the organization of care delivery was redesigned to incorporate the following key changes: 1) An open access appointment system was developed with 50% of appointments held for same or next-day availability. 2) All patients were linked with a primary clinician, with panels benchmarked and balanced. 3) Multidisciplinary clinical teams were formed and clinical and support roles were re-defined to facilitate patient flow and cross-coverage. 4) Flexible clinical schedules were developed to expand appointment hours and optimize the use of constrained clinical space. 5) Clinical information capabilities were enhanced to track clinical performance and outcomes.

FINDING TO DATE: The proportion of patients with an identified primary clinician has increased from 65% to 100%. A pilot team was developed to test and implement the new design with full department implementation planned for February 2000. For the pilot team, clinical "face-time" as a proportion of each visit has increased, on average, by 20%. Third available appointment time has improved from 8.7 days to <2 days, and for physicals from 43.6 days to <3 days. Other key measures including visit and message cycle time and patient satisfaction are being monitored.

KEY LESSONS LEARNED: Patient demand for improved access creates opportunities to develop state-of-the-art care delivery systems. Cultural change cannot be underestimated as a challenge to systems change. Key elements of success include building commitment for change within the leadership team; engagement of all levels of staff in the process; creating a high level of tension for change; accounting for the teaching mission of the academic practice; and maintaining constant attention to timely, clear and repeated communication. Utilizing the entire team to "do today's work today" in addressing patient demands for timely access is possible.

COST EFFECTIVE MANAGEMENT OF ALLERGIC RHINITIS. *KM Taylor, E Selves, SM Doyle, MT McAvoy*, Mercy Primary Care, St. Joseph Mercy Hospital, Ann Arbor, MI

The cost in treatment of allergic rhinitis ranks third in overall pharmacy costs in our network over the past 3 years exceeding costs used to treat chronic conditions such as hypertension and congestive heart failure. The control chart on the use of non-sedating antihistamines over the past 18 months demonstrates the fluctuations in costs concordant with seasonal variation. The chart also demonstrates the most recent costs exceed the statistically significant upper limit indicating a

special cause variation. The network pharmacy improvement committee applied the FOCUS-PDCA improvement model to the cost-effective management of allergic rhinitis. The committee followed the FOCUS-PDCA process, which involved bi-weekly group meetings. The team utilized brainstorming to clarify the current knowledge of the problem. Special cause variation was understood using the pharmacy utilization data collection from our PHO, a flow-chart of the office prescription refill process and a chart review of 70 patients treated for allergic rhinitis in August 1999. Using the 2 σ rule the committee selected two variables to improve: 1) Standardize the phone refill policy in physician offices and 2) disseminate allergic rhinitis practice guidelines to all providers. The committee developed a force field analysis to obtain the desired state of physician compliance with practice guidelines and office implementation of standardized refill policies.

KEY LESSONS LEARNED: The program demonstrated the importance of group process techniques necessary to identify variation in the process of care in ambulatory settings. Using the FOCUS-PDCA process the committee is able to quickly identify key areas for improvement and to identify the forces which will drive and restrain the improvement process.

A DESCRIPTION OF AN INTERACTIVE HIV/AIDS TEACHING TOOL DESIGNED FOR LOW LITERACY PATIENTS. *MG van den Berg-Wolf, S Lassiter, SBC Hamerman, A Chammaro, EM Tedaldi*, General Internal Medicine, Temple University; The Village of Arts and Humanities, Philadelphia, PA

A low health literacy status of patients poses special problems in educating HIV-infected patients about HIV.

OBJECTIVES OF PROGRAM/INTERVENTION: The objectives of this program was to use a non-paper-based educational tool to educate patients about HIV and to assess their acceptance of this model. The clinical nurse specialist in the hospital-based primary care HIV clinic designed an interactive, hands-on tool that teaches adults about HIV. She collaborated with the Village of Arts and Humanities, a community-based agency that develops innovative health promotion programs in Philadelphia, Pennsylvania, to create this tool. The tool is a life-sized soft sculpture of the human body which is able to become infected with HIV by allowing virus to attach to it. The nurse can demonstrate viral replication and CD 4 cell decline using virus and CD 4 cell attachments to the model. The tool can also teach the concepts of viral mutation and drug resistance by using color-coded virus and medication attachments.

DESCRIPTION OF PROGRAM/INTERVENTION: Random patients attending the HIV Clinic at Temple University within the Section of General Internal Medicine were invited to attend individual and group educational sessions with this visual tool when their primary care physician identified them as potentially having low literacy based on having less than a high school diploma. The nurse specialist used the model in sessions with identified patients lasting from 30-60 minutes each. Patients were given a verbal pre-session and post-session test which contained basic questions about the role of CD 4 cells and viral replication in HIV disease. Information about the highest level of education reached was obtained.

FINDING TO DATE: Ten patients so far have used the tool in patient education sessions with the clinical nurse specialist. All patients tested to date have found the model easy to use and helpful in understanding HIV disease. Pre-session and post-session testing results with the model will be analyzed.

KEY LESSONS LEARNED: This visual teaching aid is useful in teaching low literacy patients about HIV. Future areas of research will be the application of the teaching tool to larger numbers of patients and correlating its use with patient care clinical endpoints.

SHOULD GENERALISTS BIOPSY? PRELIMINARY EVALUATION OF A FLEXIBLE SIGMOIDOSCOPY SCREENING PROGRAM. *BJ Wade, D Garcia*, Medicine, Greater Los Angeles VA Health Care System, Sepulveda, CA

STATEMENT OF PROBLEM: According to colorectal cancer screening guidelines, Flexible Sigmoidoscopy (FS) by generalists, is encouraged. However, the efficacy and cost-effectiveness of a community based screening program where generalists perform FS with biopsy is not well understood.

OBJECTIVES OF PROGRAM/INTERVENTION: To evaluate a FS screening training program where generalists biopsy lesions.

DESCRIPTION OF PROGRAM/INTERVENTION: The site is a suburban, academically affiliated ambulatory care VA facility. Primary Care (PC) medicine residents referred veterans for colorectal screening by FS. Veterans underwent FS after a generalist attending physician confirmed them to be of average risk for colorectal cancer by American College of Gastroenterology (ACG) guidelines, and high risk patients were directly referred to gastroenterology. FS was done on two, half day sessions per week by two medicine residents, one nurse practitioner and an attending physician. Most polyps, especially those 5mm or larger, were biopsied. Based on the ACG guidelines, patients with sizable adenomas or multiple polyps were referred for colonoscopy. Results for all colon and rectal biopsies, and appointment history for the FS clinic, was retrieved from the Decentralized Hospital Computer Program (DHCP) database. A five year descriptive analysis is presented.

FINDINGS TO DATE: For the years 1995-1999, 5679 FS appointments were made. 2348 (41%) patients had the procedure. There were 2.4 referrals per patient visit. 384 (16.4%) patients were biopsied. There were 8 patients with suspicion of or confirmed malignancies (6 with carcinomas, 1 with suspicion for malignancy and 1 carcinoma in-situ). All were detected by gastroenterology (GI) during a definitive procedure. 2 of these 8 were biopsied during their PC FS showing tubular adenoma and hyperplastic polyp. Of interest, only 13 total patients had histologi-

cal findings of malignancy in the database. 61.5% were found as a direct referral from the program. Approximate FS cost per patient without biopsy was \$142. Estimated additional community costs for biopsy alone was \$150 per patient. This charge was incurred at our biopsy rate of 16% of patients per year, therefore the average cost of FS with biopsy was \$166 per patient.

KEY LESSONS LEARNED: The program was successful in detecting 3.4 malignancies per 1000 patients, 61.5% of all colorectal malignancies found. Because none of the biopsies submitted by PC detected malignancies and referrals to GI were primarily made because of the FS exam itself, the value and cost effectiveness in biopsing an average risk patient in our screening program is in question. Evaluation by a GI specialist following FS without biopsy may be a more cost effective strategy. Further research is needed to minimize the cancellation and no-show rates for the screening clinic, thereby maximizing practice efficiency.

THE MULTIMEDIA COMPUTER FOR LOW LITERACY PATIENT EDUCATION: A PILOT PROJECT OF CANCER RISK PERCEPTIONS. JL Wofford, D Currin, MM Wofford, R Michielutte.

STATEMENT OF PROBLEM: Inadequate reading literacy is a major barrier to better educating patients. Despite its high prevalence, practical solutions for detecting and overcoming low literacy in a busy clinical setting remain elusive.

OBJECTIVES OF PROGRAM/INTERVENTION: In exploring the potential role for the multimedia computer in improving office-based patient education, we com-

pared the accuracy of information captured from audio-computer interviewing of patients with that obtained from subsequent verbal questioning.

DESCRIPTION OF PROGRAM/INTERVENTION: A multimedia computer presentation that used audio-computer interviewing with localized imagery and voices to elicit responses to four questions on prior computer use and cancer risk perceptions.

Based on the findings below, we have developed an array of computer-based tools for routine educational tasks in the office setting.

FINDING TO DATE: Three patients refused or were unable to interact with the computer at all, and three patients required restarting the presentation from the beginning but ultimately completed the computerized survey. Of the 51 evaluable patients (72.5% African-American, 66.7% female, mean age 47.5 (+18.1)), the mean time in the computer presentation was significantly longer with older age and with no prior computer use but did not differ by gender or race. Despite a high proportion of no prior computer use (60.8%), there was a high rate of agreement (88.7% overall) between audio-computer interviewing and subsequent verbal questioning.

KEY LESSONS LEARNED: Audio-computer interviewing is feasible in this urban community health center. The computer offers a partial solution for overcoming literacy barriers inherent in written patient education materials and provides an efficient means of data collection that can be used to better target patients educational needs.